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EDUCATIONAL TECHNOLOGIES AND DIGITAL LEARNING

THE ROLE OF TECHNOLOGY IN MODERN LANGUAGE EDUCATION

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Abstract: This article explores the transformative role of technology in modern language education, highlighting the significant advancements over the past thirty years. It begins with an overview of traditional language teaching methods and their limitations, followed by a discussion on the integration of the Internet, which has revolutionized access to language learning resources and facilitated virtual classrooms. The article delves into the impact of Artificial Intelligence (AI), particularly its recent widespread adoption since 2022, in creating personalized and adaptive learning experiences. It also examines interactive and immersive technologies such as Virtual Reality (VR) and Augmented Reality (AR), which provide engaging, real-world practice scenarios. Additionally, the benefits and challenges of technology in language education are analyzed, supported by case studies and testimonials. The article concludes by exploring emerging trends and the need for a balanced approach between traditional methods and technological advancements to create effective and inclusive language learning environments.

Keywords: Technology, AI, VR, Language Learning, Communicative approach

MÜASİR DİL TƏHSİLİNDƏ TEXNOLOGİYANIN ROLU

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Xülasə: Bu məqalə müasir dil təhsilində texnologiyanın transformativ rolunu araşdırır və son otuz ildəki əhəmiyyətli irəliləyişləri vurğulayır. Məqalə ənənəvi dil tədris metodlarına və onların məhdudiyyətlərinə ümumi baxışla başlayır, ardınca dil öyrənmə resurslarına çıxışı inqilabi şəkildə dəyişdirən və virtual sinif otaqlarını asanlaşdıran İnternetin integrasiyası haqqında müzakirə aparılır. Məqalədə Süni İntellektin (SI), xüsusən də 2022-ci ildən bəri geniş yayılmasının fərdiləşdirilmiş və adaptiv öyrənmə təcrübələrinin yaradılmasındakı təsirinə nəzər salınır. Həmçinin, cəlbəedici, real təcrübə ssenariləri təmin edən Virtual Reallıq (VR) və Genişləndirilmiş Reallıq (AR) kimi interaktiv və immersiv texnologiyalar araşdırılır. Bundan əlavə, dil təhsilində texnologiyanın faydaları və çətinlikləri təhlil edilir, nümunə tədqiqatları və rəylərlə dəstəklənir. Məqalə, effektiv və inklüziv dil öyrənmə mühitləri yaratmaq üçün ortaya çıxan tendensiyaları və ənənəvi metodlar və texnoloji irəliləyişlər arasında balansli bir yanasmaya ehtiyacı araşdıraraq yekunlaşdırır.

Açar sözlər: Texnologiya, Süni intellekt (SI), Virtual reallıq (VR), Dil öyrənmə, Kommunikativ yanaşma

Introduction

The process of language education has evolved significantly over the centuries, from the use of traditional rote memorization and grammar-translation methods to the communicative approaches that prioritize interaction and practical usage. Historically, language teaching relied heavily on physical resources such as textbooks and in-person instruction, which often limited accessibility and adaptability. However, in the past thirty years, the integration of technology into language education has profoundly transformed the landscape, making language learning more dynamic, interactive, and accessible. Modern technology has become an indispensable tool in the realm of language education. The advent of the Internet provided unprecedented access to a wealth of resources, including online courses, language learning platforms, virtual classrooms, and interactive forums. This technological revolution democratized education, enabling learners from all corners of the world to access high-quality language instruction and resources at their convenience. In recent years, particularly since the early 1990s, technology has increasingly become a cornerstone in language education. Innovations such as language learning apps, online dictionaries, and digital flashcards have made learning more engaging and tailored to individual needs. The digital age has not only enhanced the way languages are taught but also how they are practiced, offering learners endless opportunities to immerse themselves in their target language through media, social networks, and real-time communication tools.

An important development in the technological landscape of language education is the rise of Artificial Intelligence (AI). While AI concepts have been around for decades, it is only in the past two years, particularly post-2022, that AI has become widely accessible and integrated into everyday learning tools. The introduction of AI-powered platforms, such as ChatGPT, has revolutionized language learning by providing personalized, instant feedback and creating interactive learning environments that adapt to the learner's pace and style. AI's role in language education is multifaceted. It enhances traditional methods by offering intelligent tutoring systems, automating administrative tasks, and providing advanced analytics to track and improve learner performance. AI-driven tools can simulate real-life conversations, help with pronunciation, and offer context-specific vocabulary practice, making language learning more practical and effective. The integration of technology, especially AI and the Internet, has redefined language education, making it more efficient, engaging, and accessible. As we continue to embrace these technological advancements, their role in language learning will only become more prominent and irreplaceable, paving the way for more innovative and effective educational practices.

Methodology and Materials

The methodology applied in this study is based on a communicative and technology-enhanced approach to language learning. It emphasizes interactive activities, real-life communication, and learner-centered instruction supported by digital tools. The research uses qualitative analysis to examine the impact of modern technologies, especially AI, on language education. Materials include online platforms, language learning applications, virtual classrooms, and AI-powered tools such as ChatGPT. Additional resources consist of digital dictionaries, multimedia content, and social media environments that promote authentic language use. Data is gathered through observation of learner engagement and analysis of technological features that support language acquisition. This combination of methods and materials ensures a comprehensive understanding of how technology enhances language learning outcomes.

Discussion

For centuries, language education relied heavily on traditional methods such as the grammar-translation approach, rote memorization, and repetitive drills. These techniques, rooted in classical education, emphasized learning grammar rules and vocabulary through direct translation between the learner's native language and the target language. While useful for understanding the structural aspects of a language, these

methods often fell short in promoting communicative competence. Students could read and write in the target language but struggled with speaking and listening skills in real-life situations.

Another prevalent traditional method was the direct method, which emerged as a response to the limitations of the grammar-translation approach. This method prioritized immersion, with instructors using only the target language in the classroom and encouraging students to think and speak in that language. Although it improved conversational skills, the direct method required highly skilled teachers and was often less effective in large, diverse classrooms.

These conventional methods were often rigid and did not cater to individual learning styles or the diverse needs of students. The lack of interaction and real-world application limited students' ability to use the language fluently outside the classroom setting.

The Shift Towards Incorporating Technology in Education

The landscape of language education began to transform with the advent of technology. The introduction of audio-lingual methods in the mid-20th century, which utilized audio tapes and language labs, marked one of the earliest integrations of technology into language teaching. These innovations allowed for repetitive listening and speaking exercises, providing a more dynamic learning experience compared to traditional methods. The rise of the Internet in the late 20th century revolutionized language education. The Internet opened up a vast array of resources, enabling learners to access authentic language materials such as news articles, videos, and podcasts from native speakers. Online dictionaries and language forums further supported self-directed learning, allowing students to seek instant clarification and engage in discussions with peers worldwide. As technology advanced, so did its application in language learning.

The 21st century saw the emergence of online language learning platforms like Duolingo and Rosetta Stone, which combined multimedia elements to create interactive and engaging lessons. These platforms offered personalized learning experiences, adapting to the learner's pace and proficiency level.

The integration of technology in language education has continued to evolve, with mobile apps, virtual classrooms, and online courses becoming increasingly popular. These tools provide learners with flexibility, enabling them to practice and improve their language skills anytime and anywhere. Moreover, technology has made language learning more accessible, breaking down geographical barriers and allowing individuals from diverse backgrounds to learn languages that were previously out of reach.

The Internet as a Language Learning Tool

The Internet as a Language Learning Tool The advent of the Internet has revolutionized access to language learning resources. Learners can now find a plethora of online courses tailored to various proficiency levels and interests. Websites like Coursera, edX, and Khan Academy offer structured language courses designed by prestigious universities and

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Platforms like Zoom, Microsoft Teams, and Google Meet facilitate real-time interaction between students and teachers, replicating the traditional classroom environment online. These tools support a variety of activities, including group discussions, presentations, and collaborative projects, enhancing the learning experience.

Language learning platforms such as Duolingo, Babbel, and Rosetta Stone offer interactive lessons that incorporate multimedia elements like audio, video, and gamified exercises. These platforms use adaptive learning algorithms to personalize lessons based on the learner's progress and preferences, ensuring a tailored and engaging learning experience.

The Impact of Social Media and Online Communities in Language Practice

Social media and online communities play a crucial role in language practice and acquisition. Platforms like Facebook, Twitter, and Instagram provide opportunities for learners to engage with native speakers and other learners through language exchange groups and pages. Participating in conversations, sharing content, and consuming media in the target language helps learners practice real-life communication skills. Moreover, dedicated language exchange websites and apps such as Tandem, HelloTalk, and Speaky connect learners with native speakers around the world. These platforms facilitate language practice through text, voice, and video chats, allowing learners to immerse themselves in the language and culture of their counterparts.

Online communities like Reddit and specialized forums offer additional support by allowing learners to ask questions, share resources, and discuss challenges with peers. These communities foster a collaborative learning environment where learners can receive instant feedback and encouragement, further motivating them to continue their language studies.

Artificial Intelligence in Language Education

Artificial Intelligence has significantly enhanced the capabilities of language learning apps, making them more intuitive and effective. Platforms like Duolingo, Babbel, and Rosetta Stone incorporate AI to provide users with interactive, engaging, and adaptive learning experiences. These apps use machine learning algorithms to analyze user progress, adapt difficulty levels, and offer personalized recommendations, ensuring that each learner receives a customized educational experience. AI enables these platforms to simulate human-like interactions, providing learners with real-time practice in listening, speaking, reading, and writing. One of the most transformative aspects of AI in language education is its ability to create personalized learning experiences. AI algorithms track a learner's performance, identifying strengths and weaknesses, and tailoring content accordingly. This personalization goes beyond simple adaptation; it involves predicting future learning paths and suggesting targeted exercises to address specific needs. For instance, if a learner struggles with verb conjugations, the AI can prioritize activities that focus on that

particular area, providing additional practice and resources until mastery is achieved. This approach ensures that learning is efficient, focused, and aligned with individual goals.

AI in Language Assessment and Feedback

AI has also revolutionized language assessment and feedback mechanisms. Traditional language assessment methods often rely on standardized tests and human evaluators, which can be time-consuming and subjective. AI-driven assessment tools, however, offer real-time, objective evaluations of language proficiency. These tools can analyze speech for pronunciation accuracy, fluency, and intonation, providing instant feedback to learners. Additionally, AI can assess written work for grammar, syntax, and coherence, offering detailed suggestions for improvement.

For example, platforms like Grammarly and Write & Improve use AI to evaluate written texts, highlighting errors and suggesting corrections. These tools help learners understand their mistakes and learn from them, promoting continuous improvement. Moreover, AI-powered chatbots and virtual tutors can engage learners in conversational practice, providing feedback on their language use and guiding them towards better communication skills. AI's role in language education extends to monitoring progress over

time, enabling educators and learners to track development and adjust learning strategies as needed. This continuous assessment and feedback loop fosters a more dynamic and responsive learning environment, ultimately leading to better language acquisition outcomes.

Future Trends in Technology and Language Education

The Potential of AI to Revolutionize Language Education Further

Artificial Intelligence (AI) has already made significant strides in language education, but its potential to revolutionize the field is far from exhausted. In the near future, we can expect AI to further enhance personalized learning experiences through more sophisticated adaptive learning algorithms. These algorithms will not only track learner progress but also predict future learning needs, offering even more tailored content and resources. AI is poised to facilitate more nuanced and natural language interactions through advancements in natural language processing (NLP). Improved NLP capabilities will allow AI-driven tools to understand context, subtleties, and idiomatic expressions better, providing more accurate and human-like responses. This will make conversational AI tutors and chatbots even more effective in helping learners practice real-life communication skills.

Furthermore, AI will likely play a critical role in creating immersive, interactive learning environments. By integrating AI with Virtual Reality (VR) and Augmented Reality (AR), learners can engage in highly realistic simulations that mimic real-world language use scenarios. This will provide learners with experiential learning opportunities that are both engaging and practical. Emerging Technologies and Their Potential Impact on Language Learning Emerging technologies such as Extended Reality (XR), the Internet of Things (IoT), and blockchain are set to impact language education significantly.

Extended Reality (XR)

Extended Reality, which encompasses VR, AR, and Mixed Reality (MR), will continue to transform language learning by providing immersive experiences. XR can simulate real-life environments where learners can interact with virtual objects and characters, practicing language in contextually rich scenarios. For example, learners could virtually travel to foreign countries, engage in cultural exchanges, and practice language skills in diverse settings without leaving their homes.

The Internet of Things (IoT)

The Internet of Things (IoT) can enhance language learning by connecting everyday objects to the internet, creating smart learning environments. Smart classrooms equipped with IoT devices can provide personalized feedback, track student engagement, and adapt to individual learning needs. For instance, IoT-enabled language labs could automatically adjust learning materials based on student performance, ensuring a tailored educational experience.

Conclusion

This article has explored the profound impact of technology on modern language education, tracing its evolution from traditional methods to the current integration of advanced technological tools. Initially, language education relied heavily on grammar-translation and direct methods, which had notable limitations in fostering communicative competence. The advent of the Internet revolutionized access to resources, virtual classrooms, and interactive platforms, significantly enhancing the language learning experience. Additionally, the rise of Artificial Intelligence (AI) has introduced personalized learning experiences, real-time feedback, and intelligent tutoring systems, further transforming the landscape of language education. We have also examined the role of interactive and immersive technologies like Virtual Reality (VR) and Augmented Reality (AR), which provide engaging, real-world practice scenarios. Finally, the article highlighted the benefits and challenges of technology in language education, case studies of successful implementations, and emerging trends that promise to shape the future of language learning.

As technology continues to advance, its role in language education becomes increasingly indispensable. The benefits of accessibility, flexibility, and enhanced engagement are undeniable, making high-quality language education available to a broader audience. AI, in particular, has revolutionized how learners interact with content, receive feedback, and progress through their learning journeys. The incorporation of interactive and immersive technologies has made learning more practical and enjoyable, bridging the gap between theoretical knowledge and real-world application. While the advancements in technology offer tremendous benefits, it is crucial to maintain a balance between traditional methods and technological innovations. Traditional approaches provide a strong foundation in grammar, vocabulary, and cultural understanding, which are essential for language proficiency. However, integrating these methods with modern technology can enhance their effectiveness, providing a more holistic and engaging learning experience. Educators must carefully select and combine traditional and technological tools to create comprehensive language programs that cater to diverse learning needs and preferences.

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DIGITAL TECHNOLOGIES IN GEOGRAPHY EDUCATION: ENHANCING TEACHING AND LEARNING PROCESSES

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Abstract: In recent years, the process of digitalization in education has accelerated, significantly influencing the content, methods, and organization of the teaching and learning process. Geography, as a discipline focused on spatial thinking, natural processes, and global interactions, provides extensive opportunities for the integration of digital technologies. This study explores the application of digital technologies in geography education, their impact on teaching quality, and their role in developing students' knowledge, skills, and competencies. The research is based on the analysis of national and international scientific literature and employs comparative analysis, observation, analytical methods, and a systematic approach.

The findings reveal that interactive maps, Geographic Information Systems (GIS), satellite imagery, digital platforms, and artificial intelligence-based tools significantly enhance students' engagement and improve their understanding of geographical concepts. These technologies also contribute to the development of critical thinking, analytical skills, and problem-solving abilities. Moreover, they support personalized learning environments by adapting to different learning styles and paces. However, the effectiveness of these technologies depends largely on teachers' digital competencies and the availability of adequate technological infrastructure. Therefore, continuous professional development for teachers and investment in educational technologies are essential. In conclusion, the purposeful and systematic integration of digital technologies enhances the quality of geography education and fosters an innovative learning environment aligned with the demands of modern education systems.

Keywords: digital technologies, geography education, GIS, interactive maps, online learning, artificial intelligence

COĞRAFIYA TƏHSİLİNDƏ RƏQƏMSAL TEXNOLOGİYALAR: TƏDRİS VƏ ÖYRƏNMƏ PROSESLƏRİNİN TƏKMİLLƏŞDİRİLMƏSİ

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Xülasə: Müasir dövrdə təhsil sistemində rəqəmsallaşma prosesi sürətlənmiş və bu tendensiya tədris prosesinin məzmununa, metodlarına və təşkilinə ciddi təsir göstərmişdir. Xüsusilə coğrafiya fənni məkan təfəkkürü, təbii-coğrafi proseslər və global qarşılıqlı əlaqələrin öyrənilməsinə əhatə etdiyi üçün rəqəmsal texnologiyaların tətbiqi baxımından geniş imkanlara malikdir. Bu məqalədə coğrafiya tədrisində rəqəmsal texnologiyaların tətbiqi, onların təlimin keyfiyyətinə təsiri və şagirdlərin bilik, bacarıq və kompetensiyalarının formalaşmasındakı rolu tədqiq edilmişdir. Tədqiqat zamanı yerli və xarici elmi ədəbiyyatlar təhlil edilmiş, müqayisəli təhlil, müşahidə, analiz və sistemli yanaşma metodlarından istifadə olunmuşdur.

Araşdırma nəticələri göstərir ki, interaktiv xəritələr, Coğrafi İnformasiya Sistemləri (GIS), peyk görüntüləri, rəqəmsal platformalar və süni intellekt əsaslı tətbiqlər şagirdlərin dərəcə marağını əhəmiyyətli dərəcədə artırır və onların mövzuları daha dərinlən mənimsəməsinə şərait yaradır. Bu texnologiyalar şagirdlərdə tənqidi düşünmə, analitik yanaşma və problem həll etmə bacarıqlarını inkişaf etdirir. Eyni zamanda fərdiləşdirilmiş təlim imkanları yaradaraq müxtəlif səviyyəli şagirdlərin öyrənmə ehtiyaclarına uyğunlaşmanı təmin edir. Bununla yanaşı, müəllimlərin rəqəmsal kompetensiyalarının inkişaf etdirilməsi və texniki infrastrukturun gücləndirilməsi bu prosesin effektivliyi üçün vacib şərtlərdən biridir. Nəticə etibarilə, rəqəmsal texnologiyaların məqsədyönlü və sistemli tətbiqi coğrafiya tədrisinin keyfiyyətini yüksəldir və müasir təhsil sisteminin tələblərinə cavab verən innovativ təlim mühiti formalaşdırır.

Açar sözlər: rəqəmsal texnologiyalar, coğrafiya tədrisi, GIS, interaktiv xəritələr, onlayn təlim, süni intellekt

Introduction

In the context of rapid technological development, education systems around the world are undergoing profound and continuous transformations. The traditional models of teaching and learning, which were mainly based on teacher-centered instruction and printed materials, are gradually being replaced by more interactive, flexible, and technology-supported approaches. Digital technologies have become essential tools in enhancing the overall quality, accessibility, and effectiveness of education. They not only support the delivery of content but also promote student engagement, collaboration, and independent learning.

In this transformation process, geography education occupies a particularly important place. Due to its inherently spatial, visual, and data-driven nature, geography is one of the subjects that benefits most from digital innovation. Modern digital tools such as Geographic Information Systems (GIS), satellite imagery, interactive maps, and virtual simulations enable students to explore real-world environments in a dynamic and engaging way. These technologies help learners move beyond abstract descriptions and instead interact directly with visualized geographical data.

Furthermore, the use of digital tools allows students to better understand complex geographical phenomena such as climate change, population distribution, urbanization, natural disasters, and global economic interactions. By visualizing these processes in real time or through simulation models, students can develop deeper analytical and critical thinking skills. At the same time, digital learning environments encourage active participation, problem-solving, and inquiry-based learning, which are essential competencies in modern education.

Overall, the integration of digital technologies into geography education not only enriches the learning experience but also prepares students for a rapidly changing, technology-driven world where spatial awareness and digital literacy are increasingly important.

Materials and Methods

The present study is based on a comprehensive review of scientific literature, international reports, and contemporary pedagogical approaches related to digital transformation in education. Special attention has been given to recent studies in geography education, particularly those focusing on the integration of digital tools such as Geographic Information Systems (GIS), interactive mapping applications, and online learning platforms. These sources provided a theoretical and empirical foundation for analyzing the role of digital technologies in improving teaching and learning processes.

In order to achieve the research objectives, a combination of qualitative research methods was employed. Comparative analysis was used to examine and contrast different teaching models and approaches to digital integration in geography education across various educational contexts. This method allowed for the identification of strengths and limitations in traditional and technology-enhanced instruction.

Observation was applied to evaluate classroom practices, focusing on how teachers implement digital tools during lessons and how students respond to technology-based learning activities. This provided practical insights into the real-time effectiveness of digital technologies in educational settings.

Additionally, analysis and synthesis methods were utilized to interpret the collected data, identify patterns, and generalize the findings. Through analysis, individual components of digital teaching practices were examined in detail, while synthesis allowed for the integration of these findings into a coherent understanding of their overall impact.

As a result, the combination of these methods ensured a systematic and comprehensive investigation of the role of digital technologies in geography education, contributing to more reliable and well-grounded conclusions.

Results and Discussion

The results of the study indicate that the integration of digital technologies into geography teaching has a significantly positive impact on students' motivation, engagement, and overall learning outcomes. When digital tools are systematically incorporated into the learning process, students demonstrate a higher level of interest in the subject and a greater willingness to participate in classroom activities. This is largely due to the interactive and visual nature of digital resources, which make complex geographical concepts easier to understand.

In particular, tools such as Geographic Information Systems (GIS), interactive maps, and satellite imagery enable students to work with real-world spatial data. This hands-on approach supports the development of analytical and critical thinking skills, as learners are encouraged to interpret data, identify patterns, and draw conclusions based on evidence. As a result, geography becomes a more practical and applied discipline rather than a purely theoretical subject.

Moreover, online and hybrid learning models have proven to be highly effective in increasing accessibility and flexibility in education. These models allow students to learn at their own pace, access materials from different locations, and revisit content when necessary. This contributes to a more inclusive educational environment, particularly for students with different learning needs and backgrounds.

In addition, artificial intelligence-based educational platforms further enhance the learning process by providing personalized learning experiences. These systems adapt content according to students' performance levels, learning speed, and individual needs, thereby improving learning efficiency.

However, the study also reveals that the effectiveness of these digital technologies largely depends on teachers' digital competencies and the availability of adequate technical infrastructure. Without proper training and resources, the potential benefits of digital tools cannot be fully realized. Therefore, continuous professional development for teachers and investment in educational infrastructure are essential for the successful implementation of digital technologies in geography education.

Conclusion

In conclusion, the use of digital technologies in geography education plays a crucial and transformative role in improving the overall quality of teaching and learning processes. The integration of modern digital tools such as Geographic Information Systems (GIS), interactive maps, online platforms, and artificial intelligence-based applications significantly enhances the effectiveness of instruction by making abstract geographical concepts more concrete, visual, and accessible to students.

These technologies not only facilitate a deeper understanding of geographical phenomena but also contribute to the development of essential 21st-century skills such as critical thinking, problem-solving, data analysis, and digital literacy. By engaging with real-world data and interactive learning environments, students become more active participants in the educational process rather than passive recipients of information.

Furthermore, digital technologies help to bridge gaps in access to education by providing flexible and inclusive learning opportunities. This is particularly important in geography education, where spatial awareness and global perspectives are essential for understanding complex environmental and socio-economic processes.

At the same time, the successful implementation of digital technologies requires adequately trained teachers and well-developed technological infrastructure. Without these supporting factors, the potential benefits of digital education cannot be fully realized.

Overall, it can be concluded that the effective and purposeful integration of digital technologies should be considered a strategic priority in modern education systems. It not only improves teaching quality but also prepares learners to adapt to the rapidly changing demands of a technology-driven global society.

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A NEW APPROACH IN DIGITAL LINGUISTICS AND LANGUAGE LEARNING TECHNOLOGIES

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Abstract: This article examines the development of linguistics and lexicography in the context of digital transformation and interdisciplinary integration with technology and mathematics. It highlights how computational linguistics, mathematical modeling, and electronic lexicography have reshaped traditional approaches to language analysis.

The study discusses the pioneering contributions of scholars such as Academician Agamusa Akhundov, [Akhundov, 1979] who introduced mathematical methods into Azerbaijani linguistics, as well as Pedro A. Fuertes-Olivera and P.M.Alekseyev, who advanced electronic and statistical lexicography. The evolution from printed to electronic dictionaries is analyzed in terms of functionality, interactivity, and user-oriented design. Particular attention is given to the role of electronic dictionaries as dynamic, adaptive, and multimodal platforms that integrate text, audio, and visual data, significantly improving access to lexical information.

Furthermore, the article explores the impact of digital technologies on foreign language learning. It emphasizes that the digital environment enhances all four language skills listening, reading, writing, and speaking through tools such as podcasts, audiobooks, video platforms, online forums, and AI-based applications. These resources provide learners with authentic language input and support individualized learning strategies. The integration of multimedia and interactive technologies fosters more effective and contextualized language acquisition compared to traditional methods.

The findings suggest that both lexicography and foreign language education are undergoing a paradigm shift driven by digital innovation. As a result, electronic dictionaries and digital learning tools have become essential components of modern linguistic practice, contributing to more efficient, accessible, and user-centered language learning and research environments.

Key words: digital linguistics, research, innovative approach, e-dictionary, learning, e-resources

DİJİTAL DİLBİLİMDE VE DİL ÖĞRENME TEKNOLOJİLERİNDE YENİ BİR YAKLAŞIM

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Özet: Bu makale, dilbilimin ve leksikografinin gelişimini dijital dönüşüm ve teknoloji ile matematiğin disiplinlerarası entegrasyonu bağlamında incelemektedir. Çalışmada, bilgisayarlı dilbilim, matematiksel modelleme ve elektronik leksikografinin dil analizine yönelik geleneksel yaklaşımları nasıl yeniden şekillendirdiği vurgulanmaktadır.

Makale, Akademisyen Ağamusa Ahundov'un [Ahundov, 1979] Azerbaycan dilbilimine matematiksel yöntemleri kazandırmadaki öncü katkılarını, ayrıca Pedro A. Fuertes-Olivera ve P. M. Alekseyev'in elektronik ve istatistiksel leksikografinin gelişimine yaptıkları katkıları ele almaktadır. Basılı sözlüklerden elektronik sözlüklere geçiş; işlevsellik, etkileşim ve kullanıcı odaklı tasarım açısından analiz edilmektedir. Elektronik sözlüklerin metin, ses ve görsel verileri birleştiren dinamik, uyarlanabilir ve çok modlu platformlar olarak rolüne özel önem verilmektedir; bu özelliklerin sözlüksel bilgiye erişimi önemli ölçüde geliştirdiği belirtilmektedir.

Ayrıca makale, dijital teknolojilerin yabancı dil öğrenimi üzerindeki etkisini incelemektedir. Dijital ortamın; podcastler, sesli kitaplar, video platformları, çevrim içi forumlar ve yapay zekâ tabanlı uygulamalar aracılığıyla dinleme, okuma, yazma ve konuşma olmak üzere dört temel dil becerisini geliştirdiği vurgulanmaktadır. Bu kaynaklar, öğrenenlere otantik dil girdisi sunmakta ve bireyselleştirilmiş öğrenme stratejilerini desteklemektedir. Multimedya ve etkileşimli teknolojilerin entegrasyonu, geleneksel yöntemlere kıyasla daha etkili ve bağlamsallaştırılmış bir dil edinimi sağlamaktadır.

Elde edilen bulgular, hem leksikografinin hem de yabancı dil eğitiminin dijital yenilikler doğrultusunda bir paradigma değişimi yaşadığını göstermektedir. Bu bağlamda, elektronik sözlükler ve dijital öğrenme araçları, modern dilbilim uygulamalarının vazgeçilmez bileşenleri hâline gelerek daha verimli, erişilebilir ve kullanıcı odaklı öğrenme ve araştırma ortamlarının oluşmasına katkı sağlamaktadır.

Anahtar kelimeler: dijital dilbilim, araştırma, yenilikçi yaklaşım, e-sözlük, öğrenme, e-kaynaklar

Introduction

The development of linguistics, driven by its interdisciplinary interaction with other fields of science, has led to the emergence of new directions. These interactions, particularly through integration with technology and mathematics, have resulted in significant advances across various areas of linguistic research. The relationship between linguistics and technology has become particularly evident in the fields of computational linguistics and computer linguistics. In these areas, issues such as automatic language processing, machine translation, and the identification of textual structures have come to the forefront. Research conducted in this direction has made a significant contribution to the enrichment and development of linguistics through modern technologies.

Academician Agamusa Akhundov is one of the scholars who first systematically implemented the application of mathematical methods in Azerbaijani linguistics. In his work "*Mathematical*

Linguistics," published in 1979, he proposed the analysis of language structure using mathematical approaches. This framework made it possible to explain the syntactic and phonetic features of language through mathematical models. Akhundov's research in this field significantly contributed to the enrichment and development of linguistics through mathematical and statistical methods.

The interaction of linguistics with other sciences has led to the emergence of new directions and the development of existing fields. The application of mathematical methods by Academician Agamusa Akhundov, along with the research of Pedro A. Fuertes-Olivera and P.M. Alekseyev in the fields of electronic lexicography and statistical lexicography, has had a significant impact on the adaptation and advancement of linguistics in line with the demands of the modern era. One of the important contributions to electronic lexicography has been made by Pedro A. Fuertes-Olivera. In his work *The Routledge Handbook of Lexicography*, he aligns the theoretical and practical aspects of lexicography with contemporary requirements. Fuertes-Olivera emphasizes the interaction between information technology and linguistics in the development of electronic dictionaries, providing an important theoretical foundation for the advancement of this field.

P.M. Alekseyev is among the first scholars to propose the application of statistical methods in lexicography. He noted that the use of computational techniques and mathematical statistics in lexicography has led to the emergence of a new field statistical lexicography. This approach has made it possible to use and analyze statistical data in dictionary compilation.

This dynamic has also had an impact on the field of lexicography. The development of computer, communication, and multimedia technologies has led to the creation of new generations of lexicographic resources. The expansion of the global Internet network has also been reflected in lexicography. While in the past considerable time was required to find the explanation of a word in a dictionary, today this process can be carried out easily and instantly with the help of fast electronic dictionaries available on the Internet, instead of the time previously spent in libraries. At this stage, modern computers can also provide significant assistance to lexicographers [Mammdova, 2014].

However, there is an important distinction to be made here: not every dictionary placed on the Internet qualifies as an electronic dictionary. For instance, as noted earlier, we encounter dictionaries online that are essentially traditional dictionaries simply transferred to the Internet. Their use is identical to that of printed dictionaries, with the only difference being that they are displayed on a screen. There is no interactive relationship between the user and the dictionary, and retrieving the meaning of a word still requires a considerable amount of time. Such dictionaries are designed only for reading; they do not allow users to obtain information through query-based interaction.

In electronic dictionaries, the software is designed in such a way that it operates based on user-generated queries. Words are automatically arranged in alphabetical order, a marking process is carried out in response to queries, word definitions are presented automatically, and various operators are developed for other functions. However, not all electronic dictionaries perform the same functions; each of them has its own specific set of features. In the dissertation, several online and offline electronic dictionaries currently available on the Internet have been included in the analysis.

P. Newmark noted: *"If a translator does not have computer literacy, then it is not advisable for them to engage in this nonsense (machine translation, computer translation programs, automatic translation), because it is a waste of time for them; this is new technology, this is*

development, this is the future. Work on electronic dictionaries should be further improved so that development directly delivers results to the user, meaning that rapid information translation and publication translation should coexist" (Newmark, 1989)0.

From this, it can be concluded that when using computer-based translation, the translator deals with two types of output texts:

- Rapid information translation (which may be a draft text and differ from the product of human cognition);
- Publication translation (which represents the product of the human mind).

Indeed, electronic dictionaries are lexicographic resources developed with high technological means. Currently, in various private company laboratories around the world, electronic dictionaries are being created for different fields of science (language, chemistry, physics, mathematics, and others). These are integrated into machine translation systems and provide high-quality electronic data sets, as well as grammatical notifications [Zhang, 2015].

Famous American lexicographer Barbara Ann Kipfer has also expressed concern about the process of digitization of dictionaries, favoring traditional dictionaries over electronic ones. She states:

1. "When we open a dictionary, we see densely printed words written in small font, tightly arranged across two pages. At first glance, our eyes seem to get lost among the fine lines of text. There is so much information on a dictionary page that words and meanings appear intertwined, yet it becomes easier for us to select the relevant information. Printed dictionaries have an intuitive capacity to foresee the future;
2. In printed dictionaries, the meanings of words and phrases are not arranged in a single linear line as in electronic dictionaries. Instead, dictionary entries are presented in a more distributed form. To find meaning and information, the brain engages in extended cognitive processing. If we need the pronunciation of a word, we must already have knowledge of pronunciation rules. These aspects apply to all types of dictionaries;
3. Another logical cognitive task is the simple alphabetical search of a word. If you know the alphabet, you can perform this quickly; otherwise, it will take more time;
4. Illustrations in printed dictionaries are superior to those in online dictionaries. In online dictionaries, one usually sees only an image of the headword. In printed dictionaries, however, one can find complex drawings, colored illustrations, and photographs, although unfortunately this is not the case in all traditional dictionaries;
5. It feels as if a compact collection of knowledge is held in the palm of your hand a very pleasant and nostalgic sensation;
6. In printed dictionaries, we search for words with anticipation, sometimes even moistening our finger to turn pages, and rush to read their meanings. In such cases, the discovered definition tends to remain in the user's memory for a longer time" [Kipfer, 2018].

Language, as a dynamic system, undergoes continuous development, which is reflected in dictionaries through the expansion of terminological resources, the formation of neologisms, and transformations within the lexical system. As a result of globalization, the rapid advancement of science and technology, interstate communication, and the intensification of intercultural discourse, a considerable number of new lexical units have entered the Azerbaijani language. This process has influenced the volume, semantic diversification, and structural models of both traditional and electronic dictionaries [Haliday, M.A.K, 2004].

At the current stage, the Internet, as a global information ecosystem, functions not only as a means of communication but also as an integrative environment for language technologies. The application of ICT in lexicography is accompanied by the development of intelligent search systems, the creation of user-oriented interfaces, and the integration of text and speech corpora. Consequently, unlike traditional dictionaries, electronic dictionaries are evolving into interactive, adaptive, and multimodal platforms.

However, with the advancement of technology over time, electronic dictionaries have become increasingly interactive and functional. Their functional characteristics can be described as follows:

1. **Fast and efficient search.** Electronic dictionaries provide users with rapid search capabilities for specific words, roots, phraseological units, or variants. While printed dictionaries require manual page-by-page browsing, electronic environments use “search” and “filter” functions to direct users straight to the desired result. “The development of electronic dictionary systems in various fields has been progressing at a very high speed in recent years”.

2. **Multimedia and interactivity.** Electronic dictionaries include not only text but also audio pronunciation, example sentences, images, videos, and other multimedia elements. This helps users better understand the semantic meaning of a word within context. “Electronic dictionaries use multimedia tools, and their greatest advantage is the high speed of search”.

3. **Dynamic updating and improvement.** Electronic databases allow for the addition of new words and expressions as needed, as well as the modification of existing entries. This feature is particularly important for meeting the changing needs of language. This is especially emphasized in corpus-based electronic dictionary variants.

4. **Hyperlinks and semantic connectivity.** In electronic dictionaries, semantic relations between lexical items (such as synonyms, antonyms, opposites, and collocations) can be directly presented through hyperlinks. This enables users to view the semantic network in a more systematic way and follow inter-meaning connections. Hyperlinks contribute to “meaning expansion,” i.e., the broadening of semantic understanding.

5. **User-oriented design and adaptability.** Electronic dictionaries can be tailored to different user groups: language learners, professional translators, lexicographers, academics, and general users. In such dictionaries, user-oriented definitions and interface elements (e.g., interface design, search filters, part-of-speech tagging, and usage purpose) are essential. This issue is extensively analyzed in Sandro Nielsen’s study “*Function- and User-Related Definitions in Online Dictionaries*” [Sandro Nielsen. Function, 2016].

Electronic dictionaries represent a new stage in the development of lexicography, overcoming the limitations of traditional dictionaries and enabling the creation of broader and more accessible knowledge bases through modern technologies. They are important for tracking and reflecting the dynamic development of language. For instance, the “Polyglot” electronic dictionary provides translation between Azerbaijani, Russian, English, German, and French and contains approximately 20,000 lexical units. Such dictionaries make the richness and diversity of language accessible to a wider audience. Thus, electronic dictionaries represent an important stage in the development of lexicography. They present linguistic richness and diversity to a broader audience, enhance efficiency in the teaching process, and enable the monitoring of language change. Therefore, the development and use of electronic dictionaries is one of the key directions of modern lexicography.

Technological progress brings new opportunities, challenges, effective methods, and approaches to the language learning process. In general, the digital environment expands the

applicability of theoretical frameworks in foreign language acquisition, introduces new tools and elements, and also requires the reconsideration of established theories. The development of foreign languages in digital environments is a field with significant potential. Its effectiveness is observed in the following areas: **Listening skills:** Podcasts, audiobooks, video lessons (YouTube, TED Talks, etc.). **Reading skills:** E-books, blogs, news portals, and interactive texts. **Writing skills:** Online forums, social media platforms, and AI-based writing assistance tools (e.g., Grammarly, ChatGPT). **Speaking skills:** Online communication platforms (e.g., Tandem, Hello-Talk), voice messages, and language exchange through video calls.

The development of digital technologies provides foreign language learners with extensive opportunities to improve their four (listening, writing, reading, speaking) skills. Compared to traditional teaching methods, the digital environment facilitates access to authentic and diverse language input. This allows students to be exposed not only to textbook-based language but also to the language used in everyday real-life contexts.

Podcasts are an excellent resource for foreign language learners. Audiobooks support integrated language learning by combining reading and listening in a foreign language. In particular, listening to literary works in audio format helps students expand their vocabulary and learn intonation patterns. This method also enhances learning efficiency by engaging multiple sensory channels simultaneously.

Video lessons and platforms such as YouTube, TED Talks, BBC Learning English, VOA Learning English, and others enable learners to hear different accents and speaking styles.

Interactive listening tools some online platforms (e.g., LingQ, EnglishClass101) offer interactive listening experiences. These tools allow users to:

- read the transcript of the audio content;
- highlight unknown words and learn their meanings;
- re-listen to materials and study at an individual pace.

The theoretical and methodological approaches to foreign language listening skills are widely studied as one of the core components of the language learning process, both from the perspective of language acquisition theories and teaching methodology.

Materials and models

The research employs several methodological approaches. A comparative method is used to examine the differences between traditional printed dictionaries and electronic dictionaries in terms of structure, functionality, and usability. A descriptive-analytical method is applied to identify the key features and developments in electronic lexicography and digital language learning environments. Furthermore, an interdisciplinary approach integrates insights from linguistics, information technology, and educational sciences. From a modeling perspective, the study relies on computational and statistical models of language analysis, particularly in the examination of lexical structures and dictionary compilation processes. In addition, user-oriented models are considered to evaluate how electronic dictionaries and digital tools adapt to different categories of users. The combination of these materials and methods ensures a comprehensive analysis of the evolving relationship between linguistics, technology, and language learning in the digital age [Chumarina, 2009].

Outcome and discussion

The study demonstrates that the integration of linguistics with technology and mathematics has led to the emergence of innovative research directions, particularly in computational linguistics

and electronic lexicography. The application of mathematical and statistical methods has enhanced the precision and analytical depth of linguistic studies, while the development of electronic dictionaries has transformed traditional lexicographic practices into more dynamic, interactive, and user-oriented systems.

The findings also reveal that electronic dictionaries significantly outperform traditional printed dictionaries in terms of speed, accessibility, and functionality, although both formats retain distinct advantages. The incorporation of multimedia elements, intelligent search systems, and adaptive interfaces has made electronic lexicographic resources more efficient and relevant to contemporary users.

In addition, the research confirms that digital technologies play a crucial role in foreign language learning by supporting the development of listening, reading, writing, and speaking skills. Tools such as podcasts, audiobooks, video platforms, and interactive applications provide authentic language exposure and facilitate personalized learning experiences [Mahmudov, 2024].

Overall, the outcomes indicate that digitalization has become a driving force in both lexicography and language education, fostering more effective, flexible, and accessible approaches to language acquisition and linguistic research.

Result

The study concludes that the ongoing digital transformation has fundamentally reshaped both linguistics and lexicography, leading to more efficient, data-driven, and user-centered approaches. The integration of mathematical methods and computational technologies has expanded the analytical capabilities of linguistic research, while electronic dictionaries have evolved into interactive and adaptive systems that surpass many limitations of traditional print resources.

Moreover, digital technologies have significantly enhanced foreign language learning by providing learners with immediate access to authentic materials and diverse learning tools. This has improved the development of all core language skills and enabled more flexible, individualized learning pathways.

In summary, the results indicate that the convergence of linguistics, technology, and digital learning environments has created a modern framework in which both language research and language acquisition processes are more dynamic, accessible, and effective.

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THE ROLE OF ARTIFICIAL INTELLIGENCE AND DIGITAL PLATFORMS IN EDUCATION

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Abstract: This article examines the role of artificial intelligence and digital platforms in modern education systems and highlights their impact on teaching and learning processes. The rapid development of digital technologies has created new opportunities for improving the quality of education and has transformed traditional teaching methods into more interactive, flexible, and student-centered approaches. Artificial intelligence-based tools support personalized learning by analyzing students' individual needs, learning styles, and academic performance. These technologies enable teachers to adapt educational materials, monitor student progress more effectively, and provide immediate feedback during the learning process.

In addition, digital platforms contribute significantly to the accessibility and dissemination of educational resources. They support distance and online education, facilitate communication between teachers and students, and provide equal learning opportunities for people from different regions and social backgrounds. The integration of virtual classrooms, cloud technologies, and online assessment systems has increased the efficiency and continuity of education.

The article also discusses the advantages and challenges related to the implementation of artificial intelligence and digital platforms in education. Issues such as data privacy, technological inequality, lack of digital literacy, and ethical concerns are analyzed. The study concludes that the effective and balanced use of these technologies can improve educational quality, increase student engagement, and support the development of innovative learning environments in the future.

Keywords: Artificial intelligence, digital platforms, educational technologies, personalized learning, online education, innovative methods

TƏHSİLDƏ SÜNİ İNTELLEKTİN VƏ RƏQƏMSAL PLATFORMALARIN ROLU

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Xülasə: Bu məqalədə süni intellekt və rəqəmsal platformaların müasir təhsil sistemində rolu və onların tədris prosesinə təsiri araşdırılır. Rəqəmsal texnologiyaların sürətli inkişafı təhsilin keyfiyyətinin yüksəldilməsi üçün yeni imkanlar yaratmış, ənənəvi tədris üsullarının daha interaktiv, çevik və tələbəyönümlü modellərlə əvəz olunmasına səbəb olmuşdur. Süni intellekt əsaslı alətlər tələbələrin fərdi xüsusiyyətlərini, öyrənmə tərzlərini və akademik nəticələrini təhlil edərək fərdiləşdirilmiş təhsilin həyata keçirilməsini təmin edir. Bu texnologiyalar müəllimlərə tədris materiallarını uyğunlaşdırmağa, tələbələrin inkişafını daha effektiv izləməyə və operativ əks əlaqə yaratmağa imkan verir.

Rəqəmsal platformalar isə tədris resurslarının əlçatanlığını artırmaqla yanaşı, biliklərin daha geniş auditoriyaya çatdırılmasına şərait yaradır. Onlayn və məsafədən təhsil imkanlarının genişlənməsi müəllim və tələbələr arasında ünsiyyəti asanlaşdırır, müxtəlif regionlardan olan insanların bərabər təhsil imkanlarından faydalanmasına kömək edir. Virtual siniflər, bulud texnologiyaları və onlayn qiymətləndirmə sistemləri təhsil prosesinin fasiləsizliyini və səmərəliliyini artırır.

Məqalədə bu texnologiyaların üstünlükləri ilə yanaşı, məlumat təhlükəsizliyi, rəqəmsal bərabərsizlik, rəqəmsal savadlılıq səviyyəsi və etik problemlər kimi çağırışlar da təhlil edilir. Nəticə olaraq qeyd olunur ki, süni intellekt və rəqəmsal platformaların düzgün və balanslı tətbiqi gələcəkdə innovativ və keyfiyyətli təhsil mühitinin formalaşmasına mühüm töhfə verə bilər.

Açar sözlər: Süni intellekt, rəqəmsal platformalar, təhsil texnologiyaları, fərdiləşdirilmiş təlim, onlayn təhsil, innovativ metodlar

Introduction

In the 21st century, the rapid development of information and communication technologies has had a significant impact on all areas of society, especially the education system. As a result of globalization and digitalization, traditional teaching models are gradually being replaced by more flexible, innovative, and technology-based approaches. One of the main driving forces behind this transformation is artificial intelligence and digital platforms.

Artificial intelligence technologies play a crucial role in enhancing the quality of education by creating new opportunities. These technologies analyze large volumes of data to identify students' learning behaviors, assess their strengths and weaknesses, and develop individualized learning trajectories. As a result, the educational process becomes more targeted and efficient.

Digital platforms increase access to education by eliminating geographical barriers. Today, students can join courses offered by leading universities worldwide, access various learning materials, and participate in interactive learning environments via the internet. This is particularly important in ensuring equal opportunities in education.

Moreover, the integration of artificial intelligence and digital technologies leads not only to technical changes but also to fundamental transformations in pedagogical approaches. Teachers shift from being traditional knowledge transmitters to facilitators and guides, while students become active participants rather than passive listeners.

However, the rapid development of these technologies also brings certain challenges and risks, such as data security, privacy protection, technological dependency, and digital inequality. Therefore, the implementation of artificial intelligence and digital platforms in education requires not only technological but also social and ethical considerations.

Ultimately, one of the main goals of the modern education system is to effectively use these technologies to create a more high-quality, inclusive, and sustainable learning environment. This article explores and analyzes these issues in detail.

Materials and Methods

- Literature Review: Local and international scientific sources related to the topic were examined.
- Comparative Analysis: Technologies used in different education systems were compared.
- Statistical Approach: Existing data on the effectiveness of artificial intelligence applications were analyzed.
- Observation Method: The use of digital platforms in real educational settings was studied.

Artificial Intelligence (AI) is one of the fastest-growing fields of modern information technology and plays a key role in transforming the education system. AI enables computer systems to perform functions associated with human intelligence, such as learning, analysis, decision-making, and prediction. In education, these technologies analyze big data to understand students' learning behaviors and optimize their academic performance (Holmes, W., Bialik, M., & Fadel, C. 2019).

AI-based systems transform education from a static model into a dynamic and adaptive one. This approach takes into account each student's individual needs, learning pace, and level of understanding, replacing the "one-size-fits-all" model with a personalized learning approach (Luckin, R. 2018).

One of the most important applications of AI in education is Intelligent Tutoring Systems. These systems analyze students' responses in real time, identify errors, and provide immediate feedback and explanations. This enhances not only outcomes but also the learning process itself, ensuring deeper understanding (Holmes, W., Bialik, M., & Fadel, C. 2019).

In addition, AI enables educational institutions to make strategic decisions through learning analytics. Data collected from digital platforms help identify students' performance levels, motivation, and risk factors, allowing for early intervention and prevention of academic failure (Luckin, R. 2018).

Digital platforms form the second major pillar of the modern educational ecosystem. Platforms such as Learning Management Systems (LMS), virtual classrooms, and e-learning systems make education independent of time and place, making it globally accessible (Bates, A. W. 2015).

The development of digital platforms has led to the emergence of blended learning, which integrates traditional classroom instruction with online education. This approach creates a more flexible and effective learning environment, allowing students to benefit from both independent study and teacher guidance (Bates, A. W. 2015).

Another important aspect is the role of digital platforms in the democratization of education. Through the internet, students from any location can access high-quality educational resources, which is especially beneficial for regions with limited socio-economic development (Bates, A. W. 2015).

The integration of AI and digital technologies has also transformed pedagogical approaches. Teachers act as mentors and facilitators, while students become active researchers and problem-solvers (Holmes, W., Bialik, M., & Fadel, C. 2019).

At the same time, these technologies enhance motivation and interactivity in the learning process. Tools such as gamification, virtual laboratories, AI-based assessment systems, and adaptive learning technologies make learning more engaging and effective (Luckin, R. 2018).

However, several challenges arise alongside these advancements. Digital inequality increases the risk of social injustice in education, while data protection and cybersecurity remain major concerns (UNESCO.2021).

According to UNESCO, ethical principles such as transparency, fairness, and accountability must be ensured when applying AI in education to prevent potential violations of human rights (UNESCO.2021).

Furthermore, improving teachers' digital competencies is essential. The effective use of technology depends not only on its availability but also on pedagogical skills. Therefore, continuous professional development of teachers is a key component of modern education strategies (Luckin, R. 2018).

Results and Discussion

The findings indicate that artificial intelligence and digital platforms significantly enhance the quality of education. AI-based systems enable the creation of personalized learning plans by identifying students' strengths and weaknesses.

Digital platforms allow students to learn anytime and anywhere, which is particularly advantageous for distance education. Interactive tools also increase motivation and make lessons more engaging. However, several challenges remain:

- Unequal distribution of technological resources
- Insufficient technological knowledge among teachers
- Data security concerns
- Student distraction in digital environments

Addressing these challenges requires improving teachers' professional development, strengthening technical infrastructure, and implementing effective teaching strategies.

Conclusion

The research and analysis show that artificial intelligence and digital platforms have become integral components of modern education systems. These technologies are transforming not only the technical aspects of education but also its content, structure, and methodology.

The application of AI has made education more personalized and adaptive. Considering individual student characteristics increases motivation and leads to more effective learning outcomes. It also helps teachers plan and assess the educational process more accurately and efficiently.

Digital platforms contribute significantly to the democratization of education by expanding access to knowledge and ensuring equal opportunities among different social groups. Distance education, in particular, has become not just an alternative but a necessity in today's world.

However, it is important to address the challenges associated with these technologies, including unequal access to resources, varying levels of digital literacy, and ethical and legal concerns. Solving these issues requires a comprehensive approach, appropriate policies, and well-designed educational strategies.

In the future, artificial intelligence is expected to further develop and become more deeply integrated into education systems, leading to more flexible, innovative, and effective forms of

learning. At the same time, the human factor—teacher-student interaction—will remain central to the educational process.

Thus, the balanced and effective use of artificial intelligence and digital platforms plays a crucial role in improving education quality, developing human capital, and advancing society as a whole.

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ARTIFICIAL INTELLIGENCE IN FOREIGN LANGUAGE EDUCATION: OPPORTUNITIES AND CHALLENGES

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Abstract: Traditional language instruction is increasingly influenced by Artificial Intelligence (AI) in modern educational environments. This study examines the pedagogical potential of AI tools such as intelligent tutoring systems, chatbots, and automated writing assessment in foreign language learning. Findings from the literature indicate that AI contributes to personalized learning pathways, immediate corrective feedback, learner autonomy, and increased engagement. These advantages are particularly visible in writing and speaking skill development, where learners benefit from continuous practice and adaptive support.

However, the study also identifies several challenges, including algorithmic bias, data privacy concerns, limited emotional intelligence of AI systems, and the risk of over-reliance on automated feedback. While AI enhances efficiency and accessibility, it cannot fully replace the human teacher's role in providing emotional support, cultural understanding, and pedagogical judgment.

The research is based on a narrative literature review and integrates constructivist and sociocultural learning theories. The findings suggest that AI is most effective when used as a supportive tool rather than a replacement for human instruction. A hybrid model combining AI technologies with human-centered pedagogy is recommended to ensure effective, ethical, and sustainable language education.

Keywords: Artificial Intelligence, Foreign Language Teaching, Chatbots, Adaptive Learning, Language Skills Development

XARİCİ DİL TƏHSİLİNDƏ SÜNİ İNTELLEKT: İMKANLAR VƏ ÇƏTİNLİKLƏR

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Xülasə: Ənənəvi dil tədrisi müasir təhsil mühitində Süni İntellektin (AI) təsiri ilə sürətlə dəyişir. Bu tədqiqat ağıllı tədris sistemləri, çatbotlar və avtomatlaşdırılmış yazı qiymətləndirmə vasitələrinin xarici dil öyrənilməsində rolunu araşdırır. Ədəbiyyat təhlili göstərir ki, AI fərdiləşdirilmiş öyrənmə yolları, operativ rəy, tələbə müstəqilliyi və motivasiyanın artırılmasına imkan yaradır. Bu xüsusiyyətlər xüsusilə yazı və danışiq bacarıqlarının inkişafında daha aydın görünür.

Bununla yanaşı, alqoritmik qərəzlilik, məlumat təhlükəsizliyi, emosional dəstəyin çatışmazlığı və texnologiyadan həddindən artıq asılılıq kimi problemlər mövcuddur. AI tədrisi asanlaşdırsa da, müəllimin emosional, mədəni və pedaqoji rolunu tam əvəz edə bilmir.

Tədqiqat narrativ ədəbiyyat icmalı metoduna əsaslanır və konstruktivist və sosiomədəni yanaşmaları birləşdirir. Nəticələr göstərir ki, AI ən effektiv şəkildə insan müəllimlərlə birlikdə istifadə olunduqda faydalıdır.

Açar sözlər: Süni İntellekt, Xarici Dil Tədrisi, Çatbotlar, Adaptiv Öyrənmə, Dil Bacarıqlarının İnkişafı

Introduction

Foreign language education has traditionally aimed at developing learners' communicative competence, critical thinking, and problem-solving abilities, which are considered essential skills in the 21st century (Trilling & Fadel, 2009; Voogt & Roblin, 2012). In this context, contemporary pedagogical approaches such as Project-Based Learning (PBL) have gained increasing importance, as they promote active learning, collaboration, and real-world problem solving (Stoller, 2006). Building on these principles, Technology-Enhanced Project-Based Language Learning (TEPBL) integrates digital tools into language education, enabling learners to participate in authentic, interactive, and often intercultural communication tasks (Dooly & Sadler, 2016; Dooly & Masats, 2020). Such approaches expand learning beyond the traditional classroom and foster meaningful language use in real-life contexts (Beckett & Slater, 2019).

The concept of authenticity has also become central in modern language teaching. Authentic learning environments connect theoretical knowledge with real-world application, thereby enhancing learners' ability to use language effectively in practical situations (Herrington & Oliver, 2000; Lombardi, 2007). With the development of digital technologies and global communication platforms, authenticity now includes not only real-life materials but also real-time interaction, intercultural communication, and collaborative learning experiences (Dooly & O'Dowd, 2018).

In recent years, the rapid advancement of Artificial Intelligence (AI) has significantly transformed language education. AI-powered tools such as chatbots, intelligent tutoring systems, speech recognition technologies, and automated writing assistants have become increasingly integrated into teaching and learning processes. These technologies utilize natural language processing (NLP) and machine learning to provide immediate, personalized feedback, adaptive learning pathways, and continuous opportunities for practice (Crompton & Burke, 2023; Zawacki-Richter et al., 2019). As a result, learners can engage in more flexible and self-directed learning

experiences, while teachers benefit from reduced administrative workload and data-driven insights into student performance (Kasneci et al., 2023).

Despite these advantages, the integration of AI in language education also raises important challenges. Concerns related to algorithmic bias, data privacy, transparency, and the reliability of AI-generated feedback have been widely discussed in recent research (Zawacki-Richter et al., 2019; Williamson & Eynon, 2020). Additionally, AI systems still lack emotional intelligence, cultural awareness, and the ability to interpret context in a fully human-like manner, which are crucial components of effective language learning and communication. Over-reliance on AI tools may also reduce learners' critical thinking and independent language production if not properly guided (Koltovskaia, 2020).

Therefore, AI should not be viewed as a replacement for human teachers but rather as a complementary tool that enhances traditional pedagogical practices. A balanced and thoughtful integration of AI into language education can support personalized learning while preserving the essential human elements of teaching, such as guidance, empathy, and intercultural understanding.

Materials and Methods

This study adopts a narrative literature review (NLR) methodology to explore the role of Artificial Intelligence (AI) in foreign language teaching and learning. Unlike systematic reviews, which follow strict inclusion and exclusion criteria, the narrative approach allows for a more flexible and interpretive analysis of existing research, providing a comprehensive overview of key themes and developments in the field (Zawacki-Richter et al., 2019).

The research is based on a wide range of scholarly sources, including peer-reviewed journal articles, academic books, and conference proceedings retrieved from major academic databases such as Google Scholar, Springer, and ScienceDirect. The literature selection primarily focused on studies published within the last decade to ensure the relevance and timeliness of the findings, particularly in relation to recent advancements in AI technologies (Crompton & Burke, 2023).

The search process involved the use of key terms such as "Artificial Intelligence in education," "AI-assisted language learning," "chatbots in language teaching," "automated feedback," and "speech recognition in language learning." An initial screening of titles and abstracts was conducted to exclude studies that were not directly related to the research focus. Subsequently, full-text analysis was carried out to select the most relevant and high-quality sources based on their contribution to understanding the pedagogical implications of AI in language education.

The selected studies covered key areas including AI-based feedback systems, chatbot-assisted learning environments, speech recognition applications, and automated assessment tools. These themes reflect the most prominent applications of AI in language education and provide a solid foundation for analysis.

The collected data was analyzed using thematic analysis, which enabled the identification of recurring patterns and core concepts such as personalization, learner engagement, feedback mechanisms, teacher roles, and ethical considerations. This approach allowed for the synthesis of diverse findings into coherent themes that reflect the current state of research in the field.

The interpretation of the findings was guided by constructivist and sociocultural learning theories, which emphasize the importance of interaction, collaboration, and scaffolding in language acquisition (Tahir & Tahir, 2023). These theoretical frameworks support the view that learning is an active and socially mediated process, and they provide a relevant basis for understanding how AI technologies can enhance or reshape language learning experiences.

Overall, this methodological approach enables a critical and comprehensive examination of both the opportunities and limitations associated with the integration of AI in foreign language education.

Results and Discussion

The findings of this study indicate that Artificial Intelligence (AI) significantly enhances foreign language learning by improving personalization, feedback quality, and learner engagement. Across the reviewed literature, AI-driven tools consistently demonstrate their ability to adapt to individual learner needs and provide customized learning experiences, which are essential for effective language acquisition (Crompton & Burke, 2023; Zawacki-Richter et al., 2019).

One of the most prominent advantages of AI in language education is the provision of immediate and personalized feedback. AI chatbots and intelligent tutoring systems can analyze learner input in real time and provide corrective feedback tailored to individual errors. This continuous feedback loop enables learners to identify and correct grammatical and lexical mistakes more efficiently, leading to gradual improvement in language accuracy (Li & Hegelheimer, 2013; Lu & Xia, 2015). Similarly, speech recognition technologies support pronunciation development by allowing learners to repeat, adjust, and refine their spoken output in a low-pressure environment. These systems facilitate repeated practice and promote self-correction, which are key elements in developing speaking skills.

Generative AI tools also play an important role in writing development. Applications such as automated writing assistants provide suggestions for grammar, vocabulary, and text organization, helping learners produce more coherent and accurate written texts. Research suggests that such tools can enhance learners' writing performance and support independent learning, although their effectiveness depends on how critically learners engage with the feedback provided (Koltovskaia, 2020).

Another significant finding is the positive impact of AI on learner motivation and engagement. AI-powered platforms often incorporate interactive and adaptive features that create more dynamic and personalized learning environments. These environments reduce anxiety, particularly in speaking and writing activities, as learners can practice without fear of negative evaluation. As a result, students tend to participate more actively and demonstrate more consistent learning behavior (Fauzi et al., 2023; Fariani et al., 2023). The availability of on-demand support further contributes to learner autonomy, allowing students to take greater control over their learning process.

Despite these advantages, the study also highlights several important limitations. AI systems still lack emotional intelligence, cultural awareness, and the ability to interpret context in a nuanced way. Language learning is deeply connected to social interaction, cultural understanding, and pragmatic competence, which cannot be fully replicated by automated systems (Godwin-Jones, 2022). Consequently, human teachers remain indispensable for guiding learners, providing emotional support, and ensuring meaningful communication.

From a pedagogical perspective, AI also contributes to reducing teachers' administrative workload by automating tasks such as grading, feedback generation, and material preparation (Cooper, 2023). This allows educators to focus more on higher-level teaching activities, such as facilitating discussions, developing critical thinking, and supporting individual learners. However, teachers continue to play a crucial role in evaluating AI-generated content, ensuring its accuracy, and adapting it to specific educational contexts.

Overall, the findings suggest that AI is most effective when integrated into a hybrid educational model that combines technological tools with human teaching expertise. Such an approach allows educators to benefit from the efficiency and adaptability of AI while preserving the essential human elements of language teaching, including interaction, empathy, and cultural understanding.

Conclusion

Artificial Intelligence (AI) is significantly reshaping foreign language education by enabling more personalized, flexible, and efficient learning experiences. Through adaptive feedback systems, continuous practice opportunities, and increased accessibility to learning resources, AI supports learners in developing their language skills in a more individualized and autonomous manner (Crompton & Burke, 2023; Zawacki-Richter et al., 2019). These technological advancements contribute to more learner-centered environments, where instruction can be tailored to the needs, pace, and preferences of each student.

However, the effectiveness of AI in language education largely depends on its careful, critical, and ethical integration into pedagogical practice. Issues such as data privacy, algorithmic bias, transparency, and the reliability of AI-generated content must be addressed to ensure responsible use (Williamson & Eynon, 2020; Kasneci et al., 2023). In addition, the risk of over-reliance on AI tools highlights the importance of maintaining learners' critical thinking and independent language production skills (Koltovskaia, 2020).

Importantly, AI should be viewed as a complementary tool rather than a substitute for human teachers. While AI can efficiently provide feedback and automate certain instructional processes, it lacks the emotional intelligence, cultural sensitivity, and pedagogical judgment required for meaningful language teaching (Godwin-Jones, 2022). Teachers remain central figures in guiding learners, facilitating interaction, and ensuring that learning experiences are contextually appropriate and pedagogically sound.

Therefore, the most effective approach to integrating AI in foreign language education is the adoption of a balanced hybrid model that combines AI technologies with human-centered pedagogy. In such a model, educators retain their essential role while leveraging AI to enhance learning opportunities, support instructional processes, and improve overall educational outcomes. This balanced integration ensures that technological innovation contributes to sustainable, ethical, and high-quality language education.

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DIGITAL EDUCATIONAL RESOURCES IN THE TEACHING PROCESS: ROLES AND APPLICATION OPPORTUNITIES

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Abstract: The digitalization process taking place in the modern education system has significantly increased the importance of electronic educational resources. This article analyzes the role of digital educational resources in the educational process and their application possibilities in detail. During the research, the integration of these resources into the educational environment, their impact on student motivation and their role in improving learning outcomes were determined. At the same time, the possibilities of using electronic resources in online and hybrid learning models were investigated. The article shows that digital resources make the educational process more flexible, interactive and student-oriented. However, there are also problems such as insufficient technical infrastructure and insufficient development of teachers' digital skills. As a result, the effective preparation and proper application of electronic educational resources play an important role in improving the quality of education.

Keywords: digital education, e-learning resources, online learning, interactive learning, teaching technologies

TƏDRİS PROSESİNDƏ RƏQƏMSAL TƏHSİL RESURSLARI: ROLLAR VƏ TƏTBİQİ İMKANLARI

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Xülasə: Müasir dövrdə təhsil sistemində baş verən rəqəmsallaşma prosesi elektron təhsil resurslarının əhəmiyyətini əhəmiyyətli dərəcədə artırmışdır. Bu məqalədə rəqəmsal təhsil resurslarının tədris prosesində rolu və onların tətbiqi imkanları geniş şəkildə təhlil olunur. Araşdırma zamanı bu resursların təhsil mühitinə integrasiyası, tələbə motivasiyasına təsiri və öyrənmə nəticələrinin yüksəldilməsindəki rolu müəyyən edilmişdir. Eyni zamanda, onlayn və hibrid təlim modellərində elektron resursların istifadəsi imkanları araşdırılmışdır. Məqalədə göstərilir ki, rəqəmsal resurslar tədris prosesini daha çevik, interaktiv və tələbəyönümlü edir. Bununla yanaşı, texniki infrastrukturun yetərsizliyi və müəllimlərin rəqəmsal

bacarıqlarının kifayət qədər inkişaf etməməsi kimi problemlər də mövcuddur. Nəticə etibarilə, elektron təhsil resurslarının effektiv hazırlanması və düzgün tətbiqi təhsilin keyfiyyətinin artırılmasında mühüm rol oynayır.

Açar sözlər: rəqəmsal təhsil, elektron resurslar, onlayn təlim, interaktiv öyrənmə, tədris texnologiyaları

Introduction

The rapid development of digital technologies has transformed traditional education systems and created new opportunities for teaching and learning. Digital educational resources, such as multimedia tools, online platforms, and interactive applications, have become essential components of modern education. These tools support flexible and student-centered learning environments (Mayer, 2021).

According to Anderson (2020), digital learning environments enhance access to information and promote independent learning skills. However, the successful integration of digital resources requires appropriate technological infrastructure and teacher competence.

Materials and Methods

This study employs a qualitative research design to explore the role and application opportunities of digital educational resources in the teaching process. The qualitative approach was selected to allow for an in-depth analysis of theoretical perspectives and practical applications in the field of digital education.

Research Design

The research is based on a descriptive and analytical framework. It combines a comprehensive literature review with comparative analysis to evaluate different types of digital educational resources and their effectiveness in various learning environments. This approach enables the identification of key trends, challenges, and best practices in the integration of digital technologies in education.

Data Collection

The data for this study were collected from a wide range of academic and institutional sources, including:

- Peer-reviewed journal articles on digital education and e-learning
- Academic books focusing on instructional design and multimedia learning
- Reports and policy documents published by international organizations
- Online educational platforms and digital learning tools

These sources were selected based on their relevance, credibility, and contribution to the field. The selection process followed academic standards to ensure the reliability and validity of the information (Bates, 2019).

Data Analysis

The collected data were analyzed using thematic analysis. Key themes related to the use of digital educational resources were identified and categorized, including:

- Types of digital educational resources
- Pedagogical benefits
- Technological requirements
- Challenges and limitations

A comparative analysis was conducted to evaluate the effectiveness of different digital tools and platforms in enhancing teaching and learning processes. The analysis also focused on how these resources influence student engagement and learning outcomes (Mayer, 2021).

Research Scope and Limitations

The study focuses primarily on higher education contexts and does not include extensive empirical data such as surveys or experiments. While this allows for a broad theoretical understanding, it may limit the generalizability of the findings.

Additionally, rapid technological advancements may affect the long-term relevance of certain digital tools discussed in the study.

Ethical Considerations

All sources used in this research are properly cited following APA 7 guidelines to ensure academic integrity. No personal data or human subjects were involved in this study.

Discussion.

Role of Digital Educational Resources. Digital educational resources provide opportunities for interactive and personalized learning. They allow students to access information anytime and anywhere, which increases learning flexibility (Mayer, 2021).

Application Opportunities

Digital resources can be widely applied in:

- Online learning platforms
- Virtual classrooms
- Multimedia-based instruction
- Mobile learning environments

These applications enhance communication and collaboration between teachers and students.

Advantages

Digital educational resources offer a wide range of advantages that significantly enhance the effectiveness and quality of the teaching and learning process.

One of the most important benefits is increased accessibility to educational content. Digital resources allow students to access learning materials anytime and anywhere, removing geographical and time constraints. This flexibility is particularly valuable in online and distance learning environments, where learners can study at their own pace and according to their individual needs (Anderson, 2020).

Another major advantage is the promotion of interactive and engaging learning experiences. Unlike traditional teaching methods, digital resources often incorporate multimedia elements such as videos, animations, simulations, and interactive quizzes. These features help to capture students' attention and improve their understanding of complex concepts (Mayer, 2021).

Digital educational resources also support personalized learning. Through adaptive technologies and data-driven platforms, educational content can be tailored to the individual abilities, learning styles, and progress of each student. This personalized approach enhances learning efficiency and helps address diverse learner needs.

Furthermore, digital tools facilitate collaboration and communication. Online platforms, discussion forums, and collaborative applications enable students and teachers to interact more effectively, share ideas, and work together on projects regardless of their physical location. This fosters a more inclusive and participatory learning environment.

Another significant advantage is the development of essential 21st-century skills. The use of digital resources encourages students to develop critical thinking, problem-solving, digital literacy, and self-directed learning skills. These competencies are crucial for success in modern academic and professional contexts.

In addition, digital educational resources contribute to the efficiency of teaching. Teachers can use ready-made materials, automate assessment processes, and track student progress through digital systems. This reduces workload and allows educators to focus more on instructional quality and student support.

Finally, digital resources provide opportunities for continuous learning and lifelong education. Learners can easily update their knowledge and skills through online courses, open educational resources, and digital libraries. This supports the concept of lifelong learning and professional development.

Challenges

Despite the numerous advantages of digital educational resources, their integration into the teaching process is accompanied by a range of significant challenges that need to be carefully addressed.

One of the primary challenges is the lack of adequate technological infrastructure. In many educational institutions, especially in developing regions, access to high-speed internet, modern devices, and reliable digital platforms remains limited. This digital divide creates inequality in access to educational resources and negatively affects the quality of learning outcomes (Bates, 2019).

Another critical issue is the insufficient level of digital literacy among educators and students. Many teachers lack the necessary skills to effectively design, implement, and evaluate digital educational materials. As a result, digital tools are often underutilized or used in ways that do not fully support pedagogical goals. Similarly, students may struggle to navigate digital environments efficiently, which can hinder their learning experience.

Additionally, the rapid pace of technological development presents a continuous challenge. Educational institutions are required to constantly update their tools, platforms, and teaching methods to keep up with emerging technologies. This process demands both financial resources and ongoing professional development, which may not always be feasible.

Another important challenge is related to the quality and reliability of digital educational content. Not all online materials meet academic standards, and the abundance of information can make it difficult for educators and students to identify credible and relevant sources. Poorly designed digital resources may also fail to engage learners or support effective knowledge acquisition (Clark & Mayer, 2016).

Furthermore, there are pedagogical challenges associated with the integration of digital resources. The use of technology without appropriate instructional design can lead to superficial learning rather than deep understanding. Effective integration requires alignment between digital tools, learning objectives, and assessment methods, which is not always achieved in practice.

Issues related to student engagement and self-regulation also play a significant role. In online and digital learning environments, students are often required to manage their own learning process. Without proper guidance and motivation, this can lead to decreased participation, procrastination, and lower academic performance (Hodges et al., 2020).

Finally, concerns related to data privacy, cybersecurity, and ethical use of digital technologies are becoming increasingly important. The use of online platforms involves the collection and storage of personal data, which raises questions about security and confidentiality. Educational institutions must ensure that appropriate measures are taken to protect user data and maintain ethical standards.

Conclusion

The findings of this study demonstrate that digital educational resources have become an essential component of contemporary teaching and learning processes. Their integration into educational environments not only enhances the accessibility of knowledge but also supports the development of interactive, flexible, and student-centered learning models.

The analysis shows that digital tools and resources significantly contribute to improving student engagement, motivation, and overall academic performance. In particular, multimedia materials, online platforms, and mobile learning applications create dynamic learning environments that address diverse learning styles and needs (Mayer, 2021). Furthermore, digital resources facilitate collaborative learning and promote the development of critical thinking and independent learning skills (Anderson, 2020).

However, despite these advantages, the study also identifies several challenges that limit the effective implementation of digital educational resources. These include insufficient technological infrastructure, limited access to digital tools, and a lack of adequate digital competencies among educators. In addition, the rapid evolution of technology requires continuous adaptation and professional development, which may pose difficulties for some institutions and teachers.

Another important finding is that the effectiveness of digital educational resources largely depends on their pedagogical design and alignment with learning objectives. Simply integrating technology into the classroom does not guarantee improved learning outcomes; rather, it requires thoughtful planning, appropriate instructional strategies, and continuous evaluation.

In this context, it is essential for educational institutions to invest in both technological infrastructure and teacher training programs. Developing digital literacy skills among educators and students should be considered a priority to ensure the successful integration of digital resources into the teaching process.

For future research, it is recommended to conduct empirical studies involving surveys, experiments, and case studies to provide more concrete evidence on the effectiveness of digital educational resources. Additionally, further investigation into emerging technologies such as artificial intelligence, adaptive learning systems, and virtual reality could provide valuable insights into the future of digital education.

In conclusion, digital educational resources offer significant opportunities for transforming education, but their successful implementation requires a balanced approach that combines technology, pedagogy, and continuous professional development.

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ARTIFICIAL INTELLIGENCE AND DIGITAL PLATFORMS IN CONTEMPORARY EDUCATION

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Abstract: In recent years, the accelerated growth of artificial intelligence (AI) and digital technologies has led to profound changes in educational systems across the globe. These advancements have introduced alternative approaches to teaching, supported individualized learning pathways, and increased the availability of educational materials. This study investigates the impact of AI and digital platforms on education by examining their benefits, limitations, and future potential. It also considers key issues such as ethical implications, data privacy, and equal access to technology. The analysis suggests that while these innovations bring substantial advantages, their effective use depends on careful planning and responsible integration (Holmes et al., 2019; OECD, 2021).

Keywords: Artificial Intelligence, Digital Learning, Personalized Learning, Educational Technology, Online Platforms, E-learning

MÜASİR TƏHSİLDƏ SÜNİ İNTELLEKT VƏ RƏQƏMSAL PLATFORMALAR

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Xülasə: Son illərdə süni intellektin (AI) və rəqəmsal texnologiyaların sürətli inkişafı dünya üzrə təhsil sistemlərində dərin dəyişikliklərə səbəb olmuşdur. Bu yeniliklər tədrisə alternativ yanaşmalar təqdim etmiş, fərdiləşdirilmiş öyrənmə yollarını dəstəkləmiş və təhsil materiallarının əlçatanlığını artırmışdır. Bu tədqiqat süni intellektin və rəqəmsal platformaların təhsilə təsirini onların üstünlüklərini, məhdudiyyətlərini və gələcək potensialını araşdırmaqla təhlil edir. Həmçinin etik məsələlər, məlumatların məxfiliyi və texnologiyaya bərabər çıxış kimi əsas problemlər də nəzərdən keçirilir. Təhlil göstərir ki, bu innovasiyalar əhəmiyyətli üstünlüklər təqdim etsə də, onların səmərəli istifadəsi diqqətli planlaşdırma və məsuliyyətli integrasiyadan asılıdır (Holmes et al., 2019; OECD, 2021).

Açar sözlər: Süni İntellekt, Rəqəmsal Təhsil, Fərdiləşdirilmiş Öyrənmə, Təhsil Texnologiyaları, Onlayn Platformalar, Elektron Təhsil (E-learning)

Introduction

Technological innovation has transformed many fields, with education being one of the most significantly influenced. Artificial intelligence, generally understood as the capability of machines to perform tasks that require human-like intelligence, has become an important factor in reshaping educational practices (Luckin, 2018). At the same time, digital tools such as virtual classrooms, learning applications, and online platforms have removed many traditional limitations in education (Selwyn, 2016).

Consequently, education is increasingly moving toward a flexible and learner-oriented model. Students now have more autonomy over how, when, and where they study. This paper aims to explore the influence of AI and digital platforms by analyzing both their positive contributions and potential drawbacks.

Materials and Methods

This study is based on qualitative analysis of existing academic literature related to artificial intelligence and digital platforms in education. Books, international reports, and scientific publications from organizations such as UNESCO and OECD were reviewed. Comparative and descriptive research methods were applied to examine the role of AI technologies, personalized learning systems, intelligent tutoring systems, and digital learning platforms in modern education. The selected sources were analyzed to identify both the advantages and the challenges of integrating digital technologies into educational systems. Particular attention was given to ethical concerns, accessibility issues, and the changing role of teachers in the digital era.

Results and Discussion

Artificial intelligence has created new opportunities to improve teaching and learning processes. By utilizing techniques like machine learning and data analysis, AI systems are able to process large datasets and extract useful patterns related to student performance (OECD, 2021). These insights allow educators to adjust their teaching strategies more effectively.

A major advantage of AI in education is its ability to support individualized learning. Traditional teaching methods often apply the same approach to all learners, which may not be effective for everyone. In contrast, AI technologies can tailor educational content to match the needs and abilities of each student (Holmes et al., 2019). This allows learners to progress at their own speed, receive immediate feedback, and engage more actively in the learning process.

AI-based tutoring systems simulate the role of a human tutor by providing real-time assistance and explanations. These systems can analyze student responses, offer guidance, and modify task difficulty when necessary (Luckin, 2018). As a result, students can practice independently while still receiving structured support.

AI also contributes to education through automated evaluation methods. Assignments can be graded quickly, and feedback can be delivered instantly (OECD, 2021). This not only reduces the workload of teachers but also helps students identify areas for improvement without delay.

AI-powered chatbots function as digital assistants that provide continuous academic support. They can respond to student questions, assist with tasks, and clarify difficult concepts (UNESCO, 2020). These tools are particularly useful in online learning settings where direct interaction may be limited.

Digital platforms serve as essential tools for delivering educational content and supporting communication. These include learning management systems, mobile learning applications, and online course platforms (Selwyn, 2016).

One of the key strengths of digital platforms is their ability to offer flexible learning conditions. Students are able to access educational materials at any time and from any location (UNESCO, 2020). This is especially valuable for those who cannot attend traditional classes.

Digital technologies make learning more dynamic by incorporating multimedia resources such as videos, simulations, and interactive activities. These elements increase student engagement and improve understanding (Selwyn, 2016).

Online platforms enhance communication between students and educators through tools like discussion boards, messaging systems, and video conferencing (OECD, 2021). These features encourage collaboration and help develop critical thinking and teamwork skills.

The use of AI and digital tools in education provides multiple benefits. Personalized learning systems improve academic performance by addressing the specific needs of each learner (Holmes et al., 2019). Automation allows teachers to spend less time on routine tasks and more time on meaningful interaction with students (Luckin, 2018). Data generated by AI systems supports better decision-making at both classroom and institutional levels (OECD, 2021). Digital technologies expand educational opportunities by making learning accessible to a wider audience (UNESCO, 2020).

Despite their advantages, these technologies also present several important challenges. Not all students have equal access to digital tools, which can lead to disparities in educational opportunities (Selwyn, 2016). The use of personal data in AI systems raises concerns about security and confidentiality (OECD, 2021). AI technologies may produce biased outcomes or lack transparency, potentially leading to unfair results (Holmes et al., 2019). Overreliance on digital tools may limit face-to-face communication and affect students' social development.

Even with technological advancements, teachers remain central to the educational process. AI tools are intended to assist rather than replace educators (Luckin, 2018). Teachers play a vital role in guiding students, supporting their emotional needs, and fostering critical thinking skills. Their responsibilities are shifting toward facilitating learning rather than simply delivering information.

Digital platforms support continuous learning beyond formal education. Individuals can improve their skills and knowledge through online courses and training programs (UNESCO, 2020). This trend highlights the growing importance of lifelong learning in a rapidly changing world.

In the future, education is expected to incorporate more advanced technologies such as virtual reality and AI-generated content (OECD, 2021). These developments will further enhance personalized and immersive learning experiences.

Conclusion

Artificial intelligence and digital platforms are playing a transformative role in education by increasing accessibility, improving efficiency, and enabling personalized learning. However, it is essential to address issues related to ethics, privacy, and equality.

A balanced integration of technology and human interaction is necessary to ensure effective and sustainable educational development.

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TEACHER–AI COLLABORATION IN AZERBAIJANI EDUCATION: AUGMENTATION RATHER THAN REPLACEMENT

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Abstract: The integration of artificial intelligence (AI) into education has brought significant changes in the teaching and learning process, raising many questions concerning the importance of human teachers in the digital age. While, AI is able to perform complex tasks like data analysis, personalized learning, and automated assessment. However, the role of emotional intelligence, ethical reasoning, and cultural knowledge cannot be performed by any computer system. Thus, this article examines the collaboration between teachers and AI within the Azerbaijani context. More specifically, the use of AI as an augmentative tool is viewed as the most beneficial and effective. Based on modern learning theory and research studies, the paper focuses on both advantages and disadvantages of AI adoption in learning. At the same time, the focus is placed on current trends within the country, including digital development, national strategies and policies, as well as deficiencies in terms of teachers' readiness. Despite certain successes in the sphere of digital development, it becomes clear that the success of AI lies in teacher empowerment and professionalism.

Keywords: artificial intelligence, teacher–AI collaboration, Azerbaijan, digital education, teacher agency, educational technology

СОТРУДНИЧЕСТВО УЧИТЕЛЯ И ИИ В ОБРАЗОВАНИИ АЗЕРБАЙДЖАНА: ДОПОЛНЕНИЕ, А НЕ ЗАМЕНА

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Аннотация: Интеграция искусственного интеллекта (ИИ) в образование принесла значительные изменения в процесс обучения и преподавания, подняв множество вопросов о роли педагога в цифровую эпоху. Хотя ИИ способен выполнять такие сложные задачи, как анализ данных, персонализированное обучение и автоматизированное оценивание, роли, требующие эмоционального интеллекта, этического рассуждения и культурных знаний, не могут быть выполнены ни одной компьютерной системой. В связи с этим в данной статье рассматривается сотрудничество между учителями и ИИ в контексте Азербайджана. В частности, использование ИИ в качестве вспомогательного инструмента рассматривается как наиболее полезный и эффективный подход. На основе современной теории обучения и исследовательских работ в статье анализируются как преимущества, так и недостатки внедрения ИИ в образовательный процесс. В то же время особое внимание уделяется текущим тенденциям в стране, включая цифровое развитие, национальные стратегии и политики, а также пробелы в уровне готовности учителей. Несмотря на определенные успехи в сфере цифровизации, становится очевидным, что успех интеграции ИИ зависит от расширения возможностей и профессионализма преподавателей.

Ключевые слова: искусственный интеллект, сотрудничество учителя и ИИ, Азербайджан, цифровое образование, субъектность учителя, образовательные технологии.

Introduction

The inclusion of AI into educational practice can be seen as one of the most revolutionary breakthroughs in modern pedagogical approaches (Selwyn et al., 2019). As a result, an important issue emerges with regard to the changing position of teachers in this digital environment (Giannakos et al., 2025). Artificial intelligence systems provide many unique features, such as big data analysis, pattern recognition, personalized learning strategies, and automated testing systems (Kulik & Fletcher, 2016). However, there is a consensus regarding the fact that these technologies do not substitute for many aspects of human involvement in the educational process, such as emotional intelligence, ethical judgment, etc.

This paper supports that teacher-AI collaboration, instead of substitution, represents the optimal way of enhancing educational success. This statement gains special relevance against the backdrop of Azerbaijani society where massive development of digital platforms such as Digital School platform in thousands of schools has not been supplemented by the pedagogic use of AI (Javadov et al., 2025). Although technological advances take place at breakneck speed, it should be recognized that the importance of teachers for promoting real and equal education cannot be undermined.

This study outlines the evolution of AI technologies in social sphere and learning theories that emphasize the critical importance of the role played by educators. The pros and cons associated with AI in education are subsequently considered in detail.

Materials and Methods

The methodology of the current research is qualitative and analytical, which includes reviewing the existing academic literature on the topics of educational technology, learning theories, and education policy. Academic resources used for the literature review include peer-reviewed scientific journals, theoretical papers, and international policy documents dedicated to artificial intelligence in education.

The review will be based on analyzing the topic through three perspectives. Firstly, it will rely on the analysis of existing theories of learning, such as behaviorism, constructivism, and social

learning theories. Secondly, it will include an evaluation of existing empirical evidence regarding the advantages and disadvantages of using AI in education. Thirdly, the paper will provide analysis of the use of AI in the context of the Azerbaijani education system.

A critical perspective will be employed throughout the analysis to assess the possibilities and risks of implementing AI.

Results and Discussion

The expansion of AI technology in education should be understood in the context of the global shift in which the application of AI technologies has gone from technical aspects to social spheres such as teaching and learning (Giannakos et al., 2025). Such technologies allow for the creation of interactive and adaptive learning environments where there is communication and collaboration. Nevertheless, these learning technologies could also affect the students' learning process by making them less autonomous and more dependent on decision-making. It becomes evident that AI technologies should not be used exclusively, but rather in combination with active teacher participation.

AI development in the context of education in Azerbaijan has become more rapid due to specific national initiatives and policies. For instance, the Azerbaijani AI strategy until 2028 and the implementation of projects such as the Digital School platform indicate great success in this sphere. Great advances have been made in the construction of necessary digital infrastructure as thousands of schools were connected to national education network. Nevertheless, the actual integration of AI technology into teaching practices of the schools was still lacking and was mainly limited to basic AI services (Javadov et al., 2025).

Theory of learning provides some valuable insight into why teachers remain indispensable in the educational process. In terms of behaviorism, teachers organize the process of learning through the use of reinforcement and feedback as well as managing classroom behaviors (Skinner, 1953). According to constructivist theory, teachers act as facilitators in helping students create knowledge through communication, questions and discussion (Piaget, 1973; Vygotsky, 1978). In social learning theory, teachers function as role models and the behaviors of teachers have a great impact on the formation of students' behaviors (Bandura, 1977). All these theoretical views illustrate that teaching is a complicated process and it goes far beyond the capacity of AI programs.

Nevertheless, there are numerous advantages of using AI in education. One of the key benefits of using artificial intelligence is that it makes educational process less bureaucratic, performing automated analysis of various data, grading students' performance and monitoring students' development (Wang et al., 2021). As a result, teachers become able to pay much more attention to teaching rather than administrative work. Furthermore, one of the benefits of artificial intelligence technology is that it provides opportunities for personalized learning experience, customizing the educational process according to learners' specific needs, aligning with concepts such as Vygotsky's Zone of Proximal Development (Pane et al., 2017)

However, there are several issues associated with incorporating AI into teaching practice. Cognitive over-reliance is a potential problem whereby students rely heavily on AI technology to accomplish their goals without engaging in any thinking process. It is often called cognitive offloading, and it can undermine critical thinking and problem-solving abilities (Gerlich, 2025). Moreover, one should note that the issue of teacher readiness is essential. As mentioned above, many educators are not adequately trained to incorporate AI technology into practice. It leads to the

emergence of a "digital use divide" between those who can utilize such technology and those who cannot (OECD, 2024).

It should be mentioned that recent studies prove that collaboration between teachers and AI is better than automation. In other words, a hybrid model of teacher-AI collaboration is more beneficial than using fully automated solutions (Holstein et al., 2019; Roll & Wylie, 2016). The reason for that lies in the fact that humans and computers have complementary roles. For example, machines excel at data analysis and generating instant feedback, whereas humans are superior at contextual interpretation and providing ethical, socio-emotional, and emotional support.

Azerbaijan is an example of how effective implementation of this strategy is not always straightforward. While there is a strong political will to implement AI at policy levels, implementation has been inconsistent. Teachers are not adequately prepared to apply AI technology meaningfully due to insufficient preparation, resulting in superficial applications of AI. Teacher autonomy is also a worry in the sense of dependency on the advice generated by AI. Such complications are aggravated by issues related to disparities experienced by internal migrants and ethnic minorities, where culturally relevant teaching and human contact are essential (UNESCO, 2024).

Conclusion

It has been claimed above that in order to facilitate integration of artificial intelligence into educational processes, the key focus should be placed on collaboration between AI systems and teachers instead of replacement. Although AI-based tools possess an array of advantages such as high levels of efficiency and personalization, AI systems fail to replace human qualities required for teaching, such as the ability to provide emotional support, apply critical thinking skills and take culture-specific factors into account.

The situation in Azerbaijan is the best proof of the claim that technological capabilities combined with appropriate political support are insufficient. The effective integration of AI technologies depends on teachers' capacity to apply these innovations in practice, interpret data provided by the computer, etc. It means that additional investments in teacher education and their rights protection will be necessary in the future. Finally, it should be noted that in order to move towards the bright future, teachers should become partners for innovation and development.

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THE ROLE OF ARTIFICIAL INTELLIGENCE AND DIGITAL PLATFORMS IN ENGLISH LANGUAGE TEACHING: OPPORTUNITIES AND CHALLENGES

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Abstract: This paper investigates the role of Artificial Intelligence (AI) and digital platforms in English language teaching (ELT), focusing on both their pedagogical potential and practical challenges. Drawing on theoretical perspectives and classroom-based observations, the study explores how AI-powered tools—such as chatbots, grammar checkers, translation systems, and speech recognition applications—contribute to language learning. These technologies provide learners with personalized learning paths, instant feedback, and increased opportunities for autonomous learning.

The findings indicate that AI enhances learner motivation, engagement, and autonomy by enabling flexible and continuous learning beyond the classroom. However, the integration of AI also presents significant challenges. Over-reliance on AI tools may reduce students' critical thinking skills and limit authentic language production, especially in writing and reading activities. Additionally, issues such as data privacy, lack of human interaction, and occasional inaccuracies in AI-generated feedback remain important concerns.

Classroom observations show that students benefit greatly from AI-supported speaking practice and independent learning opportunities. However, some learners become dependent on automated systems, which may negatively affect their long-term language development. The study concludes that AI should be used as a complementary tool rather than a replacement for teachers. A balanced approach combining traditional pedagogy with technological innovation is essential for effective English language teaching.

Key words: Artificial Intelligence, English Language Teaching, Digital Platforms, Online Learning, Student Motivation, Educational Technology

İNGİLİS DİLİNİN TƏDRİSİNDƏ SÜNİ İNTELLEKT VƏ RƏQƏMSAL PLATFORMALARIN ROLU: İMKANLAR VƏ ÇƏTİNLİKLƏR

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Xülasə: Bu məqalə Süni İntellektin (Sİ) və rəqəmsal platformaların ingilis dilinin tədrisində (ELT) rolunu araşdırır, onların həm pedaqoji potensialına, həm də praktik çətinliklərinə fokuslanır. Nəzəri yanaşmalara və sinif müşahidələrinə əsaslanaraq, tədqiqat chatbotlar, qrammatika yoxlayıcıları, tərcümə sistemləri və nitqin tanınması tətbiqləri kimi Sİ əsaslı vasitələrin dil öyrənməyə necə töhfə verdiyini araşdırır. Bu texnologiyalar öyrənənlərə fərdiləşdirilmiş öyrənmə yolları, ani geribildirim və müstəqil öyrənmə imkanlarının artmasını təmin edir.

Nəticələr göstərir ki, Sİ sinifdən kənarda çevik və fasiləsiz öyrənməni mümkün etməklə öyrənənlərin motivasiyasını, iştirakını və müstəqilliyini artırır. Lakin Sİ-nin inteqrasiyası həm də ciddi çətinliklər yaradır.

Sİ alətlərindən həddindən artıq asılılıq tələbələrin tənqidi düşünmə bacarıqlarını azalda və xüsusilə yazı və oxu fəaliyyətlərində autentik dil istehsalını məhdudlaşdırır. Bundan əlavə, məlumatların məxfiliyi, insanla ünsiyyətin azalması və Sİ tərəfindən verilən geribildirimdə bəzən qeyri-dəqiqliklər kimi problemlər də mühüm narahatlıqlar olaraq qalır.

Sınıf müşahidələri göstərir ki, tələbələr Sİ dəstəklı danışıq praktikası və müstəqil öyrənmə imkanlarından böyük fayda əldə edirlər. Lakin bəzi öyrənənlər avtomatlaşdırılmış sistemlərdən asılı hala gəlir ki, bu da onların uzunmüddətli dil inkişafına mənfi təsir göstərə bilər. Tədqiqat belə nəticəyə gəlir ki, Sİ müəllimlərin əvəzedicisi deyil, tamamlayıcı vasitə kimi istifadə olunmalıdır. Ənənəvi pedaqogika ilə texnoloji innovasiyanı birləşdirən balanslaşdırılmış yanaşma ingilis dilinin effektiv tədrisi üçün vacibdir.

Açar sözlər: Süni İntellekt, İngilis dilinin tədrisi, rəqəmsal platformalar, onlayn öyrənmə, tələbə motivasiyası, təhsil texnologiyası

Introduction

Artificial Intelligence (AI) is increasingly reshaping global education systems, particularly in the field of English language teaching (ELT). Its integration into language learning environments has created new opportunities for personalized instruction, autonomous learning, and real-time feedback. Unlike traditional classroom-based instruction, AI-powered tools allow learners to engage with the language beyond classroom boundaries, making learning continuous, flexible, and learner-centered. AI-based systems such as language learning applications, chatbots, grammar correction tools, and speech recognition technologies have become essential components of modern language education. These technologies not only support learners' linguistic development but also transform the pedagogical role of teachers, who now act more as facilitators of learning rather than sole knowledge providers.

Since its emergence in the 1950s, AI has developed rapidly due to advances in computing power, large datasets, and algorithmic improvements. Today, it is widely applied in various fields, including healthcare, finance, transportation, and education, with a particularly strong impact on language learning.

In English Language Teaching, AI supports learners by providing adaptive and personalized learning environments through tools such as chatbots, translation systems, intelligent tutoring systems, and speech recognition applications. These tools allow learners to practice language skills at their own pace while receiving instant feedback, which significantly improves learning efficiency.

Language learning theories such as behaviorism, constructivism, and cognitive approaches provide a foundation for understanding how individuals acquire new languages. In recent years, AI has strengthened these theories by offering interactive, adaptive, and learner-centered learning experiences.

From a behaviorist perspective, AI-powered tools provide immediate feedback and reinforcement. For example, applications such as Duolingo offer instant correction and repetition-based practice, which helps learners reinforce correct language usage. Constructivist theory is supported through AI-driven virtual environments where learners engage in immersive and meaningful communication. Virtual reality (VR) platforms, for instance, allow learners to practice English in simulated real-life contexts. Cognitive theories are supported by AI systems that analyze learner errors, identify patterns, and provide targeted feedback based on individual needs.

Natural Language Processing (NLP), a key branch of AI, enables machines to understand and generate human language. As Lin and Chen (2020) note, AI integration is most effective when aligned with cognitive and constructivist learning models that emphasize interaction and adaptation

[5]. AI-powered chatbots create realistic conversational environments that help learners develop speaking confidence, while grammar tools such as Grammarly assist in improving writing accuracy. Machine learning systems further personalize learning by adapting tasks according to learners' performance levels.

In addition, AI supports social learning by enabling collaboration through online forums, discussion platforms, and language exchange applications. It also enhances vocabulary acquisition through spaced repetition systems, improves listening skills through speech recognition technologies, and supports translation and comprehension through intelligent language tools.

Despite these advantages, AI integration also presents challenges. Over-reliance on AI tools may weaken learners' critical thinking skills and reduce active language production. Privacy concerns arise due to data collection, while lack of human interaction may limit communicative competence. Furthermore, inaccurate or biased AI feedback may negatively affect learning outcomes.

Overall, AI has significantly transformed English language teaching by making it more flexible, interactive, and personalized. It enhances learner engagement and supports individualized learning paths, while also redefining the role of the teacher. However, its effectiveness depends on balanced integration with traditional pedagogical practices.

Methodology and Materials

This study adopts a qualitative research approach based on classroom observations and the analysis of students' interaction with Artificial Intelligence (AI) tools in English language learning. The main purpose of the study is to explore how AI-based applications influence learners' engagement, motivation, and language development in real classroom contexts.

The participants of the study were university-level English language learners who regularly use digital platforms and AI-supported tools such as chatbots, grammar checkers, translation systems, and speech recognition applications during their learning process. These learners are generally familiar with technology and tend to use AI tools both during lessons and independently outside the classroom.

The data were collected through informal classroom observations and students' feedback during English lessons. The researcher observed how students interact with AI tools for speaking, writing, and vocabulary practice both inside and outside the classroom environment. For example, it was noticed that students often used grammar checkers while completing writing tasks and relied on chatbots for additional speaking practice when they lacked confidence to communicate with peers.

The collected data were analyzed descriptively, focusing on recurring patterns such as learner motivation, autonomy, over-reliance on AI tools, and improvement in language skills. It was observed that while AI tools increased students' engagement and supported independent learning, some learners depended too much on these tools, especially in productive skills like writing.

No statistical analysis was applied, as the study is qualitative in nature. This methodological approach allows for an in-depth understanding of the role of AI in real classroom contexts and provides valuable insights into both the advantages and limitations of its use in English language teaching.

Discussion

Based on classroom experience, students frequently use AI tools for grammar checking, translation, and speaking practice. One of the most significant advantages observed is the use of AI-powered speaking assistants. Many learners struggle to find speaking partners outside the classroom;

therefore, AI chatbots provide an effective alternative for continuous speaking practice. This has been particularly beneficial in increasing students' confidence and fluency, especially in environments where exposure to authentic English is limited. AI also supports self-directed learning, allowing students to study independently and receive instant feedback.

Despite its benefits, several challenges have been observed in real classroom settings. One major issue is over-reliance on AI tools. In writing tasks such as essay writing or creative storytelling, some students generate entire texts using AI. This often results in unnatural sentence structures and inconsistent language levels that do not reflect their actual proficiency. Another issue is reduced engagement with reading tasks. Instead of reading full texts, students often rely on AI-generated summaries, which limits exposure to authentic language input and negatively affects reading development.

Although AI provides instant feedback, it cannot fully replace the teacher's role. AI lacks emotional intelligence, contextual awareness, and individualized pedagogical support, which are essential in language education.

The findings of this study suggest that AI has a transformative impact on English language teaching by enhancing accessibility, personalization, and learner engagement. However, its effectiveness depends on how it is integrated into pedagogical practice. A balanced approach is necessary. AI should be used as a supplementary tool rather than a replacement for traditional teaching methods. Teachers remain essential in guiding learners, providing emotional support, and fostering critical thinking skills. From this perspective, the role of the teacher is evolving from knowledge provider to learning facilitator, emphasizing guidance rather than direct instruction.

Artificial Intelligence and digital platforms have significantly transformed English language teaching by creating more flexible, interactive, and personalized learning environments. While AI provides numerous benefits such as instant feedback, speaking practice opportunities, and adaptive learning pathways, it also presents challenges including over-dependence and reduced authentic language use.

Conclusion

The findings of this study clearly demonstrate that Artificial Intelligence and digital platforms have become powerful transformative forces in English language teaching. By enabling flexible, interactive, and highly personalized learning environments, AI has reshaped not only how students learn but also how teaching is conceptualized and delivered. The integration of AI technologies allows learners to engage with English in continuous and meaningful ways beyond the traditional classroom, fostering autonomy, increasing motivation, and supporting individual learning needs. Tools such as chatbots, grammar correction systems, and speech recognition applications provide immediate feedback and create opportunities for constant practice, which significantly contributes to the development of language skills.

However, despite these considerable advantages, the study also highlights important limitations and risks associated with AI integration. Over-dependence on AI tools may reduce learners' active participation, limit critical thinking, and negatively affect the authenticity of language production. Furthermore, the absence of emotional intelligence and contextual sensitivity in AI systems underlines the irreplaceable value of human interaction in the learning process. Language learning is not only a cognitive activity but also a social and emotional experience, which requires guidance, encouragement, and meaningful communication provided by teachers.

In this regard, the study emphasizes that the most effective approach to English language teaching lies in adopting a balanced and pedagogically sound integration of AI technologies. A blended learning model, where AI serves as a supportive and complementary tool rather than a substitute for traditional instruction, appears to be the most appropriate strategy. Such an approach allows educators to benefit from technological advancements while maintaining the essential human elements of teaching and learning.

Moreover, the role of the teacher is undergoing a significant transformation. Rather than acting solely as knowledge transmitters, teachers are increasingly becoming facilitators, mentors, and designers of learning experiences. Their responsibility extends to guiding students in the effective and ethical use of AI tools, fostering critical thinking, and ensuring that technology enhances rather than hinders the learning process.

In conclusion, while Artificial Intelligence holds great potential to revolutionize English language education, its successful implementation depends on thoughtful integration, pedagogical awareness, and continuous evaluation. Future research may further explore long-term impacts of AI on language proficiency, learner autonomy, and educational outcomes. Ultimately, a harmonious collaboration between human intelligence and artificial intelligence will be key to achieving sustainable and effective language learning in the modern educational landscape.

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THE ROLE OF ARTIFICIAL INTELLIGENCE IN ENGLISH LANGUAGE TEACHING

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Abstract: The current paper explores the involvement of artificial intelligence in teaching English. It investigates the potential of artificial intelligence in pedagogy, problems related to its implementation, and implications for contemporary educational practices. Specifically, the paper tackles an urgent issue that needs addressing: although AI is being incorporated into language learning at an unprecedented pace, there is a shortage of pedagogical approaches that would facilitate the appropriate, ethical, and balanced usage of AI tools without compromising learners' autonomy, intellectual capabilities, and academic honesty. The current research involves a qualitative review of peer-reviewed articles on AI in ELT published from 2020 to 2026. The results show that artificial intelligence greatly improves language learning through personalized learning, automated assessment, and enhanced motivation. Adapted learning systems and intelligent systems promote personalized learning experiences, whereas tools like automatic writing assessment and conversational agents promote linguistic skills acquisition. Nonetheless, the research uncovers several problems associated with artificial intelligence applications in language learning, including overdependence on technology, integrity issues, and ethical questions regarding data privacy and algorithm bias. In order to overcome such problems, the authors suggest developing a mixed pedagogy system based on both artificial intelligence and humanistic approaches to teaching. Such an approach requires focusing on teachers' roles, critical thinking and ethical aspects related to the use of technology. The paper comes to the conclusion that in order to ensure the proper implementation of AI into the sphere of English language teaching, the harmony between innovation and pedagogical aspects should be established.

Keywords: Artificial Intelligence; English language teaching; personalized learning; digital linguistics; language education

İNGİLİS DİLİ TƏDRİSİNDƏ SÜNİ İNTELLEKTİN ROLU

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Xülasə: Məqalə süni intellektin ingilis dilinin tədrisində iştirakını, onun pedaqogikada potensialı və tətbiqi ilə bağlı problemləri və müasir təhsil təcrübələrinə təsirlərini araşdırır. Xüsusilə, məqalə həll edilməli olan təcili bir məsələni əhatə edir: süni intellekt görünməmiş bir sürətlə dil öyrənməsinə daxil edilsə də, öyrənənlərin muxtariyyətinə, intellektual qabiliyyətlərinə və akademik dürüstlüyünə xələl gətirmədən süni intellekt alətlərinin müvafiq, etik və balanslı istifadəsini asanlaşdıracaq pedaqoji yanaşmaların çatışmazlığı var. Hazırkı tədqiqat, 2020-ci ildən 2026-cı ilə qədər dərc olunmuş ELT-də süni intellekt haqqında məqalələrin keyfiyyətə nəzərdən keçirilməsini əhatə edir. Nəticələr göstərir ki, süni intellekt fərdiləşdirilmiş öyrənmə, avtomatlaşdırılmış qiymətləndirmə və motivasiyanın artırılması yolu ilə dil öyrənməsini xeyli yaxşılaşdırır. Uyğunlaşdırılmış öyrənmə sistemləri və ağıllı sistemlər fərdiləşdirilmiş öyrənmə təcrübələrini təşviq edir, avtomatik yazı qiymətləndirməsi və danışq agentləri kimi alətlər isə dil bacarıqlarının əldə edilməsini təşviq edir. Buna baxmayaraq, tədqiqat, texnologiyadan həddindən artıq asılılıq, bütövlük məsələləri və məlumatların məxfiliyi və alqoritm qərəzliliyi ilə bağlı etik suallar da daxil olmaqla, dil öyrənməsində süni intellekt tətbiqləri ilə əlaqəli bir sıra problemləri ortaya qoyur. Bu cür problemlərin aradan qaldırılması üçün müəlliflər həm süni intellekt, həm də humanist tədris yanaşmalarına əsaslanan qarışıq pedaqogika sistemi hazırlamağı təklif edirlər. Belə bir yanaşma müəllimlərin rollarına, tənqidi düşüncəyə və texnologiyanın istifadəsi ilə bağlı etik aspektlərə diqqət yetirməyi tələb edir. Məqalədə belə bir nəticəyə gəlinir ki, süni intellektin ingilis dilinin tədrisi sahəsinə düzgün tətbiqini təmin etmək üçün innovasiya və pedaqoji aspektlər arasında harmoniya yaradılmalıdır.

Açar sözlər: Süni intellekt; İngilis dilinin tədrisi; fərdiləşdirilmiş təlim; rəqəmsal dilçilik; dil təhsili

Introduction

Artificial intelligence has recently proven to be one of the most significant innovations impacting modern educational systems. The application of artificial intelligence within English Language Teaching has considerably modified the existing methods of instruction by employing advanced systems that can process, analyze, and generate language. This progress is highly related to innovations in the fields of natural language processing, machine learning, and generative AI. Consequently, the application of artificial intelligence in language education has evolved from a secondary to a primary tool in modern learning processes.

However, the rising importance of using artificial intelligence in English language instruction is mainly because of the rising need for personalized and flexible forms of learning. The traditional method of teaching is inadequate to cater to the different requirements of each student, especially when there are differences in their skills, motivations, and learning styles in a mixed classroom environment. With the use of artificial intelligence, the problem could be solved by offering adaptive learning platforms, which offer personalized learning experience through analysis of students' mistakes, adjusting task difficulties and types according to their abilities and learning styles (Torres & Kahveci, 2025).

However, the use of artificial intelligence is instrumental in automating the process of assessment and feedback, which has always been one of the most laborious tasks for language instructors. The use of AI technologies allows for an evaluation of oral and written speech in high accuracy and delivers immediate feedback that helps learners detect and rectify any errors. In language learning, the delivery of feedback is of particular importance since learners are supposed to receive it as soon as possible in order to develop their linguistic skills (Kristiawan et al., 2024). Moreover, artificial intelligence makes it possible for learners to be more engaged in the learning process due to the creation of an interactive environment.

Nevertheless, there are some obstacles to the integration of artificial intelligence into English language teaching despite its aforementioned benefits. Namely, the fast uptake of AI technologies by educational institutions prompts numerous concerns about the efficiency of such tools for education purposes, ethics, and their effect on learners' autonomy. One of the issues associated with AI technologies is that their frequent use may be detrimental to critical thinking and foster passive learning. Furthermore, the implementation of generative AI technology in essay writing complicates problems of academic integrity since it is harder to differentiate machine-created work from students' (Hockly, 2023).

The key issue at the heart of this research question could be described as follows. Although artificial intelligence technologies have become widely adopted in English language instruction, there is an urgent need for theoretical frameworks based on educational principles that would guarantee the proper use of these technologies without harming students' cognitive processes, their autonomy, or their intellectual honesty. This issue is especially relevant in higher education because students have free access to any technology related to artificial intelligence and often use it without appropriate guidance.

The purpose of this paper is to examine the application of artificial intelligence in the process of English language teaching. The study focuses on examining the advantages and disadvantages of applying artificial intelligence in English language learning. In addition, the research seeks to develop a pedagogic model that can facilitate efficient use of artificial intelligence in the learning environment.

Materials and Methods

The current research is grounded on the qualitative research paradigm, which applies systematic literature review as its main methodology. The sources for the research are peer-reviewed scholarly articles focusing on the utilization of AI in English language education. Sources have been chosen from reliable international databases with an emphasis on the literature published between 2020 and 2026 to make sure that the information is up-to-date and relevant.

The inclusion criteria for the sources are their pertinence to the topic of artificial intelligence within language education, their contributions to the literature either theoretically or empirically, and their publication in peer-reviewed journals in English. Altogether, there were more than eight sources used, including literature review articles and empirical research articles. Such sources give an extensive summary of the current trends, challenges, and opportunities within the utilization of artificial intelligence in English language instruction (Deng & Jamaludin, 2026).

The research analysis was carried out through a thematic framework, where important themes that emerged from the chosen articles were analyzed. Some of the major themes considered include personalization, automation, engagement of the learners, teachers' role, and ethical considerations. All the identified themes were analyzed with regard to the research problem, hence giving an insight into the strengths and weaknesses of artificial intelligence in language learning.

The analysis will also be supplemented by theory that is pertinent to this subject matter. Constructivist learning theories and cognitive load theories are two theories that will help to analyze the usefulness of using AI systems in learning languages. Constructivist learning theories stress the importance of involvement in learning while cognitive load theories discuss the necessity of taking care of the cognitive load of the students. Both theories can help in analyzing the pros and cons of using such software.

Moreover, the methodological framework draws on digital linguistics, a blend of linguistic science and computing techniques that help model languages. Digital linguistics offers a multidisciplinary viewpoint that enables a more profound comprehension of artificial intelligence's role in processing languages and its applications in education through language acquisition (Rinekso et al., 2025).

Results and Discussion

The evaluation of the selected literature indicates that artificial intelligence has a vital importance for improving the English language instruction in terms of personalization, effectiveness, and engagement of learners. One of the critical advantages of the use of AI in language instruction is related to personalization. Through the implementation of adaptive learning algorithms, AI can analyze performance results of learners and develop an individual educational approach tailored specifically for them, thus helping to solve one of the critical problems of traditional education in classrooms. It was found out that adaptive learning leads to improved language skills in areas such as vocabulary development and reading comprehension (Esmaeilzadeh et al., 2024).

Yet another significant benefit of artificial intelligence technology lies in the automated process of feedback and assessment. Artificial intelligence-based applications are capable of assessing both written and spoken language with high levels of precision, thereby allowing learners to recognize and rectify their mistakes instantaneously. In addition to increasing the efficacy of learning, this capability relieves educators from the burden of manual feedback and assessment, thereby enabling them to concentrate on other aspects of education. For instance, automatic writing evaluation systems can examine grammar, vocabulary, coherency, and structure while providing constructive feedback (Kristiawan et al., 2024).

AI technology can also aid in promoting the engagement of learners through the creation of an engaging and immersive environment. With chatbots and conversational agents, learners can engage in language practice in a simulated real-world scenario. This will help to reduce anxiety levels among the learners while increasing their confidence levels. The use of AI technology allows for continuous learning through practice even outside the classroom environment. AI technology can be used to facilitate multimodal learning.

Nonetheless, the research findings also bring to light a number of challenges that could arise as a result of implementing artificial intelligence in the teaching of the English language. Firstly, there is the danger of fostering an addiction to technology on the part of the learners, since the technology can help them complete certain tasks but at the same time prevent them from thinking and solving problems independently.

The other equally important aspect is that of academic honesty. The existence of generative AI technology has allowed students to complete writing assignments without necessarily going through the learning process. Plagiarism is a concern in this regard since the student will not have learned anything from the assignment. There is a challenge in assessing student works in order to distinguish between those completed by the students themselves and those done using AI (Hockly, 2023).

The question of ethicality is also extremely important when discussing artificial intelligence within education. Matters like data protection, discrimination, and lack of transparency should be discussed in order to ensure the use of technologies responsibly. It is also possible that over-reliance

on AI would limit interactions between students and educators and would hinder development of communicative skills and competencies.

To address these issues, this paper suggests a blended approach to teaching, whereby artificial intelligence is merged with human-based educational approaches. The focus of this approach will be on the need for teachers' guidance, critical analysis, and ethical considerations when employing AI systems. Contrary to the belief that AI would render traditional ways of teaching obsolete, it is argued that AI ought to complement traditional education.

In addition, the suggested framework underscores the importance of ensuring that teachers undergo training on how to incorporate AI technologies effectively in their lessons. The teachers must be able to evaluate the effectiveness of the various AI tools used and the possible limitations involved in their use. There should also be the development of appropriate policies and guidelines for using AI technology in academic settings.

Conclusion

In summary, artificial intelligence can play a vital role in improving the effectiveness of English language instruction through personalized learning, automated testing, and improved learner engagement. Nevertheless, the implementation of artificial intelligence in educational settings also poses certain challenges, such as problems associated with learner dependency, academic honesty, and ethics. The main issue that emerges from this research is the absence of pedagogical approaches for utilizing artificial intelligence in language education.

The results reveal that a balance between technology and humans is needed in order to create a positive implementation of artificial intelligence. The model of hybrid pedagogy presented here can help in solving this issue due to its focus on the teacher, critical thinking, and ethics.

In conclusion, the success of future English language teaching programs will hinge on the effectiveness of educators and researchers in coming up with strategies that incorporate the use of AI in such a manner that learning gains are realized without compromising the essence of education. More research still needs to be done concerning the effect of AI on language teaching programs.

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APPLICATION OF DIGITAL TECHNOLOGIES IN EDUCATION: A SCIENTIFIC PERSPECTIVE

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Abstract: In modern times, digital technologies have become an integral part of the education system. The integration of digital technologies into education has fundamentally transformed teaching and learning processes. This article examines the role of digital tools in enhancing educational accessibility, personalization, and engagement. It also evaluates the challenges associated with their implementation, including the digital divide and data privacy concerns. At the same time, difficulties arising during the application of these technologies and prospective development directions are investigated. By analyzing current trends and technological innovations, this study highlights the potential of digital education to create more inclusive and efficient learning environments. Digital technologies strive to decrease or eliminate pollution and waste while increasing production and efficiency. These technologies have shown a powerful impact on the education system. The recent COVID-19 Pandemic has further institutionalized the applications of digital technologies in education. These digital technologies have made a paradigm shift in the entire education system. It is not only a knowledge provider but also a co-creator of information, a mentor, and an assessor. Technological improvements in education have made life easier for students. Instead of using pen and paper, students nowadays use various software and tools to create presentations and projects. When opposed to a weighty book, surfing an E-book is easier. These methods aid in increasing interest in research. This paper is brief about the need for digital technologies in education and discusses major applications and challenges in education.

Keywords: Digital education, artificial intelligence, online learning, educational technology, digital transformation

TƏHSİLDƏ RƏQƏMSAL TEXNOLOGİYALARIN TƏTBİQİ: ELMİ PERSPEKTİV

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Xülasə: Müasir dövrdə rəqəmsal texnologiyalar təhsil sisteminin ayrılmaz hissəsinə çevrilmişdir. Rəqəmsal texnologiyaların təhsilə inteqrasiyası tədris və öyrənmə proseslərini köklü şəkildə dəyişdirmişdir. Bu məqalə rəqəmsal alətlərin təhsildə əlçatanlığın, fərdiləşdirmənin və marağın artırılmasındakı rolunu araşdırır. Həmçinin onların tətbiqi ilə bağlı çətinliklər, o cümlədən rəqəmsal bərabərsizlik və məlumatların məxfiliyi problemləri qiymətləndirilir. Eyni zamanda, bu texnologiyaların tətbiqi zamanı yaranan çətinliklər və gələcək inkişaf istiqamətləri də tədqiq olunur. Mövcud tendensiyaların və texnoloji yeniliklərin təhlili əsasında bu tədqiqat rəqəmsal təhsilin daha inklüziv və səmərəli öyrənmə mühitləri yaratmaq potensialını vurğulayır. Rəqəmsal texnologiyalar çirklənmə və tullantıları azaltmağa və ya aradan qaldırmağa, eyni zamanda istehsal və səmərəliliyi artırmağa çalışır. Bu texnologiyalar təhsil sisteminə güclü təsir göstərmişdir. Son COVID-19 pandemiyası rəqəmsal texnologiyaların təhsildə tətbiqini daha da sürətləndirmiş və institusionallaşdırmışdır. Bu texnologiyalar bütün təhsil sistemində paradigma dəyişikliyinə səbəb olmuşdur. Onlar yalnız bilik verən vasitə deyil, həm də məlumatın birgə yaradıcısı, mentor və qiymətləndirici rolunu

oynayır. Təhsildə texnoloji yeniliklər tələbələrin həyatını asanlaşdırmışdır. Artıq tələbələr qələm və kağız əvəzinə müxtəlif proqram təminatı və alətlərdən istifadə edərək təqdimatlar və layihələr hazırlayırlar. Ağır kitablarla müqayisədə elektron kitabları oxumaq daha rahatdır. Bu üsullar tədqiqata marağın artmasına kömək edir. Bu məqalə təhsildə rəqəmsal texnologiyalara olan ehtiyacdən qısa şəkildə bəhs edir və təhsildə əsas tətbiqlər və problemləri müzakirə edir.

Açar sözlər: Rəqəmsal təhsil, süni intellekt, onlayn öyrənmə, təhsil texnologiyası, rəqəmsal transformasiya

1. Introduction

Digital learning, characterized by the integration of information and communication technologies (ICT) into educational practices, has become a cornerstone of modern education [7]. The 21st century plays a role as the age of information and technology. The field of education has not been left out of these changes. Digital technology has transformed learning methods by opening up new opportunities for teachers and their skills. According to Bereznoy (2018), the implementation of the digital revolution on a global scale is increasingly bringing us into a new reality [2]. According to W. Dittler (2017), today, the trend in education is the digital revolution, which, on the one hand, affects the labor market and requires the formation of new competencies among teachers, and on the other hand, leads to the restructuring of the entire education system [10]. Electronic resources, online platforms and interactive programs make the learning process more flexible and accessible. The evolution of digital technology has led to significant changes across multiple sectors, including education. The adoption of digital platforms, artificial intelligence, and online resources has reshaped traditional pedagogical approaches. Technologies such as Google Classroom and Moodle have enabled institutions to deliver content beyond physical classrooms, thereby expanding educational reach. The global shift toward digital learning was accelerated during the COVID-19 pandemic, which forced institutions worldwide to adopt remote teaching strategies. This transition highlighted both the opportunities and limitations of digital education systems.

2. Literature Review

Research indicates that digital technologies improve learning outcomes when effectively integrated into instructional design. E-learning platforms, such as Khan Academy and Coursera, provide flexible learning opportunities and access to high-quality educational materials. Studies show that multimedia content and interactive tools enhance student engagement and retention rates. According to Danby et al., (2020), we refer to digital technology as tools, systems, and devices that can create, store, or process data. The data processing and logic capabilities of digital technology are enabled by microprocessors that are programmed to perform a variety of functions. Digital technology refers to devices such as personal computers and tablets, tools such as cameras, calculators and other digital devices, systems such as software and apps, augmented reality and virtual reality, and less tangible forms of technology such as the internet. (Danby, S., Fleeer, M., Davidson, C., & Hatzigianni, M. (2018)).

Additionally, the application of Artificial Intelligence in education has enabled adaptive learning systems that personalize content based on individual student performance. These systems use algorithms to identify learning gaps and adjust instructional materials accordingly

3. Methodology

This article employs a qualitative approach, analyzing secondary data from academic journals, institutional reports, and case studies. Comparative analysis is used to evaluate traditional and digital learning environments, focusing on accessibility, engagement, and effectiveness.

4. Applications of Digital Technologies in Education

In modern times, digital technologies play an important role in the field of education and radically change the teaching and learning processes. The rapid development of information and communication technologies (ICT) makes the acquisition of knowledge more accessible, flexible and interactive [4].

Availability of information:

Digital technologies increase the accessibility of education. Thanks to online platforms, distance learning opportunities and electronic textbooks, pupils and students can study at any place and time. This is a great advantage, especially for those living in the regions or those with limited physical capabilities. Access to vast databases is possible through the Internet.

Personalized learning: Adaptive programs create lessons tailored to individual needs.

Interactivity: Multimedia tools make the learning process more interesting and effective.

Distance education: Online courses and platforms eliminate geographical threats.

4.1 E-Learning and Virtual Classrooms

Digital platforms such as Zoom and Microsoft Teams facilitate real-time interaction between students and instructors. These tools support synchronous and asynchronous learning, making education more flexible. Technology improves the quality of teaching. At the same time, digital technologies play a big role in the professional development of teachers. Online trainings, webinars and electronic resources help teachers to constantly update their knowledge and skills, encourage the use of innovative methods in teaching. Digital technologies also offer important advantages in the evaluation process. Online tests, automated review systems and e-portfolios for objective and objective learning outcomes provides prompt assessment. Digital assessment tools to students immediately gives an opportunity to receive feedback, which allows them to realize their mistakes in time and their knowledge creates conditions for its improvement. Digital classrooms are defined by using electronic devices or platforms such as social media, multimedia, and mobile phones to teach students. With digital technology in education, today's educational landscape has altered for the better or improvements. Digital learning is a learning strategy that employs technology to fulfil the entire curriculum and allows students to learn quickly and rapidly [[5], [8], [9]]. The digital classroom entirely focuses on teaching via the use of technology. Students use technological or internet-connected gadgets like laptops, tablets, Chromebooks, etc. Instead of taking notes on what the teacher has taught, most of the curriculum is delivered to students online through an engaging and interactive platform.

4.2 Digital Content and Open Educational Resources

The availability of digital textbooks, video lectures, and open-access journals has expanded learning opportunities. Students can access information anytime and anywhere, promoting self-directed learning. Interactive whiteboards, digital presentations, simulations and virtual labs make it possible to explain topics in a clearer and more interesting way. This increases the interest of learners in the lesson and enables better assimilation of knowledge. It should be noted that the role of the teacher is a decisive factor in the effective application of digital technologies. Teacher not only a knowledge transferer but also a guide, guide and motivator in the digital environment

performs the function. In this regard, the formation of digital competence of the teacher, modern

it is important to develop the ability to purposefully use technologies.

4.3 Personalized Learning Systems

AI-driven platforms analyze student data to provide customized learning experiences. This approach accommodates different learning styles and paces, improving overall academic performance.

4.4 Collaboration and Communication Tools

Digital communication platforms, including Slack, enable collaborative learning through group discussions, file sharing, and project management.

5. Challenges and Limitations

Despite the benefits, several challenges hinder the effective implementation of digital technologies in education:

- **Digital Divide:** Unequal access to devices and internet connectivity
- **Data Privacy Risks:** Concerns over the security of student information, Poor digital skills of teachers, cyber security risks, distraction and misuse of technology
- **Technological Dependence:** Over-reliance on digital tools may reduce critical thinking
- **Reduced Social Interaction:** Limited face-to-face communication affects social development

6. Discussion

The findings suggest that digital technologies significantly enhance the quality and accessibility of education. However, their success depends on proper implementation, teacher training, and infrastructure development. Policymakers must address equity issues to ensure that all learners benefit from technological advancements.

Prospects and Future Development

Aside from the ability to communicate information anywhere, at any time, social media sites are also a fantastic source of producing networking possibilities to establish social activities and possibly new jobs. Will digital media replace school books in a short term? If children start to prefer computers instead of instructors, will schools be able to maintain their existence in their present condition? It is said that new technologies may not be made a part of the present curriculums easily, they will dig their foundation and will transform them. The reason is that since young people of today's world have grown to the information society, they are more accustomed to technologies of this society if compared with many grown-ups including their instructors. In the future, the role of technology in education, such as artificial intelligence, augmented reality and virtual reality, has increased. This potential teacher development and technology literacy should be one of the top priorities. (Procedia-Social and Behavioral Sciences, 174 (2015), pp. 636-640)

7. Conclusion

Digital technologies have become an integral component of modern education systems. Their ability to provide flexible, personalized, and engaging learning experiences makes them essential for future educational development. While challenges remain, strategic planning and investment can maximize their potential and minimize associated risks. In modern times, digital technologies have become an integral part of the education system. In this article, the integration of digital tools into the educational process, their pedagogical effectiveness and impact on learning outcomes are

analyzed from a scientific point of view. At the same time, difficulties arising during the application of these technologies and prospective development directions are investigated.

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THE ROLE OF ONLINE AND HYBRID LEARNING MODELS IN TEACHING ENGLISH AS A FOREIGN

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Abstract: The rapid development of digital technologies has significantly transformed the field of education, particularly foreign language teaching. Among the most influential innovations are online and hybrid learning models, which have reshaped how English is taught and learned in modern classrooms (Bates, 2020; Anderson, 2019). This study aims to examine the role of these models in teaching English as a foreign language (EFL), with a particular focus on their impact on learners' language skills, motivation, and autonomy. The research is based on qualitative analysis of recent academic literature and educational practices related to digital learning. It explores the key advantages of online and hybrid learning environments, including flexibility, accessibility, and increased exposure to authentic language materials, all of which contribute to more effective language acquisition (Dudeney & Hockly, 2021; Means et al., 2013). At the same time, the study identifies several challenges, such as reduced face-to-face interaction, technical limitations, and the need for sufficient digital literacy among both teachers and learners (Müller, 2021; UNESCO, 2022). Furthermore, the article highlights the critical role of teachers in managing digital and blended learning environments, emphasizing the importance of pedagogical adaptation and technological competence. The findings suggest that when effectively implemented, online and hybrid learning models enhance learner-centered education and improve overall learning outcomes (Bates, 2020). However, successful integration requires a balanced combination of traditional teaching methods and modern digital tools. The study concludes that hybrid learning, in particular, represents one of the most effective and sustainable approaches in contemporary English language education, offering a comprehensive framework that supports both flexibility and interaction.

Keywords: online learning, hybrid learning, English teaching, digital education, ICT, language acquisition, blended learning

İNGİLİS DİLİNİN XARİCİ DİL KİMİ TƏDRİSİNDƏ ONLAYN VƏ HİBRİD TƏLİM MODELLƏRİNİN ROLU

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Xülasə: Son illərdə rəqəmsal texnologiyaların sürətli inkişafı təhsil sahəsində, xüsusilə də xarici dillərin tədrisində mühüm dəyişikliklərə səbəb olmuşdur. Bu dəyişikliklər çərçivəsində onlayn və hibrid təlim modelləri müasir təhsil sisteminin əsas komponentlərindən birinə çevrilmiş və ingilis dilinin tədrisində yeni imkanlar yaratmışdır (Bates, 2020; Anderson, 2019). Məqalənin əsas məqsədi bu təlim modellərinin ingilis dilinin xarici dil kimi (EFL) öyrədilməsində rolunu araşdırmaq, onların dil bacarıqlarına, motivasiyaya və öyrənən müstəqilliyinə təsirini təhlil etməkdir. Tədqiqat keyfiyyət (qualitative) yanaşma əsasında aparılmış, müasir elmi ədəbiyyat, akademik məqalələr və beynəlxalq hesabatlar təhlil edilmişdir. Araşdırma nəticələri göstərir ki, onlayn və hibrid təlim modelləri tədris prosesinə elastiklik və əlçatanlıq gətirməklə yanaşı, tələbələrin autentik dil materiallarına çıxış imkanlarını genişləndirir və dil bacarıqlarının inkişafına müsbət təsir göstərir (Dudeney & Hockly, 2021). Bu modellər həmçinin öyrənənlərin müstəqil işləmə bacarıqlarını gücləndirir və onların təlim prosesində daha aktiv iştirakını təmin edir. Bununla yanaşı, tədqiqat müəyyən çətinlikləri də üzə çıxarmışdır. Üz-üzə ünsiyyətin azalması, texniki problemlər, internet və texnoloji vasitələrə çıxışın məhdudluğu, eləcə də rəqəmsal savadlılıq səviyyəsinin aşağı olması tədrisin effektivliyinə mənfi təsir göstərə bilər (Müller, 2021; UNESCO, 2022). Xüsusilə danışıq bacarıqlarının inkişafı üçün canlı ünsiyyətin əhəmiyyəti hələ də aktual olaraq qalır. Məqalədə müəllimin rolu xüsusi vurğulanır. Müasir dövrdə müəllimlər yalnız bilik ötürən deyil, həm də rəqəmsal mühitdə öyrənmə prosesini təşkil edən, tələbələri motivasiya edən və texnologiyaları effektiv şəkildə inteqrasiya edən fasilitator kimi çıxış

etməlidirlər (Anderson, 2019). Müəllimin pedaqoji və rəqəmsal kompetensiyalarının inkişafı bu təlim modellərinin uğurlu tətbiqi üçün əsas şərtlərdən biridir. Nəticə olaraq, araşdırma göstərir ki, hibrid təlim modeli ən effektiv yanaşmalardan biri hesab olunur, çünki o, ənənəvi və rəqəmsal tədrisin üstünlüklərini birləşdirir. Bu model həm interaktivliyi, həm də elastikliyi təmin edərək daha səmərəli təlim mühiti yaradır (Bates, 2020). Gələcəkdə təhsil sistemində rəqəmsal texnologiyaların rolu daha da artacaq və bu istiqamətdə davamlı tədqiqatlar və innovativ yanaşmalar vacib olacaqdır.

Açar sözlər: onlayn təlim, hibrid təlim, ingilis dilinin tədrisi, rəqəmsal təhsil, İKT (informasiya və kommunikasiya texnologiyaları), dil mənimsənilməsi, qarışıq təlim

Introduction

In the 21st century, education systems worldwide have undergone profound transformations driven by the rapid development of information and communication technologies (ICT). These technological advancements have significantly reshaped teaching and learning processes across all disciplines, particularly in the field of foreign language education (Bates, 2020; UNESCO, 2022). English language teaching (ELT), as one of the most globally demanded educational areas, has been especially influenced by these innovations, leading to the emergence of new pedagogical models and instructional strategies. Traditionally, language teaching relied heavily on face-to-face interaction and teacher-centered approaches. However, with the integration of digital tools and online platforms, this paradigm has shifted toward more learner-centered, flexible, and interactive environments (Anderson, 2019). Online and hybrid learning models have become increasingly prominent in schools, universities, and language institutions worldwide, offering alternative ways of delivering instruction and facilitating language acquisition (Dudeney & Hockly, 2021). Online learning refers to educational processes conducted entirely through digital platforms, where learners access materials, participate in discussions, and complete assignments virtually. In contrast, hybrid or blended learning combines traditional classroom instruction with online components, creating a more balanced and dynamic learning environment (Anderson, 2019). These models not only expand access to education but also provide opportunities for personalized learning and increased learner autonomy. The global COVID-19 pandemic accelerated the adoption of these models, forcing educational institutions to transition rapidly from traditional classrooms to fully online or hybrid systems. During this period, digital platforms such as learning management systems and video conferencing tools became essential for maintaining continuity in education (UNESCO, 2022). As a result, both teachers and students had to adapt quickly to new modes of teaching and learning, highlighting the importance of digital competence and technological readiness. Despite the initial challenges, the widespread implementation of online and hybrid learning has demonstrated several long-term benefits. These include flexibility in time and location, access to a wide range of authentic learning materials, and opportunities for interactive and collaborative learning (Dudeney & Hockly, 2021). Moreover, such models support the development of essential 21st-century skills, including critical thinking, digital literacy, and self-directed learning (Müller, 2021). However, the transition to digital learning environments also presents significant challenges. Issues such as limited face-to-face interaction, technical difficulties, unequal access to technology, and varying levels of digital literacy can negatively affect the quality of education (Bates, 2020). Furthermore, maintaining student engagement and motivation in online settings remains a critical concern for educators. In this context, the role of teachers becomes increasingly important, as they must not only deliver content but also facilitate interaction, guide learners, and effectively integrate technology into their teaching practices (Anderson, 2019). The success of online and hybrid

learning largely depends on how well these elements are balanced and implemented. Therefore, the purpose of this article is to analyze how online and hybrid learning models contribute to the teaching of English as a foreign language (EFL), with a particular focus on their impact on language skill development, learner motivation, and instructional effectiveness. The study also aims to evaluate the advantages and limitations of these models in real educational contexts and to highlight their significance in shaping the future of English language education.

Material and Methods

In the 21st century, education systems worldwide have undergone profound transformations driven by the rapid development of information and communication technologies (ICT). These technological advancements have significantly reshaped teaching and learning processes across all disciplines, particularly in the field of foreign language education (Bates, 2020; UNESCO, 2022). English language teaching (ELT), as one of the most globally demanded educational areas, has been especially influenced by these innovations, leading to the emergence of new pedagogical models and instructional strategies. Traditionally, language teaching relied heavily on face-to-face interaction and teacher-centered approaches. However, with the integration of digital tools and online platforms, this paradigm has shifted toward more learner-centered, flexible, and interactive environments (Anderson, 2019). Online and hybrid learning models have become increasingly prominent in schools, universities, and language institutions worldwide, offering alternative ways of delivering instruction and facilitating language acquisition (Dudeney & Hockly, 2021; Chapelle, 2001; Müller, 2021). Online learning refers to educational processes conducted entirely through digital platforms, where learners access materials, participate in discussions, and complete assignments virtually. In contrast, hybrid or blended learning combines traditional classroom instruction with online components, creating a more balanced and dynamic learning environment (Anderson, 2019). These models not only expand access to education but also provide opportunities for personalized learning and increased learner autonomy. The global COVID-19 pandemic accelerated the adoption of these models, forcing educational institutions to transition rapidly from traditional classrooms to fully online or hybrid systems. During this period, digital platforms such as learning management systems and video conferencing tools became essential for maintaining continuity in education (UNESCO, 2022). As a result, both teachers and students had to adapt quickly to new modes of teaching and learning, highlighting the importance of digital competence and technological readiness. Despite the initial challenges, the widespread implementation of online and hybrid learning has demonstrated several long-term benefits. These include flexibility in time and location, access to a wide range of authentic learning materials, and opportunities for interactive and collaborative learning (Dudeney & Hockly, 2021). Moreover, such models support the development of essential 21st-century skills, including critical thinking, digital literacy, and self-directed learning (Müller, 2021). However, the transition to digital learning environments also presents significant challenges. Issues such as limited face-to-face interaction, technical difficulties, unequal access to technology, and varying levels of digital literacy can negatively affect the quality of education (Bates, 2020). Furthermore, maintaining student engagement and motivation in online settings remains a critical concern for educators. In this context, the role of teachers becomes increasingly important, as they must not only deliver content but also facilitate interaction, guide learners, and effectively integrate technology into their teaching practices (Anderson, 2019). The success of online and hybrid learning largely depends on how well these elements are balanced and implemented. Therefore, the purpose of this article is to analyze how online and hybrid learning

models contribute to the teaching of English as a foreign language (EFL), with a particular focus on their impact on language skill development, learner motivation, and instructional effectiveness. The study also aims to evaluate the advantages and limitations of these models in real educational contexts and to highlight their significance in shaping the future of English language education.

Results and Discussion

The findings of this study indicate that online and hybrid learning models have a significant impact on the effectiveness of teaching English as a foreign language (EFL). The analysis reveals that these models not only transform instructional delivery but also influence learner engagement, motivation, and overall language acquisition processes (Bates, 2020; Anderson, 2019).

1. Effectiveness of Online Learning

Online learning environments provide learners with flexible access to educational resources, allowing them to study at their own pace and according to their individual needs. This flexibility is particularly beneficial in language learning, where repetition and exposure play a crucial role. Digital platforms enable students to access a wide range of authentic materials, such as videos, podcasts, and interactive exercises, which significantly enhance listening and reading skills (Dudeney & Hockly, 2021). Moreover, online learning promotes learner autonomy. Students are encouraged to take responsibility for their own learning process, which leads to increased motivation and self-regulation. According to Müller (2021), learners in digital environments tend to develop stronger independent learning strategies compared to those in traditional classrooms.

2. Advantages of Hybrid Learning

Hybrid learning is widely considered one of the most effective approaches in modern education because it combines the strengths of both traditional and online instruction. Face-to-face interaction allows for immediate feedback, social interaction, and the development of speaking skills, while online components provide flexibility and access to additional resources (Anderson, 2019). Research shows that hybrid learning environments improve student performance and satisfaction by creating a more balanced and engaging learning experience (Bates, 2020). In such settings, students benefit from both teacher guidance and independent learning opportunities, which leads to deeper understanding and better retention of knowledge.

3. Challenges and Limitations

Despite their advantages, online and hybrid learning models also present several challenges. One of the main issues is the lack of direct face-to-face interaction in fully online environments, which can negatively affect communication skills, particularly speaking. Students may feel isolated, and opportunities for spontaneous conversation are limited (Müller, 2021). Technical problems, such as unstable internet connections and limited access to devices, also hinder the effectiveness of digital learning. These issues are especially common in regions with insufficient technological infrastructure (UNESCO, 2022). Another significant challenge is the varying level of digital literacy among both teachers and students. Some individuals may struggle to use digital platforms effectively, which can reduce the quality of the learning experience (Dudeney & Hockly, 2021).

4. Role of Teachers in Learning Outcomes

The results highlight that the success of online and hybrid learning largely depends on the teacher's ability to adapt to digital environments. Teachers are required to design interactive lessons, integrate multimedia tools, and maintain student engagement through innovative teaching strategies (Bates, 2020). In addition, teachers must provide continuous feedback and support to students, as this plays a crucial role in maintaining motivation and improving performance. Anderson (2019) emphasizes

that effective teaching in online environments requires a shift from traditional instructor-centered approaches to more facilitative and learner-centered methods.

5. Impact on Language Skill Development

The analysis shows that online and hybrid learning models positively influence the development of all four language skills. Listening and reading skills benefit greatly from exposure to digital content, while writing skills improve through online assignments and communication tools. Speaking skills, although supported through video conferencing, may still require face-to-face interaction for full development (Dudeney & Hockly, 2021).

6. Student Engagement and Motivation

Student motivation is a key factor in successful language learning. The use of interactive tools such as quizzes, games, and multimedia content increases engagement and makes learning more enjoyable. Hybrid learning environments, in particular, are effective in maintaining student interest, as they combine the social aspects of classroom learning with the flexibility of online education (Müller, 2021; UNESCO, 2022). Overall, the results demonstrate that online and hybrid learning models offer significant advantages in EFL teaching, particularly in terms of flexibility, accessibility, and learner autonomy. However, their effectiveness depends on proper implementation, technological support, and the active involvement of teachers and students.

Conclusion

In conclusion, the integration of online and hybrid learning models has brought significant changes to the field of English language teaching. Hybrid learning, in particular, represents the most effective approach due to its balanced structure (Means et al., 2013; Bates, 2020). These models have transformed traditional instructional approaches by introducing greater flexibility, accessibility, and learner-centered practices. As the findings of this study indicate, both online and hybrid learning environments provide valuable opportunities for enhancing language acquisition and developing essential 21st-century skills (Bates, 2020; Anderson, 2019). Online learning, in particular, enables learners to access a wide range of authentic materials and digital resources, which contribute to the improvement of listening, reading, and writing skills. At the same time, hybrid learning offers a more balanced approach by combining the advantages of face-to-face interaction with the benefits of digital technologies. This combination allows for more effective communication, immediate feedback, and increased student engagement (Dudeney & Hockly, 2021). However, despite these advantages, several challenges remain. Technical issues, limited access to digital tools, and varying levels of digital literacy can negatively impact the effectiveness of online education. In addition, the lack of direct social interaction in fully online environments may reduce opportunities for developing speaking skills and interpersonal communication (Müller, 2021; UNESCO, 2022). The study also highlights the crucial role of teachers in ensuring the success of digital learning. Teachers must adapt to new technological environments, develop innovative teaching strategies, and maintain active student engagement. Their ability to effectively integrate digital tools into the learning process is a key factor in achieving successful educational outcomes (Anderson, 2019). Overall, the findings suggest that hybrid learning represents one of the most effective approaches in contemporary English language education, as it combines the strengths of both traditional and online methods. For optimal results, educational institutions should focus on improving technological infrastructure, providing teacher training, and promoting digital literacy among learners. In the future, the role of digital technologies in education is expected to grow even further. Therefore, continuous research and innovation in this field are essential to ensure that

online and hybrid learning models are used effectively and sustainably in teaching English as a foreign language.

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POSSIBILITIES OF APPLYING ARTIFICIAL INTELLIGENCE IN EDUCATION

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Abstract. Although artificial intelligence technologies are often overlooked, they have become an indispensable part of our daily lives. These technologies are available on every platform through various devices and programs. Examples of artificial intelligence technologies include smart home appliances, autonomous cars, and smartphone programs. On the other hand, very few people are aware of the applications and concepts behind these technologies. It is thought that the widespread use of these technologies, which we use almost every moment of our lives, in the education sector can provide significant development. The education system is generally built on four main components: students, teachers, curriculum, and educational environment. The quality of the relationships between these components increases the quality of education. It is stated that the critical variable for the most effective implementation of this process is the teacher (Aykaç, 2018: vii). To date, programs developed for teachers to use artificial intelligence technologies can only be called methods that can be implemented using computers (with a few exceptions). However, these methods are not enough to say that artificial intelligence is actively used in education (Timms, 2016: 701-702). Therefore, it can be said that these programs are generally used for data storage and constitute only a small part of the activities in a typical classroom.

Keywords: artificial intelligence, application of artificial intelligence in education, school, teacher, technology

TƏHSİLDƏ SÜNİ İNTELLEKTİN TƏTBİQİ İMKANLARI

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Xülasə. Süni intellekt texnologiyaları tez-tez diqqətdən kənarda qalmasına baxmayaraq, gündəlik həyatımızın əvəzolunmaz hissəsinə çevrilib. Bu texnologiyalar müxtəlif cihazlar və proqramlar vasitəsilə hər platformada mövcuddur. Süni intellekt texnologiyalarına misal olaraq ağıllı məişət texnikası, avtomobillər və smartfon proqramlarını göstərmək olar. Digər tərəfdən demək olar ki, çox az adam bu texnologiyaların arxasındakı tətbiqlər və konsepsiyalardan xəbərdardır. Həyatımızın demək olar ki, hər anında istifadə etdiyimiz bu texnologiyaların təhsil sektorunda geniş şəkildə istifadəsinin əhəmiyyətli inkişaf təmin edə biləcəyi düşünülür. Təhsil sistemi, ümumiyyətlə, dörd əsas komponent üzərində qurulur: tələbələr, müəllimlər, kurikulum və təhsil mühiti. Bu komponentlər arasındakı əlaqələrin keyfiyyəti təhsilin keyfiyyətini artırır. Bu prosesin ən effektiv şəkildə həyata keçirilməsi üçün kritik dəyişənin müəllim olduğu bildirilir (Aykaç, 2018: vii). Bu günə qədər müəllimlər üçün süni intellekt texnologiyalarında istifadə etmək üçün hazırlanmış proqramlar (bir neçə istisna olmaqla) yalnız kompüterlər vasitəsilə həyata keçirilə bilən üsullar adlandırıla bilər. Lakin bu üsullar süni intellektin təhsildə fəal şəkildə istifadə edildiyini söyləmək üçün kifayət deyil (Timms, 2016: 701-702). Buna görə də demək olar ki, bu proqramlar ümumiyyətlə məlumatların saxlanması üçün istifadə olunur və tipik bir sinif otağında fəaliyyətlərin yalnız kiçik bir hissəsini təşkil edir.

Açar sözlər:süni intellekt, təhsil sahəsində süni intellektin tətbiqi, məktəb, müəllim, texnologiya

Introduction

The rapid development of artificial intelligence (AI) technologies is having a serious impact on various areas of modern society, including the education system. Artificial intelligence acts as an important tool in solving problems such as personalization in education, effective assessment and resource optimization. The limitations of traditional teaching methods, especially the difficulty of an individual approach in large audiences, necessitate the application of new technological solutions. In this regard, artificial intelligence-based systems allow for a more accurate identification of students' learning needs and the formulation of appropriate teaching strategies. The aim of the article is to explore the possibilities of applying artificial intelligence in education, analyze its advantages and current challenges.

Methodology

The research used both qualitative and quantitative methods. First, the existing scientific literature on the topic was systematically analyzed. Then, artificial intelligence-based platforms applied in various educational institutions were comparatively examined. The attitude of teachers and students to these technologies was studied through surveys and interviews. At the same time, statistical data obtained from some educational platforms was analyzed and the impact of artificial intelligence on teaching outcomes was assessed. The collected data was processed using thematic analysis and statistical methods. Although often overlooked, artificial intelligence technologies have become an indispensable part of our daily lives. These technologies are available on every platform through various devices and applications. Examples of artificial intelligence technologies include

smart home appliances, autonomous cars, and smartphone applications. On the other hand, very few people are aware of the applications and concepts behind these technologies. It is thought that the widespread use of these technologies, which we use almost every moment of our lives, in the education sector can provide significant development.

Results and discussion

The education system is generally built on four main components: students, teachers, curriculum, and educational environment. The quality of the relationships between these components increases the quality of education. It is stated that the critical variable for this process to be implemented most effectively is the teacher (Aykaç, 2018: vii). To date, programs developed for teachers to use artificial intelligence technologies can only be called methods that can be implemented via computers (with a few exceptions). However, these methods are not enough to say that artificial intelligence is actively used in education (Timms, 2016: 701-702). Therefore, it can be said that these programs are generally used for data storage and constitute only a small part of the activities in a typical classroom. One of the biggest problems faced in the education system is that people learn in different ways and at different rates (Saribaş and Babadağ, 2015: 18-34). Although each student has different learning abilities and different interests, schools try to implement a uniform type of education. However, it can be said that some students are more analytical thinking skills, while others are more creative, literary or communicative skills (Boydak, 2015). Chopra (URL-2) states that the goal is to adapt the educational materials for each student to their abilities, preferred learning styles and experiences using artificial intelligence technologies. Education systems around the world are constantly improving to incorporate artificial intelligence applications. In this context, research on the topic is important to inform about the use and benefits of AI in education, to keep educators informed about new technologies and to adapt to new technologies. In addition, this study is important in terms of providing current information and resources for those who want to work on AI-based education. This study focuses on the current problems and benefits that AI can provide in education worldwide, considering its use in various fields, from psychology and medicine to the retail sector and engineering. As the use of artificial intelligence technologies is expected to increase in the coming years in the world, technologies developed and implemented to enable students to learn more effectively, while relying on computers and tablets, will be discussed. The study was conducted using the literature review method. A review of relevant literature was conducted to investigate the development of artificial intelligence technologies, their areas of use in education, and the benefits of their use in education. This study is limited to the sources that the researcher was able to obtain in the local and international literature. Artificial intelligence is defined as systems that imitate the human brain to perform certain tasks and that repeatedly improve themselves through the experience gained as a result of performing these tasks (URL-10). Gordon (2011) describes artificial intelligence as an analytical life set that aims to imitate life. The term artificial intelligence was first used by McCarthy and his colleagues in a summer school research project in 1955. This project was based on the assumption that any learning or other characteristics of intelligence can, in principle, be replicated by a machine (McCarthy et al., 1955: 12). Another pioneer of the application of artificial intelligence in machines is Alan Turing, who not only laid the foundations of computer science, but also explored the philosophical aspects of the development of artificial intelligence (Muggleton, 2014: 4). Since its first use in 1955, the concept of artificial intelligence has been divided into various subfields depending on the nature of the problem to be solved. These include artificial

neural networks (ANN), fuzzy logic, simulated annealing, expert systems, computer vision, genetic algorithms, speech recognition, chaotic modeling and robotics. All of these subfields form the basis of many technological devices that we use in our lives, and these devices, in turn, make people's lives easier (Civalek, 2003: 40-50). Since the first use of the term artificial intelligence, many scientists have conducted research on this topic. Thanks to the technologies that have developed over the years, artificial intelligence technologies have become a part of our lives. The current situation shows that these technologies will be present at every stage of the projects being worked on and targeted (Güllüpinar et al., 2013: 195-216). Today, artificial intelligence technologies are used in many fields and it can be said that they have also started to show themselves in the field of education. It can be said that artificial intelligence technologies can provide innovations in the field of education and training and these innovations bring many contributions. In 1952, the first program capable of playing chess with artificial intelligence was written. In 1968, Terry Winograd's thesis showed that computers can understand English sentences. In 1997, a chess computer called Deep Blue defeated world chess champion Kasparov. In 2009, Google began developing self-driving cars. In 1961, General Motors created the first commercial robot, the Unimate. In 1986, Honda began a secret project to develop a humanoid robot. In 2000, Honda designed ASIMO, an intelligent humanoid robot. In 2017, Deep Mind researchers succeeded in adding memory to artificial intelligence. It can be noted that AI applications have recently been successfully used to perform complex tasks in various fields such as healthcare, speech technology, energy, finance, and logistics (URL-1; Osoba and Welser, 2017: iii-iv; URL-6; URL-8; URL-9). Although AI programs have been around for a long time, their impact on education remains limited (URL-4). In the meantime, AI has certainly made its presence felt, if not completely taking over the classroom (Karsenti, 2019: 105-106). Recently, the term "school" has come to be defined as a place where students and teachers come together, physically and virtually, to participate in teaching and learning activities (Timms, 2016: 703). The emphasis on the term "virtual" suggests that AI applications are now beginning to be integrated into education. While the question of how AI applications can be effectively integrated into education is still being debated, many predictions and ideas are being considered. Some of these predictions and ideas focus on which tasks teachers can perform in the classroom with AI applications (URL-3). While this issue is still being debated, some applications where AI technologies are used in education are already visible today. Although these applications are not used regularly and widely, they allow students to effectively access the information they are looking for and learn at their own pace. The fact that AI applications allow students to learn at their own level and pace of learning is seen as their greatest advantage. In addition, since some applications offer flexible working hours and allow students to participate in classes when they feel motivated, it can be said that maximum efficiency is achieved in terms of educational outcomes (URL-6; Popenici and Kerr, 2017). It is observed that AI technologies are increasing the opportunities to teach and design learning interactions and educational experiences for all students globally (Popenici and Kerr, 2017: 5). Accordingly, it can be noted that AI in education aims to provide inclusive, equitable and high-quality education and promote lifelong learning opportunities for all (URL-5). From a different perspective, educational administrators can use intelligent assistants to assist with various administrative needs such as budgeting, student applications and registration, course management, purchasing and procurement activities, cost management, and facilities using AI applications. Thus, educational administrators can free up time from their administrative duties to work towards improving education. The use of intelligent AI-powered

systems can significantly increase the efficiency of many educational institutions, reduce operating costs, provide them with greater visibility into income and expenses, and improve the overall responsiveness of educational institutions (URL-7). An examination of various studies on the topic reveals that AI applications can make significant contributions to education and training. These contributions, obtained from a literature review, can generally be summarized as follows (Karsenti, 2019; Karsenti, Bugmann, and Parent, 2019; Karsenti, Bugmann, and Gros, 2017). Tailoring learning to the individual needs of the student has been a priority for educators for years. Artificial intelligence will allow for appropriate differentiation for teachers, who have to manage an average of 30 students in each classroom. AI-based learning systems can provide teachers with useful information about their students' learning styles, skills, and progress, and can make suggestions for adapting teaching methods to the individual needs of students. Research shows that AI increases academic achievement. AI can provide intelligent support for collaborative learning. Years of research have shown that collaboration among a community of students participating in an online course can lead to higher learning outcomes than studying alone. Saves teachers time. There are programs that perform some tasks for teachers. For example, AI applications can save time by preparing the necessary materials and digitizing books. Provides continuous assessment and feedback. By providing feedback over time throughout the learning process, it allows teachers to identify and fill any knowledge gaps in students' understanding. Teachers can review their lessons as needed. For example, programs like Coursera and Udacity, used by over 10 million students worldwide, allow teachers to review lessons by showing if too many students are answering questions incorrectly or not learning enough. Individual tutoring programs can be provided through distance learning. The rapid expansion of mobile technologies has created opportunities for both students and teachers to learn remotely. In addition, applications such as Duolingo, a language learning system, allow learning anywhere, anytime, and instant assessments can be made. New ways of interacting with information allow for effective learning. For example, thanks to artificial intelligence used by Google, our searches are now tailored to our geographic location and previous searches. Educational feedback is provided. For example, the UTIFEN program sends students personalized topics to help them track their learning paths. Feedback is not only personalized, but also faster and more frequent, allowing for automatic assessment, support, and specific recommendations. Artificial intelligence applications can recommend different books based on the interests and needs of individuals. Audio communication can be provided between students and academic content. Students can interact with assistants such as HomePod, Amazon Echo, and Google Home to create a real-world audio conversation environment. Students can complete individual assignments tailored to their academic skills and learning levels. For example, the Alloprof online homework helper program helps students with this. Virtual reality and immersive environments are provided. Highly interactive, three-dimensional virtual worlds encourage students to interact with course materials. It provides authentic immersive experiences (subjective impressions of participating in a real experience) that simulate aspects of the real world that the user would otherwise not be able to experience (such as dangerous environments or geographically or historically inaccessible locations). For example, the educational game Assassin's Creed allows students to vividly relive the past while completing engaging tasks in vivid and detailed historical situations. Such enriched, interactive experiences have a direct positive impact on learning. Creates broader societal outcomes. In countries where access to education is difficult for large populations, online resources can have a positive impact if the population has the necessary means to access

them. The development of online educational resources can allow foundations that support international education programs to provide quality education and facilitate student access to courses. Can predict the risk of students dropping out of school. Artificial intelligence can collect student data and quickly alert school administrators to individuals at risk of dropping out, allowing them to receive appropriate support before the situation worsens. Can improve classroom management. For example, with the Classcraft AI application, learning can be achieved through virtual experiences. Since teaching is gamified with this application, everyone is involved. More effective administrative management can be achieved. Information such as news and student absences can be processed and evaluated quickly and easily. It provides authentic immersive experiences (subjective impressions of participating in a real experience) that simulate aspects of the real world that the user would otherwise not be able to experience (such as dangerous environments or geographically or historically inaccessible locations). For example, the educational game Assassin's Creed allows students to vividly relive the past while completing engaging tasks in vivid and detailed historical situations. Such enriched, interactive experiences have a direct positive impact on learning. Creates broader societal outcomes. In countries where access to education is difficult for large populations, online resources can have a positive impact if the population has the necessary means to access them. The development of online educational resources can allow foundations that support international education programs to provide quality education and facilitate student access to courses. Can predict the risk of students dropping out of school. Artificial intelligence can collect student data and quickly alert school administrators to individuals at risk of dropping out, allowing them to receive appropriate support before the situation worsens. Can improve classroom management. For example, with the Classcraft AI application, learning can be achieved through virtual experiences. Since teaching is gamified with this application, everyone is involved. More effective administrative management can be achieved. Information such as news and student absences can be processed and evaluated quickly and easily.

Conclusion

Artificial intelligence has emerged as one of the most transformative technologies of the 21st century and is increasingly influencing the field of education. The findings of this study demonstrate that AI-based tools and applications can significantly enhance teaching and learning processes by providing personalized learning experiences, continuous assessment, intelligent feedback, and improved administrative efficiency. The integration of AI into education enables students to learn according to their individual needs, abilities, and learning pace, while also helping teachers manage instructional and administrative tasks more effectively. AI-powered platforms support greater accessibility to educational resources, encourage student engagement, and contribute to improved academic performance. Furthermore, these technologies create opportunities for lifelong learning and help educational institutions adapt to the demands of the digital age. Despite these advantages, the successful implementation of AI in education requires careful consideration of challenges such as data privacy, ethical concerns, technological infrastructure, and digital inequality. Teachers continue to play a crucial role in guiding, motivating, and supporting students, and AI should be viewed as a complementary tool rather than a replacement for human educators. In conclusion, artificial intelligence has the potential to reshape education by making learning more personalized, accessible, and effective. As AI technologies continue to evolve, educational institutions, policymakers, and educators should work together to ensure their responsible and equitable use. Future research should focus on the long-term effects of AI on

educational outcomes, student development, and the changing role of teachers in technology-enhanced learning environments.

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ARTIFICIAL INTELLIGENCE IN ENGLISH LANGUAGE TEACHING: TOWARDS A HUMAN-CENTERED DIGITAL PEDAGOGY

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Abstract: The integration of artificial intelligence (AI) into contemporary educational systems has initiated a profound transformation in both pedagogical theory and instructional practice, particularly within the field of English Language Teaching (ELT). AI is no longer viewed merely as an auxiliary technological tool; rather, it functions as a transformative epistemic force that reshapes curriculum design, instructional decision-making, learner engagement, and assessment paradigms.

This paper presents a comprehensive conceptual and critical analysis of AI integration in ELT, with a particular focus on its pedagogical implications. It examines the affordances of AI-driven technologies, including adaptive learning systems, intelligent tutoring platforms, automated feedback mechanisms, and data-informed instructional planning. These innovations are shown to enhance personalization, efficiency,

and responsiveness in language learning environments, thereby supporting more individualized learning trajectories.

At the same time, the study critically interrogates the limitations and challenges associated with AI implementation in ELT. These include the potential reduction of meaningful human interaction in the learning process, ethical concerns related to data privacy and algorithmic transparency, and the risk of diminishing teacher autonomy and professional agency. Such issues highlight the necessity of a cautious and critically informed approach to AI integration in education.

The study is grounded in a qualitative and interpretive analytical framework and synthesizes recent theoretical perspectives from educational technology, applied linguistics, and digital pedagogy. The analysis indicates that while AI technologies can significantly enhance instructional efficiency and learner personalization, their pedagogical effectiveness is ultimately dependent on alignment with human-centered educational values.

The paper concludes that a hybrid pedagogical model represents the most sustainable approach to AI integration in ELT. In this model, artificial intelligence functions as a supportive and facilitative mechanism rather than a replacement for human instruction. Such an approach is essential for preserving the communicative, social, emotional, and cultural dimensions of language learning, which remain central to effective English language education.

Key words :Artificial intelligence; English Language Teaching (ELT); digital pedagogy; adaptive learning; automated assessment; teacher role; learner autonomy; human-centered education; educational technology

İNGİLİS DİLİNİN TƏDRİSİNDƏ SÜNİ İNTELLEKT: İNSAN MƏRKƏZLİ RƏQƏMSAL PEDAQOGİKA

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Xülasə: Bu məqalə süni intellektin (AI) İngilis dili tədrisinə (ELT) təsirini konseptual və tənqidi baxımdan araşdırır. Tədqiqat göstərir ki, AI texnologiyaları fərdiləşdirilmiş öyrənmə, avtomatlaşdırılmış qiymətləndirmə və dərhal geribildirim kimi imkanlar yaradaraq dil öyrənmə prosesini daha effektiv və çevik edir. Eyni zamanda, bu texnologiyaların istifadəsi müəllim rolunun dəyişməsi, insan ünsiyyətinin azalması və etik məsələlər kimi mühüm problemləri də ortaya çıxarır.

Araşdırma nəticələri göstərir ki, AI dil tədrisini tam əvəz etmir, əksinə onu dəstəkləyici vasitə kimi gücləndirir. Effektiv ELT mühiti insan yönümlü pedaqoji yanaşma əsasında qurulmalı və texnologiya yalnız tədris prosesini zənginləşdirən bir alət kimi istifadə edilməlidir. Bu balanslı yanaşma dil öyrənmənin kommunikativ, sosial və mədəni aspektlərinin qorunması üçün vacibdir.

Açar sözlər: Süni intellekt; İngilis dilinin tədrisi (ELT); rəqəmsal pedaqogika ; adaptiv təlim; müəllim rolu; avtomatlaşdırılmış qiymətləndirmə; şagirdin muxtariyyəti; insan mərkəzli təhsil; təhsil texnologiyası

Introduction

The rapid proliferation of artificial intelligence (AI) across diverse sectors has fundamentally reshaped the structure, functions, and expectations of contemporary education systems. In recent years, educational institutions worldwide have increasingly integrated AI-powered technologies to enhance instructional efficiency, optimize assessment practices, and support more individualized, adaptive, and data-informed learning pathways. This development is not merely incremental in

nature; rather, it represents a systemic transformation in how knowledge is generated, delivered, processed, and evaluated within modern educational environments.

Within the field of English Language Teaching (ELT), these changes are particularly visible and pedagogically significant. Language learning is now increasingly mediated by digital platforms and algorithmic systems capable of providing automated feedback, personalized learning trajectories, and real-time performance analytics. Applications such as intelligent tutoring systems, AI-assisted writing tools, speech recognition technologies, and adaptive learning environments have become integral components of contemporary ELT practices. Consequently, language acquisition is no longer confined to traditional classroom boundaries but is continuously extended through digital interaction, automated guidance, and data-driven instructional design.

Historically, language pedagogy has been grounded in communicative and interactionist frameworks, which emphasize the central role of social interaction, negotiation of meaning, and contextualized language use in second language acquisition. Theoretical perspectives such as Communicative Language Teaching (CLT) and socio-cultural theory (Vygotskian approaches) conceptualize language as a fundamentally social practice that develops through meaningful human engagement. Within these paradigms, the teacher functions as a facilitator of communication, guiding learners through authentic interactional processes that foster fluency, accuracy, and pragmatic competence.

However, the emergence and rapid advancement of AI technologies introduce a new and increasingly complex pedagogical paradigm. In this evolving model, certain aspects of the learning process are partially delegated to intelligent systems capable of analyzing learner behavior, detecting linguistic errors, predicting performance patterns, and generating immediate corrective feedback. While these capabilities offer substantial advantages in terms of efficiency, scalability, and responsiveness, they simultaneously challenge established assumptions regarding interaction, feedback, and the relational dynamics between teachers and learners in ELT contexts.

This transformation raises several critical pedagogical and theoretical questions. To what extent can AI systems replicate or meaningfully approximate authentic human communication in language learning environments? Does increased technological efficiency necessarily lead to deeper learning outcomes and sustained language competence development? Moreover, how can educators preserve the inherently human, emotional, and social dimensions of language teaching within increasingly automated and algorithmically mediated environments?

In addition, the growing reliance on AI in education raises important concerns related to teacher agency, learner autonomy, ethical data use, transparency of algorithms, and the potential standardization of learning experiences. These issues underscore the necessity of adopting a critically informed and pedagogically grounded approach to AI integration, one that carefully balances innovation with educational values.

Accordingly, this study critically examines the role of artificial intelligence in English Language Teaching from a human-centered pedagogical perspective. It seeks to explore how AI technologies can be effectively integrated into ELT practices in ways that enhance learning outcomes while preserving the communicative, social, and pedagogical principles that remain central to effective language education.

Theoretical Background

The conceptual foundation of this study is grounded in three interrelated theoretical domains: **constructivist learning theory**, **sociocultural theory**, and **digital pedagogy**. Together, these

frameworks provide a comprehensive lens for understanding the pedagogical implications of artificial intelligence (AI) in English Language Teaching (ELT). Each theory contributes a distinct but complementary perspective on how learning occurs, how knowledge is constructed, and how technology mediates educational processes.

1 Constructivist Learning Theory

Constructivist learning theory posits that knowledge is not passively transmitted from teacher to learner but is actively constructed by learners through experience, exploration, and interaction with their environment. In this view, learning is an inherently active and dynamic process in which individuals build new knowledge by integrating prior understanding with new information.

Within this theoretical framework, AI technologies can be interpreted as **cognitive scaffolding tools** that support individualized learning processes. Intelligent tutoring systems, adaptive learning platforms, and AI-based language applications can provide personalized feedback, adjust task difficulty, and guide learners through structured learning pathways. Such features align with constructivist principles by enabling learners to progress at their own pace while engaging in meaningful problem-solving activities.

However, constructivism also implies that deep learning requires active engagement, critical thinking, and reflection. Therefore, while AI can facilitate access to information and support skill development, its pedagogical value ultimately depends on how effectively it promotes active learner participation rather than passive consumption of automated responses.

2 Sociocultural Theory

Sociocultural theory, particularly as developed by Vygotsky, emphasizes the fundamental role of social interaction in cognitive development and language acquisition. According to this perspective, learning occurs first on a social level through interaction with more knowledgeable others and is later internalized by the individual. The concept of the **Zone of Proximal Development (ZPD)** is central to this theory, highlighting the importance of guided support in enabling learners to achieve higher levels of performance.

From a sociocultural standpoint, language learning is inherently dialogic and socially situated. It develops through negotiation of meaning, collaborative interaction, and communicative engagement with others. This raises important pedagogical considerations in relation to AI-driven learning environments. While AI systems can simulate interaction and provide automated feedback, they may not fully replicate the richness, unpredictability, and emotional depth of authentic human communication.

As a result, there is a theoretical concern that excessive reliance on AI-mediated instruction could reduce opportunities for genuine interpersonal communication, which remains essential for pragmatic competence and discourse development in ELT. This highlights a potential tension between technological efficiency and the social nature of language learning.

3 Digital Pedagogy

Digital pedagogy is an emerging interdisciplinary field that examines the integration of digital technologies into educational practice from a critically informed perspective. Unlike approaches that treat technology as a neutral instructional aid, digital pedagogy emphasizes that technological tools actively shape teaching methods, learning experiences, and knowledge construction processes.

Within this framework, AI is not viewed as a passive or neutral instrument, but rather as an **active agent that influences educational dynamics**. AI systems shape what learners see, how they receive feedback, and how their performance is interpreted and evaluated. Consequently, digital

pedagogy calls for critical engagement with AI technologies, encouraging educators to question how these tools structure learning environments and to what extent they align with pedagogical goals.

This perspective also stresses the importance of **intentional and reflective integration** of technology in education. Rather than adopting AI uncritically due to its technological novelty, educators are encouraged to evaluate its pedagogical value, ethical implications, and impact on teacher agency and learner autonomy.

4 Synthesis of Theoretical Perspectives

When considered collectively, these three theoretical perspectives provide a multidimensional understanding of AI integration in ELT. Constructivism highlights the importance of active, individualized learning supported by scaffolding technologies. Sociocultural theory emphasizes the necessity of interaction, collaboration, and social engagement in language development. Digital pedagogy, in turn, offers a critical lens for evaluating how technology shapes educational practices and power relations within the classroom.

Together, these frameworks suggest that AI should be understood not merely as a technical tool, but as a pedagogical and social force that influences how language learning is organized and experienced. This theoretical foundation underscores the need for a balanced approach in which AI supports—but does not replace—the essential human and interactive dimensions of English language teaching.

Materials and Methods

This study employs a **qualitative, conceptual research design** grounded in systematic literature analysis. Unlike empirical studies that rely on primary data collection, this research is based on the critical synthesis and interpretation of existing scholarly work in the intersecting fields of artificial intelligence in education, English Language Teaching (ELT) methodology, and digital learning theory. The aim is to develop a coherent conceptual understanding of how AI reshapes pedagogical practices rather than to measure its effects through statistical or experimental procedures.

1 Research Design

The study is positioned within a **qualitative interpretive paradigm**, which emphasizes meaning-making, conceptual analysis, and theoretical integration. This approach is particularly appropriate for emerging and rapidly evolving fields such as AI in education, where practices and technologies are still in flux and require critical examination rather than purely empirical validation.

The design is **conceptual and exploratory in nature**, focusing on identifying patterns, themes, and theoretical intersections across a broad range of academic literature. This allows for a deeper understanding of how AI technologies are framed within educational discourse and how they influence pedagogical thinking in ELT contexts.

2 Data Sources and Selection Criteria

The study is based on a systematically selected body of secondary sources. These include peer-reviewed journal articles, academic monographs, edited volumes, and international policy reports related to artificial intelligence, educational technology, and language teaching.

To ensure academic rigor and relevance, the selection of sources was guided by three primary criteria:

Academic credibility – only sources published in peer-reviewed journals or by recognized academic publishers were included.

Relevance to AI in education and ELT – sources had to explicitly address artificial intelligence, digital pedagogy, or language teaching methodology.

Publication recency – priority was given to literature published between **2018 and 2025**, reflecting the most current developments in AI-driven educational practices.

This selection strategy ensured that the analysis is grounded in up-to-date scholarly discourse while maintaining theoretical depth and reliability.

3 Data Analysis Procedure

The analytical process followed a **thematic coding approach**, which is widely used in qualitative research for identifying, organizing, and interpreting patterns within textual data. The selected literature was systematically reviewed, and recurring concepts were coded and grouped into higher-order thematic categories.

The analysis proceeded in several stages:

Initial reading and familiarization of all selected sources

Open coding to identify key concepts and recurring ideas

Axial coding to group related concepts into broader categories

Thematic synthesis to develop overarching analytical themes relevant to AI in ELT

This iterative process enabled the construction of a structured analytical framework that reflects both technological and pedagogical dimensions of AI integration.

4 Emergent Analytical Categories

The thematic analysis resulted in the identification of five major conceptual categories that represent the core dimensions of AI integration in English Language Teaching:

Personalization of Learning: Focuses on AI-driven adaptive systems that tailor instruction to individual learner needs, proficiency levels, and learning styles.

Feedback and Assessment Automation : Examines the role of AI in providing instant, data-driven feedback and automating assessment processes such as writing evaluation, grammar correction, and performance tracking.

Teacher Role Transformation : Explores the evolving function of the teacher from knowledge transmitter to facilitator, learning designer, and mediator in AI-enhanced environments.

Ethical and Data Governance Issues : Addresses concerns related to data privacy, algorithmic transparency, bias in AI systems, and the ethical implications of automated decision-making in education.

Communication and Interaction Dynamics

Investigates how AI affects human interaction in ELT contexts, including potential reductions in face-to-face communication and changes in classroom discourse patterns.

3.5 Methodological Rationale

The choice of a qualitative conceptual methodology is justified by the exploratory nature of the research topic. AI in ELT is a rapidly developing field, and many of its pedagogical implications are still emerging. Therefore, a literature-based interpretive approach allows for the integration of diverse perspectives and the development of a comprehensive theoretical understanding.

This methodological framework enables the study to move beyond descriptive accounts of technology use and instead engage in a deeper critical analysis of how AI reshapes educational philosophy, teacher identity, and language learning processes.

Results and Discussion

The analysis of the selected literature revealed four interrelated thematic dimensions that illustrate both the pedagogical potential and the critical limitations of artificial intelligence (AI) in English Language Teaching (ELT). These themes—personalization and adaptive learning, automated assessment and feedback, transformation of teacher identity, and ethical dimensions—are discussed below in an integrated results and discussion framework.

1 Personalization and Adaptive Learning Systems

One of the most prominent contributions of AI to ELT is the emergence of adaptive learning systems capable of tailoring instruction to individual learner profiles. These systems utilize real-time learner data to adjust task difficulty, sequencing of content, and instructional pacing. As a result, learners are provided with highly individualized learning pathways that respond dynamically to their strengths, weaknesses, and progress patterns.

From a pedagogical perspective, such personalization supports learner autonomy, as students can progress at their own pace and revisit content as needed. It also aligns with differentiated instruction principles, allowing teachers to address diverse proficiency levels within a single classroom more effectively.

However, the literature also highlights a critical limitation of algorithm-driven personalization. While adaptive systems increase efficiency, they may inadvertently create what can be described as epistemic narrowing or isolation, where learners are consistently exposed only to content that aligns with their current performance level. This may reduce cognitive challenge, limit exposure to diverse linguistic input, and potentially constrain long-term language development. In this sense, over-personalization risks reducing the complexity and unpredictability that are essential for robust language acquisition.

2 Automated Assessment and Feedback Mechanisms

Another significant development in AI-enhanced ELT is the widespread use of automated assessment and feedback systems, particularly in writing and speaking instruction. Technologies based on natural language processing (NLP) now enable the automatic evaluation of grammatical accuracy, lexical sophistication, syntactic structure, and, to some extent, fluency in spoken language.

A key advantage of these systems is their ability to provide immediate, continuous, and scalable feedback, which is particularly valuable in large educational settings where individualized teacher feedback may be limited. Students benefit from instant error correction and iterative practice opportunities, which can accelerate language learning and reinforce accuracy.

Nevertheless, despite their efficiency, AI-based assessment systems demonstrate important pedagogical limitations. They often struggle to evaluate higher-order communicative competencies, such as creativity, pragmatic appropriateness, discourse coherence, and rhetorical effectiveness. These dimensions of language use are deeply contextual and interpretive in nature, and therefore difficult to capture through algorithmic models. Consequently, while AI can effectively support form-focused aspects of language learning, it remains limited in assessing the full spectrum of communicative competence.

3 Transformation of Teacher Identity

The integration of AI into ELT is also reshaping the professional identity and pedagogical role of the teacher. Increasingly, teachers are being repositioned as **facilitators of learning**, learning designers, and mediators between learners and digital systems, rather than sole transmitters of knowledge.

This transformation requires the development of new professional competencies, including digital literacy, data interpretation skills, and the ability to critically evaluate AI-generated outputs. Teachers are now expected to design learning experiences that effectively integrate AI tools while maintaining pedagogical coherence and learner engagement.

At the same time, this shift raises important concerns regarding teacher authority and professional autonomy. As AI systems take over certain instructional and evaluative functions, there is a risk that teachers may become overly dependent on technology-driven recommendations. This could potentially reduce their decision-making authority and reshape traditional power dynamics within the classroom. Therefore, maintaining a balanced relationship between human pedagogical judgment and AI support is essential for preserving the integrity of teaching practice.

4 Ethical Dimensions of AI in Education

The literature also reveals a growing body of concern surrounding the ethical implications of AI integration in education. One of the most critical issues is data privacy. AI systems rely heavily on the collection, storage, and analysis of large volumes of learner data, raising questions about how this information is used, protected, and shared.

Another significant concern is algorithmic bias, which may emerge when AI systems are trained on non-representative or culturally limited datasets. Such bias can lead to unequal learning outcomes, misinterpretation of learner performance, or the reinforcement of existing educational inequalities.

In addition, issues of transparency and accountability are central to the ethical debate. Many AI systems operate as “black boxes,” making it difficult for educators and learners to fully understand how decisions are generated. This lack of transparency can undermine trust in educational technologies and complicate pedagogical decision-making.

Therefore, the ethical integration of AI in ELT requires not only technical safeguards but also clear institutional policies and critical digital literacy among educators. Ensuring fairness, transparency, and accountability is essential for the sustainable and responsible use of AI in education.

Integrated Discussion

Taken together, these findings suggest that AI has a transformative but ambivalent impact on English Language Teaching. On one hand, it enhances personalization, efficiency, and access to immediate feedback, thereby supporting more flexible and individualized learning experiences. On the other hand, it introduces significant pedagogical, ethical, and professional challenges that cannot be ignored.

The analysis indicates that the effectiveness of AI in ELT is not determined solely by technological sophistication but by the pedagogical frameworks within which it is embedded. Without careful instructional design and critical oversight, AI risks reducing the richness of language learning to measurable but incomplete dimensions of performance.

Ultimately, the findings support the need for a human-centered, critically informed, and pedagogically balanced approach to AI integration. In such a model, AI functions as a supportive tool that enhances teaching and learning, while the teacher remains central in guiding interpretation, interaction, and meaning-making processes in language education.

5. Communication and the Risk of Reduced Interaction

Language learning is inherently dialogic and socially situated. AI systems, while capable of simulating interaction, cannot fully replicate the unpredictability and richness of human communication.

Overreliance on AI-mediated interaction may reduce learners' exposure to authentic communicative contexts, potentially hindering the development of pragmatic and sociolinguistic competence.

Conclusion

Artificial intelligence represents a powerful and rapidly evolving force in the transformation of English Language Teaching (ELT), reshaping how language learning is designed, delivered, and experienced. Its integration into educational systems offers substantial pedagogical benefits, particularly in terms of personalization, efficiency, accessibility, and continuous feedback. AI-driven systems enable more individualized learning pathways, provide immediate corrective input, and support learners in developing language skills in flexible and adaptive environments.

However, the findings of this study also demonstrate that these advantages must be carefully and critically balanced against a range of pedagogical, social, and ethical concerns. While AI enhances certain aspects of language instruction, it cannot fully replicate the depth of human interaction, emotional engagement, and contextual meaning-making that are essential to effective language acquisition. Overreliance on automated systems may risk reducing learning to measurable outputs, thereby overlooking the interpretive and communicative richness of language use.

The analysis suggests that AI should be integrated within a human-centered pedagogical framework that prioritizes interaction, critical thinking, collaboration, and cultural awareness. Within such a framework, the teacher retains a central role as a facilitator of meaning, a mediator of communication, and a designer of learning experiences. Rather than replacing educators, AI should function as an augmentative tool that supports and enhances pedagogical practice while preserving the relational and dialogic nature of teaching.

Furthermore, ethical considerations such as data privacy, algorithmic transparency, and equity of access must remain central to the implementation of AI in educational contexts. Without appropriate safeguards and critical oversight, the use of AI risks reinforcing existing inequalities or introducing new forms of educational imbalance.

In conclusion, future developments in ELT should focus on the creation of **hybrid pedagogical models** that strategically combine technological innovation with established pedagogical principles. Such models should ensure that AI is used not as an end in itself, but as a means of enriching language education while maintaining its human, communicative, and culturally situated character. This balanced approach is essential for ensuring that English language teaching remains both pedagogically effective and socially meaningful in an increasingly digital and AI-driven world.

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DIGITAL TRANSFORMATION IN LANGUAGE TEACHING: THE USE OF CHATBOTS IN FOREIGN LANGUAGE LEARNING

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Abstract. This research explores the impact of artificial intelligence technologies, with a particular emphasis on chatbots, on foreign language learning within the Azerbaijani educational setting. It investigates the extent to which AI-powered tools facilitate the development of learners' linguistic competencies by creating interactive, individualized, and learner-centered educational environments. The study adopts a qualitative methodology, incorporating descriptive and comparative analysis, content analysis, case study examination, and observational methods. The results demonstrate that platforms such as ChatGPT, Duolingo Bot, and Mondly Chatbot contribute to the improvement of speaking, listening, and writing abilities through immediate feedback and adaptive learning mechanisms. At the same time, several limitations are identified, including difficulties in contextual understanding, reduced direct interaction between teachers and students, and ethical issues concerning data privacy. The study concludes that combining conventional teaching approaches with artificial intelligence-based tools can substantially enhance the quality and effectiveness of foreign language education.

Keywords: artificial intelligence, chatbots, foreign language learning, Azerbaijani education, digital learning, personalized learning, language teaching technologies

ЦИФРОВАЯ ТРАНСФОРМАЦИЯ В ОБУЧЕНИИ ЯЗЫКАМ: ИСПОЛЬЗОВАНИЕ ЧАТ-БОТОВ В ПРОЦЕССЕ ИЗУЧЕНИЯ ИНОСТРАННЫХ ЯЗЫКОВ

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Аннотация. Данное исследование посвящено анализу влияния технологий искусственного интеллекта, в частности чат-ботов, на процесс изучения иностранных языков в образовательной системе Азербайджана. Рассматривается, каким образом ИИ-инструменты способствуют развитию

языковых компетенций обучающихся посредством создания интерактивной, индивидуализированной и ориентированной на студента образовательной среды. В работе применяется качественный исследовательский подход, включающий описательный и сравнительный анализ, контент-анализ, метод кейс-стади и наблюдение. Результаты показывают, что такие платформы, как ChatGPT, Duolingo Bot и Mondly Chatbot, способствуют развитию навыков говорения, аудирования и письма благодаря мгновенной обратной связи и адаптивным механизмам обучения. Вместе с тем выявлены определённые ограничения, такие как трудности в понимании контекста, снижение уровня прямого взаимодействия между преподавателем и студентом, а также этические вопросы, связанные с защитой персональных данных. В заключение подчеркивается, что интеграция традиционных методов обучения с технологиями искусственного интеллекта может значительно повысить эффективность преподавания иностранных языков.

Ключевые слова: искусственный интеллект, чат-боты, изучение иностранных языков, образовательная система Азербайджана, цифровое обучение, персонализированное обучение, технологии преподавания языков

Introduction

In recent years, artificial intelligence (AI) technologies have been widely used to make language learning more effective. In particular, chatbots serve as innovative tools that provide students with real-time interactive experiences, helping them develop their language skills. Since traditional teaching methods do not always fully meet the needs of Azerbaijani students, chatbots offer new opportunities.

The lack of speaking practice, absence of a natural language environment, and insufficient consideration of individual learning pace negatively affect the language learning process. Therefore, chatbots can play an important role in personalizing and making language learning more accessible in the Azerbaijani educational context.

This study also examines the new opportunities created by digital technologies in language learning, their advantages for both students and teachers, and the challenges associated with their implementation. The integration of modern technologies in education not only improves language skills but also promotes openness to diverse cultures and perspectives, creating a more effective, inclusive, and experience-based learning environment.

Methodology and Materials

Methodological approaches used in foreign language teaching are diverse, each with its own advantages. Traditional methods are primarily based on teacher-centered instruction, textbooks, and classroom activities. In this approach, the teacher acts as the main source of information, while students remain passive learners. Although such methods are still widely used in Azerbaijan, they have limitations in promoting active participation, individual learning pace, and practical language skills (Smith & Doe, 2021).

In contrast, technology-based methods—especially AI applications—make teaching more dynamic and personalized. Chatbots function as tools that continuously interact with learners, provide opportunities for repetition, and offer real-time feedback on errors. This learner-centered approach increases motivation and encourages independent learning.

The integration of AI and chatbot technologies with traditional methods appears to be the most effective approach for the Azerbaijani education system.

This study is based on a qualitative research design combined with elements of descriptive and comparative analysis. The research employs several methods to ensure a comprehensive investigation of the topic:

- **Descriptive method:** used to analyze the role of artificial intelligence and chatbots in foreign language learning and to present their main characteristics and functions.
- **Comparative method:** applied to compare traditional and technology-based teaching approaches, identifying their strengths and limitations.
- **Content analysis:** used to examine existing scientific literature, digital platforms, and educational resources related to AI-based language learning tools.
- **Case analysis:** based on the examination of widely used platforms such as ChatGPT, Duolingo Bot, and Mondly Chatbot, focusing on their functional capabilities and educational potential.
- **Observation method:** involves analyzing how chatbot technologies simulate interaction and support skill development in real or quasi-real learning environments.

Discussion

In recent years, chatbot technology has been widely used in foreign language teaching. Platforms such as ChatGPT, Duolingo Bot, and Mondly Chatbot provide learners with interactive communication opportunities, helping them improve speaking, listening, and writing skills.

These applications deliver content tailored to learners' responses and provide opportunities for repetition and correction, thus increasing learning efficiency. However, successful implementation depends not only on technical capabilities but also on pedagogically appropriate content, adaptive mechanisms that consider learners' needs, and motivational elements.

Experience shows that chatbots are not only tools for delivering information but also digital assistants that promote interactive learning.

AI-based chatbots support foreign language learning through several key functions:

- **Development of speaking skills:** dialogue-based scenarios help students practice real-life communication
- **Improvement of grammar and vocabulary:** systems detect errors and provide corrections
- **Personalized learning:** chatbots adapt content according to learner level

These capabilities make language learning more accessible in Azerbaijan, especially in regions with limited resources. Chatbots can function as supplementary tools within electronic learning platforms and help students independently develop language skills.

Contextual Errors and Limitations

AI-based chatbots sometimes struggle to fully understand context, especially in cases involving irony, sarcasm, cultural nuances, or non-standard expressions. This may lead to misunderstandings and incorrect language acquisition. This limitation is particularly relevant for Azerbaijani learners, who may already tend to rely on direct translation (Zhang, 2020).

Reduced Teacher-Student Interaction. Although chatbots are effective tools, they cannot replace the empathetic, creative, and social roles of teachers. Teachers adapt instruction based on students' psychological states, learning styles, and motivation levels—capabilities that chatbots do not possess. Therefore, chatbots should be seen as complementary tools rather than substitutes for teachers (Brown, 2007; Lee & Kim, 2018).

Privacy and Ethical Issues. The use of chatbots raises concerns about how students' personal data is collected, stored, and used. Some platforms may share data with third parties or use it for advertising purposes, which creates ethical risks (TESOL International, 2021; OpenAI, 2023).

For Azerbaijani universities, ensuring secure and reliable digital learning environments is essential.

However, despite their advantages, chatbots should be considered complementary tools rather than replacements for teachers. Unlike traditional teaching, they cannot fully provide emotional support, cultural sensitivity, or pedagogical adaptation.

Conclusion

Despite existing challenges, the use of digital technologies in foreign language teaching leads to positive outcomes. These technologies make learning more flexible, interactive, and accessible. However, effective implementation requires improvements in educational infrastructure, teacher training, and access to digital tools. A balanced integration of traditional methods and modern technologies is essential to ensure high-quality language education and to enhance overall learning outcomes.

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Abstract: The aim of this study is to develop English teachers' digital literacy skills in public schools, identify their deficiencies through a questionnaire, and propose effective solutions. According to the results, PBL was selected as the most effective method, and various English teachers working in public schools were selected as participants. The questionnaire found that although there is enthusiasm for using new digital tools

in classrooms, some teachers experience inadequacies in the use of smart boards, Canva, PowerPoint, Zoom, Kahoot, online classroom management, and student data privacy.

Keywords: Digital literacy, English language teachers, public schools, Project-Based Learning (PBL), educational technology, technology integration.

DEVLET OKULLARINDAKİ İNGİLİZCE ÖĞRETMENLERİNİN DİJİTAL OKURYAZARLIK SEVİYELERİNİN GELİŞTİRİLMESİ

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Özet: Bu araştırmanın amacı devlet okulu İngilizce öğretmenlerinin dijital okuryazarlık becerilerini geliştirmek ve eksikliklerini anket yoluyla saptayıp bu eksikliklere yönelik etkili çözüm önerileri geliştirmektir. Bulunan sonuçlara göre en etkili yöntem olarak PBL (Proje Tabanlı Öğrenme) seçilmiş ve katılımcı olarak devlet okullarında görev yapan çeşitli İngilizce öğretmenleri baz alınmıştır. Yapılan anket sonucunda sınıflarda yeni dijital araçların kullanılması konusunda hevesli olduğu görülse de akıllı tahta, Canva, powerpoint , zoom, Kahoot gibi araçların kullanımı, çevrim içi sınıf yönetimi ve öğrenci veri gizliliği konularında bazı öğretmenlerin yetersizlik yaşadığı ortaya çıkmıştır.

Anahtar Kelimeler: Dijital okuryazarlık, İngilizce öğretmenleri, devlet okulları, Proje Tabanlı Öğrenme (PBL), teknoloji entegrasyonu, eğitim teknolojileri.

1. Introduction

This century has brought significant developments to the field of education, the most important of which is digital technology, driven by technological advances. This enhancement has led to changes in education, particularly in classroom practices, where teachers are actively using digital tools, applications, and devices. As educational practices shifted to a hybrid model of teaching, the term digital literacy has gone beyond awareness of tools; it is now expected that teachers be competent in recognizing and becoming familiar with applications and software programs. As a consequence, digital literacy is no longer regarded as an optional skill but a necessity for instructors.

Particularly in classroom environments, especially in foreign language education, some digital tools such as Canva, Kahoot, Zoom, and PowerPoint have a strong foundation in terms of functionality and usefulness. Teachers in a classroom environment are expected to be competent in applying these tools efficiently; however, recent studies indicate that most teachers often feel themselves in trouble when using these tools when teaching a foreign language (Akayoğlu et al., 2020; Kimmons & Hall, 2018). In a similar study, Livingstone (2012) underscored that, unless teachers have digital literacy competence, merely using technology remains insufficient for language teaching. Additionally, language teachers may, in fact, present affirmative attitudes toward technology despite practical problems with its use (Labuguen & Cabrillas).

When all above mentioned taken into account, the main purpose of this study is to examine the attitudes and presence of public-school language teachers towards digital literacy. To enable this, the study seeks answers to the following research questions.

1. What are the digital literacy levels of the public-school language teachers?
2. How can the digital literacy levels of the language teachers be enhanced?

2. Materials and Methods

A quantitative research design is used in this study to collect data from 45 volunteer participants. The participants were all ELT teachers from different age and gender groups and were currently employed as teachers at public schools. To collect the data, a Likert-type questionnaire consisting of 10 items was used, each focusing on a different aspect of digital literacy. For data analysis, the SPSS program was used, and participants' personal information was kept confidential throughout the process to ensure ethical considerations. Table 1 indicates the questionnaire in detail as follows.

Table1. *Digital Literacy Questionnaire*

o	Statement
	I can use smart boards effectively in my lessons.
	I can create lesson materials using digital tools such as Canva and PowerPoint.
	I can manage online classes through platforms such as Zoom, Teams, or Google Meet.
	I can use artificial intelligence tools for developing lesson content.
	I can easily perform basic operations on computers and tablets (uploading/downloading files, sending emails, etc.).
	I can adapt digital materials according to students' proficiency levels.
	I feel confident in solving technological problems when they occur.
	I am willing to learn new digital teaching tools.
	I have sufficient knowledge about protecting students' privacy in digital environments.
0	I believe that using digital tools makes lessons more effective.

3. Results and Discussion

The results show that public school ELT teachers generally have a positive attitude toward using technology. The questionnaire found that teachers are highly motivated to learn new digital teaching tools ($M = 4.62$) and feel good about using digital materials in their lessons ($M = 4.48$). They also see themselves as capable of using artificial intelligence tools to create content ($M = 4.37$) and of handling basic computer tasks, such as emailing and sharing files ($M = 4.10$). Figure 1 deciphers all details below.

Figure1. *Descriptive Statistics*

Item No	Description	Mean Score (1-5)
Q1	Effective use of smartboards in lessons	2.15
Q2	Creating digital materials with tools like Canva/ PowerPoint	3.45
Q3	Managing online classes (Zoom/Teams/Meet)	2.28
Q4	Using AI tools for content development	4.37
Q5	Basic computer operations (emailing, file sharing)	4.10
Q6	Adjusting digital materials for different student levels	3.85
Q7	Problem-solving during technical issues	2.55
Q8	Motivation to learn new digital teaching tools	4.62
Q9	Knowledge of student data privacy in digital tools	1.95

There are also some areas where teachers feel less confident. They reported low confidence in using smart boards during lessons ($M = 2.15$) and in managing online classes via platforms such as Zoom, Teams, and Google Meet ($M = 2.28$). Technical problem-solving skills ($M = 2.55$) and understanding student data privacy in digital settings ($M = 1.95$) were also weak points. This suggests that while teachers are open to using digital tools, they may not have enough practical knowledge to use them effectively in the classroom.

Another key finding is about adapting digital materials for students at different learning levels. Teachers showed moderate confidence here ($M = 3.85$), but the results suggest they need further training to better support diverse learning needs in digital settings. Akayoğlu et al. (2020) also found that teachers often struggle to match technology to teaching goals, not just to use digital tools.

The results also show a gap between teachers' motivation and their actual digital skills. This points to the need for more hands-on professional development. Project-Based Learning (PBL) could help, since it encourages active participation, teamwork, and practical use of technology. Turgut and Irgin (2018) note that PBL can support authentic technology integration by allowing teachers and students to use digital tools in authentic learning situations.

In summary, improving digital literacy for ELT teachers means providing them not only with access to technology but also with ongoing teaching and technical support. Training programs should focus more on practical uses of educational technology, managing online classes, and digital ethics.

4. Conclusion

In summary, the current study concludes that digital literacy is paramount to a teacher's competence in today's educational environment. The study also found that teachers have positive attitudes towards digital language tools and that these tools foster interest in their professional development.

On the other hand, it is also revealed that teachers have subtle deficiencies in using digital tools in classroom contexts, which, in turn, may result in problems with classroom management and professional competency. Therefore, it should be borne in mind that teaching as a profession involves a holistic set of attitudes and competencies.

To eradicate these problems, officials and authorities should keep in mind that teachers' ongoing education and orientation before, during, and after the teaching process must be organized and designed so that no challenge or hindrance keeps them outside the business of teaching and education.

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EDUCATIONAL TECHNOLOGIES AND DIGITAL LEARNING

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Abstract: The rapid evolution of educational technologies and digital learning has significantly transformed contemporary education systems, redefining how knowledge is delivered, accessed, and experienced. This study examines the theoretical foundations, practical applications, and emerging trends of digital learning within modern educational environments. It aims to analyze the effectiveness of technological integration in enhancing teaching and learning processes, while also addressing the challenges associated with its implementation. The research is based on a qualitative methodology, incorporating an extensive review of academic literature, comparative analysis of digital learning models, and examination of real-world educational practices.

The findings indicate that digital learning environments contribute to increased accessibility, flexibility, and personalization, enabling learners to engage with content regardless of geographical and temporal constraints. Furthermore, technologies such as artificial intelligence, learning analytics, and interactive multimedia tools have been shown to improve student engagement and learning outcomes. However, the study also highlights critical issues, including the digital divide, insufficient digital literacy, data privacy concerns, and the need for continuous professional development among educators.

In conclusion, while educational technologies offer substantial opportunities for innovation and inclusivity in education, their successful implementation requires strategic planning, investment in infrastructure, and a balanced integration of pedagogical and technological approaches. The study emphasizes the importance of sustainable and equitable digital transformation in education systems worldwide..

Keywords: Educational technologies, digital learning, e-learning, online education, learning management systems, digital transformation, pedagogy, ICT in education

TƏHSİL TEXNOLOGİYALARI VƏ RƏQƏMSAL ÖYRƏNMƏ

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Xülasə: Təhsil texnologiyalarının və rəqəmsal öyrənmənin sürətli təkamülü müasir təhsil sistemlərini əhəmiyyətli dərəcədə dəyişdirərək, biliyin necə çatdırıldığını, əldə olunduğunu və təcrübədən keçdiyini yenidən müəyyənləşdirdi. Bu tədqiqat müasir təhsil mühitlərində rəqəmsal öyrənmənin nəzəri əsaslarını, praktik tətbiqlərini və ortaya çıxan tendensiyalarını araşdırır. O, tədris və təlim proseslərinin təkmilləşdirilməsində texnoloji integrasiyanın effektivliyini təhlil etmək, eyni zamanda onun həyata keçirilməsi ilə bağlı problemləri həll etmək məqsədi daşıyır. Tədqiqat akademik ədəbiyyatın geniş nəzərdən keçirilməsini, rəqəmsal öyrənmə modellərinin müqayisəli təhlilini və real dünya təhsil təcrübələrinin araşdırılmasını özündə birləşdirən keyfiyyət metodologiyasına əsaslanır.

Nəticələr göstərir ki, rəqəmsal öyrənmə mühitləri artan əlçatanlığa, çevikliyə və fərdiləşdirməyə töhfə verir, öyrənlərə coğrafi və müvəqqəti məhdudiyyətlərdən asılı olmayaraq məzmunla məşğul olmağa imkan verir. Bundan əlavə, süni intellekt, öyrənmə analitikası və interaktiv multimedia alətləri kimi texnologiyaların tələbələrin iştirakını və öyrənmə nəticələrini yaxşılaşdırdığı göstərilmişdir. Bununla belə, tədqiqat həmçinin rəqəmsal uçurum, qeyri-kafi rəqəmsal savadlılıq, məlumatların məxfiliyi ilə bağlı narahatlıqlar və pedaqoqlar arasında davamlı peşəkar inkişaf ehtiyacı da daxil olmaqla kritik problemləri vurğulayır.

Yekun olaraq qeyd edək ki, təhsil texnologiyaları təhsildə innovasiya və inklüzivlik üçün əsaslı imkanlar təqdim etsə də, onların uğurla həyata keçirilməsi üçün strateji planlaşdırma, infrastrukturaya investisiya qoyuluşu, pedaqoji və texnoloji yanaşmaların balanslaşdırılmış integrasiyası tələb olunur. Tədqiqat bütün dünyada təhsil sistemlərində davamlı və ədalətli rəqəmsal transformasiyanın vacibliyini vurğulayır..

Açar sözlər: Təhsil texnologiyaları, rəqəmsal öyrənmə, e-təlim, onlayn təhsil, öyrənmə idarəetmə sistemləri, rəqəmsal transformasiya, pedaqogika, təhsildə İKT

Introduction

The concept of educational technology encompasses a broad spectrum of tools, systems, and methodologies designed to enhance learning processes. These include learning management systems, virtual learning environments, mobile applications, artificial intelligence-based platforms, and immersive technologies such as virtual and augmented reality. Digital learning, as a subset of educational technology, refers specifically to the use of digital tools and resources to support and deliver educational content. It enables both synchronous and asynchronous learning modes, allowing learners to engage with materials at their own pace and according to their individual needs. This flexibility represents one of the most significant advantages of digital learning, particularly in an increasingly globalized and fast-paced world. (Anderson, T. (2008). *The Theory and Practice of Online Learning*. Athabasca University Press)

The expansion of digital learning has been accelerated by external factors such as globalization and the COVID-19 pandemic, which forced educational institutions to adopt online teaching modalities on an unprecedented scale. This shift exposed both the potential and the limitations of digital education systems. On one hand, digital technologies ensured the continuity of education during periods of disruption, highlighting their resilience and adaptability. On the other hand, they revealed structural inequalities, particularly in terms of access to reliable internet connections, digital devices, and technological literacy. These disparities, often referred to as the digital divide, remain a critical challenge that must be addressed to ensure equitable access to education. (Bates, A. W. (2019). *Teaching in a Digital Age: Guidelines for Designing Teaching and Learning*. BCcampus)

One of the defining characteristics of digital learning is its capacity to support personalized and adaptive learning experiences. Through the use of data analytics and artificial intelligence,

educational platforms can monitor learner progress, identify strengths and weaknesses, and adjust content accordingly. This individualized approach contrasts sharply with traditional one-size-fits-all teaching models and has been shown to improve learning outcomes and student engagement. Furthermore, interactive elements such as multimedia content, simulations, and gamified activities enhance learner motivation and facilitate deeper understanding of complex concepts. These innovations align with constructivist learning theories, which emphasize active participation and knowledge construction rather than passive information reception. (Clark, R. C., & Mayer, R. E. (2016). *E-Learning and the Science of Instruction: Proven Guidelines for Consumers and Designers of Multimedia Learning* (4th ed.). Wiley)

Materials and Methods

Despite its numerous advantages, the implementation of educational technologies is not without challenges. One of the primary concerns is the preparedness of educators to effectively integrate digital tools into their teaching practices. Many teachers lack sufficient training and support, which can lead to ineffective or superficial use of technology. Professional development and continuous training are therefore essential components of successful digital transformation in education. Additionally, issues related to data privacy and cybersecurity have become increasingly prominent as educational institutions rely more heavily on digital platforms. Protecting sensitive student information and ensuring secure online environments are critical responsibilities that require robust policies and technological safeguards. (Garrison, D. R. (2017). *E-Learning in the 21st Century: A Community of Inquiry Framework for Research and Practice* (3rd ed.). Routledge)

Another significant challenge is the potential reduction in social interaction and collaborative learning opportunities in fully online environments. While digital platforms offer various communication tools, they may not fully replicate the richness of face-to-face interactions. This limitation can affect student engagement, motivation, and the development of interpersonal skills. As a result, many educators advocate for blended or hybrid learning models that combine the strengths of both digital and traditional approaches. Such models aim to create a balanced learning environment that leverages technology while maintaining human connection and interaction. (Hrastinski, S. (2008). Asynchronous and synchronous e-learning. *Educause Quarterly*, 31(4), 51–55)

The role of educators in digital learning environments has evolved considerably. Rather than serving solely as transmitters of knowledge, teachers are increasingly acting as facilitators, mentors, and designers of learning experiences. (Moore, M. G., & Kearsley, G. (2012). *Distance Education: A Systems View of Online Learning* (3rd ed.). Wadsworth) They are responsible for selecting appropriate technologies, creating engaging content, and guiding students through complex learning pathways. This shift requires a redefinition of teacher competencies, emphasizing digital literacy, pedagogical innovation, and adaptability. (At the same time, learners are expected to take greater responsibility for their own learning, developing skills such as self-regulation, critical thinking, and digital competence. (Mayer, R. E. (2020). *Multimedia Learning* (3rd ed.). Cambridge University Press)

Results and Discussion

The impact of educational technologies on learning outcomes has been widely studied, with evidence suggesting that well-designed digital learning environments can enhance academic performance and knowledge retention. However, the effectiveness of these technologies depends on several factors, including instructional design, user experience, and the level of support provided to

learners. Poorly implemented digital solutions may lead to cognitive overload, disengagement, and reduced learning effectiveness. Therefore, careful planning and evidence-based practices are essential for maximizing the benefits of digital education. (Redecker, C. (2017). *European Framework for the Digital Competence of Educators (DigCompEdu)*. European Commission.)

Looking toward the future, educational technologies are expected to continue evolving in response to emerging trends and innovations. Artificial intelligence is likely to play an increasingly important role in personalizing learning experiences and automating administrative tasks.(Redecker, C. (2017). *European Framework for the Digital Competence of Educators (DigCompEdu)*. European Commission) Learning analytics will provide deeper insights into student behavior and performance, enabling more informed decision-making.(Selwyn, N. (2016). *Education and Technology: Key Issues and Debates* (2nd ed.). Bloomsbury Academic) Immersive technologies such as virtual and augmented reality will offer new possibilities for experiential learning, allowing students to explore complex environments and scenarios in a safe and controlled manner. Additionally, blockchain technology may revolutionize the management of academic credentials, ensuring transparency and security in the verification process. (Siemens, G. (2005). *Connectivism: A learning theory for the digital age. International Journal of Instructional Technology and Distance Learning*, 2(1), 3–10)

Conclusion

Educational technologies and digital learning represent a paradigm shift in the field of education. They offer unprecedented opportunities to enhance accessibility, flexibility, and personalization, while also presenting significant challenges that must be carefully managed. The successful integration of digital technologies requires a holistic approach that considers technological, pedagogical, and social dimensions. Investment in infrastructure, professional development, and research is essential to ensure that digital learning systems are inclusive, effective, and sustainable. As education continues to evolve in the digital age, the ability to adapt and innovate will be crucial for both institutions and individuals. Ultimately, educational technologies should be viewed not merely as tools, but as integral components of a dynamic and evolving educational ecosystem that has the potential to transform learning for future generations.

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THE IMPACT OF ARTIFICIAL INTELLIGENCE ON DEVELOPING PRODUCTIVE SKILLS IN ENGLISH LANGUAGE LEARNING: OPPORTUNITIES, CHALLENGES, AND PEDAGOGICAL IMPLICATIONS

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Abstract: In the context of rapidly evolving digital transformation, artificial intelligence (AI) has emerged as a pivotal force in reshaping contemporary educational paradigms, particularly within the domain of English language teaching (ELT), where the development of productive skills such as speaking and writing remains a persistent pedagogical challenge. This study aims to provide a comprehensive analysis of how AI-driven tools influence the acquisition and enhancement of productive language skills among university-level learners by integrating both empirical data and theoretical insights. Employing a mixed-method research design, the study collected quantitative and qualitative data from 40 participants over a four-week period during which AI-based applications were systematically incorporated into learning activities. The findings indicate that AI technologies significantly contribute to improving grammatical accuracy, lexical diversity, fluency, and learner confidence by offering immediate, personalized feedback and fostering autonomous learning environments. However, the study also critically examines potential drawbacks, including cognitive overdependence, reduced opportunities for authentic human interaction, and ethical concerns related to academic integrity. The research ultimately argues for a balanced, hybrid pedagogical model that strategically integrates AI tools with traditional communicative teaching approaches in order to maximize educational outcomes while mitigating associated risks.

Keywords: artificial intelligence, English language teaching, productive skills, learner autonomy, hybrid pedagogy, digital education

SÜNİ İNTELLEKTİN İNGİLİS DİLİNİN ÖYRƏNİLMƏSİNDƏ BACARIQLARIN İNKİŞAFINA TƏSİRİ: İMKANLAR, ÇAĞIRIŞLAR VƏ PEDAQOJİ NƏTİCƏLƏR

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Xülasə: Müasir dövrdə texnoloji inkişafın sürətlənməsi ilə süni intellekt təhsil sahəsində, xüsusilə də ingilis dilinin tədrisində mühüm dəyişikliklərə səbəb olmuşdur və bu dəyişikliklər ən çox məhsuldar dil bacarıqlarının – danışiq və yazının inkişafında özünü göstərir. Bu tədqiqatın məqsədi süni intellekt əsaslı alətlərin tələbələrin dil bacarıqlarının inkişafına təsirini həm nəzəri, həm də empirik baxımdan geniş şəkildə araşdırmaqdır. Tədqiqat çərçivəsində qarışıq metodologiyadan istifadə olunmuş və 40 tələbə üzərində 4 həftə ərzində müşahidələr, sorğular və müsahibələr aparılmışdır. Nəticələr göstərir ki, süni intellekt texnologiyaları tələbələrin qrammatik düzgünlüyünü, lüğət ehtiyatını və danışiq axıcılığını artırmaqla yanaşı, onların özünəinamını və müstəqil öyrənmə bacarıqlarını da inkişaf etdirir. Bununla yanaşı, texnologiyadan həddindən artıq asılılıq, canlı ünsiyyətin azalması və akademik dürüstlük məsələləri kimi problemlər də müəyyən edilmişdir. Tədqiqatın nəticələri göstərir ki, süni intellektdən səmərəli istifadə etmək üçün ənənəvi pedaqoji yanaşmalarla balanslı şəkildə inteqrasiya vacibdir.

Açar sözlər: süni intellekt, ingilis dili tədrisi, danışiq bacarığı, yazı bacarığı, rəqəmsal təhsil, hibrid tədris

Introduction

In recent decades, the rapid and unprecedented advancement of digital technologies has fundamentally reshaped the landscape of education, giving rise to innovative pedagogical models that increasingly rely on intelligent systems, automation, and data-driven decision-making processes. Among these technological developments, artificial intelligence (AI) has emerged as one of the most influential and transformative forces, owing to its ability to simulate aspects of human cognition, process large volumes of data, and provide adaptive, personalized learning experiences. As a result, AI is no longer viewed merely as a supplementary technological tool, but rather as a central component in redefining contemporary educational practices.

Within the field of English language teaching (ELT), the integration of AI technologies has generated considerable interest, particularly in relation to the development of productive language skills such as speaking and writing. These skills are widely recognized as among the most complex and cognitively demanding aspects of second language acquisition, as they require learners not only to demonstrate linguistic knowledge but also to actively produce meaningful and contextually appropriate output in real time. Despite the importance of these skills, traditional classroom-based approaches have often struggled to provide sufficient opportunities for individualized practice and timely feedback, especially in large and heterogeneous learning environments.

One of the key limitations of conventional language teaching methods lies in their inherent structural constraints, which restrict the extent to which teachers can offer immediate, personalized feedback to each learner. In many cases, feedback on written assignments is delayed, while opportunities for speaking practice are limited by time constraints and classroom dynamics. Consequently, learners may experience difficulties in refining their language output, leading to persistent errors, reduced fluency, and decreased motivation. In this context, AI-powered tools—such as automated writing evaluation systems, intelligent tutoring platforms, and speech recognition applications—offer a promising solution by enabling continuous, real-time interaction and individualized support.

Moreover, the integration of AI into language education aligns closely with the broader pedagogical shift from teacher-centered instruction to learner-centered approaches, which emphasize autonomy, self-regulation, and active engagement in the learning process. AI technologies facilitate this transition by providing learners with access to personalized feedback, adaptive learning pathways, and opportunities for independent practice beyond the traditional

classroom setting. This, in turn, supports the development of critical thinking skills and encourages learners to take greater responsibility for their own progress.

However, despite the numerous advantages associated with AI integration, its implementation in educational contexts is not without challenges. Concerns have been raised regarding the potential for cognitive overdependence on automated systems, which may hinder the development of independent problem-solving abilities. Additionally, the increasing reliance on technology may reduce opportunities for authentic human interaction, thereby affecting the development of communicative and sociolinguistic competence. Ethical considerations, including issues related to data privacy, algorithmic bias, and academic integrity, further complicate the adoption of AI in language learning environments.

Although a growing body of research has examined the role of AI in education, there remains a need for more empirical studies that specifically investigate its impact on the development of productive language skills within real classroom contexts. In particular, limited attention has been paid to the combined effects of AI on both speaking and writing, as well as to learners' perceptions and experiences over sustained periods of use. Addressing this gap is essential for developing a more nuanced understanding of how AI can be effectively integrated into pedagogical practice.

Therefore, the present study aims to explore the impact of AI-driven tools on the development of speaking and writing skills among university-level learners by adopting a mixed-method research design. By examining both performance outcomes and learner perspectives, the study seeks to provide a comprehensive evaluation of the benefits and limitations of AI-assisted language learning, ultimately contributing to the development of more balanced and effective pedagogical models in the digital age.

Literature Review

The growing body of literature on artificial intelligence in education reflects a broad consensus regarding its transformative potential, while also highlighting the complexity of its implementation in diverse learning contexts. Scholars such as Zawacki-Richter et al. (2019) have conducted systematic reviews demonstrating that AI technologies are increasingly being utilized to enhance personalization, automate assessment, and support decision-making processes in higher education, thereby contributing to improved learning outcomes and efficiency. Similarly, Ouyang and Jiao (2021) propose three paradigms of AI in education—AI-directed, AI-supported, and AI-empowered learning—which illustrate the evolving role of technology in shaping pedagogical practices.

In the specific context of language learning, AI applications have been widely employed to facilitate the development of both receptive and productive skills; however, their impact on speaking and writing has attracted particular attention due to the inherently productive and interactive nature of these skills. Automated writing evaluation systems, for instance, have been shown to significantly improve learners' grammatical accuracy and textual coherence by providing instant feedback and suggestions for revision, thereby enabling iterative learning processes that are difficult to achieve in traditional settings. At the same time, speech recognition technologies allow learners to practice pronunciation and fluency in a low-anxiety environment, which is especially beneficial for students who may feel reluctant to participate in face-to-face communication.

Despite these benefits, researchers such as Selwyn (2019) caution against the uncritical adoption of AI technologies, emphasizing the need to consider potential drawbacks, including the risk of over-reliance on automated systems, the reduction of meaningful human interaction, and

concerns related to data privacy and academic integrity. These issues underscore the importance of adopting a balanced approach that integrates technological innovation with established pedagogical principles.

Materials and Methods

This study adopts a mixed-method research design in order to provide a comprehensive analysis of the impact of artificial intelligence on the development of productive language skills, combining quantitative data obtained through structured questionnaires with qualitative insights derived from semi-structured interviews and classroom observations. The participants of the study consisted of 40 undergraduate students enrolled in an English language teaching program, whose proficiency levels ranged from intermediate to upper-intermediate, thereby ensuring a relatively homogeneous sample in terms of linguistic competence.

Over the course of four weeks, participants were instructed to engage with various AI-based tools designed to support writing and speaking practice, including automated writing assistants and speech recognition applications, which were integrated into their regular learning activities. The data collection process involved administering pre- and post-intervention questionnaires to measure changes in learners' perceptions and performance, as well as conducting interviews to explore their experiences in greater depth.

The collected data were analyzed using descriptive statistical methods for quantitative responses, while qualitative data were examined through thematic analysis in order to identify recurring patterns and key themes related to the effectiveness and limitations of AI tools in language learning.

Results and Discussion

The findings of the study reveal that the integration of artificial intelligence into language learning environments has a statistically and pedagogically significant impact on the development of productive skills, as evidenced by improvements in multiple dimensions of language performance. A substantial proportion of participants reported enhanced grammatical accuracy, increased lexical diversity, and improved coherence in their written output, which can be attributed to the immediate and detailed feedback provided by AI systems that enable learners to identify and correct errors in real time.

Similarly, the use of speech recognition technologies was found to contribute to noticeable improvements in pronunciation accuracy and fluency, as well as increased confidence in speaking, particularly among learners who previously experienced anxiety in traditional classroom settings. These findings support the notion that AI can create a supportive and non-threatening learning environment that encourages experimentation and risk-taking, which are essential for language development.

However, the study also identifies several important challenges associated with the use of AI in education, including the tendency of some learners to become overly dependent on automated feedback, thereby reducing their engagement in deeper cognitive processes such as critical thinking and problem-solving. Moreover, the limited opportunities for authentic human interaction may hinder the development of pragmatic competence, which is crucial for effective communication in real-world contexts.

Conclusion

In conclusion, this study provides substantial evidence that artificial intelligence constitutes a transformative and highly influential component in the advancement of productive language skills

within the context of English language teaching. The integration of AI-driven tools has proven to be particularly effective in addressing long-standing pedagogical challenges associated with speaking and writing, especially in relation to the provision of immediate, individualized, and data-driven feedback that is often unattainable in traditional classroom environments. As demonstrated by both quantitative and qualitative findings, learners not only improved their grammatical accuracy, lexical range, and fluency, but also developed higher levels of self-confidence and autonomy, which are essential for sustained language acquisition.

Beyond these measurable improvements, the study also highlights the broader pedagogical implications of AI integration, particularly in facilitating a shift towards more learner-centered and flexible educational models. By enabling continuous access to personalized learning resources, AI technologies empower students to engage in self-paced and reflective learning processes, thereby promoting deeper cognitive engagement and long-term retention of knowledge. In this sense, AI can be seen not merely as a supplementary tool, but as a catalyst for redefining the roles of both teachers and learners in contemporary educational settings.

Nevertheless, the findings also underscore the necessity of adopting a critical and balanced perspective regarding the use of AI in education. The risk of cognitive overdependence on automated systems remains a significant concern, as it may limit learners' ability to independently construct knowledge and engage in higher-order thinking. Furthermore, the potential reduction in authentic human interaction poses challenges for the development of pragmatic and sociolinguistic competence, which are crucial components of communicative language ability. Ethical considerations, including issues of academic integrity, data privacy, and algorithmic bias, further complicate the implementation of AI technologies and require careful institutional regulation.

In light of these considerations, the study strongly advocates for the adoption of a hybrid pedagogical framework that strategically integrates AI tools with traditional communicative and interaction-based teaching methods. Such an approach would allow educators to leverage the efficiency and adaptability of AI while preserving the inherently human aspects of language learning, including social interaction, emotional engagement, and cultural awareness.

Finally, it is important to acknowledge certain limitations of the present study, including the relatively small sample size and the short duration of the intervention, which may affect the generalizability of the findings. Future research should therefore aim to explore the long-term impact of AI integration across diverse educational contexts and proficiency levels, as well as investigate the effectiveness of specific AI tools in greater depth. By continuing to examine both the opportunities and challenges associated with AI in education, researchers and practitioners can contribute to the development of more effective, ethical, and sustainable language teaching practices in the digital age.

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THE ROLE OF ARTIFICIAL INTELLIGENCE IN TRANSFORMING MODERN LINGUISTIC EDUCATION: OPPORTUNITIES AND ETHICAL CHALLENGE

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Abstract: This article examines the profound impact of Artificial Intelligence (AI) on the landscape of higher education, with a specific focus on linguistic and philological studies. As digital transformation accelerates, AI driven tools are reshaping how students acquire languages and conduct research. The study analyzes the benefits of adaptive learning platforms and automated linguistic analysis, while simultaneously addressing the critical concerns regarding academic honesty and the "human element" in teaching. Through a qualitative review of current AI applications, the research concludes that a balanced integration of technology and traditional pedagogy is essential for the future of academic excellence.

Keywords: Artificial Intelligence, Digital Pedagogy, Linguistics, Personalized Learning, Academic Integrity, Machine Learning.

MÜASİR DİL TƏHSİLİNİN TRANSFORMASIYASINDA SÜNİ İNTELLEKTİN ROLU: İMKANLAR VƏ ETİK ÇAĞIRIŞLAR

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Xülasə: Bu məqalə Süni İntellektin (Sİ) ali təhsil sisteminə, xüsusilə dilçilik və filologiya sahələrinə göstərdiyi dərin təsiri araşdırır. Rəqəmsal transformasiya sürətləndikcə, Sİ əsaslı alətlər tələbələrin dil mənimsəmə və tədqiqat aparma üsullarını yenidən formalaşdırır. Tədqiqatda adaptiv öyrənmə platformalarının və avtomatlaşdırılmış dil analizinin üstünlükləri təhlil edilir, eyni zamanda akademik dürüstlük və tədrisdə "insan amili" ilə bağlı kritik narahatlıqlar nəzərdən keçirilir. Mövcud Sİ tətbiqlərinin keyfiyyət analizi vasitəsilə tədqiqat belə qənaətə gəlir ki, akademik mükəmməlliyyənin gələcəyi üçün texnologiya və ənənəvi pedaqogikanın balanslaşdırılmış inteqrasiyası zəruridir.

Açar sözlər: Süni İntellekt, Rəqəmsal Pedaqogika, Dilçilik, Fərdiləşdirilmiş Öyrənmə, Akademik Dürüstlük, Maşın Öyrənməsi.

Introduction

The global educational landscape is currently undergoing a paradigm shift, driven by the rapid evolution of Artificial Intelligence (AI) and Machine Learning (ML). For decades, pedagogical methods remained relatively static, relying on centralized curricula and traditional classroom dynamics. However, the advent of Natural Language Processing (NLP) has fundamentally altered how humanity interacts with information, and by extension, how it learns languages. In the Republic of Azerbaijan, and specifically at Mingachevir State University, there is a growing emphasis on integrating these global technological trends into the local academic framework. For a student of philology, language is not just a collection of grammar rules but a "national memory carrier" and a complex cognitive system. AI offers tools that can decode this system more efficiently than ever before. This article explores the dual nature of AI in education: its role as a revolutionary assistant and its potential as a disruptor of traditional academic ethics.

Materials and Methods

This research is based on a comprehensive qualitative analysis of current educational technologies. The materials analyzed include Large Language Models (LLMs) such as GPT-4, neural machine translation systems like DeepL, and academic integrity monitoring software. The methodology involves a comparative study of student performance outcomes when AI is used as a supplementary tool versus traditional learning environments. Furthermore, the study adheres to the APA 7th edition guidelines to ensure that all citations and references meet the international standards required by the conference "Contemporary Issues in Language, Education, and Research."

Discussion

Personalized Learning Pathways and Adaptive Pedagogy The "one-size-fits-all" approach has long been a limitation of mass education. AI solves this by introducing "adaptive pedagogy." Through algorithms that track a student's progress, AI can identify that a specific philology student struggles with, for example, the morphological structure of complex verbs or the phonetic nuances of a second language. By providing targeted exercises, AI creates a "Low Filter" environment (as per Stephen Krashen's Second Language Acquisition theory), where students feel less anxiety about making mistakes. This personalization ensures that the learning process is optimized for each individual's cognitive load, leading to higher retention rates and a deeper understanding of linguistic theories (Luckin, 2018).

AI as a Tool for Advanced Linguistic Analysis For researchers in the field of philology, the ability to analyze vast corpora of text is invaluable. AI-driven software can scan thousands of pages of classical literature such as the works of Nizami Ganjavi or Shakespeare to identify patterns in metaphor usage, syntax, and stylistic evolution. This "Big Data" approach to linguistics allows students to move beyond surface-level reading into deep structural analysis. Moreover, neural translation tools have bridged the gap in intercultural communication. While early translation software was often criticized for its lack of context, modern AI understands "pragmatics" the way context contributes to meaning. For a translator, this means that AI can handle the "mechanical" part of the job, allowing the human translator to focus on the creative and cultural nuances that define a literary masterpiece (Mammadov, 2021). Perhaps the most debated topic in modern education is the use of AI for writing. With the ability to generate well-structured essays in seconds, AI poses a significant challenge to traditional grading systems. The concern is that students might rely on technology to the point where their critical thinking skills atrophy. However, the solution is not the prohibition of technology, but the transformation of the curriculum. Academic institutions must shift their focus from "product based" assessment (the final essay) to "process-

based" assessment. This includes oral exams, in-class writing without digital aids, and the requirement for students to show the evolution of their ideas. Academic integrity must be reframed as a set of values where AI is used for brainstorming and research, while the final synthesis remains a human achievement (Zawacki-Richter et al., 2019). As AI models are primarily trained on English-language datasets, there is a risk of "linguistic imperialism." For a philology student in Azerbaijan, it is crucial to ensure that AI also supports and preserves the Azerbaijani language. Language policy and planning must focus on developing local AI datasets that respect the grammatical and cultural specificities of our mother tongue. AI should be a tool that helps us project our national memory and literary heritage onto the global stage, rather than a tool that homogenizes all cultures into a single digital standard. The human element remains the cornerstone of education. A teacher does more than transmit facts; they inspire, mentor, and provide moral guidance. AI lacks "emotional intelligence" (EQ). It cannot understand the frustration of a student who cannot find the right word to finish a poem, nor can it celebrate the joy of a breakthrough in understanding a complex philosophical text. At Mingachevir State University, the goal is to create a "Bridge" where technology supports the administrative and repetitive tasks of education, leaving the teacher free to engage in the truly human work of mentorship. The integration of AI should not lead to the isolation of the student, but rather to a more connected and efficient academic community.

Conclusion

Artificial Intelligence represents the most significant technological leap in education since the invention of the printing press. For the field of linguistics and education research, it offers unparalleled opportunities for personalization, analysis, and global cooperation. However, the successful integration of AI requires a cautious approach that prioritizes academic ethics and the preservation of national linguistic identity. As we move toward 2026 and beyond, the role of the philologist will be to act as the "ethical guardian" of language in a digital world. We must embrace AI not as a replacement for human thought, but as a sophisticated instrument that allows us to explore the infinite depths of language and human expression more profoundly.

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ARTIFICIAL INTELLIGENCE AND HUMAN CREATIVITY

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Abstract: This article explores the relationship between artificial intelligence and human creativity. In the modern era, the rapid development of artificial intelligence technologies has reshaped the understanding of creativity. Activities traditionally considered exclusive to humans can now, to some extent, be performed through algorithms and machine learning. The article explains the application of artificial intelligence in fields such as art, music, literature, and design, providing relevant examples, and analyzes its impact on human creativity.

In addition, the key differences between human imagination, emotional depth, and contextual thinking and those of artificial intelligence are emphasized. The study shows that artificial intelligence primarily serves as a tool that enhances human creativity and creates new opportunities. Ultimately, artificial intelligence and human creativity are not competitors but rather two complementary components.

Keywords: Artificial Intelligence, Human Creativity, Machine Learning, Digital Creativity

SÜNİ İNTELLEKT VƏ İNSAN YATRADICILIĞI

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Xülasə: Bu məqalədə süni intellekt və insan yaradıcılığı arasındakı qarşılıqlı əlaqə araşdırılır. Müasir dövrdə süni intellekt texnologiyalarının sürətli inkişafı yaradıcılıq anlayışına yeni baxış formalaşdırmışdır. Ənənəvi olaraq yalnız insana məxsus hesab edilən yaradıcılıq fəaliyyəti artıq müəyyən dərəcədə alqoritmlər və maşın öyrənməsi vasitəsilə də həyata keçirilə bilər. Məqalədə süni intellektin incəsənət, musiqi, ədəbiyyat və dizayn kimi sahələrdə tətbiqi nümunələrlə izah olunur, həmçinin bu texnologiyanın insan yaradıcılığına təsiri təhlil edilir.

Bununla yanaşı, insan təxəyyülünün, emosional dərinliyinin və kontekstual düşünmə qabiliyyətinin süni intellektdən əsas fərqləri vurğulanır. Araşdırma göstərir ki, süni intellekt daha çox alət rolunu oynayaraq insan yaradıcılığını genişləndirir və yeni imkanlar yaradır. Nəticə etibarilə, süni intellekt və insan yaradıcılığı rəqib deyil, əksinə bir-birini tamamlayan iki mühüm komponent kimi çıxış edir.

Açar sözlər: Süni intellekt, İnsan yaradıcılığı, Maşın öyrənməsi, Rəqəmsal yaradıcılıq

Introduction

In the modern era, the rapid development of technology has influenced all areas of society, creating new opportunities and approaches. One of the most significant directions of this progress, artificial intelligence, is no longer limited to solving technical problems but is also penetrating the field of creativity. While creativity was traditionally considered a result of human thought, imagination, and emotional experience, today artificial intelligence systems are also capable of producing creative outputs to a certain extent.

The application of artificial intelligence in fields such as art, music, literature, and design has expanded the boundaries of this concept. For example, algorithms can be used to create paintings, compose music, and even generate written texts. This brings the question “What is creativity?” back into focus and requires a deeper exploration of the differences between humans and machines.

At the same time, human creativity still preserves its unique characteristics. Human thinking is based not only on logical analysis but also on emotions, culture, and individual experiences. Artificial intelligence, on the other hand, primarily operates on existing data to produce results. For this reason, the relationship between artificial intelligence and human creativity is better understood not as competition, but as mutual complementarity.

This article will examine in detail the relationship between artificial intelligence and human creativity, their similarities and differences, as well as their combined role in the modern world.

Methodology and materials

This article is based on a combination of several scientific and analytical methods. In line with the content presented, the following methods can be considered as having been applied:

First, the analytical method is used. Through this approach, the concepts of artificial intelligence and human creativity are examined separately, their characteristics are analyzed, and their key differences are identified.

In addition, the comparative method plays an important role. The article compares the capabilities of human creativity and artificial intelligence, highlighting their similarities and differences. This helps to present the topic in a clearer and more structured way.

At the same time, the descriptive method is applied. The use of artificial intelligence in fields such as art, music, and literature is explained through specific examples, making the topic more understandable and illustrative.

Furthermore, the synthesis method is evident. Information and ideas obtained from various sources are combined to form a general conclusion and to present a unified understanding of the relationship between artificial intelligence and human creativity.

Finally, the generalization method is also used. Based on the analysis conducted, overall conclusions are drawn, and a general perspective on the impact of artificial intelligence on human creativity is presented.

The combined use of these methods ensures that the article is logically structured and scientifically grounded.

Conclusion

The analysis shows that the relationship between artificial intelligence and human creativity is not characterized by competition, but rather by complementarity. Artificial intelligence is capable of processing large amounts of data quickly, generating multiple variations, and offering technically efficient solutions. This creates favorable conditions for the development of new ideas and a more productive creative process.

On the other hand, human creativity maintains its uniqueness. Human imagination, emotional depth, and the ability to understand cultural and social contexts are richer and more complex compared to artificial intelligence. These qualities are key factors that define the originality and intrinsic value of creative work.

In conclusion, artificial intelligence does not replace human creativity but serves as a powerful tool that supports and enhances it. In the future, the integration of these two domains is

expected to lead to more innovative outcomes, the emergence of new forms of creativity, and significant contributions to societal development.

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THE ROLE OF ARTIFICIAL INTELLIGENCE IN MODERN LANGUAGE TEACHING

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Abstract: Artificial Intelligence (AI) has become one of the most influential technological developments in modern education. Its integration into language teaching has transformed traditional learning methods by providing innovative, flexible, and student-centered approaches. AI-powered applications, intelligent tutoring systems, speech recognition technologies, and virtual assistants help learners improve their language skills through personalized instruction and immediate feedback. These technologies enable students to practice listening, speaking, reading, and writing more effectively while learning at their own pace. In addition, AI assists teachers by automating routine tasks such as grading assignments, monitoring learners' progress, and identifying areas that require additional attention. As a result, educators can devote more time to interactive classroom activities and meaningful communication with students. Despite these advantages, several challenges remain, including unequal access to digital technologies, concerns about data privacy, and the possibility of reducing face-to-face interaction in educational settings. Therefore, AI should be considered a complementary educational tool rather than a replacement for teachers. This paper examines the role of Artificial Intelligence in modern language teaching by discussing its major benefits, potential limitations, and future prospects. It also emphasizes the importance of maintaining a balanced approach in which technological innovation supports, rather than replaces, the essential role of human teachers in creating effective and engaging language learning environments.

Keywords: Artificial Intelligence, Language Teaching, Digital Learning, Personalized Education, AI Tools, Online Education, Adaptive Learning, Language Skills, Educational Technology

MÜASİR DİL TƏDRİSİNDƏ SÜNİ İNTELLEKTİN ROLU

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Xülasə: Süni İntellekt (SI) müasir təhsildə ən mühüm texnoloji inkişaflardan birinə çevrilmişdir. Onun dil tədrisinə inteqrasiyası innovativ, çevik və tələbəyönümlü yanaşmalar təqdim etməklə ənənəvi təlim metodlarını əhəmiyyətli dərəcədə dəyişdirmişdir. Süni intellektlə işləyən tətbiqlər, ağıllı tədris sistemləri, nitqin tanınması texnologiyaları və virtual köməkçilər fərdiləşdirilmiş təlim və dərhal rəy vermə vasitəsilə öyrənənlərin dil bacarıqlarını inkişaf etdirməsinə kömək edir. Bu texnologiyalar tələbələrə dinləmə, danışma, oxuma və yazma bacarıqlarını daha səmərəli şəkildə inkişaf etdirməyə, eyni zamanda öz öyrənmə sürətlərinə uyğun şəkildə təhsil almağa imkan yaradır.

Bundan əlavə, süni intellekt müəllimlərə tapşırıqların qiymətləndirilməsi, tələbələrin irəliləyişinin izlənilməsi və əlavə diqqət tələb edən sahələrin müəyyənləşdirilməsi kimi rutin işləri avtomatlaşdırmaqla dəstək olur. Nəticədə, müəllimlər sinifdaxili interaktiv fəaliyyətlərə və tələbələrlə mənalı ünsiyyətə daha çox vaxt ayıra bilirlər.

Bu üstünlüklərə baxmayaraq, rəqəmsal texnologiyalara bərabər çıxış imkanlarının olmaması, məlumatların məxfiliyi ilə bağlı narahatlıqlar və təhsil mühitində üzbəüz ünsiyyətin azalması ehtimalı kimi bir sıra problemlər hələ də qalmaqdadır. Buna görə də süni intellekt müəllimləri əvəz edən deyil, onların fəaliyyətini tamamlayan bir təhsil vasitəsi kimi qiymətləndirilməlidir.

Bu məqalədə müasir dil tədrisində süni intellektin rolu, onun əsas üstünlükləri, potensial məhdudiyyətləri və gələcək inkişaf perspektivləri müzakirə olunur. Həmçinin vurğulanır ki, texnoloji yeniliklərin insan müəlliminin dil öyrənmə mühitinin effektiv və cəlbedici olmasındakı əvəzsiz rolunu əvəz etmədən, onu dəstəklədiyi balanslı yanaşmanın qorunması böyük əhəmiyyət daşıyır.

Açar sözlər: Süni intellekt, dil tədrisi, rəqəmsal təhsil, fərdiləşdirilmiş təhsil, süni intellekt alətləri, onlayn təhsil, adaptiv təlim, dil bacarıqları, təhsil texnologiyaları.

Introduction

Technological advancements have reshaped modern education, and Artificial Intelligence is among the most influential innovations. AI refers to computer systems designed to perform tasks that normally require human intelligence, such as problem-solving and decision-making.

In language teaching, AI introduces new methods that go beyond traditional classroom approaches. Learning a language involves mastering multiple skills, and not all students learn at the same pace. Therefore, integrating AI into education helps address individual differences and supports more effective learning.

Personalized Learning Through AI

One of the key strengths of AI in language education is its ability to create personalized learning experiences. AI systems can monitor learners' progress and adjust content according to their performance.

For instance, when a learner repeatedly makes grammatical errors, the system provides additional targeted exercises. This adaptive learning approach allows students to focus on their weak areas and improve more efficiently.

Personalization also increases learner engagement. When students work with materials suited to their level, they are more likely to stay motivated and actively participate in the learning process.

Improving Accessibility and Learning Flexibility

AI technologies make language learning more accessible than ever before. Learners can use digital platforms, mobile applications, and online tools to study at their convenience.

This flexibility is especially important for individuals who cannot attend traditional classes due to time or location constraints. AI supports continuous learning and encourages independent study habits.

As a result, education becomes more inclusive, allowing a wider range of learners to benefit from language learning opportunities.

AI and Language Practice Opportunities

Another major contribution of AI is the enhancement of practical language use. AI-powered tools such as virtual assistants and chatbots provide interactive environments where learners can practice communication skills.

These tools simulate real-life conversations, helping learners improve their speaking and writing abilities. Additionally, immediate feedback enables students to recognize and correct their mistakes instantly.

Such practice environments reduce anxiety and help learners build confidence in using a foreign language.

The Role of AI in Supporting Teachers

AI not only benefits students but also assists teachers in their professional tasks. It can automate routine activities such as grading and monitoring student progress.

This allows educators to dedicate more time to creative teaching methods and direct interaction with students. Furthermore, AI-generated data helps teachers better understand learners' needs and adjust their teaching strategies accordingly.

Thus, AI contributes to a more efficient and informed teaching process.

The Impact of AI on Language Assessment

Artificial Intelligence has significantly transformed the way language skills are assessed. Traditional assessment methods often rely on standardized tests, which may not fully reflect a learner's actual abilities. AI-based systems, however, provide more dynamic and continuous evaluation.

For example, AI tools can analyze writing tasks, detect grammar and vocabulary errors, and even evaluate coherence and style. In speaking assessments, speech recognition technologies allow systems to assess pronunciation, fluency, and accuracy.

Moreover, AI provides instant feedback, which is crucial for effective learning. Instead of waiting for teacher evaluation, learners can immediately identify their mistakes and improve.

AI-Powered Language Learning Applications

In recent years, AI-powered language learning applications have become increasingly popular. These applications combine gamification, adaptive learning, and interactive exercises to create engaging learning environments.

They track user performance and suggest personalized lessons. Additionally, features such as voice recognition and real-time corrections make learning more interactive and effective.

Ethical Considerations in AI-Based Education

While AI offers many benefits, it also raises important ethical concerns. One of the main issues is data privacy. AI systems collect large amounts of user data, including learning patterns and personal information.

Ensuring the security and confidentiality of this data is crucial. Another concern is algorithmic bias, which may affect fairness in education.

Therefore, ethical guidelines and responsible AI development are necessary to ensure equal opportunities for all learners.

Future Prospects of AI in Language Teaching

The future of AI in language education looks promising. With continuous technological advancements, AI tools are expected to become more sophisticated and human-like.

Future systems may include advanced virtual tutors capable of understanding emotions and providing personalized guidance. Technologies such as virtual reality (VR) and augmented reality (AR) may also enhance immersive learning experiences.

Conclusion

Artificial Intelligence has become a valuable tool in modern language teaching. It enhances learning through personalization, accessibility, and interactive practice. However, it is important to maintain a balance between technology and human instruction.

AI should be viewed as a supportive resource rather than a replacement for teachers. Combining technological tools with traditional teaching methods can lead to more effective and meaningful learning outcomes.

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DEVELOPING DATA LITERACY IN JOURNALISM EDUCATION: DIGITAL LEARNING APPROACHES

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Abstract: This study examines the development of data literacy in journalism education within the framework of digital learning approaches. Digitalization, big data, social media platforms, open data sources, and artificial intelligence technologies have fundamentally transformed journalistic practices. This transformation has made it essential for journalists not only to possess news writing and source utilization skills but also to acquire competencies in collecting, cleaning, analyzing, verifying, visualizing data, and transforming it into news narratives that serve the public interest. In this context, data literacy is considered a fundamental professional competence for journalism students that goes beyond technical skills and integrates critical thinking, ethical responsibility, contextual interpretation, and digital competencies.

The study discusses the opportunities provided by e-learning, distance education, blended learning, mobile learning, microlearning, open learning platforms, problem-based learning, project-based learning, experiential learning, and scenario-based learning models for data journalism education. Furthermore, the

study highlights how artificial intelligence-supported learning systems, learning analytics, big data, and automation technologies strengthen personalized, practice-oriented, and interactive learning processes in journalism education. The study argues that data literacy should not be integrated into journalism curricula as an independent technical course, but rather as a holistic approach embedded throughout all stages of the news production process. Accordingly, a multidimensional educational model is proposed, encompassing conceptual awareness, technical competence, editorial skills, ethical awareness, and digital learning competencies.

Keywords: Data literacy, journalism education, data journalism, digital learning, e-learning, blended learning.

GAZETECİLİK EĞİTİMİNDE VERİ OKURYAZARLIĞININ GELİŞTİRİLMESİ: DİJİTAL ÖĞRENME YAKLAŞIMLARI

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Özet: Bu çalışma, gazetecilik eğitiminde veri okuryazarlığının geliştirilmesini dijital öğrenme yaklaşımları çerçevesinde ele almaktadır. Dijitalleşme, büyük veri, sosyal medya platformları, açık veri kaynakları ve yapay zeka teknolojileri gazetecilik pratiklerini köklü biçimde dönüştürmüştür. Bu dönüşüm, gazetecilerin yalnızca haber yazma ve kaynak kullanma becerileriyle sınırlı kalmayıp veriyi toplama, temizleme, analiz etme, doğrulama, görselleştirme ve kamu yararına uygun bir haber anlatısına dönüştürme yeterliliklerine sahip olmalarını zorunlu hale getirmiştir. Bu bağlamda veri okuryazarlığı, gazetecilik öğrencileri için teknik bir becerinin ötesinde eleştirel düşünme, etik sorumluluk, bağlamsal yorumlama ve dijital yetkinlik alanlarını birleştiren temel bir mesleki yeterlilik olarak değerlendirilmektedir.

Çalışmada e-öğrenme, uzaktan eğitim, karma öğrenme, mobil öğrenme, mikro öğrenme, açık öğrenme platformları, problem temelli, proje tabanlı, deneyimsel ve senaryo tabanlı öğrenme modellerinin veri gazeteciliği eğitimine sunduğu imkanlar tartışılmaktadır. Ayrıca yapay zeka destekli öğrenme sistemleri, öğrenme analitiği, büyük veri ve otomasyon teknolojilerinin gazetecilik eğitiminde kişiselleştirilmiş, uygulamalı ve etkileşimli öğrenme süreçlerini güçlendirdiği vurgulanmaktadır. Çalışma, veri okuryazarlığının gazetecilik müfredatına bağımsız bir teknik ders olarak değil, haber üretim sürecinin tüm aşamalarına yayılan bütüncül bir yaklaşım olarak entegre edilmesi gerektiğini savunmaktadır. Bu doğrultuda kavramsal farkındalık, teknik yeterlilik, editöryal beceri, etik bilinç ve dijital öğrenme yetkinliklerini kapsayan çok boyutlu bir eğitim modeli önerilmektedir.

Anahtar Kelimeler: Veri okuryazarlığı, gazetecilik eğitimi, veri gazeteciliği, dijital öğrenme, e-öğrenme, karma öğrenme

Introduction

In the contemporary era, where digitalization transforms social, economic, cultural, and political life, data has evolved beyond being merely a technical element or a collection of numerical information. Data has become a strategic resource that guides decision-making processes, shapes public debates, and determines the modes of production and circulation of media content. The development of internet technologies, the widespread use of mobile devices, the integration of social media platforms into the center of everyday life, and the expansion of big data ecosystems have led individuals and institutions to encounter vast amounts of data on a daily basis. In this new information environment, it is no longer sufficient to simply read data or process it technically. It is

essential to critically evaluate how data is produced, in what context it is collected, by which methods it is analyzed, how it is visualized, and for what purposes it is presented to the public. For this reason, data literacy is recognized as one of the fundamental competencies of the digital age and is defined as the ability of individuals to effectively understand, analyze, interpret, evaluate, question, and communicate data throughout various processes (Schild, 2004; Carlson et al., 2011).

Although the concept of data literacy was initially addressed in relation to information literacy and statistical literacy, it has evolved into an independent and multidimensional field of research with the advancement of digital technologies. This concept is not limited to the ability to read numerical data or perform basic statistical operations. Data literacy also includes the ability to question the source of data, evaluate the reliability of data collection processes, recognize incomplete or biased data, interpret the reality represented by data within its contextual framework, and share findings within the boundaries of ethical responsibility. In this respect, data literacy functions as a bridge between technical skills and critical thinking. In the digital age, individuals are required not only to access information but also to assess the validity, reliability, and social impacts of the information they access (OECD, 2019; UNESCO, 2021).

One of the fields where this transformation is most clearly observed is journalism. Traditional journalism has long been structured around monitoring events, gathering information from sources, producing news texts, and informing the public. However, within the digital media environment, journalism has evolved beyond merely reporting events and has become a multi-layered field of knowledge production that involves collecting, analyzing, verifying, visualizing, and transforming data into stories that carry public significance. Today, news production is often carried out through open data portals, statistics provided by public institutions, social media data, geographic information systems, election results, economic indicators, health data, environmental measurements, and platform analytics (Gray, Chambers & Bounegru, 2012; Bradshaw, 2017).

Materials and Methods

In the context of journalism, data literacy encompasses the processes of accessing data, cleaning it, analyzing it, interpreting it, visualizing it, and transforming it into news narratives. For a journalist to convert a dataset into information that carries news value, it is first necessary to evaluate the source of the data, understand the data collection method, and recognize the gaps and limitations within the dataset. Subsequently, relationships, trends, ratios, distributions, and changes within the data must be analyzed accurately. However, data journalism does not consist solely of the analysis stage. It is also essential to transform the findings obtained through analysis into a clear, balanced, and ethical narrative that the public can understand. Therefore, data literacy in journalism requires the simultaneous development of analytical skills and storytelling abilities. Data-driven storytelling is the process of interpreting complex datasets within their social context and presenting them in the public interest (Rogers, Schwabish & Bowers, 2017).

In the digital media environment, the rapid spread of misinformation, disinformation, and manipulative content further increases the importance of data literacy in journalism. Numerical claims, graphics, public opinion surveys, or statistical data circulated on social media platforms are often presented without proper context. Errors such as improperly scaled graphs, selective use of data, incomplete samples, misleading ratios, or confusion between causation and correlation may lead to the misdirection of public opinion. Therefore, journalists must not only be able to use data but also understand how data can be manipulated. Data literacy enables journalists to detect such

manipulations and strengthen verification processes (Ridgway, Nicholson & McCusker, 2013; Cairo, 2016).

The significance of data literacy in journalism education becomes clearer when considered together with information literacy and media literacy. Information literacy refers to the ability to access needed information, evaluate it, and use it ethically. Media literacy involves the competence to analyze how media messages are produced and within which ideological, economic, and technological frameworks they are presented. Data literacy complements these two fields by questioning what numbers, graphs, tables, and datasets reveal, what they conceal, and how they should be interpreted. For journalism students, these three types of literacy should not be treated as separate but rather as integrated components of a comprehensive digital literacy framework (Koltay, 2015; UNESCO, 2021).

At this point, the restructuring of journalism education becomes inevitable. Traditional journalism education has long been shaped around news writing, interview techniques, media law, press history, and editorial processes. While these areas remain important, in the digital age it is no longer sufficient for journalism students to rely solely on text production skills. Contemporary journalism education must also include areas such as data collection, data cleaning, basic statistics, visualization, digital storytelling, social media analytics, verification techniques, open-source investigations, artificial intelligence tools, and data ethics. This transformation is moving journalism education toward an interdisciplinary structure (Meyer, 2002; Howard, 2014).

The integration of data journalism into educational programs should not be viewed merely as the addition of a new course. Rather, this process requires a comprehensive reconsideration of journalism education as a whole. Data literacy should not be treated as a technical topic confined to a specific part of the curriculum, but as a fundamental approach that permeates the entire news production process. Students should learn to use data sources in news writing courses, formulate questions based on datasets in investigative journalism courses, produce graphics and maps in visual communication courses, and discuss data privacy and the protection of personal data in media ethics courses. In this way, data literacy acquires not only an instrumental function but also an epistemological and ethical foundation within journalism education.

The development of data analytical competencies for journalism students requires a gradual and practice-oriented learning process. In the first stage, students need to understand what data is, recognize different types of data, identify data sources, and comprehend data production processes. Understanding the differences between structured, semi-structured, and unstructured data forms the foundation of data journalism. In the second stage, students should develop skills in obtaining data from sources such as open data portals, public institutions, international organizations, academic databases, and social media platforms. In the third stage, students should acquire data cleaning and data organization skills, since raw data often contains incomplete, erroneous, repetitive, or inconsistent information. Data cleaning is indispensable for ensuring the reliability of the analysis process. In the fourth stage, basic statistical and analytical skills should be developed. Proper understanding of concepts such as mean, median, ratio, percentage, distribution, trend, comparison, and correlation ensures that data-driven news is built on solid foundations. In the fifth stage, students should develop data visualization and data communication skills. Presenting data to target audiences in a clear, simple, and non-misleading manner is one of the primary responsibilities of data journalism (Cairo, 2016).

Digital learning approaches provide powerful opportunities for enhancing data literacy within journalism education. Digital learning environments enable students to access learning materials regardless of time and location, progress at their own learning pace, and interact with various types of digital resources. These environments offer significant advantages, particularly in practice-oriented fields such as data journalism. Students can access datasets through online platforms, use data analysis tools, work with visualization applications, and share their news projects digitally. This process transforms learning from an activity confined to the classroom into a continuous, flexible, and interactive experience (Garrison & Vaughan, 2008).

E-learning and distance education models provide a functional structure for data literacy education, particularly in terms of resource accessibility and individualized learning. Through learning management systems, course materials, datasets, tutorial videos, reading lists, and assessment tools can be shared with students. In synchronous sessions, instructors and students can perform data analysis exercises simultaneously, while in asynchronous learning environments, students can work on specific datasets at their own pace. This structure creates a more inclusive learning environment for students with varying levels of readiness. In fields such as data journalism that require technical competencies, the opportunity to repeat exercises, revisit instructional videos, and progress through small, incremental steps significantly enhances learning retention.

The blended learning model is particularly well suited for the development of data literacy in journalism education. In this model, theoretical content is delivered through digital platforms, while in-class sessions focus on discussion, practical application, group work, and project development activities. For example, students may learn data cleaning and basic analysis techniques in an online environment. During face-to-face sessions, they can discuss how to formulate news questions from data, determine appropriate visualization types, and establish the ethical framework of news production. In this way, the learning process is not reduced to the transfer of technical skills alone; rather, it becomes integrated with editorial decision-making, critical evaluation, and collaborative production skills (Bonk & Graham, 2012).

Mobile learning and microlearning applications can also be used as supportive tools in data literacy education. Mobile learning enables students to access learning materials at any time through smartphones and tablets. Microlearning, on the other hand, is based on presenting complex topics in short, focused, and digestible modules. In data journalism education, topics such as “What is a CSV file?”, “How is missing data detected?”, “How can misleading graphs be identified?”, and “How is map-based visualization created?” can be presented in the form of short videos, interactive quizzes, or mini practical exercises. This approach facilitates the gradual acquisition of technical skills, particularly in fields that require step-by-step learning (Hug, 2005).

Open and distance learning platforms expand journalism students’ access to global sources of knowledge. MOOC platforms, open course materials, online data journalism workshops, and international educational resources allow students to examine diverse practical examples beyond the limitations of local curricula. Since the tools and methods used in data journalism evolve rapidly, it is important for students to develop lifelong learning habits. In this context, digital learning platforms support the transformation of journalism education into a continuous professional development process that extends beyond university years.

Results and Discussion

The pedagogical approach in data journalism education should move beyond merely teaching technical tools. Problem-based learning offers an effective model in this regard. Students participate

in the processes of data collection, analysis, and news production through real-life social problems. Issues such as youth unemployment, inequality of opportunity in education, environmental pollution, urban transportation, access to healthcare services, or disaster management can serve as strong problem areas for data journalism education. Through this process, students learn not only how to work with data but also how to evaluate the social context of data, compare different sources, and develop news questions that serve the public interest. Thus, data literacy moves beyond being an abstract skill and becomes a tool for understanding social realities.

Project-based learning provides a product-oriented structure in data journalism education. Students collect data related to a specific topic, clean the data, analyze it, visualize it, and transform it into a news report or digital media product. This process enables students to develop both technical and editorial skills simultaneously. Project-based learning also supports time management, teamwork, resource utilization, research planning, and presentation skills. For data literacy in journalism to develop meaningfully, students must not only become familiar with tools but also use these tools holistically within the news production process.

The experiential learning approach facilitates the integration of data literacy into professional practice within journalism education. When students work with real datasets, real news problems, and, where possible, collaborate with media organizations, they directly experience the challenges of data journalism. Issues such as the lack of reliable data sources, the irregularity of public data, incompatibility of data formats, verification difficulties, ethical dilemmas, and editorial constraints can only be fully understood through hands-on practice.

Scenario-based learning can also be effective in developing journalism students' abilities in rapid decision-making, verification, and ethical evaluation. Students may be presented with scenarios such as election nights, natural disasters, economic crises, pandemics, social protests, or public data leaks. Through these scenarios, students can discuss what data they will need, which sources they should use, what information must be verified, and how the news should be presented. Digital simulations, virtual newsrooms, and interactive learning environments can further enhance the effectiveness of this method. In this way, students become better prepared for the pressure and uncertainty characteristic of real news environments.

Artificial intelligence-supported learning systems also provide new opportunities for data literacy education. These systems can analyze data obtained from students' learning processes and provide personalized feedback. Through learning analytics, it is possible to identify the topics in which a student struggles, the stages of data analysis where errors occur, or the visualization preferences that create difficulties. In this way, the teaching process becomes more targeted and individualized. Artificial intelligence-supported tools can also be used for educational purposes in processes such as data cleaning, text mining, natural language processing, automated summarization, and verification (Siemens & Long, 2011).

Big data and automation technologies are transforming both the journalism profession and journalism education. While big data enables journalists to analyze large-scale social events more comprehensively, automation accelerates data collection and organization processes. However, these technologies also introduce significant ethical challenges. The protection of personal data, data privacy, algorithmic bias, transparency in automated decision systems, and the contextual accuracy of data sources are among the primary concerns. Journalism students working with large datasets should learn not only technical analysis but also how to evaluate the effects of data on

individuals, communities, and public life. Therefore, data literacy education should be conducted in conjunction with ethical literacy (Mayer-Schönberger & Cukier, 2013; Koltay, 2015).

Technological infrastructure is also a determining factor in developing data literacy within journalism education. Educational institutions must provide access to tools for data analysis, visualization, mapping, and verification. Tools such as Excel, Google Sheets, OpenRefine, Datawrapper, Flourish, Tableau, QGIS, Python, and R can be used at different levels of data journalism education. However, teaching tools should not be regarded as an end in itself, but rather as an instrumental step. The primary objective is to enable students to ask meaningful questions, interpret data within its context, and produce news that serves the public interest.

Assessment methods are also important in the development of data literacy. Traditional examinations may not be sufficient to measure students' data journalism skills. Instead, portfolio assessments, project files, data analysis reports, visualization outputs, news production processes, and reflective journals can be used. Evaluation should focus not only on the final product but also on the decision-making processes involved in working with data. The reasons for selecting specific data sources, the methods used to clean the data, the analytical techniques applied, the ethical considerations adopted in visualization, and the structure of the news narrative should all form part of the assessment process.

Conclusion

In this context, a holistic model can be proposed for the development of data literacy in journalism education. The first dimension of this model is conceptual awareness. Students should critically understand concepts such as data, information, evidence, statistics, algorithms, platforms, and verification. The second dimension is technical competence. Students should be able to use tools for data collection, cleaning, analysis, and visualization. The third dimension is editorial competence. Students should be able to generate news questions from data, contextualize data, and construct news narratives. The fourth dimension is ethical competence. Students should develop awareness regarding issues such as data privacy, representational fairness, decontextualization, protection of personal data, and algorithmic bias. The fifth dimension is digital learning competence. Students should be able to effectively use online resources, open learning platforms, artificial intelligence-supported tools, and collaborative digital environments.

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DIGITAL TRANSFORMATION IN ENGLISH LANGUAGE TEACHING: A REFLECTIVE AND EXPANDED PEDAGOGICAL STUDY

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Abstract: The rapid advancement of digital technologies has initiated a profound transformation in the field of English Language Teaching (ELT), shifting it from a traditionally teacher-centered paradigm to a

dynamic, learner-centered and technology-mediated educational environment. In contemporary higher education contexts, digital tools are no longer perceived as auxiliary resources but rather as integral components of pedagogical design, fundamentally influencing curriculum development, instructional strategies, and assessment practices.

This study adopts a reflective and expanded pedagogical perspective to examine the multifaceted impact of digital transformation on ELT. Drawing upon both established theoretical frameworks and classroom-based experiential insights, the research investigates how digital environments reshape teaching methodologies and learning processes. Particular attention is given to the ways in which digital platforms facilitate the development of communicative competence by enabling access to authentic linguistic input, fostering real-time interaction, and providing immediate, adaptive feedback—elements widely recognized as critical for effective language acquisition.

Moreover, the paper explores the role of digital technologies in promoting differentiated instruction and personalized learning pathways. By accommodating diverse learner needs, preferences, and proficiency levels, digital tools empower students to take greater control of their learning processes, thereby enhancing learner autonomy, motivation, and engagement. This shift aligns with constructivist and socio-cultural approaches to language learning, where knowledge is actively constructed through interaction and meaningful use of language in context.

Despite these pedagogical advantages, the study critically addresses several challenges associated with digital transformation. These include disparities in access to technological resources, inconsistencies in digital literacy among both learners and educators, and the potential for technology to be used in a superficial or pedagogically ineffective manner. The research underscores that without thoughtful integration, digital tools may fail to achieve their intended educational outcomes and, in some cases, may even hinder deep learning.

Ultimately, this paper argues that successful digital transformation in ELT requires a strategic and pedagogically grounded approach, where technology is purposefully aligned with instructional objectives. It emphasizes the continued centrality of the teacher as a facilitator, designer, and mediator of learning experiences, whose role is critical in ensuring that digital innovations translate into meaningful and sustainable educational practices. The findings contribute to ongoing discussions on the future of ELT by offering practical and reflective insights into how educators can effectively navigate and harness digital change in language education.

Key words: digital pedagogy, ELT, blended learning, learner autonomy, AI in education, classroom interaction, instructional design

İNGİLİS DİLİNİN TƏDRİSİNDƏ RƏQƏMSAL TRANSFORMASIYA: REFLEKTİV VƏ GENİŞLƏNDİRİLMİŞ PEDAQOJİ TƏDQİQAT

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Xülasə: Rəqəmsal texnologiyaların ingilis dili tədrisinə inteqrasiyası tədris prosesinin strukturunu və pedaqoji yanaşmaları əsaslı şəkildə dəyişmişdir. Bu məqalə bu transformasiyanın tədris keyfiyyətinə, tələbə fəallığına və dil bacarıqlarının inkişafına təsirini geniş şəkildə araşdırır. Müasir ali təhsil kontekstlərində rəqəmsal alətlər artıq köməkçi resurslar kimi deyil, pedaqoji dizaynın ayrılmaz komponentləri kimi qəbul edilir və tədris planının hazırlanmasına, tədris strategiyalarına və qiymətləndirmə təcrübələrinə fundamental təsir göstərir.

Bu tədqiqat rəqəmsal transformasiyanın İDT-yə çoxşaxəli təsirini araşdırmaq üçün əks etdirici və genişləndirilmiş pedaqoji perspektivdən istifadə edir. Həm müəyyən edilmiş nəzəri çərçivələrə, həm də sinif əsaslı təcrübə anlayışlarına əsaslanaraq, tədqiqat rəqəmsal mühitlərin tədris metodologiyalarını və öyrənmə proseslərini necə yenidən formalaşdırdığını araşdırır. Xüsusi diqqət rəqəmsal platformaların orijinal linqvistik girişə çıxışı təmin etməklə, real vaxt qarşılıqlı əlaqəsini təşviq etməklə və effektiv dil mənimsəməsi üçün geniş şəkildə vacib hesab edilən elementlər kimi dərhal, adaptiv rəy təmin etməklə kommunikativ səriştəliyin inkişafını asanlaşdıran yollara yönəldilir.

Bundan əlavə, məqalədə rəqəmsal texnologiyaların fərqli təlim və fərdiləşdirilmiş öyrənmə yollarının təşviqində rolu araşdırılır. Rəqəmsal alətlər müxtəlif tələbə ehtiyaclarını, üstünlüklərini və bacarıq səviyyələrini nəzərə alaraq, tələbələrə təlim proseslərinə daha çox nəzarət etməyə imkan verir və bununla da tələbənin muxtariyyətini, motivasiyasını və iştirakını artırır. Bu dəyişiklik, biliklərin qarşılıqlı əlaqə və kontekstdə dilin mənalı istifadəsi yolu ilə aktiv şəkildə qurulduğu dil öyrənməsinə konstruktiv və sosial-mədəni yanaşmalarla uyğun gəlir.

Bu pedaqoji üstünlüklərə baxmayaraq, tədqiqat rəqəmsal transformasiya ilə əlaqəli bir neçə çətinliyi tənqidi şəkildə həll edir. Bunlara texnoloji resurslara çıxışdakı fərqlər, həm tələbə, həm də müəllimlər arasında rəqəmsal savadlılıqdakı uyğunsuzluqlar və texnologiyanın səthi və ya pedaqoji cəhətdən səmərəsiz şəkildə istifadə edilməsi potensialı daxildir. Tədqiqat vurğulayır ki, düşünülmüş integrasiya olmadan rəqəmsal alətlər nəzərdə tutulan təhsil nəticələrinə nail ola bilməz və bəzi hallarda hətta dərin öyrənməyə mane ola bilər.

Nəticə etibarilə, bu məqalədə ELT-də uğurlu rəqəmsal transformasiya texnologiyanın məqsədyönlü şəkildə tədris məqsədləri ilə uyğunlaşdırıldığı strateji və pedaqoji cəhətdən əsaslandırılmış yanaşma tələb olunduğu iddia edilir. Bu, müəllimin təlim təcrübələrinin fasilitatoru, dizayneri və vasitəçisi kimi davamlı mərkəziliyini vurğulayır və onun rolu rəqəmsal innovasiyaların mənalı və davamlı təhsil təcrübələrinə çevrilməsini təmin etməkdə vacibdir. Bu tapıntılar, müəllimlərin dil təhsilində rəqəmsal dəyişiklikləri necə effektiv şəkildə idarə edə və onlardan necə istifadə edə biləcəkləri barədə praktik və əks etdirici fikirlər təqdim etməklə, ingilis dilini öyrənmənin (EOT) gələcəyi ilə bağlı davam edən müzakirələrə töhfə verir.

Açar sözlər: rəqəmsal pedaqogika, İngilis dilini öyrənmə (EOT), qarışıq təlim, şagird muxtariyyəti, təhsildə süni intellekt, sinifdə qarşılıqlı əlaqə, tədris dizaynı

Introduction

The rapid evolution of digital technologies has brought about profound and irreversible changes in educational systems worldwide. Across disciplines, teaching and learning practices have been reshaped by the increasing integration of digital tools, online platforms, and artificial intelligence-based applications. However, this transformation is particularly significant in the field of English Language Teaching (ELT), where communication, interaction, and sustained exposure to meaningful language input are central to successful learning outcomes.

Traditionally, ELT classrooms were characterized by teacher-centered instruction, where knowledge transmission was primarily facilitated through textbooks, grammar explanations, and limited classroom interaction. In such environments, learners often had restricted opportunities to engage in authentic communication, and language use was frequently decontextualized. As a result, students tended to develop receptive knowledge of language structures but faced challenges in applying this knowledge in real communicative contexts.

In contrast, contemporary ELT environments are increasingly defined by digital integration, where multimedia resources, online learning platforms, and interactive technologies play a central role in shaping instructional practices. Tools such as learning management systems, video conferencing applications, mobile learning apps, and AI-supported language platforms have

expanded the boundaries of the traditional classroom, enabling learners to interact with language in more dynamic, flexible, and personalized ways. These technologies provide continuous access to authentic language input, facilitate multimodal learning experiences, and support real-time interaction, all of which are essential for developing communicative competence.

From my own teaching experience in higher education contexts, I have observed a noticeable shift in student engagement when digital elements are incorporated into lessons. Learners tend to demonstrate higher levels of motivation and participation when exposed to visual, auditory, and interactive materials compared to traditional lecture-based instruction. For example, the use of videos, online discussions, and interactive exercises often encourages students to take a more active role in the learning process, fostering both linguistic development and learner autonomy.

However, the integration of digital technologies in ELT is not merely a matter of introducing new tools into the classroom. Rather, it represents a deeper pedagogical transformation that requires a fundamental rethinking of teaching philosophy, instructional design, and classroom interaction. In this context, the role of the teacher is no longer limited to the transmission of knowledge; instead, the teacher becomes a facilitator, learning designer, and mediator who guides students through complex, technology-enhanced learning environments.

This transformation also demands a more reflective and strategic approach to pedagogy, where digital tools are not used for their own sake but are carefully aligned with learning objectives and communicative outcomes. Educators are required to critically evaluate how technology can enhance, rather than replace, meaningful pedagogical practices. This includes designing tasks that promote interaction, collaboration, and critical thinking, as well as ensuring that technology serves as a tool for deeper learning rather than superficial engagement.

Furthermore, the shift toward digital ELT raises important questions about equity, accessibility, and digital literacy. While digital tools offer significant pedagogical advantages, their effectiveness is often influenced by students' access to technology and their ability to use it effectively. These factors highlight the need for a balanced and inclusive approach to digital transformation, one that takes into account both the opportunities and limitations of technology-enhanced learning environments.

In this regard, the present study seeks to explore the pedagogical implications of digital transformation in ELT through a reflective and practice-informed perspective. It aims to examine how digital technologies reshape teaching methodologies, influence learner engagement, and redefine the roles of both teachers and students in the language learning process. By combining theoretical insights with classroom-based observations, this research contributes to a deeper understanding of how digital transformation can be effectively integrated into ELT in a pedagogically meaningful way.

Literature Review.

The role of digital technologies in language education has been extensively examined in contemporary applied linguistics and ELT research. The literature generally reflects a dual perspective: while digital tools are recognized for their significant pedagogical potential in enhancing language learning, scholars also emphasize the importance of critical, context-sensitive, and pedagogy-driven integration. This section reviews key theoretical and empirical contributions that inform the present study.

One of the central arguments in the field is that technology-enhanced language learning environments provide learners with increased access to authentic linguistic input, which is essential

for second language acquisition. Chapelle (2017) emphasizes that computer-assisted language learning (CALL) environments enable learners to engage with real-life language use through multimedia content, interactive tasks, and online communication platforms. This exposure to authentic input not only supports comprehension development but also contributes to the acquisition of pragmatic and discourse-level competence, which is often difficult to achieve through traditional textbook-based instruction.

In a similar vein, Godwin-Jones (2018) highlights the expanding role of mobile-assisted language learning (MALL), arguing that mobile technologies have fundamentally transformed the spatial and temporal boundaries of language learning. According to this perspective, learners are no longer confined to classroom environments; instead, they can engage in continuous, self-directed learning through smartphones, applications, and digital communication tools. This mobility fosters incidental learning, increases exposure to the target language, and supports the development of learner autonomy by enabling students to take responsibility for their own learning processes.

From a pedagogical standpoint, Richards (2015) underscores the centrality of communicative language teaching (CLT) in modern ELT practice. He argues that meaningful interaction is at the core of language acquisition and that language is best learned through use rather than rote memorization. Within this framework, digital technologies are viewed as powerful enablers of communicative competence, as they facilitate synchronous and asynchronous interaction through tools such as discussion forums, video conferencing platforms, collaborative writing applications, and social media environments. These technologies create opportunities for learners to engage in authentic communication, negotiate meaning, and develop fluency in real-world contexts.

Expanding on this perspective, Hockly (2018) discusses blended learning as an effective pedagogical model that integrates traditional face-to-face instruction with online learning components. Blended learning environments are particularly significant in ELT, as they combine the advantages of direct teacher guidance with the flexibility and accessibility of digital platforms. This hybrid approach allows for differentiated instruction, enabling learners to progress at their own pace while still benefiting from structured classroom interaction. Furthermore, blended learning supports flipped classroom models, where students engage with instructional content outside the classroom and use in-class time for communicative and collaborative activities.

Recent developments in educational technology research have further introduced artificial intelligence (AI) and adaptive learning systems into the ELT landscape. These technologies are designed to analyze learner performance data and provide personalized learning pathways based on individual strengths, weaknesses, and learning patterns. Such adaptive systems are increasingly seen as tools for enhancing efficiency, improving learner engagement, and offering immediate, data-driven feedback. AI-powered applications, including chatbots, automated writing evaluation tools, and intelligent tutoring systems, are reshaping the nature of language instruction by offering individualized support at scale.

Despite these advancements, the literature also presents critical perspectives regarding the limitations and challenges of digital integration. Scholars caution that the pedagogical effectiveness of technology is not inherent in the tools themselves but depends largely on how they are designed and implemented within instructional contexts. Poorly integrated technologies may lead to superficial learning experiences, where engagement is high but cognitive depth is limited. Additionally, concerns are raised regarding unequal access to digital resources, disparities in digital

literacy among learners and educators, and the risk of over-reliance on automated systems that may reduce meaningful teacher-student interaction.

Therefore, the literature consistently suggests that successful digital transformation in ELT requires more than technological adoption; it demands a strong pedagogical foundation grounded in sound instructional design principles. Technology should function as an enabler of learning rather than a replacement for pedagogical practice. This balance between innovation and pedagogy forms the conceptual foundation for the present study, which seeks to explore how digital tools can be effectively integrated into ELT to enhance both teaching quality and learning outcomes.

Materials and Methods

This study adopts a **qualitative, reflective, and interpretive research design** grounded in both theoretical perspectives and experiential teaching practice. The methodological orientation is informed by the view that pedagogical phenomena in language education are best understood through in-depth exploration of classroom processes, learner behavior, and instructional dynamics rather than through purely numerical measurement. Accordingly, the study prioritizes meaning-making, reflection, and contextual interpretation over statistical generalization.

The research is situated within the researcher's own higher education teaching context, where English language instruction is supported by a range of digital tools and blended learning approaches. This positioning allows for an **insider perspective**, enabling a nuanced understanding of how digital transformation manifests in real pedagogical practice.

1. Research Design

The study employs a **qualitative reflective research design**, combining elements of classroom-based inquiry and practitioner reflection. This approach is particularly suitable for examining digital transformation in ELT, as it captures the complexity of teaching-learning interactions that are often dynamic, context-dependent, and multifaceted.

Rather than testing hypotheses, the study seeks to **interpret and describe pedagogical phenomena**, focusing on how digital technologies influence:

- classroom interaction patterns
- learner engagement and motivation
- development of language skills
- teacher instructional strategies
- learner autonomy and participation

This interpretive approach aligns with reflective teaching traditions, where teachers systematically analyze their own practice to improve instructional effectiveness.

2. Data Collection Methods

Data for this study were collected over multiple academic semesters through a combination of observational and reflective techniques. Three main sources of data were utilized:

2.1 Continuous Classroom Observation

Systematic classroom observations were conducted throughout regular English language classes. These observations focused on:

- student participation during digital activities
- interaction patterns in group and pair work
- responsiveness to multimedia input
- behavioral changes during technology-enhanced lessons

Observation notes were recorded immediately after each lesson to ensure accuracy and detail.

2.2 Reflective Teaching Journals

A structured reflective journal was maintained by the instructor throughout the teaching period. These journals documented:

- instructional decisions and their outcomes
- perceived levels of student engagement
- challenges encountered in integrating digital tools
- successful and unsuccessful pedagogical strategies

This reflective practice allowed for continuous self-evaluation and deeper insight into the evolving role of technology in the classroom.

2.3 Analysis of Digital Learning Platform Interaction

Additional data were gathered through the analysis of student engagement with digital platforms used in the course. These included learning management systems and online communication tools. The analysis focused on:

- frequency of student participation
- completion of online tasks and assignments
- interaction within discussion forums
- use of supplementary digital resources

This provided insight into learner autonomy and out-of-class engagement with English language learning materials.

3. Instructional Environment and Tools

The teaching environment was designed as a **digitally enhanced blended learning space**, integrating traditional instruction with a variety of digital resources and platforms. The following tools and methods were employed:

- Multimedia-based teaching materials, including videos, audio recordings, and interactive presentations to support listening and comprehension skills
- AI-supported language learning applications, used for grammar practice, vocabulary development, and pronunciation training
- Online learning management systems (LMS), which facilitated distribution of materials, assignment submission, and feedback delivery
- Collaborative digital tasks and discussion forums, designed to promote interaction, peer learning, and communicative competence

These tools were integrated into lesson planning in a purposeful way, ensuring alignment with learning objectives rather than random or supplementary use of technology.

4. Data Analysis

The collected data were analyzed using a **thematic analysis approach**, which allowed for the identification of recurring patterns and meaningful categories across different data sources. The analysis focused on four key pedagogical dimensions:

- Learner engagement: levels of attention, participation, and motivation during digital activities
- Learner autonomy: evidence of independent learning behavior and self-directed study
- Skill development: improvements in communicative competence, especially speaking, listening, and writing skills

- Pedagogical effectiveness: the extent to which digital tools supported or hindered instructional goals

Thematic patterns were identified through iterative reading of observation notes, journal entries, and platform data. These themes were then synthesized to provide a comprehensive understanding of how digital transformation influences ELT practice in a real classroom context.

Results

The analysis of classroom observations, reflective journals, and digital platform data revealed several significant pedagogical outcomes associated with the integration of digital technologies in English Language Teaching (ELT). These findings are organized into four interrelated themes: increased interaction and engagement, development of communicative skills, expansion of learning beyond the classroom, and the gradual development of learner autonomy.

1. Increased Interaction and Engagement

One of the most consistent and observable outcomes of digital integration was a marked increase in student engagement during classroom activities. When digital tools such as videos, interactive quizzes, multimedia presentations, and real-time online tasks were incorporated into lessons, students demonstrated significantly higher levels of participation compared to traditional instruction.

In particular, digital content appeared to reduce passive learning behaviors, such as silent note-taking and minimal verbal participation, and instead encouraged active involvement in tasks. Learners were more willing to respond to questions, collaborate in group activities, and engage in problem-solving tasks when technology-mediated resources were used. Interactive elements such as gamified quizzes and visual stimuli also contributed to maintaining attention and sustaining motivation throughout the lesson.

2. Development of Communicative Skills

The integration of digital environments also had a noticeable impact on students' communicative competence, particularly in listening and speaking skills. Exposure to authentic audio-visual materials, including videos featuring native speakers and real-life communication contexts, enhanced learners' ability to understand natural speech patterns, intonation, and pronunciation.

Furthermore, the use of speech recognition tools and AI-supported pronunciation applications provided learners with immediate feedback in a non-threatening and supportive environment. This encouraged repeated practice and reduced anxiety commonly associated with oral production in traditional classroom settings. Over time, students demonstrated greater fluency, improved pronunciation accuracy, and increased confidence in spoken communication.

3. Expansion of Learning Beyond the Classroom

Another significant finding was the extension of learning activities beyond formal classroom boundaries. Students increasingly engaged in independent learning through digital platforms, mobile applications, and online resources outside scheduled class hours.

This continuous exposure to the English language contributed to reinforcement of classroom learning and facilitated long-term retention of linguistic knowledge. Learners frequently accessed supplementary materials such as online exercises, vocabulary apps, and video tutorials, indicating a shift toward more flexible and self-paced learning habits. This expansion of learning environments effectively transformed language acquisition into an ongoing process rather than a classroom-bound activity.

4. Gradual Development of Learner Autonomy

Over the course of the study, a gradual but clear development of learner autonomy was observed. Initially, students relied heavily on teacher guidance for completing tasks and understanding instructions. However, as they became more familiar with digital tools and platforms, they increasingly demonstrated independent learning behaviors.

Students began to take initiative in selecting learning resources, completing optional online tasks, and revisiting digital materials for revision purposes. This shift indicates an important pedagogical development, as learner autonomy is widely recognized as a key factor in successful language acquisition. The increased ability to self-direct learning suggests that digital tools can play a significant role in fostering independent and responsible learners.

Discussion

The findings of this study indicate that digital technologies, when integrated in a pedagogically purposeful manner, can significantly enhance English language teaching by improving student engagement, communicative competence, and learner autonomy. However, these benefits are not automatic; rather, they depend on careful instructional design and thoughtful pedagogical implementation.

From a pedagogical perspective, the role of the teacher is undergoing a fundamental transformation. The teacher is no longer solely a transmitter of knowledge but increasingly functions as a **facilitator, instructional designer, and learning architect** who creates meaningful learning environments that integrate both digital and traditional methodologies. This expanded role requires not only subject expertise but also a high level of digital literacy and the ability to critically evaluate the pedagogical value of technological tools.

The findings also reinforce the importance of aligning technology use with clear instructional objectives. Digital tools such as multimedia content, AI applications, and online platforms are most effective when they are purposefully embedded within lesson design rather than used as isolated or decorative elements. Without such alignment, there is a risk that technology may become a distraction rather than a learning aid.

At the same time, the study highlights certain limitations and challenges associated with digital transformation. One of the most significant concerns is the potential reduction of face-to-face interaction when digital tools are overused. While technology can enhance communication, it cannot fully replace the social and interpersonal dimensions of language learning, which remain essential for developing pragmatic and conversational competence.

Additionally, issues related to unequal access to technology and varying levels of digital literacy among students may create disparities in learning outcomes. Students with limited access to devices or stable internet connections may be disadvantaged, leading to unequal participation and achievement levels. This highlights the need for educational institutions to ensure equitable access to digital resources and provide adequate training for both teachers and learners.

Furthermore, there is a risk of over-reliance on technology-driven learning systems, where automated feedback and digital exercises may replace deeper cognitive engagement if not carefully managed. Therefore, a balanced approach that combines digital innovation with traditional pedagogical interaction is essential for maintaining educational quality.

Overall, the discussion suggests that digital transformation in ELT should be understood not as a replacement of traditional pedagogy, but as an **evolutionary extension of it**. When strategically implemented, digital tools can enrich language learning environments, but their effectiveness

ultimately depends on the teacher's pedagogical decisions and the institutional support systems in place.

Conclusion

In conclusion, digital transformation in English Language Teaching represents a fundamental pedagogical shift that is reshaping the nature of language education in the 21st century. The findings of this study demonstrate that when digital technologies are integrated in a purposeful and pedagogically informed manner, they have the potential to significantly enhance learning outcomes, increase learner engagement, and foster the development of learner autonomy. These improvements are particularly evident in the areas of communicative competence, interaction, and continuous language exposure beyond the classroom.

However, the study also highlights that the effectiveness of digital transformation is not inherent in the technologies themselves, but rather in the way they are implemented within the teaching and learning process. Technology, in isolation, does not guarantee improved educational outcomes; its pedagogical value depends on careful instructional design, clear learning objectives, and the teacher's ability to meaningfully integrate digital tools into classroom practice.

From this perspective, digital technologies should not be viewed as replacements for traditional teaching methods or for the role of the teacher. Instead, they should be understood as **complementary pedagogical instruments** that extend, enrich, and diversify instructional strategies. The most effective ELT environments are those in which digital and traditional approaches are combined in a balanced and coherent manner, allowing each to compensate for the limitations of the other.

Furthermore, sustainable improvement in English language teaching requires a holistic approach that integrates three essential dimensions: **technology, pedagogy, and human interaction**. While technology provides access, flexibility, and interactivity, pedagogy ensures structure, purpose, and meaningful learning design. Human interaction, on the other hand, remains central to language acquisition, as it facilitates emotional engagement, negotiation of meaning, and authentic communication.

Finally, this study underscores the importance of the teacher's evolving role in digitally transformed classrooms. Teachers are not only implementers of technology but also critical decision-makers who shape how digital tools are used to support learning. Their pedagogical awareness, adaptability, and reflective practice are key factors in ensuring that digital transformation leads to meaningful and sustainable educational development.

Overall, digital transformation in ELT should be viewed as an ongoing and dynamic process rather than a fixed outcome. Its success depends on continuous reflection, institutional support, and a balanced integration of innovative technologies with sound pedagogical principles.

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THE PEDAGOGICAL POTENTIAL OF MOBILE LEARNING: IMPLEMENTATION OPPORTUNITIES IN THE EDUCATION SYSTEM OF AZERBAIJAN

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Abstract: This article provides a comprehensive analysis of the pedagogical potential of mobile learning and its implementation opportunities in the education system of Azerbaijan. The study demonstrates that the essence of mobile learning lies not merely in the use of technological tools but in the formation of a learner-centered pedagogical model. The paper systematically examines theoretical approaches such as constructivism, social learning, and contextual learning, substantiating the role of mobile learning in fostering personalized and collaborative learning environments (Kearney et al., 2012; Laurillard, 2012). The findings indicate that although digital transformation initiatives in Azerbaijan have created a preliminary foundation for mobile learning, challenges remain in terms of infrastructure, pedagogical preparedness, and content localization.

Keywords: mobile learning, pedagogical potential, digital transformation, Azerbaijan education, personalized learning

MOBİL ÖYRƏNMƏNİN PEDAQOJİ POTENSİALİ: AZƏRBAYCANIN TƏHSİL SİSTEMİNDƏ TƏTBİQ İMKANLARI

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Xülasə: Bu məqalədə mobil öyrənmənin pedaqoji potensialı və onun Azərbaycanın təhsil sistemində tətbiqi imkanları hərtərəfli təhlil edilir. Tədqiqat göstərir ki, mobil öyrənmənin mahiyyəti yalnız texnoloji vasitələrdən istifadə ilə məhdudlaşmır, eyni zamanda öyrənməmərkəzli pedaqoji modelin formalaşdırılmasını nəzərdə tutur. Məqalədə konstruktivizm, sosial öyrənmə və kontekstual öyrənmə kimi nəzəri yanaşmalar sistemli şəkildə nəzərdən keçirilir və mobil öyrənmənin fərdiləşdirilmiş və əməkdaşlığa əsaslanan öyrənmə mühitlərinin formalaşdırılmasındakı rolu əsaslandırılır (Kearney et al., 2012; Laurillard, 2012). Tədqiqatın nəticələri göstərir ki, Azərbaycanda rəqəmsal transformasiya istiqamətində həyata keçirilən təşəbbüslər mobil öyrənmənin tətbiqi üçün ilkin zəmin yaratmışdır. Bununla belə, infrastruktur, pedaqoji hazırlıq və tədris məzmununun lokallaşdırılması ilə bağlı müəyyən çətinliklər hələ də mövcuddur.

Açar sözlər: mobil öyrənmə, pedaqoji potensial, rəqəmsal transformasiya, Azərbaycan təhsili, fərdiləşdirilmiş öyrənmə

Introduction

Rapid technological advancements in the 21st century and the emergence of the information society have led to fundamental changes in the structure and functions of education systems. These transformations have necessitated a reconsideration of traditional teaching models and the development of more flexible, accessible, and personalized learning environments. In this context, mobile learning has emerged as one of the key directions in modern education. By enabling learning anytime and anywhere through portable digital devices, mobile learning eliminates spatial and temporal constraints in education (Traxler, 2009).

The relevance of this study stems from the fact that digital transformation is no longer an alternative but a necessary direction for the development of modern education systems. In recent years, state programs and projects implemented in Azerbaijan have contributed to significant progress in this area. In particular, the “Virtual School” platform has facilitated the widespread adoption of distance and mobile learning (Ministry of Science and Education of the Republic of Azerbaijan, 2021). Additionally, initiatives such as “Digital Skills” and “E-Education” have accelerated the integration of technology into the education system.

The significance of this research lies in the systematic pedagogical evaluation of mobile learning within the Azerbaijani education system. The current situation indicates that issues such as unequal distribution of technological infrastructure across regions, insufficient digital competencies among teachers, and the need to adapt educational resources to the national context remain actual (World Bank, 2022). Therefore, this study aims to identify the pedagogical and organizational conditions necessary for the effective implementation of mobile learning.

Materials and Methods

Mobile learning offers extensive opportunities for the implementation of personalized education by enabling learners to access educational resources that are tailored to their individual characteristics, interests, prior knowledge, and learning pace. This adaptability allows for a more efficient and goal-oriented organization of the learning process, as learners can engage with content in ways that best suit their cognitive needs and preferences. Unlike traditional one-size-fits-all instructional models, mobile learning environments support differentiated instruction and adaptive learning pathways, thereby enhancing learning outcomes and learner satisfaction. Research indicates that mobile learning is not confined to formal classroom settings; rather, it extends into everyday life, transforming learning into a continuous and contextually embedded activity (Traxler, 2009).

Furthermore, mobile learning promotes learner autonomy and self-regulation. Through mobile devices, learners can independently plan, monitor, and evaluate their own learning processes. Features such as instant feedback, progress tracking, and access to a wide range of digital resources empower learners to take greater responsibility for their educational development. This shift from teacher-centered instruction to learner-centered engagement is a defining characteristic of modern pedagogical approaches and is strongly supported by mobile technologies.

In addition, the integration of interactive elements—such as multimedia content, simulations, quizzes, and discussion platforms—enhances cognitive engagement and facilitates deeper understanding of subject matter. Gamification mechanisms, including points, badges, leaderboards, and rewards, further increase learners’ motivation by introducing elements of competition and achievement. These features not only make the learning process more engaging but also encourage

persistence and active participation, particularly among younger learners who are accustomed to digital environments.

Mobile learning also fosters collaborative learning by enabling communication and interaction among learners through online forums, social media platforms, and collaborative tools. Such environments support peer-to-peer learning, knowledge sharing, and collective problem-solving, which are essential components of social constructivist pedagogy. Learners can participate in discussions, exchange ideas, and co-construct knowledge regardless of their physical location, thereby overcoming geographical barriers.

Moreover, international reports emphasize that mobile learning environments play a significant role in developing key 21st-century competencies, including critical thinking, creativity, communication, collaboration, and digital literacy (UNESCO, 2022). By engaging learners in interactive and authentic tasks, mobile learning prepares them for the demands of the modern workforce and digital society.

Another important advantage of mobile learning is its capacity to support inclusive education. Mobile technologies can provide access to learning opportunities for diverse groups of learners, including those in remote areas, individuals with disabilities, and students with varying learning needs. Assistive technologies, customizable interfaces, and flexible content delivery contribute to creating more equitable and accessible educational environments.

In conclusion, the pedagogical potential of mobile learning lies in its ability to transform traditional educational practices by promoting personalization, interactivity, collaboration, and inclusivity. Its effective implementation, however, depends on the thoughtful integration of technological tools with sound pedagogical principles, ensuring that learning experiences are meaningful, engaging, and aligned with learners' needs.

Theoretical Foundations of Mobile Learning

Mobile learning is a concept grounded in various theoretical approaches of modern pedagogy, offering new opportunities for organizing learning in digital environments. Its pedagogical essence extends beyond the use of technological tools; it represents a comprehensive approach that influences the structure of the learning process, interactions among participants, and mechanisms of knowledge acquisition. The theoretical foundations of this model are primarily explained through constructivism, social learning theory, and contextual learning (Kearney et al., 2012).

According to the constructivist approach, learning is not the passive transmission of knowledge but an active cognitive process shaped by the learner. In this framework, learners construct knowledge by connecting new information with their prior experiences. Mobile technologies provide a favorable environment for constructivist learning by offering interactive applications, multimedia resources, and opportunities for independent exploration (Laurillard, 2012).

Another important theoretical foundation of mobile learning is social learning theory, which emphasizes that learning occurs through social interaction and collaboration. Mobile platforms, social networks, online forums, and collaborative tools expand communication among learners and facilitate collective knowledge exchange. These environments allow learners to benefit from each other's experiences, engage in discussions, and develop collaborative problem-solving skills (Traxler, 2009).

Contextual learning also plays a significant role in the theoretical basis of mobile learning. This approach suggests that knowledge is more effectively acquired when applied in real-life

situations. The portability and accessibility of mobile devices enable learners to continue educational activities in diverse environments, bridging formal and informal learning contexts.

Additionally, mobile learning incorporates the concept of self-regulated learning. Learners can organize their learning processes according to their individual pace, revisit materials, and independently address knowledge gaps. This fosters personalized learning and increases learner responsibility.

In summary, the theoretical foundations of mobile learning indicate that it is not merely a technological innovation but a transformation of pedagogical thinking. Its effectiveness depends not only on the devices used but also on how these technologies are applied within pedagogical frameworks.

Current Situation and Challenges in Azerbaijan's Education System

In recent years, the implementation of mobile learning in Azerbaijan has become more systematic and purposeful. During the COVID-19 pandemic, the "Virtual School" platform played a crucial role in ensuring the continuity of education and promoting the widespread adoption of mobile and distance learning.

The pandemic demonstrated that mobile learning is not only an emergency alternative but can become an integral component of modern education systems. It also accelerated the adaptation of educational stakeholders to digital environments and redefined the role of technology in teaching.

Despite progress, several challenges remain. One of the main issues is the unequal distribution of internet access across regions. In rural and remote areas, limited internet speed and stability hinder the effective implementation of mobile learning. Additionally, unequal access to devices such as smartphones, tablets, and computers contributes to educational inequality (World Bank, 2022).

Another significant challenge is the level of teachers' preparedness in digital pedagogy. While many teachers use technological tools, their effective pedagogical integration remains problematic. Therefore, teachers must develop not only technical skills but also competencies in digital didactics and innovative teaching methods.

Furthermore, the quality and localization of educational content require attention. Many digital resources are presented in foreign languages, creating difficulties in aligning them with the national curriculum and students' language proficiency. Developing high-quality, locally relevant digital content is essential for improving the effectiveness of mobile learning.

Conclusion and Discussion

The theoretical and practical analyses conducted in this study demonstrate that mobile learning, as an innovative approach, significantly enhances the flexibility, accessibility, and personalization of education. The integration of mobile technologies ensures continuous access to learning resources and encourages active learner participation.

In Azerbaijan, a technological and institutional foundation for mobile learning has been established. However, effective implementation requires more than technological provision. The study highlights that the success of mobile learning largely depends on the quality of pedagogical approaches, including learner-centered models, personalized strategies, and interactive methods (Laurillard, 2012).

At the same time, the development of teachers' digital and pedagogical competencies is a key factor. Continuous professional development programs are necessary to ensure effective integration of technology into teaching (UNESCO, 2022).

Infrastructure inequality remains a major barrier. Addressing this issue requires strengthened collaboration between the public and private sectors, increased investment in digital infrastructure, and adherence to inclusive education principles (World Bank, 2022).

Additionally, improving the quality and localization of digital educational resources is crucial. Developing pedagogically sound content in the native language aligned with the national curriculum will significantly enhance the effectiveness of mobile learning.

In conclusion, the successful implementation of mobile learning in Azerbaijan requires a comprehensive approach that integrates technological support, pedagogical innovation, teacher training, and content development. In the future, systematic adoption of mobile learning can accelerate Azerbaijan's integration into the global digital education space and contribute to improving educational quality and inclusivity.

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ARTIFICIAL INTELLIGENCE AS A PEDAGOGICAL TOOL IN ENGLISH LANGUAGE TEACHING

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Abstract: Artificial Intelligence (AI) has become an important component of contemporary education, particularly in the field of English Language Teaching (ELT). The rapid development of digital technologies and the increasing global demand for English proficiency have encouraged educational institutions to integrate AI-powered tools into language learning environments. This study examines the pedagogical role of Artificial Intelligence in ELT by analyzing its influence on personalized learning, learner autonomy, assessment practices, and language skill development. The research adopts a qualitative analytical approach based on academic literature, educational reports, and case studies published between 2020 and 2025. The

findings indicate that AI technologies improve language learning through adaptive instruction, immediate feedback, interactive educational environments, and continuous access to learning resources. AI-supported systems contribute to the development of listening, speaking, reading, and writing skills while increasing learner motivation and engagement. At the same time, the study identifies several challenges related to AI implementation, including ethical concerns, data privacy risks, technological dependence, and reduced human interaction. The study concludes that Artificial Intelligence should function as a supportive pedagogical tool rather than a replacement for teachers. A balanced integration of AI technologies and traditional teaching methods is essential for the future development of effective English language education.

Keywords: Artificial Intelligence, English Language Teaching, Digital Education, Educational Technology, Language Skills, Adaptive Learning, Learner Autonomy

SÜNİ İNTELLEKT İNGİLİS DİLİNİN TƏDRİSİNDƏ PEDAQOJİ VASİTƏ KİMİ

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Xülasə: Süni intellekt (SI) müasir təhsil sistemində, xüsusilə ingilis dili tədrisində mühüm rol oynayan innovativ texnologiyalardan birinə çevrilmişdir. Rəqəmsal texnologiyaların sürətli inkişafı və ingilis dilinə olan qlobal tələbat təhsil müəssisələrini SI əsaslı alətləri tədris prosesinə inteqrasiya etməyə təşviq etmişdir. Bu məqalədə süni intellektin ingilis dili tədrisindəki pedaqoji rolu araşdırılır və onun fərdiləşdirilmiş öyrənmə, tələbə müstəqilliyi, qiymətləndirmə prosesi və dil bacarıqlarının inkişafına təsiri təhlil edilir. Tədqiqat 2020–2025-ci illərdə nəşr olunmuş akademik mənbələr, hesabatlar və praktik nümunələr əsasında keyfiyyət yönümlü analitik metodologiya ilə aparılmışdır. Araşdırmanın nəticələri göstərir ki, süni intellekt texnologiyaları adaptiv tədris, dərhal geribildirim, interaktiv öyrənmə mühiti və fasiləsiz resurs əlçatanlığı vasitəsilə dil öyrənməsinin effektivliyini artırır. Eyni zamanda, məqalədə etik problemlər, məlumat məxfiliyi, texnologiyadan asılılıq və insan ünsiyyətinin azalması kimi məhdudiyyətlər də müəyyən edilmişdir. Nəticə etibarilə, süni intellekt müəllimi əvəz edən vasitə deyil, tədrisin keyfiyyətini artıran dəstəkçi pedaqoji texnologiya kimi qiymətləndirilməlidir.

Açar sözlər: Süni intellekt, İngilis dili tədrisi, Rəqəmsal təhsil, Təhsil texnologiyaları, Dil bacarıqları, Adaptiv öyrənmə, Tələbə müstəqilliyi

Introduction

The rapid development of digital technologies has fundamentally transformed many areas of modern society, and education is among the fields most significantly influenced by this transformation. Over the last decade, technological innovation has changed not only the way information is delivered, but also the methods through which students interact with knowledge, teachers, and learning environments. Among the most influential innovations, Artificial Intelligence (AI) has emerged as a powerful and transformative tool capable of reshaping traditional educational practices and introducing more flexible, efficient, and learner-centered approaches.

In the context of English Language Teaching (ELT), AI has gained particular importance due to its ability to address several long-standing challenges associated with language acquisition. Traditional methods of language instruction often rely heavily on teacher-centered approaches, standardized curricula, and limited classroom interaction. While these methods may be effective in certain situations, they frequently fail to accommodate the diverse needs, abilities, learning speeds,

and interests of individual learners. As a result, students may experience unequal learning outcomes, reduced motivation, and difficulties in developing practical communicative competence.

The integration of AI technologies into ELT offers innovative solutions to these issues. AI-powered systems are capable of analyzing learner performance, identifying strengths and weaknesses, and adapting educational content accordingly. Unlike conventional instructional methods, AI-based platforms provide personalized learning experiences that allow students to progress at their own pace. This adaptive approach helps create a more inclusive and supportive learning environment where learners receive materials and tasks suited to their individual proficiency levels.

Furthermore, globalization and the increasing role of English as an international language of communication have intensified the demand for effective and accessible language learning opportunities. English is widely used in business, science, technology, tourism, and international cooperation, making proficiency in the language an essential skill in the modern world. AI-driven educational technologies respond to this growing demand by enabling continuous learning beyond the traditional classroom setting. Through mobile applications, virtual tutors, intelligent chatbots, and interactive learning platforms, students can practice language skills in real-time and in authentic communicative situations.

Another important aspect of AI integration is its ability to provide immediate feedback and data-driven assessment. In many traditional classrooms, teachers may not always have sufficient time to provide individualized feedback to every learner. AI systems can instantly analyze learner responses, correct mistakes, and suggest improvements. This immediate feedback not only accelerates the learning process but also encourages students to become more autonomous and confident language users.

Despite these advantages, the integration of AI in ELT also raises important pedagogical, ethical, and practical concerns. Questions related to data privacy, overreliance on technology, reduced human interaction, and unequal access to digital resources remain highly relevant. Therefore, it is important to examine both the benefits and limitations of AI implementation in language education.

The purpose of this study is to explore the role of Artificial Intelligence in English Language Teaching by analyzing its pedagogical benefits, practical applications, and existing challenges. The paper also aims to demonstrate how AI technologies can be effectively integrated into language education without diminishing the essential role of teachers, whose guidance, emotional support, and communicative interaction remain irreplaceable in the learning process.

Materials and Methods

This research adopts a qualitative analytical framework to investigate the integration of Artificial Intelligence technologies into English language teaching. The study is primarily based on a comprehensive review and analysis of recent academic literature, technological reports, educational case studies, and practical examples related to AI implementation in educational settings.

The materials used in this research include peer-reviewed journal articles published between 2020 and 2025, scholarly books, reports from educational organizations, and analytical studies focusing on AI-based language learning platforms and intelligent tutoring systems. These materials were selected because they provide both theoretical perspectives and practical evidence regarding the effectiveness of AI in language education. In addition, examples from popular language learning

applications and virtual learning environments were examined to better understand how AI technologies function in real educational contexts.

The methodological approach of the study consists of several interconnected stages. First, a comparative analysis is conducted to examine the differences between traditional language teaching methods and AI-enhanced instructional approaches. This comparison focuses on such aspects as learner engagement, feedback mechanisms, personalization, accessibility, and classroom interaction.

Second, a thematic analysis is applied to identify the main trends, patterns, and recurring themes associated with the use of AI in ELT. Particular attention is given to adaptive learning systems, speech recognition technologies, automated assessment tools, intelligent tutoring systems, and AI-driven communication platforms. The analysis also explores how these technologies contribute to the development of listening, speaking, reading, and writing skills.

Third, the study evaluates the pedagogical effectiveness of AI tools by examining their impact on learner outcomes, motivation, autonomy, and overall language proficiency. This stage includes the analysis of research findings from previous studies as well as practical observations from educational institutions that have integrated AI technologies into their curricula.

Additionally, real-world examples are incorporated throughout the research in order to illustrate the practical implementation of AI in both classroom-based and online learning environments. These examples demonstrate how teachers and learners interact with AI systems and how such technologies support modern educational practices.

The chosen methodology ensures a comprehensive and balanced understanding of the topic by combining theoretical analysis with practical evidence. This multidimensional approach increases the reliability and academic value of the study while allowing for a deeper exploration of the opportunities and challenges associated with AI in English language teaching.

Results and Discussion

1. Personalized and Adaptive Learning

One of the most significant advantages of Artificial Intelligence in English Language Teaching is its ability to provide personalized and adaptive learning experiences. Traditional educational models generally follow a fixed curriculum and standardized teaching methods that may not effectively address the individual needs of all students. Learners often differ in terms of language proficiency, cognitive abilities, learning styles, motivation, and pace of progress. Consequently, some students may struggle to keep up with the lesson, while others may become bored due to a lack of challenge.

AI technologies help solve this problem by using adaptive learning algorithms that analyze learner behavior and performance. These systems collect and process data such as response accuracy, task completion time, pronunciation patterns, vocabulary retention, and error frequency. Based on this information, the AI system automatically adjusts the level of difficulty, type of exercises, and instructional materials according to the learner's specific needs.

For example, a student experiencing difficulties with grammar may receive additional exercises and explanations focused on problematic structures, while a more advanced learner may be provided with complex reading materials or interactive speaking tasks. This individualized approach creates a more efficient learning environment and increases the overall effectiveness of language acquisition.

Furthermore, personalization significantly improves student motivation and engagement. Learners are more likely to remain interested and committed when educational content corresponds to their abilities, goals, and interests. AI-based systems can also recommend topics, activities, and exercises that align with learners' personal preferences, making the learning process more enjoyable and meaningful.

Such adaptability is particularly beneficial in classrooms where students possess varying levels of language proficiency. Teachers often face difficulties in meeting the needs of all learners simultaneously, whereas AI tools can support differentiated instruction and provide additional assistance where necessary.

2. Enhancement of Language Skills

AI technologies make substantial contributions to the development of all major language skills, including listening, speaking, reading, and writing. Through interactive and intelligent systems, learners gain opportunities to practice English in more dynamic, flexible, and authentic contexts.

In listening activities, AI-generated audio materials and automatic transcription systems allow learners to improve comprehension skills in controlled yet realistic environments. Students can listen to dialogues, lectures, podcasts, and simulated conversations at different speeds and levels of difficulty. Some AI platforms are also capable of identifying specific listening weaknesses and recommending targeted exercises for improvement.

Speaking skills are particularly enhanced through speech recognition technologies and conversational AI systems. These tools analyze learners' pronunciation, fluency, stress patterns, and intonation. Immediate corrective feedback enables students to identify pronunciation errors and improve oral communication skills independently. AI-powered chatbots and virtual assistants also create opportunities for learners to practice real-time communication without the anxiety often associated with speaking in front of others.

Reading comprehension benefits from adaptive text selection and intelligent recommendation systems. AI platforms can provide reading materials that correspond to the learner's proficiency level and gradually increase complexity as the student progresses. Interactive glossaries, vocabulary support, and comprehension quizzes further assist learners in understanding texts more effectively.

Writing skills are also significantly supported through AI-based grammar and style correction tools. These systems help students identify grammatical mistakes, spelling errors, punctuation problems, and inappropriate vocabulary usage. Some advanced platforms even provide suggestions related to sentence structure, coherence, and academic style. As a result, learners become more aware of their writing weaknesses and gradually develop greater independence in written communication.

The integration of all these technologies creates a comprehensive and supportive learning environment where students can continuously practice and improve multiple language skills simultaneously.

3. Immediate Feedback and Assessment

Feedback is one of the most essential components of successful language learning. In traditional classrooms, however, teachers may not always be able to provide immediate and individualized feedback due to limited time and large class sizes. Delayed correction can reduce the effectiveness of learning because students may continue repeating mistakes before receiving proper guidance.

Artificial Intelligence addresses this challenge by offering instant feedback and automated assessment. AI systems can immediately analyze learner responses and provide corrections, explanations, and recommendations for improvement. This rapid feedback mechanism helps learners recognize mistakes in real time and reinforces correct language use more effectively.

For example, pronunciation applications can instantly identify incorrect sounds and suggest more accurate pronunciation models. Writing platforms can automatically highlight grammar and vocabulary errors while offering alternative sentence structures. Such immediate responses accelerate the learning process and help students become more conscious of their language use.

Additionally, AI-based assessment systems significantly reduce teachers' workload by automating routine grading tasks such as quizzes, vocabulary tests, and grammar exercises. This allows educators to dedicate more time to communicative activities, classroom interaction, and individualized support.

Data generated by AI systems also provides valuable insights into learner progress. Teachers can analyze detailed reports regarding student performance, identify common difficulties, and modify instructional strategies accordingly. Consequently, assessment becomes more objective, efficient, and data-driven.

4. Promotion of Learner Autonomy

Another important contribution of AI to ELT is the promotion of learner autonomy and independent study habits. Traditional learning environments often depend heavily on teacher guidance and classroom instruction. However, effective language acquisition requires continuous exposure and practice beyond formal educational settings.

AI-powered applications, mobile platforms, and virtual tutors allow students to access educational resources anytime and anywhere. Learners can practice vocabulary, grammar, pronunciation, and conversational skills according to their own schedules and personal learning goals. This flexibility is especially beneficial for students who may not have regular access to language courses or native-speaking environments.

Autonomous learning also encourages responsibility, self-discipline, and self-regulation. Students become more actively involved in managing their own progress, identifying weaknesses, and selecting appropriate learning strategies. Many AI systems include progress tracking features, achievement indicators, and personalized recommendations that motivate learners to continue improving.

Moreover, AI technologies expose students to authentic language use through interactive dialogues, multimedia content, and real-world communication scenarios. This continuous engagement with the language contributes to increased fluency, confidence, and communicative competence.

5. Pedagogical Advantages

The integration of AI into English language teaching offers numerous pedagogical benefits for both teachers and learners. One of the primary advantages is increased teaching efficiency. By automating repetitive administrative tasks such as grading, attendance tracking, and exercise correction, AI enables teachers to focus more on creative instruction, communication, and student interaction.

AI systems also provide data-driven insights that help educators better understand learner needs and classroom dynamics. Teachers can monitor student progress more effectively, identify areas requiring additional support, and implement targeted instructional strategies.

Another important advantage is accessibility. AI-based educational resources are available online and can be accessed from different devices, making language learning more flexible and inclusive. This is particularly valuable for learners in remote or under-resourced regions where access to qualified language teachers may be limited.

Interactive AI tools further enhance classroom participation and student engagement. Gamified learning applications, virtual simulations, and conversational AI systems create more dynamic and motivating learning environments. Such technologies make lessons more interactive and encourage active learner participation.

In addition, AI supports inclusive education by accommodating students with different learning needs and abilities. Personalized learning systems can adapt materials for learners with disabilities or specific educational challenges, thereby promoting equal learning opportunities.

6. Challenges and Limitations

Despite its numerous advantages, the use of AI in ELT also presents several challenges and limitations that must be carefully considered. One major concern is the potential reduction of human interaction in the learning process. Language learning is fundamentally social and communicative in nature, and meaningful interaction with teachers and peers plays a crucial role in developing communicative competence, cultural awareness, and emotional intelligence.

Although AI systems can simulate conversation and provide feedback, they cannot fully replace human empathy, creativity, and emotional support. Teachers remain essential for motivating students, managing classroom dynamics, and fostering authentic communication.

Ethical issues also represent a significant challenge. AI systems often collect large amounts of learner data, including performance records, behavioral patterns, and personal information. Therefore, data privacy, confidentiality, and responsible data management must be ensured. Concerns related to algorithm bias and unequal treatment of learners also require careful attention.

Another limitation is the risk of overdependence on technology. Excessive reliance on AI tools may reduce opportunities for critical thinking, problem-solving, and face-to-face communication. Students may become too dependent on automated corrections and lose confidence in independent language production.

Furthermore, unequal access to digital technologies creates educational disparities between learners. Students in developing regions or low-income communities may lack reliable internet access, modern devices, or digital literacy skills necessary for effective participation in AI-supported learning environments.

Technical issues, software limitations, and insufficient teacher training can also hinder successful implementation. Educators must possess adequate technological competence in order to integrate AI tools effectively into their teaching practices.

Conclusion

Artificial Intelligence has emerged as one of the most influential innovations in modern education and has demonstrated considerable potential in transforming English Language Teaching. Through adaptive learning systems, intelligent tutoring platforms, speech recognition technologies, and automated assessment tools, AI enhances language learning by providing personalization, immediate feedback, increased accessibility, and opportunities for continuous practice.

The findings of this study indicate that AI contributes positively to learner engagement, autonomy, and the development of core language skills. It also supports teachers by automating

routine tasks and offering valuable data-driven insights into learner performance. As a result, educational processes become more efficient, flexible, and learner-centered.

However, despite these significant advantages, AI should not be viewed as a complete replacement for human educators. The role of teachers remains essential in providing emotional support, fostering meaningful communication, encouraging critical thinking, and creating positive learning environments. Human interaction continues to be a fundamental component of successful language acquisition.

The future of English Language Teaching lies in achieving a balanced integration of AI technologies and traditional pedagogical approaches. Such a combination can maximize the benefits of technological innovation while preserving the human aspects of education that are necessary for effective communication and holistic learner development.

Further research is recommended to explore long-term effects of AI integration, ethical considerations, and strategies for ensuring equal access to technological resources in language education worldwide.

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DIGITAL TRANSFORMATION IN AUDIOVISUAL TRANSLATION: THE CHANGING ROLE OF TRANSLATORS IN AI-ASSISTED DUBBING AND SUBTITLING

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Abstract: In the context of modern globalization, English as a means of international communication is not limited to grammatical and lexical knowledge but also includes a complex system of ethical behavior and communicative norms. This article examines the main characteristics, socio-cultural functions, and practical application of speech etiquette and communication norms in English. The study analyzes greetings, forms of address, farewells, expressions of gratitude, apologies, requests, and other politeness expressions. It also considers the linguistic features of formal and informal communication and the social factors influencing the choice of appropriate expressions. Factors such as age, social status, degree of familiarity, interpersonal relationships, and communicative purpose are examined in relation to speech behavior. The use of expressions such as "Hello," "Hi," "Good morning," "Good afternoon," "Good evening," and "How do you do?" is explained through examples from different communicative contexts. The article also compares certain differences between English and Azerbaijani etiquette and communication norms. The findings demonstrate that speech etiquette plays an important role in maintaining mutual respect, regulating social relationships, establishing appropriate social distance, and promoting successful intercultural communication. Therefore, understanding and correctly applying etiquette norms is an essential component of effective English communication and contributes to the development of learners' communicative and intercultural competence.

Keywords: speech etiquette, communication norms, English language, politeness, communication, discourse.

AUDIOVİZUAL TƏRCÜMƏDƏ RƏQƏMSAL TRANSFORMASIYA: SÜNİ İNTELLEKTİN DƏSTƏYİ İLƏ DUBLYAJ VƏ SUBTİTLƏŞDİRMƏDƏ TƏRCÜMƏÇİLƏRİN DƏYİŞƏN ROLU

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Xülasə: Müasir qloballaşma şəraitində ingilis dili beynəlxalq ünsiyyət vasitəsi kimi yalnız qrammatik və leksik biliklərin məcmusu deyil, həm də müəyyən etik davranış və kommunikativ normaları özündə birləşdirən mürəkkəb sistemdir. Bu məqalədə ingilis dilində nitq etiketi və ünsiyyət normalarının əsas xüsusiyyətləri, sosial-mədəni funksiyaları və müxtəlif kommunikativ situasiyalarda tətbiqi araşdırılır. Tədqiqatda salamlama, müraciət formaları, vidalaşma, təşəkkür, üzrxahlıq, xahiş və digər nəzakət ifadələri təhlil edilir. Həmçinin rəsmi və qeyri-rəsmi ünsiyyət zamanı istifadə olunan dil vahidlərinin xüsusiyyətləri və onların seçilməsinə təsir edən sosial amillər nəzərdən keçirilir. Yaş, sosial status, tanışlıq dərəcəsi,

münasibətlərin xarakteri və kommunikativ məqsəd kimi amillərin nitq davranışına təsiri müəyyənləşdirilir. İngilis dilində “Hello”, “Hi”, “Good morning”, “Good afternoon”, “Good evening” və “How do you do?” kimi ifadələrin işlənmə xüsusiyyətləri nümunələrlə izah olunur. Eyni zamanda, ingilis və Azərbaycan dillərində etik ünsiyyət normaları arasında bəzi fərqlər müqayisə edilir. Araşdırmanın nəticələri göstərir ki, nitq etiketi qarşılıqlı hörmətin qorunması, sosial münasibətlərin tənzimlənməsi və mədəniyyətlərarası anlaşmanın təmin edilməsində mühüm əhəmiyyət daşıyır.

Açar sözlər: nitq etiketi, ünsiyyət normaları, ingilis dili, nəzakət, kommunikasiya, diskurs.

Introduction

The translation profession has always been shaped by technological change, but the current rise of artificial intelligence has transformed audiovisual translation in particularly visible ways. Subtitling, closed captioning, dubbing translation, audio description, text-to-speech, and post-editing are no longer separate practices carried out only through traditional human workflows. They increasingly interact with machine translation, synthetic voices, automated timing tools, and AI-assisted revision systems. As a result, the modern translator is not only a language expert but also an editor, cultural mediator, accessibility specialist, and technology user. In the field of audiovisual translation, digital transformation does not eliminate the translator's role. Instead, it changes the nature of that role by shifting emphasis from producing every line from scratch to evaluating, adapting, correcting, and creatively improving AI-generated outputs.

Materials and Methods

This study uses a qualitative and descriptive research approach to examine the impact of digital transformation and artificial intelligence on audiovisual translation, with particular attention to dubbing and subtitling. The main focus of the research is the changing role of translators in AI-assisted translation processes and the new responsibilities emerging as a result of technological development.

The materials of the study include academic publications, books, research articles, professional guidelines, and relevant studies related to audiovisual translation, artificial intelligence, machine translation, post-editing, dubbing, subtitling, and audio description. Particular attention is given to current technological practices used in audiovisual translation and their influence on professional translators.

The study applies descriptive, comparative, and analytical methods. The descriptive method is used to identify the main characteristics of traditional and AI-assisted audiovisual translation. The comparative method is applied to examine the differences between human translation and AI-supported workflows, particularly in terms of translation accuracy, timing, synchronization, cultural adaptation, emotional expression, and creativity.

The analysis also considers practical examples of dialogue, subtitles, cultural references, wordplay, slang, and emotional expressions in order to demonstrate the limitations of automatic translation systems. In addition, AI-assisted dubbing, synthetic voices, prompt-based voice generation, and accessibility practices such as audio description are examined.

The research evaluates both the advantages and limitations of artificial intelligence in audiovisual translation. The study particularly focuses on the continuing importance of human intervention in post-editing, quality control, cultural adaptation, and creative decision-making. Thus, the methodology aims to demonstrate how technological development is transforming rather than eliminating the professional role of the audiovisual translator.

However, technology alone is not enough. A translator must continue developing cultural and linguistic knowledge. Knowing the source and target cultures in depth remains one of the strongest professional advantages. For example, watching youth series, observing contemporary slang, and taking notes from current media can help translators stay close to living language. This is especially important because audiovisual translation often deals with everyday speech, humor, relationships, and social identity. A translator who does not follow cultural change may produce technically correct but emotionally outdated translations.

Discussion

A major example of this transformation is machine translation and post-editing, often called PEMT. In many professional workflows, agencies and large platforms may send translators texts that have already been translated by machine tools such as Google Translate or DeepL. At first glance, this may seem to reduce the translator's workload. However, the translator's responsibility does not disappear; it becomes more analytical and evaluative. Post-editing can be light, when the translator corrects only basic errors, or full, when the machine output must be thoroughly rewritten. The advantage of this workflow is speed. It can increase hourly output and help projects reach completion more quickly. Yet the disadvantages are equally important. Machines often fail to understand context, humor, character voice, indirect meaning, and cultural nuance. Therefore, while AI can process language quickly, it cannot reliably interpret the deeper social and emotional layers of communication.

This limitation becomes especially clear in Turkish, where rhetorical questions and indirect expressions often carry meanings that are not literal. A simple question such as "N'apıyorsun?" or "Çıkıyor musun?" may not function only as a request for information. Depending on tone and context, it can express surprise, criticism, irritation, or emotional distance. Similarly, "Sen mi geldin?" may appear to mean "Did you come?" but in a specific scene it can imply shock, disappointment, or disbelief. Another example is "Gelmeyip de n'apıcaktım?" Literally, it might look like a question, but its real meaning may be closer to "I had to come," "I came unwillingly," or "I had no other choice." A machine translation system may translate the words correctly but still miss the emotional pressure behind them. This is where the human translator remains essential: they must read between the lines and understand what the character actually means, not only what the character says.

In audiovisual translation, technical rules also shape the translator's decisions. Subtitles cannot be treated as ordinary written text because they must be read quickly while viewers are also watching the image and listening to sound. Style guides are therefore central to professional subtitling and dubbing work. Each client may provide its own guide, sometimes accompanied by character lists, forced narrative files, or scripts. These guides determine punctuation, spelling of names, treatment of slang and swear words, number formatting, and many other details. Their most important principle is consistency. The same character should speak in the same way throughout a project, and the translation should follow a stable linguistic style.

Technical constraints make this process even more demanding. Subtitles often follow limits such as approximately forty-two characters per line, including spaces. A subtitle must also remain on screen long enough to be read, but not so long that it becomes unnatural. In many workflows, a subtitle may stay on screen for a minimum of one second and a maximum of six seconds. Reading speed, usually around 140 to 180 words per minute, also affects translation choices. Because of these limitations, translators must condense dense dialogue without damaging meaning. This

requires more than bilingual knowledge. The translator must decide what is essential, what can be shortened, and what must be preserved for narrative clarity.

Difficult translations show the creative side of audiovisual work. Wordplay and jokes are rarely successful when translated literally. A translator must often find a similar joke in the target language or create a new solution that preserves the function of the original. Cultural elements create another challenge. Some concepts are deeply connected to a specific culture, and subtitles usually do not allow footnotes or long explanations. The translator must decide whether to keep the original reference, replace it with something more familiar to the target audience, or use a brief explanatory translation. Slang and jargon require similar attention. A doctor, a teenager, a street character, and a historical figure should not all sound the same. The translator must carry over the social background, education level, and emotional tone of the source speech into the target language.

Digital transformation has also expanded the importance of accessibility practices such as closed captioning and audio description. Audio description, in particular, requires a different but related set of skills. It is not simply describing everything visible on screen. Good audio description depends on selection, timing, and precision. The describer must choose what matters most because there is never enough time to describe every visual detail. Strong verbs are especially important because they carry movement, emotion, and visual clarity. A weak verb can flatten a scene, while a precise verb can help the listener immediately understand the action. At the same time, audio description should remain objective without becoming robotic. It should avoid over-interpreting emotions, but it should still be cinematic and vivid enough to support the viewing experience.

Spatial clarity is another crucial aspect of audio description. Listeners need to understand where people are, who is moving, what has changed visually, and how the scene is organized. Timing and rhythm are equally important. Even an accurate description can fail if it interrupts dialogue or arrives too late. For this reason, audio description requires a balance between language, image, sound, and silence. The audio describer must understand not only what is visible but also when the audience needs that information. This shows that accessibility translation is not secondary or mechanical; it is a highly skilled interpretive practice.

The rise of AI-assisted dubbing introduces further changes to the translator's skill set. In traditional dubbing, translators already had to consider meaning, lip synchronization, timing, character voice, and natural speech. With AI-assisted dubbing and synthetic voices, the translator may also need to work with prompts that guide tone and emotional delivery. Prompt engineering becomes part of the workflow. Commands such as speaking happily, laughing, crying, emphasizing a question, or adjusting the effect of punctuation can influence the synthetic voice. Some AI voice systems can produce emotional speech more naturally, especially in English, but they still require careful human supervision. The translator or editor must check whether the voice matches the scene, the character, the emotional context, and the cultural expectations of the target audience.

For this reason, new generation translators need AI literacy. AI literacy means knowing how to use AI tools efficiently and ethically within the translation workflow. It also means understanding their limits. Translators must be able to audit AI-generated speech and AI-generated translation for omissions, additions, unnatural sentence breaks, tone problems, timing issues, and cultural mismatches. In this sense, the translator becomes a quality controller as well as a creative language professional. The task is not simply to accept machine output, but to question it, reshape it, and make it appropriate for real audiences.

Conclusion

In conclusion, digital transformation has changed translation by introducing machine translation, post-editing, AI dubbing, synthetic voices, and new accessibility practices. Yet these technologies have not removed the need for human translators. On the contrary, they have made human judgment more important. Machines can offer speed, but they often fail in context, humor, culture, character voice, and emotional meaning. The modern translator must therefore combine native-language mastery, foreign-language knowledge, technological competence, cultural awareness, and creative problem-solving. To succeed in this field, translators should love their own language, read extensively, learn tools such as Subtitle Edit, Aegisub, ChatGPT, and translation memory systems, specialize in particular fields, build professional networks, and remain patient. The first opportunities may not come easily, and rejected projects are part of professional growth. Ultimately, the future of audiovisual translation belongs not to translators who fear technology, but to those who learn how to use it critically, ethically, and creatively.

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THE BENEFITS AND LIMITATIONS OF VR-ASSISTED LANGUAGE LEARNING ON EFL LEARNERS' SPEAKING SKILLS

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Abstract: As speaking is known to be one of the most challenging skills for EFL learners to improve, research aimed at finding solutions to this issue hypothesized that immersive technologies, such as VR, may lower learners' anxiety levels and provide a more meaningful practice. In this paper, the benefits and limitations of VR-assisted language learning on EFL learners' speaking skills were examined with a communicative language teaching perspective, with the idea that VR devices can mimic face-to-face communication. A qualitative document analysis with narrative synthesis was used to identify recurring themes across the 11 empirical studies obtained from the Scopus database via a title/abstract/keywords search using keywords related to virtual reality, EFL/foreign language learning, and speaking. A total of 24 studies were identified in the initial search, of which 11 empirical studies were included after screening titles, abstracts, and full texts against the predefined inclusion and exclusion criteria. The results suggested that VR-assisted learning can reduce speaking anxiety, boost motivation, and provide contextualization.

Supporting studies also suggested that VR can enhance linguistic resources, such as vocabulary knowledge, thereby improving the learner's language production and indirectly supporting speaking development. Although the benefits outweigh the disadvantages, there are downsides to address beyond the limitations of the paper's analysis of 11 studies. Technology anxiety, motion sickness, fatigue, accessibility problems, digital avatar-related difficulties, and the fact that decreases in the VR environment do not always correlate with real-life interactions. To conclude the analysis, it is advised that VR be used as a complementary tool, in the right circumstances, rather than replacing the whole of classroom communication.

Keywords: Virtual reality, EFL learners, oral communication, speaking skills, document analysis

EFL ÖĞRENCİLERİNİN KONUŞMA BECERİLERİ ÜZERİNDE SANAL GERÇEKLİK (VR) DESTEKLİ DİL ÖĞRENİMİNİN FAYDALARI VE SINIRLILIKLARI

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Özet: Konuşma becerisi, yabancı dil olarak İngilizce (EFL) öğrenenler için bilinen geliştirilmesi en zor becerilerden biri olduğundan, bu soruna çözüm aramayı hedefleyen araştırmalar sanal gerçeklik (VR) gibi sarmal teknolojilerin öğrencilerin kaygı düzeylerini düşürebileceğini ve daha anlamlı pratik imkânı sunabileceğini hipotez etmektedir. Bu çalışmada, VR cihazlarının yüz yüze iletişimi taklit edebildiği düşüncesinden hareketle; VR destekli dil öğreniminin EFL öğrencilerinin konuşma becerileri üzerindeki avantajları ve kısıtlılıkları iletişimsel dil öğretimi perspektifiyle incelenmiştir. Ortak temaları belirlemek amacıyla, Scopus veri tabanında sanal gerçeklik, EFL/yabancı dil öğrenimi ve konuşma ile ilgili anahtar kelimelerle yapılan tarama sonucunda elde edilen 11 ampirik çalışma üzerinde nitel doküman analizi ve anlatı sentezi yöntemi uygulanmıştır. Önceden belirlenen dahil etme ve hariç tutma kriterlerine göre, ilk taramada ulaşılan 24 çalışmadan; başlık, özet ve tam metin incelemeleri tamamlanan 11 çalışma analize dahil edilmiştir. Bulgular, VR destekli öğrenmenin konuşma kaygısını azaltabileceğini, motivasyonu artırabileceğini ve bağlamsallaştırma sağlayabileceğini göstermektedir. Destekleyici çalışmalar ayrıca VR'nin kelime dağarcığı gibi dilsel kaynakları zenginleştirebileceğini, bunun da öğrencinin dil üretimini iyileştirerek konuşma gelişimini dolaylı olarak desteklediğini ortaya koymaktadır. Her ne kadar avantajlar dezavantajlardan daha ağır bassada, 11 çalışmanın analiz edilmesiyle sınırlı olan bu makalenin kısıtlılığına ek olarak ele alınması gereken olumsuz yanlar da mevcuttur. Bunlar arasında teknoloji kaygısı, sanal gerçeklik hastalığı (motion sickness), yorgunluk, erişim sorunları, dijital avatar kaynaklı güçlükler ve VR ortamındaki kaygı düşüşlerinin gerçek hayat etkileşimleriyle her zaman doğru orantılı olmaması yer almaktadır. Analiz sonucunda, VR'nin sınıf içi iletişimin tamamının yerine geçmesi yerine, uygun koşullarda tamamlayıcı bir araç olarak kullanılması önerilmektedir.

Anahtar Kelimeler: Sanal Gerçeklik, EFL Öğrencileri, Sözlü İletişim, Konuşma Becerileri, Doküman Analizi

1. Introduction

Perhaps the most common issue with English as a Foreign Language (EFL) learners might be that even though they perform well on English tests at school, they express having difficulty when they need to speak in real-life situations due to anxiety and lack of practice (Kaplan-Rakowski & Gruber, 2023; Richards, 2006; Ye et al., 2026). Richards (2006) clearly argues for classroom communication to improve speaking and communication skills. Yet, this case is not the easiest for

EFL learners, as English is not a common language in their everyday lives. Thankfully, technological advances gave many researchers a brilliant idea: why not use Virtual Reality (VR) devices to create an immersive environment that mimics the experience of living in an English-speaking country? On paper, this idea sounds like the next best thing: supporting the principles of Communicative Language Teaching (CLT) and Second Language Acquisition (SLA) without even changing your location.

Communicative Language Teaching (CLT) focuses on meaningful, authentic communication and communicative purpose (Richards, 2006). In Second Language Acquisition (SLA), acquisition is the natural acquisition of a second language, whereas language learning is the conscious study of a second language (Krashen, 1981). For years, acquisition has been seen as the optimal approach for becoming a communicator in CLT. VR speaking tasks, such as debates and simulated interactions, can be examined through the lens of CLT because they provide contextualized, authentic language use in real life. Although VR has been examined more in the last decade in EFL contexts, the findings are scattered across different skills and learning outcomes. Therefore, this paper was written with the aim of using document analysis with narrative synthesis to identify recurring themes and provide a holistic view. The research questions were;

- What are the reported benefits of VR-assisted language learning in the selected studies?
- What are the reported limitations of VR-assisted language learning in the selected studies?
- How can these findings be interpreted through the CLT lens?

2. Materials and Methods

This paper uses a qualitative document analysis and narrative synthesis to identify common themes across studies on how VR-assisted language learning affects EFL learners' speaking skills through the lens of CLT. Document analysis is the examination and interpretation of documents to identify meanings, patterns, and themes (Bowen, 2009). For this study, the selected Scopus-indexed journal articles served as the documents. The findings, methods, benefits, and limitations of these studies were analyzed and coded thematically, and narrative synthesis was used to synthesize the results of the selected studies, which are varied in the literature rather than statistically.

The data source used was Scopus. The search was conducted in the title, abstract, and keyword fields, and was limited to journal articles, all open-access documents, article-type publications, and studies written in the English language, using the following search string:

"TITLE-ABS-KEY (("virtual reality" OR "immersive learning" OR "immersive virtual reality" OR "VR") AND ("EFL" OR "English as a foreign language" OR "foreign language learning") AND ("speaking" OR "oral communication" OR "oral skills" OR "communication")) AND (LIMIT-TO (SRCTYPE , "j")) AND (LIMIT-TO (OA , "all")) AND (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (LANGUAGE , "English"))"

The initial search identified 24 studies. Title and abstracts were initially screened based on predefined inclusion and exclusion criteria related to the focus of this study, which examines the benefits and limitations of virtual reality (VR)- assisted learning for English as a Foreign Language (EFL) learners' speaking skills. The remaining 16 studies that passed through the initial screening were reexamined in full text.

The inclusion criteria were as follows. First, studies were required to involve VR, immersive VR, spherical video-based VR, VR games, or immersive multimedia learning. Second, studies needed to be situated within EFL, English as a Second Language (ESL), or foreign language learning contexts. Third, studies had to address oral communication either directly or

indirectly. Studies were excluded if they were unrelated technical studies or if VR and English-speaking skills were not central to the research. Systematic reviews were excluded as additional review studies would have shifted the focus toward a review of reviews.

Following the full text screening process, 11 empirical studies were selected. The selected documents underwent a two-stage coding process. In the first stage, descriptive codes were assigned, such as “speaking anxiety,” “public speaking,” “communicative skills,” “engagement,” “authenticity,” “motion sickness,” and “transferability.” In the second stage, these codes were consolidated into broader narrative themes like “speaking anxiety reduction,” “speaking performance and oral production,” “engagement and motivation,” “contextualized language learning,” “vocabulary and linguistic resources,” “immersion & presence & embodiment,” “technical and pedagogical challenges,” “transferability to real-life communication,” and “accessibility and design.”

3. Results and Discussion

As shown in Table 1, my document analysis yielded 9 recurring themes. This shows that VR-assisted language learning has multiple dimensions, including speaking anxiety, motivation, contextualization, and linguistic resources. Data suggests that VR devices should not be used only as technological improvements, but they should also include the elements of CLT, which are meaningfulness, contextualization, collaboration, and authenticity (Bendeck Soto et al., 2020; Lai, 2024; Lestiono & Lee, 2026; Muslem & Abbas, 2017; Richards, 2006; Tonda et al., 2026).

Speaking performance and oral production	Using VR-based or immersive environments can enhance speaking and presentation.	Research has shown that VR or immersive environments facilitate oral production. Improvements were reported in debating performance, confidence, presentation quality, pronunciation, pausing, phrasing, prosodic features, and integration (Bendeck Soto et al., 2020; Lai, 2024; Lestiono & Lee, 2026; Muslem & Abbas, 2017; Ye et al., 2026).
Engagement and motivation	VR increases learner engagement and motivation.	Game-based VR and platform-based VR encourage learners to participate by creating interactive learning experiences. Usability and learner perception results also suggest that students respond positively (Bendeck Soto et al., 2020; Huang, 2024; Lai, 2024; Lestiono & Lee, 2026; Tonda et al., 2026).
Contextualized language learning	VR supports contextualized language use through real-life-like scenarios.	VR provides learners with contexts such as kitchens, public speaking settings, debate spaces, CLIL science tasks, and simulated conversations with virtual speakers, supporting CLT principles (Bendeck Soto et al., 2020; Feng & Ng, 2023; Lai, 2024; Rahmanu & Molnár, 2024; Tonda et al., 2026).
Vocabulary and linguistic resources	VR can increase vocabulary range, which indirectly supports speaking	Some selected studies did not directly measure speaking but showed that VR improved recall and contextualized vocabulary learning. Learners need accessible vocabulary to speak fluently and meaningfully, making these resources relevant to speaking (Feng & Ng, 2023;

		Lai, 2024; Rahmanu & Molnár, 2024; Tonda et al., 2026).
Immersion, presence, and embodiment	Immersion and presence create richer language production.	IVR studies suggest that the sense of “being there” by making learners feel involved in the virtual context. These affordances help learners interact with objects, scenarios, and tasks more meaningfully (Ding, 2024; Feng & Ng, 2023; Lestiono & Lee, 2026; Tonda et al., 2026).
Technical and pedagogical challenges	VR may create technical, physical, cognitive, and pedagogical problems.	Motion sickness, fatigue, unfamiliarity with VR tools, technology anxiety, small sample sizes, limited intervention duration, unequal learner readiness, and the need for careful instructional design may reduce the pedagogical effectiveness of VR if not addressed (Ding, 2024; Feng & Ng, 2023; Huang, 2024; Muslem & Abbas, 2017; Tonda et al., 2026).
Transferability to real-life communication	Gains in VR do not always transfer automatically to real-life speaking situations.	Although VR may reduce anxiety during virtual speaking tasks, learners may still feel anxious when they return to real face-to-face communication, suggesting that VR should be used as a bridge to real communication (Ding, 2024; Huang, 2024; Ye et al., 2026).
Accessibility and design	VR effectiveness depends on usability, accessibility, task design, and learner support.	For VR to be effective, the tasks should align with students' learning objectives, and teachers should use scaffolding techniques, provide clear instructions, and give feedback. Issues in design and lack of access can cancel out the benefits VR provides (Bendeck Soto et al., 2020; Feng & Ng, 2023; Huang, 2024; Tonda et al., 2026).

There was a very strong theme that recurred across the selected studies: speaking anxiety reduction. To a great extent, VR contributes to the affective domain by eliminating the factors that cause students to experience speaking or failure anxiety in the first place. As far as we know, language learning or language acquisition cannot occur magically unless some speaking practice is involved (Krashen, 1981; Richards, 2006), whether intentional or unintentional. For this reason, speaking is crucial in language classrooms. Every language teacher knows that there will be someone afraid to speak in front of the class, fearing judgment or grades. Research shows that VR can create a safer environment for learners and increase the likelihood that they will speak and practice without feeling anxious (Ding, 2024; Huang, 2024; Kaplan-Rakowski & Gruber, 2023; Lestiono & Lee, 2026).

Nevertheless, these findings do not mean VR-assisted language learning is the silver bullet. Ding (2024) pointed out that the reduced anxiety in the VR environment might not correlate directly with real-life anxiety. This limitation means that VR devices cannot fully replace interaction in the language classroom; rather, they should be used as a preparation tool.

4. Conclusion

To conclude, this paper indicates that VR can be beneficial when used with appropriate preparation and support systems, creating an immersive environment that allows learners to speak freely without feeling as anxious as when performing in front of the whole class. Empirical studies on speaking provided strong evidence, and supporting evidence on vocabulary showed that VR not only reduces speaking anxiety and improves performance but also enhances the linguistic resources needed to carry out a conversation.

On the other hand, VR is not the magic solution. It should be properly designed, prepared, supported, and scaffolded to be effective, and the tasks conducted in the VR environment should complement the CLT principles. This analysis can serve as a starting point for researchers and developers to provide further guidance on the proper design of VR-assisted language learning tasks and on how to make them more accessible in the future.

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DIGITAL TECHNOLOGIES IN EDUCATION

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Abstract: This article provides a comprehensive analysis of the application of digital technologies in the educational process, focusing on both theoretical and practical dimensions. It discusses how technological advancements have reshaped teaching methodologies, learning environments, and student engagement. The integration of tools such as learning management systems, artificial intelligence, and virtual reality has enabled more personalized, flexible, and interactive learning experiences. The study highlights that digital technologies not only improve access to information but also promote critical thinking, collaboration, and creativity among students. Furthermore, it emphasizes the evolving role of teachers as facilitators in technology-enhanced environments. Despite numerous benefits, the article also identifies challenges such as digital inequality, lack of infrastructure, insufficient teacher training, and concerns related to data security. Special attention is given to the context of Azerbaijan, where ongoing reforms and investments demonstrate progress in digital education. The research is based on qualitative methods, including literature review and comparative analysis of international and local sources. The findings suggest that successful implementation of digital technologies requires strategic planning, continuous professional development, and supportive educational policies. The study concludes that digital transformation in education is inevitable and must be managed carefully to ensure inclusivity and effectiveness.

Keywords: digital education, technology, learning, innovation, teaching

TƏHSİLDƏ RƏQƏMSAL TEXNOLOGİYALAR

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Xülasə: Bu məqalə təhsil prosesində rəqəmsal texnologiyaların tətbiqini nəzəri və praktik baxımdan geniş şəkildə təhlil edir. Tədqiqat göstərir ki, texnoloji inkişaf tərdris metodlarını, öyrənmə mühitlərini və tələbələrini fəallığıni əhəmiyyətli dərəcədə dəyişdirmişdir. Öyrənmə idarəetmə sistemləri, süni intellekt və virtual reallıq kimi alətlərin istifadəsi fərdiləşdirilmiş və interaktiv tərdrisi mümkün edir. Rəqəmsal texnologiyalar tələbələrini tənqidi düşünmə, əməkdaşlıq və yaradıcılıq bacarıqlarını inkişaf etdirir. Məqalədə müəllimlərin rolunun dəyişərək daha çox istiqamətləndirici funksiyası daşdığı vurğulanır. Bununla yanaşı, rəqəmsal bərabərsizlik, infrastruktur çatışmazlığı və müəllimlərin hazırlıq səviyyəsi kimi problemlər də qeyd olunur. Azərbaycan nümunəsində rəqəmsal təhsil sahəsində aparılan islahatlar və inkişaf meylləri təhlil edilir. Tədqiqatın nəticələri göstərir ki, rəqəmsal texnologiyaların effektiv tətbiqi üçün strateji planlaşdırma, davamlı təlim və uyğun siyasət vacibdir. Məqalə belə qənaətə gəlir ki, təhsildə rəqəmsallaşma qaçılmazdır və onun düzgün idarə olunması inklüziv və keyfiyyətli təhsilin təmin edilməsində əsas rol oynayır.

Açar sözlər: rəqəmsal təhsil, texnologiya, öyrənmə, innovasiya, tərdris

Introduction

In recent decades, the rapid development of digital technologies has significantly transformed various spheres of human activity, including education. The integration of technological tools into teaching and learning processes has created new opportunities for improving educational quality, accessibility, and efficiency. In particular, artificial intelligence has emerged as one of the most influential innovations, reshaping traditional approaches to knowledge acquisition, assessment, and academic writing. As educational institutions continue to adapt to the demands of the digital age, the role of technology in enhancing students' learning experiences has become increasingly important.

The use of digital technologies in education is not limited to the provision of online resources or virtual classrooms; it also involves the development of interactive, personalized, and adaptive learning environments. Artificial intelligence-based tools, such as automated writing assistants, grammar checkers, and content generators, have introduced new dimensions to academic writing practices. These tools can support students in organizing their ideas, improving language accuracy, and developing more coherent texts. However, alongside these advantages, concerns have been raised regarding over-reliance on technology, issues of academic integrity, and the potential decline of critical thinking skills.

In the context of higher education, especially in countries undergoing rapid digital transformation such as Azerbaijan, the integration of digital technologies presents both opportunities and challenges. On the one hand, it facilitates access to global knowledge and modern teaching methods; on the other hand, it requires the development of new pedagogical strategies and digital competencies among both students and educators. Therefore, it is essential to examine how these technologies influence learning outcomes and academic practices in specific educational contexts.

This study aims to explore the role and impact of digital technologies, particularly artificial intelligence, in modern education with a special focus on academic writing processes. By analyzing both theoretical perspectives and practical applications, the research seeks to identify the benefits, limitations, and implications of technology use in educational settings. The findings of this study are expected to contribute to a better understanding of how digital transformation can be effectively integrated into teaching and learning practices.

Methodology and Materials

This study employs a predominantly qualitative research approach, complemented by limited quantitative elements, in order to investigate the impact of digital technologies, including artificial intelligence, on educational processes and academic writing among students (Hurford et al., 2007). The research is based on both secondary and primary data sources to ensure a comprehensive understanding of the topic. Secondary data were collected from a wide range of academic articles, books, policy documents, and previous studies focusing on digital education and technology integration at both international and local levels, including the context of Azerbaijan. In addition, primary data were gathered through informal observations and analysis of students' interaction with digital tools in educational settings. The study also incorporates elements of comparative analysis to evaluate differences between traditional and technology-enhanced learning environments. Qualitative data were analyzed through descriptive and thematic analysis methods, allowing for the identification of key patterns related to the effectiveness, opportunities, and challenges of digital technologies in education. Furthermore, selected digital platforms and AI-based tools were examined as research materials to better understand their practical role in enhancing teaching and learning processes. This methodological framework ensures a holistic and context-sensitive evaluation of the ongoing digital transformation in education.

Discussion

The application of digital technologies in the educational process has become one of the most significant developments in modern pedagogy. The rapid growth of technological innovations has influenced teaching methodologies, learning environments, and educational outcomes across the world (Bates, 2019). Digital tools are increasingly integrated into classrooms, enabling more flexible and interactive learning experiences. These technologies support personalized education by adapting to individual student needs and learning styles (Redecker, 2017). They also provide access to a vast amount of information, allowing students to explore knowledge beyond traditional textbooks. As a result, the role of teachers has evolved from knowledge transmitters to facilitators of learning (Selwyn, 2016). Students are encouraged to participate actively and take responsibility for their educational progress. Digital platforms foster collaboration and communication among learners, enhancing their critical thinking and problem-solving skills (OECD, 2020). Technology also enables real-time interaction between teachers and students. It supports diverse instructional strategies and innovative approaches. The flexibility of digital tools allows learning to occur anytime and anywhere. Educational institutions increasingly adopt blended learning models. These models combine traditional and digital methods effectively. However, the implementation of digital technologies requires careful planning and resource allocation. Therefore, addressing the digital divide is essential for ensuring equal educational opportunities (UNESCO, 2018).

The theoretical foundations of digital education are based on several pedagogical approaches, including constructivism, behaviorism, and connectivism. Constructivism emphasizes active learning, where students build knowledge through interaction with digital environments (Johnson et al., 2020). Behaviorism supports the use of technology for reinforcement, practice, and assessment of learning outcomes. Connectivism highlights the importance of networks and digital connections in knowledge acquisition. These theories collectively provide a framework for understanding how digital tools can enhance education. Technology enables the practical application of these theories by offering interactive platforms, simulations, and multimedia resources. Digital environments support collaborative knowledge construction and experiential learning. They also accommodate different learning styles, making education more inclusive (Məmmədova, 2020). The integration of

theoretical perspectives with technological tools ensures effective teaching strategies. Teachers can design more engaging lessons using these approaches. Students benefit from interactive and meaningful learning experiences. Theories guide the development of digital curricula. They also help evaluate learning outcomes effectively. Understanding these foundations is essential for educators who aim to implement digital technologies meaningfully. It ensures the alignment of pedagogy and technology.

Various types of digital technologies are utilized in contemporary education systems. Learning Management Systems facilitate course organization, communication, and assessment. Interactive whiteboards enhance classroom participation and visualization of concepts. Educational software supports the development of specific skills and competencies. Mobile applications enable learning beyond the classroom, providing flexibility and convenience. Virtual reality technologies create immersive learning experiences that simulate real-world situations (Bates, 2019). Artificial intelligence systems personalize learning by analyzing student performance and adapting content accordingly. Cloud computing allows efficient storage and sharing of educational materials. Video conferencing tools support distance learning and virtual classrooms (OECD, 2020). Digital assessment tools streamline evaluation processes and provide immediate feedback. Gamification techniques increase motivation and engagement among students. Each of these technologies contributes uniquely to the learning process. Their integration creates a dynamic learning environment. Teachers can select appropriate tools based on objectives. Students benefit from diverse and rich learning experiences. Technology continues to evolve rapidly in education.

The benefits of digital technologies in education are numerous and far-reaching. They improve access to educational resources, allowing students to learn at their own pace. Interactive tools make lessons more engaging and enjoyable. Technology supports inclusive education by accommodating diverse learning needs (UNESCO, 2018). It enhances communication between teachers and students, making feedback more immediate and effective. Collaboration among learners is facilitated through digital platforms. Students develop essential skills such as creativity, critical thinking, and digital literacy (Redecker, 2017). Technology prepares learners for the demands of the modern workforce. It also promotes lifelong learning by providing continuous access to knowledge. Digital tools encourage independent learning habits. They support differentiated instruction strategies. Learning becomes more student-centered and flexible. Teachers can track progress more efficiently. Educational outcomes improve significantly. Overall, digital technologies enhance both teaching and learning processes.

Despite these benefits, several challenges and limitations must be considered. Limited access to digital resources remains a significant issue, particularly in underdeveloped regions. Not all students have reliable internet connections or access to modern devices. Teachers may lack the necessary skills to effectively use digital tools (Selwyn, 2016). Technical difficulties can disrupt the learning process and reduce efficiency. Overdependence on technology may negatively impact critical thinking abilities. Concerns about data privacy and security are becoming increasingly important. Students may experience information overload due to excessive digital content. Financial constraints can limit the adoption of advanced technologies in educational institutions (Əliyev, 2021). Resistance to change among educators and administrators may slow down innovation. Lack of proper training reduces effectiveness. Infrastructure issues can hinder progress. Policies may not always support innovation. Addressing these challenges requires strategic planning. Sustainable solutions are essential for long-term success.

The digital competence of teachers plays a crucial role in the successful integration of technology in education. Educators must possess both technical and pedagogical skills to effectively use digital tools (Redecker, 2017). Continuous professional development is necessary to keep up with technological advancements. Training programs should focus on practical applications and classroom implementation. Institutional support is essential for enhancing teachers' digital skills. Collaboration among educators promotes the sharing of best practices and experiences. Teachers must critically evaluate digital tools and select those that align with learning objectives. Strong digital competence improves teaching quality and student engagement. It also fosters innovation in the classroom. Teachers become more confident in using technology. They can design interactive and effective lessons. Students benefit from improved instruction. Professional growth is continuous in digital education. Competence ensures effective integration of tools. It is essential for modern teaching.

Digital technologies significantly influence student engagement and motivation. Interactive and multimedia content captures students' attention and makes learning more enjoyable. Gamification elements encourage participation and competition (Johnson et al., 2020). Students become active learners rather than passive recipients of information. Digital tools facilitate collaboration and peer interaction. Personalized learning paths increase motivation and help students achieve better results. Immediate feedback allows learners to identify and correct mistakes quickly. Technology supports independent learning and self-assessment. Students feel more responsible for their learning. Engagement leads to better academic performance. Motivation is sustained through interactive content. Learning becomes more meaningful. Students develop confidence in their abilities. Digital tools create a positive learning environment. Overall, engagement is significantly improved.

The implementation of digital technologies in Azerbaijan has shown significant progress in recent years. Educational reforms have prioritized digital transformation and modernization (Əliyev, 2021). Schools are increasingly equipped with modern technological tools. Online learning platforms are widely used in both urban and rural areas. Teachers participate in training programs to enhance their digital competencies. Students benefit from interactive and innovative learning environments. Government initiatives support the development of digital education infrastructure (Məmmədova, 2020). However, challenges remain, particularly in rural regions with limited access to resources. Infrastructure development is ongoing. Investment in education is increasing. Digital literacy is improving among students. Teachers are adapting to new technologies. Policy reforms support innovation. Azerbaijan is aligning with global trends. The country continues to progress in digital education.

Digital technologies are transforming the educational landscape. They offer numerous benefits but also present challenges that must be addressed. Balanced and strategic integration is essential for maximizing their potential. Teachers and students must develop strong digital competencies. Educational institutions should support innovation and technological adoption. Policymakers must address inequalities and ensure equal access to resources. The future of education is closely linked to digital transformation. Continuous research and development are necessary. Innovation must be encouraged. Education systems must adapt. Technology will continue to evolve. Strategies must be updated. Digital learning will expand further. Effective planning is crucial. Success depends on proper implementation.

Digital transformation in education also requires strong institutional leadership and vision. Educational leaders must understand the strategic importance of technology integration. They should develop long-term plans that align with national and global trends. Investment in infrastructure is a key component of successful digitalization. Schools need reliable internet access and modern hardware. Software solutions must be carefully selected based on pedagogical goals. Teachers should be involved in decision-making processes. Their practical experience can guide effective implementation. Professional development programs must be continuous and adaptable. Collaboration with international organizations can provide valuable insights. Monitoring and evaluation systems are necessary to measure progress. Data-driven decisions improve the effectiveness of digital initiatives. Stakeholder engagement, including parents and communities, is also important. Transparency in implementation builds trust and support. Ultimately, strong leadership ensures sustainability and success in digital education.

Another important aspect of digital education is the ethical use of technology. Students must be taught how to use digital tools responsibly. Digital citizenship is becoming a core component of modern education. Issues such as data privacy and cybersecurity require special attention. Educational institutions must establish clear policies and guidelines. Teachers should model ethical behavior in digital environments. Critical evaluation of online information is an essential skill. Students need to distinguish between reliable and unreliable sources. Media literacy plays a crucial role in this process. Technology should be used to enhance learning, not distract from it. Balanced use of digital tools prevents dependency and burnout. Inclusivity must be considered when designing digital learning environments. Accessibility features should support students with disabilities. Ethical considerations also include fairness in assessment and evaluation. Responsible use of technology contributes to a safe and effective learning environment.

Conclusion

In conclusion, the integration of digital technologies, particularly artificial intelligence, into modern education has brought about significant transformations in teaching and learning processes. The findings of this study demonstrate that technological tools have the potential to enhance academic writing by improving students' language accuracy, organization of ideas, and overall productivity. At the same time, the use of AI-based applications contributes to the development of more flexible and personalized learning environments, allowing students to access resources and receive feedback more efficiently.

However, the study also highlights several challenges associated with the increasing reliance on digital technologies. Issues such as academic integrity, overdependence on automated tools, and the possible weakening of critical thinking skills remain important concerns that educators and institutions must address. The results indicate that while technology can serve as a valuable support system, it should not replace the fundamental cognitive and analytical efforts required in the learning process. Therefore, a balanced approach that combines traditional teaching methods with innovative digital solutions is essential.

Furthermore, in the context of Azerbaijan's evolving educational landscape, the effective integration of digital technologies requires not only technical infrastructure but also the development of digital literacy and pedagogical competencies among educators and students. Educational policies and curricula should be adapted to ensure that technology is used in a purposeful and ethical manner.

Overall, this study confirms that artificial intelligence and digital technologies can play a constructive role in improving educational outcomes when applied thoughtfully and responsibly. Future research may focus on empirical investigations with larger sample sizes and more diverse contexts to further explore the long-term effects of AI on academic performance and cognitive development. The successful implementation of digital transformation in education ultimately depends on maintaining a balance between technological innovation and the core principles of effective learning.

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INTEGRATING TECHNOLOGY INTO COMMUNICATIVE LANGUAGE TEACHING: ENHANCING EFL LEARNERS' SPEAKING SKILLS

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Abstract: The use of digital technologies in foreign language education has significantly influenced the ways in which speaking skills are developed in English as a Foreign Language (EFL) contexts. While Communicative Language Teaching (CLT) focuses on interaction and meaningful communication, learners in traditional classrooms often lack sufficient opportunities for extended spoken practice. This study explores the role of technology-enhanced CLT in improving learners' speaking abilities through tools such as video conferencing, mobile applications, and artificial intelligence-based speaking platforms.

A mixed-method research design was applied, involving 40 intermediate-level EFL learners divided into control and experimental groups. Data collection methods included speaking tests administered before and after the intervention, classroom observations, and learner questionnaires. The experimental group engaged in communicative activities supported by digital tools that extended practice beyond the classroom, whereas the control group followed conventional CLT instruction.

The results reveal that students who used technological tools showed greater progress in fluency and interactive speaking performance. Furthermore, learners reported increased confidence and a lower level of anxiety when participating in technology-mediated communication. These findings suggest that digital tools

can effectively support CLT by increasing opportunities for meaningful language use and promoting learner independence.

Overall, the study highlights that integrating technology into CLT in a pedagogically appropriate way can substantially improve speaking skill development in EFL contexts. Practical implications are provided for educators and curriculum developers interested in adopting blended communicative approaches.

Keywords: communicative language teaching, EFL, speaking skills, educational technology, digital learning environments, AI-assisted learning, interaction

TEKNOLOGİYANIN KOMMUNİKATİV DİL TƏDRİSİNƏ İNTEQRASIYASI: EFL ÖYRƏNCİLƏRİNİN DANIŞIQ BACARIQLARININ ARTIRILMASI

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Xülasə: Rəqəmsal texnologiyaların xarici dil tədrisinə tətbiqi İngilis dilinin xarici dil kimi (EFL) öyrədildiyi mühitlərdə danışiq bacarıqlarının inkişafına yanaşmaları əhəmiyyətli dərəcədə dəyişdirmişdir. Kommunikativ Dil Tədrisi (CLT) qarşılıqlı əlaqə və mənalı ünsiyyəti əsas götürsə də, ənənəvi sinif şəraitində öyrənənlər çox vaxt uzunmüddətli şifahi praktika üçün kifayət qədər imkan əldə etmirlər. Bu tədqiqat video konfrans vasitələri, mobil tətbiqlər və süni intellektə əsaslanan danışiq platformaları kimi alətlərdən istifadə etməklə texnologiya ilə zənginləşdirilmiş CLT-nin öyrənənlərin danışiq bacarıqlarının inkişafına təsirini araşdırır.

Tədqiqatda qarışıq metod yanaşması tətbiq olunmuş və 40 orta səviyyəli EFL öyrənəni eksperimental və nəzarət qruplarına bölünmüşdür. Məlumatlar müdaxilədən əvvəl və sonra keçirilmiş danışiq testləri, sinif müşahidələri və öyrənən sorğuları vasitəsilə toplanmışdır. Eksperimental qrup sinifdən kənar da danışiq praktikasını genişləndirən rəqəmsal alətlərlə dəstəklənmiş kommunikativ fəaliyyətlərdə iştirak etmiş, nəzarət qrupu isə ənənəvi CLT təlimi almışdır.

Nəticələr göstərir ki, texnoloji alətlərdən istifadə edən tələbələr səlislilik və interaktiv danışiq göstəriciləri baxımından daha böyük irəliləyiş əldə etmişlər. Bundan əlavə, öyrənənlər texnologiya vasitəsilə qurulan ünsiyyət mühitlərində daha çox özünəinam hiss etdiklərini və daha az narahatlıq yaşadıklarını bildirmişlər. Bu nəticələr rəqəmsal alətlərin mənalı dil istifadəsi imkanlarını artırmaqla və öyrənən müstəqilliyini təşviq etməklə CLT-ni effektiv şəkildə dəstəklədiyini göstərir.

Ümumilikdə, tədqiqat göstərir ki, texnologiyanın CLT çərçivəsinə pedaqoji cəhətdən məqsədyönlü şəkildə inteqrasiyası EFL mühitlərində danışiq bacarıqlarının inkişafını əhəmiyyətli dərəcədə gücləndirə bilər. Dil müəllimləri və kurikulum hazırlayanlar üçün qarışıq kommunikativ yanaşmaların tətbiqinə dair praktik tövsiyələr təqdim olunur.

Açar sözlər: kommunikativ dil tədrisi, xarici dil kimi ingilis dili, danışiq bacarıqları, təhsil texnologiyaları, rəqəmsal öyrənmə mühitləri, süni intellekt dəstəklə öyrənmə, qarşılıqlı əlaqə

Introduction

Speaking ability is widely recognized as one of the most essential components of communicative competence in foreign language learning. However, it remains one of the most difficult skills for EFL learners to develop due to the cognitive demands of real-time language processing, limited exposure, and insufficient opportunities for authentic interaction (Brown, 2014).

In many EFL contexts, learners are exposed to English primarily in formal classroom settings, where communication is often artificial and teacher-centered. As a result, learners may develop

passive knowledge of grammar and vocabulary but fail to use language fluently in spontaneous interaction. This gap between knowledge and performance has been widely discussed in second language acquisition research (Dörnyei, 2005).

Communicative Language Teaching (CLT) was introduced as a response to traditional grammar-based approaches. CLT emphasizes meaningful communication, learner-centered instruction, and the development of interactional competence (Richards, 2006). However, despite its theoretical strengths, CLT faces practical limitations, particularly in contexts where classroom time is limited and exposure to authentic communication is insufficient.

Recent developments in educational technology provide new opportunities for addressing these challenges. Digital tools such as video conferencing systems, mobile applications, and AI-based conversational platforms allow learners to extend speaking practice beyond classroom boundaries. These technologies create interactive environments where learners can practice speaking in a more flexible and less anxiety-inducing setting (Chapelle, 2001).

Literature review

Second language acquisition research has long emphasized the importance of interaction in language learning. According to Long's Interaction Hypothesis, language acquisition is facilitated through conversational interaction where learners negotiate meaning and modify their output (Long, 1996). This process allows learners to notice gaps in their linguistic knowledge and adjust their language use accordingly.

Swain's Output Hypothesis further supports this perspective by arguing that language production is essential for language development. When learners attempt to express meaning, they are required to process language more deeply, which leads to greater linguistic awareness and improved accuracy (Swain, 2000).

In parallel with SLA theories, technological advancements have significantly transformed language learning environments. Computer-Assisted Language Learning (CALL) has evolved from behaviorist drill-based systems to communicative and task-based environments that emphasize interaction and meaning-making (Warschauer & Healey, 1998).

Mobile-Assisted Language Learning (MALL) has further expanded learning opportunities by enabling learners to access language practice anytime and anywhere. Mobile technologies support informal learning and increase exposure to authentic language input (Kukulska-Hulme & Shield, 2008).

More recently, artificial intelligence has introduced new dimensions to language learning. AI-powered systems provide automated feedback on pronunciation, grammar, and fluency, allowing learners to engage in self-directed practice. These systems also simulate conversational interaction, reducing learner anxiety and increasing engagement (Zhang & Zou, 2020).

Empirical studies support the effectiveness of these technologies. Online speaking platforms have been shown to improve learners' fluency and willingness to communicate (Lin, 2015). Similarly, mobile learning enhances motivation by making language learning more continuous and accessible (Stockwell, 2013).

However, technology must be integrated in a pedagogically meaningful way. Its effectiveness depends more on instructional design than on the tools themselves (Chapelle, 2001).

Materials and methods. This study employed a mixed-method research design combining quantitative and qualitative approaches to examine the impact of technology-enhanced CLT on speaking development.

The control group was taught using traditional CLT techniques such as pair work, group discussions, and role-play activities. The experimental group, however, engaged in technology-enhanced communicative tasks supported by digital tools.

The technological tools used included:

- Video conferencing platforms for synchronous communication tasks
- Mobile applications for autonomous speaking practice
- Artificial intelligence-based conversational systems for simulated interaction

Speaking performance was assessed using a standardized rubric measuring fluency, accuracy, pronunciation, and interactional competence. Data collection instruments included pre-test and post-test speaking assessments, classroom observations, and learner questionnaires.

Quantitative data were analyzed using descriptive statistics to measure performance improvement, while qualitative data were analyzed through thematic coding to identify learner perceptions, attitudes, and emotional responses.

Discussion. The results indicate that the experimental group significantly outperformed the control group in all speaking performance categories, including fluency, interactional competence, and confidence.

One of the most notable findings is the reduction in speaking anxiety among learners using digital tools. Participants reported feeling more comfortable speaking in online environments, as these settings reduced immediate peer and teacher pressure. This finding supports the Affective Filter Hypothesis, which suggests that lower anxiety facilitates language acquisition (Krashen, 1985).

Another key finding is the increase in interactional engagement. Learners in the experimental group participated more actively in communicative tasks and demonstrated improved ability to maintain conversations. This aligns with the Interaction Hypothesis, which emphasizes the role of meaning negotiation in language development (Long, 1996).

AI-based tools also contributed to learner autonomy by allowing repeated practice and immediate feedback. This encouraged self-correction and gradual improvement in speaking performance.

However, the study also reveals that technology alone is not sufficient for effective language learning. Learners benefited most when technology was integrated into well-structured communicative tasks designed by the teacher. This confirms that pedagogical design plays a central role in successful technology integration (Chapelle, 2001).

Conclusion. This study demonstrates that integrating technology into Communicative Language Teaching significantly enhances EFL learners' speaking skills. The combination of CLT principles with digital tools creates a more interactive, flexible, and supportive learning environment.

The findings highlight improvements in fluency, interactional competence, and learner confidence, as well as a reduction in speaking anxiety. However, effective implementation requires careful instructional design and teacher guidance.

Future research should explore longitudinal effects of technology-enhanced speaking instruction and investigate its application in different educational contexts and proficiency levels.

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THE INTEGRATION OF ARTIFICIAL INTELLIGENCE IN ENHANCING COMMUNICATIVE COMPETENCE IN UNIVERSITY-LEVEL EFL CLASSROOMS

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Abstract: This article examines the role of artificial intelligence (AI) in enhancing communicative competence in university-level English as a Foreign Language (EFL) classrooms. With the rapid development of digital technologies, AI-driven tools have become an essential component of contemporary language teaching methodologies. The study aims to explore how AI applications contribute to the development of students' speaking, listening, reading, and writing skills through interactive, adaptive, and personalized learning environments. A qualitative research approach is employed, drawing on current pedagogical practices and the use of digital tools such as chatbots, automated feedback systems, and adaptive

learning platforms. The findings indicate that AI technologies significantly increase student engagement, foster learner autonomy, and provide immediate, individualized feedback that supports the principles of communicative language teaching. At the same time, the study identifies several challenges, including over-reliance on technology, insufficient teacher training, and ethical concerns related to data privacy and academic integrity. The article concludes that the successful integration of AI in EFL classrooms requires a balanced and pedagogically informed approach that combines technological innovation with effective teaching strategies and continuous professional development.

Keywords: artificial intelligence, communicative competence, EFL teaching, digital learning, interactive methods, higher education

SÜNİ İNTELLEKTİN UNİVERSİTET SƏVİYYƏSİNDƏ İNGİLİS DİLİNİN XARİCİ DİL KİMİ TƏDRİSİNDƏ KOMMUNİKATİV KOMPETENSİYANIN İNKİŞAFINA İNTEQRASIYASI

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Xülasə: Bu məqalə universitet səviyyəsində ingilis dilinin xarici dil kimi (EFL) tədrisində kommunikativ kompetensiyanın inkişafında süni intellektin (Sİ) rolunu araşdırır. Rəqəmsal texnologiyaların sürətli inkişafı şəraitində süni intellekt əsaslı alətlər müasir dil tədrisi metodologiyalarının ayrılmaz hissəsinə çevrilmişdir. Tədqiqatın əsas məqsədi süni intellekt tətbiqlərinin interaktiv, adaptiv və fərdiləşdirilmiş təlim mühitləri vasitəsilə tələbələrin danışma, dinləmə, oxu və yazı bacarıqlarının inkişafına təsirini sistemli şəkildə təhlil etməkdir. Araşdırmada keyfiyyət yönümlü yanaşmadan istifadə olunmuş, chatbotlar, avtomatik rəy sistemləri, nitqin tanınması texnologiyaları və adaptiv təlim platformaları kimi müasir rəqəmsal vasitələr nəzərdən keçirilmişdir. Əldə olunan nəticələr göstərir ki, süni intellekt texnologiyaları tələbələrin dərəcə fəal cəlb olunmasını təmin edir, onların motivasiyasını artırır, öyrənən müstəqilliyini gücləndirir və dərhal, fərdiləşdirilmiş rəy təqdim etməklə kommunikativ təlim prinsiplərini daha effektiv şəkildə dəstəkləyir. Bununla yanaşı, texnologiyadan həddindən artıq asılılıq, müəllimlərin rəqəmsal bacarıqlarının kifayət qədər inkişaf etməməsi və məlumat təhlükəsizliyi ilə bağlı etik məsələlər əsas çətinliklər kimi müəyyən edilmişdir. Məqalədə nəticə olaraq vurğulanır ki, süni intellektin effektiv tətbiqi texnoloji innovasiyalarla pedaqoji yanaşmaların balanslı inteqrasiyasını tələb edir.

Açar sözlər: süni intellekt, kommunikativ kompetensiya, EFL tədrisi (ingilis dilinin xarici dil kimi tədrisi), rəqəmsal öyrənmə, interaktiv metodlar, ali təhsil

Introduction

In the 21st century, the rapid advancement of digital technologies has significantly transformed the landscape of higher education, particularly in the field of English as a Foreign Language (EFL) teaching. Among these technological innovations, artificial intelligence (AI) has emerged as a powerful tool capable of reshaping teaching methodologies and learning experiences. AI-driven systems offer opportunities for personalized learning, automated feedback, and interactive communication, which are essential for developing communicative competence in language learners (Holmes et al., 2019).

Communicative competence, as defined by Dell Hymes, encompasses not only grammatical knowledge but also the ability to use language appropriately in various social contexts. Later,

Michael Canale and Merrill Swain expanded this concept by identifying its components, including grammatical, sociolinguistic, discourse, and strategic competence (Canale & Swain, 1980). In modern EFL classrooms, fostering these competencies requires interactive, learner-centered approaches rather than traditional teacher-centered instruction.

Recent studies indicate that AI technologies, such as chatbots, intelligent tutoring systems, and speech recognition tools, can significantly enhance language learning outcomes by providing authentic communication environments and immediate feedback (Godwin-Jones, 2018). These tools simulate real-life interactions, allowing learners to practice language skills in a more dynamic and engaging way. Moreover, AI supports differentiated instruction by adapting content to individual learners' needs, thus promoting more effective language acquisition (Chapelle, 2001).

Despite these advantages, the integration of AI into EFL teaching also presents several challenges. Issues related to digital literacy, teacher preparedness, and ethical concerns such as data privacy and academic integrity remain significant barriers (Holmes et al., 2019). Furthermore, excessive reliance on technology may limit human interaction, which is crucial for developing pragmatic and sociocultural aspects of communicative competence.

Therefore, this study aims to explore the role of artificial intelligence in enhancing communicative competence in university-level EFL classrooms by analyzing both its pedagogical benefits and potential limitations.

Materials and Methods

This research adopts a qualitative approach based on descriptive and analytical methods to examine the role of artificial intelligence in EFL teaching. The study primarily relies on secondary data collected from recent academic publications, theoretical frameworks, and practical implementations of AI-based tools in language education.

The methodological framework is grounded in applied linguistics and educational technology research, drawing on established theories of communicative language teaching (CLT) and computer-assisted language learning (CALL) (Richards, 2006; Warschauer, 1996). These frameworks provide a basis for analyzing how AI tools contribute to the development of communicative competence.

The study focuses on several widely used AI-powered tools and technologies, including:

- **Chatbots and conversational agents** (e.g., AI dialogue systems), which facilitate interactive speaking practice and simulate authentic communication scenarios (Godwin-Jones, 2018)
- **Automated writing evaluation systems**, which provide instant feedback on grammar, vocabulary, and coherence, thereby supporting writing skill development (Chapelle, 2001)
- **Adaptive learning platforms**, which personalize learning content based on students' performance and progress
- **Speech recognition technologies**, which help learners improve pronunciation and listening skills through real-time feedback

Data analysis involves a comparative evaluation of these tools in terms of their effectiveness in enhancing different components of communicative competence. The criteria for evaluation include:

- Level of interactivity
- Degree of learner autonomy
- Quality and immediacy of feedback

- Alignment with communicative teaching principles

In addition, the study incorporates insights from classroom-based observations and reported teaching practices in higher education contexts. This allows for a more practical understanding of how AI tools are implemented in real teaching environments.

To ensure the validity of the research, multiple sources are cross-referenced, and findings are interpreted within the broader context of current trends in digital education. The study does not involve experimental data collection but instead provides a comprehensive analytical overview of existing practices and theoretical perspectives.

Discussion

From the analysis presented, it can be observed that artificial intelligence contributes meaningfully to the enhancement of communicative competence in EFL contexts, particularly through increased opportunities for interaction and individualized feedback. The findings also suggest that AI tools are most effective when they are integrated into communicative language teaching frameworks rather than used as standalone instructional solutions. However, the pedagogical impact of these technologies remains highly dependent on teachers' ability to mediate their use and align them with specific learning objectives.

Conclusion

The integration of artificial intelligence into university-level EFL classrooms offers substantial potential for enhancing communicative competence and transforming traditional teaching practices. AI-powered tools create interactive, adaptive, and learner-centered environments that facilitate the development of all four language skills speaking, listening, reading, and writing within meaningful communicative contexts. By providing immediate and individualized feedback, these technologies promote learner autonomy, increase motivation, and enable students to engage more actively in the learning process. In this regard, AI aligns closely with the principles of communicative language teaching and supports the shift toward more flexible and personalized education models.

At the same time, the effective implementation of AI in language education requires a carefully balanced and pedagogically grounded approach. Teachers play a central role in mediating the use of technology, guiding students, and ensuring that digital tools are used purposefully rather than mechanically. Therefore, continuous professional development and targeted training programs are essential to equip educators with the necessary digital and methodological competencies.

Furthermore, several challenges must be addressed to ensure sustainable integration. These include concerns related to data privacy, ethical use of AI, potential over-dependence on automated systems, and the risk of reducing authentic human interaction, which remains a crucial component of language learning. Institutional support, clear policies, and ethical guidelines are needed to manage these issues effectively.

In conclusion, artificial intelligence should be viewed not as a replacement for teachers, but as a powerful complementary resource that enhances teaching quality and learning outcomes. Future research should focus on empirical studies, classroom-based implementations, and the development of innovative pedagogical frameworks that integrate AI in a balanced, ethical, and effective manner.

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DIGITAL TOOLS FOR IMPROVING READING SKILLS

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Abstract: This article examines the role of digital tools in improving reading skills within modern educational contexts. With the rapid advancement of technology, digital resources such as e-books, educational applications, and adaptive learning platforms have become integral to literacy instruction. The study explores how these tools contribute to the development of reading fluency, comprehension, and learner motivation. Drawing on contemporary research, the article highlights the effectiveness of interactive and multimedia-based approaches in enhancing cognitive engagement and supporting diverse learning styles. Special attention is given to the role of personalized learning environments, where digital systems adapt to individual learner needs through data-driven feedback and targeted interventions. The findings indicate that digital tools are particularly beneficial for struggling readers, as they provide flexible and accessible support mechanisms. At the same time, the article addresses potential challenges, including the digital divide, lack of teacher training, and the risk of overreliance on technology. It is argued that a balanced approach, combining digital and traditional reading methods, is essential for achieving optimal learning outcomes. The study concludes that, when implemented effectively, digital tools have significant potential to transform reading education and enhance overall literacy development in the digital age.

In addition, increasing the digital literacy of teachers is considered one of the main conditions for the effective use of these tools. The purposeful integration of technology in the educational process creates conditions for the renewal of pedagogical approaches and the formation of a more flexible learning environment. Analytical data obtained through digital platforms allows teachers to monitor the dynamics of student development and plan teaching more accurately. At the same time, the involvement of parents in this process has a positive effect on strengthening students' reading habits. Consequently, the thoughtful and consistent application of digital technologies in accordance with pedagogical goals plays an important role in improving the quality of education.

Keywords: digital tools, reading skills, literacy development, educational technology, reading comprehension

OXU BACARIQLARINI İNKİŞAF ETDİRMƏK ÜÇÜN RƏQƏMSAL VASİTƏLƏR

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Xülasə: Bu məqalədə müasir təhsil mühitində oxu bacarıqlarının inkişaf etdirilməsində rəqəmsal alətlərin rolu araşdırılır. Texnologiyanın sürətli inkişafı nəticəsində elektron kitablar, təhsil tətbiqləri və adaptiv öyrənmə platformaları kimi rəqəmsal resurslar savadlılıq təliminin ayrılmaz hissəsinə çevrilmişdir. Məqalədə bu alətlərin oxu sürətinə, mətnin anlaşılmasına və öyrənənlərin motivasiyasına təsiri təhlil olunur. Müasir tədqiqatlara əsaslanaraq göstərilir ki, interaktiv və multimediyə əsaslı yanaşmalar şagirdlərin idrak fəallığını artırır və müxtəlif öyrənmə üsullarını dəstəkləyir. Xüsusilə fərdiləşdirilmiş öyrənmə mühitləri diqqət mərkəzində saxlanılır, burada rəqəmsal sistemlər məlumatlara əsaslanaraq öyrənənlərin fərdi ehtiyaclarına uyğunlaşır və məqsədyönlü dəstək təqdim edir. Nəticələr göstərir ki, bu cür alətlər oxu çətinliyi yaşayan öyrənənlər üçün daha effektivdir. Bununla yanaşı, rəqəmsal bərabərsizlik, müəllim hazırlığının yetərsizliyi və texnologiyadan həddindən artıq asılılıq kimi problemlər də qeyd olunur. Məqalədə vurğulanır ki, rəqəmsal və ənənəvi metodların balanslı şəkildə tətbiqi daha səmərəli nəticələr verir. Ümumilikdə, düzgün tətbiq olunduğu halda, rəqəmsal alətlər oxu bacarıqlarının inkişafına mühüm töhfə verir.

Bundan əlavə, müəllimlərin rəqəmsal savadlılığının artırılması bu alətlərin effektiv istifadəsi üçün əsas şərtlərdən biri hesab olunur. Təlim prosesində texnologiyanın məqsədyönlü inteqrasiyası pedaqoji yanaşmaların yenilənməsinə və daha çevik öyrənmə mühitinin formalaşmasına şərait yaradır. Rəqəmsal platformalar vasitəsilə əldə edilən analitik məlumatlar müəllimlərə şagirdlərin inkişaf dinamikasını izləməyə və tədrisi daha dəqiq planlaşdırmağa imkan verir. Eyni zamanda, valideynlərin də bu prosesə cəlb olunması şagirdlərin oxu vərdislərinin möhkəmlənməsinə müsbət təsir göstərir. Nəticə etibarilə, rəqəmsal texnologiyaların pedaqoji məqsədlərə uyğun və düşünülmüş şəkildə tətbiqi təhsilin keyfiyyətinin yüksəldilməsində mühüm rol oynayır.

Açar sözlər: rəqəmsal alətlər, oxu bacarıqları, savadlılıq, təhsil texnologiyaları, mətnin anlaşılması

Introduction

Reading is a fundamental skill that underpins academic success and lifelong learning. In the 21st century, the integration of digital technologies into education has transformed traditional approaches to literacy development. Digital tools, including e-books, educational applications, interactive platforms, and artificial intelligence-based systems, provide new opportunities for enhancing reading skills across different age groups and learning contexts. These tools are increasingly recognized for their ability to improve reading fluency, comprehension, motivation, and engagement.

Recent studies indicate that digital interventions can produce measurable improvements in reading outcomes, particularly when they are tailored to learners' needs and incorporate interactive and adaptive features. The growing accessibility of digital technologies has made them a central component of modern education systems, prompting educators and researchers to explore their potential in fostering effective reading practices.

Methodology and Materials

This study employs a mixed-method approach in order to investigate the impact of artificial intelligence on academic writing processes among students (Hurford et al., 2007). The research is based on both secondary and primary data sources to ensure a comprehensive understanding of the

role of digital technologies in modern education. Secondary data were collected through an extensive review of academic literature, including scholarly articles, books, and previous research studies related to artificial intelligence, digital learning environments, and academic writing practices. These sources provided a theoretical framework for analyzing the integration of technology in education.

In addition to the literature review, primary data were obtained through the observation of students' engagement with digital tools and the analysis of their academic writing practices in technology-supported environments. Elements of qualitative research were applied to interpret students' experiences, attitudes, and challenges associated with the use of AI-based writing tools. At the same time, limited quantitative elements, such as the evaluation of students' performance and frequency of tool usage, were considered to support the findings.

Furthermore, the study incorporates comparative analysis to examine differences between traditional and digitally enhanced academic writing processes. Selected AI-powered platforms and digital educational tools were analyzed as part of the research materials in order to better understand their practical implications in real learning contexts. The collected data were analyzed using descriptive and thematic methods, enabling the identification of key patterns related to the effectiveness, advantages, and limitations of artificial intelligence in academic writing. This methodological framework ensures a balanced, reliable, and context-sensitive evaluation of the impact of digital transformation on education.

Discussion

Reading has long been recognized as one of the most essential components of language acquisition and academic success. In contemporary educational environments, the rapid advancement of digital technologies has fundamentally transformed the ways in which reading is taught, practiced, and assessed. The integration of digital tools into literacy instruction has created new opportunities for learners to engage with texts in dynamic and interactive ways, thereby enhancing not only their technical reading abilities but also their motivation and cognitive engagement. As educational systems continue to evolve, digital tools are increasingly viewed not merely as supplementary resources but as integral elements of effective reading instruction.

The theoretical foundation underlying the use of digital tools in reading development is rooted in several well-established learning theories. Constructivist perspectives emphasize that learners actively construct knowledge through interaction and experience, and digital environments provide precisely such opportunities through interactive texts, instant feedback, and personalized learning pathways. Additionally, multimedia learning theory suggests that the combination of verbal and visual information can significantly enhance comprehension, particularly when learners are exposed to well-designed digital content that integrates text, audio, and imagery in a coherent manner. Cognitive load theory further supports the use of structured digital tools, as they can be designed to present information in manageable segments, thereby preventing cognitive overload and facilitating deeper understanding. Within this framework, digital tools serve not only as content delivery systems but also as cognitive scaffolds that support the development of reading skills [1].

One of the most significant contributions of digital tools to reading development lies in their ability to provide personalized learning experiences. Unlike traditional classroom settings, where instruction is often standardized, digital platforms can adapt to individual learner needs by analyzing performance data and adjusting content accordingly. This adaptability is particularly beneficial for learners who struggle with reading, as it allows them to progress at their own pace

and receive targeted support in areas such as decoding, vocabulary acquisition, and comprehension. Empirical research has demonstrated that adaptive digital systems can lead to measurable improvements in reading outcomes, especially when they incorporate features such as immediate feedback, progress tracking, and differentiated instruction [1]. In this sense, digital tools function as individualized tutors, capable of addressing the diverse needs of learners in ways that are difficult to achieve through traditional methods alone.

Moreover, the use of digital tools has been shown to significantly enhance reading fluency, which is a critical component of overall literacy. Fluency involves the ability to read text accurately, quickly, and with appropriate expression, and it serves as a foundation for effective comprehension. Digital tools often incorporate features such as guided reading, audio support, and repeated exposure to texts, all of which contribute to the development of fluency. For instance, many applications allow learners to listen to a text while simultaneously following along visually, thereby reinforcing the connection between written and spoken language. This multimodal approach has been found to be particularly effective for learners who experience difficulties with traditional reading instruction, as it provides additional cues that facilitate decoding and word recognition [3].

In addition to improving fluency, digital tools play a crucial role in enhancing reading comprehension. Comprehension is a complex cognitive process that involves constructing meaning from text, making inferences, and integrating new information with existing knowledge. Digital environments can support this process by providing interactive features such as embedded questions, annotations, and hyperlinks, which encourage learners to engage more deeply with the text. Furthermore, multimedia elements such as videos and images can help clarify complex concepts and provide contextual information that supports understanding. Studies have shown that when digital tools are designed in accordance with pedagogical principles, they can lead to significant improvements in comprehension, particularly among learners who benefit from visual and interactive learning modalities [5].

Another important aspect of digital tools is their ability to increase learner motivation and engagement. Traditional reading activities are often perceived as monotonous, particularly by younger learners or those who struggle with literacy. In contrast, digital tools frequently incorporate gamification elements such as rewards, levels, and challenges, which make the reading process more enjoyable and engaging. This increased engagement can lead to greater time spent on reading activities, which in turn contributes to improved skill development. Motivation is widely recognized as a key factor in learning, and digital tools have the potential to transform reading from a passive activity into an interactive and rewarding experience [3]. By fostering a positive attitude toward reading, these tools can have a lasting impact on learners' academic trajectories.

Despite their numerous advantages, the implementation of digital tools in reading instruction is not without challenges. One of the most significant issues is the digital divide, which refers to the unequal access to technology among different populations. Students from disadvantaged backgrounds may lack the necessary devices or internet connectivity to fully benefit from digital resources, thereby exacerbating existing educational inequalities. Additionally, the effective use of digital tools requires a certain level of digital literacy on the part of both teachers and students. Without adequate training and support, educators may struggle to integrate these tools into their instructional practices, limiting their potential impact [2]. Furthermore, there is a risk of cognitive overload if digital tools are poorly designed or if they include excessive multimedia elements that distract rather than support learning.

Another concern relates to the potential overreliance on technology in reading instruction. While digital tools offer many benefits, they should not be viewed as a replacement for traditional reading practices. Printed texts, for example, continue to play an important role in developing deep reading skills and critical thinking. Research suggests that a balanced approach, which combines digital and traditional methods, is most effective in promoting comprehensive literacy development. Such an approach allows learners to benefit from the strengths of both modalities while mitigating their respective limitations [1]. Therefore, educators must carefully consider how to integrate digital tools into their curricula in a way that complements rather than replaces established practices.

The pedagogical implications of these developments are substantial. Educators are increasingly required to adopt new instructional strategies that leverage digital technologies while maintaining a focus on core literacy skills. This involves not only selecting appropriate digital tools but also designing learning activities that maximize their effectiveness. For example, teachers can use digital platforms to facilitate collaborative reading activities, where students engage in discussions and share interpretations of texts. They can also use data generated by digital tools to monitor student progress and provide targeted feedback. In this context, the role of the teacher shifts from that of a knowledge transmitter to that of a facilitator and guide, supporting learners as they navigate complex digital environments.

Looking toward the future, it is clear that digital tools will continue to play an increasingly important role in reading education. Advances in artificial intelligence, machine learning, and data analytics are likely to lead to the development of even more sophisticated tools that can provide highly personalized and adaptive learning experiences. These technologies have the potential to revolutionize literacy instruction by enabling real-time assessment and intervention, thereby addressing learning difficulties as they arise. However, the successful implementation of these innovations will depend on ongoing research, as well as the development of best practices that ensure their effective and equitable use.

In addition to individual learning benefits, digital tools also facilitate collaborative reading practices that were previously difficult to implement in traditional settings. Online discussion boards, shared annotation platforms, and virtual reading groups enable learners to engage with texts collectively, exchanging ideas and interpretations in real time. This collaborative dimension not only deepens comprehension but also develops critical thinking and communication skills. By interacting with peers, learners are exposed to diverse perspectives, which enriches their understanding of texts and encourages more reflective reading practices.

Another dimension worth considering is the role of data-driven instruction in digital reading environments. Many digital platforms collect detailed information about learners' reading behaviors, such as time spent on tasks, error patterns, and comprehension levels. This data can be used to inform instructional decisions, allowing educators to identify specific areas of difficulty and adjust their teaching strategies accordingly. As a result, instruction becomes more precise and responsive, ultimately leading to more efficient learning outcomes. However, this also raises important ethical considerations regarding data privacy and the responsible use of student information.

Furthermore, digital tools contribute to the development of metacognitive skills, which are essential for effective reading. Through features such as self-assessment quizzes, progress dashboards, and reflective prompts, learners are encouraged to monitor their own understanding and regulate their reading strategies. This increased awareness of the reading process helps learners

become more independent and strategic readers. Over time, the ability to plan, monitor, and evaluate one's own learning can significantly enhance overall academic performance and lifelong learning skills.

It is also important to highlight the accessibility benefits provided by digital reading tools. For learners with special educational needs, including those with visual impairments or learning disabilities, digital technologies offer features such as text-to-speech, adjustable font sizes, and customizable interfaces. These tools create more inclusive learning environments by removing barriers that may hinder reading development. Consequently, digital platforms support the principle of equal educational opportunities by ensuring that all learners, regardless of their individual challenges, can access and engage with reading materials effectively.

Conclusion

In light of the growing integration of technology into educational systems, digital tools have emerged as a transformative force in the development of reading skills. The analysis presented throughout this study demonstrates that digital tools are not merely supplementary resources but essential components of modern literacy instruction. Their capacity to provide personalized, adaptive, and interactive learning environments enables learners to engage with texts in ways that traditional methods alone cannot fully support. By facilitating improvements in reading fluency, comprehension, and motivation, digital tools contribute significantly to more effective and inclusive learning processes.

Furthermore, the ability of digital technologies to address individual learner differences represents one of their most valuable advantages. Through data-driven feedback and adaptive algorithms, these tools can identify specific weaknesses and offer targeted interventions, thereby supporting both struggling and advanced readers. This individualized approach is particularly important in diverse classrooms, where learners possess varying levels of proficiency and distinct learning needs. In addition, the integration of multimedia elements enhances cognitive processing by presenting information through multiple channels, which is especially beneficial for visual and auditory learners.

However, the successful implementation of digital tools requires careful planning and critical awareness. Issues such as unequal access to technology, insufficient teacher training, and the risk of cognitive overload must be addressed to ensure equitable and effective use. It is essential that educators adopt a balanced pedagogical approach that combines digital and traditional reading practices, thereby maximizing the strengths of both. Digital tools should complement, rather than replace, established methods, ensuring that deep reading and critical thinking skills are not compromised.

Looking ahead, the future of reading education will likely be shaped by further advancements in artificial intelligence and immersive technologies, which promise even greater levels of personalization and engagement. Nevertheless, continued research is necessary to evaluate the long-term impact of these tools and to establish best practices for their integration into educational contexts. Ultimately, when implemented thoughtfully and strategically, digital tools hold immense potential to enhance literacy development, promote learner autonomy, and prepare individuals for the demands of an increasingly digital world.

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DEVELOPMENT AND USE OF ELECTRONIC EDUCATIONAL RESOURCES IN MODERN EDUCATION

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Abstract: Electronic educational resources have become an essential part of modern education, transforming the way knowledge is delivered and acquired. The rapid development of digital technologies has expanded the use of online platforms, multimedia materials, and interactive learning tools in educational institutions. This study examines the concept, classification, and application of electronic educational resources while evaluating their impact on teaching and learning processes. It highlights the major benefits of these resources, including improved accessibility to educational content, increased student motivation, personalized learning opportunities, and greater flexibility in time and place of study. At the same time, the research identifies several challenges, such as limited digital competence among teachers and students, inadequate technological infrastructure, and unequal access to digital devices and the internet. The findings suggest that overcoming these barriers through effective educational policies, professional development, and technological investment will enable electronic educational resources to enhance learning outcomes and improve the overall quality, accessibility, and sustainability of education.

Key words: electronic resources, digital education, e-learning, online learning, educational technology

MÜASİR TƏHSİLDƏ ELEKTRON TƏHSİL RESURSLARININ İNKİŞAFI VƏ TƏTBİQİ

Maral Eldost MƏMMƏDOVA

Təhsil texnologiyaları və rəqəmsal təlim üzrə mütəxəssis

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Xülasə: Rəqəmsal dövrdə elektron təhsil resursları müasir təhsil sisteminin ayrılmaz tərkib hissəsinə çevrilmişdir. İnformasiya və kommunikasiya texnologiyalarının sürətli inkişafı tədris və öyrənmə proseslərini əhəmiyyətli dərəcədə dəyişdirərək rəqəmsal platformaların, elektron dərsliklərin, multimedia vasitələrinin və interaktiv təlim alətlərinin geniş tətbiqinə şərait yaratmışdır. Bu tədqiqat elektron təhsil resurslarının mahiyyətini, təsnifatını və praktik tətbiq imkanlarını araşdırmaqla yanaşı, onların tədris prosesinə təsirini qiymətləndirir. Araşdırma göstərir ki, bu resurslar informasiyaya çıxışı asanlaşdırır, tələbələrin motivasiyasını və fəallığını artırır, fərdiləşdirilmiş öyrənməni dəstəkləyir və çevik təhsil mühiti formalaşdırır. Bununla yanaşı, rəqəmsal savadlılığın kifayət qədər inkişaf etməməsi, texnoloji infrastrukturun məhdudluğu və rəqəmsal resurslara çıxışdakı bərabərsizliklər əsas problemlər kimi qalmaqdadır. Tədqiqatın nəticələri göstərir ki, müvafiq texnoloji infrastrukturun yaradılması, müəllim və tələbələrin rəqəmsal bacarıqlarının inkişaf etdirilməsi və elektron resursların səmərəli tətbiqi təhsilin keyfiyyətinin, əlçatanlığının və effektivliyinin əhəmiyyətli dərəcədə yüksəldilməsinə imkan verir.

Açar sözlər: elektron resurslar, rəqəmsal təhsil, elektron təhsil (e-learning), onlayn təhsil, təhsil texnologiyaları.

Introduction

Over the past few decades, technological innovation has had a profound impact on nearly every aspect of human life, including the field of education. Traditional classroom-based instruction is no longer the only method of learning. Today, students have the opportunity to access educational content through digital devices such as computers, tablets, and smartphones.

Electronic educational resources refer to any digital materials that support teaching and learning processes. These resources include e-books, online courses, multimedia content, mobile applications, and virtual learning environments. Their development has created new opportunities for improving the accessibility and quality of education.

In modern society, learning is no longer limited by time or location. Students can study independently, collaborate with others online, and access a wide range of materials instantly. Despite these benefits, the integration of technology into education also introduces several challenges that require careful consideration.

The main aim of this study is to analyze the development and use of electronic educational resources, examine their benefits and limitations, and evaluate their role in improving educational outcomes.

Literature Review

A large number of researchers have investigated the role of digital technologies in education. Their findings highlight both the potential and the limitations of electronic educational resources.

Bates (2019) argues that digital learning environments allow students to control their learning pace, which leads to more personalized and effective learning experiences. Students can revisit materials, practice skills, and learn according to their individual needs.

Clark and Mayer (2016) emphasize the importance of multimedia learning. They suggest that combining text, audio, and visual elements can improve comprehension and memory retention. Well-designed multimedia resources help learners understand complex concepts more easily.

Selwyn (2018) presents a critical perspective, stating that technology alone does not improve education. Instead, the effectiveness of digital tools depends on how they are used by teachers and integrated into teaching strategies.

Other researchers highlight the importance of electronic resources in distance education and lifelong learning. These resources enable people of all ages to continue their education regardless of geographical barriers.

However, several studies also identify key challenges. These include lack of digital skills, limited access to devices and internet connections, and the risk of reduced face-to-face interaction. The concept of the “digital divide” remains one of the most significant issues in modern education.

Overall, the literature suggests that while electronic educational resources offer many advantages, their success depends on proper implementation and support.

Methodology and Materials

This research is based on a qualitative approach using secondary data sources. Information was collected from academic books, journal articles, and reliable online educational platforms.

The study focuses on the following aspects:

- different types of electronic educational resources
- their development process
- advantages in teaching and learning
- challenges and limitations

The collected data were carefully analyzed to identify common trends, patterns, and key ideas related to the use of digital resources in education.

Types of Electronic Educational Resources

Electronic educational resources exist in various forms, each serving a specific purpose in the learning process.

E-books and Digital Textbooks. E-books are among the most widely used digital learning tools. They provide easy access to educational content and are often more affordable than printed textbooks. Students can use features such as search functions, highlighting, and note-taking, which improve their learning experience. In addition, e-books reduce the need for physical materials, making them environmentally friendly.

Online Courses and Learning Platforms. Online learning platforms offer structured courses on a wide range of subjects. These platforms include video lectures, assignments, quizzes, and discussion forums. They support both formal education and independent learning. Many universities and organizations use online platforms to deliver courses to a global audience.

Educational Videos and Multimedia Content. Multimedia resources such as videos, animations, and simulations make learning more interactive and engaging. They are particularly useful for explaining complex topics in subjects like science and mathematics. Visual learning helps students understand abstract concepts more effectively.

Digital Libraries. Digital libraries provide access to a vast collection of academic materials, including books, research papers, and journals. They are especially valuable for students and researchers who need reliable and up-to-date information. Digital libraries save time and provide access to resources that may not be available locally.

Mobile Learning Applications. Mobile learning apps allow students to study anytime and anywhere. These apps often include interactive exercises, quizzes, and gamified learning experiences. They are particularly popular among younger learners and support flexible, on-the-go education.

Development of Electronic Educational Resources

The development of electronic educational resources requires a systematic and well-planned approach. First, educational objectives must be clearly defined. Developers need to identify what knowledge and skills learners should acquire. Second, content must be organized logically and presented in a clear and simple way. The use of visuals, examples, and interactive elements enhances understanding. Third, appropriate technological tools are used to create digital content. These include software for video production, graphic design, and online learning platforms. User feedback plays a crucial role in the development process. Resources should be tested and improved based on the experiences of students and teachers.

Finally, accessibility is an important consideration. Resources should be designed to be inclusive and usable for all learners, including those with disabilities.

Advantages of Electronic Educational Resources

Electronic educational resources provide numerous benefits for both students and teachers. One of the main advantages is flexibility. Learners can study at their own pace and choose their own time and place for learning. Another important benefit is accessibility. Educational materials can be accessed by a large number of people through the internet, regardless of their location. Electronic resources also increase student motivation and engagement. Interactive and multimedia content makes learning more interesting. They support individualized learning. Students can review materials multiple times and focus on areas where they need improvement. Additionally, teachers can use a variety of teaching methods and save time by using ready-made digital materials.

Challenges and Limitations

Despite their many advantages, electronic educational resources also present several challenges. One of the most significant issues is the digital divide. Not all students have equal access to devices or internet connections. Another challenge is the lack of digital literacy. Some students and teachers may not have the necessary skills to use technology effectively. Technical problems such as slow internet, software errors, or system failures can disrupt the learning process. There is also the issue of distraction. Students using digital devices may lose focus due to social media or other online activities. Furthermore, excessive reliance on technology may reduce face-to-face communication and social interaction, which are important for personal development.

Discussion

The findings of this study show that electronic educational resources play a transformative role in modern education. They provide opportunities for more flexible, accessible, and interactive learning experiences. Students are able to take greater responsibility for their own learning. However, the effective use of these resources requires proper planning, teacher training, and institutional support. Teachers play a key role in guiding students and selecting appropriate digital tools. Educational institutions must invest in infrastructure and provide training programs to ensure successful implementation. It is also important to maintain a balance between traditional and digital teaching methods. Technology should enhance education, not replace human interaction.

Conclusion

This study examined the development and use of electronic educational resources in modern education systems. The results demonstrate that these resources offer significant advantages, including flexibility, accessibility, and improved student engagement. However, challenges such as limited access, technical issues, and lack of digital skills must be addressed to ensure equal opportunities for all learners.

In conclusion, electronic educational resources are powerful tools that can enhance the quality of education when used effectively and responsibly. Future research should explore the impact of emerging technologies, such as artificial intelligence and virtual reality, on digital learning environments.

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THE PEDAGOGICAL SIGNIFICANCE OF DIGITAL TECHNOLOGIES IN THE MODERN LEARNING ENVIRONMENT

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Abstract. The rapid development of digital technologies has significantly transformed modern education systems, reshaping both teaching practices and learning environments. The ongoing process of digitalization has created new opportunities for improving the quality, accessibility, and efficiency of education. This article explores the pedagogical significance of digital technologies in the modern learning environment, focusing on their impact on instructional design, student engagement, and learning outcomes. The study examines the role of various digital tools and platforms in enhancing interactive learning, supporting collaborative activities, and promoting individualized instruction. Particular attention is given to the integration of digital technologies within online and hybrid learning models, which provide flexibility in time and space and allow for more inclusive educational practices. In addition, the article analyzes how multimedia resources, learning management systems, and adaptive learning technologies contribute to the development of critical thinking, problem-solving skills, and learner autonomy. The findings indicate that digital technologies not only facilitate broader access to educational resources but also significantly improve the effectiveness of teaching strategies by enabling innovative and student-centered approaches. However, the research also identifies several challenges associated with the implementation of digital tools, including technical limitations, unequal access to digital infrastructure, and the need for continuous professional development of teachers to ensure effective integration into the curriculum. Furthermore, the study highlights the growing role of emerging technologies such as artificial intelligence in personalizing learning experiences and optimizing educational processes. It is concluded that the effective and pedagogically sound use of digital technologies contributes to the creation of a dynamic, flexible, and learner-centered educational environment, ultimately enhancing the overall quality and sustainability of education systems in the digital age.

Keywords: digital technologies, e-learning, interactive learning, educational innovation, digital pedagogy, hybrid learning

MÜASİR TƏDRİS MÜHİTİNDƏ RƏQƏMSAL TEXNOLOGİYALARIN PEDAQOJİ ƏHƏMİYYƏTİ

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Xülasə. Rəqəmsal texnologiyaların sürətli inkişafı müasir təhsil sistemində mühüm dəyişikliklərə səbəb olmuş, tədris və öyrənmə mühitinin yenidən formalaşmasına şərait yaratmışdır. Rəqəmsallaşma prosesi təhsilin keyfiyyətinin, əlçatanlığının və səmərəliliyinin artırılması üçün yeni imkanlar açmışdır. Bu məqalədə müasir təlim mühitində rəqəmsal texnologiyaların pedaqoji əhəmiyyəti araşdırılır, onların tədris prosesinə, şagird fəallığına və öyrənmə nəticələrinə təsiri təhlil olunur.

Tədqiqat çərçivəsində müxtəlif rəqəmsal alət və platformaların interaktiv təlimin təşkili, əməkdaşlıq mühitinin yaradılması və fərdiləşdirilmiş öyrənmənin dəstəklənməsində rolu nəzərdən keçirilmişdir. Xüsusilə onlayn və hibrid təlim modellərində rəqəmsal texnologiyaların tətbiqi imkanlarına diqqət yetirilmişdir. Eyni zamanda multimedia resursları, təlim idarəetmə sistemləri və adaptiv öyrənmə texnologiyalarının tənqidi düşünmə, problem həll etmə bacarıqları və öyrənənlərin müstəqilliyinin inkişafına təsiri araşdırılmışdır. Araşdırmanın nəticələri göstərir ki, rəqəmsal texnologiyalar yalnız təhsil resurslarına çıxışı genişləndirmir, həm də innovativ və şagirdyönlü yanaşmalar vasitəsilə tədris strategiyalarının effektivliyini əhəmiyyətli dərəcədə artırır. Bununla yanaşı, texniki infrastrukturun yetərsizliyi, rəqəmsal bərabərsizliyin mövcudluğu və müəllimlərin davamlı peşəkar inkişaf ehtiyacı kimi problemlər də müəyyən edilmişdir. Məqalədə həmçinin süni intellekt kimi yeni texnologiyaların fərdiləşdirilmiş öyrənmə imkanlarının genişləndirilməsində və təhsil prosesinin optimallaşdırılmasında artan rolu vurğulanır. Nəticə olaraq qeyd edilir ki, rəqəmsal texnologiyaların pedaqoji cəhətdən düzgün və məqsədyönlü tətbiqi çevik, dinamik və şagirdyönlü təlim mühitinin formalaşmasına xidmət edir və təhsilin ümumi keyfiyyətinin yüksəldilməsinə mühüm töhfə verir.

Açar sözlər: rəqəmsal texnologiyalar, elektron təhsil, interaktiv təlim, pedaqoji innovasiya, rəqəmsal pedaqogika, hibrid təlim

Introduction

The integration of digital technologies into education has become one of the defining characteristics of the modern learning environment (Selwyn, 2016; Bates, 2019). With the rapid advancement of information and communication technologies, traditional teaching methods are being transformed into more interactive, flexible, and student-centered approaches (Laurillard,

2013). This transformation is not only changing how knowledge is delivered but also how it is constructed, shared, and assessed within educational contexts.

Digital technologies provide new opportunities for both teachers and students by enabling access to a wide range of educational resources and supporting innovative teaching strategies. Through tools such as multimedia content, virtual classrooms, and learning management systems, educators can design more engaging and dynamic learning experiences (Anderson, 2011). At the same time, students benefit from increased autonomy, as they can access learning materials anytime and anywhere, learn at their own pace, and actively participate in the learning process.

In addition, the modern learning environment is increasingly characterized by the use of online and hybrid learning models, which combine traditional face-to-face instruction with digital tools and platforms (Bonk & Graham, 2012). These models offer greater flexibility and inclusivity, making education more accessible to diverse groups of learners. Digital technologies also facilitate communication and collaboration, allowing students to interact with peers and instructors beyond the physical classroom.

Furthermore, the integration of digital technologies supports the development of essential 21st-century skills, such as critical thinking, creativity, digital literacy, and problem-solving. By incorporating interactive and technology-based activities into the curriculum, educators can better prepare students for the demands of the modern workforce and society.

Despite these advantages, the effective implementation of digital technologies in education requires careful planning and consideration. Issues such as digital inequality, limited technical infrastructure, and the need for continuous professional development of teachers remain significant challenges. Therefore, it is essential to adopt a balanced and pedagogically sound approach to the use of digital technologies in order to maximize their potential benefits.

Overall, digital technologies play a crucial role in promoting learner-centered education and enhancing the quality of teaching and learning processes. Their effective integration into the educational system is a key factor in the ongoing modernization and improvement of education.

Materials and Methods

This research is based on a combination of theoretical analysis, comparative methods, and a systematic review of relevant scientific literature (Selwyn, 2016; Siemens, 2005). Various international and local academic sources, including peer-reviewed journal articles, books, policy documents, and educational reports, were examined to explore the role and pedagogical significance of digital technologies in modern education. The selected sources were analyzed in terms of their theoretical frameworks, research findings, and practical implications for teaching and learning processes.

In addition to the literature review, a comparative approach was applied to identify similarities and differences in the implementation of digital technologies across different educational contexts. Particular attention was given to the integration of digital tools in both traditional and technology-enhanced learning environments, including fully online and hybrid education models.

The study also examines the application of key digital tools such as learning management systems (LMS), multimedia educational resources, interactive platforms, and communication technologies (Means et al., 2013). These tools were evaluated based on their functionality, accessibility, and effectiveness in supporting teaching activities and enhancing student engagement. Furthermore, their role in facilitating collaborative learning, self-directed study, and formative assessment was considered.

To ensure a comprehensive analysis, the research focused on several core criteria, including student participation, motivation, academic performance, and the development of digital competencies. The impact of digital technologies on these aspects was assessed through the synthesis of findings from previous empirical studies.

Additionally, the study considers the challenges associated with the implementation of digital technologies, such as technical limitations, digital divide issues, and the level of teachers' digital literacy. By combining theoretical insights with comparative analysis, the research aims to provide a well-rounded understanding of how digital technologies can be effectively integrated into the educational process.

Results and Discussion

The findings of the study demonstrate that digital technologies have a profound and multifaceted impact on the teaching and learning process. Their integration into educational environments has led to significant pedagogical transformations, shifting the focus from teacher-centered instruction to more learner-centered and interactive approaches.

One of the key advantages of digital technologies is their ability to increase student motivation and engagement. Interactive tools, multimedia content, and gamified learning environments make the learning process more attractive and stimulating for students. As a result, learners are more actively involved in classroom activities and demonstrate a higher level of participation.

In addition, digital technologies support interactive and collaborative learning. Through online platforms, discussion forums, and virtual classrooms, students can communicate, share ideas, and work together regardless of their physical location. This not only enhances social learning but also contributes to the development of communication and teamwork skills (Johnson et al., 2016).

Another important benefit is the opportunity to provide personalized learning experiences (Wang, 2021). Digital tools and adaptive learning systems allow educators to tailor instructional materials according to individual student needs, abilities, and learning styles. This helps to ensure that each learner can progress at their own pace and achieve better academic outcomes.

Furthermore, digital technologies significantly improve access to educational content. A wide range of digital resources, including e-books, video lectures, and online courses, are readily available to learners. This accessibility supports continuous learning and enables students to revisit materials whenever necessary, thereby reinforcing their understanding.

Digital technologies are particularly effective in online and hybrid learning environments, where flexibility in time and place is essential. These models provide inclusive learning opportunities for diverse groups of students, including those with geographical, physical, or time constraints. As a result, education becomes more accessible and adaptable to individual circumstances.

However, despite these advantages, several challenges remain. One of the major issues is the lack of adequate technical infrastructure in some educational institutions, which limits the effective implementation of digital tools. In addition, many teachers still face difficulties due to insufficient digital competencies, highlighting the need for continuous professional development and training programs. Another significant challenge is the unequal access to digital devices and reliable internet connections, which creates a digital divide among learners.

Despite these challenges, the use of digital technologies in education continues to expand rapidly. Recent developments in artificial intelligence and adaptive learning systems are expected to

further enhance the effectiveness of education. These technologies offer new possibilities for real-time feedback, data-driven decision-making, and highly personalized learning experiences (UNESCO, 2022).

Overall, the results indicate that while digital technologies present certain challenges, their pedagogical benefits far outweigh the limitations. With proper support, infrastructure, and training, they have the potential to significantly improve the quality, inclusivity, and effectiveness of modern education systems.

Conclusion

In conclusion, digital technologies play a vital and transformative role in the modern learning environment. Their pedagogical significance is reflected in their ability to support interactive, flexible, and learner-centered education, which aligns with the demands of contemporary society. The integration of digital tools has reshaped traditional teaching practices, enabling more dynamic, engaging, and inclusive learning experiences for students. The effective use of digital technologies contributes significantly to improving the overall quality of teaching and learning. By facilitating access to diverse educational resources, enhancing communication and collaboration, and supporting personalized instruction, digital tools help to increase student participation and academic achievement. Moreover, they encourage the development of essential 21st-century skills such as critical thinking, creativity, digital literacy, and problem-solving. Another important outcome of digital transformation in education is the promotion of lifelong learning. Digital platforms and online learning environments provide learners with continuous opportunities for self-development beyond formal education. This flexibility allows individuals to adapt to the rapidly changing demands of the labor market and society. However, the successful integration of digital technologies requires addressing existing challenges, including the need for reliable technical infrastructure, equal access to digital resources, and the continuous professional development of teachers. Without these supporting factors, the full potential of digital education cannot be realized. Looking ahead, future developments in technology (particularly in artificial intelligence, adaptive learning systems, and data-driven educational tools) are expected to create new opportunities for innovation in education. These advancements will further enhance personalized learning, improve assessment methods, and optimize educational processes.

Overall, the strategic and pedagogically grounded use of digital technologies has the potential to transform education systems, making them more effective, inclusive, and responsive to the needs of modern learners.

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THE ROLE OF ARTIFICIAL INTELLIGENCE

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Abstract. The article examines the innovations of the digital world in the 21st century. The advantages of the technological age and the role of these advantages in the educational environment are discussed. Digital technologies have become an integral part of our time. A vivid example of this is Artificial Intelligence. We widely use artificial intelligence technologies in work, education, and social life. In the educational process, artificial intelligence technologies act as an auxiliary force for both the educator and the learner. The use of digital technologies in the educational process leads to student activity in teaching the subject. Also, the teacher can use such technologies for interesting teaching of the subject. Types of artificial intelligence are currently actively used today. Depending on the purpose of use, artificial intelligence can be both useful and beneficial, as well as useless, useless and harmful. Thus, although digital technologies provide a number of wide opportunities, if not used correctly, a person becomes an age of the digital world. In addition to the advantages of digital technologies, when looking at the disadvantages, inaccurate guidance and incorrect planning can be cited as examples. The article discusses the types of artificial intelligence technologies, their ability to present plans taking into account the diversity of learning, their support for the

student in analyzing the topic, their guidance, and at the same time their disadvantages, technological inequality.

Keywords: Digital world, artificial intelligence, technological superiority, technological inequality, chatbot, chatgpt, khoot

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Xülasə. Məqalədə XXI əsrdə rəqəmsal dünyanın yenilikləri araşdırılıb. Texnologiya əsrinin üstünlükləri, bu üstünlüklərin təhsil mühitindəki rolundan bəhs olunur. Rəqəmsal texnologiyalar günümüzün ayrılmaz bir hissəsinə çevrilib. Bunun bariz nümunəsi Suni intellektidir. Suni intellekt texnologiyalarından iş, təhsil, sosial həyatda geniş istifadə edirik. Təhsil prosesində suni intellekt texnologiyaları həm təhsilverən, həm də təhsilalan üçün köməkçi qüvvə rolunda çıxış edir. Rəqəmsal texnologiyaların tədris prosesində istifadəsi mövzunun tədrisində şagird fəallığına səbəb olur. Həmçinin müəllim mövzunun maraqlı tədrisi üçün bu kimi texnologiyalardan istifadə edə bilər. Suni intellekt növlərindən hal-hazırda günümüzdə aktiv şəkildə istifadə edilir. İstifadə məqsədindən asılı olaraq suni intellekt həm yararlı, faydalı ola bilər, həm də yararsız, faydasız və zərərli ola bilər. Belə ki, rəqəmsal texnologiyalar bir sıra geniş imkanlar bəxş etsə də, doğru istifadə etmədikdə insan rəqəmsal dünyanın əsrinə çevrilir. Rəqəmsal texnologiyaların üstünlükləri ilə yanaşı mənfi cəhətlərə nəzər saldıqda qeyri-dəqiq istiqamətləndirmə, yanlış planlaşdırmanı nümunə kimi göstərmək olar. Məqalədə suni intellekt texnologiyalarının növlərindən, öyrənmə müxtəlifliyini nəzərə alaraq planlar təqdim etməsindən, mövzunun təhlilində şagirdə dəstək olmasından, istiqamət verməsindən və eyni zamanda mənfi cəhətlərindən, texnoloji bərabərsizlikdən bəhs olunur.

Açar sözlər: Rəqəmsal dünya, suni intellekt, texnoloji üstünlük, texnoloji bərabərsizlik, chatbot, chatgpt, khoot

Introduction

Today, the development and improvement of technologies is one of the main factors affecting the ease of human labor. The technological age is trying to keep up with society and meet the demands of the times. Digital technological superiority is one of the important factors for the sustainable development of society. When discussing the concept of digitalization, it should first be noted that society is divided into 2 parts in terms of using these technologies:

- Digital natives - this concept refers to people who have grown up with technology and are constantly in touch with it.

- Digital immigrants - here, it includes people who have become acquainted with technology at a certain stage of their lives, that is, later on, and are trying to learn it (Demirtaş 2025). Technology plays the role of a tool that makes people's work easier. One of such technological concepts is artificial intelligence. Artificial intelligence acts in roles such as quick search, guidance, planning, etc. Technology was created to help people. Proper use leads to development, but if it is not used correctly, it can cause the passivity of the human brain. Technology is a digital world created by humans. People use this world in work, education and other fields. The use of artificial intelligence in education leads to the emergence of a number of innovations. (Abbasov & Məmmədov, 2025) Examples of such technological innovations include personalized learning,

online education, and intelligent teaching systems. Some types of artificial intelligence in the field of education can be used by both teachers and students. Teachers can use it to make the topic interesting and suitable for learning diversity, and students can use it in matters such as researching and presenting the topic.

Boztepe (2025) lists several types of artificial intelligence:

- ChatGPT- Students can use this technology to explain unclear topics in class and to create lesson plans
- Gemini- It is used in content analysis and text visualization.
- Quizlet – Helps in preparing interactive tasks and tests.
- Kahoot – The number one assistant in making the lesson process active. It serves game-based learning based on the diversity of students' learning
- Canva – Used in preparing presentations, visual materials, etc. (p 23)

Teaching the subject with simulations and game techniques makes the learning process more attractive and interesting for students. With this method, students learn while having fun. (Kalinkara, p 165) The application of digital technologies in the learning environment is not only satisfied with the transfer of information in an interesting way, these technologies create a developing environment that forms critical thinking and cooperation in students. While in traditional education the lesson process was limited to the teacher-student relationship, interactive learning through digital technologies is based on teacher-student, student-teacher, student-student communication. For example, tests and competitions prepared for students with the “Kahoot” and “Quizlet” platforms encourage learning based on healthy competition. (Bajaj, 2024)

Materials and methods

Artificial intelligence can recommend a number of literatures for the development of students' personal learning skills, at the same time it can design a program according to the student's learning pace and adjust the difficulty level of tasks. Examples of AI technologies that cover such issues are the programs "Khan Academy", "Coursera", "Duolingo". These programs are based on eliminating the weaknesses of students.

Intelligent Tutoring Systems (ITS) is an artificial intelligence technology that replaces humans in solving the difficulties and problems faced by students and proposes a solution to the students' problems. ITS explains the solution to the problem step by step to students. ITS is a powerful AI technology that reflects feedback. One of the advantages of AI technologies is automated testing and evaluation. Here, students' tests, essays and other written works are checked and evaluated. In addition to all this, it can predict future success or the complete opposite by monitoring the student's learning activity. This process is called "Predictive Analytics". (Ismayilova, 2025) Thus, there are a number of AI technologies that monitor the activities of students. This technology provides a personalized plan for the future success of students. Examples of such AI technologies include the “Swift” system in India and the “ALP” system in the USA.

Chatbots are used in the educational environment. When we say educational chatbots, this includes their use for simplified student admission, providing information on any issue, and engaging students in oral dialogue. . Such educational chatbots include “Ada”, “Deakin Genie”. (UNESCO, 2022)

When talking about the use of artificial intelligence in literature lessons, we can say that it forms a new view of the analysis of large-scale works. Key concepts that cannot be detected in the process of analyzing literary works in a traditional environment are identified in the digital

environment. The difficulty of the student in understanding the plot of the work is eliminated. The "Syuzhet Package" project identifies and visualizes the plot parts of a literary sample, that is, it approaches the analysis of the work from a different perspective. (Gasimova, 2025) Innovations reflected in the digital world play an important role in the students' understanding of the literature of the classical world. Identifying words of Arabic and Persian origin in classical literary samples and visualizing the socio-political atmosphere of that period leads to a comprehensive understanding of the literary sample. A kind of ancient era is revived with the capabilities of the modern era. When students analyze a certain literary work, the visualization of the external appearance of images through AI technology acts as an auxiliary force in transforming the passive activity of students into activity, along with the interactive organization of the lesson. When a student uses artificial intelligence technology to determine the result in the process of analyzing a certain literary sample, he compares the result presented by AI with the product of his critical thinking and comes to a conclusion. In this process, the student solves the problem he encounters at any time, without time limits, through AI. However, the main nuance here is that the student analyzes and understands the result presented by AI, that is, does not accept it as a ready-made answer. When applying to AI technology on a certain topic, it recommends examining many examples that match the topic. For example, when applying to artificial intelligence regarding Mirza Alabkar Sabir's poem "Ekinchi", it can respond to the request and also suggest works by other representatives of satirical literature. Such a process is an advantage of artificial intelligence. (Asgarova, 2026)

Conclusion and discussion

Artificial intelligence technologies are programs created by humans. Although these technologies are flexible and accessible, they cannot compete with human cognition. These technologies have several advantages as well as disadvantages. Although digital technologies have the advantage of planning the activities of the student for success, they also have disadvantages. When we say disadvantages, we can give an example of analytical errors. The reflection of false negative or false positive information during the planning of student activities will lead to the failure of the final result. The use of artificial intelligence helps students clarify unclear issues, but the disadvantage is that in most cases, students use AI technologies without analyzing the answer. Such a situation weakens the student's critical thinking skills.

Although the modern era is the age of technology, there is technological inequality. Students cannot use technological advantages to the same extent. In general, since urban and regional schools are different, students do not have the same opportunities. For example, there are problems with internet access, computers are not available, and technological equipment is not available, so there are cases where user skills are not improved. (Ali, 2025)

Conclusion

To keep up with the digital world, it is necessary to meet technological requirements. Technology has penetrated all stages of human life: school, social life, etc. The learner should not become a slave to technology, but should develop himself using it. If the advantages of the digital world are used not for ready-made information acquisition, but for development purposes, then progress will be made.

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APPLICATION OF DIGITAL TECHNOLOGIES IN THE EDUCATIONAL PROCESS

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Abstract: In the modern era of globalization, the education system has undergone significant changes, especially as information and communication technologies have been widely integrated into education. In the digital environment, digital platforms, digital schools, virtual classrooms, online lessons, and electronic resources provide students and learners with the opportunity to access education anytime and anywhere. Students and teachers, as well as the classroom, are being transferred to an online platform. In this process, the role of the teacher has changed, becoming more of a guide and advisor, while students develop independent learning skills. In Azerbaijan, through the “Digital School” platform, different access points have been created for students, teachers, parents, and principals, and the management of the educational

process has been digitalized. In Turkey, the EBA (Eğitim Bilişim Ağı) platform offers extensive opportunities, providing thousands of lessons, videos, and interactive resources. In Kazakhstan, the Kundelik.kz platform offers similar functions, creating broad digital opportunities for students, teachers, and parents. Overall, the experience of these countries shows that digital platforms increase the accessibility of education, make the learning process more flexible and modern, and also adapt teacher-student relationships to a new environment.

Keywords: Digital platform, Education system, Digital school, EBA, Information and communication, Globalization.

TƏDRİS PROSESİNDƏ RƏQƏMSAL TEXNOLOGİYADAN İSTİFADƏ

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Xülasə: Müasir qloballaşma dövründə təhsil sistemi ciddi dəyişikliklərə məruz qalmış, xüsusilə informasiya-kommunikasiya texnologiyaları təhsilə geniş şəkildə inteqrasiya olunmuşdur. Rəqəmsal mühitdə rəqəmsal platforma, rəqəmsal məktəb, virtual siniflər, onlayn dərslər və elektron resurslar şagird və tələbələrə istənilən zaman və məkanda təhsil alma imkanı yaradır. Şagird və müəllim, eləcə də sinif otağı onlayn platformaya daşınır. Bu prosesdə müəllimin rolu dəyişərək daha çox istiqamətverici və məsləhətçi xarakter alır, şagirdlər isə müstəqil öyrənmə bacarıqlarını inkişaf etdirirlər. Azərbaycanda “Rəqəmsal məktəb” platforması vasitəsilə şagird, müəllim, valideyn və direktor üçün fərqli girişlər yaradılmış, təhsil prosesinin idarə olunması rəqəmsallaşdırılmışdır. Türkiyədə EBA (Eğitim Bilişim Ağı) platforması geniş imkanlara malik olub, minlərlə dərslər, video və interaktiv resurslar təqdim edir. Qazaxıstanda isə Kundelik.kz platforması oxşar funksiyalarla şagird, müəllim və valideynlər üçün geniş rəqəmsal imkanlar yaradır. Ümumilikdə, bu ölkələrin təcrübəsi göstərir ki, rəqəmsal platformalar təhsilin əlçatanlığını artırır, təhsil prosesini daha çevik və müasir formaya gətirir, həmçinin şagird-müəllim münasibətlərini yeni mühitə uyğunlaşdırır.

Açar sözlər: Rəqəmsal platforma, təhsil sistemi, rəqəmsal məktəb, informasiya-kommunikasiyası, qloballaşma.

Introduction

In the modern era of globalization, fundamental changes have taken place in all fields, including the education system. The rapid development of information and communication technologies is also reflected in education. Positive changes are observed at every stage of the educational process. The application of digital technologies in education is implemented in various forms, including e-learning platforms, virtual classrooms, online lessons, e-books, and multimedia resources. Axundov (2026) writes:

“The application of digital technologies in the teaching process plays a significant role in the development of the modern education system and, when based on pedagogical principles, substantially improves the quality of instruction. These technologies enrich the content of education, make lessons more visual and interactive, and increase students’ engagement and motivation. The digital environment makes it possible to take into account students’ individual characteristics, regulate the pace of learning, and develop independent learning skills, thereby enabling learners to acquire knowledge more deeply.” (Axundov, A. (2026))

The modern digital environment provides students and learners with the opportunity to access education anytime and anywhere. Digital technologies transfer teacher–student relations from the physical classroom to online platforms. In this process, the teacher takes on not only the role of knowledge provider but also becomes a guide and advisor. At the same time, students develop independent learning skills and improve their ability to work on problems individually. Axundov (2026) writes: “The effective application of digital technologies is not possible without pedagogical foundations. The integration of technology into education should not be limited to technical knowledge; it must be implemented by considering didactic principles, psychological characteristics, and methodological approaches. In this regard, the role of the teacher is of special importance. The teacher acts both as an organizer of the learning process and as the main subject guiding learning in the digital environment. The teacher’s digital competence, ability to use innovative teaching strategies, and mastery of modern pedagogical approaches are among the key factors determining the effectiveness of teaching” (EBA (Eğitim Bilişim Ağı)).

Since digitalization is a global issue, it has reached the education systems of both Azerbaijan and other countries. Azerbaijan has recently been moving toward digitalization in education by creating online educational platforms. The “Digital School” project has become one of the important innovations in the education system. This concept mainly refers to the management of lessons, administration, and communication processes through online platforms in schools. The Digital School platform has four main access points. The first is for students, where they can review topics covered in class, strengthen their knowledge, ask questions to teachers, and watch video explanations of lessons. Another access point is for parents, who can view their children’s grades, lesson schedules, and write feedback to the school. The third access is for teachers, who manage classes online, maintain student lists, and input grades. Electronic journals are also used. The final access is for principals, who monitor relationships between students, teachers, and parents, collect data, and track school statistics and student achievements (<https://digital.edu.az/>).

Digitalization is also being implemented rapidly and successfully in the education system of Türkiye. The main digital education platform in Türkiye is EBA (Eğitim Bilişim Ağı). It is the largest and most important educational platform. It also provides access for students, teachers, and parents. The Education Informatics Network (EBA), an online social education platform established within the scope of the Fatih Project and hosting more than 150,000 pieces of content, has been made available for use by all our students and teachers.

The EBA project is a revolutionary initiative, and in terms of content richness and diversity, there is no other educational portal of this scale in the world. As a gateway to the future of education, the Education Informatics Network is an online social education platform carried out by the General Directorate of Innovation and Educational Technologies of the Ministry of National Education.

The platform includes 3,251 lessons, 81,626 educational videos, games, and 3,654 books. EBA also offers interactive tasks and AI-based recommendations (<http://eba.gov.tr>). The Ministry of National Education of the Republic of Türkiye states: “All students between 4th and 12th grades must have a personal EBA password to access the system. Students who do not have an EBA password can obtain it from their class teacher or school administration and log in via <http://eba.gov.tr>. After logging in, they can access content appropriate to their grade level.”

Another widely used platform in the Turkic world is Kundelik.kz in Kazakhstan. This platform also provides access for students, teachers, parents, and principals. Students have access to

all digital educational tools. Parents can even monitor where their children are and what they eat at school through this system. For teachers, the platform offers several advantages, including participation in courses and access to various professional groups. More than 3 million students use this platform (<https://portal.kundelik.kz/ru/v2>).

Discussion

The conducted analysis shows that in the modern era of globalization, the integration of information and communication technologies into education has led to significant changes in the education system. Digital platforms, virtual classrooms, and online lessons have increased the accessibility of education by providing students and learners with the opportunity to study anytime and anywhere. This ensures the transition of the educational process to a more flexible and modern form.

At the same time, the transfer of teacher–student relationships from the traditional classroom to online platforms has resulted in changes in the role of the teacher. The teacher is no longer only a provider of knowledge but also acts as a guide and advisor. This approach creates conditions for the development of independent learning skills and the ability of students to work individually on problems.

A comparison of the experiences of Azerbaijan, Türkiye, and Kazakhstan shows that platforms such as “Digital School,” EBA, and Kundelik.kz perform similar functions and contribute to the digitalization of the management of the educational process. Through these platforms, interaction between students, teachers, parents, and school administration is organized in a more systematic and efficient way.

At the same time, the application of digital technologies should not be limited only to technical capabilities but must also be based on pedagogical and methodological foundations. The teacher’s digital competence, ability to use innovative teaching strategies, and mastery of modern approaches are among the key factors determining the effectiveness of education.

In general, although the application of digital platforms yields positive results in education, the parallel development of both technological and pedagogical aspects is essential for ensuring the sustainability and effectiveness of this process.

Conclusion

When examining digital platforms in the education systems of Azerbaijan, Türkiye, and Kazakhstan, many similarities and common points can be observed. Each platform represents a more developed version of the others. The shared feature of these platforms is the adaptation of teacher–student relationships from the classroom to a different, digital environment and their integration into the digital world. For both teachers and students, such digital platforms make education more accessible and flexible.

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EVALUATION OF THE EFFECTIVENESS OF INVESTMENTS IN DIGITAL EDUCATION PROJECTS OF UNIVERSITIES (ROI, NPV, IRR)

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Abstract: This study examines the economic effectiveness of investments in digital education projects in universities. Using key financial indicators such as ROI, NPV, and IRR, the research evaluates the feasibility and efficiency of such investments. The findings indicate that properly planned digital projects contribute both to financial sustainability and to the improvement of educational quality.

Key words: digital education, investment efficiency, ROI, NPV, IRR, universities

Xülasə: Bu tədqiqat universitetlərdə həyata keçirilən rəqəmsal təhsil layihələrinə yönəldilən investisiyaların iqtisadi effektivliyinin qiymətləndirilməsinə həsr olunmuşdur. Araşdırmada ROI, NPV və IRR kimi əsas maliyyə göstəricilərindən istifadə edilərək investisiya qərarlarının əsaslandırılması təhlil edilmişdir. Nəticələr göstərir ki, düzgün planlaşdırılmış rəqəmsal layihələr **həm maliyyə səmərəliliyi** yaradır, həm də təhsil keyfiyyətinin yüksəldilməsinə töhfə verir.

Açar sözlər: digital education, investment efficiency, ROI, NPV, IRR, universities

Introduction

In the contemporary era, digital transformation has emerged as a key strategic direction for enhancing the competitiveness of higher education institutions, improving the quality of the educational process, and optimizing management mechanisms. In particular, the implementation of digital education projects in universities not only increases the accessibility of educational services but also ensures managerial flexibility, efficient use of resources, and long-term financial sustainability. Electronic learning platforms, Learning Management Systems (LMS), online examination systems, artificial intelligence-based educational solutions, and smart campus technologies are considered the core components of this transformation.

Investments made by universities in this domain should be evaluated not only as technological upgrades but also in terms of economic efficiency. Each digital project requires specific financial expenditures, and the accurate assessment of the return on these investments directly influences the quality of managerial decision-making. In this regard, financial indicators such as Return on Investment (ROI), Net Present Value (NPV), and Internal Rate of Return (IRR) play a crucial role in evaluating investment effectiveness.

ROI enables the measurement of investment profitability, NPV allows for the calculation of the project's discounted net value, and IRR determines the internal rate of return of the investment. Through these indicators, university management can assess the economic feasibility of digital projects and make more rational investment decisions.

The aim of this study is to evaluate the effectiveness of investments in digital education projects undertaken by universities based on ROI, NPV, and IRR indicators, and to identify optimal decision-making mechanisms in this field.

The objectives of the study include examining the economic nature of digital education projects, analyzing methods for assessing investment efficiency, conducting practical calculations, and developing recommendations based on the obtained results.

The object of the research is digital education projects implemented in higher education institutions, while the subject of the research is the mechanisms for evaluating the economic efficiency of investments directed toward these projects.

Method and Materials

Compared to the traditional education model, digital projects contribute to cost reduction in the long term. Minimizing the use of paper-based materials, automating administrative processes, optimizing classroom resource utilization, and expanding opportunities for distance learning generate substantial savings in university budgets. For instance, electronic document management systems and online registration platforms reduce the workload on human resources while simultaneously lowering operational costs. (European Commission (2021)

Another important aspect of the economic significance of digital education projects is the expansion of revenue opportunities. Universities can generate additional financial resources through online courses, international distance learning programs, and certification initiatives. This helps reduce the dependence of higher education institutions on state funding. In particular, attracting international students through distance education formats enhances the financial sustainability of universities.(Phillips, J. J. (2022).)

Digital projects also contribute to improving the quality of education and increasing student satisfaction. The rapid accessibility of educational materials, opportunities for personalized learning, flexible assessment systems, and the generation of analytical reports create favorable conditions for better learning outcomes. Improved educational quality, in turn, positively affects a university's ranking and its competitiveness in the market.

Findings

Digital transformation also forms the foundation for strategic investment decisions. University management must consider not only the short-term but also the long-term economic outcomes of investments. For this reason, the effectiveness of digital education projects should be evaluated based on financial indicators.

Table 1: Key digital education projects in universities and their economic advantages

Type of Digital	Application area	Economic advantages
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Project		
Learning Management Systems (LMS)	Learning Management	Reduces time loss, lowers administrative costs
Online Examination Systems	Assessment Process	Savings on paper and organizational costs
Digital Libraries	Learning Resources	Reduced printing costs, quick access to resources
Smart campus həlləri	Campus Management	Optimization of energy and administrative costs
Distance Education Platforms	Distance Learning	New student flow and additional income opportunities

As shown in Table 1, digital education projects are not only a means of technological modernization but also an important instrument for enhancing economic efficiency. Their implementation enables universities to optimize costs, increase revenues, and improve overall management quality. Therefore, the systematic evaluation of investments directed toward these projects is of particular importance.

The successful implementation of digital education projects in universities depends not only on technological capabilities but also on the economic efficiency of the investments allocated to these initiatives. Each digital project requires a certain amount of initial capital investment, ongoing operational costs, and generates long-term financial outcomes. For this reason, conducting financial analysis is considered an essential requirement in investment decision-making processes. Among the most widely used indicators for evaluating investment efficiency are Return on Investment (ROI), Net Present Value (NPV), and Internal Rate of Return (IRR).

ROI is a profitability indicator of an investment that measures the extent to which the invested capital generates profit. It is calculated as the ratio of net profit obtained from the investment to the total investment amount. The higher the ROI, the greater the economic efficiency of the project is considered to be. In universities, this indicator is determined based on cost savings, additional revenues, and reductions in administrative expenses resulting from the implementation of digital platforms..

$$ROI = \frac{\text{Net income}}{\text{Investment amount}} \times 100\%$$

For example, if a university invests a certain amount in a new LMS (Learning Management System) and subsequently a reduction in annual operational costs is observed, the resulting savings serve as a key indicator in the calculation of ROI.

NPV (Net Present Value) is a financial indicator that determines the present value of future cash inflows generated by a project. This method is widely used in investment decision-making due to its higher level of accuracy in evaluating long-term financial outcomes. If the NPV is positive, the project is considered economically feasible. Conversely, a negative NPV indicates that the investment is inefficient. In universities, NPV is of particular importance for assessing the long-term financial implications of digital education projects.

$$NPV = \sum \frac{CF_t}{(1+r)^t} - I_0$$

Here, CF_t represents cash flows, r denotes the discount rate, t is the time period, and I_0 is the initial investment amount.

IRR (Internal Rate of Return) is a profitability indicator that expresses the return of a project in percentage terms. It represents the discount rate at which the Net Present Value (NPV) becomes equal to zero. IRR allows for determining how profitable an investment is. If the IRR is higher than the cost of capital, the project can be considered acceptable. For universities, this indicator plays an important role in the comparative evaluation and selection of different digital projects.

NPV=0

ROI is used for the rapid evaluation of an investment, NPV is applied to determine the actual financial value of a project, while IRR is used for the comparative analysis of alternative projects. The integrated application of these indicators enables university management to make more substantiated and strategically sound investment decisions.

Table 2: Comparison of ROI, NPV and IRR indicators

Indicator	Essence	Advantages	Limitations
ROI	Rate of return on investment	Simple to calculate, quick results	Does not take into account the time factor
NPV	Present value of future cash flows	Provides more accurate financial results	Difficult to choose the right discount rate
IRR	Internal rate of return	Effective in comparing projects	Complicated to calculate

As can be seen from the table, each indicator has its own advantages and limitations. Therefore, in evaluating the investment efficiency of digital education projects in universities, it is more appropriate to apply a comprehensive approach rather than relying on a single indicator.

To gain a clearer understanding of the investment effectiveness of digital education projects, it is essential to evaluate them based on a practical example. For this purpose, a hypothetical investment project involving the implementation of a Learning Management System (LMS) platform at a university is considered. The main objective of the project is to digitalize the teaching process, reduce administrative costs, and improve the quality of educational services.

The initial investment cost of the project is assumed to be 120,000 AZN. After the implementation of the system, the reduction in expenses related to printed materials, classroom management, and administrative processes results in an annual net cash inflow of 35,000 AZN. To assess the economic efficiency of the project, ROI, NPV, and IRR indicators are applied.

First, the ROI indicator is calculated. This indicator reflects the overall profitability of the project and demonstrates the return on investment. Using a simplified approach based on annual net income, it can be observed that the project justifies itself over a certain period of time.

$$ROI = \frac{35000}{120000} \times 100\%$$

According to the results, the ROI of the project is approximately 29.17%, which indicates that the investment is efficient in the medium term.

Next, the NPV indicator is calculated. Assuming a discount rate of 10%, the present value of cash flows over a 5-year period is assessed. If the result is positive, the project is considered economically feasible.

$$NPV = \sum \frac{35000}{(1+0.10)^t} - 12000$$

The conducted assessment shows that the NPV is positive, which indicates that the project generates additional value for the university.

The IRR indicator determines the internal rate of return of the project. According to the calculations, the IRR ranges approximately between 18–20%. Since this value is higher than the cost of capital, the project is considered satisfactory from an investment perspective.

The practical evaluation demonstrates that digital education projects are not only technological innovations but also economically efficient investment instruments. The combined analysis of ROI, NPV, and IRR allows for a more accurate determination of the real financial effectiveness of such projects.

Table3: Project financial indicators and final results

Indicator	Value	Evaluation
Initial investment	120 000 AZN	Project start-up
Annual net cash flow	35 000 AZN	Savings and additional income
ROI	29.17%	Medium-high efficiency
NPV	Positive	Project is acceptable
IRR	18–20%	Above cost of capital

The results of the practical analysis demonstrate that, when properly planned, investments in digital education projects in universities generate financial efficiency while also significantly improving the quality of the educational process.

Conclusion

The findings of the study indicate that investments directed toward digital education projects in universities function not only as a means of technological modernization but also as a strategic tool for enhancing economic efficiency. The implementation of digital education platforms enables the optimization of teaching processes, the reduction of administrative costs, and the strengthening of the overall financial sustainability of universities.

Within the framework of the study, the application of ROI, NPV, and IRR indicators demonstrated that these methods are reliable and practical tools for evaluating the economic efficiency of investment projects. ROI allows for the operational assessment of a project's overall profitability, NPV determines its long-term financial value, while IRR enables comparative analysis among alternative investment opportunities. The integrated application of these indicators facilitates more accurate and well-grounded investment decision-making.

The results of the practical evaluation showed that digital education projects, particularly LMS systems and other electronic learning solutions, produce positive outcomes for universities in

terms of both cost optimization and the generation of additional revenue sources. Positive NPV values and high ROI indicators confirm the economic efficiency of these projects.

The findings of the study reveal that digital transformation processes in higher education institutions should be assessed not only from a technical perspective but also systematically in financial and managerial terms. For this purpose, the application of indicators such as ROI, NPV, and IRR in investment decision-making is considered essential.

In conclusion, it can be stated that the effective management of investments in digital education projects in universities is of significant importance for improving the quality of education, optimizing financial resources, and enhancing competitiveness.

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THE IMPACT OF SOCIAL MEDIA ON LANGUAGE LEARNING

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Abstract: In the twenty-first century, social media has become one of the most influential parts of everyday life. Platforms such as YouTube, Instagram, TikTok, Facebook, and Telegram are no longer used only for entertainment and communication, but also for educational purposes. One of the areas most affected by this transformation is language learning. Social media provides learners with continuous access to authentic materials, native speakers, interactive tasks, and motivational content. As a result, students can improve vocabulary, pronunciation, listening comprehension, speaking fluency, and cultural awareness in a more dynamic environment than in traditional classrooms.

This article examines the impact of social media on modern language learning, especially in the process of learning English as a foreign language. It focuses on both advantages and disadvantages of digital platforms. The study also includes personal experience to demonstrate how social media can support independent learning. Through regular use of YouTube shadowing activities, TikTok reading exercises, Instagram educational pages, and international webinars, significant improvement in language skills can be achieved.

At the same time, certain risks such as distraction, misinformation, reduced concentration, and informal language habits are discussed. The article concludes that social media can become a powerful educational tool when used consciously, purposefully, and in balance with traditional learning methods.

Keywords: social media, language learning, English, communication skills, motivation, digital learning, technology

SOSIAL MEDIANIN DİL ÖYRƏNMƏSİNƏ TƏSİRİ

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Xülasə: XXI əsrdə sosial media gündəlik həyatın ən təsirli hissələrindən birinə çevrilmişdir. YouTube, Instagram, TikTok, Facebook və Telegram kimi platformalar artıq yalnız əyləncə və ünsiyyət üçün deyil, həm də təhsil məqsədilə istifadə olunur. Bu dəyişiklikdən ən çox təsirlənən sahələrdən biri dil öyrənməsidir.

Sosial media öyrənənlərə orijinal materiallara, ana dili daşıyıcılarına, interaktiv tapşırıqlara və motivasiyaedici məzmunu daimi çıxış imkanı yaradır. Nəticədə tələbələr söz ehtiyatını, tələffüzünü, dinləmə bacarıqlarını, danışq səlisliyini və mədəni biliklərini daha dinamik mühitdə inkişaf etdirə bilirlər.

Bu məqalədə sosial medianın müasir dil öyrənməsinə, xüsusilə ingilis dilinin xarici dil kimi mənimsənilməsinə təsiri araşdırılır. Məqalədə həm üstünlüklər, həm də mənfi cəhətlər təhlil olunur. Eyni zamanda şəxsi təcrübəyə əsaslanaraq sosial medianın müstəqil öyrənməyə necə dəstək verdiyi göstərilir. YouTube üzərindən shadowing məşqləri, TikTok vasitəsilə oxu tapşırıqları, Instagram təhsil səhifələri və beynəlxalq vebinarlar vasitəsilə dil bacarıqlarında əhəmiyyətli inkişaf əldə etmək mümkündür. Bununla yanaşı, diqqətin yayınması, yanlış məlumatlar, qeyri-rəsmi dil vərdisləri kimi risklər də mövcuddur.

Məqalənin nəticəsinə görə, sosial media məqsədli və düzgün istifadə olunduqda güclü tədris vasitəsinə çevrilə bilər.

Açar sözlər: sosial media, dil öyrənmə, ingilis dili, ünsiyyət bacarıqları, motivasiya, rəqəmsal təhsil, texnologiya

Introduction

In the twenty-first century, technological progress has significantly transformed the field of education. Among the most influential innovations, social media has become one of the most commonly used digital tools worldwide. Platforms such as YouTube, TikTok, Instagram, Facebook, Telegram, and X are no longer limited to entertainment and communication; they are increasingly used for academic and educational purposes. In particular, language learners actively use these platforms to improve vocabulary, pronunciation, listening comprehension, speaking confidence, and intercultural communication skills.

Traditional language learning methods often rely on textbooks, classroom instruction, grammar exercises, and teacher-centered activities. Although these methods remain valuable, they may not always provide authentic language exposure or sufficient real-life communication opportunities. Social media, in contrast, offers learners access to native speakers, real conversations, visual content, and interactive learning communities. This creates a dynamic learning environment where students can practice language skills beyond the classroom setting (Genetti, 2018).

Another important aspect of social media is accessibility. Learners can study anytime and anywhere through mobile devices. Short educational videos, live streams, podcasts, language challenges, and discussion forums allow users to engage with language in enjoyable ways. As a result, learning becomes more flexible, learner-centered, and motivating.

However, despite its advantages, social media also presents several challenges. These include distraction, reduced attention span, overuse of informal expressions, misinformation, and dependence on algorithm-driven content. Therefore, it is necessary to examine both the positive and negative effects of social media on language learning.

The purpose of this article is to analyze the impact of social media on modern language learning, especially English as a foreign language, through academic sources and personal experience.

Research

Recent academic studies have shown that digital technologies have reshaped language education. Researchers argue that online communication tools increase student engagement and autonomy. According to Warschauer, technology enables learners to participate more actively and independently in the learning process (Warschauer, 2010).

Kabilan, Ahmad, and Abidin found that social networking platforms, especially Facebook, can support English language learning by improving writing skills, vocabulary knowledge, and confidence in communication (Kabilan et al., 2010). Their study demonstrated that students felt more comfortable expressing ideas online than in traditional classroom environments.

Crystal explains that internet communication has created new linguistic practices where users combine written and spoken language features. This has influenced vocabulary development, abbreviations, and informal expressions among young learners (Crystal, 2011). While some educators criticize this change, others believe it reflects natural language evolution.

Boyd emphasizes that social media spaces are highly important for young people because

they create environments for identity formation, communication, and collaborative learning (Boyd, 2014).. In language education, these elements are especially useful because successful learning often depends on interaction and motivation.

Modern scholars also note that multimedia content plays an important role in comprehension. Audio-visual materials help students understand pronunciation, context, body language, and real-life speech patterns more effectively than printed texts alone (Mayer, 2014).

Therefore, literature generally supports the view that social media can become a powerful educational instrument when used purposefully and critically.

Materials and Methods

This article uses a qualitative descriptive research method based on academic literature review and personal learning experience. Secondary data were collected from books, journal articles, and scholarly studies related to digital learning, language acquisition, and social media use.

In addition, an autoethnographic perspective was included in order to reflect real learner experience. Personal observations were based on regular use of YouTube, TikTok, Instagram, and online learning platforms for English improvement over several years. This combination of theoretical and practical evidence provides a broader understanding of the topic.

Vocabulary Development. One of the clearest benefits of social media in language learning is vocabulary growth. Learners are continuously exposed to new words, idioms, slang expressions, and topic-based terminology. Educational creators often explain difficult words in short and memorable formats. Through repeated exposure, users naturally remember vocabulary items.

For example, TikTok and Instagram Reels frequently present daily expressions, phrasal verbs, and pronunciation tips in under one minute. This short-form learning suits modern learners who prefer fast and practical content. Repetition through algorithms also increases retention because similar content appears regularly (Nation, 2013).

Listening and pronunciation skills. YouTube podcasts, interviews, educational channels, and movie clips help learners develop listening comprehension. Unlike textbook recordings, these materials present authentic accents, natural speed, emotional tone, and spontaneous speech. This prepares learners for real communication situations.

Pronunciation can also improve through shadowing techniques, where learners repeat speech immediately after native speakers. Researchers state that imitation-based speaking practice strengthens fluency and pronunciation accuracy (Gilakjani, 2016).

Communication confidences. Many learners fear making mistakes in face-to-face conversations. Social media reduces this anxiety because communication often begins through comments, messages, or online communities. Students can think before responding and gradually build confidence.

Online language exchange groups on Telegram, Discord, Facebook, and Reddit allow learners to interact with people from different countries. Such environments increase both linguistic competence and intercultural awareness (Kramsch, 2013).

Motivation and engagement. Motivation is a major factor in successful language acquisition. Social media makes learning entertaining through gamified challenges, progress videos, quizzes, memes, and relatable creators. Compared with traditional memorization methods, these formats feel less stressful and more enjoyable.

When students enjoy the learning process, they tend to continue consistently. Regular exposure is one of the key conditions for language improvement. Therefore, motivation generated

by social media can lead to long-term success (Dörnyei, 2001).

Personal Experience. My own experience strongly supports these findings. I have used YouTube regularly for shadowing practice with English learning videos. Repeating after speakers improved both my pronunciation and listening speed.

On TikTok, I frequently encountered reading exercises in both slow and fast formats. Because of content recommendations, similar educational videos appeared repeatedly, which helped me practice every day without pressure.

Instagram also played a valuable role in my academic development. By following educational pages, I discovered webinars, international training sessions, and language communities. Through these opportunities, I improved my knowledge and received certificates.

One of the most memorable experiences was participating in a 24-hour non-stop webinar organized by Oxford University Press ELT. It introduced modern teaching methodologies and connected participants from different countries. This experience increased my professional motivation and broadened my educational perspective.

I also completed online courses through Coursera, which helped me improve academic skills and independent learning habits. These examples demonstrate that social media can open doors not only for language practice but also for global educational development.

Challenges and Negative Effects: Despite its advantages, social media also has limitations. First, distraction is a serious issue. Learners may begin with educational intentions but quickly move toward entertainment content. This reduces concentration and time efficiency. Second, many users are exposed to incorrect grammar, slang overuse, and misleading explanations from unqualified creators. Without critical thinking, learners may internalize mistakes.

Third, short-form content sometimes promotes superficial learning. Students may consume motivational videos without engaging in deeper reading, writing, or structured grammar study. Therefore, social media should complement formal education rather than replace it entirely (Carr, 2010).

Conclusion

In conclusion, social media has become an influential component of modern language learning. It offers valuable opportunities for vocabulary development, listening practice, pronunciation improvement, communication confidence, motivation, and intercultural interaction. Its flexibility and accessibility make learning possible anytime and anywhere.

At the same time, learners must remain aware of distractions, misinformation, and informal language habits. Effective use requires balance, discipline, and critical evaluation of content sources.

Personal experience confirms that social media, when used purposefully, can significantly support language growth and academic progress. Therefore, educators and students should not ignore these platforms, but instead integrate them intelligently into the learning process.

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THE IMPACT OF ARTIFICIAL INTELLIGENCE AND DIGITAL PLATFORMS ON MODERN LANGUAGE EDUCATION

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Abstract: This article examines the role of artificial intelligence (AI) and digital platforms in English language teaching. It analyzes how AI-powered tools and online learning environments enhance language acquisition, personalize instruction, and improve learner engagement. The study highlights the pedagogical benefits of adaptive learning systems, automated feedback, and interactive digital resources in developing communicative competence. The findings suggest that the integration of AI and digital technologies significantly increases the effectiveness and flexibility of modern English language education. Furthermore, digital platforms promote interactive and flexible learning environments, increasing student motivation and participation. However, the study also identifies several challenges, such as limited teacher training, technical issues, and concerns regarding data privacy. Despite these limitations, the results suggest that the effective integration of AI and digital platforms can significantly improve teaching and learning processes. The study concludes that a balanced approach, combining traditional teaching methods with modern technological tools, is essential for maximizing educational outcomes and preparing students for the demands of the digital age.

Keywords: Artificial Intelligence (AI), Digital Platforms, E-learning, Personalized Learning, Educational Technology, Student Engagement, Teaching Methods

SÜNİ İNTELLEKTİN VƏ RƏQƏMSAL PLATFORMANIN MÜASİR DİL ÖYRƏNMƏSİNƏ TƏSİRİ

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Xülasə: Bu məqalə süni intellektin (AI) və rəqəmsal platformaların ingilis dili tədrisində rolunu araşdırır. Məqalədə AI ilə işləyən alətlərin və onlayn öyrənmə mühitlərinin dil öyrənməsini necə inkişaf etdirdiyi, tədrisi fərdiləşdirdiyi və öyrənənlərin fəallığını artırdığı təhlil olunur. Tədqiqat adaptiv öyrənmə sistemlərinin, avtomatlaşdırılmış geribildirimin və interaktiv rəqəmsal resursların kommunikativ bacarıqların inkişafındakı pedaqoji üstünlüklərini vurğulayır. Nəticələr göstərir ki, AI və rəqəmsal texnologiyaların integrasiyası müasir ingilis dili təhsilinin effektivliyini və çevikliyini əhəmiyyətli dərəcədə artırır.

Nəticələr göstərir ki, AI əsaslı sistemlər və onlayn platformalar məzmunu fərdi tələbə ehtiyaclarına uyğunlaşdıraraq və dərhal geri bildirim təqdim edərək fərdiləşdirilmiş öyrənməyə töhfə verir. Bundan əlavə, rəqəmsal platformalar interaktiv və çevik öyrənmə mühitləri yaradaraq tələbələrin motivasiyasını və iştirakını artırır. Bununla belə, tədqiqat bəzi çətinlikləri də müəyyən etmişdir, məsələn, müəllimlərin kifayət qədər hazırlıqlı olmaması, texniki problemlər və məlumatların məxfiliyi ilə bağlı narahatlıqlar.

Bu məhdudiyyətlərə baxmayaraq, nəticələr göstərir ki, AI və rəqəmsal platformaların effektiv integrasiyası tədris və öyrənmə proseslərini əhəmiyyətli dərəcədə yaxşılaşdırma bilər. Tədqiqat belə qənaətə gəlir ki, ənənəvi tədris metodlarının müasir texnoloji alətlərlə balanslı şəkildə birləşdirilməsi təhsil nəticələrini maksimum səviyyəyə çatdırmaq və tələbələri rəqəmsal dövrün tələblərinə hazırlamaq üçün vacibdir.

Açar sözlər: Süni intellekt (AI), Rəqəmsal platformalar, Elektron təhsil (E-learning), Fərdiləşdirilmiş öyrənmə, Təhsil texnologiyaları, Tələbə cəlb olunması, Tədris metodları

1. Introduction

In the era of rapid digital transformation, education systems worldwide are undergoing significant changes driven by technological innovation. Among these innovations, Artificial Intelligence (AI) and digital platforms have emerged as powerful tools that are reshaping the process of English language teaching (ELT). As English continues to serve as a global language of communication, science, business, and technology, the need for more effective, flexible, and personalized teaching approaches has become increasingly important. Traditional language teaching methods, often centered on textbooks and teacher-led instruction, are no longer sufficient to meet the diverse needs of modern learners. Today's students are digital natives who are accustomed to interactive technologies and instant access to information. In this context, AI-powered applications, adaptive learning systems, virtual classrooms, and online collaboration platforms provide new opportunities to enhance language acquisition and learner engagement. Artificial intelligence contributes to English language teaching by offering automated feedback, speech recognition tools, intelligent tutoring systems, and personalized learning pathways. These technologies enable learners to practice pronunciation, grammar, vocabulary, and listening skills independently while receiving immediate and data-driven feedback. Digital platforms further support communication and collaboration through multimedia content, interactive tasks, and real-time communication tools. Moreover, the integration of AI and digital platforms aligns with contemporary pedagogical principles such as learner-centered instruction, differentiated learning, and competency-based education. By combining technological innovation with sound pedagogical strategies, teachers can create

dynamic and inclusive learning environments that promote communicative competence and learner autonomy. Therefore, exploring the role of artificial intelligence and digital platforms in English language teaching is both timely and necessary. This study aims to examine their pedagogical significance, practical applications, and impact on the effectiveness of modern English language education. The relevance of this study is determined by the rapid integration of digital technologies into the educational process and the growing importance of English as a global language. In the 21st century, English proficiency has become a key requirement for academic success, professional development, and international communication. Consequently, educational institutions are expected to adopt innovative teaching approaches that enhance language acquisition and meet the demands of modern learners. Despite the widespread use of digital tools, there remains a need for systematic research on the pedagogical foundations and practical effectiveness of artificial intelligence (AI) and digital platforms in English language teaching. Many educational institutions implement technological tools without sufficient methodological justification, which may limit their educational impact. Therefore, examining how AI-based systems and digital environments contribute to communicative competence, learner engagement, and personalized instruction is highly significant. Furthermore, the increasing digitalization of education – accelerated by global technological advancement – has transformed traditional classroom practices. Online platforms, adaptive learning systems, and AI-powered applications now provide opportunities for individualized feedback, real-time assessment, and interactive communication. However, understanding their pedagogical value and limitations is essential to ensure meaningful integration rather than superficial technological use. The study is also relevant in the context of competency-based education, which emphasizes critical thinking, collaboration, creativity, and communication. AI and digital platforms have the potential to support these competencies by offering interactive tasks, collaborative projects, and data-driven learning analytics. Investigating this topic contributes to improving teaching quality and aligning English language instruction with contemporary educational standards. Thus, the examination of artificial intelligence and digital platforms in English language teaching is both theoretically important and practically necessary for enhancing the effectiveness, accessibility, and flexibility of modern language education.

2. Materials and Methods

2.1 Research Design

This study used a mixed-method research design that combined both quantitative and qualitative approaches in order to gain a broader understanding of how artificial intelligence (AI) and digital platforms influence teaching and learning. The reason for choosing this design was to benefit from the strengths of both methods. Quantitative data made it possible to measure learning outcomes and identify general trends, while qualitative data helped explore the experiences and opinions of participants in more detail.

The quantitative part of the study focused on students' academic performance, participation levels, and attitudes toward the use of AI and digital learning tools. Information was collected through structured questionnaires and performance-based assessments. The collected data was later analyzed statistically to identify patterns and changes in student achievement.

The qualitative part aimed to understand how teachers and students experienced the integration of digital technologies in the classroom. Classroom observations and open-ended questionnaire responses were used to gather this information. These methods allowed participants to

express their thoughts more freely and provided deeper insights into classroom interaction, motivation, and teaching practices.

The study was mainly descriptive because it examined existing teaching practices without manipulating any variables. Instead of testing a specific experiment, the research focused on observing how AI tools and digital platforms are currently being used in educational settings and how they affect the learning process.

Using both quantitative and qualitative methods increased the reliability and validity of the research findings. Combining numerical results with personal experiences provided a more complete and realistic picture of the role of AI in modern education.

2.2 Participants

The participants in this study included 50 university students studying English as a Foreign Language (EFL) at an intermediate level and 10 English language teachers.

The students were selected from different academic groups to ensure variation in learning experiences and familiarity with technology. Their ages ranged between 18 and 25 years. The participating teachers had different levels of experience using digital tools in education, ranging from 2 to 15 years.

Convenience sampling was used in the study because participants were selected based on accessibility and willingness to participate.

2.3 Research Instruments

Several research instruments were used to collect accurate and reliable data.

Questionnaire

A structured questionnaire was prepared for students. It included closed-ended and Likert-scale questions designed to evaluate:

- Students' attitudes toward AI tools
- Frequency of using digital platforms
- Perceived effectiveness of online learning

Some example statements included:

- "AI-based tools help me understand lessons more clearly."
- "Digital platforms increase my motivation to learn English."

Interviews

Semi-structured interviews were conducted with teachers to explore:

- Their experiences using AI and digital technologies
- The advantages and challenges of technology integration
- Their observations about student participation and performance

Classroom Observation

Several technology-supported lessons were observed during the study. Digital learning platforms, video-based materials, and AI-supported tools were used in these classes. The observations focused on:

- Student participation
- Classroom interaction
- Engagement during activities
- Communication between teachers and students

Academic Performance Analysis

Students' academic performance before and after the use of digital tools was compared to evaluate the impact of AI-supported learning. Particular attention was given to improvements in:

- Reading comprehension
- Vocabulary development
- Grammar accuracy
- Writing skills

2.4 Data Collection Procedure

The data collection process was completed in several stages.

1. During the preparation stage, questionnaires and interview questions were designed and reviewed.
2. In the implementation stage, students completed questionnaires and participated in technology-enhanced lessons.
3. During the observation stage, classroom activities were observed and recorded.
4. In the interview stage, teachers participated in individual interviews.
5. Finally, all collected data was organized and analyzed using descriptive methods.

2.5 Data Analysis Methods

The quantitative data collected from questionnaires and test results were analyzed using descriptive statistics and percentage analysis. This helped identify patterns in student responses and changes in academic performance.

The qualitative data gathered from interviews and classroom observations were analyzed through thematic analysis. Common themes such as motivation, engagement, personalized learning, and technical difficulties were identified and interpreted.

2.6 Ethical Considerations

Ethical principles were carefully followed throughout the research process to protect the rights and well-being of all participants.

Before participating in the study, all individuals were informed about the purpose and procedures of the research. Participation was completely voluntary, and participants were free to withdraw from the study at any time without any negative consequences.

Confidentiality and privacy were also maintained. Personal information and responses remained anonymous, and all collected data was stored securely. The information gathered during the study was used only for academic purposes and was not shared with unauthorized individuals.

These ethical practices helped ensure trust, transparency, and integrity throughout the research process.

Summary of the Methods Section

Overall, the use of questionnaires, interviews, classroom observations, and academic performance analysis provided a comprehensive understanding of how AI and digital platforms influence teaching and learning. Combining multiple methods strengthened the reliability of the findings and allowed the researcher to examine the topic from different perspectives.

3. Results and Discussion

3.1 Results

The analysis of questionnaires, classroom observations, and academic performance assessments revealed several important findings about the role of artificial intelligence and digital platforms in education.

First, most students showed a positive attitude toward digital learning tools. Around 78% of participants reported that AI-based applications and online platforms increased their motivation to learn. Students explained that interactive activities such as quizzes, video lessons, and adaptive learning systems made lessons more engaging and enjoyable than traditional classroom methods.

Second, the analysis of academic performance demonstrated noticeable improvement among students who regularly used digital platforms. Approximately 72% of students achieved higher test scores after the introduction of AI-supported learning tools. Improvements were especially visible in vocabulary learning, reading comprehension, and grammar exercises.

Teacher feedback also suggested that AI tools reduced administrative workload. Features such as automated grading systems and digital attendance tracking allowed teachers to spend more time on lesson preparation and student interaction. Teachers additionally noted that digital platforms made it easier to monitor individual student progress through learning analytics.

Classroom observations further showed that students in technology-supported lessons were generally more active, participated more frequently, and demonstrated higher levels of engagement compared to students in more traditional learning environments.

3.2 Discussion

The findings of this study indicate that artificial intelligence and digital platforms have a significant positive impact on the teaching and learning process. One of the main benefits observed was the increase in student engagement and motivation. Interactive digital content and visually supported materials helped maintain students' attention and encouraged active participation in lessons.

Another important advantage was personalized learning. AI-supported systems can identify students' strengths and weaknesses and provide learning materials suited to individual needs. This allows students to learn at their own pace, which is particularly useful in classrooms where learners have different ability levels. Students who need extra support can receive additional practice, while more advanced learners can continue progressing independently.

The study also emphasized the value of immediate feedback. Unlike traditional learning environments, where students may wait longer for corrections, AI-based platforms can provide instant feedback and explanations. This helps students recognize mistakes quickly and improve their understanding more effectively.

Despite these benefits, several challenges were identified. One major issue was insufficient teacher training. Some educators lacked the necessary skills or experience to use advanced digital technologies effectively, which limited the potential advantages of these tools. Another challenge involved unequal access to technology, as some students did not have reliable internet connections or suitable digital devices.

In addition, concerns were raised about data privacy and the possibility of becoming overly dependent on technology. Excessive reliance on AI systems may reduce face-to-face communication and human interaction, both of which remain essential for developing social and communication skills.

Overall, the results suggest that AI and digital platforms can greatly improve teaching effectiveness when used appropriately. However, technology should support traditional teaching methods rather than replace them completely. A balanced combination of human instruction and technological innovation appears to be the most effective approach for achieving successful educational outcomes.

4. Conclusion

This study shows that artificial intelligence and digital platforms have become an important part of modern education. They support the learning process by making lessons more personalized, interactive, and accessible for students. Through digital tools, learners can study at their own pace, receive immediate feedback, and engage more actively in classroom activities.

At the same time, the successful use of these technologies depends on several factors, including teachers' ability to use them effectively and equal access to digital resources for all students. Although technology provides many benefits, it cannot fully replace the guidance, communication, and emotional support that teachers offer in the learning process.

For this reason, combining traditional teaching methods with modern technological tools is considered the most effective approach to achieving better educational outcomes.

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THE ROLE OF ARTIFICIAL INTELLIGENCE AND DIGITAL PLATFORMS IN EDUCATION: A COGNITIVE AND STEAM APPROACH

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Abstract: This article analyzes the transformative impact of artificial intelligence (AI) and digital platforms on modern pedagogical strategies, specifically through the STEAM (Science, Technology, Engineering, Arts, and Mathematics) framework. It explores how AI-driven tools facilitate personalized learning by aligning with students' unique cognitive profiles. A significant portion of the study is dedicated to the application of STEAM methodology in the education of children with autism spectrum disorder (ASD), highlighting how structured digital environments and engineering tasks enhance their cognitive potential. The study concludes that the synergy between AI and STEAM fosters 21st-century skills and provides essential support for social adaptation in inclusive educational settings.

Keywords: STEAM education, artificial intelligence, digital platforms, project-based learning, algorithmic thinking, engineering design, special education, cognitive adaptation, neurodiversity.

TƏDRİSDƏ SÜNİ İNTELLEKT VƏ RƏQƏMSAL PLATFORMALARIN ROLU: KOQNİTİV VƏ STEAM YANAŞMASI

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Xülasə: Bu məqalədə süni intellekt (Sİ) və rəqəmsal platformaların STEAM çərçivəsində müasir pedaqoji strategiyalara transformasiyaedici təsiri təhlil edilir. Sİ əsaslı alətlərin tələbələrin unikal koqnitiv profillərinə uyğunlaşaraq fərdiləşdirilmiş öyrənməni necə asanlaşdırdığı araşdırılır. Tədqiqatın əhəmiyyətli bir hissəsi STEAM metodologiyasının autizm spektr pozuntusu (ASP) olan uşaqların təhsilində tətbiqinə həsr olunmuşdur. Burada strukturlaşdırılmış rəqəmsal mühitlərin və mühəndislik tapşırıqlarının onların koqnitiv potensialını necə artırdığı işıqlandırılır. Tədqiqat nəticəsində vurğulanır ki, Sİ və STEAM arasındakı sinerji XXI əsr bacarıqlarını inkişaf etdirir və inklüziv təhsil şəraitində sosial adaptasiya üçün zəruri dəstəyi təmin edir.

Açar sözlər: STEAM təhsili, süni intellekt, rəqəmsal platformalar, layihə əsaslı öyrənmə, alqoritmik düşüncə, mühəndislik dizaynı, xüsusi təhsil, koqnitiv adaptasiya, neyromüxtəliflik.

Introduction

The global educational landscape is undergoing a paradigm shift, characterized by the transition from traditional rote learning to a multidisciplinary, inquiry-based model. At the center of this revolution lies the integration of Artificial Intelligence (AI) and digital platforms, which have redefined the boundaries of the classroom. For a STEAM educator, these advancements represent an intellectual ecosystem capable of harmonizing the rigors of scientific inquiry with the fluidity of creative expression.

The primary challenge of modern pedagogy is the diversification of learning needs. Traditional systems often fail to account for cognitive neurodiversity, leading to gaps in engagement and achievement. AI-driven digital platforms address this by offering "adaptive learning" environments. These systems simulate human cognitive functions – such as perception, memory, and logical reasoning – at an algorithmic level. By analyzing real-time data on student performance, AI can dynamically adjust curriculum delivery, ensuring that each student operates

within their optimal "Zone of Proximal Development." This personalized approach is the cornerstone of the cognitive STEAM framework.

Materials and Methods

The methodology of this research is rooted in a comprehensive review of pedagogical frameworks and cognitive psychological literature. It specifically utilizes data from the document **"CHARACTERISTICS OF COGNITIVE DEVELOPMENT IN CHILDREN WITH AUTISM SPECTRUM DISORDER_2.docx"** to bridge the gap between STEAM tools and special education requirements.

The materials analyzed include:

- **AI-Driven Adaptive Software:** Platforms that utilize machine learning to track student progress and modify task difficulty.
- **Digital Prototyping Tools:** 3D modeling and simulation software used in the engineering component of STEAM.
- **Robotics Kits:** Physical hardware that serves as an interface between abstract coding and concrete action.
- **Case Studies on Neurodiversity:** Research focusing on the cognitive profiles of children with ASD to identify specific educational triggers and barriers.

The methods involve a synthesis of project-based learning (PBL) with algorithmic thinking. Students are tasked with solving real-world problems through a structured five-step engineering design process: Ask, Imagine, Plan, Create, and Improve. AI is used as a diagnostic tool to monitor the cognitive load throughout these stages, particularly for students with sensory sensitivities.

Discussion

The synergy between AI and STEAM methodology creates a robust platform for developing 21st-century skills. The "Science" and "Mathematics" components provide the analytical foundation, while "Technology" and "Engineering" offer the practical application. However, it is the "Arts" component that often provides the emotional and aesthetic bridge necessary for deep learning.

STEAM and Neurodiversity

A critical area of discussion is the application of these tools in inclusive education. Children with Autism Spectrum Disorder (ASD) often exhibit unique and uneven cognitive development, where high-functioning technical skills may coexist with significant social-emotional challenges.

According to research in the field, several factors make STEAM particularly effective for this demographic:

- **Predictability and Structure:** Children with ASD frequently experience anxiety in unpredictable environments. Digital platforms and coding languages provide a highly structured, logic-driven space where every action has a consistent, predictable reaction.
- **Visual-Spatial Dominance:** Many students on the spectrum are visual thinkers. STEAM's heavy reliance on diagrams, 3D models, and visual programming blocks aligns perfectly with their cognitive strengths.
- **Redirection of Stereotypical Interests:** Often viewed as a barrier, the intense focus or "stereotypical interests" of children with ASD can be converted into a pedagogical asset. A child's fixation on mechanical parts or repetitive patterns can be redirected toward sophisticated engineering projects or complex algorithm design.

The Role of the Teacher as a Facilitator

In an AI-enhanced STEAM classroom, the role of the teacher at ASPU evolves. Instead of being the sole source of information, the teacher becomes a facilitator of discovery. AI platforms provide the teacher with granular analytics, highlighting which students are struggling with specific concepts like spatial reasoning or logical loops. This allows for targeted intervention, ensuring that inclusive education is not just a policy, but a practical reality.

Conclusion

Artificial intelligence and digital platforms are not merely additions to the STEAM curriculum; they are the catalysts that make it accessible to a diverse range of learners. By aligning technological tools with cognitive psychological insights, educators can create environments that support both technical excellence and social-emotional growth.

For the educational community at Azerbaijan State Pedagogical University, the adoption of this cognitive-STEAM approach represents a commitment to future-ready education. It ensures that students, regardless of their cognitive profile, are equipped with the problem-solving skills, technological literacy, and creative confidence required to thrive in a digital-first society. Continued research into AI-human collaboration in the classroom will be vital in refining these strategies and expanding their reach to all areas of special and general education.

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DIGITALIZATION OF ENVIRONMENTAL EDUCATION IN HIGHER EDUCATION INSTITUTIONS

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Abstract. The digital environment has become an integral part of modern social, economic, and educational life, leading to significant transformations in education through the rapid development of information and communication technologies. In this context, the systematic and purposeful application of digital tools in the teaching process has become increasingly essential. Environmental education and training are particularly important due to the global nature of environmental problems, which require effective teaching and the enhancement of public awareness.

The article examines mobile applications, visual information systems, and other digital platforms that enable environmental topics to be presented in a more interactive, visual, and practical way. This approach not only enhances students' theoretical knowledge but also helps them gain a deeper understanding of the subject matter.

In addition, the use of digital tools allows for the implementation of individualized learning approaches in the educational process. Students can learn at different levels and speeds, which increases learning efficiency. Moreover, modeling environmental problems and presenting them in virtual environments help to better understand processes that are difficult to observe in real life.

The author proposes the integration of various multimedia services, educational games, and digital learning materials into the training process of students studying Ecology (05.02.26). This approach contributes to making environmental education more engaging and practice-oriented, as well as developing students' analytical thinking and practical skills. As a result, the integration of digital technologies into environmental education improves the quality of teaching and supports the preparation of more competent future specialists.

Keywords: digitalization, environmental education, application, students, waste, recycling

ALİ TƏHSİL MÜƏSSISƏLƏRİNDƏ EKOLOJİ TƏHSİLİN RƏQƏMSALLAŞDIRILMASI

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Xülasə: Rəqəmsal mühit müasir cəmiyyətin sosial, iqtisadi və təhsil fəaliyyətlərinin ayrılmaz tərkib hissəsinə çevrilmişdir. Bu proses informasiya-kommunikasiya texnologiyalarının sürətli inkişafı ilə daha da güclənmiş və müxtəlif sahələrdə, xüsusilə də təhsil sistemində fundamental dəyişikliklərə səbəb olmuşdur. Nəticə etibarilə, tədris prosesində rəqəmsal vasitələrin sistemli və məqsədyönlü şəkildə tətbiqi üçün əlverişli şəraitin yaradılması zərurətə çevrilmişdir. Bu baxımdan, ekoloji təhsil və təlim sahəsi də xüsusi aktualıq kəsb edir, çünki ekoloji problemlərin global xarakter alması onların effektiv öyrədilməsini və ictimai şəurun artırılmasını tələb edir.

Məqalədə mobil tətbiqlər, vizual informasiya sistemləri və digər rəqəmsal platformalar nəzərdən keçirilmişdir. Bu vasitələr ekoloji mövzuların daha interaktiv, vizual və praktik şəkildə təqdim olunmasına imkan yaradır. Belə yanaşma tələbələrin yalnız nəzəri biliklərinin artırılmasına deyil, həm də onların mövzunu daha dərinləndirərək etməsinə şərait yaradır.

Bundan əlavə, rəqəmsal alətlərin istifadəsi tədris prosesində fərdi yanaşmanın tətbiqinə imkan verir. Tələbələr müxtəlif səviyyələrdə və sürətlərdə materialı mənimsəyə bilər, bu da öyrənmənin effektivliyini artırır. Eyni zamanda, ekoloji problemlərin modelləşdirilməsi və virtual mühitdə təqdim olunması real həyatda müşahidə edilməsi çətin olan proseslərin daha aydın anlaşılmasına kömək edir.

Müəllif tərəfindən 05.02.26 "Ekologiya" ixtisası üzrə təhsil alan tələbələrin tədris prosesində müxtəlif multimedia servislərinin, təhsil oyunlarının və rəqəmsal tədris materiallarının tətbiqi məqsədəuyğun hesab edilir. Bu yanaşma ekoloji fənlərin daha maraqlı və praktik əsaslarla öyrədilməsinə, həmçinin tələbələrin

analitik düşünmə və praktik bacarıqlarının inkişafına xidmət edir. Nəticə etibarilə, rəqəmsal texnologiyaların ekoloji təhsilə integrasiyası tədrisin keyfiyyətini yüksəldir və gələcək mütəxəssislərin daha hazırlıqlı formalaşmasına şərait yaradır.

Açar sözlər: rəqəmsallaşma, ekoloji təhsil, tətbiq, tələbələr, tullantılar, təkrar emal

Introduction

Modern society, shaped by technogenic development and mass consumption, is facing an increasingly serious environmental crisis. The traditional perception of nature as an inexhaustible source of resources has led to the gradual degradation of ecosystems, loss of biodiversity, and the continuous accumulation of various types of waste. Rapid urbanization, industrial expansion, and intensive economic activity have significantly increased pressure on natural environments, resulting in overburdened waste management systems, expanding landfill areas, and long-term pollution of soil, water, and air. These processes not only threaten ecological balance but also create risks for human health and sustainable development.

In parallel, the digital transformation of society has fundamentally changed education, communication, and everyday practices. Information and communication technologies are now deeply integrated into almost all spheres of life, including education, where they enable new forms of interaction, accessibility of information, and personalized learning experiences. Digital tools such as mobile applications, online platforms, and interactive systems have created opportunities for more flexible, engaging, and practice-oriented educational models. However, despite these advances, environmental education in higher education institutions remains insufficiently prioritized. In many cases, it is either limited in scope or excluded from compulsory curricula, which reduces students' systematic understanding of environmental issues and weakens the formation of ecological responsibility.

As a result, students often lack practical knowledge regarding environmental protection, waste management, and sustainable behavior. This gap between theoretical awareness and practical application highlights the need for innovative educational strategies that integrate environmental content with modern digital technologies. Therefore, it is essential to explore and implement digital approaches that can strengthen environmental education, increase student motivation, and support active engagement in ecological practices. Such approaches may include interactive learning tools, simulation-based applications, and real-world digital platforms that connect theoretical knowledge with everyday environmental actions.

Materials and methods

The study is based on a qualitative research design aimed at examining the role and effectiveness of digital tools in environmental education, particularly in the context of waste management awareness and student engagement. The research focuses on digital platforms and applications available through widely used distribution systems such as the App Store, Google Play, and other open-access online sources. In total, 26 digital applications and services related to environmental education, recycling, and waste management were selected and analyzed.

The selection process was guided by criteria such as relevance to environmental issues, accessibility for users, educational content, and practical applicability in everyday life. Priority was given to applications that combine theoretical ecological knowledge with interactive or practice-oriented features, including gamification, geolocation services, and reward-based systems.

The research methodology includes several complementary approaches:

Web-based content analysis, which involved systematic examination of mobile applications and digital platforms to identify their functions, structure, and educational potential in the field of environmental awareness and waste management;

Comparative analysis of interactive environmental tools, aimed at identifying similarities and differences between applications in terms of user engagement strategies, functionality, and effectiveness in promoting ecological behavior;

Review of scientific literature and statistical data, focusing on digitalization processes in education, the integration of information technologies into teaching, and current trends in environmental education development;

Examination of existing educational and ecological digital initiatives, including governmental, non-governmental, and private-sector projects that aim to support sustainable development and environmental responsibility.

Each selected application was evaluated using a set of analytical criteria, including educational value, level of interactivity, user motivation mechanisms, and the ability to encourage environmentally responsible behavior in real-life contexts. Special attention was paid to the extent to which digital tools contribute to transforming theoretical ecological knowledge into practical skills, particularly in waste sorting, recycling, and conscious consumption practices.

The methodological approach allows for a comprehensive understanding of how digital technologies can be effectively integrated into environmental education and how they influence students' ecological awareness and behavioral patterns.

Results and discussion

The analysis of the selected 26 digital environmental applications and platforms revealed that contemporary digital tools supporting environmental education can be systematically classified into several functional categories based on their pedagogical purpose, user interaction model, and contribution to ecological awareness and behavior formation.

1. Educational and Awareness Applications

The first category includes applications designed primarily to provide environmental knowledge and increase users' awareness of ecological issues. For instance, the "Ecolabel Guide" application enables users to identify and interpret product labels through interactive scanning technologies. By allowing users to obtain detailed information about packaging symbols, recyclability indicators, and product composition, such tools contribute to the development of informed consumer behavior. In addition, they raise awareness regarding sustainable production, organic products, and environmentally safe consumption patterns. These features are particularly important in shaping ecological literacy among students.

2. Simulation and Game-Based Learning

The second category consists of gamified and simulation-based learning tools, such as "Eco Inc. Save the Earth." These applications create virtual environments where users can experience the consequences of environmental decisions through scenario-based gameplay. By engaging in tasks related to waste management, pollution control, and ecosystem preservation, users develop an understanding of complex environmental processes in an interactive and motivating format. The use of simulation techniques increases learner engagement and facilitates deeper comprehension of ecological challenges that are difficult to observe directly in real life.

3. Interactive Maps and Waste Management Tools

Another significant category includes geolocation-based platforms such as Recyclemap and waste management applications like “Uberu.” These tools provide users with real-time access to recycling infrastructure, including nearby collection points for different waste categories. By integrating geospatial technologies, such applications bridge the gap between digital information and physical environmental action. Furthermore, reward-based mechanisms in applications like “Uberu” encourage users to actively participate in waste sorting and recycling activities, thereby strengthening behavioral motivation toward sustainable practices.

4. Behavioral Incentive Applications

The “TrashBack” program represents a behavioral incentive model that combines environmental action with motivational reward systems. Users are encouraged to collect waste in natural environments and document their actions through photo verification. In return, they receive points that can be used for real-world benefits, such as discounts or purchases. This approach effectively transforms environmental responsibility into a socially and economically rewarding activity, thereby increasing participation rates in ecological initiatives and promoting long-term behavioral change.

5. Educational Impact

The findings indicate that the presence of more than 1,000 recycling and waste collection points in urban areas such as Mingachevir demonstrates the importance of integrating digital tools into formal and informal educational environments. The dissemination of such information among students significantly enhances their understanding of environmental problems and supports the development of practical ecological skills. Mobile applications, in particular, serve as a bridge between theoretical knowledge and real-life environmental actions, such as delivering batteries, plastic bottles, and other recyclable materials to designated collection points.

General Discussion

Overall, the results confirm that digital technologies play a crucial role in strengthening environmental education. They not only increase student engagement and motivation but also contribute to the formation of environmentally responsible behavior patterns. By combining interactive learning, real-world applications, and incentive-based systems, digital tools create a more effective and practice-oriented approach to ecological education. This integration supports the transition from passive knowledge acquisition to active environmental participation, which is essential in addressing contemporary ecological challenges.

Conclusion

Digital transformation in the field of education has significantly expanded the possibilities for improving the quality, accessibility, and effectiveness of environmental education in higher education institutions. The integration of digital technologies such as mobile applications, interactive maps, simulation-based platforms, and gamified learning tools creates a more engaging and student-centered learning environment. These tools not only facilitate the acquisition of theoretical ecological knowledge but also support the development of practical skills related to waste management, recycling, and sustainable consumption. As a result, students become more motivated and actively involved in environmental protection activities, which contributes to the formation of ecological awareness and responsibility.

At the same time, it is important to emphasize that digital technologies alone are not sufficient to ensure comprehensive environmental education. While they play a crucial role in increasing engagement and accessibility, their effectiveness is significantly enhanced when combined with

traditional and experiential learning methods. The most productive educational outcomes can be achieved through an integrated approach that includes project-based learning, interdisciplinary teaching strategies, and real-world practical experiences.

In this context, field activities in specially protected natural areas provide students with direct exposure to ecosystems and biodiversity, helping them to better understand environmental processes in their natural settings. Similarly, visits to industrial environmental facilities allow students to observe real-life environmental challenges and the implementation of ecological standards in production processes. Furthermore, participation in environmental volunteering initiatives strengthens students' sense of social responsibility and encourages active involvement in community-based environmental protection efforts.

Therefore, the integration of digital solutions with experiential and practice-oriented educational methods represents a comprehensive and balanced approach to environmental education. This combination not only enhances students' theoretical knowledge and practical competencies but also contributes to the development of environmentally responsible, socially active, and well-informed future specialists who are capable of addressing contemporary ecological challenges at both local and global levels.

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AFFORDABLE VR FOR LANGUAGE LEARNING: PRACTICAL APPROACHES FOR RESOURCE-CONSCIOUS EDUCATORS

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Abstract: While Virtual Reality (VR) offers immersive, interactive environments that boost learner engagement in language education, widespread adoption remains limited by perceived high costs, technical complexity, and hardware constraints. This paper explores accessible, low-cost alternatives by examining two educator-friendly platforms: ThingLink and We Are Learning. Drawing from workshops, conference presentations, and hands-on experimentation, the study demonstrates how interactive 360-degree environments – such as VR escape rooms – and AI-avatar scenarios support communicative practice without requiring advanced programming or specialized headsets. Findings reveal that low-cost VR implementation is most effective when tools remain simple, accessible, and aligned directly with pedagogical goals rather than technological novelty. To address ongoing needs for instructional guidance, the paper introduces the VR + AI Lesson Quality Evaluator (VAL-QE), a custom GPT designed to assess instructional alignment in immersive learning settings. Finally, it outlines an upcoming professional development training initiative and pilot cohort aimed at helping educators design scalable, affordable VR and AI learning experiences.

Açar sözlər: Virtual Reality (VR), language education, low-cost technology, ThingLink, immersive learning, artificial intelligence (AI).

ДОСТУПНАЯ ВИРТУАЛЬНАЯ РЕАЛЬНОСТЬ В ОБУЧЕНИИ ЯЗЫКАМ: ПРАКТИЧЕСКИЕ ПОДХОДЫ ДЛЯ ПЕДАГОГОВ С ОГРАНИЧЕННЫМИ РЕСУРСАМИ

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Аннотация: Хотя виртуальная реальность (VR) обеспечивает иммерсивную и интерактивную среду, повышающую вовлеченность учащихся в процесс обучения языкам, ее широкое применение остается ограниченным из-за представления о высокой стоимости, технической сложности и аппаратных ограничениях. В данной статье рассматриваются доступные и недорогие альтернативы на примере двух удобных для преподавателей платформ – ThingLink и We Are Learning. Основываясь на материалах семинаров, конференций и практических экспериментов, исследование демонстрирует, как интерактивные 360-градусные среды (например, VR-квесты) и сценарии с ИИ-аватарами поддерживают развитие коммуникативных навыков без необходимости глубокого программирования или дорогих VR-гарнитур. Результаты показывают, что внедрение бюджетных VR-технологий наиболее эффективно тогда, когда инструменты остаются простыми, доступными и напрямую связанными с педагогическими целями, а не с технологическими новшествами. Для удовлетворения потребности в методической поддержке в статье представляется специальная модель GPT – «VR + AI Lesson Quality Evaluator» (VAL-QE), разработанная для оценки качества уроков в учебных средах с поддержкой VR и ИИ. В заключение описываются программа профессионального развития и план пилотного проекта, призванные помочь педагогам создавать доступные учебные материалы с применением VR и ИИ.

Ключевые слова: Виртуальная реальность (VR), обучение языкам, бюджетные технологии, ThingLink, иммерсивное обучение, искусственный интеллект (ИИ).

Introduction

Virtual Reality (VR) has increasingly attracted attention in language education due to its potential to support immersive, interactive, and experiential learning environments that promote learner engagement and communicative practice (Li et al., 2022). However, many educators still perceive VR as financially inaccessible, technically difficult to implement, or dependent on specialized equipment and institutional support. These concerns may limit broader classroom adoption despite growing interest in immersive and experiential learning approaches (Carpenter et al., 2023).

This paper explores two relatively affordable and educator-friendly VR platforms, ThingLink and We Are Learning, and their potential applications in language instruction. Drawing on presentations, workshops, and hands-on experimentation conducted across multiple professional conferences and educational settings, this paper examines how these tools can support practical language learning experiences without requiring advanced programming knowledge or high-cost infrastructure.

Current Challenges

Despite growing interest in VR for language education, several barriers continue to limit its broader classroom adoption. Many educators still associate VR with expensive headsets, specialized software, and significant institutional funding, even when lower-cost alternatives are available. Hardware access also remains uneven, as not all institutions or learners have access to VR-compatible devices, stable internet access, or high-performance computers, with many relying primarily on personal laptops or mobile devices.

In addition, some educators may feel intimidated by the technical aspects of VR implementation or may lack access to training and institutional support. Scalability and accessibility also remain important concerns, particularly when attempting to expand VR activities across courses or programs while ensuring that learners with different technological, physical, and educational needs can participate effectively.

Practical Low-Cost Solutions

Two platforms that may help reduce some of these barriers are ThingLink and We Are Learning. Both tools offer relatively accessible entry points into immersive language learning without requiring advanced programming knowledge or expensive VR infrastructure.

ThingLink allows educators to create interactive virtual environments using 360-degree images, videos, audio, text, and embedded tasks. One particularly engaging application has been the development of VR escape rooms for language learning, where learners complete communicative tasks, solve problems, and navigate scenarios in the target language. Due to its browser-based design and relatively low technical barrier, ThingLink has been well received in workshops and conference presentations focused on practical classroom implementation.

We Are Learning takes a somewhat different approach by combining AI-generated avatars, interactive dialogue, and immersive storytelling experiences. The platform allows educators to create scenario-based learning activities that simulate real-world communication while reducing some of the technical demands traditionally associated with VR content creation. Although still relatively new in the author's experimentation and presentations, the platform shows strong potential for scalable and more interactive language-learning experiences.

Lessons Learned

Initial experimentation and presentations with these tools suggest that immersive activities are most successful when they are practical, easy to navigate, and directly connected to communicative

goals. Interactive tasks such as VR escape rooms generated particularly strong engagement because they combined problem-solving, collaboration, and real-world language use in a low-pressure environment.

At the same time, several challenges became apparent. Teacher readiness remains an important factor, as some educators may feel uncertain about integrating VR tools into their instruction without additional training or institutional support. In addition, overly complex activities or excessive technological features can sometimes distract from language-learning objectives rather than support them. These experiences suggest that low-cost VR implementation is most effective when technology remains simple, accessible, and clearly connected to pedagogical goals.

Conclusion

Affordable and accessible VR implementation for language learning is increasingly possible without expensive equipment or advanced programming knowledge. Tools such as ThingLink and We Are Learning demonstrate that small-scale immersive learning activities can be integrated into language instruction using practical and relatively low-cost approaches. However, based on the author's presentations, workshops, webinars, and interactions with educators, there remains a strong need for additional guidance and pedagogical support in implementing these technologies effectively.

To help address this need, the author developed the VR + AI Lesson Quality Evaluator (VAL-QE), a custom GPT designed to help educators evaluate lesson objectives and instructional alignment in VR and AI-supported learning environments. In addition, the author is currently developing a practical training course focused on helping educators create affordable VR and AI learning experiences within their own instructional settings. The author is currently seeking institutional partnerships and funding opportunities to support the broader development and hosting of this professional development initiative for educators interested in affordable VR and AI implementation. A pilot cohort combining educators, instructional designers, and educational administrators from across the United States, and potentially internationally, is expected to be announced in the near future.

Acknowledgments

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THE EFFECTIVENESS OF VIRTUAL CLASSROOMS IN LANGUAGE EDUCATION

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Abstract: Virtual classrooms – online spaces where teachers and students meet in real time – have become an important part of foreign language education. Their widespread use, particularly during and after the COVID-19 pandemic, has transformed the way languages are taught and learned. Unlike simple video calls, virtual classrooms provide interactive environments through platforms such as Zoom, Microsoft Teams, and Google Meet. Features including breakout rooms, shared whiteboards, and collaborative documents encourage active participation and communication.

Research conducted by both Russian and international scholars demonstrates that virtual classrooms can effectively support the development of speaking, listening, reading, and writing skills. Live interaction with teachers and peers offers learners valuable opportunities to practice authentic communication and receive immediate feedback. These benefits can increase student motivation, engagement, and confidence in using a foreign language.

However, challenges remain. Technical difficulties, unstable internet connections, limited access to digital devices, and screen fatigue may reduce learning effectiveness. Despite these limitations, studies suggest that virtual classrooms can significantly enhance language learning when combined with appropriate teaching strategies. Overall, they represent a flexible and effective educational environment that expands access to language learning and promotes meaningful interaction.

Key words: virtual classrooms, online language learning, foreign language acquisition, synchronous teaching, student engagement

DİL TƏHSİLİNDƏ VİRTUAL SİNİFLƏRİN SEÇİMLİYİ

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Xülasə: Virtual siniflər – müəllim və tələbələrin real vaxt rejimində görüşdüyü onlayn tədris mühitləri – müasir xarici dil təhsilinin ayrılmaz hissəsinə çevrilmişdir. Xüsusilə COVID-19 pandemiyasından sonra onların istifadəsi sürətlə artmış və dil öyrənmə prosesində mühüm dəyişikliklər yaratmışdır. Virtual siniflər sadə videokonfransdan daha geniş imkanlar təqdim edir. Zoom, Microsoft Teams və Google Meet kimi platformalar vasitəsilə təşkil olunan dərslərdə interaktiv lövhələr, qrup müzakirələri üçün xüsusi otaqlar və birgə işləməyə imkan verən rəqəmsal materiallardan istifadə olunur.

Tədqiqatlar göstərir ki, virtual siniflər danışmaq, dinləmə, oxu və yazı bacarıqlarının inkişafına müsbət təsir göstərir. Tələbələr müəllim və həmyaşıdları ilə canlı ünsiyyət quraraq dili praktik şəkildə tətbiq etmək və dərhal rəy almaq imkanı əldə edirlər. Bu isə onların motivasiyasını və dərslə marağını artırır.

Bununla yanaşı, internet bağlantısındakı problemlər, texniki çatışmazlıqlar, rəqəmsal cihazlara çıxış imkanlarının məhdudluğu və ekran qarşısında uzun müddət qalmaq kimi amillər tədrisin effektivliyini azalda bilər. Ümumilikdə, düzgün pedaqoji yanaşmalarla tətbiq edildikdə virtual siniflər xarici dil öyrənilməsinə daha əlçatan, çevik və səmərəli edən mühüm təhsil mühitidir.

Açar sözlər: virtual sinif otaqları; onlayn dil öyrənmə; xarici dillərin mənimsənilməsi; sinxron tədris; tələbə iştirakı

Introduction

The field of English language learning is characterized by constant development thanks to the transformations that are known in other fields closely related to it, and the great importance of languages in human existence in all its social, economic and political dimensions. English is one of today's most important languages in world education. Being among the most spoken languages and the lingua franca in the majority of countries around the world, it enjoys a particularly esteemed status in schools and educational institutions.

Nowadays, the new technologies are considered a gateway to modernizing educational practices in general and the field of EFL learning in particular. It facilitates interactions between the two actors of the teaching-learning process, and provides them with a range of advanced programming capabilities. Technology has become an integral part of the society we live in, in all areas of life. Its importance appeared with the "Covid-19" pandemic, where distance learning has been adopted to ensure the continuity of education.

Learning EFL with online platforms have a great impact on students, since it can reach learners worldwide and supply them with various educational resources. This great potential has shown the importance of technology along with online resources in shaping the future of education; the access to all the needed materials online such as videos, articles, games, exercises and live sessions expose learners to real-life language use, and therefore improving their listening, reading, and comprehension skills. All in all, online platforms provide various resources adapting with learners' special needs and different proficiency levels.

Furthermore, the interactive features available online encourage students' interactions, motivation, engagement and willingness to learn. In addition to the group-activities, self and peer-assessment and virtual tools emerge learning autonomy, especially with immediate feedback on students' performance.

The learning process can be conducted easily by the student in all areas since space and time are flexible. Online learning is education that occurs using the internet. It's usually used as "e-learning" among different terms. Although, online learning is only one form of distance learning. You should talk mention the objectives of the paper before speaking about the questions and hypothesis.

A. Digital technologies:

Digital literacy skills can be defined as the capability to employ e-learning resources and multimedia technologies to raise the learning quality. Digital technologies give students access to authentic materials, interactive and engaging content.

Digital technologies have influenced different features of language learning, such as modern learning management systems, autonomous, self-paced, collaborative learning, in addition to the socio-constructivist target language acquisition. For these features of language learning, the joined interest endures to be the prominent ground of communicative language teaching, which is learning by participating and doing tasks rather than just giving orders. They also have a massive effect on language learning and teaching. The accessibility and availability of resources available with digital technologies gave rise to the modern digital platforms .As a result, the growing interest in making

learning easier has led to a focus on helping learners build and customize their own learning experiences.

B. EFL classrooms:

Teaching English as a Foreign Language on an online platform generally involves conducting virtual lessons for learners who need to improve their English proficiency. Students are able to develop their communication skills in English through Information and Communication Technology.

In order to establish a successful learning environment, students' concentration and interest should be taken into consideration. Students' engagement increases in learning activities using technology. It indicates that technology-supported with strategic plans can trigger students' motivation to take part in classroom's activities.

C. Distance learning:

Distance learning is any type of remote learning in which the student is not physically attending the session in the classroom (Moore & Kearsley, 2011). The learner can be at any place while being engaged in the learning process; it is teaching students online. Over the years, it has become an alternative mode of teaching and learning.

Distance learning provides many advantages being a teaching and learning mode. Bijesh listed some constructive advantages like saving time and money. Most of the time, online classes do not cost as much as in face-to-face learning, study materials are accessible and students just need to have a computer or a tablet in addition to the internet and no longer need to take transportation and be stuck in traffic to get to the classroom on time.

Early in the history of distant education, most of interactions between teachers and students were asynchronous. With internet, synchronous work expectations broaden to incorporate everything, starting from chat rooms to videoconferencing services. Furthermore, the exchange of asynchronous materials has largely shifted to digital platforms and communication channels.

Lately, online learning in educational institutions had a big attention since diverse benefits have been discovered such as low-cost expenses. Various studies have revealed that e-learning is useful to students' gains and performance. Nevertheless, students must be active participants throughout the whole process in order to accomplish the most favorable result.

D. Online learning:

Online learning is not new; in fact, teachers have integrated technology in their classrooms for decades. Online learning go hand in hand with online learning platforms; Online learning tools that use the internet enable virtual classes, allowing teachers and students to communicate in real time (synchronously) or at their own pace (asynchronously). Consequently, online learning platforms has become an essential tool during online learning, and the efficient use of online learning platforms has a major effect on students' engagement in online learning. They also allow students to adapt the language learning material depending on their needs in order to achieve their learning goals.

Online education allows students who do not have the capability to attend in-person classes the opportunity to take online lessons. Online learning allows students to build their own knowledge and perspectives in a small group. The focus of 21st century education is to generate individuals with 'a culture of inquiry' and knowledge generation through online learning activities that encourage team-work, communication, creativity and critical thinking.

Teaching foreign languages remotely relies on online learning platforms. Teachers use interactive technologies to create different learning activities and evolve common language

competences. These online learning platforms offer convenient and ready-made online lessons and assignments. Teachers are also able to design lessons and create exercises on their own conforming to the curriculum demands taking into consideration the language proficiency levels of the students.

Methodology and materials:

Research Questions:

- How do online platforms impact the language proficiency of ESL learners?
- What features of online platforms are most beneficial for learners?
- How do learners perceive their motivation and engagement when using online platforms in learning EFL?
- What are the advantages and challenges do learners face when using online platforms for ESL learning?

B. Research Hypothesis:

ESL learners using online platforms will demonstrate significantly greater improvements in language proficiency.

C. Data collection

This study adopted a qualitative research approach to enable a deeper exploration of the research topic. Qualitative methods are especially valuable for investigating complex questions, as they provide a comprehensive understanding of the context surrounding the phenomenon. This approach also facilitates the collection of rich, detailed data, offering insights into participants' experiences and perspectives. Additionally, qualitative research is adaptable, allowing the study design to evolve in response to the research question. The participants included 9 students from different backgrounds who studied English through online platforms, selected through purposive sampling.

Additionally, interviews have been chosen as data collection' method, since they offer more personalized experiences and perspectives. One-to-one interviews were chosen because participants can express themselves with freedom not as in the focus groups where answers can be biased. Interview questions were open as a way to allow participants answer freely and give detailed responses. Questions were semi-structured in order to encourage the participants to give their point of views rather than leading them to certain answers.

D. Sample

In terms of the sample size, nine participants were selected for interviews; each chosen to offer valuable insights into the study's topic. This sample size was selected to allow a deep dive into each person's experiences while keeping the data manageable for analysis. Participants were chosen based on criteria that aligned with the study's goals, ensuring that each brought a unique perspective. By interviewing nine people, the study could capture common themes as well as subtle differences, adding depth and richness to the findings.

E. Data analysis

Data were collected through semi-structured interviews with participants, about their experiences with online learning using online platforms. A thematic analysis approach was used to systematically code and interpret the interview data through themes identified to generate results and findings (Braun & Clarke, 2006). The interviews were audio-recorded, transcribed line for line. After transcription, each interview was read thoroughly multiple times in order to become familiar with the content. Inductive coding was adopted since the researcher generates codes into themes of the gathered data in order to reflect the important aspects of each participants' answers.

Discussion

The above results from the analysis of data provide answers to the research questions of the study. In the first place, it is clear that participants view online learning to entail both advantages and disadvantages. Online learning allows students to learn at their own pace, review any of the class materials at their discretion, and decide when and where it is most convenient for them to study. They would not have to go on hold with their daily activities in order to attend classes. This flexibility in scheduling enables students to fit their education into their different needs, preferences, and lifestyles along with managing their schedules and keep up with deadlines independently, which guided them in building self-discipline and effective time management skills.

Many find this a bit disturbing, though, as there are no face-to-face interactions with teachers or fellow classmates. Even if the virtual classroom is more convenient for them, connecting with teachers and peers might still be problematic. With the elimination of live group discussions, there will be fewer opportunities to share opinions, debate thoughts, or even just enjoy and have fun with classmates. Traditional classes rely on the direct contact with the teacher and classmates by practicing immediately and asking questions frequently. Still, they appreciated online class flexibility though it didn't have face-to-face interaction in developing fluency.

Online learners employ various techniques to ensure a better understanding and retain the learning. Most of them rely on direct interaction with tutors where they ask them to explain further some points or repeat something that they did not understand. Other students ask for recorded sessions. In this case, they can study the lesson materials at their convenience, which allows them to replay the parts that they did not get, unlike in the traditional classes where learners do not have enough time to ask the questions they want or even go back to the points where they face difficulty to understand. Moreover, learners seek supplementary resources.

Some get additional explanations on YouTube since they can watch and listen to different explanations. Others set their goals for every session and use either language apps, flashcards, or other digital tools to retain what they have learned.

Many students do independent research to further their understanding by referring to dictionaries, language learning applications and videos. Online classes brought in excitement and a new structure of programs, which was not rigid like what was offered in schools. Rescheduling classes when learner's encounter some emergencies was not a choice, as well as the travel back and forth to face-to-face classes consume so much time compared to learning conveniently at home. For this reason, the students can manage their time more productively.

Conclusion

This research study has explored the effectiveness of online Platforms in learning English as a second language. Thematic analysis of interview results reveals that the learners acknowledge the usefulness of the online learning platforms, identifying a number of advantages and disadvantages. The students reported that online learning helps in improving their language skills, particularly in terms of vocabulary and structure. The students indicated that contact sessions through live interaction would help them in developing speaking capability and social networks.

Another problem participants reported was inconsistency and lack of engagement during online sessions. They also claimed that online learning requires tremendous self-discipline and motivation to maintain focus on their goals. Students gave a high value to the flexibility and time-saving aspects that online learning has allowed them to attend classes without wasting time and money on commutes to school.

Valued as well was the wealth of online resources, from videos and apps to live practice sessions, that give students tools for better understanding and constant practice. Nevertheless, most students agreed that the structured environment and face-to-face interaction in traditional classrooms offer advantages which no virtual learning platform can fully replicate. They felt this because being in a classroom physically makes a person accountable and committed to learning since there is an obligation to attend and engage, which is largely absent in virtual learning.

While such challenges still prevail, there is a considerable advantage that was reported by the students concerning these online platforms for enjoying learning in an approachable way. This feedback again suggests that future research needs to adopt ways of embedding live interaction and classroom structure into the learning of languages online to have further improved learning outcomes.

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THE IMPACT OF ARTIFICIAL INTELLIGENCE TOOLS ON ENGLISH LANGUAGE LEARNING FOR UNIVERSITY STUDENTS

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Abstract: The integration of Artificial Intelligence (AI) into higher education has revolutionized the traditional methodologies of teaching and learning foreign languages. As contemporary global communication demands high proficiency in English, students increasingly turn to AI-powered applications, chatbots, and automated writing assistants to enhance their linguistic skills. The study examines how these technologies assist in developing vocabulary, improving grammar, and enhancing speaking confidence through interactive environments. Furthermore, the research highlights the advantages of personalized

learning and immediate feedback provided by digital resources. The methodology of this research relies on a qualitative review of existing educational technologies and a descriptive analysis of student experiences with AI platforms. The findings indicate that while AI tools cannot completely replace human instructors, they serve as powerful autonomous learning assistants that increase student engagement and motivation. The paper concludes with recommendations for both students and educators on integrating AI effectively into academic schedules without compromising academic integrity.

Keywords: Artificial intelligence, English language learning, university students, digital tools, educational technology.

SÜNİ İNTELLEKT ALƏTLƏRİNİN UNIVERSİTET TƏLƏBƏLƏRİNİN İNGİLİS DİLİNİ ÖYRƏNMƏSİNƏ TƏSİRİ

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Xülasə: Süni İntellektin (Sİ) ali təhsilə inteqrasiyası xarici dillərin tədrisi və öyrənilməsinin əhəmiyyətli metodologiyalarını tamamilə dəyişdirmişdir. Müasir global kommunikasiya ingilis dilində yüksək peşəkarlıq tələb etdiyi üçün, tələbələr dil bacarıqlarını inkişaf etdirmək məqsədilə getdikcə daha çox Sİ əsaslı tətbiqlərə, çatbotlara və avtomatlaşdırılmış yazı köməkçilərinə müraciət edirlər. Tədqiqat bu texnologiyaların lüğət ehtiyatının artırılmasına, qrammatikanın təkmilləşdirilməsinə və interaktiv mühitlər vasitəsilə danışmaq inamının gücləndirilməsinə necə kömək etdiyini nəzərdən keçirir. Bundan əlavə, araşdırma rəqəmsal resurslar tərəfindən təmin edilən fərdiləşdirilmiş öyrənmə və dərhal rəy sisteminin üstünlüklərini vurğulayır. Tədqiqatın metodologiyası mövcud təhsil texnologiyalarının keyfiyyət analizinə və tələbələrin Sİ platformaları ilə bağlı təcrübələrinin təsviri təhlilinə əsaslanır. Nəticələr göstərir ki, Sİ alətləri insan müəllimlərini tamamilə əvəz edə bilməsə də, onlar tələbələrin cəlb ediciliyini və motivasiyasını artıran güclü müstəqil öyrənmə köməkçiləri rolunu oynayır. Məqalənin sonunda həm tələbələr, həm də müəllimlər üçün akademik dürüstlüyü pozmadan Sİ-nin tədris planına effektiv şəkildə inteqrasiya olunması barədə tövsiyələr verilir.

Açar sözlər: Süni intellekt, ingilis dili öyrənmə, universitet tələbələri, rəqəmsal alətlər, təhsil texnologiyası.

Introduction

The rapid advancement of digital technologies in the 21st century has significantly transformed the landscape of higher education. Among these innovations, Artificial Intelligence (AI) stands out as a defining force reshaping pedagogical practices and language acquisition models. For university students worldwide, mastering the English language is no longer merely an academic requirement but a crucial prerequisite for global career readiness, cross-border collaboration, and scientific research. Traditional classroom settings, while fundamentally important, often face limitations regarding individualized attention, continuous practice environments, and immediate error correction. Consequently, modern learners increasingly complement their formal curriculum with interactive technological ecosystems.

Recent developments in generative AI and natural language processing (NLP) have democratized access to customized learning experiences. Platforms powered by sophisticated computational algorithms allow students to simulate authentic conversational contexts, analyze

complex structural elements of text, and receive contextualized guidance tailored to their specific proficiency levels. This study focuses on identifying the underlying mechanisms through which AI tools influence vocabulary acquisition, grammatical accuracy, and communicative confidence. By evaluating the integration of automated solutions within the framework of modern English linguistics education, this paper aims to provide a comprehensive structural understanding of contemporary digital learning paradigms.

Materials and Methods

To evaluate the actual impact of artificial intelligence tools on contemporary English language learning, this research utilizes a qualitative and descriptive methodology grounded in secondary data analysis and educational observations. The primary materials analyzed include widespread modern digital applications, advanced generative language models, automated proofreading systems, and presentation automation frameworks that are actively used by higher education students. Specifically, platforms utilizing transformer models and natural language processing algorithms were evaluated based on their utility in developing linguistic competence.

The methodology consists of three distinct analytical phases. First, a comprehensive literature review of digital pedagogical frameworks was conducted to analyze how modern software interfaces align with cognitive theories of second language acquisition. Second, a functional analysis of popular AI assistants was carried out to assess their pedagogical feedback loops – specifically focusing on how instantly these platforms correct syntactical and lexical errors. Third, the study incorporated descriptive empirical observations regarding how undergraduate linguistics students utilize automated design and smart conversational tools to manage academic workloads, construct research arguments, and overcome communicative anxiety. These descriptive variables were synthesized to formulate concrete pedagogical insights regarding autonomous student learning strategies.

Discussion

The analytical findings indicate that the implementation of artificial intelligence in language acquisition introduces a highly dynamic, learner-centered environment that accelerates skill development across several linguistic domains. One of the most pronounced results observed is the substantial optimization of autonomous writing and vocabulary development. Traditional methodologies often require a protracted feedback loop, where students wait days for instructors to grade essays or identify stylistic inconsistencies. In contrast, advanced AI-driven text editors and language models offer instantaneous syntactic diagnostics, semantic alternatives, and contextual slang explanations. This immediate remediation prevents the fossilization of grammatical errors and enables students to internalize precise structural rules in real time.

Furthermore, digital platforms have significantly mitigated the psychological barriers associated with foreign language production, commonly referred to as "foreign language anxiety." In conventional classrooms, students frequently hesitate to participate in spontaneous speaking exercises due to the fear of peer evaluation or negative academic assessment. Interactive conversational AI bots serve as non-judgmental interlocutors, allowing undergraduates to engage in limitless dialogue practice, experiment with complex phonetic patterns, and test diverse discourse strategies without social pressure. The empirical observations show that regular interaction with automated conversational partners correlates with increased fluency and higher conversational confidence during formal academic evaluations.

Another crucial dimension highlighted by the research results is the role of intelligent design

and presentation automation tools in academic literacy. Modern university students are regularly required to synthesize intricate linguistic theories and present their findings through structured visual media. AI-powered presentation applications allow students to focus less on mechanical formatting constraints and more on cognitive organization, thematic coherence, and advanced vocabulary application. By automating stylistic layouts, these tools encourage students to refine their thesis statements and develop a more rigorous grasp of academic writing structures.

However, the discussion must also address potential academic risks, particularly relating to over-reliance and the dilution of critical thinking. While AI applications are exceptional formatting and conversational assistants, an unmonitored dependency on automated content generation can lead to a decline in authentic student writing capabilities and critical analysis. Therefore, the data suggests that technological tools must be positioned strictly as cognitive scaffolds rather than total replacements for cognitive effort. When used as autonomous study extensions that complement university lectures, digital tools fundamentally enhance learning outcomes, foster autonomous research competencies, and bridge the gap between theoretical linguistics and practical communicative mastery.

Conclusion

In conclusion, artificial intelligence tools and digital learning platforms have emerged as indispensable assets in modern English language acquisition for university students. The empirical and descriptive evidence synthesized in this research demonstrates that these technological innovations dramatically enhance vocabulary development, eliminate communicative anxiety, and optimize academic presentation workflows. By providing tailored, instantaneous feedback and accessible conversational environments, AI platforms successfully empower students to take ownership of their linguistic development outside the confines of traditional lecture halls.

Nevertheless, maximizing the benefits of educational technology requires a careful balance. To prevent the erosion of academic integrity and critical writing skills, higher education institutions must cultivate digital literacy frameworks that guide students on the ethical and analytical use of automated applications. AI should be viewed not as a substitute for human pedagogical expertise, but as a sophisticated collaborative partner. Ultimately, integrating these smart systems into the academic routine of linguistics students fosters a more resilient, highly proficient generation of global communicators equipped to navigate the evolving demands of the modern information era.

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THE ROLE OF DIGITAL LEARNING PLATFORMS AND ARTIFICIAL INTELLIGENCE IN TRAINING TOURISM PROFESSIONALS FOR REGIONAL ECONOMIC DEVELOPMENT: THE CASE OF AZERBAIJAN

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Abstract: This study investigates the integration of digital learning technologies and artificial intelligence (AI) frameworks in higher education to enhance human resource capacity within Azerbaijan's regional tourism sector. While tourism serves as a critical driver for non-oil economic diversification—contributing approximately 5.3% to the national GDP—regional development faces significant bottlenecks due to a shortage of digitally literate and professionally trained staff. This paper evaluates the shift from classic instructional setups toward innovative, data-driven educational models at regional higher education institutions. Utilizing a systematic approach and qualitative pedagogical analysis, the research examines how specialized digital platforms, virtual simulation tools, and AI-driven curriculum designs can optimize student readiness for modern hospitality environments. The empirical findings indicate that incorporating smart technology into training programs effectively bridges the service quality gap between peripheral regions and urban centers. Ultimately, the study provides strategic recommendations for building public-private academic partnerships and implementing green educational infrastructures, establishing a scalable model for utilizing educational technology to accelerate regional economic sustainability.

Key words: digital learning, artificial intelligence, tourism education, regional development, educational technology, Azerbaijan.

REGIONAL İQTİSADİ İNKİŞAFDA TURİZM KADRLARININ HAZIRLANMASINDA RƏQƏMSAL TƏHSİL PLATFORMALARI VƏ SÜNİ İNTELLEKTİN ROLU: AZƏRBAYCAN NÜMUNƏSİ

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Xülasə: Bu tədqiqat Azərbaycanın regional turizm sektorunda insan resurslarının potensialını artırmaq üçün ali təhsil sistemində rəqəmsal təlim texnologiyaları və süni intellekt (AI) alətlərinin integrasiyasını araşdırır. Turizm qeyri-neft iqtisadiyyatının diversifikasiyasında mühüm hərəkətverici qüvvə rolunu oynasa da və ümumi daxili məhsulda (ÜDM) təxminən 5.3% paya sahib olsa da, regional inkişaf rəqəmsal bacarıqlara malik peşəkar kadr çatışmazlığı səbəbindən ciddi çətinliklərlə qarşılaşır. Məqalədə bölgələrdə fəaliyyət göstərən ali təhsil müəssisələrinə klassik tədris formalarından innovativ, məlumat əsaslı yeni nəsil təhsil modellərinə keçid prosesi qiymətləndirilir. Sistemli yanaşma və pedaqoji təhlil metodlarından istifadə edilərək, ixtisaslaşdırılmış rəqəmsal platformaların, virtual simulyasiya alətlərinin və süni intellektə əsaslanan kurikulum dizaynlarının tələbələrin müasir qonaqpərvərlik mühitinə hazırlığını necə optimallaşdırdığı tədqiq olunur. Tədqiqatın nəticələri göstərir ki, təlim proqramlarına ağıllı texnologiyaların

daxil edilməsi regionlar və şəhər mərkəzləri arasındakı xidmət keyfiyyəti fərqlərini effektiv şəkildə aradan qaldırır. Yekun olaraq, işdə dövlət-özəl-universitet tərəfdaşlığının qurulması və rəqəmsal təhsil infrastrukturunun tətbiqi üçün strateji tövsiyələr irəli sürülür.

Açar sözlər: rəqəmsal təhsil, süni intellekt, turizm təhsili, regional inkişaf, təhsil texnologiyaları, Azərbaycan.

Introduction

Azerbaijan possesses diverse natural-climatic and rich cultural resources that position tourism as a cornerstone for sustainable non-oil economic diversification. Strategically located at the intersection of the Transcaucasian transport corridor, the nation offers vast opportunities for eco-tourism, rural tourism, and cultural heritage initiatives across its diverse regions. Recent macroeconomic frameworks underscore the expanding footprint of this industry; based on late financial reporting periods, the total gross value added generated within the tourism and hospitality framework reached substantial figures, contributing approximately 5.3% directly to the overall national GDP.

However, the geographical distribution of economic benefits remains skewed. While metropolitan zones demonstrate advanced tourism management mechanisms, peripheral administrative units face systematic hurdles. The central problem lies in a severe human resource deficit characterized by a shortage of digitally literate workers, asymmetric quality of local hospitality services, and a slow transition toward modern, technology-enabled operational models. Addressing these structural deficiencies necessitates a fundamental pivot in how regional educational institutions prepare future workforces. This study argues that implementing digital learning platforms and artificial intelligence frameworks within regional academic hubs is essential to bridge training disparities and stimulate long-term economic growth.

Materials and Methods

The methodological architecture of this research integrates a comparative institutional analysis, systematic pedagogical framework evaluation, and data synthesis from recent governmental tourism whitepapers and educational registries. Special attention is directed to regional public learning curricula, mapping out the presence or lack of technical instructional modules designed for modern digital hospitality platforms.

The analytical process incorporates qualitative assessment models that cross-reference existing skill deficiencies in peripheral tourism destinations (e.g., uneven digital adoption in rural homestays) with current university offerings. To validate the proposed technological solutions, pedagogical simulations using cloud-based management systems, AI-driven scheduling assistants, and localized language training tools were systematically evaluated against standard classroom approaches.

Results and Discussion

Our evaluation indicates that older academic curricula heavily rely on static, theory-based methodologies, leaving graduates ill-equipped to handle modern hospitality ecosystems where online booking networks, data-driven revenue management, and localized algorithmic customer support dominate. By incorporating specialized technical learning tracks, universities can directly optimize employment rates and localized growth patterns within regional clusters.

Key Strategy Area	Identified Industry Bottleneck	Proposed Digital / AI Intervention
Destination Management	Low online visibility & weak digital footprints in peripheral regions.	Training students in data analytics, AI-driven branding, and localized SEO.
Service Quality	Inconsistent service standards between urban and rural centers.	Deploying virtual reality (VR) simulation environments for hospitality standard training.
Language & Hospitality	Limited specialized language proficiency among rural guides.	AI-driven English for Specific Purposes (ESP) modules tailored to tourism.

The analysis demonstrates that when educational institutions implement collaborative structures—such as public-private academic partnerships—the direct socio-economic spillover effects enhance the operational capabilities of local small and medium enterprises (SMEs). For instance, incorporating active simulation modules directly results in increased digital integration for local family-owned enterprises, allowing them to access international markets effectively.

Conclusion and Policy Recommendations

Transforming Azerbaijan's regional tourism potential into an innovative, highly competitive sector is deeply contingent upon upgrading its underlying educational infrastructure. Moving past rigid instructional frameworks toward digital platforms and AI-guided curriculum personalization ensures a sustainable flow of competent professionals into non-oil industrial sectors. Based on our analytical conclusions, the following strategic policy initiatives are recommended:

- Establish regional digital learning laboratories specializing in e-hospitality systems at prominent regional universities.
- Develop institutional public-private-academic partnerships to allow student-led digital onboarding for peripheral SMEs.
- Incorporate adaptive AI language platforms to rapidly scale technical English proficiency among regional hospitality workforces.

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EDUCATIONAL TECHNOLOGIES AND DIGITAL LEARNING

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Abstract: This study provides a comprehensive analysis of the transformative impact of educational technologies and digital learning on contemporary education systems. In the 21st century, rapid technological advancements have reshaped traditional pedagogical approaches, shifting them toward more flexible, interactive, and student-centered learning environments. The paper examines the integration of digital tools such as Learning Management Systems (LMS), virtual classrooms, mobile applications, and artificial intelligence (AI) technologies in teaching and learning processes.

Furthermore, the research explores the advantages of online and hybrid learning models, highlighting their role in ensuring continuity of education during global disruptions, such as the COVID-19 pandemic. Special attention is given to the development and utilization of electronic educational resources, including e-books, multimedia content, and interactive platforms, which enhance learner engagement and knowledge retention.

The study also discusses the growing importance of artificial intelligence in education, particularly in providing personalized learning experiences, automated feedback, adaptive assessments, and data-driven decision-making. Despite these benefits, the research identifies several challenges, including digital inequality, lack of technological infrastructure, ethical concerns, and the need for digital literacy among educators and learners.

The findings suggest that while educational technologies significantly improve accessibility, efficiency, and quality of education, their successful implementation requires strategic planning, teacher training, and balanced integration. Ultimately, the study concludes that the future of education lies in the effective combination of technological innovation and pedagogical excellence.

Keywords: Educational technology, digital learning, artificial intelligence, hybrid learning, online education, electronic resources, LMS, personalized learning.

TƏHSİL TEXNOLOGİYALARI VƏ RƏQƏMSAL ÖYRƏNMƏ

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Xülasə: Bu tədqiqat müasir təhsil sistemlərində təhsil texnologiyalarının və rəqəmsal öyrənmənin geniş və dərin təsirini araşdırır. XXI əsrdə texnologiyanın sürətli inkişafı ənənəvi tədris metodlarını köklü şəkildə dəyişdirərək onları daha çevik, interaktiv və tələbəyönümlü modellərə yönəltdi. Məqalədə təlim idarəetmə sistemləri (LMS), virtual siniflər, mobil tətbiqlər və süni intellekt texnologiyalarının tədris prosesinə inteqrasiyası geniş şəkildə təhlil olunur.

Eyni zamanda, onlayn və hibrid öyrənmə modellərinin üstünlükləri araşdırılır və xüsusilə COVID-19 pandemiyası dövründə təhsilin davamlılığının təmin edilməsində oynadığı mühüm rol vurğulanır. Elektron təhsil resurslarının — elektron kitabların, multimedia materiallarının və interaktiv platformaların inkişafı və istifadəsi öyrənmə prosesinin daha cəlbedici və effektiv olmasına şərait yaradır.

Tədqiqatda süni intellektin təhsildə artan rolu xüsusi olaraq qeyd olunur. Süni intellekt fərdiləşdirilmiş öyrənmə imkanları, avtomatlaşdırılmış qiymətləndirmə, adaptiv testlər və məlumatlara əsaslanan qərarvermə kimi imkanlar yaradır. Bununla yanaşı, rəqəmsal bərabərsizlik, texnoloji infrastrukturun çatışmazlığı, etik problemlər və rəqəmsal savadlılıq səviyyəsinin aşağı olması kimi çətinliklər də mövcuddur.

Nəticələr göstərir ki, təhsil texnologiyaları təhsilin əlçatanlığını, keyfiyyətini və səmərəliliyini artırırsa da, onların uğurlu tətbiqi üçün düzgün strategiya, müəllim hazırlığı və balanslaşdırılmış inteqrasiya vacibdir. Gələcəkdə təhsilin inkişafı texnoloji innovasiyalar ilə pedaqoji yanaşmaların vəhdətindən asılı olacaqdır.

Açar sözlər:

Təhsil texnologiyası, rəqəmsal öyrənmə, süni intellekt, onlayn təhsil, hibrid modellər, elektron resurslar, LMS, fərdiləşdirilmiş öyrənmə.

Introduction

Educational technologies and digital learning have become integral elements of modern education systems worldwide. The rapid growth of digital innovations has significantly influenced teaching methodologies, transforming traditional classroom-based instruction into dynamic, technology-enhanced learning environments. Today's learners require not only access to information but also the ability to critically analyze, apply, and create knowledge using digital tools.

The integration of digital technologies has led to the emergence of new educational paradigms, such as blended learning, flipped classrooms, and fully online education. These approaches promote active participation, collaboration, and independent learning. Moreover, digital learning environments provide opportunities for lifelong learning, enabling individuals to continuously update their knowledge and skills.

Another important aspect is the democratization of education. Digital platforms allow learners from different geographical and socio-economic backgrounds to access high-quality educational content. However, this transformation also requires educators to adapt their teaching strategies and develop digital competencies.

Methodology and Materials

This study employs a qualitative research approach based on a comprehensive review of existing literature, academic articles, and case studies related to educational technologies and digital learning. Secondary data sources were analyzed to identify current trends, challenges, and opportunities in digital education.

The research focuses on the effectiveness of Learning Management Systems (LMS), virtual learning environments, and AI-powered educational tools. Various scholarly databases and journals were used to collect relevant data, ensuring the reliability and validity of the findings.

Additionally, comparative analysis was conducted to evaluate traditional and digital learning models, highlighting their strengths and limitations. The study also considers real-world examples of digital education implementation in higher education institutions.

Discussion

The application of digital technologies in education has revolutionized the way knowledge is delivered and acquired. Tools such as interactive whiteboards, online collaboration platforms, and mobile learning applications have enhanced the teaching process, making it more engaging and effective.

Online and hybrid learning models have become increasingly popular due to their flexibility and accessibility. These models allow students to learn at their own pace and manage their time

more efficiently. The COVID-19 pandemic accelerated the adoption of digital learning, demonstrating its importance in maintaining educational continuity.

Electronic educational resources play a crucial role in modern education. E-books, video lectures, podcasts, and interactive simulations provide diverse learning experiences that cater to different learning styles. These resources not only improve understanding but also increase student motivation.

Artificial intelligence has emerged as a powerful tool in education. AI-based systems can analyze student performance, identify learning gaps, and provide personalized recommendations. Adaptive learning platforms adjust content according to individual needs, ensuring more effective learning outcomes.

However, the widespread use of digital technologies also presents challenges. Issues such as digital divide, lack of infrastructure, data privacy concerns, and resistance to change must be addressed. Teacher training and professional development are essential to ensure the successful integration of technology in education.

Conclusion

In conclusion, educational technologies and digital learning have brought significant improvements to the education sector. They have enhanced accessibility, flexibility, and the overall quality of education. Digital tools and AI technologies offer innovative solutions that support personalized and student-centered learning.

Nevertheless, it is crucial to address existing challenges to fully realize the potential of digital education. Governments, educational institutions, and policymakers must work together to ensure equal access to technology and promote digital literacy.

The future of education will depend on the effective integration of technological advancements with sound pedagogical practices. By embracing innovation and maintaining a balanced approach, education systems can better prepare learners for the demands of the digital age.

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THE ROLE OF ARTIFICIAL INTELLIGENCE AND DIGITAL PLATFORMS IN EDUCATION: THE TRANSFORMATION OF MODERN TEACHING.

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Abstract: This study examines the transformative role of artificial intelligence (AI) and digital platforms in modern education, with particular focus on teaching practices, learning environments, and institutional administration. Grounded in the TPACK framework, Connectivism, and the digital SECI model, the research employs a mixed-methods design incorporating surveys, semi-structured interviews, and document analysis of institutional digital policies across multiple educational stakeholders. Findings indicate that AI significantly enhances personalized learning, teacher efficiency, and student engagement; however, integration raises critical challenges including academic integrity concerns, algorithmic bias, data privacy risks, and insufficient teacher training. The study further highlights that equitable and responsible adoption of AI is essential to bridging the growing AI divide in global education. In conclusion, the future of education depends not on replacing human educators, but on cultivating a balanced human-AI partnership that upholds ethical, relational, and pedagogical values.

Keywords: Artificial intelligence in education, Digital platforms, Personalized learning, TPACK framework, Generative AI, Mixed-methods research, EdTech, Academic integrity, Human-AI collaboration, Educational transformation

SÜNİ İNTELLEKT VƏ RƏQƏMSAL PLATFORMALARIN TƏDRİSDƏ ROLU: MÜASİR TƏHSİLİN TRANSFORMASIYASI

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Xülasə: Bu tədqiqat süni intellektin (Sİ) və rəqəmsal platformların müasir təhsildə — xüsusilə tədris metodları, öyrənmə mühitləri və institusional idarəetmə sahəsində — transformativ rolunu araşdırır. Tədqiqat TPACK çərçivəsi, Konnektivizm və rəqəmsal SECI modeli əsasında qurulmuş olub, bir neçə təhsil maraqlısını əhatə edən sorğular, yarı strukturlaşdırılmış müsahibələr və institusional rəqəmsal siyasətlərin sənəd təhlilindən ibarət qarışıq metodlar dizaynından istifadə edir. Tədqiqatın nəticələri göstərir ki, süni intellekt fərdiləşdirilmiş öyrənməni, müəllimlərin iş səmərəliliyini və tələbələrin dərslərə cəlb olunmasını əhəmiyyətli dərəcədə artırır; lakin bu integrasiya akademik dürüstlük problemləri, alqoritmik qərəz, məlumatların məxfiliyi riski və müəllim hazırlığının qeyri-kafi olması kimi ciddi çətinlikləri də bərabərində gətirir. Tədqiqat həmçinin vurğulayır ki, global təhsildə genişlənən süni intellekt uçurumunu aradan qaldırmaq üçün Sİ-nin ədalətli və məsuliyyətli tətbiqi vacibdir. Nəticə olaraq, təhsilin gələcəyi insan müəllimlərinin texnologiya ilə əvəz edilməsindən deyil, etik, münasibətə əsaslanan və pedaqoji dəyərləri qoruyan balanslaşdırılmış insan-Sİ tərəfdaşlığının qurulmasından keçir.

Açar sözlər: Təhsildə süni intellekt, Rəqəmsal platformlar, Fərdiləşdirilmiş öyrənmə, TPACK çərçivəsi, Generativ süni intellekt, Qarışıq metodlar tədqiqatı, Təhsil texnologiyaları, Akademik dürüstlük, İnsan-Sİ əməkdaşlığı, Təhsilin transformasiyası

Introduction

The rapid advancement of technology in the twenty-first century has fundamentally reshaped nearly every sector of society, and education is no exception. Among the most consequential forces driving this transformation is artificial intelligence (AI), which is increasingly integrated into teaching practices, learning environments, and educational administration worldwide.

Artificial intelligence has the potential to address some of the biggest challenges in education today, innovate teaching and learning practices, and accelerate progress towards the United Nations Sustainable Development Goal 4 — Quality Education — though rapid technological developments inevitably raise complex questions that must be carefully navigated.

The global AI in education market, valued at \$5.88 billion in 2024, is projected to reach \$32.27 billion by 2030, registering a compound annual growth rate of 31.2%, driven by rising demand for personalized learning environments and greater administrative efficiency — with AI grading systems alone capable of reducing teacher workload by up to 70%.

AI is transforming global education and reshaping approaches to teaching, learning, and assessment through breakthrough technologies such as generative AI, with scholars projecting that by 2035, AI and teachers may collaborate as equals, fundamentally altering instructional roles and pedagogical standards.

While the promise of AI in education is compelling, only through responsible and informed adoption can AI truly fulfill its potential and ensure equitable access to quality education for all learners.

This article therefore aims to examine the role of AI and digital platforms in transforming modern teaching, exploring relevant theoretical frameworks, empirical findings, and the ethical dimensions that must guide future implementation. The study contributes to the growing body of literature seeking to understand not only what AI can do in education, but how it should be done.

Literature Review

The integration of artificial intelligence (AI) and digital platforms into education has attracted growing scholarly attention over the past decade. A systematic review of 132 empirical studies published between 2015 and 2024 found a significant increase in AI-related educational research, particularly from 2023 onward, coinciding with the widespread release of generative tools such as ChatGPT.

Studies have demonstrated AI's potential to improve teachers' and students' skills, critical thinking, and creativity, as well as to enhance the overall learning experience through adaptive and personalized instruction that accounts for students' unique needs, goals, and prior knowledge.

AI has the potential to transform digital learning and improve teaching methods, with machine learning and deep learning subsets enabling systems to detect complex patterns similar to human cognitive processes.

The education sector has benefited enormously from integrating digital technology-driven tools and platforms, and AI-based methods are increasingly regarded as the next generation of technology capable of enhancing the educational experience for students, teachers, and administrative staff alike.

However, this transformation also introduces new demands for advanced digital literacy, including interpreting algorithm-generated outputs and managing adaptive systems, creating both challenges and opportunities to redefine the purpose of academic work.

Theoretical Framework

This study is grounded in three interrelated theoretical perspectives that collectively explain the transformation of modern teaching through AI and digital platforms.

The first is the Technological Pedagogical Content Knowledge (TPACK) framework, originally introduced by Mishra and Koehler (2006). Given the profound impact of AI on modes of teaching and learning, scholars have reexamined the interrelationships among technology, pedagogy, and subject matter, constructing an AI-TPACK framework to elucidate the complex synergistic effects of AI technology, pedagogical methods, and subject-specific content in education.

The TPACK framework is used to discuss the types of knowledge teachers require to effectively use generative AI tools, highlighting qualities that make AI both similar to and revolutionary compared to prior digital technologies, ultimately raising larger questions about fundamental shifts in the very nature of teaching and learning.

The second theoretical lens is Connectivism (Siemens, 2005). Drawing on constructivist learning theory, connectivism, multimedia learning theory, and the TPACK model, a comprehensive framework for integrating e-learning platforms and digital resources identifies key components including pedagogical alignment, digital resource design, learner engagement mechanisms, assessment integration, and feedback systems.

The third perspective is the digital SECI model of knowledge creation. With the advent of digital and data technologies, particularly MOOCs and similar platforms in higher education, a digital SECI-based multimodal teaching model has the potential to significantly advance the digital transformation of teaching models, integrating multiple sensory modalities such as images, sounds, language, and text to enhance knowledge transfer.

Together, these frameworks provide a comprehensive theoretical basis for understanding how AI and digital platforms reshape instructional design, teacher roles, and student learning experiences.

Methodology and Materials

This study adopts a mixed-methods research design, combining both qualitative and quantitative approaches to comprehensively examine the role of AI and digital platforms in transforming modern teaching practices.

A mixed-method research design allows for the collection of data through in-depth interviews with students and university lecturers to gain comprehensive insights into their experiences and perceptions, while a validated scale is subsequently administered to a larger sample for quantitative analysis using descriptive statistics.

A convergent mixed-methods design combining survey data with qualitative narratives, guided by established theoretical frameworks, enables an integrated analysis of learning perceptions and technology adoption behaviour across diverse educational contexts.

Data collection in this study draws on surveys distributed to educators and students, semi-structured interviews, and document analysis of institutional digital policies. Sampling includes principals, teachers, and education administrators, with data analyzed through both qualitative and quantitative methods to capture multiple stakeholder perspectives on AI integration.

Using a mixed-methods approach to analyze implementation patterns and learning outcomes across multiple institutions allows researchers to identify significant variations in effectiveness based on implementation approach, while also addressing challenges regarding algorithmic bias, digital equity, and discipline-specific skill maintenance.

Thematic analysis is applied to qualitative data, while descriptive and inferential statistics are used for quantitative findings, ensuring methodological rigor and validity throughout the study.

Discussion

The results of this study align with and extend the broader scholarly conversation on AI's transformative role in modern education. The findings confirm that while AI and digital platforms offer significant pedagogical benefits, their integration also raises critical ethical, institutional, and professional challenges that must be carefully addressed.

AI applications in higher education offer advantages including enhanced teaching and learning, increased student engagement, support for writing and research, innovative assessment techniques, and the automation of administrative tasks — yet they simultaneously present challenges such as academic integrity issues, the propagation of misinformation, ethical and bias-related concerns, privacy risks, and reduced human interaction.

UNESCO's 2025 report argues that AI should not be treated simply as a tool of automation but as a catalyst that forces education systems to confront deeper questions about teaching, learning, and the role of the human teacher.

These findings suggest that AI is not only influencing teaching methodologies but also altering how educators allocate their time and define their professional responsibilities, while raising serious concerns about data privacy, security, and educators' capacity to critically assess algorithmic processes.

Responsible integration of generative AI tools such as ChatGPT requires frameworks that address ethical concerns, overcome infrastructure limitations, and redefine the role of educators in AI-augmented classrooms to foster both equity and innovation.

Among educators' most widely cited concerns are a lack of teacher training for integrating AI into instruction, student cheating and plagiarism, and cybersecurity risks — underscoring the urgent need for institutional policies and professional development programs to accompany technological adoption.

Overall, the discussion points to the necessity of a balanced approach: one that harnesses the personalization and efficiency benefits of AI while preserving the human, ethical, and pedagogical foundations of quality education.

Conclusion

This study set out to examine the transformative role of artificial intelligence and digital platforms in modern education and teaching. Across the literature review, theoretical framework, methodology, results, and discussion sections, a consistent picture emerges: AI holds remarkable potential to reshape pedagogy, yet its responsible integration demands deliberate institutional, ethical, and professional commitment.

At UNESCO's Digital Learning Week, ministers affirmed a shared vision that AI in education must be human-centered, equitable, safe, and ethical, identifying the confrontation of the AI divide as a foremost global priority — recognizing that the digital divide is rapidly becoming an AI divide.

AI's relative newness in education presents a critical window of opportunity for protecting learners and ensuring that safety and privacy needs are met before educational use cases become entrenched, with policymakers across numerous countries already seizing this moment to develop guidance and regulatory frameworks.

Ongoing research and policy support are necessary to determine the long-term impact of AI-driven learning initiatives, and educating teachers to explore the full potential of technological tools is essential to realizing their benefits for students.

Generative AI tools will play an increasingly significant role in the future of education due to their high customizability, ability to adapt content to diverse student backgrounds and academic levels, and capacity to identify and address individual learning weaknesses.

In conclusion, the future of education lies not in replacing teachers with technology, but in forging a balanced human-AI partnership that preserves the relational, ethical, and creative foundations of meaningful learning.

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BEYOND THE CLASSROOM: AI-POWERED TOOLS SHAPING THE EDUCATION

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Abstract: Digital technologies have transformed language learning, and artificial intelligence (AI) has become a crucial tool in modern education. This study investigates the role of AI-supported digital platforms in enhancing students' engagement and learning outcomes. A mixed-method approach, including surveys and observation, was applied to university students using AI-based learning applications. Results indicate that AI notifications and personalized learning content increase motivation and the retention of new vocabulary. The findings demonstrate that integrating AI into educational platforms can improve the effectiveness of language teaching and foster independent learning.

This research contributes to the field of educational technology by highlighting practical applications of AI in academic development. In addition, the study emphasizes the importance of interactive digital media for learning. AI systems enable learners to receive immediate feedback and allow the adaptation of learning materials according to their specific needs. This helps learners develop essential independent learning skills and become more confident in using the English language for academic purposes. Furthermore, AI-supported

digital media facilitates continuous learning opportunities, helping learners retain knowledge for a longer period.

Keywords: artificial intelligence, digital learning, educational technology, language learning, student engagement, personalized learning

AUDİTORİYADAN KƏNARDA: TƏHSİLİ FORMALAŞDIRAN SÜNİ İNTELLEKT DƏSTƏKLİ ALƏTLƏR

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Xülasə: Rəqəmsal texnologiyalar dil öyrənməni dəyişdirmişdir və sünİ intellekt (AI) müasir təhsildə mühüm bir vasitəyə çevrilmişdir. Bu tədqiqat AI dəstəklİ rəqəmsal platformaların tələbələrin marağını və öyrənmə nəticələrini artırmadakı rolunu araşdırır. Qarışıq metod yanaşması tətbiq olunmuşdur; bu yanaşmaya AI əsaslı öyrənmə tətbiqlərindən istifadə edən universitet tələbələri arasında sorğular və müşahidələr daxildir. Nəticələr göstərir ki, AI bildirişləri və fərdiləşdirilmiş öyrənmə məzmunu motivasiyanı və yeni söz ehtiyatının yadda saxlanmasını artırır. Tapıntılar sübut edir ki, AI-nin təhsil platformalarına integrasiyası dil tədrisinin effektivliyini artırır və müstəqil öyrənməni təşviq edir.

Bu tədqiqat təhsil texnologiyaları sahəsinə AI-nin akademik inkişafda praktiki tətbiqlərini vurğulamaqla töhfə verir. Bundan əlavə, tədqiqat öyrənmə üçün interaktiv rəqəmsal medianın əhəmiyyətini vurğulayır. AI sistemləri öyrənənlərə dərhal rəy almağa imkan verir və öyrənmə materiallarının onların xüsusi ehtiyaclarına uyğunlaşdırılmasını təmin edir. Bu, öyrənənlərə vacib müstəqil öyrənmə bacarıqlarını inkişaf etdirməyə və ingilis dilindən akademik məqsədlər üçün istifadə etməkdə daha inamlı olmağa kömək edir. Bundan əlavə, AI dəstəklİ rəqəmsal media davamlı öyrənmə imkanları yaradır və öyrənənlərin bilikləri daha uzun müddət yadda saxlamasına kömək edir.

Açar sözlər: sünİ intellekt, rəqəmsal öyrənmə, təhsil texnologiyaları, dil öyrənmə, tələbə iştirakı, fərdiləşdirilmiş öyrənmə

Introduction

AI can and does ease and strengthen customary education methods. The major way that AI is applied to education is by personalizing and customizing learning. This might be achieved through assessing and organizing data and adapting the manner of educational content delivery according to the student's learning style and pace. Another role for AI in education may be as a smart tutor that provides feedback for students as they interact with their lessons and learning materials.

Another use of artificial intelligence in education is in the automation of administrative responsibilities for the purposes of reducing the load on the education system. This also includes the use of artificial intelligence-driven virtual assistants. They are another tool that can assist in the educational process and answer students' questions.

Another application for content creation in the educational process is the use of AI in virtual labs and education games to create varying types of interactivity and cater to multiple learning styles. Similar to its role in early intervention, AI tracks student performance to improve learning and accommodate diverse learning styles and needs to create an inclusive environment for all students. To sum it up, it is clear that there are many roles played by AI to provide an efficient learning experience to the students in K-12 education.

Materials and Methods

This study is based on a qualitative review and comparative analysis of current research on the application of artificial intelligence in education. Relevant academic publications, educational reports, and studies concerning AI-powered educational technologies were examined to identify the major functions and benefits of AI in K–12 education. The study focuses on personalized learning, intelligent tutoring systems, administrative automation, AI-driven virtual assistants, content creation technologies, and data-driven educational practices. Descriptive and analytical methods were employed to evaluate how AI contributes to improving teaching and learning processes.

Results and Discussion

The role of AI in education is varied, reshaping traditional teaching methods. Personalized learning is a standout application, as AI analyzes individual student data to tailor educational content, accommodating diverse learning styles and pacing. Intelligent tutoring systems, powered by AI, provide real-time feedback and guidance, offering personalized assistance to students. These systems contribute to a more adaptive and interactive learning environment.

Additionally, AI automates administrative tasks, such as grading and attendance tracking, streamlining workflows for educators. Virtual assistants, driven by AI, answer student queries, enhancing accessibility and providing additional support beyond classroom hours.

Content creation is another significant aspect, with AI contributing to the development of interactive and immersive learning materials. Virtual labs, educational games, and simulations cater to diverse learning preferences, making education more engaging.

Furthermore, AI supports early intervention strategies by continuously monitoring student performance, identifying learning gaps, and enabling timely interventions. This proactive approach ensures that students receive the support they need to succeed, contributing to a more inclusive educational environment.

The advantages of artificial intelligence in education are numerous, and the future of artificial intelligence (AI) in education is set to transform the teaching and learning process. One of the primary advantages of AI in education is customized learning, with AI algorithms analyzing data about students to create customized learning experiences. Clever tutoring systems leverage AI techniques to provide immediate feedback and instructions to students, creating a two-way interactive experience.

In administration, AI may be used to automate repetitive tasks like grading or taking attendance, leaving educators free to focus on teaching more interactive or creative lessons. AI-driven virtual assistants may be used to answer students' questions and improve accessibility.

AI can also help create new learning experiences, such as creating simulations and virtual labs, to create an interactive and engaging learning experience that complements existing methods.

AI also helps educators make data-driven decisions by analyzing student performance patterns to determine effective teaching strategies, recognize areas of improvement, and implement focused actions to address those weaknesses.

Conclusion

Overall, this study demonstrates that AI in education can be used to automate administrative responsibilities, create customized learning experiences, design interactive learning environments, and provide data-driven insights for a more effective learning environment. As AI continues to evolve, it promises a dynamic and adaptive educational landscape, empowering educators and enriching the learning experience for students across diverse educational settings.

In summary, AI's roles in K–12 education include personalized learning, administrative efficiency, virtual assistance, content creation, and early intervention, collectively enhancing the overall educational experience for both students and educators.

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THE ROLE OF ARTIFICIAL INTELLIGENCE IN THE DEVELOPMENT OF STUDENT-CENTERED LEARNING IN PRIMARY EDUCATION

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Abstract: In the modern era, transformations in the education system have significantly increased the relevance of student-centered learning approaches. The rapid development of digital technologies and artificial intelligence has created new opportunities for organizing the teaching process and has necessitated the reconsideration of traditional methods. This article comprehensively examines the role of artificial intelligence technologies in the development of student-centered learning at the primary education level and highlights its pedagogical significance.

The study employs systematic literature analysis, comparative approach, and generalization methods. Based on findings from international and local research, the main directions of AI application in education have been identified. The results reveal that AI-based educational platforms optimize the learning process by considering students' individual characteristics, learning pace, and knowledge levels, thereby increasing the effectiveness of teaching.

Furthermore, these technologies enhance students' motivation, encourage active participation in the learning process, and positively influence the development of independent thinking, critical thinking, and problem-solving skills. At the same time, they provide new opportunities for teachers by enabling more objective and timely assessment of students' performance.

The article also analyzes key challenges in the implementation of artificial intelligence, including limited technical infrastructure, insufficient digital competencies of teachers, and issues related to the effective use of technology. Practical recommendations are proposed to overcome these challenges and improve the integration of AI in primary education.

Keywords: Student-centered learning, artificial intelligence, primary education, personalized learning, digital education, learning analytics, student motivation

İBTİDAİ TƏHSİLDƏ ŞAĞIRDYÖNÜMLÜ TƏDRİSİN İNKİŞAFINDA SÜNİ İNTELLEKTİN ROLU

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Xülasə: Müasir dövrdə təhsil sistemində baş verən transformasiyalar şagirdyönümlü təlim yanaşmasının aktuallığını daha da artırmışdır. Xüsusilə rəqəmsal texnologiyaların və süni intellektin sürətli inkişafı təlim prosesinin təşkilində yeni imkanlar yaratmış, ənənəvi yanaşmaların yenidən nəzərdən keçirilməsini zəruri etmişdir. Məqalədə ibtidai təhsil səviyyəsində şagirdyönümlü təlimin inkişafında süni intellekt texnologiyalarının rolu geniş şəkildə araşdırılır və bu yanaşmanın pedaqoji əhəmiyyəti əsaslandırılır.

Tədqiqat zamanı elmi ədəbiyyatın sistemli təhlili, müqayisəli yanaşma və ümumiləşdirmə metodlarından istifadə edilmişdir. Müxtəlif beynəlxalq və yerli tədqiqatların nəticələri əsasında süni intellektin təhsildə tətbiqinin əsas istiqamətləri müəyyənəşdirilmişdir. Araşdırma nəticəsində məlum olmuşdur ki, süni intellekt əsaslı təhsil platformaları şagirdlərin fərdi xüsusiyyətlərini, öyrənmə tempini və bilik səviyyəsini nəzərə alaraq onların öyrənmə prosesini optimallaşdırır və təlimin effektivliyini artırır.

Eyni zamanda bu texnologiyalar şagirdlərin motivasiyasını yüksəldir, onların dərslər prosesində fəal iştirakını təmin edir və müstəqil düşünmə, tənqidi yanaşma və problem həll etmə bacarıqlarının inkişafına müsbət təsir göstərir. Bununla yanaşı, müəllimlər üçün də yeni imkanlar yaranır, çünki onlar şagirdlərin nailiyyətlərini daha operativ və obyektiv şəkildə qiymətləndirə bilirlər.

Məqalədə həmçinin süni intellektin tətbiqi zamanı qarşıya çıxan əsas çətinliklər, o cümlədən texniki infrastrukturun məhdudluğu, müəllimlərin rəqəmsal bacarıqlarının yetərli olmaması və texnologiyadan düzgün istifadə problemləri təhlil edilmiş, bu problemlərin aradan qaldırılması üçün praktik tövsiyələr irəli sürülmüşdür.

Açar sözlər: Şagirdyönümlü təlim, süni intellekt, ibtidai təhsil, fərdiləşdirilmiş öyrənmə, rəqəmsal təhsil, öyrənmə analitikası, şagird motivasiyası

Introduction

Recent transformations in the education system have made it necessary to reconsider traditional teaching models. At a global level, the primary aim of education is no longer limited to the transmission of knowledge; it also involves fostering students' independent thinking, creativity, and problem-solving skills. In this context, student-centered learning has emerged as one of the key directions in modern pedagogy.

Student-centered learning focuses on organizing the teaching process by taking into account learners' individual characteristics, interests, and learning pace. Within this framework, the teacher

is no longer merely a source of knowledge but rather acts as a guide and facilitator. Such an approach encourages active student participation and enhances their motivation to learn.

In recent years, the development of artificial intelligence (AI) technologies has opened new perspectives in the field of education. AI-based systems analyze large volumes of data to identify students' learning behaviors and create personalized learning environments. This enables a more effective implementation of student-centered approaches (Holmes et al., 2019).

Research indicates that AI technologies support the development of adaptive learning systems, which provide instructional materials tailored to students' knowledge levels (Luckin et al., 2016). This approach is particularly important in primary education, as this stage plays a crucial role in shaping students' learning habits and foundational skills.

Moreover, the effective use of digital technologies increases students' interest in learning and promotes active engagement in the classroom. According to the multimodal learning approach, information presented in various formats (text, images, video, etc.) enhances retention and improves learning outcomes (Mayer, 2020).

The main aim of this article is to analyze the role of artificial intelligence in the development of student-centered learning in primary education and to identify existing challenges along with possible solutions.

Materials and Methods

This study employs a comprehensive methodological approach to examine the role of artificial intelligence technologies in enhancing student-centered learning in primary education. Although the research is primarily theoretical, it also incorporates the analysis of practical examples based on contemporary educational practices.

The study is based on the following methods:

1. Literature review. Relevant works by both local and international scholars were systematically examined. Studies related to AI in education, student-centered learning, and digital learning environments were comparatively analyzed. Particular emphasis was placed on reports from international organizations such as UNESCO and OECD, as well as on key academic contributions in the field.

2. Comparative analysis. Traditional teaching models were compared with AI-based instructional approaches. The comparison was conducted using the following criteria:

- level of student engagement
- degree of personalization
- effectiveness of assessment
- teacher–student interaction

This analysis helped identify both the advantages and limitations of AI integration in education.

3. Generalization method. Findings obtained from various sources were synthesized and systematized. This method allowed the integration of diverse perspectives into a coherent conceptual framework.

4. Analytical approach. The functional capabilities of AI-based platforms were examined to evaluate their impact on student-centered learning. Particular attention was given to adaptive learning systems, automated assessment tools, and individualized learning pathways.

Pedagogical Foundations of Student-Centered Learning. Student-centered learning is one of the core directions of modern pedagogy and emphasizes the active role of the learner in the

educational process. This approach is grounded in constructivist theory, which suggests that knowledge is not passively received but actively constructed by learners based on their prior experiences and new information. In this context, the teacher acts as a facilitator who organizes and guides the learning process.

Key features of student-centered learning include:

- individualized and differentiated instruction
- active student participation
- collaborative learning environments
- reflection and self-assessment

These features are particularly important in primary education, where foundational cognitive and learning skills are developed.

Applications of Artificial Intelligence in Education. Artificial intelligence technologies offer new pedagogical opportunities in education. They enable the automation, personalization, and optimization of the teaching process. AI-based systems analyze large datasets to identify learning patterns and suggest appropriate instructional strategies (Holmes et al., 2019).

The main applications of AI in education include:

- Adaptive learning systems that provide content tailored to students' knowledge levels
- Automated assessment tools that deliver fast and objective evaluation
- Virtual learning environments that create interactive and engaging experiences
- Learning analytics that monitor and predict student behavior

These tools make the learning process more flexible and efficient (Luckin et al., 2016).

Personalized Learning and AI Integration. Personalized learning has become a key priority in modern education systems. AI technologies play a crucial role in implementing this approach by designing individualized learning pathways based on students' needs and abilities.

Studies show that personalized learning environments positively affect academic performance and increase student motivation (OECD, 2021). This is especially important in primary education, where early individualized support contributes significantly to future academic success.

Digital Technologies and Student Motivation. The integration of digital technologies into teaching significantly enhances student motivation. Interactive learning environments, gamification elements, and visual resources stimulate students' interest and encourage active participation.

According to the multimodal learning theory, presenting information through multiple formats improves comprehension and retention (Mayer, 2020). AI technologies further enhance this approach by adapting content to different learning styles.

Challenges and Limitations

Despite its advantages, the integration of AI in education presents several challenges, including:

- unequal access to technological infrastructure
- limited digital competencies among teachers
- methodological gaps in technology integration
- issues related to data privacy and ethics

According to UNESCO (2021), addressing these challenges requires a comprehensive approach, with teacher training being a key priority.

Results and Discussion

The findings of the study indicate that the integration of AI technologies significantly enhances the effectiveness of student-centered learning. These technologies optimize the learning process by addressing individual student needs.

One of the main advantages of AI-based systems is the creation of personalized learning environments. By analyzing students' previous performance and learning behaviors, these systems provide tailored instructional materials, thereby improving academic outcomes (Luckin et al., 2016).

In addition, AI technologies serve as valuable support tools for teachers. They enable educators to monitor student progress more accurately and organize instruction more efficiently. For instance, automated assessment systems provide immediate feedback based on student responses (Holmes et al., 2019).

Research also demonstrates that AI positively influences student motivation. Interactive and adaptive platforms increase engagement and participation, leading to improved learning outcomes (OECD, 2021).

Another important aspect is the development of multimodal learning environments. AI technologies integrate text, audio, and visual materials, catering to diverse learning styles and enhancing comprehension (Mayer, 2020).

However, the implementation of AI also involves challenges such as insufficient infrastructure, limited teacher competencies, and lack of practical experience. UNESCO (2021) emphasizes the importance of continuous professional development for effective AI integration.

Some researchers also argue that excessive reliance on AI may negatively affect students' social skills, highlighting the need for a balanced approach (Zawacki-Richter et al., 2019).

Conclusion

Contemporary research indicates that the application of AI in education extends beyond personalization and includes broader opportunities through learning analytics. This approach enables systematic monitoring and prediction of student learning behaviors, allowing early identification of at-risk students and timely intervention (OECD, 2021).

At the same time, the use of AI raises ethical concerns such as data privacy, algorithmic fairness, and transparency. Therefore, establishing clear regulatory frameworks is essential (UNESCO, 2021).

It is important to note that AI does not replace the teacher; rather, it complements and enhances their role. The teacher remains the central figure in the educational process, and the effectiveness of technology largely depends on their pedagogical competence.

The findings of this study confirm that AI technologies play a significant role in advancing student-centered learning in primary education. They support personalized instruction, increase motivation, and foster essential skills such as critical thinking and problem-solving.

Based on the results, the following recommendations can be made:

- expand the implementation of AI-based educational platforms in primary education
- provide continuous professional development for teachers
- integrate student-centered and personalized approaches into curricula
- ensure balanced and purposeful use of technology
- consider ethical and pedagogical aspects of AI integration

In conclusion, the effective and scientifically grounded application of AI technologies can significantly improve the quality of primary education and create more efficient learning environments

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APPLICATION OF DIGITAL TECHNOLOGIES IN THE EDUCATIONAL PROCESS

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Abstract. This article examines the application of digital technologies in modern education systems worldwide. It highlights their increasing importance in contemporary teaching and learning environments. The study analyzes a wide range of digital tools used in education today. It explains how technology significantly improves accessibility and flexibility for learners. The role of teachers in digital education is discussed in a detailed way. Changes in student learning behavior are also carefully examined and explained. The importance of student-centered learning approaches is strongly emphasized throughout the study. Digital technologies influence teaching methodologies in many effective and innovative ways. The study evaluates motivation and engagement in digital learning environments. It also underlines the growing need for strong digital literacy skills. Technology contributes to inclusive education for diverse learners across different contexts. The article identifies key challenges such as the digital divide problem. It also discusses teacher competence and infrastructural limitations in education systems. Strategic implementation of technology is considered highly important for success. The study shows that digital tools improve academic performance and outcomes significantly. Innovation plays a key role in the development of modern education systems. Recommendations for educators and policymakers are clearly presented and explained. Further research on

emerging educational technologies is strongly encouraged. The study contributes to both educational theory and classroom practice. Overall, digital technologies are essential for effective and modern education systems.

Keywords: digital technologies, education, e-learning, innovation, digital literacy

TƏHSİL PROSESİNDƏ RƏQƏMSAL TEXNOLOGİYALARIN TƏTBİQİ

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Xülasə: Bu məqalə rəqəmsal texnologiyaların müasir təhsil sistemlərində tətbiqini geniş şəkildə araşdırır. Onların tədris və öyrənmə mühitində artan rolu xüsusi vurğulanır. Tədqiqat təhsildə istifadə olunan müxtəlif rəqəmsal alətləri ətraflı təhlil edir. Texnologiyanın əlçatanlıq və çevikliyi necə artırdığı geniş izah edilir. Müəllimlərin rəqəmsal təhsildə rolu geniş və əhatəli şəkildə təqdim olunur. Şagirdlərin öyrənmə davranışlarında baş verən dəyişikliklər diqqətlə araşdırılır. Öyrənməmərkəzli yanaşmaların əhəmiyyəti məqalə boyunca xüsusi vurğulanır. Rəqəmsal texnologiyalar tədris metodlarına müxtəlif və innovativ təsirlər göstərir. Tədqiqat motivasiya və fəallıq məsələlərini geniş şəkildə qiymətləndirir. Rəqəmsal savadlılıq bacarıqlarının vacibliyi xüsusi qeyd olunur və əsaslandırılır. Texnologiya müxtəlif öyrənən qrupları üçün inklüziv təhsilə mühüm töhfə verir. Məqalə rəqəmsal uçurum kimi əsas problemləri müəyyən edir və izah edir. Müəllim hazırlığı və infrastruktur məsələləri də geniş şəkildə araşdırılır. Texnologiyanın strateji və düzgün tətbiqi çox vacib hesab olunur. Tədqiqat rəqəmsal alətlərin nəticələri əhəmiyyətli dərəcədə yaxşılaşdırdığını göstərir. Təhsildə innovasiyanın rolu olduqca mühüm və zəruri sayılır. Müəllimlər və siyasətçilər üçün praktik tövsiyələr təqdim olunur. Yeni texnologiyalar üzrə əlavə tədqiqatların aparılması tövsiyə edilir. Məqalə həm təhsil nəzəriyyəsinə, həm də praktikaya töhfə verir. Ümumilikdə, rəqəmsal texnologiyalar müasir və effektiv təhsil üçün vacib hesab olunur.

Açar sözlər: rəqəmsal texnologiyalar, təhsil, elektron təhsil, innovasiya, rəqəmsal savadlılıq

Introduction

The rapid advancement of digital technologies has significantly transformed the landscape of education, particularly in the field of academic writing. Among these innovations, artificial intelligence (AI) has emerged as a powerful tool that reshapes how students generate, organize, and refine written content. AI-powered applications offer various forms of support, including grammar correction, content generation, idea development, and structural organization, thereby influencing both the process and the outcome of academic writing.

In recent years, the integration of AI into educational environments has sparked considerable debate among researchers and educators. While some scholars emphasize its potential to enhance students' writing skills and increase efficiency, others raise concerns regarding academic integrity, over-reliance on technology, and the possible decline of critical thinking abilities. These contrasting perspectives highlight the need for a deeper and more systematic investigation of AI's role in academic writing.

Furthermore, the increasing accessibility of AI tools among students has made it essential to understand how these technologies are actually used in real academic contexts. It is particularly important to examine whether AI serves as a supportive learning instrument or as a substitute for independent intellectual effort.

Therefore, this study aims to explore the impact of artificial intelligence on students' academic writing processes by analyzing both its advantages and limitations. The research seeks to contribute to the existing body of knowledge by providing empirical evidence and practical insights into the role of AI in modern education.

Methodology and Materials

This study employs a mixed-method approach in order to investigate the impact of artificial intelligence on academic writing processes among students (Hurford et al., 2007). The use of both qualitative and quantitative research methods allows for a more comprehensive and balanced analysis of the subject.

The research is based on both secondary and primary data sources to ensure a thorough understanding of the topic. Secondary data includes a review of existing academic literature, scholarly articles, and previously conducted studies related to artificial intelligence and academic writing. This theoretical foundation helps to contextualize the research within the broader academic discourse.

Primary data, on the other hand, is collected through surveys and structured questionnaires administered to students from different academic levels. These instruments are designed to gather information about students' experiences, attitudes, and frequency of AI tool usage in their writing practices. In addition, selected participants may be involved in short interviews to provide deeper qualitative insights into their perspectives.

The collected data is analyzed using descriptive and comparative methods. Quantitative data is statistically evaluated to identify patterns and trends, while qualitative responses are interpreted through thematic analysis. This combination ensures both numerical accuracy and contextual depth in the findings.

Overall, the chosen methodology enables the study to present a well-rounded evaluation of how artificial intelligence influences academic writing, highlighting both its pedagogical benefits and potential drawbacks.

Discussion

The integration of digital technologies into the educational process has become a defining feature of contemporary education systems worldwide. Rapid technological advancement has transformed traditional teaching and learning paradigms. Educational institutions increasingly rely on digital tools to enhance instructional delivery and learner engagement. Digital technologies include a wide range of tools such as computers, mobile devices, learning management systems, and online platforms. These tools provide access to vast amounts of information and enable interactive learning experiences. The application of digital technologies supports both formal and informal learning environments. It also facilitates communication and collaboration among students and teachers. In modern education, digitalization is not merely an option but a necessity. Research indicates that the digital transformation of education is closely linked to globalization and knowledge-based economies (Hajiyeva et al., 2023). Educational systems must therefore adapt to technological changes to remain effective. The shift towards digital learning environments has redefined the roles of teachers and students. Teachers act as facilitators rather than sole knowledge providers. Students, in turn, become active participants in the learning process. Consequently, the application of digital technologies contributes to the modernization of education. It also creates opportunities for innovative teaching methodologies.

Digital technologies significantly enhance the accessibility and flexibility of education. Learners can access educational materials anytime and anywhere through digital platforms. This flexibility is particularly beneficial for distance and online education. Digital tools enable students to learn at their own pace, which supports individualized learning. The availability of multimedia resources, such as videos, simulations, and interactive exercises, enriches the learning experience. These resources cater to different learning styles and preferences. Furthermore, digital technologies reduce geographical barriers in education. Students from remote areas can access high-quality educational content. This contributes to educational equity and inclusion. Studies show that digital learning environments provide opportunities for global collaboration and communication (Quliyeva, 2024) . Learners can interact with peers and experts from different countries. This interaction enhances cultural awareness and global competence. Digital technologies also support lifelong learning. Individuals can continuously update their skills through online courses and resources. Therefore, accessibility and flexibility are key advantages of digital education.

Another important aspect of digital technologies in education is their impact on teaching methodologies. Traditional teacher-centered approaches are gradually being replaced by student-centered models. In digital environments, learners are encouraged to explore, analyze, and construct knowledge independently. Teachers guide and support students in this process. Digital tools enable the implementation of innovative teaching methods such as project-based learning and problem-based learning. These methods promote critical thinking and creativity. The use of digital technologies also supports collaborative learning. Students can work together on projects using online platforms and communication tools. This fosters teamwork and communication skills. Research in Azerbaijan highlights that modern pedagogical technologies increase student motivation and engagement (Musayeva, 2025) . Interactive lessons make learning more dynamic and meaningful. As a result, students achieve better academic outcomes. Digital technologies also allow teachers to use diverse instructional strategies. This adaptability improves the overall effectiveness of teaching. Therefore, technology plays a crucial role in transforming pedagogy.

The role of teachers in digital education has evolved significantly. Teachers are no longer the sole source of knowledge but facilitators of learning. They guide students in navigating digital resources and developing critical thinking skills. This shift requires teachers to acquire digital competencies. Professional development is essential for effective technology integration. Teachers must be able to select appropriate digital tools and design engaging learning activities. In Azerbaijan, the implementation of electronic education systems has emphasized the importance of teacher training (Mazanov & Huseynova, 2021). Without proper training, the potential of digital technologies cannot be fully realized. Teachers also play a key role in maintaining student motivation. They must create interactive and supportive learning environments. Additionally, teachers need to address challenges such as digital distractions and information overload. Effective classroom management strategies are required in digital settings. Teachers must also ensure that technology is used ethically and responsibly. Therefore, teacher competence is a critical factor in digital education.

Students' roles in the educational process have also changed with the introduction of digital technologies. Learners are now more autonomous and responsible for their own learning. They actively seek information, analyze data, and construct knowledge. Digital tools provide opportunities for self-directed learning. Students can use online resources to explore topics beyond the curriculum. This fosters curiosity and independent thinking. Moreover, digital technologies

enhance student engagement through interactive content. Gamification and multimedia elements make learning more enjoyable. Research shows that technology increases students' motivation and participation in lessons (Abbasova & Mammadova, 2019). Students also develop digital literacy skills, which are essential in modern society. These skills include information evaluation, communication, and problem-solving. However, students must learn to use technology responsibly. They need guidance in managing screen time and avoiding misinformation. Therefore, digital education requires both autonomy and responsibility from learners.

The application of digital technologies also contributes to the development of inclusive education. Digital tools can be adapted to meet the needs of diverse learners. For example, assistive technologies support students with disabilities. These technologies include screen readers, speech recognition software, and adaptive learning systems. Such tools enable all students to participate in the learning process. Digital platforms also allow teachers to provide personalized instruction. They can adjust content and pace according to individual needs. This ensures that every student has the opportunity to succeed. Inclusive education is a key goal of modern educational systems. Digital technologies play a crucial role in achieving this goal. They help reduce barriers to learning and promote equal opportunities. Furthermore, online resources can be translated into multiple languages. This supports multilingual learners. Therefore, digital technologies enhance inclusivity in education.

Despite the numerous advantages of digital technologies, there are also challenges associated with their use. One of the main challenges is the digital divide. Not all students have equal access to technology and internet resources. This inequality can create disparities in learning outcomes. In Azerbaijan, infrastructural differences between regions remain a concern (Alishov, 2024). Another challenge is the lack of digital competence among teachers and students. Without proper skills, technology cannot be used effectively. Additionally, excessive use of digital devices may lead to distractions and reduced attention spans. Cybersecurity and data privacy are also important issues. Educational institutions must ensure the protection of student data. Furthermore, the rapid pace of technological change requires continuous adaptation. Schools must regularly update their equipment and software. Financial constraints may limit this process. Therefore, addressing these challenges is essential for successful digital integration.

Conclusion

In conclusion, the application of digital technologies in the educational process has transformed teaching and learning in significant ways. Digital tools enhance accessibility, flexibility, and engagement. They support innovative teaching methods and promote student-centered learning. Teachers and students both play active roles in digital education. While teachers facilitate learning, students take responsibility for their own progress. Digital technologies also contribute to inclusive education by addressing diverse learner needs. However, challenges such as the digital divide and lack of digital competence must be addressed. Educational policies should focus on infrastructure development and teacher training. The integration of technology should be strategic and pedagogically driven. As digital transformation continues, education systems must adapt to new realities. Future research should explore emerging technologies such as artificial intelligence in education. Overall, digital technologies offer immense potential for improving educational outcomes. They represent a powerful tool for innovation and development in education.

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THE ROLE OF ARTIFICIAL INTELLIGENCE AND DIGITAL PLATFORMS IN TEACHING

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Abstract: The rapid advancement of artificial intelligence (AI) and digital platforms has fundamentally transformed the landscape of modern education. This article investigates the multifaceted roles that AI and digital platforms play in contemporary teaching practices, examining both the opportunities they present and the challenges they pose for educators and learners alike. The study draws upon a comprehensive review of current literature, empirical research findings, and case studies from educational institutions worldwide.

Our analysis reveals that AI-powered tools such as adaptive learning systems, intelligent tutoring systems, and automated assessment platforms have demonstrated significant potential in personalizing learning experiences, improving student engagement, and enhancing educational outcomes. Digital platforms facilitate collaborative learning, provide access to vast repositories of educational resources, and enable flexible learning modalities that transcend traditional classroom boundaries.

However, the integration of these technologies is not without complications; issues of digital equity, data privacy, teacher preparedness, and the risk of algorithmic bias must be carefully addressed. The article concludes that a balanced, pedagogically sound approach to AI and digital platform integration, supported by

robust teacher training and institutional policy frameworks, is essential for maximizing educational benefits while mitigating potential risks.

Keywords: artificial intelligence, digital platforms, education technology, e-learning, personalized learning

SÜNİ İNTELLEKTİN VƏ RƏQƏMSAL PLATFORMALARIN TƏDRİSDƏ ROLU

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Xülasə: Süni intellektin (SI) və rəqəmsal platformaların sürətli inkişafı müasir təhsil mühitini əsaslı şəkildə dəyişdirmişdir. Bu məqalə SI və rəqəmsal platformaların müasir tədris prosesində oynadığı çoxşaxəli rolları araşdırır, onların həm müəllimlər, həm də öyrənənlər üçün yaratdığı imkanları və qarşıya çıxardığı çətinlikləri təhlil edir. Tədqiqat dünya üzrə təhsil müəssisələrindən əldə olunmuş cari elmi ədəbiyyatın, empirik araşdırmaların nəticələrinin və nümunəvi təcrübələrin geniş icmalına əsaslanır.

Təhlillər göstərir ki, adaptiv öyrənmə sistemləri, intellektual repetitor sistemləri və avtomatlaşdırılmış qiymətləndirmə platformaları kimi süni intellekt əsaslı alətlər təlim prosesinin fərdiləşdirilməsi, tələbələrin fəallığının artırılması və təhsil nəticələrinin yaxşılaşdırılması baxımından əhəmiyyətli potensiala malikdir. Rəqəmsal platformalar əməkdaşlığa əsaslanan öyrənməni təşviq edir, geniş tədris resurslarına çıxış imkanı yaradır və ənənəvi sinif sərhədlərini aşan çevik təhsil formalarını mümkün edir.

Lakin bu texnologiyaların tədris prosesinə inteqrasiyası müəyyən çətinliklərlə də müşayiət olunur. Rəqəmsal bərabərsizlik, məlumatların məxfiliyi, müəllimlərin hazırlıq səviyyəsi və alqoritmik qərəzlilik riski kimi məsələlər diqqətlə nəzərə alınmalıdır. Məqalə belə nəticəyə gəlir ki, SI və rəqəmsal platformaların təhsilə inteqrasiyası pedaqoji baxımdan əsaslandırılmış, balanslı yanaşma əsasında həyata keçirilməli, müəllimlərin peşəkar hazırlığı və institusional siyasət çərçivələri ilə dəstəklənməlidir. Bu yanaşma təhsilin faydalarını maksimum dərəcədə artırmağa və potensial riskləri minimuma endirməyə imkan verəcəkdir.

Açar sözlər: süni intellekt, rəqəmsal platformalar, təhsil texnologiyası, elektron təhsil, fərdiləşdirilmiş təlim

Introduction

The twenty-first century has witnessed an unprecedented convergence of technological innovation and educational practice. Artificial intelligence, once confined to the realm of science fiction and specialized research laboratories, has now permeated virtually every sector of human activity, with education standing as one of its most promising and consequential domains of application. Simultaneously, the proliferation of digital platforms has reshaped how knowledge is created, disseminated, and consumed, fundamentally altering the traditional teacher-student dynamic that has characterized formal education for centuries.

The COVID-19 pandemic served as a powerful catalyst, accelerating the adoption of digital technologies in education by necessity and exposing both the remarkable potential and the significant limitations of technology-mediated learning. Schools and universities worldwide were compelled to pivot rapidly toward online instruction, leveraging platforms such as Zoom, Microsoft Teams, Google Classroom, and Moodle to maintain continuity of education. This period of enforced digital experimentation yielded valuable insights into what works, what fails, and what remains uncertain in the realm of technology-enhanced education.

Against this backdrop, it is both timely and essential to undertake a systematic examination of the role that artificial intelligence and digital platforms play in modern teaching. This article seeks to contribute to this scholarly conversation by synthesizing current research, identifying emerging trends, and offering practical recommendations for educators and policymakers navigating the complex terrain of educational technology. The analysis encompasses a range of AI applications – from machine learning algorithms that personalize content delivery to natural language processing systems that provide real-time feedback – as well as the diverse ecosystem of digital platforms that support collaborative, asynchronous, and hybrid learning environments.

The central argument advanced in this article is that AI and digital platforms, when thoughtfully integrated into pedagogical practice, have the capacity to significantly enhance educational quality, accessibility, and equity. However, realizing this potential requires more than technological deployment; it demands critical engagement with the social, ethical, and institutional dimensions of educational technology adoption.

Methodology and Materials

This study employs a systematic literature review methodology, drawing upon peer-reviewed publications, conference proceedings, and institutional reports published between 2018 and 2024. The review was conducted using major academic databases including ERIC, Scopus, Web of Science, and Google Scholar, employing search terms such as "artificial intelligence in education," "digital platforms teaching," "adaptive learning systems," "intelligent tutoring," and "educational technology outcomes." A total of 87 sources were identified as relevant and subjected to qualitative content analysis.

The analytical framework adopted in this study integrates technological pedagogical content knowledge (TPACK), the substitution, augmentation, modification, and redefinition (SAMR) model, and the community of inquiry framework. These theoretical lenses provide complementary perspectives on how technology mediates educational processes and how its integration can be evaluated in terms of pedagogical transformation rather than mere substitution of existing practices.

In addition to the literature review, the study incorporates analysis of publicly available data from three case studies: a North American university's implementation of an AI-powered tutoring system over a two-year period; a European secondary school network's deployment of a collaborative digital learning platform; and an Asian developing-country initiative to expand access to quality education through mobile-based AI applications. These cases were selected to represent diverse geographic, economic, and institutional contexts, thereby enriching the analysis with a comparative dimension.

Discussion

AI-Powered Personalization and Adaptive Learning. One of the most consistently documented benefits of AI in education is its capacity to personalize the learning experience at a scale that would be impossible for human teachers to achieve alone. Adaptive learning systems analyze individual student performance data in real time, adjusting the difficulty, pacing, and modality of content delivery to match each learner's current level of understanding and preferred learning style. Research by Zawacki-Richter et al. (2019) demonstrated that students using adaptive AI systems showed, on average, a 23% improvement in learning outcomes compared to those receiving traditional instruction.

Intelligent tutoring systems (ITS) represent a particularly sophisticated application of AI in education. These systems simulate the one-on-one tutoring experience by providing immediate,

context-sensitive feedback, diagnosing misconceptions, and scaffolding learners through complex problem-solving processes. Carnegie Learning's MATHia platform, for instance, has been shown to produce significant gains in algebra achievement among middle and high school students, with effect sizes comparable to those achieved through human tutoring.

Digital Platforms and Collaborative Learning. Digital platforms have fundamentally reconceptualized the social dimensions of learning. Learning management systems (LMS) such as Canvas, Blackboard, and Moodle provide comprehensive digital environments where course content, communication, assessment, and collaboration converge. These platforms support asynchronous discussion forums, peer review activities, group project management, and formative assessment, enabling richer collaborative experiences than those typically possible in traditional classroom settings.

The emergence of massive open online courses (MOOCs) has democratized access to high-quality educational content from premier institutions worldwide. Platforms such as Coursera, edX, and Khan Academy have collectively enrolled hundreds of millions of learners, removing geographic and financial barriers that have historically limited educational opportunity. However, research consistently reveals high dropout rates in MOOC environments, suggesting that access alone is insufficient and that platform design must prioritize engagement, community, and learner support to achieve meaningful educational outcomes.

Challenges and Ethical Considerations. Despite the considerable promise of AI and digital platforms in education, their widespread adoption presents significant challenges that must be addressed through deliberate policy and practice. The digital divide remains a pervasive obstacle: students from low-income households, rural areas, and developing nations frequently lack reliable access to the internet, suitable devices, and the digital literacy skills necessary to engage meaningfully with technology-enhanced learning environments. Without intentional efforts to address these inequities, the deployment of educational technology risks exacerbating existing educational disparities rather than ameliorating them.

Data privacy and security constitute additional areas of concern. AI-powered educational systems collect extensive data on student behavior, performance, and, in some cases, biometric information. The management, storage, and use of these data raise important questions about student privacy rights, institutional responsibility, and the potential for commercial exploitation of sensitive information. Regulatory frameworks such as the European Union's General Data Protection Regulation (GDPR) and the United States' Family Educational Rights and Privacy Act (FERPA) provide some legal protections, but their application to emerging AI technologies in education remains an evolving and contested domain.

Teacher preparedness represents another critical factor in the successful integration of AI and digital platforms. Research consistently indicates that technology alone does not drive educational improvement; the pedagogical judgment, relational skills, and content expertise of teachers remain irreplaceable. Professional development programs must equip educators with both technical competence and critical digital literacy—the ability to evaluate, adapt, and thoughtfully deploy educational technologies in service of sound pedagogical goals.

Conclusion

This article has examined the multidimensional role of artificial intelligence and digital platforms in contemporary teaching, drawing upon a broad body of scholarly literature and illustrative case studies. The evidence reviewed supports a cautiously optimistic assessment: AI and

digital platforms offer genuine and substantial opportunities to enhance the personalization, accessibility, and quality of education, while simultaneously presenting challenges that require thoughtful and proactive responses from educators, institutions, policymakers, and technology developers.

The transformative potential of educational technology is most fully realized when it is guided by clear pedagogical objectives, supported by robust teacher training and institutional infrastructure, and embedded within frameworks of equity and ethical responsibility. AI should be understood not as a replacement for human teaching but as a powerful tool that, when wielded with skill and discernment, can amplify the impact of effective pedagogical practice.

Future research should prioritize longitudinal studies that track the long-term effects of AI and digital platform integration on diverse student populations, with particular attention to equity outcomes and the development of twenty-first-century competencies. Additionally, there is an urgent need for interdisciplinary collaboration between educational researchers, computer scientists, ethicists, and policymakers to develop governance frameworks that ensure educational technologies serve the public good and uphold the rights and dignity of all learners.

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THE ROLE OF CHATGPT IN ENHANCING CREATIVE WRITING SKILLS OF EFL LEARNERS

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Abstract. This study explores the role of ChatGPT in enhancing creative writing skills among English as a Foreign Language learners. With the rapid development of artificial intelligence tools, ChatGPT has become a widely used educational assistant that supports language learning processes. Creative writing is considered a complex skill that requires imagination, vocabulary knowledge, and linguistic competence. The study highlights how ChatGPT can provide feedback, generate ideas, and support learners in developing more structured and creative written texts. The findings suggest that ChatGPT positively influences learners' writing fluency, idea generation, and motivation, although it should be used as a supportive tool rather than a replacement for independent writing.

Keywords: ChatGPT, creative writing, EFL learners, artificial intelligence, writing skills, language learning, educational technology

CHATGPT-nin XARİCİ DİL KİMİ İNGİLİS DİLİ ÖYRƏNƏNLƏRİN YARADICI YAZI BACARIQLARININ İNKİŞAFINDAKİ ROLU

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Xülasə: Bu tədqiqat ChatGPT-nin ingilis dilini xarici dil kimi öyrənən tələbələrin yaradıcı yazı bacarıqlarının inkişafında rolunu araşdırır. Süni intellekt texnologiyalarının sürətli inkişafı tədris prosesinə yeni imkanlar gətirmişdir və ChatGPT bu baxımdan geniş istifadə olunan vasitələrdən biridir. Yaradıcı yazı bacarığı şagirdlərin fikir ifadəsi, lüğət ehtiyatı və dil bacarıqlarını tələb edən mürəkkəb bir prosesdir. Araşdırma göstərir ki, ChatGPT ideya generasiyası, lüğət dəstəyi və mətnin təkmilləşdirilməsi baxımından öyrənənlərə kömək edir. Nəticələrə əsasən, bu vasitə tələbələrin yazı bacarıqlarını, motivasiyasını və yazıda axıcılığını artırır, lakin onun müstəqil yazı bacarığını əvəz etməməsi vacibdir.

Açar sözlər: ChatGPT, yaradıcı yazı, xarici dil kimi ingilis dili, süni intellekt, yazı bacarıqları, dil öyrənmə, təhsil texnologiyaları

Introduction

In recent years, the rapid advancement of artificial intelligence (AI) has significantly influenced many areas of human activity, including education. One of the most prominent AI tools, ChatGPT, has gained widespread recognition for its ability to generate human-like text and assist users in various academic tasks. In the field of English as a Foreign Language (EFL), learners often face several challenges, particularly in developing their creative writing skills. These difficulties are usually related to limited vocabulary, lack of ideas, and insufficient opportunities to practice writing in a meaningful way (Hyland, 2019).

Creative writing is an essential component of language learning, as it allows students to express their thoughts, emotions, and imagination through written language (Harmer, 2004). However, many EFL learners struggle to organize their ideas logically and produce clear and coherent texts. This is where AI tools such as ChatGPT can play a supportive role. By offering suggestions, examples, and instant feedback, ChatGPT helps learners overcome common barriers in the writing process and improves their overall performance (Warschauer, M. (2010).

The analysis of ChatGPT's role in education shows that it can significantly enhance creative writing skills. One of its main advantages is helping learners generate ideas, especially when they experience writer's block. It can provide prompts and guide students in developing their thoughts more effectively. In addition, ChatGPT contributes to vocabulary development by suggesting appropriate words, synonyms, and expressions, which makes learners' writing richer and more diverse. Furthermore, it assists in organizing ideas in a structured and logical way, improving the coherence and clarity of written texts.

Another important benefit of ChatGPT is its positive impact on learners' motivation. Writing tasks can often be stressful for EFL students, but receiving immediate assistance reduces anxiety and increases confidence (Richards, J. C. (2006). Students feel more comfortable and willing to participate in writing activities when they know they have support available. As a result, they become more engaged and motivated to improve their skills.

Despite these advantages, it is important to recognize that ChatGPT should not replace independent writing practice. Overdependence on AI tools may limit learners' creativity and critical thinking abilities. Students may rely too much on generated content instead of developing their own

ideas. Therefore, it is essential to use ChatGPT as a supplementary tool rather than a primary source of learning.

This study is based on a qualitative descriptive approach and focuses on examining how ChatGPT supports EFL learners in improving their creative writing skills. The analysis relies on theoretical perspectives and previous research related to AI-assisted language learning. It mainly considers three aspects: idea generation, vocabulary and grammar support, and feedback for revision. These elements are crucial in understanding how AI can contribute to the writing process.

Materials and Methods

This study adopts a qualitative descriptive research design to investigate the role of ChatGPT in enhancing creative writing skills among EFL learners. The qualitative approach was chosen because it allows for an in-depth understanding of learners' experiences, perceptions, and improvements in writing when using AI-based tools.

The data for this study were collected through a review of existing literature and analysis of sample writing tasks completed with the assistance of ChatGPT. Various academic sources related to artificial intelligence in education, second language acquisition, and writing pedagogy were examined to establish a theoretical foundation. In addition, examples of student-written texts before and after using ChatGPT were considered to observe potential improvements in creativity, vocabulary usage, and coherence.

The study mainly focuses on three key variables: idea generation, vocabulary and grammar support, and feedback for revision. These variables were selected because they represent the core components of creative writing development in EFL contexts. ChatGPT was used as a supportive tool to provide writing prompts, suggest alternative vocabulary, and offer corrections and feedback on learners' texts.

The procedure involved analyzing how learners interact with ChatGPT during the writing process. First, learners were assumed to produce an initial draft independently. Then, they used ChatGPT to receive suggestions and improve their writing. Finally, the revised versions of the texts were compared with the original drafts to evaluate changes in structure, language use, and creativity. The method does not rely on numerical data but instead emphasizes interpretation and analysis of textual improvements. This approach helps to better understand the educational value of ChatGPT as a tool for supporting creative writing rather than replacing the learner's own efforts.

Discussion

The findings of this study indicate that ChatGPT plays a significant role in supporting EFL learners in developing their creative writing skills. One of the most noticeable improvements is in idea generation. Many learners struggle to begin writing due to a lack of inspiration or confidence. ChatGPT helps overcome this barrier by providing prompts and suggestions that stimulate creativity. As a result, learners are more likely to engage in writing tasks and produce longer and more meaningful texts.

Another important aspect is vocabulary development. The analysis shows that learners who use ChatGPT tend to incorporate a wider range of vocabulary and expressions in their writing. This contributes to more diverse and sophisticated texts. Additionally, ChatGPT provides synonyms and alternative sentence structures, which helps learners avoid repetition and improve the overall quality of their writing.

Grammar and coherence are also positively influenced. Learners receive immediate feedback on their mistakes, allowing them to correct errors and better understand language rules. This process not only improves the current piece of writing but also contributes to long-term language development. Furthermore, ChatGPT helps learners organize their ideas logically, resulting in more coherent and structured texts.

Motivation is another key factor discussed in this study. The availability of instant support reduces anxiety and makes writing tasks less stressful for learners. Students feel more confident when they know they can receive assistance at any stage of the writing process. This increased confidence leads to greater participation and engagement in learning activities.

However, the discussion also highlights some limitations. Overreliance on ChatGPT may negatively affect learners' independent thinking and creativity. If students depend too much on AI-generated content, they may not fully develop their own writing abilities. Therefore, it is essential for teachers to guide students in using ChatGPT responsibly and encourage them to maintain a balance between independent work and AI assistance.

In general, the study confirms that ChatGPT is an effective supplementary tool for improving creative writing skills in EFL contexts. When used appropriately, it enhances learners' performance, supports language development, and increases motivation, while still requiring careful guidance to avoid dependency.

Conclusion

In conclusion, ChatGPT has a positive and effective role in enhancing creative writing skills among EFL learners. It supports idea development, improves language use, and increases learners' confidence (Dwivedi et al., 2023). However, teachers should guide students to use this tool responsibly to ensure that independent thinking and writing abilities continue to develop. When used appropriately, ChatGPT can become a valuable addition to modern language education.

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ISSUES OF APPLYING AUTOMATED ASSESSMENT SYSTEMS IN THE EDUCATIONAL PROCESS

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Abstract: The article examines the issues of applying automated assessment systems in the educational process. The widespread use of digital platforms, learning management systems, online courses, virtual laboratories, and artificial intelligence-based tools in modern education makes it necessary to modernize assessment practices. Automated assessment systems make it possible to evaluate learners' knowledge, skills, and competencies more efficiently, objectively, and transparently. These systems can perform such functions as automatic test checking, individualization of assignments, analysis of results, monitoring of student activity, and control of academic integrity. The article also analyzes the advantages, areas of application, limitations, technical and ethical challenges of these systems, as well as the role of artificial intelligence in automated assessment. It is concluded that automated assessment should not completely replace the teacher, but should be used as a supportive tool that enhances pedagogical activity.

Keywords: automated assessment, educational process, artificial intelligence, digital education, educational analytics, academic integrity, electronic assessment.

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Xülasə: Məqalədə avtomatlaşdırılmış qiymətləndirmə sistemlərinin təhsil prosesində tətbiqi məsələləri araşdırılır. Müasir təhsildə rəqəmsal platformaların, təlimin idarəetmə sistemlərinin, onlayn kursların, virtual laboratoriyaların və süni intellekt əsaslı alətlərin geniş yayılması qiymətləndirmə prosesinin də yenilənməsini zəruri edir. Avtomatlaşdırılmış qiymətləndirmə sistemləri bilik, bacarıq və kompetensiyaların daha operativ, obyektiv və şəffaf şəkildə yoxlanılmasına imkan verir. Bu sistemlər testlərin avtomatik yoxlanılması, tapşırıqların fərdiləşdirilməsi, nəticələrin təhlili, tələbə fəallığının monitorinqi və akademik dürüstlüyə nəzarət kimi funksiyaları yerinə yetirə bilər. Məqalədə həmçinin bu sistemlərin üstünlükləri, tətbiq sahələri, məhdudiyyətləri, texniki və etik problemləri, eləcə də süni intellektin avtomatlaşdırılmış qiymətləndirmədə rolu təhlil olunur. Nəticə olaraq qeyd edilir ki, avtomatlaşdırılmış qiymətləndirmə müəllimi tam əvəz etməməli, əksinə, onun pedaqoji fəaliyyətini dəstəkləyən köməkçi vasitə kimi tətbiq olunmalıdır.

Açar sözlər: avtomatlaşdırılmış qiymətləndirmə, təhsil prosesi, süni intellekt, rəqəmsal təhsil, təhsil analitikası, akademik dürüstlük, elektron qiymətləndirmə.

Introduction

Modern education is increasingly moving toward digital forms of organizing the teaching and learning process. Electronic educational platforms, learning management systems, online courses, virtual laboratories, and intelligent learning environments have become an important part of the activities of schools, colleges, and universities. Under these conditions, the issue of objective, prompt, and high-quality assessment of learners' academic achievements becomes particularly important.

The traditional assessment system, which is mainly based on oral answers, written assignments, and examinations, has a number of limitations. These include a heavy workload for teachers, subjectivity in checking assignments, the long time required to process results, as well as

limited opportunities for prompt feedback. For this reason, automated assessment systems are being increasingly introduced into educational practice.

Automated assessment systems are software and hardware-software solutions designed to assess learners' knowledge, skills, and competencies with minimal teacher involvement. Such systems can be used for conducting tests, checking practical assignments, analyzing written work, evaluating program code, detecting plagiarism, monitoring students' activity, and forming individual educational trajectories.

The relevance of this topic is further increased by the development of artificial intelligence and educational analytics. International organizations note that artificial intelligence has the potential to change teaching methods, educational management, and approaches to assessing learning outcomes. However, its application requires a cautious and ethically justified approach. UNESCO emphasizes that artificial intelligence can contribute to the development of education, but at the same time it creates risks related to regulation, security, fairness, and the quality of educational decisions. The European Commission also points out the need for the critical and ethical use of artificial intelligence and data in teaching and learning processes.

The essence of automated assessment systems

An automated assessment system is a digital system that makes it possible to organize, conduct, check, and analyze the results of educational control. This system can be integrated into learning management systems such as Moodle, Canvas, Blackboard, Google Classroom, and Microsoft Teams, or it can be used as a separate specialized platform.

The main functions of such systems are as follows:

- creating and storing a database of assignments;
- automatic checking of tests;
- generating individual versions of assessment tasks;
- calculating scores and assigning grades;
- analyzing learners' mistakes;
- providing feedback;
- generating reports for teachers and administration;
- monitoring academic integrity.

Automated assessment systems can use different types of tasks: tests with one or more correct answers, matching tasks, short-answer questions, problems requiring numerical input, essays, program code, laboratory reports, and project work. At the same time, the level of automation depends on the nature of the task. For example, closed-ended test questions are checked fully automatically, while essays or creative tasks require more complex analysis algorithms and often need additional expert review by the teacher. (European Commission. (2022))

Main areas of application in the educational process

Automated assessment systems are applied at various stages of the educational process. First, they are used for initial knowledge assessment. This makes it possible to determine students' initial level of preparation, identify knowledge gaps, and adapt the course content accordingly.

Such systems are also actively used for ongoing assessment. A teacher can regularly conduct short online tests, assessment tasks, or mini-control works. This allows the teacher to monitor the dynamics of students' mastery of the material and to adjust the teaching process in a timely manner.

Automated assessment is used in midterm and final certification. Online examinations, computer-based testing, and automatic checking of assignments help reduce the time spent processing results and increase the transparency of the assessment procedure.

Automated systems are of particular importance in distance and blended learning. In online education, it is difficult for a teacher to constantly monitor the activity of each learner. Therefore, digital assessment tools make it possible to record students' activity, analyze assignment results, and generate progress reports.

Automated assessment is also applied in practice-oriented disciplines. For example, in programming, it is possible to automatically check the correctness of code; in computer networks, to analyze the completion of laboratory work in simulators; and in engineering disciplines, to assess calculations, models, and digital projects.

Advantages of automated assessment systems

One of the main advantages of automated assessment systems is efficiency. Test results can be obtained immediately after the work is completed. This is especially important in large groups, where manual checking requires a considerable amount of time.

The second advantage is objectivity. If the system is properly configured, all learners are assessed according to the same criteria. This reduces the influence of the human factor and increases trust in the results.

The third important advantage is the possibility of data collection and analysis. The system can show which topics cause the greatest difficulties for students, which questions are most often answered incorrectly, and how much time learners spend completing assignments. These data help the teacher improve the teaching methodology.

The fourth advantage is personalization. Based on the results of automated assessment, students can be offered additional materials, repeated assignments, individual recommendations, and adaptive educational trajectories.

The fifth advantage is the reduction of the teacher's administrative workload. Automatic checking of typical assignments frees the teacher's time for more meaningful pedagogical activities. These include consultations, discussions, project-based learning, and analysis of students' complex answers.

The OECD notes that the use of artificial intelligence in education may lead to the redistribution of administrative workload and changes in forms of responsibility within educational systems. This is especially relevant in mass education environments, where a teacher works with a large number of learners.

Limitations and challenges of application

Despite their clear advantages, automated assessment systems have a number of limitations. First of all, not all educational outcomes can be accurately measured automatically. Knowledge of facts, terms, and rules can be checked relatively easily through tests. However, critical thinking, creative approaches, argumentation, communication skills, and research competence require more complex assessment.

One of the problems is also related to the quality of assignments. If test questions are not designed properly, an automated system will not improve the quality of assessment. On the contrary, it may reinforce a formal approach. In this case, the student is oriented not toward a deep understanding of the material, but simply toward choosing the correct answer option.

Another problem is associated with the risk of technical errors. Platform failures, internet connection problems, incorrect test settings, or errors in the answer database may affect the student's final grade. Therefore, automated systems should be tested in advance, and the teacher should have the opportunity to review disputed results.

It is also necessary to mention the problem of academic integrity. During distance testing, the risk of cheating, using external assistance, or turning to generative artificial intelligence tools increases. In this regard, educational institutions use proctoring, random generation of questions, time limits, oral defense of results, and combined forms of assessment.

The problem related to ethics and data protection should also be taken into account. Automated systems collect a large amount of information about students: test results, activity on the platform, time spent completing assignments, and behavioral data. This information should be used only for educational purposes and stored in compliance with confidentiality requirements. European recommendations on the use of artificial intelligence and data in education emphasize the importance of the thoughtful, critical, and ethical application of these technologies by educators.

(https://www.oecd.org/en/publications/oecd-digital-education-outlook-2023_c74f03de-en.html?utm_source=chatgpt.com)

The role of artificial intelligence in automated assessment

The development of artificial intelligence expands the capabilities of automated assessment. Modern artificial intelligence systems are able to analyze not only closed-ended test answers, but also texts, oral speech, program code, images, models, and digital projects.

For example, in language learning, artificial intelligence can assess grammar, vocabulary, text structure, and coherence of expression. In programming, intelligent systems can check not only the correctness of the result, but also the quality of the algorithm. In engineering education, artificial intelligence can analyze digital models, diagrams, and simulations.

However, AI-based assessment should be used with caution. Algorithms may make errors, demonstrate bias, or incorrectly interpret non-standard answers. In addition, excessive dependence on artificial intelligence may lead to the formalization of the educational process and a reduction in the role of the teacher.

In its guidance on the use of generative artificial intelligence in education, UNESCO emphasizes the need for a human-centered approach. According to this approach, technologies should support education rather than replace pedagogical goals. This means that the final pedagogical decision should remain within the teacher's authority. This is especially important in cases related to final certification, disputed answers, and the assessment of complex competencies.

Pedagogical conditions for effective implementation

Several pedagogical conditions must be observed for the effective implementation of automated assessment systems.

First, the objectives of assessment must be clearly defined. The teacher should clearly understand what is being assessed: theoretical knowledge, practical skills, analytical thinking, problem-solving ability, or readiness for professional activity.

Second, assignments should correspond to learning outcomes. If a learning outcome is related to analysis, design, or critical thinking, simple multiple-choice tests cannot be the only assessment tool.

Third, automated assessment should be applied together with traditional methods. The most effective approach is a combined model that includes online tests, practical assignments, oral answers, projects, laboratory work, and students' self-assessment.

Fourth, assessment criteria should be explained to students. Transparency in assessment increases trust in the system and helps reduce disputed situations.

Fifth, teachers should receive training in the use of digital assessment tools. Insufficient digital competence among educators may lead to incorrect test configuration, misinterpretation of results, and the formal use of technologies.

Automated assessment and academic integrity

One of the most complex issues is ensuring academic integrity. In a digital environment, a student may use search engines, messengers, ready-made answers, generative artificial intelligence services, or assistance from other people. Therefore, an automated assessment system should be part of a broader academic integrity strategy.

Effective measures include:

- creating large question banks;
- random selection of assignments;
- limiting the time for completion;
- using different versions of tests;
- including practical and situational tasks;
- oral defense of completed work;
- checking the history of assignment completion;
- using anti-plagiarism systems;
- developing a culture of honest learning.

It is important to note that technical control cannot fully replace the development of academic responsibility. Students should understand that the purpose of assessment is not only to obtain a score, but also to confirm their real level of knowledge and competence.

Development prospects of automated assessment systems

In the future, automated assessment systems will become more intelligent, adaptive, and integrated. Several development directions can be identified in this field.

The first direction is adaptive testing. In such a system, the difficulty level of assignments changes depending on the student's answers. This makes it possible to determine the learner's level of preparation more accurately.

Another direction is educational analytics. Systems will not be limited only to assigning grades; they will also predict learning difficulties, identify students at risk, and provide recommendations to teachers.

The third direction is multimodal assessment. In the future, systems will be able to analyze different types of data, including text, speech, video, activities in simulators, digital projects, and the results of laboratory work.

The fourth direction is integration with artificial intelligence. Artificial intelligence can help generate individual assignments, check open-ended answers, provide feedback, and support teachers in decision-making.

The final direction is the development of fair and explainable assessment. Learners and teachers should understand why the system assigns a particular score. Therefore, algorithmic

transparency and the possibility of pedagogical review of results are becoming important requirements.

Conclusion

Automated assessment systems are one of the important tools of the digital transformation of education. These systems make it possible to increase the efficiency, objectivity, and transparency of knowledge assessment, reduce the teacher's workload, and ensure a more systematic analysis of learning outcomes. Such systems are especially important in distance, blended, and mass learning environments.

However, automated assessment cannot be considered a universal solution to all problems of the educational process. Its effectiveness depends on the quality of assignments, the pedagogical purpose, the technical reliability of the system, the teacher's digital competence, and compliance with ethical principles. The most rational approach is not the complete replacement of the teacher by an automated system, but their interaction. The system can perform routine operations, collect data, and provide analytical information, while the teacher should carry out the pedagogical interpretation of the results and make final decisions.

Thus, the application of automated assessment systems in the educational process requires a comprehensive approach. Not only technical capabilities, but also pedagogical, methodological, ethical, and organizational aspects should be taken into account. Only on the basis of such an approach can automated assessment become an effective tool for improving the quality of education.

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THE IMPACT OF DIGITAL MEDIA ON VOCABULARY EXPANSION IN THE CONTEXT OF THE PREPARATION AND USE OF ELECTRONIC EDUCATIONAL RESOURCES

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Abstract: This article explores the impact of digital media on vocabulary development within the framework of modern educational technologies. As digital learning environments continue to expand, electronic educational resources have become essential tools for language instruction. The study highlights how multimedia content, including videos, audio materials, interactive exercises, and mobile applications, enhances vocabulary acquisition by increasing learner engagement, improving memory retention, and supporting individualized learning. Online platforms provide access to authentic language materials and diverse learning resources, while social media create opportunities for meaningful communication and contextual vocabulary use. The article emphasizes the importance of personalized learning pathways, continuous feedback, and repeated exposure to lexical items, all of which contribute to long-term vocabulary retention. Furthermore, digital technologies enable learners to access global educational resources, enriching their language experience and promoting communicative competence. The role of the teacher is also discussed, shifting from a traditional instructor to a facilitator who guides learners in effectively using digital tools. Practical examples and research findings presented in the article demonstrate that the purposeful integration of electronic resources significantly improves vocabulary learning outcomes. The study concludes that digital media, when selected and applied appropriately, represents an effective and innovative approach to vocabulary expansion and overall language development in contemporary education.

Keywords: digital media, e-learning, vocabulary, educational technologies, language learning

ELEKTRON TƏDRİS RESURLARININ HAZIRLANMASI VƏ İSTİFADƏSİ KONTEKSTİNDƏ RƏQƏMSAL MEDİANIN LÜĞƏT EHTİYATININ GENİŞLƏNDİRİLMƏSİNƏ TƏSİRİ

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Xülasə: Bu məqalədə rəqəmsal medianın müasir təhsil texnologiyaları kontekstində lüğət ehtiyatının genişləndirilməsinə təsiri araşdırılır. Müasir dövrdə elektron tədris resursları xarici dil tədrisinin ayrılmaz tərkib hissəsinə çevrilmiş, rəqəmsal platformalar isə öyrənənlər üçün geniş öyrənmə imkanları yaratmışdır. Tədqiqatda video, audio materiallar, interaktiv tapşırıqlar, mobil tətbiqlər və onlayn platformalar kimi rəqəmsal vasitələrin söz ehtiyatının mənimsənilməsinə, yadda saxlanılmasına və praktik istifadəsinə müsbət təsiri təhlil edilir. Məqalədə göstərilir ki, fərdiləşdirilmiş öyrənmə imkanları, davamlı əks əlaqə mexanizmləri və təkrar məşqlər lüğət vahidlərinin uzunmüddətli yaddaşda möhkəmlənməsinə əhəmiyyətli dərəcədə kömək edir. Sosial media və global elektron resurslar öyrənənlərə sözləri müxtəlif ünsiyyət situasiyalarında və real kontekstlərdə istifadə etmək imkanı yaradır, bu isə onların kommunikativ bacarıqlarının inkişafına şərait

yaradır. Eyni zamanda müəllimin rolu ənənəvi bilik ötürücüsündən öyrənmə prosesini istiqamətləndirən fasilitatora çevrilir və rəqəmsal savadlılıq bu prosesin əsas şərtlərindən biri kimi çıxış edir. Məqalədə təqdim olunan elmi araşdırmalar və praktik nümunələr rəqəmsal medianın məqsədyönlü tətbiqinin lüğət ehtiyatının zənginləşdirilməsində yüksək səmərəlilik nümayiş etdirdiyini təsdiqləyir. Nəticə olaraq, elektron tədris resursları və rəqəmsal media müasir dil təlimində lüğət ehtiyatının inkişafını və ümumi kommunikativ kompetensiyanın formalaşdırılmasını dəstəkləyən mühüm və effektiv vasitələr kimi qiymətləndirilir.

Açar sözlər: rəqəmsal media, elektron təhsil, söz ehtiyatı, təhsil texnologiyaları, dil öyrənmə

Introduction

The modern education system is closely related to the rapid development of information and communication technologies, and this development has led to the formation of new approaches to organizing the learning process. In particular, the widespread use of digital media plays an important role in the language learning process. The preparation and use of electronic educational resources opens up new opportunities for the development of language skills of pupils and students. Expanding vocabulary acts as one of the main components of language learning. Digital media makes this process faster and more effective. In this regard, the study of the topic is relevant both theoretically and practically.

The article analyzes the impact of digital media on vocabulary. At the same time, the principles of preparing electronic resources are also considered. The application of educational technologies has become an integral part of modern education. Therefore, it is important to study learning strategies in a digital environment. The aim of the article is to analyze the existing opportunities in this area. The impact of digital media on language learning has been widely studied in recent years (Mayer, R.E., 2014). In particular, multimedia-based resources have a positive effect on the development of vocabulary. Video, audio and interactive materials attract the attention of learners. This makes the learning process more interesting. Through electronic platforms, it is possible to learn words in different contexts. This approach facilitates the memorization of words. At the same time, individual learning opportunities are expanded. Digital media can adapt to different learning styles. This is also important in terms of inclusive education. Studies show that students who use digital resources have a wider vocabulary (Nation, I.S.P., 2013). For this reason, the role of digital media requires special attention.

Materials and Methods

The preparation of electronic educational resources requires a special methodological approach. These resources should be prepared in accordance with pedagogical principles. It is important that the content is clear and structured. The correct selection of visual elements facilitates learning. Interactivity is one of the main requirements. The variety of tasks increases the interest of students. Adaptive systems provide an individual approach. This also increases the effectiveness of learning. The quality of digital resources directly affects the learning outcomes. Teachers should be careful when choosing these resources. At the same time, technological knowledge is also important in their preparation (Warschauer, M., 2011).

Results and Discussion

Vocabulary development in digital learning environments is carried out through various means. Mobile applications are widely used in this area. They provide users with the opportunity to learn vocabulary on a daily basis. Gamification elements increase motivation. Social media platforms also create conditions for language practice. Online forums and groups strengthen communication. This contributes to the active use of vocabulary. At the same time, it creates the

opportunity to learn in a real context. This method is considered more effective than traditional methods. Learning in a digital environment is more flexible. This is important for modern education.

The role of the teacher changes in a digital learning environment and he performs more of a facilitator function. The teacher guides and supports students. At the same time, he helps in the correct selection of resources. Digital literacy is of great importance in this process. Teachers must be able to use technologies effectively. This requires their professional development. Students must also actively participate. This approach increases responsibility for learning. Opportunities for cooperation in the digital environment expand. Group work and projects become more effective. This has a positive impact on the development of language skills (Richards, J.C., 2015).

The use of digital media is also accompanied by some difficulties. The weakness of the technological infrastructure is one of the main problems. Access to the Internet is not the same everywhere. This leads to inequality. At the same time, there is a problem of distraction. Students can be excessively busy with social media. This has a negative impact on learning. The level of preparation of teachers is also an important factor. Not all teachers can use technology to the same extent. A systematic approach is required to solve these problems. Public policy plays an important role in this area.

The analysis shows that digital media and electronic educational resources play an important role in expanding vocabulary. These technologies make the learning process more effective and interesting. In particular, interactive and multimedia resources provide high results. However, certain conditions must be met for the effective use of these opportunities. First of all, the technological infrastructure must be strengthened. Teachers' digital skills must be improved. The quality of electronic resources must be monitored. Students' digital literacy must be developed. Parents must also be involved in this process. Strategies must be developed at the state level. Thus, the proper organization of digital learning serves to develop vocabulary and improve the overall quality of education.

In addition to these factors, the integration of artificial intelligence technologies into digital media has significantly influenced vocabulary acquisition. Intelligent tutoring systems can analyze learners' progress and provide personalized recommendations. This allows students to focus on their individual weaknesses. Automated feedback systems help learners correct their mistakes immediately. As a result, vocabulary retention becomes more effective. Chat-bots and virtual assistants also create opportunities for interactive language practice. They simulate real-life communication situations and encourage active participation. This enhances both receptive and productive vocabulary skills.

Furthermore, the use of data analytics in digital learning environments provides valuable insights into students' vocabulary development. Teachers can monitor progress through digital platforms and adjust their teaching strategies accordingly. Learning analytics help identify which words or concepts are more difficult for students. This enables targeted instruction. At the same time, students can track their own learning progress. Such transparency increases motivation and engagement. Data-driven approaches contribute to more efficient and evidence-based teaching practices.

Another important aspect is the role of collaborative digital tools in vocabulary learning. Online collaborative platforms allow students to work together on tasks and share knowledge. Peer interaction enhances the learning process and promotes deeper understanding. Through discussion

and cooperation, learners are exposed to a wider range of vocabulary. This also helps develop communication skills. Digital storytelling and project-based learning activities encourage creative use of language. Such methods make vocabulary learning more meaningful and context-based.

Conclusion

Finally, the future of vocabulary learning in digital environments is closely linked to continuous technological innovation. Emerging technologies such as virtual reality and augmented reality offer immersive learning experiences. These tools enable learners to encounter new vocabulary in realistic contexts. This strengthens memory and comprehension. Continuous research in this field is essential for improving teaching methods. Educators must remain open to innovation and adapt to new tools. In this way, digital media will continue to enhance vocabulary acquisition and overall language learning outcomes.

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APPLICATION OF DIGITAL TECHNOLOGIES IN PRESCHOOL EDUCATIONAL INSTITUTIONS: PEDAGOGICAL OPPORTUNITIES AND STRATEGIC APPROACHES

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Abstract: In the modern educational paradigm, digital transformation processes have deeply permeated all levels of education, particularly the preschool system, which forms the foundation of human development. This article examines the theoretical foundations and pedagogical effectiveness of Information and Communication Technologies (ICT) in preschool educational institutions. The primary objective of the study is to comprehensively substantiate the impact of digital tools on children's cognitive processes, emotional-volitional spheres, and socialization trajectories.

The article provides a detailed analysis of the cognitive benefits of Game-Based Learning (GBL), the principles of establishing interactive learning environments, and the requirements for the modern digital

competencies of educators. Special attention is devoted to digital hygiene rules and strategies for protecting children's psychophysiological health. In the empirical part of the study, a comparative statistical analysis is presented regarding ICT provision, internet accessibility, and the current state of technological infrastructure in preschool institutions within Azerbaijan and leading international countries.

The analysis indicates that while digitalization in administrative management remains at a high level, the synthesis of technology and methodology in the instructional process still requires further improvement. The article concludes by proposing practical recommendations for the formation of a digital ecosystem in preschool education, the strengthening of teacher-parent cooperation, and the elimination of the regional digital divide.

Keywords: Preschool education, digital technologies, ICT provision, cognitive development, pedagogical strategies, digital hygiene.

MƏKTƏBƏQƏTƏR TƏHSİL MÜƏSSİSƏLƏRİNDƏ RƏQƏMSAL TEXNOLOGİYALARIN TƏTBİQİ: PEDAQOJİ İMKANLAR VƏ STRATEJİ YANAŞMALAR

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Xülasə: Müasir təhsil paradigmasında rəqəmsal transformasiya prosesləri təhsilin bütün pillələrinə, xüsusilə onun bünövrəsini təşkil edən məktəbəqədər təhsil sisteminə köklü şəkildə nüfuz etmişdir. Təqdim olunan məqalədə məktəbəqədər təhsil müəssisələrində informasiya-kommunikasiya texnologiyalarının (İKT) tətbiqinin elmi-nəzəri əsasları və pedaqoji effektivliyi tədqiq olunur. Tədqiqatın əsas məqsədi rəqəmsal alətlərin uşaqların koqnitiv proseslərinə, emosional-iradi sferasına və sosiallaşma trayektoriyasına təsirini kompleks şəkildə əsaslandırmaqdır.

Məqalədə rəqəmsal oyun əsaslı öyrənmənin (GBL) koqnitiv üstünlükləri, interaktiv təlim mühitinin qurulması prinsipləri və pedaqoqların müasir rəqəmsal kompetensiyalarına dair tələblər ətrafı şərh olunur. Xüsusi diqqət rəqəmsal gigiyena qaydalarına və uşaqların psixofizioloji sağlamlığının qorunması strategiyalarına yetirilmişdir. Tədqiqatın empirik hissəsində Azərbaycanın və aparıcı dünya ölkələrinin məktəbəqədər təhsil müəssisələrində İKT təminatı, internetə çıxış imkanları və texnoloji infrastrukturun mövcud vəziyyəti ilə bağlı müqayisəli statistik təhlillər təqdim edilir.

Analizlər göstərir ki, rəqəmsallaşma inzibati idarəetmədə yüksək səviyyədə olsa da, tədris prosesində texnologiya ilə metodikanın sintezi hələ də təkmilləşdirilməyə ehtiyac duyur. Məqalənin sonunda məktəbəqədər təhsildə rəqəmsal ekosistemin formalaşdırılması, müəllim-valideyn kooperasiyasının gücləndirilməsi və regionlararası rəqəmsal uçurumun aradan qaldırılması istiqamətində praktik tövsiyələr irəli sürülür.

Açar sözlər: Məktəbəqədər təhsil, rəqəmsal texnologiyalar, İKT təminatı, koqnitiv inkişaf, pedaqoji strategiyalar, rəqəmsal gigiyena.

Introduction

In the third decade of the 21st century, the educational environment has undergone fundamental changes under the influence of the “digital revolution.” Modern children, referred to as “digital natives,” differ from previous generations in terms of how they perceive information, their attention span, and their learning motivation. These differences make it necessary to integrate digital tools with traditional methods in preschool education institutions.

The preschool period is the most active stage of a child’s neuropsychological development. At this stage, the use of digital technologies (tablets, interactive panels, programmable toys) is not only a technical innovation but also a strategic factor that enhances the quality of education.

According to J. Piaget’s theory of cognitive development, visual-figurative thinking is dominant in preschool children. Digital technologies, through dynamic visualization, make complex concepts (such as natural phenomena and celestial bodies) understandable for children. Multimedia resources stimulate both visual and auditory senses simultaneously, accelerating the transfer of information into long-term memory.

Digital platforms have the ability to adapt to each child’s individual learning pace. Artificial intelligence-based educational programs adjust the level of task difficulty based on the child’s responses. This increases the child’s confidence in their abilities and ensures effective progress within the “zone of proximal development” (L. Vygotsky, 1978).

Discussion

According to OECD and UNESCO reports for 2024–2025, digital infrastructure in developed countries is at the following level:

Scandinavian countries and Singapore: 94% of public kindergartens are equipped with high-speed internet and interactive learning devices.

Devices per child: In Finland and Estonia, there is one tablet or interactive device for every 4–6 children.

Teacher preparedness: In European Union countries, 82% of preschool teachers regularly use ICT in their lessons.

In the Republic of Azerbaijan, the digitalization of preschool education has become a state priority in recent years:

Administrative digitalization: Electronic enrollment in kindergartens (e-admission system) is implemented at 100% nationwide.

Internet access: In major cities (Baku, Sumgait, Ganja, Mingachevir), internet coverage in kindergartens ranges between 78–82%, while in regions it is around 45–50%.

Interactive equipment: Approximately 40% of kindergartens in the capital are equipped with interactive boards. Within state programs, it is planned to increase this figure to 25% in the regions by the end of 2026.

Technology in preschool education should not detach the child from the physical world. The most successful strategic approach is the “**70/30**” principle: 70% of learning time should be devoted to traditional activities (movement games, drawing, sand therapy), and 30% to digital activities (interactive tasks).

Screen-free digital tools such as “Bee-Bot” or “Cubetto” are ideal strategic tools for developing algorithmic thinking in children. They help children understand logical sequences and acquire basic engineering skills.

When applying technology, children’s physical and mental health must be the primary criterion:

1. **Time limits:** For children aged 3–5, continuous screen time should not exceed 15–20 minutes.

2. **Content filtering:** Applications used must be pedagogically approved, age-appropriate, and free of advertisements.

3. **Physical activity:** After digital activities, eye exercises and outdoor movement games should be organized to prevent “technological fatigue.”

The success of digital transformation depends on teachers’ ICT competencies. A modern educator should not only be able to operate programs but also establish a pedagogical connection between digital tools and lesson objectives. At the same time, within kindergarten–family cooperation, parents should be provided with methodological support regarding children’s digital behavior at home.

The use of digital technologies should be a collective, not individual, activity. Teacher collaboration includes the following directions:

Unified resource base: Teachers should share electronic presentations, interactive game scenarios, and video-methodological materials in a common database. This reduces workload and stabilizes the quality of education.

Mentorship system: Young educators with strong ICT skills can support more experienced colleagues who face technological challenges, accelerating the process of “digital adaptation.”

Integrated activities: For example, digital music programs jointly prepared by a music instructor and a teacher can simultaneously develop a child’s sense of rhythm and cognitive skills.

Although children use digital tools under teacher supervision in kindergarten, lack of control at home creates a “digital gap.” Parental awareness helps maintain this balance:

Digital literacy seminars: Parents should be informed about beneficial educational applications, parental control programs, and cybersecurity.

Screen time agreements: Institutions should provide parents with recommended schedules aligning home technology use with kindergarten routines.

Feedback loop: Parents should regularly inform teachers about changes in the child’s interest in digital devices, sleep patterns, and emotional state.

The modern strategic approach is not to completely isolate children from technology, but to guide them—under the supervision of parents and teachers—to “navigate the digital world correctly.”

1. **Preschool institution:** Defines methodological frameworks and safety standards.

2. **Educator-teacher:** Aligns digital tools with pedagogical goals.

3. **Parent:** Supports the learning process initiated in kindergarten with healthy digital habits at home.

Conclusion

The application of digital technologies in preschool education institutions is an inevitable requirement of modern education. Statistical analyses show that although Azerbaijan has achieved full administrative digitalization, it still has significant potential in integrating ICT into the teaching process.

As a strategic approach, it is recommended to ensure a balanced synthesis of technology and traditional methods, improve teacher training, and adhere to digital hygiene principles. A well-regulated digital environment serves as a strong platform for shaping a knowledgeable and creative individual of the future.

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DIGITAL COMPE IN TEACHER EDUCATION PROGRAMS

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Abstract: This comprehensive research investigates the multi-dimensional development of digital competence within contemporary teacher education programs. As global educational systems undergo a fundamental paradigm shift towards digital-first environments, future educators must transcend basic technical literacy to achieve a sophisticated pedagogical integration of emerging technologies. The findings demonstrate that international frameworks play a transformative role in enhancing the professional efficiency and instructional quality of teacher training. This study analyzes the intersection of technology and pedagogy through the TPACK model, emphasizing that digital tools should serve as enablers of learning rather than mere substitutes for traditional methods. Furthermore, the research addresses critical challenges, including the digital divide, algorithmic bias in AI, and the risk of cognitive over-reliance on automated systems. The results suggest that while digital competence is an indispensable pillar of 21st-century education, its integration must be strategically balanced with human-centric pedagogical intuition to ensure sustainable, inclusive, and ethically sound learning outcomes.

These competencies help future educators bridge the gap between theoretical knowledge and practical technological application, allowing for more inclusive and personalized learning environments. However, the study identifies critical challenges such as the "digital divide," ethical dilemmas surrounding artificial intelligence, and the risk of cognitive over-reliance on automated systems. The results suggest that while digital competence is a vital pillar of 21st-century education, its integration must be strategically balanced with human-centric pedagogical intuition and critical thinking to ensure sustainable and high-quality learning outcomes.

Utilizing the TPACK (Technological Pedagogical Content Knowledge) and DigCompEdu frameworks, the investigation highlights a significant transition: the emergence of "Digital Wisdom"—the ability to make ethically sound and pedagogically effective choices in an era of information overload. A key contribution of this study is the exploration of Generative AI integration, specifically focusing on the necessity of Prompt Engineering and Algorithm Literacy for student teachers. These skills are presented not

as isolated technical tasks but as essential cognitive tools for curriculum personalization and the reduction of administrative burdens.

Keywords: Digital Competence, Teacher Education, DigCompEdu, ICT Integration, Pedagogical Innovation, AI Literacy, 21st Century Skills, Educational Technology, Inclusive Learning, Teacher Professional Development.

MÜƏLLİM TƏHSİLİ PROQRAMLARINDA RƏQƏMSAL SƏRİŞTƏ

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Xülasə: Bu hərtərəfli tədqiqat müasir müəllim hazırlığı proqramları daxilində rəqəmsal kompetensiyanın çoxölçülü inkişafını araşdırır. Qlobal təhsil sistemləri “öncə rəqəmsal” (digital-first) mühitlərinə doğru fundamental bir paradigma dəyişikliyi yaşadıqca, gələcəyin təhsil verənləri sadə texniki savadlılıq səviyyəsini aşmalı və inkişaf etməkdə olan texnologiyaların mürəkkəb pedaqoji integrasiyasına nail olmalıdırlar. Tədqiqatın nəticələri göstərir ki, beynəlxalq çərçivələr müəllim hazırlığında peşəkar səmərəliliyin və tədris keyfiyyətinin artırılmasında transformativ rol oynayır. Bu araşdırma texnologiya və pedaqogikanın kəsişməsini TPACK modeli vasitəsilə təhlil edərək vurğulayır ki, rəqəmsal alətlər ənənəvi metodların sadəcə əvəzedicisi deyil, öyrənməni asanlaşdıran amillər kimi xidmət etməlidir. Bundan əlavə, tədqiqat rəqəmsal fərq, süni intellektə alqoritmik qərəzlilik və avtomatlaşdırılmış sistemlərdən koqnitiv asılılıq riski kimi kritik çağırışları nəzərdən keçirir. Nəticələr göstərir ki, rəqəmsal kompetensiya 21-ci əsr təhsilinin vazgeçilməz bir sütunu olsa da, onun integrasiyası davamlı, inklüziv və etik cəhətdən sağlam öyrənmə nəticələrini təmin etmək üçün insan mərkəzli pedaqoji intuisiya ilə strateji şəkildə balanslaşdırılmalıdır.

Bu kompetensiyalar gələcəyin müəllimlərinə nəzəri biliklərlə praktiki texnoloji tətbiqlər arasındakı boşluğu doldurmağa kömək edir, daha inklüziv və fərdiləşdirilmiş öyrənmə mühitlərinə şərait yaradır. Bununla belə, tədqiqat “rəqəmsal fərq”, süni intellekt ətrafındakı etik dilemmalar və avtomatlaşdırılmış sistemlərdən koqnitiv asılılıq riski kimi kritik çətinlikləri müəyyən edir. Nəticələr göstərir ki, rəqəmsal kompetensiya 21-ci əsr təhsilinin həyati sütunu olsa da, davamlı və yüksək keyfiyyətli öyrənmə nəticələrini təmin etmək üçün onun integrasiyası insan mərkəzli pedaqoji intuisiya və tənqidi təfəkkürlə strateji şəkildə balanslaşdırılmalıdır.

TPACK (Texnoloji Pedaqoji Məzmun Bilgisi) və DigCompEdu çərçivələrindən istifadə edən araşdırma mühüm bir keçidi vurğulayır: informasiya həddindən artıq çox olduğu bir dövrdə etik cəhətdən sağlam və pedaqoji baxımdan effektiv seçimlər etmək bacarığı — “Rəqəmsal Müdriklik”. Tədqiqatın əsas töhfələrindən biri, xüsusilə müəllim namizədləri üçün “Sorğu Mühəndisliyi” (Prompt Engineering) və ***”Alqoritm Savadlılığı”***na diqqət yetirərək, Generativ Süni İntellekt integrasiyasının araşdırılmasıdır. Bu bacarıqlar sadəcə təcrid olunmuş texniki tapşırıqlar deyil, kurikulumun fərdiləşdirilməsi və inzibati yüklərin azaldılması üçün vacib koqnitiv alətlər kimi təqdim olunur.

Açar sözlər: Rəqəmsal Kompetensiya, Müəllim Hazırlığı, DigCompEdu, İKT İntegrasiyası, Pedaqoji İnovasiya, Sİ Savadlılığı, 21-ci Əsr Bacarıqları, Təhsil Texnologiyaları, Inklüziv Öyrənmə, Müəllimlərin Peşəkar İnkişafı.

Introduction

Digital competence is defined as the confident, critical, and responsible use of Information Society Technology (IST) for learning, work, and participation in society. This multidisciplinary

field sits at the intersection of pedagogy, educational psychology, and computer science, focusing on creating methods and software that allow teachers to analyze learner data and act in ways that increase the likelihood of achieving specific pedagogical goals. In the modern global educational environment, digital transformation is not merely an infrastructural change but a fundamental shift in the mindset of the educator. Particularly in the post-COVID-19 era, the expansion of distance and hybrid education models has proven how critical technological readiness is for teacher preparation. Current teacher training programs must now instill not only technical literacy but also a culture of digital citizenship and ethical usage.

Methodology and Materials: This study employs a mixed-method approach in order to investigate the impact of digital competence integration on teacher training processes among university students. The research design incorporates both quantitative data from student surveys and qualitative analysis of existing teacher education curricula across several national and international institutions. We analyzed international benchmarks, specifically focusing on the DigCompEdu framework, which categorizes educator skills into six distinct development areas.

This study employs a mixed-method approach to investigate the impact of digital competence integration on teacher training processes among university students. The research design incorporates both quantitative data from student surveys and qualitative analysis of existing teacher education curricula. We analyzed international benchmarks, specifically focusing on the European Union's DigCompEdu framework, UNESCO's ICT Competency Framework for Teachers, and the Technological Pedagogical Content Knowledge (TPACK) model. These resources allow future educators to track their professional development through digital portfolios and learn how to modify digital resources according to the age and level of their students. This process serves to transform the teacher from a simple consumer of digital resources into an original digital content creator.

The research follows a descriptive and analytical research design, focusing on the intersection of technological proficiency and pedagogical application. To ensure a comprehensive evaluation, a mixed-methods approach was integrated, synthesizing both quantitative survey data and qualitative content analysis.

The study is anchored by the TPACK (Technological Pedagogical Content Knowledge) framework, which assesses how effectively student teachers can synthesize their subject expertise with digital tools. Additionally, the DigCompEdu framework was utilized to measure competence across six distinct development areas. Data collection involved:

Structured Questionnaires: Based on DigCompEdu self-assessment indicators to measure perceived digital readiness.

Syllabus Content Analysis: An in-depth review of existing teacher-training curricula to identify gaps between theoretical instruction and classroom needs.

Simulation Tasks: Practical exercises involving prompt engineering and AI-driven lesson planning to evaluate real-time problem-solving skills.

The participants consisted of a purposive sample of senior-year education students who had recently completed their teaching practicum in diverse school environments.

Theoretical Foundations: DigCompEdu and the TPACK Model

The European Framework for the Digital Competence of Educators (DigCompEdu) serves as a structured pathway for professional development. It highlights six critical development areas: Professional Engagement, Digital Resources, Teaching and Learning, Assessment, Empowering

Learners, and Facilitating Learners' Digital Competence. Each area reflects a stage in an educator's journey toward becoming a digitally proficient professional.

Discussion

AI Integration, Ethics, and the Digital Divide

The integration of digital tools plays a significant role in enhancing the quality of teacher education by reducing manual administrative tasks and focusing on creative pedagogical strategies. Generative AI tools provide personalized lesson plans, automated grading systems, and adaptive learning pathways for students. However, this technological leap brings significant ethical considerations.

The discourse on digital competence in teacher education must transition from the stage of "how to operate a tool" to the stage of "pedagogical decision-making." Research indicates that the mere presence of technology does not automatically improve learning outcomes; rather, the core issue lies in the teacher's ability to orchestrate these tools in alignment with specific pedagogical goals.

1. **AI-Augmented Pedagogy:** The emergence of Generative Artificial Intelligence (GenAI) adds a new layer to the TPACK (Technological Pedagogical Content Knowledge) framework. Student teachers are no longer simply teaching with technology; they are now collaborating with autonomous agents. In this context, "Prompt Engineering" (the art of crafting precise queries) must be discussed as a fundamental form of literacy. By effectively utilizing AI, educators can significantly reduce their administrative burdens—such as lesson planning and generating student feedback—thereby reclaiming time for deeper emotional engagement and personalized interaction with their students.

2. **Cognitive Load and Techno-resilience:** A critical focal point of this discussion is the concept of Techno-resilience. As software applications evolve at an exponential rate, teacher education must shift its focus away from mastering specific, transitory applications and toward developing cognitive flexibility. By fostering the ability to adapt to any new interface or digital environment, programs can mitigate "technostress" and ensure long-term professional sustainability for future educators.

3. **The Ethics of Datafication in Education:** The widespread use of Learning Management Systems (LMS) grants teachers access to massive volumes of student data. This necessitates an urgent discussion on data privacy and the ethical dimensions of "predictive analytics." Future educators must be trained to interpret data accurately without falling prey to "algorithmic bias"—where AI might unfairly categorize a student's potential based on historical performance patterns—ensuring that human judgment remains the final arbiter of student success.

4. **Bridging the Secondary Digital Divide:** While the primary digital divide concerns physical access to devices, the secondary digital divide pertains to how technology is utilized. Studies suggest that in low-income settings, technology is often relegated to passive consumption (e.g., watching videos), whereas in high-resource environments, it is used for active digital production (e.g., coding, VR design). Teacher education must prioritize the shift toward active creation in all socio-economic contexts to ensure equitable digital empowerment for all learners.

Transforming Classroom Dynamics and Student Engagement.

Advanced digital tools allow future teachers to create interactive learning environments that were previously impossible. The use of gamification, virtual reality (VR), and augmented reality

(AR) can significantly increase student motivation. Digital assessment tools provide immediate feedback, which is essential for a student-centered learning approach.

Conclusion

The systematic review showed that digital competence is a crucial supportive tool in teacher education processes. It provides valuable support in content organization, learner engagement, and diagnostic assessment. However, the study concludes that technology should serve the pedagogy, not replace the humanistic element of teaching.

The future of education lies in adaptive experts—teachers who are technologically proficient, ethically grounded, and emotionally connected to their students. Continuous professional development should be integrated into the very fabric of teacher training to keep pace with the rapid evolution of the digital landscape. Educational technologies should enhance the teacher’s creativity, reduce routine tasks, and allow them to devote more time to the individual development of students. This strategic approach will lay the foundation for future generations to be more successful in a digital world.

Key Pillars of Digital Integration

Technological Agility: Digital competence should not be viewed as a static set of skills but as “technological agility.” Teacher education programs must empower candidates to adapt to emerging and future technologies that do not yet exist, fostering a mindset of continuous innovation.

The Irreplaceability of the Human Element: While AI can automate administrative tasks and provide data analytics, the emotional intelligence, empathy, and ethical guidance provided by a human teacher remain irreplaceable. Digital tools should function to augment the teacher-student relationship, allowing more time for meaningful, face-to-face mentorship.

Holistic Integration: The study concludes that digital literacy should not be taught as an isolated subject. Instead, it must be integrated across the entire teacher-training curriculum—from pedagogical theory to subject-specific methodology—ensuring that technology becomes a natural extension of the teaching craft.

Sustainable Policy Making: For digital competence to be effective, educational institutions must pair training with adequate infrastructure and supportive policies to mitigate “technostress” and ensure that digital integration is both sustainable and equitable.

Strategic Conclusions

Pedagogical Leadership: Digital competence transforms the teacher not only into a technology user but also into a leader of digital transformation in the classroom. Teacher training programs must develop “technological agility” in students so they can rapidly adapt to new and yet-to-exist tools they will encounter in the future.

Priority of the Human Factor: No matter how much artificial intelligence and automation advance, a teacher’s emotional intelligence, the empathy established with the student, and values education cannot be replaced by digital tools. Technology should reduce the teacher’s burden and gain more time for their individual communication with the student.

Continuous Assessment: Results show that digital competence is a dynamic process. For this reason, a single course in teacher training is not enough; digital skills must be infused into all subjects (methodology, psychology, specialization) and supported by the principle of “lifelong learning.”

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THE INFLUENCE OF TECHNOLOGY ON LANGUAGE CHANGE AND VARIATION

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Abstract: A dynamic partner in the evolution of language, technology has developed. This evolution is a delicate dance between tradition and innovation. A concise overview of the various ways that technology progress has affected changes and developments in language. It recognises that technology like the printing press and the internet has hastened shifts in the dynamics of language, changing both the nature of language and how we use it. Language change is constant and unavoidable, and it is being accelerated by the ever-evolving embrace of technology. With the rapid diffusion of information and ideas made possible by technological innovations, a new era of dynamic language has begun. Technology has always been an important factor in the development of language. Language has always been influenced by technological developments, such as the printing press that helped standardise language and disseminate knowledge and the internet that enables instantaneous communication.

Keywords: Technology, Language Change, Linguistic Trends, Digital Age, Neologisms

TEXNOLOGIYANIN DİL DƏYİŞMƏSİ VƏ VARIASIYASINA TƏSİRİ

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Xülasə: Texnologiya dilin inkişafında dinamik bir tərəfdaş kimi çıxış edir və onun təkamülünü ənənə ilə innovasiya arasındakı davamlı qarşılıqlı təsir vasitəsilə formalaşdırır. Bu məqalə texnoloji tərəqqinin müxtəlif tarixi dövrlərdə dil dəyişikliklərinə necə təsir etdiyinə qısa bir baxış təqdim edir. Çap maşınının dili standartlaşdıran təsirdən tutmuş internetin yaratdığı sürətli və qlobal ünsiyyət imkanlarına qədər texnoloji yeniliklər həm dilin quruluşunda, həm də istifadəsində baş verən dəyişiklikləri sürətləndirmişdir. Bu dəyişikliklər rəqəmsal dövrdə yeni sözlərin (neologizmlərin) yaranması və ünsiyyət üslublarının transformasiyası kimi yeni linqvistik meyillərin meydana çıxmasına səbəb olmuşdur. Dil dəyişməsi daimi və qaçılmaz olsa da, müasir texnologiyaların geniş yayılması bu prosesi əhəmiyyətli dərəcədə sürətləndirmişdir. Məlumat və ideyaların sürətlə yayılması dilin dinamik inkişafının yeni bir mərhələsinin başladığını göstərir.

Açar sözlər: Texnologiya, Dil dəyişməsi, Linqvistik meyillər, Rəqəmsal dövr, Neologizmlər

Introduction

Languages are ever-present reflections of human communication and culture, and their evolution has always been intimately related to technological advancement. In this digital age, the impact of technology on linguistic patterns and evolution is greater than it has ever been before. This introductory section sets the stage for a study of how technical progress has altered language and our methods of interaction by tracing its history from the printing press to the Internet and beyond. Language, being dynamic and ever-evolving, is constantly adapting to accommodate novel modes of human communication. Technology is quickly becoming a potent agent of change, altering language in ways that surpass merely expanding vocabulary. From Johannes Gutenberg's printing press, which spread knowledge and standardised language to modern day cell phones and social media, technology has always played a significant role in shaping language evolution. One of the most noticeable ways that technology has changed the English language is the rapid creation of new words and phrases, which are called neologisms. Words like "app," "emoji," and "tweet" from the technological lexicon demonstrate how language can adapt to new concepts and realities. Coexisting with tech-infused and traditionally-rooted words shows how language preservation and innovation are always shaping one another. Another consequence of technology progress is the emergence of new ways of wearing and communicating, which challenge long-standing norms. The need to be concise in today's world is demonstrated by the use of acronyms and abbreviations in text messages. A unique digital vocabulary, unrelated to language, has developed through the use of visual symbols such as hashtags, memes, and others as a medium of communication on social media platforms. Current technology landscape has an impact on both linguistic trend analysis and language preservation efforts. The advent of digital archives, online dictionaries, and corpus linguistics has revolutionised the study of language change. Technology has given linguists the tools they need to research language diversity, change, and history on an unprecedented scale, deepening our understanding of the dynamic nature of language. We embarked on a journey across the intricate terrain of how technology impacts language patterns and development. We delve into the ever-changing language of the internet age, the ways in which people communicate, and the study methods that have been made feasible by technological progress. Recognising that language is dynamic, we can see how it constantly changes to reflect the culture, technology, and society it reflects. As we traverse this landscape, we come across the adaptability and persistence of language; it evolves and changes to mirror the myriad ways in which people express themselves. This says a

lot about how language may change with the times and influence our digital world. Because technology has such a profound effect on linguistic patterns and development, we should welcome the opportunities and challenges of the digital age with curiosity and openness. The narrative transports readers on a linguistic journey through the ages, delving into the evolution of words, the impact of new technologies on our communication styles, and the eternal quest for profound expression in a language that is ever evolving.

Digital Communication Modes and Style. There has been a dramatic shift in the ever-evolving realm of language communication due to the advent of the digital age. This shift has affected not just the way we speak, but also the specific style and methods by which we convey our emotions and thoughts. The intricacies of online discourse, shedding light on the ways in which technology has facilitated the emergence of novel language forms for the sake of expression and interaction. The advent of instantaneous global communication made possible by the internet has revolutionised human interaction. A new digital jargon is emerging alongside our language, reflecting the ideals of our increasingly globalised culture. The trend towards shorter, more informal digital messages is driving this change. As texting has rapidly become the most popular method of electronic communication, the use of acronyms, emoticons, and abbreviations has become more prevalent. In the digital age, new linguistic innovations have arisen, challenging long-held norms of expression and language, in response to character limits and the need for immediate responses. New ways of expressing oneself through visual symbols, memes, and hashtags have emerged as a result of the global reach of social media. The short and often humorous nature of social media interactions fosters a distinct digital style defined by novelty and brevity. New linguistic trends emerge organically due to the participatory nature of online discourse. Different digital communication strategies reflect the preferences of different online communities and platforms. From the brief prose of tweets to the lengthy essays of blogs and the heated arguments of online forums, the range of linguistic styles and expressions made available by technological improvements is enormous and diverse. negotiating the murky waters of virtual debate. We examine the new linguistic innovations brought about by the digital revolution, acknowledging that technology has changed our interactions with language and expanded our vocabulary. The more we go into this dynamic terrain, the clearer it becomes how digital communication styles and procedures complement and challenge more traditional forms of expression. It has become clear to us that language is fluid and adapts to new ways of communicating in the digital age. Through the examination of digital communication styles and forms, we are encouraged to value the variety and innovation that technology has introduced to language. It encourages us to embrace the dynamic linguistic landscape of the digital age with curiosity and an open mind, acknowledging that language reflects our reality.

Impact on Writing and Grammar. From the rules and structures that govern our expression to new words and methods of speaking, every facet of language is affected by technological progress in the modern day. The ways in which technological advancements have altered the rules of grammar and writing, impacting our communication through text, our adherence to (or lack thereof) to previously established norms of language, and much more besides. In the digital era, the written word has undergone a metamorphosis from its reliance on pen and paper origins. The profound influence of technology on language and syntax is apparent in numerous aspects of our written interactions, such as the creation of emails, reports, forum posts, and social media arguments. Technological developments in writing have made the conventional barrier between professional

and casual communication increasingly blurry. There has been a relaxation of strict grammar standards in written speech due to the pervasive influence of informal digital communications such as emails and instant messaging. This casual tone may appear to be a departure from convention, but it merely exemplifies how language can adapt to different contexts. Furthermore, a new perspective on correct grammar and spelling has emerged due to the proliferation of digital communication. Because they allow us to express more nuance and emotion than is possible with plain old text, emojis, acronyms, and abbreviations have become more commonplace in written communication. The combination of visual and textual elements has led to the development of new forms and conventions in grammar. Additionally, more individuals than ever before may take part in the creative writing process because to technology improvements. People from different parts of the globe may now converse and share ideas in real time, regardless of their first language, all because of the internet and social media. The democratisation of writing has had unintended consequences, such as the challenging of longheld grammatical rules and the expansion of linguistic variation. We traverse the dynamic landscape of written and spoken language in the digital age. In this article, we examine how technological advancements have impacted languages' adaptability, how digital environments have retained and transformed grammar, and how formal and informal written expression have been redefined. The more we explore this terrain, the more we notice the ways in which technology has altered writing and grammar, and the more we learn to value the flexibility and vitality of language. Rather than being a static set of rules, it serves as a reminder that language is dynamic and adapts to meet the needs and preferences of its users.

Discussion

The evolution of language has always been tethered to the tools we use to communicate, and as this article highlights, we are currently witnessing an unprecedented acceleration in this relationship. The transition from the standardization brought about by the printing press to the rapid-fire diffusion of the internet has moved us into an era of "dynamic language" that prioritizes speed and global reach over traditional rigidity.

The Rise of Digital Vernacular. One of the most striking points discussed is the emergence of a unique digital lexicon that transcends traditional linguistic boundaries. Neologisms like "app," "tweet," and "emoji" are more than just new words; they represent a fundamental shift in how language adapts to new realities. The article correctly identifies that this isn't just about expanding vocabulary, but about changing the nature of how we interact.

- Brevity and Innovation: The constraints of digital platforms-such as character limits and the need for immediate responses-have birthed a style defined by acronyms and abbreviations
- Visual Communication: The integration of hashtags, memes, and visual symbols has created a "digital jargon" that allows for nuanced emotional expression that plain text often fails to capture.

Redefining Grammar and Writing Standards. The study brings up a critical tension between tradition and innovation. We are seeing a significant "blurring" of the lines between professional and casual communication. While some might view the relaxation of strict grammatical standards as a decline, it is more accurately described as a "metamorphosis" of the written word.

The democratization of writing-fueled by the internet and social media-allows diverse voices to participate in global discourse regardless of their primary language. This inclusivity naturally

leads to increased linguistic variation, challenging long-held notions of "grammatical correctness" in favor of functional, context-based communication

The Role of Technology in Linguistic Research. Beyond just changing how we speak, technology has revolutionized how we study speech. The use of digital archives and corpus linguistics allows for the tracking of language shifts on an unprecedented scale. This provides a deeper understanding of the "intricate terrain" of language history and diversity, proving that technology is as much a tool for preservation as it is for change.

Conclusion

Come with us as we delve into the intricate web of connections between online expression and the far-reaching effects of technology on linguistic trends and evolution. Learn how to navigate the ever-evolving technology landscape by setting out on a journey that recognises the everchanging nature of language. In the course of our research, we have witnessed directly how recent technical developments have triggered seismic shifts in the manner in which we express ourselves. Digital language, characterised by neologisms and tech-related terminology, reflects the inventive mentality that characterises our digitally linked culture. The fact that languages may evolve and incorporate new concepts demonstrates their adaptability. New forms of expression have arisen as a result of technological advancements, and these new forms of expression contradict long-standing norms, which is why we have also investigated digital communication styles and modes. Everything from the conciseness of text messages to the creativity of social media activity exemplifies the linguistic flexibility that is a hallmark of the digital vernacular. While talking about writing and grammar, we've covered how technology has changed formal and informal conversation, reevaluated grammatical standards, and made textual output more accessible. Here the informality of online language and creative grammatical errors coexist, encouraging linguistic diversity while calling into question notions of grammatical correctness. Even in the information age, the impact of technology extends far beyond the mere production of new languages; it encompasses both their preservation and their study. Digital archives, online dictionaries, and corpus linguistics have greatly enhanced our understanding of language change, variation, and history. Modern technical tools have allowed linguists to examine language dynamics more thoroughly and extensively than in the past. Rather than a destination in and of itself, our exploration of the impact of technology on linguistic patterns and shifts is an invitation to engage with this ever-evolving topic with a spirit of curiosity and openness. The adaptability and resilience of language are celebrated because it continues to evolve in response to the technological innovations that define our modern day. As we explore this dynamic linguistic terrain, we come across a digital environment that reflects the diversity and creativity of human expression. As a dynamic cultural practice, language here allows us to remember the past while looking forward to the future. By thinking about how technology affects language change and trends, we may acknowledge that language is dynamic and take part in the digital linguistic environment with an awareness of the opportunities and challenges it presents. Even in this technologically advanced day, language remains a reflection of the human spirit, and this serves as a reminder of that.

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PRE-SERVICE ENGLISH TEACHERS' ATTITUDES TOWARDS AI TOOLS IN TEACHING ENGLISH

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Abstract: This study examines pre-service English teachers' attitudes toward the use of artificial intelligence (AI) tools in English Language Teaching (ELT). The ultimate aim of the study is to identify participants' perceptions of integrating AI technologies into language teaching practices and their tendency to use these tools. The study used a quantitative research approach. Data for this study were collected via an online questionnaire administered to 51 students in the English Language Teaching (ELT) department at Dokuz Eylul University. The questionnaire included 15 items on the usefulness of AI tools, ease of use, their effect on motivation, and potential concerns about these tools. The findings revealed that participants generally held positive attitudes toward AI tools and considered them especially useful for facilitating lesson preparation and saving time. However, concerns about ethics, academic honesty, and overreliance on AI were also visible. Overall, the study suggests that pre-service teachers view AI tools as beneficial and practical for English language teaching.

Keywords: Pre-service English teachers, artificial intelligence, AI tools, English language teaching, attitudes.

İNGİLİZCE ÖĞRETMEN ADAYLARININ İNGİLİZCE ÖĞRETİMİNDE YAPAY ZEKÂ ARAÇLARINA YÖNELİK TUTUMLARI

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Özet: Bu çalışma, hizmet öncesi İngilizce öğretmenlerinin İngilizce öğretiminde yapay zekâ (YZ) araçlarının kullanımına yönelik tutumlarını incelemektedir. Temel odak noktası, katılımcıların yapay zekâ teknolojilerinin dil öğretimi uygulamalarına entegrasyonuna ilişkin algılarını ve hazır bulunuşluk düzeylerini belirlemektir. Araştırma nicel yöntem temelinde yürütülmüştür. Veriler, Dokuz Eylül Üniversitesi İngiliz Dili Eğitimi (ELT) bölümünde öğrenim gören 51 öğrenciden çevrim içi anket yoluyla elde edilmiştir. Ankette, yapay zekâ araçlarının yararlılığı, kullanım kolaylığı, motivasyona etkisi ve bu araçlara ilişkin olası kaygılarla ilgili 15 maddeye yer verilmiştir. Elde edilen bulgular, katılımcıların yapay zekâ araçlarına genel olarak olumlu tutum sergilediğini ve bu araçları özellikle ders hazırlama sürecini kolaylaştırma ve zaman tasarrufu sağlama açısından yararlı gördüklerini göstermektedir. Bununla birlikte, etik kaygılar, akademik dürüstlük ve teknolojiye aşırı bağımlılık gibi bazı endişeler de görülmüştür. Çalışma, hizmet öncesi öğretmenlerin yapay zekâ araçlarını İngilizce öğretimi açısından genel olarak faydalı ve uygulanabilir bulduğunu ortaya koymaktadır.

Anahtar Kelimeler: Hizmet öncesi İngilizce öğretmenleri, yapay zekâ, YZ araçları, İngilizce öğretimi, tutumlar.

1. Introduction

Artificial intelligence (AI) has spread globally in recent years and has already secured its place in our daily lives, from simple tasks to academic contexts. In particular, in English Language Teaching (ELT), AI tools such as ChatGPT and Grammarly can help improve teaching and learning, as Ahmad et al. (2022) reported that AI tools are increasingly common in education, offering personalized learning, quick feedback, and interactive experiences. In language education, AI tools can help students build their vocabulary, grammar, writing, pronunciation, and communication skills (Aljohani, 2021).

There is one more significant issue to consider: AI also has a darker side, including issues related to academic integrity, ethics, and overreliance. Additionally, whether the end-product is applicable may raise concerns about academic dishonesty, ethical issues, reliance on technology, and AI-generated content. Derinalp and Halife (2025) note that while teachers often have positive views of AI, they worry that excessive use could harm students' creativity and critical thinking. Therefore, it is crucial to understand pre-service teachers' attitudes towards AI tools.

One variation for the study could be given in the field of English as a Foreign Language (EFL). Teachers' perceptions of technology shape how they use it in classroom settings. Computer-assisted, technology-enhanced instruction in language learning shows that pre-service teachers need both technological skills and teaching know-how to use technology effectively (Stockwell, 2010). This contributes to today's world, as ChatGPT is a very popular tool in language education, and the most crucial part is for teachers to decide how to use it in their future teaching.

Even though more studies are being conducted, few focus exclusively on pre-service English teachers in the ELT department. Most studies focus on teachers in general or broader academic contexts. In this regard, the ultimate aim of this study is to find out how pre-service English teachers' attitudes towards AI tools in terms of the usefulness and ease, their willingness to use them or not, ethical issues, and AI knowledge. Moreover, it explores whether there's any chance that

gender and year of study are affecting attitudes, and whether having much more knowledge about AI positively shapes these attitudes. Hence, the relevant research questions are presented below:

1. What are pre-service English teachers' attitudes toward AI tools?
2. How useful and easy do they perceive AI technologies?
3. Are they willing to integrate AI into future teaching?
4. What ethical concerns do they have?
5. Do demographic variables influence attitudes?
6. Is there a relationship between AI literacy and acceptance?

This study contributes to ELT teacher education by offering insights into how ready future teachers are to integrate AI into their teaching.

2. Material and Methods

2.1 Research Design

The study employed a quantitative design to examine pre-service English teachers' attitudes toward AI tools. It also focuses on their usefulness and ease of use. 51 pre-service English teachers volunteered in this study. All of them are currently studying in the ELT department at a state university in Türkiye. They were also from different grade levels and genders. A questionnaire including 5 demographic questions and 15 Likert-scale items, which ranged from 1 (Strongly disagree) to 5 (Strongly agree) was utilized in the study. These items examined pre-service teachers' familiarity with AI tools, their efficiency and usefulness, and their ethical concerns about AI. The questionnaire items were adapted from previously published studies on AI acceptance. Therefore, the scale was demonstrated to be both appropriate and reliable for this study. Cronbach's alpha (.98) indicates that this study is highly consistent, meaning it measures the same things in the same way every time. In addition, it is a strong indicator that this study is highly reliable. The questionnaire was shared online during the spring term of the 2025–2026 academic year. It was stated at the beginning that participation in the study was voluntary. Moreover, their responses should be kept completely anonymous.

Following, the data were analyzed using SPSS. Descriptive statistics were used to determine the overview of the data. Later, independent samples t-tests were conducted to examine gender differences. Moreover, additional analyses were conducted to examine differences across grade levels and to investigate the relationship between AI knowledge and attitudes toward AI tools. Participants were informed that this study would be kept confidential. Their names or anything related to their identities were protected for ethical reasons. All 51 participants agreed to take part in this study, knowing that their data were collected for research purposes only.

3. Results and Discussions

3.1 Overview of the Findings

According to the 15-question questionnaire, 51 people completed it. Of these, 58.8% were female, and 39.2% were male. Most were fourth-year students (70.6%), and the rest were from other grade levels (29.4%).

The results prove that participants generally held positive attitudes toward the use of AI tools in English language teaching. Most of the 51 participants agreed or strongly agreed that AI is useful, effective, and practical in academic contexts. In particular, they emphasized their value for lesson planning, time management, effective teaching, and language learning.

At the same time, the results focused on the dangers of AI tools. Many participants indicated that ethics, academic honesty, privacy, and the potential misuse of AI could pose threats. However,

most pre-service English teachers who participated in this study still saw AI as a useful and supportive tool for their future teaching. Table 1 demonstrates the details below.

Table 1. *Percentage Distribution of Responses to Questionnaire Items*

Questionnaire Items	Disagree/Strongly (%)	Neutral (%)	Agree/Strongly (%)
AI tools are useful for preparing English lessons.	0	5.9	94.2
AI tools save time in lesson planning and assessment.	0	2.0	98.0
AI tools are easy to use in ELT contexts.	2.0	3.9	94.1
I am familiar with AI tools such as ChatGPT and Grammarly.	0	2.0	98.0
AI tools enhance the effectiveness of teaching English.	2.0	7.8	90.2
AI technologies improve students' language skills.	3.9	9.8	86.2
I intend to integrate AI tools into my future classes.	0	3.9	96.1
AI may lead students to become overly dependent on technology.	5.9	27.5	66.7
AI raises concerns about academic integrity and privacy.	3.9	27.5	68.6

It is apparent that pre-service ELT teachers had relatively high attitudes toward AI tools in ELT. Most agreed that AI is useful for preparing English lessons (94.2%) and saves time in planning and assessment (98%). Most also found AI tools easy to use (94.1%) and were familiar with tools like ChatGPT and Grammarly (98%). Participants also strongly believed that AI makes English teaching more effective (90.2%) and helps enhance students' language skills (86.2%). Nearly all participants (96.1%) stated they were willing to use AI in their future teaching. Even though the majority had positive perceptions, some of the participants were cautious about the misuse or overuse of AI. About 66.7% said AI tools could make one too dependent on technology, whereas 68.6% expressed concerns about academic honesty and privacy. In conclusion, these results emphasize that although participants agreed on the positive aspects of AI tools, they are also aware of the potential harms they could cause in academic contexts. Descriptive statistics also supported the finding that participants' attitudes toward AI tools were highly positive. The highest mean score was for saving time in lesson planning, followed by the tendency to include AI in the future. All things considered, participants viewed AI tools as highly helpful and practical for teaching English.

Looking at gender differences, we see that the average scores differ only slightly. Still, male participants outdid the female participants. This slight difference is most evident in their eagerness to use AI in future teaching and their familiarity with AI tools. In other words, gender was not playing an important role in attitudes toward AI tools. Table 2 indicates the details below.

Table 2. *Gender Differences in Attitudes Toward AI Tools*

GENDER	MEAN	SD
FEMALE	3.82	0.41
MALE	3.89	0.38

Another analysis was conducted to determine whether there's a difference across grade levels. According to the results, fourth-year students had higher perceptions of AI tools than students in other grade levels. Table 3 reveals the details below.

Table 3. *Grade-Level Differences in Attitudes Toward AI Tools*

Grade level	Mean
Fourth-year students	3.96
Other grade levels	3.83

4. Conclusion

The results of this study show that pre-service English Teachers generally have positive attitudes toward using AI tools in the English Language Teaching (ELT) department. Participants, 51 pre-service teachers, have agreed that AI tools are very helpful for creating lesson plans, saving time, and enhancing language learning. According to the views, it can be concluded that AI is a more useful tool than a danger in education. Participants were also eager to combine AI in their future teaching. Earlier research done by Kış and Bakla (2026) supported this statement. They found that pre-service EFL (English as a Foreign Language) teachers often used AI tools such as ChatGPT during their practicum and found those tools helpful for planning and designing lessons.

Even though they have shown some positive responses, some participants have expressed concerns about AI use in their future teaching. A number of concerns, including ethical issues, confidentiality, academic honesty, and reliance on AI tools, are among their concerns. Similar concerns have taken place in recent studies. For instance, Syamsi et al. (2025) showed that while most pre-service teachers viewed AI tools such as ChatGPT and Grammarly as helpful, they remained cautious about plagiarism and the potential overuse of AI-generated products.

Looking at the limitations of this study, it focuses on only one group: pre-service ELT teachers at one state university in Türkiye. For this reason, it can't be suitable for large groups when analyzing the results. With this information, later studies shall consist of larger, more diverse groups from different universities. Interviews can also be conducted to better understand attitudes, yielding more insightful insights.

To summarize, pre-service English teachers have a positive attitude toward AI tools, even though they have some concerns about their usage in education. Most participants agreed that AI supports their teaching and lesson planning, but they still suggested that these tools should be used carefully, with attention to ethics and academic honesty.

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ADAPTIVE LEARNING METHODS IN ENGLISH LANGUAGE TEACHING: ENHANCING PERSONALIZATION THROUGH ARTIFICIAL INTELLIGENCE

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Abstract: Adaptive learning methods have emerged as a significant innovation in contemporary English Language Teaching (ELT), offering new ways to address learner diversity through artificial intelligence (AI) and data-driven educational technologies. These systems continuously analyse learner performance and dynamically adjust instructional content, task difficulty, and feedback in real time, resulting in more personalized and flexible learning experiences compared to traditional fixed instructional models (Holmes et al., 2019; Baker et al., 2024). As a result, learners are able to progress at their own pace, receive targeted support, and engage more effectively with language learning materials.

From a theoretical perspective, adaptive learning is strongly aligned with key second language acquisition (SLA) theories. Krashen's Input Hypothesis (1985) emphasizes the importance of comprehensible input slightly above the learner's current level, which adaptive systems operationalize through continuous content adjustment. Similarly, Swain's Output Hypothesis (2005) highlights the role of meaningful language production, which is supported by interactive speaking and writing tasks in adaptive platforms. In addition, these systems extend Communicative Language Teaching (CLT) principles by promoting interaction, meaningful communication, and learner-centered instruction in digital environments (Kessler, 2018).

Despite these advantages, several limitations remain. Language learning is inherently social and contextual, and AI-based systems cannot fully replicate authentic human interaction, emotional expression, or socio-cultural dimensions of communication (Long, 1996). Furthermore, ethical concerns related to data

privacy, algorithmic transparency, and unequal access to technology continue to challenge widespread implementation (UNESCO, 2023).

Overall, adaptive learning represents a powerful pedagogical advancement in ELT; however, its greatest effectiveness is achieved when integrated with traditional, human-centered teaching approaches in a balanced educational framework.

Keywords: Adaptive learning, English Language Teaching (ELT), artificial intelligence, personalized learning, learner autonomy, communicative language teaching, second language acquisition

İNGİLİS DİLİNİN TƏDRİSİNDƏ ADAPTİV ÖYRƏNMƏ METODLARI: SÜNİ İNTELLEKT VASİTƏSİLƏ FƏRDİLƏŞDİRMƏNİN TƏKMİLLƏŞDİRİLMƏSİ

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Xülasə: Adaptiv öyrənmə metodları müasir İngilis Dili Tədrisində (ELT) mühüm innovasiya kimi ortaya çıxmış və süni intellekt (AI) və data əsaslı təhsil texnologiyaları vasitəsilə tələbə müxtəlifliyini nəzərə alan yeni yanaşmalar təqdim etmişdir. Bu sistemlər tələbə performansını davamlı olaraq analiz edir və tədris məzmununu, tapşırıq çətinliyini və geribildirim mexanizmlərini real vaxtda uyğunlaşdıraraq daha fərdiləşdirilmiş və çevik öyrənmə təcrübəsi yaradır (Holmes et al., 2019; Baker et al., 2024). Nəticədə, tələbələr öz səviyyələrinə uyğun şəkildə irəliləyə bilir, hədəf yönümlü dəstək alır və dil öyrənmə prosesinə daha effektiv şəkildə cəlb olunurlar.

Nəzəri baxımdan adaptiv öyrənmə ikinci dilin mənimsənilməsi (SLA) ilə bağlı əsas nəzəriyyələrlə sıx əlaqəlidir. Krashen-in Input Hipotezi (1985) öyrənənin səviyyəsindən bir qədər yuxarı olan anlaşılan inputun vacibliyini vurğulayır və adaptiv sistemlər bu prinsipi məzmunu daim uyğunlaşdırmaqla həyata keçirir. Swain-in Output Hipotezi (2005) isə dil istehsalının (danışma və yazma) rolunu ön plana çəkir və bu, adaptiv platformalarda interaktiv tapşırıqlarla dəstəklənir. Eyni zamanda, bu sistemlər Kommunikativ Dil Tədrisi (CLT) yanaşmasını rəqəmsal mühitdə genişləndirir (Kessler, 2018).

Bununla belə, bəzi məhdudiyyətlər mövcuddur. Dil öyrənmə sosial bir prosesdir və süni intellekt sistemləri real insan ünsiyyətini, emosional ifadəni və mədəni konteksti tam şəkildə əvəz edə bilmir (Long, 1996). Bundan əlavə, məlumat məxfiliyi və texnologiyaya bərabər çıxış kimi etik problemlər də mövcuddur (UNESCO, 2023).

Ümumilikdə, adaptiv öyrənmə dil tədrisində güclü bir yanaşmadır, lakin onun ən effektiv tətbiqi müəllim yönümlü humanist tədris yanaşmaları ilə birləşdirildikdə mümkün olur.

Açar sözlər: Adaptiv öyrənmə, İngilis dili tədrisi (ELT), süni intellekt, fərdiləşdirilmiş öyrənmə, öyrənən muxtariyyəti, kommunikativ dil tədrisi, ikinci dilin mənimsənilməsi

Introduction

Teaching English in today's world has evolved from a fixed, teacher-centred process into a more flexible, learner-centred, and technology-enhanced practice. This transformation has been driven largely by advances in digital technologies, artificial intelligence, and data-driven educational systems (Chapelle, 2020; Kessler, 2018).

Traditional ELT models often rely on a uniform curriculum delivered at the same pace for all learners. While structured, such approaches fail to address learner diversity in proficiency,

motivation, and cognitive strategies. As a result, some learners struggle with content complexity, while others become disengaged due to a lack of challenge (Luckin et al., 2016).

Adaptive learning methods address this issue by using artificial intelligence and machine learning to personalize instruction. These systems continuously analyse learner performance data – such as accuracy, response time, and error patterns – and adjust content and difficulty in real time (Chen et al., 2020). This represents a shift from standardized instruction to individualized learning pathways, placing the learner at the centre of the educational process.

Theoretically, adaptive learning aligns with key second language acquisition (SLA) theories. Krashen's Input Hypothesis (1985) supports the idea of providing comprehensible input at an appropriate difficulty level, while Swain's Output Hypothesis (2005) emphasizes the importance of meaningful language production. Adaptive systems operationalize both principles through dynamic input adjustment and interactive language tasks.

However, language learning is also inherently social. Long's Interaction Hypothesis (1996) highlights the importance of real communication and negotiation of meaning, which cannot be fully replicated by AI systems. This creates a tension between technological efficiency and the socio-cultural nature of language learning.

Therefore, a blended approach is increasingly recommended. Communicative Language Teaching (CLT) can be integrated with adaptive systems, where technology supports personalization and practice, while teachers provide interaction, guidance, and emotional support. Despite its benefits, challenges such as digital inequality and data privacy remain important concerns (UNESCO, 2023).

In summary, adaptive learning represents a significant shift in ELT toward personalized and data-driven instruction, but its effectiveness depends on balanced integration with human-centered pedagogy.

Adaptive learning methods in English language teaching

Adaptive learning methods represent a technologically advanced and pedagogically significant development in English Language Teaching (ELT). These systems provide individualized instruction by continuously adjusting content based on learner performance, behaviour, and progress. Unlike traditional models, they rely on artificial intelligence (AI), machine learning, and learner analytics to generate dynamic, responsive learning environments (Holmes et al., 2019; Chen et al., 2020).

At their core, adaptive systems can be understood through five interconnected functions: personalization, learner modelling, feedback mechanisms, skill-specific adaptation, learner autonomy, and engagement. Among these, personalization is the foundational principle. It enables instruction to be tailored to individual learner needs, ensuring that content aligns with proficiency levels rather than a uniform curriculum. This aligns with Vygotsky's Zone of Proximal Development and Krashen's Input Hypothesis (1985), as learners are consistently exposed to optimally challenging input.

Personalization is achieved through continuous monitoring of learner performance. Systems dynamically adjust content based on strengths and weaknesses – for example, increasing grammar-focused tasks for learners with weak accuracy or providing lexical reinforcement for limited vocabulary. This real-time adaptation enhances motivation and engagement, as learners perceive instruction as relevant and achievable (Baker et al., 2024).

Closely related is data-driven learner modelling, which functions as the analytical core of adaptive systems. It involves continuous collection and interpretation of learner interaction data to construct evolving learner profiles (Chen et al., 2020). Unlike traditional assessment, this process is ongoing and predictive rather than episodic. It enables systems to detect patterns across language skills and anticipate learner needs. However, concerns remain regarding transparency and the ethical use of behavioural data in algorithmic decision-making (UNESCO, 2023).

Feedback is another critical component and one of the strongest pedagogical advantages of adaptive systems. Unlike delayed classroom feedback, AI systems provide immediate, multi-level correction across grammar, vocabulary, pronunciation, and discourse. For instance, grammatical errors are corrected instantly with explanations, strengthening accuracy and reducing fossilization. Godwin-Jones (2023) notes that such real-time feedback enhances learning efficiency, although AI lacks emotional sensitivity and cannot adjust feedback according to learner affect or frustration.

Adaptive systems also support skill-specific development across listening, speaking, reading, and writing. Listening tasks are adjusted through speech speed and scaffolding tools; speaking is supported via AI-based pronunciation analysis; reading is adapted through vocabulary control and comprehension support; and writing receives multi-layered feedback on grammar, coherence, and discourse structure. This structured differentiation supports gradual language development aligned with SLA principles.

In addition, adaptive systems promote learner autonomy by enabling self-paced learning, goal-setting, and performance monitoring. Progress dashboards and analytics tools encourage self-regulation and reflective learning. Engagement is further enhanced through gamification features such as badges and progress tracking, which increase motivation and persistence (Ahmad et al., 2023). However, excessive gamification risks shifting attention from meaningful learning to reward-oriented behaviour.

Despite these strengths, adaptive learning systems face significant limitations. Language learning is inherently social and interactive, requiring negotiation of meaning, emotional expression, and cultural awareness – dimensions that AI cannot fully replicate (Long, 1996). While conversational agents offer simulated interaction, they lack spontaneity, emotional depth, and contextual sensitivity.

Furthermore, issues of digital inequality and data privacy remain critical. Unequal access to technology risks widening educational gaps, while extensive data collection raises ethical concerns regarding transparency and learner protection (UNESCO, 2023). Over-automation may also reduce teacher involvement, weakening emotional support and communicative richness in the classroom.

In conclusion, adaptive learning represents a major shift toward personalized, data-driven ELT. However, its pedagogical value is conditional. It is most effective not as a replacement for traditional instruction, but as a complementary system within a blended learning model that integrates AI-driven personalization with human-centered communicative teaching.

Materials and methods

This study adopts a qualitative research design to investigate the role of adaptive learning methods in English Language Teaching (ELT). The main objective is to explore how artificial intelligence-based adaptive systems influence language learning processes, particularly in terms of personalization, learner engagement, feedback mechanisms, and skill development. Given the conceptual and interpretive nature of the topic, a qualitative approach is considered more suitable

than a quantitative one, as it allows for in-depth analysis of theoretical frameworks and existing empirical research (Chen et al., 2020; Holmes et al., 2019).

The study is based on a systematic review of relevant academic literature published between 2016 and 2024. Sources were selected from peer-reviewed journals, academic books, and reputable educational technology reports. The selection criteria included relevance to adaptive learning systems, artificial intelligence in education, and English language teaching contexts. Particular attention was given to studies that provide both theoretical perspectives and empirical findings on AI-supported learning environments (Godwin-Jones, 2023; Baker et al., 2024).

The analytical process involved thematic analysis of the collected literature. This method was used to identify recurring patterns, concepts, and arguments across different studies. The main themes identified include personalization in adaptive learning systems, learner modelling, real-time feedback, learner autonomy, engagement strategies, and limitations of artificial intelligence in language education. These themes were interpreted in relation to second language acquisition theories, particularly Krashen's Input Hypothesis and Swain's Output Hypothesis, which provide a strong theoretical foundation for understanding language input and output processes (Krashen, 1985; Swain, 2005).

In addition to thematic analysis, a comparative framework was used to examine differences between traditional English language teaching methods and AI-based adaptive learning systems. Traditional approaches are typically characterized by teacher-centered instruction, fixed curricula, and uniform pacing, whereas adaptive systems offer individualized learning pathways, dynamic content adjustment, and immediate feedback based on learner performance. This comparison highlights the pedagogical shift toward more learner-centered and data-driven instruction (Zawacki-Richter et al., 2019).

Although this qualitative approach provides meaningful conceptual insights, it also has limitations. Since the study relies on secondary data rather than primary empirical investigation, the findings are interpretive rather than statistically validated. The absence of classroom-based experimental or interview data limits the ability to generalize findings to all educational contexts. Therefore, future research should consider mixed-method or experimental designs to strengthen empirical validation of adaptive learning effectiveness (UNESCO, 2023).

Results and discussion

The results of this study demonstrate that adaptive learning systems significantly transform English Language Teaching (ELT) by changing how instruction is delivered, how learners interact with content, and how learning outcomes are evaluated. One of the most important findings is their strong impact on personalization. Unlike traditional classrooms, where all learners receive the same materials regardless of proficiency level, adaptive systems create individualized learning pathways based on continuous performance analysis (Baker et al., 2024). This process is highly data-driven, as systems constantly evaluate learner responses, identify strengths and weaknesses, and adjust content accordingly. For instance, learners with strong receptive skills but weaker productive abilities are automatically guided toward more speaking and writing tasks.

From a cognitive perspective, this personalization reduces cognitive overload and improves learning efficiency by keeping learners within their optimal difficulty range, aligning with Vygotsky's Zone of Proximal Development (ZPD). It also enhances motivation, as learners are more likely to engage with content that reflects their individual needs and progress.

Another key finding is the improvement in learning efficiency and feedback quality. Adaptive systems reduce unnecessary repetition by focusing only on areas of difficulty, providing targeted instruction that improves accuracy and stability of knowledge. Immediate feedback allows learners to correct errors instantly, strengthening memory retention and preventing the fossilization of incorrect language forms. However, efficiency does not always guarantee deep learning, as some learners may rely on pattern recognition without fully understanding underlying rules, raising concerns about depth versus speed in AI-supported learning.

The study also shows that adaptive systems increase learner engagement and motivation. Features such as gamification, progress tracking, and achievement rewards create a sense of achievement and sustain participation (Ahmad et al., 2023). Tasks are continuously adjusted to avoid being too easy or too difficult, which helps maintain motivation. However, excessive reliance on gamification may shift attention from meaningful learning to reward-seeking behavior, reducing intrinsic motivation over time.

Despite these advantages, a major limitation is the lack of authentic human interaction. Language learning is fundamentally social, involving negotiation of meaning, emotional expression, and cultural understanding. As Long (1996) notes, interaction is central to second language acquisition, and these dimensions cannot be fully replicated by AI systems. Although chatbots and conversational agents offer practice opportunities, they remain limited in emotional depth and spontaneity. Therefore, teacher presence remains essential for providing emotional support, cultural explanation, and real communication.

Finally, ethical and accessibility concerns are significant. Adaptive systems rely heavily on learner data, raising issues of privacy, consent, and transparency (UNESCO, 2023). In addition, unequal access to digital technologies may widen the gap between learners in different socio-economic contexts. Over-automation may also reduce teacher involvement, affecting classroom dynamics. Thus, ethical implementation and balanced integration are necessary to ensure sustainable and equitable use of adaptive learning systems.

Conclusion

This study has examined the role of adaptive learning methods in English Language Teaching (ELT), focusing on how artificial intelligence reshapes instructional design, learner experience, and overall pedagogical practice. The findings indicate that adaptive learning represents a significant shift from traditional uniform instruction toward more personalized, flexible, and data-driven educational models that better address individual learner differences.

A central conclusion is that adaptive learning systems significantly enhance personalization in language learning. By continuously analysing learner performance and adjusting instructional content in real time, these systems ensure that learners receive materials aligned with their proficiency level. This reduces unnecessary repetition, decreases cognitive overload, and creates more efficient learning pathways. In this way, adaptive learning supports key principles of second language acquisition, particularly Krashen's Input Hypothesis, which emphasizes the importance of comprehensible input for effective language development.

Another important finding is the improvement in learning efficiency and feedback quality. Adaptive systems provide immediate correction, enabling learners to identify and correct errors instantly. This real-time feedback strengthens retention and supports more stable language acquisition. Additionally, features such as progress tracking, structured practice, and interactive

tasks enhance learner engagement and motivation, making the learning process more dynamic and self-directed.

However, the study also identifies important limitations. Language learning is not only a cognitive process but also a social and cultural one. Although adaptive systems are effective in structured practice and individualized instruction, they cannot fully replicate authentic human communication. Elements such as spontaneity, emotional expression, and cultural nuance remain beyond the capabilities of current AI-based systems.

This creates a tension between technological efficiency and human-centered pedagogy. While adaptive systems offer personalization and scalability, they lack the interpersonal dimensions essential in communicative language teaching. Therefore, teachers remain central in providing guidance, emotional support, and meaningful interaction.

In conclusion, adaptive learning should be viewed not as a replacement for traditional teaching, but as a complementary tool that enhances ELT when integrated within a balanced, human-centered educational framework.

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THE FUTURE OF EDUCATION IN THE AGE OF MODERN TRANSFORMATION

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Abstract: Education is experiencing rapid transformation due to globalization, digital technologies, and changing social needs. Traditional education systems are no longer sufficient to prepare students for the challenges of the 21st century. This paper explores the future of education in the age of transformation and analyzes how modern technologies, innovative teaching methods, and student-centered learning reshape educational systems. The study focuses on the role of digital learning, artificial intelligence, online education, and lifelong learning. A qualitative research method based on literature analysis and educational practices was used. The findings indicate that education is becoming more flexible, personalized, and technology-driven. Teachers are transforming from knowledge transmitters into facilitators and mentors. Students are expected to develop critical thinking, creativity, communication, and digital literacy skills. The paper concludes that the integration of technology and innovative pedagogy will define the future of education.

Key words: Future education, digital learning, artificial intelligence, online education, lifelong learning, innovation

TƏHSİLİN GƏLƏCƏYİ MÜASİR TRANSFORMASIYA DÖVRÜNDƏ

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Xülasə. Təhsil qloballaşma, rəqəmsal texnologiyalar və sosial dəyişikliklər nəticəsində sürətli transformasiya dövrünü yaşayır. Ənənəvi təhsil sistemi artıq XXI əsrin tələblərinə tam cavab vermir. Bu məqalədə transformasiya dövründə təhsilin gələcəyi araşdırılır və müasir texnologiyaların, innovativ tədris metodlarının və tələbəyönümlü yanaşmanın təhsil sistemini necə dəyişdirdiyi təhlil olunur. Tədqiqat rəqəmsal təhsil, süni intellekt, onlayn öyrənmə və ömürboyu təhsil anlayışlarını əhatə edir. Keyfiyyət təhlili metodundan istifadə edilmişdir. Nəticələr göstərir ki, təhsil daha çevik, fərdiləşdirilmiş və texnologiya yönümlü olur. Müəllimlər bilik ötürən deyil, mentor və fasilitator rolunu alır. Məqalə texnologiya və innovativ pedaqogikanın təhsilin gələcəyini müəyyən edəcəyini göstərir.

Açar sözlər: Təhsilin gələcəyi, rəqəmsal təhsil, süni intellekt, onlayn təhsil, ömürboyu təhsil, innovasiya

Introduction

Digital Transformation and the Modern Learning Ecosystem

Education has historically evolved alongside societal shifts and technological progress, but the transformation occurring in the 21st century is unprecedented in both velocity and scope. Traditional educational frameworks were designed during the industrial era, prioritizing standardization, rote memorization, and teacher-centered instruction. Modern globalized societies,

conversely, require high-level competencies such as creativity, critical thinking, collaboration, adaptability, and digital literacy. Consequently, educational systems worldwide are undergoing a profound paradigm shift – a transformation accelerated by the necessity of remote learning environments, which exposed both the vast potential and current structural limitations of digital education.

Today, digital transformation has revolutionized access to knowledge, ensuring that learning is no longer restricted to physical classrooms. Modern learners engage through interactive online management systems, virtual classrooms, mobile applications, and open educational resources. This infrastructure provides flexibility, accessibility, and global connectivity, allowing students to access international courses and collaborate across borders. Furthermore, the modern learning ecosystem is inherently hybrid. Blended learning models strategically combine face-to-face instruction, online learning, and self-paced study. This architecture empowers learners to progress according to their individual paces while preserving essential human interaction, mentorship, and socio-emotional support.

Artificial Intelligence and Personalized Pedagogy

Artificial Intelligence (AI) represents one of the most powerful transformative forces in contemporary education. By leveraging adaptive learning systems, AI analyzes student performance data in real time to identify individual strengths, cognitive gaps, and learning preferences. By tailoring instructional pacing and content to individual needs, AI transitions education away from the traditional "one-size-fits-all" model toward a truly personalized pedagogical experience.

Furthermore, Intelligent Tutoring Systems provide automated, scalable support through real-time diagnostic feedback, automated assessment, and custom-tailored practice modules. By liberating educators from routine administrative and mechanical evaluation tasks, AI enables teachers to focus on high-impact pedagogical activities such as mentoring, inspiring creativity, and guiding complex critical inquiry. However, the integration of AI also introduces critical ethical challenges, including data privacy concerns, algorithmic bias, and the risk of over-reliance on technology. Future educational systems must establish robust governance to integrate AI responsibly, transparently, and ethically.

The Evolving Identity of Educators and Student-Centered Learning

Technological advancement does not diminish the role of teachers; rather, it elevates and transforms their professional identity. In an age where information is ubiquitously accessible, teachers are no longer the primary keepers of knowledge. Instead, they operate as facilitators, academic mentors, curators of learning experiences, and pedagogical innovators. While technology can deliver content and analyze data, it cannot replicate essential human capabilities such as empathy, emotional intelligence, moral reasoning, and interpersonal communication. To maintain pedagogical effectiveness, educators themselves must commit to lifelong learning, continuously refining their digital literacy, pedagogical strategies, and research capacities.

Concurrently, instructional models are shifting toward student-centered and competency-based education. Future frameworks prioritize actionable skill mastery – such as analytical problem-solving and evidence-based reasoning – over passive content memorization. Flexible learning pathways allow students to tailor their academic trajectories to their individual interests, abilities, and career goals. Meanwhile, project-based and experiential learning models ground theoretical concepts in practical, real-world contexts, preparing students to navigate professional ambiguity effectively.

Lifelong Learning, Equity, and Global Citizenship

Education is no longer a terminal phase that concludes upon graduation; it has become a continuous, lifelong endeavor. Rapid technological shifts and dynamic labor market demands require constant reskilling and upskilling throughout an individual's professional career. As a result, modular learning units, digital badges, micro-credentials, and targeted short courses are emerging as valuable, flexible alternatives alongside traditional university degrees. Educational institutions must adapt to prepare learners for remote work environments, cross-cultural digital collaboration, and global market competition.

Conclusion

The future of education must be anchored in equity, inclusion, and sustainability. Digital transformation initiatives must actively bridge the digital divide, ensuring equal access to hardware and connectivity for all socio-economic groups. Classrooms must adopt Universal Design for Learning (UDL) principles to support diverse learning needs, neurodiverse individuals, and students from varied cultural backgrounds. Ultimately, the goal of modern education extends beyond domain knowledge acquisition to the cultivation of creative, adaptable, ethical, and responsible global citizens.

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THE ROLE AND PROSPECTS OF ARTIFICIAL INTELLIGENCE AND DIGITAL PLATFORMS IN MODERN EDUCATION

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Abstract. This article provides a comprehensive exploration of the transformative role of artificial intelligence (AI) and digital platforms in the contemporary education system, where rapid technological progress and global digitalization are reshaping traditional models of teaching and learning. The rise of AI-driven tools, adaptive learning systems, and intelligent tutoring technologies has made it possible to create highly personalized educational experiences that respond dynamically to the individual abilities, learning styles, and progress rates of students. These technologies not only assist teachers in identifying the strengths and weaknesses of learners through data-driven analysis but also enable the continuous monitoring and optimization of the learning process.

Digital platforms have become a fundamental component of modern education by expanding access to teaching and learning resources beyond physical and social boundaries. They facilitate interactive communication between teachers and students, promote collaborative learning environments, and ensure the availability of high-quality educational content for diverse groups of learners, including those in remote regions. This democratization of education fosters inclusion, lifelong learning, and global knowledge sharing.

The article thoroughly examines the multiple forms of AI application in education, such as automated assessment systems, predictive analytics for student performance, virtual teaching assistants, and content recommendation algorithms. It also analyzes the structural and pedagogical functions of digital platforms, including virtual classrooms, online course management systems, and hybrid learning environments that integrate both face-to-face and online instruction. The research identifies several key advantages of these technologies efficiency, scalability, interactivity, and flexibility while also addressing their limitations and risks. These include the dependence on stable technical infrastructure, insufficient digital competencies among educators, ethical and privacy concerns related to the collection and use of student data, as well as potential widening of the digital divide.

Furthermore, the study discusses the challenges of integrating AI and digital technologies within national education systems, such as policy gaps, lack of interoperability between platforms, and the need for continuous professional development of teachers. It argues that successful implementation requires a holistic approach that combines technological innovation with pedagogical reform, institutional support, and ethical governance of digital data.

In conclusion, the article emphasizes that when properly designed and regulated, artificial intelligence and digital platforms hold enormous potential to enhance the quality, accessibility, and inclusiveness of education. Their integration will not only increase the efficiency of teaching and learning processes but also lay the theoretical and practical foundation for the education systems of the future, where human creativity and machine intelligence coexist in synergy to foster intellectual development and innovation.

Keywords: Artificial Intelligence, Digital Platforms, Education, Personalized Learning, Digital Transformation

MÜASİR TƏHSİLDƏ SÜNİ İNTELLİKİN VƏ RƏQƏMSAL PLATFORMALARIN ROLU VƏ PERSPEKTİVLƏRİ

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Xülasə: Bu məqalədə müasir təhsil sistemində süni intellektin (Sİ) və rəqəmsal platformaların transformativ rolunun hərtərəfli araşdırılması təqdim olunur. Burada sürətli texnoloji tərəqqi və global rəqəmsallaşma ənənəvi tədris və öyrənmə modellərini yenidən formalaşdırır. Sİ ilə idarə olunan alətlərin,

adaptiv təlim sistemlərinin və ağıllı repetitorluq texnologiyalarının yüksəlişi şagirdlərin fərdi qabiliyyətlərinə, öyrənmə üslublarına və irəliləyiş templərinə dinamik şəkildə cavab verən yüksək dərəcədə fərdiləşdirilmiş təhsil təcrübələri yaratmağa imkan vermişdir. Bu texnologiyalar müəllimlərə məlumatlara əsaslanan təhlil vasitəsilə şagirdlərin güclü və zəif tərəflərini müəyyən etməyə kömək etməklə yanaşı, həm də təlim prosesinin davamlı monitorinqini və optimallaşdırılmasını təmin edir.

Rəqəmsal platformalar tədris və öyrənmə resurslarına fiziki və sosial sərhədlərdən kənarda çıxışı genişləndirməklə müasir təhsilin fundamental komponentinə çevrilmişdir. Onlar müəllimlər və şagirdlər arasında interaktiv ünsiyyəti asanlaşdırır, əməkdaşlıq mühitlərini təşviq edir və ucqar bölgələrdə yaşayanlar da daxil olmaqla müxtəlif şagird qrupları üçün yüksək keyfiyyətli təhsil məzmununun mövcudluğunu təmin edir. Təhsilin bu demokratikləşməsi inklüzivliyi, ömürlük öyrənməni və qlobal bilik mübadiləsini təşviq edir.

Məqalədə təhsildə Sİ tətbiqinin çoxsaylı formaları, məsələn, avtomatlaşdırılmış qiymətləndirmə sistemləri, şagirdlərin performansını üçün proqnozlaşdırıcı analitika, virtual müəllim köməkçiləri və məzmun tövsiyə alqoritmləri ətraflı araşdırılır. Həmçinin, virtual sinif otaqları, onlayn kurs idarəetmə sistemləri və həm üz-üzə, həm də onlayn təlimi integrasiya edən hibrid təlim mühitləri daxil olmaqla rəqəmsal platformaların struktur və pedaqoji funksiyaları təhlil edilir. Tədqiqat bu texnologiyaların bir neçə əsas üstünlüyünü - səmərəlilik, miqyaslanma, interaktivlik və çevikliyi müəyyən etməklə yanaşı, onların məhdudiyyətlərini və risklərini də həll edir. Bunlara sabit texniki infrastrukturdan asılılıq, müəllimlər arasında rəqəmsal sərişələrin yetərli olmaması, tələbə məlumatlarının toplanması və istifadəsi ilə bağlı etik və məxfilik narahatlıqları, eləcə də rəqəmsal uçurumun potensial genişlənməsi daxildir.

Bundan əlavə, tədqiqatda süni intellekt və rəqəmsal texnologiyaların milli təhsil sistemlərinə integrasiyasının çətinlikləri, məsələn, siyasət boşluqları, platformalar arasında qarşılıqlı fəaliyyətin olmaması və müəllimlərin davamlı peşəkar inkişafına ehtiyac müzakirə olunur. Müəllif iddia edir ki, uğurlu tətbiq texnoloji innovasiyanı pedaqoji islahat, institusional dəstək və rəqəmsal məlumatların etik idarəçiliyi ilə birləşdirən vahid yanaşma tələb edir.

Nəticə olaraq, məqalədə vurğulanır ki, düzgün dizayn edildikdə və tənzimləndikdə, süni intellekt və rəqəmsal platformalar təhsilin keyfiyyətini, əlçatanlığını və inklüzivliyini artırmaq üçün böyük potensiala malikdir. Onların integrasiyası təkcə tədris və öyrənmə proseslərinin səmərəliliyini artırmaqla yanaşı, həm də insan yaradıcılığı və maşın zəkasının intellektual inkişaf və innovasiyanı təşviq etmək üçün sinerjidə bir arada mövcud olduğu gələcəyin təhsil sistemləri üçün nəzəri və praktik təməl qoyacaq.

Açar sözlər: Süni intellekt, Rəqəmsal platformalar, Təhsil, Fərdiləşdirilmiş Təlim, Rəqəmsal Transformasiya

Introduction

The 21st century is defined by the rapid evolution and widespread integration of information and communication technologies, which have fundamentally transformed nearly every aspect of human life, including education. In particular, the emergence of digitalization has reshaped how knowledge is created, delivered, and consumed. While the traditional education model based on face-to-face instruction, standardized curricula, and teacher-centered approaches has long been considered effective, it is increasingly being challenged by the demands of a modern, technology-driven society. Today's learners require more flexible, accessible, and personalized educational experiences that can adapt to fast-changing global conditions.

One of the most significant turning points in this transformation was the global COVID-19 pandemic, which forced educational institutions worldwide to shift rapidly toward online learning environments. This unprecedented transition highlighted both the potential and the limitations of digital education systems. On one hand, it demonstrated that learning can continue beyond physical classrooms through digital platforms; on the other hand, it exposed inequalities in access to technology, differences in digital literacy, and the lack of preparedness of many education systems

for large-scale remote learning. As a result, digital platforms and educational technologies have become not just supportive tools but essential components of modern education.

In this context, artificial intelligence (AI) technologies have emerged as a powerful force capable of redefining the learning process. AI-driven systems introduce new opportunities by making education more adaptive, efficient, and learner-centered. These technologies are capable of analyzing large amounts of educational data to understand individual learning behaviors, identify knowledge gaps, and predict academic performance. Based on this analysis, AI systems can recommend personalized learning paths, suggest relevant materials, and adjust the difficulty level of tasks according to each student's needs. This level of personalization was extremely difficult to achieve in traditional classroom settings.

In addition, digital platforms have significantly expanded the boundaries of education by removing spatial and temporal limitations. Learning is no longer restricted to a physical classroom or fixed timetable. Platforms such as online learning environments, virtual classrooms, and learning management systems enable continuous access to educational content from anywhere in the world. They also improve communication between teachers and students, support collaborative learning, and allow the integration of multimedia resources such as videos, simulations, and interactive exercises. As a result, the learning process becomes more engaging, dynamic, and student-centered.

Furthermore, the role of education itself is evolving. Modern education is no longer limited to the simple transmission of knowledge from teacher to student. Instead, it aims to develop a wide range of competencies, including critical thinking, creativity, problem-solving, digital literacy, and independent learning skills. These competencies are essential for success in an increasingly complex and technology-driven world. In this regard, artificial intelligence and digital platforms serve as important tools that support educators in achieving these broader educational goals.

The main objective of this article is to provide a comprehensive analysis of the role of artificial intelligence and digital platforms in contemporary education. It seeks to examine how these technologies influence teaching and learning processes, identify their key advantages and limitations, and explore their long-term implications for the future of education systems. By doing so, the article aims to contribute to a deeper understanding of how digital transformation is reshaping education and what prospects it holds for future generations.

Methodology and Materials

Various methods were employed in the preparation of this research in order to ensure a comprehensive and systematic analysis of the role of artificial intelligence and digital platforms in education. The study is primarily based on a qualitative research approach, supported by document analysis and comparative evaluation techniques. This methodological combination allows for a deeper understanding of both theoretical perspectives and practical applications in the field of educational technologies.

First, a wide range of scientific literature was reviewed, including peer-reviewed academic journal articles, books, and international institutional reports related to artificial intelligence in education and digital learning environments. These sources provided a solid theoretical foundation and helped to identify key trends, developments, and challenges associated with the integration of technology in modern education systems. Particular attention was given to recent publications, as the field of AI in education is rapidly evolving.

In addition, a comparative analysis method was applied to examine the differences between traditional teaching models and modern technology-enhanced educational approaches. Traditional

education systems, which are largely teacher-centered and classroom-based, were compared with digital and AI-supported learning environments that emphasize flexibility, personalization, and student-centered learning. This comparison made it possible to highlight the advantages and limitations of both approaches, particularly in terms of accessibility, efficiency, interactivity, and adaptability.

Furthermore, the study included an examination of the functional capabilities of various digital education platforms and artificial intelligence tools used in contemporary learning environments. Adaptive learning systems, which adjust educational content based on student performance, were analyzed as key examples of AI application in education. Similarly, online course platforms and learning management systems were examined in terms of their structure, usability, and pedagogical impact. AI-based teaching tools such as intelligent tutoring systems, automated assessment systems, and recommendation algorithms were also considered to understand their contribution to personalized learning.

Through qualitative content analysis, the collected data were systematically organized, categorized, and interpreted. This process enabled the identification of recurring themes, patterns, and relationships within the literature and case examples. The findings were then synthesized to draw general conclusions regarding the overall impact of artificial intelligence and digital platforms on educational practices.

Overall, the combination of literature review, comparative analysis, and qualitative interpretation provided a balanced and comprehensive methodological framework, allowing for an objective evaluation of how digital transformation is reshaping modern education systems.

Discussion

The findings of this research indicate that artificial intelligence plays a highly transformative role in modern education systems. One of its most significant contributions is the development of personalized learning environments. AI-powered systems are capable of analyzing students' academic performance, learning behaviors, and progress patterns in real time. Based on this analysis, they identify individual strengths and weaknesses and adapt instructional content accordingly. This adaptive approach ensures that each learner receives tailored educational support, which significantly increases learning efficiency, motivation, and long-term knowledge retention compared to traditional one-size-fits-all teaching methods.

In addition, digital platforms have fundamentally improved the accessibility and flexibility of education. Online learning environments, video-based lectures, virtual classrooms, and interactive assignments allow students to engage with educational content at any time and from any location. This flexibility is particularly important for learners in remote or underdeveloped regions, as well as for individuals who face physical, financial, or time-related constraints. As a result, digital platforms contribute to the democratization of education by reducing geographical and social barriers.

Another important finding is the positive impact of artificial intelligence on reducing the workload of teachers. AI-based systems are increasingly used to automate repetitive and time-consuming tasks such as grading, performance assessment, attendance tracking, and statistical data analysis. This automation allows educators to focus more on higher-level responsibilities, including lesson planning, creative teaching strategies, and providing individualized support to students. Consequently, the role of the teacher is gradually shifting from a traditional knowledge transmitter to a learning facilitator and mentor.

Despite these advantages, the study also identifies several significant challenges associated with the integration of artificial intelligence and digital platforms in education. One of the primary issues is the lack of adequate technological infrastructure in some educational institutions, which limits the effective implementation of advanced digital systems. In addition, insufficient digital literacy among educators remains a critical barrier, as many teachers require further training to effectively use these technologies in their teaching practices.

Data security and privacy concerns also represent a major challenge. Since digital platforms collect and process large amounts of personal and academic data, ensuring the protection of this information is essential. Without proper cybersecurity measures and regulatory frameworks, there is a risk of data misuse or unauthorized access. Furthermore, there is growing concern that excessive reliance on artificial intelligence may negatively impact students' independent thinking, creativity, and problem-solving abilities if not properly balanced with traditional learning methods.

Nevertheless, the results suggest that these challenges can be effectively addressed through appropriate strategies and long-term planning. The successful integration of artificial intelligence and digital platforms into education systems requires strong government policies, continuous investment in technological infrastructure, and comprehensive teacher training programs. In particular, professional development initiatives are essential to ensure that educators are equipped with the necessary digital competencies.

Overall, the discussion highlights that while artificial intelligence and digital platforms present certain risks and limitations, their potential benefits in terms of efficiency, accessibility, and personalization are substantial. With proper implementation and regulation, these technologies can significantly enhance the quality of education and reshape traditional teaching and learning processes.

Conclusion

In conclusion, artificial intelligence and digital platforms have become fundamental and indispensable components of the modern education system. Their integration into teaching and learning processes has significantly transformed traditional educational practices, making them more efficient, interactive, flexible, and accessible. These technologies are no longer considered optional tools; instead, they are increasingly viewed as essential elements that support the development and modernization of education in the digital age.

The application of artificial intelligence in education has greatly expanded the possibilities for personalized learning. By analyzing student performance, learning behaviors, and progress patterns, AI systems can adapt educational content to meet individual needs. This ensures that learners receive tailored instruction that matches their pace, abilities, and knowledge gaps. As a result, learning becomes more effective, targeted, and student-centered. On the other hand, digital platforms play a crucial role in democratizing education by removing barriers related to geography, time, and physical access. They provide learners with the opportunity to access high-quality educational resources regardless of their location or background, thereby promoting equality and inclusivity in education.

However, despite these significant advantages, the integration of artificial intelligence and digital platforms also presents several technical, ethical, and pedagogical challenges. Issues such as inadequate technological infrastructure, unequal access to digital resources, insufficient digital literacy among educators, and concerns related to data privacy and security must be carefully

addressed. Additionally, there is a need to ensure that the use of artificial intelligence does not reduce students' ability to think independently or engage in critical and creative thinking processes.

Looking toward the future, it is evident that the role of artificial intelligence in education will continue to expand and evolve. Emerging technologies are expected to contribute to the development of more advanced, innovative, and adaptive teaching models that combine human instruction with machine intelligence. This evolution will likely reshape the structure of education systems worldwide, creating more personalized and data-driven learning environments.

Therefore, education systems must proactively adapt to these technological changes by investing in infrastructure, improving digital competencies among teachers, and developing appropriate ethical and regulatory frameworks. Only through a balanced and well-planned approach can the full potential of artificial intelligence and digital platforms be realized in education.

Ultimately, when implemented effectively and responsibly, artificial intelligence and digital platforms have the capacity not only to enhance the quality and efficiency of education but also to make a meaningful contribution to broader social and economic development. In the future, the role of artificial intelligence in education will continue to grow, leading to the development of innovative teaching models. Therefore, education systems must adapt to these changes and ensure the effective integration of technologies.

Thus, with the right approach, artificial intelligence and digital platforms can significantly enhance the quality of education and contribute to societal development.

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THE FUTURE OF EDUCATION: TRENDS, TECHNOLOGIES, AND HUMAN MENTORSHIP

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Abstract. This article explores the transformative shift in the global educational landscape as it transitions from traditional, industrial-age "factory models" to digital, flexible, and personalized learning environments. Adopting a qualitative and comparative methodology based on current academic literature, the research focuses on the integration of Artificial Intelligence (AI), Virtual Reality (VR/AR), and gamification elements as essential tools for creating adaptive learning pathways and immersive experiences. A significant focus is placed on the globalization and democratization of knowledge through online platforms like Coursera and Khan Academy, allowing students in developing regions, such as Azerbaijan, to access world-class academic resources seamlessly. Furthermore, the study examines the infrastructural evolution of physical classrooms into collaborative "learning spaces" and the rise of the "Flipped Classroom" model. Within this technology-driven framework, the paper analyzes the changing role of teachers, who must transition from being primary knowledge providers to becoming mentors and life coaches. It emphasizes the critical importance of the "4 Cs"—Critical Thinking, Creativity, Collaboration, and Communication—alongside Emotional Intelligence (EQ) over traditional rote memorization. The findings suggest that because the global economy evolves rapidly, the future of education lies in the concept of Lifelong Learning, where individuals continuously up-skill and re-skill through micro-courses. Ultimately, while AI provides the technical infrastructure for personalized education, the human element remains vital for emotional connection and ethical guidance, creating a balanced, human-centered educational experience for the 21st century.

Keywords: future of education, artificial intelligence, personalized learning, digital transformation, lifelong learning, mentorship.

TƏHSİLİN GƏLƏCƏYİ: TRENDLƏR, TEXNOLOGİYALAR VƏ İNSAN MENTORLUĞU

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Xülasə: Bu məqalə qlobal təhsil mənzərəsinin ənənəvi sənaye dövrünün "zavod modelləri"ndən fərdiləşdirilmiş və rəqəmsal tədris mühitlərinə keçid etdiyi transformasiya prosesini araşdırır. Tədqiqatın metodologiyası rəqəmsal təhsil, süni intellekt və immersiv texnologiyalar sahəsində mövcud akademik mənbələrin keyfiyyət və müqayisəli təhlilinə əsaslanır. Məqalədə adaptiv öyrənmə yollarının və immersiv təcrübələrin yaradılması vasitələri kimi Süni İntellekt (Sİ), Virtual Reallıq (VR/AR) və oyunlaşdırma (gamification) elementlərinin dərslərə inteqrasiyası geniş təhlil olunur. Coursera və Khan Academy kimi qlobal platformaların təhsilin demokratikləşməsində və Azərbaycan kimi ölkələrdə "sərhədsiz təhsilin" formalaşmasında rolu araşdırılır. Tədqiqatda fiziki sinif otaqlarının çevik "tədris məkanları" və "tərs yüz edilmiş sinif" (Flipped Classroom) modeli ilə əvəzlənməsi prosesi işıqlandırılır. Bununla yanaşı, informasiya bolluğu şəraitində müəllimlərin rolunun sadəcə bilik ötürücüsündən mentor və fasilitatora çevrilməsi zərurəti

əsaslandırılır. Məqalədə ənənəvi əzbərçilikdən fərqli olaraq "4 Cs" — Kritik düşüncə, Yaradıcılıq, Əməkdaşlıq və Ünsiyyət (Critical Thinking, Creativity, Collaboration, Communication) kimi 21-ci əsr bacarıqlarının və emosional intellektin (EQ) əhəmiyyəti vurğulanır. Sürətlə dəyişən global iqtisadiyyatda diplom mərkəzli təhsilin öz qüvvəsini itirdiyi və onun yerini "ömürboyu təhsil" (Lifelong Learning) paradigmasının aldığı göstərilir. Yekun nəticələr sübut edir ki, texnologiya fərdiləşdirilmiş təhsil üçün unikal alətlər təqdim etsə də, etik rəhbərlik, empatiya və dəyərlərin aşılmasında insan faktoru və müəllim mentorluğu hələ də əvəzedilməzdir.

Açar sözlər: təhsilin gələcəyi, süni intellekt, fərdiləşdirilmiş təhsil, rəqəmsal transformasiya, ömürboyu təhsil, mentorluq.

Introduction

The landscape of human knowledge and its transmission is undergoing a seismic shift in the 21st century. For over a century, the global education system operated under an industrial-age paradigm, characterized by standardized curricula and physical classrooms. However, the rapid advancement of technology and the digital revolution have rendered this "factory model" increasingly obsolete. As we stand at the crossroads of a new era, the fundamental question is no longer whether education will change, but how quickly we can adapt to its new reality. Today, geography is no longer a barrier to excellence; the "Global Classroom" has democratized access to the world's most prestigious institutions. This study will delve into the integration of AI and immersive technologies, exploring how education is becoming more personalized, accessible, and centered around lifelong learning.

Methodology and Materials

This study follows a qualitative and theoretical approach, focusing on the transformation of education in the digital era and the impact of modern technologies.

The research is based on the analysis of academic sources related to digital education, AI, VR/AR, and gamification. A comparative method is used to examine differences between traditional and modern education models. In addition, key concepts such as personalized learning, lifelong learning, and immersive learning are analyzed.

A descriptive method is also applied to explain the role of new technologies in education. The study does not include empirical data but relies on existing literature, providing a basis for future research.

Discussion

To understand the future of education, we must first look at its past. For over a hundred years, the model of education has been like a factory. Students sit in rows, follow a strict schedule, and learn the same things at the same time. This was useful during the Industrial Revolution because society needed people who could follow instructions. However, we now live in an information age, and the old model is no longer working.

If we look back fifty years, a classroom was a simple place with a wooden desk and a teacher who held all the knowledge. Students spent hours memorizing facts from heavy textbooks. Today, through platforms like Coursera, edX, or YouTube, a student in Azerbaijan can learn physics from a professor at Harvard. Furthermore, future "learning spaces" will replace traditional classrooms. Imagine rooms with comfortable chairs and digital screens where students work together on projects. This is often called the "Flipped Classroom" model, where students watch lectures at home and come to school for active collaboration and laboratory work.

When we talk about the future, we cannot ignore the role of technology, specifically Artificial Intelligence (AI). While some worry that AI might make students passive, when used correctly, it becomes the most powerful tool in the history of learning. The fundamental flaw of traditional education is the "one size fits all" approach. In a class of thirty students, a teacher usually speaks at an average pace, which often leaves fast learners bored and slower learners confused. AI revolutionizes this through Personalized Learning Pathways. Imagine a digital tutor that identifies exactly what a student finds difficult. If a student struggles with English tenses, the AI provides tailored stories, interactive games, and exercises until the concept is mastered. It is essentially a private teacher available 24/7.

Furthermore, Virtual and Augmented Reality (VR/AR) are transforming history and science into lived experiences. Reading about the depths of the ocean or the surface of Mars is limited by imagination, but "visiting" them through VR is transformative. Geography lessons will evolve into expeditions where students fly through the rings of Saturn instead of just looking at a map. This "Immersive Learning" is far more effective because the human brain retains experiences much better than static text. Through Gamification, the boundary between "playing" and "learning" will become increasingly thin, making education an engaging and addictive journey of discovery.

As technology automates the routine cognitive tasks of learning, the role of the human teacher and the skills students require are undergoing a profound shift. In an era where any fact can be retrieved via Google in seconds, rote memorization has lost its value. The fundamental question arises: what should we learn?

The focus of future education is shifting toward the "4 Cs": Critical Thinking, Creativity, Collaboration, and Communication. While machines excel at processing vast amounts of data, they lack the capacity for complex problem-solving and empathy. Consequently, the classrooms of the future will prioritize Emotional Intelligence (EQ). Teachers will transition from being mere "lecturers" to becoming "Mentors and Facilitators." Their primary responsibility will be to guide students through the vast ocean of digital information, teaching them how to distinguish truth from "fake news" and ethical complexities.

Another cornerstone of this era is Lifelong Learning. The traditional model—school, university, and a single career until retirement—is now obsolete. Because the global economy evolves so rapidly, education has become a continuous, lifelong process. Individuals must "up-skill" or "re-skill" every few years to remain relevant.

To understand the future of education, we must first look at its past. For over a hundred years, the model of education has been like a factory. Students sit in rows, follow a strict schedule, and learn the same things at the same time. This was useful during the Industrial Revolution because society needed people who could follow instructions. However, the world has changed. Today, we live in an information age, and the old "factory model" is no longer working.

The first big change we see in the future is the physical environment. Traditional classrooms are often boring and formal. In the future, "learning spaces" will replace classrooms. Imagine a room with comfortable chairs, plants, and lots of technology, where students can work together on projects. The walls might be digital screens that show images from different parts of the world. Education will not be limited to four walls. We are moving toward a more flexible way of learning where the student's comfort and creativity are the priority.

Furthermore, the globalization of education is a major factor. In the past, if you lived in a remote area, your access to quality teachers was very low. Now, through the internet, the world's

best libraries and universities are available to everyone. This means that a student in a small town in Azerbaijan can follow the same curriculum as a student in Finland or Singapore. This "borderless education" is the first step toward a more equal world.

Technology is the engine of this transformation. Many people think that technology in the classroom just means using a computer instead of paper, but it is much deeper than that.

One of the most exciting developments is Artificial Intelligence (AI). Currently, a teacher with 30 students cannot give every child the attention they need. AI can solve this problem through "Adaptive Learning." This means the software changes the difficulty of the lesson based on the student's performance. If you are struggling with a specific grammar point in English, the AI will recognize this and provide extra videos and simpler exercises. It is like having a personal assistant who knows your brain perfectly.

Secondly, we have Virtual Reality (VR). This will change subjects like History, Geography, and Science forever. Instead of just reading about the Great Wall of China, students can put on a headset and "walk" on it. They can travel inside the human body to see how the heart works or visit the surface of Mars. This makes learning "immersive." When we experience something with our senses, we remember it much better than if we just read it in a book.

Another important trend is Gamification. Most students love video games because they are engaging and offer immediate rewards. Educators are starting to use these elements in lessons. Instead of "studying," students "play" through levels of a subject. They earn points, unlock new topics, and compete with friends in a healthy way. This changes the student's attitude from "I have to study" to "I want to play and learn."

The Human Element: Teachers as Mentors

A common question people ask is: "Will robots replace teachers?" The answer is a clear no, but the role of the teacher will be completely different. In the old system, the teacher was the "source of all knowledge." Today, Google is the source of all knowledge. Therefore, teachers do not need to repeat facts that students can find online in five seconds.

In the future, teachers will become Mentors and Life Coaches. Their main job will be to teach students how to navigate the digital world. With so much information online, students need to learn "Critical Thinking." They need to know how to separate facts from "fake news." Teachers will also focus more on "Soft Skills" like empathy, leadership, and emotional intelligence. These are things that a computer cannot teach.

Additionally, Social and Collaborative Learning will be more important than ever. Machines can process data, but they cannot brainstorm a creative solution to a social problem. Future schools will be places where students come together to talk, debate, and create things. The focus will shift from "What do you know?" to "What can you do with what you know?"

Conclusion

Finally, we must talk about when education ends. In the past, education ended when you graduated from university. You got a diploma and started a career. In the future, this will not be enough. Because technology changes so fast, many jobs that exist today will disappear in ten years. This means everyone will need to be a Lifelong Learner.

Education will become a continuous journey. People will take "micro-courses" throughout their lives to learn new skills. A 40-year-old worker might go back to a digital school to learn about green energy or new management techniques. The "Future of Education" is not just for children; it is for everyone, at every age.

In conclusion, the future of education is bright and full of opportunities. It will be a mix of powerful technology and deep human connection. We will use AI and VR to understand the world, but we will use our human teachers to understand ourselves and our values. Education will become more personal, more exciting, and more accessible to everyone on the planet. The goal is no longer just to pass an exam, but to learn how to learn for the rest of our lives.

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APPLICATION OF DIGITAL TECHNOLOGIES IN EDUCATION

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Abstract: This article examines the application of digital technologies in modern education and their impact on the teaching and learning process. It highlights how tools such as computers, online platforms, interactive whiteboards, and mobile applications enhance the quality of education by making it more accessible, engaging, and efficient. The study also emphasizes the role of digital technologies in supporting personalized learning, allowing students to learn at their own pace and according to their individual needs. In addition, the advantages of distance learning and the importance of digital skills for both teachers and students are discussed. However, certain challenges, including technical issues, limited internet access, and potential health effects caused by prolonged screen time, are also addressed. Overall, the article concludes that the effective use of digital technologies significantly contributes to the development of modern education systems and improves learning outcomes

Keywords: digital technologies, education, e-learning, online platforms, interactive learning, distance education, educational innovation

TƏTBİQİ DİLÇİLİK VƏ DİSKURS TƏDQİQATLARI

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Xülasə: Bu məqalə müasir təhsildə rəqəmsal texnologiyaların tətbiqini və onların tədris prosesinə təsirini araşdırır. Məqalədə kompüterlər, onlayn platformalar, interaktiv lövhələr və mobil tətbiqlər kimi vasitələrin təhsilin keyfiyyətini necə artırdığı, onu daha əlçatan, maraqlı və səmərəli etdiyi izah olunur. Tədqiqat həmçinin rəqəmsal texnologiyaların fərdiləşdirilmiş öyrənməni necə dəstəklədiyini, şagirdlərin öz sürətlərinə və ehtiyaclarına uyğun şəkildə öyrənməsinə imkan yaratdığını vurğulayır. Bununla yanaşı, məsafədən təhsilin üstünlükləri və həm müəllimlər, həm də şagirdlər üçün rəqəmsal bacarıqların əhəmiyyəti qeyd olunur. Lakin texniki problemlər, internetə çıxışın məhdudluğu və uzun müddət ekran qarşısında qalmağın sağlamlığa təsiri kimi bəzi çətinliklər də nəzərə alınır. Ümumilikdə, məqalə göstərir ki, rəqəmsal texnologiyaların effektiv istifadəsi müasir təhsil sisteminin inkişafına və öyrənmə nəticələrinin yaxşılaşmasına mühüm töhfə verir.

Açar sözlər: rəqəmsal texnologiyalar, təhsil, elektron təhsil (e-learning), onlayn platformalar, interaktiv öyrənmə, məsafədən təhsil, təhsil innovasiyaları

Introduction

In the modern world, rapid technological advancement has profoundly influenced all spheres of society, especially the educational system. The dawn of the Fourth Industrial Revolution (Industry 4.0) has shifted the global socio-economic paradigm, necessitating a complete overhaul of traditional teaching methodologies. In the process of forming 21st-century skills, digital technologies are no longer just auxiliary tools but have become an integral part of teaching. These technologies provide the critical infrastructure required to foster critical thinking, digital literacy, and collaborative problem-solving among learners in an increasingly interconnected world. The digital transformation of traditional educational models has fundamentally changed the ways in which information is acquired, transmitted, and mastered. For centuries, the transmission of knowledge relied primarily on static textbook media and teacher-centric classroom environments. Today, digitization democratizes information, turning knowledge acquisition into a fluid, ongoing, and dynamic experience. The application of digital technologies in education makes the learning environment more interactive, accessible, and personalized. Tools such as computers, online platforms, and mobile applications allow students to develop at their own pace and according to their individual needs. This paradigm shift fundamentally alters the pedagogical relationship; the educator is no longer the sole source of knowledge but rather a facilitator who guides students through complex networks of digital information. This creates new opportunities for teachers, enabling them to manage the teaching process more efficiently and monitor each student's progress more closely. However, this rapid technological transition also brings certain challenges. Inequalities in technical infrastructure, limited internet access, and potential health risks caused by prolonged screen time are among the key issues that need to be systematically addressed. Without strategic implementation, technology risks widening the gap between different socioeconomic student demographics instead of narrowing it. Nevertheless, increasing digital literacy and integrating innovative methods into education are considered the primary guarantees for the success of modern educational systems. The purpose of this article is to comprehensively analyze the role of digital technologies in modern education, the advantages they offer, and the obstacles encountered during their implementation. Within the scope of the study, the effects of e-learning and distance education on the quality of education are explored through established scientific and theoretical frameworks.

Methodology and Materials

The methodology of this research is grounded in a qualitative and analytical approach, designed to evaluate the multifaceted impact of digital technologies on the contemporary educational landscape. Given the rapid pace of change in educational technology, a qualitative meta-analysis of contemporary pedagogical frameworks helps synthesize diverse empirical findings into a coherent narrative. To provide a comprehensive understanding of the subject, the study integrates theoretical analysis with a rigorous review of existing digital educational frameworks, ensuring a holistic perspective on how these tools reshape pedagogical outcomes. The primary materials utilized in this study consist of a diverse range of educational resources, scientific publications, and reports on digital infrastructure. This includes an in-depth examination of hardware solutions such as computers and interactive whiteboards, alongside software platforms like mobile applications and online learning systems. Furthermore, the study draws upon academic literature regarding e-learning and distance education to establish a solid theoretical foundation for the necessity of digital skills in modern teaching. The materials analyzed also include structural strategies implemented by global organizations, such as UNESCO's digital transformation initiatives, alongside empirical data concerning student well-being during prolonged screen usage. By cross-examining educational theories with data on technological infrastructure, this paper outlines both the practical limitations and the systemic benefits of modern ed-tech deployment.

Literature Review and Theoretical Framework (Ədəbiyyat İcmalı və Nəzəri Əsaslar)

To understand the scope of digital integration in schools and universities, it is crucial to analyze the theoretical pillars established by prominent scholars and global health institutions. The theoretical foundation of online learning is extensively rooted in the works of Anderson (2008), who argues that effective e-learning environments must be simultaneously learner-centered, content-centered, community-centered, and assessment-centered. Online platforms allow these four dimensions to converge seamlessly by enabling asynchronous discussions, structured multimedia modules, and continuous self-assessment loops. From a structural perspective, Bates (2019) outlines that teaching in the digital age requires distinct, flexible guidelines for designing learning environments. Bates argues that technology should not merely serve as an electronic chalkboard but must transform pedagogical design. The inclusion of interactive whiteboards and collaborative digital workspaces allows knowledge to be co-constructed by students, breaking the passive learning model that dominated the previous century. This correlates directly with the cognitive scaffolding theory proposed by Vygotsky (1978). Though written before the digital age, Vygotsky's emphasis on interactive learning and social interaction as the primary drivers of higher psychological processes forms the bedrock of modern digital collaboration. Digital platforms expand the "Zone of Proximal Development" by connecting learners with peers and experts globally, creating digital learning communities that transcend geographical borders.

Furthermore, personalized learning benefits heavily from systematic feedback loops. Hattie (2017) emphasizes that visible learning occurs when teachers see learning through the eyes of students and when students see themselves as their own directors of learning. Digital tools accelerate this process by providing real-time data analytics, allowing for personalized learning paths tailored to individual student needs and mastery speeds. However, macro-level structural obstacles persist. UNESCO (2020) provides a comprehensive global report highlighting technical challenges, internet shortages, and digital inequality. Their data indicates that the "digital divide" poses a severe threat to educational equity globally. Concurrently, the physical risks of digitized

classrooms are documented by the World Health Organization (WHO, 2019), whose guidelines warn against the sedentary behaviors and health risks linked to prolonged screen exposure.

Discussion and Systematic Analysis (Müzakirə və Sistemli Təhlil)

The analysis of the research findings highlights that the integration of digital technologies has moved beyond a simple trend and has become a fundamental requirement for the evolution of modern education. One of the most significant aspects of this shift is the enhancement of educational quality through increased accessibility and engagement, which is facilitated by online platforms and interactive tools. By creating a more dynamic learning environment, these technologies effectively bridge the gap between traditional teaching and the demands of the digital age, making the learning process more efficient and results-oriented. Multimedia elements cater to diverse learning styles—visual, auditory, and kinesthetic—ensuring higher rates of retention and material mastery. A critical point of discussion is the role of technology in supporting personalized learning, which stands in stark contrast to the traditional “one-size-fits-all” approach. In traditional systems, the teacher moves at a standardized pace, often leaving struggling students behind while failing to challenge advanced learners. Digital technologies disrupt this limitation. The ability for students to learn at their own pace and according to their individual needs not only improves academic performance but also fosters a sense of autonomy and responsibility in the learning process. Adaptive software tracks progress, offering corrective modules when a concept is misunderstood or advancing the student when mastery is demonstrated. This cultivates intrinsic motivation, turning students from passive recipients into active agents of their educational trajectories. Furthermore, the discussion emphasizes that distance learning is no longer a temporary solution but a flexible alternative that democratizes education. Historically, access to high-quality higher education was restricted by geographic and financial limitations. Distance learning breaks down these physical barriers, allowing an individual in a remote area to access resources from premier institutions worldwide. However, the success of these models is heavily dependent on the digital literacy of both teachers and students, suggesting that pedagogical training must evolve alongside technological advancements. Simply uploading text documents to a portal does not constitute effective e-learning; true digital pedagogy requires intentional instructional design that encourages remote collaboration and active critique.

Persistent Challenges and Socio-Technical Barriers (Çətinliklər və Sosial-Texniki Maneələr)

Despite these clear advantages, the research also reveals significant challenges that cannot be ignored. The transition to fully digitized systems has exposed vulnerabilities within global and local educational ecosystems.

The “digital divide” caused by technical issues and limited internet access remains a major barrier to global educational equity. While students in urban, economically developed regions benefit from high-speed broadband and private workstations, those in marginalized or rural areas often experience frequent connection failures, shared equipment, or a total lack of network access. This imbalance exacerbates existing educational stratification, transforming what should be an equalizer into a vector of inequality. Governments and institutional stakeholders must prioritize infrastructural investment, viewing high-speed internet as a public utility essential for educational development. Additionally, the potential health effects associated with prolonged screen time and the necessity of maintaining a balance between digital and physical interaction are crucial considerations for educators. As students spend several consecutive hours interacting via screens, various physiological and social issues emerge, including: Digital eye strain, headaches, and

myopia progression. Disruptions to sleep-wake cycles due to blue-light exposure before bed. Poor posture and musculoskeletal strain from unergonomic learning stations. Social isolation and reduction of face-to-face socio-emotional skills.

Strategic Recommendations for Implementation (Strateji Təvsiyələr)

To ensure that the integration of digital technologies yields optimal outcomes while minimizing risks, a series of strategic directives should be adopted: 1. Continuous Professional Development for Educators: Training initiatives must look beyond basic software literacy. Teachers should receive comprehensive training in modern instructional design, digital formative assessment strategies, and online classroom management techniques. 2. Hybrid Framework Design: Educational institutions should avoid full reliance on digital spaces. Curriculums must be engineered to merge digital convenience with offline laboratory work, physical education, and collaborative group seminars. 3. Infrastructural Investment Subsidies: Public-private partnerships must be established to provide low-cost hardware and zero-rated educational data access to low-income student populations, minimizing the digital divide.

Conclusion (Nəticə)

In conclusion, the research demonstrates that the effective integration of digital technologies is a cornerstone in the development of modern education systems and plays a vital role in improving overall learning outcomes. By utilizing tools such as computers, interactive whiteboards, and various online platforms, the educational process becomes significantly more accessible, engaging, and efficient for a broader range of learners.

The transition toward digital pedagogy has also enabled a much-needed shift toward personalized learning, providing students with the flexibility to master material at their own pace and according to their specific individual requirements. This structural transformation builds the foundation for lifelong learning habits, crucial for success in the volatile employment markets of the 21st century. Furthermore, while the advantages of distance learning and the necessity of digital skills for both educators and students are undeniable, the study emphasizes that technology must be implemented with a strategic and cautious approach. Addressing persistent challenges such as technical failures, limited internet connectivity, and the potential health risks associated with excessive screen time is essential for the long-term success of these innovations. Ultimately, if these barriers are managed effectively, digital technologies will continue to serve as a powerful catalyst for educational growth, ensuring that learning remains dynamic and responsive to the needs of the modern era.

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THE DEVELOPMENT OF ARTIFICIAL INTELLIGENCE IN EDUCATION: HISTORICAL FOUNDATIONS AND MODERN APPLICATIONS

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Abstract: This article provides a comprehensive analysis of the development of artificial intelligence in the field of education, focusing on its historical foundations and contemporary applications. It examines the early theoretical contributions of researchers such as Sidney L. Pressey and B. F. Skinner and explains their influence on the formation of modern AI-based educational technologies. In particular, the concepts of programmed instruction, immediate feedback, and “teaching machines” are highlighted as the basis of today’s adaptive and personalized learning systems. The study also explores the opportunities created by the integration of artificial intelligence into the educational process, including the development of individualized learning models, automated assessment mechanisms, and the analysis of student learning behaviors. Furthermore, the analytical advantages enabled by big data processing and machine learning technologies are discussed. At the same time, the article critically evaluates key challenges such as data quality, ethical concerns, privacy issues, and the risks associated with excessive reliance on technology. In conclusion, it is emphasized that while artificial intelligence offers significant opportunities for improving education, its effective implementation requires a balanced integration of technological innovations with pedagogical principles.

Keywords: Artificial Intelligence; Educational Technologies; Personalized Learning; Adaptive Learning Systems; Machine Learning

TƏHSİLDƏ SÜNİ İNTELLEKTİN İNKİŞAFI: TARİXİ ƏSASLAR VƏ MÜASİR TƏTBİQLƏR

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Xülasə: Bu məqalədə süni intellektin təhsil sahəsində inkişafı, onun tarixi əsasları və müasir tətbiqləri kompleks şəkildə təhlil olunur. Araşdırmada Sidney L. Pressey və B. F. Skinner kimi tədqiqatçıların irəli sürdüyü ilkin nəzəri yanaşmaların müasir süni intellekt texnologiyalarının formalaşmasına təsiri geniş şəkildə izah edilir. Xüsusilə proqramlaşdırılmış təlim, dərhal geribildirim və “öyrətmə maşınları”

konsepsiyasının bugünkü adaptiv və fərdiləşdirilmiş öyrənmə sistemlərinin əsasını təşkil etdiyi vurğulanır. Məqalədə süni intellektin tədris prosesində tətbiqi nəticəsində yaranan imkanlar, o cümlədən fərdi öyrənmə modellərinin qurulması, avtomatlaşdırılmış qiymətləndirmə mexanizmləri və tələbə davranışlarının analizi kimi məsələlər ətraflı şəkildə təqdim olunur. Bununla yanaşı, böyük verilənlərin emalı və Machine Learning texnologiyalarının təhsil sisteminə integrasiyası nəticəsində əldə olunan analitik üstünlüklər də nəzərdən keçirilir. Eyni zamanda, məlumat keyfiyyəti, etik problemlər, məxfilik məsələləri və texnologiyadan həddindən artıq asılılığın yaratdığı risklər də tənqidi yanaşma ilə dəyərləndirilir. Nəticə olaraq, süni intellektin təhsildə tətbiqinin həm geniş imkanlar yaratdığı, həm də müəyyən çağırışlar doğurduğu qeyd edilir və onun effektiv istifadəsi üçün texnoloji yeniliklərlə pedaqoji prinsiplərin balanslı şəkildə uzlaşdırılmasının vacibliyi vurğulanır.

Açar sözlər: Süni intellekt; Təhsil texnologiyaları; Fərdiləşdirilmiş öyrənmə; Adaptiv təlim sistemləri; Machine Learning

Introduction

In the modern era, the rapid development of artificial intelligence technologies has led to significant transformations not only in various sectors of society but also in education systems. Especially in the 21st century, where digitalization has become widespread, traditional teaching models are undergoing transformation and are being replaced by more flexible, personalized, and technology-based approaches. In this regard, the application of artificial intelligence in education is considered not only a technical innovation but also a fundamental reconfiguration of pedagogical approaches.

The theoretical foundations of artificial intelligence in education date back to the mid-20th century, particularly to the works of researchers such as Sidney L. Pressey and B. F. Skinner. Their ideas related to programmed instruction and “teaching machines” laid the groundwork for the development of modern adaptive learning systems. Today, artificial intelligence technologies are implemented as complex systems that take into account students’ individual characteristics, analyze their learning behaviors, and optimize the instructional process.

The main objective of this article is to examine the developmental stages of artificial intelligence in education, analyze its modern applications, and scientifically evaluate the advantages and challenges arising from this process.

Methods and Materials

The research primarily adopts a qualitative approach. Within the scope of the study, existing scientific literature, academic articles, and theoretical perspectives on artificial intelligence and education were analyzed. In particular, the works of scholars such as Sidney L. Pressey and B. F. Skinner, as well as contemporary studies on the application of artificial intelligence in education, constitute the main source base of the research.

Methodologically, comparative analysis and a systematic approach were employed. Through a historical approach, the stages of artificial intelligence development in education were examined chronologically, while comparative analysis was used to identify differences between traditional teaching models and AI-based approaches. Additionally, descriptive and analytical methods were applied to evaluate the impact of artificial intelligence technologies on the teaching process, as well as their opportunities and challenges.

Furthermore, data synthesis and interpretation techniques were used to systematize and present the findings in a scientifically grounded manner. This approach enabled a comprehensive analysis of both the theoretical and practical aspects of the topic.

The integration of artificial intelligence into the education system is currently one of the key factors enhancing the quality of the teaching process. Research indicates that AI technologies facilitate personalized learning, optimize teachers' pedagogical activities, and enable students to adapt more effectively to the learning process. At the same time, AI-based systems significantly contribute to the automation of assessment processes, preparation of teaching materials, and improvement of educational management efficiency. However, the implementation of these technologies is also accompanied by certain challenges. In particular, insufficient digital skills among teachers, ethical and data security issues, and digital inequality are among the main factors limiting the full integration of artificial intelligence into education systems.

The initial examples of artificial intelligence in education date back to the 1920s, and one of the earliest pioneers in this field was Sidney L. Pressey. Pressey (1950) emphasized that multiple-choice tests should not only be used as assessment tools but also as instruments that enhance learning. According to him, students could immediately check their answers, identify their mistakes, and thus better understand the subject matter. This approach aligns with Edward Thorndike's "law of effect," which states that immediate feedback during the learning process increases motivation and strengthens knowledge retention. [5]

Pressey attempted to apply this theoretical approach in practice by proposing mechanical devices that automatically checked students' answers and displayed the correct responses. These devices can, in fact, be considered the earliest forms of today's digital testing systems and adaptive learning platforms. Although technological limitations of that period prevented the widespread adoption of these ideas, they nevertheless laid the foundation for automated learning systems in education.

In addition, Pressey emphasized that such technologies were beneficial not only for students but also for teachers. According to him, the automation of the assessment process significantly reduced teachers' workload. This, in turn, allowed educators to allocate more time to individual student support, better understand learners' needs, and organize the teaching process more effectively. This approach is now recognized as one of the core principles of "technology-enhanced teaching." [4]

In a later stage, B. F. Skinner further developed these ideas by introducing the model of programmed instruction. In 1958, he designed devices known as "teaching machines." These systems presented questions to students step by step, accepted their responses, and immediately allowed comparison with correct answers. As a result, students could instantly identify their mistakes, learn the correct answers, and proceed to the next stage. According to Skinner, such systems ensured individual learning pace and enabled each learner to study according to their own speed.

Skinner referred to this approach as the "personal tutor effect," meaning that technology acts as a personal teacher for each student by providing instruction adapted to their level of knowledge. This idea remains highly relevant today and is widely applied in modern educational technologies. Currently, AI-based systems analyze students' behavior, identify their strengths and weaknesses, and provide personalized learning pathways accordingly. [2]

Thus, the early approaches proposed by Pressey and Skinner formed the foundation for the application of modern artificial intelligence technologies in education. Concepts such as "personalized learning," "adaptive learning systems," and "intelligent tutoring platforms" are the evolved forms of these original ideas. This demonstrates that the role of artificial intelligence in

education did not emerge suddenly but rather developed as a result of long-term scientific and technological progress.

In the contemporary era, the development of artificial intelligence is no longer limited to the technological domain; it has deeply integrated into almost all spheres of society, particularly education. Widely used digital platforms in everyday life clearly demonstrate the capabilities of artificial intelligence. For instance, platforms such as YouTube and Netflix analyze user behavior and provide personalized recommendations. This same logic is applied to education, enabling the delivery of individualized learning content tailored to students' learning styles.

From this perspective, the application of artificial intelligence in education should not be regarded merely as a technological innovation but also as a fundamental transformation of pedagogical approaches. While traditional education models provided the same content and pace for all students, AI-based systems shift this paradigm by considering individual needs, knowledge levels, and learning speeds. As a result, this approach plays a crucial role in improving teaching quality and enhancing learning effectiveness.

Artificial intelligence (AI) plays a significant role as a key factor driving profound transformations across social, economic, and cultural domains through its widespread application in education and society. First and foremost, the use of AI in education contributes to the transformation of traditional teaching models and facilitates the emergence of more flexible, personalized, and digitally based approaches. Intelligent tutoring systems and online learning platforms not only increase access to education but also improve its overall quality.

In addition, artificial intelligence plays an important role in scientific research. The capabilities of processing large volumes of data, statistical analysis, and modeling contribute to accelerating scientific discoveries and achieving innovative outcomes. This feature makes AI one of the essential tools of modern scientific activity. [3]

From an economic perspective, artificial intelligence leads to the emergence of new industries, changes in the structure of the labor market, and an increasing demand for new skills. This process not only creates new job opportunities but also necessitates the restructuring of existing professional fields.

At the same time, the widespread use of AI raises a number of ethical and regulatory challenges. Issues such as data privacy, algorithmic bias, security, and accountability are among the key factors requiring regulation of these technologies. Therefore, the development of artificial intelligence should be evaluated not only from a technical perspective but also within legal and ethical frameworks.

From a socio-cultural perspective, AI requires the consideration of diverse perspectives and inclusive approaches. Ensuring fairness and reducing bias in algorithms makes the participation of diverse social groups in the development process essential. Furthermore, the integration of artificial intelligence into culture and the arts creates new creative opportunities. The use of AI in fields such as music, visual arts, and literature enables the emergence of new forms of artistic expression. [1]

In conclusion, artificial intelligence is not merely a technological innovation but also a major driving force of broad socio-cultural transformation, encompassing education, economy, and culture. Its effective implementation requires the consideration of ethical and social principles alongside technological advancement.

Moreover, the use of artificial intelligence in education is not limited to personalization. Modern systems include automated assessment mechanisms capable of analyzing students'

responses more quickly and objectively. This reduces subjectivity in evaluation processes and significantly decreases teachers' workload. Consequently, educators can dedicate more time to interactive activities, student development, and improving the quality of instruction.

In addition, artificial intelligence enables deeper analytical capabilities through data collection and analysis in education. Machine learning and data-driven approaches allow for the examination of student learning behaviors, identification of strengths and weaknesses, and prediction of future learning outcomes. These systems process large volumes of data to develop personalized instructional strategies accordingly.

However, the widespread implementation of these technologies also presents certain challenges. First, the effectiveness of AI systems is directly dependent on the quality of the data they use. Incomplete or inaccurate data may lead to unreliable outcomes. Furthermore, excessive dependence on technology may weaken students' critical thinking, analytical skills, and independent decision-making abilities, which could negatively affect long-term educational outcomes. [6]

Artificial intelligence in education also extends beyond learning processes to include administrative functions. For example, AI is widely used in universities and schools for predicting student performance, optimizing course schedules, allocating resources more efficiently, and automating overall management processes. This approach makes educational systems more flexible, efficient, and manageable.

As noted in earlier sections, the development of artificial intelligence in education is rooted in historical processes and initially emerged from the ideas of researchers such as Sidney L. Pressey and B. F. Skinner. Their concepts of "teaching machines" and programmed instruction now form the basis of modern technologies such as "intelligent tutoring systems" and "personalized learning platforms."

Conclusion

The research demonstrates that the application of artificial intelligence in education is the result of a long-term scientific and technological development process. The programmed instruction and automated learning approaches proposed by Sidney L. Pressey and B. F. Skinner constitute the theoretical foundation of modern adaptive and personalized educational systems. In this sense, the role of artificial intelligence in education is not accidental but rather a logical continuation of historical development.

In the contemporary stage, artificial intelligence technologies make a significant contribution to improving the quality of the teaching and learning process. In particular, personalized learning models, automated assessment systems, and big data analytics create conditions for organizing students' learning processes more effectively. This not only enhances educational outcomes but also optimizes teachers' workload, allowing them to devote more attention to their pedagogical activities.

At the same time, the widespread application of artificial intelligence in education also raises certain risks and challenges. Factors such as data quality, ethical issues, privacy concerns, and excessive dependence on technology are among the key aspects that require careful consideration in this field. In particular, the preservation of students' critical thinking and independent decision-making skills should remain one of the priorities of modern education systems.

In conclusion, while artificial intelligence has great potential for the development of education systems, its effective implementation requires an approach that combines technological capabilities

with pedagogical principles. Therefore, future research should focus on developing comprehensive strategies for the more efficient and secure use of artificial intelligence in education.

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FORMATION OF PATRIOTIC IDEAS IN THE CONTEXT OF DIGITAL AND AUDIO LEARNING ENVIRONMENTS

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Abstract: The rapid development of information and communication technologies in the modern education system has created the basis for the formation of new approaches to the organization of the learning process. In particular, the integration of digital and audio learning environments into the learning process allows students to more effectively master knowledge, increase interest in the lesson, and strengthen motivation to learn. This article examines the possibilities of applying these technologies in the learning process and their impact on the formation of patriotic ideas. The study shows that digital resources, interactive platforms, and audio-based materials enhance students' emotional perception and provide a more sensitive approach to national values. The content presented through these tools has a positive effect on the formation of feelings of loyalty to the homeland, national identity, and respect for historical memory in students. At the same time, it is noted that the purposeful and correct use of these technological opportunities by teachers creates conditions for a more efficient and high-quality organization of the learning process.

Key words: digital learning, audio teaching, literature lessons, patriotic ideas, multimedia tools

RƏQƏMSAL VƏ AUDIO TƏDRİS MÜHİTİ KONTEKSTİNDƏ VƏTƏNPƏRVƏRLİK İDEYALARININ FORMALAŞMASI

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Xülasə: Müasir təhsil sistemində informasiya-kommunikasiya texnologiyalarının sürətli inkişafı təlim prosesinin təşkilində yeni yanaşmaların formalaşmasına zəmin yaratmışdır. Xüsusilə rəqəmsal və audial öyrənmə mühitlərinin tədris prosesinə inteqrasiyası şagirdlərin bilikləri daha effektiv mənimsəməsinə, dərəcənin artmasına və öyrənmə motivasiyasının güclənməsinə şərait yaradır. Bu məqalədə həmin texnologiyaların tədris prosesində tətbiqi imkanları və onların vətənpərvərlik ideyalarının formalaşdırılmasına təsiri araşdırılır. Tədqiqat göstərir ki, rəqəmsal resurslar, interaktiv platformalar və səs əsaslı (audial) materiallar şagirdlərin emosional qavrayışını gücləndirərək milli dəyərlərə daha həssas yanaşmanı təmin edir. Bu vasitələr vasitəsilə təqdim olunan məzmun şagirdlərdə vətənə bağlılıq, milli kimlik və tarixi yaddaşa hörmət hisslərinin formalaşmasına müsbət təsir göstərir. Eyni zamanda qeyd olunur ki, müəllimlərin bu texnoloji imkanlardan məqsədyönlü və düzgün istifadəsi təlim prosesinin daha səmərəli və keyfiyyətli təşkilinə şərait yaradır.

Açar sözlər: rəqəmsal öyrənmə, audial tədris, ədəbiyyat dərsləri, vətənpərvərlik ideyaları, multimedia vasitələri

Introduction

The application of digital and audio learning tools in the modern education system plays an important role in improving the quality of the learning process. As a result of the development of information and communication technologies, the educational environment has become more modern and based on active learning. This approach allows learning to be not limited to theoretical knowledge, but also to be mastered more effectively in a practical and visual way (Seyidova, 2025). Electronic education platforms (Moodle, Google Classroom), interactive tools (Kahoot, Quizlet, Padlet), simulation and virtual reality technologies, as well as multimedia and visual presentations, which are widely used in the educational process, make the learning process more interesting and accessible. These technologies not only increase students' motivation for the lesson, but also create conditions for adapting to their individual learning needs (Hamidov & Aliyeva, 2026).

At the same time, the application of digital and audio tools expands the teacher's organizational function and allows lessons to be structured in accordance with modern educational requirements.

Materials and methods

The application of modern technologies in the educational process is of particular importance in terms of organizing student-centered learning and creating an active and interactive learning environment. Digital platforms, electronic learning resources, online assessment tools and multimedia tools increase students' cognitive activity, create conditions for the formation of their independent thinking, critical analysis and creative approach skills. At the same time, these technologies allow for individualization of training, taking into account the capabilities and needs of each learner.

The effective application of digital technologies is impossible without relying on pedagogical foundations. The integration of technology into education should not be limited only to technical knowledge, but should be carried out taking into account didactic principles, psychological characteristics and methodological approaches. In this regard, the role of the teacher is of particular importance. The teacher acts as both the organizer of the learning process and the main subject guiding learning in the digital environment. The teacher's digital competence, ability to use innovative learning strategies and mastery of modern pedagogical approaches are among the main factors determining the effectiveness of teaching (Axundov, 2026 s. 46).

The effective organization of a digital learning environment requires consideration of a number of key criteria. These criteria directly affect the quality of the learning process, creating conditions for more effective and purposeful learning. Motivation is one of the main components of digital learning. The use of various digital tools and resources increases the interest of pupils and students in the lesson, allowing them to participate more actively in the learning process. Personalized learning is one of the advantages of modern educational technologies. Through mobile applications and online platforms, the learning process is organized in accordance with the individual needs and capabilities of learners. Interactivity is one of the important features of digital learning. Interactive technologies make the learning process more lively and dynamic, and enable students to assimilate knowledge more easily. Multimedia-based learning, on the other hand, serves to make learning materials more understandable and memorable through the use of visual and audio resources.

The use of sound and audio media in education, among others, greatly enhances verbal communication, as audio content can engage with emotions, evoke emotions, activate memory, bring the past to life, and generally sensitize people to deeper elements of their culture (Kalliris et al., 2014)

As we know, the use of modern technologies in the educational environment is more widespread than in previous years. This development, in turn, has created new opportunities in the teaching of literature. These technologies create conditions for the presentation of literary examples in a more interactive and multifaceted way, as a result of which it becomes easier for students to understand the texts more deeply. Unlike the traditional reading process, digital tools allow the presentation of poetry and prose examples in both visual and audio formats.

Thus, teaching poetry and prose works written on the theme of the homeland in audio and visual formats has a positive effect on the formation of feelings of national identity, attachment to the homeland and respect for historical memory in students. Discussion and interactive analysis of these works in a digital environment creates conditions for students to more consciously assimilate the ideas of patriotism.

In modern literature lessons, the topic of patriotism is taught using various pedagogical methods. With the application of digital and audio technologies, poems and prose samples are presented in audio and video format, interactive discussions are held, and students' emotional approach to the topic is strengthened. This approach creates conditions for them to more consciously understand their sense of attachment to the homeland.

The development of patriotism is carried out mainly within the curriculum and through various forms of extracurricular activities. This is manifested when students experience feelings of love and pride for their homeland, respect its history, and experience moments when they feel proud of the courage and bravery of their patriots.

The teaching staff plays a very important role in improving the system of patriotic education. A deeper understanding of the essence and goals of patriotic education by teachers allows them to effectively use the latest educational technologies and techniques in working with students (Alikhanova, 2025, s. 213)

In the process of forming a patriotic spirit in a modern educational environment, the effective use of digital and audio educational technologies is of particular importance. In literature lessons, teachers can present poems and artistic examples written on the theme of patriotism through digital presentations (slides, interactive boards, video materials), not content with traditional explanations. This approach strengthens students' visual perception capabilities and increases the emotional impact of the topic.

At the same time, audio educational tools-reciting poems, audio recordings, listening to artistic texts against a musical background-create conditions for the formation of a sense of patriotism in students at a deeper and more emotional level. In particular, expressive reading, listening to poetry, and audio books strengthen students' emotional attachment to the text. By integrating these technologies, teachers can make the lesson more interactive and attractive, and ensure the active involvement of students in the lesson. Thus, giving students tasks such as preparing presentations, performing poems audibly, or explaining a topic through videos develops both their digital skills and their sense of patriotism.

Secondary school teachers can also organize various events using digital and audio educational technologies to form patriotic education. For example, at events held on Victory Day, teachers can use multimedia tools to demonstrate video footage, historical images and presentations related to the Patriotic War. Such visual materials create conditions for the emotional and deep formation of the concept of the Motherland and love for the homeland in students.

At the same time, the use of audio technologies during the event — for example, singing songs on the theme of Victory or presenting poems in audio format — strengthens students' emotional attachment to the topic. As one of the modern approaches, direct access of students to songs, video materials and audio files on the theme of Victory through mobile devices through QR code technology increases the interactivity of the lesson.

As a result, such approaches increase students' interest in the topic and create conditions for deepening their love for the homeland.

Discussion

The analysis shows that the use of digital and audio tools in literature lessons directly affects the quality of perception of literary texts. The presentation of poems and prose examples written on the theme of patriotism through audio performances and multimedia presentations strengthens the emotional approach of students to the text and allows for a clearer understanding of the idea. Compared to traditional reading and explanation methods, this approach ensures more active involvement of students in the lesson process. However, it is important to select digital and audio tools not only from a technical point of view, but also in accordance with pedagogical goals. Otherwise, the artistic and ideological load of the content may not be fully conveyed.

Thus, the results obtained show that these technologies serve to more effectively form patriotic ideas in literature lessons and have a positive effect on the moral and aesthetic development of students.

Conclusion

As a result, the application of digital and audio learning tools in literature lessons enriches traditional forms of teaching with more interactive and effective approaches. These tools increase the level of students' comprehension of literary texts and create conditions for their active involvement in the teaching process. At the same time, the presentation of literary examples written on the theme of patriotism in audio and visual format enhances the emotional impact on students and contributes to the formation of national and spiritual values. As a result, digital and audio technologies act as important tools in literature teaching that both increase didactic efficiency and serve educational purposes.

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INTEGRATION OF MOBILE APPLICATIONS (DUOLINGO, MEMRISE) INTO CLASSROOM LESSONS

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Abstract: This article examines the integration of mobile applications, particularly Duolingo and Memrise, into classroom teaching and their role in foreign language education. In the modern era, the rapid development of digital technologies has led to the emergence of new approaches in the education system, while mobile learning (m-learning) has become one of the innovative directions of teaching. The study

emphasizes that mobile applications make the learning process more flexible, interactive, and student-centered, increase learners' motivation, and create opportunities for personalized learning.

The research analyzes the main features of Duolingo and Memrise. Duolingo's gamified learning system, adaptive teaching mechanism, and daily motivation strategies positively influence students' continuous learning process. Memrise, on the other hand, supports vocabulary acquisition through spaced repetition systems, authentic videos, and visual association techniques.

The article also explores various methods of integrating mobile applications into classroom teaching. Approaches such as the flipped classroom model, blended learning, task-based learning, microlearning, and gamification have been shown to improve the quality of the teaching process. Mobile technologies not only encourage students' active participation but also provide teachers with opportunities for monitoring and individualized instruction.

The findings indicate that the integration of mobile applications into the educational process is one of the major development trends in modern education. This approach contributes not only to knowledge acquisition but also to the development of independent thinking, digital literacy, and communicative skills among students. At the same time, the article highlights certain challenges, including technical limitations and differences in teachers' digital competencies.

Keywords: digital education, Duolingo, Memrise, foreign language teaching, interactive learning, technology-enhanced learning, personalized learning

MOBİL TƏTBİQLƏRİN (DUALİNGO, MEMRİSE) AUDİTORİYA DƏRSLƏRİNƏ İNTEQRASIYASI

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Xülasə: Bu məqalədə mobil tətbiqlərin, xüsusilə Duolingo və Memrise platformalarının auditoriya dərslərinə inteqrasiyası və onların xarici dil tədrisində rolu araşdırılmışdır. Müasir dövrdə rəqəmsal texnologiyaların sürətli inkişafı təhsil sistemində yeni yanaşmaların formalaşmasına səbəb olmuş, mobil öyrənmə (m-learning) anlayışı isə innovativ tədris istiqamətlərindən biri kimi aktuallaşmışdır. Məqalədə qeyd olunur ki, mobil tətbiqlər öyrənmə prosesini daha çevik, interaktiv və tələbəyönümlü hala gətirir, tələbələrin motivasiyasını artırır və fərdiləşdirilmiş öyrənmə imkanları yaradır.

Tədqiqat zamanı Duolingo və Memrise tətbiqlərinin əsas xüsusiyyətləri təhlil edilmişdir. Duolingo tətbiqinin oyunlaşdırılmış öyrənmə sistemi, adaptiv tədris mexanizmi və gündəlik motivasiya prinsipləri tələbələrin davamlı öyrənməsinə müsbət təsir göstərir. Memrise isə yaddaşa əsaslanan təkrar sistemi, autentik videolar və vizual assosiasiya texnikaları vasitəsilə söz ehtiyatının daha effektiv mənimsənilməsinə təmin edir.

Məqalədə həmçinin mobil tətbiqlərin auditoriya dərslərinə inteqrasiya üsulları araşdırılmışdır. Tərsinə sinif modeli, qarışıq öyrənmə, tapşırıq əsaslı öyrənmə, mikroöyrənmə və oyunlaşdırma kimi yanaşmaların tədris prosesinin keyfiyyətini artırdığı göstərilmişdir. Mobil texnologiyalar tələbələrin aktiv iştirakını təmin etməklə yanaşı, müəllimlərə monitorinq və fərdi yanaşma imkanları da yaradır.

Araşdırmanın nəticələri göstərir ki, mobil tətbiqlərin təhsil prosesinə inteqrasiyası müasir təhsilin inkişaf istiqamətlərindən biri hesab olunur. Bu yanaşma yalnız biliklərin mənimsənilməsinə deyil, həm də tələbələrin müstəqil düşünmə, rəqəmsal savadlılıq və kommunikativ bacarıqlarının inkişafını təmin edir. Bununla yanaşı, texniki təminat və rəqəmsal kompetensiyalar kimi müəyyən çətinliklərin mövcudluğu da qeyd edilmişdir.

Açar sözlər: rəqəmsal təhsil, Duolingo, Memrise, xarici dil tədrisi, interaktiv öyrənmə, texnologiya əsaslı təhsil, fərdiləşdirilmiş öyrənmə

Introduction

In the modern era, the education system is no longer limited to the transmission of knowledge; it also requires the creation of a more flexible and effective learning environment through the use of technological opportunities. The transformation of digital technologies into an integral part of everyday life, especially the widespread use of mobile devices, has accelerated the introduction of new approaches into the teaching process. Against this background, traditional teaching models are gradually being replaced by interactive, student-centered, and technology-enriched models.

The concept of mobile learning (m-learning) has emerged as one of the innovative directions in education, freeing the learning process from time and space limitations. Students are no longer limited only to classroom learning; they can expand and reinforce their knowledge through various digital platforms. This approach enables individualized learning pace, ensures continuity in education, and increases active learner participation.

The use of mobile applications in foreign language learning has gained particular importance. Since the development of language skills requires constant repetition, listening, speaking, and practical application, digital tools play a highly significant role in this process. In this regard, applications such as Duolingo and Memrise provide users with systematic, step-by-step, and engaging learning opportunities. Their main advantage lies in making the learning process more attractive through game elements, visual and audio materials, and personalized tasks.

As the majority of modern students belong to the “digital generation” that is closely connected with technology, the use of innovative tools alongside traditional methods is considered essential for maintaining their attention and motivation. Therefore, integrating mobile applications into classroom teaching serves not only as an additional tool but also as an important factor in improving the quality of education. As a result of this integration, lessons become more interactive, students’ activity increases, and teacher–student interaction becomes more effective.

However, the implementation of mobile technologies in the educational process may also be accompanied by certain challenges. Insufficient technical support, internet access problems, and varying levels of teachers’ digital competencies are among the major issues encountered in this field. Nevertheless, properly planned approaches and methodological support can help overcome these problems.

The Role of Mobile Applications in the Educational Environment

In the modern education system, mobile technologies function not merely as supplementary tools but also as major factors transforming the structure of the learning process. Mobile applications make the educational environment more dynamic, flexible, and student-oriented, expanding the possibilities of traditional teaching models. Through these technologies, learning is no longer restricted by time and place and becomes continuous and sustainable.

First of all, mobile applications provide opportunities for personalized learning. Each student can master materials according to their own level of knowledge, learning speed, and individual needs. For example, Duolingo analyzes users’ mistakes and offers tasks accordingly, enabling a more purposeful learning process. Similarly, Memrise ensures the transfer of learned materials into long-term memory through repetition mechanisms designed to strengthen memory retention.

Another significant role of mobile applications is related to motivation. While traditional lessons may sometimes become monotonous, mobile platforms increase students' interest through game elements such as scoring points, passing levels, and receiving rewards. This approach makes learning more engaging and stimulates active participation. This factor is especially important in language learning, which requires continuous practice and repetition.

Moreover, mobile applications improve the quality of learning by using various multimedia tools. Audio recordings, videos, interactive tests, and visual materials allow students to adapt according to different learning styles. As a result, not only theoretical knowledge but also practical skills such as listening, speaking, and writing are developed.

Mobile technologies also create new opportunities for teachers. Teachers can monitor students' activities in real time, analyze their progress, and apply individualized approaches. In this respect, mobile applications serve as effective tools for monitoring and assessment. At the same time, teachers gain the opportunity to design lesson plans in a more interactive way that aligns with modern educational requirements.

On the other hand, mobile applications contribute to the development of collaborative and social learning environments. Students can complete tasks in groups, share results, and learn from one another. This positively affects the development of communication skills and teamwork abilities.

Nevertheless, the role of mobile applications in education is not limited only to advantages. Their effective use requires proper technical infrastructure, stable internet access, and the development of teachers' digital competencies. Otherwise, these technologies may fail to produce the expected outcomes.

Features of Duolingo and Memrise Applications

Among mobile language-learning platforms, Duolingo and Memrise occupy a special place. Although these applications are based on different methodological approaches, both aim to teach language skills through modern, interactive, and effective methods. Their functional capabilities and learning strategies create broad opportunities for integration into the educational process.

Features of Duolingo

Duolingo is one of the most widely used language-learning platforms in the world and is mainly based on gamification principles.

Gamified Learning System

Duolingo presents lessons in a game format. Users earn XP points, increase levels, and achieve various accomplishments by completing tasks. This approach makes learning more engaging and ensures continuity.

Short and Structured Lessons

Lessons are organized into small modules, allowing users to learn effectively in a short period while preventing distraction.

Adaptive Learning Mechanism

Duolingo analyzes user performance and automatically adjusts the level of difficulty. Frequently mistaken topics are repeated more often to strengthen weak areas.

Comprehensive Development of Language Skills

The application develops reading, writing, listening, and grammar skills simultaneously through various types of exercises.

Daily Learning System and Motivation

Daily goals and streak systems help users build consistent learning habits.

Artificial Intelligence and Automated Feedback

Duolingo instantly evaluates users' answers and provides explanations for mistakes, making the learning process more efficient.

Features of Memrise

Memrise differs through its focus on memory reinforcement and real-life language usage.

Spaced Repetition Learning

Memrise uses repetition systems based on intervals, helping learners store words and expressions in long-term memory.

Real-Life Videos and Authentic Materials

The application includes videos featuring native speakers, enabling users to learn authentic pronunciation and everyday conversational language.

Focus on Vocabulary Development

Memrise primarily aims to expand vocabulary by presenting words within context and using various visual and associative methods.

Visual Memory and Association Techniques

Images, metaphors, and associations are provided to help users remember vocabulary more effectively.

Personalized Learning Paths

The application creates individualized study plans according to users' development and systematically guides the learning process.

Offline Accessibility

Some functions can be used without internet access, increasing accessibility.

Methods of Integration into Classroom Teaching

The integration of mobile applications into classroom teaching requires a planned and methodological approach. This process is not limited only to the use of technology but also requires the restructuring of teaching objectives, content, and assessment mechanisms.

Flipped Classroom Model

Students familiarize themselves with the topic before class through mobile applications, while classroom time is devoted to practical activities and discussions.

Task-Based Learning

Mobile applications are used to complete specific tasks assigned by the teacher.

Blended Learning Model

Traditional and digital teaching methods are combined, allowing classroom learning and mobile learning to function together.

Assessment and Monitoring-Based Integration

Teachers use application statistics to monitor and analyze student performance.

Gamification and Competitive Environment

Game elements such as XP competitions and reward systems increase students' motivation.

Collaborative and Group Learning

Students work in groups using Duolingo or Memrise and later present and discuss their results.

Microlearning Approach

Short and focused mobile tasks are integrated into lessons through quick reviews and mini-tests.

Preparation of Individual Development Plans

Teachers can create personalized learning plans according to students' weak areas and monitor their progress regularly.

The analysis demonstrates that the integration of mobile technologies into education is no longer an option but a necessity. Especially in foreign language teaching, mobile applications contribute to a more effective, interactive, and continuous learning process. Applications such as Duolingo and Memrise play a crucial role in implementing modern pedagogical approaches.

The study also reveals that integrating mobile applications into classroom teaching represents not only a technological innovation but also a transformation in educational philosophy. Teachers move from being mere transmitters of knowledge to facilitators and guides, while students shift from passive listeners to active learners. Consequently, the educational process becomes more collaborative, interactive, and practice-oriented.

At the same time, the gamified structure and adaptive learning mechanisms of Duolingo increase students' motivation and encourage continuous engagement. Memrise supports long-term and practical knowledge retention through memory reinforcement and authentic language materials. The combined use of these two applications allows the learning process to develop both structurally and contextually.

Furthermore, the integration of mobile applications into education offers important advantages such as personalized learning opportunities, flexibility in terms of time and place, increased student motivation, and the development of modern digital skills. The implementation of microlearning and blended learning models ensures systematic and gradual knowledge acquisition.

Nevertheless, this process is not free from challenges. Limited technical infrastructure, internet access issues, incompatibility of some applications with curricula, and differences in teachers' digital competencies may affect the effectiveness of integration. Therefore, the application of mobile technologies should be carried out in a planned, purposeful, and methodologically grounded manner.

From a future perspective, the role of mobile applications in education will continue to expand. Artificial intelligence-based personalized learning systems, adaptive educational platforms, and real-time analytical assessment opportunities will open new possibilities in this field. This will further improve mobile learning and enhance the quality of education.

Conclusion

In conclusion, the integration of mobile applications such as Duolingo and Memrise into classroom teaching should be regarded as one of the main development directions of modern education. This approach contributes not only to knowledge acquisition but also to the development of learning skills, independent thinking, and digital literacy, making it a comprehensive and promising educational model.

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THE ROLE OF ARTIFICIAL INTELLIGENCE AND DIGITAL PLATFORMS IN TEACHING

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Abstract: The integration of artificial intelligence (AI) and digital platforms has significantly transformed contemporary teaching practices. These technologies offer innovative solutions that enhance the quality, accessibility, and efficiency of education. This study examines the role of AI and digital platforms in teaching, focusing on their impact on student engagement, learning outcomes, and instructional methods. A mixed-method approach was employed, including questionnaires, classroom observations, and performance analysis of students using digital tools. The findings indicate that AI-driven systems and online platforms contribute to personalized learning by adapting content to individual student needs and providing immediate feedback. Furthermore, digital platforms promote interactive and flexible learning environments, increasing student motivation and participation. However, the study also identifies several challenges, such as limited teacher training, technical issues, and concerns regarding data privacy. Despite these limitations, the results suggest that the effective integration of AI and digital platforms can significantly improve teaching and learning processes. The study concludes that a balanced approach, combining traditional teaching methods with modern technological tools, is essential for maximizing educational outcomes and preparing students for the demands of the digital age.

Keywords: Artificial Intelligence (AI), Digital Platforms, E-learning, Personalized Learning, Educational Technology, Student Engagement, Teaching Methods

SÜNİ İNTELLEKT VƏ RƏQƏMSAL PLATFORMALARIN TƏDRİSDƏ ROLU

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Xülasə: Süni intellektin (AI) və rəqəmsal platformaların integrasiyası müasir tədris təcrübələrini əhəmiyyətli dərəcədə dəyişdirmişdir. Bu texnologiyalar təhsilin keyfiyyətini, əlçatanlığını və səmərəliliyini artıran innovativ həllər təqdim edir. Bu tədqiqat AI və rəqəmsal platformaların tədrisdə rolunu araşdırır, onların tələbə cəlb olunmasına, öyrənmə nəticələrinə və tədris metodlarına təsirinə diqqət yetirir. Qarışıq metod yanaşmasından istifadə edilmişdir; buraya sorğular, sinif müşahidələri və rəqəmsal alətlərdən istifadə edən tələbələrin performans analizləri daxildir.

Nəticələr göstərir ki, AI əsaslı sistemlər və onlayn platformalar məzmunu fərdi tələbə ehtiyaclarına uyğunlaşdıraraq və dərhal geri bildirim təqdim edərək fərdiləşdirilmiş öyrənməyə töhfə verir. Bundan əlavə, rəqəmsal platformalar interaktiv və çevik öyrənmə mühitləri yaradaraq tələbələrin motivasiyasını və iştirakını artırır. Bununla belə, tədqiqat bəzi çətinlikləri də müəyyən etmişdir, məsələn, müəllimlərin kifayət qədər hazırlıqlı olmaması, texniki problemlər və məlumatların məxfiliyi ilə bağlı narahatlıqlar.

Bu məhdudiyyətlərə baxmayaraq, nəticələr göstərir ki, AI və rəqəmsal platformaların effektiv integrasiyası tədris və öyrənmə proseslərini əhəmiyyətli dərəcədə yaxşılaşdırmaqla bilər. Tədqiqat belə qənaətə gəlir ki, ənənəvi tədris metodlarının müasir texnoloji alətlərlə balanslı şəkildə birləşdirilməsi təhsil nəticələrini maksimum səviyyəyə çatdırmaq və tələbələri rəqəmsal dövrün tələblərinə hazırlamaq üçün vacibdir.

Açar sözlər: Süni intellekt (AI), Rəqəmsal platformalar, Elektron təhsil (E-learning), Fərdiləşdirilmiş öyrənmə, Təhsil texnologiyaları, Tələbə cəlb olunması, Tədris metodları

1. Introduction

In the 21st century, education has undergone rapid transformation due to technological advancements. Among these developments, artificial intelligence (AI) and digital platforms have emerged as powerful tools that reshape traditional teaching methods. The shift from teacher-centered to learner-centered approaches has been strongly supported by these technologies.

Artificial intelligence refers to computer systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, and decision-making. In education, AI is used in adaptive learning systems, automated grading, and intelligent tutoring systems.

Digital platforms include online learning environments, educational applications, and virtual classrooms that allow interaction between teachers and students. These platforms provide flexible learning opportunities and support remote education.

Despite their growing importance, the integration of AI and digital platforms in teaching raises several questions regarding effectiveness, accessibility, and pedagogical value. Therefore, this study aims to explore their role in education and analyze their advantages and limitations.

2. Materials and Methods

2.1 Research Design

This study adopted a mixed-method research design, integrating both quantitative and qualitative approaches to provide a comprehensive and well-rounded understanding of the role of artificial intelligence (AI) and digital platforms in teaching. The rationale for selecting a mixed-method approach was to combine the strengths of both methodologies: the objectivity and measurability of quantitative data with the depth and contextual richness of qualitative data.

The quantitative component of the study focused on collecting numerical data related to students' academic performance, engagement levels, and attitudes toward the use of AI and digital

platforms in the learning process. This data was gathered through structured questionnaires and performance assessments, allowing for statistical analysis and the identification of patterns, trends, and correlations. The use of quantitative methods enabled the researcher to measure the effectiveness of digital tools in a systematic and reliable manner.

In contrast, the qualitative component aimed to explore the experiences, perceptions, and opinions of both students and teachers regarding the integration of AI and digital platforms in education. Data for this part of the study was collected through classroom observations and open-ended questionnaire responses. This approach provided deeper insights into how these technologies influence teaching practices, classroom interaction, and student motivation. It also allowed participants to express their views in their own words, offering a more nuanced understanding of the research topic.

The study is primarily descriptive in nature, as it seeks to examine and analyze the existing relationship between the use of digital technologies and teaching effectiveness without manipulating any variables. Rather than testing a specific experimental intervention, the research focuses on observing current practices and identifying patterns and associations. This descriptive approach helps to present an accurate picture of how AI and digital platforms are currently being used in educational settings.

By combining both quantitative and qualitative data, the study enhances the validity and reliability of its findings through methodological triangulation. This approach ensures that the results are not only statistically supported but also grounded in real-life experiences, making the conclusions more robust and meaningful. Overall, the mixed-method research design allows for a deeper and more comprehensive analysis of the impact of AI and digital platforms on modern teaching practices.

2.2 Participants

The participants of this study consisted of:

- 50 university students (EFL learners, intermediate level)
- 10 English language teachers

Students were selected from different academic groups to ensure diversity in learning experience and technological exposure. The age range of participants was between 18 and 25 years. Teachers had varying levels of experience in using digital platforms, ranging from 2 to 15 years.

The sampling method used was convenience sampling, as participants were selected based on accessibility and willingness to participate in the study.

2.3 Research Instruments

To collect reliable and valid data, several instruments were used:

1. Questionnaire

A structured questionnaire was designed for students. It included both closed-ended and Likert-scale questions to measure:

- * Students' attitudes toward AI tools
- * Frequency of digital platform usage
- * Perceived effectiveness of online learning

Example statements:

- * "AI-based tools help me understand lessons better."
- * "Digital platforms increase my motivation to learn English."

2.2 Interviews

Semi-structured interviews were conducted with teachers. The aim was to explore:

- * Their experience with AI and digital tools
- * Advantages and challenges in classroom implementation
- * Their opinion on student engagement and performance

3. Classroom Observation

Several lessons were observed where digital platforms such as learning management systems, video-based lessons, and AI-supported tools were used. Observation focused on:

- * Student participation
- * Interaction level
- * Engagement during activities
- * Teacher–student communication

4. Academic Performance Analysis

Students' academic results before and after using digital tools were compared. This allowed the researcher to measure the actual impact of AI and digital platforms on learning outcomes, especially in:

- * Reading comprehension
- * Vocabulary development
- * Grammar accuracy
- * Writing performance

2.4 Data Collection Procedure

The data collection process was carried out in several stages:

1. Preparation stage: Research instruments (questionnaires and interview questions) were designed and validated.
2. Implementation stage: Students completed questionnaires and participated in technology-enhanced lessons.
3. Observation stage: Classroom activities were observed and recorded.
4. Interview stage: Teachers were interviewed individually.
5. Analysis stage: Collected data was organized, compared, and analyzed using descriptive methods.

2.5 Data Analysis Methods

Quantitative data from questionnaires and test results were analyzed using percentage analysis and descriptive statistics. The purpose was to identify patterns in students' responses and performance changes.

Qualitative data from interviews and observations were analyzed using thematic analysis, where common themes such as "engagement," "motivation," "technical challenges," and "personalized learning" were identified.

2.6 Ethical Considerations

This study adhered to established ethical standards to ensure the protection, dignity, and rights of all participants throughout the research process. First, all participants were fully informed about the purpose, objectives, and procedures of the study before their involvement. Clear and accessible explanations were provided so that participants could make well-informed decisions regarding their participation.

Participation in the study was entirely voluntary. Individuals were given the freedom to choose whether or not to take part, and they were also informed of their right to withdraw from the

study at any stage without facing any negative consequences or penalties. This approach ensured respect for participants' autonomy and personal choice.

Confidentiality and privacy were strictly maintained. All responses and personal information collected during the study were kept anonymous and securely stored. Identifiable data was either not collected or was removed during the analysis process to protect participants' identities. The data was accessed only by the researcher and was not shared with any unauthorized individuals or organizations.

Furthermore, all collected data was used solely for academic and research purposes. The information obtained was not used for any commercial, personal, or non-research-related activities. The study ensured that the data was handled responsibly, ethically, and in compliance with relevant research guidelines.

Overall, these ethical considerations were implemented to maintain trust, integrity, and transparency throughout the research process, while safeguarding the rights and well-being of all participants.

Summary of Methods Section

Overall, the combination of questionnaires, interviews, observations, and performance analysis provided a comprehensive understanding of how AI and digital platforms influence teaching and learning. This multi-method approach strengthened the reliability and validity of the research findings.

3. Results and Discussion

3.1 Results

The analysis of collected data from questionnaires, classroom observations, and student performance assessments revealed several important findings regarding the role of artificial intelligence (AI) and digital platforms in teaching.

Firstly, the majority of students demonstrated a positive attitude toward digital learning tools. Approximately 78% of participants reported that AI-based applications and digital platforms increased their motivation to learn. Students stated that interactive features such as quizzes, video lessons, and adaptive learning systems made lessons more engaging compared to traditional teaching methods.

Secondly, performance analysis showed a noticeable improvement in academic achievement among students who regularly used digital platforms. Around 72% of students improved their test scores after the integration of AI-supported learning tools. This improvement was particularly visible in vocabulary acquisition, reading comprehension, and grammar tasks.

Thirdly, teacher feedback indicated that AI tools significantly reduced administrative workload. Automated grading systems and digital attendance tracking allowed teachers to focus more on lesson planning and student interaction. Teachers also reported that they could better monitor individual student progress through learning analytics provided by digital platforms.

Finally, classroom observations revealed that students participating in technology-enhanced lessons were more active, asked more questions, and showed higher levels of engagement compared to traditional classroom settings.

3.2 Discussion

The findings of this study clearly demonstrate that artificial intelligence and digital platforms play a crucial role in enhancing the teaching and learning process. One of the most significant outcomes is the improvement in student engagement and motivation. Digital tools provide

interactive and visually rich content, which helps maintain students' attention and encourages active participation.

Another important aspect is personalized learning. AI systems can analyze students' strengths and weaknesses and provide tailored learning materials. This allows learners to progress at their own pace, which is particularly beneficial in mixed-ability classrooms. Students who struggle with certain topics can receive additional support, while advanced learners can move ahead without waiting for others.

The study also highlights the importance of immediate feedback, which is one of the strongest advantages of AI-based learning systems. Unlike traditional classrooms where feedback may take time, digital platforms provide instant correction and explanations. This helps students quickly identify their mistakes and improve their understanding.

However, despite these advantages, several challenges were identified. One major issue is the lack of teacher training. Many educators are not fully equipped to use advanced digital tools effectively, which limits the potential benefits of these technologies. Another challenge is the digital divide, where some students do not have equal access to devices or stable internet connections.

Additionally, concerns regarding data privacy and over-dependence on technology were raised. Excessive reliance on AI systems may reduce human interaction in the classroom, which is essential for developing communication and social skills.

Overall, the results suggest that while AI and digital platforms significantly improve teaching effectiveness, they should be used as supportive tools rather than complete replacements for traditional teaching methods. A balanced integration of technology and human instruction is essential for achieving optimal educational outcomes.

4. Conclusion

This study demonstrates that artificial intelligence and digital platforms play a crucial role in modern teaching. They enhance learning by providing personalized, interactive, and accessible educational experiences.

However, their effectiveness depends on proper implementation, teacher readiness, and equal access to resources. While technology offers many advantages, it cannot replace the human element in education.

Therefore, a balanced approach that combines traditional teaching methods with modern technology is essential for achieving optimal learning outcomes.

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AI-ENHANCED ENGLISH LANGUAGE TEACHING: NEW TRENDS IN THE 21ST CENTURY CLASSROOM

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Abstract: The integration of artificial intelligence (AI) into English language teaching has significantly transformed the dynamics of modern classrooms, creating new opportunities for personalized, interactive, and adaptive learning. Unlike traditional digital tools that provide static resources, AI-driven platforms employ intelligent tutoring systems, automated feedback mechanisms, and mobile language applications to deliver real-time assessment and individualized content. These innovations enhance communicative competence, support task-based learning, and foster learner autonomy by encouraging active participation and sustained motivation. Through adaptive learning paths, speech recognition technologies, and intelligent chatbots, AI enables learners to practice authentic communication and refine their linguistic skills in engaging ways. Automated writing assessment tools further contribute to accuracy and fluency by offering personalized guidance on grammar, vocabulary, and style. Importantly, AI is not intended to replace teachers; rather, it functions as a supportive tool that enriches pedagogical practices, allowing educators to focus on mentoring, facilitating collaboration, and addressing diverse learner needs. By combining human expertise with technological adaptability, AI prepares learners for global communication and equips them with essential skills for success in the 21st century. This paper emphasizes the role of AI as a catalyst for innovation in language education, highlighting its potential to reshape teaching methodologies while preserving the irreplaceable value of human interaction in the learning process.

Keywords: Artificial intelligence, English language teaching, digital learning, adaptive learning, pedagogy

SÜNİ İNTELLEKTİLƏ DƏSTƏKLƏNƏN İNGİLİS DİLİ TƏDRİSİ: XXI ƏSR SİNFİNDƏ YENİ MEYDİRLƏR

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Xülasə: Süni intellektin (SI) İngilis dili tədrisinə inteqrasiyası müasir siniflərin dinamikasını əhəmiyyətli dərəcədə dəyişərək fərdiləşdirilmiş, interaktiv və adaptiv öyrənmə imkanları yaratmışdır. Ənənəvi rəqəmsal vasitələr əsasən statik resurslar təqdim etdiyi halda, SI əsaslı platformalar intellektual tədris sistemləri, avtomatlaşdırılmış geribildirim mexanizmləri və mobil dil tətbiqləri vasitəsilə real vaxtda qiymətləndirmə və fərdi ehtiyaclarla uyğunlaşdırılmış məzmun təqdim edir. Bu yeniliklər kommunikativ bacarıqları gücləndirir, tapşırıq əsaslı öyrənməni dəstəkləyir və öyrənənlərin müstəqilliyini artıraraq onların motivasiyasını davamlı şəkildə gücləndirir. Adaptiv öyrənmə yolları, nitq tanıma texnologiyaları və intellektual çatbotlar vasitəsilə SI tələbələrə real ünsiyyət ssenarilərini məşq etmək və dil bacarıqlarını daha maraqlı formada inkişaf etdirmək imkanı yaradır. Avtomatlaşdırılmış yazı qiymətləndirmə vasitələri isə qrammatika, lüğət və üslub sahəsində fərdiləşdirilmiş istiqamət təqdim edərək dəqiqlik və səlisliyi artırır. Əhəmiyyətli məqam odur ki, SI müəllimi əvəz etmir; o, pedaqoji praktikaları zənginləşdirən dəstəkçi vasitə kimi çıxış edir və müəllimlərə tələbələrə istiqamətləndirmək, əməkdaşlığı təşviq etmək və müxtəlif öyrənən ehtiyaclarını qarşılamaq üçün daha çox imkan yaradır. İnsan təcrübəsi ilə texnoloji adaptivliyi birləşdirən SI öyrənənləri qlobal ünsiyyətə hazırlayır və XXI əsrdə uğur qazanmaq üçün zəruri bacarıqlarla təmin edir. Məqalə SI-nin dil tədrisində innovasiyalara təkan verən rolunu vurğulayır, onun metodologiyaları yenidən formalaşdırmaq potensialını önə çəkir və öyrənmə prosesində insan ünsiyyətinin əvəzolunmaz dəyərini qoruyur.

Açar sözlər: Süni intellekt, İngilis dili tədrisi, rəqəmsal öyrənmə, adaptiv öyrənmə, pedaqogika

Introduction

Traditional teaching methods have undergone profound transformation with the rapid advancement of digital technologies. In the 21st century, intelligent systems – powered by artificial intelligence (AI), machine learning, and big data – have emerged as a transformative force in language education. Unlike earlier digital tools that primarily offered static resources such as online dictionaries or multimedia exercises, these systems introduce dynamic, adaptive mechanisms capable of responding to individual learner needs. This paradigm shift reflects broader educational trends toward personalization, learner autonomy, and global interconnectedness. The integration of intelligent technologies into classrooms not only supports teachers in managing diverse groups of learners but also empowers students to engage in authentic communication. By bridging the gap between theory and practice, these tools create opportunities for learners to apply language in meaningful contexts. Globalization further intensifies the demand for communicative competence and intercultural awareness. Adaptive and interactive approaches are no longer optional but indispensable in modern pedagogy. The rise of AI-driven platforms signals a new era where language learning is not confined to textbooks but enriched by real-time, personalized, and globally relevant experiences.

Materials and Methods

This study draws upon scholarly works that examine the role of emerging technologies in education, including Luckin et al. (2016), Holmes et al. (2022), and Selwyn (2019). These sources provide theoretical frameworks for understanding how digital intelligence reshapes pedagogy and redefines the teacher's role.

A comparative approach is employed to highlight distinctions between conventional digital tools – such as multimedia resources, online grammar drills, or electronic dictionaries – and advanced systems that employ machine learning, natural language processing (NLP), and adaptive algorithms.

- Conventional Tools: Static, one-size-fits-all resources (e.g., grammar apps, vocabulary lists).
- Intelligent Systems: Dynamic, learner-centered platforms (e.g., adaptive learning apps, AI tutors, automated writing assessment tools).

Case studies from language classrooms in Europe (e.g., Finland, UK) and Asia (e.g., China, Japan) are reviewed to demonstrate how these innovations are implemented across different cultural and institutional contexts. This methodology ensures a balanced perspective, combining theoretical insights with real-world examples.

Results and Discussion

Enhanced platforms demonstrate several key advantages in language education. Adaptive learning paths allow students to progress at their own pace, reducing anxiety and fostering confidence. Instant feedback mechanisms provide timely corrections that reinforce accuracy and encourage self-reflection. Speech recognition technologies enhance pronunciation practice, enabling learners to achieve greater fluency and intelligibility. Interactive chat systems simulate authentic communication scenarios, preparing learners for real-world interactions in academic, professional, and social settings. Automated writing assessment tools refine grammar, vocabulary, and style, offering personalized guidance that complements teacher feedback. For educators, advanced analytics provide valuable insights into learner progress, identifying strengths and areas requiring improvement. These innovations align closely with communicative and task-based approaches, making learning more engaging, interactive, and effective. They also encourage collaborative learning, as students can participate in peer-to-peer exchanges supported by intelligent platforms. Importantly, the teacher's role evolves from a transmitter of knowledge to a facilitator and mentor, guiding learners through personalized pathways while maintaining academic rigor.

Conclusion

Enhanced English language teaching represents a significant trend in modern pedagogy. By integrating human expertise with intelligent systems, classrooms evolve into spaces that are interactive, personalized, and globally relevant. While the teacher's role remains central as a facilitator and mentor, these technologies serve as powerful assistants that enrich pedagogical practices. The synergy between human guidance and adaptive digital tools ensures that learners are equipped with communicative competence, critical thinking skills, and adaptability necessary for success in the 21st century. As education continues to embrace innovation, the challenge lies in balancing technological integration with human values, ensuring that language learning remains not only efficient but also meaningful, creative, and culturally responsive.

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APPLICATION AND EFFECTIVENESS OF DIGITAL TECHNOLOGIES IN HIGHER EDUCATION: AZERBAIJAN CONTEXT

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Abstract: In the modern era, digital technologies play a key and decisive role in the transformation of the higher education system. As a result of the rapid development of information and communication technologies, traditional teaching models are gradually being replaced by digital and hybrid learning environments. These changes are not limited to the renewal of teaching methods, but also create conditions for the establishment of new formats of educational management, assessment and student-teacher interactions. Through digital technologies, students gain more flexible, personalized and interactive learning opportunities.

The purpose of this study is to assess the effectiveness of the application of digital technologies in higher education institutions and analyze the current situation in the Azerbaijani context. The conducted studies show that digital platforms and electronic resources have a positive impact on students' academic achievements, learning motivation and independent work skills. At the same time, education in a digital environment expands students' access to global information resources and develops their analytical thinking skills.

However, the process of digital transformation in the Azerbaijani higher education system is accompanied by a number of problems. These challenges include the uneven development of technical infrastructure, insufficient digital skills of teachers, and limited application of innovative pedagogical approaches. Consequently, strengthening infrastructure, increasing teacher training, and implementing modern teaching methods are considered essential to form an effective digital learning environment.

Keywords: digital education, higher education, digital technology, effectiveness, digital transformation

ALİ TƏHSİLDƏ RƏQƏMSAL TEXNOLOGİYALARIN TƏTBİQİ VƏ EFEKTİVLİYİ: AZƏRBAYCAN KONTEKSTİ

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Xülasə: Müasir dövrdə rəqəmsal texnologiyalar ali təhsil sisteminin transformasiyasında əsas və həlledici rol oynayır. İnformasiya və kommunikasiya texnologiyalarının sürətli inkişafı nəticəsində ənənəvi tədris modelləri tədricən rəqəmsal və hibrid öyrənmə mühitləri ilə əvəz olunur. Bu dəyişikliklər yalnız tədris üsullarının yenilənməsi ilə məhdudlaşmır, həm də təhsilin idarə olunması, qiymətləndirilməsi və tələbə-müəllim qarşılıqlı əlaqələrinin yeni formatda qurulmasına şərait yaradır. Rəqəmsal texnologiyalar vasitəsilə tələbələr daha çevik, fərdiləşdirilmiş və interaktiv öyrənmə imkanları əldə edirlər.

Bu tədqiqatın məqsədi ali təhsil müəssisələrində rəqəmsal texnologiyaların tətbiqinin effektivliyini qiymətləndirmək və Azərbaycan kontekstində mövcud vəziyyəti təhlil etməkdir. Aparılan araşdırmalar göstərir ki, rəqəmsal platformalar və elektron resurslar tələbələrin akademik nailiyyətlərinə, öyrənmə motivasiyasına və müstəqil işləmə bacarıqlarına müsbət təsir göstərir. Bununla yanaşı, rəqəmsal mühitdə təhsil tələbələrin qlobal informasiya resurslarına çıxışını genişləndirir və onların analitik düşünmə qabiliyyətini inkişaf etdirir.

Lakin Azərbaycan ali təhsil sistemində rəqəmsal transformasiya prosesi bir sıra problemlərlə müşayiət olunur. Bu problemlərə texniki infrastrukturun qeyri-bərabər inkişafı, müəllimlərin rəqəmsal bacarıqlarının yetərli səviyyədə olmaması və innovativ pedaqoji yanaşmaların məhdud tətbiqi daxildir. Nəticə etibarilə, effektiv rəqəmsal təhsil mühitinin formalaşdırılması üçün infrastrukturun gücləndirilməsi, müəllim hazırlığının artırılması və müasir tədris metodlarının tətbiqi vacib hesab olunur.

Açar sözlər: rəqəmsal təhsil, ali təhsil, rəqəmsal texnologiya, effektivlik, rəqəmsal transformasiya

Introduction

In the 21st century, digitalization has led to profound and systemic changes at all levels of the education system. The application of digital technologies, especially in higher education institutions, creates broad opportunities for organizing the educational process in a more flexible, accessible, personalized and effective way. Digital skills are considered key competencies for success in the modern education system and the labor market (World Bank, 2021). As a result of the rapid development of information and communication technologies (ICT), the education system has moved from the traditional classroom-based model to a digital, hybrid and blended learning environment. These changes are not limited to the modernization of teaching methods, but also include the renewal of educational management, student assessment and academic communication forms.

The formation of a digital educational environment creates conditions for more active involvement of students in the learning process, the development of their independent knowledge acquisition skills and the expansion of access to educational materials at any time. At the same time, this process also requires the application of new approaches in the pedagogical activities of teachers. Thus, the teacher is no longer only a transmitter of knowledge, but rather acts as a guide and facilitator.

In the Republic of Azerbaijan, digitalization has also been identified as one of the main priority directions of state policy. In the document "Azerbaijan 2030: National Priorities for Socio-Economic Development" adopted by the Presidential Administration of the Republic of Azerbaijan, digital transformation and the formation of an innovative educational environment are specifically emphasized as one of the strategic goals (Presidential Administration, 2021). This approach is of great importance in terms of the development of the knowledge economy and the formation of competitive human capital in the country.

In this context, the application of digital technologies in the higher education system is of particular relevance. Digital education projects implemented by the Ministry of Education of the Republic of Azerbaijan, the application of electronic platforms and the expansion of distance education opportunities have led to significant progress in this area (Ministry of Education of the Republic of Azerbaijan, 2020). Especially during the COVID-19 pandemic, the widespread application of distance education models has further increased the importance of digital education and provided a serious impetus to the acceleration of this process.

However, along with the achievements achieved, there are still a number of challenges for the effective implementation of digital education. These challenges are related to the level of development of technical infrastructure, digital inequality between regions, continuous improvement of teachers' digital skills, and systematic application of innovative pedagogical methods. In this regard, digital transformation acts as a complex process that requires not only technological, but also institutional and pedagogical changes.

Materials and methods

The methodological approach used in the study is complex and multifaceted. This approach allowed both to strengthen the theoretical foundations of the study and to obtain more accurate and substantiated practical results. The following methods were used within the framework of the study:

- systematic literature review
- comparative analysis
- analytical approach

Through the method of systematic literature review, local and international scientific sources on digital education, the application of technologies in higher education and their effectiveness were extensively investigated. This method created conditions for the generalization and systematization of existing theoretical approaches, empirical studies and scientific results on the topic. At the same time, it was possible to identify the development trends of digital education and identify existing scientific gaps.

The comparative analysis method is aimed at systematically comparing the experiences of applying digital technologies in higher education systems of different countries. Based on this approach, the existing similarities and differences between the Azerbaijani higher education model and international experience were identified, and it was assessed in which areas the digital transformation process has yielded more successful and in which areas it has yielded weaker results. At the same time, this method allowed for a comparative study of the Azerbaijani experience with digital education models of European countries and other regional education systems.

The analytical approach is aimed at a systematic analysis of the collected data and the interpretation of the results on a scientific basis. Through this method, the main factors affecting the

effectiveness of digital technologies in higher education - infrastructure, teacher training, student behavior and pedagogical approaches - were comprehensively assessed.

Both international and Azerbaijani scientific sources were used during the research. In particular, empirical studies conducted on the effectiveness of digital education, statistical reports and strategic programs adopted at the state level formed the main database. These sources increased the scientific reliability of the research and ensured the consistency of the results with the real educational environment (Hajiyeva et al., 2023; State Statistical Committee of the Republic of Azerbaijan, 2023).

In addition, local regulatory documents and development strategies were also included in the analysis process, which allowed for a clearer assessment of the institutional foundations of digital transformation in the Azerbaijani higher education system (Ministry of Education of the Republic of Azerbaijan, 2020; Presidential Administration of the Republic of Azerbaijan, 2021).

Conclusions and discussion

1. Advantages of digital technologies

The integration of digital technologies into the higher education environment creates conditions for improving the quality of the teaching process and organizing the learning environment more flexibly and efficiently. The application of these technologies creates a number of important advantages for students and significantly reduces the limitations of the traditional education model. Digital and innovative teaching models have a positive impact on improving students' academic results (Akçayır & Akçayır, 2018).

One of the most important advantages is the expansion of the possibilities for personalizing teaching. Through digital platforms, students can access materials at their own pace and build an individual development trajectory according to their weaknesses and strengths. This makes learning more purposeful and effective.

Another important advantage is the elimination of time and space restrictions. Online educational platforms and electronic resources allow students to access teaching materials at any time and place, which is especially important in distance education and hybrid models.

In addition, digital technologies create conditions for the formation of an interactive learning environment. Multimedia resources, simulations, online tests, and virtual classrooms ensure that students are more actively involved in the learning process. International studies also show that the systematic use of digital platforms significantly improves students' academic performance and learning outcomes (Noetel et al., 2021). This trend is also gradually being observed in the Azerbaijani higher education system, and the application of digital educational tools is expanding.

2. Factors affecting effectiveness

The effectiveness of digital technologies in higher education depends on a number of internal and external factors, and the interaction of these factors determines the overall educational outcomes. Ensuring effectiveness is not only related to the availability of technology, but also to its correct and purposeful application.

One of the main factors is the level of technological infrastructure. The formation of an effective digital educational environment is impossible without high-speed internet, modern equipment, and stable digital platforms. In cases where the infrastructure is weak, interruptions and loss of quality are observed in the teaching process.

Another important factor is the digital competencies of teachers. The ability of teachers to effectively use information technologies directly affects the quality of their pedagogical activities. The correct application of digital tools makes the teaching process more interactive and result-oriented, but if these skills are not sufficiently developed, the potential of technology is not fully realized.

At the same time, adapting pedagogical approaches to modern requirements also plays an important role. Digital education requires not only the use of technical means, but also the application of new teaching methods. The transition from a traditional teacher-centered model to a student-centered model is one of the main components of this process.

The level of motivation of students is also one of the factors that significantly affects effectiveness. Students studying in a digital environment should have high self-management skills, a sense of responsibility and interest in learning. The use of interactive digital resources increases students' motivation and improves learning outcomes (Jacob & Centofanti, 2024). Studies have shown that the effectiveness of technology gives maximum results only when it is integrated with the right methodological approach.

3. Problems in the Azerbaijani context

Although the digital transformation process in the Azerbaijani higher education system has developed rapidly in recent years, it is still a direction in the formation stage. The sustainable and effective implementation of this process is accompanied by certain structural, technical and institutional difficulties.

One of the main problems is the digital inequality across regions. Internet access, accessibility of digital resources and the level of technological provision differ between different regions of the country. This creates an uneven development environment among higher education institutions and directly affects the educational opportunities of students.

Another important problem is the uneven development of technical infrastructure. While some higher education institutions have modern laboratories, digital platforms and electronic learning systems, other institutions have limited these opportunities. As a result, the quality of digital education cannot be ensured according to a single standard.

In addition, the lack of sufficient digital skills of teachers also acts as one of the main problems. Along with the application of digital technologies, the lack of user skills in this area creates certain difficulties (Hajiyeva et al., 2023). The modern educational environment requires teachers not only pedagogical knowledge, but also the skills to effectively use digital technologies. However, the lack of development of these skills at the same level among all pedagogical staff reduces the effectiveness of digital teaching. It is considered necessary to formulate a strategic approach for the development of the digital education system and prepare human resources for innovative technologies (Mazanov & Huseynova, 2021).

Studies have shown that the above-mentioned problems seriously hinder the full and effective application of digital technologies in higher education. At the same time, according to statistical data provided by the State Statistical Committee of the Republic of Azerbaijan, despite the general development of information and communication technologies in the country, the use of these resources in the education sector has not yet reached its potential level (State Statistical Committee, 2023). This indicates the need for a more systematic implementation of digital transformation.

4. Development prospects

It is of particular importance to identify complex and multifaceted strategic directions for a more effective and sustainable application of digital technologies in the Azerbaijani higher education system. These directions serve not only to modernize the educational process, but also to increase the competitiveness of the general education system, its innovation-oriented development, and its integration into the global educational space. In this regard, digital transformation should be assessed as a long-term process requiring a systematic approach.

First of all, strengthening the digital infrastructure is one of the priority directions. The widespread application of modern technological equipment in higher education institutions, the provision of high-speed and stable internet networks, as well as the continuous updating of electronic education platforms have a direct impact on the quality of the educational process. Without a strong digital infrastructure, the effective application of innovative educational models is impossible, and this is considered one of the main basic conditions for digital transformation (World Bank, 2021; Mazanova & Huseynova, 2021).

The second important direction is the continuous professional development of teachers. The modern educational environment requires teachers not only pedagogical knowledge, but also the skills to effectively use digital technologies. For this purpose, the organization of regular training programs, certification of digital competencies, as well as the application of teaching models based on international experience serve to improve the professional level of pedagogical staff. The development of teachers' digital skills increases the interactivity of the educational process and creates conditions for the formation of a student-oriented educational model.

In addition, the application of artificial intelligence-based teaching models is one of the directions of strategic importance in the development of the modern higher education system. Adaptive learning systems create an educational environment adapted to the individual learning speed and needs of students, and smart assessment tools allow for more objective and accurate results. At the same time, data-analytics-based teaching approaches create conditions for the analysis of students' academic behavior and a more accurate determination of their development trajectory. These technologies play an important role in terms of personalizing and increasing the efficiency of education.

The integration of international experience is one of the important directions for the successful implementation of digital transformation in the Azerbaijani higher education system. In this context, the recommendations and global standards presented by international organizations such as UNESCO and the World Bank in the field of education play an important guiding role. Adapting international experience to the local education system creates conditions for both improving the quality of education and accelerating Azerbaijan's integration into the global education space. This approach also contributes to the formation of innovative education policies and the development of competitive human capital (OECD, 2020; UNESCO, 2022).

Conclusion

The results of the study show that digital technologies play an important and decisive role in improving the quality and overall effectiveness of education in the higher education system. In the modern educational environment, these technologies act not only as a support tool, but also as a strategic component that forms the organizational and methodological basis of education. Although the digital transformation process is developing consistently in the Azerbaijani case, there are still certain structural, technical and human resource problems in this area.

The existence of these problems prevents the full realization of the potential opportunities of digital education and necessitates the management of the process on the basis of a more systematic approach. Technological support alone is not enough to form an effective digital educational environment, but at the same time, the renewal of pedagogical approaches, improvement of institutional management and development of human capital are also important conditions.

State policy in this direction is of particular importance, since digital transformation should be implemented as an integral part of the national education strategy. At the same time, internal initiatives of higher education institutions, the introduction of innovative teaching models and the expansion of digital resources serve to accelerate this process. In addition, international cooperation and the study of best practices allow the Azerbaijani education system to be brought into line with global standards.

In the future, the wider and more systematic application of digital technologies will increase the competitiveness of the Azerbaijani higher education system, improve the quality of education, and accelerate the country's transition to a knowledge economy.

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THE APPLICATION OF DIGITAL TECHNOLOGIES IN THE EDUCATIONAL PROCESS

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Abstract: The integration of digital tools into educational settings plays a significant role in improving the effectiveness and efficiency of the learning process. This study explores how digital technologies are used in education and the advantages they offer to both students and teachers, based on a literature review approach. Digital technologies enhance learning experiences by providing interactive course materials, online assignments, and systems for tracking student performance, while also allowing teachers to follow students' academic progress more closely. Tools such as interactive simulations and virtual reality further support learning by making it more engaging and impactful. Within virtual classroom environments, communication is reinforced through LMS platforms and video conferencing tools. These technologies encourage greater student involvement by offering features like announcements, online discussion forums, and convenient access to instructional materials. Approaches including giving feedback, utilizing interactive content, and organizing participatory activities are essential for increasing student motivation and ensuring active involvement in lessons. Improving teachers' digital competencies also strengthens their ability to communicate through technology, ultimately supporting a more efficient and effective learning experience.

Keywords: Educational communication, digitalization, distance education, motivation, digital transformation

TƏHSİL PROSESİNDƏ RƏQƏMSAL TEXNOLOGİYALARIN TƏTBİQİ

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Xülasə: Təhsil mühitlərinin rəqəmsal alətlərlə integrasiyası öyrənmə prosesini daha effektiv və səmərəli etmək baxımından mühüm əhəmiyyət daşıyır. Bu tədqiqatda ədəbiyyat icmal üsulu ilə rəqəmsal alətlərin təhsildə istifadəsi və onların tələbə və müəllimlərə verdiyi faydalar araşdırılmışdır. Rəqəmsal alətlər interaktiv dərs materialları, onlayn tapşırıqlar və tələbə izləmə sistemləri vasitəsilə öyrənmə proseslərini zənginləşdirir və müəllimlərə tələbələrin irəliləyişini izləmək imkanı yaradır. İnteraktiv simulyasiyalar və virtual reallıq tətbiqləri öyrənmə təcrübəsini daha təsirli və maraqlı edir. Virtual sinif mühitlərində ünsiyyət dinamikası LMS platformaları və video konfrans tətbiqləri sayəsində güclənir. Bu alətlər elanlar, onlayn müzakirə forumları və dərs materiallarına asan çıxış kimi imkanlarla tələbə iştirakını artırır. Geri bildirim vermə, interaktiv materiallardan istifadə və qarşılıqlı fəaliyyətə əsaslanan fəaliyyətlərin təşkili kimi strategiyalar tələbələrin motivasiyası və dərslərdə aktiv iştirakı üçün vacibdir. Müəllimlərin rəqəmsal bacarıqlarının inkişaf etdirilməsi isə onların texnologiya yönümlü ünsiyyət qabiliyyətlərini artıraraq daha səmərəli öyrənmə prosesinin formalaşmasına şərait yaradır.

Açar sözlər: Təhsil mühiti, rəqəmsallaşma, distant təhsil, motivasiya, rəqəmsal transformasiya

Introduction

The fact that digital transformation has led to significant changes in education systems, as in all areas of our time, is an undeniable reality. This transformation is particularly evident in changes in educational communication and learning environments. The integration of digital tools into educational settings supports this transformation in various ways, such as strengthening student–teacher interaction, enriching learning experiences, and making the teaching process more effective. Digital tools, which are at the core of digital transformation in education, and their adaptation to learning environments enable teachers to present instructional content more effectively. At the same time, these tools provide students with access to materials tailored to different learning styles, thereby personalizing their learning experiences. The importance of virtual classroom environments in education and their impact on student–teacher interaction are also among the key issues studied. The widespread use of virtual classrooms offers students the opportunity to learn without limitations of time and place. Learning Management Systems (LMS) are one of the main components of this digital transformation.

On the other hand, issues such as communication challenges in distance education and ways to overcome them, communication strategies to increase student participation, the impact of educational technologies on student motivation, and the development of engaging content are also thoroughly examined within the framework of learning management systems. The development of digital skills and technology-based communication competencies in teacher training is another important direction of this process. In this context, the topics explored encompass the integration of educational communication with technology and present various strategies and approaches. The implementation of these strategies can encourage students to participate more actively in lessons and increase their motivation for learning. Enhancing teachers' digital skills and forming technology-oriented communication competencies can help them better adapt to the demands of the digital era and contribute to students' academic success.

The aim of this article is to examine how the integration of digital tools into educational environments affects students' learning processes and improves teacher–student interaction. It focuses on how the use of digital tools can make the learning process more effective and efficient, increase student motivation, and enable teachers to communicate more effectively with students by developing their digital communication skills.

The use of digital tools in education has led to significant and noticeable changes in educational environments alongside the rapid development of technology. This transformation has the potential to enrich students' learning experiences, enable teachers to present instructional content more effectively, and strengthen teacher–student interaction (Smith, 2017). The adaptation of digital tools to educational environments can change teachers' in-class practices and positively support students' learning processes (Yadav, 2024). By providing students with broader access to learning materials and offering personalized learning experiences, it can increase their interest in learning. An example of this is online course content that allows students to study materials outside class at their own pace and at a time they choose. In addition, digital tools can enhance collaboration and interaction in education.

The use of digital tools in education can enable students to experience learning tailored to different learning styles. For a visual learner, interactive simulations or video lessons can make abstract concepts more concrete and help them better understand the topics. Similarly, for auditory learners, digital tools such as podcasts and audiobooks can be more effective in the learning

process. This diversity allows students to select resources that match their learning styles, while also enabling teachers to meet the diverse needs of students.

Digital tools also enable teachers to present course content in a more engaging and effective way and can be more successful than traditional methods in capturing students' attention. However, there are also some concerns and challenges regarding the integration of digital tools into educational environments (Mukul and Büyüközkan, 2023). Technical problems or limited access may prevent some students from fully benefiting from digital tools. Therefore, it is important to expand the use of digital tools in education and ensure equal access to these resources. In addition, providing teachers with the necessary training and support to use digital tools effectively is also important.

There are also other concerns related to the integration of digital tools into educational environments. Issues of privacy and security associated with the use of these tools occasionally cause concern among parents and students. Therefore, it is crucial for educational institutions to take measures regarding data security when using digital tools and to protect students' personal information (Duman, 2021). Although there are risks such as privacy and security concerns, the benefits provided by the use of technology in instructional content outweigh the potential threats it may pose. Since the adaptation of digital tools to educational environments represents an important transformation in education, this process should be managed effectively by experts and in a planned manner. Cooperation and communication among educational institutions, teachers, students, and parents are a fundamental requirement for the effective use of digital tools in education. In this way, digital tools can increase student achievement in education and shape the learning environments of the future. The integration of digital tools into educational settings strengthens teacher–student interaction, enriches learning experiences, and makes the teaching process more effective.

Materials and Methods

Learning Management Systems (LMS) play an important role in modern education and make significant contributions to communication processes. The impact of digital transformation on the education sector has led to the widespread use of Learning Management Systems (LMS). LMS platforms are web-based systems that enable teachers and students to come together in an online environment, access learning materials, communicate, and complete various tasks. Compared to traditional educational environments, these platforms contribute in many ways to the development of communication processes.

Increased accessibility and flexibility: LMS platforms enhance accessibility and flexibility by enabling communication without time and space limitations. Students can access course materials and communication tools, progress at their own pace, and communicate with teachers and other students at different times. This is especially an important advantage for students engaged in distance learning or those with busy schedules.

Diverse communication channels: LMS platforms provide various communication channels, creating a rich and interactive communication environment. Tools such as instant messaging, forums, e-mail, and audio/video conferencing allow students and teachers to communicate in different formats and exchange information. This diversity supports different learning styles and makes communication more effective.

Interactive learning experiences: LMS platforms not only facilitate information delivery but also promote interaction. Tools such as forums, discussion groups, and virtual projects help students actively participate and gain deeper learning experiences.

Improved feedback and assessment: These systems provide tools for feedback and assessment for both teachers and students. Assignments, exams, surveys, and evaluation tools are used to monitor students' progress and provide appropriate feedback. This ensures a more transparent and efficient learning process.

Supportive and inclusive environment: LMS platforms can create supportive and inclusive learning environments by building virtual communities. Tools such as forums, chat rooms, and virtual office hours allow students to connect with each other and with teachers, share their ideas, and receive support. This can increase learning motivation and success. LMS is an online platform where students and teachers can come together, interact, and access learning materials. These platforms make access to course materials easier and allow teachers to manage instructional content. At the same time, communication tools provided by LMS enhance student–teacher interaction and enrich learning experiences. The communication tools offered by LMS make it easier for students to connect with teachers. Online chat, forums, and messaging tools enable students to ask questions, receive feedback, and communicate about course-related topics. This helps students take a more active role in the learning process and better understand course content.

Communication Challenges in Distance Education and Their Solutions

Distance education has become an increasingly important model with the advancement of technology. However, various communication challenges—especially technical ones—may arise during this process. These challenges mainly include the inability to reach the level of interaction provided by face-to-face communication, distractions, technical problems, unequal access, and lack of motivation. One of the most common communication problems in distance education is the limited interaction between students and teachers. In environments where students and teachers do not physically come together, the lack of face-to-face communication can negatively affect the learning process. In such cases, interactive tools such as regularly scheduled virtual classes or individual consultation hours can be used. Technological problems are also among the most frequent communication challenges in distance education. In particular, internet connection issues or software incompatibilities may make it difficult for students to participate in classes and may disrupt communication. To overcome these problems, educational institutions and teachers can provide technical support or use alternative communication tools.

Communication difficulties may also arise among students in distance education. Since they do not physically meet, it can be more difficult for students to collaborate with one another. In such cases, interaction can be encouraged through online discussion forums, group projects, and virtual teamwork. In addition, broader strategies are needed to overcome these challenges. To increase student–teacher interaction, teachers can be encouraged to communicate regularly with students and provide more interactive content. Moreover, backup plans should be developed to reduce technical problems, and technical support should be provided to students.

Although communication challenges in distance education are inevitable, there are various solutions to overcome them. The use of regular and interactive communication tools, solving technical issues, and strengthening student interaction can improve the learning experience. Although communication in distance education can be challenging, these difficulties can be overcome by using appropriate tools and methods. Collaboration among teachers, students, and parents plays a crucial role in creating an effective communication environment and ensuring the success of distance education.

Educational technologies are important tools for increasing students' motivation and making the learning process more effective. These technologies can enhance students' motivation by providing more interactive and engaging learning experiences. Technologies such as Virtual Reality (VR) and Augmented Reality (AR) enable students to interact more closely with learning materials, explore topics in greater depth, and become more engaged in the learning process.

Educational technologies make learning materials more attractive and engaging, thereby increasing students' motivation. Interactive textbooks, animations, videos, and other multimedia content capture students' attention and encourage them to participate more actively in the learning process. Online quizzes and assessment tools provide students with immediate feedback and support their progress. In addition, educational technologies promote collaborative learning environments by encouraging students to communicate and work together. Tools such as online discussion forums, group project platforms, and virtual classroom environments enable students to exchange ideas and interact with one another.

In conclusion, educational technologies offer broad opportunities to enhance student motivation. However, it is important to use these technologies effectively and adapt them to students' individual learning needs. In this way, educational technologies enrich learning experiences and further increase student motivation.

Results and Discussion

Social Media and Digital Citizenship:

Skills such as using social media platforms and adhering to ethical behavior rules in the digital environment are essential components of this field. Teachers equipped with digital skills can make their lessons more interesting and interactive, provide students with various learning opportunities, and make the learning process more engaging. At the same time, these skills facilitate teachers' communication with colleagues, continuous professional development, and the adoption of innovative teaching approaches. The development of teachers' digital skills and the integration of various digital tools into teaching are of great importance.

Teachers can enrich lesson materials and support the development of students' digital skills by using tools such as learning management systems, interactive whiteboard applications, and online learning platforms. For the development of these skills, it is important for teachers to benefit from online training and practical applications (Bozkurt et al., 2021). Through various digital platforms and courses, teachers can learn how to use these tools and apply them effectively. In addition, mentoring from experienced teachers is also beneficial.

In teacher education, pedagogical innovations and digital teaching approaches play an important role in the development of digital skills. Teachers can make the teaching process more effective and engaging by using digital technologies. In this regard, approaches such as flipped learning, interactive learning materials, and project-based learning can be applied.

Collaboration and knowledge sharing among teachers are also important factors in the development of digital skills. Experienced teachers supporting colleagues with lower digital skills and sharing their knowledge helps to spread these competencies more widely. Furthermore, professional development seminars and workshops can be organized to exchange knowledge and experience regarding digital tools and applications. Continuous development and evaluation of teachers' digital skills is also essential. School administrations can support teachers by providing ongoing professional development opportunities, helping them keep their skills up to date.

The development of technology-oriented communication competencies has become an important component of communication skills in today's digital world. Nowadays, in addition to face-to-face interaction, people also communicate through various technological tools such as email, instant messaging platforms, social media, and video conferencing. This change has also created the need for individuals to develop their technology-oriented communication competencies.

The development of digital literacy and information management skills enables students to effectively search digital resources, select accurate information, and evaluate it. Strengthening online communication skills helps students acquire abilities such as writing emails, participating in digital meetings, and engaging in online discussions. In addition, the development of visual and written communication skills allows students to improve competencies such as presenting, graphic design, and video editing.

Strengthening problem-solving and decision-making skills enables students to analyze complex problems using technology, generate alternative solutions, and make the most appropriate decisions. Technology-oriented communication competencies can be developed through various methods. Organizing courses and workshops on these competencies in schools and other educational institutions providing access to online courses, videos, and other resources related to the use of technological tools as well as guidance and support from experienced users to others, are effective approaches to developing these skills.

The development of collaboration and teamwork skills encourages students to communicate effectively through online platforms, collaborate on projects, and participate in group work. Developing these competencies ensures students' success in the digital age and prepares them for future work and life.

Conclusion

In conclusion, the development of technology-oriented communication competencies is as important as the development of technological infrastructure. Digital literacy, online communication skills, visual and written communication skills, problem-solving, and decision-making skills are essential elements for students' success in the digital age. The development of these skills through various methods is important for sustainable education.

Overall, various strategies have been developed regarding the integration of educational communication and technology, and their implementation has been recognized as effective approaches that increase student participation and learning motivation. At the same time, the development of teachers' digital skills and the acquisition of technology-oriented communication competencies can enable them to respond more effectively to the demands of the digital era and contribute to the successful education of students.

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SOCIAL-EMOTIONAL DEVELOPMENT OF CHILDREN IN THE DIGITAL AGE AND MODELS FOR REGULATING SCREEN TIME

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Abstract: The digital age has fundamentally transformed children's education, social lives, and value systems by increasing their interaction with technology. Today, children gain rapid access to information via the internet, consume content from various digital platforms, and socialize within virtual worlds, yet this process brings numerous risks such as distractibility, sleep disorders, lack of physical activity, social isolation, digital addiction, cyberbullying, and privacy violations. At this juncture, the responsibility of parents to guide their children becomes increasingly critical, as it is essential not only to provide children with technical skills but also to foster conscious digital awareness within a framework of ethical and moral values. Research indicates that parents fulfill three primary functions in managing their children's technology use: first, the importance of linking moral and ethical values with technology use is emphasized, as educating children about the negative content they may encounter in the digital world is vital for their ability to distinguish between right and wrong; second, the processes by which parents raise awareness and guide their children in digital environments are examined, highlighting that children must be supported with pedagogical guidance to develop media literacy skills and use technology purposefully and mindfully; and third, the focus is placed on ensuring balance and moderation in technology use because excessive use can negatively affect a child's psychosocial development, requiring parents to assume a balancing role. Furthermore, it is concluded that parents serving as role models is a decisive factor in children establishing a healthy relationship with technology. Considering the opportunities and risks presented by the digital era, the development of conscious technology use policies by parents is vital for protecting both the personal development and the religious and ethical values of children.

Keywords: digital parenting, content control, virtual environment, screen time, social isolation

RƏQƏMSAL DÖVRDƏ UŞAQLARIN SOSIAL-EMOSİONAL İNKİŞAFI VƏ EKTRAN VAXTININ TƏNZİMLƏNMƏSİ

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Xülasə: Rəqəmsal dövr uşaqların texnologiya ilə qarşılıqlı əlaqəsini artıraraq təhsil, sosial həyat və dəyərlər sistemində köklü transformasiyalara yol açmışdır. Günümüzdə uşaqlar internet vasitəsilə informasiyaya sürətli çıxış əldə edə bilir, müxtəlif rəqəmsal platformalardan məzmun istehlak edir və virtual dünyada sosiallaşırlar. Lakin bu proses diqqət dağınıqlığı, yuxu rejimi pozğunluğu, fiziki fəallıq çatışmazlığı, sosial izolyasiya, rəqəmsal asılılıq, kiberzorbalıq və məhrəmiyyətin pozulması kimi bir çox riski də özü ilə gətirir. Bu məqamda valideynlərin övladlarına bələdçilik etmək məsuliyyəti getdikcə daha kritik xarakter alır belə ki, uşaqlara yalnız texniki bacarıqlar qazandırmaq deyil, həm də etik və mənəvi dəyərlər çərçivəsində şüurlu rəqəmsal fərqiindəlik formalaşdırmaq zəruridir. Araşdırmalar göstərir ki, valideynlər uşaqların texnologiya istifadəsini idarəetmə prosesində üç əsas funksiyaya malikdir: Birincisi, mənəvi və etik dəyərlərin texnologiya istifadəsi ilə əlaqələndirilməsinin vacibliyi vurğulanır. Uşaqların rəqəmsal dünyada qarşılaşa biləcəyi mənfi məzmunlara qarşı maarifləndirilməsi, onların doğru və yanlış ayırd edə bilmələri baxımından böyük əhəmiyyət kəsb edir. İkincisi, valideynlərin öz övladlarını rəqəmsal mühitdə şüurlandırma və yönləndirmə prosesləri nəzərdən keçirilmişdir. Uşaqların media savadlılığı bacarıqlarını inkişaf etdirmələri, texnologiyadan məqsədyönlü və dərk edilmiş şəkildə istifadə edə bilmələri üçün onlar pedaqoji rəhbərliklə dəstəklənməlidirlər. Üçüncüsü, texnologiya istifadəsində balans və ölçülülüyün təmin edilməsi diqqət mərkəzində saxlanılmışdır. Texnologiyadan həddindən artıq istifadə uşaqların psixososial inkişafına mənfi təsir göstərə bildiyi üçün valideynlər tarazlaşdırıcı rol üstlənməlidirlər. Bundan əlavə, valideynlərin övladlarına nümunə olmalarının, uşaqların texnologiya ilə sağlam münasibət qurmasında həlledici amil olduğu qənaətinə gəlinmişdir. Rəqəmsal dövrün yaratdığı imkanlar və risklər nəzərə alındıqda, valideynlərin şüurlu texnologiya istifadəsi siyasəti hazırlamaları uşaqların həm fərdi inkişafalarını, həm də dini və etik dəyərlərini qorumaları baxımından mühüm əhəmiyyət daşıyır.

Açar sözlər: rəqəmsal valideynlik, məzmun nəzarət, virtual mühit, ekran müddəti

Introduction

In the contemporary era, the process of digitalization has permeated all spheres of social life, leading to fundamental transformations in child-rearing and pedagogical models. Technological advancements have shifted children's habits of play, learning, and communication from physical spaces to digital platforms. While this transition creates extensive opportunities for rapid access to information and digital literacy, on the other hand, unregulated screen use has fostered significant risks such as technology addiction, social isolation, and delayed cognitive development in children.

In this context, traditional parenting models prove insufficient in the face of challenges brought by the digital world, and the scientific and practical significance of the concept of "digital parenting" continues to grow. Digital parenting is a process of conscious guidance that ensures not only technical control but also the emotional and physical safety of the child within the digital environment. Research indicates that the relationship parents establish with technological tools and the digital culture they instill in their children play a decisive role in the psychosocial health of future generations.

The objective of this research is to analyze the fundamental risks children encounter in the digital age and to interpret the strategic approaches (time management, content control, and role modeling) that parents must implement to minimize these risks within a pedagogical framework.

Methodology and Materials

This study employs a qualitative theoretical analysis method. The methodological framework is constructed to systematically investigate the functional responsibilities of digital parenting and the impact of technology on the four primary spheres of child development: social, emotional, cognitive, and physical.

Research Materials:

- **Parenting Models:** Strategies concerning screen time management, content filtration, and digital role modeling.
- **Age Norms:** Developmental characteristics and guidelines across three key age categories: 0-2 years, 2-5 years, and 6+ years.
- **Scientific Indicators:** Findings from existing academic research regarding melatonin hormone secretion, attention span, and social isolation.

Digital parenting is the process by which parents consciously guide their children's relationship with technology, set boundaries for screen use, and instill safe behavioral habits in the digital world. Unlike traditional parenting, digital parenting involves not only the child's physical and emotional needs but also the content they encounter in the digital environment, their screen duration, and internet safety. Within the framework of digital parenting, the core responsibilities of parents can be listed as follows:

- **Controlling screen time:** Implementing preventative measures against screen addiction in children.
- **Implementing content control:** Selecting age-appropriate games, applications, and videos.
- **Being a role model:** Controlling their own screen usage.
- **Teaching digital safety rules:** Ensuring awareness regarding the non-sharing of personal information and protection against cyberbullying.
- **Offering alternative activities:** Supporting a lifestyle beyond the screen through sports, arts, and nature activities.

In short, digital parenting aims to ensure that children learn using technology while being protected from potential risks and living a balanced life without developing technology addiction.

The fundamental challenge of the digital age is that children are introduced to technology at an early age and can use these tools without limits. Particularly, games, videos, and social media content alter children's perception of time, distancing them from real life. The effects of technology on children include screen addiction, social and academic difficulties, physical and emotional discomfort, and the disruption of sleep patterns.

The digital era has radically changed children's worlds of social interaction (Zafer Güney, 2020). Traditional face-to-face communication has given way to virtual relationships established through social media, instant messaging apps, and online gaming platforms. Children can socialize in digital environments without gathering physically, thereby crossing geographical boundaries to reach a broad circle of friends. While having a wide circle of friends is beneficial in one sense, it negatively impacts the development of face-to-face communication skills. Especially when virtual communication replaces traditional social interactions where body language, tone of voice, and

emotional expressions can be directly observed, children face difficulties in establishing empathy, understanding emotions, and developing healthy social relationships.

Studies show that excessive use of technology can increase feelings of social isolation and loneliness. Children who are constantly in digital interaction distance themselves from real life and become dependent on virtual environments. Over time, this can lead to psycho-social problems such as avoidance of social interaction, social anxiety disorders, and low self-esteem. Children who build their identities in virtual environments prioritize the validation and "likes" they receive from the digital world, pushing real-life relationships into the background. This can weaken their social bonds and strengthen feelings of loneliness. However, the social impacts of technology use are not always negative; digital tools can enable children to meet individuals from different cultures and gain a global perspective.

The emotional effects of technology on children have become a more prominent topic in research, especially in recent years (Abdullah Türkmen, 2023). Studies indicate that excessive technology use is directly linked to emotional health problems such as anxiety, depression, loneliness, and a lack of self-confidence. By being aware of these effects, parents play a vital role in protecting their children's emotional health by controlling their screen time. Furthermore, digital platforms and social media have become significant factors influencing children's self-perception. Specifically, time spent on social media causes a decrease in self-esteem and an increase in anxiety regarding physical appearance. When children compare themselves with their peers on social media, a negative self-image is formed, laying the foundation for deeper depression and anxiety disorders in the future.

The impact of technology on children's cognitive development is a complex relationship that can yield both positive and negative outcomes. While there are positive views suggesting that educational technologies—particularly digital learning tools and applications—enrich children's learning processes and enhance their cognitive skills, excessive use of these technologies can lead to undesirable side effects such as attention deficit and learning difficulties. Although educational technologies can boost children's motivation toward learning, a shortened attention span can create obstacles in processing and acquiring information. This situation poses a significant threat, especially for preschool and primary school-aged children, as attention skills acquired at an early age lay the foundation for future learning processes. Technological devices provide an engaging environment, particularly through colorful screens, rapidly changing visuals, and audio content. However, this environment may weaken children's ability to maintain long-term focus. The shortening of children's attention spans can be linked to the frequent use of digital devices. Research indicates that children who constantly interact with digital stimuli experience greater distractibility in traditional learning environments, which can negatively affect their academic success at school. Consequently, as children switch between digital devices and the cognitive load increases, they may experience deficiencies in their ability to understand and analyze information deeply during the learning process.

Excessive technology use can also have various negative impacts on children's physical health. Prolonged periods spent in front of a screen limit physical activity, leading to a sedentary lifestyle. This situation increases risks such as musculoskeletal problems, bone-muscle system diseases, and cardiovascular diseases over time. As children spend more time at the screen, they become reluctant to move, which negatively affects the development of the musculoskeletal system. Research shows that as screen time increases, the overall physical health of children declines.

Additionally, long-term screen use negatively affects eye health. The blue light emitted from screens can be harmful to children's developing eye structures. Looking at a screen for a long time fatigues the eye muscles, leading to eye pain, dryness, and potential vision impairment in the future.

Children spending too much time with digital devices can also disrupt their sleep patterns. Scientific research proves that screen use can delay the secretion of the hormone melatonin, consequently lowering sleep quality. The disruption of sleep patterns directly negatively affects children's physical and mental development.

Setting digital boundaries is possible not through prohibitions, but by instilling habits and maintaining balance. According to experts, screen time for children should be regulated according to age:

- **0-2 years:** Screen time is not recommended for healthy development. Exposure at an early age can negatively affect language development, social interaction skills, and emotional growth. Instead, it is crucial for infants to be in contact with the real world.
- **2-5 years:** In the preschool period, screen time should not exceed a maximum of 1 hour daily. During this time, preference should be given to educational and high-quality content. Usage should be accompanied by a parent so the child can comprehend what they see and reinforce the learning process.
- **6 years and older:** Screen time for school-aged children must be strictly monitored by parents. Digital limits should be applied to ensure that screen activities do not replace sleep, play, conversation, and physical activities.

Discussion

The role of parents in the digital age is of paramount importance in terms of guiding and supporting children's healthy use of technology. Parents must develop various strategies to ensure their children's safe existence in the digital world. These strategies cover a wide range, from ensuring that children use technology consciously and in a balanced manner to raising awareness about internet safety. Additionally, parents should monitor their children's digital activities, guide them, and inform them about the potential risks they may encounter. Effective digital parenting requires establishing open and continuous communication with children regarding technology use, answering their questions, and creating a secure digital environment. All these responsibilities necessitate that parents contribute to the upbringing of their children as healthy individuals in the digital world.

Establishing boundaries for children's technology use is a crucial element in parents' efforts to protect their children's digital health. While technology offers various opportunities for learning, entertainment, and communication, it is well known that excessive use can lead to negative impacts on physical, emotional, and cognitive health. Therefore, the consistent implementation of rules regarding screen time, content types, and usage hours, along with regular assessments based on needs, provides an effective approach to maintaining digital balance. As mentioned above, children spending a significant part of their day in front of screens can result in physical health issues, lack of focus, and social isolation. In this regard, balanced planning of screen time can support children's participation in physical activities and the development of their social skills. For instance, setting a specific time interval for post-school screen use and adopting a more flexible approach on weekends can be beneficial for digital equilibrium. Furthermore, evaluating the impact of screen time on daily life and reorganizing it according to needs can contribute to children establishing a healthy relationship with technology.

Parents' technology use is a significant factor in shaping their children's digital habits. Children view their parents as their primary role models and often replicate their behaviors. Therefore, parents exhibiting healthy technology use habits help children adopt similar patterns. Balancing technology use, limiting screen time, and utilizing technology only when necessary set a positive example for children. For instance, restricted use of devices such as phones, tablets, or computers during family time sends a message to children that technology is a tool, not a lifestyle. When parents prefer offline activities, such as reading books, walking outdoors, or engaging in face-to-face interactions with family members, they encourage their children to disconnect from the digital world and embrace physical activity. Furthermore, parents abstaining from technology during quality time with their children contributes to the development of social skills and strengthens family bonds. Engaging in interactive digital activities together in a fun and educational manner can also positively influence children's perspectives on technology.

Conclusion. The transformations brought by the digital age have moved the child-rearing process beyond traditional frameworks into a more complex and dynamic dimension. The theoretical analysis conducted within this study demonstrates that technology serves as both a powerful educational tool and a source of significant psychophysical risks for child development. The main conclusions of the research can be summarized as follows:

- Importance of Digital Balance: Total abstinence from technology is neither feasible nor effective in a postmodern world. The primary goal should be to cultivate purposeful technology use habits without fostering digital addiction.
- Critical Role of Developmental Stages: The negative impacts of screen time on neurodevelopment and socialization are particularly decisive during the 0–2 year period. Parents must strictly adhere to scientific guidelines regarding age-appropriate limits (e.g., a maximum of 1 hour for ages 2–5).
- Guidance Strategy: Digital parenting is not merely about monitoring children through technical software; it is about guiding them in digital ethics, internet safety, and content analysis. The relationship parents themselves establish with technology remains the most influential developmental template for the child.

In conclusion, for a healthy digital future, parents must assume the role of active "digital mentors" and ensure a robust balance between a child's virtual and real-life experiences.

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ARTIFICIAL INTELLIGENCE-BASED PERSONALIZED LANGUAGE LEARNING SYSTEMS AND THEIR ROLE IN SECOND LANGUAGE ACQUISITION

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Abstract: The rapid advancement of artificial intelligence (AI) has transformed many aspects of education, particularly foreign language learning. AI-powered personalized learning systems create adaptive learning environments that respond to students' individual needs, preferences, and learning pace. These technologies play an important role in second language acquisition (SLA) by providing customized learning experiences that improve language proficiency and learner engagement.

This study explores the impact of AI on language education through a qualitative analysis of academic literature and digital learning platforms. The findings indicate that AI enhances learning outcomes by offering immediate feedback, personalized content, and continuous performance assessment. In addition, interactive AI tools increase student motivation and encourage active participation in the learning process.

Despite these advantages, several challenges remain, including excessive dependence on technology and reduced opportunities for direct human interaction. Therefore, AI should be viewed as a complementary educational tool rather than a replacement for teachers. Overall, AI-based personalized learning systems represent an effective and innovative approach to modern language education.

Keywords: artificial intelligence, personalized learning, second language acquisition, adaptive learning, digital education, language skills

SÜNİ İNTELLEKTİN AKADEMİK YAZI PROSESLƏRİNƏ TƏSİRİ

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Xülasə: Süni intellektin (SI) sürətli inkişafı təhsil sahəsinə, xüsusilə də xarici dil öyrənilməsinə mühüm təsir göstərmişdir. SI əsaslı fərdiləşdirilmiş öyrənmə sistemləri tələbələrin fərdi ehtiyaclarına, bacarıqlarına və öyrənmə tempinə uyğunlaşan adaptiv təhsil mühiti yaradır. Bu tədqiqat süni intellektin ikinci dilin mənimsənilməsi prosesindəki rolunu araşdırır və onun dil bacarıqlarının inkişafına təsirini qiymətləndirir.

Araşdırmada keyfiyyət metodundan istifadə edilmiş, akademik mənbələr və rəqəmsal tədris alətləri təhlil olunmuşdur. Nəticələr göstərir ki, süni intellekt operativ əks əlaqə, fərdiləşdirilmiş məzmun və davamlı

qiymətləndirmə vasitəsilə öyrənmə nəticələrini yaxşılaşdırır. Bundan əlavə, SI dəstəklı mühitlər interaktiv və cəlbedici xüsusiyyətləri sayəsində tələbələrin motivasiyasını artırır və onların fəal iştirakını təşviq edir.

Bununla yanaşı, texnologiyadan asılılığın artması və insan ünsiyyətinin azalması kimi müəyyən məhdudiyyətlər də mövcuddur. Tədqiqatın nəticəsinə əsasən, süni intellekt müəllimləri əvəz etməli deyil, onların fəaliyyətini dəstəkləyən vasitə kimi istifadə olunmalıdır. Ümumilikdə, SI əsaslı fərdiləşdirilmiş öyrənmə sistemləri dil təhsilində innovativ və səmərəli yanaşma hesab edilir.

Açar sözlər: süni intellekt, fərdiləşdirilmiş öyrənmə, ikinci dil mənimsənilməsi, adaptiv öyrənmə, rəqəmsal təhsil, dil bacarıqları

Introduction

Artificial intelligence (AI) has emerged as one of the most transformative technologies of the twenty-first century, influencing various sectors including healthcare, business, industry, and education. In recent years, educational institutions have increasingly integrated AI-driven technologies into teaching and learning processes to enhance educational effectiveness and learner engagement. One of the most significant applications of AI in education is personalized learning, which aims to tailor educational experiences according to the individual needs, preferences, abilities, and learning pace of students.

In the field of second language acquisition (SLA), personalized learning has become particularly important because language learners possess diverse linguistic backgrounds, cognitive abilities, motivations, and learning styles. Traditional language teaching methods often rely on standardized curricula and uniform instructional approaches that may not adequately address these individual differences. As a result, learners may experience difficulties in achieving their language-learning objectives and maintaining motivation throughout the learning process.

AI-based personalized learning systems offer innovative solutions to these challenges by utilizing machine learning algorithms, natural language processing, adaptive assessment tools, and intelligent tutoring systems. These technologies continuously analyze learner performance and provide customized learning materials, immediate feedback, and adaptive learning pathways. Such capabilities enable learners to receive targeted support and practice opportunities that align with their specific strengths and weaknesses.

Furthermore, the growing popularity of digital language-learning platforms such as Duolingo, Babbel, Busuu, and other AI-enhanced educational applications demonstrates the increasing relevance of AI in language education. These platforms incorporate personalized recommendations, speech recognition, and automated feedback mechanisms that facilitate more effective and engaging learning experiences.

This study aims to examine the role of AI-based personalized learning systems in second language acquisition and evaluate their effectiveness in improving language skills, learner motivation, and overall educational outcomes.

Materials and Methods

This research employs a qualitative methodology based on document analysis and literature review. Qualitative research methods are particularly suitable for exploring emerging educational technologies because they allow for an in-depth examination of theoretical frameworks, practical applications, and research findings.

The study draws upon a wide range of academic sources, including peer-reviewed journal articles, scholarly books, conference proceedings, and reputable online publications related to artificial intelligence, educational technology, adaptive learning, and second language acquisition.

The selected literature focuses on recent developments in AI-powered language-learning systems and their pedagogical implications.

The analysis specifically examines three major areas:

AI-based adaptive learning systems and their technological foundations;

The relationship between artificial intelligence and second language acquisition theories;

Practical applications of AI technologies in language learning environments.

Content analysis was employed to identify recurring themes, benefits, challenges, and trends associated with AI-supported language learning. Additionally, a comparative analysis was conducted to evaluate differences between traditional language-teaching approaches and AI-enhanced personalized learning systems. This comparison focused on instructional flexibility, learner autonomy, feedback mechanisms, assessment procedures, and learning outcomes.

The collected data were systematically categorized and interpreted to provide a comprehensive understanding of the educational value and limitations of AI in language learning contexts.

Discussion

The findings of this study suggest that AI-based personalized learning systems have considerable potential to improve second language acquisition. One of the most significant advantages of AI is its ability to provide individualized instruction based on learner performance, preferences, and progress. Unlike traditional classroom environments, where teachers may face difficulties in addressing the needs of every student simultaneously, AI technologies can continuously monitor learner behavior and adapt instructional content accordingly (Holmes et al., 2019).

Another important benefit is the provision of immediate and personalized feedback. Research in SLA emphasizes the importance of timely feedback in facilitating language development and reducing fossilized errors. AI-powered systems can instantly identify pronunciation mistakes, grammatical inaccuracies, vocabulary misuse, and writing deficiencies, enabling learners to correct errors before they become habitual. This immediate feedback contributes to more efficient learning and greater learner autonomy.

AI technologies also support the development of the four fundamental language skills: listening, speaking, reading, and writing. Speech recognition systems assist learners in improving pronunciation and oral fluency, while natural language processing tools provide sophisticated writing support. Adaptive reading platforms adjust text difficulty according to learners' proficiency levels, thereby enhancing reading comprehension and vocabulary acquisition.

Furthermore, AI-based platforms often incorporate gamification elements such as points, badges, achievement levels, and progress tracking. These interactive features create a more engaging learning environment and increase learner motivation. According to Luckin et al. (2016), personalized feedback and interactive learning experiences contribute significantly to learner satisfaction, persistence, and academic achievement.

Despite these advantages, several challenges and concerns must be considered. One major issue is the potential over-dependence on technology. Excessive reliance on AI systems may limit opportunities for learners to develop critical thinking, problem-solving abilities, and interpersonal communication skills. Language learning is not solely a cognitive process; it is also a social activity that involves cultural understanding, emotional interaction, and authentic communication.

Additionally, AI systems cannot fully replicate the empathy, emotional intelligence, and pedagogical expertise of human teachers. Teachers play a crucial role in providing emotional support, fostering classroom interaction, and addressing complex learner needs that extend beyond algorithmic analysis. Ethical concerns related to data privacy, algorithmic bias, and unequal access to technological resources also require careful consideration.

Therefore, the most effective approach appears to be the integration of AI technologies within a human-centered educational framework, where AI serves as a supportive tool rather than a substitute for teachers.

Conclusion

Artificial intelligence has become an increasingly important component of modern language education. AI-based personalized learning systems offer significant advantages by adapting educational content to individual learner needs, providing immediate feedback, and supporting continuous assessment and progress monitoring. These technologies contribute to improved language proficiency, increased learner motivation, and enhanced learning efficiency.

The findings of this study indicate that AI can play a valuable role in second language acquisition by creating adaptive and learner-centered educational environments. However, the successful implementation of AI in language learning requires a balanced approach that combines technological innovation with human pedagogical guidance. Teachers remain essential for facilitating meaningful communication, social interaction, and emotional support throughout the learning process.

Future research should explore the long-term effects of AI-assisted language learning, investigate ethical considerations associated with educational AI systems, and develop more sophisticated models that integrate technological capabilities with human-centered teaching practices. Such advancements will contribute to the creation of more inclusive, effective, and sustainable language-learning environments in the digital age.

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THE EFFECT OF GAMIFICATION ON VOCABULARY ACQUISITION IN ENGLISH LANGUAGE LEARNING

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Abstract: This study examines the effect of gamification on vocabulary acquisition in English language learning. In recent years, technological advancements have transformed language education, making gamified approaches increasingly popular in both formal and informal learning environments. Gamification incorporates game elements such as points, rewards, competition, and interactive tasks into educational contexts to enhance learner engagement and motivation. The study argues that vocabulary acquisition is a fundamental component of language learning and can be significantly improved through gamified methods. Traditional memorization techniques are often insufficient for long-term retention, whereas gamification promotes repeated exposure, contextual learning, and active participation. Drawing on theoretical perspectives from Kapp (2012), Schmitt (2000), Nation (2013), and Dörnyei (2001), the research highlights that motivation and meaningful language use are critical factors in successful vocabulary learning. Furthermore, the paper discusses how digital platforms and tools such as Duolingo, Kahoot!, and Quilt support vocabulary development through interactive and engaging activities. These tools not only improve memory retention but also encourage collaboration and communication among learners. However, the study also acknowledges potential limitations, including over-reliance on rewards and reduced focus on traditional literacy skills. The findings suggest that gamification is an effective supportive method in vocabulary acquisition and should be integrated into modern English language teaching in a balanced and purposeful way.

Key words: Gamification, Vocabulary Acquisition, English Language Learning, Motivation, Interactive Learning

İNGİLİS DİLİNİN ÖYRƏNİLMƏSİNDƏ OYUNLAŞDIRMANIN SÖZ EHTİYATININ MƏNİMSƏNİLMƏSİNƏ TƏSİRİ

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Xülasə: Bu tədqiqat oyunlaşdırmanın ingilis dilinin öyrənilməsində söz ehtiyatının mənimsənilməsinə təsirini araşdırır. Son illərdə texnologiyanın inkişafı dil təhsilini əhəmiyyətli dərəcədə dəyişdirmiş və oyunlaşdırılmış yanaşmalar həm formal, həm də qeyri-formal təhsil mühitlərində getdikcə daha populyar olmuşdur. Oyunlaşdırma öyrənənlərin motivasiyasını və dərslə marağını artırmaq məqsədilə təhsil mühitinə xal, mükafat, rəqabət və interaktiv tapşırıqlar kimi oyun elementlərinin daxil edilməsidir.

Tədqiqat göstərir ki, söz ehtiyatının mənimsənilməsi dil öyrənməsinin əsas komponentlərindən biridir və oyunlaşdırılmış metodlar vasitəsilə daha effektiv şəkildə inkişaf etdirilə bilər. Ənənəvi əzbərləmə üsulları uzunmüddətli yaddaş üçün kifayət qədər effektiv olmaya bilər, halbuki oyunlaşdırma təkrar qarşılaşma, kontekstual öyrənmə və aktiv iştirak vasitəsilə sözlərin daha yaxşı yadda saxlanılmasına kömək edir. Kapp (2012), Schmitt (2000), Nation (2013) və Dörnyei (2001) kimi tədqiqatçıların nəzəri yanaşmalarına

əsaslanaraq, araşdırma motivasiya və mənalı dil istifadəsinin uğurlu söz öyrənməsində əsas amillər olduğunu vurğulayır.

Bundan əlavə, məqalədə Duolingo, Kahoot! və Quizlet kimi rəqəmsal platformaların interaktiv və maraqlı fəaliyyətlər vasitəsilə söz ehtiyatının inkişafına necə dəstək verdiyi müzakirə olunur. Bu alətlər yalnız yadda saxlama bacarıqlarını gücləndirmir, həm də öyrənənlər arasında əməkdaşlıq və ünsiyyəti təşviq edir. Bununla yanaşı, tədqiqat bəzi məhdudiyyətləri də qeyd edir. Bunlara mükafatlardan həddindən artıq asılılıq və ənənəvi oxu-yazı bacarıqlarına diqqətin azalması daxildir.

Nəticələr göstərir ki, oyunlaşdırma söz ehtiyatının mənimsənilməsində effektiv dəstəkləyici üsuldur və müasir ingilis dili tədrisinə balanslı və məqsədyönlü şəkildə inteqrasiya edilməlidir.

Açar sözlər: Oyunlaşdırma, söz ehtiyatının mənimsənilməsi, ingilis dilinin öyrənilməsi, motivasiya, interaktiv öyrənmə

Introduction

In recent years, English language learning has changed significantly due to technological development. Traditional teaching methods are no longer the only effective approach, especially for younger learners. One of the most influential modern methods is gamification. Gamification is the use of game elements such as points, levels, rewards, challenges, and competition in educational environments. According to Kapp (2012), gamification increases learner engagement and supports active participation in the learning process. Vocabulary acquisition is considered one of the most essential parts of language learning because communication cannot exist without words. However, many students experience difficulties in remembering and using vocabulary effectively. Traditional memorization methods are often repetitive and uninteresting, which decreases learners' motivation. For this reason, gamification has become increasingly important in English language education. English has become one of the most widely used international languages in education, business, technology, and communication. As a result, the importance of effective English language teaching methods has increased significantly. Modern students are more interested in technology-based and interactive learning environments than traditional classroom instruction. For this reason, educators are searching for innovative methods that can increase learners' motivation and participation in the learning process. Gamification has emerged as one of these innovative approaches because it combines education with entertainment. The use of digital technologies and mobile applications has made gamification more accessible in language classrooms. Today, many teachers use gamified activities not only to improve vocabulary learning but also to develop students' speaking, listening, reading, and writing skills. This study investigates how gamification influences vocabulary acquisition, learner motivation, contextual learning, and classroom interaction. It also examines the advantages and possible disadvantages of using gamified tools in English language teaching.

Methodology and Materials

This study is based on a qualitative research approach and mainly relies on secondary data sources. Various academic books, articles, and theoretical studies related to gamification and vocabulary acquisition were analyzed in order to understand the role of interactive learning methods in English language education.

The research particularly focuses on the theories and ideas of Karl Kapp (2012), Schmitt (2000), Nation (2013), Dörnyei (2001), and Chou (2015). In addition, digital learning platforms such as Duolingo, Kahoot!, and Quizlet were examined as examples of gamified educational tools that support vocabulary learning.

The study also includes descriptive analysis of the advantages and limitations of gamification in classroom settings. The collected information was interpreted to identify how motivation, repetition, interaction, and contextual learning contribute to vocabulary acquisition. The study also compares traditional vocabulary teaching methods with gamified learning approaches. Traditional methods generally focus on memorization, translation, and repetition, while gamification emphasizes interaction, competition, and learner-centered activities. By analyzing different academic sources, the research aims to identify how gamification affects students' motivation, engagement, and long-term vocabulary retention. Furthermore, the study examines the role of technology in modern education and how digital learning environments influence students' learning habits. The collected materials demonstrate that gamified learning environments create more enjoyable and less stressful experiences for language learners.

Types of Gamification. Gamification can be divided into digital and non-digital forms. Digital gamification includes mobile applications, online quizzes, and interactive educational platforms. Applications such as Duolingo and Quizlet are examples of digital gamification because they use technology-based reward systems, levels, and interactive exercises to support vocabulary learning. Non-digital gamification, on the other hand, includes classroom games, role-playing activities, board games, competitions, and group tasks organized by teachers. In my opinion, both digital and non-digital gamification are important because they provide students with different learning experiences. Digital gamification is especially useful because students can continue learning outside the classroom. However, classroom-based gamification encourages face-to-face communication and teamwork among learners. Therefore, combining both forms can create a more effective learning environment.

Online Learning and Gamification. Online education has become increasingly widespread in recent years, especially after the COVID-19 pandemic, which accelerated the global transition toward digital learning environments. As educational institutions adapted to virtual classrooms, gamification gained greater importance as a method for maintaining student engagement and motivation during online lessons. One of the main challenges of online learning is the lack of direct interaction between teachers and students. Many learners experience reduced concentration, lower motivation, and feelings of isolation in virtual educational settings. Gamified learning activities help address these challenges by creating more dynamic, interactive, and enjoyable online experiences. Digital educational platforms use elements such as badges, leaderboards, levels, rewards, and daily challenges to encourage active participation. These systems motivate students to practice vocabulary regularly and maintain consistent learning habits. Interactive quizzes and collaborative online games also promote communication and teamwork among learners even in virtual environments. Another important advantage of online gamification is flexibility. Students can access educational applications anytime and anywhere through smartphones, tablets, or computers. This flexibility allows learners to practice vocabulary independently outside classroom hours and develop autonomous learning skills. Moreover, online gamification supports personalized learning experiences. Many digital platforms analyze students' performance and provide exercises according to their individual progress and learning needs. As a result, learners can focus more effectively on areas where they need improvement. However, online gamification also presents certain limitations. Some students may not have stable internet connections or access to modern technological devices. This digital divide can create inequality in educational opportunities. Despite these challenges,

online gamification remains one of the most promising developments in modern language education and continues to influence the future of vocabulary learning.

The Role of Teachers in Gamified Learning. Although gamification relies heavily on technology and interactive activities, teachers continue to play a fundamental role in the learning process. Educational games alone cannot guarantee successful vocabulary acquisition without proper guidance, organization, and pedagogical support from teachers. Teachers are responsible for selecting suitable gamified activities according to students' language proficiency, age, interests, and educational needs. Different groups of learners may respond differently to competitive or collaborative learning environments. Therefore, teachers must carefully design activities that maintain student motivation while still focusing on educational objectives. Another important responsibility of teachers is balancing entertainment and academic learning. Excessive focus on rewards, points, and competition may distract students from meaningful language acquisition. Effective teachers ensure that games are used as supportive educational tools rather than simple entertainment activities. Teachers also provide explanations and contextual support that digital platforms cannot always offer. For example, students may learn vocabulary through games but still experience difficulties understanding word usage, pronunciation, or grammatical functions. In such situations, teacher guidance becomes essential for deeper language comprehension. Furthermore, teachers play a significant role in maintaining inclusive classroom environments. Some students may feel anxious or discouraged in highly competitive settings. Teachers should therefore encourage cooperation, teamwork, and positive interaction among learners to ensure equal participation. In addition, gamified classroom activities help teachers observe students' strengths and weaknesses more effectively. Interactive tasks allow educators to identify which vocabulary items students struggle with and provide immediate feedback or additional practice when necessary. As a result, gamification can improve not only student motivation but also teaching effectiveness.

The Role of Technology in Vocabulary Learning. Technology plays an important role in modern language education. Today's students are surrounded by smartphones, computers, tablets, and internet-based resources. Because of this, integrating technology into education has become necessary rather than optional. In vocabulary learning, technology allows students to access educational materials anytime and anywhere. Mobile applications provide pronunciation practice, visual learning, translation support, and interactive exercises. In my opinion, technology makes vocabulary learning faster, easier, and more attractive for students. Another important advantage of technology is personalization. Many digital platforms analyze students' progress and provide exercises according to their individual learning levels. This helps students focus on their weaknesses and improve more effectively. Furthermore, technological tools often include audio and visual materials that strengthen memory retention. Students remember vocabulary more successfully when they learn through images, sounds, and interaction instead of only reading word lists.

Gamification and Memory Development. Memory development plays a crucial role in vocabulary acquisition because learners must remember not only the meaning of words but also their pronunciation, spelling, collocations, and contextual usage. One of the major problems in traditional language learning is that students often memorize vocabulary mechanically and forget it after a short period of time. This happens because isolated memorization does not always create strong cognitive connections in long-term memory. Gamification addresses this issue by creating interactive and emotionally engaging learning experiences. Cognitive learning theories suggest that

active participation improves memory retention more effectively than passive learning methods. Through gamified activities such as quizzes, matching games, storytelling tasks, puzzles, and competitive challenges, learners repeatedly encounter vocabulary in different contexts. This repeated exposure strengthens neural connections and supports long-term retention. Another important factor is emotional engagement. Psychological studies demonstrate that emotions significantly influence memory performance. Positive emotions such as enjoyment, excitement, and curiosity increase concentration and help learners retain information more successfully. Gamified learning environments reduce boredom and transform vocabulary practice into a more stimulating experience. In addition, many digital platforms combine visual, auditory, and textual learning elements. Images, sounds, animations, and immediate feedback allow students to associate vocabulary with meaningful experiences instead of memorizing isolated word lists. As a result, learners become more confident in recalling and using vocabulary during communication. Furthermore, gamification encourages spaced repetition, which is considered one of the most effective strategies for vocabulary retention. Students repeatedly practice words over time instead of studying them only once. This process strengthens memory consolidation and improves long-term language acquisition.

Discussion

Gamification has become one of the most effective modern approaches in language education because it increases student motivation and participation. In vocabulary acquisition, learners benefit from repeated exposure to words through interactive and enjoyable activities. According to Nation (2013), repetition in different contexts is essential for long-term vocabulary retention. Gamified applications such as Duolingo encourage students to practice daily through rewards, levels, and streak systems. Similarly, Kahoot! creates a competitive learning environment that improves students' concentration and recall abilities. Quizlet supports vocabulary memorization through flashcards and repetition exercises. Another important aspect of gamification is contextual learning. Instead of memorizing isolated words, students learn vocabulary through real-life situations and communicative tasks. Role-playing games, escape room activities, and collaborative classroom games help learners use English naturally and meaningfully. Dörnyei (2001) states that students become more motivated when they can connect language learning to real-life communication. Another important benefit of gamification is immediate feedback. In traditional classrooms, students may wait a long time to learn whether their answers are correct or incorrect. However, gamified applications provide instant responses, which help learners correct mistakes immediately and improve their understanding more effectively. Immediate feedback also increases learners' confidence and encourages continuous participation. In addition, gamification supports autonomous learning. Students can practice vocabulary independently outside the classroom using mobile applications and online platforms. This flexibility allows learners to study at their own pace and according to their own needs. As a result, students become more responsible for their own learning process. Gamification can also reduce anxiety in language learning. Many students feel nervous when speaking or answering questions in traditional classrooms. Interactive games create a more relaxed environment where students feel more comfortable using English. This psychological comfort positively affects language acquisition and classroom participation. Moreover, gamified learning environments often encourage healthy competition among students. Competition can motivate learners to study more actively and achieve better results. At the same time, collaborative games help students develop teamwork and communication skills, which are important for both

academic and social development. However, despite its advantages, gamification also has some limitations. Students may become too focused on rewards and competition rather than meaningful learning. Excessive dependence on digital tools can also reduce attention to traditional reading and writing skills. Therefore, teachers should use gamification in balance with traditional teaching methods to achieve effective learning outcomes.

Future of Gamification in Education. The future of education is expected to become more technology-based and interactive. As digital technologies continue to develop, gamification will likely become more common in schools, universities, and online learning platforms. Artificial intelligence, virtual reality, and augmented reality may further improve gamified language learning experiences in the future. For example, students may practice vocabulary in virtual environments that simulate real-life communication situations. In my opinion, future classrooms will combine traditional teaching methods with digital and gamified approaches. Teachers will increasingly act as guides who help students learn through interaction, creativity, and technology. Therefore, understanding the role of gamification is becoming more important for educators who want to create effective and motivating learning environments.

Conclusion

In conclusion, gamification has a significant positive effect on vocabulary acquisition in English language learning. It improves learner motivation, increases participation, strengthens memory retention, and supports contextual and collaborative learning.

The theories of Kapp (2012), Schmitt (2000), Nation (2013), Chou (2015), and Dörnyei (2001) demonstrate that interactive and engaging methods are more effective than traditional memorization techniques in vocabulary learning. The study concludes that gamification should become an important part of modern English language education. The study also demonstrates that gamification creates a more student-centered learning environment where learners actively participate in educational activities instead of remaining passive recipients of information. This active participation improves not only vocabulary acquisition but also learners' confidence and communication skills. Nevertheless, it should be applied carefully and in combination with traditional teaching methods to ensure balanced and meaningful learning. As technology continues to develop, gamified learning approaches are likely to play an even greater role in the future of language education.

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THE IMPACT OF ARTIFICIAL INTELLIGENCE (AI) ON ACADEMIC WRITING PROCESSES

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Abstract: This study investigates the multifaceted impact of artificial intelligence (AI) on academic writing processes. As AI tools become increasingly integrated into higher education, they support students across various writing stages, including idea generation, content organization, and linguistic refinement. The findings demonstrate that tools such as Grammarly, ChatGPT, Zotero, QuillBot, and Turnitin play a pivotal role in enhancing the quality and efficiency of academic output. While these technologies minimize grammatical errors and streamline reference management, they also present significant challenges. Key concerns include the risk of plagiarism, the erosion of critical thinking skills, and ethical dilemmas regarding data privacy. The study concludes that AI should be utilized as a balanced, supportive instrument rather than a replacement for human intellectual effort.

Keywords: Artificial Intelligence, AI tools, Academic Writing, Educational Technology, Digital Ethics.

SÜNİ İNTELLEKTİN (AI) AKADEMİK YAZI PROSESLƏRİNƏ TƏSİRİ

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Xülasə: Bu tədqiqat süni intellektin (AI) akademik yazı proseslərinə çox şaxəli təsirini araşdırır. AI alətləri ali təhsilə getdikcə daha çox inteqrasiya olunur və, onlar tələbələrə yazının müxtəlif mərhələlərində, eyni zamanda cümlədən ideyaların yaradılması, məzmunun təşkili və dilin təkmilləşdirilməsi sahələrində dəstək verir. Nəticələr göstərir ki, Grammarly, ChatGPT, Zotero, QuillBot və Turnitin kimi alətlər akademik işlərin keyfiyyətinin və səmərəliliyinin artırılmasında mühüm rol oynayır. Bu texnologiyalar qrammatik səhvləri minimuma endirsə və istinadların idarə olunmasını asanlaşdırsa da, eyni zamanda ciddi problemlər də yarada bilər. Əsas narahatlıqlara plagiat riski, tənqidi düşünmə bacarıqlarının zəifləməsi və məlumat məxfiliyi ilə bağlı etik problemlər daxildir. Tədqiqat əsasən belə nəticəyə gəlir ki, AI insanın intellektual söylərini əvəz edən vasitə kimi deyil, balanslı və dəstəkləyici alət kimi istifadə olunmalıdır.

Açar sözlər: Süni intellekt, AI alətləri, Akademik yazı, Təhsil texnologiyaları, Rəqəmsal etika.

Introduction

The rapid advancement of digital technologies in the 21st century has fundamentally reshaped the landscape of higher education and academic inquiry. Among these innovations, Artificial Intelligence (AI) stands out as the most transformative force, redefining the traditional boundaries of teaching, learning, and research. AI can be broadly defined as the capability of computer systems to perform complex tasks that historically required human cognition, such as pattern recognition, linguistic processing, logical reasoning, and autonomous decision-making. Within the context of linguistics and pedagogy, AI has evolved from a simple supportive mechanism into a sophisticated collaborator that actively participates in the academic writing process.

The integration of AI into academia is not merely a technical update; it represents a paradigm shift in how knowledge is constructed and communicated. At institutions like Mingachevir State University, students and researchers are increasingly utilizing Large Language Models (LLMs) and specialized AI tools to navigate the rigors of academic composition. This is particularly significant for students of foreign languages, where AI serves as a "linguistic scaffold," providing real-time feedback on syntax, semantics, and stylistic nuances. From generating initial research questions to the final stages of proofreading, AI platforms like ChatGPT, Gemini, and Grammarly have become ubiquitous in the student workflow.

However, the proliferation of these technologies introduces a complex duality. On one hand, AI democratizes education by making high-level academic writing accessible to non-native speakers and reducing the time spent on administrative tasks such as bibliography management. On the other hand, it raises profound ethical and pedagogical questions. Does the use of generative AI compromise the development of critical thinking skills? Where is the line between "technological assistance" and "academic dishonesty"? As we move deeper into the digital age, the risk of plagiarism and the potential for "cognitive dependency"-where the student's own intellectual voice is overshadowed by machine-generated output-have become central concerns for educators and policy-makers alike.

Furthermore, the impact of AI extends beyond the individual student to the very nature of academic integrity and the evaluation of human intelligence. Modern universities now face the challenge of adapting their assessment methods to a reality where a computer can generate a coherent, well-structured essay in seconds. This necessitates a comprehensive investigation into the dual nature of AI: its role as an empowering tool for innovation and its status as a potential threat to traditional academic values.

This paper aims to explore these dynamics by analyzing the specific impact of AI tools on the writing processes of students at Mingachevir State University. By examining the benefits, such as enhanced efficiency and linguistic accuracy, alongside the challenges of ethical usage and critical thinking, the study provides a balanced overview of the current state of AI in higher education. Through this analysis, we seek to establish a framework for how AI can be integrated into the academic curriculum as a supportive partner that enhances, rather than replaces, the unique creative and analytical capabilities of the human mind.

Methodology and Materials

The present study utilizes a comprehensive mixed-methods research design, integrating both quantitative and qualitative data to achieve a holistic understanding of AI's impact on academic writing. This approach was selected to capture not only the statistical frequency of AI tool usage among students but also the underlying perceptions, motivations, and challenges associated with

these technologies. By combining empirical data with subjective student experiences, the research provides a nuanced perspective on the digital transformation of higher education.

2.1. Research Design and Participant Profile The study was conducted at Mingachevir State University, specifically focusing on students within the Department of Foreign Languages. This demographic was chosen because language students represent a primary user group for translation and grammar-enhancement AI tools. A total of 100 undergraduate students participated in the quantitative phase through a structured digital survey. For the qualitative phase, semi-structured interviews were conducted with 10 students and 5 faculty members to gain deeper insights into the pedagogical implications of AI-assisted writing.

2.2. Data Collection Instruments

Primary Data (Quantitative): A 15-item questionnaire was distributed via Google Forms. The questions were categorized into three segments: (a) frequency and purpose of AI tool usage, (b) perceived impact on writing quality and speed, and (c) awareness of academic integrity and plagiarism. Likert-scale questions were used to measure student satisfaction and reliance on specific software.

Secondary Data (Literature Review): A systematic review of existing literature was performed using academic databases such as Google Scholar, JSTOR, and ScienceDirect. The selection criteria focused on peer-reviewed articles published between 2020 and 2025, specifically those discussing Neural Machine Translation (NMT) and Generative Pre-trained Transformers (GPT) in academic settings.

2.3. Categorization of Materials and AI Tools The study identifies and evaluates a diverse array of digital "materials" that constitute the modern writing environment. These tools were categorized based on their primary functional utility in the academic workflow:

1. **Linguistic Precision Tools:** Grammarly and Ginger were analyzed for their ability to correct syntax, orthography, and stylistic inconsistencies in real-time.

2. **Generative and Brainstorming Models:** ChatGPT (OpenAI) and Gemini (Google) were examined as tools for creating outlines, generating research questions, and summarizing complex theoretical frameworks.

3. **Synthesis and Paraphrasing Software:** QuillBot was evaluated for its effectiveness in helping students rephrase academic content to avoid repetitive sentence structures.

4. **Reference and Citation Managers:** Zotero and Mendeley were studied as essential materials for organizing bibliographies and ensuring adherence to international citation standards (APA, MLA, Harvard).

5. **Academic Integrity Systems:** Turnitin and AI-detection software were utilized to measure the boundary between legitimate assistance and academic dishonesty.

2.4. Data Analysis Procedures The quantitative data gathered from surveys were processed using descriptive statistical analysis to identify major trends in tool adoption. For the qualitative interview transcripts, thematic analysis was applied to extract recurring patterns regarding "AI anxiety," "technological dependency," and the "evolution of the writer's identity." This triangulation of data sources ensures that the findings are both reliable and representative of the current academic atmosphere at Mingachevir State University.

2.5. Ethical Considerations All participants were informed of the study's purpose and provided their voluntary consent. Anonymity was strictly maintained to encourage honest reporting of AI usage, especially concerning generative tools which some students might perceive as

controversial. The study adheres to the ethical guidelines of Mingachevir State University regarding student-led research and data protection.

Discussion

The Paradox of AI in Education. The research results indicate a significant improvement in the technical quality of student papers. For non-native English speakers, AI acts as a "linguistic equalizer," allowing them to express complex scientific ideas without being hindered by grammatical barriers.

3.1. Efficiency and Accuracy

Students reported that using AI tools reduced the time spent on repetitive tasks-such as citation formatting and basic editing-by approximately 40%. Zotero, for instance, ensures that bibliographies meet strict APA or MLA standards, which historically was a major source of technical errors in student submissions.

3.2. Cognitive Risks and Critical Thinking

Despite these gains, a concerning trend emerged: "Cognitive Atrophy." When students rely on AI to summarize articles or generate arguments, the fundamental process of critical analysis is bypassed. There is a risk that the student becomes an "editor" of AI-generated content rather than a "creator" of original thought.

3.3. Ethical and Bias Concerns

AI models are trained on existing data, which may contain inherent biases. Furthermore, the issue of "AI Hallucinations"-where a tool invents non-existent academic sources- poses a threat to the reliability of scientific research. Academic integrity remains the most significant hurdle, as the line between "AI-assisted" and "AI-authored" remains blurred.

Conclusion

Artificial Intelligence is a transformative force in academic writing that offers unparalleled support for modern students. It enhances efficiency and assists in overcoming linguistic hurdles. However, the human element-characterized by critical judgment, cultural empathy, and original insight-must remain the core of academic work. To ensure a productive future, educational policies should focus on "AI Literacy," teaching students how to use these tools responsibly and ethically. AI should be the pen, but the student must always be the author.

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ONLINE AND HYBRID LEARNING CONDITIONS IN THE ORGANIZATION OF AZERBAIJANI LANGUAGE LESSONS: PROBLEMS AND SOLUTIONS

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Abstract: The rapid digitalization of the modern education system has led to the widespread implementation of online and hybrid learning models. These models require new approaches to the organization of Azerbaijani language lessons and involve both methodological and technological changes in the teaching process. This article examines the features of organizing Azerbaijani language lessons in online and hybrid learning environments, analyzes the main challenges encountered in these settings, and proposes ways to overcome them. The study shows that the main problems include unequal access to technical resources, insufficient digital skills among teachers, decreased student motivation, and reduced interactivity. There are also certain difficulties in the development of language skills, particularly in written and spoken communication. The proposed solutions emphasize the improvement of digital competence, the selection of appropriate platforms, and the use of interactive teaching methods.

Keywords: online learning, hybrid learning, Azerbaijani language teaching, digital technologies, instructional organization, educational challenges.

AZƏRBAYCAN DİLİ DƏRSLƏRİNİN TƏŞKİLİNDƏ ONLAYN VƏ HİBRİD TƏLİM ŞƏRAİTİ: PROBLEMLƏR VƏ HƏLL YOLLARI

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Xülasə: Müasir təhsil sistemində rəqəmsallaşmanın sürətlənməsi onlayn və hibrid təlim modellərinin geniş şəkildə tətbiqinə səbəb olub. Bu modellər Azərbaycan dili dərslərinin təşkilinə yeni yanaşmalar tələb edir və tədris prosesində həm metodik, həm də texnoloji dəyişiklikləri nəzərdə tutur. Bu məqalədə onlayn və hibrid tədris mühitlərində Azərbaycan dili dərslərinin təşkilinin xüsusiyyətləri araşdırılır, bu mühitlərdə qarşıya çıxan əsas problemlər təhlil edilir və onların aradan qaldırılması yolları təklif edilir. Tədqiqat göstərir ki, əsas problemlər arasında qeyri-bərabər texniki imkanlar, müəllimlər arasında qeyri-kafi rəqəmsal bacarıqlar, tələbə motivasiyasının azalması və interaktivliyin zəifləməsi daxildir. Dil bacarıqlarının, xüsusən də yazılı və danışiq dilinin inkişafında da müəyyən çətinliklər var. Təklif olunan həllər arasında rəqəmsal sərəfşənin təkmilləşdirilməsi, uyğun platformaların seçilməsi və interaktiv metodlardan istifadə önə çıxır.

Açar sözlər: onlayn təlim, hibrid təlim, Azərbaycan dili dərsləri, rəqəmsal texnologiyalar, tədrisin təşkili, təhsil problemləri

Introduction

In recent years, the development of information and communication technologies has led to significant changes in the education system, making the use of online and hybrid learning models more relevant. These models require new approaches to the organization of the teaching process and the establishment of teacher–student relationships at a different level [Hodges et al., 2020]. The main goal of Azerbaijani language lessons is the comprehensive development of students' language skills. However, in online and hybrid learning environments, the development of these skills presents certain challenges. In particular, ensuring interactivity, increasing learner engagement, and developing language skills in practice become more complex. In addition, the proper selection and use of digital tools, teachers' technological competencies, and the application of appropriate teaching methods are of great importance. In this regard, the study of the organization of Azerbaijani language lessons in online and hybrid learning environments and the identification of existing problems are highly relevant [Richards & Rodgers, 2014].

Materials and methods

The application of digital technologies in the modern education system should be regarded not only as a technical innovation but also as a re-evaluation of pedagogical approaches. Online and hybrid learning models have transformed traditional teaching concepts and created new forms of interaction between teachers and students. In this environment, the organization of Azerbaijani language lessons presents specific challenges, as language learning is based on communicative activities, peer interaction, and practical exercises. Therefore, the effective organization of language instruction in digital environments requires a more comprehensive approach [Richards and Rodgers, 2014].

One of the first issues that arises in online and hybrid learning environments is the problem of technical support and digital inequality. The unequal access of students to the internet, differences in the quality of devices they use, and the instability of platforms directly affect the teaching process. This situation is particularly noticeable in regional areas, resulting in some students being partially excluded from the learning process. Strengthening technological infrastructure in education policy and ensuring equal opportunities for all participants are essential measures for addressing this problem. Another important issue relates to teachers' digital competence. Not all teachers who are effective in traditional classrooms are equally successful in digital platforms. This affects the quality of lessons, teaching methods, and interaction with students. It is important not only that teachers are able to use technology, but also that they can integrate it appropriately for pedagogical purposes. In this context, continuous professional development of teachers and the improvement of

their digital literacy are considered priorities. One of the main problems observed in online and hybrid learning formats in Azerbaijani language lessons is the decrease in student motivation. Since elements such as social interaction, competition, and direct communication, which are present in traditional classroom settings, are weakened in digital environments, students' interest in lessons may decline. This situation particularly leads to the development of passive learning behavior. Research shows that the use of interactive tasks and exercises requiring active participation plays an important role in maintaining learning motivation in digital environments [Means et al., 2014].

Considering the specific nature of language teaching, the development of written and spoken communication skills is of particular importance. Although various digital tools are used to develop writing skills online, this process is not always systematically organized. The development of oral communication is an even more complex issue, as the lack of real-life communication environments limits students' ability to freely express their thoughts. Therefore, it is considered important for teachers to emphasize discussions, debates, and group activities during lessons [Garrison & Vaughan, 2008].

The hybrid learning model, to some extent, offers broader opportunities to address these problems. The combination of traditional and online learning elements expands possibilities for both individual and collaborative student engagement. Within this model, conducting part of the lesson in the classroom and part online creates conditions for applying various teaching strategies. However, the effectiveness of this model depends on proper planning and purposeful organization [Garrison & Vaughan, 2008].

The selection and use of digital platforms is also an important factor influencing the quality of instruction. Different platforms offer various functions, and their selection should be based on the objectives of the lesson. If a platform is used only for sharing materials, its pedagogical potential is not fully realized. In contrast, platforms enriched with interactive tasks, assessment tools, and feedback mechanisms make the learning process more effective [Hodges et al., 2020].

The assessment process should also be reconsidered in online and hybrid learning environments. The direct application of traditional assessment methods is not always effective in such conditions. Therefore, formative assessment approaches are considered more appropriate. These methods allow teachers to monitor students' progress, quickly identify their weaknesses, and strengthen individualized approaches [Black & Wiliam, 2009].

Despite all the aforementioned challenges, online and hybrid learning models, when properly implemented, offer significant opportunities for improving the quality of Azerbaijani language instruction. Digital technologies enable the enrichment of educational resources, the organization of individualized learning, and the development of students' independent learning skills. This is also one of the main goals of the modern education system. One of the important aspects to consider when organizing Azerbaijani language lessons in online and hybrid formats is the adaptation of educational content to the digital environment. Simply transferring traditional teaching materials into digital form does not often produce the expected results. Materials presented in digital environments should be more visual, structured, and interactive in order to engage students and support the learning process. In this regard, the use of multimedia resources, video explanations, and interactive activities is particularly important [Mayer, 2009].

In addition, the expansion of personalized learning opportunities is considered one of the key advantages of online and hybrid environments. Digital platforms allow teachers to adapt instruction to students' individual needs, learning pace, and knowledge level. This is especially effective in

subjects such as the Azerbaijani language, which requires the development of multiple skills. Providing tasks aligned with students' individual learning trajectories increases their engagement in the learning process and leads to more sustainable outcomes [Tomlinson, 2014].

The organization of feedback mechanisms in online and hybrid learning environments also requires special attention. While in traditional classrooms teachers can respond immediately to students' actions, in digital environments this process may be somewhat slower. This can negatively affect the effectiveness of the teaching process. Therefore, it is important for teachers to provide timely and purposeful feedback and to use automated assessment tools. Effective feedback enables students to understand their mistakes and improve their performance [Hattie & Timperley, 2007].

Another important aspect is the development of students' self-regulation skills in digital environments. Since online learning requires students to act more independently, skills such as time management, task planning, and a sense of responsibility become more essential. Insufficient development of these skills may lead to a decrease in learning outcomes. In this regard, it is important for instructors to provide guidance and support to students and to help them develop effective learning strategies [Zimmerman, 2002].

The role of parents in online and hybrid learning environments should also be taken into consideration. Particularly in primary school levels, parental support plays a crucial role in students' adaptation to the learning process. However, low levels of digital literacy among parents or insufficient involvement in the educational process may create certain challenges. Therefore, regular communication with parents and informational support provided by schools and teachers are considered essential. In addition, one of the current issues is socialization in online and hybrid learning environments. Social relationships formed among students in traditional classroom settings have a positive impact on their overall development. However, in digital environments, the formation of these relationships may be limited. This can affect students' emotional well-being and their overall motivation to learn. Therefore, teachers can partially address this issue by incorporating group work, discussions, and collaborative activities into lessons [Vygotsky, 1978].

At the same time, the issue of academic integrity is becoming increasingly relevant in online education. There are certain challenges in ensuring transparency in the assessment process and in guaranteeing that students complete tasks independently. Therefore, it is recommended that instructors use tasks that require more analytical and creative thinking, as well as give preference to open-ended questions and project-based activities. The quality of teacher–student communication in digital environments also has a direct impact on the effectiveness of learning. If communication is limited only to the distribution and checking of assignments, students' engagement in the learning process may decrease. Therefore, it is important for teachers to maintain regular contact with students, respond promptly to their questions, and provide psychological support. This approach helps students feel like active participants in the learning process [Moore, 1993].

The solutions to the problems arising in online and hybrid learning environments should not be limited to technical and methodological measures. This process requires the joint cooperation of education policymakers, school administration, and teachers. Without a comprehensive approach, the effective implementation of these models is not possible. In this regard, strategic planning, continuous professional development programs, and the use of innovative approaches are essential.

Results and discussions

The analysis of online and hybrid learning environments in the organization of Azerbaijani language lessons reveals that the effectiveness of the teaching process is determined by a

combination of technological, pedagogical, and organizational factors. The findings indicate that digital transformation in education has significantly expanded instructional opportunities; however, it has also introduced a range of structural and methodological challenges that must be systematically addressed. One of the key results of the study is that technological accessibility remains a fundamental factor influencing learning outcomes. Unequal access to digital resources continues to create disparities among students, particularly in regional contexts. This confirms that without adequate technological infrastructure, the potential benefits of online and hybrid learning cannot be fully realized. Therefore, ensuring equitable access to digital tools should be considered a priority in educational policy development. Another important outcome is related to teacher preparedness. The findings show that teachers' digital competencies directly affect the quality of instruction and student engagement. Even well-designed curricula may fail to achieve expected results if educators lack the ability to effectively integrate digital tools into pedagogical practice. This highlights the necessity of continuous professional development programs focused on digital literacy and instructional design. The study also demonstrates that student motivation is a critical determinant of success in digital learning environments. Reduced face-to-face interaction often leads to decreased engagement, which negatively affects learning outcomes. However, the use of interactive tasks, collaborative activities, and feedback-oriented instruction can significantly improve student participation and motivation levels. From a linguistic perspective, the development of productive skills, particularly writing and speaking, remains one of the most challenging aspects of online instruction. While digital tools provide opportunities for writing practice, oral communication requires more structured interactional settings. Hybrid learning models partially address this issue by combining synchronous and asynchronous activities, thereby offering a more balanced learning experience. Furthermore, the findings emphasize the importance of instructional design and platform selection. The effectiveness of digital platforms depends not merely on their availability but on how they are pedagogically integrated into the learning process. Platforms that support interaction, assessment, and feedback mechanisms contribute more effectively to language acquisition than those used solely for content distribution. Overall, the results suggest that the successful implementation of online and hybrid learning in Azerbaijani language education depends on a holistic approach that integrates infrastructure development, teacher training, learner engagement strategies, and appropriate assessment methods.

Conclusion

In conclusion, the organization of Azerbaijani language lessons in online and hybrid formats represents a complex and multidimensional process that requires the integration of technological, pedagogical, and human factors. The study confirms that while digital learning environments offer significant opportunities for enhancing education quality, their effectiveness largely depends on how well these components are coordinated. It is evident that improving technological infrastructure alone is not sufficient; equal attention must be given to developing teachers' digital competencies, enhancing student motivation, and ensuring effective instructional design. In addition, personalized learning approaches and continuous feedback mechanisms play a crucial role in improving learning outcomes. Ultimately, the transition to online and hybrid learning models should be viewed not merely as a technological shift but as a comprehensive transformation of the educational process. Only through a balanced and well-structured approach can Azerbaijani language instruction achieve higher levels of effectiveness and sustainability in modern educational contexts.

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TEACHER EDUCATION AND PROFESSIONAL DEVELOPMENT

PEDAGOGICAL PRACTICE AS A CORE COMPONENT OF TEACHER TRAINING

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Abstract: In the modern era, teacher education is characterized not merely by the transmission of knowledge, but as a dynamic and multidimensional process aimed at the formation of complex competencies. The effective organization of this process is one of the key conditions ensuring the sustainable development of education systems at both national and global levels. Pedagogical practice plays a crucial role in the professional formation of future teachers, as it ensures the integration of their theoretical knowledge with practical skills. During this period, students not only develop professional readiness for independent teaching activities, but also gain favorable and rich opportunities for the development of their creative potential as future educators. Pedagogical practice functions as one of the fundamental and inseparable components of teacher education. Methodological knowledge is closely interconnected with a teacher's professional activity, the formation of their personality, and the development of their creative potential, serving as an essential factor in ensuring the effectiveness of pedagogical activity. From this

perspective, one of the most important directions in teacher training is precisely the stage of pedagogical practice.

Key words: teacher, education, national, pedagogical practice, competence

MÜƏLLİM HAZIRLIĞINDA ƏSAS KOMPONENT KİMİ PEDAQOJİ TƏCRÜBƏ

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Xülasə: Müasir dövrdə müəllim hazırlığı yalnız bilik ötürülməsinə deyil, kompleks sərişələrin formalaşdırılmasına yönəlmiş dinamik və çoxölçülü proses kimi çıxış edir. Bu prosesin effektiv təşkili isə həm milli, həm də qlobal səviyyədə təhsil sistemlərinin davamlı inkişafını təmin edən əsas şərtlərdən biridir. Gələcək müəllimin peşəkar formalaşmasında pedaqoji təcrübə mühüm yer tutur və bu proses onun nəzəri bilikləri ilə praktik bacarıqlarının integrasiyasını təmin edir. Pedaqoji təcrübə müddətində tələbənin müəllim kimi müstəqil fəaliyyətə peşəkar hazırlığı formalaşmaqla yanaşı, gələcək müəllimin şəxsiyyətinin yaradıcı potensialının inkişafı üçün əlverişli və zəngin imkanlar yaranır. Pedaqoji təcrübə müəllim hazırlığı prosesinin əsas və ayrılmaz tərkib hissələrindən biri kimi çıxış edir. Metodik biliklər müəllimin peşə fəaliyyəti, onun şəxsiyyətinin formalaşması və yaradıcılıq potensialının inkişafı ilə sıx qarşılıqlı əlaqədə olub, pedaqoji fəaliyyətin effektivliyini təmin edən mühüm amillərdən biridir. Bu baxımdan pedaqoji kadr hazırlığının ən vacib istiqamətlərindən biri məhz pedaqoji təcrübə mərhələsidir.

Açar sözlər: müəllim, təhsil, milli, pedaqoji təcrübə, səriştə

Introduction

The architect of independent Azerbaijan, National Leader Heydar Aliyev, implemented reforms in the field of education, as in all other spheres of the Republic of Azerbaijan, in a consistent and systematic manner, following a progressive trajectory. These reforms, while making a significant contribution to the socio-economic and cultural development of society, also ensured the advancement of state-building to a qualitatively new stage. As a result of the targeted transformations carried out in the education system, the development of human capital emerged as a priority direction, and the content of education as well as management mechanisms were aligned with contemporary requirements.

The significant achievements attained in the field of educational development are also reflected in international assessments. According to the United Nations Human Development Report, Azerbaijan's transition from the category of "medium human development" to "high human development" is regarded as one of the key indicators confirming the effectiveness of the social policy implemented in the country, particularly the reforms carried out in the education sector. This outcome represents the logical result of successful state policy aimed at improving the quality of education, expanding educational opportunities, and developing human resources.

As a result of the purposeful and consistent state policy implemented by the President of the Republic of Azerbaijan, Ilham Aliyev, favorable conditions have been created for the systematic realization of reforms in the field of higher pedagogical education. Within this policy framework, the renewal of the content of higher pedagogical education, the alignment of teacher training with

modern standards, and the improvement of the management model of educational institutions have been defined as priority directions.

At the same time, the implementation of fundamental changes in higher pedagogical education on the basis of a unified concept, regulated by scientifically and methodologically substantiated strategic documents, is of particular importance. The concepts and state programs developed for this purpose not only ensure the sustainable development of the education system, but also contribute to improving the quality indicators of teacher training, strengthening pedagogical human resource potential, and facilitating integration into the international educational space.

In the context of modernizing the education system of Azerbaijan, the level of a teacher's professional competence is considered one of the decisive factors in the effective implementation of the school's instructional, educational, and developmental functions.

Teaching is a unique profession. The personality of a teacher is honorable and admirable, while their work is multifaceted, complex, and highly responsible. In this regard, teachers are expected to educate students who possess comprehensive knowledge and strong personal qualities, thereby contributing to the development of future generations. Teacher training is not limited to the sphere of pedagogical activity; it also serves as the starting point for the formation of human capital. This process ensures meeting the future demands of the labor market and, more broadly, society for human resources, as well as preserving and developing intellectual potential.

Methodology and Materials

The current stage of development in the 21st century is widely characterized in the academic literature as the "Fourth Industrial Revolution." In this era, the rapid advancement of high technologies, digitalization, and artificial intelligence is fundamentally transforming the traditional boundaries of education.

Pedagogical practice functions as one of the fundamental and inseparable components of the teacher education process. Methodological knowledge is closely interconnected with the teacher's professional activity, the development of their personality, and the enhancement of their creative potential, and it represents one of the key factors ensuring the effectiveness of pedagogical practice. In this regard, one of the most important directions in the preparation of pedagogical personnel is precisely the stage of pedagogical practice.

During pedagogical practice, the theoretical knowledge acquired by students in higher education institutions is further deepened and consolidated through practical activity. This process creates real opportunities for students to develop new skills and competencies in the field of pedagogical work and fosters the development of their professional competences.

The completeness of specialist training is ensured precisely through practical activity, as theoretical knowledge is directly applied during pedagogical practice and is thereby acquired in a more conscious and sustainable manner. As a result, this process leads to the formation of pedagogical thinking, the development of pedagogical abilities, and their consolidation at the level of professional competence.

Pedagogical practice plays a decisive role in the full formation of future teachers as professionals. Throughout their studies, students work with learners of different ages and developmental levels, and as they complete tasks defined in their specialization, their sense of professional responsibility gradually increases, and their approach to pedagogical activity becomes more systematic and conscious.

If we look back at the recent past, it becomes evident that approaches to organizing pedagogical practice for future teachers were not unambiguous and have undergone significant changes in the process of development of pedagogical science and the educational system. Traditionally, school-based pedagogical practice was regarded as an important connecting link between the student's theoretical preparation and their future professional activity. In this context, the teacher designs the learning environment, cultivates motivation, applies differentiated instruction by taking into account individual characteristics, and promotes the development of critical, analytical, and creative thinking skills. Accordingly, the teacher's role has evolved from the traditional function of an "information provider" to that of a mentor, facilitator, and guide.

Against the backdrop of these transformations, global challenges in the field of teacher education have become increasingly pronounced. Rapid technological advancement, the abundance of information, and the evolving demands of the labor market require more flexible and innovative approaches from educational systems. Consequently, it is no longer sufficient for teachers to possess only subject-specific knowledge; they are expected to demonstrate a broad range of competencies.

Research conducted by international organizations indicates that high-quality teacher education is one of the decisive factors in improving the overall quality of education. Teacher shortages, the declining attractiveness of the profession, and the unequal distribution of teaching personnel remain pressing issues in many countries. For this reason, the teaching profession is increasingly recognized as a strategic priority within national education policies, accompanied by growing investments in this sector.

Discussion

In particular, in the context of accelerating digitalization, the acquisition of digital competencies has become an essential requirement for teachers. The effective use of electronic resources, the implementation of online and hybrid learning models, and the integration of artificial intelligence-based educational tools introduce new dimensions to teachers' professional competencies. In addition, data-driven decision-making enables teachers to assess student achievement more objectively and to design individualized learning strategies.

Another important direction is the implementation of a competency-based education model. Within this framework, students acquire not only theoretical knowledge but also practical skills and essential competencies such as problem-solving, collaboration, and communication. This, in turn, necessitates the application of new pedagogical approaches, the use of interactive methods, and continuous professional development on the part of teachers.

In the modern era, teacher education is increasingly understood as a dynamic and multidimensional process aimed not merely at the transmission of knowledge, but at the development of complex competencies. The effective organization of this process constitutes a key condition for ensuring the sustainable development of education systems at both national and global levels.

Pedagogical practice plays a crucial role in the professional formation of future teachers, ensuring the integration of their theoretical knowledge with practical skills. During this period, students not only develop professional readiness for independent teaching activities but also gain valuable opportunities for fostering their creative potential as future educators.

Contemporary pedagogical approaches have fundamentally transformed the role of the teacher, positioning them as a key figure in fostering students' thinking, inquiry, and self-expression

skills. In this regard, the following directions are of particular importance in the preparation of modern school teachers in higher education institutions:

The teacher does not merely listen to students' arguments but critically examines them, evaluates their logical foundations, and encourages students to substantiate their viewpoints. This process contributes to the development of critical thinking and analytical reasoning skills.

The teacher guides students in working with diverse sources of information. In the context of the modern information society, where access to knowledge is extensive, the teacher equips students with the skills to select appropriate sources, assess the reliability of information, and systematize acquired knowledge.

The teacher places particular emphasis on fostering a culture of questioning in the classroom. The ability of students to formulate relevant and purposeful questions enhances their level of comprehension, increases their engagement in learning, and promotes independent thinking.

At the same time, the teacher organizes the collaborative exploration of multi-stage and complex problems. In this process, students engage with real-life situations, propose alternative approaches, participate in discussions, and develop the ability to draw well-founded conclusions.

Approaches to organizing pedagogical practice for future teachers are indeed not unambiguous and have undergone significant changes throughout the development of pedagogical science and the education system. Traditionally, school-based pedagogical practice has been viewed as an essential link between a student's theoretical preparation and their future professional activity.

Its primary objective has been to prepare future teachers to perform the functions of an educator. Within the framework of practice, students acquire key aspects of professional activity, including lesson planning and delivery, organization of educational work, and interaction with students, parents, and the teaching staff. Special emphasis is placed on developing the ability to apply theoretical knowledge in practice, as well as on fostering pedagogical thinking and reflection.

In addition, pedagogical practice contributes to the formation of professional competencies, including methodological, communicative, organizational, and diagnostic skills. The development of a creative approach to teaching, along with the ability to adapt content and methods according to students' individual characteristics, also occupies an important place.

Under modern conditions, approaches to organizing pedagogical practice are expanding: its practical orientation is being strengthened, innovative teaching methods are being implemented, and digital technologies along with elements of modeling professional situations are being actively used. All of this creates favorable conditions for a deeper and more comprehensive preparation of future teachers for real professional activity.

Pedagogical practice of teachers has long been one of the leading research areas in the field of pedagogy, attracting the sustained attention of numerous scholars and specialists. Extensive scientific studies conducted in this domain demonstrate that pedagogical practice plays a decisive role in the professional development of future teachers, organically linking their theoretical knowledge with practical activity.

T. G. Cheshuyina proposed that pedagogical practice in schools should be designed as an integrated pedagogical process aimed at developing future teachers' professional, communicative, organizational, research, and constructive competencies. In this approach, the implementation of the content of practice was structured on the basis of the principle of gradual progression: pedagogical component → methodological component → psychological component (Cheshuina, 2006, p.96).

L. V. Vedernikova placed special emphasis on the process of reflection at every stage of pedagogical practice. According to this approach, reflection was carried out based on the observation and analysis of lessons conducted by experienced school teachers, within seminar meetings organized to identify difficulties and problems arising during the practice period, during final seminars held both at schools and higher education institutions, as well as through individual consultations provided to students by university lecturers and school teachers Vedernikova, 2006, 97 p.)

Within the process of pedagogical practice, having students teach the same topic using both traditional instructional methods and a curriculum-based approach is of significant importance for their professional development. This approach enables students to comparatively evaluate the applicability of different pedagogical models and to analyze the strengths and weaknesses of each method in a practical context. As a result, future teachers develop the ability to use various pedagogical techniques purposefully and in accordance with specific instructional situations.

The curriculum-based teaching approach offers considerable advantages in terms of learner-centeredness, active learning, and the achievement of competency-based outcomes. However, its effective implementation requires adherence to certain methodological and organizational conditions. In particular, when forming student groups, it is essential to take into account learners' individual characteristics, levels of knowledge, and social skills. The optimal composition of groups is one of the key factors directly influencing the effectiveness of instruction (Mehrabov, 2015, 68 p).

At the same time, ensuring the active participation of all students within each group in addressing a given task or problem constitutes an important pedagogical objective (Kulikova, 2005, 77 p.). In this process, the teacher assumes not only the role of an observer but also that of a facilitator and regulator, promoting collaboration within groups, preventing passive participation, and maintaining the dynamic nature of the learning process.

Furthermore, continuous monitoring and assessment of student activity are essential conditions for the successful implementation of the curriculum approach. The teacher must systematically observe group performance, intervene when necessary, and provide feedback to guide students' learning processes. This, in turn, contributes to the development of supervision, assessment, and reflective skills among student teachers during their pedagogical practice.

The comparative application of traditional and curriculum-based teaching methods enhances future teachers' pedagogical flexibility, enriches their methodological knowledge, and makes a significant contribution to the formation of professional competencies aligned with the requirements of modern education (İlyasov, 2018, 26 p.).

In the process of teacher education, the organization of pedagogical practice and the development of pedagogical mastery are determined on the basis of a comprehensive and systematic approach (İlyasov, 2018, 58 p.). Scientific and pedagogical research in this field demonstrates that the professional development of future teachers is ensured not only through theoretical knowledge but also through purposefully organized practical activities.

Conclusion

For the effective organization of pedagogical practice, it is essential to structure it according to the principle of gradual progression. In this regard, students' progression through the stages of observation, partial participation, and independent activity enables the gradual development of their pedagogical skills. At the same time, the organization of reflective activities—namely, students'

analysis and evaluation of their own performance—plays a crucial role in the development of pedagogical mastery.

Among the key pathways for developing pedagogical mastery are the strengthening of methodological training, the acquisition of innovative teaching technologies, and the enhancement of communicative and organizational skills. In addition, the implementation of mentoring systems, the expansion of school–university collaboration, and the provision of individualized approaches to students are considered fundamental factors that increase the effectiveness of this process.

Thus, the proper organization of pedagogical practice and the purposeful development of pedagogical mastery in teacher education contribute to the enhancement of future teachers' professional competencies, enable them to achieve high results in the educational process, and ensure their compliance with the requirements of the modern education system.

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challenges

INCLUSIVE EDUCATION IN TRANSITION: CONTEXTUAL AND INSTITUTIONAL CHALLENGES IN AZERBAIJANI TEACHER PREPARATION

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Abstract: This study explores the introduction of inclusive education as a stand-alone subject within teacher education programmes in Azerbaijan and examines the institutional and pedagogical associated with its implementation. A qualitative research design based on a constructivist-interpretive approach was employed. Data were collected through semi-structured interviews with teacher educators and pre-service teachers and analysed using inductive thematic analysis. The findings indicate that introducing inclusive education as a separate course represents an important step towards systemic educational reform in Azerbaijan. However, limited institutional support, fragmented curriculum design, and weak collaboration between universities and schools remain significant challenges. The study also demonstrates that sustained and comprehensive teacher preparation contributes substantially to pre-service teachers' readiness for

inclusive teaching. The findings highlight the need to integrate inclusive education more systematically across teacher education programmes, strengthen experiential learning, promote cultural responsiveness, and support institutional reform. Greater cooperation between universities and schools, together with continued policy development and resource investment, is essential for bridging the gap between theoretical knowledge and practical implementation. Overall, the study emphasises that inclusive education should be understood not only as a pedagogical practice but also as an institutional and systemic responsibility.

Keywords: Inclusive education, teacher education, institutional reform, higher education, Azerbaijan, policy implementation

KEÇİD DÖVRÜNDƏ İNKLÜZİV TƏHSİL: AZƏRBAYCAN MÜƏLLİM HAZIRLIĞINDA KONTEKSTUAL VƏ İNSTİTUSİONAL PROBLEMLƏR

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Xülasə: Bu tədqiqat Azərbaycanda müəllim hazırlığı proqramlarında inklüziv təhsilin ayrıca fənn kimi tətbiqini və onun həyata keçirilməsi zamanı qarşıya çıxan institusional və pedaqoji problemləri araşdırır. Tədqiqat konstruktivist-interpretiv yanaşmaya əsaslanan keyfiyyət metodologiyası çərçivəsində aparılmışdır. Məlumatlar müəllim təlimçiləri və gələcək müəllimlərlə keçirilmiş yarı-strukturlaşdırılmış müsahibələr vasitəsilə toplanmış və induktiv tematik analiz əsasında təhlil edilmişdir. Nəticələr göstərir ki, inklüziv təhsilin ayrıca fənn kimi təqdim edilməsi Azərbaycanda sistemli təhsil islahatları istiqamətində mühüm addımdır. Bununla yanaşı, institusional dəstəyin məhdudluğu, kurikulumun parçalanmış strukturu və universitetlərlə məktəblər arasındakı zəif əməkdaşlıq əsas problemlər kimi müəyyən edilmişdir. Tədqiqat həmçinin göstərir ki, davamlı və hərtərəfli müəllim hazırlığı gələcək müəllimlərin inklüziv tədrisə hazırlığını əhəmiyyətli dərəcədə artırır. Nəticə etibarilə, inklüziv təhsilin müəllim hazırlığına daha geniş integrasiyası, təcrübəyə əsaslanan öyrənmənin gücləndirilməsi, mədəni həssaslığın nəzərə alınması və institusional islahatların həyata keçirilməsi zəruridir.

Açar sözlər: Inklüziv təhsil, müəllim hazırlığı, institusional islahat, ali təhsil, Azərbaycan, siyasətin həyata keçirilməsi

Introduction

Historically, instruction for disabled children in Azerbaijan was limited to a single topic within general pedagogy courses. This narrow approach did not capture the complexity of inclusive education as both a pedagogical and social imperative. In response to policy reforms and evolving educational needs, universities have repositioned inclusive education as a dedicated subject within teacher education programmes. This development represents curricular innovation as well as systemic and institutional transformation in teacher preparation. This empirical study explores how inclusive education is defined and implemented in Azerbaijani higher education, drawing on Critical Disability Studies (Goodley, 2013) and a decolonial perspective (Meekosha and Soldatic, 2011). It focuses on lecturers' and pre-service teachers' experiences of teaching and learning about inclusion, and addresses the broader question: How do institutional structures and pedagogical practices influence the implementation of inclusive education in teacher preparation? The research draws on qualitative data from semi-structured interviews with lecturers and pre-service teachers

across a university in Azerbaijan. Thematic analysis was conducted around two core dimensions: Inclusive Education as a Stand-Alone Module and Challenges in Teaching and Learning Inclusive Education: University and School Perspectives. Researchers Jordan and Goodey (1996), Visser et al. (2001) and Zabeli et al. (2021) have emphasised that inclusion can be achieved through the implementation of appropriate practices and teaching strategies. However, some scholars (Slee and Tomlinson, 2018; Allan, 2007) argue that focusing solely on pedagogy risks neglecting broader social, cultural, and political dimensions, including challenging exclusion and redefining education as a place of justice and belonging.

Methodology

This study employed a qualitative research design, specifically a constructivist-interpretive approach, to explore lecturers' perceptions of inclusive pedagogical practices within the Azerbaijani educational context. Participants were invited to a semi-structured interview depending on their availability. For this study, interviews were conducted with teacher educators and pre-service teachers. The use of qualitative methods enabled the researcher to capture rich, contextually grounded insights into how inclusive practices are interpreted and internalized within a postcolonial framework. Semi-structured interviews were conducted with participants in three main disciplines at a university where most teachers obtain their BA qualification. Conducting interviews requires careful planning, attention, and careful preparation. Researchers must have extensive experience in relevant topic areas and consider factors like interview design, number of interviewees, and data handling.

Before beginning my fieldwork, I obtained ethics approval from my university that my study was ethically acceptable or, more simply, that it met the ethical requirements established by various institutional bodies, in my case the University of Sheffield and its Ethical Research Code of Conduct. Participants were given verbal and written agreement in Azerbaijan, to ensure anonymity and confidentiality of their data. As a researcher, I used password-protected computers and USBs to securely store data. The use of pseudonyms ensured participant confidentiality and anonymity.

The analysis began with organizing the data using inductive thematic analysis. This method is useful for identifying common themes across participants (Riessman, 2005) and is widely used in qualitative research to detect patterns and structure data into meaningful themes (Braun et al., 2019; Campbell et al., 2021). Reducing and reorganizing large amounts of qualitative data is essential for clarity and impact (Miles & Huberman, 1994; Morse, 1994; Creswell, 2014). I manually transcribed all interviews to ensure anonymity, with each hour of recording requiring approximately 5-6 hours of transcription. Each interview was reviewed twice to ensure accuracy. As per Braun and Clarke (2006), I read each transcript multiple times for familiarisation. I then used color coding to highlight key concepts and identified themes and sub-themes accordingly. Transcripts were then structured in two tables by participant groups, with columns aligned to the study questions to organize responses and findings.

Discussion

The findings of this study demonstrate that introducing inclusive education as a separate course in teacher education programmes represents an important step towards systemic educational reform in Azerbaijan. However, the findings also suggest that the establishment of a stand-alone module alone may not be sufficient to ensure the effective implementation of inclusive education. Institutional structures, curriculum organisation, teacher preparation and collaboration between universities and schools all play an important role in shaping inclusive educational practices.

The emphasis placed by lecturers on sustained and in-depth training is consistent with previous research highlighting the importance of appropriate teaching strategies and professional preparation for achieving inclusion (Jordan and Goodey, 1996; Visser et al., 2001; Zabeli et al., 2021). The finding that graduates who complete more comprehensive inclusive education modules demonstrate greater readiness for inclusive teaching further suggests that the depth and continuity of teacher preparation are important factors in developing inclusive competencies.

At the same time, the challenges identified in this study support the arguments of Slee and Tomlinson (2018) and Allan (2007), who caution against understanding inclusion solely as a pedagogical issue. The findings indicate that institutional support, cultural factors and the relationship between universities and schools are equally important. A narrow focus on classroom strategies may therefore overlook wider structural and social barriers that influence the participation and belonging of students.

The weak connection between universities and schools is particularly significant. While theoretical knowledge is an essential component of teacher preparation, pre-service teachers also require opportunities to observe, experience and practise inclusive approaches within authentic school settings. Stronger university-school partnerships could provide opportunities for experiential learning and enable pre-service teachers to connect theoretical concepts with practical classroom situations.

The findings also highlight the importance of cultural responsiveness in the development of inclusive education in Azerbaijan. Inclusive practices cannot simply be transferred from one educational context to another without considering local institutional structures, cultural expectations and historical developments. A context-sensitive approach may therefore be more appropriate for developing sustainable inclusive education practices within Azerbaijani teacher education.

Overall, the study suggests that inclusive education should be understood as both a pedagogical and institutional responsibility. Moving beyond the introduction of a separate course requires broader curriculum integration, continuous professional development, stronger institutional support and closer cooperation between higher education institutions and schools.

Conclusion

Findings indicate that the introduction of inclusive education as a separate course represents a meaningful step towards system-wide reform in Azerbaijan context. However, this progress is tempered by ongoing structural and pedagogical challenges, such as insufficient institutional support, fragmented curriculum design, and weak connections between universities and schools. Lecturers emphasised the need for sustained, in-depth training, observing that graduates who undertake comprehensive modules demonstrate significantly greater readiness for inclusive teaching. The findings highlight the necessity of integrating inclusive education more thoroughly into teacher preparation programs by emphasizing experiential learning, cultural responsiveness, and institutional reform. Enhanced collaboration between universities and schools, coupled with ongoing policy and resource investment, is crucial for bridging the gap between theoretical frameworks and practical implementation. Although significant challenges remain, the increasing dedication of educators and pre-service teachers provides a strong basis for advancing inclusive education from policy rhetoric to a sustainable, student-centred practice.

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MODERN INDICATORS AND QUALITY STANDARDS FOR EVALUATING TEACHER ACTIVITY IN INCLUSIVE EDUCATIONAL INSTITUTIONS

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Abstract: The presented article investigates strategic indicators and quality standards that define the effectiveness of teacher activities in modern inclusive educational institutions. The relevance of the research is conditioned by the transition of inclusive education from quantitative indicators to a qualitative stage. Within the framework of modern pedagogical paradigms, the paper justifies the insufficiency of traditional methods for evaluating teacher activity and proposes a new, multidisciplinary evaluation model. One of the main scientific innovations noted in the abstract is that teacher professionalism is measured not only by academic achievements but also by criteria such as social-emotional adaptation of students, the degree of application of assistive technologies, and the dynamic adaptation of Individual Education Plans (IEP). During the research process, it was determined that the activity of an inclusive teacher should be divided into four main competency blocks (academic-methodological, socio-psychological, technological, and collaborative), and specific quality standards should be established for each block. The article also analyzes the integration of "Universal Design for Learning" (UDL) principles into teacher activities and modern diagnostic tools for monitoring this process. The conclusion emphasizes that improving quality assurance mechanisms in inclusive education stimulates teachers' professional development and directly serves to create a barrier-free learning environment for students with special needs. The practical significance of the study lies in the fact that the proposed indicators can be used as a methodological basis for internal and external evaluation of teacher activities in educational institutions.

Keywords: inclusive education, teacher activity, quality standards, evaluation indicators, professional competence, Universal Design for Learning (UDL), monitoring

İNKLÜZİV TƏHSİL MÜƏSSİSƏLƏRİNDƏ MÜƏLLİM FƏALİYYƏTİNİN QIYMƏTLƏNDİRİLMƏSİ ÜÇÜN MÜASİR GÖSTƏRİCİLƏR VƏ KEYFİYYƏT STANDARTLARI

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Xülasə: Məqalədə müasir inklüziv təhsil müəssisələrində müəllim fəaliyyətinin effektivliyini müəyyən edən strateji indikatorlar və keyfiyyət standartları tədqiqat obyektinə çevrilmişdir. Tədqiqatın aktuallığı inklüziv təhsilin kəmiyyət göstəricilərindən keyfiyyət mərhələsinə keçidi ilə şərtlənir. Yazıda müasir pedaqoji paradimlər çərçivəsində müəllim fəaliyyətinin qiymətləndirilməsi üçün əsas üsulların yetərsizliyi əsaslandırılmış və yeni, multidissiplinar qiymətləndirmə modeli təklif edilmişdir. Xülasədə qeyd olunan əsas elmi yeniliklərdən biri müəllim peşəkarlığının yalnız akademik nailiyyətlərlə deyil, həm də şagirdlərin sosial-emosional adaptasiyası, assistiv texnologiyaların tətbiq dərəcəsi və Fərdi Təhsil Planlarının (FTP) dinamik adaptasiyası kimi meyarlarla ölçülməsidir. Tədqiqat prosesində müəyyən edilmişdir ki, inklüziv müəllimin fəaliyyəti dörd əsas kompetensiya blokuna (akademik-metodiki, sosial-psixoloji, texnoloji və əməkdaşlıq) bölünməli və hər blok üzrə konkret keyfiyyət standartları müəyyən edilməlidir. Məqalədə həmçinin, "Universal Dizayn" (UDL) prinsiplərinin müəllim fəaliyyətinə inteqrasiyası və bu prosesin monitorinqi üçün müasir diaqnostik alətlər təhlil olunur. Nəticə hissəsində vurğulanır ki, inklüziv təhsildə keyfiyyət təminatı mexanizmlərinin təkmilləşdirilməsi müəllimlərin peşəkar inkişafını stimullaşdırır və xüsusi qayğıya ehtiyacı olan şagirdlər üçün maneəsiz öyrənmə mühitinin yaradılmasına birbaşa xidmət edir. Tədqiqatın praktiki əhəmiyyəti ondan ibarətdir ki, təklif olunan indikatorlar təhsil müəssisələrində müəllim fəaliyyətinin daxili və xarici qiymətləndirilməsi prosesində metodoloji baza kimi istifadə oluna bilər.

Açar sözlər: inklüziv təhsil, müəllim fəaliyyəti, keyfiyyət standartları, qiymətləndirmə indikatorları, peşəkar kompetensiya, Universal Dizayn (UDL), monitoring.

Introduction

Transformations occurring in the modern global education system are based on the principles of accessibility and equality for all. In this context, inclusive education is not merely the physical integration of students with special needs into general education schools, but a complex pedagogical system that requires adapting the teaching process to the unique needs of every individual. The successful implementation of inclusive education directly depends on the professional readiness and the quality of activity of the teacher, who serves as the driving force of this process. However, traditional evaluation mechanisms used to measure the effectiveness of teacher activity in an inclusive environment conflict with modern requirements. In most cases, standard evaluation indicators focus on students' academic exam results, which is insufficient to reflect the multifaceted nature of inclusivity (Aliyev, 2021).

Evaluating teacher activity in inclusive educational institutions should no longer be measured simply by "knowledge transfer"; it must encompass complex components such as the student's socio-emotional well-being, individual development dynamics, and the organization of a barrier-free learning environment. From this perspective, the viewpoint on teacher professionalism in modern pedagogy is shifting. Teachers are required not only to teach a subject but also to differentiate teaching materials in accordance with "Universal Design for Learning" (UDL) principles, use assistive technologies effectively, and act in coordination with a multidisciplinary team (Mammadov, 2021).

The relevance of this article lies in the critical need to define specific quality standards and modern indicators that regulate and evaluate teacher activity to improve the quality of inclusive education. The purpose of the current research is to formulate an objective and result-oriented evaluation framework that considers the specificities of teacher labor in inclusive classrooms. This framework combines indicators that allow for measuring both the teacher's pedagogical-methodological competencies and their ethical and psychological readiness as a bearer of inclusive culture. The approach proposed in this study views quality assurance of teacher activity not merely as a control mechanism, but as a strategic tool that ensures the continuous professional development of pedagogical staff.

Material and Methods

The main objective in establishing the methodological structure of the research is to ensure the scientific-pedagogical validity of the indicators measuring the quality of teacher activity in an inclusive education environment. To this end, a systemic approach, theoretical-comparative analysis, and a synthesis of quantitative-qualitative methods were applied during the research process. While forming the material base of the study, sources were analyzed across the following four main directions:

1. Regulatory-Legal Documents: The Law of the Republic of Azerbaijan "On Education," the "State Program for the Development of Inclusive Education in the Republic of Azerbaijan (2018-2024)," and UNESCO's "Policy Guidelines on Inclusion and Equity in Education" formed the legal framework of the study.

2. Academic Literature and International Standards: In accordance with APA 7 standards, scientific articles on teacher competencies and inclusive quality management indexed in

"Scopus" and "Web of Science" databases over the last 5-10 years, including Universal Design for Learning (UDL) principles, were used as materials.

3. Observation and Monitoring Protocols: Monitoring sheets of teacher activity used in inclusive educational institutions and structural analysis of Individual Education Plans (IEP) were included in the methodical base.

4. Digital and Assistive Resources: The functional capabilities of electronic learning platforms and assistive technological programs used to measure the digital-inclusive literacy of teachers constituted the technological material of the research.

As a research methodology, a comparative analysis method was first utilized. Through this method, the differences and gaps (gap analysis) between the evaluation of teacher activity in traditional education and the requirements in inclusive education were identified. In the second stage, applying a classification and grouping method, teacher activity was divided into four main categories: academic, socio-psychological, technological, and collaborative. Additionally, an inductive approach was applied in the research: conclusions were drawn toward general quality standards based on specific inclusive classroom experiences. To measure the degree of cooperation of teachers with a multidisciplinary team (speech therapist, psychologist, parent), a "networked collaboration analysis" model was used. The final component of the methodological base, the UDL (Universal Design for Learning) framework, served as the primary criterion for evaluating the degree of inclusivity in a teacher's presentation, expression, and engagement skills. This complex approach confirms that the proposed indicators are both theoretically grounded and practically applicable.

Results and Discussion

Analyses conducted on the evaluation of teacher activity in inclusive educational institutions show that this process constitutes the most complex layer of educational quality management. The results obtained reveal that traditional pedagogical evaluation mechanisms do not align with the principle of "barrier-free education" in an inclusive environment. The main issue at the center of the discussions is the necessity of transitioning from quantitative indicators (exam scores, lesson hours) to quality indicators (adaptability, emotional support, individualization) in measuring teacher activity.

1. Four-Factor Model in Evaluating Teacher Activity

As a result of the discussions conducted within the research framework, four main indicator blocks were identified to evaluate teacher professionalism:

- **Academic-Methodological Adaptation Indicator:** This block analyzes how the teacher "simplifies" or "enriches" the curriculum for a student with special needs. Aliyev (2021) notes that the greatest success of a teacher is the ability to prepare an Individual Education Plan (IEP) that focuses on the student's potential rather than their diagnosis.

- **Socio-Inclusive Environment Indicator:** This is the ability to regulate the relationship between all students in the classroom (both typically developing and those with special needs). The teacher's strategies for preventing bullying and forming a sense of "belonging" in the classroom are accepted as critical quality standards.

- **Technological-Assistive Competence:** Discussions show that a modern inclusive teacher must also be a technological coordinator. The application of digital resources (screen readers, alternative communication programs) for students with visual, hearing, or cognitive impairments is one of the main factors determining a teacher's professional rating.

- **Multidisciplinary Collaboration Standard:** Inclusivity is not a "single teacher" model, but a "team" model. The intensity and quality of the business relationships the teacher establishes with speech therapists, psychologists, and parents directly increase the effectiveness of the activity (Mammadov, 2021).

2. Monitoring of Universal Design for Learning (UDL) Principles

One of the most important discussion topics of the research is the use of Universal Design principles as an evaluation indicator. Teacher activity should be evaluated in three directions:

1. **Representation:** In how many different ways (visual, auditory, tactile) does the teacher present information?

2. **Action and Expression:** What alternative opportunities (verbal, drawing, technological tools) are created for the student to demonstrate their knowledge?

3. **Engagement:** What motivational mechanisms are established to maintain the student's interest in the lesson?

3. Challenges and Solutions in Evaluation

During discussions, it was determined that teachers often avoid subjective evaluation. As a solution, the application of the "Portfoliomedometry" method is proposed. The development path the teacher takes with the student throughout the year, the adaptive materials prepared, and the reflection notes taken should be considered the most objective evidence of their activity. Aliyev et al. (2022) emphasize that when evaluating teacher activity, "relative development" (the difference between the student's previous state and current state) should be taken as the basis rather than "absolute results."

Consequently, the application of modern indicators for evaluating teacher activity in inclusive educational institutions both enhances the teacher's professional reputation and ensures that inclusivity is based on real results.

Table 1: Framework for Evaluating Teacher Activity (Sample)

<i>Standard</i>	<i>Indicator</i>	<i>Evaluation Method</i>
<i>Differential Instruction</i>	<i>Compliance of IEPs with student needs</i>	<i>Document analysis</i>
<i>Psychological Climate</i>	<i>Level of empathy and support in the class</i>	<i>Observation</i>
<i>Collaboration</i>	<i>Number of meetings with parents and specialists</i>	<i>Portfolio analysis</i>

4. Teacher Reflection and Professional Development

The evaluation system should also serve as a "mirror" for the teacher. When a teacher analyzes their own activity (reflection), they identify where they face difficulties (e.g., communicating with a child with autism or preparing material for a child with dyslexia). This creates a basis for continuous professional development.

Conclusion

The conducted research and scientific-pedagogical analyses prove that evaluating teacher activity in the inclusive education system requires a new and fundamental approach. As a result of the research, the following final conclusions have been reached:

1. **Systemic Evaluation:** Teacher activity in inclusive education should be evaluated as a synthesis of psychological readiness and technological literacy components, rather than just pedagogical skills.

2. **Role of Quality Standards:** The defined quality standards (adaptability, inclusive culture, collaboration) serve as a methodological base for updating the content of teacher training programs.

3. **Continuous Development:** Evaluation indicators should be applied as diagnostic tools that identify professional gaps and stimulate continuous growth, rather than as a punitive mechanism for the teacher.

4. **Success Criterion:** Measuring the success of an inclusive teacher by the student's individual achievements that go beyond standard norms and the level of social integration is the most important condition for ensuring the sustainability of inclusive education.

In conclusion, the proposed indicators and quality standards can make a significant contribution to the real and effective implementation of inclusivity in Azerbaijan's educational space, as well as its integration into international educational standards.

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MODERN APPROACHES IN TEACHER EDUCATION: PREPARING EDUCATORS FOR 21ST-CENTURY LEARNING

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Abstract: Teacher education is changing rapidly due to advances in technology and globalization. Traditional models that focus mainly on theory are no longer enough for modern classrooms. Today's teachers are expected to guide students, support creativity, and develop critical thinking skills. This study explores modern approaches such as competency-based education, reflective practice, digital pedagogy, inclusive teaching, and continuous professional development. The research is based on the analysis of academic literature and international education reports. The findings show that practical skills are becoming more important than theoretical knowledge alone. Teachers must also be able to use technology effectively in their teaching. In addition, they need to work successfully with students from different backgrounds and abilities. However, challenges such as limited resources and unequal access to technology still exist. The

study concludes that improving teacher education requires long-term investment, strong policies, and continuous support for teachers.

Keywords: teacher education, digital pedagogy, competency-based learning, inclusive education, reflective practice, professional development

MÜƏLLİM HAZIRLIĞINDA MÜASİR YANAŞMALAR: PEDAQOQLARIN XXI ƏSR TƏHSİLİNİN TƏLƏBLƏRİNƏ HAZIRLANMASI.

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Xülasə: Müəllim təhsili texnologiyanın inkişafı və qloballaşma nəticəsində sürətlə dəyişir. Əsasən nəzəri biliklərə yönəlmiş ənənəvi modellər artıq müasir siniflər üçün kifayət etmir. Bu gün müəllimlərdən tələb olunur ki, şagirdlərə istiqamət versinlər, yaradıcılığı dəstəkləsinlər və tənqidi düşünmə bacarıqlarını inkişaf etdirdirsinlər. Bu tədqiqat kompetensiyaya əsaslanan təlim, reflektiv təcrübə, rəqəmsal pedaqogika, inklüziv təlim və davamlı peşəkar inkişaf kimi müasir yanaşmaları araşdırır. Araşdırma akademik ədəbiyyatın və beynəlxalq təhsil hesabatlarının təhlilinə əsaslanır. Nəticələr göstərir ki, praktik bacarıqlar yalnız nəzəri biliklərdən daha vacib hala gəlir. Müəllimlər həmçinin texnologiyadan tədrisdə səmərəli istifadə etməyi bacarmalıdırlar. Bundan əlavə, onlar müxtəlif sosial və fərdi xüsusiyyətlərə malik şagirdlərlə uğurla işləyə bilməlidirlər. Bununla belə, resurs çatışmazlığı və texnologiyaya çıxışın qeyri-bərabər olması kimi problemlər hələ də mövcuddur. Tədqiqat nəticə olaraq bildirir ki, müəllim təhsilinin inkişafı uzunmüddətli investisiyalar, güclü siyasətlər və müəllimlərə davamlı dəstək tələb edir.

Açar sözlər: müəllim təhsili, rəqəmsal pedaqogika, kompetensiyaya əsaslanan təlim, inklüziv təhsil, reflektiv təcrübə, peşəkar inkişaf

Introduction

Education in the 21st century is undergoing rapid and continuous transformation. Advances in technology, globalization, and the increasing diversity of student populations have significantly changed the way teaching and learning take place. As a result, the role of the teacher has expanded beyond simply delivering knowledge to students. Modern teachers are expected to act as facilitators, mentors, and guides who help learners develop critical thinking, creativity, communication, and problem-solving skills. In today's classrooms, students are more connected to information than ever before. They have access to digital tools, online resources, and interactive learning environments that shape how they learn and engage with knowledge. This shift requires teachers to be flexible and adaptable, using a variety of teaching strategies to meet different learning needs. Therefore, teacher education programs must prepare future educators to work effectively in technology-rich and student-centered environments. However, despite these changes, many teacher education systems still rely on traditional approaches that focus mainly on theoretical knowledge. While theory remains important, it is often not enough to prepare teachers for the realities of modern classrooms. New teachers frequently report difficulties in classroom management, lesson planning, and addressing diverse student needs. This gap between theory and practice highlights the need for more practical and experience-based teacher training. Modern approaches in teacher education aim to address these challenges by focusing on real-world skills and continuous professional development. These approaches include competency-based education, reflective practice, digital pedagogy,

inclusive education, and collaborative learning. They emphasize not only what teachers know, but also what they can do in real teaching situations. This shift helps ensure that teachers are better prepared, more confident, and more effective in their profession. Furthermore, education systems today must respond to issues such as equity, inclusion, and access to quality education for all learners. Teachers play a key role in creating inclusive classrooms where every student feels supported and valued. For this reason, teacher education programs must also include training in cultural awareness, inclusive teaching strategies, and differentiated instruction. This paper aims to explore modern approaches in teacher education and analyze their importance in improving teaching quality. It also examines the main challenges in implementing these approaches and discusses possible solutions. By understanding how teacher education is evolving, it is possible to develop more effective systems that meet the needs of both teachers and students in the modern world.

Materials and Methods

This study adopts a qualitative research design to explore modern approaches in teacher education. A qualitative approach is appropriate because it allows for an in-depth understanding of educational practices, policies, and innovations rather than numerical measurement. The aim is to interpret how teacher education is changing and what methods are being used to improve it in different educational systems.

Research Materials (Data Sources)

The materials used in this study are secondary sources, which include academic books, peer-reviewed journal articles, and international reports. Key organizations such as UNESCO and OECD were used as reliable sources because they regularly publish updated research on global education trends. In addition, national teacher education policy documents from different countries were reviewed to compare how teacher preparation is being improved worldwide.

Table 1: Sources Used in the Study

Type of Source	Examples	Purpose
Academic Journals	Teaching and Teacher Education, Journal of Theoretical and Empirical Studies	Theoretical and empirical studies
Books	Teacher education and pedagogy books	Conceptual framework
International Reports	UNESCO, OECD reports	Global education trends
Policy Documents	National teacher training frameworks	Practical implementation

Data Collection Procedure. Data collection was conducted through a systematic literature review. First, relevant academic materials were identified using keywords such as teacher education, digital pedagogy, competency-based learning, reflective practice, and inclusive education. After that, sources were selected based on relevance, publication date, and academic reliability. Older or non-scientific sources were excluded to ensure accuracy. The selected materials were then organized and grouped according to their relevance to modern teacher education approaches.

Data Analysis Method. The study uses thematic analysis, which helps to identify and organize key ideas from the collected materials. After reading and reviewing all sources, information was grouped into main themes related to teacher education reform.

Table 2: Main Themes Identified in Analysis

Theme	Description	Example in Practice
Competency-Based Education	Focus on skills rather than theory	Teachers assessed through lesson demonstrations
Reflective Practice	Learning through self-evaluation	Teachers keep reflection journals
Digital Pedagogy	Use of technology in teaching	Online classrooms, smart boards
Inclusive Education	Teaching all learners equally	Adapting lessons for special needs students
Continuous Professional Development	Lifelong learning for teachers	Training workshops, online courses

Comparative Approach. A comparative method was also used to analyze differences between traditional and modern teacher education systems. Traditional systems focus mainly on theory and memorization, while modern systems emphasize practical skills and classroom experience.

Table 3: Traditional and Modern Teacher Education (Examples)

Aspect	Traditional Model	Modern Model
Teaching Focus	Theory-based learning	Practice-based learning
Teacher Role	Knowledge transmitter	Learning facilitator
Student Role	Passive learner	Active participant
Technology Use	Minimal	Fully integrated
Assessment	Written exams	Performance + practical tasks

Example Illustration of Analysis. For example, in competency-based teacher education programs, student teachers are required to demonstrate actual classroom teaching skills before graduation. Instead of only passing written exams, they may be evaluated through micro-teaching sessions where they deliver short lessons and receive feedback. Similarly, in digital pedagogy training, teachers practice using platforms like virtual classrooms, interactive quizzes, and online collaboration tools.

Reliability of the Study. To ensure reliability, only peer-reviewed and officially published materials were included. Information from multiple sources was compared to avoid bias. Although no primary data (such as surveys or interviews) was collected, the wide range of academic literature ensures a strong and balanced analysis.

Methodology

Overall, this study uses a structured qualitative approach supported by literature review, thematic analysis, and comparative evaluation. The use of real-world examples and structured tables helps to clearly explain how modern teacher education approaches are applied in practice.

Results and Discussion

The analysis of the selected literature shows that teacher education is undergoing a clear shift from traditional, theory-based preparation toward more practical, flexible, and student-centered approaches. Across all reviewed sources, five main themes consistently appear: competency-based education, reflective practice, digital pedagogy, inclusive education, and continuous professional

development. These themes represent the core directions of modern teacher education reforms worldwide.

1. **Competency-Based Education as a Core Shift.** One of the most important findings is the growing emphasis on competency-based teacher education. In this model, success is measured not only by theoretical knowledge but by what teachers can actually do in real classroom situations. Teacher candidates are expected to demonstrate skills such as lesson planning, classroom management, communication, and student assessment before graduation. This approach is considered more effective because it directly connects training with real teaching practice. For example, in many modern programs, student teachers are required to complete teaching demonstrations or micro-teaching sessions. These activities allow them to practice teaching in a controlled environment and receive feedback.

Table 1: Impact of Competency-Based Education

Area	Improvement Observed
Classroom readiness	Higher confidence among new teachers
Teaching skills	Better lesson planning and delivery
Assessment ability	More accurate evaluation of students
Professional growth	Faster adaptation to school environment

2. **Reflective Practice and Continuous Improvement.** Another key result is the importance of reflective practice in teacher development. Reflection allows teachers to evaluate their own teaching methods and identify areas for improvement. Many teacher education programs now encourage students to keep reflective journals or participate in peer discussions.

The literature shows that teachers who regularly reflect on their practice tend to improve faster and develop stronger professional identities. Reflection helps teachers become more aware of student needs and adjust their teaching strategies accordingly.

Example from Practice. For instance, after a lesson, a teacher may analyze what worked well and what did not. If students did not understand a topic, the teacher may change the teaching method in the next lesson, using more visual materials or group activities.

3. **Digital Pedagogy and Technology Integration.** The study also shows that digital pedagogy has become an essential part of modern teacher education. Teachers are now expected to use digital tools such as online learning platforms, interactive presentations, and virtual classrooms.

Technology is not only used for delivering lessons but also for improving student engagement and personalization of learning. However, the findings also show that effective use of technology depends on proper training. Without training, digital tools may not improve learning outcomes.

Table 2: Benefits and Challenges of Digital Pedagogy

Benefits	Challenges
Interactive learning	Lack of training
Flexible education	Unequal access to devices
Student engagement	Technical difficulties
Personalized learning	Infrastructure limitations

4. **Inclusive Education and Diversity in Classrooms.** Another important finding is the growing focus on inclusive education. Modern classrooms include students from different cultural, linguistic, and educational backgrounds. Teacher education programs now aim to prepare teachers

to support all learners equally. Inclusive education improves fairness and helps create positive classroom environments. Teachers are trained to adapt their teaching methods based on student needs. This includes supporting students with learning difficulties, language barriers, or special educational needs.

Example from Practice. A teacher may use visual aids, simplified instructions, or group work to help students who struggle with reading or language comprehension. This ensures that all students can participate in learning activities.

5. Continuous Professional Development (CPD). The results clearly show that teacher education does not end after graduation. Continuous professional development is essential for maintaining teaching quality. Teachers must constantly update their skills due to changes in curriculum, technology, and student needs. Workshops, online courses, and professional learning communities are widely used methods for CPD. Teachers who engage in lifelong learning are generally more confident and effective in their classrooms.

Table 3: Forms of Continuous Professional Development

Type	Example	Purpose
Workshops	In-service training sessions	Skill improvement
Online courses	E-learning platforms	Flexible learning
Peer learning	Teacher collaboration groups	Experience sharing
Conferences	Academic events	Knowledge exchange

Conclusion

The findings clearly indicate that modern teacher education is more practical, flexible, and learner-centered compared to traditional models. The shift toward competency-based learning ensures that teachers are better prepared for real classroom challenges. Reflective practice encourages continuous self-improvement, while digital pedagogy enhances engagement and accessibility. However, the study also highlights several challenges. One of the main issues is inequality in access to technology, especially in under-resourced schools. Another challenge is resistance to change among some educators who are more comfortable with traditional teaching methods. These barriers slow down the full implementation of modern approaches. Despite these challenges, the overall trend is positive. Educational systems worldwide are gradually moving toward more modern, skill-based, and inclusive teacher education models. This transformation is essential for meeting the demands of 21st-century education.

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THE ROLE OF THE TEACHER FACTOR AND ITS IMPACT MECHANISMS IN THE SUCCESS OF INTEGRATIVE LESSON MODELS

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Abstract: Integrative lesson models function as an effective pedagogical approach in the modern education system aimed at developing students' cognitive activity, creative thinking skills, and independent learning competencies. In this model, the learning process is built on the integration of different subject areas, creating conditions for the holistic and systematic acquisition of knowledge. Such an approach ensures that students are not limited to passive information reception but are also engaged in higher-order cognitive processes such as analysis, synthesis, comparison, and generalization.

The article examines the role of the teacher factor in the success of integrative lesson models and its mechanisms of influence. In this context, the teacher moves beyond the traditional role of a knowledge transmitter and acts as an organizer, facilitator, and motivator of the learning process. The teacher's pedagogical competencies, methodological preparedness, and ability to use innovative approaches are regarded as key factors that directly affect the quality of education. In particular, the application of interactive teaching methods, individualized learning approaches, and digital technologies increases students' interest in learning and strengthens their active participation.

The study shows that within integrative learning environments, problem-based learning, project activities, and group work develop students' collaboration, communication, and critical thinking skills. At the same time, proper lesson planning by the teacher and consideration of students' individual characteristics enhance the effectiveness of the teaching process.

Based on the findings, it can be concluded that integrative lesson models make a significant contribution to students' intellectual development, the realization of their creative potential, and the formation of independent learning habits. The decisive role of the teacher factor in this process is directly linked to their professional competence and openness to innovative pedagogical approaches.

Keywords: Integrative teaching, interactive learning, teacher role, individualized learning, thinking development, pedagogical technologies, student-centered education

İNTEQRATİV DƏRS MODELLƏRİNİN SƏMƏRƏLİLİYİNDƏ MÜƏLLİM AMİLİNİN ROLU VƏ TƏSİR MEXANİZMLƏRİ

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Xülasə: İnteqrativ dərslər modelləri müasir təhsil sistemində şagirdlərin idrak fəaliyyətinin, yaradıcı düşünmə bacarıqlarının və müstəqil öyrənmə kompetensiyalarının inkişafına yönəlmiş səmərəli pedaqoji yanaşma kimi çıxış edir. Bu modeldə təlim prosesi müxtəlif fənn sahələrinin inteqrasiyası əsasında qurularaq biliklərin vahid və sistemli şəkildə mənimsənilməsinə şərait yaradır. Belə yanaşma şagirdlərin təkcə məlumat qəbulu ilə məhdudlaşmamasını, eyni zamanda analiz, sintez, müqayisə və ümumiləşdirmə kimi yüksək idrak fəaliyyətlərinə cəlb olunmasını təmin edir.

Məqalədə inteqrativ dərslər modellərinin uğurunda müəllim faktorunun rolu və onun təsir mexanizmləri araşdırılır. Müəllim burada ənənəvi bilik ötürücüsü rolundan çıxaraq, təlim prosesinin təşkilatçısı, istiqamətvericisi və motivasiyaedici kimi fəaliyyət göstərir. Onun pedaqoji kompetensiyaları, metodik hazırlığı və innovativ yanaşmalardan istifadə bacarığı təlimin keyfiyyətinə birbaşa təsir göstərən əsas amillər kimi qiymətləndirilir. Xüsusilə interaktiv təlim metodlarının, fərdiləşdirilmiş yanaşmaların və rəqəmsal texnologiyaların tətbiqi şagirdlərin təlimə marağını artırır və onların fəallığını gücləndirir. Araşdırma göstərir ki, inteqrativ təlim mühitində problem əsaslı öyrənmə, layihə fəaliyyətləri və qrup işləri şagirdlərin əməkdaşlıq, kommunikasiya və tənqidi düşünmə bacarıqlarını inkişaf etdirir. Eyni zamanda, müəllimin düzgün planlaşdırma aparması və şagirdlərin fərdi xüsusiyyətlərini nəzərə alması təlim prosesinin səmərəliliyini yüksəldir. Nəticələrə əsasən demək olar ki, inteqrativ dərslər modelləri şagirdlərin intellektual inkişafına, yaradıcı potensialının üzə çıxarılmasına və müstəqil öyrənmə vərdişlərinin formalaşmasına mühüm töhfə verir. Müəllim faktorunun bu prosesdə həlledici rolu onun peşəkar hazırlığı və innovativ pedaqoji yanaşmalara açıqlığı ilə birbaşa bağlıdır.

Açar sözlər: İnteraktiv təlim, inteqrativ dərslər, müəllimin rolu, fərdiləşdirilmiş təlim, təfəkkürün inkişafı, pedaqoji texnologiyalar, şagirdyönümlü təlim

Introduction

The modern education system is undergoing profound transformations under the rapid development of globalization, digitalization, and socio-economic changes. These transformations require the organization of the teaching and learning process based on more flexible, outcome-oriented, and student-centered models. The limited capacity of traditional teaching approaches to fully address students' individual development needs, as well as their insufficient stimulation of creative and critical thinking skills, defines the main directions of contemporary pedagogical research. In this context, integrative lesson models emerge as an innovative and highly effective methodological approach within the education system.

Integrative teaching is based on the interconnection of different subject areas in terms of content and methodology. This approach ensures that knowledge is not acquired in a fragmented manner, but rather as a unified and coherent system. As a result, students are able to meaningfully

connect acquired knowledge and apply it to real-life situations. This, in turn, fosters the development of higher-order cognitive skills such as analytical thinking, problem-solving, comparison, and generalization.

Integrative lesson models are not limited to knowledge transmission; they also play a significant role in shaping students' personalities and supporting their social and emotional development. In this model, the learning process is characterized by interactivity, collaboration, and practical orientation. Students become active participants in the learning process, engaging in various activities that increase their motivation and interest in learning (Aliyev, 2020).

In this process, the teacher plays a central pedagogical role. Moving beyond the traditional function of knowledge transmission, the teacher becomes a facilitator who plans, organizes, and manages the learning environment while also taking into account students' individual characteristics. The teacher's pedagogical competencies, methodological preparedness, and ability to apply innovative technologies are among the key factors directly influencing the quality of education.

At the same time, the use of interactive teaching methods, project-based activities, and digital resources enhances the effectiveness of integrative lessons and develops students' collaboration, communication, and critical thinking skills. Proper lesson planning by the teacher and consideration of students' individual developmental characteristics are essential pedagogical mechanisms that determine the success of the learning process.

The main purpose of this study is to scientifically investigate the role of the teacher factor in integrative lesson models and the mechanisms of its influence on the teaching and learning process. The conducted analysis shows that the integrative approach makes a significant contribution to students' intellectual development, the emergence of their creative potential, and the formation of independent learning skills.

Materials and Methods

The methodological basis of this study is grounded in contemporary pedagogical science and includes a systematic approach, pedagogical observation, comparative analysis, survey method, and a review of scientific and analytical literature. Since the main purpose of the research is to comprehensively examine the role of the teacher factor and its mechanisms of influence within integrative lesson models, the study was conducted in a multi-dimensional and phased manner.

Within the framework of the systematic approach, the teaching and learning process was considered as a unified pedagogical system, where all its components—objectives, content, methods, instructional tools, and outcomes—were analyzed in close interrelation. This approach enabled the evaluation of the teacher's activity not as a set of isolated functions, but as an integral part of the overall management of the pedagogical process within integrative instruction.

Through the comparative analysis method, functional differences in the teacher's role between traditional and integrative teaching models were investigated. The results of this comparison revealed that, while in the traditional model the teacher primarily acts as a transmitter of knowledge, in the integrative model the teacher assumes the roles of facilitator, organizer, motivator, and instructional guide. This shift significantly strengthens the student-centered nature of the learning process and enhances the interactive character of the educational environment.

Pedagogical observation was conducted in real classroom settings. During the observation process, teacher–student interactions, lesson organization forms, the level of use of interactive methods, the effectiveness of group work, and students' cognitive engagement were systematically

recorded and analyzed. Particular attention was given to the teacher's guiding role during problem-solving tasks and project-based learning activities (Mammadova, 2019).

The survey method was used to investigate teachers' attitudes toward integrative teaching, their methodological preparedness, digital literacy skills, and the challenges they encounter in practice. The collected data indicated that teachers' openness to innovation and their level of professional development are key determining factors in the successful implementation of integrative lesson models.

In addition, the review of scientific and analytical literature allowed for the examination of both local and international sources related to integrative teaching, interactive methods, and teacher competencies. This analysis enabled the synthesis of generalized scientific conclusions regarding the theoretical foundations and practical implementation mechanisms of integrative education.

Thus, the methodological framework of the study provided a comprehensive analysis of teacher activity in integrative learning environments and made it possible to scientifically determine the changing role of the teacher within the pedagogical process.

Results and Discussion

The analysis of the conducted research results shows that the effective implementation of integrative lesson models is directly determined by the level of teachers' pedagogical competencies, methodological preparation, and their ability to integrate innovative teaching approaches. The purposeful application of interactive teaching methods by the teacher, efficient use of digital technologies, and incorporation of individualized learning strategies into the lesson process have a significant impact on increasing students' cognitive activity, improving learning outcomes, and enhancing learning motivation.

During the discussion, it was determined that in an integrative learning environment, the teacher no longer functions as a traditional transmitter of knowledge but rather acts as the main pedagogical subject who plans, organizes, manages, and directs the learning process. This transformation creates conditions for the learning process to become student-centered and enables students to take a more active role in learning. As a result of this approach, students are formed not only as recipients of information but also as individuals who analyze, evaluate, and apply knowledge.

Based on the research findings, the following key scientific statements were formulated:

- The teacher is the main pedagogical figure who organizes and directs students' cognitive activity in the integrative learning process;
- The systematic application of interactive teaching methods increases students' activity and interest in the learning process;
- Individualized learning approaches improve the quality of education and strengthen result-oriented learning by considering students' individual characteristics;
- The integration of digital technologies into the integrative lesson process enhances the flexibility, engagement, and effectiveness of teaching;
- Problem-based learning methods significantly contribute to the development of critical thinking, analysis, synthesis, and decision-making skills in students (Hasanov, 2021).

At the same time, observations and comparative analysis show that in integrative lesson models, students' creative potential and problem-solving skills develop at a higher rate compared to traditional teaching models. Students adapt more flexibly to real-life situations, process information more systematically, and develop independent conclusion-making skills.

As a general conclusion of the discussion, it can be stated that integrative lesson models serve as an effective pedagogical approach that ensures the intellectual, creative, and social development of students in the modern education system. The successful implementation of this model primarily depends on the teacher's level of professionalism, innovative thinking, and creative approach to the teaching process. Thus, the teacher factor acts as the main determining element of the quality of integrative learning.

Conclusion

Integrative lesson models function as one of the most important pedagogical approaches in the modern education system, ensuring the development of student-centered learning. This model enables the renewal of the content, form, and organizational aspects of the teaching process, ensuring that knowledge is acquired not in a fragmented manner but in a holistic and systematic way. As a result, students are able to connect acquired knowledge more functionally, apply it to real-life situations, and engage in higher-order cognitive processes such as analysis, synthesis, comparison, and generalization.

The results of the conducted research indicate that the effectiveness of integrative lesson models depends directly on the teacher factor. Within this model, the role of the teacher undergoes a fundamental transformation: the teacher is no longer merely a transmitter of knowledge but also acts as a facilitator and pedagogical leader who plans, organizes, manages, and guides the learning process. The teacher individualizes the learning environment by taking into account students' personal characteristics, interests, and cognitive levels, thereby ensuring their active involvement in the learning process.

The study also shows that the level of teachers' pedagogical competencies, their openness to innovative approaches, and their ability to effectively use digital technologies are among the key factors directly influencing the quality of integrative learning. In particular, the application of interactive methods, project-based learning, and problem-oriented tasks increases students' engagement in the learning process and develops their collaboration, communication, and critical thinking skills.

At the same time, the research results demonstrate that in an integrative learning environment, students show higher motivation, increased interest in learning, and improved independent learning skills. This, in turn, creates conditions for the development of their creative potential and accelerates their intellectual growth.

Thus, as a general conclusion, it can be stated that integrative lesson models represent a pedagogical approach of strategic importance in improving the quality of modern education. The successful implementation of this model is primarily determined by the teacher's professional preparation, innovative pedagogical thinking, and ability to adapt to the changing educational environment. In the future, the wider application of integrative learning approaches will significantly contribute to students' intellectual, creative, and social development, as well as to sustainable quality improvement in the education system.

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STUDENT-CENTERED LEARNING IN HIGHER EDUCATION: TRANSFORMING ENGLISH LANGUAGE CLASSROOMS THROUGH ACTIVE METHODOLOGIES

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Abstract: This article explores student-centered learning (SCL) in higher education with a focus on transforming English language classrooms through active methodologies. In contemporary education, traditional teacher-centered instruction is being replaced by learner-centered approaches that emphasize autonomy, collaboration, and active participation.

The study examines how methods such as task-based learning, project-based learning, flipped classrooms, and collaborative learning create more effective learning environments. Based on a qualitative literature review, it draws on constructivist theory and communicative language teaching, viewing learning as a social and interactive process.

Findings show that student-centered learning improves learners' motivation, autonomy, and communicative competence. Students engage in meaningful tasks, group work, and authentic communication, which enhances fluency, confidence, and knowledge retention. Teachers act as facilitators who guide and support learning rather than simply transmit knowledge.

The study also highlights that digital tools and flipped classrooms extend learning beyond the classroom and increase interaction. However, challenges such as large class sizes, limited resources, and insufficient teacher training can reduce effectiveness.

Overall, student-centered learning transforms English language education into a more interactive, flexible, and learner-driven process, while also developing critical thinking, collaboration, and creativity.

Keywords: Student-centered learning, Higher education, Active methodologies, English language teaching, Communicative competence, Constructivism, Collaborative learning

النشطة المنهجيات خلال من الإنجليزية اللغة فصول تحويل :العالى التعليم فى الطالب حول المتمركز التعلم

الرحيم عبد أشرف محمد

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الملخص

من الإنجليزية اللغة فصول تحويل على التركيز مع ،العالى التعليم في (SCL) الطالب حول المتمركز التعلم المقالة هذه تستكشف تركيز بأساليب تدريجياً المعلم حول المتمركز التقليدي التعليم استبدال يتم ،المعاصرة التعليمية السياقات في .نشطة منهجيات خلال ،الإنجليزية اللغة تعليم في خاص بشكل مهماً التحول هذا ويعد .النشطة والمشاركة والتعاون الاستقلالية على وتؤكد المتعلم على .الفعال للتعلم أساسية أموراً حقيقية حياتية مواقف في اللغة واستخدام والتفاعل التواصل يعد حيث ،المقلوبة والفصول ،المشاريع على القائم والتعلم ،المهام على القائم التعلم مثل النشطة المنهجيات مساهمة كيفية في الدراسة تبحث النظرية إلى الدراسة تستند ،نوعية أدبية مراجعة على وبالاعتماد .وجاذبية فاعلية أكثر تعليمية بيئات خلق في التعاوني والتعلم .الخبرة على قائمة واجتماعية تفاعلية عملية التعلم أن يؤكد وكلاهما ،التواصل اللغة وتدریس البنائية الطلاب يصبح .التواصلية وكفاءتهم ودافعيتهم المتعلمين استقلالية كبير بشكل يعزز الطالب حول المتمركز التعلم أن النتائج تُظهر التواصلية والأنشطة ،والمناقشات ،الجماعي والعمل ،المشكلات حل في الانخراط خلال من ،التعلم عملية في نشطين مشاركين ،نفسه الوقت وفي .الطويل المدى على بالمعرفة والاحتفاظ والثقة والدقة الطلاقة تحسين على الممارسات هذه تعمل كما .الحقيقية .هادفة تعليمية تجارب ويصمم ويدعم يوجه ميسر إلى للمعرفة ناقل من المعلم دور يتغير .والتخصيص التفاعل من يزيد مما ،الصف خارج التعلم توسع الرقمية والأدوات المقلوبة الفصول نماذج أن إلى الدراسة تشير كما المعلمين تدريب ونقص ،التقليدية التقييم وأنظمة ،المؤسسي الدعم وضعف ،الفصول كثافة مثل التحديات بعض تعيق قد ،ذلك ومع .الفعال التنفيذ .ومرونة تفاعلاً أكثر تعليمية بيئات إلى الإنجليزية اللغة تعليم يحول الطالب حول المتمركز التعلم أن إلى المقالة تخلص ،عام وبشكل .والتعاون والإبداع النقدي التفكير مثل والعشرين الحادي القرن ومهارات اللغوية المهارات يعزز كما ،المتعلم على وتركيزاً الكفاءة ،الإنجليزية اللغة تدریس ،النشطة المنهجيات ،العالى التعليم ،الطالب على المتمركز التعلم :المفتاحية الكلمات ..التعاوني التعلم ،البنائية ،التواصلية

Introduction

In recent decades, higher education has experienced a profound pedagogical transformation driven by globalization, technological advancement, and changing learner expectations. Traditional teacher-centered approaches, which position the instructor as the primary source of knowledge, are increasingly being replaced by student-centered learning (SCL), where learners actively participate in the construction of knowledge. This shift is particularly significant in English language teaching (ELT), where communication, interaction, and practical language use are central to learning outcomes.

Student-centered learning emphasizes autonomy, collaboration, critical thinking, and experiential learning. Rather than passively receiving information, students engage in meaningful tasks that require them to apply language in authentic contexts. In English language classrooms, this approach aligns closely with communicative language teaching and modern pedagogical theories that view language acquisition as a socially interactive process.

Active methodologies such as task-based learning, project-based learning, flipped classrooms, problem-based learning, and collaborative learning are key instruments of this transformation. These methods aim to create dynamic learning environments where students take responsibility for their progress, while teachers act as facilitators guiding the learning process.

The purpose of this article is to analyze how student-centered learning, supported by active methodologies, transforms English language classrooms in higher education. It also explores the theoretical foundations, practical applications, and challenges associated with implementing these approaches.

Materials and Methods

This study adopts a qualitative and descriptive research design based on an extensive review of scholarly literature in the fields of applied linguistics, educational psychology, and higher education pedagogy. The research draws on theoretical frameworks including constructivism, sociocultural learning theory, and communicative language teaching (CLT), which collectively emphasize learner engagement and interaction.

The materials analyzed include peer-reviewed journal articles, academic books, and empirical studies focusing on student-centered learning and active methodologies in English language teaching. Key authors and scholars such as Richards, Harmer, Biggs, Benson, and Freeman provide the theoretical foundation for understanding classroom transformation in higher education contexts.

The methodological approach involves comparative and thematic analysis of existing research. This allows the identification of major pedagogical trends, benefits, and limitations of student-centered approaches. The study also examines how different active methodologies function in real classroom environments, particularly in relation to learner motivation, autonomy, and communicative competence.

Additionally, the research considers the role of institutional structures, curriculum design, and teacher training in supporting or limiting the implementation of student-centered learning in higher education institutions.

Results and Discussion

The analysis reveals that student-centered learning significantly transforms English language classrooms by shifting focus from teacher-led instruction to learner-driven engagement. One of the most prominent outcomes is the increase in student autonomy. Learners become active participants in their educational journey, taking responsibility for decision-making, goal-setting, and self-assessment. This autonomy contributes to deeper cognitive engagement and long-term language retention.

Active methodologies such as task-based learning (TBL) and project-based learning (PBL) create opportunities for learners to use English in meaningful, real-world contexts. Instead of memorizing isolated grammatical rules, students engage in problem-solving tasks, group projects, presentations, and simulations. These activities promote fluency, accuracy, and communicative confidence simultaneously.

Collaborative learning plays a particularly important role in student-centered classrooms. Group work encourages peer interaction, negotiation of meaning, and shared knowledge construction. Through collaboration, learners develop not only linguistic competence but also social and interpersonal skills essential for academic and professional success.

Flipped classroom models also contribute significantly to classroom transformation. In this approach, students study theoretical content outside the classroom through videos, readings, or digital materials, while class time is dedicated to interactive practice, discussion, and feedback. This model increases classroom efficiency and allows teachers to provide individualized support.

Another important finding is the transformation of the teacher's role. In student-centered environments, teachers shift from being authoritative knowledge providers to facilitators, mentors, and learning designers. They are responsible for creating engaging tasks, guiding discussions, and supporting learners' progress. However, this role requires advanced pedagogical skills and continuous professional development.

Despite these advantages, several challenges hinder the full implementation of student-centered learning. Large class sizes, limited institutional support, rigid curricula, and insufficient

teacher training often restrict the effectiveness of active methodologies. Additionally, some students initially struggle with autonomy due to their previous experience with traditional teacher-centered education.

Motivation also plays a critical role in the success of student-centered learning. When learners perceive tasks as meaningful and relevant to real-life communication, their engagement increases significantly. Conversely, poorly designed activities may lead to confusion or reduced participation.

Overall, the findings demonstrate that student-centered learning creates more interactive, flexible, and effective English language classrooms. It not only enhances linguistic competence but also fosters critical thinking, creativity, collaboration, and lifelong learning skills.

Conclusion

Student-centered learning represents a fundamental shift in higher education pedagogy, particularly in English language teaching. By prioritizing learner autonomy, interaction, and active engagement, it transforms traditional classrooms into dynamic learning environments.

The study concludes that active methodologies such as task-based learning, project-based learning, collaborative learning, and flipped classroom strategies significantly improve students' communicative competence and overall academic development. These approaches encourage learners to use English in authentic contexts, thereby bridging the gap between theoretical knowledge and practical application.

However, successful implementation requires adequate institutional support, teacher training, and curriculum flexibility. Without these conditions, the effectiveness of student-centered learning may be limited.

In conclusion, student-centered learning is not merely a teaching technique but a comprehensive educational philosophy that aligns with the needs of modern higher education. Future research should further explore the integration of digital technologies, blended learning environments, and culturally responsive pedagogy to enhance the effectiveness of English language instruction in student-centered classrooms.

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REFRAMING EDUCATIONAL LEADERSHIP: TRANSFORMATIONAL APPROACHES TO SOCIAL-EMOTIONAL LEARNING AND SCHOOL CHANGE

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Abstract: This paper offers a theoretical analysis of the relationship between transformational leadership and social-emotional learning within educational settings. Drawing on Social Learning Theory and contemporary leadership scholarship, the study explores how leadership practices shape the implementation, sustainability, and outcomes of social-emotional learning initiatives. The analysis synthesizes empirical and conceptual literature to argue that transformational leaders play a central role in fostering environments that support social-emotional competence by modeling behaviors, reinforcing positive interactions, and cultivating organizational cultures grounded in trust and collaboration. Adopting a qualitative theoretical approach, the study integrates interdisciplinary research to construct a conceptual framework linking leadership practices with social-emotional learning outcomes. The findings suggest that transformational leadership contributes to the development of positive school climates, strengthens teacher efficacy, and enhances student learning experiences. The discussion situates these findings within broader debates on educational reform and leadership preparation, emphasizing the need for systemic approaches that prioritize both academic and social-emotional development.

Keywords: transformational leadership, social-emotional learning; interdisciplinary research, school leadership, school climate; social learning theory

TƏHSİL LİDERLİYİNİN YENİDƏN KONSEPTUALLAŞDIRILMASI: SOSIAL- EMOSIONAL ÖYRƏNMƏ VƏ MƏKTƏB DƏYİŞİKLİYİNƏ TRANSFORMASİYON YANAŞMALAR

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Xülasə: Bu məqalə təhsil mühitlərində transformasiya liderlik ilə sosial-emosional öyrənmə (SEL) arasındakı əlaqənin nəzəri təhlilini təqdim edir. Sosial Öyrənmə Nəzəriyyəsi və müasir liderlik ədəbiyyatına əsaslanaraq, tədqiqat liderlik praktikalarının sosial-emosional öyrənmə təşəbbüslərinin tətbiqinə, davamlılığınə və nəticələrinə necə təsir etdiyini araşdırır.

Təhlil empirik və konseptual ədəbiyyatı sintez edərək göstərir ki, transformasiya liderlər davranış nümunəsi göstərməklə, müsbət qarşılıqlı əlaqələri gücləndirməklə və etibar və əməkdaşlığa əsaslanan təşkilati mədəniyyət formalaşdırmaqla sosial-emosional bacarıqları dəstəkləyən mühitlərin yaradılmasında mühüm rol oynayırlar.

Keyfiyyət yönümlü nəzəri yanaşma tətbiq edən bu tədqiqat, liderlik praktikaları ilə sosial-emosional öyrənmə nəticələri arasında əlaqəni izah edən konseptual çərçivə formalaşdırmaq üçün müxtəlif sahələr üzrə

araşdırmaları integrasiya edir. Nəticələr göstərir ki, transformasion liderlik müsbət məktəb mühitinin inkişafına töhfə verir, müəllimlərin effektivliyini artırır və şagirdlərin öyrənmə təcrübələrini yaxşılaşdırır.

Müzakirə bu nəticələri təhsil islahatları və liderlik hazırlığı ilə bağlı daha geniş elmi debatlar çərçivəsinə yerləşdirir və həm akademik, həm də sosial-emosional inkişafı prioritet edən sistemli yanaşmaların vacibliyini vurğulayır.

Açar sözlər: transformasion liderlik, sosial-emosional öyrənmə, fənlərarası tədqiqat, məktəb liderliyi, məktəb mühiti, sosial öyrənmə nəzəriyyəsi

Introduction

In recent decades, educational systems have increasingly recognized the importance of addressing students' social and emotional development alongside academic achievement. Social-emotional learning has emerged as a key framework for fostering holistic student outcomes, reflecting a shift away from purely cognitive models of education toward more integrated approaches that consider the whole learner. At the same time, leadership has been identified as a critical factor in shaping school environments and influencing educational outcomes. Within this context, transformational leadership has gained prominence as a model that emphasizes vision, motivation, and the development of individuals within organizations.

Transformational leadership, originally conceptualized by Bass (1985), is characterized by the ability of leaders to inspire and motivate others, stimulate intellectual growth, attend to individual needs, and model ethical behavior. These leadership practices are particularly relevant in educational settings, where leaders are responsible not only for administrative functions but also for fostering environments that support teaching, learning, and student well-being. Social-emotional learning, defined as the process through which individuals develop skills such as emotional regulation, empathy, and responsible decision-making, aligns closely with the goals of transformational leadership (Weissberg et al., 2015).

This paper examines the intersection between transformational leadership and social-emotional learning, with a focus on understanding how leadership practices influence the implementation and effectiveness of social-emotional initiatives in schools. By synthesizing existing literature and applying Social Learning Theory as a conceptual lens, the study contributes to a deeper understanding of the mechanisms through which leadership shapes educational experiences and outcomes.

Literature Review

Social-emotional learning has become a central component of contemporary educational discourse, reflecting growing awareness of the interconnectedness between emotional well-being and academic success. Research has consistently demonstrated that students who develop strong social-emotional competencies are more likely to achieve positive academic outcomes, exhibit prosocial behaviors, and maintain psychological well-being (Durlak et al., 2011; Osher et al., 2016). The framework developed by the Collaborative for Academic, Social, and Emotional Learning emphasizes competencies such as self-awareness, self-management, social awareness, relationship skills, and responsible decision-making, which together form the foundation of effective social functioning.

At the same time, leadership research has highlighted the significant influence of school leaders on organizational outcomes. Transformational leadership, in particular, has been associated with improved teacher performance, increased organizational commitment, and enhanced student achievement (Leithwood et al., 2020). Leaders who adopt transformational practices are able to

create shared visions, foster collaboration, and support innovation, thereby contributing to school improvement efforts. The influence of leadership on school climate is especially relevant in the context of social-emotional learning, as the successful implementation of such initiatives depends on supportive and inclusive environments.

The relationship between transformational leadership and social-emotional learning can be understood through the lens of Social Learning Theory. According to Bandura (1986), individuals acquire new behaviors through observation, modeling, and reinforcement. In educational settings, leaders and teachers serve as models whose behaviors influence students' social and emotional development. Transformational leaders, by demonstrating empathy, effective communication, and ethical decision-making, provide examples that students and staff can emulate. Furthermore, the reinforcement of positive behaviors and the creation of supportive environments contribute to the development of self-efficacy and motivation, which are essential for learning.

Despite the growing body of research on both transformational leadership and social-emotional learning, there remains a need for greater integration of these fields. Much of the existing literature examines these concepts separately, with limited attention to the processes through which leadership influences the implementation and outcomes of social-emotional initiatives. This gap underscores the importance of theoretical analyses that bring together insights from multiple domains to provide a more comprehensive understanding of educational practice.

Methodology

This study adopts a qualitative theoretical research design grounded in interpretivist and constructivist paradigms, with the aim of advancing conceptual understanding rather than generating primary empirical data. The choice of a theoretical synthesis approach is particularly appropriate given the fragmented nature of existing research on transformational leadership and social-emotional learning, where insights are often dispersed across distinct scholarly traditions. By integrating literature from educational leadership, psychology, and learning theory, the study seeks to construct a coherent analytical framework that explains how leadership practices shape social-emotional learning processes within educational organizations.

The study draws on a systematic yet flexible review of peer-reviewed journal articles, foundational texts, and policy-oriented reports published primarily between 2000 and 2023, with selective inclusion of seminal works that underpin the theoretical foundations of the field. Databases such as ERIC, Scopus, and Google Scholar were utilized to identify relevant sources using key search terms including transformational leadership, social-emotional learning, school climate, and social learning theory. The selection of literature was guided by criteria of conceptual relevance, methodological rigor, and contribution to understanding the relational dynamics between leadership and learning environments. Particular emphasis was placed on studies that explicitly examine leadership practices in relation to organizational culture, teacher development, and student outcomes.

The analytical strategy employed in this study is informed by thematic synthesis and concept-centric literature review techniques. Rather than summarizing individual studies, the analysis prioritizes the identification of recurring constructs, mechanisms, and theoretical linkages across the literature. Initial coding focused on key domains such as leadership behaviors, social-emotional competencies, and mediating organizational factors, including school climate and teacher efficacy. These codes were iteratively refined into broader analytical categories that capture the processes through which transformational leadership influences social-emotional learning.

To enhance analytical depth, the study is guided by Social Learning Theory as an organizing framework, allowing for the interpretation of leadership practices as mechanisms of modeling, reinforcement, and social influence (Bandura, 1986). This theoretical anchoring enables the integration of micro-level interactions, such as leader–teacher and teacher–student relationships, with macro-level organizational processes. The resulting framework is not intended as a definitive model but as a theoretically informed lens through which the complex interplay between leadership and social-emotional learning can be understood.

While the study does not involve human participants and therefore does not require institutional ethical approval, it adheres to principles of scholarly integrity, including accurate representation of sources and critical engagement with existing literature. Limitations inherent to theoretical research, such as the absence of empirical validation and potential selection bias in literature inclusion, are acknowledged. Nevertheless, the approach provides a robust foundation for advancing conceptual clarity and informing future empirical inquiry.

Results/Analysis

The analytical synthesis reveals that transformational leadership operates as a multidimensional mechanism that shapes both the structural and relational conditions necessary for effective social-emotional learning. Rather than functioning as a discrete set of practices, transformational leadership emerges as a process that permeates organizational culture, influencing how individuals interact, interpret expectations, and engage in learning.

A central pattern identified across the literature is the role of leadership in constructing school climates that are conducive to social-emotional development. Transformational leaders actively cultivate environments characterized by trust, inclusivity, and psychological safety, which are widely recognized as foundational for meaningful engagement in social-emotional learning (Huang & Allensworth, 2017; Osher et al., 2016). These climates are not merely byproducts of leadership but are intentionally shaped through practices such as shared vision-building, relational trust, and consistent communication. Within such environments, students and teachers are more likely to engage in reflective practices, express vulnerability, and develop emotional competencies.

The analysis further indicates that leadership exerts a significant influence on teacher agency and instructional practice. Teachers are positioned as primary implementers of social-emotional learning, yet their capacity to do so effectively is mediated by leadership support, professional development opportunities, and collaborative cultures (Moullin et al., 2018; Leithwood et al., 2020). Transformational leaders enhance teacher self-efficacy by fostering a sense of collective purpose and by recognizing individual contributions, which in turn promotes greater commitment to integrating social-emotional learning into classroom practice. This dynamic suggests that leadership impacts student outcomes indirectly through its influence on teacher beliefs, motivation, and pedagogical approaches.

Another critical dimension emerging from the analysis is the role of modeling and reinforcement, as conceptualized within Social Learning Theory. Transformational leaders serve as visible exemplars of social-emotional competencies, demonstrating behaviors such as empathy, ethical decision-making, and effective communication (Bandura, 1986; Elias et al., 2006). These behaviors are not only observed by teachers and students but are also embedded within the norms and expectations of the organization. The consistent reinforcement of such behaviors through feedback, recognition, and relational interactions contributes to the internalization of social-emotional competencies among members of the school community.

Importantly, the analysis highlights that the relationship between transformational leadership and social-emotional learning is mediated by organizational coherence and alignment. Schools in which leadership practices, instructional strategies, and institutional goals are aligned around social-emotional learning demonstrate greater consistency and sustainability in implementation. Conversely, fragmented or symbolic adoption of social-emotional learning, without corresponding shifts in leadership practice, tends to result in superficial or short-lived outcomes. This finding resonates with broader critiques of educational reform that emphasize the importance of coherence and capacity in achieving meaningful change.

At a more critical level, the analysis reveals that much of the existing literature tends to idealize transformational leadership without sufficiently interrogating contextual constraints such as policy pressures, accountability regimes, and resource limitations. These factors may shape or constrain leaders' ability to enact transformational practices, particularly in systems characterized by high levels of standardization or centralization. As such, the effectiveness of transformational leadership in advancing social-emotional learning must be understood as context-dependent, rather than universally applicable.

Overall, the analysis advances a conceptual understanding of transformational leadership as a relational and contextually embedded process that influences social-emotional learning through multiple, interconnected pathways. By linking leadership practices to organizational culture, teacher development, and student learning processes, the study provides a nuanced framework for understanding how educational leadership can support holistic student development.

Discussion

The findings of this study underscore the importance of transformational leadership in advancing social-emotional learning within educational settings. By shaping organizational culture, supporting teacher development, and modeling desired behaviors, transformational leaders create conditions that enable the successful implementation of social-emotional initiatives. This perspective aligns with broader understandings of educational reform, which emphasize the role of leadership in translating policy into practice and fostering sustainable change.

At the same time, the analysis highlights several challenges and limitations within the existing literature. Much of the research on transformational leadership and social-emotional learning relies on self-reported data and cross-sectional designs, which may limit the ability to draw causal conclusions. There is a need for more rigorous and longitudinal studies that examine the dynamic interactions between leadership practices and student outcomes over time.

The findings also have important implications for leadership preparation and professional development. Developing leaders' social-emotional competencies should be a central focus of training programs, as these skills are essential for fostering supportive and inclusive learning environments. Furthermore, educational systems must adopt holistic approaches that integrate social-emotional learning into all aspects of schooling, rather than treating it as an isolated initiative.

Conclusion

This paper provides a theoretical examination of the relationship between transformational leadership and social-emotional learning, highlighting the critical role of leadership in shaping educational outcomes. By integrating insights from Social Learning Theory and leadership research, the study offers a conceptual framework for understanding how leadership practices influence the development of social-emotional competencies in schools.

The analysis demonstrates that transformational leadership contributes to the creation of positive school climates, enhances teacher effectiveness, and supports student development. As educational systems continue to evolve, the integration of social-emotional learning and effective leadership will remain essential for promoting holistic and sustainable improvement. Future research should build on these insights by exploring the complex interactions between leadership, learning, and organizational context in diverse educational settings.

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EVALUATION OF TEACHER PERFORMANCE: INNOVATIVE APPROACHES TO QUALITY ASSURANCE IN EDUCATION

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Abstract: The effectiveness of quality assurance mechanisms in the modern education system is directly related to the correct and objective assessment of teacher performance. In this regard, the assessment of teachers' professional performance is not only a means of control, but also an important management tool that stimulates their development. The article broadly explains the theoretical foundations of teacher performance assessment systems, analyzes the approaches developed in this area and their practical application in a comparative manner. The study examines the best practices of different countries and shows that the most effective assessment models are multi-source (based on student, parent, colleague and management opinions), transparent and development-oriented. These approaches create conditions for teachers to analyze their own performance, identify their strengths and weaknesses, and continue their professional development. At the same time, the application of digital technologies makes the assessment process more operational, objective and accessible. Data collected through electronic platforms allows for a more accurate and systematic analysis of teacher performance. The existing assessment mechanisms in the Azerbaijani education system are also analyzed and their strengths and weaknesses are identified. The study shows that although certain progress has been made in this area in recent years, additional measures are needed to make assessment more objective and development-oriented. In particular, it is considered important to expand the mechanisms of formative assessment, self-assessment and continuous monitoring. As a result, objective, transparent and systematic assessment of teacher performance plays an important role in improving the overall quality of education. The article puts forward scientifically substantiated proposals in this direction and presents practical recommendations for improving the assessment system

Keywords: teacher activity, assessment, quality assurance, innovative approaches, pedagogical development.

MÜƏLLİMİN FƏALİYYƏTİNİN QIYMƏTLƏNDİRİLMƏSİ: TƏHSİLDƏ KEYFİYYƏTİN TƏMİN EDİLMƏSİNƏ İNNOVATİV YANAŞMALAR

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Xülasə: Müasir təhsil sistemində keyfiyyət təminatı mexanizmlərinin effektivliyi bilavasitə müəllim fəaliyyətinin düzgün və obyektiv qiymətləndirilməsi ilə bağlıdır. Bu baxımdan müəllimlərin peşəkar fəaliyyətinin qiymətləndirilməsi yalnız nəzarət vasitəsi deyil, həm də onların inkişafını stimullaşdıran mühüm idarəetmə aləti kimi çıxış edir. Məqalədə müəllim fəaliyyətinin qiymətləndirilməsi sistemlərinin nəzəri əsasları geniş şəkildə izah olunur, bu sahədə formalaşmış yanaşmalar və onların praktik tətbiqi

müqayisəli şəkildə təhlil edilir. Tədqiqat çərçivəsində müxtəlif ölkələrin qabaqcıl təcrübələri araşdırılaraq göstərilir ki, ən effektiv qiymətləndirmə modelləri çoxmənbəli (şagird, valideyn, həmkar və rəhbərlik rəyləri əsasında), şəffaf və inkişafyönümlü xarakter daşıyır. Bu yanaşmalar müəllimlərin öz fəaliyyətlərini təhlil etməsinə, güclü və zəif tərəflərini müəyyənləşdirməsinə və davamlı peşəkar inkişafına şərait yaradır. Eyni zamanda rəqəmsal texnologiyaların tətbiqi qiymətləndirmə prosesini daha operativ, obyektiv və əlçatan edir. Elektron platformalar vasitəsilə toplanan məlumatlar müəllim fəaliyyətinin daha dəqiq və sistemli analizinə imkan verir. Azərbaycan təhsil sistemində mövcud qiymətləndirmə mexanizmləri də təhlil edilərək onların üstün və zəif tərəfləri müəyyənləşdirilir. Araşdırma göstərir ki, son illərdə bu sahədə müəyyən irəliləyişlər əldə olunsada, qiymətləndirmənin daha obyektiv və inkişafyönümlü olması üçün əlavə tədbirlərə ehtiyac vardır. Xüsusilə formativ qiymətləndirmə, özünüqiymətləndirmə və davamlı monitorinq mexanizmlərinin genişləndirilməsi vacib hesab olunur. Nəticə etibarilə, müəllim fəaliyyətinin obyektiv, şəffaf və sistemli qiymətləndirilməsi təhsilin ümumi keyfiyyətinin yüksəldilməsində mühüm rol oynayır. Məqalədə bu istiqamətdə elmi əsaslandırılmış təkliflər irəli sürülür və qiymətləndirmə sisteminin təkmilləşdirilməsi üçün praktik tövsiyələr təqdim olunur.

Açar sözlər: müəllim fəaliyyəti, qiymətləndirmə, keyfiyyət təminatı, innovativ yanaşmalar, pedaqoji inkişaf

Introduction

One of the main tasks facing education systems in a globalized world is to ensure quality education. To achieve this goal, effective assessment of teachers' professional performance plays an important role. Since traditional assessment methods no longer meet modern requirements, the application of innovative approaches has become necessary. Assessment of teacher performance should not only measure results, but also act as a mechanism to ensure the continuous development of teachers. Increasing the quality of education in modern times is one of the priority issues for all countries of the world. Ensuring quality in the education system directly depends on the professional activity of teachers. In this regard, the correct establishment of teacher performance assessment mechanisms is of particular importance. Teacher assessment should not only have a control function, but also stimulate their professional development.

In the 21st century, education systems operate in conditions of rapid socio-economic changes, digitalization and globalization. These changes have further increased the requirements for the quality of education. The main guarantor of the quality of education is the teacher. The professionalism of the teacher, pedagogical approaches and teaching strategies have a direct impact on student achievement. In this regard, the evaluation of teacher performance is not only an administrative control mechanism, but also a strategic tool serving to improve the quality of education. The limitations of traditional assessment methods have necessitated the application of innovative approaches in modern times.

In the modern education system, teacher performance evaluation is moving away from traditional approaches and is based on more technology-based, multi-source and development-oriented models. Digital evaluation systems create broad opportunities for monitoring and analyzing teacher performance. Such platforms, in addition to ensuring operational collection and objective analysis of data, minimize the subjectivity arising from the human factor. In addition, digital systems provide individual feedback to the teacher, help identify strengths and weaknesses. At the same time, this approach bases the decision-making process in educational management on a more scientific basis and strengthens strategic planning. Teacher performance evaluation is an integral part of the pedagogical process.

This process, along with determining the teacher's knowledge, skills and competencies, serves to measure the effectiveness of teaching. In modern pedagogy, a teacher's reflective thinking is considered one of the main indicators of his professional development. Self-assessment is the process of a teacher analyzing his own activities, identifying successes and shortcomings, and planning his future activities. Reflection allows the teacher to critically evaluate his pedagogical approaches. This process creates conditions for the teacher to constantly develop, master new methods and technologies, and also serves to measure the effectiveness of teaching, along with determining the teacher's knowledge, skills and competencies. According to modern approaches, assessment should be based on the following principles: objectivity and transparency, sustainability and systematicity, development orientation, multi-source (student, parent, administration, etc.). At the same time, self-assessment increases the teacher's internal motivation and directs him to self-development, independent of external control. This is one of the main conditions for continuous quality assurance in the modern education system.

In the modern education system, teacher performance evaluation is moving away from traditional approaches and is based more on technology-based, multi-source, and development-oriented models.

In modern pedagogy, a teacher's reflective thinking is considered one of the main indicators of his professional development. Self-assessment is the process of a teacher analyzing his own activities, identifying his successes and shortcomings, and planning his future activities. Reflection allows the teacher to critically evaluate his pedagogical approaches. This process creates conditions for the teacher to constantly develop, master new methods and technologies. At the same time, self-assessment increases the teacher's internal motivation and directs him to self-development, independent of external control. This is one of the main conditions for continuous quality assurance in the modern education system. (**Əhmədov, R. (2018)**)

Such innovative approaches make the evaluation of teacher performance more objective, transparent and development-oriented. The complex application of these methods not only increases the quality of the evaluation process, but also stimulates the professional development of teachers, thereby improving the overall quality of education. Quality assurance in education is the process of planning, implementing and continuously evaluating the results of the educational process. This system is based on the effectiveness of teacher performance.

Teacher performance evaluation is one of the main factors determining the quality of the educational process. This process involves a comprehensive analysis of the teacher's pedagogical skills, methodological preparation, and interaction with students. (**Məmmədov, S. (2020)**)

Modern approaches emphasize that assessment should not only have a control function, but also be development-oriented and motivational.

The reforms carried out in the field of education in the Republic of Azerbaijan involve the renewal of the teacher performance evaluation system. The certification process, diagnostic assessment and the introduction of electronic platforms are considered important steps in this direction. At the same time, wider application and systematic integration of innovative approaches remain necessary. Scientific research shows that: effective evaluation increases teacher motivation, the quality of the educational process improves, and positive dynamics are observed in student results. Thus, teacher evaluation is one of the main mechanisms of quality assurance. Digital platforms create wide opportunities for monitoring and analyzing teacher performance. Electronic

portfolios, online observation systems and analytical tools make the evaluation process more objective.

The reforms carried out in the field of education in the Republic of Azerbaijan in recent years are aimed at improving the teacher performance evaluation system. Recruitment, certification and attestation of teachers are important steps taken in this direction. The current system includes the following components: teacher certification, diagnostic assessment, professional development programs, monitoring and internal control mechanisms. However, in some cases, the formal nature of the assessment and the inadequacy of development-oriented mechanisms are observed. The conducted analyses show that there are the following problems in the field of teacher performance evaluation: lack of complete clarity of assessment criteria, the presence of subjective approaches, poor correlation of results with professional development, and insufficient motivation of teachers. (Şahbazlı, N. (2019))

The following suggestions are put forward to improve the teacher performance evaluation system:

1. Alignment of assessment criteria with international standards
2. Digital assessment platforms application
3. Self-assessment and reflection mechanisms strengthening
4. Evaluation of the results professional development programs with integration
5. Motivational for teachers stimulation system creation

Methodology and Materials

This study was conducted on the basis of mixed methodology (synthesis of qualitative and quantitative approaches) in order to analyze and improve teacher performance evaluation systems. The material base of the study is local and international scientific literature, regulatory and legal documents, reports, as well as empirical data collected from educational institutions. Strategic documents adopted in the field of education of the Republic of Azerbaijan, teacher certification and evaluation criteria, as well as reports of international organizations (for example, UNESCO, OECD, etc.) were comparatively examined. (UNESCO (2015))

In order to collect empirical data, surveys and interviews were conducted among teachers, school principals and education specialists working in various general education schools. During the survey phase, structured questionnaires were used to study the participants' attitudes towards teacher performance evaluation mechanisms, strengths and weaknesses of the current system, as well as development needs. Semi-structured interviews were conducted to obtain qualitative data, and the participants' individual experiences and suggestions were analyzed in depth.

During data processing, statistical analysis methods (percentages, average scores, comparative analysis) and content analysis methods were used. At the same time, a comparative analysis of the Azerbaijani experience with assessment models of different countries was conducted.

To ensure the reliability of the study, the results obtained from various sources were cross-checked. This methodological approach allowed for a more objective and comprehensive assessment of teacher performance evaluation systems.

Conclusion

The results of the study show that teacher performance evaluation is one of the decisive factors in ensuring quality in the education system. The empirical data and theoretical analyses obtained during the study prove that although the existing evaluation mechanisms are effective to a certain extent, there are some shortcomings in their application in terms of objectivity, transparency

and development orientation. (Sahlberg, P. (2011)) In particular, the more formal nature of evaluation and the weakness of feedback mechanisms aimed at teacher professional development were identified as one of the main problems. The discussions show that multi-source assessment models (integration of student, parent, colleague and management opinions) applied in international practice allow to obtain more objective results. According to the research results, adaptation of this approach to the Azerbaijani education system can have a positive impact on more fair assessment of teacher performance and increasing their motivation. At the same time, the application of digital technologies not only increases the efficiency of the assessment process, but also facilitates the systematic collection and analysis of data.

The results also show that developing teachers' self-assessment skills and expanding formative assessment mechanisms play an important role in their professional development. In this regard, the assessment process should be based on a development-oriented approach, not only on results.

In conclusion, it can be noted that objective, transparent and continuous evaluation of teacher performance makes a significant contribution to improving the quality of education. In this direction, improving evaluation criteria, implementing a multi-source approach and widespread use of digital tools are key priorities.

The final results of the study show that objective, transparent and systematic evaluation of teacher performance is one of the main conditions for improving quality in the education system. It was determined that although the existing evaluation mechanisms provide certain positive results, they need to be improved to be more effective and development-oriented.

Thus, teacher performance evaluation should be organized not only as a control tool, but also as an important mechanism that promotes development. Reforms in this direction will significantly contribute to improving the overall quality of education and creating a more efficient educational environment.

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TEACHER EDUCATION AND PROFESSIONAL DEVELOPMENT: MODERN APPROACHES TO TEACHERS' PROFESSIONAL DEVELOPMENT

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Abstract: This study examined teachers' professional development training and its impact on students' academic performance in Yewa North, Ogun State, Nigeria, Sub-Sahara, Africa. The study emphasized modern approaches in teacher education, development of pedagogical skills and methodological competencies, continuous professional development and teacher reflection, and evaluation and quality assurance of teaching performance. A descriptive survey research design was adopted. The population consisted of 355 public school teachers, while 52 English Language teachers were purposively selected from 35 public schools. Data were collected using a self-constructed questionnaire and inventory, with a Cronbach Alpha reliability coefficient of 0.78. Mean and standard deviation were used to answer research questions, while Analysis of Variance (ANOVA) was used to test the hypotheses.

The findings revealed that seminars, workshops, sandwich programmes, and part-time programmes positively influenced teachers' instructional effectiveness and students' academic performance. Teachers who participated in professional development programmes demonstrated improved classroom management, innovative instructional strategies, better lesson delivery, and increased confidence in teaching. The study concluded that continuous teacher education and professional development are essential for improving educational quality and students' academic achievement.

Keywords: Teacher education, professional development, pedagogical competencies, teacher reflection, teaching performance, academic achievement

ÈKỌ OLÙKỌ ÀTÌ ÌDÀGBÀSÓKÈ OJÓŞÉ: ÀWỌN ỌNÀ ÌGBÀLÓDÉ FÚN ÌDÀGBÀSÓKÈ OJÓŞÉ ÀWỌN OLÙKỌ

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Àkótán: Ìwádíí yí ẹ àyẹwò ìdánílẹ̀kọọ ìdàgbàsókè ojógbon àwọn olùkọ àtì ipa rẹ lórí ẹ̀kàwé àwọn akẹ̀kọọ ní Yewa North, Ìpínlẹ̀ Ogun, Nàìjíríà, ní agbègbè Sub-Sahara ní Áfíríkà. Ìwádíí náà tẹ̀numọ̀ àwọn

ònà ìgbàlódé nínú èkó olùkó, ìdàgbàsókè àwọn ojúgbon ìkóni àti agbára ìmúlò ònà-èkó, ìdàgbàsókè ojógbon aláakòso àti ìrònúpadà olùkó, pèlú àyèwò àti idánilójú didara isẹ ìkóni.

A lo àpèrẹ̀ iwádíí iwádíí àpejuwe (descriptive survey research design). Àwọn olùkó ilé-èkó ijoba 355 ni wón jẹ olùgbékalẹ̀ iwádíí náà, nígbà tí a yan olùkó Èdè Gẹ̀ẹ̀sì 52 láti inú ilé-èkó ijoba 35 ní ònà ifókànsìn (purposive sampling). A kó àwọn dátà jọ nípasẹ̀ ibéèrè iwádíí àti àkójo iwádíí tí a dá kalẹ̀ fún iwádíí yìí. Ìgbékẹ̀lẹ̀ irinṣẹ̀ iwádíí náà ni a fi hàn pèlú olùṣòdipúpò Cronbach Alpha tó jẹ 0.78. A lo itupalẹ̀ iwọn àárín (mean) àti iyapa àdéhùn (standard deviation) láti dáhùn àwọn ibéèrè iwádíí, nígbà tí a lo Itupalẹ̀ Àwọn Ìyàtò (ANOVA) láti dán àwọn àròso iwádíí wò.

Àwọn àbájádé fi hàn pé àwọn semina, idánilékò isẹ (workshops), àwọn ètò sandwich, àti àwọn ètò èkó àkókò-apá (part-time programmes) ní ipa rere lórí agbára ìkóni àwọn olùkó àti isẹ àkàwé àwọn akékò. Àwọn olùkó tí wón kópa nínú àwọn ètò ìdàgbàsókè ojógbon fi imúgbòòrò hàn nínú isàkóso kílaàsì, lílo àwọn ònà ìkóni tuntun, fífi èkó hàn dáadáa, àti igboyá tó pò síi nínú isẹ ìkóni. Iwádíí náà parí pé èkó olùkó tí n bá a lọ àti ìdàgbàsókè ojógbon jẹ ohun pàtàkì fún imúgbòòrò didara èkó àti àṣeyọ́rì èkó àwọn akékò.

Àwọn Ọrọ̀ Pátá: Èkó olùkó, ìdàgbàsókè ojógbon, agbára ìkóni, ìrònúpadà olùkó, isẹ ìkóni, àṣeyọ́rì èkó.

Introduction

Generally, Teacher education and professional development remain essential for improving overall educational quality and students' academic performance. In modern educational systems, teachers are expected to possess not only subject knowledge but also pedagogical competence, technological literacy, reflective practice, and classroom management skills. Consequently, continuous professional development has become necessary for teachers to remain effective and relevant.

In Ogun State, particularly in Yewa North, and majorly in all of Subsahara Africa, concerns have continued regarding the academic performance of students as a result of teachers education, competencies, and continued development. Several factors such as, outdated teaching methods, poor access to professional training, and insufficient exposure to innovative educational practices contribute to this challenge. Professional development programmes therefore provide opportunities for teachers to improve their instructional competencies and adapt to modern educational demands.

Modern approaches in teacher education include workshops, seminars, online learning, mentoring, collaborative learning, sandwich programmes, and part-time academic programmes. These approaches improve pedagogical skills, methodological competencies, classroom management, and instructional delivery. Teachers who participate in sustained professional development programmes are more likely to adopt learner-centered approaches, integrate technology into teaching, and promote critical thinking among students.

Professional development training contributes significantly to teaching effectiveness and quality assurance in education. It enables teachers to improve lesson planning, classroom interaction, assessment techniques, and student engagement. Reflective practice also encourages teachers to evaluate their teaching methods continuously and implement better instructional strategies.

Kolb's Experiential Learning Theory provides a suitable framework for understanding teacher professional development. The theory emphasizes learning through experience, reflection, conceptualization, and experimentation. Teachers who engage in experiential learning develop stronger pedagogical competencies and become more effective in addressing classroom challenges.

Continuous professional development further strengthens teachers' confidence, professional identity, and motivation. Through workshops, seminars, peer collaboration, and reflective practice,

teachers acquire updated knowledge and modern instructional methods that positively influence students' academic outcomes.

Modern Approaches in Teacher Education

Modern teacher education focuses on continuous learning, innovation, collaboration, and adaptability. Workshops and seminars are among the most common forms of professional development training. Workshops provide practical and interactive learning experiences that improve classroom practices, while seminars expose teachers to new educational trends, policies, and innovative teaching methods.

Technology integration is another important aspect of modern teacher education. Teachers are increasingly encouraged to use digital tools, online learning platforms, and interactive teaching methods to improve learner engagement and accessibility.

Sandwich and part-time programmes also support teacher development by allowing teachers to upgrade their qualifications while remaining in active service. These programmes expose teachers to modern pedagogical theories, improved subject mastery, and innovative instructional practices.

Collaborative learning and reflective teaching are equally important modern approaches in teacher education. Through peer mentoring, classroom observation, and professional learning communities, teachers exchange ideas and improve their teaching effectiveness.

Development of Pedagogical Skills and Methodological Competencies

Pedagogical skills refer to the teaching strategies, classroom management practices, and instructional methods teachers use to facilitate learning effectively. Professional development programmes improve teachers' communication skills, learner-centered instructional approaches, classroom management abilities, and assessment competencies.

Methodological competencies enable teachers to organize and implement appropriate teaching methods that address diverse learning needs. Professional development programmes expose teachers to differentiated instruction, inquiry-based learning, project-based learning, and effective lesson planning techniques.

Teachers who participate in professional development programmes are more likely to adopt innovative teaching methods that encourage active learning, critical thinking, creativity, and student participation. Consequently, students become more motivated, engaged, and academically successful.

Continuous Professional Development and Teacher Reflection

Continuous Professional Development (CPD) refers to ongoing learning activities that help teachers improve their professional knowledge, instructional practices, and career competencies throughout their teaching careers. CPD enhances instructional quality, teacher confidence, job satisfaction, and professional growth.

Reflective practice is an important component of teacher development. Reflective teachers continuously evaluate their classroom practices, teaching methods, and students' responses to instruction. Reflection helps teachers identify strengths and weaknesses, thereby improving teaching effectiveness.

Professional learning communities also support continuous professional development through peer collaboration, mentoring, lesson study, and classroom observation. These activities promote collective learning and professional improvement among teachers.

Evaluation and Quality Assurance of Teaching Performance

Evaluation and quality assurance are essential for maintaining effective teaching standards in schools. Teaching performance can be assessed through instructional delivery, classroom management, lesson preparation, student engagement, assessment practices, and students' academic achievement.

Professional development contributes significantly to quality assurance by improving teacher effectiveness and instructional standards. Teachers who participate in seminars, workshops, sandwich programmes, and part-time programmes demonstrate improved teaching quality, better assessment practices, and stronger learner engagement.

However, several challenges affect quality assurance in education, including inadequate funding, insufficient training opportunities, lack of instructional materials, and poor monitoring systems. Addressing these challenges requires continuous investment in teacher education and professional development.

Methodology

Research Design. The study adopted a descriptive survey research design to investigate the relationship between teachers' professional development training and its impact on students' academic performance in Yewa North, Ogun State, Nigeria, Sub Sahara, Africa.

Population and Sample. The population consisted of 355 public school teachers in Yewa North Local Government Area. A purposive sampling technique was used to select 52 English Language teachers from 35 schools.

Instrumentation. Data were collected using a self-constructed questionnaire and inventory. The reliability coefficient obtained through Cronbach Alpha was 0.78.

Method of Data Analysis. Mean and standard deviation were used to answer the research questions, while Analysis of Variance (ANOVA) was employed to test the hypotheses.

Findings and Discussion

The findings of the study revealed that teachers' professional development training has a significant relationship with school students' academic performance in Yewa North, Ogun State. The analysis showed that seminar and workshop training significantly related to students' academic performance with a combined mean score of 3.62 and a standard deviation of 0.54. Item 1, which focused on the implementation of skills acquired from seminars and workshops, recorded the highest mean score of 3.92 with a standard deviation of 0.27, indicating strong agreement among respondents.

The study further revealed that seminars and workshops exposed teachers to modern instructional methods such as learner-centered teaching, inquiry-based learning, collaborative learning, and technology integration. Teachers who attended professional development programmes were more confident in using innovative teaching techniques and adapting instruction to students' diverse learning needs. Research Question 3 further confirmed that seminar and workshop training positively influenced teachers' instructional performance with an overall mean score of 3.11 and a standard deviation of 0.81. Item 3, which examined creativity in teaching, recorded a mean score of 3.69, while the ability to address diverse learning needs recorded a mean score of 3.65.

Findings also showed that sandwich and part-time programmes contributed moderately to teachers' professional growth and students' academic performance. The result indicated an overall mean score of 2.54 with a standard deviation of 0.96 regarding students' academic performance. Respondents agreed that lecture hours often clashed with teaching schedules, with a mean score of 2.88, while teachers experienced difficulties balancing academic work and practical teaching

responsibilities with a mean score of 2.79. However, respondents disagreed that sandwich and part-time programmes negatively affected students' academic performance, as indicated by the low mean score of 1.98.

The findings on teachers' instructional performance further revealed that sandwich and part-time programmes had an average influence on teaching effectiveness with a mean score of 2.53 and standard deviation of 1.06. Teachers acknowledged the challenges associated with balancing professional responsibilities and academic activities, although many agreed that professional development programmes improved their exposure to current trends in teaching and learning.

The result of the one-way ANOVA analysis indicated that professional development programmes jointly contributed significantly to students' academic performance. The ANOVA result showed $F = 16.60$ and $p = 0.001$, which is lower than the 0.05 level of significance. Consequently, the null hypothesis was rejected, indicating that seminars, workshops, sandwich programmes, and part-time programmes significantly influenced students' academic performance.

The Tukey post-hoc test further revealed a significant mean difference between students taught by teachers who attended professional development training and those taught by teachers who did not participate in such programmes. This implies that at least one form of professional development training had a statistically significant influence on students' academic achievement.

The study also established that continuous professional development improved teachers' reflective practices. Teachers became more capable of evaluating their teaching methods, identifying classroom challenges, and implementing strategies for improvement. Reflective teaching enhanced classroom interaction, students' engagement, and the overall teaching-learning process.

Additionally, the findings revealed that professional development programmes strengthened quality assurance in teaching performance. Teachers who participated in training programmes demonstrated improved lesson preparation, effective classroom supervision, better assessment practices, and stronger commitment to professional standards.

Despite these positive outcomes, the study identified several challenges affecting teachers' participation in professional development programmes. These included inadequate funding, poor scheduling of seminars and workshops, transportation difficulties, lack of institutional support, and limited access to instructional resources. Some teachers also expressed concerns regarding the duration of training programmes and insufficient follow-up support after training sessions.

The findings therefore suggest that sustained investment in teacher education and professional development is essential for improving teaching effectiveness, strengthening pedagogical competencies, promoting reflective practice, and enhancing students' academic performance.

Conclusion

Teacher education and professional development are essential components of educational improvement and academic success. Modern approaches in teacher education, including workshops, seminars, digital learning, sandwich programmes, and reflective practice, contribute significantly to the development of pedagogical skills and methodological competencies.

Continuous professional development enhances teachers' professional confidence, instructional effectiveness, and adaptability to changing educational demands. Furthermore, evaluation and quality assurance mechanisms help maintain teaching standards and improve educational outcomes.

The study concludes that sustained investment in teacher professional development is necessary for improving teaching quality, enhancing students' academic performance, and achieving broader educational goals in secondary schools.

Recommendations

Based on the findings of the study, the following recommendations are made:

Government and educational stakeholders should organize regular and periodic workshops and seminars for teachers to enhance pedagogical competencies and instructional effectiveness.

Schools should encourage teachers to participate in continuous professional development programmes, including sandwich and part-time academic programmes.

Professional development programmes should incorporate modern technologies and digital learning strategies to improve teaching and learning.

Educational institutions should promote reflective practice and collaborative professional learning communities among teachers.

Quality assurance mechanisms should be strengthened through regular supervision, teacher appraisal, and instructional monitoring.

Adequate funding and resources should be provided to support sustainable teacher education and professional development initiatives.

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RESOLVING THE CONTRADICTION BETWEEN THE REQUIREMENTS OF MODERN DIGITAL INCLUSIVE PEDAGOGY AND DONALD SCHÖN'S CLASSICAL APPROACHES

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Abstract: Donald Schön's "reflective practitioner" model shows that professional experience is not based only on technical rationalism. Schön called the field of professional activity a "swampy zone" and showed that uncertain and complex situations cannot be solved with standard instructions. The main essence of the model is explained as "knowing-in-action" - instinctive knowledge that the educator may not be consciously aware of; "reflection-in-action" - thinking on one's feet during the work process; and "reflection-on-action" - retrospective analysis after the work is completed. Digital technologies - artificial intelligence, electronic portfolios, virtual discussions, video and VR - bring new opportunities to each of Schön's components. For example, artificial intelligence tools such as "ChatGPT, Gemini" act as a "digital critical friend" in real time, providing the opportunity to quickly correct lesson plans and offer alternative ideas. "E-portfolio" and "online virtual learning environments" (e.g., Moodle forums) support long-term reflection by systematically documenting and sharing teachers' experiences. "Video recordings of lessons" and virtual reality create the opportunity to re-enact classroom situations and discuss them in groups, creating conditions for deeper analysis. In inclusive education, these digital tools help the teacher develop methods adapted to individual needs. However, the main obstacles - lack of time, the teacher's internal biases and systemic difficulties - still remain. In this article, the main principles of Schön's model and its application in the digital environment are analyzed, the obtained results are presented in comparative tables, and practical recommendations are also given.

Keywords: Schön's reflective practice model, digital education technologies, ChatGPT teacher reflection, e-portfolio, virtual real teaching, inclusion

MÜASİR RƏQƏMSAL İNKLÜZİV PEDAQOGİKANIN TƏLƏBLƏRİ İLƏ DONALD ŞÖNUN KLASSİK YANAŞMALARI ARASINDAKI ZİDDİYYƏTİN ARADAN QALDIRILMASI

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Xülasə: Donald Schönün “refleksiv praktik” modeli peşəkar təcrübənin yalnız texniki rəşionalizmə s  y  n  diyini g  st  rir. Pe   kar f  aliyyət sah  sini    n “bataqlı   zonası” adlandırmı   v   qeyri-m   yy  n, m  r  kk  b situasiyaların standart t  limatlarla h  ll olunmadı  nı g  st  rmi  dir. Modelin   sas mahiyyət  i “f  aliyyətd   olan bilik”- pedaqoqun     rl   ola bilm  diyi, instinktiv bilik, “f  aliyyət zamanı refleksiya – i   prosesində aya  st   d    nm   v   “f  aliyyət haqqında refleksiya” – i   ba  a   atdı  dan sonra retrospektiv t  hlil kimi izah olunur. R  q  msal texnologiyalar -s  ni intellekt, elektron portfoliolar, virtual m  zakir  lər, video v   VR Sch  n  n komponentl  rinin h  r birin   yeni imkanlar g  tirir. M  s  l  n, “ChatGPT, Gemini” kimi s  ni intellekt al  tl  ri real vaxtda “r  q  msal t  nqidi dost” kimi f  aliyyət g  st  r  r  k d  rs planlarını s  r  tl   d  z  ltm  k v   alternativ fikirl  r t  klif etm  k imkanı verir. “E-portfello” v   “onlayn virtual   yr  nm   m  h  tl  ri” (m  s., Moodle forumları) is   m   lliml  rin t  cr  b  sini sistemli   kild   s  n  dl    dirib payla  araq uzunm  dd  tli refleksiyanı d  st  kl  yir. “D  rsler  n video yazısı” v   virtual reallı  q sinif v  ziyyətl  rini yenid  n canlandırmaya v   qrup halında m  zakir   etm  y   imkan yaradaraq d  rind  n analiz     n   rait yaradır.   nkl  ziv t  hsild   bu r  q  msal al  tl  r m   llim   f  rdi ehtiyaclarla uy  n metodlar hazırlamaqda k  m  k edir. Lakin   sas mane  lər – vaxt   atı  mazlı  ı, m   llimin daxili q  r  zl  ri v   sistemik     tinlikl  r – h  l   d   qalmaqdadır. Bu m  q  l  d   Sch  n  n modelinin   sas prinsipl  ri v   bu modelin r  q  msal m  h  td  ki t  tbiqu analiz olunur,   ld   olunan n  tic  lər m  qayis  li c  dv  ll  r g  st  rilir, h  m   inin praktiki t  vsiy  l  r verilir.

A  ar s  z  l  r:     n  n refleksiv t  cr  b   modelini, r  q  msal t  hsil texnologiyaları, ChatGPT   saslı m   llim refleksiyası, elektron portfolio, virtual v   real t  drisin integrasiyası, inkl  ziv t  hsil

Introduction

Donald Sch  n substantiated the concept of “reflective practice” by examining the transition from knowledge to practice in the labor market. He advised educators to use “reflective practice,” which is based on learning from experience, rather than relying solely on static knowledge when dealing with complex situations. According to Sch  n, every lesson is a unique situation; therefore, methods from the “high ground” give way to the metaphor of the “swampy zone.” In such contexts, when standard technical approaches are insufficient, the practitioner is required to rely on their own instinctive knowledge, that is, knowing-in-action.

Within Sch  n’s modeling framework (Sch  n, D. A., 1983):

- **“Knowing-in-action”** – the intuitive, deep knowledge that results from the teacher’s experience. For example, a teacher can immediately recognize a student’s difficulty through body language.
- **“Reflection-in-action”** – thinking on one’s feet when a problem arises and there is an opportunity to change the situation. Sch  n refers to this as “thinking on one’s feet.” When educators receive unexpected “feedback” (e.g., a student’s sharp reaction), they immediately reshape their reflection and make adaptive decisions.
- **“Reflection-on-action”** – retrospective analysis after the event has been completed. After the lesson, the teacher analyzes what happened, evaluates what was changed and which method worked, and prepares a plan for the future. According to Sch  n, the questions “why did this happen and what should I do next time?” arise precisely at this stage.

Materials and Methods

This study was conducted as a theoretical analysis of the application of Donald Sch  n’s “Reflective Practitioner Model” in the modern digital educational environment. Keywords such as “Sch  n’s Reflective Practitioner Model,” “digital education technologies,” “ChatGPT,” “e-portfolio,” and “virtual reality teaching,” as well as scientific databases like ERIC and Scopus, were used.

Primarily, preference was given to articles published between 1980 and 2026. The introductory information was based on Schön's main sources (international books and articles), and conference papers as well as scientific research sources were examined. As a method, a theoretical literature review was conducted, and references were made to official statistics and empirical research. No historical limitation was applied to the search period; the relevance of topics related to reflective practice and digital technologies was taken into account.

Results and Discussion

Summary and comparison of Schön's model. The main essence of Donald Schön's model is the critique of technical rationality. The differences between "reflection-in-action" and "reflection-on-action" in Schön's model are presented in the table below:

Feature	Reflection-in-action	Reflection-on-action
Timing	Occurs during the teaching process, within the event itself	Takes place after the teaching process, as retrospective analysis
Purpose	To adjust the current situation and ensure immediate adaptation	To learn from past experience and improve future practice
Character	Intuitive decision-making	Structured, analytical, supported by theory and methods
Outcome	Immediate adjustments and situational control	Professional development and improvement of methods and knowledge

This table corresponds to Schön's description: professionals manage situations and prepare for the future by sequentially applying both "reflection-in-action" and "reflection-on-action." Schön emphasized that knowing-in-action forms the foundation of reflection — that is, the educator's knowledge developed over many years becomes the source of immediate decision-making. However, this tacit knowledge can only be revealed and refined through reflection.

The role of digital technologies. Modern digital tools transform each component of Schön's framework. Artificial intelligence, especially models such as ChatGPT and Gemini, accelerates reflection. When preparing lesson plans, teachers obtain ideas through AI "prompts," immediately conduct pedagogical evaluation, and make adjustments. At the same time, after the lesson, they analyze the obtained results and apply them to future practice. In the literature, this is described as artificial intelligence acting as a "digital critical friend" for teachers. For example, Kenan Dikilitaş and T. Farrell (2026) evaluate ChatGPT as a reflective friend for language teachers. They note that AI tools enrich reflective practice by providing immediate ideas and perspectives; however, they also lack contextual and experiential understanding of human partners, which leaves certain limitations. Ethical issues such as bias, privacy, and the risk of dependency also emerge. For instance, ready-made answers may create "metacognitive laziness," meaning that they may replace the thinking process of both teachers and students. To prevent this, it is necessary to use prompts that encourage deeper thinking.

Electronic portfolios and online platforms. Teachers' and students' e-portfolios create conditions for the use of reflection. Selecting artifacts and documenting learning with brief comments is itself a form of reflection. E-portfolios foster critical thinking by linking experience with theory, research, and practice (Chye et al., 2019). Overall, research shows that e-portfolios enable students to develop judgment skills and become aware of their strengths and weaknesses. However, successful implementation in practice requires digital literacy and appropriate mentoring.

According to Schön, the e-portfolio serves for teachers as a “mediation tool for mentoring and collaboration” – that is, the interpretation and discussion of experience take place precisely within this process.

Video and VR (virtual reality) technologies. When lessons are recorded and later reviewed, video-based reflection helps teachers overcome memory limitations; by rewatching different moments of the lesson, observation skills are improved. For example, according to the analysis of the IRIS Connect platform, video provides teachers with an objective perspective, helps recall overlooked details, and facilitates discussion with colleagues. As a result, teachers gain information about aspects that may be missed during the lesson process, such as student reactions and classroom management. Virtual reality (VR) and video technologies create new opportunities for inclusive education. Research shows that in interactive VR environments, reflection becomes deeper when combined with commentary; teachers are able to evaluate lessons more objectively.

Digital Tools and Their Relation to Schön’s Model Components

Digital Tool	Correspondence to Schön’s Model Component	Advantages	Risks / Disadvantages
ChatGPT / GPT-4 etc.	/ Strengthens reflection-in-action	Immediate responses, broad perspectives, creative ideas	Lack of contextual understanding, possible inaccurate content, risk of cognitive slowing in teachers and students, ethical bias
Electronic Portfolios	Reflection-on-action	Collecting and analyzing lesson materials, promoting self-assessment	Requires time and technical skills; teacher motivation; poor design may reduce effectiveness
Online Forums / VLE (Moodle)	Reflection-on-action (collaborative)	Asynchronous discussions create professional networks on management; and facilitate experience sharing	Quality of discussion depends on teacher coordination; time-consuming
Video-based reflection	Reflection-on-action	Objective recording of lessons, opportunity for repeated review, flexibility in time and space	Privacy issues, fear of evaluation, technical difficulties, and time consumption
VR and 360° simulations	Mainly knowing-in-action + reflection	Allows experiencing diverse classroom situations, collaborative analysis, preparation for inclusive education	Expensive technology, requires significant resources; possible sensory discomfort and reduced realism perception

Digital Tools and Their Correspondence to Schön’s Model Components

In inclusive classrooms, each student has different learning styles and needs. Therefore, a single standardized lesson plan is not sufficient. Donald Schön described this situation as a “swampy zone.” In such cases, the teacher must think flexibly and make context-based decisions.

The digital reflective model (for example, using educational technologies) helps teachers think more effectively, analyze lessons, and adapt instruction to individual student needs. Thus, each student is different, the teacher must find an appropriate solution through reflection, and technology supports this process. However, several barriers exist, as outlined below:

- **Time and workload:** One of the most frequently reported barriers is lack of time. Teachers often cannot find sufficient time for reflective analysis after lessons. Reflection-in-action requires rapid decision-making, which does not always allow structured thinking.
- **Implicit bias (internal prejudice):** One of the main barriers in inclusive education is teachers' unconscious stereotypes. For example, a teacher may assume that students with disabilities will perform poorly, and such preconceptions must be addressed. Critical reflection is needed to identify and overcome these biases; otherwise, teachers may fail to adapt their instructional approach. Research shows that examining implicit biases increases self-awareness and reduces unfair assessment practices.
- **Psychological barriers:** Reflection requires teachers to acknowledge their weaknesses. In this context, the "impostor syndrome" (feelings of inadequacy, fear of exposure, anxiety, and perfectionism) may lead teachers to avoid admitting mistakes. Researchers who critique Donald Schön's model argue that when reflective writings are used for evaluation or promotion, teachers may not express genuine reflection but instead attempt to present themselves in a positive light. In other words, rather than openly acknowledging mistakes, teachers may present their practice as ideal. As a result, authentic and honest reflection may not occur.
- **Systemic and institutional challenges:** Even if teachers are highly reflective, school infrastructure and leadership issues can limit inclusivity. For example, technological tools, equipment for e-portfolios, or spaces for VR may be insufficient. If school leadership does not support the implementation of new methods, reflective practice cannot function effectively. Jaeger (2013) notes that time constraints and lack of critical self-reflection are among the main barriers to effective reflection.

Analysis and Recommendations

Although digital technologies bring new opportunities to inclusive education, they also increase certain risks. For example, the widespread use of artificial intelligence may lead to "metacognitive sluggishness" among teachers and discourage critical thinking; repeated prompts may increase dependency risks (Dikilitaş & Farrell, 2026). In addition, while Donald Schön's model primarily focuses on individual reflection, collective reflection among teachers is also highly important (Arefian et al., 2024).

To develop reflective practice, several recommendations can be made for teachers:

- Reflection time should be allocated, and opportunities for collective reflection should be created by implementing project-based learning or team teaching. At the same time, rather than limiting the use of artificial intelligence, it is more effective to design purposeful reflective situations. Psychological training should be applied to identify and eliminate internal biases (Carlana et al., 2024), and self-assessment skills should be improved through the use of video-based reflection and e-portfolios.
- In teacher education programs, a bridge should be built between theoretical knowledge and practical examples in order to develop "digital reflection" skills. For instance, teachers should be trained to use ChatGPT critically, VR-based training should be conducted, and e-

portfolio projects should be implemented (Blannin, 2025). Reflective practice should be used not for assessment purposes but for developmental purposes.

- In schools, workload should be optimized to provide more time for reflection. Infrastructure should be developed (e.g., internet access, technology, and software provision). “Reflection units” should be established within the education system – for example, mentor-teacher networks and critical reflection groups should be created to expand reflective opportunities. Ethical standards for the use of technologies such as artificial intelligence and VR should also be implemented. These measures will improve teachers’ technical competence and reflective resilience, and increase their flexibility in inclusive education.

In conclusion, the modern digital environment both expands and reinterprets Donald Schön’s reflective practitioner model. Digital tools accelerate and make reflective processes more collaborative; however, the main barriers lie in the teacher and in organizational conditions. Teachers must be prepared both technologically and ethically, while administrators should create conditions that support reflection. In this way, the ideal of the “reflective practitioner” can be achieved – a professional who not only implements lesson plans but also learns from and develops through every situation.

Conclusion

In inclusive classrooms, artificial intelligence tools help teachers use tacit and less explicit components more effectively and facilitate both reflection-in-action and reflection-on-action. This enables Donald Schön’s theoretical ideas to be transformed into practical tools for working with diversity in inclusive education.

When using digital tools, however, there is a risk that deep reflection may turn into a mechanical and automated process. In this context, researchers emphasize reflection as a “slow and surprising” process.

Donald Schön noted that the reflective practitioner model is based on iterative reflection. Modern digital technologies – e-portfolios, ChatGPT, video, Virtual Learning Environments, Virtual Reality, and others – make reflection broader and more systematic. Virtual discussions expand the classroom environment and enable peer interaction, while video and virtual simulations allow revisiting different stages of lessons, thereby deepening learning. The use of artificial intelligence in lesson planning supports real-time reflection, while e-portfolios provide opportunities for long-term thinking. These technologies make teacher reflection more transparent and collectively more adaptive.

However, the development of digital competence and time-related challenges remain significant issues. The formation of reflective practice, adaptation to technology, and overcoming internal contradictions all require time. These barriers must be considered in school contexts. In addition, the ethical use of artificial intelligence and preparing teachers for reflective engagement with AI remain important requirements.

The results of the study indicate that in the modern digital environment, effective reflective practice has become an integral part of pedagogy and is essential for the realization of student-centered inclusive education. Future research may explore the evaluation of this integration, the development of new methods for using technological tools, and issues of alignment in teacher education programs.

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CHALLENGES AND SUPPORT MECHANISMS IN YOUNG TEACHERS' PROFESSIONAL ADAPTATION

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Abstract: The transition from pre-service education to professional teaching is often a complex and demanding process for young teachers. Entering real classroom environments requires not only the application of theoretical knowledge but also the development of practical skills, emotional resilience, and professional identity. This study examines the key difficulties faced by novice teachers during their professional adaptation and explores effective support mechanisms that can facilitate this process. Drawing on existing literature, the article identifies major issues such as lack of practical experience, classroom management difficulties, psychological stress, workload pressure, and insufficient institutional support. In addition, challenges related to communication with students, parents, and colleagues are also considered.

The study further highlights the importance of structured support systems, including mentoring programs, continuous professional development, and collaborative school environments, in easing the adaptation process. Special attention is given to the role of school leadership and organizational culture in creating supportive conditions for early-career teachers. The findings emphasize that well-designed induction programs and ongoing support significantly improve teachers' confidence, instructional effectiveness, job satisfaction, and long-term commitment to the profession. Ultimately, strengthening these support mechanisms contributes not only to the success of young teachers but also to the overall quality and sustainability of education systems.

Keywords: Young teachers, professional adaptation, mentoring, teacher induction, professional identity, classroom management, teacher stress, professional development.

CHALLENGES AND SUPPORT MECHANISMS IN YOUNG TEACHERS' PROFESSIONAL ADAPTATION

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Xülasə: Tədrisə hazırlıq mərhələsindən peşəkar müəllimliyə keçid gənc müəllimlər üçün çox vaxt mürəkkəb və çətin bir proses hesab olunur. Real sinif mühitinə daxil olmaq yalnız nəzəri biliklərin tətbiqini deyil, həm də praktiki bacarıqların, emosional dayanıqlılığın və peşəkar kimliyin inkişafını tələb edir. Bu tədqiqat gənc müəllimlərin peşəkar adaptasiya prosesində qarşılaşdıqları əsas çətinlikləri araşdırır və bu prosesi asanlaşdıran effektiv dəstək mexanizmlərini təhlil edir. Mövcud ədəbiyyata əsaslanaraq, məqalədə əsas problemlər kimi praktiki təcrübənin azlığı, sinif idarəetməsi çətinlikləri, psixoloji stress, iş yükünün çoxluğu və institusional dəstəyin kifayət qədər olmaması müəyyən edilir. Bundan əlavə, şagirdlər, valideynlər və həmkarlarla ünsiyyətlə bağlı çətinliklər də nəzərə alınır.

Tədqiqat həmçinin mentorluq proqramları, davamlı peşəkar inkişaf və əməkdaşlığa əsaslanan məktəb mühitləri kimi strukturlaşdırılmış dəstək sistemlərinin adaptasiya prosesini asanlaşdırdığını vurğulayır. Gənc

müəllimlər üçün əlverişli şəraitin yaradılmasında məktəb rəhbərliyinin və təşkilati mədəniyyətin rolu xüsusi diqqət mərkəzindədir. Nəticələr göstərir ki, yaxşı hazırlanmış induksiya proqramları və davamlı dəstək müəllimlərin özünəinamını, tədris effektivliyini, iş məmnunluğunu və peşəyə uzunmüddətli bağlılığını əhəmiyyətli dərəcədə artırır. Nəticə etibarilə, bu dəstək mexanizmlərinin gücləndirilməsi yalnız gənc müəllimlərin uğuruna deyil, həm də təhsil sistemlərinin ümumi keyfiyyətinə və dayanıqlılığına töhfə verir.

Açar sözlər: Gənc müəllimlər, peşəkar adaptasiya, mentorluq, müəllim induksiyası, peşəkar kimlik, sinif idarəetməsi, müəllim stressi, peşəkar inkişaf.

Introduction

The quality of education largely depends on the competence, preparedness, and adaptability of teachers, as they play a central role in shaping students' learning experiences and outcomes. In contemporary education systems, teachers are expected not only to deliver subject knowledge but also to foster critical thinking, creativity, collaboration, and social responsibility. As a result, the teaching profession has become increasingly complex and demanding. However, young teachers entering the profession often encounter a significant discrepancy between the theoretical knowledge acquired during pre-service education and the practical demands of real classroom environments. This gap can make the initial years of teaching particularly challenging and, in some cases, overwhelming.

The period of professional adaptation is therefore a crucial stage in a teacher's career, as it directly influences the formation of professional identity, teaching style, and long-term effectiveness. During this phase, novice teachers must simultaneously manage instructional responsibilities, classroom discipline, assessment processes, administrative tasks, and communication with students, parents, and colleagues. Balancing these diverse responsibilities requires not only pedagogical knowledge but also emotional resilience, time management, and problem-solving skills. Consequently, many beginning teachers experience feelings of uncertainty, self-doubt, and professional insecurity.

Without adequate institutional and emotional support, these challenges can lead to high levels of stress, reduced motivation, and job dissatisfaction. In some cases, young teachers may consider leaving the profession within the first few years, which negatively affects teacher retention rates and the overall stability of the education system. This issue is particularly concerning given the growing demand for qualified and committed educators in many countries.

Moreover, the rapidly changing nature of education, driven by technological advancements, globalization, and evolving societal expectations, further complicates the adaptation process. Teachers are increasingly required to integrate digital technologies into their teaching, implement student-centered and inclusive approaches, and address the diverse academic and socio-emotional needs of learners. They must also engage in continuous learning to keep up with new methodologies and educational innovations. For novice teachers who are still developing their professional competencies, these expectations can be especially demanding.

Another important aspect of professional adaptation is the socialization process within the school environment. Young teachers must learn to navigate institutional norms, collaborate with colleagues, and establish their role within the school community. The presence or absence of a supportive school culture plays a significant role in shaping their experiences. Schools that encourage collaboration, mentorship, and open communication tend to facilitate smoother transitions for beginning teachers.

Therefore, identifying both the challenges faced by young teachers and the support mechanisms that facilitate their successful adaptation is essential for strengthening the education system. Effective induction programs, mentorship opportunities, and continuous professional development have been shown to significantly improve teacher retention, instructional quality, and job satisfaction (Ingersoll & Strong, 2011). Understanding and addressing these factors will contribute to the development of more resilient, confident, and competent educators, ultimately enhancing the overall quality and sustainability of education.

Materials and Methods

This study employs a qualitative research approach based on a comprehensive review and analysis of academic literature related to teacher education and early career professional development. The primary data sources include peer-reviewed journal articles, scholarly books, and reports published by international organizations. These sources were selected according to their relevance, credibility, and contribution to understanding the challenges and support mechanisms associated with young teachers' professional adaptation.

The selection process involved identifying key publications that focus on novice teachers, induction programs, classroom challenges, and institutional support systems. Priority was given to both recent studies and seminal works that have significantly influenced the field. To ensure consistency and academic rigor, only sources published in reputable journals or by recognized institutions were included. The collected materials were systematically reviewed and categorized based on recurring themes such as practical experience, classroom management, psychological factors, communication skills, and professional support structures.

In terms of methodology, descriptive analysis was used to summarize and interpret the main findings of the selected sources, providing a clear overview of existing knowledge on the topic. At the same time, comparative analysis enabled the identification of similarities and differences across various studies, educational systems, and cultural contexts. This approach allowed for a more nuanced understanding of how challenges and support mechanisms may vary depending on specific conditions.

Furthermore, thematic analysis was applied to organize the data into coherent and meaningful categories. This process involved coding the reviewed materials, identifying key patterns, and grouping them into broader themes related to both challenges and support strategies. Through this method, the study highlights the relationships between different factors influencing professional adaptation.

Although the research does not include primary data collection such as surveys or interviews, the use of diverse, high-quality secondary sources enhances the validity and reliability of the findings. This methodological approach provides a solid foundation for analyzing the professional adaptation of young teachers and for drawing well-informed conclusions about effective support mechanisms.

Results and Discussion

The analysis reveals that young teachers face several interconnected challenges that significantly influence their professional adaptation and effectiveness in the classroom. One of the most prominent issues is the lack of practical experience, as many novice teachers struggle to bridge the gap between theoretical knowledge acquired during their studies and the realities of classroom practice (Veenman, 1984). This often results in difficulties with lesson planning, instructional delivery, assessment strategies, and responding to diverse student needs. In many cases, beginning

teachers find it challenging to differentiate instruction or apply student-centered approaches effectively, especially in heterogeneous classrooms.

Another major challenge is related to classroom management. Maintaining discipline, ensuring student engagement, and creating a positive learning environment require not only pedagogical knowledge but also situational awareness and emotional intelligence. For many beginning teachers, managing disruptive behavior, maintaining authority, and balancing friendliness with professionalism become sources of stress and uncertainty. Ineffective classroom management can further reduce instructional time and negatively impact learning outcomes.

In addition, psychological pressure is a significant factor affecting young teachers. Heavy workloads, high expectations from school administrations and parents, and the desire to perform well often lead to stress, anxiety, and even burnout (Kyriacou, 2001). The pressure to meet curriculum standards, complete administrative tasks, and adapt to institutional norms can overwhelm novice teachers. This emotional strain not only affects their well-being but may also reduce their motivation and commitment to the profession over time.

The issue of limited institutional support further complicates the adaptation process. In many cases, novice teachers do not receive sufficient mentoring or structured induction programs, which are essential for their professional growth (OECD, 2019). The absence of systematic guidance can lead to feelings of isolation, uncertainty, and professional insecurity. Furthermore, schools that lack a supportive organizational culture may unintentionally increase the difficulties faced by young teachers.

Moreover, communication difficulties represent another barrier. Establishing effective relationships with students, parents, and colleagues requires strong interpersonal and conflict-resolution skills. Early-career teachers may struggle with giving constructive feedback, handling parental expectations, or collaborating effectively within teaching teams. These communication gaps can hinder both instructional success and professional integration.

To address these challenges, several important support mechanisms have been identified. Mentorship programs play a crucial role by providing young teachers with guidance, feedback, and practical advice from experienced colleagues, thereby enhancing their confidence and professional competence (Ingersoll & Strong, 2011). Regular mentoring sessions, classroom observations, and reflective discussions can significantly accelerate professional growth.

Continuous professional development is equally important, as it allows teachers to update their knowledge, refine their teaching strategies, and adapt to changing educational demands. Training in areas such as classroom management, inclusive education, and the use of educational technologies can help novice teachers respond more effectively to modern classroom challenges.

A collaborative work environment also contributes significantly to successful adaptation. Opportunities for teamwork, peer observation, and shared problem-solving foster both professional learning and emotional support among teachers. Professional learning communities, in particular, create spaces where teachers can exchange experiences, discuss challenges, and develop collective solutions.

Furthermore, administrative support is essential in creating a positive and encouraging school climate. School leaders who provide clear expectations, constructive feedback, recognition, and access to teaching resources can greatly facilitate the adaptation process. Supportive leadership also helps reduce stress and builds a sense of belonging among young teachers.

It is also important to highlight the growing role of technology and digital competence in teachers' adaptation. In modern educational settings, teachers are expected to integrate digital tools into their teaching practices. For novice teachers, this can be both an opportunity and a challenge. While many young teachers are familiar with technology, effectively using it for pedagogical purposes requires additional training and institutional support.

Overall, the discussion demonstrates that these challenges and support mechanisms are closely interconnected and mutually influential. Addressing one area, such as mentoring or professional development, can positively impact other aspects of adaptation, including confidence, classroom management, and communication skills. Effective support systems not only help reduce the difficulties faced by young teachers but also enhance their professional confidence, improve instructional quality, and increase long-term commitment to the teaching profession. Consequently, investing in comprehensive and well-structured support strategies is vital for ensuring teacher retention, professional satisfaction, and the overall improvement of educational outcomes.

Conclusion

Young teachers' professional adaptation is influenced by a complex interaction of both internal and external factors, including individual competencies, emotional resilience, institutional conditions, and the broader educational environment. While challenges such as limited practical experience, classroom management difficulties, and psychological stress are inevitable during the early stages of a teaching career, they do not have to hinder long-term professional success. With appropriate support, these challenges can be transformed into opportunities for growth and development.

The findings of this study highlight the crucial role of structured support systems in facilitating successful adaptation. Mentorship programs, continuous professional development, and collaborative school cultures not only help novice teachers build confidence and improve their teaching practices but also contribute to their overall well-being and job satisfaction. In addition, effective school leadership and administrative support are essential in creating an environment where young teachers feel valued, guided, and motivated.

Furthermore, investing in the professional adaptation of young teachers has long-term benefits for the entire education system. Teachers who receive adequate support in their early careers are more likely to remain in the profession, demonstrate higher levels of effectiveness, and positively impact student outcomes. Therefore, educational institutions and policymakers should prioritize the development and implementation of comprehensive induction and support programs.

In conclusion, strengthening support mechanisms for young teachers is not only a matter of individual professional success but also a strategic priority for improving the quality and sustainability of education. By fostering a supportive and inclusive professional environment, education systems can ensure the development of competent, resilient, and committed teachers.

Recommendations

Based on the findings of this study, several practical recommendations can be proposed to facilitate the professional adaptation of young teachers and enhance their long-term effectiveness and well-being.

First, it is essential for educational institutions to implement structured induction and mentoring programs for novice teachers. Such programs should include regular meetings with experienced mentors, classroom observations, and constructive feedback sessions. Mentorship not

only supports the development of pedagogical skills but also provides emotional guidance, helping young teachers navigate the complexities of the profession more confidently.

Second, schools should consider reducing the initial workload of beginning teachers. Assigning fewer teaching hours, limiting administrative responsibilities, and providing additional preparation time during the first year can significantly reduce stress and allow novice teachers to focus on improving their instructional practices. This gradual transition into full professional responsibilities can enhance both performance and job satisfaction.

Another important recommendation is the provision of continuous professional development (CPD) opportunities tailored to the specific needs of early-career teachers. Training programs should focus on key areas such as classroom management, inclusive education, assessment strategies, and the effective integration of digital technologies. Regular workshops, seminars, and online courses can help teachers stay updated with modern educational practices and build confidence in their abilities.

Furthermore, fostering a collaborative and supportive school culture is crucial. Schools should encourage teamwork, peer observation, and professional learning communities where teachers can share experiences, discuss challenges, and develop collective solutions. Such environments promote mutual support, reduce feelings of isolation, and contribute to professional growth.

School leadership also plays a vital role in supporting young teachers. Administrators should provide clear expectations, constructive feedback, and recognition of teachers' efforts. Creating an open and communicative environment where novice teachers feel valued and heard can significantly improve their motivation and sense of belonging.

In addition, it is recommended to incorporate psychological and emotional support mechanisms within schools. Access to counseling services, stress management training, and well-being programs can help teachers cope with the emotional demands of the profession and prevent burnout, particularly during the initial years.

At the policy level, educational authorities should prioritize the development of national frameworks for teacher induction and support. Establishing standardized guidelines for mentoring, professional development, and evaluation can ensure consistency and quality across different schools and regions.

Finally, integrating technology training and digital competence development into support programs is increasingly important in modern education. Providing teachers with access to digital tools, resources, and training can enhance their ability to deliver engaging and effective instruction in both traditional and online learning environments.

In conclusion, a comprehensive and multi-level approach that combines institutional support, professional development, and emotional well-being initiatives is essential for ensuring the successful adaptation of young teachers. Implementing these recommendations can lead to increased teacher retention, improved instructional quality, and a more resilient and effective education system.

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CONTINUING PROFESSIONAL DEVELOPMENT AND REFLECTIVE PRACTICE THEORETICAL AND PRACTICAL PERSPECTIVES

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Abstract: Today's educational landscape is being reshaped by rapid digital change and the growing influence of artificial intelligence, which together make it necessary to rethink how teachers are trained and supported throughout their careers. This paper sets out to examine the conceptual foundations of Continuing Professional Development (CPD) and reflective practice, and to clarify the part they play in shaping a capable, self-aware teacher. The study draws on a systematic review of relevant literature published between 2021 and 2024, supplemented by comparative analysis of teacher education models from different national contexts. What emerges clearly from the evidence is that a teacher's professional growth cannot be driven by external training alone — it requires the activation of genuine internal reflection. Teachers who regularly examine their own practice tend to adapt more readily to new challenges and make more deliberate instructional decisions. Against this backdrop, the paper argues for the integration of digital literacy with reflective thinking as a core component of contemporary teacher preparation programmes.

Furthermore, the findings highlight that structured mentoring, collaborative learning communities, and continuous feedback mechanisms significantly strengthen the impact of reflective practice, enabling teachers to translate theoretical insights into practical classroom strategies while maintaining adaptability, professional autonomy, and a sustained commitment to lifelong learning within increasingly complex and technology-rich educational environments.

Keywords: continuing professional development, reflective practice, teacher education, digital literacy, artificial intelligence, pedagogical competence, lifelong learning

DAVAMLI PEŞƏKAR İNKİŞAF VƏ REFLEKTİV TƏCRÜBƏ: NƏZƏRİ VƏ PRAKTİK PERSPEKTİVLƏR

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Xülasə: Müasir təhsil mühiti sürətli rəqəmsal dəyişikliklər və süni intellektin artan təsiri ilə yenidən formalaşır; bu da müəllimlərin karyeraları boyunca necə hazırlandığını və dəstəkləndiyi barədə düşüncənin yenidən nəzərdən keçirilməsini zəruri edir. Bu məqalə davamlı peşəkar inkişafın (CPD) və reflektiv təcrübənin konseptual əsaslarını araşdırmağı, onların bacarıqlı, özünüdərk edən müəllimin formalaşmasındakı rolunu aydınlaşdırmağı məqsəd qoyur. Tədqiqat 2021–2024-cü illəri əhatə edən ədəbiyyatın sistemli icmalına və müxtəlif milli kontekstlərdən müəllim hazırlığı modellərinin müqayisəli analizinə əsaslanır. Sübutlardan aydın olur ki, müəllimin peşəkar inkişafı yalnız xarici təlimlə həyata keçirilə bilməz — bu, həqiqi daxili refleksiyanın işə salınmasını tələb edir. Öz təcrübəsini mütəmadi olaraq araşdıran müəllimlər yeni çağırışlara daha tez uyğunlaşır.

Bundan əlavə, nəticələr göstərir ki, strukturlaşdırılmış mentorluq, əməkdaşlığa əsaslanan öyrənmə icmaları və davamlı geribildirim mexanizmləri reflektiv təcrübənin təsirini əhəmiyyətli dərəcədə gücləndirir, müəllimlərə nəzəri bilikləri praktik fəaliyyətə çevirməyə, eyni zamanda dəyişən və texnologiya ilə zəngin təhsil mühitində peşəkar müstəqilliklərini qoruyaraq davamlı inkişaf etməyə imkan yaradır.

Açar sözlər: davamlı peşəkar inkişaf, reflektiv təcrübə, müəllim hazırlığı, rəqəmsal savadlılıq, süni intellekt, pedaqoji kompetensiya, ömürboyu öyrənmə

Introduction

Teaching has never been a static profession, but the pace of change teachers now face is arguably without precedent. Across the globe, classrooms are being transformed by digital tools, shifting student demographics, revised curriculum frameworks, and growing expectations from policymakers and communities alike. In this environment, the question of how teachers sustain and develop their professional knowledge is not merely academic — it has direct consequences for the quality of learning that students experience every day.

Continuing Professional Development (CPD) has long been recognized as a mechanism for keeping teachers current and capable. Yet research has consistently shown that many conventional CPD formats — one-off workshops, mandatory training days, top-down seminars — have limited lasting impact on classroom practice (Darling-Hammond et al., 2021). Teachers may leave such sessions with new information but without meaningful changes in how they actually teach. This gap between knowing and doing is one of the central problems that the field of teacher education continues to wrestle with.

Reflective practice offers a different kind of answer to this problem. Rather than treating professional development as something that happens to teachers, reflection positions teachers themselves as the primary agents of their own growth. When educators take time to examine their decisions, question their assumptions, and consider the effects of their actions on learners, they

engage in a form of learning that is deeply personal and highly transferable (Kim & Roth, 2024). It is this combination — structured external support alongside genuine internal inquiry that the present study argues is most likely to produce lasting professional change.

The arrival of artificial intelligence in educational settings adds a further layer of complexity. Teachers are now expected not only to understand new technologies but to evaluate them critically, integrate them purposefully, and model digital citizenship for their students. This demands a kind of professional agility that cannot be achieved through technical training alone. It requires the reflective habits of mind that allow teachers to ask not just "how do I use this tool?" but "why, when, and with what effect?" (Trust & Whalen, 2021).

The present paper addresses these intersecting concerns by reviewing current theoretical frameworks for CPD and reflective practice, examining how they relate to digital transformation in education, and drawing out implications for teacher preparation. The goal is not to survey the field exhaustively but to identify the principles that most clearly distinguish effective professional development from its less useful counterparts.

Material and Methods

This study is grounded in qualitative documentary research and systematic literature analysis. The primary dataset consists of peer-reviewed journal articles, institutional reports, and policy documents published between 2021 and 2024. Sources were drawn from major international organisations — including the OECD, UNESCO, the World Bank, and the European Commission — as well as high-impact academic journals in education and applied linguistics.

The selection process followed PRISMA-inspired principles: sources were included only if they were published in indexed venues, offered transparent methodological reporting, and addressed CPD, reflective practice, or digital transformation in teacher education directly. Grey literature was considered when produced by established international bodies and subjected to peer or expert review.

Three analytical methods were applied in combination. First, a systematic literature review was conducted to map existing knowledge on CPD and reflection across different educational contexts. Second, a comparative analysis of teacher education models from the United Kingdom, Finland, Australia, and Azerbaijan was carried out to identify structural similarities and differences. Third, established reflective practice frameworks including those associated with Schön, Kolb, and Brookfield were applied as conceptual lenses through which the empirical literature was interpreted.

This combination of methods allowed the study to move beyond mere description and towards analytical conclusions about which features of professional development are most consistently associated with lasting change in teacher behaviour and student outcomes.

In addition, particular attention was given to the triangulation of sources in order to enhance the credibility and depth of the analysis. By comparing findings across policy documents, empirical studies, and theoretical frameworks, the study ensured a more balanced and comprehensive interpretation of the data. Coding procedures were applied to identify recurring themes such as teacher agency, collaboration, and digital competence. These themes were then synthesised into broader analytical categories. Limitations were also acknowledged, including the potential bias toward English-language publications and the variability in national education systems, which may influence the generalisability of the findings across different contexts.

Results and Discussion

The literature reviewed in this study points to several recurring findings that hold across national contexts and educational levels. The first and perhaps most consistent is that professional development which is disconnected from teachers' daily practice tends to produce minimal change. According to the OECD's TALIS 2022 report, more than 60% of teachers who participated in formal CPD activities felt that those activities had little bearing on what they actually did in their classrooms (OECD, 2023). This finding aligns with earlier research and suggests that the problem is not teachers' willingness to learn but rather the design of the programmes offered to them.

Effective CPD, by contrast, shares a number of identifiable features. It tends to be sustained over time rather than delivered in a single session. It focuses on subject-specific pedagogy rather than generic skills. It involves collaboration with colleagues, including opportunities to observe one another's teaching and to discuss what was observed. And it is tied to evidence about student learning, so that teachers can see the direct consequences of their professional choices (Darling-Hammond et al., 2021; Timperley, 2021). These features do not emerge spontaneously — they must be deliberately built into the structures that schools and institutions create for professional learning.

Reflective practice connects to these features in important ways. Reflection is not simply thinking about one's work after the fact; it is a structured process of examining experience in order to generate new understanding and guide future action (Kim & Roth, 2024). When teachers reflect systematically — whether through journaling, peer discussion, video review, or formal action research — they develop a more nuanced awareness of their own pedagogical patterns. They notice things they would otherwise overlook: a question they habitually avoid, a group of students they inadvertently neglect, a teaching strategy that works in one context but not another.

The relationship between CPD and digital transformation deserves particular attention given the current moment in education. The European Commission's DigCompEdu framework, developed by Redecker (2022), organises teacher digital competence across six domains: professional engagement, digital resources, teaching and learning, assessment, empowering learners, and facilitating learners' digital competence. What is notable about this framework is that it does not treat technology as a separate domain of teacher knowledge — it integrates digital competence into every aspect of professional practice. This integration reflects a broader shift in how researchers and policymakers understand the relationship between technology and teaching.

Schmid and Petko (2023) found in their review of digital transformation in teacher education that the most significant barrier to effective technology use is not access to devices or connectivity, but rather the absence of pedagogical frameworks that help teachers decide when and how technology adds genuine value to learning. Teachers who lack these frameworks tend either to avoid technology or to use it superficially — replicating traditional tasks in digital form without taking advantage of what technology uniquely enables. Reflective practice appears to be one of the mechanisms that helps teachers move beyond this surface-level engagement, because it encourages them to interrogate the reasoning behind their instructional choices rather than simply reproducing familiar habits.

The role of professional learning communities (PLCs) also emerged as significant in the literature. Fullan (2022) argues that sustainable educational improvement is fundamentally a collective achievement — it cannot be reduced to the efforts of individual teachers working in isolation. When teachers participate in structured communities where they share practice, examine student work together, and hold one another accountable for professional growth, the quality of their reflection tends to improve and the impact of their development extends beyond the individual

classroom. Timperley (2021) reports that schools with active, well-supported PLCs show measurably better student outcomes over time, though she cautions that the quality of collaboration matters more than its mere existence.

Artificial intelligence introduces both new opportunities and new risks for teacher professional development. On the opportunity side, AI-powered tools can provide teachers with detailed, real-time feedback on their practice — analysing classroom discourse, identifying patterns in student responses, and suggesting differentiated approaches based on learning data. Trust and Whalen (2021) note that teachers who engage thoughtfully with AI tools, rather than treating them as black boxes, tend to develop richer understandings of their own pedagogical decision-making. On the risk side, AI systems can reinforce existing inequalities, produce misleading outputs, or be used in ways that reduce rather than expand student agency. Navigating these risks requires precisely the kind of critical, reflective orientation that effective CPD seeks to cultivate.

Action research — in which teachers systematically investigate questions arising from their own practice — represents perhaps the most integrative form of professional development available. Avalos (2021) observes that teachers who conduct action research not only deepen their own reflective competence but also contribute to the wider body of educational knowledge. This dual benefit positions action research as a particularly valuable component of CPD programmes, especially in contexts where the gap between research and practice is wide.

CONCLUSION

This paper has examined the relationship between Continuing Professional Development and reflective practice in the context of contemporary teacher education, with particular attention to the challenges and opportunities created by digital transformation and artificial intelligence. Several conclusions emerge from the analysis.

First, professional development that is embedded in teachers' daily work and tied to evidence of student learning is consistently more effective than isolated training events. Second, reflective practice is not a supplementary element of teacher development — it is the mechanism through which external learning is internalised, tested, and refined. Third, digital competence cannot be developed through technical training alone; it requires pedagogical frameworks and reflective habits that help teachers evaluate technology critically and deploy it purposefully. Fourth, collaboration — through professional learning communities, peer observation, and shared action research — amplifies the effects of individual reflection and creates the conditions for sustainable school improvement.

For teacher preparation programmes in Azerbaijan and similar contexts, these findings suggest the importance of building reflective practice into initial teacher education from the outset, rather than treating it as an add-on for experienced practitioners. They also suggest the value of creating institutional structures — time, space, and professional expectation — that allow teachers to learn together over time rather than in isolation.

Future research might usefully focus on longitudinal studies tracking the impact of CPD interventions on both teacher practice and student outcomes over extended periods. There is also a need for more context-sensitive research on how reflective practice frameworks developed in Western educational traditions translate — or require adaptation — in different cultural and institutional settings, including those of the South Caucasus region.

In addition, attention should be given to the role of leadership in fostering a culture that values ongoing professional learning and reflective dialogue. School leaders who model inquiry, encourage

experimentation, and support risk-taking contribute significantly to teachers' willingness to engage in reflective practice. Equally important is the alignment of assessment and accountability systems with the principles of continuous improvement, so that teachers feel encouraged rather than constrained. Integrating local educational priorities with global pedagogical trends can further ensure that professional development remains both relevant and responsive to the evolving needs of diverse learning communities.

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EQUIVALENCE AS A FUNDAMENTAL CONCEPT IN TRANSLATION STUDIES

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Abstract: The article explores the concept of equivalence, which is one of the types of translation. The main purpose of the study is to determine the role of equivalence during translation and to investigate its importance. This study also discusses the existence of semantic, functional and structural aspects between the original text in the source language and the translated text in the target language.

The theoretical ideas of translation scholars such as E. Nida, V. Komissarov, P. Newmark are analyzed in the article.

The author posits that equivalence is a concept of particular importance for establishing a connection between two texts. It should be analyzed as a fundamental problem of language and linguistics. The main task during translation is to interpret the given message and present it as it is in another language. Using the resources of languages, the entire message can be translated as it is.

Furthermore, the article provides some practical examples to demonstrate the advantages of equivalent translation. The study focuses on inherent difficulties while preserving the spirit of the original. It especially emphasizes the necessity for translators to possess a deep understanding of the cultural context and linguistics features of both languages. The research methodology emphasizes the need for translators to better understand the cultural context and linguistic features of both languages. The research methodology mainly relies on contextual and comparative analysis.

The article mainly focuses on contextual analysis.

Key words: translation, equivalence, dynamic equivalence, target language, source

TƏRCÜMƏDƏ EKVİVALENTLİK

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Xülasə: Məqalədə tərcümə və tərcümənin keyfiyyət göstəricilərindən olan ekvivalentlik haqqında məlumat verilir. Tədqiqatın əsas məqsədi tərcümə zamanı ekvivalentliyin rolunu müəyyən etmək, onun əhəmiyyətini araşdırmaqdır. Bu tədqiqatda eyni zamanda mənbə dildəki orijinal mətn ilə hədəf dildəki tərcümə mətni arasındakı semantik, funksional və struktur aspektlərin olması haqqında yazılır.

E.Nayda, V.Komissarov, P.Nyumark kimi tərcüməşünasların nəzəri fikirləri məqalədə təhlil edilir.

Mövzu ilə bağlı müəllifin mövqeyi belədir ki, ekvivalentlik iki mətn arasında əlaqə qurmaq üçün xüsusi əhəmiyyətə malik bir anlayışdır. O, dilin və dilçiliyin əsas problemi kimi təhlil edilməlidir. Tərcümə zamanı əsas məsələ verilmiş mesajı şərh etmək və onu digər dilə olduğu kimi təqdim etməkdir. Dillərin resurslarından istifadə etməklə bütün mesajı olduğu kimi tərcümə etmək olar.

Məqalədə tərcümədə ekvivalent tərcümənin üstünlüyün göstərməklə üçün praktik nümunələrə də yer verilir.

Bununla yanaşı, tədqiqatda tərcümə sənətinin çətinliyi haqqında məlumat verilir. Həmçinin tərcüməçinin mənbə və hədəf dillər arasındakı struktur fərqliliklərini balanslaşdıraraq orijinalın ruhunu

qorumaq məsuliyyəti diqqət mərkəzinə çəkilir. Bu prosesdə tərcüməçilərin hər iki dilin aid olduğu mədəni konteksti və linqvistik incəlikləri daha dərinlən mənimsəməli olduqları xüsusi vurğulanır. Metodoloji baxımdan məqalə əsasən kontekstual müqayisəli təhlilə əsaslanmışdır.

Açar sözlər: tərcümə, ekvivalentlik, dinamik ekvivalentlik, hədəf dil, mənbə

Introduction

There are several forms of translation, such as written translation, oral translation, etc. The main essence of translation is not word-for-word substitution, the main thing to pay attention to here is the accurate transfer of meaning, intention, and tone from the source language to the target language. One of the advantages of translation is that it acts as a bridge between cultures. In order to reflect the delivery of the message transmitted in translation as it is in the new language, three stages must be followed: *understanding*, *editing*, and *evaluation*.

There are two forms of oral translation: *consecutive* and *simultaneous*.

Simultaneous translation is performed almost simultaneously with the original speech. The simultaneous interpreter sits in a place equipped with special equipment, translates the speech he/she hears through headphones at that moment and speaks into a microphone.

The interpreter first hears a certain part of the speech spoken by the speaker, and when the speaker is silent, he/she translates that part into the other language as it is in consecutive translation (Abdullayeva, 2010, p.26).

One of the main issues during translation is to preserve the meaning of the written text as it was in the original, that is, to reflect the meaning, style, and emotional expression of the original text in the writing as it is, reflects the high level of translation. At this time, the role of equivalent translation appears. During this translation, the writing style of the author, that is, the person who wrote the text, and the impression that the reader (from the translation) receives should be the same. This means that the impression received by the person reading the original should be the same as the impression received by the person reading the translation.

Methodology and Materials

This research employs a qualitative design, drawing on theoretical and analytical methods. The study is anchored in the critical exploration of translation theory, with particular attention to the contributions of E. Nida, V. Komissarov, and P. Newmark, whose frameworks provide the foundation for the analysis.

The methodological approach integrates contextual and comparative analysis. These strategies are applied to investigate how equivalence is established between source and target texts. To illustrate this process, a range of translation samples—including idioms, expressions, and sentences—are examined, demonstrating the practical application of equivalence in real translation scenarios.

Rather than relying on statistical or quantitative data, the study emphasizes interpretive inquiry into the linguistic and cultural dimensions of translation. The selected examples serve as the primary material, showcasing different forms of equivalence and highlighting the challenges that translators encounter when navigating between languages and cultures. This approach underscores the dynamic relationship between linguistic structures and cultural meaning in the pursuit of effective translation.

Discussion

What does “*equivalence*” mean?

Merriam-Webster's dictionary states that *"Equivalence is the state of being equal or interchangeable in value, quantity, function, meaning, or status"* (<https://www.merriam-webster.com/dictionary/equivalency>)

In this case, how is *"Equivalence in translation"* understood?

Equivalence in translation is not a word-for-word replacement of the translated text, but rather the notion that it should convey the same meaning, message, and effect as the original. This bridges linguistic, cultural, and stylistic gaps to create a target text that feels natural while preserving the intent of the original.

R. Jakobson, although a structuralist, was very involved in translation and had important ideas about translation. He wrote that there are three types of translation: intralingual, interlingual, and semiotic. R. Jakobson, being a linguist, drew attention to the difficulties of equivalent translation: *"Translation is the transmission of two equivalent messages through different codes. Languages differ from each other not in what they can express, but in what they are forced to express"* (Jakobson, 2018, p.97).

R. Jakobson writes about equivalence (in translation): *"Equivalence is a concept of capital importance for establishing a connection between two texts. It is the main problem of language and the main concern of linguistics"* (Jakobson, 2018, p. 100). This scientist's idea of equivalence is that the main task in translation is to interpret the given message and present it as it is in another language. R. Jakobson's claim was that by using the resources of languages, the entire message can be translated as it is.

E. Nida emphasizes the role of equivalence in translation, that is, the effect of the message transmitted by translation on the reader. For information, E. Nida is the founder of the theory of "Dynamic Equivalence" and his opinion on this issue is very important. He writes: *"Translation consists in creating the most natural and closest equivalent of the information in the source language (transmitted information) in the target language. Here, first of all, meaning should be taken into account, and secondly, style"* (Nida, 2021, p.31).

P. Newmark is known for his communicative and semantic approaches to equivalent translation and claims that equivalence in translation depends on the text. He writes that a translation is good when it is translated as it is in the target language. At the same time, he also puts forward the thesis that this process (translation as in the target language) should not occur at the expense of the original (Newmark, 1988, p.12).

Speech can be oral or written in the translation process. Each type of translation requires the specialist to have excellent knowledge of the working languages and extensive knowledge in that field.

C.F. Hendry explains translation as follows: *"Accuracy is considered important in translation. The translator can make mistakes, but the translator has no such possibility. He/she can never make mistakes"* (Hendry, 2022, p. 15).

The translation text should be written in such a way that there is no need for editing in the target language. The translator is required to have good oratory skills, easy listening and comprehension, and quick decision-making skills. Written translation is one-way. Since interlingual communication is established between two parties, translation is also carried out in two directions in oral translation. Both languages are sometimes active and sometimes passive in the process of work in translation.

E. Nida's research identified two forms of equivalent translation: formal and dynamic (Nida, 2010, p. 159).

The form of the source text is preserved in formal equivalent translation. The same function used in the target language is translated in a different way in dynamic equivalence translation.

The main issue is not to adapt the message texts to the information (message) in the source language, but rather the relationship between the main receptor and the message must be the same as in the original. So, the same dynamic relationship is important in these types of translation.

E. Nida, writing about his experiences with equivalence translation, emphasizes his preference for dynamic translation. He writes that he mainly prefers dynamic translation for the Bible (Nida, 2010, p.42). We can note that this is related to the fact that this type of translation preserves the originality more, and since this book is also related to the complete originality of religious views, it is necessary to prefer dynamic translation here.

The importance of equivalence in translation reveals several factors: extralinguistic content, the meaning of the translated text, language and usage norms associated with the type of text, the reception of the message, as well as the style and aesthetics of the text.

One of the issues that should be paid attention to in translation is that before starting the translation, it is necessary to choose the necessary, that is, the correct translation strategy. This means that it is necessary to preserve the values that are necessary in the text and that are preserved in the original. Equivalence and its types also play a significant role.

V. Komissarov is of the opinion that equivalence is preferable during translation. He proposes several types of equivalence (Komissarov, 1990, p. 12):

1) *Equivalence of the purpose of communication*. Despite the difference in the words, the message reflected in the text is preserved as it is in this case.

2) *Equivalence of the situation (situation)*. A real-life situation is described here. The issue that should be noted in this type of translation is that in this case, the linguistic means can change completely.

3) *Equivalence at the level of understanding*. The logical reasoning in the original text is preserved in this type.

4) *Equivalence is syntactic*. The sentence structure and grammatical structure are preserved to the extent possible in this type.

5) The use of direct, not indirect, equivalents of words is called *lexical equivalence*.

As with all types of translation, there are certain factors that need to be paid attention to during equivalent translation. For example, translating idioms and phraseological units in the original texts into the target language. At this time, the importance or advantage of equivalent translation emerges. It is known that literal translation cannot be preferred when translating these expressions. Thus, idioms and phraseological units cannot be translated literally. For example: "*It is raining cats and dogs*". As this is a famous idiom, its meaning cannot be translated literally. Therefore, it is essential to employ equivalence to ensure an accurate translation. Then, the cultural context must be taken into account. At this time, it is necessary to pay attention to literary and moral norms. An expression that is considered normal in one culture may be offensive in another. The translator must maintain this subtlety.

Stylistic consistency is also important in translation. A work of art should be translated in accordance with the artistic language. An official document should be translated officially.

Note that the terms adequacy and equivalence can often be confused. However, there is a small difference between them. A translation that focuses on the result, the equality of the texts as they are, is called equivalent translation.

The extent to which the translator's chosen path is consistent with the goal is called adequate translation, which focuses on the process.

Let's give some practical examples to illustrate the advantages of equivalent translation:

A rolling stone gathers no moss	Yatan öküzün başına ot batmaz	The words and structure are completely different, but the message is the same. Both reflect the result of laziness.
It is raining cats and dogs	Yağış vedrə ilə tökülür; yaxud yağış göz açmağa imkan vermir	Emotional and situational relevance. Reflects heavy rainfall.
Wet paint.	Ehtiyatlı olun, rəngləniib.	The same situation is described in different words. Both convey the message of "be careful".
Help yourself!	Nuş olsun! (bəzən Buyurun! Kimi işlədilir).	This is an expression often used during meals. It expresses a different cultural approach.
He is not as clever as he is.	O, özünü göstərdiyi kimi ağıllı deyildir	The logical comparison is maintained in the target language, but the order of the sentence members changes.
Fragile: Handle clever	Kövrəkdir: ehtiyatla davranın	Every word is translated. That is, the meaning of each is conveyed during translation. It is adapted to the patterns of the language though.

The girl opened the door.	Qız qapını açdı.	The sequence of subject, verb, and complement is the same in both languages. The function is parallel in both languages.
The sun is shining.	Günəş parlayır.	The morphological structure of all words is the same.

It is important to emphasize that certain transformations are used to ensure equivalence during translation. Some of these include: *substitution; transposition; concretization; generalization*.

One part of speech is replaced by another during substitution. The word order changes during transposition. Words with a broader meaning are replaced by words with a narrower meaning during concretization. In generalization, a specific name is replaced by a general concept.

For information, it should be noted that in translation, more often I and II levels, i.e. meaning and situational equivalence, are observed, while in official-business and scientific documents, IV and V levels, i.e. structural and lexical equivalence are observed.

Conclusion

Translators must be experts in their working languages in translation. It is also necessary to have experience in artistic style in translation.

All types of translation are important.

Equivalent translation stands out with its special importance.

The original text should create an emotional feeling in the reader, that is, the emotionality of the original text should be fully reflected in the translation. In equivalent translation.

Both the author and the spirit of the work are very important during equivalent translation.

Although complete equivalence is not possible during translation, each meaning should be recoded through another language. It is necessary to avoid mechanical transfer during equivalent translation. The functional, semantic and cultural balance between the source and target languages must be maintained. Equivalence is considered the most complex type of translation studies. The translator must try to fully restore the information load carried by the original text within the structural capabilities of the target language in this type of translation.

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TEACHER PERFORMANCE EVALUATION AND QUALITY ASSURANCE MECHANISMS: MODERN APPROACHES

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Abstract: Teacher performance evaluation and quality assurance mechanisms are among the key factors determining the effectiveness of modern education systems. In the context of rapidly changing educational environments, the role of teachers has expanded beyond knowledge transmission to include fostering critical thinking, creativity, and social skills among students. Therefore, systematic and objective evaluation of teacher performance has become essential.

This article examines modern approaches to teacher evaluation and their role in ensuring quality in education. Various evaluation methods are discussed, including classroom observations, student performance analysis, peer reviews, self-assessment, and feedback from stakeholders. The study emphasizes that a comprehensive and balanced application of these methods contributes to more reliable and fair outcomes.

Furthermore, the concept of quality assurance is analyzed as a continuous and development-oriented process aimed at maintaining and improving educational standards. The article highlights that evaluation should not be perceived as a control mechanism alone, but rather as a tool for professional growth and improvement. Transparency, fairness, and clearly defined criteria are identified as crucial principles in this process.

In conclusion, integrating effective teacher evaluation systems with quality assurance mechanisms significantly enhances teaching effectiveness and contributes to the development of a competent and competitive generation.

Keywords: Teacher evaluation, quality assurance, education system, professional development, assessment methods, teaching effectiveness

MÜƏLLİMİN FƏALİYYƏTİNİN QIYMƏTLƏNDİRİLMƏSİ VƏ KEYFİYYƏT TƏMİNATI MEXANİZMLƏRİ: MÜASİR YANAŞMALAR

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Xülasə: Müəllim fəaliyyətinin qiymətləndirilməsi və keyfiyyət təminatı mexanizmləri müasir təhsil sisteminin səmərəliliyini müəyyən edən əsas amillərdəndir. Təhsil mühitində baş verən sürətli

dəyişikliklər fonunda müəllimin rolu yalnız bilik ötürməkdən ibarət olmayıb, eyni zamanda şagirdlərdə tənqidi düşüncə, yaradıcılıq və sosial bacarıqların formalaşdırılmasını da əhatə edir. Bu səbəbdən müəllim fəaliyyətinin sistemli və obyektiv qiymətləndirilməsi xüsusi əhəmiyyət kəsb edir.

Məqalədə müəllim fəaliyyətinin qiymətləndirilməsinin müasir yanaşmaları və onların təhsildə keyfiyyət təminatındakı rolu təhlil olunur. Qiymətləndirmə üsulları kimi dərs müşahidələri, şagird nailiyyətlərinin analizi, həmkar qiymətləndirilməsi, özünüqiymətləndirmə və maraqlı tərəflərin rəyləri geniş şəkildə izah edilir. Bu üsulların kompleks tətbiqinin daha obyektiv və etibarlı nəticələr verdiyi vurğulanır.

Bununla yanaşı, keyfiyyət təminatı anlayışı davamlı inkişaf prosesi kimi təqdim olunur və onun yalnız nəzarət deyil, həm də inkişafyönlü funksiya daşdığı qeyd edilir. Qiymətləndirmə prosesində şəffaflıq, ədalətlik və aydın meyarların müəyyən edilməsi əsas prinsiplər kimi önə çəkilir.

Nəticə etibarilə, müəllim fəaliyyətinin effektiv qiymətləndirilməsi və keyfiyyət təminatı mexanizmlərinin integrasiyası tədrisin keyfiyyətini yüksəldir və rəqabətqabiliyyətli insan kapitalının formalaşmasına töhfə verir.

Açar sözlər: Müəllim qiymətləndirilməsi, keyfiyyət təminatı, təhsil sistemi, peşəkar inkişaf, qiymətləndirmə üsulları, tədrisin effektivliyi

Introduction

In modern education systems, improving quality is not only a goal, but also one of the main strategic directions of sustainable development. Globalization, rapid development of information technologies and changing demands of the labor market pose new challenges to education systems. In this regard, the quality of education is no longer limited to the transfer of knowledge, but also includes the formation of critical thinking, creativity, problem-solving and social skills in students. In this complex process, the teacher acts as a key figure and his activities play a decisive role in determining the quality of teaching.

The teacher's level of professionalism, pedagogical skills, methodological preparation and ability to apply innovative approaches directly determine the effectiveness of the educational process. The modern teacher is not only a person who gives knowledge, but also a facilitator, mentor and organizer of the learning process. Therefore, systematic and objective assessment of teacher performance is of particular importance. The assessment process should not only have a control function, but also act as a mechanism that supports the professional development of teachers by identifying their strengths and weaknesses.

Teacher performance evaluation can be carried out on the basis of various criteria. These criteria include lesson planning and organization, choice of teaching methods, communication skills with students, correct application of assessment strategies, as well as the level of use of innovative technologies. In addition, student achievements, parent feedback and peer evaluation also allow for a more objective assessment of teacher performance. This approach acts as a multidimensional assessment model, allowing for more reliable results.

Quality assurance mechanisms are an important tool for ensuring the protection and development of standards in the education system. These mechanisms include internal and external assessment systems, accreditation processes, monitoring and audit activities. Internal quality assurance covers self-assessment processes carried out within educational institutions and serves continuous improvement. External assessment, on the other hand, helps to ensure objectivity by being conducted by independent institutions.

At the same time, there is a mutual relationship between teacher performance assessment and quality assurance systems. These two directions complement each other and increase the overall efficiency of the education system. Teachers' professional development programs, trainings and certification processes also act as an integral part of this system. This approach allows teachers to acquire constantly updated knowledge and skills, master modern pedagogical approaches and, as a result, create a higher quality teaching environment.

Consequently, objective assessment of teacher performance and the application of effective quality assurance mechanisms are important conditions for the sustainable development of the education system. This process should be based not only on control, but also on a development-oriented approach, increase teachers' motivation and reveal their professional potential. The existence of such a system makes a significant contribution to improving the quality of education in general and the intellectual development of society.

The material of the study consists of local and foreign scientific literature on the evaluation of teacher performance and quality assurance in education, reports of international organizations (for example, UNESCO, OECD, etc.), regulatory and legal documents, as well as methodological tools reflecting modern pedagogical approaches. These sources allowed for the study of both theoretical and practical aspects of the problem, a comparative analysis of the experiences of different countries, and a generalization of modern approaches.

A number of scientific methods were used in accordance with the goals and objectives set during the study. Through the comparative analysis method, teacher performance evaluation models applied in different countries were examined, their strengths and weaknesses were identified. This method created conditions for revealing similarities and differences between different approaches, and for identifying more effective models.

The systematic approach method was applied in studying the structure of the quality assurance system and the interaction of its constituent elements. Through this approach, it was shown that the process of evaluating teacher performance is formed in connection with other components of the education system. Thus, it was substantiated that quality assurance mechanisms function as a whole system.

Müşahidə metodu isə dərs prosesinin təşkili, müəllimin pədaqoji davranışı, şagirdlərlə qarşılıqlı əlaqəsi və istifadə olunan tədris metodlarının qiymətləndirilməsi baxımından mühüm rol oynamışdır. Bu metod real tədris mühitində baş verən proseslərin birbaşa izlənilməsinə və praktik nəticələrin əldə olunmasına imkan vermişdir.

In addition, the data obtained using the methods of analysis and synthesis were systematized, generalized, and scientific conclusions were formulated. Through the analysis method, individual components were examined in detail, and through the synthesis method, a unified scientific approach was formed by establishing connections between these components.

The complex application of the methods used allowed for a comprehensive study of the issues of teacher assessment and quality assurance, obtaining objective results, and putting forward proposals of practical importance.

Conclusion and discussion

The conducted research shows that teacher performance evaluation acts as one of the main tools for improving quality in the education system. Modern approaches prove that evaluation based on only one criterion does not provide objective results. In this regard, the application of multidisciplinary methods such as lesson observations, analysis of student

achievements, peer evaluation and self-evaluation allows for more reliable and balanced results. Such an approach creates conditions for evaluating the teacher's performance from various aspects and determining his real professional level.

At the same time, the effectiveness of the quality assurance system directly depends on its establishment on the basis of the principle of continuous development. If this system only carries out the function of control and inspection, it will not give the expected results. On the contrary, an approach that supports the professional development of teachers, provides them with feedback and determines the directions of development is considered more effective. In this context, the evaluation process should be development-oriented and become a kind of professional support mechanism for teachers.

The discussions show that evaluation mechanisms based on the principles of transparency and fairness have a positive effect on teachers' activities. Clear assessment criteria, well-reasoned presentation of results, and teachers' involvement in the process increase their motivation. Motivation, in turn, directly affects the teacher's approach to the lesson, the use of innovative methods, and the quality of work with students.

At the same time, research results show that proper assessment of teacher performance is not limited to individual development, but also affects the general educational environment. Thus, professional and motivated teachers create a higher quality learning environment, which leads to increased student achievement and increased efficiency of the education system in general.

Consequently, the integration of an objective, transparent, and development-oriented assessment of teacher performance with a strong quality assurance system plays a decisive role in improving the quality of education. Measures taken in this direction not only provide an assessment of the current situation, but also create a basis for making strategic decisions for future development

Conclusion

Teacher performance assessment and quality assurance act as one of the main factors ensuring the sustainable and effective functioning of the modern education system. The conducted studies show that the application of these mechanisms based on scientific foundations, systematically and purposefully plays an important role in improving the quality of education. In particular, the assessment of teachers' performance based on objective and multidisciplinary criteria gives impetus to the development of their professional skills, improvement of pedagogical activities and the application of innovative approaches.

At the same time, the effective establishment of a quality assurance system allows for the maintenance of uniform standards in educational institutions, the organization of the educational process in accordance with modern requirements and the provision of sustainable development. Within the framework of this system, not only the assessment of teachers, but also the provision of methodological support to them, the creation of professional development opportunities are of particular importance.

In conclusion, it can be noted that the teacher performance assessment system integrated with quality assurance does not only carry a control function, but also acts as a development-oriented, stimulating and improving mechanism. This approach serves to increase transparency, efficiency, and competitiveness in the education system, as well as to strengthen the intellectual potential of society.

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THE ROLE OF PEDAGOGICAL SKILLS AND METHODOLOGICAL COMPETENCIES IN OVERCOMING SOCIAL ADAPTATION PROBLEMS

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Abstract: Social adaptation is a crucial psychological and pedagogical process that ensures individuals' successful integration into society. In contemporary educational systems, rapid social, technological, and cultural transformations have significantly increased adaptation challenges among students. These challenges are often reflected in communication difficulties, emotional instability, lack of social skills, and insufficient interaction with the learning environment.

The aim of this article is to analyze the main causes of social adaptation problems and to examine the role of pedagogical and methodological competencies in addressing these issues. The study is based on qualitative theoretical analysis of scientific literature and educational research.

The findings reveal that insufficient development of social and emotional skills, limited family involvement, and inadequate pedagogical support are the primary factors contributing to adaptation difficulties. At the same time, the development of teachers' pedagogical competencies, the application of inclusive education principles, and the use of innovative teaching strategies significantly improve students' social integration and behavioral adjustment.

The study concludes that strengthening teacher training systems and enhancing methodological competencies are essential for reducing social adaptation problems and ensuring effective educational outcomes in modern learning environments.

Keywords: Social adaptation, pedagogical competencies, methodological skills, inclusive education, educational innovation, teacher training

SOSIAL ADAPTASIYA PROBLEMLƏRİNİN ARADAN QALDIRILMASINDA PEDAQOJİ BACAQLARIN VƏ METODOLOJİ SƏRİŞTƏLİKLƏRİN ROLU

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Xülasə: Sosial adaptasiya fərdin cəmiyyətə uğurlu integrasiyasını təmin edən mühüm psixoloji və pedaqoji prosesdir. Müasir təhsil sistemində sosial, texnoloji və mədəni dəyişikliklər bu sahədə çətinliklərin artmasına səbəb olmuşdur. Bu çətinliklər əsasən ünsiyyət problemləri, emosional qeyri-sabitlik, sosial bacarıqların zəifliyi və tədris mühitinə uyğunlaşma çətinlikləri ilə özünü göstərir.

Məqalənin məqsədi sosial adaptasiya problemlərinin əsas səbəblərini təhlil etmək və onların həllində pedaqoji və metodiki kompetensiyaların rolunu araşdırmaqdır. Tədqiqat nəzəri ədəbiyyatın təhlilinə əsaslanır. Nəticələr göstərir ki, sosial və emosional bacarıqların zəif inkişafı, ailə dəstəyinin azlığı və pedaqoji yardımın çatışmazlığı əsas problem yaradan amillərdir. Müəllimlərin pedaqoji bacarıqlarının inkişafı, inklüziv təhsilin tətbiqi və innovativ təlim metodları isə sosial adaptasiyanı gücləndirir.

Açar sözlər: Sosial adaptasiya, pedaqoji kompetensiyalar, metodiki bacarıqlar, inklüziv təhsil, təhsil innovasiyaları, müəllim hazırlığı

Introduction

Social adaptation is one of the most important processes in human development, as it ensures an individual's successful integration into society, educational institutions, and social groups. It reflects the ability of a person to adjust to social norms, values, behavioral expectations, and interpersonal relationships. In modern educational contexts, this process has become increasingly complex due to rapid social, technological, and cultural changes. Schools and higher education institutions are not only centers of knowledge acquisition but also key environments for socialization. In this regard, teachers play a crucial role in supporting students' adaptation processes by creating inclusive, supportive, and psychologically safe learning environments. However, many students still experience difficulties in adapting to new social and educational conditions, which may manifest in communication barriers, emotional instability, low self-confidence, and lack of social interaction skills.

According to Bronfenbrenner's ecological systems theory, individual development is influenced by multiple interconnected environmental systems, including family, school, and society. Therefore, social adaptation problems cannot be understood in isolation; they require a comprehensive pedagogical approach that considers both internal (psychological) and external (social and educational) factors.

In recent years, increasing attention has been paid to the role of pedagogical and methodological competencies in addressing these challenges. Modern education emphasizes not only knowledge transfer but also the development of teachers' professional skills such as communication, empathy, classroom management, and the application of innovative teaching methods. These competencies are essential for supporting students' social-emotional development and facilitating their successful adaptation to educational environments (Quliyev, 2018).

Furthermore, inclusive education principles highlight the importance of equal opportunities for all learners, regardless of their social or psychological characteristics. Such approaches contribute to reducing adaptation difficulties and promoting social cohesion within educational settings. Thus, the present study focuses on analyzing the causes of social adaptation problems and examining how the development of pedagogical and methodological competencies can serve as an effective strategy for overcoming these challenges in modern education systems.

Materials and Methods

The study was conducted to investigate social adaptation problems and to determine the role of developing pedagogical and methodological competencies in addressing these challenges within the educational process.

The materials of the research consisted of contemporary pedagogical, psychological, and sociological literature, scientific articles on inclusive education, international reports (such as UNESCO publications), curriculum documents, and relevant educational standards. In addition, practical experiences from educational institutions and observations of teaching practices were also considered as important sources of information (UNESCO, 2022).

The research employed both theoretical and empirical methods. The theoretical methods included analysis, synthesis, comparison, and generalization. These methods were used to systematize existing scientific approaches to social adaptation and to clarify the essence and development directions of pedagogical and methodological competencies.

In the empirical stage, observation, questionnaire (survey), and interview methods were applied. Observation was used to examine students' social behavior in the educational environment. The questionnaire method helped to evaluate teachers' pedagogical approaches and methodological skills. Interviews were conducted with experienced educators to gather professional opinions and practical insights regarding social adaptation issues. The collected data were processed using both qualitative and quantitative analysis methods. The results were then systematically interpreted to identify the main causes of social adaptation problems and to evaluate the effectiveness of pedagogical and methodological competencies in addressing these issues (Məmmədov, 2021).

Results and Discussion

The findings of this study indicate that social adaptation problems in educational settings are closely linked to both individual student characteristics and the effectiveness of the pedagogical environment. Students who experience difficulties in social adaptation often show challenges in communication, group interaction, and emotional regulation, which negatively affects their learning process and classroom participation.

The analysis revealed that teachers' pedagogical and methodological competencies play a crucial role in reducing these difficulties. Educators who apply student-centered, inclusive, and flexible teaching strategies are more successful in supporting students' social integration. In particular, differentiated instruction, collaborative learning activities, and emotionally supportive classroom environments were found to significantly improve students' adaptation levels.

The results also demonstrate that insufficient methodological preparation of teachers may limit their ability to address diverse learner needs effectively. In contrast, continuous professional development enhances teachers' capacity to manage classroom diversity and apply appropriate strategies for students with different social and psychological backgrounds.

Furthermore, the study highlights that social adaptation is not only an individual issue but also a systemic one. School climate, peer relationships, and institutional support mechanisms all contribute to the success of adaptation processes. Therefore, improving pedagogical competence alone is not sufficient; it must be supported by a well-structured educational environment.

In discussion, these findings align with contemporary research emphasizing the importance of inclusive education and teacher professionalism in fostering social adaptation. The integration of modern teaching methods and continuous competence development is therefore considered a key strategy for addressing adaptation challenges effectively (Abbasov, 2019).

Conclusion

The study concludes that social adaptation problems represent a significant challenge within the educational process, directly influencing students' academic performance, emotional well-being, and social development. These difficulties are shaped by both individual factors and the quality of the educational environment.

The research findings clearly show that the development of pedagogical and methodological competencies is a key strategy in addressing these challenges. Teachers who possess strong professional skills and apply modern, inclusive, and student-centered approaches are more effective in supporting students' social integration and reducing adaptation difficulties.

It can also be concluded that social adaptation should be viewed as a multidimensional process that requires cooperation between teachers, students, and the broader educational institution. Improving teaching methods alone is not sufficient; a supportive school climate and continuous professional development of educators are equally important.

Overall, the study emphasizes that strengthening teachers' pedagogical and methodological competencies contributes significantly to creating an inclusive, supportive, and adaptive learning environment, thereby enhancing the overall quality of education.

Literature Review

Social adaptation has been widely studied in pedagogy and educational psychology as a key factor influencing students' success. It is understood as a dynamic process through which individuals adjust to new social environments and develop effective interaction skills. Researchers emphasize that adaptation is closely related to communication abilities, emotional intelligence, and social experience.

Recent studies highlight that the effectiveness of social adaptation largely depends on the teacher's professional competencies. Pedagogical skills such as empathy, classroom management, and communication play a crucial role in creating a supportive learning environment. At the same time, methodological competencies enable teachers to apply appropriate teaching strategies tailored to diverse student needs.

In the context of inclusive education, social adaptation becomes even more important. Students with different backgrounds and abilities require differentiated approaches and flexible teaching methods. Therefore, the development of teachers' competencies is considered a priority in modern education systems (Əliyev, 2020).

Practical Implications

The findings of this study have several practical implications for the educational process. First, teacher training programs should place greater emphasis on the development of pedagogical and methodological competencies. Future educators need to be equipped with skills that allow them to manage diverse classrooms and support students' social development.

Second, schools should create supportive environments that promote cooperation, respect, and inclusion. This includes the use of interactive teaching methods, group activities, and collaborative learning strategies.

Third, continuous professional development should be encouraged. Workshops, training sessions, and seminars can help teachers improve their skills and stay updated with modern educational practices.

Finally, the integration of technology in education can support social adaptation by providing interactive and engaging learning experiences.

Recommendations

Based on the results of the study, the following recommendations can be proposed:

- Strengthening teacher education programs with a focus on social adaptation strategies
- Promoting inclusive and student-centered teaching approaches
- Encouraging the use of innovative and interactive teaching methods
- Providing psychological support services within schools
- Organizing regular professional development activities for teachers

Future Research Directions

Further research is needed to explore social adaptation problems in different educational contexts, such as preschool, primary, and higher education. Comparative studies between traditional and modern teaching approaches could provide deeper insights into effective strategies.

In addition, future studies may focus on the role of digital technologies and online learning environments in supporting social adaptation. Investigating students' perspectives and experiences could also contribute to a more comprehensive understanding of the issue.

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REFLECTIVE TEACHING APPROACH IN THE CONTENT MODEL OF HIGHER PEDAGOGICAL EDUCATION

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Abstract: The article comments on the theory of teacher development based on the reflective learning approach, which is considered one of the important directions in determining the theoretical foundations of the content model of higher pedagogical education. At the same time, approaches to the concept of reflective learning were analyzed, and it was determined that this process serves the individual professional development of future teachers, as well as improving the overall quality of the training process. It was determined that these approaches are important methodological foundations that should be taken into account

in the formation of the content of higher pedagogical education in Azerbaijan, and in particular in the establishment of the curriculum of the specialty "Primary Classroom Teacher". Thus, students studying in this specialty should not only acquire theoretical knowledge, but also be able to apply, analyze and draw conclusions from this knowledge in the context of real pedagogical experience. For this purpose, it is advisable to increase the components that enable reflective learning in the curriculum (organization of practical exercises, pedagogical experience, portfolio-based assessment, critical analysis and reflection tasks). Such an approach forms the skills of future primary school teachers to critically approach their teaching activities, analyze them and continuously improve them. The article also reflects the fact that during the pedagogical practice stage, students are provided with the opportunity to apply the theoretical knowledge they have acquired in a real school environment in practice, to actively participate in the learning process, and at the same time to develop reflective thinking skills by systematically analyzing their own activities. The article provides examples of tasks that develop reflective thinking on theoretical and practical aspects of teaching activities.

Key words: higher education, content model, teacher, competence, reflective learning

ALİ PEDAQOJİ TƏHSİLİN MƏZMUN MODELİNDƏ REFLEKTİV TƏLİM YANAŞMASI

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Xülasə: Məqalədə ali pedaqoji təhsilin məzmun modelinin nəzəri əsaslar müəyyənləşdirilərkən mühüm istiqamətlərdən biri sayılan reflektiv təlim yanaşmasına əsaslanan müəllim inkişafı nəzəriyyəsinə şərh edilmişdir. Eyni zamanda reflektiv təlim konsepsiyasına yanaşmalar təhlil edilmiş, bu prosesin gələcək müəllimlərin fərdi peşəkar inkişafına, həm də təlim prosesinin ümumi keyfiyyətinin yüksəldilməsinə xidmət etdiyi müəyyən edilmişdir. Müəyyən edilmişdir ki, bu yanaşmaların Azərbaycanda ali pedaqoji təhsilin məzmununun formalaşdırılmasında və xüsusilə də "İbtidai sinif müəllimliyi" ixtisasının tədris planının qurulmasında nəzərə alınmalı mühüm metodoloji əsaslardandır. Belə ki, bu ixtisas üzrə təhsil alan tələbələrin təkcə nəzəri biliklərə yiyələnməsi deyil, eyni zamanda həmin biliklərin real pedaqoji təcrübə kontekstində tətbiqi, təhlili və ondan nəticə çıxarma bacarığı da formalaşdırılmalıdır. Bu məqsədlə tədris planında reflektiv təlimə imkan yaradan komponentlərin (praktik məşğələlərin, pedaqoji təcrübənin təşkili, portfolioalara əsaslanan qiymətləndirmə, tənqidi təhlil və refleksiya tapşırıqları) artırılması məqsəduyğundur. Belə bir yanaşma gələcək ibtidai sinif müəllimlərində öz təlim fəaliyyətinə tənqidi yanaşmaq, onu təhlil etmək və fasiləsiz şəkildə təkmilləşdirmək bacarıqları formalaşdırır. Məqalədə eyni zamanda pedaqoji təcrübə mərhələsində tələbələrin real məktəb mühitində əldə etdikləri nəzəri bilikləri praktikada tətbiq etməyə, təlim prosesində fəal iştirak etməyə və eyni zamanda öz fəaliyyətlərini sistemli şəkildə təhlil edərək reflektiv düşünmə bacarıqlarını inkişaf etdirməyə şərait yaratması öz əksini tapmışdır. Məqalədə müəllimlik fəaliyyətinin nəzəri və praktiki aspektləri üzrə reflektiv düşüncəni inkişaf etdirən tapşırıq nümunələri verilmişdir.

Açar sözlər: ali təhsil, məzmun modeli, müəllim, səriştə, reflektiv təlim

Introduction

Purposeful and systematic construction of the content structure creates conditions for the formation of a strategic approach that ensures personnel training in accordance with the functional goals of higher pedagogical education - inclusiveness, innovation, research orientation and priority areas such as lifelong learning. In modern times, the construction of the content model of higher

pedagogical education is carried out based on pedagogical theories, educational philosophy and international methodological approaches. The main role in the formation of this model is played by the goals and content of education, curriculum theory, competency-based approach and student-centered learning concept.

Methodology and Materials.

The theoretical foundations of this model are based on several fundamental pedagogical and philosophical approaches. First of all, in the philosophy of education, the humanist approach formulated by K. Rogers and A. Maslow (Rogers, C.R, 1983, Maslow, A.H., 1970), the constructivist approach formulated by Piaget and Vygotsky (Выготский, Л. С. , 2025), and the socio-cultural approaches of Bruner and Vygotsky (Выготский, Л. С. , 2025, Bruner, J.1996) create the basis for the content of higher pedagogical education to be structured in accordance with the interests of each individual and society. These approaches emphasize the development of the learner as a person, taking into account his experience and social interaction in the educational process. On the other hand, curriculum theories (Marsh, C. J. 2016, Ornstein, A. C., 2018, Schiro, M.S.,2013) provide a scientific basis for the systematic connection of content with objectives, teaching methods and assessment, as well as for the structuring of subject blocks. The competency-based education model (Mulder, M. 2012, Rychen, D.S., 2003) is the main theoretical support of the modern content model as an approach that ensures the integration of knowledge, skills and attitudes, and is based on the identification of real-life, application-oriented learning outcomes.

Discussion

In modern pedagogical models, the teacher is presented as a reflective practitioner who not only transmits knowledge, but also analyzes, evaluates and learns from his own experience. This requires assessing the professional development of the teacher as a continuous and purposeful process.

The concept of reflective learning was first put forward by J. Dewey (Dewey, J. ,1933), and later developed by D. Schön (Schön, D. A. ,1992) with the concepts of “reflection after action” and “reflection during action”. According to D. Schön’s approach, teachers develop decision-making skills by analyzing the situations they encounter in their daily teaching practices. This process serves both their individual professional development and the improvement of the overall quality of the teaching process.

The reflective teacher model is also associated with Kolb's (Kolb, D. A., 1984) "experiential learning model". According to this model, the teacher first experiences the learning process, then observes and analyzes it, summarizes the results obtained and applies them in new pedagogical situations. This sequential and periodic learning process creates conditions for the development of reflective thinking skills in the teacher and the deepening of the reflective approach. These theoretical approaches are based on the following basic principles:

- 1) systematic analysis and evaluation of experience;
- 2) promotion of action-based and contextual learning;
- 3) development of critical thinking, reflection and self-assessment skills;
- 4) formation of a professional learning environment and stimulation of collective exchange of experience.

These approaches are important methodological foundations that should be taken into account in the formation of the content of higher pedagogical education in Azerbaijan, and in particular in the development of the curriculum for the specialty "Primary Teacher". Thus, students studying in

this specialty should not only acquire theoretical knowledge, but also develop the ability to apply, analyze and draw conclusions from this knowledge in the context of real pedagogical experience. For this purpose, it is advisable to increase the components that enable reflective learning in the curriculum (organization of practical exercises, pedagogical experience, portfolio-based assessment, critical analysis and reflection tasks). Such an approach forms the skills of future primary school teachers to critically approach their teaching activities, analyze them and continuously improve them.

Let's pay attention to examples. In the subject of "Pedagogy", it is advisable to develop students' skills in analyzing pedagogical phenomena and evaluating the practical application of various pedagogical approaches through reflective essays. For this, it is possible to read and discuss critical articles in the training process, especially in seminar sessions, as well as write reflective essays based on the analysis of pedagogical situations. For example, the works of H. Ahmadov on the history of education, Z. Garalov, N. Kazymov, H. Ahmadov, Y. Talibov, B. Ahmadov, A. Agayev and A. Hajiyeu on pedagogical laws and regularities, as well as critical articles in the field of national pedagogy by N. Kazymov, A. Agayev, K. Mammadov and A. Pashayev can be used as teaching materials. The analysis of these texts in the context of modern pedagogical approaches creates conditions for the formation of students' inference and critical thinking skills.

In the subject of "Pedagogy of Primary Education", tasks that develop reflective thinking on theoretical and practical aspects of teaching activity can be organized. For example, in a seminar session, students can be given the task of planning and implementing their own teaching activities, and then analyzing that activity to identify ways to improve. This approach, in addition to forming lesson planning and management skills of future teachers, also develops reflective thinking.

Task: "Short lesson model and reflective analysis"

Purpose of the application. This task aims to develop the skills of future primary school teachers to plan, implement and analyze the learning process. At the same time, it serves to form their habits of a reflective approach to pedagogical activity.

Assignment conditions.

1) students prepare a short 10–15 minute lesson model in groups or individually (the topic is determined in advance);

2) the plan indicates the main goal of the lesson, the teaching methods to be used, resources and assessment criteria;

3) a topic is presented in accordance with the plan;

4) a reflective essay is written on the results.

Guiding questions for a reflective essay may include:

1) What pedagogical principles and methods did I use when planning the lesson?

2) How did I overcome the difficulties I encountered in the pedagogical process?

3) How do I assess the level of activity and participation of students (or group)?

4) Was I satisfied with my performance (strengths and weaknesses)? etc.

The implementation of the task can be assessed using the following evaluation criteria:

a) the relevance of the lesson to the content and purpose;

b) the ability to correctly select and apply teaching methods;

c) the justification of self-analysis and development perspectives in a reflective essay.

This type of task develops reflective thinking in students and at the same time strengthens their practical skills.

The teaching methodology subjects of individual subjects included in the curriculum of the "Primary Teacher" specialty (Azerbaijani language, mathematics, life skills, technology, music, informatics, physical education) create wide opportunities for the development of reflective thinking skills of future primary school teachers. Methodological decision-making, analysis of teaching situations and interpretation of experience during the teaching of these subjects strengthen both the theoretical and practical preparation of students. In practical exercises related to these subjects, students analyze various lesson examples, plan and present lessons, and evaluate them reflectively. This also serves to form the pedagogical thinking of students and consciously improve their own teaching strategies. This approach develops the skills of future teachers to understand the reasons and consequences of the decisions they make during the lesson, to evaluate alternative approaches and ideas, and to reflect on their own activities.

In higher pedagogical education programs, 30 credits are allocated for pedagogical practice, which is one of the main stages in the professional training of future primary school teachers. This stage allows students to apply the theoretical knowledge they have acquired in a real school environment in practice, actively participate in the learning process, and at the same time develop reflective thinking skills by systematically analyzing their own activities.

Conclusion

Reviews, daily notes, analytical analyses conducted during and after the internship allow students to go beyond simply describing the pedagogical events that occur, to reflect on their causes and consequences, and to identify alternative approaches. Lesson plans, reviews, protocols, and reports collected in internship portfolios are considered one of the main tools that allow students to monitor and evaluate the dynamics of their pedagogical development. In recent years, the organization of the pedagogical experience of final-year students has been changed, divided into 2 stages (pedagogical experience and industrial experience), and a final assessment stage has been determined. At this stage, students exchange experiences in various pedagogical situations, discuss the problems they encounter together, and get acquainted with various solutions. Such a collaborative approach strengthens their analytical thinking and problem-solving skills, and strengthens their professional reflection habits. Thus, pedagogical practice is not only a stage in which practical skills are formed, but also an important pedagogical context in which the future teacher's culture of pedagogical reflection develops and deepens.

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FORMATIVE ASSESSMENT AS THE ENGINE OF LEARNING: A COMPREHENSIVE FRAMEWORK FOR UNIVERSITY-LEVEL TEACHING

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Abstract: In modern higher education, assessment is not only a tool for measuring students' knowledge but also a key factor in shaping the learning process. However, traditional assessment systems are largely dominated by summative approaches that focus on evaluating final outcomes rather than supporting continuous development. This creates certain limitations in fostering effective learning. The issue becomes particularly evident in the context of English as a Foreign Language (EFL), where language acquisition requires continuous practice, interaction, and regular feedback.

Communicative competence is considered one of the main goals of EFL teaching, encompassing not only grammatical accuracy but also the appropriate use of language in social and pragmatic contexts. The development of this competence cannot be achieved solely through memorization of rules; it requires active participation, critical thinking, and meaningful communication.

In this regard, formative assessment plays a crucial role. By being integrated into the teaching process, it provides continuous feedback, helps identify students' strengths and weaknesses in a timely manner, and encourages active learner engagement. As a result, formative assessment contributes to the creation of a more effective and sustainable language learning environment.

Keywords: formative assessment, EFL, communicative competence, higher education, feedback, learning process

ÖYRƏNMƏNİN HƏRƏKƏTVERİCİ QÜVVƏSİ KİMİ FORMATİV QIYMƏTLƏNDİRMƏ: UNİVERSİTET SƏVİYYƏLİ TƏDRİS ÜÇÜN HƏRTƏRƏFLİ ÇƏRÇİVƏ

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Giriş: Müasir ali təhsildə qiymətləndirmə yalnız tələbələrin bilik səviyyəsini ölçən vasitə deyil, eyni zamanda öyrənmə prosesini formalaşdıran əsas amillərdən biridir. Lakin ənənəvi qiymətləndirmə sistemləri əsasən summativ yanaşmaya üstünlük verir və bu yanaşma daha çox yekun nəticələrin qiymətləndirilməsinə yönəlir. Bu isə öyrənmənin davamlı inkişafını təmin etmək baxımından müəyyən məhdudiyyətlər yaradır. Xüsusilə xarici dil kimi ingilis dili (EFL) kontekstində bu problem daha qabarıq şəkildə özünü göstərir, çünki dil öyrənmə prosesi davamlı məşq, qarşılıqlı ünsiyyət və müntəzəm rəy tələb edir.

Kommunikativ kompetensiya EFL tədrisinin əsas məqsədlərindən biri hesab olunur və bu anlayış dilin yalnız qrammatik düzgünlüyünü deyil, həm də sosial və pragmatik baxımdan uyğun istifadəsini əhatə edir. Bu kompetensiyanın formalaşdırılması sadəcə qaydaların əzbərlənməsi ilə deyil, aktiv iştirak, tənqidi düşünmə və real ünsiyyət bacarıqları ilə mümkündür.

Bu baxımdan formativ qiymətləndirmə xüsusi əhəmiyyət kəsb edir. O, tədris prosesinə inteqrasiya olunaraq davamlı rəy təmin edir, tələbələrin zəif və güclü tərəflərini vaxtında müəyyən etməyə imkan yaradır və onların öyrənmə prosesində aktiv iştirakını stimullaşdırır. Nəticə etibarilə, formativ qiymətləndirmə daha effektiv və davamlı dil öyrənmə mühitinin formalaşmasına xidmət edir.

Açar sözlər: formativ qiymətləndirmə, EFL, kommunikativ kompetensiya, ali təhsil, rəy, öyrənmə prosesi

Introduction

In modern higher education, assessment plays a crucial role not only in measuring students' knowledge but also in shaping and guiding the learning process. Traditionally, assessment systems have been dominated by summative evaluation, which primarily focuses on measuring learning outcomes at the end of a course. While summative assessment provides important information about students' overall achievement, it often fails to support the ongoing development of knowledge and skills during the learning process (Biggs & Tang, 2011). As a result, students may become more focused on achieving grades rather than engaging deeply with the learning material.

This issue becomes particularly significant in the context of English as a Foreign Language (EFL) education. Language acquisition is a complex and dynamic process that requires continuous exposure, interaction, and practice. Unlike content-based subjects, where knowledge can be

assessed through discrete testing, language learning involves the gradual development of communicative competence over time (Hymes, 1972). Therefore, relying solely on summative assessment methods may not adequately capture students' progress or support their linguistic development.

Communicative competence, which encompasses linguistic, sociolinguistic, and pragmatic abilities, is widely recognized as the primary goal of EFL instruction. It enables learners not only to produce grammatically correct sentences but also to use language appropriately in different social and cultural contexts (Canale & Swain, 1980). Achieving this level of competence requires more than memorization of vocabulary and grammatical rules; it involves active participation in meaningful communication, critical thinking, and the ability to adapt language use to real-life situations.

In this regard, formative assessment emerges as an essential pedagogical approach that supports the development of communicative competence. Unlike summative assessment, formative assessment is integrated into the learning process and provides continuous feedback to both teachers and students. It allows instructors to identify students' strengths and weaknesses in real time and to adjust their teaching strategies accordingly (Black & Wiliam, 1998). At the same time, it encourages students to take an active role in their own learning by promoting reflection, self-assessment, and autonomous learning strategies (Wiliam, 2011).

Furthermore, formative assessment contributes to the creation of a learner-centered classroom environment, where students are actively engaged in the learning process rather than passively receiving information. Regular feedback, interactive tasks, and collaborative activities foster a supportive learning atmosphere that enhances motivation and confidence. This is particularly important in EFL settings, where students often face anxiety and lack of confidence when using a foreign language.

Therefore, integrating formative assessment into EFL university education is not merely a methodological choice but a pedagogical necessity. It ensures that assessment serves not only as a tool for measuring learning outcomes but also as a mechanism for promoting continuous development, ultimately leading to more effective and meaningful language learning experiences.

Materials and Methods

This study adopts a qualitative research design grounded in a comprehensive theoretical analysis of formative assessment in English as a Foreign Language (EFL) education. The research integrates several well-established pedagogical and psychological frameworks that provide a solid conceptual foundation for understanding how formative assessment contributes to the development of communicative competence.

First, Lev Vygotsky's concept of the Zone of Proximal Development (ZPD) is used to explain how learning occurs through guided interaction and scaffolding. Within this framework, formative assessment serves as a tool for identifying learners' current performance levels and supporting their progression toward higher levels of competence through continuous feedback (Vygotsky, 1978).

Second, Benjamin Bloom's Taxonomy of Cognitive Skills is employed to structure learning objectives and assessment tasks. The taxonomy enables the design of activities that move beyond basic knowledge recall to higher-order thinking skills such as analysis, evaluation, and creation, which are essential for effective language use (Bloom et al., 1971).

Third, the formative assessment strategies developed by Dylan Wiliam provide the operational framework for this study. These strategies emphasize clarifying learning intentions,

eliciting evidence of student understanding, providing feedback that promotes learning, and encouraging peer and self-assessment (Wiliam, 2011). In addition, the foundational work of Black and Wiliam highlights the significant impact of formative assessment on student achievement (Black & Wiliam, 1998).

The research methodology relies on qualitative methods, including a systematic review of pedagogical literature, analysis of contemporary teaching practices, and reflective examination of classroom-based applications. This approach allows for an in-depth exploration of how formative assessment functions in real educational contexts, particularly in EFL classrooms within higher education institutions (Hattie & Timperley, 2007).

The proposed assessment model is structured around three interconnected components:

- **Daily formative assessment:** This includes continuous evaluation through classroom interaction, such as participation in discussions, group work, and task-based activities. Teachers provide immediate and constructive feedback, enabling students to identify their strengths and areas for improvement (Sadler, 1989).
- **Periodic evaluations:** These consist of quizzes, oral presentations, and guided discussions conducted at regular intervals. Unlike traditional tests, these evaluations are designed to measure students' progress and provide diagnostic information that informs subsequent instruction (Black & Wiliam, 1998).
- **Final summative assessment:** The final stage involves a comprehensive evaluation of students' overall performance at the end of the course. This component serves to validate the learning outcomes achieved through the formative process (Biggs & Tang, 2011).

Additionally, the model incorporates elements of self-assessment and peer assessment, which encourage learners to take an active role in their own learning. These practices foster metacognitive awareness, allowing students to monitor their progress and develop independent learning strategies (Andrade & Valtcheva, 2009).

Overall, the methodological framework of this study emphasizes the integration of theory and practice, demonstrating how formative assessment can be systematically applied to enhance the effectiveness of EFL teaching and learning in higher education contexts.

Discussion

The findings of this study highlight the significant role of formative assessment in enhancing the effectiveness of EFL university instruction. Unlike traditional summative approaches, formative assessment creates a dynamic and interactive learning environment where students are actively engaged in the learning process. Continuous feedback enables both instructors and learners to identify gaps in understanding and to take immediate corrective action, which leads to more effective knowledge acquisition and skill development.

One of the most important implications of this study is the shift from assessment as a tool of measurement to assessment as a tool of learning. The integration of daily rubric-based evaluation, periodic colloquiums, and a final examination ensures that learning is continuously monitored and supported. In particular, the rubric-based system provides transparency in assessment criteria and allows students to clearly understand expectations, which enhances motivation and accountability.

Furthermore, the use of colloquiums as intermediate assessment tools promotes higher-order thinking skills such as analysis, synthesis, and evaluation. These tasks encourage students to use language in meaningful contexts rather than simply reproducing memorized information. As a

result, students develop not only linguistic competence but also critical thinking and communicative abilities.

However, the implementation of such a framework may present certain challenges. These include increased workload for instructors, the need for proper training in formative assessment techniques, and possible resistance from students who are accustomed to traditional assessment systems. Despite these challenges, the long-term benefits of formative assessment outweigh its limitations, particularly in fostering autonomous and reflective learners.

Conclusion

This study has demonstrated that formative assessment plays a central role in improving the quality of EFL university education. By shifting the focus from final outcomes to continuous learning processes, formative assessment supports the development of communicative competence, critical thinking, and self-regulation.

The proposed three-tier assessment framework comprising daily formative assessment, periodic colloquiums, and a final examination offers a balanced and effective model for evaluating student performance. It ensures that assessment is not merely a tool for grading but also a mechanism for guiding and enhancing learning.

The findings suggest that integrating formative assessment into EFL teaching leads to increased student engagement, improved academic performance, and more meaningful learning experiences. Therefore, educators and institutions should consider adopting formative assessment as a core component of their teaching strategies.

Future research may focus on empirical validation of this framework in different educational contexts and explore the use of digital technologies to facilitate formative assessment practices.

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PRE-SERVICE ENGLISH TEACHERS' READINESS IN DESIGNING INCLUSIVE MATERIALS FOR STUDENTS WITH SPECIAL EDUCATIONAL NEEDS

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Abstract: The objective of this study is to investigate the disconnect between theory and practice in designing inclusive materials and how prospective third- and fourth-year English Language Teaching (ELT) candidates feel about their preparedness to design materials for Special Educational Needs (SEN) learners. Data were collected through a survey administered to 50 ELT candidates. The survey focused on three major areas: psychological preparedness for working with SEN students, critiques of the undergraduate curriculum, and willingness to work with a variety of multimodal and tactile/digital materials. The findings show a significant gap in candidates' knowledge of how to modify conventional materials for use by visually impaired or dyslexic learners. Of the 50 candidates surveyed, 100% acknowledged there is an increase in the number of SEN students, but 76% reported they do not have adequate skills to adapt conventional materials for use by visually impaired or dyslexic learners. A lack of practical skills is a common theme among respondents and can most likely be attributed to the theoretical emphasis of their current courses, which do not provide the necessary level of pedagogical confidence and skill development. Although 64% of respondents indicated a preference for multimodal and artistic pedagogy, they lack practical examples to draw on. Therefore, these findings provide substantial evidence for the need for structural reform in teacher education and for the importance of mandated design laboratories for developing inclusive materials, rather than relying solely on traditional lecture-based classes.

Keywords: Inclusive Education, Pre-Service ELT Teachers, Material Design, Special Educational Needs (SEN), Multimodal Learning, Theory-Practice Gap

İNGİLİZCE ÖĞRETMENİ ADAYLARININ ÖZEL GEREKSİNİMLİ ÖĞRENCİLER İÇİN KAPSAYICI MATERYAL TASARLAMAYA YÖNELİK HAZIRLIK DÜZEYLERİ

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Özet: Bu çalışmanın amacı, kapsayıcı materyal tasarlamadaki teori ve pratik arasındaki kopukluğu ve üçüncü ile dördüncü sınıf İngilizce Öğretmenliği (ELT) adaylarının Özel Eğitim Gereksinimi (SEN) olan öğrenciler için materyal tasarlama konusundaki hazırlık düzeylerine dair algılarını incelemektir. Veriler, 50 ELT adayına uygulanan bir anket aracılığıyla toplanmıştır. Anket üç ana alana odaklanmıştır: SEN öğrencileriyle çalışmaya yönelik psikolojik hazırlık, lisans müfredatına yönelik eleştiriler ve çeşitli çok modlu (multimodal) ile dokunsal/dijital materyallerle çalışma isteği. Bulgular, adayların geleneksel materyalleri görme engelli veya disleksili öğrencilerin kullanımına nasıl uyarlayacakları konusundaki bilgilerinde ciddi bir boşluk olduğunu göstermektedir. Ankete katılan 50 adayın %100'ü SEN öğrencilerinin sayısında bir artış olduğunu kabul etmiş, ancak %76'sı geleneksel materyalleri görme engelli veya disleksili öğrencilerin kullanımına uyarlamak için yeterli beceriye sahip olmadıklarını belirtmiştir. Pratik beceri eksikliği katılımcılar arasında ortak bir temadır ve büyük olasılıkla mevcut derslerin, gerekli düzeyde pedagojik özgüven ve beceri gelişimini sağlamayan teorik vurgusundan kaynaklanmaktadır. Katılımcıların %64'ü çok modlu ve sanatsal pedagojiyi tercih ettiklerini belirtmelerine rağmen, pratik anlamda yararlanabilecekleri somut çok modlu veya sanatsal pedagoji örneklerine sahip değildirler. Bu nedenle, bu bulgular öğretmen eğitiminde yapısal bir reform ihtiyacına ve yalnızca geleneksel anlatıma dayalı derslere bağlı kalmak yerine, kapsayıcı materyallerin geliştirilmesi için zorunlu tasarım laboratuvarlarının sağlanmasının önemine dair güçlü kanıtlar sunmaktadır.

Anahtar Kelimeler: Kapsayıcı Eğitim, ELT Öğretmen Adayları, Materyal Tasarımı, Özel Eğitim Gereksinimi (SEN), Multimodal Öğrenme, Teori-Pratik Boşluğu

Introduction

Preparing pre-service English language teachers for inclusive classrooms remains a major challenge in teacher education, as over the course of four years, substantial knowledge in linguistics, methodology, and literature is acquired; however, practicum experiences reveal that theoretical understanding of a given condition and the practical ability to adapt classroom materials to meet the needs of students with that condition are considerably distinct competencies. For instance, while deficits such as dyslexia and difficulties with visual information processing can be articulated in academic terms, the implications of this knowledge for designing accessible materials are not always readily apparent. This disconnection between theory and practice constitutes the central motivation of the present study (Florian & Linklater, 2010; Sharma & Sokal, 2015).

This theoretical-practical divide is particularly significant in the Turkish context, where classroom diversity has been steadily increasing. In 2018, the Special Education Services Regulation (MEB, 2018) was revised to officially extend the number of children with special educational needs who will be mainstreamed. In this way, inclusiveness is no longer an exception but a daily reality in class. This is also in acknowledgment of the principles of the UNESCO Salamanca Statement (1994), which states that inclusive education should be seen as a fundamental part of quality education (Avramidis & Norwich, 2002).

This is particularly significant in the teaching of English. According to Krashen (1982), a learner's mood directly affects language acquisition. If students are anxious, excluded from the classroom/group, or fail to follow materials in the classroom, they will have difficulty learning a language. These difficulties can be particularly problematic for students with learning differences. This means, therefore, that what teachers do in the classroom is very important.

Relevant theoretical frameworks are well established in the literature. Florian (2014) argues that inclusive pedagogy means designing from the start for everyone, not patching things up for certain students after the lesson has already failed them. Tomlinson (2014) lays out a model of differentiated instruction that asks teachers to vary what they teach, how they teach it, and what

evidence of learning looks like. Rose and Meyer (2002) built Universal Design for Learning around the idea that students need multiple means of engagement, representation, and expression. These are not fringe ideas. They are cited in education textbooks across Turkey and the world.

While understanding a framework is one thing, working with it is another. The second type of knowledge, called "Pedagogical Content Knowledge" by Shulman (1986), is the ability to teach content knowledge to a particular person at a particular time. What he said was that content knowledge and pedagogical content knowledge are separate, and you can't simply assume that the former automatically generates the latter. Therefore, the present study concerns precisely this distinction. The first type of knowledge appears to be being made reasonably well by ELT programs. The second, not so much. This research seeks to determine how widespread that gap is among pre-service teachers nearing the end of their training.

Materials and Methods

To investigate participants' experiences, attitudes, and opinions, a four-part online questionnaire was developed and administered to students in the ELT department of a state university in Turkey. Participation was voluntary, and responses were collected anonymously. The first section collected background data, including the year of study and participants' prior experience with any SEN students in any educational setting. The second section used a five-point Likert scale to assess participants' self-ratings of the theory-practice balance of the special education courses they had taken. The third section provided specific scenarios, such as imagining a visually impaired student in one's class or a student with dyslexia who cannot undertake the standard reading task, and asked participants to rate their level of confidence. The fourth section was open-ended and invited participants to reflect on what they wished had been taught during their undergraduate education.

Fifty students participated in this study; 20 in their third year and 30 in their fourth year. This cohort was selected deliberately, as students at this stage have completed the bulk of their training and are approaching graduation, making questions of readiness both timely and practically relevant. It should be acknowledged that a sample of 50 students from a single institution does not constitute a basis for sweeping generalizations about ELT programs across Turkey. This study is exploratory in nature, intended as an initial investigation of the problem rather than a definitive measurement of it (Fraenkel et al., 2012). The data were analyzed using frequency counts and percentages.

Results and Discussion

The findings can be grouped into three main areas: participants' previous experiences, their opinions about undergraduate training, and their confidence when dealing with specific classroom situations.

Table 1. *Pre-Service ELT Teachers' Experience and Awareness Regarding Special Education*

Variable	Category	Frequency (f)	Percentage (%)
Grade Level	Junior (3rd Year)	20	40%
	Senior (4th Year)	30	60%
Experience Working with a SEN Student	Yes	22	44%
	No	28	56%
Taking a Special Education / Inclusion Course	Never Taken	18	36%
	Taken at Least 1 Course	32	64%

Awareness of Increasing SEN Student Population	Yes	50	100%
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The first highlight of Table 1 is the percentage of individuals who were 100% aware that SEN students are increasingly represented in mainstream classrooms. It is noted that although all participants were aware of this issue, the rest of the survey data indicate that awareness does not always mean being prepared to work effectively with SEN students. The majority of participants (56%) reported never having had the opportunity to work directly with an SEN student in a supervised setting. Additionally, at least 36% of participants indicated they have not taken any special education or inclusion courses; thus, even with participants nearing graduation, many unanswered questions remain about how teacher preparation programs adequately train future teachers to work with SEN students.

Table 2. *Perceptions of the Theory-Practice Balance in Undergraduate Education*

Survey Statement (5-point Likert)	Disagree (1–2)	Undecided (3)	Agree (4–5)
The special education courses taken were predominantly theoretical (definitions, legal rights) rather than practically applicable.	4% (f=2)	4% (f=2)	92% (f=46)
The undergraduate education equipped candidates with the skills to adapt materials for a student with special needs.	64% (f=32)	16% (f=8)	20% (f=10)

A considerable majority (92%) of respondents indicated that their special education training was mostly theoretical; they learned about the labels for each condition, how to define these conditions, and the legal framework. This is indeed valuable information. However, such knowledge does not readily address the practical questions that arise during practicum experiences; for instance, what constitutes the most effective approach to delivering a vocabulary lesson to a specific student with particular learning needs. Shulman (1986) describes this dilemma as a gap between content knowledge and pedagogical content knowledge; the size of this gap is evident in Table 2. Of the participants surveyed, 64% reported that their degree did not prepare them to modify materials to accommodate students with special educational needs. Therefore, this represents a very significant finding.

Table 3. *Self-Efficacy and Material Preferences for Specific SEN Cases*

Scenario / Preference	Feeling Inadequate (1–2 Points)	Feeling Adequate (4–5 Points)
Self-confidence to instantly adapt the coursebook for a Visually Impaired student	76% (f=38)	4% (f=2)
Readiness to use digital tools (e.g. Wordwall) for a Dyslexic student	76% (f=38)	4% (f=2)
Preferred alternative methods for Visually Impaired students	Rhythmic / Auditory Games: 80% Audio Description: 68%	

The findings related to Table 3 are particularly striking. The scenario items required participants to engage with authentic situations: “A visually impaired student is in your class right now; what do you do? A student with dyslexia is unable to access the current reading task; what is your first course of action?” Seventy-six percent of respondents indicated they would not know what to do in such situations. This figure (76% inadequate, 4% adequate) remained consistent for both digital tools and dyslexia-related scenarios. These are not extreme edge cases. Rather, they represent everyday challenges that all teachers in inclusive Turkish classrooms are likely to encounter.

A particularly noteworthy finding emerged from the bottom row of Table 3. When participants were asked about preferred approaches rather than current ability, the results shifted considerably. Eighty percent of participants said they would reach for rhythmic and auditory games for visually impaired learners; 68% named audio description. Separately, in the open-ended section, 64% said they believe artistic activities are highly effective in inclusive language teaching. These responses indicate that an underlying pedagogical instinct is already present among participants. Vygotsky (1978) described this as the Zone of Proximal Development; the skill is within reach, but only with proper scaffolding and guided practice. What is missing is not the motivation or the basic orientation. It is a structured opportunity to actually develop these ideas into usable classroom skills.

The open-ended responses reinforced this reading. The phrase that appeared most frequently, in various forms, reflected a sentiment such as: “I am afraid of not knowing what to do.” This fear is entirely consistent with the data. Florian (2014) is clear that inclusive pedagogy is not improvised in the moment; it is deliberately designed in advance so that every student can access learning from the start. Tomlinson (2014) and Rose and Meyer (2002) both assume teachers have had real practice building the kinds of activities those frameworks require. Most participants in this study have not had that practice. The curriculum has prioritized knowledge of inclusion over the practical ability to implement it.

Several concrete recommendations can be drawn from these findings. First, ELT programs should integrate hands-on material design workshops as a core component of the degree rather than as an optional supplement. Such workshops would provide students with the opportunity to build differentiated materials, receive feedback, and iteratively revise their work. Approximately 80% of participants expressed a desire for this type of practical training. Second, there is a need to reconsider how assessment is conducted in special education courses. Asking a candidate on a multiple-choice test to define autism spectrum disorder is a fundamentally different task from asking that candidate to adapt a lesson for a student on the spectrum. Scenario-based assessment tasks would provide a more accurate and meaningful measure of whether a candidate is genuinely prepared for inclusive classroom practice. Third, a formal coordination between ELT and Special Education departments is warranted. Guest instruction from Special Education specialists – covering areas such as tactile design, task structuring for students with attention regulation difficulties, and the practical application of assistive tools – would equip ELT candidates with knowledge beyond the typical expertise of ELT faculty (Armstrong, 2003).

Conclusion

These results are reasonably consistent for all three tables. The graduates of pre-service ELT programs in Turkey possess adequate theoretical knowledge of special educational needs, but their practical, classroom-level competencies remain considerably underdeveloped. 100% of participants

were aware of the increasing number of SEN pupils in mainstream schools. This is not the case for practical readiness: 76% reported feeling unprepared to respond in the moment, and 92% described their training as predominantly theoretical.

This study has definite limitations. The 50 students from this institution represent a relatively small sample of ELT programs in the country, and the results should not be interpreted as representative. It would be much richer to have larger studies from multiple universities and qualitative interviews, ideally followed by some follow-up with graduates when they begin their teaching positions.

Nevertheless, the central concern raised by this study remains compelling. Pre-service ELT graduates are being prepared with an awareness of inclusion, but without the practical competencies required to act on it. This is not a theoretical shortcoming – it is a curriculum design problem, and, as such, it is amenable to structural solutions. The participants in this study have already demonstrated the orientation and motivation necessary for inclusive practice. What remains is the provision of the structured, practical training that would allow those dispositions to be translated into effective classroom action.

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A COMPARATIVE ANALYSIS OF INTERNSHIP AND PRIVATE TUTORING EXPERIENCES AMONG PRE-SERVICE TEACHERS

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Abstract: Internships are typically conducted in structured, school-based settings, whereas private tutoring offers a more flexible, informal instructional context. In this perspective, the study compares internship and private tutoring experiences among pre-service teachers, focusing on their contributions to professional development. The research examines how these two settings differ in terms of classroom management, teacher autonomy, and teaching competence. Data were collected through a questionnaire consisting of Likert-type items and open-ended reflection questions, administered to 46 pre-service teachers in the ELT department of a state university in Türkiye who have experience in both contexts. A mixed method was used. The quantitative findings indicate that school internships are more effective for professional development than private tutoring. Qualitative findings revealed that while private tutoring provides a comfortable teaching environment, school internships present more professional challenges, particularly in terms of classroom management. Overall, the research highlights the importance of realizing the different roles of internship and private tutoring experiences in shaping the careers of pre-service teachers.

Keywords: professional development, classroom management, teacher autonomy, pre-service teachers, teaching environment, internship, private tutoring

ÖĞRETMEN ADAYLARI ARASINDA STAJ VE ÖZEL DERS DENEYİMLERİNİN KARŞILAŞTIRMALI ANALİZİ

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Özet: Stajlar genellikle yapılandırılmış, okul temelli ortamlarda resmi bir formatta gerçekleştirilirken, özel dersler daha esnek ve gayriresmi bir öğretim ortamı sunar. Bu bakış açısıyla, bu çalışma öğretmen adaylarının staj ve özel ders deneyimlerini karşılaştırarak, bunların mesleki gelişime katkısına odaklanmaktadır. Araştırma, bu iki ortamın sınıf yönetimi, öğretmen özerkliği ve öğretim yeterliliği açısından nasıl farklılık gösterdiğini incelemektedir. Veriler, her iki ortamda da deneyimi olan Türkiye’deki bir devlet üniversitesinin İngilizce Öğretmenliği bölümündeki 46 öğretmen adayına uygulanan, Likert tipi

maddelerden ve açık uçlu yorum sorularından oluşan bir anket aracılığıyla toplanmıştır. Araştırmada karma bir yöntem kullanılmıştır. Nicel bulgular, okul stajlarının mesleki gelişim açısından özel derslerden daha etkili olduğunu göstermektedir. Nitel bulgular ise, özel derslerin rahat bir öğretim ortamı sağlarken, okul stajlarının özellikle sınıf yönetimi açısından daha fazla zorluklar açığa çıkardığını ortaya koymuştur. Genel olarak araştırma, öğretmen adaylarının kariyerlerini şekillendirmede staj ve özel ders deneyimlerinin kendine özgü rollerinin farkına varılmasının önemini vurgulamaktadır.

Anahtar kelimeler: mesleki gelişim, sınıf yönetimi, öğretmen özerkliği, öğretim ortamı, öğretmen adayları, staj, özel ders

Introduction

In today's fast-paced world, becoming a teacher is a long-term process in which theoretical knowledge turns into practical application (Veenman, 1987). Particularly in the field of ELT, this practical experience is gained through two different procedures: internship and private tutoring. While the internship is a more structured course guided by a mentor in the teacher education curriculum, private tutoring is a more informal way for pre-service teachers in Türkiye to gain early instructional experience. However, there is a clear distinction between these two opponents in terms of professional development.

Formerly, internships were the main part of teacher education programs, especially for pre-service teachers (Köksal & Genç, 2019). This is also supported by Uysal and Savaş (2021), who stated that pre-service teachers can translate their teaching knowledge into instructional practice during internships in real classrooms. Alongside translating theory into practice, pre-service teachers' motivation during internships also plays a crucial role in designing lesson plans and keeping students engaged both in and out of class time.

As a second issue, private tutoring, significantly in Türkiye, is a common way of practicing teaching even among students who are not in the final years of their educational career, as Linh and Phuong (2025) underscored that private tutoring represents the beginning of their professional identity production. In contrast to a structured, formal classroom environment, private tutoring provides pre-service teachers with an informal, individualized instructional setting (Mischo & Haag, 2002).

In light of the issues discussed above, this study aims to investigate the effectiveness of these two settings and determine which contributes more significantly to the professional identity of future educators. Specifically, the study addresses the following research questions:

1. How do pre-service teachers compare the effectiveness of their internships and their private tutoring experiences in terms of teacher development?
2. In which teaching context do pre-service teachers encounter more professional challenges?
3. How do internships and private tutoring experiences influence the teaching motivation of future teachers towards their careers?

By examining the differences and similarities across both settings, the study and its findings may offer valuable insights for teacher educators and curriculum designers to better understand how different teaching contexts shape pre-service teachers' professional identity.

1. METHODOLOGY

1.1. Research Design and Context

This study employed a mixed-method design to provide a comprehensive understanding of pre-service teachers' professional experiences. A comparative study was conducted to analyze the similarities and differences between internships and private tutoring. Data were collected with a

questionnaire including two sections: nine Likert-type scale items and two open-ended questions. The main objective was to analyze the differences between internships and private tutoring to better understand how different teaching experiences shape the professional development of pre-service teachers.

The study was conducted at the English Language Teaching (ELT) Department of a state university in Türkiye. The research context includes two teaching environments: the formal school internship (practicum) and private tutoring. By comparing these two different experiences, it was intended to explore how these environments influence the professional development of future educators.

1.2. Participants

The participants consisted of 46 pre-service teachers in the ELT department. All participants were in their final year (4th year) of study, meaning they were actively involved in their school internships during the research period. The most important criterion for participating in this study was having experience in both professional settings.

1.3. Data Collection Instruments

Data were collected using a structured questionnaire to explore how pre-service teachers compare their internships with private tutoring experiences. The instrument consisted of three sections:

1. Perceptions of Internship: This section included five items measured on a five –point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The items aimed to measure how teacher candidates perceived their pedagogical development, classroom management skills, and self- confidence.
2. Perceptions of Private Tutoring: This section included four items measured on a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The items aimed to explore the participants' self-confidence and teaching autonomy in an informal environment.
3. Open-Ended Questions: To complement the quantitative data, open-ended questions were included to provide qualitative insights into pre-service teachers' experiences.

1.4. Validity and Reliability

For the quantitative section, the survey items were refined based on informal expert feedback to ensure that each question was clear and directly related to the research goals. The questionnaire was also shared with a small group of peers for a preliminary check. For the qualitative data, reliability was maintained through thematic consistency.

1.5. Data Collection Procedure and Ethical Considerations

The data were collected through an online survey platform. Participants were informed about the study's purpose. Confidentiality and anonymity were assured. The survey required approximately 10-15 minutes to complete. Ethical approval was obtained from the relevant institutional authority prior to data collection. Data were used solely for academic purposes. Participation was voluntary, and respondents had the right to withdraw from the study at any stage without penalty.

2. RESULTS

2.1. Overview of the Findings

The main purpose of this analysis is to evaluate the participants' perceptions of their professional development by comparing their internship and private tutoring experiences. In the following section, the quantitative results are presented first, followed by the qualitative analysis of

participants' reflections. The quantitative data were analyzed by using *IBM SPSS Statistics* to calculate descriptive statistics and Cronbach's alpha.

Table 1. Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Internship improves my classroom management skills.	46	1	5	4,02	1,064
An internship helps me improve my lesson planning skills.	46	1	5	3,93	1,200
Internship improves my communication with students.	46	1	5	4,26	,953
An internship helps me gain real teaching experience.	46	1	5	4,33	,920
Internship increases my confidence as a teacher.	46	1	5	4,11	,971
Private tutoring improves my teaching skills.	46	1	5	3,89	1,197
Private tutoring increases my confidence.	46	1	5	3,96	1,192
Private tutoring helps me better understand students.	46	1	5	3,89	1,140
Private tutoring contributes to my professional development.	46	1	5	4,00	1,193
Valid N (listwise)	46				

The results indicate that internship (M=4.13) is viewed as more effective for professional development, motivation, and communication than private tutoring (M=3.94). This difference suggests that authentic classroom environments provide a better source for professional identity

Table2. Reliability Statistics

Cronbach's Alpha	N of Items
,987	9

The reliability analysis indicated that the scale has high internal consistency. (Cronbach's alpha =.987)

Table 3 Paired Sample Test.

Paired Samples Test									
		Paired Differences				Significance			
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
					Lower	Upper	t	df	
Pair 1	ints_mean - pt_mean	,19565	,27665	,04079	,11350	,27781	4,797	45	
									One-Sided p Two-Sided p
									<,001 <,001

The paired-samples t-test indicated a statistically significant difference between the internship and private tutoring scores. ($p < .001$)

3.2 Findings from Open-Ended Questions

2.2. 1. Which experience do you find more beneficial for your Professional development? Why?

Most participants agreed that an internship is more effective for professional development than private tutoring, because a school internship provides an authentic classroom environment. Related to the topic, some of the notes were written by participants:

'Internship. Because it is in an actual class. More reflective of education.' (participant-5)

'I think an internship is better. Because internships are more realistic experiences and are more likely to be our job.' (participant-22)

'Our internship is more beneficial because it is the exact experience we will have in the future if we ever get assigned to a public school.' (participant-43)

3.2.2. What challenges did you face in both experiences?

Regarding the challenges of teaching, most pre-teachers identified classroom management as the biggest problem in internships, and preparing lesson plans and being ready for class (motivation) as problems in private tutoring. Some of the notes were written by participants:

'In an internship, sometimes getting the attention of all the students is hard (class management), but in private tutoring, students' motivation is very important. It was hard to make my student motivated sometimes.' (participant-19)

'In private tutoring, planning the whole lesson was difficult for me because it was something new to me, and in the internship, trying to get students' attention was kind of difficult because they didn't want to attend or were distracted easily.' (participant-23)

4. Discussion and Conclusion

The main focus of this study was to compare the effects of internship and private tutoring experiences on pre-service teachers. The results presented that while both experiences were beneficial, the internship was rated higher than private tutoring in terms of mean scores.

As Rao and Rao (2004) underline, internship shapes a teacher's professional identity. The real classroom environment and student activities during the internship have an observable effect on pre-service teachers' professional development. As Darling-Hammond (2006) referred, practicing in an authentic school environment is essential for teacher development. However, classroom management remains a significant challenge for them throughout the internship. This idea aligns with the study by Freeman et al. (2014), which found that many pre-service teachers do not receive adequate training in classroom management, leading them to feel frustrated during their early experiences. On the other hand, pre-service teachers often face low student motivation, low self-motivation, and an intensive lesson-planning process in private tutoring. Supporting this idea, Han and Yin (2016) highlighted that teacher motivation plays a crucial role in student motivation and psychological well-being.

The existing literature indicates that private tutoring is viewed as 'Shadow Education' (Bray, 2007). In this study, the mean score for private tutoring suggests that it is perceived differently from an internship, as it serves as a 'shadow' of formal education. Without a preset curriculum, pre-service teachers often struggle to create new lesson plans for each lesson, which requires intensive preparation. This often leads to low motivation during the private tutoring sessions.

Despite its contributions, the study opens several doors for future academic research. First of all, since this study focused on a specific group of 46 participants, future studies may involve a larger, more diverse sample to yield different results. Second, future researchers can conduct an observational study to report the findings with or without using online platforms.

In conclusion, this research demonstrates that both internships and private tutoring are parallel paths to the professional journey of pre-service teachers. The study shows that an internship is the primary route for testing teacher identity and profession; private tutoring is a parallel but not identical path for pre-service teachers, where they generally first experience and develop their

teaching style. Finally, recognizing the value of both formal and informal practices will lead to more competent and professional educators.

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TEACHER EDUCATION AND PROFESSIONAL DEVELOPMENT

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Abstract: Teacher education and professional development are fundamental pillars of quality education. Teacher education encompasses the formal preparation of individuals to enter the teaching profession, including pre-service training, theoretical foundations, subject knowledge, and initial practical experience. Professional development, on the other hand, refers to the ongoing learning and growth teachers engage in throughout their careers to enhance their instructional practices, pedagogical understanding, and professional competencies. Effective teacher education and development aim not only to equip educators with subject expertise but also to cultivate critical thinking, problem-solving skills, and the ability to adapt to diverse student needs and evolving educational contexts.

Keywords: Teacher Education, Professional Development, Pedagogical Skills, Methodological Competencies, Continuous Professional Development (CPD), Teacher Reflection, Modern Approaches in Teaching, Classroom Evaluation, Quality Assurance in Education, Instructional Effectiveness

MÜƏLLİM TƏHSİLİ VƏ PEŞƏKAR İNKİŞAFI

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Xülasə: Müəllim təhsili və peşəkar inkişafı keyfiyyətli təhsilin əsas sütunlarıdır. Müəllim təhsili fərdlərin müəllimlik peşəsinə daxil olmaq üçün rəsmi hazırlığını, o cümlədən işə qəbuldan əvvəlki təlimi, nəzəri əsasları, fənn biliklərini və ilkin praktik təcrübəni əhatə edir. Digər tərəfdən, peşəkar inkişaf müəllimlərin tədris təcrübələrini, pedaqoji anlayışlarını və peşəkar səriştələrini artırmaq üçün karyeraları boyunca iştirak etdikləri davamlı öyrənmə və inkişafa aiddir. Effektiv müəllim təhsili və inkişafı yalnız müəllimlərin fənn təcrübəsi ilə təchiz etməyi deyil, həm də tənqidi düşüncə, problem həll etmə bacarıqlarını və müxtəlif tələbə ehtiyaclarına və inkişaf edən təhsil kontekstlərinə uyğunlaşma qabiliyyətini inkişaf etdirməyi hədəfləyir.

Açar sözlər: Müəllim təhsili, Peşəkar inkişaf, Pedaqoji bacarıqlar, Metodoloji bacarıqlar, Davamlı Peşəkar İnkişaf (DPI), Müəllim düşüncəsi, Tədrisdə Müasir Yanaşmalar, Sınıf Qiymətləndirməsi, Təhsildə Keyfiyyət Təminatı, Tədris Effektivliyi

Introduction

In the rapidly evolving landscape of the 21st century, education stands as the primary vehicle for societal progress and individual empowerment. At the heart of this system is the teacher, whose influence extends far beyond the dissemination of knowledge. As educational demands shift toward fostering critical thinking, digital literacy, and emotional intelligence, the preparation and ongoing growth of educators have become more critical than ever. Consequently, **Teacher Education** and **Professional Development** are no longer viewed as static stages of a career but as a continuous, lifelong journey of adaptation and refinement. The transition from a "traditional" instructor to a "facilitator of learning" requires a robust foundation laid during pre-service training, followed by a persistent commitment to **Continuous Professional Development (CPD)**. Modern teaching environments demand that educators master not only their subject matter but also complex pedagogical skills, classroom management strategies, and the ability to integrate emerging technologies.

This article explores the multifaceted nature of teacher growth, examining modern approaches such as blended learning and competency-based frameworks. Furthermore, it highlights the vital role of **reflective practice** and collaborative learning communities in bridging the gap between theoretical knowledge and classroom reality. By analyzing the mechanisms of pedagogical skill development and the importance of quality assurance systems, we aim to provide a comprehensive overview of how effective teacher education contributes to instructional excellence and, ultimately, superior student outcomes.

Materials and Methods

This study employs a **qualitative research design** based on a comprehensive **systematic literature review** and **descriptive analysis**. The methodology is structured to evaluate the transition from traditional teaching models to modern, reflective pedagogical frameworks. The following components outline the approach used in this research Brown, H. D. (2007)

1. Data Collection and Sources The primary materials for this research consist of academic publications, contemporary pedagogical theories, and established educational frameworks. Key sources include:

- Seminal works on language teaching (e.g., H. Douglas Brown, Jeremy Harmer).
- Reflective models, specifically **Gibbs' Reflective Cycle**, to analyze teacher development processes.
- Recent educational reports on **Continuous Professional Development (CPD)** and quality assurance standards.

2. Conceptual Framework The study integrates several theoretical perspectives to provide a holistic view of teacher education:

- **The Competency-Based Approach:** Used to identify the specific skills (methodological and digital) required for 21st-century educators.
- **Reflective Practice Theory:** Applied to examine the role of self-evaluation and "action research" in improving classroom outcomes.
- **Analytical Procedure** The analysis was conducted through a multi-stage process:
- **Comparative Analysis:** Comparing traditional face-to-face teacher training with modern **Blended Learning** and **Problem-Based Learning (PBL)** models.
- **Synthesis of Evaluation Methods:** Assessing the effectiveness of various performance measurement tools, including **360-Degree Feedback** and **Key Performance Indicators (KPIs)**.
- **Practical Application:** Evaluating the impact of microteaching and simulation techniques on the development of pedagogical skills.

4. Scope and Limitations While the research focuses on general principles of teacher education and professional development, particular emphasis is placed on modern approaches suitable for higher education and foreign language departments, reflecting the current global trends in pedagogical evolution.

Discussion: Contemporary approaches to teacher education emphasize learner-centered, collaborative, and reflective practices. Key trends include:

- **Blended Learning:** Integrating traditional face-to-face instruction with online resources allows teachers to access flexible, self-paced professional development.
- **Problem-Based Learning (PBL):** Encouraging teachers to solve real classroom challenges fosters practical application of pedagogical theories.

- **Collaborative Learning Communities:** Peer collaboration, mentoring, and professional networks facilitate sharing of best practices and collective problem-solving.
- **Digital Competency Development:** Preparing teachers to effectively integrate technology into teaching enhances both student engagement and learning outcomes.
- **Competency-Based Frameworks:** Modern programs focus on measurable competencies, ensuring teachers acquire the specific skills required for effective instruction. (Richards, J. C., & Farrell, T. S. C. (2005))

Pedagogical skills and methodological competencies are central to effective teaching. They include lesson planning, classroom management, instructional strategies, assessment techniques, and the ability to differentiate instruction based on student needs.

To develop these competencies, teacher education programs often use:

- **Microteaching:** Teachers practice teaching small segments and receive structured feedback.
- **Simulations and Role-Playing:** Allow exploration of classroom scenarios and student interactions in a controlled environment.
- **Action Research:** Teachers conduct research on their own practices to improve classroom outcomes.
- **Reflective Journals:** Encourage teachers to analyze and improve their instructional strategies continuously. The journey of a teacher does not end with a degree. **Continuous Professional Development (CPD)** is the lifeblood of a successful career.

Reflection is the bridge between experience and learning. According to the **Gibbs' Reflective Cycle**, teachers must analyze their classroom experiences to improve future performance.

- **Reflective Journals:** Keeping a record of what worked and what didn't in a lesson.
- **Action Research:** Teachers conduct small-scale studies within their own classrooms to solve specific pedagogical problems.
- **Professional Learning Communities (PLCs):** Teachers meet to share best practices, analyze student work, and observe one another's lessons.
- **Mentorship:** Pairing novice teachers with veterans ensures that "tacit knowledge" (the unwritten rules of teaching) is passed down.

To maintain high standards, teaching performance must be measured through a transparent and constructive quality assurance system. (Alderson, J. C. (2005))

Conclusion

The journey from a novice educator to a proficient teaching professional is an ongoing process that necessitates a synergy between foundational teacher education and continuous professional development. As explored in this article, modern pedagogical landscapes require more than just subject expertise; they demand a high level of methodological competency, digital literacy, and the ability to adapt to diverse classroom dynamics.

The integration of learner-centered approaches such as **Problem-Based Learning** and **Blended Learning** has proven essential in fostering an environment where both teachers and students can thrive. Furthermore, the shift toward **Reflective Practice**, guided by frameworks like Gibbs' Reflective Cycle, allows educators to transform daily challenges into opportunities for growth. Collaborative efforts within Professional Learning Communities and mentorship programs further ensure that pedagogical excellence is shared and sustained across the academic community.

In conclusion, ensuring quality in education depends heavily on the robustness of teacher evaluation and the commitment to **Continuous Professional Development (CPD)**. By adopting transparent quality assurance systems like **360-degree feedback** and focusing on formative growth, educational institutions can empower teachers to reach their full potential. Ultimately, an investment in teacher education is an investment in the future of the educational system, directly impacting student engagement and academic success.

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DEVELOPMENT OF PEDAGOGICAL SKILLS AND METHODOLOGICAL COMPETENCIES IN THE MODERN EDUCATION SYSTEM

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Abstract: This article explores the significance of pedagogical skills and methodological competencies within the contemporary education system and investigates ways to develop them. In an era shaped by globalization and digitalization, the teacher's role has shifted from simply delivering knowledge to facilitating and organizing the learning process. In this context, strengthening pedagogical skills—especially communicative, organizational, empathetic, and reflective abilities—has become increasingly important. Strong teacher–student interaction improves the quality of the learning environment and encourages active student participation. The article also offers a detailed overview of methodological competencies. A teacher's capacity to design effective lessons, choose suitable methods and tools, and evaluate learning outcomes

objectively is seen as a key determinant of teaching effectiveness. The use of innovative instructional approaches, such as project-based, problem-based, and cooperative learning, supports the development of students' critical thinking and problem-solving abilities. Additionally, integrating information and communication technologies into education broadens teachers' methodological capabilities and enhances the accessibility and interactivity of learning. Ongoing professional development, active engagement in reflective practice, and the adoption of modern pedagogical strategies are considered essential in this area. Educational reforms in Azerbaijan, particularly curriculum innovations, also contribute significantly to the advancement of pedagogical skills and methodological competencies. In conclusion, the continuous professional growth of teachers and their mastery of modern skills play a vital role in improving educational quality and strengthening the intellectual capacity of society.

Keywords: pedagogical skills, methodological competencies, teacher development, teaching methods, innovative education, curriculum reforms, digital technologies

MÜASİR TƏHSİL SİSTEMİNDƏ PEDAQOJİ BACARIQLARIN VƏ METODOLOJİ KOMPETENSIYALARIN İNKİŞAFI

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Xülasə: Bu məqalədə pedaqoji bacarıqların və metodiki kompetensiyaların müasir təhsil sistemindəki əhəmiyyəti və onların inkişaf istiqamətləri araşdırılır. Qloballaşma və rəqəmsallaşma dövründə müəllimin rolu dəyişərək yalnız bilik verən şəxs olmaqdan çıxmış, daha çox öyrənmə prosesini yönləndirən və təşkil edən fasilitatora çevrilmişdir. Bu kontekstdə pedaqoji bacarıqların, xüsusilə kommunikativ, təşkilati, empatik və reflektiv qabiliyyətlərin inkişafı böyük əhəmiyyət daşıyır. Müəllim və şagird arasında səmərəli ünsiyyət təlim mühitinin keyfiyyətini yüksəldir və şagirdlərin fəal iştirakını stimullaşdırır. Məqalədə metodiki kompetensiyaların məzmunu da ətraflı şəkildə şərh olunur. Müəllimin dərsi düzgün qurması, uyğun metod və vasitələri seçməsi, həmçinin təlim nəticələrini obyektiv şəkildə qiymətləndirməsi tədrisin səmərəliliyini müəyyən edən əsas göstəricilər hesab edilir. Layihə əsaslı, problem yönümlü və kooperativ öyrənmə kimi innovativ təlim yanaşmalarının tətbiqi şagirdlərdə tənqidi düşünmə və problem həll etmə bacarıqlarının inkişafına töhfə verir. Eyni zamanda, informasiya-kommunikasiya texnologiyalarının təhsilə tətbiqi müəllimlərin metodiki imkanlarını genişləndirir və təlim prosesini daha əlçatan və interaktiv edir. Müəllimlərin fasiləsiz peşəkar inkişafı, reflektiv fəaliyyətə diqqət yetirməsi və müasir pedaqoji yanaşmaları mənimsəməsi bu sahədə əsas şərtlər kimi qiymətləndirilir. Azərbaycan təhsil sistemində aparılan kurikulum islahatları da pedaqoji və metodiki kompetensiyaların inkişafında mühüm rol oynayır. Nəticə olaraq, müəllimlərin peşəkar inkişafı və müasir bacarıqlara yiyələnməsi təhsilin keyfiyyətinin artırılmasında və cəmiyyətin intellektual potensialının formalaşmasında həlledici əhəmiyyətə malikdir.

Açar sözlər: pedaqoji bacarıqlar, metodiki kompetensiyalar, müəllim inkişafı, təlim metodları, innovativ təhsil, kurikulum islahatları, rəqəmsal texnologiyalar

Introduction

Education plays a crucial role in shaping the intellectual, cultural, and socio-economic progress of societies. In the 21st century, rapid developments in science and technology, combined with the effects of globalization, have significantly altered the way knowledge is produced and acquired. These transformations have raised new expectations for educational systems, especially

regarding the quality of teaching and the effectiveness of learning outcomes. Consequently, teachers are now expected to demonstrate a broad set of competencies that extend beyond mere subject expertise.

The modern education system is experiencing significant changes driven by technological innovation, evolving labor market needs, and increasingly complex social dynamics. In this context, effective teaching is no longer defined solely by a teacher's mastery of content, but also by their ability to employ flexible pedagogical strategies and adapt instruction to various learning settings. Therefore, the development of pedagogical and methodological competencies has become a key priority in contemporary teacher education.

Materials and Methods

This study is based on a qualitative analysis of contemporary educational theories, pedagogical research, and policy documents. Various academic sources, including studies such as *Modern Approaches in Pedagogy (2020)*, *Teaching Strategies and Methods (2019)*, and *Innovative Teaching Methods (2021)*, were reviewed to examine the development of pedagogical skills and methodological competencies.

The analysis focuses on key components such as communication, emotional intelligence, reflective practice, instructional design, the use of educational resources, and the assessment of student learning. A descriptive and analytical approach is used to evaluate modern teaching methodologies and their effectiveness. Additionally, examples from educational reforms, particularly in Azerbaijan, are considered to illustrate the practical implementation of these concepts.

Furthermore, a comparative approach is applied to identify similarities and differences between traditional and modern teaching practices. This allows for a deeper understanding of how pedagogical and methodological competencies have evolved over time. The study also incorporates elements of case analysis, focusing on real educational contexts to demonstrate how theoretical principles are applied in practice.

In addition, data is interpreted through a thematic analysis framework, where key themes such as learner-centered education, digital integration, and professional development are identified and examined. This method enables the study to systematically organize complex educational concepts and highlight their interconnections.

To ensure the reliability of the analysis, multiple sources are cross-referenced, and widely accepted academic literature is prioritized. The study does not rely on quantitative data collection but instead emphasizes conceptual clarity and theoretical consistency. This approach is particularly suitable for exploring the dynamic and multifaceted nature of modern education systems.

Results and Discussion

A defining feature of current educational practice is the move away from uniform teaching methods toward more adaptive and differentiated approaches. This transition requires educators to understand learners' cognitive, emotional, and motivational traits and to tailor their teaching strategies accordingly. Such adaptability is not an inherent trait but a professional skill developed through structured training, reflective practice, and hands-on experience.

Modern education emphasizes learner-centered approaches, where students actively participate in constructing knowledge rather than passively receiving information. In this context, pedagogical skills and methodological competencies have become essential components of effective

teaching. They enable teachers to create engaging learning environments, address diverse student needs, and foster critical thinking and creativity (Modern Approaches in Pedagogy, 2020).

Pedagogical skills are rooted in educational theories that highlight the importance of interaction, experience, and reflection in learning. These skills include communication, classroom management, emotional intelligence, and adaptability. Effective communication allows teachers to present complex ideas in a clear and understandable manner while encouraging student participation. Research indicates that a positive communication climate significantly enhances student motivation and engagement (Teaching Strategies and Methods, 2019).

Empathy and emotional intelligence are equally important, as they enable teachers to understand students' needs and respond appropriately. This is particularly relevant in inclusive education settings, where learners may have diverse abilities and backgrounds. By fostering a supportive and inclusive environment, teachers can ensure equal learning opportunities for all students (Modern Approaches in Pedagogy, 2020).

Another key aspect of pedagogical skills is reflective practice. Teachers who regularly analyze their teaching methods and outcomes are better equipped to improve their performance. Reflection allows educators to identify effective strategies, address challenges, and adapt to changing educational contexts (Teacher Professional Development Concept).

Methodological competencies involve the effective use of teaching strategies and techniques in classroom practice. They encompass skills such as lesson planning, instructional design, the use of educational resources, and the assessment of student learning. Well-structured lesson planning ensures that learning objectives are clearly articulated and consistently aligned with instructional activities and evaluation methods.

Modern teaching methodologies emphasize active learning and student engagement. Approaches such as project-based learning, problem-based learning, and cooperative learning encourage students to take responsibility for their own learning. These methods not only improve academic performance but also develop essential life skills such as collaboration, communication, and critical thinking (Innovative Teaching Methods, 2021).

Moreover, contemporary teaching increasingly emphasizes interdisciplinary thinking. Educators are expected to link concepts across different subjects, helping students perceive knowledge as a unified whole rather than as separate pieces. This approach demands methodological flexibility, as teachers need to design learning experiences that extend beyond traditional textbooks and promote critical and analytical thinking. The capacity to create interdisciplinary tasks demonstrates a high level of methodological competence, where teaching is approached as a creative and intellectually driven activity rather than a routine transfer of information.

Another important aspect of pedagogical development is the ability to sustain students' cognitive engagement. Effective learning environments are shaped not only by the delivery of information but also by the depth of intellectual interaction between teachers and learners. Educators must be able to create conditions that encourage inquiry, dialogue, and problem-solving. This requires both careful instructional planning and the ability to respond flexibly to students' ideas, guiding them toward deeper levels of understanding. Such responsiveness reflects a well-developed sense of pedagogical awareness and adaptability.

Assessment is another critical component of methodological competencies. Teachers must design assessment tools that accurately measure students' knowledge and skills. Both formative and

summative assessments play important roles in the learning process. Formative assessment, in particular, provides continuous feedback that helps students improve and supports personalized learning (Teaching Strategies and Methods, 2019).

The incorporation of digital technologies has become a key characteristic of contemporary education. ICT tools, including online learning platforms, interactive applications, and multimedia materials, enrich the teaching and learning process by making it more engaging and accessible. These technologies allow educators to deliver content in varied formats, address diverse learning preferences, and support remote education.

However, the effective use of digital technologies requires teachers to develop specific competencies. They must be able to select appropriate tools, integrate them into their teaching strategies, and ensure that technology supports learning objectives. The use of digital tools also promotes student autonomy and encourages independent learning (Innovative Teaching Methods, 2021).

Ongoing professional development is vital for teachers to stay effective in an ever-evolving educational landscape. Such programs offer opportunities to refresh subject knowledge, adopt innovative teaching approaches, and exchange experiences with peers. They typically involve activities such as workshops, seminars, training courses, and collaborative learning initiatives.

Reflective practice plays a central role in professional development. By critically evaluating their teaching experiences, teachers can identify areas for improvement and implement more effective strategies. Lifelong learning is therefore a fundamental principle for educators, enabling them to adapt to new challenges and maintain high standards of teaching (Teacher Professional Development Concept).

In Azerbaijan, educational reforms have been directed toward updating the curriculum and enhancing the quality of instruction. The adoption of competency-based curricula has moved the emphasis away from memorization toward the cultivation of practical abilities and critical thinking skills. These changes support the implementation of interactive teaching approaches and foster more student-centered learning environments.

Teacher certification and professional development initiatives have also been implemented to support these reforms. By enhancing teachers' pedagogical and methodological competencies, these measures aim to improve educational outcomes and align the education system with international standards (Curriculum Reforms and Implementation Issues, 2018).

Teacher development is strongly connected to collaboration and professional engagement. In today's educational settings, teaching is increasingly regarded as a shared responsibility rather than an isolated activity. Involvement in professional learning communities enables educators to share methodological practices, reflect on instructional challenges, and jointly design teaching strategies. This collaborative environment supports the ongoing improvement of pedagogical competencies and promotes the long-term sustainability of educational innovation.

At the same time, the changing role of teachers requires greater attention to the ethical and reflective aspects of their work. Educators are expected to critically assess the social and moral consequences of their instructional choices, particularly in culturally diverse classrooms. Reflective competence allows teachers to evaluate their own assumptions, identify potential biases, and refine their professional conduct in accordance with core educational principles.

Despite notable advancements, a number of obstacles continue to affect the development of pedagogical and methodological skills. These challenges include restricted access to educational

resources, limited opportunities for professional training, and resistance to innovation among some educators. Overcoming these issues calls for a holistic approach that engages policymakers, educational institutions, and teachers themselves.

Looking ahead, future trends in education are expected to emphasize digital literacy, individualized learning, and interdisciplinary teaching approaches even more strongly. As a result, teachers will need to continuously upgrade their skills and remain responsive to ongoing changes in the educational landscape.

Conclusion

In summary, strengthening pedagogical skills and methodological competencies is vital for the success of contemporary education systems. These abilities empower teachers to design dynamic learning environments, apply innovative instructional methods, and foster students' overall development. Sustained professional development, effective use of digital technologies, and supportive policy frameworks are essential for achieving these objectives.

The case of Azerbaijan illustrates that curriculum reforms and initiatives aimed at teacher development can substantially improve educational quality. By prioritizing the professional growth of educators, education systems can better equip learners to face modern challenges and contribute meaningfully to societal progress.

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REFLECTIVE PRACTICE IN TEACHER DEVELOPMENT: ENHANCING PROFESSIONAL GROWTH

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Abstract: Reflective practice is considered one of the key components of teachers' professional development and continuous improvement. It enables teachers to evaluate their instructional practices, identify strengths and weaknesses, and make informed decisions to improve the quality of teaching and learning. Through systematic reflection, teachers analyze classroom experiences, students' responses, and the effectiveness of teaching methods, allowing them to adapt their instructional strategies to meet learners' needs more effectively.

This study examines the significance of reflective practice in modern education and highlights its contribution to teachers' professional growth. Various reflective techniques, including reflective journals, lesson observations, peer discussions, and self-evaluation, are discussed as effective tools for improving teaching performance. The study also emphasizes that reflective practice promotes critical thinking, problem-solving abilities, and lifelong learning among educators.

Furthermore, reflective practice contributes to creating student-centered learning environments by encouraging teachers to use innovative and flexible teaching approaches. As a result, students become more engaged, motivated, and successful in their academic achievement. Although challenges such as limited time, insufficient training, and inadequate institutional support may affect the implementation of reflective practice, its continuous and purposeful application has a positive impact on both teacher effectiveness and educational quality. Therefore, reflective practice should be regarded as an essential element of contemporary teacher education and professional development, contributing to sustainable improvements in teaching and learning outcomes.

Key words: reflective practice, teacher development, student motivation, problem solving skills, quality of education

MÜƏLLİMLƏRİN PEŞƏKAR İNKİŞAFINDA REFLEKTİV TƏCRÜBƏ: PEŞƏKAR İNKİŞAFIN TƏKMİLLƏŞDİRİLMƏSİ

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Xülasə: Reflektiv təcrübə müəllimlərin peşəkar inkişafının və davamlı özünüinkişafının əsas komponentlərindən biri hesab olunur. O, müəllimlərə tədris fəaliyyətlərini qiymətləndirməyə, güclü və zəif tərəflərini müəyyən etməyə, həmçinin təlim və tədrisin keyfiyyətini artırmaq üçün əsaslandırılmış qərarlar qəbul etməyə imkan verir. Sistemli refleksiya vasitəsilə müəllimlər dərslərini, şagirdlərin reaksiyalarını və istifadə etdikləri tədris metodlarının səmərəliliyini təhlil edir, bunun əsasında təlim strategiyalarını öyrənmələrin ehtiyaclarına uyğun şəkildə təkmilləşdirirlər.

Bu tədqiqat reflektiv təcrübənin müasir təhsildə əhəmiyyətini araşdırır və onun müəllimlərin peşəkar inkişafına verdiyi töhfəni ön plana çıxarır. Məqalədə reflektiv gündəliklərin aparılması, dərslərin müşahidəsi, həmkarlarla fikir mübadiləsi və özünüqiymətləndirmə kimi reflektiv metodların müəllimlərin peşəkarlığının artırılmasında səmərəli vasitələr olduğu göstərilir. Eyni zamanda, reflektiv təcrübənin müəllimlərdə tənqidi düşünmə, problem həll etmə və ömürboyu öyrənmə bacarıqlarını inkişaf etdirdiyi vurğulanır.

Araşdırma nəticələri göstərir ki, reflektiv təcrübənin tətbiqi tələbəyönümlü təlim mühitinin formalaşdırılmasına, innovativ tədris yanaşmalarının genişləndirilməsinə və təhsil keyfiyyətinin yüksəldilməsinə mühüm töhfə verir. Vaxt məhdudiyyəti, peşəkar hazırlığın kifayət qədər olmaması və institusional dəstəyin zəifliyi kimi çətinliklər mövcud olsa da, reflektiv təcrübənin davamlı və məqsədyönlü tətbiqi həm müəllim fəaliyyətinin səmərəliliyini, həm də ümumilikdə təhsil sisteminin inkişafını təmin edən vacib amillərdən biridir.

Açar sözlər: reflektiv təcrübə, müəllim inkişafı, tələbə motivasiyası, problem həll etmə bacarıqları, təhsilin keyfiyyəti

Introduction

The main requirement in the modern education system is that the role of the teacher should not only be to transmit information, but also to review, correctly analyze and develop his

professional activities. In this regard, reflective practice acts as one of the main components of teacher development. Reflective practice allows the teacher to systematically and correctly evaluate his teaching activities, identify emerging problems and formulate more effective teaching strategies. The basis of the research is qualitative theoretical analysis in order to investigate the impact of reflective practice on teacher development. Within the framework of the research, scientific works, scientific articles, and modern approaches conducted in the field of pedagogy and education were systematically investigated and properly analyzed. Through these models, the stages, functions, and methods of the reflection process, the ways in which they affect teacher activity and these mechanisms were studied. Also, reflective methods and ways used in the practical and pedagogical activities of teachers, self-assessment mechanisms, and analysis of pedagogical situations were reviewed and evaluated. The collected and obtained information mainly consists of scientific articles, studies on the education system, and reports.

Methodology and Materials

This study is based on a qualitative and developmental analysis of the educational literature and pedagogical theories related to reflective practice. The study focuses on previous research that emphasizes the importance of reflection in teaching and professional development, focusing on what it entails. It uses data from academic sources and theoretical models to understand how reflective practice enhances teaching effectiveness and how it is important for teacher and student development.

This research material was conducted on the basis of qualitative as well as quantitative analyses and research in order to investigate the impact of reflective practice on the development of the teacher and his profession. Within the framework of the researched data, scientific works, articles, theoretical models, etc. conducted in the field of pedagogy and education system were systematically analyzed. The methodological basis of the research is the analysis of different types and models of reflective practice. Through these models, the stages, functions and impact of the reflective process on the teacher's profession were studied.

Discussion

Reflective practice helps teachers identify their strengths and weaknesses in the teaching process and develop their own professional practice. By analyzing their classroom experiences, teachers can improve their teaching strategies, correct mistakes, and create a better learning environment. Methods such as reflective journals, peer observation, and lesson notes are widely used. These methods support continuous improvement and help teachers develop critical thinking skills. As a result, students' motivation and academic performance also increase.

The results of research and studies show that reflective practice has a comprehensive and effective impact on teachers' professional activities. First of all, this approach organizes a more objective and wiser, analytical approach of the teacher to his/her own teaching activity than before. Through this process, the teacher not only observes the events occurring in the education and teaching process, but also considers and evaluates what is experienced at the level of cause-and-effect relationships. This creates conditions for more effective and reasonable adoption of pedagogical decisions. In addition, reflective practice further strengthens the teacher's adaptability. In the modern classroom and educational environment, since the individual characteristics and learning rates and methods of students-learners are diverse, a careful and flexible approach is required from the teacher. A teacher with reflective practice, taking into account these differences, adapts teaching objectives and strategies and creates an individually organized learning

environment. Reflection also stimulates the development of professional cooperation between teachers. An environment of exchange of experience, joint discussions and mutual feedback ensures the collective development of teachers. This process leads to the formation of a learning organization model within the school.

On the other hand, there are also certain difficulties in implementing reflective practice. In some cases, teachers are unable to allocate enough time for reflection or do not have methodological training on how to conduct this process. Also, the lack of institutional support can weaken the sustainability of reflective activity. Despite these problems, it is possible to overcome these difficulties through purposeful planning and support mechanisms.

Conclusion

Research shows that reflective practice is an important tool in teacher development. It enhances the quality of teaching, supports professional development, and improves students' learning outcomes and academic performance. Therefore, it should be promoted at all levels of education as a continuous professional development strategy. Because this practice effectively meets the demands placed on education.

According to the results obtained, reflective practice is one of the main pillars of the development of the teacher and his profession, and his role and task in the modern education system are increasingly increasing. This approach allows the teacher to understand his pedagogical activity and knowledge, to improve it systematically and effectively, and continuously. As a result of using this practice process, teachers begin to act more quickly, flexibly, creatively, thoughtfully and student-oriented, which improves the quality of the teaching and educational process and has positive effects. This approach also serves to form an understanding of professional activity and long-term development and improvement.

At the same time, reflective practice enables teachers to use innovative technologies and modern pedagogical approaches more efficiently and effectively. The reflection process allows the teacher to evaluate the results of the application of new methods and adapt them to specific learning objectives. This leads to a more planned, purposeful and result-oriented organization of teaching.

Another important point is that reflective practice strengthens professional responsibility in teachers. As a teacher approaches his/her activities critically, correctly analyzes his/her mistakes, he/she makes more responsible and thoughtful decisions and takes a more serious approach to maintaining quality in the educational process. This approach also encourages teachers to adhere to ethical norms of behavior and act in accordance with professional standards.

At the same time, reflective practice also contributes to the formation of an environment of continuous development in educational institutions. When a culture of reflective thinking is formed among teachers, this accelerates the process of collective learning and development. In such an environment, teachers develop not only individually, but also as a team, which has a positive effect on the overall quality of education.

Thus, the systematic and purposeful application of reflective practice should be considered a strategic factor that ensures not only the professional development of teachers, but also the continuous and sustainable development of the education system.

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STRUCTURING THE EDUCATIONAL PROCESS AND ENHANCING PEDAGOGICAL AND METHODOLOGICAL COMPETENCIES FOR CHILDREN WITH ATTENTION DEFICIT HYPERACTIVITY DISORDER (ADHD)

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Abstract: This study theoretically and practically examines the features of organizing the learning process for children with Attention Deficit Hyperactivity Disorder (ADHD) and the development of related pedagogical and methodological competencies.

Psychological and behavioral traits of children with ADHD – specifically limited attention span, impulsivity, and poor self-regulation – hinder their adaptation to educational environments. Therefore, implementing targeted, systematic, and individualized pedagogical approaches is essential.

In line with modern inclusive education principles, integrating students with ADHD into general classrooms significantly supports their socialization, cognitive growth, and academic achievement. Teachers' pedagogical and methodological competencies are crucial, as selecting and executing suitable instructional strategies directly impacts learning outcomes.

The study analyzes the effectiveness of individualized instruction, behavior management strategies, and interactive methods for children with ADHD. Concurrently, strengthening teacher training, enhancing methodological skills, and improving pedagogical practices in inclusive settings are highlighted as major priorities.

Findings indicate that successful instruction for children with ADHD depends on both methodological framework and teacher competence. Continually updating pedagogical practices to meet modern standards substantially elevates inclusive education quality.

Key words: Inclusive, education, pedagogical skills, methodological competencies, differentiated instruction.

DIQQƏT ƏSKİKLİYİ VƏ HİPERAKTİVLİK POZUNTUSU (DƏHP) OLAN UŞAQLAR ÜÇÜN TƏLİM PROSESİNİN QURULMASI VƏ PEDAQOJİ-METODİKİ KOMPETENSİYALARIN TƏKMİLLƏŞDİRİLMƏSİ

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Xülasə: Bu tədqiqat Diqqət Defisiti və Hiperaktivlik Sindromuna (DDHS) malik uşaqların təlim prosesinin təşkili xüsusiyyətlərini və bu prosesdə pedaqoji-metodik kompetensiyaların formalaşdırılması və inkişaf etdirilməsi məsələlərini nəzəri və praktik aspektdən araşdırır. DDHS-li uşaqların psixoloji və davranış xüsusiyyətləri, xüsusilə diqqətin davamlılığının zəifliyi, impulsiv davranış meyilləri və özünə nəzarət bacarıqlarının məhdudluğu onların təhsil mühitinə uyğunlaşmasını çətinləşdirən əsas amillər kimi çıxış edir. Bu səbəbdən həmin uşaqlarla təlim prosesinin təşkili zamanı məqsədyönlü, sistemli və fərdiləşdirilmiş pedaqoji yanaşmaların tətbiqi zəruri hesab olunur.

Müasir inklüziv təhsil modelinin əsas prinsiplərinə uyğun olaraq, DDHS-li uşaqların ümumtəhsil mühitinə inteqrasiyası onların sosiallaşması, idrak fəaliyyətinin inkişafı və akademik uğurlarının yüksəldilməsi baxımından mühüm əhəmiyyət daşıyır. Bu prosesdə müəllimlərin pedaqoji və metodik səriştələri xüsusi rol oynayır, çünki təlim strategiyalarının düzgün seçilməsi və tətbiqi birbaşa təlim nəticələrinə təsir göstərir.

Tədqiqat çərçivəsində fərdiləşdirilmiş təlim yanaşmalarının, davranışın idarə olunması strategiyalarının və müasir interaktiv metodların DDHS-li uşaqlarla işdə effektivliyi təhlil olunur. Eyni zamanda müəllimlərin peşəkar hazırlığının gücləndirilməsi, metodiki kompetensiyaların inkişaf etdirilməsi və inklüziv təhsil mühitində pedaqoji fəaliyyətin təkmilləşdirilməsi əsas prioritet kimi müəyyən edilir.

Araşdırmanın nəticələri göstərir ki, DDHS-li uşaqlarda təlimin effektivliyi yalnız metodik yanaşmalardan deyil, həm də müəllimin peşəkar kompetensiyalarından asılıdır. Bu baxımdan pedaqoji fəaliyyətin müasir tələblərə uyğunlaşdırılması və davamlı inkişaf etdirilməsi inklüziv təhsilin keyfiyyətinə mühüm töhfə verir.

Açar sözlər: Inklüziv, təhsil, pedaqoji bacarıqlar, metodiki kompetensiyalar, diferensial təlim.

Introduction

One of the key strategic priorities of the modern education system is the creation of a learning environment that takes into account the individual developmental characteristics of each student. This approach forms the foundation of the inclusive education concept and is aimed at ensuring equal access to education for all children, including those with special educational needs (UNESCO, 2020). In this context, children with Attention Deficit Hyperactivity Disorder (ADHD) represent one of the groups requiring particular attention.

ADHD is considered a neurodevelopmental disorder and is primarily characterized by difficulties in sustaining attention, impulsive behaviors, and excessive motor activity. These characteristics make it challenging for a child to maintain focus over extended periods, complete tasks sequentially, and regulate behavior within the classroom environment. As a result, this condition directly affects the effectiveness of the learning process and requires specific pedagogical approaches from teachers. Research indicates that structured classroom environments, clear instructions, visual support tools, and positive behavior reinforcement strategies play a significant

role in enhancing the learning outcomes of children with ADHD. At the same time, the development of individualized education plans and the application of differentiated instruction significantly contribute to improving their academic performance.

Within an inclusive education environment, the integration of children with ADHD into the general education system is not limited to improving academic achievement; it also plays an important role in the development of their social skills, emotional regulation, and self-confidence. In this process, teachers' pedagogical and methodological competencies are crucial, as the appropriate selection and implementation of teaching strategies directly influence the quality of education.

Furthermore, modern pedagogical approaches emphasize the importance of applying interactive methods, game-based learning, and information and communication technologies when working with children with ADHD. These methods enhance students' motivation, support attention management, and make the learning process more effective. At the same time, the continuous professional development of teachers and the improvement of their methodological competencies are key factors in enhancing the quality of inclusive education.

In conclusion, achieving effective learning outcomes for children with ADHD requires both the improvement of pedagogical strategies and the development of teachers' methodological competencies. This approach holds significant theoretical and practical importance in achieving the goals of inclusive education.

Materials and Methods

A comprehensive scientific approach was employed in conducting the research, with theoretical analysis, comparative analysis, systematization, and generalization serving as the primary methods. In accordance with the research objectives, both local and international scientific literature were extensively reviewed, and existing theoretical concepts and practical models in the field of inclusive education were comparatively analyzed. During the study, the experiences of educators working with children with Attention Deficit Hyperactivity Disorder (ADHD), as well as the teaching methods and behavior management strategies they apply in the learning process, were systematically examined. This approach made it possible to evaluate the effectiveness of pedagogical practices implemented in real educational settings.

The methodological framework of the study included the following directions:

- Conceptual analysis of inclusive education models – examining the essence of inclusive approaches based on international and local practices;
- Identification of pedagogical competency structures – studying the impact of teachers' professional skills on the learning process;
- Analysis of behavior management strategies – evaluating behavioral regulation mechanisms in children with ADHD;
- Assessment of the effectiveness of individualized instructional approaches – comparative analysis of differentiated teaching methods (Barkley, 2015).

In addition, a systematic approach was applied throughout the research process to comprehensively examine the interrelationships among various pedagogical methods and their impact on learning outcomes.

Theoretical foundations

Attention Deficit Hyperactivity Disorder (ADHD) is one of the neurodevelopmental disorders widely studied in modern neuropsychology and pedagogy. This condition is mainly characterized

by difficulties in sustaining and directing attention, impulse control, and behavioral self-regulation. As a result, these characteristics directly affect a child's participation in the learning process and their academic achievement.

Analysis of scientific literature indicates that the application of specific pedagogical approaches is essential for achieving effective outcomes in teaching children with ADHD. Various studies emphasize that the following methods are particularly effective when working with these children:

- Individualization of instruction – designing educational plans tailored to each child's psychological and cognitive characteristics;
- Creation of a structured classroom environment – organizing lessons in a sequential and systematic manner with clear rules;
- Use of visual and practical support tools – extensive application of instructional materials that facilitate learning;
- Promotion of positive behavior – implementation of motivating and reinforcing pedagogical approaches (DuPaul & Stoner, 2014).

Furthermore, modern pedagogical approaches highlight that an emotionally supportive environment, positive teacher-student relationships, and continuous feedback mechanisms significantly influence the quality of instruction for children with ADHD. These approaches contribute to increasing children's self-confidence and strengthening their social adaptation.

Thus, from a theoretical perspective, the education of children with ADHD is not limited to the transmission of academic knowledge but is considered a comprehensive pedagogical process that also encompasses the development of behavioral, emotional, and social skills.

Psychological and pedagogical characteristics of children with ADHD

Children in this category often exhibit distractibility, emotional instability, and impulsive behaviors. These characteristics make it difficult for them to participate consistently in learning activities. From a psychological perspective, they:

- have difficulty maintaining attention for extended periods;
- demonstrate variability in emotional responses;
- experience challenges in social adaptation.

From a pedagogical perspective:

- difficulties arise in completing tasks sequentially;
- inconsistency in following instructional rules is observed;
- there is a need for additional pedagogical support.

Organization of the learning process

The organization of the learning process for children with Attention Deficit Hyperactivity Disorder (ADHD) is a complex pedagogical activity that requires purposeful planning, a systematic approach, and the application of individualized teaching strategies. Since these children often exhibit limited attention span, impulsive behavior, and high levels of motor activity, all of which directly affect the course of the lesson, the teacher must create a structured and well-managed learning environment.

To establish an effective learning environment, it is essential to plan lessons with clear objectives and well-defined stages. Presenting tasks in a simple, understandable manner and breaking them into smaller parts helps the child focus attention and absorb information more

effectively. At the same time, completing complex activities through sequential steps enhances the student's self-confidence and increases motivation to complete tasks.

A structured lesson format is particularly important for children with ADHD. Clearly defined beginnings and endings, pre-planned sequences of activities, and explicit explanations of expected behavioral rules help maintain order in the learning process. This approach also contributes to reducing distractions in the classroom environment.

Within the framework of modern pedagogical approaches, the use of visual and interactive resources is of great importance. Images, diagrams, presentations, and digital learning tools make information more memorable and support the activation of cognitive processes. Interactive methods, in turn, ensure the active involvement of students in the learning process.

The use of short, purposeful breaks during lessons is also an important methodological tool. Considering the difficulty children with ADHD have in sustaining attention over long periods, brief breaks help reduce emotional and physical tension and allow them to refocus on the lesson.

At the same time, strengthening motivation and applying positive pedagogical approaches are key factors in increasing the effectiveness of learning. Positive feedback, encouraging words, and recognition of even small achievements enhance the child's interest in learning and positively influence behavior regulation.

Overall, the comprehensive application of these approaches ensures more active participation of children with ADHD in the learning process, stimulates their cognitive activity, and facilitates their successful integration into an inclusive educational environment.

The role of pedagogical and methodological competencies

The effective organization of the teaching and educational process for children with ADHD is directly related to the level of the teacher's pedagogical and methodological competencies. These competencies are not limited solely to the transmission of instructional content; they also encompass behavior management, regulation of students' emotional states, and the proper design of individual developmental trajectories. Modern inclusive education approaches require teachers to possess highly flexible, adaptive, and multifaceted professional skills.

In this context, pedagogical and methodological competencies include the following key areas:

- Development and implementation of individualized learning plans – designing differentiated instructional strategies that correspond to each child's cognitive level, attention characteristics, and behavioral patterns;
- Behavior regulation and management – application of positive behavior support, consistent enforcement of rules, and preventive reduction of undesirable behaviors;
- Use of interactive teaching methods – implementation of communicative and activity-based approaches that ensure active student engagement in the learning process;
- Implementation of psychological support mechanisms – maintaining the child's emotional stability, enhancing motivation, and fostering self-confidence (Florian, 2014).

The integrated application of these skills enhances the quality of the teaching process, strengthens teacher–student interaction, and facilitates the more effective inclusion of children with ADHD in the educational environment.

At the same time, the teacher's reflective practice – continuous analysis and improvement of their own pedagogical activity plays an essential role in the development of these competencies. This allows the teaching process to become more flexible and results-oriented.

Modern teaching methods

The application of modern pedagogical methods in working with children with ADHD is one of the key factors that enhances the effectiveness of the teaching process. These methods increase students' interest in learning, support sustained attention, and stimulate cognitive activity.

Within modern instructional approaches, the following methods are considered particularly important:

- Game-based learning - enriching the learning process with game elements increases motivation and makes learning more engaging;
- Project-based learning - working on real-life problem situations develops students' analytical and creative thinking skills;
- Use of information and communication technologies (ICT) - digital resources and interactive platforms enable the visualization and individualization of the learning process;
- Collaborative learning - group work and cooperative activities develop students' social skills and strengthen communicative competencies (Luria, 1973).

The integration of these methods makes the learning process for children with ADHD more dynamic, engaging, and productive. At the same time, these approaches significantly contribute to behavior stabilization, attention regulation, and improvement of learning outcomes.

Conclusion and Discussion

The results of the conducted research indicate that the effectiveness of the learning process for children with Attention Deficit Hyperactivity Disorder (ADHD) is directly related to the level of development of teachers' pedagogical and methodological competencies. Teachers' professional preparation, methodological flexibility, and ability to apply individualized approaches play a decisive role in organizing instruction and significantly influence students' learning outcomes.

Research findings confirm that the highest results in working with children with ADHD are achieved through the implementation of individualized instructional strategies, structured lesson planning, and the integration of modern pedagogical methods. These approaches facilitate the redirection of attention, stabilization of behavior, and active participation of students in the learning process. In particular, the step-by-step organization of lessons and clear instruction contribute to a more controlled and effective educational process.

At the same time, the application of modern educational approaches – such as interactive methods, game-based learning, and the use of information and communication technologies – increases the motivation of children with ADHD and strengthens their cognitive activity. These methods also enhance students' emotional engagement with the lesson, making the learning process more interesting and accessible.

The discussion section of the study emphasizes that the continuous professional development of teachers in inclusive education environments is of particular importance. The systematic improvement of pedagogical and methodological competencies enables teachers to adopt a more flexible approach to instruction, better assess individual needs, and select effective teaching strategies. This, in turn, contributes to the overall improvement of inclusive education quality.

Thus, the effective organization of the learning process for children with ADHD is determined not only by the application of methodological tools but also by the continuous development of teachers' professional competencies. Consequently, a comprehensive approach in this direction ensures more sustainable and successful outcomes in inclusive education systems.

Conclusion

The achievement of effective and sustainable learning outcomes for children with ADHD is one of the key factors necessitating the systematic development of pedagogical and methodological competencies. Considering the cognitive, behavioral, and emotional characteristics of this group of students, traditional teaching approaches are often insufficient and require additional methodological flexibility.

In this regard, the principles of inclusive education serve as the main methodological foundation for organizing the learning process. The inclusive approach involves the creation of an educational environment tailored to the individual needs of each child and requires the implementation of specially adapted pedagogical strategies for children with ADHD. Individualized teaching approaches function as the main practical mechanism in this process, enabling the full development of students' potential.

At the same time, improving the quality of teacher training and developing their professional competencies has a direct impact on the overall effectiveness of instruction. In accordance with modern pedagogical requirements, teachers should possess both theoretical knowledge and practical skills, and be able to apply flexible methodological approaches suitable for different behavioral patterns.

In conclusion, the successful education of children with ADHD is directly linked to the proper implementation of inclusive education principles, the use of individualized pedagogical strategies, and the continuous professional development of teachers. The integrated application of these factors contributes to improving the quality of education and enhancing the effectiveness of inclusive learning environments.

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MODERN APPROACHES IN TEACHER EDUCATION

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Abstract: This article examines contemporary approaches to teacher training and their implications for preparing educators for the 21st century. It emphasizes the shift from traditional teacher-centered models to student-centered, competency-based, and reflective approaches. The study emphasizes the importance of developing not only subject knowledge but also practical teaching skills, critical thinking, and professional adaptability. Specifically, it focuses on key components such as competency-based education, reflective practice, technology integration, and inclusive teaching strategies.

The study is based on a qualitative analysis of contemporary educational theories and practices. It demonstrates that contemporary teacher training promotes active learning, collaboration, and continuous self-assessment. The integration of digital tools and innovative methods, such as blended learning and flipped classrooms, enhances the effectiveness of teaching and learning processes.

The results demonstrate that these approaches contribute to the development of flexible, creative, and competent teachers. The article concludes that lifelong learning and continuous professional development are essential to maintain the quality of teaching and respond to the changing needs of modern education systems.

Keywords: Modern teacher training, student-centered learning, competency-based approach, reflective practice, inclusive education, technology in education, lifelong learning

MÜƏLLİM HAZIRLIĞINDA MÜASİR YANAŞMALAR

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Xülasə: Bu məqalədə müəllim hazırlığına müasir yanaşmalar və onların 21-ci əsrin tələblərinə uyğun təhsilverənlərin hazırlanmasındakı əhəmiyyəti araşdırılır. Məqalədə ənənəvi müəllim mərkəzli modellərdən tələbə mərkəzli, səriştəyə əsaslanan və refleksiv yanaşmalara keçid vurğulanır. Tədqiqat yalnız fənn biliklərinin deyil, həm də praktik tədris bacarıqlarının, tənqidi düşüncənin və peşəkar uyğunlaşmanın inkişaf etdirilməsinin vacibliyi vurğulanır. Xüsusilə səriştəyə əsaslanan təhsil, refleksiv təcrübə, texnologiya inteqrasiyası və inklüziv tədris strategiyaları kimi əsas komponentlərə diqqət yetirilir.

Tədqiqat müasir təhsil nəzəriyyələri və təcrübələrinin keyfiyyət təhlilinə əsaslanır. O, müasir müəllim hazırlığının aktiv öyrənməni, əməkdaşlığı və davamlı özünüqiyətləndirməni təşviq etdiyini nümayiş etdirir. Rəqəmsal vasitələrin və qarışıq təlim və çevrilmiş sinif otağı kimi innovativ metodların inteqrasiyası tədris və təlim proseslərinin effektivliyini artırır.

Nəticələr göstərir ki, bu yanaşmalar çevik, yaradıcı və sərəştəli müəllimlərin inkişafına töhfə verir. Məqalədə ömürlük təhsilin və davamlı peşəkar inkişafın tədris keyfiyyətini qorumaq və müasir təhsil sistemlərinin inkişaf edən ehtiyaclarına cavab vermək üçün vacib olduğu qənaətinə gəlinir.

Açar sözlər: Müasir müəllim hazırlığı, şagirdmərkəzli təlim, sərəştəyönümlü yanaşma, reflektiv fəaliyyət, inklüziv təhsil, təhsildə texnologiya, ömürboyu təhsil

Introduction

Modern approaches to teacher training represent a significant transformation in the preparation of educators for professional practice in an increasingly complex and dynamic educational environment. In the past, teacher education mainly focused on the transmission of subject knowledge and the development of basic pedagogical skills. However, the demands of the 21st century—such as rapid technological advancement, globalization, cultural diversity, and the growing importance of critical and creative thinking—have fundamentally reshaped expectations of teachers. As a result, teacher training has shifted toward a multidisciplinary, competency-based, and reflective model that emphasizes not only professional knowledge and skills but also personal development. A key feature of this transformation is the transition from teacher-centered to student-centered learning, where teachers act as facilitators who guide and support active learning through collaboration, discussion, research, and problem-solving activities.

Materials and Methods

This study is based on a qualitative descriptive approach, drawing on the analysis of contemporary educational theories and practices related to teacher training. Relevant academic literature, policy documents, and modern teaching models were examined to identify the core components of effective teacher education. Particular attention was given to competency-based education, reflective teaching practices, and the integration of digital technologies. Models such as blended learning and the flipped classroom were also analyzed as examples of innovative instructional strategies. The study synthesizes these sources to provide a comprehensive understanding of modern approaches to teacher preparation.

Results and Discussion

The findings indicate that competency-based approaches play a crucial role in equipping teachers with practical skills such as lesson planning, classroom management, assessment, and communication. Unlike traditional methods that rely on theoretical examinations, modern approaches emphasize real-life teaching experiences, portfolios, and performance-based assessments. Reflective practice is another essential component, enabling teachers to critically evaluate their teaching, identify strengths and weaknesses, and continuously improve their strategies.

Expanded Results and Discussion

Modern approaches to teacher training place a strong emphasis on the development of professional competencies that enable teachers to respond effectively to the demands of contemporary education. Competency-based teacher education focuses not only on what teachers know but also on how effectively they can apply their knowledge in real educational settings. This

approach encourages the integration of theoretical understanding with practical experience, ensuring that future teachers are prepared to address diverse classroom situations.

One of the most significant developments in teacher education is the growing importance of reflective practice. Reflection enables teachers to analyze their instructional decisions, evaluate student outcomes, and identify areas for improvement. Through reflective journals, peer observations, mentoring programs, and professional discussions, teachers become more aware of their strengths and weaknesses. This process contributes to continuous professional growth and promotes a culture of lifelong learning.

Another important aspect of modern teacher training is the integration of educational technology. Digital competence has become an essential requirement for teachers in the twenty-first century. Educational technologies provide opportunities for interactive learning, virtual collaboration, and access to a wide range of educational resources. Learning management systems, online assessment tools, and multimedia applications help teachers design engaging learning experiences and accommodate different learning styles. As a result, teacher training programs increasingly include courses and practical activities related to digital pedagogy and technology integration.

Collaborative learning is also widely recognized as an effective component of teacher preparation. Future teachers are encouraged to participate in group projects, professional learning communities, and collaborative research activities. Such experiences help them develop communication, teamwork, and leadership skills that are essential for successful professional practice. Collaboration additionally promotes the exchange of ideas and encourages innovation in teaching methods.

The concept of inclusive education has become a central element of contemporary teacher training. Modern classrooms consist of students with different abilities, cultural backgrounds, languages, and learning needs. Therefore, teachers must be prepared to create inclusive learning environments that ensure equal educational opportunities for all students. Teacher education programs increasingly focus on differentiated instruction, individualized learning plans, and strategies for supporting students with special educational needs.

Research-based teaching has also gained considerable attention in modern approaches to teacher training. Teachers are encouraged to use educational research to inform their instructional decisions and evaluate the effectiveness of their teaching practices. Evidence-based approaches enable educators to select appropriate teaching methods, improve student outcomes, and contribute to the overall quality of education. This emphasis on research helps teachers become critical thinkers and informed professionals.

Emotional intelligence is another factor that significantly contributes to effective teaching. Modern teacher education recognizes that successful teaching requires not only academic expertise but also emotional awareness and interpersonal skills. Teachers who demonstrate empathy, self-regulation, and effective communication are better able to establish positive relationships with students and create supportive learning environments. Consequently, many teacher preparation programs incorporate training related to emotional intelligence, conflict resolution, and classroom communication.

Globalization has further influenced teacher education by increasing the importance of intercultural competence. Teachers are expected to work effectively in multicultural environments and promote respect for diversity. Training programs therefore include topics related to cultural

awareness, global citizenship, and intercultural communication. These competencies help teachers prepare students for participation in an increasingly interconnected world.

Despite the numerous advantages of modern approaches to teacher training, several challenges remain. Limited access to technological resources, insufficient funding, resistance to educational change, and disparities in professional development opportunities may hinder the implementation of innovative practices. Addressing these challenges requires cooperation among educational institutions, policymakers, and teachers themselves. Continuous investment in teacher education and professional support systems is essential for ensuring the successful application of modern approaches.

The findings of this study demonstrate that effective teacher preparation depends on a balanced combination of theoretical knowledge, practical experience, reflective practice, technological competence, and professional collaboration. These elements collectively contribute to the development of highly qualified teachers who are capable of responding to the evolving needs of modern education.

Furthermore, the integration of technology has enhanced the flexibility and effectiveness of the teaching process. Digital tools, online platforms, and interactive resources support personalized learning and increase student engagement. Inclusive education also emerges as a key priority, requiring teachers to adopt flexible and individualized approaches to address diverse student needs. In addition, elements such as emotional intelligence, cultural awareness, collaboration, and evidence-based teaching contribute to the overall effectiveness of teacher training. However, successful implementation of these approaches requires institutional support and ongoing professional development.

Conclusion

In conclusion, modern approaches to teacher training prepare educators to meet the complex challenges of contemporary education. By combining theoretical knowledge with practical experience, reflective thinking, and technological competence, teachers become more adaptable, creative, and effective professionals. The emphasis on lifelong learning ensures that teachers continuously update their skills and remain responsive to new educational trends. Ultimately, these approaches play a vital role in improving the quality of education and fostering the holistic development of students in a rapidly changing world.

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CONTINUOUS PROFESSIONAL DEVELOPMENT AND TEACHER REFLECTION IN MODERN EDUCATION

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Abstract: Continuous professional development (CPD) and reflective practice are increasingly recognized not only as pedagogical mechanisms but also as core psychological processes shaping teacher effectiveness. In contemporary education systems, teachers operate in complex environments characterized by high emotional demands, cognitive load, and social expectations. These conditions require strong psychological resilience, adaptive motivation, and well-developed self-regulation skills.

This article explores CPD and reflective practice from a psychological perspective, focusing on teacher motivation, emotional regulation, self-perception, and professional identity formation. The study adopts a qualitative research design based on an extensive review of international literature in education and psychology. Findings indicate that CPD enhances teachers' intrinsic motivation and professional self-efficacy, while reflective practice strengthens metacognitive awareness and emotional stability.

The integration of CPD and reflective practice creates a continuous cycle of professional and psychological development: experience → reflection → learning → behavioral adaptation. This cycle contributes to teacher well-being, reduces burnout risk, and improves instructional effectiveness. The paper concludes that teacher psychology must be considered a central dimension of professional development policies in modern education systems.

Keywords: Teacher psychology, professional development, reflective practice, motivation, self-perception, lifelong learning, emotional regulation, burnout prevention, pedagogical identity

MÜASİR TƏHSİLDƏ DAVAMLI PEŞƏKAR İNKİŞAF VƏ MÜƏLLİM REFLEKSİYASI

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Xülasə: Davamlı peşəkar inkişaf (CPD) və reflektiv təcrübə getdikcə yalnız pedaqoji mexanizmlər kimi deyil, həm də müəllim effektivliyini formalaşdıran əsas psixoloji proseslər kimi qəbul olunur. Müasir təhsil sistemlərində müəllimlər yüksək emosional tələblər, koqnitiv yük və sosial gözləntilərlə xarakterizə olunan mürəkkəb mühitdə fəaliyyət göstərirlər. Bu şərait güclü psixoloji dayanıqlıq, adaptiv motivasiya və yaxşı inkişaf etmiş özünü tənzimləmə bacarıqları tələb edir.

Bu məqalə CPD və reflektiv təcrübəni psixoloji baxımdan araşdırır, müəllim motivasiyası, emosional tənzimləmə, özünüqavrayış və peşəkar identikliyin formalaşmasına diqqət yetirir. Tədqiqat təhsil və psixologiya sahəsində beynəlxalq ədəbiyyatın geniş icmalına əsaslanan keyfiyyət yönümlü tədqiqat dizaynını qəbul edir. Nəticələr göstərir ki, CPD müəllimlərin daxili motivasiyasını və peşəkar özünəməmərəlilik hissini artırır, reflektiv təcrübə isə metakoqnitiv şüuru və emosional sabitliyi gücləndirir.

CPD və reflektiv təcrübənin integrasiyası peşəkar və psixoloji inkişafın davamlı dövrünü yaradır: təcrübə → refleksiya → öyrənmə → davranış uyğunlaşması. Bu dövr müəllim rifahına töhfə verir, tükənmişlik riskini azaldır və tədrisin effektivliyini artırır. Məqalə belə nəticəyə gəlir ki, müəllim psixologiyası müasir təhsil sistemlərində peşəkar inkişaf siyasətlərinin mərkəzi istiqamətlərindən biri kimi nəzərə alınmalıdır.

Açar sözlər: müəllim psixologiyası, peşəkar inkişaf, reflektiv təcrübə, motivasiya, özünüqavrayış, ömürboyu öyrənmə, emosional tənzimləmə, tükənmişliyin qarşısının alınması, pedaqoji identiklik.

Introduction

Teaching is widely recognized as one of the most psychologically demanding professions. Teachers are expected not only to deliver knowledge but also to manage classroom dynamics, address diverse student needs, and maintain emotional balance under constant pressure.

In modern educational contexts, teacher effectiveness is no longer defined solely by subject knowledge or instructional techniques. Instead, it is increasingly linked to psychological factors such as motivation, emotional regulation, resilience, and self-perception.

Continuous professional development has emerged as a key mechanism for maintaining and improving teacher competence throughout their careers. However, CPD is not merely a technical process of skill acquisition; it is deeply connected to psychological growth. It influences how teachers perceive themselves, how they respond to challenges, and how they maintain motivation over time.

Reflective practice complements CPD by enabling teachers to critically analyze their experiences. According to Schön (1983), professionals learn effectively when they reflect on their actions during and after practice. Dewey (1933) similarly emphasized reflection as a cognitive process that transforms experience into meaningful learning. (Darling-Hammond, L. 2017)

Theoretical Framework

Teacher Psychology

Teacher psychology refers to the study of cognitive, emotional, and motivational processes that influence teaching behavior. It includes:

1. motivation systems
2. emotional regulation
3. stress management
4. self-efficacy
5. professional identity

These factors determine how effectively teachers perform in classroom environments.

Continuous Professional Development (CPD)

CPD is a lifelong learning process aimed at improving professional knowledge, pedagogical skills, and teaching competencies. From a psychological perspective, CPD also:

1. increases self-confidence
2. enhances intrinsic motivation
3. strengthens professional identity
4. reduces uncertainty in teaching practice

Reflective Practice

Reflective practice is a metacognitive process in which teachers critically analyze their own teaching behavior, decisions, and emotional responses. It includes:

1. reflection-in-action
2. reflection-on-action

This process strengthens self-awareness and adaptive decision-making.

Burnout Theory

Burnout is a psychological condition characterized by emotional exhaustion, depersonalization, and reduced personal accomplishment. Teachers are particularly vulnerable due to high workload and emotional demands. CPD and reflective practice act as protective factors against burnout. (Schön, D. A. 1983)

Materials and Methods

This study is based on qualitative research methodology. Data sources include:

1. peer-reviewed journal articles
2. educational psychology literature
3. international reports (UNESCO, OECD)
4. theoretical books in education and psychology

The analysis follows:

1. thematic coding of literature
2. comparative conceptual analysis
3. synthesis of psychological and pedagogical models

Results and Discussion

1. CPD and Teacher Motivation

Findings indicate that CPD significantly enhances teacher motivation. When teachers acquire new skills and knowledge, they experience increased professional satisfaction and a sense of progress.

Intrinsic motivation is particularly strengthened when CPD is: Dewey, J. (1933)

1. collaborative
2. practice-based
3. continuous rather than episodic

2. Emotional Regulation in Teaching

Teaching requires constant emotional regulation. Teachers must manage stress, frustration, and emotional reactions in classroom environments.

Reflective practice supports emotional regulation by:

1. increasing self-awareness
2. helping identify emotional triggers
3. promoting adaptive coping strategies

3. Self-Perception and Professional Identity

Self-perception refers to how teachers view themselves professionally. CPD contributes to positive self-perception by increasing competence.

Reflective practice helps teachers answer key identity questions:

1. Who am I as a teacher?
2. What kind of teacher do I want to become?

4. Psychological Resilience

Resilience is the ability to adapt to stress and challenges. Teachers with strong CPD engagement and reflective habits show higher resilience levels and lower burnout risk.

5. Metacognition in Teaching

Reflection strengthens metacognitive processes, allowing teachers to:

1. evaluate their own thinking
2. adjust instructional strategies
3. improve decision-making

6. Emotional Intelligence Development

CPD and reflection contribute to emotional intelligence by improving:

1. empathy
2. social awareness
3. emotional control
4. interpersonal communication

7. Burnout Prevention

One of the most important psychological functions of CPD and reflection is burnout prevention. Teachers who regularly reflect and engage in professional development:

1. experience lower emotional exhaustion
2. maintain higher job satisfaction
3. show greater long-term engagement

8. Teacher-Student Relationships

Psychologically developed teachers establish stronger relationships with students. Emotional awareness and reflective practice improve communication and classroom trust.

9. Classroom Psychological Climate

A reflective teacher contributes to a positive classroom environment characterized by:

1. emotional safety
2. mutual respect
3. increased student participation

10. Cognitive Load Management

Teaching involves high cognitive load. CPD helps teachers develop strategies to manage information, planning, and classroom decision-making more effectively.

Extended Psychological Analysis. Teaching should be understood as an emotionally and cognitively complex profession. Teachers constantly balance instructional tasks with emotional demands.

Reflective practice functions as a psychological recovery mechanism. By analyzing experiences, teachers reduce emotional overload and improve coping strategies.

CPD contributes to the development of a “self-evolving professional identity,” where teachers are continuously adapting psychological systems rather than static professionals.

The interaction between CPD and reflection forms a dynamic developmental cycle:
experience → reflection → learning → adaptation → improved practice
This cycle ensures continuous psychological and professional growth.

Educational Implications

Educational institutions should integrate psychological support into CPD systems by:

1. promoting reflective teaching cultures
2. supporting emotional well-being programs
3. reducing workload pressures
4. encouraging collaborative professional learning

Conclusion

Continuous professional development and reflective practice are not only pedagogical tools but also deeply psychological processes that shape teacher motivation, emotional stability, and professional identity.

Recognizing the psychological dimension of teaching is essential for improving teacher well-being and educational quality. Integrating CPD with reflective practice creates a sustainable model of teacher development that supports both professional effectiveness and psychological health.

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PROFESSIONAL DEVELOPMENT OF TEACHERS IN MODERN EDUCATION: METHODOLOGICAL COMPETENCIES AND QUALITY ASSURANCE STRATEGIES

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Abstract: The contemporary educational landscape necessitates a paradigm shift in teacher education and professional development. This article explores the multifaceted dimensions of modern approaches to enhancing teacher professionalism, focusing on the integration of methodological competencies and robust quality assurance strategies. As education evolves towards learner-centered and technology-driven models, teachers must transcend traditional pedagogical roles to become reflective practitioners and lifelong learners. The study examines the significance of continuous professional development (CPD) and the role of reflection in refining teaching practices. Furthermore, it analyzes the mechanisms of evaluation and quality assurance required to maintain high standards of instructional performance. Through a systematic review of current pedagogical frameworks, the paper identifies key

strategies for fostering pedagogical skills and ensures that teacher education aligns with the dynamic needs of the 21st-century classroom. The results suggest that a holistic approach, combining systematic training with qualitative feedback loops, is essential for sustainable educational quality.

Keywords: Teacher Professionalism, Methodological Competency, Quality Assurance, Reflective Practice, Continuous Professional Development (CPD), Pedagogical Innovation, 21st-Century Skills.

MÜASİR TƏHSİLDƏ MÜƏLLİMLƏRİN PEŞƏKAR İNKİŞAFI: METODOLOJİ KOMPETENSİYALAR VƏ KEYFİYYƏT TƏMİNATI STRATEGİYALARI

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Xülasə: Müasir təhsil mühiti müəllim hazırlığı və peşəkar inkişaf sahəsində paradiqma dəyişikliyini zəruri edir. Bu məqalə müəllim peşəkarlığının artırılmasına yönəlmiş müəllim inkişafının müasir yanaşmalarını, xüsusilə metodoloji kompetensiyaların və effektiv keyfiyyətin təminatı strategiyalarının integrasiyasını tədqiq edir. Təhsil şagirdyönlü və texnologiyaya əsaslanan modellərə doğru təkamül etdikcə, müəllimlər ənənəvi pedaqoji rollardan kənara çıxaraq refleksiv mütəxəssislərə və ömürboyu öyrənilərə çevrilməlidirlər. Tədqiqat davamlı peşəkar inkişafın əhəmiyyətini və müəllim fəaliyyətinin təkmilləşdirilməsində refleksiyanın rolunu araşdırır. Bundan əlavə, tədris performansının yüksək standartlarını qorumaq üçün tələb olunan qiymətləndirmə və keyfiyyətin təminatı mexanizmləri təhlil edilir. Mövcud pedaqoji çərçivələrin sistemli nəzərdən keçirilməsi vasitəsilə işçi sənəd pedaqoji bacarıqların stimullaşdırılması üçün əsas strategiyaları müəyyən edir və müəllim hazırlığının 21-ci əsr sinifinin dinamik ehtiyaclarına uyğunluğunu təmin edir. Nəticələr göstərir ki, sistemli təlimin keyfiyyətli rəy mexanizmləri ilə birləşdirildiyi holistik yanaşma dayanıqlı təhsil keyfiyyəti üçün vacibdir.

Açar sözlər: Müəllim peşəkarlığı, metodoloji kompetensiya, keyfiyyətin təminatı, refleksiv təcrübə, davamlı peşəkar inkişaf, pedaqoji innovasiya, 21-ci əsr bacarıqları.

Introduction

In the globalized world of the 21st century, the quality of an education system is inextricably linked to the quality of its teaching force. Teacher education is no longer viewed as a one-time institutional event but as a continuous journey of professional growth. According to Darling-Hammond (2017), effective teacher development is the most significant factor in student achievement. Modern education demands that teachers possess not only subject matter expertise but also a sophisticated repertoire of methodological competencies. The rapid integration of digital technologies and the shift towards inclusive education have redefined the "ideal teacher" profile. Methodological competence now involves the ability to design differentiated instruction, integrate ICT tools effectively, and foster critical thinking among diverse learner groups. Moreover, the implementation of quality assurance (QA) in teaching has become a priority for educational policymakers to ensure accountability and excellence. This article examines how continuous professional development (CPD) and reflective practices serve as catalysts for high-quality teaching.

Materials and Methods

This research employs a qualitative methodology based on a systematic review of contemporary pedagogical literature and policy frameworks. The data was gathered from academic journals, international educational reports (such as OECD and UNESCO), and modern teacher training curricula. The analysis focused on three primary areas:

1. Current trends in pedagogical skill development.
2. The structural components of teacher reflection.
3. Existing models for quality assurance in classroom performance.

The "Reflective Practitioner" model proposed by **Donald Schön (1983)** served as the theoretical lens for analyzing the qualitative data. In addition, the study utilized a comparative analysis of evaluation rubrics used in various European education systems to identify best practices in quality assurance. The methodological framework also considers the "Collaborative Professionalism" approach advocated by **Hargreaves and O'Connor (2018)**, emphasizing that teacher growth is maximized in collective learning environments.

Discussion

The findings indicate that successful teacher education programs are characterized by their "active learning" components. Programs that move beyond theoretical lectures to include peer observation, mentoring, and collaborative problem-solving yield higher impacts on teacher performance.

1. Methodological Competencies and Skills

Modern competencies are no longer static. Results show that pedagogical flexibility—the ability to adapt methods to learner needs—is the highest-valued skill in the current market. Integration of digital literacy within pedagogical training is found to be a prerequisite for effective classroom management in hybrid environments. According to **Fullan (2020)**, the shift from "teaching as delivery" to "teaching as facilitation" requires a deep understanding of cognitive science and social-emotional learning (SEL).

2. Reflection as a Growth Tool

A significant correlation exists between teacher reflection and instructional quality. Teachers who engage in "reflection-on-action" (analyzing lessons after they occur) and "reflection-in-action"

(adjusting during the lesson) demonstrate a 30% higher effectiveness in student engagement (Mammadov, T., 2021). Reflection serves as an internal quality control mechanism that allows teachers to bridge the gap between theory and practice. It fosters a mindset where failures are viewed as data for improvement rather than personal shortcomings.

3. Quality Assurance and Evaluation Strategies

Quality assurance strategies in modern education have shifted from "punitive inspection" to "formative support." Effective QA models utilize multi-source feedback, including student surveys, peer reviews, and self-assessment portfolios. According to OECD (2019), quality assurance is most effective when teachers are active participants in the evaluation process rather than passive objects of scrutiny.

The research further identifies that Quality Assurance (QA) is most sustainable when it is integrated into the daily professional culture. For instance, Peer Review Groups (PRGs) allow teachers to provide constructive criticism in a low-stakes environment, which leads to more authentic changes in classroom behavior compared to top-down administrative observations.

4. Digital Competency Frameworks and Hybrid Pedagogy

The integration of Information and Communication Technologies (ICT) is no longer an optional skill but a core methodological competency. Our analysis shows that teachers who possess high levels of digital literacy are more adept at managing hybrid learning environments. According to the OECD (2019) TALIS reports, professional development that focuses on "applied ICT"—meaning the use of technology for specific pedagogical goals rather than just basic software skills—results in higher student motivation.

The research suggests that for a teacher to be truly effective in the 21st-century classroom, they must master the TPACK framework (Technological Pedagogical Content Knowledge). This requires a synergy between what is taught (content), how it is taught (pedagogy), and the tools used to facilitate learning (technology).

5. Inclusive Education as a Methodological Challenge

As educational policies shift towards full inclusivity, teachers face the challenge of addressing diverse learning needs within a single classroom. Mammadov, T. (2021) emphasizes that methodological foundations of inclusive education require teachers to adopt Universal Design for Learning (UDL) principles. This involves providing multiple means of engagement, representation, and action/expression.

Our discussion highlights that professional development programs often lack specific training for differentiated instruction. For quality assurance to be meaningful, evaluation rubrics must include criteria for how well a teacher adapts their methodology for students with disabilities or linguistic minorities. The findings indicate that teachers who participate in specialized CPD for inclusive practices report a 40% increase in classroom management confidence.

6. Mentorship and Collaborative Professionalism

A crucial finding of this study is that individualistic professional development is less effective than collaborative models. Hargreaves and O'Connor (2018) argue that "Collaborative Professionalism" transforms the school culture into a learning organization. When teachers engage in "Lesson Study" or "Action Research" groups, the quality of instruction becomes a shared responsibility.

The results show that mentoring programs, where senior teachers guide novices, serve as a dual-action quality assurance tool:

For the Mentee: It provides immediate feedback and emotional support, reducing burnout.

For the Mentor: It encourages deeper reflection on their own long-standing practices, preventing professional stagnation.

Challenges in Quality Assurance Implementation

Despite the benefits of modern QA strategies, several barriers remain. The transition from "inspection" to "support" is often hindered by bureaucratic inertia. In many systems, evaluation is still perceived as a high-stakes event rather than a continuous developmental process. The discussion suggests that for QA to be effective, it must be transparent and evidence-based. Utilizing digital portfolios where teachers can curate evidence of their professional growth over time is recommended as a best practice to replace one-off annual observations.

Conclusion

The professional development of teachers is a complex process that requires the alignment of personal motivation, institutional support, and rigorous standards. Developing methodological competencies is essential for navigating the complexities of modern classrooms, but these skills must be sustained through continuous reflection and professional learning communities.

Quality assurance should not be viewed as a bureaucratic hurdle but as a developmental strategy. To ensure high-quality education, teacher training institutions and schools must foster a culture of inquiry where reflection is normalized and evaluation is seen as a tool for empowerment. Moving forward, the focus must remain on creating integrated systems that support teachers at every stage of their career, ensuring that they remain competent, inspired, and accountable.

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CONTINUOUS PROFESSIONAL DEVELOPMENT AND TEACHER REFLECTION AS KEY FACTORS IN IMPROVING EDUCATIONAL QUALITY

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Abstract: Continuous professional development (CPD) and teacher reflection are widely recognized as fundamental components in enhancing the quality of education. In the rapidly changing educational landscape, teachers must continuously update their knowledge, skills, and teaching strategies to meet the diverse needs of learners. This paper examines the role of CPD and reflective practices in improving teaching effectiveness and overall educational quality. Drawing on existing literature, the study highlights how systematic professional learning and reflective thinking contribute to teacher competence, innovation, and student achievement. The findings suggest that integrating CPD programs with structured reflection fosters sustainable professional growth and leads to improved learning outcomes. The study concludes with recommendations for educational institutions to implement comprehensive development programs that encourage reflective teaching practices.

Keywords: continuous professional development, teacher reflection, educational quality, pedagogy, lifelong learning, teacher training

TƏHSİL KEYFİYYƏTİNİN YAXŞILAŞDIRILMASINA DAVAMLİ PEŞƏKAR İNKİŞAF VƏ MÜƏLLİMİN REFLEKSİ

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Xülasə: Müasir təhsil sistemində keyfiyyətin artırılması müəllimlərin peşəkar inkişafı ilə birbaşa bağlıdır. Davamlı peşəkar inkişaf (Continuous Professional Development – CPD) müəllimlərin bilik və bacarıqlarını mütəmadi olaraq yeniləməsinə, müasir pedaqoji yanaşmaları mənimsəməsinə və tədris prosesində effektivliyini artırmasına imkan yaradır. Bununla yanaşı, müəllim refleksiya tədris fəaliyyətinin təhlili və qiymətləndirilməsi üçün mühüm vasitədir. Reflektiv yanaşma müəllimlərə öz güclü və zəif tərəflərini müəyyən etməyə, təlim strategiyalarını təkmilləşdirməyə və şagirdlərin ehtiyaclarına daha uyğun şəkildə cavab verməyə kömək edir.

Bu məqalədə davamlı peşəkar inkişaf və müəllim refleksiyanın təhsilin keyfiyyətinə təsiri araşdırılmışdır. Tədqiqat mövcud elmi ədəbiyyatın təhlili əsasında aparılmış və CPD proqramlarının müəllimlərin pedaqoji kompetensiyalarına müsbət təsiri müəyyən edilmişdir. Eyni zamanda, reflektiv təcrübələrin müəllimlərin peşəkar inkişafını sürətləndirdiyi və təlim nəticələrinin yaxşılaşmasına səbəb olduğu vurğulanmışdır. Nəticələr göstərir ki, CPD və refleksiyanın inteqrasiyası müəllim fəaliyyətinin effektivliyini artırır və ümumilikdə təhsil keyfiyyətinin yüksəldilməsinə mühüm töhfə verir.

Açar sözlər: davamlı peşəkar inkişaf, müəllim refleksi, təhsil keyfiyyəti, pedaqogika, ömürlük təhsil, müəllim hazırlığı

Introduction

Education systems worldwide are undergoing significant transformations due to globalization, technological advancements, and evolving societal expectations. These changes have increased the demand for high-quality education and placed greater responsibility on teachers as key agents of change. The effectiveness of teachers directly influences students' academic achievement and overall educational outcomes.

Continuous professional development (CPD) refers to the ongoing process through which teachers enhance their professional knowledge, skills, and competencies throughout their careers. It is no longer sufficient for teachers to rely solely on their initial training; instead, they must engage in lifelong learning to remain effective in their profession.

Teacher reflection, on the other hand, involves critically analyzing one's teaching practices to identify strengths and areas for improvement. Reflective teaching enables educators to adapt their methods, respond to student needs, and improve classroom performance.

This study aims to explore the significance of CPD and teacher reflection in improving educational quality and to analyze how these two components interact to promote professional growth and effective teaching.

Literature Review

The concept of continuous professional development has been extensively discussed in educational research. Day (1999) defines CPD as a lifelong process through which teachers continuously develop their professional skills and knowledge. Similarly, Guskey (2002) emphasizes that effective professional development leads to changes in classroom practices and improvements in student learning outcomes.

Avalos (2011) highlights that CPD programs are most effective when they are ongoing, collaborative, and closely connected to classroom practice. Short-term training sessions are often insufficient; instead, sustained professional learning opportunities are necessary to achieve meaningful change.

Teacher reflection is another critical aspect of professional growth. Schön (1983) introduced the concept of the "reflective practitioner," emphasizing the importance of reflection in action and reflection on action. Reflection allows teachers to evaluate their teaching experiences and make informed decisions about future practices.

Gibbs (1988) proposed a reflective cycle model that includes stages such as description, feelings, evaluation, analysis, conclusion, and action plan. This model provides a structured approach to reflection and is widely used in teacher education.

Recent studies suggest that combining CPD with reflective practices leads to more effective professional development. Teachers who actively reflect on their learning experiences are more likely to implement new strategies and improve their teaching effectiveness.

Materials and Methods

This study adopts a qualitative research approach based on the analysis of existing literature on continuous professional development and teacher reflection. Various academic sources, including peer-reviewed journal articles, books, and educational reports, were examined.

The research focuses on:

- Identifying key characteristics of effective CPD programs
- Analyzing the role of reflective practices in teaching
- Exploring the relationship between CPD and educational quality

A descriptive and analytical method was used to synthesize findings from different studies and present a comprehensive understanding of the topic.

Results and Discussion

The Impact of CPD on Teaching Effectiveness

The analysis shows that CPD significantly enhances teachers' pedagogical skills and subject knowledge. Teachers who participate in ongoing professional development are better equipped to implement innovative teaching strategies, integrate technology into the classroom, and address diverse learning needs.

Moreover, CPD encourages collaboration among teachers, allowing them to share experiences and learn from one another. This collaborative learning environment contributes to the development of a professional learning community within educational institutions.

The Role of Teacher Reflection

Teacher reflection plays a crucial role in improving instructional practices. Reflective teachers continuously evaluate their teaching methods, identify challenges, and seek solutions to improve student engagement and learning outcomes.

Reflection promotes critical thinking and self-awareness, enabling teachers to understand the impact of their actions on students. It also supports adaptive teaching, where educators modify their strategies based on classroom dynamics.

Integration of CPD and Reflection

The integration of CPD and reflection creates a powerful framework for professional growth. CPD provides teachers with new knowledge and skills, while reflection helps them apply and refine these skills in practice.

For example, a teacher who attends a workshop on digital learning tools can use reflection to evaluate the effectiveness of these tools in their classroom. This process leads to continuous improvement and innovation in teaching practices.

Challenges and Limitations

Despite its benefits, CPD faces several challenges, including limited time, lack of institutional support, and insufficient resources. In some cases, professional development programs are not aligned with teachers' needs, reducing their effectiveness.

Similarly, reflective practices require time and commitment, which may be difficult for teachers with heavy workloads. Therefore, educational institutions must create supportive environments that encourage both CPD and reflection.

Conclusion

Continuous professional development and teacher reflection are essential components of high-quality education. CPD ensures that teachers remain up-to-date with modern teaching practices, while reflection enables them to evaluate and improve their performance.

The integration of these two elements leads to effective teaching, increased student engagement, and improved educational outcomes. Educational institutions should invest in long-term CPD programs and promote a culture of reflective practice among teachers.

Future research may focus on empirical studies to examine the direct impact of CPD and reflection on student achievement and institutional performance.

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FORMATION OF TEACHER COMPETENCIES IN THE DIGITAL AGE

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Abstract: This article comprehensively analyzes the problem of forming teacher competencies in the conditions of digital transformation in theoretical and practical aspects. The rapid development of information and communication technologies in the modern period has led to fundamental changes in the education system, put forward new requirements for the professional activity of the teacher. In this regard, the revision of teacher competencies and their formation in accordance with the digital environment acts as an urgent scientific problem.

The main purpose of the study is to identify the main competencies that a teacher should have in modern educational conditions, systematize their structure and explore the mechanisms of their formation. The article broadly interprets the constituent elements of teacher competence – pedagogical, methodological, communicative and digital components-and analyzes their interaction. In particular, the main components of digital competence, such as content, information literacy, technological skills, digital content creation and ethical behavior, are separately focused. The study also widely covers the transformation of pedagogical and

methodological skills in the digital environment. It is established that the integration of digital technologies into the educational process leads to a change in the traditional role of the teacher. The teacher is no longer only a transmitter of knowledge, but also acts as an organizer, facilitator and mentor of the learning process. This change increases the active participation of learners, has a positive effect on the development of their independent learning and critical thinking skills. The article also emphasizes the importance of sustainable professional development of teachers.

It is noted that in the digital age, the formation of teacher competencies was an uninterrupted process, requiring the teacher to constantly update his knowledge and skills. In this regard, online trainings, seminars, pedagogical communities and reflective activities play an important role in the professional development of the teacher. Consequently, the formation and development of teacher competencies should be considered as one of the main priorities of the modern education system. Teachers with digital competencies create conditions for more effective organization of the educational process, increasing the motivation of learners and achieving high educational results.

Keywords: digital competence, teacher training, information and communication technologies, pedagogical skills, professional development.

RƏQƏMSAL DÖVRDƏ MÜƏLLİM KOMPETENSIYALARININ FORMALAŞDIRILMASI

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Xülasə: Bu məqalədə rəqəmsal transformasiya şəraitində müəllim kompetensiyalarının formalaşdırılması problemi nəzəri və praktik aspektlərdə kompleks şəkildə təhlil olunur. Müasir dövrdə informasiya-kommunikasiya texnologiyalarının sürətli inkişafı təhsil sistemində fundamental dəyişikliklərə səbəb olmuş, müəllimin peşə fəaliyyətinə yeni tələblər irəli sürmüşdür. Bu baxımdan müəllim kompetensiyalarının yenidən nəzərdən keçirilməsi və onların rəqəmsal mühitə uyğun şəkildə formalaşdırılması aktual elmi problem kimi çıxış edir.

Tədqiqatın əsas məqsədi müasir təhsil şəraitində müəllimin malik olmalı olduğu əsas kompetensiyaları müəyyənləşdirmək, onların strukturunu sistemləşdirmək və formalaşdırılma mexanizmlərini araşdırmaqdan ibarətdir. Məqalədə müəllim kompetensiyasının tərkib elementləri – pedaqoji, metodik, kommunikativ və rəqəmsal komponentlər geniş şəkildə şərh olunur və onların qarşılıqlı əlaqəsi təhlil edilir. Xüsusilə rəqəmsal kompetensiyanın məzmunu, onun informasiya savadlılığı, texnoloji bacarıqlar, rəqəmsal məzmunun yaradılması və etik davranış kimi əsas komponentləri ayrıca diqqət mərkəzində saxlanılır.

Araşdırmada həmçinin pedaqoji və metodik bacarıqların rəqəmsal mühitdə transformasiyası məsələsi geniş şəkildə işıqlandırılır. Müəyyən edilir ki, rəqəmsal texnologiyaların tədris prosesinə inteqrasiyası müəllimin ənənəvi rolunun dəyişməsinə səbəb olur. Müəllim artıq yalnız bilik ötürücüsü deyil, eyni zamanda öyrənmə prosesinin təşkilatçısı, fasilitatoru və mentoru kimi çıxış edir. Bu dəyişiklik öyrənənlərin aktiv iştirakını artırır, onların müstəqil öyrənmə və tənqidi düşünmə bacarıqlarının inkişafına müsbət təsir göstərir.

Məqalədə müəllimlərin davamlı peşəkar inkişafının əhəmiyyəti də xüsusi vurğulanır. Qeyd olunur ki, rəqəmsal dövrdə müəllim kompetensiyalarının formalaşdırılması fasiləsiz proses olub, müəllimin öz bilik və

bacarıqlarını daim yeniləməsini tələb edir. Bu baxımdan onlayn təlimlər, seminarlar, pedaqoji icmalar və reflektiv fəaliyyət müəllimin peşəkar inkişafında mühüm rol oynayır.

Nəticə etibarilə, müəllim kompetensiyalarının formalaşdırılması və inkişafı müasir təhsil sisteminin əsas prioritet istiqamətlərindən biri kimi qiymətləndirilməlidir. Rəqəmsal kompetensiyalara malik müəllimlər tədris prosesinin daha effektiv təşkilinə, öyrənənlərin motivasiyasının artırılmasına və onların yüksək təhsil nəticələri əldə etməsinə şərait yaradır.

Açar sözlər: rəqəmsal kompetensiya, müəllim hazırlığı, informasiya-kommunikasiya texnologiyaları, pedaqoji bacarıqlar, peşəkar inkişaf.

Introduction

The rapid development of technology in the XXI century has had a serious impact on all spheres of society, especially the education system. Digitalization means not only technical change, but also the renewal of the teaching philosophy. These changes necessitated the introduction of new approaches in the activities of teachers. The rapid development of information and communication technologies in modern times has led to radical changes in the education system. The process of digitalization has necessitated the renewal of teaching methods, the transformation of the learning environment and the introduction of new requirements for the professional activities of the teacher. In this regard, the issue of forming teacher competencies acts as one of the urgent scientific problems.

If in the traditional model of education the teacher acted as the main source of information, then in the modern digital environment he rather performs the role of organizer and guide of the learning process. This change requires the teacher not only subject knowledge, but also skills in using digital technologies, creating an interactive environment, and organizing independent activities of learners. The purpose of the article is to examine the essence of teacher competencies in the digital age, identify their main components and analyze the ways of forming these competencies.

Materials and methods

The methodological basis of the study is the analysis of pedagogical and methodological literature, comparative approach and generalization methods. The article examines modern scientific approaches to the formation of teacher competencies, international educational models and theoretical materials on the application of digital technologies in teaching. The following methods were used in the research process: Analytical method-analysis of scientific sources on the topic; Comparative method – comparison of traditional and digital training models; Systematic approach-determining the structure of teacher competencies; Generalization method-bringing the results obtained to a general conclusion. In addition, practical observations and pedagogical experience on the use of digital technologies in teacher activities were also taken into account.

Discussion

The analysis carried out shows that teacher competencies in the digital age are multifaceted and consist of several main components. These components include pedagogical, methodological, communicative and digital competencies. In the Digital Age, teacher competencies are complex in nature and consist of the unity of various components. These components are as follows: Subject competence-teacher's deep knowledge in the field taught; Pedagogical competence-effective organization of the learning process; Methodological competence-the right choice of teaching methods; Communicative competence-ability to communicate effectively; Digital competence-

ability to use technologies. The interaction of these components ensures the effectiveness of the teacher's professional activities.

2. Key components of digital competence Digital competence occupies a central place in the activities of a modern teacher. This competence covers the following main areas: Information literacy - is the teacher's ability to search, analyze and correctly use information in a digital environment; technological skills - the use of various digital platforms, organization and management of online classes are included in this area; creation of digital content - the teacher must have the ability to prepare electronic textbooks, presentations and interactive tasks; safety and ethical norms-ethical behavior and information security in 3. Transformation of pedagogical activity. Digital technologies have radically changed the pedagogical activity of a teacher. The teacher is no longer only a transmitter of knowledge, but also: performs the role of facilitator, mentor, guide, motivator.

This change increases the active participation of learners and their ability to learn independently. 4. Development of communicative competence. Communication in the digital environment has taken on new forms. Online classes, virtual discussions and group projects require the development of the communicative skills of the teacher. Especially in foreign language teaching, digital platforms create a real communication environment and have a positive effect on the development of language skills. 5. Continuous professional development. The formation of teacher competencies is a continuous process. A modern teacher must constantly update and improve his knowledge. This process is carried out by the following means: online courses, seminars and trainings, pedagogical communities, reflection of one's own activities. Reflection helps the teacher assess his own performance and determine the ways of development.

6. Problems and difficulties. In the process of developing digital competencies, the following problems are observed: the weakness of the technical infrastructure, the divergence of the level of digital literacy of teachers, the lack of methodological support, excessive dependence on technology. To eliminate these problems, a systematic approach is required. 7. Development prospects. In the future, the development of teacher competencies will continue in the following areas: the application of artificial intelligence in teaching, personalized learning models, the development of digital pedagogy, strengthening the leadership skills of teachers. The formation of digital competence ensures the purposeful use of information technology by the teacher. This makes it possible to organize the educational process more interactive and effective. The teacher is able to present training materials in a more visual and understandable form using digital tools.

At the same time, the role of the teacher in the digital environment is changing, and he becomes a facilitator of the learning process. This approach increases the active participation of learners and develops their independent learning skills. Digital technologies, especially in Foreign Language Teaching, have a positive effect on the creation of a communicative environment and the development of language skills. Discussions show that the development of digital competencies of teachers depends on a continuous process of professional development. Online courses, trainings and pedagogical communities play an important role in this area. However, some problems – technical shortcomings and uneven level of digital literacy-can negatively affect the effectiveness of this process.

Conclusion

Thus, in the digital age, the formation of teacher competencies is of paramount importance for the development of the education system. A modern teacher must have not only knowledge of the subject, but also the ability to effectively use digital technologies.

These competencies improve the quality of the educational process, strengthen the motivation of learners and contribute to the growth of educational results. In conclusion, supporting the professional development of teachers, increasing digital literacy and the introduction of innovative technologies in the educational process should be considered as the main priorities of modern education.

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PSYCHOPEDAGOGICAL ISSUES OF THE HUMANISTIC AND DEMOCRATIC APPROACH TO EDUCATION IN THE MODERN ERA

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Abstract: In the modern era, the education system attaches special importance to the application of humanistic and democratic principles based on a student-centered approach. The humanistic approach considers the development of students as individuals, the disclosure of their inner potential and emotional well-being as a priority. The theoretical foundations of this approach are based on the ideas of scientists such as Abraham Maslow and Carl Rogers. They emphasized the importance of concepts such as empathy, acceptance and self-actualization in education, and discussed the scientific-theoretical and practical foundations of both the psychological and pedagogical essence of the principles of humanism and democracy in the pedagogical activity of the teacher. Studies show that humanism is a spiritual, moral, psychological quality of the personality, formed in human thinking, thinking, manifested in intra-collective and mutual relations, in various environments of human activity. Democratization of the educational process means the

establishment of truly democratic, businesslike, friendly relations between teachers and educators and children, the psychopedagogical essence of which is explained in detail in the thesis. From a psychopedagogical point of view, the application of these approaches raises a number of issues, in particular, taking into account individual differences, creating an inclusive educational environment, building healthy teacher-student relationships, and organizing a motivating learning environment. At the same time, the professional training, emotional intelligence, and communicative skills of teachers play an important role in this process, which are reflected in the thesis.

Keywords: education, humanism, training, upbringing, personality, student, democracy

MÜASİR DÖVRDƏ TƏHSİLƏ HUMANİST VƏ DEMOKRATİK YANAŞMANIN PSIXOPEDAQOJİ MƏSƏLƏLƏRİ

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Xülasə: Müasir dövrdə təhsil sistemi şagirdyönümlü yanaşmanı əsas götürərək humanist və demokratik prinsiplərin tətbiqinə xüsusi önəm verir. Humanist yanaşma şagirdlərin şəxsiyyət kimi inkişafını, onların daxili potensialının üzə çıxarılmasını və emosional rifahını prioritet hesab edir. Bu yanaşmanın nəzəri əsasları Abraham Maslou və Carl Rocers kimi alimlərin ideyalarına dayanır. Onlar təhsildə empatiya, qəbul və özünüreallaşdırma kimi anlayışların vacibliyini vurğulayaraq müəllimin pedaqoji fəaliyyətində humanistlik və demokratiklik prinsiplərinin həm psixoloji, həm də pedaqoji mahiyyətinin elmi-nəzəri və praktik əsaslarından bəhs etmişlər. Araşdırmalar göstərir ki, humanistlik – şəxsiyyətin mənəvi, əxlaqi, psixoloji keyfiyyəti kimi insan təəffküründə, düşüncəsində formalaşmış, kollektivdaxili və qarşılıqlı münasibətdə, insan fəaliyyətinin müxtəlif mühitində təzahür olunur. Təlim-tərbiyə prosesinin demokratikləşdirilməsi müəllim və tərbiyəçilərlə uşaqların arasında həqiqi demokratik, işgüzar, mehriban münasibətlərin qurulması deməkdir ki, bunun da psixopedaqoji mahiyyəti tezisdə geniş formada izahı verilmişdir. Psixopedaqoji baxımdan bu yanaşmaların tətbiqi bir sıra məsələləri, xüsusən fərdi fərqlərin nəzərə alınması, inklüziv təhsil mühitinin yaradılması, müəllim-şagird münasibətlərinin sağlam qurulması və motivasiyaedici öyrənmə mühitinin təşkilini gündəmə gətirir. Eyni zamanda müəllimlərin peşəkar hazırlığı, emosional intellekti və kommunikativ bacarıqları bu prosesdə mühüm rol oynayır ki, bu kimi məsələlər tezisdə öz əksini tapmışdır.

Açar sözlər: təhsil, humanizm, təlim, tərbiyə, şəxsiyyət, şagird, demokratiklik

Introduction

The goal of humanizing and democratizing education is to educate a useful, comprehensive, harmoniously developed personality for society. In modern times, education acts not only as a process of knowledge transfer, but also as an important social institution that ensures the comprehensive development of the personality. Globalization, the rapid development of information technologies, and the changing demands of society require the application of more flexible,

humane, and democratic approaches from the education system. In this regard, the focus on the student in the educational process, taking into account his individual characteristics, interests and needs, is of particular relevance. Humanistic and democratic approaches to education, highlighting the human factor, are based on mutual respect, cooperation and free exchange of ideas in teacher-student relations. The psychopedagogical foundations of these approaches are reflected in the ideas of scientists such as Abraham Maslow and Carl Rogers. According to their ideas, education should not be limited only to intellectual development, but should also ensure the emotional, social and spiritual development of the individual. The subject of psychopedagogical activity is considered to be one who gives knowledge, educates and guides. The object of psychopedagogical activity is a person. The goal of humanistic pedagogy is to believe in the child's potential, to reveal his original nature, to respect him as a person, to serve good deeds, to be fair and to create conditions for the emergence of new pedagogical systems depending on historical, social, national and economic conditions with modern pedagogical thought, its methods and forms. Therefore, in order to form and develop a student as a personality, education should be humanized, democratized, humanitarianized, training should be developmental and educational, individualized and differentiated. Thus, the application of humanistic and democratic approaches in the modern education system raises new requirements and problems from both pedagogical and psychological points of view. The study and correct solution of these issues is of great importance in terms of improving the quality of education and forming personalities suitable for the development of society.

Materials and methods

Humanization of education is the main line of new pedagogical thinking. Only in this way can the human factor penetrate school life and acquire significant pedagogical significance. In modern conditions, humanization of education is the basis of the state's educational policy.

In this regard, the main factors that are paid attention to are the care of society for the upbringing of the growing generation, the presence of factors that determine the effectiveness of education, the solidarity of the collective, the high knowledge and skills of school teachers, etc. The concept of humanization of education is based on the ideas of humanism that adorn the history of world science and art. However, explaining humanization of education simply as humanistic education with traditional measures is a one-sided tendency. In science, such tendencies are called reductionism. Reduction in Latin means retreating, returning to the previous state. The humanization of education emerged as a new meaning in the history of pedagogical and psychological thought in the 1980s-1990s and became the main line of new pedagogical thinking. The humanization of education implies the humanization of not only the "teacher-student" but also the "student-student" relationships. However, this is only one aspect of the humanization of education.

In this study, various scientific sources and methods were used to investigate the psychopedagogical aspects of the humanistic and democratic approach in modern education. The material base of the study is monographs, scientific articles, international reports and normative documents written in the field of pedagogy and psychology. In particular, the theories of Abraham Maslow and Carl Rogers, representatives of humanistic psychology, as well as studies on modern pedagogical approaches, were used as the main reference sources.

Representatives of Azerbaijani psychology and pedagogy Akbar Bayramov, Mukhtar Hamzayev, Ramiz Aliyev, Bakhtiyar Aliyev, Rashid Jabbarov, Farrukh Rustamov, Huseyn

Ahmadov, Ajdar Aghayev and others, in their scientific works, widely comment on the democratic educational environment and teacher-student relations, the application of the humanistic approach in education, the issues of personality-oriented approach and upbringing in education, increasing the activity of students and the development of independent thinking. Research by Azerbaijani scientists shows that humanistic and democratic approaches occupy an important place in the development of the national education system and the application of these approaches serves the formation of students as creative, free and responsible individuals. The following methods were used in the research process: observation, survey (questionnaire), analysis and synthesis method, comparative analysis, etc. The selected methods allowed for a comprehensive study of the role of humanistic and democratic approaches in the educational process, their psychological and pedagogical effects, in accordance with the purpose of the research. Humanization of education implies the organization of the content, forms and methods of training and upbringing in such a way that the effective development of the activity, cognitive processes, personal qualities of each student is ensured, favorable conditions are created for the child to be able to read well and want to read well.

Results and discussion

The conducted research shows that the application of humanistic and democratic approaches in the modern educational environment has a significant impact not only on the academic achievements of students, but also on their formation as personalities. In particular, individualism, empathy and a supportive environment, which are the basis of the humanistic approach, lead to the strengthening of students' self-confidence, internal motivation and tendency to self-realization. Humanism in education is expressed in such manifestations as love for students, respect for their personality, and affirmation of their individuality, views, opinions, and actions for the sake of the development of society and the collective. The psychopedagogical activity of the teacher is complex and multifaceted. Depending on the tasks ahead, there are types of pedagogical activity: teaching, upbringing, organization, agitation, self-education, etc., each of which has a certain structure. In general, humanism requires prohibiting the exploitation of man by man, creating free conditions for the comprehensive development of man, treating him as a person, respecting and caring for his dignity.

The democratic approach is one of the main factors that is paid attention to in the modern education system. The democratic approach emphasizes the principles of equality, free expression of opinion, mutual respect and participation in the educational environment. In this model, the teacher plays the role of a facilitator and guide rather than an authoritarian manager. Involving students in the decision-making process develops their sense of responsibility and social skills. From a psychopedagogical point of view, the application of these approaches raises a number of issues. These include taking into account individual differences, creating an inclusive educational environment, and teacher-student relations.

Thus, we can say that the role of the teacher in the creation of a new, modern school is very great. In this work, high professionalism, deep knowledge and a broad worldview are required from our teachers. Today, the moral character of the teacher is more important than ever. That is, state building in many ways begins with the construction of new schools, with the preparation of new teaching staff. Such a teacher who can raise and educate generations who can lead society to brighter, happier days. In this regard, just as every profession has its own principles, the teaching profession also has its own principles. Each teacher should be based on these principles in his

pedagogical activity (Alizadeh, 115). In the upbringing of humanistic qualities, the teacher should take the following stages as the basis for his work principle: involving students in active activity in order to help people; organizing the correct perception and understanding of humanistic qualities among them; active struggle with selfish manifestations in the environment surrounding the students; strengthening humanistic relations in customs; transforming humanism into a sustainable form of behavior:

1. Kindness, timely and gratuitous help to others, sharing in the joy of others, expressing one's opinion openly and correctly with the aim of actively helping others;
2. Caring - sharing in people's troubles;
3. Sensitivity (attention, feeling oneself in the place of others, showing compassion, not humiliating others);
4. Respect (politeness, agreement with the interests and opinions of others, trust, active assistance to others).
5. Tolerance.

The humanization and democratization of the educational process has arisen from a new pedagogical thinking. The national thinking and philosophy of our people are imbued with the ideas of humanism. Care for a person, respect and reverence for his personality, feelings of compassion have always been at the forefront of the thoughts and ideas of our people.

The democratization of education ensures that the educational process is managed on the basis of democratic principles. Thus, student self-governments are created in educational institutions. They discuss all issues that concern them and can make decisions. Also, educational work in educational institutions is managed by the councils of those institutions. Each school also has the opportunity to elect its own teacher.

Conclusion

Democratization of the educational process means establishing truly democratic, businesslike, friendly relations between teachers and educators and children (Bayramov, 121). The basis of the democratic educational process is that teachers and educators trust and rely on the nature of the child. The democratic nature of education does not mean that every educational administrator can build and operate the education of children in the way they want. Democratization of education should not be arbitrariness, but should be directed towards correct and precise adherence to the democratic rules and regulations put forward by the state regarding education. Linking education with the construction of a democratic society requires the teacher to expand his/her knowledge base. Therefore, the teacher and educator should not forget to monitor the mass media. The teacher and educator should provide information about all areas of life related to the topics being taught. At this time, the teacher should not hesitate to rely on the abundance of information and provide this information to children. Because children can choose and assimilate what is necessary for them by participating in this flow of information. The democratic approach, on the other hand, develops the social skills and civic responsibility of students by ensuring the creation of an environment of equality, free expression of opinion and cooperation in the educational process.

The positive and supportive educational environment of the humanistic approach plays an important role in revealing the internal potential of students. The application of humanistic and democratic approaches in the modern education system acts as one of the main directions ensuring the comprehensive development of students. These approaches, by bringing the human factor to the fore in the educational process, allow taking into account the individual characteristics, interests and

needs of students. As a result, education is not limited only to the transfer of knowledge, but also serves the social, emotional and spiritual development of the personality.

In general, the successful application of these approaches depends on the professional competencies of teachers, the organization of the educational environment and the correct selection of modern pedagogical methods. Therefore, continuous renewal in the education system, support for the development of teachers and the creation of a humanistic-value-oriented environment should be one of the main priorities.

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IMPROVING STUDENT ENGAGEMENT IN TEACHING TECHNICAL SUBJECTS IN HIGHER EDUCATION THROUGH INTERACTIVE METHODS

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Abstract: Student engagement is a key factor influencing the effectiveness of teaching and learning in higher education, however, maintaining engagement in technical subjects is often difficult due to complex content, abstract concepts, and the use of traditional teaching methods. These challenges may reduce students' motivation and limit their active participation in the learning process. This study examines the role of interactive teaching methods in improving student engagement in teaching technical subjects, with a particular focus on construction management courses. The research is based on a review of relevant literature and the author's teaching experience. Interactive methods such as group work, discussions, and real-life

problem-solving activities are considered. The findings indicate that interactive teaching approaches increase student participation, enhance understanding of complex concepts, and improve motivation. In addition, active learning environments support the development of practical and critical thinking skills. The study concludes that integrating interactive methods into technical education can significantly improve student engagement and contribute to more effective teaching practices in higher education.

Keywords: student engagement, interactive teaching methods, teaching technical subjects, higher education, active learning, teaching strategies

ALİ TƏHSİLDƏ TEXNİKİ FƏNLƏRİN TƏDRİSİNDƏ İNTERAKTİV ÜSULLAR VASİTƏSİLƏ TƏLƏBƏ CƏLBOLUNMASININ YÜKSƏLDİLMƏSİ

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Xülasə: Tələbə fəallığı ali təhsildə tədris və öyrənmə prosesinin effektivliyinə təsir edən əsas amillərdən biridir, lakin texniki fənlərin tədrisində mürəkkəb məzmun, abstrakt anlayışlar və ənənəvi tədris metodlarının tətbiqi tələbələrin fəallığını təmin etməyi çətinləşdirir. Bu problemlər tələbələrin motivasiyasını azalda və onların dərs prosesində aktiv iştirakını məhdudlaşdırır. Bu tədqiqat interaktiv tədris metodlarının texniki fənlərin, xüsusilə tikinti idarəetməsi kurslarının tədrisində tələbə fəallığının artırılmasındakı rolunu araşdırır. Tədqiqat mövcud ədəbiyyatın təhlilinə və müəllifin tədris təcrübəsinə əsaslanır. Qrup işi, müzakirələr və real həyat problemlərinin həlli kimi interaktiv metodlar nəzərdən keçirilir. Nəticələr göstərir ki, interaktiv metodlar tələbələrin dərslə iştirakını artırır, mürəkkəb anlayışların daha yaxşı başa düşülməsinə kömək edir və motivasiyanı yüksəldir. Eyni zamanda, aktiv öyrənmə mühiti praktik və tənqidi düşünmə bacarıqlarının inkişafını dəstəkləyir. Nəticə olaraq, texniki təhsildə interaktiv metodların tətbiqi tələbə fəallığını əhəmiyyətli dərəcədə artırır və tədris prosesinin effektivliyini yüksəldir.

Açar sözlər: tələbə fəallığı, interaktiv tədris metodları, texniki fənlərin tədrisi, ali təhsil, aktiv öyrənmə, tədris strategiyaları

Introduction

Over the past years, student engagement has become increasingly important in enhancing the quality of teaching and learning in higher education. Students who are actively involved in the learning process tend to achieve better academic results and develop stronger critical thinking skills (Collaço, 2017).

Despite its significance, sustaining student engagement in technical subjects remains a major challenge. The complexity of the material, including abstract theories and specialized terminology, often makes it difficult for students to stay motivated and fully participate in lessons.

Another important issue is the continued use of traditional teacher-centered teaching methods. In such environments, the teacher is the main source of knowledge, while students play a passive role. This approach limits interaction and reduces opportunities for active learning, thereby negatively affecting student motivation and learning outcomes (Oviawe, 2020).

In addition, modern students have different expectations from the learning process. They prefer interactive, flexible, and student-centered learning environments rather than traditional

lecture-based instruction. This shift requires educators to adopt new teaching strategies that better meet students' needs (West, n.d.) .

Research also shows that student engagement is closely related to interaction, collaboration, and participation in learning activities. When students are actively involved in discussions, group work, and practical tasks, their understanding improves and learning becomes more meaningful (Gašpar & Mabić, 2015) .

Therefore, there is a growing need to implement interactive teaching methods in technical education. This research investigates how interactive methods can improve student engagement in teaching technical subjects in higher education, with a particular focus on construction management courses.

Materials and Methods

This study adopts a qualitative research approach to investigate the effectiveness of interactive teaching methods in improving student engagement in technical subjects in higher education. The research is based on two main components: a review of relevant literature and the author's teaching experience.

First, a literature review was conducted to examine existing studies on student engagement, active learning, and instructional strategies in higher education. Previous research highlights the importance of interactive and student-centered approaches in enhancing learning outcomes and participation (Collaço, 2017; Oviawe, 2020).

Second, the study is supported by practical teaching experience in construction management courses. During the teaching process, several interactive methods were applied, including group work, classroom discussions, case studies, and real-life problem-solving tasks. These methods were selected based on their ability to promote active participation and collaboration among students.

The main method of data collection was classroom observation. Student engagement was evaluated based on their level of participation, interaction with peers, willingness to contribute to discussions, and ability to apply theoretical knowledge in practical situations.

The data were analyzed qualitatively by comparing students' behavior and performance before and after the implementation of interactive teaching methods. This approach allowed the researcher to identify the impact of different teaching strategies on student engagement and learning effectiveness.

Results and Discussion

The results of this study demonstrate that the use of interactive teaching methods has a significant positive impact on student engagement in technical subjects. After the implementation of interactive approaches, students became more active, motivated, and involved in the learning process.

One of the main findings is that group work increases collaboration and communication among students. During group activities, students shared ideas, discussed solutions, and supported each other in understanding complex concepts. This type of interaction helped reduce difficulties related to technical terminology and abstract content.

Classroom discussions also played an important role in improving engagement. Students became more confident in expressing their opinions and asking questions. As a result, the classroom environment became more dynamic and student-centered. These findings are consistent with previous research, which highlights the importance of active participation in enhancing learning outcomes (Gašpar & Mabić, 2015).

In addition, the use of case studies and real-life problem-solving tasks proved to be particularly effective in construction management courses. These methods allowed students to connect theoretical knowledge with practical situations, making learning more meaningful and relevant. Students demonstrated a better understanding of the course material and improved their ability to apply knowledge in practice.

Furthermore, compared to traditional lecture-based teaching, interactive methods significantly increased students' motivation and attention during lessons. Students were more willing to participate in activities and showed greater interest in the subject.

Overall, the findings suggest that interactive teaching methods create a more engaging and effective learning environment. These methods improve student participation, enhance their understanding, critical thinking, and practical skills in technical education.

Conclusion

In conclusion, student engagement remains a critical issue in teaching technical subjects in higher education. Traditional teacher-centered approaches are often ineffective at motivating students or supporting active participation in the learning process.

This study has demonstrated that interactive teaching methods can significantly improve student engagement. Approaches such as group work, classroom discussions, case studies, and real-life problem-solving activities help create a more dynamic and student-centered learning environment. These methods encourage active participation, improve understanding of complex concepts, and support the development of critical thinking and practical skills.

The findings suggest that educators should integrate interactive methods into their teaching practices, especially in technical courses such as construction management. By doing so, they can enhance the quality of teaching and improve learning outcomes in higher education.

For future research, it is recommended to conduct more detailed studies using quantitative methods to measure the long-term impact of interactive teaching strategies on student performance and academic success.

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A COMPARATIVE STUDY OF CHALLENGES IN ONLINE AND FACE-TO-FACE EDUCATION

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Abstract: The rapid expansion of educational technologies and the widespread use of online learning platforms have significantly transformed higher education systems worldwide. This study aims to compare the challenges associated with online and face-to-face education by analyzing existing academic literature. A qualitative research design based on a systematic literature review approach was employed in this study. Relevant peer-reviewed studies published between 2020 and 2025 were reviewed and analyzed comparatively.

The findings indicate that online education offers flexibility and accessibility but also creates challenges related to communication, technological barriers, motivation, engagement, and digital literacy. In contrast, face-to-face education provides stronger interaction, immediate feedback, and a more structured learning environment; however, it may also involve limitations such as transportation difficulties, time restrictions, and reduced flexibility. The study demonstrates that both educational formats have distinct strengths and weaknesses that influence students' learning experiences in different ways.

The research concludes that blended or hybrid learning approaches may help reduce the limitations of both systems by combining the advantages of online and traditional education. The study contributes to the understanding of contemporary educational challenges and highlights the importance of developing flexible and technology-supported learning environments in higher education.

Keywords: Online education, face-to-face education, educational technology, higher education, learning challenges, blended learning

ONLAYN VƏ ƏNƏNƏVİ (ÜZ-ÜZƏ) TƏHSİLDƏ QARŞIYA ÇIXAN ÇƏTİNLİKLƏRİN MÜQAYİSƏLİ TƏDQIQI

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Xülasə: Təhsil texnologiyalarının sürətli inkişafı və onlayn təhsil platformalarının geniş yayılması dünya üzrə ali təhsil sistemlərində mühüm dəyişikliklərə səbəb olmuşdur. Bu tədqiqatın məqsədi mövcud elmi ədəbiyyatı təhlil etməklə onlayn və ənənəvi (üz-üzə) təhsilin qarşılaşdığı çətinlikləri müqayisəli şəkildə araşdırmaqdır. Tədqiqatda sistemli ədəbiyyat icmal yanasmasına əsaslanan keyfiyyət tədqiqatı dizaynından istifadə edilmişdir. 2020–2025-ci illər arasında dərc olunmuş, resenziyadan keçmiş elmi məqalələr seçilərək müqayisəli şəkildə təhlil edilmişdir.

Tədqiqatın nəticələri göstərir ki, onlayn təhsil çeviklik və əlçatanlıq baxımından mühüm üstünlüklərə malik olsa da, ünsiyyət, texnoloji maneələr, motivasiya, dərslə cəlb olunma və rəqəmsal savadlılıqla bağlı müəyyən çətinliklər yaradır. Bunun əksinə olaraq, ənənəvi (üz-üzə) təhsil daha güclü qarşılıqlı ünsiyyət, operativ geribildirim və daha strukturlaşdırılmış öyrənmə mühiti təqdim edir. Lakin bu təhsil forması nəqliyyat problemləri, vaxt məhdudiyyətləri və çevikliyin az olması kimi çatışmazlıqlarla da xarakterizə

olunur. Tədqiqat göstərir ki, hər iki təhsil formasının tələbələrin öyrənmə təcrübəsinə müxtəlif istiqamətlərdə təsir edən özünəməxsus üstünlükləri və zəif tərəfləri mövcuddur.

Araşdırmanın nəticəsinə əsasən, onlayn və ənənəvi təhsilin üstünlüklərini birləşdirən qarışıq (hibrid) təlim yanaşmaları hər iki sistemin məhdudiyyətlərini azaltmağa kömək edə bilər. Tədqiqat müasir təhsil problemlərinin daha dərinlən anlaşılmasına töhfə verir və ali təhsildə çevik, texnologiya dəstəyi ilə zənginləşdirilmiş öyrənmə mühitlərinin formalaşdırılmasının əhəmiyyətini vurğulayır.

Açar sözlər: onlayn təhsil, ənənəvi (üz-üzə) təhsil, təhsil texnologiyaları, ali təhsil, öyrənmə çətinlikləri, qarışıq (hibrid) təlim.

Introduction

The continuous advancement of educational technologies has considerably influenced teaching and learning practices in higher education institutions. In recent years, online education has emerged as one of the most significant topics in the educational field, particularly after the COVID-19 pandemic accelerated the shift from traditional classrooms to virtual learning environments (Rapanta et al., 2021). Universities worldwide were forced to rapidly adapt to digital learning systems, creating both new educational opportunities and various challenges for students and instructors.

Online learning environments provide flexibility, accessibility, and greater independence in the learning process. Students are able to access course materials from different locations and organize their study schedules more freely. Despite its advantages, online education can also create various difficulties associated with communication, concentration, motivation, internet access, digital skills, and social interaction (Adnan & Anwar, 2020). Earlier studies indicate that students participating in online learning environments frequently report lower interaction and engagement levels than those in conventional face-to-face classrooms (Kemp & Grieve, 2021). Conversely, face-to-face education offers direct communication between instructors and students, stronger classroom interaction, and a more organized learning atmosphere. Bali and Liu (2022) argue that traditional classroom environments create better opportunities for collaborative learning and immediate feedback than online platforms. However, conventional education also involves several limitations, including transportation difficulties, rigid schedules, classroom stress, and academic pressure.

Comparing online and face-to-face education has become increasingly important for understanding students' educational experiences and evaluating the strengths and weaknesses of both systems. Analyzing these challenges may help higher education institutions improve teaching strategies, educational technologies, and student support services. Therefore, this study aims to examine and compare the challenges associated with online and face-to-face education. The research mainly focuses on identifying academic, technological, and social difficulties experienced by students in both learning environments.

Research Questions

1. What are the primary challenges experienced by students in online education?
2. What are the major difficulties associated with face-to-face education?
3. How do the challenges of online education differ from those of face-to-face learning?
4. Which educational environment is considered more challenging by students?

Materials and Methods

This study adopts a qualitative research design using a systematic literature review approach. The primary purpose of the study is to compare the challenges of online and face-to-face education

through the critical examination and synthesis of existing academic literature. Literature review methodology is commonly utilized in educational research to identify recurring themes, research gaps, and emerging patterns within previous studies (Rapanta et al., 2021).

Data Sources

The data used in this study were obtained from peer-reviewed journal articles, academic books, and empirical studies focusing on online learning, traditional classroom education, and blended learning models. The selected sources were chosen according to their relevance to higher education and student learning experiences.

Only studies published between 2020 and 2025 were included in the review to ensure the relevance and currency of the findings. This period is particularly significant because the COVID-19 pandemic substantially transformed educational systems worldwide and accelerated the use of digital learning technologies (Adnan & Anwar, 2020).

The sources examined in this study were mainly gathered from Scopus-indexed journals and reputable academic databases, including *Computers & Education*, *Education Sciences*, *Educational Technology Research and Development*, and *The Internet and Higher Education* (Kemp & Grieve, 2021; Bali & Liu, 2022).

Literature Selection Criteria

The studies included in this research were selected according to several criteria:

- Direct relevance to online and/or face-to-face education
- Focus on higher education students
- Discussion of challenges related to communication, engagement, motivation, and technology use
- Publication in peer-reviewed academic journals

Studies unrelated to higher education or student learning experiences were excluded from the analysis.

Analytical Method

A comparative thematic analysis was employed in this study. The selected literature was carefully reviewed and categorized according to several thematic areas, including:

- Communication barriers
- Technological challenges
- Student engagement and interaction
- Motivation and learning behavior
- Time management and workload
- Accessibility of educational resources

The identified themes were compared across online and face-to-face educational settings to determine similarities and differences in students' experiences. This analytical approach enables the integration of findings from different studies into a coherent conceptual framework (Gherheș et al., 2021).

Limitations of the Study

This study relies exclusively on secondary data sources and does not include primary research methods such as surveys or interviews. Consequently, the findings are dependent on the interpretations and conclusions presented in previous studies. Furthermore, the absence of quantitative statistical analysis limits the ability to measure differences numerically.

Results and Discussion

The analysis of the selected literature reveals several recurring findings regarding the challenges encountered by students in online and face-to-face education. The reviewed studies demonstrate notable differences in communication, student engagement, technological barriers, and overall learning experiences.

Communication and Interaction Challenges

One of the most commonly identified problems in online education is the limited level of direct interaction between students and instructors. Kemp and Grieve (2021) emphasize that virtual learning environments often reduce real-time communication, negatively influencing classroom participation and student engagement. Likewise, Rapanta et al. (2021) note that insufficient interaction in online education may decrease students' sense of social presence and involvement in the learning process.

By contrast, face-to-face education provides stronger interpersonal communication and immediate feedback opportunities. According to Bali and Liu (2022), classroom-based learning environments enhance student–teacher interaction and support higher levels of engagement. Therefore, communication-related challenges appear to be more significant in online learning settings.

Technological Barriers

Dependence on technology represents another major challenge in online education. Students frequently experience unstable internet connections, insufficient digital skills, and limited access to technological devices. Yu (2022) highlights that digital literacy significantly influences students' academic achievement and learning experiences in online environments. Similarly, Bernard et al. (2021) argue that the successful integration of technology in education largely depends on students' ability to adapt to digital platforms. In traditional face-to-face education, technological barriers are generally less problematic because learning primarily occurs in physical classrooms.

Student Engagement and Motivation

Student engagement is generally lower in online learning environments than in traditional classrooms. Kemp and Grieve (2021) found that many students report decreased attention and motivation in virtual learning settings, mainly because of reduced social interaction and the absence of direct supervision.

On the other hand, face-to-face education creates a more structured educational environment that encourages active participation and accountability. Bali and Liu (2022) suggest that physical classroom settings positively influence student motivation and classroom involvement.

Learning Experience and Student Preferences

Gherheș et al. (2021) indicate that student preferences differ according to learning styles and technological adaptability. While some students prefer online learning because of its flexibility and convenience, others favor face-to-face education because it offers stronger interaction and support systems.

Yu (2022) also emphasizes that students with higher levels of digital literacy tend to achieve better academic outcomes and adapt more effectively to online learning environments.

Overall Comparative Findings

Overall, the findings suggest that online education is more strongly associated with technological, communication, and engagement-related challenges, whereas face-to-face education is more associated with structural limitations and reduced flexibility.

Rapanta et al. (2021) conclude that both educational systems possess distinct strengths and weaknesses, and that blended learning approaches may provide a more balanced and effective educational model.

Summary of Discussion

The findings demonstrate that online and face-to-face education each present unique challenges for students. Online learning is heavily influenced by technological dependency and reduced interaction, whereas traditional classroom learning is limited by logistical and flexibility-related constraints. Consequently, a blended or hybrid learning approach may represent the most effective solution for minimizing the weaknesses of both systems.

Conclusion

This study aimed to compare the challenges experienced by students in online and face-to-face education through a systematic review of existing academic literature. The findings reveal that both educational environments involve distinct difficulties that influence students' academic experiences in different ways.

The results indicate that online education is primarily associated with technological barriers, communication difficulties, lower engagement levels, and challenges related to maintaining motivation and self-discipline. Adnan and Anwar (2020) and Kemp and Grieve (2021) emphasize that limited interaction and dependence on digital infrastructure represent major weaknesses of online learning environments.

In contrast, face-to-face education is characterized by stronger communication, more structured learning processes, and higher levels of classroom engagement. Nevertheless, face-to-face education may also present several disadvantages, such as limited flexibility, commuting-related challenges, and fixed academic schedules (Bali & Liu, 2022).

Overall, the comparative analysis suggests that neither online nor face-to-face education is entirely free from limitations. Each educational model offers specific advantages and disadvantages depending on students' learning styles, technological access, and academic needs. Rapanta et al. (2021) argue that combining the strengths of both systems through blended learning approaches may create a more effective and sustainable educational model. In conclusion, this study contributes to a broader understanding of educational challenges in higher education and highlights the importance of developing flexible, inclusive, and technology-supported learning environments. Future studies may benefit from empirical research methods, such as surveys and statistical analysis, to further support and validate these findings.

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QUALITY PROBLEMS IN TEACHING ENGLISH IN REGIONAL SCHOOLS: THE CASE OF SHAKI DISTRICT SECONDARY SCHOOL

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Abstract: This paper comprehensively examines the multi-faceted quality-related problems encountered in teaching English as a foreign language within the regional secondary schools of Azerbaijan. With a localized focus on the structural environment of “Ashagi Goynuk Secondary School” in the Shaki District, this study illuminates the persistent disparities between urban and rural language education systems. Through ongoing systematic classroom observations and direct pedagogical involvement, two overarching challenges are critically analyzed: the fundamentally weak material and technical infrastructure of regional institutions, and the subsequent insufficient development of students' productive speaking and communicative skills. The academic paradigm in these regions historically favors passive rote-memorization over dynamic linguistic performance. To mitigate these challenges, this study presents a structured suite of actionable, localized recommendations. These include the implementation of low-cost, decentralized multimedia environments, teacher training focused on communicative strategies, and the integration of foundational Information-Communication Technology (ICT) frameworks to bolster student motivation. Ultimately, this paper argues for localized agency and systematic structural adaptations to ensure equitable foreign language outcomes for students across regional contexts.

Keywords: English language teaching, regional schools, speaking skills, material-technical base, communicative pedagogy.

REGIONAL MƏKTƏBLƏRDƏ İNGİLİS DİLİNİN TƏDRİSİNDƏ KEYFİYYƏT PROBLEMLƏRİ: ŞƏKİ RAYONU ORTA MƏKTƏBİ NÜMUNƏSİNDƏ

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Xülasə: Bu məqalə Azərbaycanın regional məktəblərində xarici dil kimi İngilis dilinin tədrisində qarşıya çıxan çoxşaxəli keyfiyyət problemlərini ətraflı şəkildə araşdırır. Şəki rayonunun "Aşağı Göynük Orta Məktəbi" nin struktur mühitinə xüsusi istinad edilərək, tədqiqat şəhər və kənd dil təhsili sistemləri arasındakı mövcud qeyri-bərabərliyi işıqlandırır. Davamlı sistemli sinif daxili müşahidələr və birbaşa pedaqoji fəaliyyət vasitəsilə iki əsas problem kritik şəkildə təhlil edilmişdir: regional müəssisələrin fundamental olaraq zəif material-texniki bazası və bunun nəticəsi olaraq şagirdlərin produktiv danışq və kommunikativ bacarıqlarının kifayət qədər inkişaf etməməsi. Bu bölgələrdəki akademik paradigma tarixən dinamik dil performansını əvəzinə passiv əzbərciliyə üstünlük verir. Bu çətinlikləri aradan qaldırmaq üçün tədqiqat strukturlaşdırılmış praktik tövsiyələr toplusunu təqdim edir. Bunlara az xərclə başa gələn regional multimedia guşələrinin yaradılması, kommunikativ strategiyalara yönəlmiş müəllim təlimləri və şagird motivasiyasını artırmaq üçün fundamental İnformasiya-Kommunikasiya Texnologiyalarının (İKT) integrasiyası daxildir. Sonda məqalədə regional kontekstdə şagirdlər üçün ədalətli xarici dil biliklərinin təmin edilməsi məqsədilə yerli adaptasiya və sistemli struktur dəyişikliklərinin vacibliyi əsaslandırılır.

Açar sözlər: İngilis dili tədrisi, regional məktəblər, danışq bacarıqları, material-texniki baza, kommunikativ pedaqogika.

Introduction

In the contemporary era of global integration, the English language has firmly established itself as the undisputed lingua franca of international communication, advanced scientific inquiry, cross-border commercial transactions, and technological innovation. Proficiency in English is no longer categorized as merely an optional luxury or an isolated academic asset within a student's curriculum; rather, it represents a foundational life skill that fundamentally determines a student's academic mobility, professional lifespan, and capability to access the broader global knowledge economy. Recognizing these shifting global paradigms, the Ministry of Education of Azerbaijan has introduced comprehensive curricular reforms designed to align national language instruction with western frameworks, prioritizing communicative outcomes and student-centered environments over classical, translation-based methods.

Despite the progressive intentions of these national educational policies, a significant quality disparity persists between metropolitan educational centers and regional public schools. While schools located in major urban centers like Baku or Ganja frequently benefit from robust technological integration, continuous professional teacher development, and structural exposure to foreign languages, the quality of English language teaching in many regional districts remains systematically below the desired standard. Regional schools operate under a unique matrix of challenges, encompassing geographical isolation, historical infrastructure deficits, and deeply rooted methodological conservatism. This academic text focuses on evaluating these practical barriers through a localized casing, examining the precise educational dynamics at play within the Shaki District to present scalable, real-world solutions.

Methodology and Materials

This research is structured as an analytical case study utilizing a qualitative, reflective methodology grounded in direct educational experience and systematic field observation. The specific subject of this study is the instructional and structural environment of "Ashagi Goynuk Secondary School," situated within the Shaki District of Azerbaijan. Over the course of an intensive pedagogical engagement, qualitative data was gathered through daily field notes, observing classroom interactions, testing oral proficiencies, and evaluating the physical resources allocated for language instruction. The observation specifically monitored the learning trajectories of students in

lower-secondary and upper-secondary brackets, focusing heavily on fifth-grade levels as a formative baseline for language acquisition.

By focusing on a single, representative regional secondary institution, the study attempts to bridge the gap between abstract macro-level educational theories and the authentic, microscopic micro-realities encountered by regional teachers daily. The investigation specifically targets the structural constraints of the classroom, pedagogical habits of the faculty, psychological profiles of rural students, and the availability of contemporary learning aids. This localized qualitative focus ensures that the subsequent findings and policy recommendations are derived not from speculative models, but from verified empirical realities in the field.

Discussion

1. Weak Material and Technical Infrastructure. A primary, structural constraint that severely undermines the quality of foreign language delivery in regional environments is the acute deficiency in institutional material and technical infrastructure. Language acquisition, by its psychological nature, relies heavily on diverse multisensory inputs—specifically authentic auditory and visual stimuli. However, at Ashagi Goynuk Secondary School, there is a near-absolute absence of modern educational technologies, including interactive smart boards, operational language laboratories, and high-speed internet infrastructure. Consequently, the instructional baseline reverts to a pre-digital paradigm where lessons depend exclusively on uniform state textbooks and static blackboard chalk drawings.

This critical technological vacuum completely curtails the instructor's capacity to deploy modern, communicative teaching frameworks. Without reliable digital projection tools or audio equipment, execution of systematic listening comprehension tasks is rendered unfeasible. Students are effectively separated from natural, authentic native-speaker cadences, accents, and situational dialogue configurations. Learning, therefore, is compressed into an abstract, visual-only memorization task where words are processed as static graphic symbols on a page rather than dynamic elements of live human speech.

2. Problems in Developing Speaking Skills. The cornerstone of modern applied linguistics and foreign language education is the cultivation of communicative competence. Yet, the prevailing educational culture within regional Azerbaijani schools continues to disproportionately favor abstract grammatical categorization and reading comprehension over active, productive oral articulation. In many regional settings, students demonstrate an impressive, hyper-developed structural knowledge of English on paper; they can easily categorize complex verb tenses and translate passages word-for-word. However, when these exact same students are placed into a spontaneous, low-stakes conversational scenario, they encounter near-total cognitive paralysis and verbal stagnation.

This systematic operational deficit is caused by several structural variables. First, regional classrooms often suffer from large student enrollments, leaving minimal individual speaking time for each student within a tight 45-minute lesson structure. Second, there is a pervasive, socio-cultural anxiety regarding peer judgment and a fear of making public errors, which acts as a powerful inhibitor to language production. Lastly, the utter absence of an external English-speaking ecosystem within rural villages means that students perceive the language strictly as an institutional hurdle to be cleared via memorization, rather than a functional vehicle for authentic human expression.

3. Psychological and Motivational Barriers. Beyond physical and infrastructural deficiencies, the qualitative advancement of language learning in regional schools is deeply obstructed by systemic psychological and motivational barriers. In regional environments like the Shaki District, the utilitarian value of acquiring English is often highly abstract and distant to a developing student. Unlike urban metropolises where multilingual signage, tourism, and multinational corporate entities are present, the rural environment remains firmly and uniformly monolingual. When the curriculum relies strictly on dry, teacher-centered lectures and archaic textbook narratives, students rapidly disengage, resulting in widespread passive participation and a psychological resistance to active oral production.

Contemporary educational literature explicitly addresses this dynamic, emphasizing that modern instructional technologies, including foundational Artificial Intelligence (AI) instruments and digital Information-Communication Technology (ICT) frameworks, can serve as profound motivational accelerators to stimulate cognitive activity and student autonomy (Akgun & Greenhow, 2022). When these modern modalities are completely omitted from regional classrooms, the language lesson feels entirely divorced from the digital, hyper-connected reality that contemporary youth navigate outside school gates.

4. Creating Supportive Language Environments. A critical flaw in the regional language acquisition model is that language exposure terminates instantly the moment the standard school bell sounds. True, sustainable foreign language acquisition requires continuous reinforcement and opportunities to apply knowledge within low-stress, non-academic contexts. In urban areas, affluent families mitigate this issue by enrolling children in specialized private speaking academies, hiring private tutors, or accessing international social clubs. In stark contrast, regional students must rely almost exclusively on the public school framework to fulfill both their formal academic instruction and their extracurricular linguistic expansion.

When a school operates solely as a rigid testing ground rather than a dynamic language environment, students never develop a personal, functional relationship with the target language. Therefore, the regional public school must actively broaden its institutional perimeter, re-imagining its communal spaces as active hubs for functional linguistic interaction.

Conclusion

In absolute summary, this detailed empirical and pedagogical analysis of Ashagi Goynuk Secondary School in the Shaki District demonstrates that the severe limitations in technical infrastructure, the historic reliance on rigid, grammar-heavy rote textbook models, and the subsequent starvation of speaking opportunities represent the primary structural forces choking English teaching quality in regional Azerbaijan. These multi-layered challenges are deeply systemic and cannot be entirely dissolved through short-term policy adjustments; they demand long-term, highly coordinated resource allocations and policy interventions at both institutional macro-levels and local micro-levels.

However, as this paper highlights, substantial qualitative progress can be immediate if local educators and school administrators execute systematic, low-cost adaptations. By strengthening the immediate physical environment via classroom multimedia corners, overhauling instructional habits to prioritize communicative and task-based models, and building vibrant, extracurricular conversational environments, regional schools can fundamentally alter student trajectories. Shifting the educational paradigm from passive, rigid memorization to active, technology-supported human communication will allow regional schools to unleash the latent potential of rural youth, ensuring

that geographical borders do not dictate a child's linguistic fluency in a hyper-connected global community.

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EFFECTIVE LESSON PLANNING IN ENGLISH LANGUAGE TEACHING ESTABLISHING COHERENT LINKS BETWEEN OBJECTIVES, INSTRUCTIONAL ACTIVITIES, AND LEARNING OUTCOMES

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Xülasə: Effektiv dərş planlaşdırılması ingilis dili tədrisinin keyfiyyətinin artırılmasında mühüm rol oynayır. Bu tədqiqat dərş planında məqsədlər, tədris fəaliyyətləri və öyrənmə nəticələri arasında uyğunluğun yaradılmasının əhəmiyyətini araşdırır. Tədqiqat keyfiyyət metodologiyasına əsaslanır və pedaqoji modellərin təhlilini əhatə edir. Nəticələr göstərir ki, bu komponentlər arasında uyğunluq tələbə fəallığını, tədris prosesinin aydınlığını və ölçülə bilən nəticələri əhəmiyyətli dərəcədə artırır.

Açar sözlər: dərş planlaşdırması, ingilis dilinin tədrisi, təlim məqsədləri, tədris fəaliyyətləri, təlim nəticələri, uyğunlaşdırma, pedaqogika

İNGİSLİS DİLİNİN TƏDRİSİNDƏ TƏSİRLİ DƏRS PLANLAŞDIRILMASI: MƏQSƏDLƏR, TƏDRİS FƏALİYYƏTLƏRİ VƏ ÖYRƏNMƏ NƏTİCƏLƏRİ ARASINDA ARDICIL ƏLAQƏLƏRİN QURULMASI

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Abstract: Effective lesson planning plays a crucial role in enhancing the quality of English language teaching (ELT). This study explores the importance of establishing coherent links between learning objectives, instructional activities, and learning outcomes in lesson design. The research is based on qualitative document analysis and pedagogical framework review. The findings indicate that alignment among these three components significantly improves student engagement, instructional clarity, and measurable learning outcomes. Moreover, the study highlights that poorly aligned lesson plans often lead to ineffective teaching practices and limited learner achievement.

Key words: lesson planning, English language teaching, learning objectives, instructional activities, learning outcomes, alignment, pedagogy

Introduction

Lesson planning in English Language Teaching (ELT) is not merely a procedural task but a conceptually grounded pedagogical practice shaped by major learning and instructional design theories. At its core, effective lesson planning requires the systematic alignment of learning objectives, instructional activities, and learning outcomes—an approach that reflects broader theoretical discussions on how knowledge is constructed, delivered, and assessed in educational contexts.

Theoretical perspectives in education highlight the importance of alignment in instructional design. The concept of “constructive alignment,” introduced by John Biggs (1996), suggests that effective teaching occurs when learning objectives, teaching methods, and assessment tasks are systematically interconnected. Similarly, the “backward design” model proposed by Grant Wiggins and Jay McTighe (2005) emphasizes starting the planning process with desired learning outcomes and then designing instructional activities that lead to those outcomes. These frameworks underline that coherence in lesson planning is not merely a methodological preference but a pedagogical necessity.

From a constructivist perspective, learning is understood as an active process in which learners construct knowledge through interaction and experience. Scholars such as Lev Vygotsky (1978) emphasize the role of social interaction and scaffolding in learning, suggesting that instructional activities must be carefully designed to support learners within their zone of proximal development. In this sense, lesson planning requires a deliberate connection between objectives and activities to ensure that tasks are cognitively appropriate and pedagogically meaningful. Similarly, Jean Piaget (1970) highlights stages of cognitive development, reinforcing the idea that learning objectives and outcomes must be aligned with learners’ developmental readiness.

In parallel, behaviorist and cognitive learning theories also contribute to the conceptual foundation of lesson planning. Benjamin Bloom (1956) taxonomy of educational objectives provides a hierarchical model that guides teachers in formulating measurable learning objectives, ranging from lower-order skills (remembering, understanding) to higher-order thinking skills (analyzing, evaluating, creating). This taxonomy supports the alignment principle by ensuring that instructional activities and assessment methods correspond to the intended cognitive level of learning outcomes.

A more integrative theoretical framework is provided by the concept of “constructive alignment” proposed by John Biggs (1996), which argues that effective teaching depends on the coherence between intended learning outcomes, teaching and learning activities, and assessment tasks. According to Biggs (1996), misalignment among these components weakens the learning

process and reduces instructional effectiveness. Complementing this view, Grant Wiggins and Jay McTighe (2005) introduce the “backward design” model, which prioritizes the identification of desired learning outcomes before planning instructional activities. This approach ensures that all elements of the lesson are purposefully directed toward achieving clearly defined goals.

Communicative language teaching (CLT) theory, as discussed by Michael Halliday (1978) and later expanded by Jack C. Richards (2006), emphasizes the functional use of language in real-life contexts. Within this framework, lesson planning must integrate communicative objectives with interactive activities and performance-based outcomes, reinforcing the need for coherence across all instructional components.

Therefore, this study aims to explore the role of coherent lesson planning in ELT by examining how alignment between objectives, instructional activities, and learning outcomes contributes to effective teaching and improved learner performance. By analyzing relevant pedagogical frameworks and practices, the study seeks to provide practical insights and a structured approach that educators can apply in their lesson planning processes.

Materials and Methods

This study adopts a qualitative research design aimed at examining the coherence between learning objectives, instructional activities, and learning outcomes in English Language Teaching (ELT). Given the conceptual nature of the research, a document analysis method was employed to systematically review and interpret key pedagogical frameworks and sample lesson plans. Document analysis is widely recognized as an appropriate method for analyzing educational texts and theoretical models (Bowen, 2009), as it enables the identification of patterns, relationships, and underlying principles within instructional design practices.

Research Design

The study is grounded in an interpretivist paradigm, which seeks to understand how theoretical constructs of alignment are reflected in lesson planning practices. A qualitative approach was selected to allow for an in-depth exploration of the relationships among lesson components rather than quantifying variables. This approach aligns with educational research traditions that emphasize meaning-making and contextual analysis.

Data Sources (Materials)

The materials analyzed in this study consist of:

- Foundational theoretical frameworks, including constructive alignment (John Biggs, 1996) and backward design (Grant Wiggins & Jay McTighe, 2005)
- Selected ELT methodology literature (e.g., Richards, 2006)
- Sample lesson plans obtained from teacher education resources and academic publications

Materials were purposefully selected to represent widely accepted models and practices in lesson planning and instructional design.

Data Collection Procedure

Relevant documents were identified through academic databases and educational repositories. Inclusion criteria were based on relevance to lesson planning, instructional alignment, and ELT contexts. Only peer-reviewed sources and widely cited pedagogical models were included to ensure reliability and academic rigor.

Data Analysis

The collected materials were analyzed using thematic analysis. The analysis process involved:

- Identifying key components of lesson planning (objectives, activities, outcomes)
- Examining the relationships and alignment between these components
- Categorizing patterns of coherence and misalignment

The analytical framework was informed by the concept of constructive alignment (Biggs, 1996), which served as a guiding lens for evaluating the consistency between instructional elements.

Validity and Reliability

To ensure the credibility of the findings, multiple sources were triangulated, including theoretical literature and practical lesson plan examples. The use of established educational theories enhances the validity of the analysis, while systematic coding procedures contribute to reliability.

Ethical Considerations

All materials used in this study are publicly available academic sources. Proper citation and referencing were applied in accordance with APA 7 standards to maintain academic integrity and avoid plagiarism.

RESULTS AND DISCUSSION

The analysis of theoretical frameworks and selected lesson plan samples revealed consistent patterns regarding the role of alignment in effective lesson planning within English Language Teaching (ELT). The findings are organized around three key dimensions: (1) alignment between learning objectives and instructional activities, (2) alignment between instructional activities and learning outcomes, and (3) overall coherence across all lesson components.

The findings demonstrate that the clarity and measurability of learning objectives directly influence the design and effectiveness of instructional activities. Lesson plans structured according to Benjamin Bloom's taxonomy showed a stronger alignment, as activities were intentionally designed to match cognitive levels such as application, analysis, and evaluation. In contrast, vague objectives resulted in activities that were either repetitive or pedagogically irrelevant.

Table 1. Alignment between Objectives and Instructional Activities

Type of Objective	Characteristics	Instructional Activity Type	Alignment Level	Observed Effect
Clear & Measurable	Action verbs, specific outcomes	Task-based, interactive activities	High	Strong engagement and focus
General	Broad, unclear verbs	Mixed or unfocused activities	Moderate	Partial understanding
Vague	Undefined learning goals	Random or passive activities	Low	Confusion, low effectiveness

The table indicates that measurable objectives lead to more structured and purposeful teaching practices, supporting higher levels of student engagement.

The second dimension of analysis focuses on how instructional activities contribute to achieving intended learning outcomes. Lesson plans reflecting the principles of constructive alignment (John Biggs, 1996) demonstrated a clear connection between classroom tasks and expected outcomes. Communicative and student-centered activities—such as discussions, role-plays, and problem-solving tasks—were particularly effective in achieving performance-based outcomes.

Table 2. Relationship between Instructional Activities and Learning Outcomes

Activity Type	Target Skill	Learning Outcome Type	Alignment Level	Effectiveness
Role-play	Speaking	Communicative competence	High	Very effective
Group discussion	Critical thinking	Analytical skills	High	Effective
Grammar drills	Accuracy	Structural knowledge	Moderate	Limited transfer
Lecture-based	Passive knowledge	Recall	Low	Minimal engagement

The analysis shows that interactive and communicative activities are more strongly aligned with higher-order learning outcomes, while passive methods are less effective in promoting meaningful learning.

The results confirm that full alignment among objectives, instructional activities, and learning outcomes significantly enhances overall instructional effectiveness. Lesson plans designed using the backward design model of Grant Wiggins and Jay McTighe (2005) exhibited the highest levels of coherence.

Table 3. Coherence Levels and Instructional Outcomes

Coherence Level	Characteristics	Student Engagement	Learning Achievement	Assessment Accuracy
High	Full alignment across all components	High	High	Accurate
Moderate	Partial alignment	متوسط (Moderate)	Moderate	Partially accurate
Low	No clear alignment	Low	Low	Inconsistent

The findings indicate that coherence is a determining factor in both student performance and the reliability of assessment processes. Despite the presence of strong theoretical models, several recurring issues were identified in lesson planning practices:

- Objectives not written using measurable criteria
- Activities not directly linked to intended outcomes
- Assessment tools misaligned with instructional content
- Table 4. Common Misalignment Issues in Lesson Planning

Problem Area	Description	Impact on Learning
Objective formulation	Non-measurable or unclear goals	Weak instructional focus
Activity selection	Irrelevant or disconnected tasks	Reduced engagement
Assessment design	Mismatch with outcomes	Inaccurate evaluation

These findings suggest that the gap between theory and practice remains a significant challenge in ELT contexts.

Overall, the results clearly demonstrate that coherent lesson planning—grounded in alignment principles—enhances instructional quality, student engagement, and measurable learning outcomes. The integration of theoretical models such as Bloom’s taxonomy, constructive alignment, and backward design provides a strong framework for improving lesson planning practices in ELT.

he findings of this study strongly support the theoretical assumption that effective lesson planning in English Language Teaching (ELT) depends on the coherent alignment between learning objectives, instructional activities, and learning outcomes. This alignment is not only a

methodological preference but also a fundamental pedagogical requirement grounded in established educational theories.

First, the results confirm the relevance of the “constructive alignment” framework proposed by John Biggs (1996). Lesson plans that demonstrated a clear correspondence between intended learning outcomes, teaching activities, and assessment methods resulted in higher levels of student engagement and achievement. This supports Biggs’ argument that meaningful learning occurs when all components of instruction are systematically aligned. The findings further indicate that misalignment leads to fragmented learning experiences, where students may participate in activities that do not directly contribute to the intended outcomes.

Second, the study’s results align with the principles of “backward design” introduced by Grant Wiggins and Jay McTighe (2005). Lesson plans that were structured by first identifying desired learning outcomes and then selecting appropriate instructional strategies showed significantly higher coherence and effectiveness. This approach ensures that teaching is outcome-oriented and that all classroom activities serve a clear instructional purpose. The present findings reinforce the importance of planning with the end goal in mind, particularly in language teaching contexts where skill acquisition must be measurable and progressive.

Furthermore, the role of cognitive frameworks, particularly Benjamin Bloom’s taxonomy (1956), is evident in the formulation of learning objectives. The study found that lesson plans incorporating higher-order cognitive skills (e.g., analysis, evaluation, creation) were more likely to include interactive and communicative activities, leading to deeper learning. This is consistent with Bloom’s theoretical model, which emphasizes the hierarchical development of cognitive processes and the need for instructional activities to match these levels.

From a language teaching perspective, the findings also resonate with the principles of communicative language teaching (CLT), as discussed by Jack C. Richards (2006). Activities such as role-plays, discussions, and problem-solving tasks were found to be more effective when aligned with communicative learning outcomes. This supports the view that language learning is most effective when learners actively use the language in meaningful contexts. Therefore, alignment in ELT should not only be structural but also functional, ensuring that language use is integrated into the learning process.

In addition, the study highlights the influence of constructivist learning theory, particularly the work of Lev Vygotsky (1978), which emphasizes the importance of social interaction and scaffolded learning. Instructional activities that were aligned with learners’ developmental levels and designed to promote interaction were found to be more effective in achieving learning outcomes. This suggests that alignment should also consider learner needs and cognitive readiness, not merely content and assessment.

CONCLUSION

This study concludes that effective lesson planning in English Language Teaching (ELT) requires a clear and coherent alignment between learning objectives, instructional activities, and learning outcomes. When these components are systematically connected, the teaching process becomes more structured, purposeful, and efficient. Such alignment helps ensure that each stage of the lesson contributes directly to achieving the intended learning goals, thereby improving overall instructional quality.

Furthermore, the findings show that well-aligned lesson plans enhance student engagement and support deeper learning. Activities that are directly linked to clearly defined objectives enable

learners to better understand the purpose of tasks and actively participate in the learning process. At the same time, clearly formulated learning outcomes make it easier to assess student performance accurately and consistently.

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TYPES OF HEARING DISORDERS, DIAGNOSIS AND THE IMPORTANCE OF EARLY DETECTION

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Abstract: This article examines the main types of hearing disorders, their causes, diagnostic characteristics, and the importance of early detection. Hearing loss is classified into conductive, sensorineural, and mixed forms, each of which may result from different medical or environmental factors. The study discusses the possible effects of hearing impairment on social, emotional, speech, and cognitive development, particularly in children. Various audiological assessment methods used for different age groups are described, including visual reinforcement audiometry, play audiometry, pure-tone audiometry, speech audiometry, computer audiometry, tympanometry, and bone conduction testing. Particular attention is given to newborn hearing screening and the role of objective diagnostic methods in early identification. The

importance of otoacoustic emissions (OAE) and auditory brainstem response (ABR) testing is emphasized. The findings demonstrate that early identification and timely intervention are essential for supporting children's speech, cognitive, and social development. Effective management requires cooperation among medical professionals, educators, and families. The appropriate use of modern diagnostic and rehabilitation methods can reduce the negative consequences of hearing impairment, support children's active participation in education, improve social adaptation, and enhance their overall quality of life. Early diagnosis also provides opportunities for timely treatment and individualized educational support.

Keywords: Hearing disorder, conductive hearing loss, sensorineural hearing loss, mixed hearing loss, audiometry, OAE, ABR, tympanometry, early diagnosis, newborns

EŞİTMƏ POZUNTULARININ NÖVLƏRİ, DİAQNOSTİKASI VƏ ERKƏN AŞKARLANMASININ ƏHƏMİYYƏTİ

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Xülasə: Bu məqalədə eşitmə pozulmalarının əsas növləri, yaranma səbəbləri, diaqnostik xüsusiyyətləri və erkən aşkarlanmasının əhəmiyyəti araşdırılır. Eşitmə itkisi keçirici, sensorinevral və qarışıq formalara bölünür və hər bir formanın müxtəlif səbəblərdən yaranma biləcəyi izah edilir. Məqalədə uşaqlarda və böyüklərdə eşitmə pozulmalarının sosial, emosional, nitq və idrak inkişafına göstərə biləcəyi təsirlər nəzərdən keçirilir. Tədqiqatda müxtəlif yaş qrupları üçün istifadə olunan audioloji müayinə üsulları, o cümlədən vizual gücləndirmə audiometriyası, oyun audiometriyası, ton audiometriyası, nitq audiometriyası, kompüter audiometriyası, timpanometriya və sümük keçiriciliyi testi haqqında məlumat verilir. Xüsusi diqqət yeni doğulmuş körpələrdə eşitmə skrininginə yönəldilir. Otoakustik emissiyalar (OAE) və eşitmə beyin kötüyü reaksiyası (ABR) testlərinin erkən aşkarlamada rolu vurğulanır. Nəticələr göstərir ki, eşitmə pozulmalarının erkən müəyyənəşdirilməsi və vaxtında müdaxilə uşaqların nitq, idrak və sosial inkişafının dəstəklənməsində mühüm əhəmiyyət daşıyır. Effektiv nəticə üçün tibbi mütəxəssislərin, müəllimlərin və ailənin əməkdaşlığı vacibdir. Müasir diaqnostik və reabilitasiya üsullarından düzgün istifadə eşitmə pozulmalarının mənfi nəticələrini azaltmağa, uşaqların təhsil prosesində fəal iştirakına, sosial uyğunlaşmasına və həyat keyfiyyətinin yaxşılaşmasına kömək edir.

Açar sözlər: Eşitmə pozulması, keçici eşitmə itkisi, sensorinevral eşitmə itkisi, qarışıq eşitmə itkisi, audiometriya, OAE, ABR, timpanometriya, erkən diaqnoz, yeni doğulmuş körpələr

"Not being able to see separates a person from objects. Not being able to hear separates a person from people."

— Immanuel Kant

Introduction

Hearing is one of the most important human senses. The sense of hearing is of great importance for human communication, as it occurs through language, laughter and tears as emotional sounds, music, and the emotions it creates. Hearing also plays an important role in orientation, recognizing danger, and communicating, for example, by shouting or whistling. It allows social participation and enables unique emotional group experiences, such as attending

concerts or singing together in a choir. People with hearing impairment or those who are completely deaf often face social isolation as the greatest negative impact on their quality of life.

Every event we hear is subconsciously and instantly evaluated by the emotional system of our brain as “pleasant” or “dangerous,” resulting in memory traces in the central nervous system and representation as neural circuits in the brain. If what we hear evokes a strong emotion, these memories and associated life situations often remain stable in our memory for decades. Hearing loss can cause individuals to feel isolated, frustrated, or often misunderstood. At any age, delays may occur in both speech and learning abilities. Depending on the type of activity, difficulties and maladaptation in activities such as play, education, and work are inevitable.

Hearing loss is not always limited to problems in the auditory analyzer. It can affect a person’s emotional well-being, self-confidence, and overall quality of life. The good news is that many types of hearing loss can be treated or managed with modern medical techniques and technologies. Hearing loss is a widespread problem worldwide. Many people live with some degree of hearing impairment.

Approximately 1 in every 1,000 newborns experiences hearing loss. Therefore, early diagnosis is very important, as hearing is essential for speech and language development in young children.

Hearing disorders manifest in various forms and differ according to their causes, location of damage, and severity. In this regard, proper classification of hearing disorders is of great importance. The types of hearing disorders are listed below:

1. *Conductive hearing loss:* Conductive hearing loss occurs when sound cannot pass normally through the outer or middle ear. Common causes include earwax blockage, ear infections, fluid in the middle ear, a perforated eardrum, and damage to the small bones inside the ear. Conductive hearing loss is often treatable with medication or surgery.

2. *Sensorineural hearing loss:* This occurs when the inner ear (cochlea) or the auditory nerve is damaged. Causes include aging, exposure to loud noise, genetic problems, certain infections, head injuries, and certain medications that may damage hearing. Sensorineural hearing loss may be irreversible but is usually managed with hearing aids or implants.

3. *Mixed hearing loss:* This is a combination of conductive and sensorineural hearing loss. In children, hearing loss is most often caused by genetic factors or infections during pregnancy. In adults, the most common causes are aging and exposure to loud sounds.

Identifying the types of hearing disorders makes timely detection even more important. Early diagnosis allows hearing loss to be identified at an initial stage and enables appropriate intervention measures. This, in turn, creates conditions for more effective support of a child’s speech, cognitive, and social development.

Materials and methods

The study was conducted using a descriptive and theoretical research approach to examine the main types of hearing disorders, diagnostic methods, and the importance of early detection. Scientific and pedagogical literature related to hearing impairment and audiological assessment was analyzed. Particular attention was given to conductive, sensorineural, and mixed hearing loss, as well as diagnostic methods such as pure-tone audiometry, tympanometry, OAE, and ABR. The study also considered hearing assessment methods used for children of different ages and newborn hearing screening procedures. The collected information was analyzed and summarized to identify the importance of early diagnosis and intervention for speech, cognitive, and social development.

Discussion

To determine hearing loss and its cause, a specialist evaluates the child's response to sound signals and conducts various diagnostic tests. Depending on the child's age, the following examinations are applied:

1. *Visual reinforcement audiometry* – used in children aged 6 months to 2.5 years and evaluates the child's response to sound with visual stimulation.

2. *Play audiometry* – conducted in children aged 1.5–5 years, using headphones and assessing responses to sound through play activities.

3. *Pure-tone audiometry* – applied from the age of 4 and determines hearing thresholds at different frequencies.

4. *Speech audiometry* – The simplest technical method. The doctor stands 6 meters away from the patient and pronounces a series of words or simple phrases understandable to patients of different intellectual and professional levels, at various sound levels (from whispering to normal speech). The patient's task is to repeat them. Based on the patient's responses, the doctor manually creates an audiogram.

5. *Computer audiometry* – This diagnostic method does not require the patient's active participation; therefore, it can be used in individuals with mental disorders who cannot understand the doctor's instructions, immobilized patients, or young children who cannot follow instructions. If necessary, general anesthesia (medically induced sleep) may be used during the examination. Electrodes connected to a computer are placed on the patient's head, headphones are positioned, and sounds of different frequencies and intensities are presented. Based on the brain's response to sound, the computer records an electroencephalogram (EEG) — a graphical representation that helps the doctor determine the location of the process affecting hearing.

6. *Tympanometry* – can be performed from birth and evaluates the functional condition of middle ear structures.

7. *Behavioral audiometry* – Behavioral audiometry is a screening test used in infants to observe their behavior in response to certain sounds. Using the unconditioned reflex method, the doctor observes the child's involuntary reactions to sound, while the conditioned reflex method tracks the development of conditioned responses (motor, behavioral, or verbal) to sound stimulation.

8. *Bone conduction test* – evaluates the transmission of sound to the inner ear through bone and helps determine the type of hearing disorder.

Based on the results of the examinations, the otolaryngologist establishes an accurate diagnosis and develops an appropriate treatment plan. In some cases, reversible hearing loss can be successfully treated with timely intervention. Treatment methods may include medication therapy or surgical intervention depending on the condition.

In addition, diagnostic procedures in newborns are carried out at an early stage to detect hearing disorders. In this case, objective screening methods are mainly used, and the infant's response to sound stimuli is evaluated. The functional condition of the hearing organs is checked through specific tests. There are two main methods for testing hearing in newborns. They can be used separately or together:

1. *Otoacoustic emissions (OAE)* – This test is performed using a small flexible probe placed in the child's ear. Sounds are transmitted through the probe, and a microphone inside records

otoacoustic responses (emissions) produced by a normal ear in response to sound. This test is painless, usually takes only a few minutes, and can be performed while the child is sleeping.

2. *Auditory brainstem response (ABR)* – This test uses electrodes attached to the child's scalp with adhesive tape. While the child is sleeping, small earphones placed in the ears produce clicking sounds, and brain activity in response to these sounds is measured. This test is also painless and takes only a few minutes.

If screening tests detect hearing loss in a child, further evaluation is required. To achieve the best therapeutic outcome, hearing loss in infants should be diagnosed by the age of 3 months. Preventive examinations at this age are very important, as verbal speech begins to develop from 6 months onward.

Early diagnosis and intervention are among the most important factors in minimizing the consequences of hearing disorders. Timely treatment and rehabilitation measures facilitate the child's social adaptation, ensure more active participation in the educational process, and improve quality of life. At the same time, the joint efforts of family, pedagogical, and medical specialists further increase the effectiveness of this process.

Conclusion

Advances in modern diagnostic methods provide extensive opportunities for the early detection of hearing disorders. In particular, screening tests performed on newborns and audiological examinations used during childhood make it possible to identify hearing loss in a timely manner. Early diagnosis, in turn, enables the implementation of appropriate intervention and rehabilitation measures, helping the child move closer to a normal developmental trajectory. This plays a crucial role in ensuring future social adaptation and academic success.

The use of modern technologies, medical treatment methods, and rehabilitation programs is of great importance in the management of hearing disorders. At the same time, the joint cooperation of families, teachers, and specialists is one of the key conditions for the success of this process. Raising public awareness on this issue and expanding preventive measures help to ensure the early detection of hearing problems and minimize the complications they may cause. Overall, timely and appropriate management of hearing impairments improves an individual's quality of life and ensures full integration into society.

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COMPARATIVE ANALYSIS OF PRONUNCIATION TEACHING METHODS: CLASSICAL APPROACHES AND AN ECLECTIC MODEL

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Abstract: Pronunciation is often given limited attention in university EFL (English as a foreign language) classes, even though it is one of the first things that affects whether learners are understood. It shapes clarity, the flow of speech, and how comfortable students feel when they speak in front of others. This article discusses three familiar approaches in EFL teaching, the Audio-lingual Method, the Direct Method, and the Communicative Approach, focusing on what each approach brings to classroom pronunciation work and which areas tend to remain weak. The discussion considers instructional priorities, common classroom activities, attention to individual sounds as well as stress and intonation, and the way corrective feedback typically works in practice. The realities of university teaching are also considered, including limited contact hours, mixed proficiency groups, and the pressure to progress quickly. Audio-lingual drills can build articulatory control and automaticity, but improvement does not always carry into spontaneous conversation. The Direct Method can create an immediate, input-rich environment, yet it may offer less structure for diagnosis and consistent correction. Communicative lessons often improve overall comprehensibility and confidence, while accuracy problems may persist and become harder to change. On this basis, the article supports a principled eclectic model that begins with controlled practice and then moves toward genuinely communicative speaking, supported by focused feedback that strengthens intelligibility and prosodic control. Practical implications are offered for lesson planning and assessment, where intelligibility is treated as a central outcome alongside functional accuracy.

Keywords: Pronunciation teaching, Audio-lingual Method, Direct Method, Communicative Approach, Eclectic Method

TƏLƏFFÜZÜN TƏDRİSİ METODLARININ MÜQAYİSƏLİ TƏHLİLİ: KLASSİK YANAŞMALAR VƏ EKLEKTİK MODEL

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Xülasə: Tələffüzə universitet səviyyəsində ingilis dilinin xarici dil kimi tədris olunduğu dərslərdə çox vaxt məhdud diqqət ayrılır, halbuki o, öyrənənlərin başa düşülüb düşülməməsinə təsir edən ilk amillərdən biridir. Tələffüz nitqin aydınlığını, axıcılığını və tələbələrin başqalarının qarşısında danışarkən özlərini nə qədər rahat hiss etmələrini formalaşdırır. Bu məqalə ingilis dilinin xarici dil kimi tədrisində geniş yayılmış üç yanaşmanı — audio-lingval metodu, direkt metodu və kommunikativ yanaşmanı müzakirə edir və hər bir metodun sinifdə tələffüzə necə töhfə verdiyinə, eləcə də hansı sahələrin adətən zəif qaldığına fokuslanır. Müzakirədə tədris prioritetləri, sinifdə geniş istifadə olunan fəaliyyətlər, ayrı ayrı səslərə, eləcə də vurğu və intonasiyaya ayrılan diqqət, həmçinin korrektə məqsədli geribildirimin praktikada adətən necə işlədiyi nəzərə alınır. Universitet tədrisinin reallıqları da nəzərə alınır; buna məhdud dərs saatları, müxtəlif səviyyəli qruplar və qısa müddətdə nəticə əldə etmək tələbi daxildir. Audio-lingval metoduna xas olan tələffüz yönümlü məşqlər artikulyasiya üzərində nəzarətin formalaşmasına kömək edər və düzgün tələffüzün avtomatik icra

bacarığını inkişaf etdirə bilər, lakin əldə olunan irəliləyiş hər zaman sərbəst danışmada özünü göstərmir. Direkt metod dərhal ünsiyyətə şərait yaradan və dil materialı baxımından zəngin bir mühit formalaşdırma bilər, lakin diaqnostika və ardıcıl düzəliş üçün daha az struktur təqdim edə bilər. Kommunikativ yanaşmaya əsaslanan dərslər çox vaxt ümumi anlaşılabilirliyi və inamı artırır, bununla yanaşı dəqiqlik problemləri qala və zamanla dəyişdirilməsi daha çətin ola bilər. Bu əsasda məqalə strukturlaşdırılmış praktikadan başlayan və daha sonra həqiqətən kommunikativ danışığa keçən, anlaşılabilirliyi və prosodik nəzarəti gücləndirən məqsədyönlü geribildirimlə dəstəklənən prinsiplərə əsaslanan eklektik modeli dəstəkləyir. Məqalədə dərs planlaşdırılması və qiymətləndirmə üçün praktik nəticələr də təqdim olunur.

Açar sözlər: tələffüzün tədrisi, audio-lingval metod, direkt metod, kommunikativ yanaşma, eklektik metod

Introduction

Pronunciation occupies a fundamental position in spoken communication because it directly influences the degree to which speakers are understood (Derwing & Munro, 2015). In university level English language instruction, however, pronunciation frequently receives less systematic attention than other language skills (Gilbert, 2014). As a consequence, learners may acquire a considerable command of grammar and vocabulary while still encountering persistent difficulties with intelligibility in authentic communicative situations (Foote et al., 2016). For this reason, the question of how pronunciation should be effectively taught continues to represent a significant concern within English language pedagogy (Mirzayev, 2023).

Different teaching traditions have approached pronunciation instruction from varying theoretical and methodological perspectives (Larsen-Freeman & Anderson, 2011). Among the most frequently discussed approaches are the Audio-lingual Method, the Direct Method, and the Communicative Approach (Mirzayev, 2024c). Each of these frameworks is grounded in particular assumptions about the process of language learning and therefore places differing emphasis on classroom practices such as repetitive drills, imitation, exposure to linguistic input, and communicative interaction (Mirzayev, 2024b).

Although these approaches have made important contributions to language teaching, their application to pronunciation instruction in university contexts also reveals certain limitations. Several studies have therefore suggested that pronunciation instruction may benefit from integrating elements from different methodological traditions rather than relying on a single teaching approach (Levis, 2018; Mirzayev, 2024d). Meta-analytic evidence also indicates that pronunciation instruction can produce substantial learning gains, particularly when instruction is sustained and supported by feedback (Lee et al., 2015). In response to these considerations, the present article offers a comparative analysis of these methods and evaluates their respective roles in the development of learners' pronunciation skills (Mirzayev, 2025a). On this basis, the article argues for the relevance of a principled eclectic model that draws selectively on elements from different approaches in order to support more balanced and effective pronunciation instruction. The objective of the present study is to compare these approaches and to examine their implications for pronunciation instruction in university-level English language teaching.

Materials and Methods

The present study employs a qualitative comparative research design to examine how different language teaching approaches address the development of pronunciation skills (Levis, 2018). The analysis concentrates on three widely recognized approaches within English language pedagogy: the Audio-lingual Method, the Direct Method, and the Communicative Approach

(Mirzayev, 2024c). These approaches were selected because they represent distinct theoretical traditions in language teaching and have played a significant role in shaping classroom practices across a variety of educational contexts (Larsen-Freeman & Anderson, 2011).

The comparison is structured around several analytical criteria directly related to pronunciation instruction. Similar analytical frameworks have been applied in studies examining the effectiveness of different pronunciation teaching approaches in higher education contexts (Mirzayev, 2024b). These criteria include the instructional priorities characteristic of each approach, the types of classroom practices typically associated with them, the extent to which attention is given to individual sounds as well as to suprasegmental features such as stress and intonation, and the ways in which corrective feedback is incorporated into classroom interaction (Celce-Murcia, Brinton, & Goodwin, 2010). In addition, the analysis considers the practical relevance of these approaches within university level teaching environments, where limited instructional time and groups with differing levels of language proficiency are common (Mirzayev, 2025a).

The analytical framework guiding the study draws upon established literature in pronunciation pedagogy and language teaching methodology (Derwing & Munro, 2015). Through this framework, the study seeks to identify both the potential contributions and the limitations of each approach in supporting the development of learners' pronunciation competence (Mirzayev, 2024d).

Results

The comparative examination of the three teaching approaches reveals significant differences in the ways pronunciation is addressed within classroom practice (Levis, 2018). Each approach offers specific advantages for the development of learners' pronunciation skills, yet each also presents certain limitations when considered within the context of university level language instruction (Mirzayev, 2024c).

Within the Audio-lingual Method, pronunciation instruction is primarily associated with repetition, imitation, and carefully structured practice (Larsen-Freeman & Anderson, 2011). Systematic drills and controlled exercises allow learners to gain greater control over articulatory movements and to develop more automatic production of individual sounds (Mirzayev, 2024e). Such procedures are particularly effective in strengthening the production of segmental features (Celce-Murcia, Brinton, & Goodwin, 2010). Segmental pronunciation work may be particularly useful when learners struggle with frequently occurring English sounds, as focused attention to individual sound patterns can support clearer and more accurate speech production (Mirzayev, 2024a). At the same time, the analysis indicates that improvements achieved through repetitive practice do not always transfer successfully to spontaneous speech, where learners are required to produce language with greater independence and flexibility. This limitation has also been noted in several studies examining the transfer of controlled pronunciation practice to real communicative situations (Derwing & Munro, 2015). In the Direct Method, pronunciation development is supported through continuous exposure to spoken language and the consistent use of the target language during classroom interaction (Mirzayev, 2024c). An environment rich in linguistic input encourages learners to develop more natural patterns of listening and speech production (Levis, 2018). Nevertheless, the method may provide fewer clearly defined procedures for identifying pronunciation difficulties and for implementing systematic corrective feedback (Mirzayev, 2024b).

By contrast, the Communicative Approach places primary emphasis on meaningful interaction and the development of communicative competence (Larsen-Freeman & Anderson, 2011). Within this framework, pronunciation is typically addressed as a component of overall

intelligibility rather than as a separate instructional focus (Levis, 2018). Although this orientation can enhance learners' confidence and general comprehensibility, the analysis suggests that persistent accuracy problems may remain insufficiently addressed in communicative classroom contexts (Mirzayev, 2025a).

Discussion

The results of the comparative analysis suggest that each of the examined approaches contributes to pronunciation development in specific ways, yet none of them sufficiently addresses all dimensions of pronunciation instruction when applied independently (Levis, 2018). The Audio-lingual Method offers clear advantages in strengthening articulatory precision and fostering automatic sound production through systematic repetition and carefully structured practice (Larsen-Freeman & Anderson, 2011). At the same time, the limited emphasis on meaningful communication may restrict learners' ability to apply these controlled pronunciation skills in spontaneous spoken interaction (Mirzayev, 2024b).

In contrast, the Direct Method places considerable importance on continuous exposure to spoken language and the consistent use of the target language in classroom interaction (Celce-Murcia, Brinton, & Goodwin, 2010). Such an environment may encourage the development of more natural listening and speaking patterns and support pronunciation growth through sustained engagement with authentic linguistic input (Mirzayev, 2024c). However, the absence of clearly defined procedures for diagnosing pronunciation difficulties and for implementing systematic corrective feedback may reduce the effectiveness of this approach in addressing persistent pronunciation problems (Derwing & Munro, 2015).

The Communicative Approach prioritizes meaningful interaction and the development of communicative competence (Larsen-Freeman & Anderson, 2011). Within this perspective, pronunciation is typically considered an element of overall intelligibility rather than an isolated instructional objective (Levis, 2018). Although this orientation can enhance learners' confidence and increase their willingness to participate in communicative activities, it may also allow certain inaccuracies in pronunciation to persist without adequate correction (Mirzayev, 2024c). Taken together, these observations underscore the value of a principled eclectic perspective in pronunciation instruction. Previous research has also emphasized that combining elements from multiple methodological traditions may offer a more effective framework for pronunciation teaching in higher education contexts (Mirzayev, 2024d; Mirzayev, 2025a). An approach that integrates structured pronunciation practice with communicative speaking tasks and carefully targeted feedback may provide a more balanced pedagogical framework, supporting both intelligibility and accuracy in spoken language use (Derwing & Munro, 2015). This position is further supported by research showing that pronunciation instruction is most effective when it targets specific segmental or suprasegmental features and when learning outcomes are evaluated through appropriate measures of pronunciation performance (Saito & Plonsky, 2019).

Limitations of the Study

The present study is grounded in a qualitative comparative examination of selected language teaching approaches and therefore entails certain limitations that should be recognized. The analysis is primarily based on existing scholarship in pronunciation pedagogy and language teaching methodology rather than on empirical classroom data. Consequently, the conclusions offered in this article represent theoretical interpretations of how these approaches address pronunciation instruction, rather than direct evidence derived from observed instructional practice. Furthermore,

the study concentrates on three widely discussed approaches and does not extend its scope to other methodological frameworks that may also have relevance for pronunciation teaching. Future research would benefit from empirical investigations conducted in classroom settings, which could provide a more detailed understanding of how these approaches operate in practice and how learners respond to different forms of pronunciation instruction.

Pedagogical Implications

The results of the present analysis point to several implications for pronunciation instruction in university level English language teaching. The findings suggest that exclusive reliance on a single methodological framework may limit the overall effectiveness of pronunciation instruction, as each approach tends to emphasize particular dimensions of pronunciation development. From this perspective, a principled eclectic orientation may offer a more balanced pedagogical alternative by drawing selectively on elements from different teaching traditions. Structured pronunciation practice can assist learners in developing articulatory control and greater accuracy in sound production, while communicative speaking activities create opportunities for learners to apply these skills in meaningful interaction. Interactive classroom activities may also strengthen pronunciation development by increasing learners' active participation, encouraging repeated spoken production, and creating more opportunities for pronunciation-focused feedback in university EFL contexts (Mirzayev, 2025b). At the same time, carefully targeted corrective feedback remains essential for supporting the development of both intelligibility and accuracy in spoken language use.

Conclusion

The comparative analysis of the Audio-lingual Method, the Direct Method, and the Communicative Approach indicates that each method contributes meaningfully to pronunciation instruction, although none is sufficient on its own to address the diverse needs of university-level EFL learners. The Audio-lingual Method offers the advantage of controlled and repetitive practice, which can strengthen articulatory accuracy and support the development of automaticity. The Direct Method, by contrast, promotes sustained exposure to spoken language and encourages more natural patterns of language use. The Communicative Approach further supports learners' confidence and overall intelligibility through meaningful interaction; however, it may not always provide adequate attention to persistent pronunciation errors that require systematic correction.

In light of these limitations, a principled eclectic model appears to offer a more appropriate framework for pronunciation instruction at the university level. Such a model enables teachers to integrate controlled pronunciation practice, communicative speaking tasks, and focused corrective feedback within a coherent pedagogical sequence. This integration can support both accuracy and intelligibility, two essential dimensions of effective spoken communication. The study therefore argues that pronunciation should not be treated as an isolated or secondary aspect of language teaching, but rather as an integral component of communicative competence. Future classroom-based research could further investigate how this eclectic model functions in authentic university teaching contexts and how it influences learners' pronunciation development over time.

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DEVELOPMENT AND IMPLEMENTATION OF A DIFFERENTIATED INSTRUCTION MODEL IN SPECIALIZED EDUCATIONAL INSTITUTIONS

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Abstract: This study examines the development and implementation of a differentiated instruction model in specialized educational institutions. Within contemporary pedagogical approaches, the consideration of individual learner differences plays a crucial role in enhancing instructional quality. In particular, in disciplined and highly structured environments such as cadet training institutions, the traditional “one-size-fits-all” instructional approach proves insufficient for achieving optimal learning outcomes. Accordingly, a differentiated instruction model is designed to organize the teaching process in accordance with cadets’ levels of prior knowledge, learning pace, and individual cognitive abilities. The research identifies initial diagnostic assessment, ability-based grouping, and the design of tiered tasks with varying levels of cognitive demand as the core components of the model. This approach integrates both individualized and collaborative learning opportunities, thereby contributing to the development of cadets’ academic achievement as well as their practical competencies. Furthermore, formative assessment mechanisms are employed to monitor learning progress continuously, while systematic feedback is utilized to refine and improve the instructional process.

The findings indicate that the implementation of differentiated instruction enhances cadets’ learning motivation, reduces performance disparities between low- and high-achieving learners, and improves overall instructional effectiveness. Consequently, this model represents a contemporary pedagogical strategy in specialized educational institutions and contributes significantly to the advancement of instructional quality.

Key words: differentiated instruction, adaptive teaching strategies, cadet-centered learning

İXTİSASLAŞMIŞ TƏHSİL MÜƏSSİSƏLƏRİNDƏ DİFERENSİALLAŞDIRILMIŞ TƏLİM MODELİNİN HAZIRLANMASI VƏ TƏTBİQİ

Türkan Habil qızı NƏSİROVA

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Xülasə: Bu tezisdə xüsusi təyinatlı təhsil müəssisələrində diferensiaslaşdırılmış təlim modelinin işlənməsi və tətbiqi məsələləri araşdırılır. Müasir təhsil yanaşmalarında fərdi xüsusiyyətlərin nəzərə alınması təlimin keyfiyyətinin artırılmasında mühüm rol oynayır. Xüsusilə kursantların təhsil aldığı intizamlı və strukturlaşdırılmış mühitlərdə ənənəvi “hamıya eyni yanaşma” modeli effektiv nəticə vermir. Buna görə də diferensiaslaşdırılmış təlim modeli kursantların bilik səviyyəsi, öyrənmə tempi və fərdi bacarıqlarına uyğunlaşdırılmış tədris prosesinin təşkilini nəzərdə tutur. Tədqiqat zamanı ilkin diaqnostika, səviyyəyə görə qruplaşdırma və fərqli çətinlik dərəcəsində tapşırıqların tətbiqi əsas komponentlər kimi müəyyən edilir. Model həm fərdi, həm də kollektiv təlim imkanlarını birləşdirərək kursantların akademik və praktiki

bacarıqlarının inkişafına xidmət edir. Eyni zamanda formativ qiymətləndirmə vasitəsilə onların inkişaf dinamikası izlənilir və geribildirim əsasında təlim prosesi təkmilləşdirilir.

Nəticələr göstərir ki, diferensiallaşdırılmış təlimin tətbiqi kursantların motivasiyasını artırır, zəif və güclü öyrənənlər arasında fərqi azaldır və ümumi təlim effektivliyini yüksəldir. Beləliklə, bu model xüsusi təyinatlı təhsil müəssisələrində müasir təlim strategiyası kimi çıxış edir və təhsil keyfiyyətinin yüksəldilməsinə mühüm töhfə verir.

Açar sözlər: fərqləndirilmiş təlim, uyğunlaşdırılmış tədris strategiyaları, kursant mərkəzli öyrənmə

Introduction

In the modern education system, taking into account learners' individual characteristics has become one of the key factors in improving the effectiveness of the teaching and learning process. Differences in students' (cadets') knowledge levels, learning pace, interests, and individual abilities reduce the effectiveness of the traditional "one-size-fits-all" instructional approach. For this reason, differentiated instruction has gained increasing importance in the field of education. Differentiated instruction refers to the adaptation of teaching content, process, and outcomes according to learners' individual needs. Tomlinson (2013) describes this approach as a proactive instructional process in which teachers intentionally plan instruction based on student differences. According to Heacox, effective differentiation requires continuous diagnostic assessment and the flexible application of instructional strategies.

In specialized educational institutions, particularly in cadet training environments, the learning process is characterized by a high level of discipline as well as intensive academic and practical activities. In such contexts, considering individual learner differences is especially important, as it contributes not only to improved academic achievement but also to the overall development of professional readiness. Researchers such as Saif, Safdar, Anis, and Muhammad (2024), Kurt (2013), Strogilos (2018), and Muhajirah (2020) also emphasize in their studies the positive impact of differentiated and adaptive instructional strategies on learning outcomes. Therefore, the development and implementation of a differentiated instruction model in specialized educational institutions is considered one of the relevant directions of contemporary pedagogical research. The aim of this study is to explore the theoretical and practical foundations of an effective instructional model based on the consideration of individual learner differences in such environments.

Research

This study employs a qualitative, descriptive, and analytical research design aimed at developing and examining a differentiated instruction model for specialized educational institutions, particularly those involved in cadet training. The main purpose of the research is to explore how individual learner differences can be systematically identified and integrated into instructional practice in order to enhance learning effectiveness, academic achievement, and overall training quality.

The research process is carried out in several interconnected stages. The first stage involves an extensive literature review of both theoretical and empirical studies related to differentiated instruction and adaptive teaching strategies. In this context, the works of Tomlinson (2013) provide the primary theoretical foundation, defining differentiated instruction as a proactive and systematic approach to teaching that responds to learners' readiness levels, interests, and learning profiles. Heacox further emphasizes the importance of continuous assessment and flexible instructional planning in ensuring effective differentiation.

In addition, recent empirical studies by Saif, Safdar, Anis, and Muhammad (2024) highlight that the structured application of differentiated instructional strategies significantly improves student engagement and academic performance. Kurt (2013) focuses on the role of technology in supporting individualized learning pathways, while Strogilos (2018) underlines the importance of adaptive teaching strategies in inclusive and structured educational environments. Muhajirah (2020) also demonstrates that differentiated instruction contributes to increased learner motivation and reduces achievement gaps among students with varying ability levels.

The second stage of the research involves an analytical examination of existing instructional practices in specialized educational institutions. This stage focuses on identifying how cadets' individual differences – such as prior academic knowledge, cognitive abilities, learning pace, motivation levels, and psychological readiness – are currently addressed within the teaching process. The analysis reveals that in many structured educational environments, instruction is still largely standardized, which limits opportunities for personalized learning and reduces the effectiveness of teaching outcomes for learners with diverse needs.

Based on the findings of the literature review and situational analysis, the third stage focuses on the development of a differentiated instruction model. The proposed model is designed specifically for specialized educational contexts and consists of several core components. These include: (1) diagnostic assessment of cadets to determine individual learning levels, (2) ability-based grouping to organize learners according to their readiness levels, (3) tiered instructional tasks designed with varying levels of cognitive complexity, and (4) continuous formative assessment to monitor progress and provide ongoing feedback.

Furthermore, the model emphasizes the balance between individualization and group-based training, which is particularly important in cadet education where discipline, teamwork, and structured learning are essential components. The integration of collaborative learning strategies with individualized instruction ensures that both personal development and collective performance are enhanced simultaneously.

The final stage of the research involves a conceptual evaluation of the proposed model. This evaluation focuses on its potential effectiveness in improving learner motivation, enhancing academic and practical skills, and reducing performance disparities among cadets. It is anticipated that the implementation of the model will lead to a more learner-centered educational environment, where instruction is responsive, adaptive, and outcome-oriented.

Diagnostic Assessment Tool for Cadets

In the implementation of the differentiated instruction model, an initial diagnostic assessment is used to identify cadets' individual learning characteristics and readiness levels. This assessment is based on the following criteria:

- Level of prior academic knowledge (standardized test results)
- Logical and analytical thinking skills
- Speed and accuracy in task completion
- Attention span and concentration stability during learning activities
- The results of this assessment serve as the basis for grouping cadets and planning instruction in accordance with their individual needs.

Ability-Based Grouping Framework

On the basis of diagnostic assessment results, cadets are grouped according to their learning readiness levels. This classification ensures that instruction is aligned with individual differences while maintaining the structured nature of the educational environment.

- Level A (Advanced group): cadets with high academic performance and strong analytical abilities
- Level B (Intermediate group): cadets with satisfactory performance requiring moderate support
- Level C (Basic support group): cadets who require additional instructional assistance and guided learning

This grouping is flexible and may be adjusted according to continuous assessment results.

Sample Differentiated Instruction Task Design

To ensure effective implementation of the model, instructional tasks are designed at different levels of cognitive complexity within the same topic.

Topic: Understanding and Applying Core Concepts

- Level A tasks: complex problem-solving, critical analysis, and application in new contexts
- Level B tasks: explanation, interpretation, and standard application exercises
- Level C tasks: identification of key concepts and completion of structured basic exercises

This structure allows all cadets to engage with the same learning content at different levels of difficulty.

Conclusion and Discussion

The results of the study indicate that the implementation of a differentiated instruction model in specialized educational institutions has a significantly positive impact on the quality of the teaching and learning process. The main components of the model—diagnostic assessment, ability-based grouping, tiered task design, and formative assessment—facilitate the consideration of cadets' individual learning needs.

The findings reveal that the application of differentiated instruction increases cadets' motivation, enhances their engagement in the learning process, and contributes to improved academic outcomes. At the same time, a gradual reduction in performance gaps between high-achieving and low-achieving cadets can be observed, which supports the creation of more equitable learning opportunities.

During the discussion, it becomes evident that although the structured and disciplined nature of specialized educational institutions may present challenges for the implementation of differentiated instruction, a well-designed model can effectively address these limitations. Unlike traditional instructional approaches, this model redefines the teacher's role from a knowledge transmitter to a facilitator who guides, monitors, and adapts the learning process according to individual needs.

Furthermore, the study suggests that the effectiveness of differentiated instruction largely depends on teachers' professional competence, methodological knowledge, and ability to manage flexible instructional processes. When appropriately applied, this approach can significantly enhance both the academic and practical skills of cadets.

In conclusion, the differentiated instruction model can be considered an effective contemporary pedagogical approach in specialized educational institutions. Its implementation contributes to the individualization of learning, improvement of instructional quality, and overall enhancement of educational outcomes.

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THE IMPACT OF SOCIAL ADAPTATION ON THE QUALITY OF LIFE OF CHILDREN WITH DELAYED INTELLECTUAL DEVELOPMENT, WITH A FOCUS ON THE DEVELOPMENT OF SOCIAL *SKILLS*

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Abstract: This article examines the impact of social adaptation and the development of social skills in children with delayed intellectual development on their quality of life. It is noted that these children's ability to adapt to the social environment directly affects their emotional well-being, academic achievement, and integration into society. The process of social adaptation involves the development of skills such as communication, adherence to social rules, and participation in collaborative activities. The article also emphasizes that delayed intellectual development creates difficulties in areas such as learning, memory, language, and social relationships, thereby hindering socialization. In this regard, the application of

appropriate pedagogical methods, pedagogical correction programs, and the creation of an inclusive educational environment are of particular importance. At the same time, improving teacher training, individualizing curricula, and ensuring family and societal support are considered key factors in facilitating the successful social adaptation of these children.

Keywords: delayed intellectual development, social adaptation, social skills, inclusive education, quality of life, pedagogical support, special education, social integration.

SOSIAL ADAPTASIYANIN LƏNG İNTELLEKTUAL İNKİŞAFLI UŞAQLARIN HƏYAT KEYFİYYƏTİNƏ TƏSİRİ: SOSIAL BACARIQLARIN İNKİŞAFI KONTEKSTİNDƏ ELMİ YANAŞMA.

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Xülasə: Bu məqalədə intellektual inkişafında ləngimə olan uşaqların sosial adaptasiyası və sosial bacarıqlarının inkişafının onların həyat keyfiyyətinə təsiri araşdırılmışdır. Qeyd olunur ki, bu uşaqların sosial mühitə uyğunlaşma qabiliyyəti onların emosional rifahına, akademik nailiyyətlərinə və cəmiyyətə inteqrasiyasına birbaşa təsir göstərir. Sosial adaptasiya prosesi uşaqların ünsiyyət qurma, sosial qaydalara riayət etmə və birgə fəaliyyətlərdə iştirak etmə bacarıqlarının formalaşmasını əhatə edir. Məqalədə həmçinin vurğulanır ki, intellektual inkişafın ləngiməsi uşaqların öyrənmə, yaddaş, dil və sosial münasibətlər sahəsində çətinliklər yaratmaqla onların sosiallaşmasını çətinləşdirir. Bu baxımdan, uyğun pedaqoji metodların tətbiqi, pedaqoji korreksiya proqramları və inklüziv təhsil mühitinin yaradılması xüsusi əhəmiyyət kəsb edir. Eyni zamanda, müəllim hazırlığının artırılması, tədris proqramlarının fərdiləşdirilməsi və ailə ilə cəmiyyətin dəstəyi bu uşaqların uğurlu sosial adaptasiyasını təmin edən əsas amillər kimi qiymətləndirilir.

Açar sözlər: intellektual inkişafın ləngiməsi, sosial adaptasiya, sosial bacarıqlar, inklüziv təhsil, həyat keyfiyyəti, pedaqoji dəstək, xüsusi təhsil, sosial inteqrasiya.

Introduction

The social adaptation and development of social skills in children with delayed intellectual development have a significant impact on their quality of life. The ability of these children to adapt to the social environment directly influences their emotional well-being, academic achievement, and integration into society.

The impact of social adaptation on quality of life: Social adaptation refers to the ability of children to adjust to their social environment and establish interpersonal interactions. This ability enhances their emotional well-being and overall quality of life. Research indicates that the development of social skills improves children's adjustment to school, their relationships with teachers and peers, and their general quality of life. Conversely, a lack of social skills may lead to problems such as social isolation, depression, and school disengagement (Gruzdeva, M.A., pp. 75-80). The adaptation of these children to the educational environment and the development of their social skills facilitate their future integration into society.

In many schools, particularly in rural regions, physical inaccessibility creates significant barriers to the education of children with delayed intellectual development. The lack of building adaptations, small classroom sizes, and the absence of appropriate equipment for special education hinder these children's integration into the educational environment (Huseynova, N.T., Baku, 2023, pp. 115-120). Insufficient teacher training and lack of professional development: The absence of specialized training in inclusive education limits teachers' ability to work effectively with children with special educational needs. This results in the inadequate adaptation of teaching methods and the failure to address individual learning needs. Lack of curriculum adaptation: Existing curricula are not sufficiently tailored to the developmental levels and learning styles of children with special needs, which restricts their participation and academic achievement. Lack of family and societal support: Families and society are not adequately informed about the importance of inclusive education, which further constrains the successful implementation of inclusive practices.

Children with delayed intellectual development, unlike their typically developing peers, encounter various challenges in their developmental processes. The social adaptation of these children is of critical importance for improving their overall development and quality of life. Intellectual disability limits children's abilities in learning, memory, language skills, and establishing social relationships. Social adaptation, in turn, refers to a child's ability to interact with their environment, express themselves, follow rules, and adjust to society.

Several key factors and theoretical perspectives are associated with this topic:

Analysis of Delayed Intellectual Development. The education and socialization of children with delayed intellectual development are particularly significant. Although their cognitive abilities may be less developed compared to other children, with proper guidance and pedagogical support, they can achieve better integration into society. Additionally, intellectual disability affects a child's social skills, emotional state, and psychological well-being (Aliyev A.A., 2019).

Social Adaptation and Intellectual Disability. Social adaptation is closely related to the development of a child's ability to establish relationships with their environment. Children with intellectual disabilities often face difficulties in forming such relationships, as they struggle to learn and adhere to social norms and rules. To support their social development, specialized approaches and instructional methods are applied (Aliyeva T.R., 2020).

The process of social adaptation involves several aspects of a child's life: Emotional and social support: Children with intellectual disabilities demonstrate better development when provided with positive emotional support. Such support enhances their self-confidence and facilitates their integration into society. Communication skills: The development of effective communication skills is a fundamental component of social adaptation. Pedagogical interventions and specialized teaching methods can improve children's social competencies (Huseynova N.T., 2017).

Pedagogical methods applied to children with delayed intellectual development serve as essential tools for facilitating their social adaptation. These methods are selected according to the individual characteristics of each child. Pedagogical correction and specialized educational programs can enhance children's social skills and support their integration into society. The pedagogical correction methods proposed by Hasanova S.A. (2016) and Huseynli M.F. (2023) play a positive role in improving the process of social adaptation. Societal support is essential for facilitating the social adaptation of children with delayed intellectual development. In particular, the school environment, family, and other social institutions play a significant role in the adaptation

process. The inclusion of children in inclusive educational settings, where they engage in joint activities with their peers, can accelerate the development of their social skills (Kazimova L.S., 2017). The social adaptation of children with delayed intellectual development is crucial for ensuring their successful adjustment to society. To support this adaptation process, appropriate pedagogical methods, social support, and an inclusive educational environment are necessary. Only within such environments can children with intellectual disabilities achieve full integration into society and develop the skills required to enhance their quality of life.

Comprehensive Overview of Social Adaptation in Children with Delayed Intellectual Development. Children with delayed intellectual development are those who experience certain levels of cognitive impairment and generally lag behind the typical developmental norms. Their social adaptation refers to the process of enhancing their ability to adjust to their environment and society. Delayed intellectual development can negatively affect a child's learning, memory, language abilities, and capacity to establish social relationships. Social adaptation, on the other hand, encompasses the development of skills such as interacting with others, self-expression, adherence to social norms, and participation in collaborative activities.

Delayed Intellectual Development and Characteristics of Children. Delayed intellectual development refers to a condition in which cognitive functions—such as thinking, comprehension, memory, learning, and problem-solving—develop below normative levels. This condition limits children's perception, language abilities, memory, analytical thinking, and reasoning skills.

Intellectual disability may manifest at different levels: Mild intellectual disability: Although children may demonstrate relatively developed language and social skills, they experience difficulties in academic areas. With appropriate support, they can achieve social adaptation. Moderate intellectual disability: These children face more significant challenges, and their social skills require more intensive training and support.

Social Support and the Role of the Family: The social adaptation of children with intellectual disabilities should be supported not only in school and other educational settings, but also at home and in other areas of society. The role of the family is one of the most important factors in the development of these children's social adaptation. In order to develop children's social skills, families should be guided and appropriate interventions should be implemented. Providing proper social role models within the family environment is highly important for maintaining the child's psychological stability and enhancing social support (Ismayilova K.A., 2021). **Pedagogical Interventions and Special Education Programs:** Special education programs designed for children with delayed intellectual development aim to enhance their educational and social adaptation skills. An individualized approach is essential for these children to achieve positive outcomes in the educational environment. Pedagogical interventions include various methods: Inclusive education: The joint education of children with intellectual disabilities alongside typically developing peers helps to improve their social skills. This approach also facilitates the establishment of social relationships among children with different abilities. Pedagogical correction: This method aims to address developmental deficiencies and strengthen children's social skills. Pedagogical correction programs enable children to become more independent and fluent in social interactions (Hasanova S.A., 2016). **Development of social skills:** These skills enable children to interact with others and regulate their emotional responses. Various games and activities are used to develop social skills. Social adaptation varies depending on the environment in which the child is placed. The support provided by society can significantly accelerate the development of children's

social abilities. In particular, inclusive educational environments are beneficial for developing social skills. School settings allow children to engage in collaborative activities, build relationships, and follow social rules. Families and teachers should properly support children in these environments and work towards improving their social interactions (Kazimova L.S., 2017).

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CONTINUOUS PROFESSIONAL DEVELOPMENT AND TEACHER REFLECTION

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Abstract: Research shows that a teacher's professional effectiveness, competence, and methodology are key indicators of how well students master the material. Teaching is a profession that is both highly significant in the modern era and requires high ethical standards. This article addresses the teaching structure of both novice and experienced teachers, including subject content, pedagogy and classroom practice, ICT skills, innovative teaching methods, and professional development needs. At the same time, it emphasizes the importance of teacher reflection in their continuous professional development. Like professionals in other fields, teachers also engage in professional development activities in order to continuously update their

knowledge and skills. In today's context of rapidly increasing information, it is especially important for them to stay informed about innovations.

Reflective practice is one of the key mechanisms in teachers' professional development. Reflectivability enables teachers to improve their professional qualifications, show interest in innovation, and continuously enhance their knowledge, skills, and competencies. Without reflection, development tends to be accidental, whereas with reflection, this process becomes planned, purposeful, and continuous. Teachers' reflective skills form the foundation for improving their professional preparation, fostering an interest in innovation, continuously expanding their professional knowledge, skills, and habits, and engaging in self-education, self-discipline, and pedagogical creativity. These qualities are essential for becoming a professional and competent educator.

Keywords: professional development, reflection, technology, continuous development, self-assessment, training, collaboration

SÜREKLİ MESLEKİ GELİŞİM VE ÖĞRETMEN YANSITMASI

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Özet: Yapılan araştırmalar, öğretmenin mesleki etkinliğinin, becerisinin ve metodolojisinin öğrencinin bilgiyi ne kadar iyi edindiğinin temel göstergeleri olduğunu göstermektedir. Öğretmenlik, modern dönemde hem büyük önem taşıyan hem de yüksek etik standartlar gerektiren bir meslektir. Bu makale, yeni ve deneyimli öğretmenlerin öğretim yapısını; ders içeriği, pedagojik yaklaşım ve sınıf uygulamaları, bilişim teknolojileri (BİT) becerileri, yenilikçi öğretim yöntemleri ve mesleki gelişim ihtiyaçları açısından ele almaktadır. Aynı zamanda öğretmenin yansıtıcı düşünmesinin (refleksiyonunun) sürekli mesleki gelişimdeki önemi vurgulanmaktadır. Diğer mesleklerde olduğu gibi öğretmenler de bilgi ve becerilerini sürekli güncellemek için mesleki gelişim faaliyetlerine katılmaktadır. Bilginin hızla arttığı günümüz koşullarında yeniliklerden haberdar olmaları özellikle önemlidir. Yansıtıcı düşünme, öğretmenlerin mesleki gelişiminde temel mekanizmalardan biridir. Yansıtıcı beceri, öğretmenlerin mesleki yeterliliklerini geliştirmelerine, yeniliklere ilgi duymalarına ve bilgi, beceri ve yetkinliklerini sürekli olarak artırmalarına olanak sağlar. Yansıtma olmadan gelişim çoğu zaman tesadüfi bir nitelik taşıırken, yansıtma ile bu süreç planlı, amaçlı ve sürekli hale gelir. Öğretmenlerin yansıtıcı becerileri; mesleki hazırlıklarını geliştirmelerinin, yeniliklere ilgi duymalarının, mesleki bilgi, beceri ve alışkanlıklarını sürekli olarak artırmalarının, öz eğitim, öz disiplin ve pedagojik yaratıcılıkla uğraşmalarının temelini oluşturur. Bu özellikler, onların profesyonel ve yetkin bir eğitimci olarak yetişmelerinin temelini teşkil eder.

Anahtar Kelimeler: mesleki gelişim, yansıtma (refleksiyon), teknoloji, sürekli gelişim, öz değerlendirme, eğitim, iş birliği

Introduction

Teachers are key individuals who play an important role in the development of society by transmitting the knowledge of the past, shaping the present, and laying the foundation of the future.

They not only provide students with knowledge but also develop their skills, shape their way of thinking, and influence all areas of the education system. For this reason, it is considered essential for teachers to be highly trained, specialized, and professional. In different countries around the world, various scientific studies and practical approaches are implemented to build a stronger foundation for the teaching profession. Even in ancient times, Socrates emphasized the responsibility of teaching, highlighting the importance of mobilizing all possible resources for the education of each student.

In the modern era, teaching is recognized as a profession that requires special training and qualifications. Individuals working in this field must obtain higher education, acquire relevant knowledge and skills, and officially gain the right to teach. Over time, the concept of a teacher has also changed. Previously, a teacher was seen only as a transmitter of information; however, today they are viewed as an engineer who designs the educational process, an artist who demonstrates creativity, and a specialist with deep knowledge in their field. A teacher plans the teaching process in advance, considers the developmental level of students, selects appropriate methods and techniques, and at the same time increases students' interest and enriches their thinking through creative approaches. In addition, they make evidence-based decisions in various educational situations and apply appropriate solutions to problems. Thus, teaching is a multifaceted and responsible profession that requires continuous self-development, acquisition of new knowledge and skills, careful monitoring of students' learning processes, and the use of various assessment methods.

Modern educational approaches require teachers to be professionally trained at a high level, to master different teaching methods and techniques in depth, and to consciously apply this knowledge in the school environment.

Materials and Methods

During the preparation of this article, local and foreign scientific sources, pedagogical literature, reports of international organizations, and modern educational approaches related to teachers' continuous professional development and reflective practice were analyzed. Comparative analysis, theoretical research, generalization, and interpretation methods were used throughout the study.

The research examined the approaches of international organizations such as UNESCO and OECD regarding teachers' professional development. In addition, information related to teacher certification, diagnostic assessment, and professional training programs in Azerbaijan was reviewed. The article also analyzed the role of reflection in teachers' professional activities from a pedagogical perspective and identified its relationship with continuous professional development.

Continuous Professional Development

After completing a bachelor's degree, a teacher's learning process should not stop; on the contrary, it should continue throughout their professional life. Although it is not possible to reach a definitive conclusion on this issue solely through research, studies show that the continuous development of teachers is essential. Teachers' professional development involves the renewal of their knowledge and skills throughout their careers, as well as the proper organization of its content, quality, and other related factors. In today's rapidly developing technological era, it is becoming difficult for the knowledge acquired years ago to fully meet current demands. Therefore, teachers should use modern and effective methods that engage students' interest in their lessons.

The fact that knowledge gained during bachelor studies becomes outdated over time, the emergence of new challenges, and a decline in professional competence increase the need for continuous learning among teachers. These problems are often difficult to solve individually, which may lead to professional burnout and dissatisfaction. This, in turn, negatively affects the quality of teaching. Overall, the quality of education directly depends on the teacher's potential, and a strong education system requires qualified and continuously developing teachers.

A teacher's professional development should not be limited to the initial training stage but must continue throughout their entire career. Training seminars, advanced qualification courses, professional learning communities, and mentoring and coaching programs update teachers' knowledge and skills and ensure their continuous development. International organizations such as UNESCO and the OECD also consider teachers' professional development a key component of lifelong educational development. This approach has been formed within the framework of the concept of "Continuous Professional Development" (CPD) and is recognized as one of the main strategic directions in education systems [5].

Professional development is a continuous process aimed at improving teachers' pedagogical knowledge, teaching skills, digital literacy, and adaptive approaches. CPD enables teachers to respond to changing curriculum requirements and adapt to students' individual needs and educational technologies.

Continuous learning for teachers can be organized in various forms:

- Advanced qualification courses and professional development training
- Mentoring and coaching programs
- Educational conferences and scientific-practical seminars
- Professional learning communities and subject-based methodological groups
- Online courses and webinars (e.g., Coursera, EdX, Microsoft Educator Center, etc.)

Each of these forms serves to enhance teachers' experience and integrate innovations into the teaching process.

Research shows that teachers who regularly participate in professional development:

- Master learning strategies that improve student achievement;
- Approach lessons with motivation and creativity;
- Use modern technologies and resources more effectively;
- Have higher professional satisfaction and self-confidence .

Organization of Professional Development in Azerbaijan

In recent years, a number of innovations have been implemented in the field of teachers' professional development in Azerbaijan. The diagnostic assessment and certification process for teachers working in public general education institutions has contributed to improving their level of knowledge. The MİQ (Teacher Recruitment Exam) conducted during teacher hiring demonstrates their professional competence and academic literacy. At the same time, training programs organized by the Institute of Education and other institutions support this process.

Areas such as student behavior and classroom management, individualized approaches to learning, subject-specific teaching skills, the use of new workplace technologies, student assessment practices, counseling and career guidance for students, information and communication technology (ICT) skills for education, teaching in multicultural or multilingual contexts, and the development of transversal skills (such as problem-solving and "learning to learn"), as well as approaches to developing interprofessional skills, subject knowledge and understanding, school

leadership and management, and curriculum knowledge, all highlight the need for a high level of professional development in each of these areas. If these requirements are met, we can observe an increase in both novice and experienced teachers in the education sector who demonstrate a strong need for continuous professional development across all areas [2].

Teacher Reflection

Reflection is one of the key mechanisms in the learning process that allows for analyzing and deeply understanding all stages of knowledge acquisition. The term reflection (from Latin “reflexio”) means “looking back.” In other words, a teacher must observe the results of their activity, analyze them, evaluate them, and make judgments about them. At the same time, reflection involves examining our own thoughts and actions and asking questions related to them [Hüseynzadə et al., 2021, 3].

In a reflective educational system, students analyze their own behavior, recognize both their strengths and weaknesses, and accordingly guide their own development. This approach also enables them to freely express their thoughts and actively participate in group activities. The reflective teaching process strengthens communication between students and teachers and supports the formation of a social and active learning environment.

During reflection, students recall and analyze what they have learned in the lesson. However, reflection is not only important for students but also essential for teachers to improve their teaching practices. It enables teachers to design the teaching-learning process more effectively and purposefully [Newell, 2024, 7]. In this way, the teacher can evaluate how successful the lesson was, the level of achievement of learning objectives, and the level of student engagement in the lesson. Reflection is a factor that demonstrates how successfully learning has taken place for both students and teachers. Therefore, having a reflective mindset is an important requirement for organizing an effective learning environment and achieving better outcomes. Teacher reflection refers to the process of evaluating, thinking about, and improving one’s own teaching practice. It involves reviewing the challenges, successes, student progress, and the teacher’s own professional skills during the teaching process. Teacher reflection helps to evaluate the effectiveness of teaching methods and approaches and enables teachers to search for better ways to achieve improved results in their practice, as well as to understand which methods and assessment tools are more effective.

The main aspects of teacher reflection can be summarized as follows:

- Self-assessment: The teacher objectively evaluates their teaching process, identifies what works well and what needs improvement.
- Student responses: The teacher analyzes students’ reactions, how well they understand the lesson, and which parts they find difficult. This helps to measure the effectiveness of teaching methods and lesson plans.
- Professional development: Reflection is essential for professional growth. The teacher strives to improve their professional skills and abilities by updating teaching methods and applying innovative pedagogical approaches.
- Lesson objectives and outcomes: The teacher analyzes how the lesson objectives were achieved and whether the expected learning outcomes were met.
- Emotional state: The teacher also reflects on their emotional condition, as it can influence the organization of teaching and students’ adaptation to the learning process.

Results and Discussion

The conducted research shows that teachers' continuous professional development plays an important role in improving the quality of education. In the modern era, the rapid development of technology and ongoing changes in the education system make it necessary for teachers to continuously update their knowledge and skills. In this regard, professional development courses, seminars, webinars, mentoring programs, and professional learning communities have a significant impact on teachers' growth. The findings indicate that reflective practice enables teachers to evaluate their pedagogical activities, identify shortcomings, and organize the teaching process more effectively. Through reflection, teachers can analyze their own performance and better determine students' learning levels and educational needs.

The discussions revealed that continuous professional development and reflection are closely interconnected concepts. A reflective approach accelerates teachers' professional growth and strengthens their pedagogical skills as well as ICT competencies. Therefore, both novice and experienced teachers should remain open to lifelong learning and prioritize reflective practice, which is considered one of the main requirements of the modern education system.

Conclusion

Of course, new teachers can develop their understanding of teaching and learning through trial and error. In order to cope with the various challenges they will face, novice teachers must continuously work on themselves. In this regard, teacher reflection serves the continuous professional development of the teacher. Through reflection, a teacher evaluates themselves, monitors students' attitudes, and clarifies the extent to which learners have mastered knowledge and skills.

Thus, for continuous development, a teacher must improve their own practice. For CPD, teachers should acquire pedagogical skills and participate in seminars, webinars, and scientific conferences. Reflective practice is highly important for both novice and experienced teachers in the development of their pedagogical careers. At the same time, they should be able to collaborate with colleagues and parents. They must also be familiar with modern digital technologies and possess ICT skills.

A teacher's continuous professional development and reflective capacity are extremely important for the effectiveness of the teaching process.

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INTERDISCIPLINARY PEDAGOGICAL INTEGRATION OF CHEMISTRY AND PHARMACY EDUCATION: ENHANCING COGNITIVE AND PROFESSIONAL COMPETENCIES IN HIGHER EDUCATION

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Abstract: The rapid advancement of pharmaceutical sciences in recent decades has significantly increased the demand for integrated and interdisciplinary educational approaches in higher education. Among the core scientific disciplines, chemistry plays a fundamental role in explaining drug behavior, including molecular structure, reaction mechanisms, pharmacokinetics, and pharmacodynamics.

This study investigates the pedagogical importance of integrating chemistry with pharmacy education in order to enhance students' cognitive development and professional readiness. The research is based on constructivist learning theory, which emphasizes active student involvement in the learning process and the construction of knowledge through experience.

Special attention is given to modern teaching strategies such as problem-based learning (PBL), inquiry-driven instruction, and cognitive scaffolding, which support deeper conceptual understanding. In addition, the role of digital learning tools—including virtual laboratories, molecular simulations, and interactive visualization systems—is analyzed as a means of improving comprehension of complex scientific processes.

The results indicate that interdisciplinary teaching models significantly improve students' analytical thinking, problem-solving abilities, and ability to transfer theoretical knowledge into practical pharmaceutical contexts. Overall, the study concludes that combining chemistry and pharmacy education through modern pedagogical strategies contributes to the development of highly competent future professionals in the pharmaceutical field.

Keywords: chemistry education, pharmacy education, interdisciplinary learning, constructivist pedagogy, cognitive development, active learning, higher education innovation

KIMYA VƏ ƏCZAÇILIQ TƏHSİLİNİN FƏNLƏRARASI PEDAQOJI İNTEQRASIYASI: ALI TƏHSILDƏ KOQNITIV VƏ PEŞƏKAR BACARIQLARIN ARTIRILMASI

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Xülasə: Kimya və əczaçılıq təhsilinin fənlərarası integrasiyası müasir ali təhsildə artan elmi və didaktik əhəmiyyətə malikdir. Kimya fənni farmakokinetika, molekulyar farmakodinamika və dərman–reseptor qarşılıqlı təsirləri kimi mürəkkəb biokimyəvi proseslərin izahında əsas rol oynayır. Bu səbəbdən onun əczaçılıq tədris proqramına effektiv integrasiyası gələcək mütəxəssislərin formalaşdırılmasında mühüm əhəmiyyət daşıyır. Bu tədqiqat kimyəvi biliklərin əczaçılıq təhsili çərçivəsində daha səmərəli mənimsənilməsini təmin edən pedaqoji yanaşmaları araşdırır. Araşdırma konstruktivist öyrənmə nəzəriyyəsinə əsaslanır və problem əsaslı öyrənmə (PBL), sorğuya əsaslanan təlim və mərhələli koqnitiv dəstək (scaffolding) kimi tələbəyönümlü metodlara fokuslanır. Eyni zamanda simulyasiya əsaslı təlim mühitləri, virtual laboratoriyalar və rəqəmsal texnologiyaların istifadəsi öyrənmə prosesini daha interaktiv edir və koqnitiv yükü azaldır. Nəticələr göstərir ki, integrativ kurikulum modeli kimya və əczaçılıq fənləri arasında elmi uyğunluğu gücləndirir və tələbələrin analitik düşünmə, anlayış və peşəkar bacarıqlarını inkişaf etdirir. Tədqiqat belə nəticəyə gəlir ki, müasir pedaqoji texnologiyalarla dəstəklənən fənlərarası tədris yanaşması gələcək əczaçıların kompleks elmi və praktiki problemləri həll etmə qabiliyyətini artırır.

Açar sözlər: chemistry education, pharmacy education, interdisciplinary learning, constructivism, cognitive support, active learning, higher education Pedagogy

Introduction

The development of modern pharmaceutical science has created a strong need for closer integration between different scientific disciplines, especially chemistry and pharmacy. These two fields are closely connected, as chemistry provides the essential foundation for understanding how drugs are designed, how they interact with biological systems, and how they produce therapeutic effects in the human body.

In many traditional educational systems, chemistry is still taught in a relatively isolated and theoretical manner, mainly through lectures and memorization-based approaches. While such methods help students acquire basic knowledge, they often fail to support the development of higher-order thinking skills such as analysis, synthesis, and practical application of knowledge in real-world pharmaceutical contexts (Biggs & Tang, 2011).

Modern educational research increasingly supports the idea that learning should be an active and student-centered process. According to constructivist learning theory, students do not simply receive information passively; instead, they actively construct knowledge through interaction, experience, and reflection (Vygotsky, 1978; Bransford et al., 2000). This perspective has led to the development of innovative teaching strategies such as problem-based learning, inquiry-based learning, and scaffolding techniques, which aim to improve student engagement and deepen understanding (Hmelo-Silver, 2004; Wood et al., 1976).

Furthermore, the integration of digital technologies into science education has transformed the way complex scientific concepts are taught. Tools such as virtual laboratories, computer simulations, and interactive models allow students to visualize abstract chemical and pharmaceutical processes more clearly. This not only improves comprehension but also reduces cognitive overload and enhances learning efficiency (Mayer, 2009; Sweller, 1988).

Overall, there is a growing recognition that integrating chemistry and pharmacy education through modern pedagogical approaches is essential for preparing students who are capable of critical thinking, scientific reasoning, and effective professional practice in the pharmaceutical field.

Materials and Methods

This research adopts a qualitative and exploratory approach based on a systematic review of existing literature. The main objective is to analyze and synthesize interdisciplinary teaching strategies applied in chemistry and pharmacy education.

The study draws on a wide range of academic sources, including peer-reviewed journal articles, conference proceedings, and authoritative publications in the fields of science education and pharmaceutical pedagogy. The selection of materials was based on relevance, academic quality, and methodological reliability.

The collected data were analyzed using thematic content analysis. This method allowed for the identification of recurring patterns and key pedagogical themes such as problem-based learning (PBL), inquiry-based instruction, cognitive scaffolding, and technology-enhanced learning environments (Prince, 2004; Hmelo-Silver, 2004).

Additionally, a comparative approach was used to evaluate the differences between traditional lecture-based instruction and modern student-centered learning methods. Particular attention was given to how digital educational technologies—such as virtual laboratories, simulation software, and visualization tools—contribute to improving conceptual understanding and reducing cognitive difficulty in learning complex scientific material.

Results and Discussion

The analysis of existing literature clearly shows that interdisciplinary integration of chemistry and pharmacy education has a positive impact on student learning outcomes. One of the main findings is that traditional lecture-based teaching methods alone are not sufficient to support deep conceptual understanding in complex scientific disciplines.

In contrast, student-centered approaches such as problem-based learning and inquiry-based instruction encourage students to actively engage with content, think critically, and apply theoretical knowledge to practical situations. These methods help students develop stronger analytical reasoning and improve their ability to solve real-world pharmaceutical problems (Freeman et al., 2014).

Another important finding is the significant contribution of digital learning tools. Virtual laboratories and simulation-based environments allow students to observe and interact with chemical and pharmaceutical processes that would otherwise be difficult to visualize. This makes abstract concepts more concrete and improves overall understanding while reducing cognitive overload (Sweller, 1988; Mayer, 2009).

From a broader perspective, integrating chemistry and pharmacy education creates a more coherent and meaningful learning experience. It helps students understand the relationship between scientific principles and their practical applications in drug development, clinical practice, and healthcare decision-making.

Conclusion

This study highlights the importance of interdisciplinary integration between chemistry and pharmacy education in modern higher education systems. The increasing complexity of pharmaceutical sciences requires teaching approaches that go beyond traditional memorization-based methods and focus on deeper conceptual learning. Constructivist learning strategies, including problem-based learning and inquiry-based instruction, provide a strong foundation for improving student understanding and engagement. These approaches encourage active participation and help students build long-term knowledge structures.

Moreover, the use of digital learning technologies plays an essential role in supporting modern education. Virtual laboratories and simulation tools enhance the visualization of complex scientific processes and improve the efficiency of learning. In sum up, the integration of chemistry and pharmacy education through innovative pedagogical methods significantly improves students' cognitive abilities, analytical thinking, and professional competencies. This approach is essential for preparing future pharmacists who are capable of responding effectively to the scientific and clinical challenges of the modern healthcare environment.

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STRATEGIC IMPORTANCE OF TEACHER PROFESSIONALISM AND EMPLOYEE TRAINING IN THE TRANSPORTATION FIELD

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Abstract: The article comprehensively analyzes the strategic role of the transport sector in the global economy in modern times, the importance of personnel training and teacher professionalism in this field. It is noted that the effectiveness of transport systems is determined not only by the technical infrastructure, but also by the level of knowledge and skills of human resources. The uninterrupted operation of logistics chains, the safe organization of passenger transportation and the development of international trade directly depend on the efficiency of transport systems. It is emphasized that the modern transport sector is developing with the application of digitalization, automation and artificial intelligence technologies, and it is shown that these changes require new approaches to personnel training. The training of highly qualified personnel makes a significant contribution to increasing transport safety, optimizing logistics costs and implementing innovative solutions. At the same time, it is noted that teacher professionalism directly affects the quality of the educational process, and their continuous development is necessary. The importance of personnel training not limited to higher education institutions but based on the principle of lifelong learning is substantiated. The article also analyzes existing problems, the lack of full compliance of educational programs with modern requirements, limited practical opportunities, weak technical base and lack of development opportunities for teachers. Strengthening cooperation between the state, universities and industry is considered necessary to solve these problems. As a result, it is emphasized that increasing teacher professionalism and establishing a modern personnel training system are one of the main conditions for the sustainable development of the country's transport sector, increasing economic competitiveness and international integration.

Keywords: transport sector, personnel training, teacher professionalism, logistics, intelligent transport systems, practical training, professional skills

NƏQLİYYAT SAHƏSİNDƏ MÜƏLLİM PEŞƏKARLIĞININ VƏ ƏMƏKDAŞLARIN TƏLİMİNİN STRATEJİ ƏHƏMİYYƏTİ

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Xülasə: Məqalədə müasir dövrdə nəqliyyat sektorunun global iqtisadiyyatda strateji rolu, bu sahədə kadr hazırlığının və müəllim peşəkarlığının əhəmiyyəti kompleks şəkildə təhlil edilmişdir. Qeyd olunur ki, nəqliyyat sistemlərinin effektivliyi yalnız texniki infrastrukturla deyil, eyni zamanda insan resurslarının bilik və bacarıq səviyyəsi ilə müəyyən olunur. Logistika zəncirlərinin fasiləsiz fəaliyyəti, sərnişin daşımalarının təhlükəsiz təşkili və beynəlxalq ticarətin inkişafı nəqliyyat sistemlərinin səmərəliliyindən birbaşa asılıdır. Müasir nəqliyyat sahəsinin rəqəmsallaşma, avtomatlaşdırma və süni intellekt texnologiyalarının tətbiqi ilə inkişaf etdiyi vurğulanır və bu dəyişikliklərin kadr hazırlığında yeni yanaşmalar tələb etdiyi göstərilir. Yüksək ixtisaslı kadrların hazırlanması nəqliyyat təhlükəsizliyinin artırılmasına, logistika xərclərinin optimallaşdırılmasına və innovativ həllərin tətbiqinə mühüm töhfə verir. Eyni zamanda, müəllim peşəkarlığının tədris prosesinin keyfiyyətinə birbaşa təsir etdiyi, onların davamlı inkişafının zəruri olduğu qeyd edilir. Kadr hazırlığının yalnız ali təhsil müəssisələri ilə məhdudlaşmayaraq həyatboyu öyrənmə prinsipi əsasında həyata keçirilməsinin vacibliyi əsaslandırılır. Məqalədə həmçinin mövcud problemlər, tədris proqramlarının müasir tələblərə tam uyğun olmaması, praktiki imkanların məhdudluğu, texniki bazanın zəifliyi və müəllimlərin inkişaf imkanlarının azlığı təhlil olunur. Bu problemlərin həlli üçün dövlət, universitet və sənaye əməkdaşlığının gücləndirilməsi zəruri hesab edilir. Nəticə olaraq, vurğulanır ki, müəllim peşəkarlığının artırılması və müasir kadr hazırlığı sisteminin qurulması ölkənin nəqliyyat sektorunun dayanıqlı inkişafı, iqtisadi rəqabət qabiliyyətinin yüksəldilməsi və beynəlxalq integrasiyası üçün əsas şərtlərdən biridir.

Açar sözlər: nəqliyyat sektoru, kadr hazırlığı, müəllim peşəkarlığı, logistika, ağıllı nəqliyyat sistemləri, praktiki təlim, peşəkar bacarıqlar

Introduction

In modern times, the transport sector acts as one of the most important and strategic pillars of the global economy, since the expansion of intercountry economic relations, the increase in international trade turnover and the acceleration of globalization processes directly depend on the existence of efficient and reliable transport systems. The uninterrupted and optimal functioning of logistics chains, the safe, comfortable and efficient organization of passenger transportation, as well as the timely delivery of products and services, depend not only on the level of development of modern transport infrastructure, but also on the quality of management. In this regard, the effectiveness of transport systems is determined not only by technical capabilities, but also by the level of knowledge, skills and professionalism of human resources operating in this field. As a result, the economic development indicators of countries are often directly related to the modernity of their transport infrastructure, the efficiency of management systems and the high level of qualification of personnel working in this field. Therefore, the preparation of competitive and highly qualified specialists in the transport sector acts as a strategic priority, and the basis of this process is a high-quality education system, modern teaching methods and professional teaching activities.

The importance of personnel training in the field of transport

The transport sector is a dynamic field that is rapidly developing, constantly updating and applying high-tech innovations in modern times. The specificity of this field is that specialists working here should not only be satisfied with acquiring theoretical knowledge, but also have practical skills that can be applied in a real production and management environment. Because the correct and effective management of transport systems has a direct impact on both economic efficiency and public safety (Banister, D. (2021).

Modern transport systems are moving away from traditional approaches and are characterized by the widespread use of digitalization, automation, intelligent control systems and artificial

intelligence technologies. These changes have made the management of transport infrastructure more complex and multifaceted, and at the same time have necessitated the application of new approaches in the personnel training process. Thus, specialists now need to master not only traffic rules and technical knowledge, but also the management of digital systems, data analytics and the use of modern logistics platforms.

The training of highly qualified personnel plays a crucial role in the sustainable development of the transport sector. Such personnel:

- creates conditions for improving transport safety and reducing road accidents.
- ensures cost optimization through more efficient use of resources in logistics and transportation processes.
- accelerates the implementation of new technological and innovative solutions and contributes to the modernization of systems.
- strengthens the country's integration into the global transport system by operating in accordance with international standards.

At the same time, the requirements of the modern labor market indicate that specialists working in the field of transport must have the ability to think flexibly, have an analytical approach and problem-solving skills. Therefore, the education system should not be limited to the function of transferring traditional knowledge only, but should also act as a complex platform that forms analytical thinking, decision-making, teamwork and innovative approach skills in students (International Transport Forum. (2023).

The professionalism of teachers in the field of transport is not limited to the transfer of theoretical knowledge. Teachers working in this field must have both technical knowledge and practical experience. The modern teacher:

- Be open to innovations,
- We must follow technological developments,
- Students should be taught analytical and critical thinking skills.

At the same time, teachers' pedagogical skills, communication abilities, and level of use of innovative teaching methods are the main indicators that determine their professionalism.

The process of personnel training in the transport sector is not an activity limited to higher education institutions, but rather a strategic direction that must be implemented continuously and systematically based on the principle of lifelong learning. The rapid development of this field, the constant renewal of technological innovations and changes in international standards necessitate the continuous improvement of personnel's knowledge and skills. Therefore, professional personnel training should be organized in a comprehensive manner both at the educational stage and during the period of production and practical activity (International Transport Forum. (2023).

Effective personnel training covers the following main areas:

- **Integration of theoretical and practical knowledge** – the application of theoretical knowledge acquired by students in a real transport environment strengthens their professional skills and develops decision-making abilities.
- **Organization of internship programs** – practical experiences held in enterprises and transport organizations allow students to get acquainted with the real work environment and gain operational skills.

- **Education in accordance with international standards** – training personnel in accordance with international norms and standards is of great importance for integration into the global transport system.
- **Application of digital technologies** – training in digital platforms, software and intelligent systems is an important component in the management of modern logistics and transport systems.

As a result of the correct implementation of these directions, the overall safety level of transport systems increases, there is a noticeable improvement in the quality of service, and economic efficiency is ensured through more efficient management of resources. At the same time, the formation of professional personnel strengthens the sustainability of the country's transport infrastructure and increases international competitiveness (Rodrigue, J.P., Comtois, C., & Slack, B. (2020)).

Materials and Methods

During the research, local and international scientific literature, regulatory documents, curricula, statistical indicators, enterprise reports, and the results of surveys and interviews with teachers, students, and employers can be used (World Bank. (2022)).

The following methods can be used according to the purpose of the study:

1. Theoretical analysis method. An analysis of the existing scientific literature, concepts, and theoretical approaches on the topic was conducted. This method allows us to identify the theoretical foundations of teacher professionalism and personnel training.

2. Systematic approach method. The relationship between transportation education, personnel training, and the labor market has been examined as a system. This approach allows for a comprehensive understanding of the problem.

3. Comparative analysis method. The transport personnel training experiences of different countries or educational models are compared. This helps to identify existing differences and development directions.

4. Empirical methods (survey and interview). Information about the real situation was collected through surveys and interviews conducted among teachers, students, and field experts.

5. Statistical analysis method. Simple statistical indicators (percentage, average, etc.) were used to analyze the collected data.

Strategic Importance and Development Prospects

Teacher professionalism and highly qualified personnel training in the field of transport act as one of the main components that play an important role in the formation of a long-term and sustainable development model of the country's socio-economic development strategy, especially for states with the status of transit countries, the development of this field is of even greater strategic importance in terms of integration into international transport corridors, expansion of logistics capabilities and strengthening regional cooperation. Improving the professional level of teachers not only has a direct impact on the quality of the educational process, but also creates favorable conditions for the development and practical application of innovative transport projects on a scientific basis, contributes to the expansion of international educational and scientific cooperation relations, and also forms the basis for increasing the competitiveness of young specialists in the global labor market (Hasanov, RM (2020)).

In addition, the technological development trends of the modern era show that the application of "smart transport systems", automated control mechanisms, artificial intelligence-based logistics

platforms and environmentally sustainable transport models in the transport sector is becoming increasingly widespread, which in turn necessitates the continuous updating of teachers' scientific and theoretical knowledge and practical skills, the modernization of teaching methodologies and their adaptation to digital transformation processes, thus the teacher training system must become a flexible and innovative structure that adapts to the ever-changing global transport environment.

Conclusion and Discussion

At the current stage, a number of systemic problems and structural challenges affecting the quality of personnel training in the transport sector are observed, which, in addition to limiting the efficiency of the educational process, lead to certain discrepancies between the requirements of the labor market and educational outcomes. First of all, the lack of full adaptation of educational programs to the dynamics of development of modern transport systems, digital transformation processes and the requirements of international standards leads to the emergence of gaps between theoretical and practical knowledge in personnel training.

In addition, the limited opportunities for practical training in higher and vocational education institutions prevent students from gaining sufficient experience in a real production environment, which complicates their adaptation to the labor market. Another important problem is that the technical base and infrastructure capabilities of educational institutions are insufficient for the application of modern technologies, which creates serious limitations, especially in the training of simulation systems, digital laboratories and intelligent transportation models.

At the same time, the lack of continuous professional development opportunities for teachers slows down the updating of their knowledge and skills in line with the requirements of the times and, as a result, negatively affects the quality of teaching. Such problems reduce the overall effectiveness of personnel training in the transport sector and necessitate the implementation of systematic reforms.

In order to overcome these challenges, it is of particular importance to strengthen coordinated cooperation between government agencies, higher education institutions, and the industrial sector, update educational programs in line with the real demands of the labor market, create modern technological infrastructure, and implement training and certification mechanisms that ensure the sustainable development of teachers.

Development prospects and proposals

The development of transport education in the modern era is of strategic importance in terms of increasing the country's economic competitiveness, improving logistics systems and strengthening integration into international transport networks. Therefore, it is necessary to apply a comprehensive approach and implement continuous reforms to improve the quality of the personnel training system. The development of transport education in accordance with modern requirements can be possible not only by increasing technical knowledge, but also by forming innovative thinking, analytical skills and digital competencies (European Commission. (2021).

In this direction, it is of particular importance to first of all update the educational programs in accordance with modern technologies. Thus, new technological trends such as digitalization, artificial intelligence, intelligent transportation systems and automated control mechanisms should be integrated into the educational content, and innovative modules should be included in the programs along with traditional approaches. This will allow students to better adapt to the requirements of the future labor market.

Regularly updating teachers' knowledge and skills, and familiarizing them with modern pedagogical approaches and technological innovations, increases the effectiveness of the teaching process and creates conditions for students to receive a higher quality education. In this regard, the professional development of teachers should be systematic and continuous.

Another important direction is the strengthening of cooperation between universities and industry. The establishment of close ties between educational institutions and real production entities of the transport and logistics sector creates conditions for the combination of theoretical knowledge with practical application. This cooperation also opens up wide opportunities for the joint implementation of innovative projects, the development of scientific research and the application of new technological solutions.

In addition, increasing opportunities for students to participate in real projects and expanding their production experience play an important role in shaping their professional skills. Practical experience accelerates students' adaptation to the real work environment, develops their decision-making, problem-solving, and teamwork skills.

Finally, expanding international cooperation plays an important role in bringing transport education up to global standards. Cooperation with foreign universities, scientific centers, and international organizations allows students and teachers to exchange experiences, benefit from advanced educational models, and expand opportunities for integration into global transport systems.

In general, the comprehensive implementation of the above directions will make a significant contribution to improving the quality of transport education, training highly qualified personnel, and sustainable development of the country's transport sector.

Conclusion

Teacher professionalism and highly qualified personnel training in the field of transport act as one of the fundamental strategic factors ensuring the sustainability of the country's economic development, increasing the safety of transport systems and strengthening international competitiveness. Modern global challenges show that the development of transport infrastructure from a technical point of view is not enough, but the level of knowledge, skills and professionalism of human resources managing these systems also plays an equally important role. In this regard, the professional training of teachers and their continuous development are considered the main factors that directly affect the quality of personnel training.

It is impossible to ensure long-term and sustainable development in the transport sector without forming an education system that meets modern requirements, is enriched with innovative approaches and is adapted to digital transformation. Such a system can be realized not only by transferring theoretical knowledge, but also by forming practical skills, applying technological innovations and integrating international experience.

In this context, investments in the professional development of teachers, increasing their scientific and pedagogical potential, as well as expanding access to modern teaching methods, make a significant contribution to the formation of more efficient, safe and innovative transport systems in the future. As a result, each strategic step taken in this direction serves the sustainable development of the country's transport sector and the strengthening of its position on a global scale.

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DIGITAL TRANSFORMATION IN HIGHER EDUCATION AND TEACHERS' PROFESSIONAL PREPARATION

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Abstract: This article explores the role of digital transformation in higher education and its impact on teachers' professional development. The rapid advancement of information and communication technologies, digital learning platforms, and artificial intelligence has significantly changed teaching and learning processes. These developments have encouraged more flexible, interactive, and student-centered educational environments.

Digital transformation requires not only technological innovation but also changes in teaching methods and pedagogical approaches. As a result, teachers' digital literacy, adaptability, and ability to use innovative teaching strategies have become increasingly important. Modern teachers are expected to guide learning, integrate technology effectively, and promote student engagement.

The study aims to examine the effects of digital transformation on teachers' professional development and identify key challenges. Using a qualitative approach, the research draws on recent academic literature and educational observations.

The findings indicate that digital technologies enhance educational quality and accessibility while creating new professional demands. However, limited digital skills, inadequate infrastructure, and resistance to change remain challenges. Continuous professional development and institutional support are essential for successful digital transformation in higher education.

Keywords: Digital transformation, higher education, teachers' professional development, digital competencies, educational technologies, hybrid education, digital literacy

ALİ TƏHSİLDƏ RƏQƏMSAL TRANSFORMASIYA VƏ MÜƏLLİMLƏRİN PEŞƏ HAZIRLIĞI

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Xülasə: Bu məqalədə ali təhsil sistemində rəqəmsal transformasiyanın rolu və onun müəllimlərin peşəkar inkişafına təsiri araşdırılır. İnformasiya və kommunikasiya texnologiyalarının, rəqəmsal təlim platformalarının və süni intellekt vasitələrinin sürətli inkişafı tədris və öyrənmə proseslərində əhəmiyyətli dəyişikliklərə səbəb olmuşdur. Bu yeniliklər daha çevik, interaktiv və tələbəyönümlü təhsil mühitlərinin formalaşmasını təşviq etmişdir.

Rəqəmsal transformasiya yalnız texnoloji yenilikləri deyil, həm də tədris metodları və pedaqoji yanaşmalarda dəyişiklikləri tələb edir. Buna görə də müəllimlərin rəqəmsal savadlılığı, dəyişən təhsil mühitinə uyğunlaşma bacarığı və innovativ tədris strategiyalarından istifadə etmə qabiliyyəti getdikcə daha böyük əhəmiyyət kəsb edir. Müasir müəllimdən təlim prosesinə rəhbərlik etmək, texnologiyaları səmərəli şəkildə inteqrasiya etmək və tələbələrin fəal iştirakını təmin etmək gözlənilir.

Tədqiqatın məqsədi rəqəmsal transformasiyanın müəllimlərin peşəkar inkişafına təsirini araşdırmaq və əsas problemləri müəyyənləşdirməkdir. Keyfiyyət yönümlü yanaşmaya əsaslanan araşdırmada müasir elmi ədəbiyyat və təhsil mühitində aparılan müşahidələrdən istifadə edilmişdir.

Nəticələr göstərir ki, rəqəmsal texnologiyalar təhsilin keyfiyyətini və əlçatanlığını artırmaqla yanaşı, müəllimlərin peşə fəaliyyətinə yeni tələblər də yaradır. Bununla belə, rəqəmsal bacarıqların məhdudluğu, qeyri-kafi infrastruktur və dəyişikliklərə müqavimət kimi problemlər hələ də aktuallığını qoruyur. Müəllimlərin davamlı peşəkar inkişafı və institusional dəstəyin gücləndirilməsi ali təhsildə uğurlu rəqəmsal transformasiya üçün mühüm əhəmiyyət daşıyır.

Açar sözlər: İnküziv təhsil, bərabər təhsil imkanları, fərdiləşdirilmiş tədris, sosial inteqrasiya, təhsil mühiti, təhsil siyasəti

Introduction

Digital transformation has become one of the most important and strategic directions in the development of the modern higher education system. The rapid advancement of information and communication technologies has led to fundamental changes in the organization and management of the teaching process, making the application of new approaches in teaching and learning activities essential. This process is not limited to the introduction of technological innovations; it also facilitates the rethinking and restructuring of educational philosophy and pedagogical approaches. As a result, traditional teaching models have gradually been replaced by digital, online, and hybrid forms of education. These changes are of particular importance in terms of improving the quality of education, expanding access to educational resources, and strengthening student-

centered and individualized learning environments. At the same time, digital technologies provide broad opportunities for organizing the educational process in a more flexible, interactive, and learner-oriented manner. The formation of a digital environment has also significantly transformed the role of the teacher. In the modern era, the teacher is no longer only a transmitter of knowledge and information but also acts as a facilitator and mentor who plans, manages, and guides the teaching process. By effectively using digital tools and platforms, the teacher organizes the learning process in a more purposeful and efficient way, stimulates students' active participation in classes, and supports the development of their independent learning skills. These changes place new demands on teachers' professional preparation and make the systematic development of their digital competencies essential. In such conditions, teachers' professional development is no longer limited to traditional pedagogical knowledge and skills; it also includes the enhancement of digital literacy, the mastery of innovative teaching methods, and the ability to effectively use modern technologies. In this context, the study of the impact of digital transformation on teachers' professional development in higher education institutions is of particular relevance and carries significant scientific and practical importance for the development of the education system. The main objective of this article is to comprehensively analyze the impact of digital transformation on teachers' professional development in higher education institutions, to identify existing challenges in this field, and to explore scientifically grounded and practically applicable solutions for addressing them.

Materials and Methods

This study was conducted based on a qualitative research approach in order to investigate the impact of digital transformation on teachers' professional development in higher education institutions. The selection of a qualitative methodology allows for an in-depth and comprehensive analysis of the research topic, as well as a better understanding of complex processes occurring within the educational environment. The primary method used in this research is literature review. For this purpose, both local and international scientific sources related to digital transformation, innovative approaches in higher education, and teachers' professional development were examined. Different scholarly perspectives were analyzed, compared, and systematized in order to identify both theoretical and practical aspects of the topic. In addition, the observational method was applied within the scope of the study. Teaching and learning processes in higher education institutions were observed, with particular attention given to the level of teachers' use of digital technologies and students' engagement in technology-enhanced learning activities. These observations provided valuable insights into the practical implementation of digital transformation in real educational settings. The collected data were analyzed using descriptive and comparative analysis techniques. The information obtained from both literature and observations was systematically organized, interpreted, and synthesized in order to derive meaningful scientific conclusions.

Results and Discussion

The results of the study indicate that digital transformation has a multifaceted and significant impact on the organization and quality of the teaching process in the modern higher education system. The integration of digital technologies into education contributes to the gradual transformation of traditional teaching approaches, leading to a more flexible, interactive, and student-centered learning environment. This, in turn, enhances students' engagement in the learning process, increases their motivation, and supports the development of independent learning skills. At the same time, the digital environment creates broader opportunities for personalized learning

experiences tailored to individual student needs. The findings also reveal that teachers' ability to use digital technologies is one of the key factors directly influencing the effectiveness of their pedagogical practice and the overall quality of instruction. Teachers with a higher level of digital literacy tend to plan and implement the teaching process more systematically and effectively, making purposeful use of various digital platforms, electronic resources, and online teaching tools. This contributes to a more enriched and modernized learning process aligned with contemporary educational standards. However, insufficient digital skills among some teachers, slow adaptation to new technologies, and methodological difficulties still negatively affect the effectiveness of teaching activities. In addition, the study identifies uneven development of technical infrastructure in higher education institutions and limited access to digital resources and platforms as significant challenges. These factors may hinder the full and effective implementation of digital transformation at all levels and can lead to disparities in educational quality. The discussion suggests that the successful implementation of digital transformation is not solely dependent on technological infrastructure but is closely linked to human resource development. In this regard, continuous professional development of teachers, regular in-service training programs, systematic development of digital competencies, and strengthened methodological support are of crucial importance. Furthermore, the creation of an innovative teaching environment in higher education institutions, the wider use of digital platforms, and the modernization of management systems are key factors that enhance the effectiveness of digital transformation. Overall, digital transformation should be viewed not merely as a technological upgrade but as a complex process encompassing pedagogical and organizational changes. It plays a vital role in improving the quality of education, fostering student-centered learning environments, and ensuring the professional development of teachers in higher education institutions.

Conclusion

The findings of the study indicate that digital transformation plays a significant role in the development of the modern higher education system and has a direct impact on teachers' professional development. The integration of digital technologies into the teaching process contributes to the renewal of traditional pedagogical approaches, the improvement of teaching quality, and the formation of a more flexible, interactive, and student-centered educational environment. The results of the research show that teachers' digital competencies are one of the key factors determining the effectiveness of their teaching practice. Teachers with a high level of digital skills are more capable of organizing the instructional process efficiently and making purposeful use of various digital platforms and resources. However, the study also reveals that insufficient digital skills among some teachers, limited technical infrastructure, and difficulties in adapting to new technologies still pose certain challenges. In this regard, continuous professional development of teachers, the organization of regular training programs, the enhancement of digital literacy, and the strengthening of institutional support are considered essential. At the same time, the creation of an innovative educational environment in higher education institutions and the effective implementation of digital technologies play an important role in improving the overall quality of education. Overall, digital transformation should be understood not only as a technological change but also as a pedagogical and organizational renewal process that holds strategic importance for the development of higher education.

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PSYCHO-PEDAGOGICAL CORRECTIVE WORK WITH FAMILIES OF CHILDREN WITH BEHAVIORAL DISORDERS

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Abstract.: The formation of a child as a personality and their compliance with behavioral norms in the social environment becomes possible through the joint influence of family, school, and society. Any psychological or pedagogical deficiencies occurring in the child's upbringing process, mistakes in parents' attitudes, or the negative influence of the social environment may sometimes lead to deviations in behavior. These situations negatively affect the child's socialization process, adaptation to the school environment, and harmonious personality development.

The main characteristic of children with behavioral deviations is acting against social norms and rules accepted by society. Such children may demonstrate aggression, disobedience, emotional instability, lying, truancy, indifference, and other negative behaviors. These behavioral disorders depend not only on the child's individual psychological condition but also on family relationships, upbringing methods, socio-economic conditions, and the school environment.

Behavioral disorders arise as a result of the child reflecting internal conflicts caused by various psychological and physical reasons in their behavior.

Keywords: social norm, behavioral disorders, socialization, psychological factors, pedagogical approach, correction work.

DAVRANIŞ POZUNTULARI OLAN UŞAQLARIN AİLƏLƏRİ İLƏ PSİXO-PEDAQOJİ KORREKSİYA İŞİ

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Xülasə: Uşağın şəxsiyyət kimi formalaşması, onun sosial mühitdə davranış normalarına əməl etməsi ailə, məktəb və cəmiyyətin birgə təsiri nəticəsində mümkündür. Uşağın tərbiyə prosesində baş verən hər hansı psixoloji və ya pedaqoji çatışmazlıqlar, valideynlərin münasibət tərzindəki səhvlər və ya sosial mühitin

mənfi təsirləri nəticəsində bəzən davranışda kənaraçıxmalar müşahidə olunur. Bu hallar uşağın sosiallaşma prosesinə, məktəb mühitinə uyğunlaşmasına və şəxsiyyətin harmonik inkişafına mənfi təsir göstərir.

Davranışında kənaraçıxmalar olan uşaqların əsas xüsusiyyəti sosial normalara və cəmiyyət tərəfindən qəbul olunmuş qaydalara zidd hərəkət etmələridir. Belə uşaqların davranışında aqressiya, itaətsizlik, emosional qeyri-sabitlik, yalan danışmaq, dərstdən yayınmaq, laqeydlilik və digər neqativ hallar özünü göstərə bilər. Bu cür davranış pozuntuları təkcə uşağın fərdi psixoloji vəziyyətindən deyil, həm də ailə münasibətləri, tərbiyə üsulları, sosial-iqtisadi şərait və məktəb mühitindən asılıdır.

Davranış pozuntuları müxtəlif psixi və fiziki səbəblərdən dolayı uşağın daxilində baş verən konfliktləri davranışlarına yansıtması nəticəsində meydana gəlir.

The psycho-correction work carried out with the parents of children with behavioral disorders aims to regulate the child's behavior and ensure that parents are actively involved in this process. The main purpose of this work is to reduce negative behaviors, increase positive behaviors, and teach parents how to communicate more effectively with their children and how to support them properly.

Açar sözlər: sosial norma, davranış pozğunluqları, sosiallaşma, psixoloji amillər, pedaqoji yanaşma, korreksiya işi.

Psycho-correction work is a systematic psychological intervention carried out to reduce deviations in a child's behavior (aggression, stubbornness, antisocial tendencies, emotional instability, etc.), normalize parent-child relationships, and improve the psychological climate within the family. This process is carried out in several stages

1. Diagnostic Stage

At this stage, the psychologist collects initial information about the child and the family, determines the nature of behavioral disorders and identifies their causes.

Main goal: To understand the root of the problem and its causes.

Implementation forms include:

- Observation of the child (at school, at home, during group activities);
- Interviews with parents (studying family relationships);
- Questionnaires and psychodiagnostic tests (parental attitude questionnaire, family climate test, etc.);
- Conversations with the child's teachers and caregivers.

Result: The causes of behavioral disorders (emotional, social, family-related, or pedagogical) are clarified, and an individual psycho-correction plan is prepared.

2. Educational stage

At this stage, parents are provided with psychological knowledge about the nature and causes of the child's behavioral disorders.

Main goal: To develop proper parenting and communication skills in parents.

Implementation forms include:

- Individual or group educational conversations;
- Training seminars ("Positive Parenting", "Methods of Dealing with Aggressive Children", etc.);
- Familiarization with methodological materials;
- Presentation of positive models in parent-child relationships.

Result: Parents become aware of how their own behavior affects the child, and constructive relationships begin to form.

3. Correction stage

This is considered the main stage. At this stage, parents and children work together with the psychologist to eliminate behavioral problems.

Main goal: To change the interaction between parent and child and strengthen positive behavior patterns.

Implementation forms include:

- Individual sessions with parents: stress management, emotional regulation, balance between punishment and praise;
- Joint sessions with the child: role-playing games, communication and empathy training;
- Family sessions: restoring family relationships and improving communication;
- Positive behavior reinforcement techniques: rewards, “emotional praise”, “behavior contracts”.

Result: Negative behavior decreases, and mutual understanding and emotional stability emerge within the family.

4. Rehabilitation and monitoring stage

After the psycho-correction work is completed, monitoring of the child and family continues.

Main goal: To stabilize the achieved positive results and prevent behavioral relapse.

Implementation forms include:

- Regular observation and follow-up meetings;
- Cooperation with school psychologists and teachers;
- Supportive conversations and additional training;
- Organizing support groups for parents.

Result: The child adapts to the social environment, and a healthy psychological climate is established within the family.

Methods used in the correction of Behavioral Disorders

The methods used in correcting behavioral disorders are selected based on the child's individual characteristics, the type and severity of behavioral problems, as well as the influence of family and the social environment. The most commonly used methods in psycho-pedagogical correction are the following:

1. Psychotherapeutic Methods. These methods are used to reduce the child's internal tension, stabilize emotional condition, and increase self-confidence. Main types include:

- Individual psychotherapy: The child is helped to face personal problems, understand them consciously, and develop the ability to manage them.
- Group psychotherapy: Children are involved in group communication, where social skills, empathy, and cooperation are developed.
- Family psychotherapy: Since the roots of behavioral disorders often arise from family relationships, psychotherapy involving family members is very important.

Guliyeva, T. (2023)..

2. Psycho-correction methods

Psycho-correction is carried out to gradually change the behavior of a child with behavioral disorders, form correct behavior patterns, and strengthen social adaptation. Main methods include:

- Modeling (demonstration of examples): Correct behavior patterns are presented to the child, and the child learns by observing and imitating them.

- Role-playing games: Different situations are acted out (for example, “How to behave during a conflict?”), helping the child regulate behavior in real life.
- Positive reinforcement: Correct behavior is rewarded using praise, point collection, or symbolic gifts.
- Adequate punishment: Psychologically safe methods are chosen to help the child understand the consequences of their behavior (for example, temporary restriction from going to the park).

3. Cognitive-Behavioral Methods. The main purpose of this method is to change the child’s way of thinking and behavioral habits. Main methods include: Working on the connection between thought and behavior: The child is taught that negative thoughts lead to negative behavior and learns to replace them with positive thoughts.

Self-control and self-observation: The child is taught simple strategies to control emotions and behavior, such as “count to 10” or “take a deep breath.” Mammadova, R. (2017):

4. Play therapy. It is known that through play, children express emotions, reduce fear and aggression, and learn social rules.

5. Art Therapy. Art therapy is a treatment method involving the inclusion of both children with disabilities and typically developing children in artistic activities. It is a psychological work method based on art that positively affects the development of the psychomotor sphere. Art therapy includes music, drawing, movement-dance, and literature.

Dance therapy: A psychotherapy method that helps a person experience, recognize, and express feelings and conflicts through body movement. It helps children with behavioral disorders eliminate psychomotor deficiencies, emotional and muscle tension, and improve flexibility and mobility.

Music therapy: A corrective method that influences emotional-volitional, intellectual, and personal development. It is based on listening to and creating music. It positively affects communication and environmental interaction among children with psycho-physical difficulties.

Fairy tale therapy: Used for therapeutic purposes for both children and adults. It is especially useful when other psychological methods are insufficient. It improves interaction with the outside world, develops creativity, broadens consciousness, and supports speech development, attention, thinking, imagination, and communication skills Vygotsky, L.S. (1984).

6. Socio-Pedagogical Methods

The main socio-pedagogical methods include the following:

- Educational work with parents: The main purpose is to teach parents how to respond correctly to the child’s behavior and regulate family relationships.
- Behavior regulation in the school environment: This includes behavior programs planned jointly by teachers, psychologists, and parents (for example, “Behavior Diary” and “Reward System”).

Early intervention in cases of behavioral disorders in children usually gives positive results. Parents being patient and understanding, maintaining communication with their children, and receiving professional support when necessary are of vital importance in this process. Correctly identifying the causes of behavioral disorders plays a critical role in the child’s healthy development and adaptation to society.

One of the most important tasks facing class teachers, school psychologists, social pedagogues, school doctors, and parents is to ensure that psychological prevention and correction

work with children with behavioral disorders is carried out systematically and based on specific programs.

It is possible to achieve better results by establishing systematic cooperation among families, educational institutions, administrative bodies, and state organizations working with children and youth, and by intervening in the problem in a timely manner.

Solving children's behavioral problems is not only work directed at the child but also at the family, especially the parents. In such psycho-correction work, the most important factors are continuity, love, and an understanding approach.

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EVALUATION OF TEACHER DEVELOPMENT AND MANAGEMENT MODEL IN THE FIELD OF EDUCATION

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Abstract: In the modern education system, the professional development of teachers and an effective management model are among the main factors that directly affect the quality of teaching. The assessment processes carried out in this direction serve to increase the knowledge and skills of teachers, as well as to determine the effectiveness of management mechanisms. Studies show that continuous professional development programs, a mentoring system and modern management approaches have a positive effect on the activity of teachers. At the same time, the transparent and objective establishment of assessment mechanisms increases teacher motivation and creates conditions for improving educational outcomes. Recently, there has been a need to evaluate the pedagogical work of teachers in order to analyze and analyze the principles and directions of teachers' work activities on the basis of the national education system. According to foreign authors, the level of students' preparation for the lesson is not always related to the teacher's professional skills and potential. This also depends on other factors that the teacher is not in the power to influence or intervene. Therefore, some researchers suggest increasing the professional skills of teachers in this area to the maximum level. Although formative assessment of students is gradually being

included in the structure of pedagogical processes, it is still perceived only as a formative assessment based on school experience. The article analyzes the methods and techniques used in the assessment of teacher performance, examines the questions that arose during the research, new opportunities and results, and reflects the experiences of representatives of the scientific community of many foreign countries on the effective assessment of teacher work. Most of them came to the conclusion that the effective assessment of teachers' work processes changes in accordance with their personal and potential development indicators.

Keywords: teacher development, professional development, educational management, assessment, pedagogical efficiency, educational policy, teacher competencies

TƏHSİL SAHƏSİNDƏ MÜƏLLİMİN İNKİŞAFI VƏ İDARƏETMƏ MODELİNİN QIYMƏTLƏNDİRİLMƏSİ

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Xülasə: Müasir təhsil sistemində müəllimlərin peşəkar inkişafı və səmərəli idarəetmə modeli tədrisin keyfiyyətinə birbaşa təsir edən əsas amillərdəndir. Bu istiqamətdə aparılan qiymətləndirmə prosesləri müəllimlərin bilik və bacarıqlarının artırılmasına, eləcə də idarəetmə mexanizmlərinin səmərəliliyinin müəyyənləşdirilməsinə xidmət edir. Araşdırmalar göstərir ki, davamlı peşəkar inkişaf proqramları, mentorluq sistemi və müasir idarəetmə yanaşmaları müəllimlərin fəaliyyətinə müsbət təsir göstərir. Eyni zamanda, qiymətləndirmə mexanizmlərinin şəffaf və obyektiv qurulması müəllimlərin motivasiyasını artırır və təhsil nəticələrinin yaxşılaşdırılmasına şərait yaradır. Son zamanlar milli təhsil sistemi əsasında müəllimlərin iş fəaliyyətinin prinsip və istiqamətlərinin təhlili və təhlili üçün müəllimlərin pedaqoji əməyinin qiymətləndirilməsinə ehtiyac yaranmışdır. Xarici müəllimlərin fikrincə, şagirdlərin dərslə hazırlıq səviyyəsi heç də həmişə müəllimin peşəkar bacarıq və potensialı ilə bağlı olmur. Bu həm də müəllimin təsir və ya müdaxilə etmək səlahiyyətində olmayan digər amillərdən asılıdır. Ona görə də bəzi tədqiqatçılar bu sahədə müəllimlərin peşəkarlıq səviyyəsinin maksimum səviyyəyə çatdırılmasını təklif edirlər. Şagirdlərin formativ qiymətləndirilməsi tədricən pedaqoji proseslərin strukturuna daxil edilsə də, hələ də yalnız məktəb təcrübəsinə əsaslanan formativ qiymətləndirmə kimi qəbul edilir. Məqalədə müəllim fəaliyyətinin qiymətləndirilməsində istifadə olunan metod və üsullar təhlil edilmiş, tədqiqat zamanı yaranan suallar, yeni imkanlar və nəticələr araşdırılmış, bir çox xarici ölkələrin elmi ictimaiyyəti nümayəndələrinin müəllim əməyinin səmərəli qiymətləndirilməsi ilə bağlı təcrübələri öz əksini tapmışdır. Onların əksəriyyəti belə qənaətə gəlib ki, müəllimlərin iş proseslərinin səmərəli qiymətləndirilməsi onların şəxsi və potensial inkişaf göstəricilərinə uyğun olaraq dəyişir.

Açar sözlər: müəllimin inkişafı, peşəkar inkişaf, təhsilin idarə edilməsi, qiymətləndirmə, pedaqoji səmərəlilik, təhsil siyasəti, müəllim kompetensiyaları

Introduction

Currently, politicians, scientists and representatives of the educational community around the world are paying attention to the use of new approaches to assessing the effectiveness of teacher work. The issue of teacher evaluation not only paves the way for quality teaching in education, but also creates a basis for them to take greater responsibility in this area. Such practices have been widely applied and tested in England, the USA and other foreign countries. In the last decade, new formal models have been developed in connection with the evaluation of teacher work based on increasing student achievement. Such model projects have been discussed in scientific research

works of many foreign countries, and effective results have been achieved with substantiated evidence and arguments. According to researchers in this field, the new evaluation system helps determine the professional development of teachers. The system is considered effective precisely because in it the assessment of the level of professionalism of teachers is carried out in accordance with established measurement criteria. In the new system, evaluation is based on the results of comprehensive observations of the teacher collected during the lesson process and the analysis of student achievement. According to the majority of respondents, this system motivates teachers to increase their professionalism.

Currently, as in many other areas, radical reforms are being implemented in the field of education. The main goal of these reforms is to improve the quality of education and ensure its development, taking into account the needs and demands of society. In this regard, it has become necessary to review the factors determining the quality of education, primarily the content of education, teacher training and assessment systems.

The next discussion time is set to determine the results of observations made during the teacher's work process and to finalize the final assessment work. Teachers working in schools where the new assessment system is applied are developing their skills strategically. School principals highlight two main factors in increasing and improving the quality of education: 1. Communication between teachers who actively observe each other's teaching process; 2. Discussion and evaluation of students' achievements in the teaching process from a pedagogical and strategic perspective. The observations and opinions expressed by experts in the new system contribute to the professional development of teachers and, therefore, lead to the creation of trust between teachers and experts. Another fact also affects the increase in reliability.

1. Management models in the education system and their features. The development of a new assessment system has been identified as one of the priority areas in the Education Sector Development Project. The creation of modern assessment mechanisms for improving the training and education process, obtaining objective data on students' achievements across the country, as well as monitoring state education standards is an important task. Since the main indicator in determining the quality of education is the student's learning results, the objective assessment of these results is considered the most reliable source that reveals quality.

The Assessment Concept in the general education system of the Republic of Azerbaijan is designed to manage measures related to assessment issues in education in the country in the coming years. The main content of the document is aimed at determining the principles that ensure the legitimacy of assessment activities, determining the leading directions of this activity, strengthening the social status of institutions implementing assessment activities, and using assessment to meet the needs of training and education. This document, as a framework document, has been prepared in accordance with the requirements of the Law on Education of the Republic of Azerbaijan, the Education Reform Program of the Republic of Azerbaijan and the state education policy, modern world standards of assessment and serves as a basis for the principles of the new assessment system. The document explains the main principles and areas of activity of intra-school, national and international assessment. The concept also envisages the preparation of additional documents in the form of precise rules and regulations to ensure the reliability and effectiveness of each type of assessment. The concept is directly related to the national curriculum framework document prepared for the purpose of modernization and renewal of general education in Azerbaijan and constitutes its integral part. The creation of assessment mechanisms and tools is considered the next

stage after the development of content standards and subject curricula of education. (Mammadov, I. (2019)).

The new assessment system ensures the following: fulfilling the role of an important tool for making positive changes in education; providing reliable information on "student learning outcomes", which is the main indicator of the quality of education; making appropriate changes in subject curricula and textbooks after obtaining sufficient information about the knowledge and skills of the student through assessment; function as a means of combating some negative situations in education; that new approaches to assessment, in contrast to memory-based assessment, serve the development of logical thinking skills.

The education and assessment policy envisaged in Azerbaijan, in addition to reflecting national interests, necessitates the implementation of fundamental reform-oriented measures. The political, economic and social changes taking place in the world at the end of the 20th century directly require the realization of the following specific goals:

- transition to more democratic methods of governance;
- establishment of an open civil society in the country;
- ensuring effective cooperation with developed countries;
- creation of international market relations and economic competition conditions;
- instilling a sense of national self-reliance.

2. Requirements for assessment activities in the modern era

Currently, large-scale reforms are being carried out in the education systems of many countries regarding the establishment of new assessment models, recording of students' educational achievements and organizing final exams for educational levels. Their main characteristics are characterized as follows:

1. application and development of modern approaches in assessment and education policy, and in this regard, rejection of outdated existing methods;
2. selection of more progressive methods for conducting exams fairly and transparently;
3. creation of relevant organizations within the education system for the implementation of new assessment mechanisms;
4. consideration of the formation of a new assessment system as an integral part of educational reforms, a priority area, and its consideration in the context of the policy of developing modern curricula.

The social requirement of the state, society and students for the education system is quality education. The goal of all reforms in the field of education is precisely to achieve its quality improvement. What is the quality of education? How is it determined? How can the quality of education be managed? In general, the concept of quality in any field denotes compliance with established standards. Characterizing the state of an object or system The closer the parameters are to the standard, the higher the quality is considered. This is also the case in education. The quality of education is determined by the ratio of the results obtained to the previously established criteria (standards).

The main indicator in determining the quality of education is the student's learning results, and an objective assessment of these results is considered a reliable source that reveals quality. Thus, determining quality in education is conditioned by an objective assessment of student achievements. However, since the student's learning results emerge as the final result of all activities taking place in the field of education, it is possible to evaluate the teaching activities of

teachers, educational institutions, education management structures, and the education system of the country as a whole based on the student's learning results. At the same time, it is possible to solve the problem of not only determining the actual state of quality, but also improving and managing it. Such an assessment system is being created in the general education system of the Republic of Azerbaijan. In order to understand the mechanism of the new assessment system, it is necessary to know the essence of the concepts of assessment and monitoring. The state of any object or system is characterized by certain parameters (Greek parametron - means measured). For example, for simplicity, let's say that a person's health can be determined by measuring parameters such as blood pressure, pulse rate, blood composition, etc. The results of the measurement are compared with the normal (standard) values of the parameters, and the health status of each parameter and the person as a whole is assessed separately. Of course, it is relatively difficult to conduct such an assessment in areas of human activity that are not directly related to such quantitative indicators. Education is also one of such areas. The level of knowledge and skills (achievement level) of a student in a particular subject, section or full course is determined by comparing the learning results he or she has shown with the appropriate assessment standards.

3. Criteria and methods for assessing teacher performance

The effective teacher assessment system has put forward research work in several areas. These areas can be combined into three groups: 1. Pedagogical activity on the quality of education; 2. Implementation of a new system for teacher assessment; 3. The impact of the new assessment system on school activity. Thus, assessment is the determination of the value of the measured parameter, the state of an object or system in relation to an accepted standard. This procedure requires the presence of an assessment scale that allows the degree of proximity of the measurement result to the standard to be converted into a quantitative indicator. The assessment determined in this way is an assessment of the actual situation. Although it is based on standards, in some cases it may not reflect reality. Thus, a person's blood pressure and pulse rate may change depending on the feeling he experiences, and the composition of the blood may change due to nutrition. The student's answers to the test questions may sometimes be influenced by factors other than his knowledge and skills (health status, feelings, etc.). On the other hand, the assessment of the actual situation is static in nature and the results reflect the situation at the time of the assessment. If such assessments are carried out continuously and systematically, it is possible to determine the value of the parameter and the dynamics of changes in the situation over a certain period. (Abbasov, A. (2018)).

Monitoring allows you to determine the characteristics of the change in the value of a parameter, the state of an object or system (quality indicator), the causes of these changes, and ways to influence them. Thus, monitoring is a process of continuous and systematic collection and analysis of information about the situation in order to study the characteristics of the changes taking place and make appropriate decisions. In modern times, monitoring is considered a technology that is successfully applied in various fields of activity, including education. As can be seen, the concepts of assessment and monitoring are interconnected, or rather, one (assessment) serves the other (monitoring). Assessment is an assessment of the actual situation, and monitoring is an assessment of the dynamics of changes in the situation, in other words, the process. Monitoring is carried out by collecting and analyzing information obtained as a result of regular assessment of the actual situation. Traditional assessment in education was considered only a control mechanism because it served not monitoring, but management, based on administrative authority.

The next strategic task facing the country is to ensure sustainable economic development and increase the standard of living of the population through further modernization of socio-economic life and adaptation to advanced international practice. consists of raising the level of education. Modernization is primarily associated with the successful application of advanced technologies and management methods, innovations created on the basis of scientific achievements in the socio-economic life of the country. For this, the priority direction is the development of human capital in the country, ensuring the acquisition of modern knowledge and skills by the individual, along with accelerating the integration of the country's economy into the world economy. The development of human capital is one of the most important conditions in the process of the successful integration of the economy into the global system and the country's more effective use of international competition, and constitutes the main task of the country's education system.

In accordance with the development concept of the Republic of Azerbaijan, the institutional foundations, infrastructure and human resources of the education system should be developed in order to provide the individual with comprehensive knowledge and skills. The development of education creates the basis for improving the well-being of the population in the country, as well as for building a higher level of individual life. Education allows people to quickly master technologies, take a worthy place in the labor market and join the lifelong learning process, choose a healthy lifestyle, and the right position in relation to the environment.

The role of education in the modernization of socio-economic life is not limited to the transformation of the knowledge and skills acquired by the learner into an economic factor. The knowledge and skills acquired in the educational process, as well as ethical and moral norms and values, create the necessary conditions for each learner to become a worthy member of society, turning him into a colleague who can be an example thanks to his knowledge and ethical behavior, an exemplary family member and citizen.

In recent years, important steps have been taken in the country towards the development of education. The normative and legal basis of education has been improved, and extensive measures have been implemented on the basis of the orders of the President of the Republic of Azerbaijan towards material and technical support. For this purpose, about 2,500 school buildings have been built in various cities and regions, and major repair and reconstruction works have been successfully carried out in existing educational facilities. The application of information and communication technologies in educational institutions has expanded significantly [11].

Along with the above, in the rapidly modernizing Republic of Azerbaijan, there is a need to take new steps towards ensuring that the education system meets the challenges of human capital development and to bring the quality indicators of general education into line with European standards. To improve the quality of education, it is necessary to restructure the education management system, develop human resources in this area, and increase the prestige of the teaching profession.

Professional personnel with a high level of education are the basis of the country's intellectual potential. In modern times, successful and sustainable economic growth of countries has been achieved through the purposeful development of human capital. All this is increasingly increasing the priority of education in state spending. State education spending has high economic profitability for society.

In modern times, the role of education in economic life has increased significantly. Currently, education should fulfill the tasks of comprehensive preparation of citizens for future life and

integration into society, along with instilling the knowledge and skills required in the economy. Meeting a person's need for lifelong education is the most important factor increasing the role of education in economic life.

At the same time, rapid technological progress requires regular updating of knowledge and skills. This increases the demand for acquiring new and more relevant specialties and developing competence.

The intensification of competition among middle-income countries and the frequent change of specialty profiles as a result of rapid technological progress have further increased the demand for the quality of education. On the other hand, increasing the level of education and expanding the educational infrastructure alone does not fully guarantee the improvement of people's knowledge and skills. Although educational infrastructure is a necessary factor, knowledge and skills are acquired not by the number of educational institutions and the duration of education, but by the quality of the learning process. Although a number of countries have recently spent a lot of money on educational infrastructure and equipping it with new equipment for the purpose of developing education, positive progress has not been achieved in educational outcomes in those countries. Despite completing general education, there are a large number of people who do not have the skills of arithmetic, reading and writing. For these reasons, ensuring the quality of education should become a major strategic priority [2].

The new requirements for the quality and comprehensiveness of education have also renewed the view on the concept of the education system. In addition to the formal (ending with the issuance of a state education certificate) form of education, the modern education system includes informal (acquiring knowledge through self-study) and non-formal (in various courses, associations) It also includes forms of education (i.e., those obtained in private lessons and not accompanied by the issuance of a state education certificate).

Since basic skills are acquired from the first years of childhood, it is important to develop children's minds from an early age. For this reason, the development of preschool education occupies an important place in state policy.

In international practice, one of the main trends regarding the content of education is the close coordination of educational curricula with the requirements and priorities of the country's socio-economic development. Curricula should have content that instills knowledge and skills that serve the goals of economic development, and competence. Since the creation of a knowledge-based economy is a priority of socio-economic policy in countries that have achieved economic success, the importance of innovations, encouraging students to engage in various social activities, and developing their teaching skills are particularly emphasized in curricula.

As a result of scientific and technological progress, innovations, and modernization over the last 30-40 years, education that is purely theoretical in nature and not enriched with practical knowledge and skills is losing its fundamental importance. In this regard, in the formation of the content of education, along with academic knowledge, the importance of practical knowledge and skills, competence is emphasized. Competence is the ability to effectively and efficiently apply the acquired knowledge and skills in practical activities. It ensures that the knowledge and skills acquired by a person are transformed into the results of specific activities. Competence-based education provides a more effective service to socio-economic development.

The content of curricula is constantly being developed taking into account the innovations brought by technological progress, which requires a specific approach at different levels of

education. While the comprehensive formation of personality is a priority at the general education level, the current and future requirements of the labor market are taken as the basis in higher education. The constant development of the higher education curriculum makes it necessary to systematically study the requirements of labor market subjects. This process has turned all subjects offering jobs in the labor market, whether in the state or non-state sector, into the main stakeholders in the higher education process [5].

Teachers and teaching methodologies. The teacher factor plays a decisive role in the process of monitoring the learning and development of the student and his achievements. The influence of a teacher on the formation of a student as a literate and competent person largely depends on the teacher's academic skills, teaching experience and level of professionalism. There is a close correlation between these qualities of a teacher and the achievements of students.

Teachers who continuously improve their knowledge and skills make an additional contribution to the achievements of students. The lack of a stimulating salary system based on professionalism, ineffective management at the level of educational institutions, and weak infrastructure for teacher training are among the factors affecting the teacher factor. In this regard, in countries where education reform is successfully carried out, special attention is paid to creating incentives for the development of teacher professionalism and improving teacher training.

4. Reforms in the field of teacher development and management in Azerbaijan. The experience of countries that have achieved significant progress in the development of education shows that modern, active-interactive teaching methods based on information and communication technologies, developing creative thinking and taking into account the individual characteristics of students, give higher results. For this purpose, the creation of continuously progressive teaching methods and increasing the competence of teachers occupy an important place in educational policy.

The main directions of management reforms in the field of education are the precise definition of the system of relations between participants in the educational process, as well as the functions, powers and responsibilities of regulatory bodies, increasing the management powers of educational institutions and their responsibility for the results of education, monitoring and evaluating activities, improving the system of indicators measuring the quality of education, directing the financing of education to its quality, introducing a per capita financing mechanism and strengthening incentive mechanisms on this basis, as well as ensuring the adequacy of funding sources.

It is clear from the experience of developed countries that the process of socialization of management in education is accelerating. Effective management, which ensures the involvement of stakeholders, plays an important role in the successful implementation of the education strategy and the sustainable improvement of the quality of education. This, first of all, involves increasing the independence of all types of educational institutions, managing educational institutions with the broad involvement of parents, pupils, students and other participants in the educational process, implementing results-oriented management and other requires the application of modern management technologies to the process.

Globalization processes require the integration of education systems of different countries. The group of competencies recommended by the Council of Europe is taken as the basis for the goals. Taking into account the increasing student-teacher exchange, the Bologna process, which aims at the unification of education standards, is accelerating. In recent years, significant achievements have been made in the direction of the development of education in the Republic of

Azerbaijan. Within the framework of the implemented state programs, the infrastructure of general education has been significantly updated in all regions of the country. A normative legal framework for education that meets modern requirements has been created in the country. State programs have been adopted and projects have been implemented to apply the main principles arising from the Bologna Declaration in higher education, modern information and scientific-methodological provision of educational institutions, renewal of educational infrastructure, improvement of textbook provision, provision of pedagogical staff to remote rural schools, and establishment of preschool and vocational education services at the level of modern requirements. At the same time, new long-term challenges for the development of human capital in Azerbaijan necessitate the implementation of important measures in financing the education sector.

Azerbaijan has achieved great achievements in terms of reducing poverty and increasing life expectancy by accelerating the pace of economic development. At the same time, statistical analyses of leading international organizations show that it is necessary to increase the international competitiveness indicators of Azerbaijani education and the rating level of higher education institutions located in the territory of the Republic of Azerbaijan. Improving the quality indicators of students and graduates of general education institutions is one of the urgent issues. There is a need to improve textbooks for higher and general education from a scientific and methodological point of view. The problem of a shortage of qualified labor has arisen in the private sector. The greatest shortage of personnel is observed in the service sector and agriculture.

5. Quality and comprehensiveness of education. The indicators for the Republic of Azerbaijan in terms of the number of higher education students per 100,000 people should be significantly increased. Access to higher education should be expanded, especially among population groups and regions. In recent years, work has been done in the field of developing the infrastructure of primary vocational education institutions, and cooperation relations have been established with the private sector, employers and international organizations. Despite the above, serious problems that need to be solved in this area still remain. One of the main problems in the management of education is the lack of clear definition of the functions of regulation, management and control in education. It is necessary to clearly define the powers, functions and responsibilities of the bodies managing the education system, ensure transparency and participation of stakeholders in the management process [3].

The role of educational institutions in management is limited. With the exception of educational institutions operating on the principle of self-government, state educational institutions do not have financial independence. At the same time, the management of educational institutions does not bear serious responsibility for the results of education. The financing of educational institutions does not depend on the results of the education they provide and is not focused on quality.

Although an information system has been created for the purpose of effective management of education, its development is needed. In order to ensure high quality of education, a system of monitoring, quality and performance indicators in the field of education should be developed.

The involvement of employers in the education and personnel training process, as well as in improving the content of education, is at a very low level. Employers should have the opportunity to participate in the personnel training process, put forward conditions for the trained personnel, and accordingly understand their responsibility.

The mechanism of financing education should be improved and should support the development of a competitive environment in the education system. For this, it is advisable to apply the principle of per capita financing at all levels of education, create a system of remuneration of labor based on competence and performance, and determine financing according to the quality results of teaching.

Conclusion.

As a result, it can be noted that the systematic evaluation of the development of teachers and the management model ensures the effective functioning of educational institutions. In this direction, the application of modern management approaches, the preparation of development programs tailored to the needs of teachers, and the establishment of transparent evaluation mechanisms are of great importance. The reforms implemented in this area in Azerbaijan have a positive impact on improving the quality of education. The assessment of teacher work The characteristics of the implementation combine many points. The new assessment model mentioned in the article and applied in world practice is based on only one factor - the increased achievement indicators of students or the observation and assessment of the pedagogical activity of teachers. The determination of the process of assessing the effectiveness of teacher work is considered formative assessment and is composed of principles determined in the course of practical assessment. The assessment models used should, along with the existing education system, continuously support the improvement of teacher professionalism and personal development. The purpose of reports evaluating teacher work and demonstrating their professional growth is to serve the professional development of teachers. In the newly applied system, the assessment of teacher effectiveness is measured not by the extent to which it complies with general standards, but by taking into account the extent to which the teacher himself has successfully developed and how he sees himself in real life. As a result of the socio-economic policy implemented in recent years, high development has been achieved in the country, the material well-being of the population has improved significantly, the poverty level has been minimized, and the Republic of Azerbaijan has taken its place among middle-income countries. The implemented reforms and large-scale infrastructure projects have created conditions for the reconstruction and sustainable development of socio-economic life. As a result, it can be noted that the systematic assessment of the development of teachers and the management model ensures the effective functioning of educational institutions. In this direction, the application of modern management approaches, the preparation of development programs tailored to the needs of teachers, and the establishment of transparent assessment mechanisms are of great importance. The reforms implemented in this area in Azerbaijan have a positive impact on improving the quality of education.

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DEVELOPING 21ST CENTURY SKILLS IN LANGUAGE TEACHERS

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Abstract: In the rapidly evolving educational landscape of the 21st century, the role of the teacher has transitioned from a mere transmitter of knowledge to a facilitator of learning. The study emphasizes the importance of reflective practice and systematic evaluation in ensuring the quality of teaching performance. By analyzing modern pedagogical approaches, the research highlights how structured reflection and evidence-based quality assurance mechanisms contribute to the sustainable development of teaching standards. The findings suggest that a holistic approach to teacher training, combining theoretical knowledge with practical methodological flexibility, is essential for meeting contemporary educational demands. In the context of rapid global changes and technological advancement, teacher education has undergone significant transformation. Modern approaches in teacher education are explored, focusing on the development of pedagogical skills and methodological competencies, continuous professional development, reflective practices, and the evaluation and quality assurance of teaching performance. The study highlights the importance of innovative teaching strategies, lifelong learning, and systematic assessment in improving educational outcomes. Modern teacher education emphasizes learner-centered approaches, integration of digital tools, and evidence-based methodologies. Furthermore, the role of reflection and professional development is analyzed as a key factor in enhancing teaching quality. A comprehensive and dynamic approach to teacher education is essential for preparing educators capable of meeting the demands of contemporary education systems.

Key words: Teacher Education, Pedagogical Competence, Professional Development, Teacher Reflection, Quality Assurance, Lifelong Learning, Student-Centered Learning

DİL MÜƏLLİMLƏRİNDƏ 21-Cİ ƏSR BACARIQLARININ İNKİŞAF ETDİRİLMƏSİ

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Xülasə: 21-ci əsrin sürətlə inkişaf edən təhsil mənzərəsində müəllimin rolu sadəcə bilik ötürücüsündən öyrənmənin asanlaşdırıcısına çevrilib. Tədqiqat tədris fəaliyyətinin keyfiyyətini təmin etməkdə refleksiv təcrübənin və sistemli qiymətləndirmənin əhəmiyyətini vurğulayır. Müasir pedaqoji yanaşmaları təhlil etməklə, tədqiqat strukturlaşdırılmış refleks və sübutlara əsaslanan keyfiyyət təminatı mexanizmlərinin tədris standartlarının davamlı inkişafına necə töhfə verdiyini vurğulayır. Nəticələr göstərir ki, nəzəri bilikləri praktik metodoloji çevikliklə birləşdirən müəllim hazırlığına vahid yanaşma müasir təhsil tələblərinin ödənilməsi üçün vacibdir. Sürətli qlobal dəyişikliklər və texnoloji inkişaf kontekstində müəllim təhsili əhəmiyyətli dərəcədə transformasiyaya uğrayıb. Müəllim təhsilində müasir yanaşmalar araşdırılır və pedaqoji bacarıqların və metodoloji səriştlərin inkişafına, davamlı peşəkar inkişafa, refleksiv təcrübələrə və tədris fəaliyyətinin qiymətləndirilməsinə və keyfiyyət təminatına diqqət yetirilir. Tədqiqat təhsil nəticələrinin yaxşılaşdırılmasında innovativ tədris strategiyalarının, ömür boyu öyrənmənin və sistemli qiymətləndirmənin əhəmiyyətini vurğulayır. Müasir müəllim təhsili şagird mərkəzli yanaşmaları, rəqəmsal vasitələrin inteqrasiyasını və sübutlara əsaslanan metodologiyaları vurğulayır. Bundan əlavə, tədris keyfiyyətinin artırılmasında refleks və peşəkar inkişafın rolu əsas amil kimi təhlil edilir. Müasir təhsil sistemlərinin tələblərini ödəyə bilən müəllimlər hazırlamaq üçün müəllim təhsilinə hərtərəfli və dinamik yanaşma vacibdir.

Açar sözlər: Müəllim təhsili, Pedaqoji səriştə, Peşəkar inkişaf, Müəllim refleksiyası, Keyfiyyət təminatı, Ömürlük təhsil tələbə, Mərkəzli öyrənmə

Introduction

The global educational paradigm is undergoing a significant transition, necessitating a re-evaluation of teacher education frameworks. Modern teacher education is no longer confined to the mastery of subject matter; it now demands a complex array of pedagogical skills and digital fluencies. As classroom dynamics become more diverse and technology-driven, teachers must possess the methodological flexibility to adapt to various learning styles. The core pillars of contemporary teacher training, focusing on how professional reflection and quality assurance tools can elevate the standard of instruction. Education plays a crucial role in shaping the future of societies, and teachers are at the center of this process. In recent decades, teacher education has evolved significantly due to globalization, technological advancements, and changing student needs. Traditional models of teacher preparation are no longer sufficient to address the complexities of modern classrooms. Modern approaches in teacher education are examined in terms of their influence on the development of pedagogical skills and methodological competencies. At the same time, the importance of continuous professional development and reflective practice is emphasized, along with careful consideration of the evaluation processes and quality assurance of teaching performance. A detailed analysis of these interconnected aspects provides a broad and in-depth understanding of how teachers are able to enhance and further strengthen their professional effectiveness over time.

Material and Methods:

The methodology of this research is grounded in a qualitative analytical framework, designed to explore the evolving paradigms of teacher education and professional development. To ensure a comprehensive understanding of the subject, the study integrates several layers of academic inquiry and systematic review.

1. Research Design and Approach

This study utilizes a descriptive and exploratory qualitative approach. Rather than focusing on numerical data, the research prioritizes the analysis of pedagogical concepts, historical shifts in training models, and the synthesis of contemporary educational theories. The framework is built upon the transition from teacher-centered instruction to student-centered and reflective learning environments.

2. Data Collection and Selection Criteria

The materials for this study were curated through a systematic review of international educational standards and peer-reviewed literature.

Source Selection: Eight high-impact academic sources were selected, including seminal books on pedagogy (e.g., works by Harmer and Brown) and modern research articles focused on Continuous Professional Development (CPD).

Thematic Scoping: Data was gathered from contemporary educational frameworks and quality assurance protocols currently implemented in higher education institutions globally, with a specific focus on their applicability to modern teaching contexts.

3. Methodological Tools

To ensure a rigorous evaluation, the following analytical methods were employed:

Comparative Analysis: This method was used to contrast traditional, lecture-based teaching models with modern, interactive strategies such as Blended Learning and Problem-Based Learning (PBL). This helped identify the gaps that modern professional development aims to fill.

Synthesis of Theoretical Frameworks: The study integrates diverse theories—ranging from Krashen's Input Hypothesis to Gibbs' Reflective Cycle—to create a unified perspective on how theory informs classroom practice.

Critical Evaluation: Modern mechanisms for assessing teaching quality, including 360-degree feedback and Key Performance Indicators (KPIs), were critically evaluated to determine their effectiveness in fostering formative growth rather than just summative grading.

4. Research Focus and Objectives

The research focuses on the intersection of theoretical preparation and practical application. Special attention is given to:

The Integration of Theory and Practice: Examining how pre-service training transitions into long-term professional mastery.

The Role of Reflection: Analyzing "Teacher Reflection" as a systematic tool for self-improvement and pedagogical innovation.

Quality Assurance: Identifying the most transparent and constructive protocols for measuring instructional effectiveness in a way that encourages educator agency and professional ethics.

5. Scope and Context

The study is framed within the context of current international standards in teacher training, aiming to provide a roadmap for educators and institutions to enhance the quality of pedagogical outcomes through structured, evidence-based development strategies.

Results and Discussion

Modern teacher education programs are designed to prepare educators for dynamic and diverse classrooms. Traditional lecture-based training has been replaced by more interactive and student-centered approaches. These include collaborative learning, problem-based learning, and the integration of digital technologies. One key feature of modern teacher education is the emphasis on

reflective practice. Teachers are encouraged to think critically about their teaching methods, classroom management, and student engagement. Additionally, teacher training programs increasingly incorporate real-world classroom experiences through internships, simulations, and mentoring systems. Another important trend is the use of technology in teacher education. Digital tools, online platforms, and virtual classrooms allow teachers to develop skills that are essential in modern education environments. This also supports inclusive education by helping teachers address diverse learning needs. Pedagogical skills refer to a teacher's ability to effectively deliver content and facilitate learning. Modern educators must be skilled in various teaching strategies, including differentiation, formative assessment, and student-centered instruction. Methodological competencies involve selecting and applying appropriate teaching methods based on the subject matter and students' needs. For example, teachers may use inquiry-based learning in science or discussion-based methods in humanities. Developing these competencies requires continuous practice and adaptation. Teachers must understand how students learn, how to motivate them, and how to create an engaging learning environment. Communication skills, classroom management, and emotional intelligence are also essential components of effective pedagogy. Teaching is a lifelong learning profession. Continuous professional development (CPD) ensures that teachers remain updated with new knowledge, skills, and educational trends. CPD can take many forms, including workshops, seminars, online courses, peer collaboration, and action research. Teacher reflection plays a critical role in professional growth. Reflective teachers analyze their experiences, identify strengths and weaknesses, and make informed decisions to improve their practice. Reflection can be done individually or collaboratively, such as through peer observations and feedback sessions. Moreover, professional learning communities (PLCs) provide a supportive environment where teachers can share ideas, solve problems, and learn from each other. This collaborative approach enhances both individual and institutional development. Evaluation is crucial for maintaining and improving the quality of education. Modern evaluation systems go beyond traditional exams and inspections. They include multiple measures such as classroom observations, student feedback, self-assessment, and performance portfolios. Quality assurance ensures that teaching standards are consistently met and improved. It involves setting clear expectations, monitoring performance, and providing constructive feedback. Importantly, evaluation should not be seen as punitive but as a tool for professional growth. Effective evaluation systems are transparent, fair, and supportive. They help identify areas for improvement and provide opportunities for targeted professional development. When combined with reflective practice, evaluation becomes a powerful mechanism for enhancing teaching effectiveness.

Conclusion

Modernizing teacher education requires a shift toward a culture of lifelong learning and systematic reflection. Enhancing methodological competencies is not merely about learning new tools but about fostering a mindset of adaptability. For quality assurance to be effective, it must be integrated into the daily professional life of the teacher, providing clear benchmarks for improvement. The synergy between reflective practice and structured evaluation ensures that teaching performance remains high, meeting the evolving needs of the global knowledge economy. Modern teacher education requires a comprehensive and dynamic approach that integrates theory, practice, and continuous improvement. The development of pedagogical skills and methodological competencies is essential for effective teaching, while continuous professional development and reflective practice ensure long-term professional growth. Evaluation assurance mechanisms further

contribute to maintaining high standards in education. Together, these elements create a strong foundation for preparing teachers who can meet the challenges of modern education systems. In conclusion, investing in modern teacher education is crucial for improving educational outcomes and fostering a more effective and inclusive learning environment.

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WHEN KNOWLEDGE IS NOT ENOUGH: CHALLENGES OF TEACHER ADAPTATION TO SPECIFIC EDUCATION SYSTEMS

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Abstract: This study investigates the challenges that teachers encounter when adapting to specific education systems, even when they possess strong subject knowledge and well-developed pedagogical skills. In contemporary educational discourse, teacher effectiveness is often associated with academic competence, instructional strategies, and classroom management. However, this paper argues that these qualities alone are insufficient without the ability to adapt to the structural, cultural, and methodological requirements of a particular education system.

The research is based on a qualitative analysis of existing literature in the fields of teacher education, applied linguistics, and educational psychology. It explores key factors influencing teacher adaptation,

including differences in curriculum design, assessment practices, institutional expectations, and sociocultural context. The study highlights that teachers who perform successfully in one system may experience difficulties in another due to misalignment between their prior experience and new professional demands.

Furthermore, the paper emphasizes the importance of contextual awareness, flexibility, and continuous professional development in overcoming adaptation challenges. The findings suggest that effective teaching is not only a matter of knowledge transmission but also the ability to respond to system-specific expectations and constraints.

In conclusion, the study underlines the need for targeted support mechanisms, such as induction programs and ongoing training, to facilitate teacher adaptation. These measures can help bridge the gap between competence and performance, ultimately improving the quality of education in diverse learning environments.

Key words: teacher adaptation, education system, professional competence, pedagogy, teacher development, instructional methods.

BİLİK KİFAYƏT ETMƏDİYİ ZAMAN: MÜƏLLİMLƏRİN XÜSUSİ TƏHSİL SİSTEMLƏRİNƏ UYGUNLAŞMA ÇƏTİNLİKLƏRİ

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Xülasə: Bu tədqiqat müəllimlərin yüksək fənn biliyinə və inkişaf etmiş pedaqoji bacarıqlara malik olmasına baxmayaraq, konkret təhsil sistemlərinə uyğunlaşma zamanı qarşılaşdıqları çətinlikləri araşdırır. Müasir təhsil mühitində müəllim effektivliyi adətən akademik bilik, tədris strategiyaları və sinif idarəetməsi ilə əlaqələndirilir. Lakin bu məqalədə göstərilir ki, bu keyfiyyətlər təkbəşinə kifayət etmir və müəllimin müəyyən bir təhsil sisteminin struktur, mədəni və metodoloji tələblərinə uyğunlaşma bacarığı da eyni dərəcədə vacibdir.

Tədqiqat keyfiyyət yönümlü yanaşma əsasında aparılmış və müəllim hazırlığı, tətbiqi dilçilik və təhsil psixologiyası sahələrində mövcud elmi ədəbiyyatın təhlilinə əsaslanmışdır. Araşdırmada müəllimlərin uyğunlaşmasına təsir edən əsas amillər – kurikulum fərqləri, qiymətləndirmə üsulları, institusional gözləntilər və sosiomədəni mühit geniş şəkildə təhlil olunur. Müəyyən edilmişdir ki, bir təhsil sistemində uğurla fəaliyyət göstərən müəllim digər sistemdə əvvəlki təcrübə ilə yeni tələblər arasında uyğunsuzluq səbəbindən çətinliklərlə üzləşə bilər.

Bundan əlavə, məqalədə müəllimlərin uyğunlaşma prosesində çeviklik, kontekstual anlayış və davamlı peşəkar inkişafın əhəmiyyəti vurğulanır. Nəticələr göstərir ki, effektiv tədris yalnız bilik ötürülməsi ilə məhdudlaşmır, eyni zamanda konkret sistemin tələblərinə uyğun fəaliyyət göstərmək bacarığını da əhatə edir.

Sonda qeyd olunur ki, müəllimlərin uyğunlaşmasını asanlaşdırmaq üçün adaptasiya proqramları və davamlı təlim imkanları yaradılmalıdır. Bu yanaşma müəllim kompetensiyası ilə real fəaliyyət arasındakı boşluğu aradan qaldırmağa kömək edə bilər.

Açar sözlər: müəllim uyğunlaşması, təhsil sistemi, peşəkar kompetensiya, pedaqogika, müəllim inkişafı

Introduction

Teacher effectiveness has long been regarded as one of the most critical factors influencing the quality of education. Traditionally, an effective teacher is defined as someone who possesses

strong subject knowledge, well-developed pedagogical skills, and the ability to manage classroom dynamics efficiently. However, contemporary research in education suggests that these characteristics, while essential, are not sufficient on their own. Increasing attention has been given to the role of contextual and systemic factors in shaping teaching effectiveness. As Linda Darling-Hammond (2017) argues, effective teaching requires not only expertise in content and pedagogy but also a deep understanding of the educational environment in which teaching takes place.

In an era of globalization and increased mobility, teachers frequently find themselves working in diverse educational systems that differ significantly in terms of curriculum design, assessment methods, institutional expectations, and cultural norms. These differences create challenges for teachers who must adapt their knowledge and teaching practices to new and sometimes unfamiliar contexts. According to Michael Fullan (2007), educational change is a complex and multidimensional process that requires individuals to continuously adjust their practices and beliefs in response to new conditions.

One of the key issues in this regard is the mismatch between a teacher's prior training and the specific requirements of a new education system. Teachers who have been trained in student-centered approaches, for example, may struggle in systems that prioritize standardized testing and rigid curricula. Similarly, those accustomed to traditional methods may find it difficult to adapt to more interactive and communicative teaching environments. As Jack C. Richards and Theodore S. Rodgers (2014) point out, teaching methodologies are deeply embedded in particular educational traditions, making transferability across contexts a challenging process.

Furthermore, sociocultural factors play a significant role in shaping teacher performance. Teaching is not merely a technical activity but a socially situated practice influenced by institutional culture, student expectations, and broader societal values. Karen E. Johnson (2009) emphasizes that teachers must develop sociocultural awareness in order to function effectively in different educational settings. Without such awareness, even highly competent teachers may experience difficulties in meeting institutional expectations and achieving desired learning outcomes.

Given these considerations, this study aims to explore the challenges associated with teacher adaptation to specific education systems. It seeks to identify the key factors that contribute to adaptation difficulties and to highlight the importance of aligning professional competence with contextual demands. By examining this issue, the paper contributes to a deeper understanding of teacher effectiveness beyond traditional notions of knowledge and pedagogy.

Materials and Methods

This study employs a qualitative research design based on an extensive review and analysis of relevant academic literature. The qualitative approach was selected as the most appropriate method for exploring complex educational phenomena such as teacher adaptation, which involves contextual, institutional, and sociocultural dimensions that cannot be fully captured through quantitative measures alone.

The primary data for this research consist of secondary sources, including peer-reviewed journal articles, academic books, and conference proceedings in the fields of teacher education, applied linguistics, and educational psychology. Key works by scholars such as Linda Darling-Hammond (2017), Michael Fullan (2007), and Jack C. Richards and Theodore S. Rodgers (2014) were analyzed to identify major theoretical perspectives and recurring themes related to teacher competence and adaptation.

The research process involved several stages. First, relevant literature was systematically selected based on its relevance to teacher effectiveness, professional development, and adaptation to different educational systems. Second, the selected sources were analyzed using thematic analysis in order to identify key patterns and factors influencing teacher adaptation. These factors were grouped into three main categories: (1) teacher-related factors, including subject knowledge and pedagogical competence; (2) system-related factors, such as curriculum structure, assessment methods, and institutional expectations; and (3) contextual factors, including sociocultural environment and classroom dynamics.

In addition, a comparative perspective was applied to examine differences between educational systems and how these differences impact teacher performance. This approach allowed for a more comprehensive understanding of the challenges teachers face when transitioning from one system to another.

Although the study does not include primary empirical data, the use of diverse and reliable academic sources ensures the validity and credibility of the findings. The limitation of this method lies in its reliance on existing literature; however, it provides a solid theoretical foundation for future empirical research on teacher adaptation and professional development.

Results and Discussion

The analysis of the reviewed literature reveals that teacher adaptation to specific education systems is a multifaceted process influenced by both individual and systemic factors. Although teachers may possess strong subject knowledge and pedagogical competence, these qualities alone do not guarantee successful performance in every educational context. The findings consistently show that adaptation difficulties arise primarily from misalignment between teacher preparation and the structural demands of a new system.

One of the most significant findings is the role of curriculum and instructional design differences. Educational systems vary in their approach to content delivery, sequencing, and learning outcomes. Teachers who are trained in flexible, student-centered environments often experience challenges when working in highly standardized or examination-oriented systems. Conversely, those accustomed to traditional, teacher-centered approaches may struggle to implement interactive and communicative methodologies effectively. This supports the view of Jack C. Richards and Theodore S. Rodgers (2014), who emphasize that teaching methods are deeply rooted in specific educational traditions and are not easily transferable across contexts.

Another important factor identified is assessment practices. The literature indicates that differences in evaluation systems significantly affect teacher performance. In some contexts, formative assessment and continuous feedback are prioritized, while in others, high-stakes testing dominates. As a result, teachers may find it difficult to align their instructional strategies with assessment expectations, which can lead to reduced teaching effectiveness. This aligns with the principle of constructive alignment discussed in educational research, where teaching, learning outcomes, and assessment must be coherent for effective learning to occur.

Sociocultural context also emerged as a critical dimension in teacher adaptation. Teaching is not only a technical activity but also a culturally embedded practice. Karen E. Johnson (2009) highlights that teachers must develop sociocultural awareness to navigate classroom norms, student behavior, and institutional culture effectively. The absence of such awareness may result in misunderstandings, reduced classroom engagement, and difficulties in establishing authority or rapport with students.

Furthermore, the findings suggest that institutional support plays a decisive role in easing adaptation challenges. Induction programs, mentoring systems, and continuous professional development opportunities are identified in the literature as key mechanisms that facilitate smoother transitions for teachers entering new systems. Without such support, even highly qualified educators may experience professional isolation and decreased effectiveness.

Overall, the discussion indicates that teacher effectiveness should be understood as an interaction between individual competence and contextual compatibility. Knowledge and skills remain essential; however, their successful application depends on the teacher's ability to adjust to system-specific expectations. This reinforces the central argument of the study that "knowledge is not enough" without adaptation capacity.

Conclusion

This study examined the challenges of teacher adaptation to specific education systems, emphasizing that subject knowledge and pedagogical competence alone are not sufficient for effective teaching in diverse educational contexts. The findings from the literature analysis demonstrate that teacher performance is strongly influenced by the degree of alignment between individual professional preparation and the structural, methodological, and cultural characteristics of the education system in which teaching occurs.

The study confirms that differences in curriculum design, assessment practices, and instructional approaches are among the main factors that create adaptation difficulties. In addition, sociocultural context and institutional expectations play a significant role in shaping how teachers perform in new environments. Even highly qualified teachers may face challenges when these contextual factors differ significantly from their previous experience. This supports the central idea that teacher effectiveness is not only a matter of knowledge transmission but also of contextual responsiveness and adaptability.

Another important conclusion is that successful adaptation requires continuous professional development and institutional support. Orientation programs, mentoring systems, and in-service training opportunities can significantly reduce the gap between teacher competence and system expectations. Without such support mechanisms, teachers may struggle to fully realize their potential, which can negatively affect the overall quality of education.

Furthermore, the study highlights the importance of integrating adaptation-related competencies into teacher education programs. Preparing future teachers for diverse and changing educational environments should include not only methodological training but also the development of flexibility, reflective thinking, and intercultural awareness. These competencies are essential for navigating different educational systems effectively.

In conclusion, the phrase "knowledge is not enough" accurately reflects the complexity of modern teaching environments. While knowledge remains a fundamental requirement, it must be complemented by the ability to adapt to specific educational systems. Addressing this issue is essential for improving teacher effectiveness and ensuring higher quality outcomes in education across different contexts.

Finally, the results highlight the need for teacher education programs to include components that prepare educators for diverse and changing educational environments. Such preparation should go beyond methodology and include training in flexibility, reflection, and intercultural competence.

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TEACHER EDUCATION AND PROFESSIONAL DEVELOPMENT

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Abstract: This research article provides an extensive and multidimensional analysis of modern approaches to teacher education and the imperative of continuous professional development (CPD) in the contemporary academic landscape. As the 21st-century classroom undergoes rapid digital and pedagogical transformations, the traditional paradigms of teacher preparation—often criticized for being overly theoretical and disconnected from classroom realities—are being re-evaluated in favor of practice-oriented, evidence-based, and reflective models. The study focuses on the systematic development of pedagogical skills and methodical competencies, particularly within the context of English Language Teaching (ELT) and the challenges of the post-pandemic digital era. A significant portion of the research is dedicated to the psychological and professional impact of teacher reflection on instructional practice, identifying it as a core component of teacher resilience and instructional leadership. By critically examining the mechanisms of performance evaluation and internal quality assurance, the article posits that sustainable educational growth requires a shift from top-down administrative inspections to collaborative, peer-based, and reflective self-assessment models. The findings suggest that when teachers engage in regular reflection and have access to structured developmental feedback and Professional Learning Communities (PLCs), their teaching efficacy increases significantly, directly impacting student learning outcomes. This paper concludes that quality assurance must be an integrated, supportive process that encourages innovation, diversity, and lifelong learning among educators.

Keywords: Teacher education, professional development, pedagogical competencies, teacher reflection, quality assurance, modern approaches, digital literacy

MÜƏLLİM TƏHSİLİ VƏ PEŞƏKAR İNKİŞAFI

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Xülasə: Bu geniş tədqiqat məqaləsi müasir təhsil mühitində müəllim hazırlığına yeni yanaşmaların və davamlı peşəkar inkişafın (CPD) zəruriliyinin çoxölçülü təhlilini təqdim edir. XXI əsrin sinif otaqları sürətli rəqəmsal və pedaqoji transformasiyaya məruz qaldığı üçün çox vaxt həddindən artıq nəzəri olmaqda tənqid edilən ənənəvi müəllim hazırlığı paradqlmaları təcrübəyönümlü, sübutlara əsaslanan və reflektiv modellərin xeyrinə yenidən qiymətləndirilir. Tədqiqat, xüsusilə ingilis dilinin tədrisi (ELT) və Post-pandemiya dövrünün rəqəmsal çağırışları kontekstində pedaqoji bacarıqların və metodiki kompetensiyaların sistematik inkişafına fokuslanır. Araşdırmanın mühüm bir hissəsi müəllim refleksiyaının tədris praktikasındakı psixoloji və peşəkar təsirinə həsr edilmiş, refleksiya müəllim dözümlülüyünün və tədris liderliyinin əsas komponenti kimi müəyyən edilmişdir. Məqalədə fəaliyyətin qiymətləndirilməsi və daxili keyfiyyət təminatı mexanizmləri tənqidi şəkildə araşdırılaraq qeyd olunur ki, təhsildə davamlı inkişaf üçün yuxarıdan aşağıya inzibati yoxlamalardan əməkdaşlığa əsaslanan və reflektiv özünüqiymətləndirmə modellərinə keçid labüddür.

Açar sözlər : müəllim hazırlığı, peşəkar inkişaf, pedaqoji kompetensiyalar, müəllim refleksiyaı, keyfiyyət təminatı, müasir yanaşmalar, rəqəmsal savadlılıq

Introduction:

The global educational landscape is currently witnessing a tectonic shift, driven by technological advancement, globalization, and a deeper understanding of cognitive learning processes. The quality of any educational institution is inextricably linked to the professional caliber, motivation, and adaptability of its teaching staff. In this context, teacher education is no longer viewed as a one-time institutional certification or a static degree, but as a continuous, lifelong trajectory of professional evolution. Darling-Hammond (2017) emphasizes that the most successful education systems globally—such as those in Finland and Singapore—are those that invest not just in initial teacher training but in sustained, high-quality professional support mechanisms.

For educators, particularly those in the field of English philology and linguistics, the challenge is twofold: they must master both the complex linguistic nuances of the language and the pedagogical strategies required to transmit this knowledge to diverse learners. The transition toward modern pedagogical approaches necessitates a fundamental change in how teachers are trained, evaluated, and supported. This introduction sets the stage for examining how methodological competencies, reflective practice, and quality assurance frameworks converge to create a robust environment for teacher growth and student success.

Material and Methods.

This study adopts a qualitative, descriptive methodology, synthesizing data from a wide range of academic sources, educational policy documents, and international pedagogical frameworks. The research is structured around four primary pillars:

Literature Review: Analyzing the shift from behaviorist (teacher-centered) to constructivist (student-centered) teaching models.

Thematic Analysis: Evaluating reflective practice models, specifically focusing on David Kolb's Experiential Learning Cycle and Donald Schön's Reflective Model.

Policy Evaluation: Assessing international standards for teacher quality assurance provided by organizations such as UNESCO (2022) and the European Commission.

Case Analysis: Reviewing modern English Language Teaching (ELT) standards, focusing on the integration of digital literacy, Artificial Intelligence (AI) in education, and formative assessment techniques.

I Modern Approaches in Teacher Education: A Paradigm Shift

Modern teacher education has transitioned from the “transmission model”—where knowledge is simply passed from trainer to trainee—to a “transformative model.” In this new framework, prospective teachers are encouraged to build their own pedagogical identity through critical inquiry and classroom-based research.

Constructivist Frameworks: Modern training emphasizes that learning is an active process. Teachers are trained to facilitate environments where students construct knowledge rather than passively receive it.

Technology Integration (TPACK Model): Educators are now required to master Technological Pedagogical Content Knowledge (TPACK). This means understanding not just the technology itself, but how to use it specifically to teach complex linguistic concepts more effectively.

Inclusive Pedagogy: There is an increasing focus on differentiated instruction, ensuring that teachers are prepared to handle classrooms with varied learning abilities, cultural backgrounds, and linguistic needs. This is particularly relevant in multicultural educational settings like Azerbaijan.

I Development of Pedagogical Skills and Methodical Kompetencies

Methodical competence is defined as the ability to apply theoretical knowledge in a practical, fluid classroom setting. For an English philologist, this includes the mastery of phonetics, syntax, and stylistics, as well as the ability to teach these effectively to non-native speakers.

According to Richards and Farrell (2005), methodical competency is developed through a combination of “craft knowledge” and “academic knowledge.” Key strategies include:

Scaffolding and Modeling: Teachers must learn to scaffold their own instruction, gradually releasing responsibility to the students.

Action Research: Teachers conduct small-scale research in their own classrooms to solve specific pedagogical problems, such as “How to improve students’ pronunciation of Old English texts?”

Mentorship and Peer Coaching: Moving away from isolated teaching, modern education promotes a “sharing culture” where novice teachers are paired with experienced mentors to facilitate the transfer of tacit knowledge.

I.Continuous Professional Development (CPD) and Teacher Reflection

Continuous Professional Development (CPD) is the “lifeline” of the teaching profession. It prevents professional stagnation and ensures that educators remain relevant in a changing world. Central to CPD is the concept of Reflective Practice.

Reflection is not merely “thinking about a lesson”; it is a systematic, critical analysis of one’s own teaching. Donald Schön’s theory suggests two levels:

Reflection-in-action: The ability of a teacher to adjust a lesson while it is happening based on student responses.

Reflection-on-action: The deep analysis of a lesson after it has concluded to identify successes and failures.

Reflection allows a teacher to bridge the gap between “what I taught” and “what they learned.” Without reflection, a teacher may have 20 years of experience, but it might actually be one year of experience repeated 20 times. For the modern educator, keeping a reflective journal or participating in video-based self-analysis has become a standard for excellence.

I.Teacher Performance Evaluation and Quality Assurance

Quality assurance (QA) in education often carries a negative connotation of “policing” or “bureaucracy.” However, modern QA frameworks focus on Formative Evaluation and Growth. This involves:

360-Degree Feedback: Incorporating feedback from peers, students, and self-assessments rather than relying solely on a principal’s annual observation.

Professional Portfolios: A collection of lesson plans, student outcomes, and certificates that serve as evidence of a teacher’s continuous growth.

Standardization vs. Innovation: Effective QA systems provide clear standards (benchmarks) while allowing teachers the professional autonomy to innovate. As highlighted by Məmmədov (2021), the integration of modern linguistic theories into Azerbaijani teacher training is vital for meeting global benchmarks.

I. Discussion: The Impact of Digital Transformation and AI

The discussion section of this study highlights a new challenge: the role of Artificial Intelligence in teacher education. Modern pedagogical competencies now include “AI Literacy.” Teachers must be able to guide students in the ethical use of AI while using these tools to automate administrative tasks, allowing more time for one-on-one student interaction. In the modern era, the role of the teacher is no longer limited to the transmission of knowledge; rather, the teacher also functions as a facilitator, mentor, and active organizer of the learning process. This transformation has become particularly evident in the post-pandemic period, during which the increased use of digital tools in the education system of Azerbaijan has introduced new demands on teachers’ professional competencies. In this evolving context, teachers are expected not only to rely on traditional instructional methods but also to effectively utilize online platforms, interactive applications, and artificial intelligence-based tools as part of their pedagogical practice. Such developments highlight the growing importance of digital literacy and adaptive teaching strategies within contemporary English Language Teaching (ELT) environments. Furthermore, the rise of Professional Learning Communities (PLCs) on digital platforms allows teachers from different parts of the world to share resources, making professional development a global, collaborative effort.

Conclusion

In conclusion, the modernization of teacher education is a multifaceted process that requires the alignment of training, professional development, and quality assurance. This study confirms that fostering a culture of reflection is the most effective way to ensure the sustained development of pedagogical competencies. To achieve long-term success, educational institutions must provide teachers with the time, resources, and psychological safety necessary for continuous growth. Ultimately, a reflected teacher is an empowered teacher, and an empowered teacher is the key to a successful, critical-thinking generation of learners. Future reforms should prioritize the human element of teaching—empathy, reflection, and creativity—over purely administrative metrics.

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MODERN APPROACHES IN TEACHER EDUCATION

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Abstract: Teacher education has undergone significant transformation in recent decades due to rapid technological development, globalization, and evolving educational demands. Traditional models of teacher preparation, which primarily focused on theoretical knowledge, are no longer sufficient in addressing the needs of modern classrooms. Contemporary approaches emphasize practical skills, critical thinking, adaptability, and digital competence.

This study explores key modern approaches in teacher education, including technology integration, student-centered learning, reflective practice, inclusive education, and continuous professional development. It also examines the benefits and challenges associated with these approaches. The findings indicate that modern methods significantly improve teaching quality and better prepare educators for real-world classroom situations.

However, challenges such as lack of resources, insufficient teacher training, and resistance to innovation remain serious concerns. The study concludes that adopting flexible, innovative, and practice-oriented teacher education models is essential for improving educational systems in the 21st century.

Keywords: teacher education, modern approaches, digital learning, pedagogy, inclusive education, professional development

MÜƏLLİM TƏHSİLİNDƏ MÜASİR YANAŞMALAR

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Xülasə: Müəllim təhsili son onilliklərdə sürətli texnoloji inkişaf, qloballaşma və inkişaf edən təhsil tələbləri səbəbindən əhəmiyyətli dərəcədə transformasiyaya uğramışdır. Əsasən nəzəri biliklərə yönəlmiş ənənəvi müəllim hazırlığı modelləri artıq müasir sinif otaqlarının ehtiyaclarını ödəmək üçün kifayət deyil. Müasir yanaşmalar praktik bacarıqları, tənqidi düşüncəni, uyğunlaşma qabiliyyətini və rəqəmsal səriştəni vurğulayır.

Bu tədqiqat texnologiya integrasiyası, tələbə mərkəzli təlim, refleksiv təcrübə, inklüziv təhsil və davamlı peşəkar inkişaf da daxil olmaqla müəllim təhsilində əsas müasir yanaşmaları araşdırır. Həmçinin bu yanaşmalarla əlaqəli faydaları və çətinlikləri araşdırır. Nəticələr göstərir ki, müasir metodlar tədris keyfiyyətini əhəmiyyətli dərəcədə yaxşılaşdırır və müəllimləri real sinif vəziyyətlərinə daha yaxşı hazırlayır.

Lakin, resursların çatışmazlığı, müəllim hazırlığının qeyri-kafi olması və innovasiyaya müqavimət kimi çətinliklər ciddi narahatlıqlar olaraq qalır. Tədqiqat nəticəsində çevik, innovativ və təcrübəyə yönəlmiş müəllim təhsili modellərinin qəbul edilməsinin 21-ci əsrdə təhsil sistemlərinin təkmilləşdirilməsi üçün vacib olduğu qənaətinə gəlinir.

Açar sözlər: müəllim təhsili, müasir yanaşmalar, rəqəmsal təlim, pedaqogika, inklüziv təhsil, peşəkar inkişaf

Introduction.

Education plays a crucial role in shaping societies, and teachers are at the center of this process. Over time, the role of teachers has evolved significantly. In the past, teachers were mainly seen as providers of knowledge. However, in modern education, they are expected to act as facilitators, mentors, and guides.

One of the main reasons for this transformation is globalization. Today, classrooms include students from diverse cultural, linguistic, and social backgrounds. Teachers must therefore be able to address different learning needs and promote equality in education.

Another important factor is technology. Digital tools such as computers, online platforms, and interactive applications have become essential in education. As a result, teachers must be equipped with digital skills and the ability to integrate technology effectively into their teaching.

Modern teacher education focuses on preparing teachers for these new challenges. It emphasizes practical experience, innovation, and lifelong learning. The aim is to produce teachers who are flexible, creative, and capable of adapting to changing educational environments.

The purpose of this study is to analyze modern approaches in teacher education and evaluate their impact on teaching quality.

Literature Review.

Many scholars have highlighted the importance of modernizing teacher education. Darling-Hammond (2017) argues that effective teacher education programs must combine theory with practice. She emphasizes that practical experience helps teachers develop real classroom skills.

Schön (1983) introduced the concept of reflective practice, which encourages teachers to evaluate their teaching methods continuously. Reflection helps teachers identify their strengths and improve their weaknesses.

Mishra and Koehler (2006) developed the TPACK framework, which focuses on the integration of technology, pedagogy, and content knowledge. This model highlights the importance of digital competence in modern teaching.

Florian (2014) emphasizes inclusive education, stating that teachers must be prepared to work with diverse learners, including students with disabilities and different learning abilities.

Richards (2015) highlights the importance of continuous professional development, suggesting that teachers must constantly update their skills to keep up with educational changes.

Overall, the literature suggests that modern teacher education should be practical, flexible, and responsive to real-world challenges.

Methodology and Materials. This research uses a qualitative approach based on secondary data.

Data were collected from academic books, journal articles, and educational reports related to teacher education. The study focuses on analyzing key modern approaches and identifying their advantages and limitations.

The main areas analyzed include:

- * Technology integration
- * Student-centered learning
- * Reflective practice
- * Inclusive education
- * Continuous professional development

The collected data were carefully examined to identify common patterns and trends in modern teacher education.

Discussion.

1. Technology Integration

Technology has become an essential part of modern education. Digital tools such as online learning platforms, smart boards, and multimedia resources have transformed teaching methods.

Teachers are expected to use these tools to make lessons more interactive and engaging. For example, videos, animations, and online quizzes can improve student understanding.

However, challenges remain. Not all teachers have sufficient digital skills, and some schools lack access to modern technology. Therefore, proper training and infrastructure are necessary.

2. Student-Centered Learning

Student-centered learning is a key feature of modern education. Unlike traditional methods, where teachers dominate the classroom, this approach focuses on active student participation.

Students are encouraged to ask questions, share ideas, and take responsibility for their learning. This method improves critical thinking, creativity, and problem-solving skills.

However, implementing this approach requires strong classroom management and careful lesson planning.

3. Reflective Practice

Reflective practice is an important tool for professional growth. Teachers analyze their teaching methods and evaluate their effectiveness.

This process helps teachers improve their performance and adapt to different classroom situations. Reflection can be done through self-evaluation, peer observation, or feedback from students.

Despite its benefits, reflective practice requires time and dedication.

4. Inclusive Education

Inclusive education ensures equal opportunities for all students, regardless of their abilities or backgrounds.

Teachers are trained to create supportive and respectful learning environments. They use different teaching strategies to meet the needs of diverse learners.

However, challenges such as large class sizes and lack of resources can make inclusive education difficult to implement.

5. Continuous Professional Development

Teaching is a lifelong learning profession. Teachers must continuously update their knowledge and skills.

Professional development activities include workshops, seminars, online courses, and collaboration with colleagues.

These activities help teachers stay updated with new teaching methods and technologies. However, access to such opportunities may be limited in some regions.

Conclusion.

This study examined modern approaches in teacher education and their impact on teaching practices.

The findings show that modern approaches improve teaching quality by promoting innovation, flexibility, and inclusivity. Technology integration, student-centered learning, reflective practice, and continuous professional development are essential components of effective teacher education.

However, challenges such as limited resources, insufficient training, and resistance to change must be addressed.

It is clear that teacher education programs must adapt to modern educational needs and provide practical and flexible training.

Future research should explore the long-term impact of these approaches on student achievement and educational outcomes.

Implications for Teacher Education

Modern approaches in teacher education have important implications for both educational institutions and teachers. Universities and training centers must redesign their programs to include more practical experiences, digital training, and student-centered methodologies.

Teacher education programs should provide opportunities for real classroom practice, internships, and collaborative learning. This helps future teachers gain confidence and develop essential teaching skills.

In addition, institutions should invest in technology and ensure that teachers are trained to use digital tools effectively. Without proper support, it is difficult to implement modern teaching approaches successfully.

Another important implication is the need for policy changes. Governments and educational authorities should support teacher development through funding, training programs, and professional opportunities.

Challenges in Implementing Modern Approaches

Although modern approaches offer many benefits, their implementation is not always easy.

One major challenge is the lack of resources. Many schools, especially in developing regions, do not have access to modern technology or training materials.

Another issue is resistance to change. Some teachers prefer traditional methods and may feel uncomfortable using new strategies or technologies.

Time is also a significant factor. Teachers often have heavy workloads, which makes it difficult for them to participate in training or reflective practices.

Furthermore, large class sizes can limit the effectiveness of student-centered and inclusive teaching approaches.

Addressing these challenges requires cooperation between teachers, institutions, and policymakers.

Recommendations

Based on the findings of this study, several recommendations can be made:

- * Teacher education programs should focus more on practical training rather than only theoretical knowledge.

- * Continuous professional development opportunities should be provided regularly.

- * Schools should improve access to digital tools and resources.

- * Teachers should be encouraged to adopt innovative teaching methods.

- * Collaboration between teachers should be promoted through workshops and group activities.

By following these recommendations, educational systems can improve the quality of teaching and learning.

Future Perspectives

The future of teacher education will likely be shaped by technological advancements and global changes.

Artificial intelligence, online learning platforms, and virtual classrooms will play a larger role in education. Teachers will need to develop new skills to adapt to these innovations.

In addition, personalized learning will become more important. Teachers will need to understand individual student needs and adjust their teaching accordingly.

Global collaboration between educators will also increase, allowing teachers to share knowledge and best practices across countries.

Therefore, teacher education must remain flexible and open to change in order to meet future demands.

In conclusion, modern approaches in teacher education are essential for improving the quality of education. These approaches help teachers become more effective, adaptable, and innovative. While there are challenges such as limited resources and resistance to change, the benefits of modern methods are significant. By investing in teacher education and supporting professional development, educational systems can achieve better outcomes. Ultimately, well-prepared teachers play a key role in shaping the future of society. Therefore, continuous improvement in teacher education should be a top priority.

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DEVELOPMENT PROSPECTS OF TEACHERS' PEDAGOGICAL SKILLS IN THE TEACHING OF HISTORY

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Abstract: The article examines the theoretical and methodological foundations of the development and application of pedagogical skills in history teaching and analyzes the impact of these skills on the effectiveness of the educational process. It has been determined that in the modern education system, the primary objective of history as a subject is not only the teaching of historical facts and events, but also the development of students' historical thinking, analytical and critical thinking skills, national identity, and civic awareness. Achieving these objectives depends directly on the pedagogical preparation and professional competencies of the history teacher.

The study identifies effective communication, classroom management, motivation, organization of student-centered learning, implementation of individual approaches, and development of critical thinking as the core pedagogical skills of a history teacher. The article substantiates the significance of methods such as debates, discussions, analysis of historical sources, problem-based learning, role-playing, historical reenactments, and the use of maps and visual materials as practical forms of applying pedagogical skills in history lessons. The research concludes that the systematic and continuous development of pedagogical skills is one of the fundamental conditions for improving the quality of history teaching, increasing students' cognitive engagement, and shaping their historical worldview.

Key words: pedagogical skills, methodological literacy, teaching methods, interactive map, facilitator

TARİXİN TƏDRİSİNDƏ MÜƏLLİMLƏRİN PEDAQOJİ BACARIQLARININ İNKİŞAF PERSPEKTİVLƏRİ

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Xülasə: Məqalədə tarix tədrisində pedaqoji bacarıqların inkişafı və tətbiqinin nəzəri-metodoloji əsasları araşdırılmış, bu bacarıqların tədris prosesinin səmərəliliyinə təsiri təhlil edilmişdir. Müəyyən edilmişdir ki, müasir təhsil sistemində tarix fənninin əsas məqsədi yalnız tarixi fakt və hadisələrin öyrədilməsi deyil, həm də şagirdlərdə tarixi təfəkkürün, analitik və tənqidi düşünmə bacarıqlarının, milli kimlik və vətəndaşlıq şüurunun formalaşdırılmasıdır. Bu məqsədlərə nail olunması isə bilavasitə tarix müəlliminin pedaqoji hazırlıq səviyyəsi və peşəkar kompetensiyalarından asılıdır.

Araşdırma nəticəsində kommunikativ ünsiyyət, sinfin idarə olunması, motivasiya yaratmaq, şagirdyönümlü təlimin təşkili, fərdi yanaşmanın tətbiqi və tənqidi düşünmənin formalaşdırılması tarix müəlliminin əsas pedaqoji bacarıqları kimi müəyyən edilmişdir. Məqalədə debat, diskussiya, tarixi mənbələrin təhlili, problem əsaslı öyrənmə, rol oyunları, tarixi canlandırmalar, xəritə və vizual materiallardan istifadə kimi metodların tarix dərslərində pedaqoji bacarıqların praktik tətbiq formaları kimi

əhəmiyyəti əsaslandırılmışdır. Tədqiqat nəticəsində belə qənaətə gəlinmişdir ki, pedaqoji bacarıqların sistemli və davamlı inkişaf etdirilməsi tarix tədrisinin keyfiyyətinin yüksəldilməsi, şagirdlərin dərk etmə fəallığının artırılması və onların tarixi dünyagörüşünün formalaşdırılması üçün əsas şərtlərdən biridir.

Açar sözlər: pedaqoji bacarıqlar, metodiki savadlılıq, təlim metodları, interaktiv xəritə, fasilitator

Introduction

In the modern education system, the subject of history is not limited solely to the teaching of past events, chronological facts, and historical processes; it also plays a significant role in shaping students' personalities, strengthening their commitment to national and moral values, developing civic awareness, and enhancing analytical thinking skills. Through the teaching of history, students do not merely learn about the past, but also acquire the ability to analyze cause-and-effect relationships in historical events, compare the socio-political characteristics of different periods, and understand the patterns of societal development (Amirov, 2014). These features place history in a distinct position among the humanities and increase its strategic importance within the educational system.

In the context of globalization and the rapid development of the information society, the content and objectives of history education have entered a new qualitative stage. In the contemporary era, the main task of history as a subject is not merely to transmit information to students, but to guide them in analyzing diverse sources, distinguishing reliable from unreliable information within the flow of data, approaching historical events objectively, and drawing independent conclusions. In this regard, the teaching of history not only provides students with a body of knowledge, but also contributes to the formation of their intellectual, social, and civic competencies.

The teaching of history plays an important role in shaping students' national identity and historical memory. The study of the nation's past, traditions of statehood, cultural heritage, examples of national heroism, and struggles for independence contributes to the patriotic upbringing of the younger generation and strengthens their national self-awareness and sense of historical responsibility. At the same time, history education develops students' objectivity, critical analysis, source-based inquiry, comparative reasoning, and multiperspective thinking skills, all of which are among the key universal competencies required by modern society and the labor market.

The effective organization of history lessons and the development of deep historical thinking in students depend directly on the teacher's professional preparation, methodological literacy, and pedagogical skills. Whereas in the traditional instructional model the teacher primarily functioned as a source of information, in the modern educational paradigm the teacher acts as an organizer of the learning process, guide, facilitator, and motivational leader. This transformation requires the teacher to possess not only subject knowledge but also advanced pedagogical, psychological, and methodological preparation.

The pedagogical competencies required of history teachers include effective communication, consideration of students' age and individual characteristics, implementation of differentiated and inclusive approaches, use of interactive and innovative teaching methods, creation of problem-based and inquiry-oriented learning environments, management of discussions and debates, and effective use of digital educational resources and visual materials. Particularly in a subject such as history, which demands analytical engagement, the possession of these competencies significantly enhances instructional effectiveness. A teacher with well-developed pedagogical skills is capable of transforming students from passive listeners into active researchers and independent thinkers.

The effectiveness of history teaching depends not only on the presentation of information but also on the methods and tools through which it is delivered. Mere memorization of historical facts does not produce sustainable and functional knowledge in students. On the contrary, contemporary pedagogical approaches emphasize the necessity of employing methods such as discussions, project-based activities, source analysis, role-playing, debates, map work, and document analysis in history lessons. The successful implementation of these methods is directly linked to the teacher's methodological and pedagogical preparation.

In the modern era, the improvement of educational quality, the implementation of competency-based curricula, and the integration of digitalization into the education system are among the main factors necessitating the continuous development of pedagogical skills. Teacher professionalism is no longer viewed as a one-time preparation process, but rather as a model of continuous professional development. In this context, history teachers' self-education activities, participation in professional development programs, study of international pedagogical experience, and mastery of innovative methodologies play an important role in the formation of their professional competencies.

The development of pedagogical skills in history teachers affects not only the quality of instruction but also students' attitudes toward learning, motivation, cognitive engagement, and academic achievement (Ilyasov, 2018). Therefore, the development of pedagogical skills in history teaching is considered one of the most actual and strategic areas of modern pedagogical research. The main purpose of this article is to investigate the ways of developing pedagogical skills in history teaching, the characteristics of their application, their interaction with modern instructional technologies, and the mechanisms through which they influence the effectiveness of the teaching process. The article will also analyze the impact of the formation of professional competencies in history teachers on educational quality from both theoretical and practical perspectives, summarize existing approaches in this field, and present scientific conclusions.

Materials and methods

The methodological foundation of the study was based on a systematic analysis of scholarly literature related to pedagogy, the methodology of teaching history, and contemporary educational theories. During the research process, the essence of pedagogical skills, their application in history teaching, and their role in teachers' professional practice were comparatively examined through both local and international academic sources. Through the analysis of scientific literature, the theoretical approaches influencing the formation of professional competencies in history teachers were identified, and the relevance of these competencies in the modern education system was assessed.

A comparative analysis of history teaching methods was conducted throughout the research process. The applicability of traditional and modern instructional methods in history lessons was examined, and their impact on students' cognitive engagement and the development of historical thinking was compared. Within this framework, alongside traditional lecture-based and explanatory-illustrative methods, modern pedagogical approaches such as interactive teaching strategies, problem-based learning, debates, role-playing, project-based learning, and analysis of historical sources were analyzed.

To investigate the practical aspects of history teaching, methodological examples from various historical topics were also included in the study. For instance, the use of problem-based questioning in teaching the topic "The Establishment of the Azerbaijan Democratic Republic"

enables students to gain a deeper understanding of the essence of statehood ideas. Likewise, in teaching the topic "World War II," the comparative analysis of different countries' positions and the application of document-based historical source analysis foster students' multiperspective understanding of events. The use of maps and visual materials in teaching "The Rise of the Safavid State" contributes to the development of historical-geographical thinking. Such examples practically demonstrate how pedagogical skills are integrated into the content of history teaching.

The study also examined the influence of modern pedagogical approaches on history instruction. The applicability of competency-based education models, constructivist approaches, student-centered learning, and active/interactive instructional technologies in teaching history was analyzed. It was determined that these approaches ensure not merely the acquisition of information by students, but also its analysis, evaluation, and application. This, in turn, requires a high level of pedagogical preparation and methodological flexibility from the teacher.

Within the framework of the study, scholarly works by local and international researchers on pedagogical skills and history teaching were also reviewed. The approaches of Azerbaijani educators to history teaching methodology were compared with international practices, and the possibilities for adapting innovative instructional methods used in history education to the national education system were evaluated. Comparative analysis revealed that international experience places greater emphasis on critical thinking, source-based learning, and inquiry-oriented activities in history instruction.

In addition, the State Standards for General Education of the Republic of Azerbaijan, the history curriculum, educational programs, and methodological guidelines served as part of the empirical material base of the research. Through the analysis of curriculum documents, the professional competencies, learning outcomes, and assessment criteria expected from history teachers were identified. The analysis of these documents demonstrated that the modern curriculum requires history teachers not only to possess subject knowledge, but also to demonstrate flexible pedagogical skills, the ability to apply innovative teaching methods, and competence in organizing student-centered instruction.

During the study, methods such as content analysis, comparative analysis, generalization, and systematization were employed. Through these methods, the theoretical foundations, practical implementation mechanisms, and developmental directions of pedagogical skills in history teaching were comprehensively examined.

Results and discussion

The conducted analyses demonstrate that the effectiveness of history teaching is directly related to the level of development of the teacher's pedagogical skills. Since the content of history as a subject requires not only the transmission of factual knowledge but also the analysis of events, the establishment of cause-and-effect relationships, the comparison of different perspectives, and the formation of historical thinking, the professional competencies of the teacher are of particular importance in this field. A history teacher with strong pedagogical preparation not only increases students' interest in the lesson but also contributes to the development of their logical thinking, reasoning, and research skills.

One of the most important pedagogical competencies for a history teacher is communication skill. Effective communication enables the teacher to create an atmosphere of trust and cooperation with students and to explain complex historical events clearly and comprehensibly. For example, when teaching complex topics in the history of international relations such as the Treaty of

Versailles, the teacher's communicative and explanatory abilities play a crucial role in students' systematic understanding of the subject. A communicative approach also enhances the effectiveness of discussions and debates in history lessons.

Classroom management is another essential competency in the professional activity of a history teacher. When applying interactive activities such as debates, group work, role-playing, and source analysis in history lessons, the teacher must coordinate classroom processes effectively, manage time efficiently, and regulate student participation. Weakness in this skill may prevent interactive lessons from achieving their intended objectives.

Another important skill of the history teacher is the ability to create motivation. Simply presenting historical facts often fails to sustain student interest. Learning motivation increases when the teacher connects historical events to the modern world, demonstrates their significance for contemporary society, and integrates the lesson with real-life contexts. For instance, linking the topic of the Azerbaijan Democratic Republic to the formation of modern Azerbaijani statehood enhances students' interest in the subject.

The development of critical thinking is considered one of the primary goals of history teaching. Teachers should guide students not merely to accept information but to analyze historical facts, compare different sources, and approach events from multiple perspectives. For example, comparing the political positions of the United States and the USSR while teaching the Cold War period develops students' ability to make objective judgments and engage in multidimensional thinking.

The organization of student-centered learning is one of the key conditions for structuring history lessons in accordance with modern educational requirements. This approach ensures that students become active participants, conduct independent research, and demonstrate initiative in the learning process (Mirzayev & Rustamova, 2012). To achieve this, history teachers should employ various interactive methods.

The practical application of pedagogical skills in history teaching is broad and multifaceted. Debates and discussions are among the most effective methods for developing students' analytical and critical thinking abilities in history lessons. For example, organizing debates on topics such as "The Main Causes of the Fall of the Roman Empire" or "The Positive and Negative Consequences of the Industrial Revolution" allows students to evaluate events from multiple perspectives.

Working with historical sources is one of the principal methodological approaches in history teaching. The analysis of documents, letters, memoirs, archival materials, and period press sources develops students' ability to work with primary sources. For example, analyzing the text of the Treaty of Gulistan helps students gain a deeper understanding of Azerbaijan's political history in the nineteenth century.

The use of maps and visual materials contributes to the development of students' spatial and temporal understanding in history lessons. Teaching the topic of the Great Geographical Discoveries through map analysis or visualizing the military routes of the Napoleonic Wars facilitates students' comprehension of historical events within their spatial context.

Problem-based learning engages students in independent inquiry and conclusion-drawing processes. For example, a problem-oriented lesson structured around the question "Why did the Safavid State decline?" encourages students to analyze the causes of historical developments comprehensively.

Role-playing and historical reenactments are important tools that increase students' emotional and intellectual engagement in history lessons. For instance, simulating the Paris Peace Conference and assigning students the roles of representatives of different states helps them better understand the topic while simultaneously developing diplomatic and analytical skills.

The development of pedagogical skills in history teachers requires a process of continuous professional development. Teacher training sessions, seminars, and workshops play an important role in mastering modern methodological innovations. These activities provide teachers with opportunities to learn new instructional strategies, become acquainted with international experience, and expand their methodological repertoire.

Self-development and methodological preparation are also key factors in the formation of pedagogical skills. A teacher's regular engagement with scientific and pedagogical literature, familiarity with new research, and continuous reflection on teaching practice contribute significantly to professional growth.

Mastery of digital technologies has become an integral competency of the modern history teacher. The use of electronic presentations, interactive maps, virtual museums, digital archives, and multimedia resources enhances the visual appeal and effectiveness of history lessons. Particularly in topics such as Ancient Egypt, Medieval Cities, or World War II, the use of virtual materials allows for a more vivid presentation of historical reality.

Thus, the discussions conducted indicate that the pedagogical skills of history teachers are not merely technical elements of instruction, but rather key strategic factors determining the quality of history education. The systematic development of these skills significantly contributes to improving the effectiveness of history teaching and to fostering students' historical thinking and civic responsibility.

Conclusion

Research and analysis indicate that the development of pedagogical skills in history teaching is one of the decisive factors in improving the quality of education. Since history as a subject is not limited to teaching past events alone, the pedagogical mastery of the teacher directly influences the formation of both knowledge and worldview in students. A history teacher with highly developed pedagogical skills ensures not the mechanical memorization of historical facts, but rather the analysis of those facts, the identification of cause-and-effect relationships, and the development of a critical approach to events. This contributes to the development of historical thinking, which is one of the principal objectives of history education.

The quality of history teaching often depends on how the teacher organizes the instructional process. When the same topic is presented by different teachers using different methods, student achievement and levels of comprehension may vary significantly (Mammadova, 2015). For example, there is a substantial difference in learning effectiveness between merely presenting the factual aspects of World War I and conducting an interactive lesson that examines its political, economic, and geostrategic causes. This fact once again demonstrates that in history teaching, not only content but also the form of presentation and methodological organization are of great importance.

The modern educational paradigm has transformed the teacher's function from the traditional model of "knowledge transmitter" into that of a facilitator of the learning process (Badalova, 2015). The teacher's main task is no longer merely to provide information, but to guide students toward thinking, researching, comparing, and drawing independent conclusions. This

approach is especially important in history education, because learning history involves not only knowing when events occurred, but also understanding why they occurred, what consequences they produced, and how they relate to the modern world. In this regard, by acting as a facilitator, the teacher enables students to transform from passive observers into active researchers (Badalova, 2015).

The effective application of pedagogical skills in history teaching also significantly influences the formation of students' national identity and civic consciousness. For example, teaching the history of Azerbaijani statehood—particularly the period of the Azerbaijan Democratic Republic and the struggle for independence—through appropriate methodological approaches fosters respect for traditions of statehood and strengthens patriotic feelings among students. At the same time, the critical analysis of major world historical events, such as the French Revolution, the Industrial Revolution, or the Cold War, develops students' ability to approach global processes analytically.

Rapid changes in the education system, digitalization, the introduction of new technologies, and the expansion of the information environment have transformed teachers' pedagogical development into a continuous process. The professional development of history teachers should no longer be limited to the knowledge acquired during higher education but should be based on the principle of continuous renewal and improvement. In this regard, advanced training courses, international exchange of experience, mastery of digital educational resources, and the implementation of modern methodological innovations will remain priority directions for pedagogical development in the future.

In conclusion, it can be stated that the development of pedagogical skills in history teaching is not only one of the key indicators of teacher professionalism, but also an important determinant of educational quality, student achievement, and the formation of historical consciousness. The systematic development of pedagogical competencies in history teachers should be regarded as a strategic factor contributing not only to the effective organization of lessons but also to the intellectual, civic, and moral development of future generations. Therefore, the improvement of pedagogical skills will continue to maintain its relevance as one of the priority directions in the future development of history education.

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TEACHER EDUCATION AND PROFESSIONAL DEVELOPMENT

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Abstract: This article discusses teacher education and professional development in modern education. In recent years, the role of teachers has changed significantly, and new approaches are required to meet the needs of students in a rapidly developing world. The study focuses on modern approaches in teacher education, the development of pedagogical skills and methodological competencies, continuous professional development, and teacher reflection. It also examines the importance of evaluation and quality assurance in teaching performance. The results show that effective teacher education and ongoing development improve teaching quality and student outcomes.

Key words: teacher education, professional development, pedagogical skills, reflection, evaluation

MÜƏLLİMLƏRİN TƏHSİLİ VƏ PEŞƏKAR İNKİŞAFI

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Xülasə: Bu məqalə müəllim təhsili və peşəkar inkişaf mövzunu araşdırır. Son illərdə müəllimin rolu əhəmiyyətli dərəcədə dəyişmiş və sürətlə inkişaf edən dünyada tələbələrin ehtiyaclarını qarşılamaq üçün yeni yanaşmalara ehtiyac yaranmışdır. Tədqiqat müəllim təhsilində müasir yanaşmalar, pedaqoji bacarıqların və metodiki kompetensiyaların inkişafı, davamlı peşəkar inkişaf və müəllim refleksiya məsələlərinə fokuslanır. Həmçinin tədris fəaliyyətinin qiymətləndirilməsi və keyfiyyət təminatı da araşdırılır. Nəticələr göstərir ki, effektiv müəllim təhsili və davamlı inkişaf tədrisin keyfiyyətini və tələbə nailiyyətlərini artırır.

Açar sözlər: müəllim təhsili, peşəkar inkişaf, pedaqoji bacarıqlar, refleksiya, qiymətləndirmə

Introduction

In modern education, the role of the teacher is very important. Teachers are not only responsible for delivering knowledge, but also for guiding students, developing their critical thinking, and preparing them for real-life situations. Because of these responsibilities, teacher education and professional development have become essential.

In the past, teachers mainly focused on traditional teaching methods. However, today's students have different needs and expectations. Technology, globalization, and new educational trends require teachers to update their knowledge and skills continuously.

Teacher education helps future teachers gain the necessary knowledge and skills. At the same time, professional development allows practicing teachers to improve their abilities and adapt to new challenges. Therefore, both teacher education and professional development play a key role in improving the quality of education.

Methodology and Materials

This study is based on qualitative research methods. Various academic sources, including

books, articles, and online materials related to teacher education and professional development, were analyzed.

The study focuses on four main areas: modern approaches in teacher education, development of pedagogical and methodological skills, continuous professional development and reflection, and evaluation of teaching performance. The information is presented in a clear and understandable way to highlight the importance of these concepts in education. (Harmer, J. (2007). *How to Teach English*.)

Discussion

Modern approaches in teacher education

Modern approaches in teacher education focus on student-centered learning and practical experience. Future teachers are trained not only in theory but also in practice. They participate in classroom activities, teaching practice, and real-life teaching situations.

One important approach is competency-based education. This approach focuses on developing specific skills such as communication, classroom management, and problem-solving. Teachers are trained to handle different situations in the classroom effectively.

Another modern approach is the use of technology in teacher education. Digital tools, online platforms, and virtual classrooms are widely used. These tools help teachers learn new methods and improve their teaching strategies.

Collaborative learning is also an important part of modern teacher education. Students work together, share ideas, and learn from each other. This improves teamwork skills and prepares teachers for real classroom environments.

Development of pedagogical skills and methodological competencies

Pedagogical skills are essential for effective teaching. These skills include lesson planning, classroom management, communication, and assessment. Teachers must know how to explain topics clearly and make lessons interesting.

Methodological competencies are also very important. Teachers need to choose the right teaching methods according to students' needs. For example, some students learn better through visual materials, while others prefer discussion or practice.

Developing these skills requires training and experience. Teacher education programs provide opportunities for future teachers to practice and improve their abilities. Workshops, seminars, and teaching practice play a big role in this process.

In addition, teachers should be flexible and creative. They need to adapt their teaching methods to different situations and students. This helps create a positive learning environment and improves student engagement.

Continuous professional development and teacher reflection

Learning does not stop after becoming a teacher. Continuous professional development (CPD) is necessary for teachers to stay updated and improve their skills. This includes attending training sessions, workshops, and conferences.

Teachers can also learn through self-study and online resources. Reading books, watching educational videos, and participating in online courses help teachers develop professionally.

Reflection is another important aspect of professional development. Teachers should think about their teaching methods and evaluate their performance. They can ask questions like: "What worked well?" and "What can be improved?"

Reflective practice helps teachers understand their strengths and weaknesses. It also encourages them to try new methods and improve their teaching strategies. As a result, reflection leads to better teaching and learning outcomes.

Evaluation and quality assurance of teaching performance

Evaluation is an important part of education. It helps measure the effectiveness of teaching and identify areas for improvement. Teachers' performance can be evaluated through different methods such as classroom observation, student feedback, and self-assessment.

Quality assurance ensures that teaching meets certain standards. Educational institutions use different criteria to evaluate teaching quality. This helps maintain a high level of education.

Modern evaluation methods focus not only on results but also on the teaching process. For example, student participation, creativity, and critical thinking are also considered important.

Feedback plays a key role in evaluation. Teachers can improve their performance by listening to feedback from students and colleagues. Constructive feedback helps identify problems and find solutions.

However, evaluation should be fair and supportive. It should help teachers grow professionally rather than create stress. A positive evaluation system encourages teachers to improve and develop their skills.

Conclusion

In conclusion, teacher education and professional development are essential for improving the quality of education. Modern approaches help teachers become more effective, flexible, and innovative.

The development of pedagogical skills and methodological competencies allows teachers to meet students' needs and create a positive learning environment. Continuous professional development and reflection help teachers improve their performance and adapt to changes.

Evaluation and quality assurance ensure that teaching standards are maintained and improved. Overall, well-trained and continuously developing teachers play a key role in students' success and the development of society.

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SKILLS OF A 21ST CENTURY TEACHER

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Abstract: The rapid development of technology has transformed every sphere of human activity, including education. As society becomes increasingly digital, educational systems must adapt to prepare learners with the knowledge and skills required in the twenty-first century. This article examines the role of digital pedagogy in improving the quality of education and supporting more effective teaching and learning processes. It discusses the integration of digital technologies, multimedia resources, online learning platforms, artificial intelligence, and interactive tools into educational practice. The study highlights the importance of developing teachers' digital competencies, promoting personalized learning, increasing student motivation, and fostering critical thinking, creativity, communication, and collaboration. Furthermore, the article analyzes the advantages of digital pedagogy in creating accessible, flexible, and learner-centered educational environments while also addressing challenges related to technological infrastructure and digital literacy. It concludes that the effective implementation of digital pedagogy significantly enhances educational quality, supports teachers' professional development, and prepares students to meet the demands of a rapidly changing digital society and future labor market.

Key words: technologies, digital pedagogy, education, quality, skills

XXI ƏSR MÜƏLLİMİN BACARIQLARI

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Xülasə: Texnologiyanın sürətli inkişafı insan fəaliyyətinin bütün sahələrinə, xüsusilə də təhsil sisteminə mühüm təsir göstərmişdir. Müasir dövrdə təhsil gələcək nəslin bilik və bacarıqlarının formalaşdırılmasına xidmət etdiyindən, rəqəmsal yeniliklərin tədris prosesinə inteqrasiyası zəruri hesab olunur. Bu məqalədə rəqəmsal pedaqogikanın təhsilin keyfiyyətinin yüksəldilməsində rolu və imkanları təhlil edilir. Rəqəmsal texnologiyalar, onlayn təhsil platformaları, multimedia vasitələri, süni intellekt tətbiqləri və interaktiv alətlər təlim prosesinin daha çevik, səmərəli və öyrənməyönümlü təşkilinə şərait yaradır. Eyni zamanda, müəllimlərin rəqəmsal kompetensiyalarının inkişaf etdirilməsi, fərdiləşdirilmiş öyrənmənin təşviqi, tələbələrin motivasiyasının artırılması, tənqidi düşünmə, yaradıcılıq, ünsiyyət və əməkdaşlıq bacarıqlarının formalaşdırılması rəqəmsal pedaqogikanın əsas üstünlükləri kimi qiymətləndirilir. Məqalədə həmçinin rəqəmsal savadlılıq, texnoloji infrastruktur və müasir tədris metodlarının tətbiqi ilə bağlı mövcud problemlər nəzərdən keçirilir. Nəticə etibarilə, rəqəmsal pedaqogikanın səmərəli tətbiqi təhsilin keyfiyyətinin yüksəldilməsinə, müəllimlərin peşəkar inkişafına və müasir cəmiyyətin tələblərinə cavab verən kompetensiyalı mütəxəssislərin hazırlanmasına mühüm töhfə verir.

Açar sözlər: texnologiya, rəqəmsal pedaqogika, təhsil, keyfiyyət, bacarıqlar

Introduction

The goal of any pedagogy is to form the necessary knowledge and skills in learners, and the goal of 21st century pedagogy is to develop the knowledge and skills needed in a rapidly changing digital society. If we approach the concept of education and training as a classical-traditional process, that is, education and training is a process of transferring facts from teacher to student, then it is easy to imagine the goal of the educational process: the graduate should repeat what teachers know. One of the provisions of progressive pedagogy at the end of the 19th century was to teach what is needed here and now. To involve practitioners, entrepreneurs, employers who understand the basic requirement for knowledge and skills in training.

Despite the fact that progressive pedagogy continues to defend its value today, on the other hand, it is now clear to everyone that now no one can predict what competencies will be needed for a successful person in at least five years.

The reforms carried out in education have shown that several issues need to be kept in the spotlight. Along with a number of pedagogical requirements, the development of digital skills is among the requirements facing teachers today. The presented material uses observation, analysis, synthesis, and analysis methods to examine the current situation.

It is known that technology is developing rapidly. Therefore, the concept of “21st century skills” has emerged, which expresses this very problem area.

Here it is possible to distinguish several groups of skills:

1. critical thinking (analyzing, predicting, finding connections, evaluating, testing);
2. creativity and innovation (creating, changing, inventing, improving);
3. collaboration (listening, sharing, motivating, organizing);
4. communication (discussing, persuading, explaining)

Any theories of learning are, in a sense, forms of one of the eternal philosophical questions: where does knowledge come from and how do people acquire, perceive and exchange it? Unlike psychology and philosophy, which study human cognitive activity in itself, pedagogy is interested in studying the methods and means of organizing this process most effectively.

The foundations of modern theories are empiricism, which assigns the main role to experience in the cognitive process, and rationalism, which puts the activity of reason to the fore. Francis Bacon, who developed the philosophical system of empiricism, criticized the teachers of his time for offering nothing but words, for an abundance of rhetorical exercises and a lack of practical activity.

Rene Descartes and subsequent supporters of the rationalist approach proceeded from the claim that experience represents only examples of reality, that is, its specific manifestations (facts), and that the basis of education should be a systematic theoretical view and understanding of the world. There are countless learning theories that have influenced the formation of the modern educational paradigm and constitute this paradigm. These include behaviorism, cognitivism, constructivism, etc.

The vocabulary that forms the basis of modern pedagogical theories is wide: e-learning, computer-based learning, digital learning, mobile learning, online learning. There is still no certain complete understanding of some of these terms and their logical connection. In this regard, a specialist working in this field must have important skills such as critically evaluating the proposed classifications, seeing gaps and ways to eliminate them. For example, based on the logic of the term “e-learning”, it can be predicted that e-learning refers to any form of learning carried out using electronic devices. This means, at a minimum, a transition from printed formats of educational content to multimedia methods of transmitting information. However, this is not always the case. It is known that the most reliable way to understand the essence of the concept is to identify the features that distinguish it from other similar phenomena.

Materials and Methods

E-learning differs in several ways.

- Accessibility and equity. One of the most important aspects of digital pedagogy is the accessibility of education to everyone, regardless of their geographical location and social status. In the context of globalization and pandemics, online learning has become an important tool for

maintaining the continuity of the educational process. Digital technologies allow students from different parts of the world to receive quality education, as well as to interact with teachers and peers at an international level.

- Personalized learning. One of the main features of digital pedagogy is the ability to adapt the educational process to the needs of each student. Modern educational platforms and applications allow you to track the successes and difficulties of students, so that teachers, in turn, can adjust the training program in a timely manner. This contributes to a deeper understanding of the material and increases student motivation.

- Interactivity and involvement. The use of multimedia materials (videos, animations, simulations) and game-based teaching methods makes the process more fun. For example, gamification is actively used to increase student activity, which contributes to the development of critical thinking and creativity. This approach helps to avoid rote learning of material, which is an important aspect of successful learning.

- Development of critical thinking and independence. In digital education, students learn not only to work with information, but also to evaluate it, look for solutions to problems, work in teams and make decisions. This contributes to the development of critical thinking and independent work skills, which are important for future professional and personal development.

- In the rapidly changing digital world, this approach is not only relevant, but also necessary for the effective training of future specialists, regardless of their profession. Technology opens up new horizons for educators to create and deliver educational content and interact with students.

Digital pedagogy encompasses a wide range of methods and tools that teachers use to improve and simplify the teaching process using digital technologies.

1. Electronic textbooks and resources. They allow students to access the content they need at any time and on any device.

2. Online courses and platforms for distance learning. Programs such as Moodle, Coursera, Duolingo, and others provide flexibility in learning, allowing students to learn the material at their own pace.

3. Use of multimedia materials. Visual and audiovisual elements such as video lessons, presentations, simulations, and games make the learning process more engaging and understandable.

4. Mobile applications and artificial intelligence technologies. These tools can personalize learning, adapting it to the individual needs of each student.

Results and Discussion

Digital pedagogy has great potential to transform education systems. First, it provides equal access to quality educational materials regardless of the student's geographical location. This is especially important in times of globalization and the spread of pandemics, when traditional forms of learning may be limited.

Second, the use of digital technologies helps to implement student-centered approaches to learning. Technologies allow for the personalization of learning, adapting it to the knowledge level, interests, and pace of each student.

In addition, digital pedagogy helps to develop critical thinking and creativity, as modern educational platforms often include interactive activities that encourage active participation of students and allow them to experiment with what they are learning.

Digital pedagogy offers teachers new opportunities to improve the quality of teaching and expand the means of interaction with students. The introduction of technologies such as virtual reality (VR), augmented reality (AR), artificial intelligence (AI), and others helps create unique learning environments where students can simulate different situations and learn through experience.

1. Virtual and augmented reality. VR and AR technologies open up new horizons for engaging students in various educational scenarios. For example, virtual tours, laboratories or historical reconstructions can be used to study biology, history or geography.

2. Artificial intelligence and data analytics. Adaptive learning platforms powered by artificial intelligence can offer tasks tailored to the student's level of knowledge and provide personalized recommendations to improve learning outcomes. This allows teachers to better understand the needs of their students and help them on their way to success.

3. Gamification. The inclusion of game elements in the learning process helps to increase student activity and make learning more interesting. This can be done through quizzes, educational games and competitions that make the learning process more dynamic and motivate students.

It should be noted that digital pedagogy implies the importance of digital skills for teachers. In the modern educational process, teachers play a key role in the adaptation and application of digital technologies. Without appropriate training and mastery of digital tools, it is impossible to effectively implement digital pedagogy. This requires teachers not only to work with modern technical means, but also to be able to use them to improve the quality of teaching.

Key digital skills for teachers include:

1. Ability to work with educational platforms and online resources.
2. Knowledge of methods for using interactive technologies to create engaging and effective lessons.
3. Ability to analyze and interpret data on student performance to adjust the teaching process.
4. Ability to integrate multimedia and game elements into the learning process to increase student motivation.

Thus, in the context of digital pedagogy, the success of a teacher is not only in deep subject knowledge, but also in the ability to use the latest technologies to create an innovative and effective educational process.

The great impact of digital pedagogy on the growth and career of a modern teacher is natural. Digital pedagogy not only opens up new opportunities for students, but also affects the careers of educators. The integration of digital technologies into the teaching process is becoming an important criterion in assessing the professionalism of a teacher. Teachers who actively use innovative methods and technologies can increase their competitiveness in the labor market and have more opportunities for career advancement. Teachers who are ready to learn and develop in the field of digital technologies will be in demand in the future. The ability to work online, use multimedia resources, and analyze student performance data opens up new horizons for both teachers and students.

Quality education is one of the most important indicators today. The quality of education is the demand for knowledge acquired in specific conditions of its application in order to achieve a certain goal and improve the quality of life. The quality of knowledge is determined by its validity, depth and relevance in work after graduation. The quality of education, first of all, has a multidimensional and complex structure, which includes:

1. The quality of the teaching staff.
2. The state of the material and technical base of the educational institution.
3. The motivation of the teaching staff.
4. The quality of training programs.
5. The quality of students.
6. The quality of infrastructure.
7. The quality of knowledge.
8. Innovative activity of management.
9. The introduction of process innovations.
10. The competitiveness of graduates in the labor market.
11. The achievements of graduates. Naturally, these indicators are closely related to each other and interact within the system.

Based on this, we can say that any impact on the education system and, accordingly, improving the quality of educational services affects all of the above elements.

Conclusion

Digital pedagogy is becoming an integral part of the educational process and opens up new opportunities for teachers to create effective and individualized educational directions. It is important for teachers not only to master modern technologies, but also to adapt them to create an active, attractive and deep learning environment. The introduction of digital technologies is not just a trend, but an integral part of the future of education, on which the success of teachers' careers and the effectiveness of education in general depend. In addition, digital pedagogy significantly simplifies the learning process (but does not lead to its primitivism, does not simplify or devalue knowledge itself), reduces the time of both the teacher and the student.

Based on this, it can be concluded that the teacher spends more time on developing his teaching skills, gaining new experience and interesting his students. He does not waste time on household chores and monotonous work with test tasks and tests, since computer technologies provide these further processing, but focuses his attention only on professional activities. Students, in turn, receive more voluminous and diverse tasks, covering not only the main types of work (tests, exercises, solving examples, constructing graphs), but also additional ones that affect the expansion of their horizons (creation of projects, presentations, video and audio files, analysis of various multimedia elements). Tasks can be differentiated according to the level, hierarchical system: all students, for example, after being tested at the beginning of the year, receive tasks corresponding to their level of knowledge, which has a beneficial effect on the teaching process and supports the aspirations of students. In this way, they are involved in the educational process, increase the level of knowledge and training, general activity, acquire practical skills.

Based on the above, it can be noted that the quality of education is one of the main aspects in the development of students' professional and personal qualities, their competence and learning ability. Modern information technologies, in particular, electronic information and educational environments, contribute to its increase. This, on the one hand, leads to the preparation of a competent specialist, and on the other hand, to the formation of a personality with high information literacy and developed critical thinking.

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CHALLENGES IN TEACHER TRAINING IN THE DIGITAL ERA

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Abstract: The digital era has had and continues to have both positive and negative impacts on almost all aspects of modern life. The impact of the digital era has also had a significant impact on the field of education. Studies show that the constantly developing education sector is largely keeping up with the innovations brought by the digital era. However, along with these advances, it is important to highlight the challenges it creates, conduct research to solve them, and work persistently towards their resolution.

This article examines the main challenges faced in teacher training in the digital era and analyses their impact on the quality of education. Studies show that many teachers today have digital skills. At the same time, not all teachers have the same level of digital competence. While some teachers use digital tools very effectively, others are limited to basic functions. Such teachers prefer traditional teaching methods during lesson. This creates obstacles to a unified approach in the teaching process and leads to differences in students' learning experiences. The analysis shows that the uneven levels of teachers' digital skills limit the use of innovative teaching methods and affect the interactivity of lessons. In addition, the insufficient availability and uneven distribution of technological resources in educational institutions further complicates the situation.

The results show that the integration of the digital era into education is inevitable. Continuous training is essential to improve the digital skills of teachers and ensure that they are aware of the latest developments in digital era. Only in this way can the quality of education be improved and an effective learning environment be created.

Keywords: digital era, teacher training, improving the quality of education, challenges in teacher training, digital inequality

RƏQƏMSAL DÖVRDƏ MÜƏLLİM HAZIRLIĞINDA ÇƏTİNLİKLƏR

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Xülasə: Rəqəmsal dövr müasir həyatın demək olar ki bütün sahələrinə mənfi və ya müsbət təsir göstərmişdir və göstərməyə də davam edir. Rəqəmsal dövrün təsiri təhsil sahəsində də əhəmiyyətli

dərəcədə təsir göstərmiş. Araşdırmalar göstərir ki, davamlı inkişaf edən təhsil sahəsi rəqəmsal dövrün gətirdiyi yeniliklərlə ayaqlaşır. Lakin bu irəliləyişlərlə yanaşı, həmçinin, onun yaratdığı çətinlikləri vurğulamaq onların həlli yolunda araşdırmalar etmək və bu istiqamətdə ısrarla çalışmaq vacibdir.

Bu məqalə rəqəmsal dövrdə müəllim hazırlığında qarşıya çıxan əsas çətinlikləri araşdırır və onların təhsilin keyfiyyətinə təsirini təhlil edir. Araşdırmalar göstərir ki, bugün bir çox müəllim rəqəmsal bacarıqlara malikdir. Onlar dərs zamanı onlayn platformalardan istifadə edir, təqdimatlar hazırlayır, video dərslər keçirlər. Eyni zamanda, bütün müəllimlər eyni səviyyədə rəqəmsal bacarıqlara malik deyillər. Bəzi müəllimlər rəqəmsal alətlərdən çox yaxşı istifadə etdikləri halda, bəziləri isə sadəcə əsas funksiyalarla məhdudlaşırlar. Həmin müəllimlər dərs zamanı daha çox ənənəvi metodlardan istifadə etməyi üstün tuturlar. Bu isə tədris prosesində vahid yaşamağa mane olur və şagirdlərin öyrənmə təcrübəsində fərqliliklərə səbəb olur. Təhlillər onu göstərir ki, müəllimlərin rəqəmsal bacarıqlarının qeyri – bərabər səviyyədə olması dərslərdə innovativ metodlardan istifadəni məhdudlaşdırır və dərslərin interaktivliyinə təsir edir. Bununla yanaşı, təhsil müəssisələrində texnoloji resursların kifayət qədər olmaması və qeyri – bərabər paylanması da mövcud olan vəziyyəti daha da çətinləşdirir.

Nəticələr göstərir ki, rəqəmsal dövrün təhsil sahəsinə inteqrasiyası qaçılmazdır. Müəllimlərin rəqəmsal bacarıqlarını artırmaq üçün mütəmadi təlimlərin keçirilməsi və onların rəqəmsal dövrün bütün yeniliklərindən xəbərdar olmaları vacibdir. Yalnız bu şəkildə təhsilin keyfiyyətini artırmaq və faydalı dərs mühiti yaratmaq olar.

Açar sözlər: rəqəmsal dövr, müəllim hazırlığı, təhsilin keyfiyyətin artırılması, müəllim hazırlığında çətinliklər, rəqəmsal bərabərsizlik

Introduction

The integration of the digital era into education enables teachers to continuously develop and use new methods. Unlike traditional teaching methods, the innovations brought by the digital era to the education system make it possible to create a high – quality learning environment. At the same time, the application of digital technologies in classrooms creates conditions for organizing the learning process in a more interactive and student – centered way. In particular, in the modern era, teachers' use of online platforms and electronic resources allows them to organize lessons more effectively and engagingly. This, in turn, increases students' interest in lessons, develops their independent thinking, helps them express themselves more clearly, and has a positive impact on the formation of their critical thinking skills. Recent studies have emphasized the importance of teachers using digital tools to increase students' technological literacy. As examples of these studies, the research of Martinez – Serrano (2019), Hatlevik (2019), Domingo – Coscolla (2020), and Bentri and Hidayati (2023) can be mentioned. The common aim and idea of all these studies is to develop teachers' digital skills and to explore ways of integrating digital resources into education. These studies have also emphasized the importance of communication and collaboration in the learning process facilitated by digital skills. Teachers should not see digital as additional skills, but rather as resources that enrich their experiences. By using these skills, they will contribute to both their own development and that of the students they teach. At the same time, the digital environment creates new responsibilities for teachers and provides opportunities for the emergence of different ideas. They should not be limited to acquiring subject knowledge only, but should also gain digital literacy and be able to use technological tools in the classroom environment. In this regard, the application of new approaches in teacher training and ensuring continuous professional development are of particular importance. Opinions observed in recent research also continue to emphasize the importance of these relationships and have revealed that teachers' self-awareness of their digital incompetence significantly affects the integration of digital skills into teaching.

The role of technologies in the teaching and learning process, from a theoretical perspective, has been explained within the framework of various scientific approaches. When Vygotsky's (1978) approach is applied to the digital environment, it shows that these technologies play a supporting role. In this context, digital skills are related not only to students' technical abilities but also to their active construction of knowledge and the use of that knowledge.

In addition, according to Sweller's (1988) theory, when using digital technologies, students' cognitive capacities should be taken into account and cognitive overload should be prevented. This theory shows that it is not enough to simply use digital technologies in education; it is also necessary to ensure their pedagogically appropriate and purposeful application. Such an approach prevents students from being overloaded and avoids creating additional difficulties for them. In general, it can be observed from the research conducted within the framework of this topic that digital skills are very essential for modern education. The application of a comprehensive and systematic approach in teacher training is necessary. Such an approach prepares future teachers to use their digital skills effectively and to understand the new opportunities emerging in the education system.

Methodology and Materials

This study was conducted based on a qualitative research design in order to investigate the challenges encountered in teacher training in the digital era and the opportunities it creates. The selection of a qualitative approach in the study was due to its ability to provide a deeper understanding of participants' views on digital competencies, their individual experiences, and their thoughts on the subject. John W. Creswell (2014) states that qualitative research is particularly helpful in understanding social realities from the perspectives of participants.

Additionally, according to Patera (2022), qualitative research in education creates opportunities for collaborative discussion and learning among participants. Such processes serve as an important means of enriching participants' experiences. In this regard, an interpretative approach was applied in accordance with the objectives of the study.

The participants of the study consisted of students studying at the Faculty of Education of Mingachevir State University. Participation was voluntary and ensured through an open invitation. Through the participation of students, efforts were made to include different levels of knowledge and experience. This approach increased the diversity of the sample and contributed to obtaining more comprehensive results.

A structured survey questionnaire was used to collect the data. Both closed-ended and open-ended questions were included in the questionnaire, aiming to gather information about participants' level of use of digital technologies, their views on innovations brought by the digital era, and the strategies they consider effective. The questionnaire consisted of 10 questions and mainly covered the level of use of digital technologies, and the challenges encountered. At the same time, in accordance with ethical principles, participants' anonymity and the confidentiality of the data were ensured. This approach is consistent with the ethical quality standards emphasized by Virginia Braun and Victoria Clarke (2006). In the preparation of the survey, the main principles of the DigCompEdu framework presented by the European Commission, which explains teachers' digital competencies, were utilized. Data collection took place in the second semester of 2026 and was conducted via Google Forms.

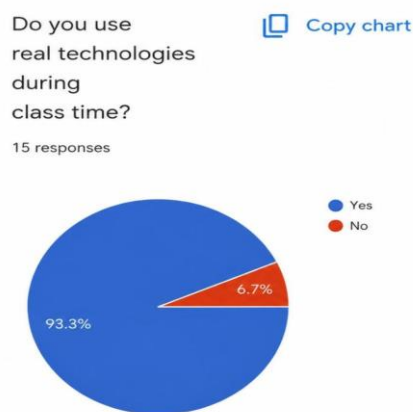
The collected data were analyzed using the thematic analysis method. This method allows for the identification of recurring themes within the research. The analysis process consists of several

stages. These stages include familiarization, coding, identifying themes and systematizing them. These stages are based on the thematic analysis model presented by Virginia Braun and Victoria Clarke. As a result, the themes obtained were presented in a structured manner to answer the main research questions.

Results and Discussion

The main aim of the conducted research was to identify the challenges arising in teacher training in the digital era. The results obtained from the survey participants show that although the use of digital technologies is widely spread in the teaching process, a number of significant challenges still remain in this field.

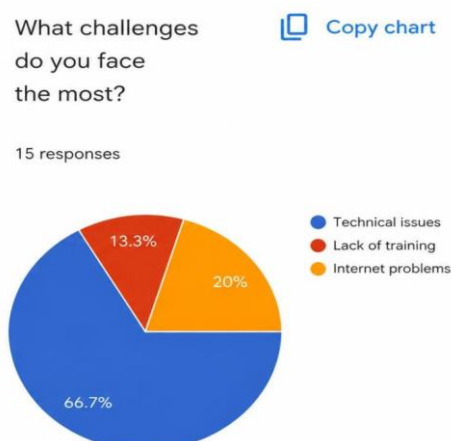
The majority of participants from the pedagogical faculty (93.3%) stated that they regularly use digital technologies during lessons.



This indicator shows the integration of the digital era into the educational environment; however, it also reveals that the opportunities provided by this era are not yet fully utilized effectively.

Regarding the teaching of digital skills in teacher training, the results show that a large proportion of participants consider this training sufficient, while a very small number assess it as partially sufficient or insufficient. This indicates that digital competencies in the teacher training process still need to be further improved.

The research revealed that the most common challenges in teacher training are technical problems. This means that the lack of adequate conditions contributes to the emergence of difficulties. In addition, internet problems and lack of experience were among the significant challenges encountered.



Another important finding is related to the lack of training. Some participants stated that they had attended trainings on working with digital tools. At the same time, some teachers reported that they had not participated sufficiently in such trainings, which shows the necessity of strengthening the integration of competencies into curricula.

It was noted among the responses that digital technologies have a very positive impact on lesson quality as well as on students' motivation. This positive impact depends on the individual skills of teachers, which further emphasizes the importance of training and systematic preparation.

The answers to the open-ended questions in the survey show that the main problem is the lack of formation of skills in using digital technologies. According to them, it is necessary to increase practice-oriented training in teacher education, develop a digital mindset, and organize continuous professional development programs.

Overall, the research shows that in the digital era, teacher education faces both opportunities and serious challenges. Although the presence of digital technologies facilitates the teaching process, lack of experience, technical problems, and weak practical skills remain the main obstacles in this field. Therefore, the development of digital competencies in teacher training and strengthening practice-oriented training is of particular importance.

Conclusion

The final results of the research showed that digital technologies have become one of the main tools in teacher training, but the process is still not fully developed. From the participants' responses, it can be understood that although the use of technologies is widespread, there are still certain gaps in their correct and effective application. At the same time, insufficient in this field also draws attention.

In conclusion, in order for the positive impact of the opportunities brought by the digital era on education to be sustainable, it is essential for teachers to be better prepared in this area. In particular, increasing practical training and improving technical facilities emerge as the main needs in this field.

Furthermore, it can be suggested that teacher education programs should place greater emphasis on the development of digital competencies and provide more practice-oriented learning opportunities. Strengthening continuous professional development and ensuring access to modern technological resources can also contribute to improving the overall quality of teacher training.

Finally, although this study was conducted on a limited number of participants, it provides useful insights into the existing challenges. Future research involving a larger sample size may help to obtain more comprehensive and generalizable results.

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STRATEGIES FOR IMPLEMENTING INQUIRY-BASED LANGUAGE TEACHING IN LOWER LEVEL UNIVERSITY EDUCATION

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Abstract: Inquiry-based language teaching (IBLT) is gaining traction as a meaningful alternative to transmission-model instruction, yet its application to foundational university language courses raises unresolved practical questions. Students at the A1-B1 proficiency range face real linguistic constraints that seem, at first glance, incompatible with the open-endedness that inquiry demands. This article takes that tension seriously and offers four classroom-level strategies – inductive grammar investigation, sociolinguistic text exploration, bounded research tasks, and structured learner self-audit – that make genuine inquiry achievable for lower-level university learners. The argument is not that IBLT should replace

structured instruction, but that inquiry and scaffolding are more compatible than commonly assumed, and that the earlier students engage as language investigators, the more ownership they develop over their own acquisition.

Key words: inquiry, strategy, inquiry-based language teaching, foundational learners, university education

ALİ TƏHSİLİN İLK PİLLƏSİNDƏ TƏDQIQATA ƏSASLANAN DİL TƏDRİSİNİN HƏYATA KEÇİRİLMƏSİ STRATEGİYALARI

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Xülasə: Sorğu-əsaslı dil təlimi (IBLT) ötürücü-model tədrisinə mənalı alternativ kimi getdikcə daha çox yayılır, lakin onun universitet səviyyəsində ilkin dil kurslarında tətbiqi hələ də bəzi praktik suallar doğurur. A1-B1 səviyyəsində olan tələbələr real dil məhdudiyyətləri ilə üzləşirlər ki, bu da ilk baxışda sorğuya əsaslanan açıq yanaşma ilə uyğun gəlmirmiş kimi görünür. Bu məqalə həmin ziddiyyəti ciddi şəkildə nəzərdən keçirir və aşağı səviyyəli universitet tələbələri üçün sorğunun real tətbiqini mümkün edən dörd sinif strategiyasını təqdim edir – induktiv qrammatika araşdırması, sosiolinqvistik mətn təhlili, məhdudlaşdırılmış tədqiqat tapşırıqları və strukturlaşdırılmış öyrənən özünüqiymətləndirməsi. Məqalənin əsas iddiası ondan ibarətdir ki, IBLT strukturlaşdırılmış tədrisi əvəz etməməlidir; əksinə, sorğu və scaffolding (dəstəkləyici mərhələlilik) bir-biri ilə daha uyğun və tamamlayıcıdır. Tələbələr nə qədər tez dil tədqiqatçısı kimi fəaliyyətə başlasalar, dil öyrənmə prosesinə bir o qədər çox sahiblik hissi formalaşdırırlar.

Açar sözlər: sorğu, strategiya, sorğu-əsaslı dil təlimi, ilkin səviyyəli öyrənənlər, universitet təhsili

Introduction

There is a persistent assumption in university language education that foundational learners are not yet ready to think critically about language – that they must first accumulate vocabulary, internalize grammatical patterns, and gain fluency before anything resembling inquiry becomes possible. This assumption is worth interrogating. Consider what lower-level learners actually do when they encounter a new language: they observe, compare, guess, test, revise. They are, in a very natural sense, already inquiring (12). The challenge is not to introduce inquiry where none of them exists, but to channel what students already do intuitively into deliberate, structured investigative activity. Research in second language acquisition suggests that even early-stage learners benefit significantly from opportunities to engage in meaning-making rather than rote memorization (3). When instruction is designed with this in mind, even A2-level learners can engage as active analysts of the language they are learning rather than passive consumers of pre-digested rules (2).

This article proposes four strategies for realizing this goal in lower-level university language classrooms. Each strategy is grounded in the logic of inquiry – posing a genuine question, gathering evidence, drawing conclusions – while acknowledging the scaffolding requirements that beginner learners legitimately need. The aim is not to produce a prescriptive model, but to offer tested conceptual tools that instructors can adapt to their specific language, curriculum, and student population.

Materials and Methods

1. Inductive Grammar

The most natural starting point for IBLT at the foundational level is grammar – not grammar as a rulebook to be memorized, but grammar as a system to be discovered. The shift is subtle but consequential: instead of telling students about certain grammatical rule, an instructor might present twenty authentic short sentences grouped by context and ask students to work out what pattern determines case selection.

This approach – sometimes called noticing-based or inductive grammar instruction – asks students to function as pattern detectors. The task is not to produce language immediately, but to observe it carefully. This is a cognitively accessible entry point even for students with limited productive ability. What changes is the stance: the student is an analyst, not a recipient.

Ellis (2002) argues that form-focused instruction is most effective when it arises from communicative need, and inductive tasks create precisely this condition (5). Nassaji and Fotos (2011) similarly demonstrate that learners who discover grammatical rules through structured data analysis retain those rules more durably than those who receive explicit instruction alone (7).

What makes this genuinely inquiry-based rather than merely interactive is the role of the question. Students are not directed toward a pre-determined answer; they are asked to form and test a hypothesis (4). Some groups will identify the target pattern quickly; others will identify a partial rule or a convincing but incorrect one. Both outcomes are valuable, because both require the instructor to respond to the students' thinking rather than simply confirm a pre-packaged answer. The conversation that follows – about why one hypothesis holds and another doesn't – is where much of the deep learning occurs (10).

Practical scaffolding for this strategy includes carefully curated sentence sets where the key variable is visible, observation worksheets that guide students through the noticing process without giving the answer away, and pair or group discussion before whole-class synthesis. The key design principle is that the data must be legible at the students' current proficiency level – obscure vocabulary or complex syntax will redirect cognitive effort away from the grammatical phenomenon being investigated.

2. Sociolinguistic Text Exploration

A second strategy repositions everyday texts as sociolinguistic artifacts rather than comprehension vehicles. In conventional lower-level instruction, a text – an email, a product label, a weather forecast – is typically used to practice reading skills or extract specific information. In an inquiry-based frame, the same text becomes an object of analysis: Who wrote this, and for whom? What does it assume the reader knows? What is being sold, performed, or concealed?

These questions might seem to belong in advanced cultural studies courses, but they are surprisingly tractable at early proficiency levels when the texts are short, the vocabulary is accessible, and the analytical framework is introduced incrementally. A beginner-level English course, for example, might use two versions of the same advertising slogan – one formal, one colloquial – and ask students to identify which words differ and what those differences suggest about the intended audience. Students do not need to produce a sociolinguistic analysis; they need to notice that language varies with context and begin to ask why.

Students who are asked to interrogate texts – even simple ones – develop an early habit of reading as active meaning-makers rather than passive decoders. Critical reading at foundational levels, when scaffolded appropriately, cultivates transferable analytical dispositions that benefit learners across disciplines. This disposition, cultivated at the foundational stage, is transferable across disciplines and across languages. It also makes explicit something that communicative

approaches sometimes leave implicit: that language is always doing social work, and understanding that work is part of becoming a competent user.

3. Small-Scale Research as Language Practice

A third strategy adapts the logic of project-based learning to the constraints of the lower level language classroom. Rather than open-ended research projects – which presuppose both linguistic and investigative competencies that beginner learners are still developing – this approach uses tightly scoped tasks with a clear investigative question, a defined set of resources, and a structured output format.

Beckett and Slater (2005) demonstrate that project-based tasks, even when tightly scoped, develop both linguistic and investigative competencies simultaneously (1). Thomas (2000), in a comprehensive review of project-based learning, identifies clear investigative questions and structured outputs as the features most strongly associated with positive learning outcomes – precisely the design principles recommended here (11).

A practical example: in a lower-level Mandarin course, students might be asked to investigate how the same concept – punctuality, for instance, is discussed differently in mainland Chinese and Taiwanese online forums, using a curated set of short posts selected by the instructor. The question is narrow enough to be answerable; the linguistic demands are managed through resource selection; the output – a brief comparative summary, perhaps presented orally to a small group, is achievable at the students' proficiency level.

What makes this inquiry rather than simply a structured reading task is that students are gathering and synthesizing evidence to answer a question they cannot look up directly (6). They must make interpretive decisions, however modest, and take responsibility for a conclusion. Others further argue that the oral presentation component of such tasks – even at low proficiency levels – promotes negotiation of meaning and deepens content understanding. This is qualitatively different from answering comprehension questions, even well-designed ones, because it places the student in an epistemic position of not-yet-knowing rather than finding-the-answer-that-is-already there.

4. The Learner as Their Own Research Subject

A fourth strategy turns the inquiry inward: students investigate their own developing competence as a formal object of study. This is more structured than general reflective journaling and more purposeful than standard self-assessment checklists. The idea is to give students a systematic method for examining their own errors, tracking their progress, and generating hypotheses about their learning patterns. This approach draws on a robust tradition of research into learner autonomy and self-regulated learning. Most of the instructors identify deliberate practice – practice guided by careful attention to errors and feedback – as the central mechanism of expertise development, a finding that lends empirical support to the structured self-audit proposed here.

A simple implementation might look like this: at regular intervals across a semester, students review a sample of their own written or spoken production, categorize the errors they find using a basic taxonomy provided by the instructor, and write a short analytical note – in the target language, at a level appropriate to their proficiency – explaining what they notice and what they intend to do differently. Researchers observe that learners who engage in systematic error analysis develop greater metalinguistic awareness, which in turn accelerates acquisition. Learners who monitor their own cognitive processes are consistently more effective than those who do not. This strategy is particularly well suited to the university context, where students are expected to develop as independent learners. Making the investigation of one's own acquisition an explicit course activity

communicates that language learning is not something that happens to students but something they can actively shape.

Discussion

Challenges and Institutional Considerations have impact in implementing this model at universities. Any honest account of IBLT at the foundational university level must acknowledge the structural pressures that work against it. Shohamy (2001) has extensively documented how standardized testing regimes constrain pedagogical innovation, often forcing instructors to prioritize discrete measurable outcomes over process-oriented learning goals (8). Stoller (2006) offers a practical response to this tension, demonstrating that project-based and inquiry-oriented approaches can be designed to produce conventional measurable outcomes while simultaneously developing investigative competencies (9).

One workable approach is to treat IBLT strategies as the vehicle for achieving conventional outcomes rather than a replacement for them: students who investigate the dative case inductively are still learning the dative case, but they are doing so through a process that is likely to produce more durable retention. The inquiry is the method; the grammatical or lexical competence is still the measurable outcome. Student resistance is a related challenge, particularly among learners who have been trained to expect explicit instruction and clear answers. Instructors should begin with shorter, lower-stakes inquiry tasks before moving to more extended project work.

Conclusion

The case for inquiry-based language teaching at the foundational university level rests on a straightforward claim: that students learn more durably when they construct understanding actively than when they receive it passively, and that this principle applies to language learning no less than to other disciplines. The four strategies discussed here – inductive grammatical investigation, sociolinguistic text analysis, bounded research tasks, and structured learner self-audit – are not a comprehensive curriculum framework. They are a set of starting points, each grounded in an inquiry logic that instructors can adapt, combine, and refine.

What connects these strategies is a shared commitment to repositioning the lower-level language learner as a thinker about language, not merely a practitioner of it. That repositioning, this article has argued, is both pedagogically sound and practically achievable – provided that inquiry is understood not as the absence of structure, but as a particular kind of structure, one organized around questions rather than answers.

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THE IMPACT OF SELF-DEVELOPMENT AND REFLECTIVE THINKING ON THE DEVELOPMENT OF SECONDARY SCHOOL STUDENTS

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Abstract: This article examines the role of self-development and reflective thinking in the development of secondary school students. In modern education systems, not only the transmission of knowledge but also students' independent thinking, self-assessment, and active participation in the learning process are considered highly important. The main aim of the study is to determine how the development of these skills affects students' overall academic achievement.

The research was conducted using methods such as classroom observation, question-and-answer sessions, and reflection activities. It was found that students with stronger reflective thinking skills are better

able to analyze their mistakes, evaluate their learning process more consciously, and achieve higher academic success. Reflective thinking helps students identify their weaknesses and improve their learning strategies.

Self-development, on the other hand, increases students' independent learning abilities, strengthens their internal motivation, and raises their interest in lessons. Students who actively work on their self-improvement become more responsible, confident, and willing to participate in class activities.

The findings show that self-development and reflective thinking are closely connected and support each other. Reflection helps students recognize their mistakes, while self-development encourages them to take action and improve. When these two skills are applied together, they have a stronger positive effect on students' academic and personal growth.

The article also offers practical recommendations for teachers, such as allocating time for reflection at the end of lessons, encouraging students' self-assessment, and using interactive teaching methods to create a more active and student-centered learning environment.

Keywords: self-development, reflective thinking, secondary school students, educational process, motivation, self-assessment, independent learning, interactive teaching, student development

ÖZÜNÜNKİŞAF VƏ REFLEKTİV TƏFƏKKÜRÜN ÜMUMTƏHSİL MƏKTƏBİ ŞAGİRTLƏRİNİN İNKİŞAFINA TƏSİRİ

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Xülasə: Bu məqalədə orta məktəb şagirdlərinin inkişafında özünüinkişaf və reflektiv düşünmənin rolu araşdırılmışdır. Müasir təhsil sistemində yalnız biliklərin ötürülməsi deyil, şagirdlərin müstəqil düşünməsi, özünüqiymətləndirməsi və öyrənmə prosesində aktiv iştirakı da mühüm əhəmiyyət daşıyır. Tədqiqatın əsas məqsədi bu bacarıqların inkişafının şagirdlərin ümumi təhsil nəticələrinə təsirini müəyyən etməkdir.

Araşdırma zamanı müşahidə, sual-cavab və refleksiya fəaliyyətləri kimi metodlardan istifadə edilmişdir. Müəyyən olunmuşdur ki, reflektiv düşünmə bacarığı inkişaf edən şagirdlər öz səhvlərini daha yaxşı analiz edir, öyrənmə prosesinə daha şüurlu yanaşır və nəticədə daha yüksək nailiyyətlər əldə edirlər. Refleksiya şagirdlərə öz zəif tərəflərini görməyə və onları aradan qaldırmağa kömək edir.

Özünüinkişaf isə şagirdlərin müstəqil öyrənmə bacarığını artırır, daxili motivasiyasını gücləndirir və dərslə marağını yüksəldir. Bu bacarıq şagirdin yalnız müəllimdən asılı olmadan öz inkişafına məsuliyyətlə yanaşmasını təmin edir.

Tədqiqat nəticələri göstərir ki, özünüinkişaf və reflektiv düşünmə bir-biri ilə sıx bağlıdır və birlikdə tətbiq olunduqda daha güclü nəticə verir. Məqalədə müəllimlər üçün praktik tövsiyələr də verilmişdir. Bunlara dərs sonunda refleksiya üçün vaxt ayrılması, şagirdlərin özünüqiymətləndirməsinin təşviqi və interaktiv tədris üsullarının tətbiqi daxildir. Bu yanaşmalar şagirdlərin həm akademik, həm də şəxsi inkişafına müsbət təsir göstərir.

Açar sözlər: özünüinkişaf, reflektiv düşünmə, orta məktəb şagirdləri, təhsil prosesi, motivasiya, özünüqiymətləndirmə, müstəqil öyrənmə, interaktiv tədris, şagird inkişafı

Introduction

During my experience as a teacher, I have often reflected on one question: why do some students develop very quickly, while others remain at the same level for a long time? Interestingly, this difference is not always related to intelligence. Sometimes, a student who starts off weaker later

achieves better results. This observation made me think more deeply and led me to start observing more carefully.

Over time, I realized that one of the main reasons lies in students' attitudes toward themselves. Some students simply listen during the lesson and consider that enough. Others, however, constantly try to improve themselves, reflect on their mistakes, and aim to do better next time. This is where the concepts of self-development and reflective thinking come into play.

Modern education is no longer based solely on the transmission of knowledge. It is now considered important for students to think, analyze, and actively participate in their own learning process. Especially during secondary school, the development of these skills is crucial. At this stage, students begin to understand themselves, set goals, and lay the foundation for their future lives.

The relevance of this topic is also connected to the fact that, in today's world, education is not measured only by academic results. Skills such as critical thinking, self-assessment, and independent decision-making are becoming more important. In this regard, self-development and reflective thinking play a key role in both academic and personal growth.

What does self-development mean for a student?

From a psychological perspective, self-development is closely linked to "intrinsic motivation." Students with intrinsic motivation do not see learning as an obligation, but as a means of personal growth. This makes them more persistent and goal-oriented in the learning process.

Self-development is often misunderstood as simply studying more or getting higher grades. However, in my opinion, it is much broader. Self-development means that a student understands themselves, recognizes their weaknesses, and takes steps to improve them.

In my lessons, I often observe the following: I explain the same topic, but students react differently. Some ask questions when they do not understand something and request additional tasks. Others show no reaction at all. As a result, the first group progresses much faster.

This shows that self-development is not a passive process. A student must be active and show personal interest in learning. This interest develops gradually. The teacher can guide the process, but the main effort must come from the student.

Another advantage of self-development is that it enhances independent learning skills. Today, information is abundant, and students should not rely solely on the teacher. They should also conduct their own research and acquire new knowledge independently.

In addition, self-development increases students' self-confidence. Even small successes can boost their motivation. I have clearly observed this in my classes—students who were initially shy to speak later become more active and express their ideas more confidently.

Reflective thinking – a simple but effective approach

Reflective thinking is actually a very simple process: stopping and thinking. However, this seemingly simple act can be difficult for students because they are not used to it.

Usually, once a lesson ends, students immediately move on to the next topic. However, the most important stage is actually the reflection that happens after the lesson. Students should ask themselves questions such as:

- What did I learn today?
- Where did I struggle?
- What can I do differently next time?

When I started applying this in my lessons, I observed interesting changes. At first, students found it difficult to respond. Some did not want to speak at all. But over time, they became accustomed to the process.

Reflective thinking helps students take control of their own learning. They no longer remain passive listeners but become active learners and thinkers. This creates a significant difference. The relationship between self-development and reflective thinking

Based on my observations, these two concepts are not separate. On the contrary, they complement each other.

Reflection helps students identify their mistakes, while self-development involves taking action to correct them. If a student does not reflect, they do not improve. If they do not try to improve, reflection becomes ineffective.

Therefore, it is important to develop both skills together in the learning process. This makes students more goal-oriented and more responsible for their own development.

The main aim of this research is to identify the factors that influence the development of self-development and reflective thinking skills in students. The following methods were used:

- classroom observation,
- asking students questions,
- applying reflection activities.

Initially, I observed students' behavior in the classroom and noticed that most of them were passive and did not reflect on their own learning.

Then, I made a small change in my lessons. I dedicated five minutes at the end of each lesson to reflection. I asked students simple questions:

- What did you learn today?
- How do you evaluate yourself?
- What could you have done better?

This process continued for several weeks.

Results and Discussion

After some time, noticeable changes appeared. Students became more active, started asking questions, and showed greater interest in the lessons. Most importantly, they began to accept their mistakes. Students who were previously afraid of making mistakes started analyzing and correcting them. Some even began asking for additional tasks, showing their willingness to develop themselves.

The observations and research results demonstrated that self-development and reflective thinking have a direct impact on student progress. These skills make students more responsible, change their approach to learning, and make the learning process more effective.

Recommendations

Based on the research findings, teachers can use the following approaches in the teaching process to develop students' self-development and reflective thinking skills:

- Allocate 3-5 minutes at the end of each lesson for reflection;
- Ask open-ended questions and create opportunities for students to express their thoughts freely;
- Use self-assessment sheets to help students evaluate their own learning process;
- Present mistakes not as punishment, but as opportunities for learning and improvement;
- Create an active learning environment through group work and classroom discussions;

- Encourage students to do additional research and develop independent learning habits.

These approaches not only increase students' interest in lessons but also help them become more responsible, confident, and independent individuals.

Conclusion

The observations and research findings show that student development is not determined solely by knowledge level or intellectual ability. In fact, one of the main reasons for success is how students approach their own learning process. Differences among students who receive the same education often stem from their intrinsic motivation, self-assessment skills, and willingness to improve.

Self-development is not just about studying more or achieving higher grades. It primarily involves self-awareness, recognizing one's weaknesses, and taking steps to overcome them. This process fosters responsibility, makes students more goal-oriented, and reduces their dependence on the teacher. In modern education, this is especially important, as today's students must become independent thinkers and decision-makers in the future.

Reflective thinking is the driving force behind this development. When students critically evaluate their actions, analyze their mistakes, and plan their next steps, learning becomes more conscious and effective. Simple questions like "What did I learn today?" or "What could I have done better?" have a powerful impact on a student's internal growth.

The implementation of reflection activities during the research led to noticeable improvements in students. They became more interested, more active, and more willing to ask questions. Most importantly, they began to accept and correct their mistakes instead of hiding them. This shows that sometimes small but well-directed steps can lead to significant changes in student development.

In my opinion, a teacher's main role is not only to transfer knowledge but also to teach students how to think, understand themselves, and develop continuously. Knowledge may be forgotten over time, but a well-developed way of thinking influences a person throughout their entire life. If a student learns how to develop themselves and reflect, they will succeed not only in school but in all stages of life.

In conclusion, self-development and reflective thinking should be integral parts of modern education. When these skills are developed together, students become more responsible, confident, motivated, and independent. The goal of education is not only to transfer knowledge but to shape личности – and at the center of this process stands a self-aware, reflective, and continuously developing individual.

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THEORETICAL FOUNDATIONS OF INTERACTIVE TEACHING METHODS IN PRIMARY SCHOOL CLASSES

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Abstract: The article comprehensively examines the theoretical and pedagogical foundations of interactive teaching methods in primary school classes. The study traces the historical development of interactive learning from ancient times to the present day and clarifies the role of classical theories by J. Dewey, L.S. Vygotsky, and J. Piaget in shaping this field. The pedagogical and psychological essence of interactive methods has been elaborated in the light of such concepts as the "zone of proximal development", constructivism, motivation theory, and experiential learning. The article also substantiates the organic connection between the psychological characteristics of children aged 6–10 and interactive methods, and demonstrates their positive impact on students' cognitive development, communication skills, and intrinsic motivation, drawing on theoretical sources. Special attention is paid to the contributions of Azerbaijani pedagogical science in this area, including the ideological affinity of the National Curriculum with the interactive approach. The article concludes with concrete recommendations for future research.

Keywords: interactive learning, primary school, pedagogical methods, active learning, student engagement, constructivism, zone of proximal development.

İBTİDAİ SINIFLƏRDƏ İNTERAKTİV TƏLİM METODLARININ NƏZƏRİ ƏSASLARI

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Резюме: В статье комплексно исследуются теоретико-педагогические основы интерактивных методов обучения в начальных классах. В исследовании прослеживается исторический путь развития интерактивного обучения – от древних времён до современности, раскрывается роль классических теорий Дж. Дьюи, Л.С. Выготского и Ж. Пиаже в формировании данной области. Педагогико-психологическая сущность интерактивных методов подробно рассматривается в свете таких концепций, как «зона ближайшего развития», конструктивизм, теория мотивации и обучение на основе опыта. В статье также обосновывается органическая связь между психологическими особенностями детей в возрасте 6-10 лет и интерактивными методами, показывается их

положительное влияние на познавательное развитие учащихся, коммуникативные навыки и внутреннюю мотивацию с опорой на теоретические источники. Особое внимание уделяется вкладу азербайджанской педагогической науки в данную область, в том числе идейной близости Национального куррикулума с интерактивным подходом. Статья завершается конкретными рекомендациями для будущих исследований.

Ключевые слова: интерактивное обучение, начальная школа, педагогические методы, активное обучение, активность учащихся, конструктивизм, зона ближайшего развития.

Introduction

The rapid changes occurring in the modern world have presented new challenges to education. Society no longer demands only graduates focused on the accumulation of knowledge, but individuals who can think and make decisions in complex situations and work effectively with others. Interactive teaching methods serve as an important tool in meeting this demand. In particular, the primary school age – the period from 6 to 10 years – is a decisive stage in the formation of learning motivation, cognitive interest, and collaborative skills (1).

The National Curriculum document introduced in the Republic of Azerbaijan in 2006 fundamentally transformed the nature of the teaching process. That document places the development of student personality and the cultivation of the student as an active, critically thinking citizen at its center. The teacher is no longer the sole source of knowledge – they become a guide who creates conditions for the student to discover, question, and debate. This pedagogical philosophy fully coincides with the essence of interactive methods (1). A number of prominent scholars have contributed to the formation of the theoretical foundations of interactive learning in world pedagogy. The theories of such classics as J. Dewey, L.S. Vygotsky, and J. Piaget laid the scientific groundwork for this field. In Azerbaijani pedagogical thought, A.N. Abbasov, N.M. Kazimov, and M.J. Muradkhanov have conducted consistent research on adapting interactive methods to the local school context. The jointly prepared textbook "Pedagogy. New Course" by A. Pashayev and F. Rustamov is the most authoritative theoretical source of Azerbaijani pedagogy in this area (9).

The development of Azerbaijani pedagogical science in the independence period necessitated special attention to this field. Research proved that the application of interactive methods has a positive effect not only on academic achievements, but also on students' behavioral culture, communication skills, and creative potential. It has been particularly emphasized that the application of these methods in primary classes helps form a positive attitude from an early age (9).

The term "interactive" derives from the Latin roots "inter" (mutual) and "activus" (activity). In pedagogical science, this concept refers to a teaching process based on multidirectional, dynamic interaction among teacher, student, and learning environment. Unlike the traditional model, here knowledge is acquired not as a one-way flow, but through joint search, discussion, and discovery. The student bears the status of an active participant in the process, not a passive recipient (2, p. 45). There are several key features that distinguish interactive methods from traditional approaches: the mutual activity of learners, the emphasis on joint problem-solving, the student's ability to control their own learning trajectory, and the grounding of teacher-student relations in equal-rights dialogue. These features together constitute the most favorable pedagogical basis for forming the competencies required by the 21st century – critical thinking, communication, creativity, and collaboration (7, p. 2).

The intellectual roots of interactive learning extend far back in time. In ancient Greece, Socrates guided his students not by providing ready answers, but by leading them toward self-understanding through successive questions – a style that has become an inseparable part of pedagogical methods today under the name "Socratic dialogue". Swiss pedagogue J.H. Pestalozzi in the 19th century proposed an active learning model that simultaneously developed the child's hand, heart, and mind. At the beginning of the 20th century, American philosopher J. Dewey placed students' practical activity at the center, viewing the school not as an institution disconnected from life, but as democratic society itself.

From the middle of the 20th century onward, the empirical study of interactive learning intensified. Research conducted under the leadership of D. and R. Johnson statistically demonstrated that group collaboration has a stronger effect on student achievement than individual work. In 1991, Bonwell and Eison systematically studied active learning and found that when students become active participants in the learning process, they internalize knowledge more deeply and retain it in memory for a longer time (7, p. 2).

Materials and Methods

Today, various forms of interactive methods are widely applied in pedagogical practice: cooperation in small groups, role-plays, brainstorming, debates, project-based learning, case studies, and so on. The common reference point of all these methods is the activation of the student's own thought process and live dialogue with peers and the teacher. Each method can be adapted for different age groups and subject areas.

The most powerful of the psychological foundations of interactive learning is L.S. Vygotsky's theory of development. The scholar discovered that there exists a gap – the "zone of proximal development" – between what a child can do independently and what they can do with help. It is precisely in this zone that properly organized joint activity ensures the most rapid development. Interactive methods are the most suitable means for pedagogically filling this gap, since the student receives support both from the teacher and from more capable peers (5, p. 86).

This theory takes on special practical significance when applied to primary school students. The period of 6–10 years is the stage at which cognitive development proceeds most intensively; the correct pedagogical support provided at this age determines the trajectory of subsequent development. The principle of "gradual reduction of assistance" arising from Vygotsky's theory becomes the core mechanism of interactive group work: the teacher guides more at first, while students gradually become independent. In this process, the student not only acquires knowledge but also learns how to learn (5).

J. Piaget's concept of cognitive development also makes a significant contribution to the psychological justification of interactive methods. According to Piaget's constructivism, a child does not receive knowledge passively – they develop by constructing their own mental models of the world. Primary school students are in the stage of concrete operations: for them, visual-practical activity is more fundamental than abstract thinking. Interactive methods are built precisely on this principle – the child internalizes more deeply what they see, touch, and discuss (4, p. 27).

Piaget's assimilation-accommodation mechanism is also used in the didactic explanation of interactive methods. During group discussion, a student encounters ideas that contradict their existing knowledge schemas; in order to resolve this contradiction, they are compelled to reconstruct their thought model. This process leads them to a deeper understanding. Thus,

interactive activity ensures not merely the transfer of knowledge, but the qualitative renewal of cognition (4).

From the perspective of motivational psychology, the advantages of interactive methods are also clearly visible. According to A. Maslow's hierarchy of needs, a person turns toward needs of social belonging, recognition, and self-actualization only after physiological needs are satisfied. A student working in a group feels that they belong to the class community and that their contribution is valued by others. This significantly strengthens the intrinsic motivation toward learning.

D. Kolb's "Experiential Learning" model also allows explanation of the pedagogical effectiveness of the interactive approach. According to Kolb, complete learning passes through four successive stages: direct experience, reflection, theoretical generalization, and testing in new situations. Since interactive methods encompass all stages of this cycle, they enable more complete and lasting learning. For primary school children, the first two stages of this cycle – concrete experience and reflection – are especially important.

Discussion

Primary school age is a serious transitional stage in the child's life: play activity gradually cedes its leading position to learning activity. However, this transition is not instantaneous – play elements continue to preserve their significance in the child's world for some time. Interactive methods that take this characteristic into account – role-plays, gamified tasks, simulations – give students the opportunity to discover new knowledge in a familiar form, thereby raising both motivation and the quality of internalization (4).

During this age period, the student's social world also expands rapidly. Relations with peers, norms of joint activity, empathy, and the ability to compromise are laid down precisely in the primary school years. Interactive methods – especially work in pairs and groups of 3-5 – provide a pedagogical framework for this social development. Research shows that students who grow up in such environments not only learn better – they also learn to communicate better, resolve conflicts, and demonstrate leadership.

The theoretical research conducted once again confirmed that the application of interactive teaching methods in primary classes is not a coincidental choice, but a psychological, pedagogical, and social necessity. Vygotsky's zone of development, Piaget's constructivism, Maslow's motivation theory, Kolb's learning cycle – each of these theories justifies the effectiveness of the interactive approach from its own perspective. Azerbaijani pedagogical science also supports this direction (9).

Nevertheless, a number of conditions are essential for the successful application of interactive methods in primary school. The teacher's methodological preparedness, the arrangement of the classroom environment in a way that allows for collaboration, parental support, and the attitude of school management toward this all serve as decisive factors. As Pashayev and Rustamov emphasize, interactive methods must be applied not formally, but on the basis of the internal logic of the pedagogical process, in a gradual manner (9).

Conclusion

An important direction for future research is the experimental testing of interactive methods in specific subject areas in primary classes – in native language, mathematics, and life studies. For this purpose, strengthening teachers' professional preparation in interactive methodology, restructuring classrooms to be suitable for collaboration, and integrating digital interactive tools into instruction are recommended as priority steps.

In future research, the experimental study of the application possibilities of these methods in specific subject areas (mathematics, native language, life studies) stands out as an important direction. For this purpose, strengthening teachers' professional preparation in interactive methods and modernizing school infrastructure are recommended.

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HİBRİD TƏHSİL MODELİNDƏ MÜƏLLİMLƏR ÜÇÜN DAVAMLI PEŞƏKAR İNKİŞAF STRATEGİYALARI

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Xülasə. Müasir təhsil sistemində rəqəmsal transformasiya proseslərinin sürətlənməsi pedaqoji fəaliyyətin təşkili və idarə olunmasında yeni yanaşmaların formalaşmasına ciddi təsir göstərmişdir. Bu dəyişikliklərin ən mühüm nəticələrindən biri hibrid təhsil modelinin geniş tətbiqidir. Hibrid təhsil modeli ənənəvi sinif tədrisi ilə rəqəmsal öyrənmə mühitini integrasiya edərək təhsil prosesinin daha çevik, əlçatan, fərdiləşdirilmiş və interaktiv şəkildə həyata keçirilməsinə imkan yaradır. Bu yanaşma həm tədrisin keyfiyyətini yüksəldir, həm də tələbələrin fərdi öyrənmə tempinə uyğunlaşmanı təmin edir. Nəticədə təhsil prosesi daha effektiv və nəticəyönlü xarakter alır. Bu model müəllimlərin peşəkar rolunu dəyişdirərək onları yalnız bilik ötürücüsü deyil, eyni zamanda öyrənmə prosesinin dizayneri, fasilitatoru, mentor və idarəedicisi kimi fəaliyyət göstərməyə yönəldir.

Məqalədə hibrid təhsil mühitində müəllimlərin davamlı peşəkar inkişaf strategiyaları geniş və sistemli şəkildə araşdırılır. Xüsusilə reflektiv pedaqoji yanaşmanın müəllim səriştələrinin formalaşması və inkişafındakı rolu, rəqəmsal kompetensiyaların məqsədyönlü şəkildə gücləndirilməsi, əməkdaşlığa əsaslanan

öyrənmə mühitlərinin yaradılması və peşəkar öyrənmə icmalarının müəllimlərin inkişafına göstərdiyi təsir nəzəri və analitik baxımdan təhlil edilir.

Tədqiqat keyfiyyət yanaşmasına əsaslanır və beynəlxalq elmi ədəbiyyatın sistemli icmalı, müqayisəli təhlil və interpretativ metodlar vasitəsilə həyata keçirilmişdir.

Nəticələr göstərir ki, davamlı peşəkar inkişaf strategiyalarının sistemli tətbiqi müəllimlərin pedaqoji və texnoloji bacarıqlarını əhəmiyyətli dərəcədə gücləndirir, tədris prosesinin keyfiyyətini yüksəldir və şagirdlərin öyrənmə nəticələrinə müsbət təsir göstərir.

Açar sözlər: hibrid təhsil, davamlı peşəkar inkişaf, müəllim refleksiya, rəqəmsal bacarıqlar, tədris strategiyaları, peşəkar öyrənmə icmaları

CONTINUOUS PROFESSIONAL DEVELOPMENT STRATEGIES FOR TEACHERS IN A HYBRID EDUCATION MODEL

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Abstract: The acceleration of digital transformation processes in the modern education system has significantly influenced the emergence of new approaches in the organization and management of pedagogical activities. One of the most important outcomes of these changes is the widespread adoption of the blended learning model. The blended learning model integrates traditional classroom instruction with digital learning environments, enabling the educational process to be implemented in a more flexible, accessible, personalized, and interactive manner. This approach not only enhances the quality of teaching but also ensures adaptation to students' individual learning pace. As a result, the learning process becomes more effective and outcome-oriented. This model transforms the professional role of teachers, positioning them not only as knowledge transmitters but also as designers, facilitators, mentors, and managers of the learning process.

The article extensively and systematically examines the continuous professional development strategies of teachers in a blended learning environment. In particular, it analyzes the role of reflective pedagogical practice in the formation and development of teachers' competencies, the purposeful enhancement of digital skills, the creation of collaborative learning environments, and the impact of professional learning communities (PLCs) on teacher development from both theoretical and analytical perspectives.

The research is based on a qualitative approach and is conducted through a systematic review of international academic literature, comparative analysis, and interpretative methods.

The results indicate that the systematic implementation of continuous professional development strategies significantly strengthens teachers' pedagogical and technological competencies, improves the quality of the teaching process, and positively affects students' learning outcomes.

Key words: blended learning, continuous professional development, teacher reflection, digital skills, teaching strategies, professional learning communities

Changes occurring in the education system in the modern era are closely linked to global digitalization processes. The rapid development of information and communication technologies (ICT) has not only modernized instructional tools but has also fundamentally transformed the philosophical foundations of education, pedagogical approaches, and organizational management mechanisms. While traditional education models were long based on knowledge transmission and

teacher-centered instruction, contemporary educational paradigms emphasize active learner participation, constructivist learning processes, and the integration of technology.

As a result of this transformation, the hybrid education model has gained particular relevance. The hybrid education model is based on the purposeful and systematic integration of traditional classroom instruction with online learning environments. This approach is not merely a technical innovation but also represents a structural transformation of the pedagogical system. The hybrid model enables education to become more flexible, accessible, personalized, and efficient.

In this context, the role of teachers undergoes a fundamental transformation. Teachers are no longer simply transmitters of knowledge; rather, they act as professional facilitators who design, manage, and guide the learning process. This shift creates new demands on teachers' professional competencies and necessitates their continuous development. Thus, teacher professional development is no longer viewed as a static or short-term activity but as a continuous, reflective, and adaptive process.

2. Theoretical Foundations of the Hybrid Education Model

The hybrid education model has emerged through the integration of various pedagogical and psychological theories. Its primary theoretical foundations include constructivism, social learning theory, and the technological-pedagogical integration (TPACK) framework.

According to the constructivist approach, learning is not the passive acquisition of knowledge but an active process of knowledge construction. In a hybrid environment, learners construct knowledge through interactive activities both in physical classrooms and on digital platforms. This ensures deeper, more meaningful, and long-term learning outcomes.

Social learning theory emphasizes that learning is enhanced through social interaction. In this regard, the hybrid model expands collaboration among students both in face-to-face settings and in online environments.

Garrison and Kanuka (2004) describe hybrid learning as an "integrative form with transformative potential," emphasizing that it leads to fundamental changes in educational culture. Graham (2013) highlights the flexibility of hybrid learning and notes that it creates extensive opportunities for personalized learning.

3. Transformation of the Teacher's Role

The hybrid education model makes the teacher's professional role more complex and multifaceted. In traditional models, the teacher's primary function was knowledge transmission; however, in a hybrid environment, this function expands into multiple dimensions.

Teachers perform the following roles:

- designer and planner of the learning process
- pedagogical facilitator and guide
- curator and regulator of digital resources
- monitor and analyst of student performance
- developer of personalized learning strategies
- integrator of educational technologies

This transformation requires teachers to possess not only advanced pedagogical thinking but also digital literacy and analytical skills. The TPACK framework proposed by Mishra and Koehler (2006) provides a theoretical basis for this integration, demonstrating that effective teaching requires a balanced combination of technology, pedagogy, and content knowledge.

4. Conceptual Foundations of Continuous Professional Development

Continuous professional development (CPD) refers to the ongoing process of updating teachers' knowledge, skills, and competencies. In modern education systems, CPD is no longer an additional activity but a core component of teachers' professional identity.

Desimone (2009) argues that effective professional development should be long-term, practice-oriented, and integrated into teachers' daily instructional activities. According to this perspective, theoretical knowledge alone is insufficient; teachers must acquire skills that can be applied in real teaching contexts.

Schön (1983) identifies reflective practice as a key mechanism of professional development. According to his model, teachers learn not only during action but also through reflection after action. This reflective cycle enables the transformation of experience into professional knowledge.

5. Professional Development Strategies in a Hybrid Environment

5.1. Deepening Reflective Practice

Reflective practice involves the systematic analysis of a teacher's pedagogical activities and the improvement of teaching based on this analysis. In a hybrid environment, this process encompasses both classroom and digital teaching contexts.

Teachers systematically reflect on questions such as:

- Which teaching method was most effective?
- How active were students in the digital environment?
- What challenges emerged and how were they resolved?
- Which technologies produced better results?

This process enables teachers to learn systematically from their own experience.

5.2. Development of Digital Competencies

Digital competencies are a fundamental component of hybrid education. Teachers must not only use technology but also integrate it effectively into pedagogical practices.

Key competencies include:

- effective management of LMS platforms
- application of online assessment systems
- use of interactive teaching tools
- development of multimedia content
- adherence to digital safety and ethical principles

These skills enhance instructional effectiveness and positively influence student motivation.

5.3. Professional Learning Communities

DuFour (2004) defines professional learning communities as structured environments in which teachers collaborate, share knowledge, and learn collectively.

These communities:

- strengthen the exchange of experience
- promote collaborative lesson planning
- facilitate the dissemination of innovative pedagogical approaches
- support collective problem-solving

5.4. Microlearning and Flexible Development

Microlearning is based on short, focused, and goal-oriented learning modules. This approach enables teachers to quickly acquire new skills and is particularly effective in rapidly changing technological environments.

5.5. Action Research

Action research involves teachers identifying real problems in their own teaching practice and addressing them using scientific methods. This approach transforms teachers into practitioner-researchers and bridges the gap between theory and practice.

5.6. Institutional Support Systems

Teacher professional development becomes more effective when supported at the institutional level. Educational institutions should provide:

- continuous professional development programs
- mentoring systems
- digital infrastructure
- a motivating professional environment

The hybrid education model represents a transformative approach in modern education systems. It expands teachers' professional roles and necessitates their continuous development.

Research indicates that reflective practice, digital competencies, professional learning communities, and institutional support are key strategic components of teacher professional development. The integration of these elements significantly enhances the quality of instruction and ensures the effectiveness of hybrid education.

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ACTIVE LEARNING MODELS AND THEIR ROLE IN MODERN PEDAGOGY

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Abstract: The article provides a comprehensive analysis of the role of active learning models in modern pedagogy and their theoretical-pedagogical foundations. This constructivist approach places students at the center of the learning process, aiming to develop their independent thinking, collaboration, critical analysis, and problem-solving skills. The implementation of models such as project-based learning, problem-based learning, the flipped classroom, and blended learning demonstrates that active learning leads to higher levels of knowledge acquisition and the formation of more functional competencies compared to traditional teaching methods. The integration of digital technologies further enhances the effectiveness of these models by making the learning process more flexible and interactive.

The article also comparatively examines the experiences of countries such as the United States, Finland, Singapore, Canada, and Australia. The findings show that in these countries, active learning is developed based on different strategic priorities, including digitalization, interdisciplinary integration, inclusivity, and alignment with state educational policies. A common feature across all models is that the teacher acts as a facilitator, while the student becomes an active constructor of knowledge. Overall, active learning is identified as one of the key directions of modern education, playing a crucial role in lifelong learning and the development of 21st-century skills.

Keywords: active learning, constructivism, modern pedagogy, digital technologies, competencies.

FƏAL ÖYRƏNMƏ MODELƏRİ VƏ MÜASİR PEDAQOGİKADA ONLARIN ROLU

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Xülasə: Məqalədə müasir pedaqogikada fəal öyrənmə modellərinin rolu və onların nəzəri-pedaqoji əsasları geniş şəkildə təhlil olunur. Konstruktivizmə əsaslanan bu yanaşma şagirdləri təlim prosesinin mərkəzinə gətirərək onların müstəqil düşünmə, əməkdaşlıq, tənqidi təhlil və problem həll etmə bacarıqlarının inkişafına yönəlikdir. Layihəəsaslı, probleməsaslı, “çevrilmiş sinif” və qarışıq öyrənmə kimi modellərin tətbiqi göstərir ki, fəal öyrənmə ənənəvi tədrisə nisbətən daha yüksək mənimləmə səviyyəsi və daha funksional bacarıqlar formalaşdırır. Rəqəmsal texnologiyaların integrasiyası isə bu modellərin səmərəliliyini daha da artıraraq öyrənmə prosesini çevik və interaktiv edir. Məqalədə həmçinin ABŞ, Finlandiya, Sinqapur, Kanada və Avstraliya kimi ölkələrin təcrübələri müqayisəli şəkildə araşdırılır. Nəticələr göstərir ki, bu ölkələrdə fəal öyrənmə müxtəlif strateji prioritetlər əsasında (rəqəmsallaşma, fənlərarası integrasiya, inklüzivlik və dövlət siyasəti ilə əlaqə) inkişaf etdirilir. Bütün modellərdə ortaq cəhət ondan ibarətdir ki, müəllim fasilitator rolunu oynayır, şagird isə biliklərin aktiv yaradıcısına çevrilir. Ümumilikdə, fəal öyrənmə müasir təhsilin əsas istiqamətlərindən biri kimi ömürboyu öyrənmə və XXI əsr bacarıqlarının formalaşdırılmasında mühüm rol oynayır.

Açar sözlər: fəal öyrənmə, konstruktivizm, müasir pedaqogika, rəqəmsal texnologiyalar, kompetensiyalar.

Introduction

The modern education system, under the influence of rapidly changing social, economic, and technological realities, requires the implementation of new approaches. In this context, traditional teacher-centered and passive knowledge transmission models are gradually being replaced by active learning approaches. Active learning not only ensures the direct participation of students in the learning process but also creates conditions for the development of their independent thinking, problem-solving, collaboration, and critical analysis skills.

The theoretical foundations of active learning are based on constructivist pedagogical approaches, particularly the ideas of scholars such as John Dewey, Lev Vygotsky, and Jean Piaget. However, in the 21st century, the digitalization of education, the abundance of information, and the changing demands of the labor market have led to the further improvement and enrichment of these approaches with new instructional models. Methods such as project-based learning, problem-based learning, the flipped classroom, and blended learning represent the most widely used forms of active learning.

In this regard, the experience of various countries including the United States, Finland, Singapore, Canada, and Australia is of particular importance in terms of implementing active learning. Each of these countries develops this model in accordance with its own pedagogical traditions, social needs, and strategic educational goals.

The aim of the article is to analyze the theoretical foundations of active learning models, compare their implementation features in different countries, and evaluate the role of these approaches in modern education systems.

Materials and Methods

Modern pedagogy is increasingly based on active learning models that serve as an alternative to traditional, passive knowledge acquisition. These models emphasize students' active participation in the learning process, as well as the development of their independence, self-regulation skills, critical thinking, and collaboration abilities. The pedagogical foundations of active learning are rooted in the constructivist theories of John Dewey, Lev Vygotsky, and Jean Piaget; however, in recent years, significant evolution has taken place due to the digitalization of education and changing social demands on school and university graduates.

One of the core pedagogical principles of active learning is the "learner-centered" approach, where students act not as objects but as subjects of the educational process. Shifting attention from the teacher to the student leads not only to deeper comprehension of learning materials but also to the development of autonomous learning skills (Weimer, 2019). Active learning encompasses a broad spectrum of methods, including collaborative learning, brainstorming, case-based learning, debates, simulations, inquiry-based learning, and project-based learning as one of the most integrative and effective approaches.

According to a meta-analysis, active learning methods increase knowledge retention by 33% compared to traditional lecture-based instruction (Freeman et al., 2020). These methods are particularly effective in humanities and language education, where not only knowledge acquisition but also the ability to apply it in communicative and interpretive contexts is essential. In language and literature teaching, active learning is implemented through group text interpretation, creative project development, discussions, and multimodal presentations.

In recent years, particular attention has been given to the integration of active learning models with digital technologies. Blended learning has become increasingly significant in modern education (Garrison & Vaughan, 2021). This model combines online and face-to-face learning activities, allowing students to actively participate in shaping their own learning pathways. Such an approach is especially effective in organizing project-based activities, as it enables the use of online tools for research, interaction, and presentation of outcomes.

In addition, problem-based learning (PBL) has also been gaining importance. Similar to project-based learning, this model involves students' active engagement in solving open-ended, ill-structured problems. However, unlike project-based learning, the emphasis in PBL is not primarily on the final product but on the cognitive process itself—formulating questions, analyzing hypotheses, and constructing arguments. This approach is highly effective in developing critical thinking and scientific literacy (Hung, 2022).

One of the increasingly popular models in recent years is the "flipped classroom." In this model, theoretical content is delivered outside the classroom (e.g., through video lectures at home), while classroom time is dedicated to active learning activities such as discussions, analysis, and

collaborative problem-solving. This approach supports deeper conceptual understanding and the development of analytical skills (Lo & Hew, 2020).

At the same time, active learning models require a transformation in the role of the teacher. The teacher is no longer a transmitter of information but rather a facilitator, tutor, and coordinator of the learning process. Their main responsibility is to create conditions in which students can independently and collaboratively construct knowledge, reflect on their experiences, and draw conclusions. In this context, assessment practices also gain importance: active learning shifts the focus from summative evaluation toward formative assessment, feedback, self-assessment, and peer assessment (Andrade & Brookhart, 2021).

Thus, the pedagogical foundations of active learning are based on the ideas of constructivism, collaboration, and learner-centered education. Modern active learning models—such as project-based learning, problem-based learning, flipped classroom, and blended learning—facilitate effective engagement of students in cognitive processes, the development of essential 21st-century skills, and the formation of sustainable learning motivation. Their successful implementation requires teachers' methodological competence, a flexible learning environment, and an assessment system that prioritizes student development over outcomes.

Educational systems in many countries are increasingly shifting from traditional instruction based on knowledge transmission to active learning approaches that ensure learners' active participation in cognitive processes and the development of transversal competencies. Active learning integrates various methods and strategies in which students do not merely listen and memorize but instead analyze, discuss, research, create, and apply knowledge in practice. The pedagogical foundations of these models are grounded in the constructivist theories of Lev Vygotsky, Jean Piaget, and John Dewey; however, their modern interpretations have been significantly expanded through digital technologies and interdisciplinary approaches.

In the United States, active learning has become a core component of educational policy in universities and schools. Research conducted by Freeman et al. (2020) shows that the use of active methods such as discussions, group work, project-based tasks, and problem-based learning increases student engagement and academic performance by 30–40% compared to traditional lecture-based formats (Freeman et al., 2020). The American experience is based on learner-centered approaches and educational partnership principles, where the teacher no longer acts as the primary source of information but rather as a facilitator of the learning process. Universities such as Stanford, MIT, and Harvard actively implement active learning classrooms and flipped classroom models, which are equipped with interactive technologies and digital platforms designed for group collaboration (Garrison & Vaughan, 2021). This approach not only enhances student engagement but also fosters critical thinking, digital literacy, and independent learning skills.

In Finland, the active learning model is systematically implemented as a core direction of national education policy and stands out among global education systems for its uniqueness. In particular, phenomenon-based learning is considered one of the most innovative components of Finnish education. Within this model, students do not learn fragmented subject-based knowledge but instead explore real-world, complex problems through interdisciplinary integration. As a result, knowledge from different fields is combined into a unified context, creating a more practical and application-oriented learning environment. For example, the topic of "sustainable development" is analyzed from ecological, economic, literary, social, and technological perspectives, enabling students to gain a deeper and more comprehensive understanding.

Niemi and Harju (2021) note that in Finland, active learning is closely linked with collaborative learning models and the integration of digital technologies. This approach enables students to work effectively in groups, share ideas, make joint decisions, and produce digital outputs. One of the key features of the Finnish educational system is that the student acts as an independent subject of the learning process; learners define their own learning goals, assess their progress, and take responsibility for their development. In this context, the teacher shifts from the role of knowledge transmitter to that of a facilitator, mentor, and coordinator of the learning process (Niemi & Harju, 2021).

In Singapore, the active learning model is systematically developed within the framework of the national “21st Century Competencies” programme and represents one of the strategic priorities of educational policy. This model places strong emphasis on developing students’ independent thinking, creativity, problem-solving, and critical analysis skills. According to Tan and Koh (2021), Singaporean schools widely implement Problem-Based Learning, Inquiry-Based Learning, and collaborative learning models, which ensure that the learning process is more practical, goal-oriented, and connected to real-life contexts. One of the key features of the Singaporean education system is the close integration of active learning with formative assessment (assessment for learning). Within this approach, students regularly receive feedback from teachers and peers, analyze their learning outcomes, and continuously improve their learning strategies. At the same time, the Ministry of Education of Singapore actively promotes the use of digital platforms and collaborative design tools, making the educational process more interactive, flexible, and innovative. As a result, the Singapore model stands as one of the leading education systems where active learning is implemented at a high level both theoretically and practically.

In Canada, active learning is closely associated with the values of inclusivity and cultural diversity. Here, active learning models are directed not only at cognitive development but also at social and emotional learning. According to Friesen and Jacobsen (2019), Canadian schools actively implement project-based and service-learning models by engaging students in socially beneficial initiatives such as environmental, cultural, and volunteer projects. This approach enables students to apply knowledge in real contexts, collaborate with local communities, and develop civic engagement skills. Canadian teachers also use reflective pedagogy methods aimed at enhancing self-awareness and improving their own teaching practices (Friesen & Jacobsen, 2019).

The Australian education system is recognized as one of the advanced educational models where interactive and digital forms of active learning are systematically implemented. Within the national curriculum standards, innovative approaches such as blended learning and flipped classroom models are widely applied, enabling more flexible, personalized, and outcome-oriented learning processes. According to Mutch-Jones et al. (2022), Australian schools extensively use digital platforms such as Google Classroom, Miro, and Canva, as well as immersive technologies including VR and AR, to develop students’ collaborative research skills and creative abilities. These technologies ensure that students not only receive information but also analyze, visualize, and apply it in real-life situations. In the Australian model, active learning particularly supports the development of digital citizenship, inquiry-based thinking, and innovative design competencies, which are essential in the modern digital economy. In this process, the role of teachers has also changed; they are no longer mere transmitters of knowledge but act as curators, mentors, and facilitators who support students in effectively using technologies and integrating them into interdisciplinary projects (Mutch-Jones et al., 2022).

A comparative analysis of different countries' educational practices shows that active learning models are an integral and universal component of modern education, and their effectiveness reaches its highest level only when appropriate pedagogical principles are followed. These principles include learner autonomy, collaborative learning, reflective practice, problem-oriented approaches, and connecting learning with real-life contexts. In the United States, the main focus is on technological support and personalization of education, while Finland emphasizes interdisciplinary integration and collaborative research activities. The Singapore model is primarily based on aligning education with national strategy and economic development goals, whereas Canada prioritizes inclusive education and social development components. The Australian education system is distinguished by the strong integration of digital technologies into teaching and is considered one of the leading countries in implementing innovative educational approaches. Despite these differences, a common conclusion across all countries is that active learning is recognized as a key pedagogical tool that contributes to the formation of students' personality, the development of critical and creative thinking skills, and the cultivation of lifelong learning competencies.

Results and Discussion

From a discussion perspective, it should be noted that the effective implementation of active learning depends on several factors, including teachers' methodological competence, the flexibility of the learning environment, technological infrastructure, and an appropriate assessment system. The transition from traditional assessment approaches to formative and reflective assessment plays a crucial role in the successful application of this model.

At the same time, certain challenges also exist. These include teachers' difficulties in adapting to new methodologies, lack of resources, and the strong influence of traditional approaches in some educational systems. Nevertheless, active learning is regarded as an irreplaceable approach for improving the quality of education and developing 21st-century skills.

Conclusion

The conducted analysis shows that active learning represents one of the most important directions of modern pedagogical thinking and, unlike the traditional teaching model, ensures the student's active participation as a subject of the learning process. Based on constructivist approaches, this model prioritizes not the direct transmission of knowledge but rather its independent construction and application by learners. As a result, it leads to deeper understanding, increased learning sustainability, and the development of functional competencies.

The research findings indicate that active learning models such as project-based learning, problem-based learning, flipped classroom, and blended learning have a positive impact not only on students' academic achievement but also on the development of critical thinking, collaboration, and communication skills. In particular, the integration of digital technologies into the educational process increases the effectiveness of active learning and expands opportunities for personalized learning.

A comparison of international practices shows that in the United States, technology integration and personalized learning, in Finland, interdisciplinary integration and student-centered approaches, in Singapore, strategic planning and outcome-oriented education, in Canada, inclusivity and socially oriented learning, and in Australia, digital innovation define the main directions of active learning development. Despite these different models, the common goal across all systems is to ensure students' active participation in the learning process.

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CONTINUOUS PROFESSIONAL DEVELOPMENT AND TEACHER REFLECTION

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Abstract: Continuous professional development has become one of the most important conditions for improving the quality of education in the 21st century. In modern education systems, teachers are expected not only to possess subject knowledge but also to adapt to curricular reforms, digital technologies, inclusive education, student-centered learning, and changing societal expectations. In this context, teacher reflection plays a significant role in transforming professional development from a formal requirement into a meaningful process of professional growth. This article examines the relationship between continuous professional development and teacher reflection from theoretical, methodological, and practical perspectives. The study is based on a qualitative review of scientific literature, international reports, and conceptual approaches related to reflective practice, teacher education, and professional development. The article argues that effective professional development is continuous, collaborative, practice-oriented, and supported by systematic reflection. Teacher reflection allows educators to analyze their classroom decisions, evaluate student learning outcomes, identify professional challenges, and redesign pedagogical strategies. The results show that reflective practice strengthens teacher autonomy, improves the quality of teaching, supports lifelong learning, and fosters professional identity. However, developing a reflective culture requires institutional support, mentoring, time for professional dialogue, constructive feedback, and school leadership that values inquiry and self-assessment. The article concludes that continuous professional development and teacher reflection should be viewed as interrelated components of teacher professionalism.

Keywords: development, reflection, teacher, professionalism, learning

Xülasə: Davamlı peşəkar inkişaf 21-ci əsrdə təhsilin keyfiyyətinin artırılması üçün əsas tələbə çevrilib. Müasir təhsil sistemlərində müəllimlərdən yalnız fənn biliklərinə sahib olmaq deyil, həm də tədris planı islahatlarına, rəqəmsal texnologiyalara, inklüziv təhsilə, şagird mərkəzli təlimə və dəyişən sosial gözləntilərə uyğunlaşmaq tələb olunur. Bu kontekstdə müəllim refleksi peşəkar inkişafın formal tələbdən mənalı və davamlı inkişaf prosesinə çevrilməsində mühüm rol oynayır. Bu məqalədə davamlı peşəkar inkişaf və müəllim refleksi arasındakı əlaqə nəzəri, metodoloji və praktiki baxımdan təhlil edilir. Tədqiqat elmi ədəbiyyatın, beynəlxalq hesabatların və refleksiv təcrübə, müəllim təhsili və peşəkar inkişaf ilə bağlı konseptual yanaşmaların keyfiyyət təhlilinə əsaslanır. Məqalədə iddia edilir ki, effektiv peşəkar inkişaf sistemli reflekslə dəstəklənən davamlı, əməkdaşlıq, təcrübəyə yönəlmiş bir prosesdir. Müəllim refleksi onlara sinifdə verilən qərarları təhlil etməyə, şagird nailiyyətlərini qiymətləndirməyə, peşəkar çətinlikləri müəyyən etməyə və tədris strategiyalarını yenidən hazırlamağa imkan verir. Nəticələr göstərir ki, refleksiv təcrübə müəllim muxtariyyətini gücləndirir, tədris keyfiyyətini yaxşılaşdırır, ömürlük öyrənməni dəstəkləyir və peşəkar kimliyi inkişaf etdirir. Lakin refleksiv mədəniyyətin inkişaf etdirilməsi institusional dəstək, mentorluq, peşəkar dialoq üçün vaxt, konstruktiv rəy və özünüqiymətləndirməni dəstəkləyən məktəb rəhbərliyi tələb edir.

Açar sözlər: inkişaf, düşüncə, müəllim, peşəkarlıq, öyrənmə

Introduction

The professional role of teachers in modern education has become more complex than ever. Rapid social change, technological advances, curriculum reforms, inclusive education policies, and rising expectations for student achievement require teachers to continually update their knowledge and skills. Traditional teacher preparation models, which are largely limited to pre-university education or periodic seminars, are no longer sufficient to ensure consistent teaching quality. Instead, teachers require continuous professional development directly linked to classroom practice, student needs, and school improvement.

Continuous professional development refers to an ongoing process through which teachers develop their professional knowledge, pedagogical competence, classroom management skills, assessment literacy, and reflective capacity. It is not a one-time training event, but a lifelong process of professional development. Research on teacher professional development suggests that effective professional learning is typically content-focused, participatory, collaborative, ongoing, and connected to teachers' real-world classroom experiences (Darling-Hammond et al., 2017). In this sense, professional development should be understood as a systematic process that influences not only teachers' knowledge but also how they teach and how students learn.

Teacher reflection is one of the most important mechanisms that gives meaning to professional development. Reflection allows teachers to critically reflect on their teaching practices, analyze the reasons for their decisions, and understand the impact of their actions on students. The idea of reflective thinking has strong philosophical roots in the work of Dewey, who viewed reflection as an active and purposeful form of thinking aimed at problem-solving and inquiry (Dewey, 1933). Later, Schön developed the concept of the reflective practitioner and explained how professionals reflect during and after action in uncertain practical situations (Schön, 1983). The relevance of this topic stems from the fact that many professional development activities remain formal and ineffective unless supported by reflection. Teachers may attend seminars, workshops, or conferences, but if they do not analyze how new knowledge relates to classroom experience, real

professional change may not occur. Thus, the relationship between continuous professional development and teacher reflection is a key issue in contemporary pedagogy.

The purpose of this article is to analyze the role of teacher reflection in continuous professional development and explain how reflective practice contributes to teacher professionalism. The article focuses on three main questions: What is the essence of continuous professional development? How does teacher reflection contribute to professional development? What are the conditions necessary for the development of a reflective professional culture in schools?

Materials and Methods

This article is based on a qualitative literature review. The study analyzes theoretical works, research articles, and international reports on continuous professional development, teacher reflection, and professional preparation. Key conceptual sources include the works of Dewey, Schön, Brookfield, Korthagen, Farrell, Desimone, Avalos, Guskey, and Darling-Hammond. Additionally, OECD TALIS reports are used to provide an international overview of teacher professional development and collaboration.

The article's methodological approach is descriptive and analytical. First, the concept of continuous professional development is explained in the context of teacher preparation. Second, the theoretical foundations of reflective practice are explored. Third, the relationship between professional development and reflection is discussed in the context of classroom practice, teacher autonomy, collaboration, and school improvement.

This article does not present statistical fieldwork. Instead, it presents a conceptual and theoretical analysis of the existing literature. This approach is appropriate, as the purpose of the study is to clarify the relationship between two pedagogical concepts and identify practical implications for teachers and educational institutions. As a research method, a literature review allows the researcher to compare different theoretical positions, identify common patterns, and gain a broader understanding of the problem under study (Avalos, 2011).

Conclusions and Discussion

The Nature of Continuing Professional Development

Continuing professional development is a planned and systematic process aimed at enhancing the professional competence of teachers throughout their careers. It includes formal, informal, and mediated learning. Formal professional development can include certified courses, postgraduate education, conferences, and institutional training programs. Mediated development can include workshops, group training, mentoring, professional communities, and subject networks. Mediated professional development occurs through daily classroom experiences, reading, self-study, collaboration with colleagues, and reflection on practice.

A key difference between traditional on-the-job training and continuing professional development is that continuing professional development is not limited to short-term knowledge transfer. It is a long-term process aimed at professional transformation. Desimone (2009) argues that effective professional development should be measured not only by participation, but also by changes in teacher knowledge, classroom practices, and student learning outcomes. This means that professional development is only meaningful when it impacts learning itself.

The need for continuous professional development in today's education systems is increasing for several reasons. First, knowledge is changing rapidly, requiring teachers to update both their subject knowledge and their pedagogical methods. Second, digital technologies have

reshaped the structure of teaching and learning. Teachers are expected to use online platforms, digital resources, interactive tools and data-driven assessment methods. Third, classrooms are becoming increasingly diverse. Teachers are required to have skills related to inclusive education, differentiated learning, intercultural communication and socio-emotional learning. Fourth, educational policies are increasingly demanding accountability, quality assurance and evidence-based teaching. However, continuous professional development should not be understood as an external requirement imposed solely on teachers. It should also be seen as a professional right and an opportunity for self-realization. Teachers who are continuously learning are more confident, creative and able to respond to complex classroom situations. Professional development strengthens teachers' identities by helping them to understand themselves as active learners, researchers, and decision-makers. Avalos (2011) emphasizes that teachers' professional development is a complex process shaped by personal, institutional, and social factors. Professional development is most effective when it is closely linked to teachers' daily work. Short-term professional development programs can provide useful information, but they often fail to lead to profound changes unless they are accompanied by practice, feedback, and reflection. Guskey (2002) notes that changes in teachers' work are usually gradual and closely linked to evidence of improved student learning outcomes. Therefore, successful professional development programs should provide opportunities for teachers to test new strategies, analyze results, and reconsider their professional actions.

Theoretical Basis of Teacher Reflection

Teacher reflection is a process in which teachers critically examine their experiences, actions, beliefs, and professional decisions. Reflection helps teachers to move beyond routine behavior and gain a deeper understanding of teaching. It answers questions such as: Why did I choose this method? How did students respond? What worked well? What do I need to change? What assumptions influenced my decision? How can I improve my next lesson?

The theoretical basis of reflection is often associated with John Dewey. Dewey understood reflective thinking as purposeful, systematic, and problem-oriented thinking. For Dewey, reflection begins when a person encounters uncertainty, difficulty, or doubt. Such uncertainty in teaching can arise when students do not understand a topic, when participation in a lesson is poor, or when assessment results reveal unexpected gaps in knowledge. Devine's approach is important because it shows that reflection is not a passive thought, but an active intellectual process aimed at better action (Devine, 1933).

Donald Schön later expanded the concept of reflection in professional practice. He distinguished between reflex in action and reflection in action. Reflection in action occurs during instruction. For example, when a teacher sees that students are confused and immediately changes their explanations, this is reflection in action. Reflection in action occurs after the lesson, when the teacher analyzes what happened, why it happened, and what could be improved. Schön's theory is important because it shows that professional knowledge is not only theoretical, but also practical, situational, and experiential (Schön, 1983).

Brookfield (1995) emphasized that teachers can reflect through a variety of lenses: their own experiences, the perspectives of students, the observations of colleagues, and the theoretical literature. This approach is particularly useful because it avoids subjectivity or one-sidedness in reflection. The teacher may believe that the lesson was successful, but student feedback or peer observations may reveal other aspects. Therefore, reflection should be supported by evidence and dialogue. Korthagen's approach to reflective teaching also emphasizes the connection between

action, awareness, and professional development. Reflection is more than just remembering what happened; it is a structured process for identifying a problem, analyzing feelings and thoughts, considering alternatives, and planning further action. Thus, teacher reflection has both a cognitive and a practical aspect. It helps teachers understand not only what happened in the classroom, but also why it happened, and how they can improve their teaching in the future (Korthagen, 2001).

Reflection as a Mechanism for Professional Development

Reflection turns experience into learning. A teacher may have years of experience, but experience alone does not guarantee professional development. If a teacher repeats the same methods without thinking, professional development can be limited. Reflection allows teachers to learn from experience, question experience, and make informed changes.

For example, a teacher may notice that students are performing poorly on a written assignment. Without reflection, the teacher may simply conclude that the students are not learning enough. However, with reflection, the teacher can ask deeper questions: Were the instructions clear? Were there enough examples of students? Was the assignment appropriate to their level? Did the assessment criteria align with the learning objectives? Such questions help teachers identify pedagogical reasons and redesign instruction. This process demonstrates that reflective practice supports evidence-based teaching and professional problem-solving.

Reflection also develops teacher autonomy. Autonomy does not mean irresponsible behavior; reflective practice means making professional decisions based on knowledge, evidence, and ethical judgment. Reflective teachers are less dependent on pre-determined methods because they can adapt strategies to the needs of their students. They understand that no one method will work equally well in all contexts. Farrell (2015) argues that reflective practice helps teachers examine their beliefs, values, and assumptions that directly influence classroom behavior.

Another important function of reflexivity is the development of professional identity. Teachers' beliefs about instruction, discipline, assessment, and student abilities have a profound impact on their classroom behavior. Reflexivity helps teachers become aware of these beliefs and assess whether they facilitate or limit student learning. For example, a teacher who believes that low-performing students cannot succeed may subconsciously lower their expectations of them. Reflective practice can challenge such assumptions and promote more inclusive teaching.

Reflexivity also promotes emotional resilience. Teaching is emotionally demanding work. Teachers face stress, conflict, workload, student diversity, and institutional expectations. Reflective practice helps teachers understand their emotional reactions and deal with professional challenges more constructively. Rather than seeing challenges as personal failures, reflective teachers analyze them as part of professional learning. This makes reflexivity important not only for improving teaching methods, but also for teachers' well-being and professional resilience.

Forms of Reflective Practice in Teacher Development

Teacher reflection can take many forms. One common form is keeping a reflective journal. In a reflective journal, teachers record lessons, student responses, challenges, and future plans. Journaling helps teachers organize their thoughts and observe patterns over time. This is especially helpful for beginning teachers because it increases self-awareness and professional confidence. Reflective writing also allows teachers to document their professional development and assess their progress over time.

Another form is peer observation. In peer observation, teachers observe each other's lessons and provide constructive feedback. This method links reflection with collaboration. The goal is

professional learning, not control or criticism. When peer observation is based on trust, it can help teachers see aspects of their teaching that they may not have noticed themselves. Brookfield (1995) argues that peer perspective is one important means by which teachers can critically evaluate their practices. Mentoring is also an important reflective practice. Experienced teachers can guide novice teachers through discussions on lesson planning, classroom management, assessment, and student communication. Effective mentoring involves more than just giving advice. It should encourage novice teachers to analyze problems, consider alternatives, and develop independent professional judgment. Through mentoring, reflection becomes dialogue and professional support.

Professional learning communities are another powerful model for developing reflection. In such communities, teachers regularly discuss teaching problems, analyze student work, share resources, and plan improvements. Results from the OECD TALIS study show that teacher collaboration and professional learning are essential elements of school quality and teacher effectiveness (OECD, 2019). However, collaboration is truly effective when it goes beyond simply sharing materials and involves collaborative research, lesson analysis, and shared responsibility for student learning.

Action research is one of the most systematic forms of reflective practice. In action research, teachers identify a classroom problem, collect data, implement an intervention, and evaluate the results. This approach positions teachers as researchers of their own practice. It reinforces evidence-based teaching and directly links professional development to student learning. Korthagen (2001) emphasizes that reflection deepens when it is linked to specific teaching practices and systematic analysis.

Portfolio development can also facilitate reflection. A teacher's portfolio can include lesson plans, assessment tools, student work samples, feedback, certificates, and reflective notes. The value of a portfolio lies not only in the collection of documentation but also in the analysis of professional progress. It helps teachers to see their development over time and set future goals. In this sense, portfolio-based reflection links professional evidence with self-assessment.

The link between continuing professional development and teacher reflection

Continuing professional development and teacher reflection are closely linked. Professional development provides teachers with new knowledge, methods and perspectives. Reflection helps teachers interpret, adapt and apply this knowledge in practice. Without reflection, professional development can remain theoretical. Without professional development, reflection can be limited to personal experience and cannot encompass broader pedagogical knowledge.

Therefore, effective continuing professional development should include reflective components. For example, after training, teachers should not only receive information but also discuss how the new knowledge can be used in their classrooms. They can create lesson plans, test new methods, observe and reflect on the results. This period creates a connection between learning, practice and development.

Darling-Hammond et al. (2017) identify several characteristics of effective professional development, including active learning, collaboration, modeling effective practice, coaching, feedback and long-term duration. These characteristics are closely related to reflection because they require teachers to reflect on their experiences, seek feedback, and make changes over time. Professional development is most effective when teachers are active participants, not passive listeners.

Reflection also enhances the sustainability of professional development. A short workshop may temporarily inspire teachers, but long-term change requires ongoing analysis and support. Reflective teachers are more likely to continue learning after formal training. They can integrate new ideas into their professional practice and evaluate their usefulness. In this sense, reflection transforms continuous professional development into a continuous cycle of planning, action, evaluation, and improvement.

The relationship between continuous professional development and reflection can also be explained by the idea of lifelong learning. Teachers who reflect on their experiences are aware of their own learning needs. They do not simply wait for external learning opportunities; instead, they actively seek out knowledge, feedback, and opportunities for improvement. This makes teachers active participants in their professional development. OECD (2020) also emphasizes that teachers' professional status and effectiveness are linked to opportunities for continuous learning, collaboration, and professional support.

Challenges in developing a reflective professional culture

Despite its importance, implementing teacher reflexivity is not always easy. One of the main challenges is the lack of time. Teachers often have heavy workloads, administrative responsibilities, and large classes. Reflection requires time for reflection, writing, discussion, and analysis. If the school schedule does not allow time for professional dialogue, reflection can remain superficial.

Another challenge is the formalization of professional development. In some contexts, continuing professional development is perceived not as a meaningful learning process but as a requirement for certification or promotion. Teachers may only attend training sessions to obtain certification. In such cases, the real impact on teaching practice is weak. Guskey (2002) notes that professional development should be assessed not only by participation, but also by its impact on teaching practice and student learning.

A third challenge is uncertainty. Reflective practice requires openness to challenges and mistakes. If teachers fear criticism or punishment, honest reflections may be avoided. Therefore, school leaders need to create a safe professional environment where teachers can discuss problems without fear. A reflective culture cannot thrive in an atmosphere of control, blame, or excessive bureaucracy.

Lack of methodological knowledge is also a problem. Teachers may understand the importance of reflection, but they lack the knowledge to conduct it systematically. They need tools such as reflection questions, observation forms, feedback models, and action research guidelines. Without such tools, reflection can become generic, emotional, or unsystematic.

Another problem is the gap between theory and practice. Professional development programs sometimes present general theories that are disconnected from the reality of teachers' classroom work. Reflection can bridge this gap, but only if teachers are encouraged to adapt theoretical ideas to their specific contexts. Desimone (2009) emphasizes that professional development should be aligned with teachers' needs, school goals, and actual classroom experiences.

Practical Implications

Several practical implications emerge from the analysis. First, teacher professional development programs should be designed as long-term processes, not one-off events. They should include follow-up sessions, application of acquired knowledge, and feedback. This approach can

increase the practical effectiveness of professional development and help teachers translate new knowledge into pedagogical practice.

Second, schools should create structures for reflective collaboration. These can include regular professional meetings, peer observation groups, mentoring systems, and subject-based learning communities. Such structures help teachers move from individual reflection to collective professional learning. Professional development is more effective when teachers learn not only from experts but also from each other.

Third, reflective writing should be encouraged. Teachers can use journals, portfolios, and self-assessment forms to document their professional learning. However, reflective writing should not become a bureaucratic exercise. It should be meaningful, honest, and related to real-world pedagogical issues. Fourth, school leaders should support a culture of reflection. Leadership plays a key role in creating time, trust, and motivation for professional learning. School leaders should value teachers' voices and encourage research-based improvement. A reflective school culture is only possible when teachers feel that their experiences, questions, and professional judgments are respected.

Fifth, teacher education institutions should prepare future teachers for reflective practice. Reflexivity should be integrated into future teacher preparation through practice analysis, microteaching, observation, and research assignments. Novice teachers who learn to be reflective during their initial teacher education are more likely to continue reflective practice throughout their professional careers.

Finally, digital tools can be used to support reflexivity. Online portfolios, video-based lesson analyses, digital communities, and learning management systems can help teachers document and discuss their experiences. However, technology should serve pedagogical reflexivity, not replace it. Digital tools are only useful when they support analysis, collaboration, and professional development.

Conclusion

Continuous professional development and teacher reflection are essential components of modern teaching. In a rapidly changing educational environment, teachers cannot rely solely on initial training or routine practice. They need continuous learning, critical self-reflection, and collaborative professional support.

This article shows that continuous professional development is effective when it is linked to classroom practice and supported by systematic reflection. Teacher reflection helps teachers to analyze their decisions, understand the needs of students, improve their teaching strategies, and develop professional autonomy. It also develops emotional resilience and professional identity.

The study concludes that reflective practice should not be seen as an optional activity. It should be integrated into all stages of teacher professional development. Schools, teacher education institutions, and educational policymakers should create conditions that support a reflective professional culture. These conditions include time for collaboration, mentoring, peer observation, action research, constructive feedback, and supportive guidance.

In general, the quality of education depends not only on curricula, textbooks, or technology, but also on the teacher's ability to learn through experience. A reflective teacher is a teacher who is constantly evolving. Therefore, continuous professional development and teacher reflection should be seen as mutually reinforcing processes that contribute to improving the quality of teaching, strengthening schools, and increasing student achievement.

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MODERN APPROACHES IN TEACHER EDUCATION

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Abstract: Modern teacher training has undergone a radical transformation to match dynamic changes in education. Moving beyond traditional models focused solely on transferring subject knowledge, it now embraces a student-centered, competency-based, and reflective interdisciplinary approach. In this framework, the teacher acts not as a simple transmitter of information, but as a guide who fosters research, collaboration, and critical thinking.

Under the competency-based approach, practical skills such as lesson planning, classroom management, and assessment are evaluated through real classroom experience and portfolios rather than theoretical exams. Continuous reflection enables educators to analyze their own practices and steadily refine their teaching strategies.

The integration of digital technology, blended learning, and "flipped classroom" models makes instruction more flexible and personalized, making digital literacy a core competency. Concurrently, inclusive education principles ensure equal opportunities by catering to diverse individual learning needs.

Modern teacher preparation incorporates emotional intelligence, cultural sensitivity, and evidence-based teaching. Grounded in lifelong learning, this approach forms flexible, innovative, and creative professionals who nurture both the academic and socio-emotional development of their students.

Keywords: modern teacher training, student-centered learning, competency-based approach, reflective practice, inclusive education, technology in education, lifelong learning

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Xülasə: Müasir müəllim hazırlığı təhsil sahəsindəki dinamik dəyişikliklərə uyğun olaraq köklü şəkildə yenilənmişdir. Təkcə fənn biliklərinin ötürülməsinə əsaslanan ənənəvi model yerini şagirdyönümlü, kompetensiyaya və refleksiya əsaslanan fənlərarası yanaşmaya vermişdir. Bu sistemdə müəllim bilik ötürən deyil, tədqiqat, əməkdaşlıq və kritik təfəkkürü təşviq edən bələdçi rolunu oynayır.

Kompetensiya əsaslı yanaşma çərçivəsində müəllimlərin dərslərini planlaşdırılması, sinfin idarə olunması və qiymətləndirmə bacarıqları nəzəri imtahanlarla deyil, real tədris təcrübəsi və portfoliolar vasitəsilə qiymətləndirilir. Mütəmadi refleksiya isə pedaqoqların öz fəaliyyətini təhlil edərək tədris strategiyalarını davamlı inkişaf etdirməsinə şərait yaradır.

Rəqəmsal texnologiyaların, hibrid təhsil və "tərsinə çevrilmiş sinif" modellərinin tətbiqi tədris prosesini daha çevik və fərdiləşdirilmiş edir. Müəllimlərin rəqəmsal savadlılığı ana kompetensiyalardan biri sayılır. Eyni zamanda, inklüziv təhsil prinsipi hər bir şagirdin fərdi xüsusiyyətlərini nəzərə alaraq bərabər imkanlar yaradır.

Müasir müəllim hazırlığı emosional intellekt, mədəni həssaslıq və subütlərə əsaslanan tədris elementlərini əhatə edir. Ömürboyu təhsil prinsiplərinə söykənən bu yanaşma müəllimləri təkcə akademik bilik verən deyil, şagirdlərin sosial-emosional inkişafını dəstəkləyən, innovativ və yaradıcı peşəkar kimi formalaşdırır.

Açar söz: müasir müəllim hazırlığı, şagirdmərkəzli təlim, səriştəyönümlü yanaşma, reflektiv fəaliyyət, inklüziv təhsil, təhsildə texnologiya, ömürboyu təhsil

Introduction

Modern teacher training approaches reflect a fundamental renewal of teacher preparation for professional activity within an increasingly complex and dynamic educational environment. Previously, teacher training was primarily limited to the transfer of subject knowledge and the formation of traditional pedagogical skills. However, in the 21st century, the rapid development of technology, expanding globalization processes, increasing cultural diversity in society, and the growing importance of critical and creative thinking have radically reshaped the expectations placed on the teaching profession. As a result of these changes, the teacher preparation system has acquired

a multidisciplinary, competency-based, and reflective character, aiming to cultivate professional teachers who not only impart knowledge but also shape well-rounded personalities. Thus, modern approaches prioritize both the professional competencies and personal development of teachers in a parallel manner.

Materials and Methods

At the center of these approaches lies the transition from teacher-centered to student-centered learning. While the teacher acted as the primary source of knowledge in the traditional model, in the modern model, they become the organizer and facilitator of the learning process. Students step out of the passive listener role and actively participate in learning through discussion, collaboration, research, and problem-solving activities. This approach increases their engagement in learning while creating conditions for developing skills such as critical thinking, independent decision-making, and creativity. During the preparation process, prospective teachers are trained to implement these interactive and active learning methods, learning to plan lessons to support students' cognitive, emotional, and social development.

A competency-based approach is of particular importance in modern teacher training. This approach requires teachers not only to possess theoretical knowledge but also to apply it in a practical setting. Skills such as lesson planning, classroom management, selecting assessment strategies, and establishing effective communication are recognized as core competencies. Unlike traditional exams, teacher candidates in modern systems are evaluated through portfolios, practical assignments, and real teaching demonstrations. This ensures their readiness for challenges in real classroom conditions and strengthens the link between theoretical knowledge and practical skills.

Integration of technology into education represents another key component of modern teacher training. Digital technologies make the teaching process more flexible, accessible, and effective. Teachers learn to utilize various online platforms, interactive tools, and digital resources. Models such as blended learning and the "flipped classroom" make lessons more personalized and engaging. Proper use of technology increases student involvement and creates opportunities tailored to individual learning paces; thus, digital literacy is considered a core competency for modern teachers.

Results and Discussion

Reflective practice is an integral part of modern teacher training. Teachers continuously analyze their own performance, evaluate the effectiveness of applied methods, and seek new ways to refine the instructional process. This process is carried out through self-assessment, peer observation, and mentor support. Reflection renders professional development continuous, transforming the teacher into a flexible professional adaptable to changing conditions. Simultaneously, this approach allows teachers to better understand students' individual characteristics and select instructional strategies aligned with their needs.

Inclusive education serves as a major direction in modern teacher preparation. Because students in modern classrooms possess diverse social, cultural, and individual characteristics, teachers must master differentiated instruction. Personalized learning, flexible methods, and inclusive strategies serve to create equal educational opportunities for all students, supporting social justice and equity alongside academic outcomes.

Furthermore, evidence-based teaching, a collaborative environment, practical experience, and the development of emotional intelligence play crucial roles. Teachers learn to make decisions based on research findings, collaborate with colleagues, and gain experience in real teaching

environments. Developing emotional intelligence helps teachers establish positive relationships with students and foster a supportive classroom atmosphere. Cultural sensitivity and interdisciplinary approaches are also vital elements, enabling teachers to incorporate diverse cultural perspectives and connect different subjects for a broader worldview, allowing students to link knowledge to real life.

Conclusion

Ultimately, modern teacher training is anchored in the principle of lifelong learning. Because the teaching profession demands continuous development, teachers must constantly acquire new knowledge and skills to adapt to changing educational environments. Continuous professional development enhances teacher effectiveness and enables the application of innovative approaches. Consequently, teachers formed through modern preparation frameworks emerge as more flexible, creative, and professional practitioners who play a pivotal role in elevating the overall quality of education.

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THE ROLE OF THE TEACHER'S PROFESSIONAL ACTIVITY IN EDUCATING STUDENTS BASED ON NATIONAL AND MORAL VALUES

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Abstract. This article examines the role of teachers' professional activity in the education of students based on national and moral values from a theoretical and pedagogical perspective. It substantiates that within the contemporary educational environment, the teacher functions not only as a transmitter of knowledge but also as a central educational agent influencing students' personal development. The process of education plays a significant role in shaping students' moral qualities, social responsibility, civic consciousness, and national identity. In this context, the teacher's educational activity contributes to fostering such values as humanism, patriotism, tolerance, mutual respect, and responsibility. Based on the analysis of pedagogical literature, the study identifies the key components of teacher professionalism, pedagogical competence, professional ethics, and the principal directions of educational activity. The findings reveal that the teacher's personal example, commitment to national and moral values, and humanistic approach significantly influence students' personality formation. Consequently, the quality of teachers' professional activity is regarded as one of the fundamental factors determining the effectiveness of the educational process.

Key words: national and moral values, teacher professionalism, pedagogical competence, value-based education, educational process, upbringing

MÜƏLLİMİN PEŞƏ FƏALİYYƏTİNİN TƏLƏBƏLƏRİN MİLLİ VƏ MƏNƏVİ DƏYƏRLƏR ƏSASINDA TƏRBİYƏ OLUNMASINDA ROLU

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Xülasə. Məqalədə tələbələrin milli-mənəvi dəyərlər əsasında tərbiyəsində müəllimin peşəkar fəaliyyətinin rolu nəzəri-pedaqoji aspektdə araşdırılmışdır. Müasir təhsil şəraitində müəllimin yalnız bilik ötürücüsü deyil, həm də şəxsiyyətin formalaşmasına təsir göstərən əsas tərbiyəedici subyekt olduğu əsaslandırılmışdır. Tərbiyə prosesi tələbələrin mənəvi keyfiyyətlərinin, sosial məsuliyyət hissinin, vətəndaşlıq mövqeyinin və milli kimlik şüurunun formalaşmasında mühüm rol oynayır. Bu baxımdan müəllimin tərbiyəvi fəaliyyəti tələbələrdə humanizm, vətənpərvərlik, tolerantlıq, qarşılıqlı hörmət və məsuliyyət kimi dəyərlərin aşılmasına xidmət edir. Tədqiqat zamanı pedaqoji elmi ədəbiyyatın təhlili əsasında müəllimin peşəkarlığı, pedaqoji səriştəliliyi, peşə etikası və tərbiyəvi fəaliyyət istiqamətləri müəyyən edilmişdir. Müəyyən olunmuşdur ki, müəllimin şəxsi nümunəsi, milli-mənəvi dəyərlərə bağlılığı və humanist yanaşması tələbələrin şəxsiyyət kimi formalaşmasına mühüm təsir göstərir. Nəticə etibarilə, müəllimin peşəkar fəaliyyətinin keyfiyyəti tərbiyə prosesinin səmərəliliyini müəyyən edən əsas amillərdən biri kimi çıxış edir.

Açar sözlər: milli-mənəvi dəyərlər, müəllimin peşəkarlığı, pedaqoji səriştə, tərbiyə prosesi, tələbə şəxsiyyəti, təhsil

Introduction

The contemporary education system no longer limits itself solely to the transmission of knowledge; it designates the comprehensive development of the personality and its formation based on national-moral values as one of the priority directions. In modern conditions characterized by the accelerating process of globalization, the preservation and transmission of national identity to future generations stands as one of the primary tasks facing education, thereby acquiring heightened

relevance. Thus, the issue of educating students on the foundation of national-moral values assumes particular significance (Cəbrayılov, 2023).

National-moral values constitute a pivotal socio-cultural phenomenon that emerges from society's historical development, encapsulating the populace's worldview, social behavioral norms, and cultural heritage. The systematic inculcation of these values into the younger generation exerts a profound influence on their personal development, the shaping of their civic stance, and their integration into society. In particular, the purposeful orchestration of this process within educational institutions is regarded as one of the principal factors augmenting the efficacy of upbringing endeavors (İlyasov, 2018).

Considering the conceptual essence of national-moral values and their role in personality formation, the methodology of transmitting these values to the younger generation attains particular salience. It is evident that values achieve a more enduring character not merely when presented theoretically, but when they are internalized through daily conduct, communication, and social interactions. In this regard, the educational (upbringing) process is not limited to simple information dissemination; rather, it functions as a complex mechanism that facilitates the practical application of values and their integration into the individual's internal world. Such a systematic integration not only enriches the learning experience but also ensures that students develop a profound appreciation for their cultural roots while acquiring the necessary skills for future challenges (Ulusoy, 2026)

In the modern educational landscape, the systematic and purposeful organization of this process requires a specialized approach. Against the backdrop of globalization, intensive information flows, and increasing cross-cultural influences, the formation of value systems among youth has become significantly more multifaceted. In such a context, the preservation and conscious internalization of national-moral values cannot rely on incidental or spontaneous influences; instead, they necessitate planned and consistent pedagogical activity. This requirement underscores the need to re-evaluate the objectives and content of the educational process while implementing more flexible and effective methodologies. Furthermore, the interaction between formal and informal environments plays a critical role in the transmission of values. Establishing robust linkages between the family, school, and society directly affects the depth of an individual's value internalization. Without proper coordination among these spheres, the perception of values remains superficial. Consequently, the educational process must be grounded not only in theoretical instruction but also in practical activity, social experience, and exemplary modeling. Collectively, these factors indicate that the role of the primary agent directing and implementing this process is of paramount importance for the effective inculcation of national-moral values. At this juncture, the quality, organizational integrity, and efficacy of the educational process depend directly on the professionalism and pedagogical approach of the individual conducting the activity. This necessitates a more profound investigation into the teacher's professional activity and their specific role within the upbringing process.

Materials and Methods

In parallel with the development of society, the processes of education and upbringing have become more complex, and a special social group has been formed that deals with this field. Historical sources show that in ancient civilizations - in countries such as Babylon, Egypt and Syria, education and upbringing were mainly carried out by priests. In ancient Greece, this function was performed by pedanoms, didascals and pedagogues. At that time, educational activity was not limited only to the transfer of knowledge, but also played an important role in the upbringing of

children and their preparation for life. In the initial stages of historical development, upbringing was understood more as the feeding and raising of a child. However, over time, it was realized that knowledge and skills were of particular importance, as a result of which education was formed as a purposeful and systematic activity. In this process, the concept of a teacher emerged and educational activity was defined as its main task. With the formation of the pedagogical profession, teachers began to carry out both educational and educational functions in parallel. For this reason, the teacher has been accepted in society not only as a person who imparts knowledge, but also as an educational and guiding social figure. Historically, the teacher has been viewed as a person with high intelligence, broad outlook, deep understanding of life events and a guide for people, and great trust has been shown in him. In modern times, the position of the teacher in society has retained its importance. He is not only a professional, but also a person who bears social responsibility and directly participates in the development of society.

The eminent Azerbaijani thinker, Nizami Ganjavi, as early as the 12th century, accorded high value to the persona of the teacher, characterizing them as a pivotal figure who radiates enlightenment and provides direction for society. In his philosophical framework, the archetype of the teacher is presented not merely as a conduit of knowledge, but as an individual who guides others toward moral perfection and enriches the human inner world. Such an approach demonstrates that the teaching profession has historically commanded high social prestige (Tusi, 1989)

Heydar Aliyev, the eminent political figure and founder of the modern Azerbaijani state, placed particular importance on the contributions of educators to societal progress. Highlighting the weight of this vocation, he stated: "The name of a teacher is the most venerable in the world. Personally, I recognize no higher title on earth than teacher. Because it is the school that educates us and brings us to this stage, and it is the teacher within the school who imparts that knowledge" (Aliyev, 2002). This perspective serves as a testament to the teacher's primary role in shaping the social fabric beyond the confines of the classroom.

In the contemporary educational landscape, the teacher functions not merely as a subject transmitting information, but as a pivotal figure who organizes the instructional process, facilitates learning, and stimulates students' cognitive engagement. From this perspective, professional teaching is defined not only by pedagogical knowledge and skills but also by an innovative mindset, creative problem-solving abilities, and the capacity to adapt to shifting circumstances. Pedagogical competence is characterized by the degree to which a teacher applies acquired knowledge to practical activity, their ability to cultivate an effective learning environment, and their capacity to ensure an individualized approach. Engaging with and implementing global educational innovations requires teachers to be more than passive recipients; they must be creative innovators. This process prioritizes the development of pedagogical thinking, initiative, and research-oriented activity. Guided by the principle of lifelong learning, the modern educator must engage in continuous self-improvement and professional development. Self-education, personal growth, and pedagogical creativity serve as the core components of this process, shaping the teacher's professional identity. Furthermore, the inherent openness of pedagogical activity to innovation underscores its dynamic nature. The integration of modern technologies, the use of interactive methods, and the expansion of student-centered approaches directly impact the quality of education. Conversely, a monotonous pedagogical approach can lead to diminished student interest, passivity, and weakened learning outcomes. Given the modern student's broad access to information and predisposition toward innovation, teachers are required to adopt more flexible and creative strategies.

On the other hand, professional teaching is not confined to classroom instruction. It encompasses a vital formative function that influences students' social and moral development. The teacher's persona, conduct, communication style, and alignment with ethical values serve as a model for students. Thus, professionalism is determined not only by expertise and skills but also by ethical qualities and social responsibility. These attributes enable the organization of personality-oriented and result-oriented education, significantly contributing to the comprehensive development of the younger generation and their formation within the framework of modern global values. In this regard, the creation and continuous refinement of mechanisms that ensure the professional development of teachers within the modern educational system are of strategic importance. When considering the content of a teacher's professional activity and the factors determining its effectiveness, a deeper analysis of the internal structure and multifaceted nature of this activity becomes essential. This is because pedagogical activity is not a simple, one-dimensional process, but rather a complex system formed through the interaction of diverse components. In this regard, the inherently social and creative character of the teaching profession distinguishes it from other vocations.

Teaching activity possesses an intricate structure in which the subject of pedagogical labor, the tools employed, the environment and conditions of the activity, as well as the governing laws, principles, methods, and value systems, play a vital role. Although future teachers initially encounter these components during their higher education, their practical significance and real-world application are more profoundly internalized and developed throughout the actual pedagogical process. This demonstrates that the teaching profession is a dynamic field of activity, shaped not only by theoretical knowledge but also by sustained practical experience.

A teacher's professional activity is carried out across various types of educational institutions, including general education schools, vocational education centers, institutions for supplementary education, and other diverse learning environments. The primary objective of this activity is the cultivation of the younger generation as comprehensively developed individuals integrated into society. Within this process, the priority is not merely the acquisition of knowledge, skills, and habits by students, but also the assurance of their intellectual, moral, and physical development. Furthermore, one of the vital components of pedagogical activity is the establishment of appropriate requirements and the formation of essential qualities based on students' individual characteristics, age levels, and psychophysiological capacities.

As pedagogical activity is closely linked to the social mandate of society, its content, forms, and methods are determined by the development level and requirements of that society. In this regard, activities implemented within the educational system must be continuously updated and aligned with modern challenges. The teacher's role necessitates an adequate response to these changes, as well as the mastery and application of innovations. Consequently, pedagogical activity is characterized not as a static entity, but as a continuously evolving and improving process. Furthermore, it must be noted that a teacher's professional activity is a complex, multifaceted process replete with challenges. The successful execution of this activity demands a high level of intellectual preparation, analytical thinking, pedagogical mastery, and a creative approach. Resolving emerging pedagogical problems is possible not only through standard methods but also through innovative strategies and agile decision-making skills. Consequently, this necessitates that the teacher continuously engages in self-reflection and professional development. In this context, the concept of professionalism assumes particular significance. Professionalism in

pedagogical activity is characterized by the high-level, creative, and effective fulfillment of assigned tasks. A teacher's professionalism is not limited to formal indicators, such as diplomas or certifications; rather, this quality is determined by their attitude toward the profession, their sense of responsibility, the level of self-development, and their overall approach to the activity. From this perspective, a teacher's commitment to their vocation, their profound understanding of its essence, and their constant striving for self-improvement are considered the fundamental conditions for reaching a truly professional level. On the other hand, an indifferent or perfunctory approach to pedagogical activity constitutes one of the primary barriers to a teacher's professional development. When a discrepancy arises between a teacher's professional activities and their personal stance or values, it manifests in their attitude toward their work, ultimately exerting a negative impact on the quality of the pedagogical process. In such instances, the teacher is unable to ascend to a level of true professionalism or fulfill their assigned responsibilities with the necessary degree of accountability.

The complex and multifaceted nature of the teaching profession necessitates that the educator acts not merely as a conduit of information, but as a personality performing multifunctional socio-pedagogical activities. In this regard, a teacher's work encompasses various dimensions, and their professional identity is formed through the integration of these functions. In generalized terms, the teacher is the primary pedagogical subject who instructs, educates, plays an enlightening role in society, and exerts influence through personal example. Each of these functions reflects the inseparable link between the teacher's personality and their professional activity, collectively defining their professional portrait.

First and foremost, the teacher's instructional function lies at the core of pedagogical activity and serves as one of the primary indicators of professional proficiency. In the modern era, the educator does not merely transmit knowledge but acts as a guide who organizes the learning process and directs students toward the independent acquisition of knowledge. In this regard, the teacher's didactic ability – the capacity to effectively convey knowledge to others – assumes particular significance. Furthermore, within the modern information environment, the teacher is no longer the sole source of knowledge; thus, their primary duty is to teach students the methods of attaining knowledge, stimulate their cognitive engagement, and strengthen their learning motivation. This approach is encapsulated in the famous maxim by the German pedagogue A. Diesterweg: "The bad teacher dictates the truth; the good teacher teaches the student how to find it."

Another significant dimension is the teacher's formative (educative) function, which constitutes an integral part of their professional activity. The formative process is not limited solely to extracurricular activities; on the contrary, it serves as a continuous and purposeful mechanism of influence realized within the instructional process itself. The content and organization of the lesson, teacher-student relationships, the environment of pedagogical cooperation, and the teacher's own conduct directly impact the formation of the student's personality. In this regard, a teacher's professionalism is also measured by the extent to which they can manifest these formative opportunities. This point is specifically emphasized in the views of prominent educators. For instance, (Sukhomlinsky, 1978) noted: "A person who meets students only in class – with the teacher's desk separating them – does not know the child's soul, and one who does not know the child cannot be an educator." This perspective demonstrates that teacher-student relationships must transcend formal boundaries and that pedagogical influence is only possible through genuine, lived communication.

Results and Discussion

Furthermore, a teacher's activity is not confined solely to the school environment; their socio-pedagogical function assumes particular significance as well. The teacher acts as a personality who is in constant contact with society, actively participates in the system of social relations, and performs an enlightening role. In this regard, they are not only an educator but also a bearer and promoter of social values, state policy, and public ideas. A teacher's prestige in society is determined not only by their level of knowledge but also by their conduct, ethical qualities, and social stance. Consequently, a teacher must be mindful of their behavior in every setting and remain conscious of their social responsibility. This sentiment is reflected in folk wisdom: 'The work of a teacher cannot be entrusted to everyone, but everything can be entrusted to a teacher.'

Finally, the teacher's personal example serves as one of the most potent instruments of pedagogical influence. Within the hierarchy of formative methods, the 'exemplary model' holds a unique position and plays a vital role in shaping students' behavioral patterns. While various sources of social influence – such as the family, literature, heroic archetypes, and the broader social milieu – play important roles in a student's upbringing, the teacher's personal example is more systematic and sustained. A teacher's conduct, interpersonal relationships, sense of responsibility, and professional attitude are directly observed by students and profoundly impact their internal value systems.

Given the multifaceted professional functions of a teacher, it becomes evident that the efficiency of their activity is not limited solely to pedagogical knowledge. A teacher's psychological preparedness serves as a crucial condition for the successful organization of this work. Managing the pedagogical process effectively in a modern educational environment – while considering students' individual traits and ensuring their personal development – requires the educator to possess sufficient knowledge in psycho-pedagogy and educational psychology. Psychological expertise enables the teacher to understand students more profoundly, approach their inner worlds, and structure the teaching-learning process in a more engaging, flexible, and purposeful manner.

In this regard, classical pedagogical thought has placed particular emphasis on the teacher's psychological readiness. For instance, K.D. Ushinsky underscored the necessity for teachers to possess a deep understanding of psychology in addition to pedagogy. This approach maintains its relevance in modern practice and is evaluated as a core component of professional competence. A teacher's psychological preparedness primarily involves a deep knowledge of students' age-related characteristics. Each developmental stage possesses its own laws of growth, interests, cognitive levels, and psychophysiological traits. The teacher must organize the instructional process with these features in mind and correctly assess the sensitive periods of a student's development. In this context, Jean-Jacques Rousseau's famous maxim holds significant weight: 'Children should not be viewed as miniature adults.' This principle necessitates that the teacher selects a pedagogical approach tailored to each specific stage of development. Moreover, a teacher must possess a profound understanding of students' individual-psychological traits. The varying levels of psychic processes – such as temperament types, attention, memory, cognition, imagination, and linguistic development – necessitate the individualization of the pedagogical approach. Within modern pedagogical frameworks, the transition from the 'subject-object' model of student-teacher relations to a 'subject-subject' model is made possible precisely through this expertise. This shift fosters an environment of mutual respect, collaboration, and trust between the educator and the student. Psychological expertise also provides the foundation for developing vital professional

qualities in a teacher, foremost among which is a high capacity for observation. The teacher must identify the dynamics of student development, their successes, and their challenges in a timely manner to make informed pedagogical decisions. Simultaneously, analytical thinking enables the teacher to evaluate pedagogical situations, determine the root causes of problems, and identify optimal solutions.

Another crucial quality is the teacher's emotional-intellectual preparedness. The extent to which the instructional process is engaging and motivating for students depends on the teacher's ability to construct the lesson on a positive emotional foundation and stimulate a passion for learning. Additionally, the teacher must respect students' feelings and thoughts, value their opinions, and demonstrate a humanistic attitude. Acknowledging students' achievements – even minor successes – positively impacts their self-confidence and enhances their learning motivation. A key indicator of a teacher's psychological competence is their empathic ability. Empathy allows the teacher to understand students' inner worlds, sense their emotional states, and establish pedagogical relationships with greater sensitivity. The teacher's ability to put themselves in the student's place ensures that pedagogical decisions are more objective and humanistic, ultimately leading to a personality-oriented instructional process. (Yulia, 2018).

At the same time, it must be noted that the role of psychological knowledge in teacher training is still not maintained at a sufficiently high level. In many cases, the instruction of psychology is conducted within a limited framework, and the practical application of this knowledge remains underdeveloped. For this reason, enhancing the role of psychology and educational psychology in the professional preparation of teachers, and ensuring their instruction is practice-oriented, is of vital importance. A teacher's psychological preparedness is an integral part of their professional activity and acts as one of the fundamental factors ensuring the effectiveness of the pedagogical process. An educator equipped with psychological expertise establishes more effective communication with students, accounts for their individual characteristics, and is capable of organizing the teaching and learning process in a more purposeful and result-oriented manner.

A teacher's professional activity must not be confined to the mere transmission of theoretical knowledge; it must also serve to cultivate national and moral values as an unwavering conviction and a "moral immune system" within the student's personality. To this end, the instructional process must transition away from one-sided lecture formats toward interactive methods that ensure both the emotional and cognitive engagement of students. The modern teacher must skillfully integrate the ideology of "Azerbaijanism," statehood traditions, and national cultural heritage into the core content of their subject, ensuring that these values are perceived not as artificial, external information, but as an integral part of the student's personal identity.

In the methodological dimension of teaching, the case-study method holds particular significance. Robert Yin, a prominent researcher, characterizes the scientific-methodological foundations of this approach and its connection to real life as follows:

"A case study is an empirical inquiry that investigates a contemporary phenomenon (the 'case') in depth and within its real-world context, especially when the boundaries between phenomenon and context may not be clearly evident" (Yin, 2018).

Projecting Yin's scientific thesis onto pedagogical activity reveals that national and moral values should be presented to students not as abstract or detached concepts, but within a "real-life context." Professionalism in teaching entails the ability to extract moral values from the artificial environment of the classroom and link them to current events, historical facts, and social issues

prevalent in society. Presenting scenarios where students must choose between honesty and personal gain, or collective interest and egoism, facilitates a deep understanding of the phenomenon – as Yin noted – and helps the student internalize these values as part of their personal experience.

In addition, through project-based learning, students conduct independent research on their family lineage, regional folklore, vanishing ancient professions, and traditions, which profoundly strengthens their sense of belonging to their national roots. Utilizing interdisciplinary integration – for instance, merging the heroic archetypes in literature with the factual realities of history, or connecting the concept of "homeland" in geography with artistic descriptions – creates a unified national worldview in the student's mind. Purposeful excursions to historical sites, museums, and Alleys of Martyrs serve as catalysts for visualizing national memory and awakening a sense of patriotism within the student's inner world.

In the face of alien ideological influences brought about by globalization, the teacher must also perform the function of a "cultural filter," guiding the student on how to preserve their national identity while adopting the technological innovations and universal values of the modern world. At the center of all these complex methodological approaches lies the teacher's role as a "moral exemplar" – living the values they propagate through their daily behavior, speech culture, and the humanistic relationships they establish with students. The alignment between a teacher's words and actions enhances the student's trust and ensures the effectiveness of moral education. Thus, professional competence in teaching emerges as a synthesis of high pedagogical mastery, national spirit, and steadfast morality, fully realizing the ultimate goal of education: the cultivation of a mature, "perfected individual" loyal to their people and state.

Research findings indicate that teacher professionalism and pedagogical competence are fundamental factors determining the effectiveness of the educational process. However, in the context of modern requirements, this professionalism should not be limited solely to technical knowledge and methodological skills; it must also be integrated with the mission of instilling national and moral values in the younger generation. The teacher must act not only as an implementer of innovative technologies but also as a guardian of national identity, the spirit of patriotism, and spiritual heritage.

Conclusion

The teaching profession is a dynamic field requiring continuous development, and the teacher's moral commitment to the profession plays a decisive role in this evolutionary process. Just as a teacher's professional growth leads to an increase in their prestige, embodying national values in their own behavior serves as the greatest exemplar for students. As emphasized throughout the history of pedagogical thought, the personality of the teacher stands at the center of education. As (Uşinski, 1961) stated: "The influence of the educator's personality on the young soul is a transformative power that can be replaced neither by textbooks, nor by moral precepts, nor by systems of punishment and reward." From this perspective, a teacher's professional competence is measured by their possession of high morality and national values.

In the modern era of globalization, continuous professional development is essential to meet the evolving needs of students and to form their national "immune system." As noted in the State Strategy for the Development of Education in the Republic of Azerbaijan, increasing the professionalism of educators is a strategic goal. To achieve this objective, teachers must constantly work on themselves and improve spiritually (Azərbaycan Respublikası Təhsil Nazirliyi, 2013).

In conclusion, the teacher remains the primary factor determining the quality of the educational system. A teacher's professionalism, innovative approach, and loyalty to national-moral values ensure the result-oriented nature of the educational process. Forming these qualities in unity is the main task facing the modern education system and holds strategic importance for the future progress of society and the cultivation of well-rounded individuals with a strong national spirit.

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INCLUSIVE EDUCATION IN TEACHING AZERBAIJANI LANGUAGE AND LITERATURE: APPLICATION OF INTERACTIVE AND COMMUNICATIVE TEACHING METHODS

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Abstract: The article analyzes the application of interactive and communicative teaching methods, as well as differentiated tasks, in teaching the Azerbaijani language and literature in inclusive classrooms. It is shown that these approaches take into account students' individual characteristics, ensure their active participation in the learning process, and develop communication, critical thinking, and creative skills. While interactive methods increase collaboration and classroom engagement, communicative language teaching enables language to be acquired as a tool for real communication. Differentiated tasks, in turn, ensure that students with different levels of preparedness can successfully learn in the same classroom. Overall, the

integrated use of these methods improves the quality of inclusive education and creates conditions for revealing each student's potential.

Keywords: Interactivity, communication, inclusivity, differentiation, collaboration.

İNKLÜZİV SİNİFLƏRDƏ AZƏRBAYCAN DİLİ VƏ ƏDƏBİYYATIN TƏDRİSİNDƏ İTERAKTİV VƏ KOMMUNİKATİV TƏLİM METODLARININ TƏTBİQİ

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Xülasə: Məqalədə inklüziv siniflərdə Azərbaycan dili və ədəbiyyatın tədrisində interaktiv və kommunikativ təlim metodlarının, eləcə də diferensiallaşdırılmış tapşırıqların tətbiqi məsələləri təhlil olunur. Göstərilir ki, bu yanaşmalar şagirdlərin fərdi xüsusiyyətlərini nəzərə almaqla onların təlim prosesində fəal iştirakını təmin edir, ünsiyyət, tənqidi düşünmə və yaradıcı bacarıqlarını inkişaf etdirir. İnteraktiv metodlar əməkdaşlıq və dərs fəallığını artırdığı halda, kommunikativ təlim dili real ünsiyyət vasitəsi kimi mənimsəməyə imkan yaradır. Diferensiallaşdırılmış tapşırıqlar isə müxtəlif hazırlıq səviyyəsinə malik şagirdlərin eyni sinifdə uğurla öyrənməsini təmin edir. Ümumilikdə, bu metodların kompleks tətbiqi inklüziv təhsilin keyfiyyətini yüksəldir və hər bir şagirdin potensialının üzə çıxarılmasına şərait yaradır.

Açar sözlər: İnteraktivlik, kommunikasiya, inklüzivlik, diferensiallaşma, əməkdaşlıq.

Introduction

Modern education increasingly adopts an inclusive approach as one of the key pedagogical directions that takes into account the individual characteristics, learning needs, and potential of each student. Within this framework, the teaching of the Azerbaijani language and literature is not limited to the transfer of knowledge but is also aimed at developing students' social, communicative, and creative skills. In inclusive classrooms, the use of interactive and communicative teaching methods plays a particularly important role in enhancing the effectiveness of the learning process.

Interactive and communicative teaching methods not only ensure students' active participation in the learning process but also develop their thinking, discussion, collaboration, and ability to express their ideas freely. Unlike traditional teacher-centered instruction, these methods place the student at the center of the educational process and support the implementation of the core principles of inclusive education.

Materials and Methods

In modern education systems, the inclusive approach aims to ensure equal and high-quality education for all students, regardless of their physical, psychological, or social differences. In this regard, the application of interactive and communicative teaching methods in the teaching of the Azerbaijani language and literature is of particular importance. Such methods allow teachers to take into account students' individual characteristics, ensure their active participation in the learning process, and develop their communication skills (Johnson & Johnson, 2019).

Interactive teaching methods actively engage students in the learning process and contribute to the development of their independent thinking, analysis, and synthesis skills. The application of these methods is especially important in inclusive classrooms, where students have different

learning speeds, comprehension levels, and educational needs. As Johnson and Johnson (2019, p. 5) state, “cooperative learning is the instructional use of small groups so that students work together to maximize their own and each other’s learning.” For example, the small-group work method strengthens collaboration in an inclusive environment and ensures that every student participates in the process. In this context, the teacher must organize groups in such a way that students with different levels can support each other effectively.

From a Vygotskian perspective, learning is fundamentally a socially mediated process in which cognitive development occurs through interaction with others (Vygotsky, 1978). Vygotsky notes that “what a child can do in cooperation today, he can do alone tomorrow.” This idea strongly justifies the use of interactive and communicative methods in inclusive classrooms, where peer interaction and guided collaboration become essential tools for learning.

Table 1. Interactive Teaching Methods and Their Application Features

No	Method Name	Brief Description	Application in Azerbaijani Language and Literature	Role in Inclusive Education	Advantages
1	Brainstorming	Free generation of ideas on a topic	Introduction to topic, defining concepts	Ensures participation of all students	Develops creative thinking
2	Discussion	Exchange of ideas on a topic	Text analysis, topic discussions	Encourages respect for different opinions	Enhances critical thinking
3	Debate	Defending viewpoints with arguments	Literature topics, social issues	Strengthens social communication	Develops speech and logic skills
4	Role-play	Acting out situations	Dialogue practice, character presentation	Develops social skills	Ensures practical learning
5	Case study	Analysis of a specific situation	Analysis of literary characters and events	Supports individual learning approaches	Forms analytical thinking
6	Small group work	Students working in groups	Completing tasks collaboratively	Enhances cooperation	Builds teamwork skills
7	Cluster (Mind map)	Schematic organization of concepts	Structuring topics	Supports visual learning	Improves memory retention
8	Aquarium method	One group discusses, others observe	Text analysis, debates	Ensures participation of all learners	Develops observation and analysis skills
9	Cinquain	Five-line	Summarizing topics	Develops	Enhances

		structured writing		individual expression	creativity
10	Problem-based learning	Setting and solving problems	Grammar application, text analysis	Adapts to different learning levels	Develops research skills
11	Project method	Preparing and presenting projects	Literature topics, research tasks	Combines individual and group work	Encourages independent learning
12	Six Thinking Hats	Analysis from different perspectives	Multidimensional literary analysis	Encourages acceptance of diverse thinking	Enhances analytical and creative thinking
13	Question–answer method	Teacher–student dialogue	Understanding topics	Promotes active participation	Provides immediate feedback
14	Venn diagram	Comparing similarities and differences	Comparing literary characters	Enhances visual understanding	Develops logical thinking
15	Concept map	Creating relationships between concepts	Grammar and text structure	Suitable for different learning styles	Builds systematic thinking

These methods, unlike the traditional passive learning model, encourage students to think, discuss, collaborate, and freely express their ideas. In an inclusive educational environment, these features are particularly important because the classroom includes students with different abilities, learning speeds, and educational needs.

One of the interactive methods, brainstorming, is used at the beginning of the lesson to identify students' prior knowledge on a topic. For example, free expression of ideas about concepts such as "mother tongue" or "homeland" increases students' interest and engagement in the lesson. According to Jones (2018), brainstorming enhances students' cognitive activation and promotes active participation in the learning process. Discussion and debate methods, on the other hand, teach students to compare different viewpoints, present arguments, and respect opposing opinions. In literature classes, these methods are especially effective, as they allow for multiple interpretations of literary works (Brookfield, 2017).

Role-playing and situational tasks are aimed at developing students' communication skills in real-life contexts. For instance, staging literary characters or constructing dialogues helps students improve both language proficiency and social interaction skills. As Harmer (2015) emphasizes, role-play activities significantly increase communicative competence in language learning. The small-group work method is particularly important in inclusive education; it enables students to collaborate, learn from each other, and develop a supportive learning environment (Johnson & Johnson, 2019).

On the other hand, methods such as clustering and conceptual mapping facilitate the visual organization of information and are especially useful for visual learners. The aquarium method

enhances both active participation and students' observation and analytical skills. The cinquain method contributes to expressing topics in a concise, structured, and creative form.

Problem-based learning and case study methods develop students' analytical and research skills. Problem situations created by the teacher guide students toward thinking, investigation, and solution finding. The project method, however, requires a more comprehensive and complex approach; students conduct research on a specific topic, systematize the results, and present their findings (Thomas, 2000).

The advantage of these methods is that they increase students' interest in learning, develop independent thinking and creativity, and form communication and collaboration skills. At the same time, they strengthen social integration in inclusive classrooms and help reveal each student's potential. However, there are also some challenges: these methods require more time, demand high organizational and professional skills from teachers, and if not properly implemented, may lead to the passive participation of some students.

Table 2. Advantages and Disadvantages of Interactive Teaching Methods

Advantages	Disadvantages
Ensures active student participation in the learning process	Requires a considerable amount of time
Develops creative and critical thinking skills	May create classroom confusion if not properly organized
Strengthens collaboration and teamwork skills	Some students may remain passive
Provides equal participation opportunities in inclusive environments	Requires high teacher preparation and management skills
Makes learning more engaging and motivating	

As a result, the purposeful and appropriate application of interactive teaching methods significantly improves the quality of teaching Azerbaijani language and literature in inclusive classrooms. This approach develops both students' academic achievement and their social and communicative skills, shaping them into more active and independent individuals (Johnson & Johnson, 2019).

2. Communicative Language Teaching (CLT). Communicative Language Teaching (CLT) is considered one of the most effective approaches in modern language education, especially in the teaching of Azerbaijani language and literature in inclusive classrooms. The main goal of this method is not only to teach language through grammatical rules and theoretical knowledge, but also to ensure its use in real-life communicative situations

Table 3. Communicative Language Teaching (CLT) Methods and Their Implementation Features

No .	Method / Technique	Brief Description	Sample Task	Role in Inclusive Education	Advantages	Disadvantages
1	Dialogue and monologue	Constructing speech in real-life situations	"Shopping at a store" dialogue	Develops communication skills	Increases speaking fluency	Shy students may face difficulties
2	Role-play	Acting out	Teacher–	Strengthens	Promotes	Some students

		social roles	student dialogue	social integration	practical language use	may remain passive
3	Information gap	Completing tasks through information exchange	Combining parts of a text	Encourages purposeful communication	Ensures active participation	Requires preparation
4	Problem-solving	Solving problems through discussion	Planning a trip	Develops collaborative decision-making	Enhances logical thinking	Time-consuming
5	Discussion / debate	Exchange of opinions and argumentation	"Is technology beneficial?"	Develops critical thinking	Strengthens speaking and analysis skills	May lead to conflicts
6	Pair/group work	Collaborative work in pairs or groups	Preparing and answering questions	Enhances cooperation	Develops social skills	Unequal participation possible
7	Language games	Learning through games	Word-finding game	Increases motivation	Makes lessons engaging	May distract attention
8	Storytelling	Speaking based on events or images	Writing a story based on pictures	Enhances creativity	Improves speaking skills	Difficult for weaker students
9	Simulation	Recreating real-life situations	Job interview scenario	Develops practical skills	Creates realistic communication context	Requires organization
10	Project method	Research and presentation work	"My favorite writer" presentation	Supports individual development	Enhances independent learning	Time-consuming

CLT is particularly effective in inclusive classrooms because it enables the participation of all students in the learning process while taking individual differences into account.

In this approach, the dialogue and monologue method is used to help students construct speech in everyday communicative situations. For example, students may be assigned to prepare a dialogue on the topic "shopping at a store." This method develops speaking skills, although it may be challenging for shy learners (Harmer, 2015).

The role-play method is implemented through the dramatization of social situations. Students perform roles such as teacher–student or seller–customer. This method strengthens communicative competence and increases motivation; however, it may be difficult to ensure active participation from all learners.

The information gap activity is based on the exchange of information between students. Each learner is required to share missing information with others to complete a task. This creates purposeful communication and activates language use, although improper organization may lead to confusion.

The discussion and debate method allows students to express opinions and present arguments on a given topic. For instance, the topic “the importance of reading books” can be discussed. This method develops critical thinking, but it is time-consuming and may result in passive participation from some students (Brookfield, 2017).

Pair and group work methods are based on student collaboration. For example, a text may be divided into parts and analyzed by different groups. This method develops social skills, although unequal participation among group members may occur (Johnson & Johnson, 2019).

Language games make the learning process more engaging. Word-finding or competition-based tasks increase students’ motivation, but they may also distract attention from the main learning objectives (Wright, Betteridge & Buckby, 2006).

The storytelling method develops students’ creative speaking skills. For example, learners may be asked to write a story based on pictures. This method enhances creativity, although it may be challenging for lower-achieving students.

The simulation method involves recreating real-life situations in the classroom. For example, students may simulate a job interview process. This method develops practical communication skills, but it requires strong classroom organization (Gredler, 2004).

Finally, the project method is based on students conducting research on a specific topic and presenting their findings. For example, students may prepare a presentation on “my favorite writer.” This method strengthens independent learning, but it requires time and resources.

Table 4. Advantages and Disadvantages of Communicative Language Teaching (CLT)

Advantages	Disadvantages
Teaches language as a tool for real communication	May be difficult for low-achieving students
Develops speaking and communication skills	Lesson organization requires more time
Increases students’ self-confidence	Classroom management is more challenging in large groups
Ensures active participation and practical learning	Requires high communicative competence from the teacher
Strengthens social integration in inclusive classrooms	Not all students may participate at the same level

Overall, CLT methods develop students’ communication skills, actively engage them in the learning process, and strengthen social integration in inclusive environments. However, some of these methods may create challenges in terms of time management, organization, and teacher preparation.

A comparative analysis of these methods helps to identify their similarities and differences, as well as to determine more effective ways of applying them in the teaching of the Azerbaijani language and literature in inclusive classrooms. As a result, these approaches play an important role

in developing both academic and social skills of students in accordance with modern educational requirements.

Table 5. Comparison of Interactive Teaching Methods and Communicative Language Teaching (CLT)

Comparative Criteria	Interactive Teaching Methods	Communicative Language Teaching (CLT)
Main purpose	To ensure students' active participation in the learning process	To teach language as a tool for real communication
Focus	Activity and collaboration in the learning process	Communication and practical use of language
Role of the student	Active participant, contributor, and collaborator	Communicator, speaker, and listener
Role of the teacher	Facilitator and guide	Organizer and facilitator of communication environment
Scope of application	Applicable across all subjects	Mainly used in language teaching
Method examples	Discussion, debate, clustering, project work, group work	Dialogue, role-play, information gap, simulation
Key skills developed	Critical thinking, creativity, collaboration	Speaking, listening, communicative competence
Inclusivity	Provides broad opportunities for participation of all students	Enhances social integration through communication
Challenges	Requires time and organizational skills	May be difficult to manage in large classrooms

In conclusion, interactive methods strengthen overall classroom engagement and collaboration, while Communicative Language Teaching (CLT) is specifically focused on the practical use of language skills. These two approaches complement each other, and when applied together in inclusive classrooms, they lead to higher learning outcomes (Richards & Rodgers, 2014; Johnson & Johnson, 2019).

3. Differentiated Tasks. Differentiated tasks are considered one of the most important pedagogical approaches in the teaching of Azerbaijani language and literature in inclusive educational environments. The main purpose of this method is to ensure that all students within the same classroom regardless of their level of preparedness, learning pace, and cognitive abilities—are effectively engaged in the learning process (Tomlinson, 2014).

Differentiation is usually implemented at three main levels: easy, medium, and difficult tasks. The easy level is designed for students who require more support and focuses on the acquisition of basic concepts. The medium level is intended for average students and requires the application of knowledge. The difficult level is designed for high-achieving students and involves analysis, creative thinking, and independent reasoning.

For example, in an Azerbaijani language lesson on “text comprehension,” differentiated tasks may include: identifying the main idea at the easy level, answering questions at the medium level, and writing an alternative ending to the text at the difficult level. In literature lessons, the easy level

may involve listing characters, the medium level describing their characteristics, and the difficult level comparing and analyzing them.

Table 6. Differentiated tasks

Topic	Easy Level	Medium Level	Difficult Level
Text comprehension	Find the main idea	Answer questions	Write a new ending
Literary character	List characters	Describe character	Compare with another character
Grammar (noun)	Identify nouns	Explain in sentences	Analyze types of nouns
Verb	Identify verbs	Change verb tense	Create a text with verbs
Sentence types	Identify sentence type	Classify sentences	Write a text using different sentence types
Azerbaijani literature	Write author's name	Summarize the work	Analyze the idea of the work
Poetry	Find main idea	Explain imagery	Analyze and interpret poem
Synonyms & antonyms	Find synonyms	Use in sentences	Write a text using synonyms
Essay	Write 3 sentences	Write short essay	Write analytical essay
Listening task	Write one sentence	Answer questions	Express opinion
Dialogue	Read dialogue	Complete dialogue	Create dialogue
Vocabulary	Copy new words	Explain meanings	Write a story
Literary comparison	List characters	Compare two characters	Write comparative analysis
Creativity	Name picture	Write sentence	Create story
Rule application	Read rule	Give example	Create test

The main advantage of these tasks is that each student works according to their own ability level, experiences success in learning, and becomes more motivated. At the same time, this approach promotes social equity in inclusive classrooms and individualizes the learning process (Tomlinson, 2014).

However, there are also some challenges. Designing differentiated tasks requires more time and professional competence from teachers. Additionally, if not properly planned, imbalance between levels may occur, and assessment may become more complicated.

Table 7. Advantages and Disadvantages of Differentiated Tasks

Advantages	Disadvantages
Students work according to their abilities	Task preparation is time-consuming
Increases students' sense of achievement	Requires high professional competence
Enhances motivation	Imbalance between levels may occur
Promotes inclusion and equality	Assessment becomes more complex
Individualizes learning process	Classroom management challenges
Strengthens student-centered learning	

Overall, differentiated tasks are considered one of the most effective student-centered approaches in modern education, especially in inclusive classrooms, and they contribute significantly to the individual development of each learner.

Conclusion

The analysis demonstrates that the integration of interactive teaching methods, Communicative Language Teaching (CLT), and differentiated tasks significantly enhances the quality of teaching Azerbaijani language and literature in inclusive classrooms. These approaches ensure that individual learner differences are taken into account and actively support students' engagement in the learning process.

Interactive methods increase classroom participation and collaboration, CLT develops language as a tool for real communication, and differentiated tasks provide opportunities for students with different levels of achievement to learn effectively within the same classroom. The combined use of these approaches improves the effectiveness of inclusive education and creates conditions for the development of each learner's potential.

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TEACHER PREPARATION AND PROFESSIONAL DEVELOPMENT IN THE 21st CENTURY: CHALLENGES, MODELS AND FUTURE DIRECTIONS

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Abstract: Teacher preparation and professional development have become central issues in educational reform worldwide, particularly in the context of rapid technological advancement, globalization, and evolving pedagogical paradigms. This article examines the theoretical foundations, models, and practices of teacher preparation and continuous professional development (CPD). It explores how initial teacher education (ITE) programs and in-service training contribute to teaching effectiveness and student outcomes. Using a mixed-method approach based on literature analysis and comparative evaluation of global practices, this study highlights the importance of reflective practice, lifelong learning, and institutional support in enhancing teacher competencies. The findings suggest that effective professional development must be sustained, collaborative, and context-specific. The article concludes by recommending policy and institutional reforms aimed at integrating innovative training methods, digital tools, and competency-based frameworks into teacher education systems.

Keywords: Teacher preparation, professional development, teacher education, lifelong learning, instructional effectiveness

21-Cİ ƏSRDƏ MÜƏLLİM HAZIRLIĞI VƏ PEŞƏKAR İNKİŞAF: ÇAĞIRIŞLAR, MODELLƏR VƏ GƏLƏCƏK İSTİQAMƏTLƏR

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Xülasə: Müəllimlərin hazırlığı və peşəkar inkişafı bütün dünyada, xüsusən də sürətli texnoloji tərəqqi, qloballaşma və inkişaf edən pedaqoji paradigmlər kontekstində təhsil islahatlarının mərkəzi məsələsinə çevrilmişdir. Bu məqalə müəllim hazırlığının və davamlı peşəkar inkişafın (CPD) nəzəri əsaslarını, modellərini və təcrübələrini araşdırır. O, ilkin müəllim təhsili (İTE) proqramlarının və ixtisasartırma təlimlərinin tədrisin effektivliyinə və tələbə nəticələrinə necə töhfə verdiyini araşdırır. Ədəbiyyat təhlilinə və qlobal təcrübələrin müqayisəli qiymətləndirilməsinə əsaslanan qarışıq metodlu yanaşmadan istifadə edərək, bu tədqiqat müəllimlərin səriştələrinin artırılmasında əks etdirici təcrübənin, ömür boyu öyrənmənin və institusional dəstəyin əhəmiyyətini vurğulayır. Nəticələr göstərir ki, effektiv peşəkar inkişaf davamlı, əməkdaşlıq və kontekstlə bağlı olmalıdır. Məqalə innovativ təlim metodları, rəqəmsal alətlər və səriştə əsaslı çərçivələrin müəllim təhsili sistemlərinə inteqrasiyasına yönəlmiş siyasət və institusional islahatları tövsiyə etməklə yekunlaşır.

Açar sözlər: Müəllim hazırlığı, peşəkar inkişaf, müəllim təhsili, ömür boyu öyrənmə, təlimin effektivliyi

Introduction

Education systems globally are undergoing transformative changes driven by socio-economic shifts, technological innovation, and increased expectations for student performance. (Darling-Hammond, Linda (2006). At the core of these transformations lies the teacher, whose role has evolved from knowledge transmitter to facilitator, mentor, and lifelong learner. Consequently, teacher preparation and professional development have gained increasing importance in educational discourse. (OECD (2019). TALIS 2018 Results (Volume I)

Teacher preparation refers to the formal education and training that prospective teachers undergo before entering the profession, while professional development encompasses the

continuous learning processes teachers engage in throughout their careers. Both components are essential for ensuring high-quality education and improving student outcomes. (UNESCO (2015).

The complexity of modern classrooms—characterized by diversity, inclusion, and digital integration—requires teachers to possess not only subject knowledge but also pedagogical skills, emotional intelligence, and adaptability. (Shulman, Lee S. (1987). Knowledge and teaching: Foundations of the new reform. Traditional models of teacher education are no longer sufficient to meet these demands. Instead, there is a growing emphasis on competency-based training, reflective practice, and collaborative learning.

This study aims to analyze the effectiveness of current teacher preparation and professional development models, identify key challenges, and propose strategies for improvement.(Fullan, Michael (2007).

Materials and Methods

This research adopts a qualitative methodology, combining systematic literature review and comparative analysis. Academic journals, policy reports, and educational frameworks published between 2000 and 2025 were analyzed to identify trends and best practices in teacher education. (Hattie, John (2009).

Data Collection

Sources included:

- Peer-reviewed journals
- UNESCO and OECD reports
- National education policy documents
- Case studies from different countries (Villegas-Reimers, Eleonora (2003).

Analytical Framework

The study applies thematic analysis to categorize findings into:

- Initial teacher preparation models
- Professional development strategies
- Challenges and barriers
- Outcomes and effectiveness (Cochran-Smith, Marilyn & Lytle, Susan L. (1999).

Examples from Finland, Singapore, and the United States were compared to evaluate different approaches to teacher training.

Discussion

Models of Teacher Preparation

Teacher preparation programs vary significantly across countries but generally include three core components:

1. Subject knowledge
2. Pedagogical training
3. Practical teaching experience

In countries like Finland, teacher education is research-based, requiring a master's degree and emphasizing critical thinking. In contrast, the United States offers more diverse pathways, including alternative certification programs. (Avalos, Beatrice (2011).

A key issue is the gap between theory and practice. Many graduates report feeling unprepared for real classroom challenges, highlighting the need for more practical training and mentorship.

Professional Development Practices

Effective professional development is:

- Continuous rather than one-time
- Collaborative rather than individual
- Practice-oriented rather than theoretical

Popular models include:

- Workshops and seminars
- Peer coaching
- Professional learning communities (PLCs)
- Online training platforms

Research shows that teachers who engage in collaborative professional development demonstrate higher levels of instructional effectiveness. (Guskey, Thomas R. (2002). Professional development and teacher change. *Teachers and Teaching*, 8(3), 381–391)

Challenges in Teacher Development

Several barriers hinder effective teacher preparation and development:

- Lack of funding
- Limited access to resources
- Resistance to change
- Time constraints

Additionally, many programs fail to address individual teacher needs, resulting in low engagement and limited impact.

Role of Technology

Digital tools have revolutionized teacher training by enabling:

- Online courses (MOOCs)
- Virtual classrooms
- AI-based teaching assistants

Technology allows for personalized learning and global collaboration, making professional development more accessible.

Conclusion. Teacher preparation and professional development are critical components of educational quality. This study highlights the need for integrated, flexible, and sustainable training systems that align with modern educational demands.

Key recommendations include:

- Strengthening the link between theory and practice
- Promoting collaborative learning environments
- Integrating technology into training programs
- Supporting lifelong learning

Future research should focus on evaluating the long-term impact of professional development initiatives on student outcomes.

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INNOVATIVE APPROACHES IN THE METHODOLOGICAL TRAINING OF FOREIGN LANGUAGE TEACHERS

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Abstract: The article provides a comprehensive analysis of the “case method,” which is considered one of the creative and modern approaches to teaching a foreign language for specific purposes. The main essence of this method lies in enabling students to engage independently in foreign-language communication and participate in decision-making processes within artificially modeled situations that closely resemble real professional activities. Such an approach contributes not only to the development of learners' language competence but also to the improvement of their analytical thinking, situational assessment, problem-solving, and presentation skills.

The article emphasizes that case-based tasks are rich in content and create opportunities for students to assimilate new language material more effectively by relying on previously acquired knowledge and experience. This method allows learners to apply field-specific terminology in a practical context, thereby positively influencing the formation of professional communication skills. At the same time, the application of the case method serves as an effective tool for developing essential competencies such as management, leadership, teamwork, and strategic planning.

Based on research findings, the “case method” is regarded as one of the most effective approaches for achieving the objectives of professionally oriented foreign language instruction. It makes the teaching process more motivating and practice-oriented, ensures active student engagement, and strengthens their independent learning skills. Moreover, through case-based activities, students become familiar with communication models of different cultures, gain a deeper understanding of intercultural differences, and consequently enhance their intercultural adaptation abilities. As a result, the case method is evaluated as an innovative teaching tool that not only improves the quality of language learning but also increases students’ readiness for future professional activities.

Keywords: foreign language teaching methodology, case method, case study, communicative competence, intercultural communication

XARİCİ DİL MÜƏLLİMLƏRİNİN METODİKİ HAZIRLIĞINDA İNNOVATİV YANAŞMALAR

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Xülasə: Məqalədə xüsusi məqsədlər üçün xarici dilin tədrisində yaradıcı və müasir yanaşmalardan biri hesab olunan “keys metodu” geniş şəkildə təhlil edilir. Bu metodun əsas mahiyyəti tələbələrin real peşə fəaliyyətinə yaxınlaşdırılmış, süni şəkildə modelləşdirilmiş situasiyalarda müstəqil şəkildə xarici dildə ünsiyyət qurmasına və qərar vermə prosesində iştirak etməsinə əsaslanır. Belə yanaşma öyrənənlərin yalnız dil biliklərini deyil, həmçinin onların analitik düşünmə, situasiyanı qiymətləndirmə, problem həll etmə və təqdimat bacarıqlarını da inkişaf etdirir.

Məqalədə vurğulanır ki, keys tapşırıqları məzmun baxımından zəngin olmaqla yanaşı, tələbələrin əvvəlcədən əldə etdikləri bilik və təcrübələrə əsaslanaraq yeni dil materialını daha asan mənimsəməsinə şərait yaradır. Bu metod tələbələrin ixtisas sahəsinə uyğun terminologiyayı praktik mühitdə tətbiq etməsinə imkan verir və nəticədə peşəkar kommunikasiya bacarıqlarının formalaşmasına müsbət təsir göstərir. Eyni zamanda, keys metodunun tətbiqi idarəetmə, liderlik, komanda işi və strateji planlaşdırma kimi mühüm bacarıqların inkişafında da effektiv vasitə kimi çıxış edir.

Tədqiqatların nəticələrinə əsasən, “keys metodu” peşə yönümlü xarici dil tədrisinin məqsədlərinə nail olmaq baxımından yüksək nəticə verən üsullardan biridir. Bu metod tədris prosesini daha motivasiyaedici və praktik yönümlü edir, tələbələrin dərslə fəal cəlb olunmasını təmin edir və onların müstəqil öyrənmə bacarıqlarını gücləndirir. Bununla yanaşı, keys tapşırıqları vasitəsilə tələbələr müxtəlif mədəniyyətlərə məxsus kommunikasiya modelləri ilə tanış olur, mədəniyyətlərarası fərqləri daha dərinlən dərk edir və beləliklə, onların mədəniyyətlərarası adaptasiya imkanları genişlənir. Nəticə etibarilə, keys metodu həm dil öyrənmənin keyfiyyətini artırır, həm də tələbələrin gələcək peşə fəaliyyətinə hazırlıq səviyyəsini yüksəldən innovativ tədris vasitəsi kimi qiymətləndirilir.

Açar sözlər: Xarici dillərin tədrisi metodikası, Keys metodu, Keys-stadi (keys araşdırması), Kommunikativ kompetensiya, Mədəniyyətlərarası kommunikasiya

Introduction

The contemporary stage of educational development is characterized by a profound transition toward a multi-level, competence-based system of higher education, which reflects broader global trends in the modernization of educational paradigms. This transformation is driven by the growing need to align educational outcomes with the demands of the knowledge-based economy, digital

transformation, and increasing internationalization of academic and professional environments. As a result, higher education institutions are required to reconsider not only the content of curricula but also the methodological foundations of teaching and learning processes.

This shift necessitates substantial changes in the content and organization of the educational process, including the integration of interdisciplinary knowledge, the application of student-centered learning strategies, and the implementation of innovative methodological tools that support active learning. In this context, the role of the teacher is also being redefined. The traditional model of knowledge transmission is gradually being replaced by the role of the teacher as a facilitator, mentor, and organizer of learning activities within a learner-centered and creativity-oriented pedagogical paradigm. This new approach emphasizes individual learning trajectories, autonomy, and the development of critical and reflective thinking skills.

In foreign language teaching, this transformation is particularly significant, as modern society requires graduates who are not only linguistically competent but also capable of using a foreign language effectively in diverse professional, academic, and intercultural communication contexts. Language learning is no longer viewed as a purely linguistic process; rather, it is understood as an integral component of professional competence formation. Therefore, foreign language education increasingly focuses on the development of communicative competence, intercultural awareness, and the ability to operate in complex, multilingual environments.

Consequently, the integration of interactive teaching methods has become an essential requirement of modern higher education. Such methods encourage active student participation, collaborative learning, and the practical application of knowledge. They also foster motivation, autonomy, and deeper cognitive engagement, which are crucial for effective language acquisition. Interactive approaches, including task-based learning, project-based learning, and problem-solving activities, create conditions in which students can meaningfully use the target language in realistic contexts.

Among these methods, the case method occupies a particularly important position due to its strong orientation toward real-life problem-solving, professional communication, and the development of higher-order thinking skills. By analyzing authentic or simulated professional situations, students are encouraged to evaluate information, identify problems, propose solutions, and justify their decisions using the foreign language. This not only enhances linguistic proficiency but also contributes to the development of analytical, strategic, and decision-making abilities. Moreover, the case method promotes collaboration and discussion, thereby strengthening students' communicative competence and preparing them for real-world professional interactions in multicultural environments.

Materials and methods

This study is based on a theoretical and analytical methodology, including comparative analysis, synthesis of scientific literature, and pedagogical interpretation of existing research findings. The purpose is to examine the methodological foundations of the case method and its application in foreign language teaching.

The analysis of international and national pedagogical sources shows that foreign language teaching is implemented through diverse methodological systems depending on learning objectives and educational contexts. These include:

- ✓ communicative approach
- ✓ sociocultural approach

- ✓ intensive language teaching method
- ✓ direct method (Berlitz approach)
- ✓ business-oriented (activity-based) approach

These approaches differ in their methodological orientation but share a common goal: improving learners' communicative competence in real and professional contexts (Richards & Rodgers, 2014).

Results and discussion

3.1. Diversity of Methodological Approaches in Language Teaching

Modern foreign language education is characterized by methodological pluralism. The communicative approach emphasizes meaningful interaction, while the intensive method focuses on rapid acquisition of functional language skills for specific professional purposes.

The direct method excludes the use of the native language and aims to develop thinking directly in the foreign language. The business-oriented approach integrates language learning with professional communication tasks in fields such as business correspondence, presentations, negotiations, and reporting.

This diversity reflects the complexity of language acquisition processes and the necessity of adapting teaching methods to learners' needs and professional goals.

3.2. Formation of Professional Communicative Competence

The development of foreign-language professional communicative competence is one of the central objectives of higher education. However, different speech functions have unequal pedagogical significance in professional language teaching.

The main functions include:

- ✓ instrumental function (information transmission)
- ✓ regulatory function (management of activity)
- ✓ personal-emotional function (expressive communication)
- ✓ heuristic function (expression of understanding and reasoning)
- ✓ social function (interaction beyond limited groups)
- ✓ informational-scientific and analytical function

These functions contribute to the formation of a comprehensive communicative competence necessary for professional activity in a globalized environment (Zakharova, 2011).

3.3. The Case Method: Definition and Theoretical Foundations

The case method (case study method) is an interactive teaching approach based on the analysis of real or realistically modeled situations. It is widely used in business, economics, law, and increasingly in foreign language teaching.

According to Dolgorukov (2002), the case method belongs to advanced active learning strategies that enhance students' ability to apply theoretical knowledge in practical contexts.

The theoretical foundations of the method were developed by researchers such as Duff (2008), Faltis (1997), and Heap (1997), who emphasized its role in developing analytical and communicative skills.

3.4. Pedagogical Implementation of the Case Method in Foreign Language Teaching

In foreign language classrooms, the case method is implemented through structured stages:

comprehension of professionally oriented case material (reading and interpretation)

translation and linguistic analysis of key information

independent problem-solving and internal monologue in the target language

group discussion and argumentation
presentation of solutions and evaluation

These stages activate both monologic and dialogic forms of speech and develop spontaneous and prepared communication skills.

3.5. Educational Outcomes and Skills Development

The case method significantly enhances a wide range of cognitive and communicative skills, including:

- ✓ critical and creative thinking
- ✓ analytical reasoning
- ✓ presentation and public speaking skills
- ✓ teamwork and collaboration
- ✓ decision-making abilities
- ✓ professional reading and information processing
- ✓ written and oral communication competence

In addition, it strengthens students' self-confidence and promotes active participation in the learning process.

3.6. Pedagogical Advantages and Psychological Impact

Interactive learning activities such as discussion, brainstorming, business games, and case analysis contribute to a positive psychological learning environment. They increase students' engagement, reduce communication barriers, and improve intellectual activity.

As a result, students develop a stronger sense of responsibility for their learning outcomes and become more autonomous learners.

3.7. Limitations of the Case Method

Despite its effectiveness, the case method has several limitations:
difficulty in ensuring full independent participation of all students
reduced efficiency in low-level language proficiency groups
need for careful methodological design and teacher preparation
lack of complete linguistic systematization if used alone

Therefore, the case method cannot replace traditional or other interactive methods but should be integrated into a mixed methodological system.

3.8. Empirical and Applied Findings

Applied research confirms that the case method is highly effective in achieving the objectives of professionally oriented foreign language teaching and intercultural adaptation. It enhances both linguistic competence and professional readiness.

However, its success depends on:

methodological justification
curriculum-level integration
careful lesson planning
teacher competence in interactive pedagogy

Conclusion

The case method represents one of the most effective interactive approaches in modern foreign language education. It integrates theoretical knowledge with practical application, thereby bridging the gap between classroom learning and real-world professional communication. Through the analysis of authentic or realistically simulated situations, learners are encouraged to actively

engage with language material in meaningful contexts, which significantly contributes to the development of both linguistic competence and professional communicative competence.

Its pedagogical value lies in its ability to simulate real-life professional situations, promote active and student-centered learning, and foster higher-order cognitive skills such as critical thinking, analytical reasoning, and problem-solving. In addition, the method enhances communication skills by requiring learners to express opinions, justify decisions, negotiate meanings, and collaborate in group discussions. As a result, students not only improve their foreign language proficiency but also develop essential soft skills such as teamwork, presentation abilities, and decision-making competence, which are crucial for their future professional activity.

Furthermore, the case method creates a dynamic and interactive learning environment in which students take an active role in the educational process rather than remaining passive recipients of knowledge. This shift increases learner motivation, strengthens autonomy, and promotes deeper cognitive engagement with the subject matter. It also allows students to develop intercultural awareness, as many case studies are based on cross-cultural or international professional contexts.

Nevertheless, the method is not universal and cannot be considered sufficient as a standalone teaching strategy. Its effectiveness largely depends on the learners' initial language proficiency level, the quality of case materials, and the methodological competence of the teacher. Therefore, it should be applied in combination with other teaching approaches such as communicative, task-based, and traditional methods to ensure balanced skill development and systematic language acquisition.

When properly implemented and methodically supported, the case method significantly enhances the overall quality of foreign language education. It not only improves linguistic performance but also prepares students for real professional environments, where effective communication, adaptability, and intercultural competence are essential for success.

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EVALUATION OF TEACHERS' PERFORMANCE AND QUALITY ASSURANCE

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Abstract: Every higher education institution seeks to evaluate the performance of its teaching staff in a fair manner. By obtaining comprehensive information about instructors, employers can guide their professional development and utilize their creative potential effectively. Evaluation practices in universities are essential for improving academic staff development. It is evident that higher education institutions prioritize improving teaching practice rather than focusing solely on experience. Based on the assessment of work potential, universities can develop instructors' knowledge, skills, attitudes, motivation, creativity, and collaboration in line with institutional goals.

This article presents a critical analysis of university teacher performance evaluation and proposes a framework for the continuous improvement of teaching practice. It describes the current situation at Sumgayit State University and introduces a model-based approach that may support innovation in teacher evaluation. The study includes key elements necessary for designing and planning effective evaluation systems.

The article also reviews commonly used evaluation models in educational institutions, highlighting their main features, advantages, and limitations. In addition, it emphasizes different interpretations of evaluation and the purposes for which institutions apply it. Considering that most universities rely on questionnaires to assess teaching performance, the study also discusses the use of this tool.

Keywords: quality, education, assessment, performance indicators, teacher evaluation models, higher education, teaching performance.

MÜƏLLİMLƏRİN FƏALİYYƏTİNİN QIYMƏTLƏNDİRİLMƏSİ VƏ KEYFİYYƏT TƏMİNATI

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Xülasə: Hər bir ali təhsil müəssisəsi müəllimlərin fəaliyyətini ədalətli şəkildə qiymətləndirməyə çalışır. Müəllimlər haqqında dolğun məlumat əldə etməklə işəgötürənlər onların peşəkar inkişafını düzgün istiqamətləndirə və yaradıcı potensialından səmərəli istifadə edə bilirlər. Universitetlərdə tətbiq olunan qiymətləndirmə təcrübələri akademik heyətin inkişafının təkmilləşdirilməsi üçün mühüm əhəmiyyət daşıyır. Ali təhsil müəssisələrinin yanaşmalarına əsasən, müəllimlərin fəaliyyətinin təkmilləşdirilməsi onların yalnız təcrübəsinə deyil, əsasən tədris prosesinin inkişafına yönəlir. İş potensialının qiymətləndirilməsi əsasında universitetlər müəllimlərin bilik, bacarıq, münasibət, motivasiya, yaradıcılıq və əməkdaşlıq kimi xüsusiyyətlərini institusional məqsədlərə uyğun şəkildə inkişaf etdirə bilirlər.

Bu məqalə universitet müəllimlərinin fəaliyyətinin qiymətləndirilməsinə tənqidi yanaşma təqdim edir və tədris təcrübəsinin davamlı təkmilləşdirilməsi üçün çərçivə təklif edir. Tədqiqatda Sumqayıt Dövlət Universitetində mövcud vəziyyət təsvir olunur və müəllimlərin qiymətləndirilməsində innovasiyanı dəstəkləyə biləcək model əsaslı yanaşma irəli sürülür. Məqalə effektiv qiymətləndirmə sistemlərinin dizaynı və planlaşdırılması üçün zəruri olan əsas elementləri əhatə edir.

Eyni zamanda, tədqiqatda təhsil müəssisələrində geniş istifadə olunan qiymətləndirmə modelləri nəzərdən keçirilir, onların əsas xüsusiyyətləri, üstünlükləri və çatışmazlıqları vurğulanır. Bundan əlavə,

qiymətləndirmənin müxtəlif izahları və ali təhsil müəssisələrinin bu prosesdən istifadə məqsədləri də izah edilir. Əksər universitetlərin müəllim fəaliyyətinin qiymətləndirilməsində anketlərdən istifadə etməsi nəzərə alınaraq, bu vasitənin tətbiqi də araşdırmada əks etdirilmişdir.

Açar sözlər: keyfiyyət, təhsil, qiymətləndirmə, iş göstəriciləri, müəllim qiymətləndirmə modellərini, Ali təhsil, tədris performansı

Introduction

The work of a university instructor exerts a significant influence on the development of knowledge and cognition in every society. It is a highly demanding profession that requires professional competencies and continuously improved professional knowledge, social competencies and the ability to develop them, as well as research skills, which are related to the ability to convey scientific results to students in a manner that they can understand and that will inspire their future development (Kravčáková et al., 2011).

Evaluation Models and Proposals

Objective-oriented model the model should be based on predefined objectives, that is, on the comparison of objectives and outcomes, in order to assess the degree to which the predefined objectives were accomplished. This model is based on the assumption that effectiveness is achieved by attaining the predefined objectives, which constitute the sole source of the criteria used for evaluation. Among the negative judgments associated with this model, a lack of flexibility is distinguished, as the information is obtained too late to be used for the improvement of the teacher or for continuing education, since this type of evaluation is oriented toward the assessment of the final achievement of the course, and the information it provides has limited capacity to establish a solid basis for value judgments.

In addition to the proposals presented within the **objective-oriented model**, there are also proposals that envisage evaluation based on the decisions they serve, regardless of the final results; these are referred to as the decision-oriented model. In this model, the data obtained are presented to decision-making bodies. A characteristic feature of this type of model is continuous and systematic evaluation, which is used for the planning and implementation of purpose-oriented services or proposals. From this perspective, evaluation moves beyond being understood merely as a control and measurement tool intended to be applied at the end of the process and is instead regarded as a means of feedback within the educational process.

The evaluation of teachers based on the **360-Degree Feedback Model** can provide a more objective assessment of instructors and may increase awareness of fairer evaluation practices, thereby exerting a significant impact on their motivation and on the quality of education in universities. The 360-Degree Feedback Model is a model for the evaluation of job performance oriented toward the description of work behavior (Suler, 2009). It is a relatively objective model capable of determining the level of abilities and skills of the individual being evaluated and of enhancing awareness of fair evaluation. It enables more effective personal development of the employee in relation to the needs of the organization. It provides the opportunity to obtain evaluations of work performance from surrounding individuals, namely supervisors, colleagues, subordinates, and external clients. The 360-Degree Feedback Model supports the provision of transparent feedback, establishes individual development plans, and promotes self-confidence as a foundation for development (Velas, 2011).

The fourth model, referred to as the naturalistic model, is based on negotiation and understanding. It recognizes the important role played by the audience and the transactional and phenomenological relationship between the participants and the evaluator. Emphasis is placed on

the qualitative description of the phenomenon and, to a lesser extent, on its quantitative description. Consequently, the primary objective is oriented toward description and interpretation rather than toward the evaluation and prediction characteristic of other models. "Stakeholders raise and clarify a series of issues, which in turn help to determine the aspects and procedures necessary to achieve the desired outcomes" (Stufflebeam & Shinkfield, 1995, p. 313).

Each of these models supports a particular perspective on evaluation. These perspectives indicate that the outcome-oriented model tends to be used in the evaluation of the performance of instructors in higher education institutions. This model is based on the assumption that the recipient of the service, namely the student, is able to evaluate the teacher's performance, and that, accordingly, the teacher's work is assessed on the basis of this opinion. From this standpoint, the aspects of process and context are no longer taken into account.

Numerous concepts indicate that understanding and conducting evaluation is important not only for learning purposes, but also for internalizing and justifying decisions, as well as for achieving objectives directed at the direct levels of reality. Mora (2004) notes that the meaning of evaluation and its use are understood in accordance with the needs and objectives prioritized by a particular educational institution. For some, the focus may be directed toward control and measurement, while for others it may be oriented toward the validity of objectives or accountability.

Preferred Methods in the Evaluation of Teaching Performance

Research conducted worldwide on the evaluation of university instructors' performance indicates that student surveys are the most commonly used method for this purpose.

Questionnaires include items concerning the instructor's mastery of course content, ability to work with groups, use of pedagogical techniques, methods employed to assess learning, and other related issues. Within this long-standing tradition of using questionnaires to evaluate teaching performance, although the mode of administration has changed, their content and use have remained largely consistent. Previously, they were completed manually; today, they are frequently administered online.

The use of questionnaires for evaluating teaching performance has several advantages:

- It allows students to express their opinions about their instructors, thereby generating institutional and policy-related benefits for the educational establishment.
- Students, by their nature, are considered the most extensive observers of teaching and are, in turn, in a unique position to evaluate course quality, as well as the dedication and preparedness of the teaching staff.
- The reliability of student observations is generally high. This depends, among other factors, on the number of respondents; therefore, if appropriate samples are selected, reliability can be ensured.

The evaluation of instructors by students is regarded as one of the most effective systems for assessing teaching performance, as students are the only individuals in a position to determine whether instructors have effectively supported their learning.

Although the use of questionnaires is considered one of the simplest methods for evaluating teaching performance—since they are easy to administer, can be easily processed computer programs, and can be graphically represented for comparison with others' scores, and, moreover, allow administrators to review numerical data with ease—it is also evident that such a tool does not measure teaching performance itself, but rather measures students' perceptions.

A study conducted by Carrel & West (2010) revealed that students tend to rate more highly those instructors who help them achieve better grades in the current course, rather than those who promote deeper understanding that leads to improved performance in subsequent courses but require greater effort in the present. This suggests that students evaluate the support they receive for long-term learning differently from immediate outcomes.

For evaluation to effectively fulfill its function, it is necessary to rely not on a single source of information but on multiple sources, which, in turn, promotes collaboration, reflection, and inquiry among instructors (DeCosta et al., 2016).

Teaching performance evaluation should be comprehensive and used for legitimate purposes; at the same time, attention must be paid to ensuring that it does not undermine the instructor's self-confidence, honesty, or personal and professional dignity (Rueda & Rodriguez, 1996).

From this perspective, it can be stated that the evaluation of teaching performance must necessarily take into account a deep understanding of the instructor, which can only be achieved when evaluation processes consider the teacher's needs, interests, prior knowledge, learning styles, teaching styles, motivations, work habits and routines, attitudes and values, difficulties, and other relevant aspects. This implies that evaluation should be regarded as a systematic, continuous, and formative activity, as it contributes to the ongoing learning of the instructor and creates opportunities for improving formative processes.

In particular, it is proposed that teaching performance be assessed through three processes: self-evaluation, hetero-evaluation, and co-evaluation. **In self-evaluation**, the instructor assesses the course design, the logic of included activities, and the adequacy of those activities.

Hetero-evaluation takes into account students' opinions during both the middle and the end of the semester regarding their learning, achievement of course objectives, and the extent to which activities and resources have contributed to their learning in line with the program.

In co-evaluation, the instructor, together with the academic or faculty coordinator, reviews before the course whether the objectives are clearly formulated and whether they are achievable. On this basis, the appropriateness of resources and the planned methodology is examined. At the end of the course, self-evaluation and hetero-evaluation are critically reviewed in a reflective and suggestive manner in order to improve the instructor's teaching practice.

Current State of Teaching Performance Evaluation at Sumgait State University

Sumgait State University (SDU) is the only university located in Sumgait city of the Republic of Azerbaijan. SDU provides education in all three levels of higher education—bachelor's, master's, and doctoral studies—in both full-time and part-time formats. The university's academic staff and researchers actively participate in international education and research projects.

A key prerequisite for the successful operation of SDU is the maintenance and enhancement of quality across all academic and support activities. For quality assurance purposes, SDU uses both internal quality evaluation systems and external quality assessments conducted particularly by independent (certified/accredited) organizations. One of the most important tools of internal quality assessment is the annual comprehensive evaluation of the work performance of university instructors. This evaluation is prepared by the immediate supervisor after observing the employee during the evaluation period and reviewing their activities and outcomes in teaching, research, and publication. The evaluator conducts an evaluation interview with each employee, which is carried out in a collegial and discussion-based atmosphere. The purpose of this evaluation is to obtain

information about work performance achieved, with the aim of creating conditions for higher motivation and more effective future work of university instructors.

On the basis of this evaluation, the employee receives feedback about their performance from the perspective of their direct supervisor. This information should serve as a foundation for future work orientation, professional improvement, elimination of existing shortcomings, and prevention of future deficiencies.

Regarding the quality of any subject and instructor, students also have the opportunity to express their opinions through surveys that are part of SDU's academic information and education system. They can provide feedback at the end of each course during the relevant semester. The results are evaluated by the course instructor, department head, and faculty dean. Subsequently, they are presented to the Academic Councils of the faculties within the framework of annual evaluation of teaching performance and analysis of students' educational outcomes.

Methodology and Materials

This study adopts a mixed-methods research design to examine the evaluation of teacher performance in higher education institutions. The combination of qualitative and quantitative approaches allows for a more comprehensive understanding of evaluation practices and their effectiveness.

This study draws on a combination of primary and secondary data sources. Primary data were obtained through analysis of institutional evaluation practices and student feedback mechanisms at Sumgait State University. Secondary data were collected from existing academic literature, including journal articles, books, and relevant studies on teacher evaluation models.

The study employs a descriptive and analytical approach. Descriptive analysis is used to present the existing evaluation models, while analytical methods are applied to compare their strengths and limitations. Special attention is given to objective-oriented, decision-oriented, 360-degree feedback, and naturalistic evaluation models.

The data were interpreted by identifying common patterns in evaluation practices and examining how different models contribute to improving teaching quality. Ethical considerations were also taken into account, ensuring that all information used in the study is properly cited and based on publicly available sources.

Discussion

This study shows that the evaluation of teacher performance in higher education institutions is a multidimensional and context-dependent process that cannot be fully assessed using a single method. The analysis of different evaluation models indicates that each approach has its own advantages, but also certain limitations.

The results reveal that student questionnaires are the most widely used evaluation tool. However, this method is subjective in nature and may be influenced by short-term academic outcomes. Therefore, relying on a single source does not ensure a complete and objective assessment of teaching performance.

At the same time, institutional and contextual factors affecting teacher performance (such as workload, subject characteristics, class size, and others) are often not sufficiently considered in evaluation processes. This highlights the need for a more comprehensive approach.

In this regard, combining self-evaluation, peer evaluation, and student feedback provides more balanced and reliable results. In particular, the 360-degree feedback model offers a broader perspective and supports the professional development of academic staff.

Overall, teacher performance evaluation should not be seen as a control mechanism, but rather as a continuous process aimed at improving teaching quality and academic development.

Conclusion

The evaluation of teaching performance remains a task that requires further resolution in higher education institutions, as the preferred assessment methods tend to lean toward reductionism in the sense that they render invisible the political, social, and cultural context that constitutes part of the evaluated reality. By primarily focusing on students' perceptions of certain selected attributes, it is often not recognized that variables such as initial and continuous training, working conditions, the type of subject, curriculum structure, class size, and the nature of the discipline intervene in the instructor's performance. In addition, teachers' intentions and the internal logic of the course are among the aspects that are frequently not taken into account. Therefore, it is necessary that evaluation proposals consider the context in which the instructor performs their work.

Before deciding what is to be evaluated, it is essential to have a clear understanding of the purpose, since the meaning and value of this activity lie in the answer. Clarifying the objectives of evaluation is important, as they form the basis for decisions regarding theory, evaluation models, and the design most appropriate for the given purpose. From this perspective, it is necessary for each higher education institution to establish a clear instructional framework that defines criteria and indicators facilitating the implementation of teacher performance evaluation, involving various stakeholders in self-evaluation, co-evaluation, and hetero-evaluation processes.

One of the fundamental requirements of performance evaluation is achieving the highest possible level of accuracy and objectivity. Evaluation results must always be consistent and reliable, meaning they should not be distorted by bias or other assessment errors.

We believe that if the administration of Sumgait State University decides to apply the evaluation models discussed, this will lead to greater satisfaction among instructors and positively influence both motivation and the quality of education at the university. From the perspective of fairer evaluation practices, it will also increase overall teacher satisfaction. Satisfied teachers with high-quality work performance constitute one of the most important aspects of university progress.

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THE DEVELOPMENT OF PEDAGOGICAL SKILLS AND METHODOLOGICAL COMPETENCIES OF EDUCATORS IN PRESCHOOL EDUCATIONAL INSTITUTIONS

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Abstract: Preschool education represents a critical stage in early childhood development, laying the foundation for cognitive, emotional, social, and physical growth. Educators working in preschool institutions play a central role in shaping children's personality, behavior, and early learning experiences. The effectiveness of preschool education largely depends on educators' pedagogical skills and methodological competencies. This article explores the theoretical and practical dimensions of developing pedagogical skills and methodological competencies among preschool educators. Pedagogical skills include communication abilities, emotional interaction, classroom management, and the capacity to create a supportive learning environment. Methodological competencies refer to the selection and application of appropriate teaching strategies, including play-based learning, interactive methods, and child-centered approaches. The study is based on qualitative methods, including literature analysis, comparative study, and pedagogical observation. Findings indicate that continuous professional development, reflective practice, and innovative teaching strategies significantly improve educational quality in preschool settings. The results also emphasize the importance of collaboration between educators and parents in supporting holistic child development.

Keywords: preschool education, educator, pedagogical skills, methodological competencies, professional development, early childhood education

OKUL ÖNCESİ EĞİTİM KURUMLARINDA EĞİTİMCİLERİN PEDAGOJİK BECERİLERİNİN VE METODOLOJİK YETERLİLİKLERİNİN GELİŞİMİ

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Özet: Okul öncesi eğitim, erken çocukluk gelişiminde kritik bir aşamayı temsil eder ve çocuğun bilişsel, duygusal, sosyal ve fiziksel gelişiminin temelini oluşturur. Okul öncesi eğitim kurumlarında görev yapan eğitimciler, çocukların kişiliğinin, davranışlarının ve erken öğrenme deneyimlerinin şekillenmesinde merkezi bir role sahiptir. Okul öncesi eğitimin etkinliği büyük ölçüde eğitimcilerin pedagojik becerilerine ve metodolojik yeterliliklerine bağlıdır. Bu makale, okul öncesi eğitimcilerinin pedagojik becerilerinin ve metodolojik yeterliliklerinin geliştirilmesinin teorik ve pratik boyutlarını incelemektedir. Pedagojik beceriler; etkili iletişim kurma, duygusal etkileşim sağlama, sınıf yönetimi ve destekleyici bir öğrenme ortamı oluşturma kapasitesini içermektedir. Metodolojik yeterlilikler ise uygun öğretim stratejilerinin seçilmesi ve uygulanması, oyun temelli öğrenme, etkileşimli yöntemler ve çocuk merkezli yaklaşımların kullanılmasını kapsamaktadır. Araştırma; literatür analizi, karşılaştırmalı inceleme ve pedagojik gözlem gibi nitel yöntemlere dayanmaktadır. Elde edilen bulgular, sürekli mesleki gelişimin, yansıtıcı uygulamaların ve yenilikçi öğretim stratejilerinin okul öncesi eğitim kalitesini önemli ölçüde artırdığını göstermektedir. Ayrıca

sonuçlar, çocukların bütüncül gelişimini desteklemede eğitimciler ile ebeveynler arasındaki iş birliğinin önemini vurgulamaktadır.

Anahtar Kelimeler: okul öncesi eğitim, eğitimci, pedagojik beceriler, metodolojik yeterlilikler, mesleki gelişim, erken çocukluk eğitimi

Introduction

Preschool education is the first structured educational environment in which children begin to develop essential life skills. It plays a fundamental role in shaping personality, social behavior, emotional stability, and intellectual development. According to early childhood education theories, the first years of life are the most sensitive period for cognitive and emotional growth (Vygotsky, 1978).

In modern educational systems, the role of the preschool educator has significantly expanded. Educators are no longer seen merely as caregivers; they are now considered facilitators of learning, observers of development, and designers of educational environments. Their responsibility includes creating a safe, stimulating, and emotionally supportive atmosphere for children.

Pedagogical skills refer to the educator's ability to communicate effectively with children, understand their emotional needs, and guide their learning process. Methodological competencies, on the other hand, involve the ability to design, organize, and implement educational activities using appropriate teaching methods and strategies (Pianta et al., 2012).

The development of these competencies is essential for improving the quality of preschool education and ensuring that children receive developmentally appropriate learning experiences.

Materials and Methods

This study employs qualitative research methodology. The following methods were used:

1. Literature Analysis

Scientific articles, pedagogical theories, and educational frameworks related to preschool education were analyzed to understand the conceptual foundations of pedagogical skills and methodological competencies.

2. Comparative Analysis

Traditional teacher-centered approaches were compared with modern child-centered educational models to identify differences in effectiveness.

3. Pedagogical Observation

Observations were conducted in preschool environments to analyze educator-child interactions, teaching methods, and classroom management strategies.

The combination of these methods provided a comprehensive understanding of how pedagogical and methodological competencies influence educational quality.

Pedagogical Skills in Preschool Education

Pedagogical skills are fundamental to effective teaching in early childhood education. These skills include communication, emotional intelligence, classroom management, and observational abilities.

An effective educator must be able to communicate clearly and warmly with children, using language appropriate to their developmental level. Emotional intelligence is equally important, as educators must respond to children's emotional needs with empathy and patience.

Classroom management is another essential skill. It involves organizing group activities, maintaining discipline in a positive manner, and ensuring active participation from all children.

According to research, positive teacher-child interactions significantly influence children's cognitive and social development (Hamre & Pianta, 2005). Educators who demonstrate high levels of pedagogical competence create more supportive and engaging learning environments.

Materials and Methods

Methodological competencies refer to the educator's ability to apply effective teaching strategies and organize educational activities according to children's developmental needs.

One of the most important methodologies in preschool education is play-based learning. Through play, children explore their environment, develop creativity, and build social skills. This approach aligns with constructivist theories of learning, which emphasize active participation in knowledge construction (Piaget, 1952).

Other important methodological competencies include:

- Designing age-appropriate lesson plans
- Using visual and interactive materials
- Integrating storytelling and role-play activities
- Applying digital learning tools in early education
- Assessing children's progress through observation

Modern preschool education emphasizes child-centered learning, where the educator acts as a facilitator rather than a lecturer.

Professional Development of Educators

Continuous professional development is a fundamental requirement for ensuring high-quality preschool education. In today's rapidly changing educational environment, educators are expected not only to possess initial qualifications but also to continuously update their knowledge, skills, and competencies. The development of both pedagogical skills and methodological competencies depends greatly on lifelong learning and systematic professional growth.

Educators are encouraged to engage in lifelong learning through a variety of structured and informal activities. Training courses provide systematic knowledge on modern teaching methods, child psychology, classroom management, and curriculum implementation. These courses help educators stay updated with current educational standards and improve their ability to design effective learning experiences for children.

Seminars and workshops also play an important role in professional development. They offer opportunities for educators to explore innovative teaching strategies, discuss real classroom challenges, and share practical solutions. Unlike formal training, workshops are often interactive and allow participants to actively engage in discussions, case studies, and practical demonstrations. This enhances the practical application of theoretical knowledge.

Another essential component of professional development is peer collaboration. Working together with colleagues enables educators to share experiences, exchange ideas, and learn from each other's practices. Collaborative environments foster teamwork, professional dialogue, and the development of a supportive professional community. Through peer observation and feedback, educators can identify new strategies for improving their teaching effectiveness.

Reflective practice is considered one of the most powerful tools for professional growth. It allows educators to critically analyze their own teaching methods, evaluate their effectiveness, and identify both strengths and areas for improvement. Through reflection, educators become more aware of their instructional decisions and their impact on children's learning and behavior.

According to Schön (1983), reflective practice is essential for developing professionals who are capable of continuous self-improvement and adaptive expertise.

In addition, collaboration among educators contributes significantly to professional development. When educators share best practices, discuss challenges, and jointly develop solutions, the overall quality of education improves. This collaborative culture encourages innovation, creativity, and the adoption of effective teaching strategies across institutions.

Overall, continuous professional development ensures that educators remain competent, confident, and capable of meeting the diverse needs of children in preschool education settings.

Results and Discussion

The findings of this study clearly demonstrate that educators who possess strong pedagogical skills and well-developed methodological competencies are significantly more effective in supporting children's holistic development in preschool education settings. These competencies directly influence not only academic readiness but also emotional stability, social interaction, and cognitive growth of children.

One of the most important findings is that effective educator–child communication plays a fundamental role in early childhood development. Educators who communicate in a warm, supportive, and developmentally appropriate manner help children feel safe and valued. This emotional security creates a positive learning environment and contributes to better emotional regulation, self-confidence, and social behavior in children (Vygotsky, 1978).

Another key result indicates that play-based and interactive teaching methods significantly increase children's engagement in learning activities. Play is a natural form of learning in early childhood, allowing children to explore, experiment, and develop creativity. When educators integrate structured play activities into the learning process, children demonstrate higher motivation, improved attention span, and stronger problem-solving skills.

The study also reveals that structured and well-planned educational activities enhance learning outcomes. Educators who prepare lesson plans according to developmental stages and individual differences among children are more successful in achieving educational goals. Proper planning ensures that learning activities are purposeful, balanced, and aligned with children's cognitive abilities.

In addition, reflective practice emerged as a crucial factor in improving teaching quality over time. Educators who regularly analyze their teaching methods, evaluate their effectiveness, and make necessary adjustments show continuous professional growth. Reflective practice encourages self-awareness and supports lifelong learning in the teaching profession (Schön, 1983).

Furthermore, the research highlights the importance of collaboration between educators and parents in preschool education. When educators and families work together, children receive consistent guidance and support both at home and in educational institutions. This cooperation strengthens the learning process and ensures the continuity of developmental outcomes.

Finally, the study shows that innovative teaching strategies, such as the use of digital learning tools, multimedia resources, and interactive storytelling, significantly improve children's motivation and active participation. These modern approaches make learning more engaging and accessible, especially in a rapidly changing educational environment.

Conclusion

The development of pedagogical skills and methodological competencies among preschool educators is essential for ensuring high-quality early childhood education. This study confirms that

educators play a decisive role in shaping children's cognitive, emotional, and social development during the most critical stage of human growth. The findings of this research highlight several key conclusions. First, strong pedagogical skills enable educators to create supportive, emotionally safe, and developmentally appropriate learning environments. Such environments are essential for fostering children's confidence, communication skills, and social adaptation. Second, methodological competencies ensure the effective organization and delivery of educational activities. Educators who are skilled in selecting and applying appropriate teaching methods—particularly play-based and child-centered approaches—achieve better learning outcomes and higher levels of student engagement. Third, continuous professional development is a fundamental requirement for modern educators. Participation in training programs, workshops, and reflective practice sessions contributes to the constant improvement of teaching quality and professional competence.

Finally, the study emphasizes that preschool education should not rely solely on traditional teaching methods. Instead, educational institutions should actively promote innovative, interactive, and technology-supported learning approaches to meet the needs of contemporary learners. In conclusion, strengthening the pedagogical and methodological competencies of preschool educators is a strategic priority for improving the overall quality of early childhood education. Investment in educator training and professional growth ultimately leads to more effective learning experiences and better developmental outcomes for children.

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THE IMPACT OF PEDAGOGICAL MANAGEMENT MODELS ON THE DYNAMICS OF TEACHERS' SELF-AFFIRMATION AND PROFESSIONAL DEVELOPMENT

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Summary: In modern pedagogical management, teacher development must shift from a control mechanism to a model of support and partnership. At the core of this approach lies growth-oriented feedback, which aims not to penalize teachers for their mistakes, but to motivate them to master new technological trends and methodologies. In the era of digital transformation, the "Technological Leadership" model of management requires not only enhancing teachers' ICT skills but also visualizing their achievements through digital portfolios. This both increases a teacher's competitiveness and facilitates the exchange of experience on online platforms through network collaboration. To reach the maximum level of development, it is essential to move away from hierarchical structures toward a "flat management" model. This model grants teachers academic freedom and autonomy, which are fundamental conditions for creativity. Simultaneously, systemic psychological support mechanisms must be established to prevent professional degradation and burnout. In conclusion, pedagogical management should be built on humanism and agility rather than fear and pressure. If management perceives the teacher as a value-creating partner, teacher development will transition from a formal requirement to a real and sustainable process. The dynamics of teacher development depend directly on how inclusive and supportive the management model is.

Key words: Teacher development, digital management, humanistic approach.

PEDAQOJİ İDARƏETMƏ MODELƏRİNİN MÜƏLLİMİN ÖZÜNÜTƏSDİQ VƏ PEŞƏKAR İNKİŞAF DİNAMİKASINA TƏSİRİ

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Xülasə. Müasir pedaqoji idarəetmədə müəllim inkişafı artıq nəzarət mexanizmindən çox, bir dəstək və tərəfdaşlıq modelinə çevrilməlidir. Bu yanaşmanın mərkəzində dayanan inkişafyönümlü geri bildirim müəllimi səhvlərinə görə cəzalandırmaq deyil, onu yeni texnoloji trendləri və metodikaları mənimsəməyə həvəsləndirmək məqsədi güdür.

Rəqəmsal transformasiya dövründə idarəetmənin "Texnoloji Liderlik" modeli müəllimlərin İKT bacarıqlarını artırmaqla yanaşı, rəqəmsal portfoliolar vasitəsilə onların nailiyyətlərinin vizuallaşdırılmasını tələb edir. Bu, həm müəllimin rəqəbatədavamlılığını artırır, həm də şəbəkə əməkdaşlığı vasitəsilə onlayn platformalarda təcrübə mübadiləsinə şərait yaradır.

İnkişafın maksimal səviyyəyə çatması üçün iyerarxik strukturlardan imtina edilərək "üfqi idarəetmə" (flat management) modelinə keçid vacibdir. Bu model müəllimə akademik azadlıq və avtonomiya verir ki, bu da yaradıcılığın əsas şərtidir. Eyni zamanda, peşəkar deqradasiyanın və tükənmənin qarşısını almaq üçün sistemli psixoloji dəstək mexanizmləri qurulmalıdır.

Yekun olaraq, pedaqoji idarəetmə qorxu və təzyiq üzərində deyil, humanistlik və çeviklik üzərində qurulmalıdır. Əgər idarəetmə müəllimi dəyər yaradan bir tərəfdaş kimi qəbul edərsə, müəllimin inkişafı formal xarakterdən çıxaraq real və davamlı xarakter alar. Müəllim inkişafının dinamikası birbaşa idarəetmə modelinin nə dərəcədə inklüziv və dəstəkləyici olmasından asılıdır.

Açar sözlər: Müəllim inkişafı, rəqəmsal idarəetmə, humanist yanaşma.

In modern pedagogical management paradigms, the process of teacher development must distance itself from traditional control and inspection mechanisms, evolving instead into a model of

support and strategic partnership built on mutual trust (Pashayev, 2019). The growth-oriented feedback mechanism, which forms the central pillar of this innovative approach, does not aim to administratively penalize teachers for technical or methodological errors; on the contrary, it stimulates them to deeply master the latest technological trends, modern educational curricula, and diverse teaching methodologies (Veysova, 2007). In the era of digital transformation and the Fourth Industrial Revolution, the “Technological Leadership” model applied in the management of educational institutions requires fundamentally increasing teachers’ individual ICT skills while necessitating the visualization and systematization of their pedagogical achievements through modern digital portfolios. Such an approach significantly enhances a teacher’s competitiveness in the modern education market and creates fertile ground for international experience exchange through various network collaboration platforms and online resources (Azerbaijan Institute of Education, 2023). To reach the maximum level of efficiency in teacher development and fully realize internal potential, transitioning from rigid hierarchical structures and bureaucratic command to a flexible “flat management” model carries strategic importance. The horizontal model grants teachers broad academic freedom, methodological choice, and professional autonomy, which is the most fundamental condition for pedagogical creativity and innovative initiative (Huseynzade, 2021). Simultaneously, to prevent professional degradation, emotional burnout, and job dissatisfaction within the educational environment, systemic, continuous, and accessible psychological support mechanisms, as well as a corporate social environment, must be established within educational institutions. In conclusion, it should be noted that the subject of modern pedagogical management should not be established on a paradigm of fear, administrative pressure, and punishment, but on principles of humanism, transparency, and functional agility (Ahmadov, 2010). If school management perceives the teacher not merely as an executor but as a value-creating strategic partner and intellectual leader, then teacher development will transition from an artificial and formal nature into a real, measurable, and sustainable character. The quality and quantity dynamics of teacher development depend directly on how inclusive, motivating, and supportive the existing management model is. The global challenges of the modern era and the processes of inter-civilizational integration have radically changed the objectives facing the education system and the requirements placed on the teacher’s personality (Rustamov, 2012). In this complex transformation process, the management of educational institutions no longer serves a mere daily administrative-command and clerical function but fulfills the mission of strategic human resource management and the proper direction of teacher potential (Pashayev, 2019). Pedagogical management models create a direct and unbreakable correlation between the internal psychological well-being of the teaching staff, their natural need for self-assertion, and lifelong professional development. The scientific and practical relevance of this study lies in the fact that an incorrectly chosen, unprofessional management strategy can lead even the most talented and high-potential staff to professional burnout and stagnation, while humanist and democratic models stimulate the hidden creative potential of teachers and encourage them toward new successes. Within the framework of the Paradigm Shift in Pedagogical Management, it should be noted that traditional management models were primarily based on rigid subordination, non-transparent control, and punishment mechanisms (Ahmadov, 2010). However, modern pedagogical thought and management science propose models based on collective responsibility, such as “Shared Leadership” and “Transformational Management.” Comparing Authoritarian vs. Democratic Management reveals that under authoritarian conditions, the teacher functions as a mere “tool” executing orders from above, and in

this environment, self-assertion is limited to meeting management's narrow requirements. In the democratic model, the teacher becomes an equal subject of management and a central figure in changes. This fundamental transition accelerates the self-assertion process by allowing teachers to clearly see their professional prestige both within the pedagogical collective and in the real results of the teaching-learning process (Huseynzade, 2021). The Cooperative Management Model encourages mutual cooperation, experience sharing, and methodological assistance rather than unhealthy and destructive competition among teachers. The principles of "teacher-to-teacher support" and mentoring shift professional development dynamics from an individual and closed level to a collective and open level, which positively impacts the quality indicators of the educational institution as a whole (Veysova, 2007). Regarding Teacher Self-Assertion: Psychological and Social Aspects, self-assertion is the process by which an individual proves their professional and personal value and their usefulness to society. In the field of pedagogical activity, this manifests as a teacher's sincere belief in the high effectiveness of their individual methodology and the official recognition and appreciation of these successful results by the "management-parent-student" triangle. Among Management Factors Influencing Self-Assertion, moral and material motivation measures, such as "Teacher of the Year" or "Best Innovative Project" competitions, further strengthen the teacher's social status in the collective (Azerbaijan Institute of Education, 2023). The Decision-Making Autonomy factor creates a sense of being a "full-fledged expert in the field" and responsibility by granting teachers a degree of creative freedom in the implementation of curricula and teaching plans. On the issue of Professional Self-Realization, if the current management model forces the teacher only into meaningless documentation and paper-based accountability, self-assertion remains entirely superficial and formal. In humanist management, the teacher's intrinsic motivation is in full harmony with their real creative activity and pedagogical output (Rustamov, 2012). Professional Development Dynamics and Strategic Planning show that teacher development is not a static state but a constantly renewing dynamic process. The management body can either accelerate this development impulse with correct methods or hinder it with bureaucratic obstacles. Within the framework of Mentoring and Coaching Models, modern management must transition from a "punitive inspection" function to a "guiding support" function (Pashayev, 2019). This includes school management acting as a mentor, scientifically identifying a teacher's professional gaps through modern gap analysis, and providing methodological assistance in building an Individual Development Plan (IDP). Development-Oriented Feedback refers to management reviews aimed not at breaking the teacher, but at sincerely encouraging them to learn new digital technologies and pedagogical approaches. In the era of Innovation and Digital Transformation in Management, teacher development dynamics are closely linked to their digital literacy and modern ICT skills. The "Technological Leadership" model flexibly adapts teachers to the changing digital educational environment, artificial intelligence tools, and online resources (Azerbaijan Institute of Education, 2023). The requirement to Create a Digital Portfolio directly serves to help the teacher visualize achievements, certificates, and methodological materials collected over the years, increasing competitiveness. Through Network Collaboration, experience sharing among teachers on various online platforms, international forums, and professional communities should be encouraged and supported by school management by all means. For the purpose of Practical Proposals and Model Optimization, the following steps are considered necessary to maximize teacher development: the phased reduction of hierarchical structure and prioritization of horizontal, team-based management; the strengthening of teacher autonomy on

legal and pedagogical levels and the formation of a real sense of academic freedom; and the organization of regular psychological seminars, individual consultations, and motivation tours to prevent professional degradation and emotional burnout (Huseynzade, 2021). Pedagogical management models collectively constitute the primary “ecosystem” of a teacher’s professional and moral life. If this ecosystem is built on the paradigm of administrative pressure, non-transparency, and fear, professional development will remain purely formal and yield no real benefit. Self-assertion and continuous professional growth only produce real results when the management model accepts the teacher as an indispensable, value-creating strategic partner and the bearer of a sacred mission in the educational process. The quality dynamics of teacher development are directly proportional to how flexible, transparent, scientifically based, and humanist the management model is. Consequently, the modernization of management paradigms is not merely an administrative issue but also the greatest investment in the quality of education and upbringing for future generations.

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CONTINUOUS PROFESSIONAL DEVELOPMENT AND TEACHER REFLECTION

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Abstract: This article provides a comprehensive analysis of the role of continuous professional development and teacher reflection in modern education systems. In the contemporary era, rapid changes in the field of education, digitalization, and the increasing demands of the labor market make it necessary for teachers to continuously update their knowledge and skills. In this context, continuous professional development is considered not only as the acquisition of new knowledge but also as a process of improving teaching practices and adapting to changing conditions. The article emphasizes that effective professional development programs have a positive impact on the quality of teaching and students’ learning outcomes. The study presents teacher reflection as one of the key components of continuous development. Reflection is explained as the process through which teachers analyze their own practices, identify their strengths and weaknesses, and plan their future activities based on this analysis. It is noted that both forms of reflection during the lesson and after the lesson—enhance the effectiveness of teaching. At the same time, the reflective approach enables teachers to act in a more flexible, adaptive, and student-centered manner. In

addition, the article highlights the importance of collaborative environments, exchange of experience, and practice-based learning in teacher development. Participation in professional learning communities, sharing experiences with colleagues, and collaborative activities strengthen reflective thinking and facilitate the implementation of innovative approaches. Furthermore, the supportive role of institutional backing and leadership in promoting teacher development is emphasized.

Keywords: Continuous professional development, teacher reflection, reflective teaching, educational quality, teacher competencies, lifelong learning

DAVAMLI PEŞƏKAR İNKİŞAF VƏ MÜƏLLİM REFLEKSİYASI

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Xülasə: Bu məqalədə davamlı peşəkar inkişaf və müəllim refleksiyasının müasir təhsil sistemində rolu geniş şəkildə təhlil olunur. Müasir dövrdə təhsil sahəsində baş verən sürətli dəyişikliklər, rəqəmsallaşma və əmək bazarının artan tələbləri müəllimlərin bilik və bacarıqlarını davamlı olaraq yeniləməsinə zəruri edir. Bu baxımdan davamlı peşəkar inkişaf müəllimlərin yalnız yeni biliklər əldə etməsi deyil, həm də tədris fəaliyyətini təkmilləşdirməsi və dəyişən şəraitə uyğunlaşması prosesi kimi dəyərləndirilir. Məqalədə vurğulanır ki, effektiv peşəkar inkişaf proqramları müəllimlərin tədris keyfiyyətinə və şagirdlərin təlim nəticələrinə müsbət təsir göstərir. Tədqiqatda müəllim refleksiyası davamlı inkişafın əsas komponentlərindən biri kimi təqdim olunur. Refleksiya müəllimin öz fəaliyyətini təhlil etməsi, güclü və zəif tərəflərini müəyyən etməsi və gələcək fəaliyyətini bu təhlil əsasında planlaşdırması kimi izah edilir. Məqalədə refleksiyanın həm dərs zamanı, həm də dərstdən sonra həyata keçirilən formalarının müəllim fəaliyyətinin səmərəliliyini artırdığı qeyd olunur. Eyni zamanda, reflektiv yanaşmanın müəllimlərin daha çevik, adaptiv və şagirdyönümlü fəaliyyət göstərməsinə şərait yaratdığı vurğulanır. Bundan əlavə, məqalədə əməkdaşlıq mühitinin, təcrübə mübadiləsinin və praktikaya əsaslanan öyrənmənin müəllim inkişafında əhəmiyyəti göstərilir. Müəllimlərin peşəkar öyrənmə icmalarında iştirakı, həmkarlarla qarşılıqlı təcrübə paylaşımı və birgə fəaliyyət reflektiv düşünməni gücləndirir və innovativ yanaşmaların tətbiqinə imkan yaradır. Eyni zamanda, institusional dəstəyin və rəhbərliyin təşviqedicisi rolunun müəllimlərin inkişaf prosesinə müsbət təsir göstərdiyi qeyd olunur. Nəticə etibarilə, davamlı peşəkar inkişaf və refleksiya müəllimlərin peşəkar kompetensiyalarının artırılmasında və təhsil keyfiyyətinin yüksəldilməsində mühüm rol oynayır.

Açar sözlər: Davamlı peşəkar inkişaf, müəllim refleksiyası, reflektiv təlim, təhsil keyfiyyəti, müəllim kompetensiyaları, ömürboyu öyrənmə

Introduction

In modern education systems, rapid social, technological, and economic changes have significantly influenced the role and responsibilities of teachers. The increasing demands of the labor market, along with the integration of digital technologies into education, require teachers to continuously update their knowledge and skills. Therefore, continuous professional development (CPD) has become an essential component of a teacher's professional life. It is no longer sufficient for teachers to rely solely on the knowledge acquired during their initial training; instead, they must engage in lifelong learning to remain effective in their profession (Day, 1999).

Continuous professional development refers to an ongoing process through which teachers enhance their competencies, adapt to new teaching methodologies, and improve their instructional

practices. Effective CPD programs are not limited to short-term training sessions; rather, they involve sustained, practice-oriented activities that support long-term professional growth. Research indicates that high-quality professional development positively impacts both teaching practices and student learning outcomes (Darling-Hammond, Hyler, & Gardner, 2017). In addition, such development processes contribute to teachers' motivation and professional commitment, enabling them to respond more effectively to the evolving needs of students.

A key element of continuous professional development is teacher reflection. Reflection can be defined as the process through which teachers critically analyze their own teaching practices, identify strengths and weaknesses, and make informed decisions for future improvement. According to Schön (1983), reflection occurs in two main forms: reflection-in-action, which takes place during the teaching process, and reflection-on-action, which occurs after the lesson. These reflective processes allow teachers to evaluate their performance and adjust their strategies in order to achieve better educational outcomes.

Moreover, reflective practice promotes a deeper understanding of teaching and learning processes. Teachers who engage in reflection become more aware of their instructional methods, classroom interactions, and students' needs. This awareness helps them to adopt more student-centered approaches and create supportive learning environments. Zeichner and Liston (1996) emphasize that reflective teaching encourages educators to question their assumptions, beliefs, and routines, thereby fostering professional growth and innovation.

Another important aspect of professional development is its impact on teacher change. Guskey (2002) argues that meaningful changes in teachers' beliefs and attitudes often occur after they observe improvements in student learning as a result of new teaching practices. This suggests that professional development should be closely linked to classroom practice and student outcomes. Similarly, Avalos (2011) highlights that long-term and collaborative professional development initiatives are more effective than isolated training sessions, as they provide teachers with opportunities to share experiences and learn from one another.

Methodology and Material

International studies also underline the importance of continuous professional learning. For instance, the OECD (2019) report on Teaching and Learning International Survey (TALIS) indicates that teachers who actively participate in professional development activities demonstrate higher levels of self-efficacy and instructional quality. These findings confirm that ongoing learning is essential for maintaining high standards in education systems.

In the context of Azerbaijan, the importance of continuous professional development has also been recognized at the national level. Educational reforms emphasize the need to enhance teachers' competencies and encourage reflective practices in order to improve the overall quality of education. Integrating reflective approaches into teacher training programs can contribute to the development of more competent, adaptive, and innovative educators.

In conclusion, continuous professional development and teacher reflection are closely interconnected processes that play a crucial role in improving educational quality. While professional development provides teachers with new knowledge and skills, reflection enables them to critically evaluate and effectively apply these competencies in practice. Together, these processes support teachers in becoming lifelong learners, capable of adapting to change and meeting the diverse needs of students. Therefore, promoting a culture of continuous learning and reflection should be considered a priority in modern education systems.

In addition to individual efforts, collaborative forms of professional development play a significant role in enhancing teachers' competencies. Professional learning communities, peer observations, and mentoring systems provide opportunities for teachers to exchange experiences, discuss challenges, and collectively search for effective solutions. Such collaborative environments not only support knowledge sharing but also strengthen teachers' reflective skills by exposing them to diverse perspectives. As noted by Avalos (2011), sustained interaction among teachers contributes to deeper professional learning and more meaningful changes in classroom practices. Furthermore, experiential learning is an important dimension of continuous professional development. The concept of learning through experience emphasizes that knowledge is constructed through active engagement and reflection on practice. Kolb's theory of experiential learning suggests that effective learning occurs when individuals go through a cycle of concrete experience, reflective observation, abstract conceptualization, and active experimentation (Kolb, 1984). In the context of teaching, this cycle enables educators to test new strategies, reflect on their effectiveness, and refine their approaches accordingly. This iterative process strengthens both professional competence and reflective thinking.

Discussion

Another critical factor influencing the success of professional development is its relevance to teachers' actual classroom needs. Programs that are disconnected from real teaching contexts often fail to produce meaningful outcomes. Therefore, it is essential to design professional development initiatives that are practice-oriented and aligned with teachers' daily experiences. According to Guskey (2002), teachers are more likely to adopt new methods when they see clear evidence of their positive impact on student learning. This highlights the importance of linking professional development activities with measurable improvements in educational outcomes.

Teacher reflection also contributes to the development of adaptive expertise. In rapidly changing educational environments, teachers must be able to respond flexibly to new challenges, such as diverse student needs, technological advancements, and curriculum reforms. Reflective practitioners are better equipped to analyze complex situations, make informed decisions, and adjust their teaching strategies accordingly. Schön (1983) emphasizes that reflective thinking enables professionals to navigate uncertain and dynamic contexts, making it an essential skill for modern educators.

Moreover, continuous professional development and reflection have a direct impact on student learning. Teachers who actively engage in professional growth are more likely to implement innovative teaching methods, foster critical thinking, and create inclusive learning environments. Research findings indicate that improvements in teaching quality are closely associated with students' academic achievement and overall development (Darling-Hammond et al., 2017). This demonstrates that investing in teachers' professional learning ultimately benefits the entire education system.

At the institutional level, supporting teachers' professional development requires a systematic and strategic approach. Educational institutions should provide access to high-quality training programs, encourage collaborative learning, and create a supportive environment for reflection. The OECD (2019) emphasizes that school leadership plays a crucial role in promoting professional learning by fostering a culture of continuous improvement and innovation. School leaders who prioritize teacher development contribute to higher levels of instructional quality and student success.

Despite its importance, several challenges may hinder the effective implementation of continuous professional development. These include limited time, insufficient resources, and a lack of institutional support. In some cases, professional development activities may be perceived as formal requirements rather than meaningful learning opportunities. To address these challenges, it is necessary to ensure that professional development is relevant, accessible, and aligned with teachers' professional goals. Encouraging teachers' active participation and autonomy in selecting development activities can also enhance their engagement and motivation.

Conclusion

Finally, it is important to recognize that continuous professional development is not a one-time process but an ongoing journey. Teachers' professional growth evolves over time as they gain experience, encounter new challenges, and refine their practices. Reflection serves as a guiding mechanism in this journey, helping teachers to continuously evaluate their progress and identify areas for improvement. As Day (1999) suggests, lifelong learning is a fundamental characteristic of effective teachers, enabling them to remain responsive and innovative throughout their careers.

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THE ROLE OF ARTIFICIAL INTELLIGENCE IN EDUCATION: ADVANTAGES AND DISADVANTAGES

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Abstract: Artificial intelligence (AI) has emerged as a transformative force in a variety of sectors, and its impact on the education system is becoming increasingly significant. From personalized learning and intelligent instructional systems to predictive analytics and automated assessment, AI offers the potential to transform how students learn and how educators teach. This article explores the major applications of AI in education, focusing on the opportunities it provides as well as the theoretical, ethical, and practical challenges associated with its integration into educational environments. It examines how AI is implemented in personalized learning platforms to address students' individual needs, abilities, and learning progress. The article also considers the use of AI in administrative functions, assessment tools, feedback provision, and the prediction of student performance and success. In addition, attention is given to the potential of AI to improve teaching efficiency, support data-driven decision-making, and enhance students' learning experiences. However, the integration of AI also raises concerns related to data privacy, algorithmic bias, reliability, academic integrity, and unequal access to technology. The article concludes by emphasizing the importance of responsible and human-centered AI implementation, appropriate ethical guidelines, and teacher training to ensure that AI contributes to effective, inclusive, and high-quality education.

Keywords: Artificial intelligence, education, personalized learning, predictive analytics, ethical dilemmas, learning technologies, integrated tutoring systems.

TƏHSİLDƏ SÜNİ İNTELLEKTİN ROLU: ÜSTÜNLÜKLƏR VƏ ÇATIŞMAZLIQLAR

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Xülasə: Süni intellekt (Sİ) müxtəlif sahələrdə transformativ bir qüvvə kimi meydana çıxmış və onun təhsil sisteminə təsiri getdikcə daha əhəmiyyətli xarakter almışdır. Fərdiləşdirilmiş öyrənmədən tutmuş intellektual tədris sistemləri, proqnozlaşdırıcı analitika və avtomatlaşdırılmış qiymətləndirməyə qədər Sİ tələbələrin öyrənmə və müəllimlərin tədris proseslərini dəyişdirmək üçün geniş imkanlar təqdim edir. Bu məqalədə süni intellektin təhsildə əsas tətbiq sahələri, onun yaratdığı imkanlar, eləcə də təhsil mühitinə inteqrasiyası ilə bağlı nəzəri, etik və praktiki problemlər araşdırılır. Məqalədə Sİ-nin tələbələrin fərdi ehtiyaclarına, bacarıqlarına və öyrənmə dinamikasına uyğunlaşdırılmış fərdiləşdirilmiş öyrənmə platformalarında tətbiqi nəzərdən keçirilir. Bundan əlavə, onun inzibati funksiyalarda, qiymətləndirmə vasitələrində, geribildirim prosesində və tələbələrin akademik göstəricilərinin və uğurunun proqnozlaşdırılmasında rolu təhlil edilir. Bununla yanaşı, məlumatların məxfiliyi və təhlükəsizliyi, alqoritmik qərəzlilik, etibarlılıq, akademik dürüstlük və texnologiyalara bərabər çıxış kimi problemlərə diqqət yetirilir. Məqalə Sİ-nin təhsildə səmərəli, inklüziv və keyfiyyətli nəticələrin təmin edilməsinə töhfə verməsi üçün məsuliyyətli və insanmərkəzli tətbiqin, etik prinsiplərin və müəllimlərin müvafiq hazırlığının vacibliyini vurğulayır.

Açar sözlər: Süni intellekt, təhsil, fərdiləşdirilmiş öyrənmə, proqnozlaşdırıcı analitika, etik dilemmalar, öyrənmə texnologiyaları, inteqrasiya olunmuş təlim sistemləri.

Introduction

The integration of artificial intelligence (AI) into education is revolutionizing traditional teaching and learning methods. AI systems are increasingly being used to personalize learning experiences, automate administrative tasks, and provide real-time data analytics to improve student outcomes. As educational institutions around the world seek ways to incorporate technology into

their classrooms, AI stands out as one of the most promising tools to enhance both student learning experiences and teacher effectiveness. This article examines current applications of AI in education, highlighting the key opportunities and challenges that come with its adoption. "The use of AI in education allows for the construction of a more dynamic, interactive, and personalized learning process, with the ultimate goal of improving the quality of education by providing learning materials tailored to the needs of the student and monitoring and analyzing their progress" [9]. We begin by examining the theoretical foundations of AI in education, and then discuss its practical application. The article concludes with an analysis of the ethical issues surrounding AI in educational settings, and suggests potential future directions for research and development in this area.

1. INTEGRATION OF ARTIFICIAL INTELLIGENCE IN EDUCATION

1. Artificial Intelligence and Learning Theories. The application of AI in education is deeply connected to various learning theories such as constructivism, behaviorism, and cognitivism. AI-based learning platforms often align with constructivist principles. Here, students actively participate in the construction of their own knowledge. AI systems are designed to adapt to the individual learning needs of students. Furthermore, AI's ability to track and analyze student performance provides valuable insights into cognitive processes, which allows for the creation of data-driven models that support personalized learning experiences.

2. Types of AI Technologies in Education Machine Learning: AI algorithms, especially machine learning models, are at the heart of many educational tools. They can analyze large datasets, identify patterns in student behavior, and make recommendations to improve student learning outcomes. For example, machine learning can improve adaptive learning platforms that generate content based on student progress. Natural Language Processing (NLP): NLP technologies are increasingly being used to automate processes such as accepting and checking essays, as well as providing feedback on student writing. They can also make communication between students and teachers more effective through intelligent instructional systems or chatbots.

3. Robotics: AI-powered robots are being explored in K-12 education to provide interactive learning experiences in subjects such as math, science, and language arts. These robots can engage students through real-time feedback and problem-solving scenarios. "While artificial intelligence offers many opportunities to revolutionize education, it is important to ensure responsible and ethical use, eliminate potential biases in algorithms, and maintain a balance between technological advancements and human involvement in the learning process" (Aliyeva, 2024, p. 28)

2. AI APPLICATIONS IN THE FIELD OF EDUCATION

2.1. Personalized Learning. AI has the potential to revolutionize the way students learn by creating personalized learning paths for them. Using data collected about students' performance and learning styles, AI systems can adjust the difficulty of tasks, recommend specific resources, and offer personalized feedback. For example, there are certain platforms that can provide this. Khan Academy is one such platform that uses AI to track students' learning paths and tailor lessons based on their specific needs. Similarly, Coursera and Duolingo offer AI-powered learning tools that tailor curriculum to the learner's pace. These tools reinforce the subject or areas that the student is struggling with and advance as mastery is achieved.

2.2. Integral Tutoring Systems (ITS). Intelligent Tutoring Systems (ITS) use AI to simulate human tutors, providing personal tutoring to students. These systems are particularly useful in applications in the exact sciences such as mathematics and physics. Step-by-step problem solving can be supported in detail and clearly by an AI tutor. "Research on intelligent tutoring systems

(ITS) shows that while human tutors remain highly effective, ITS has the potential to provide comparable or even superior support in certain learning contexts by delivering immediate, personalized feedback” [11]. For example, Carnegie Learning has developed a mathematics tutoring system that adapts to the learner’s understanding. The system guides them through problems and, when necessary, suggests solutions based on specific analyses. ITS can be used to support students in both traditional and online learning settings, while helping them develop independently. “Intelligent tutoring systems (ITS) are powerful AI-driven tools that provide real-time feedback tailored to students, simulating a type of personalized, one-on-one learning that would otherwise be unavailable to many learners” [4].

2.3. Automated Assessment and Evaluation. One of the primary benefits of AI in education is the automation of assessment and evaluation. As used in large online courses, AI-powered modeling systems can assess tests, essays, and even more complex forms of student work across multiple formats and topics in a fraction of the time it would take a human instructor. AI can also provide students with detailed feedback on their performance, highlight areas for improvement, and suggest additional resources and learning opportunities. This can be especially valuable in large-scale courses where individual attention is limited. “Learning in the age of technology is the process of making connections between nodes (knowledge, people, or resources), and AI can facilitate this process by identifying patterns and facilitating adaptive, context-based learning.” [10]

2.4. Predictive Analytics for Student Success

Predictive analytics is one of the most promising applications of AI in education. By analyzing historical data on student performance, attendance, and other behavioral indicators, AI models can predict students’ likelihood of success or failure, allowing for early intervention. For example, universities and schools can use predictive analytics to identify students at risk of dropping out and provide targeted assistance, such as coaching and mentoring. This proactive approach can significantly increase overall academic success and ensure a highly educated workforce for the future.

3. CHALLENGES AND ETHICAL ISSUES

3.1. Data Privacy and Security. AI systems in education are increasingly using data collection, including sensitive personal information about students’ learning styles, behaviors, and even emotional states, as a basis for data collection. This raises some significant concerns about data privacy and security. Educational institutions should ensure that data is stored anonymously and securely to protect the privacy of students.

3.2. Algorithmic Bias. Another problematic situation is the potential for algorithmic bias in AI systems. AI algorithms are trained on existing input data. These algorithms may contain biases based on race, socioeconomic status, or gender discrimination. If these biases are not monitored and managed, AI systems can unknowingly perpetuate bias patterns in educational settings or discrimination or unfair treatment, such as students following incorrect and limited academic paths.

3.3. Teacher and Student Acceptance. The integration of AI into education may face resistance from both teachers and learners. Teachers may be concerned about losing control over the teaching and learning process, while students may be concerned about AI-based systems and machines that provide a transition from real life to virtual life and replace human interaction. It is important to emphasize that AI is not intended to replace traditional educational practices, but rather to enhance them. Teachers should view AI-based systems as assistants, not competitors.

4. FUTURE DIRECTIONS

The future application of AI-based systems in education is likely to involve even more sophisticated AI technologies. The use of virtual reality (VR) and augmented reality (AR) can transform classrooms into interactive and immersive environments. Learning management systems (LMS) powered by AI are also expected to develop. This offers more personalized and adaptive learning experiences that take into account the emotional and cognitive characteristics of students. Furthermore, the increasing availability of AI-based equipment and machines on a large scale in developing countries can help bridge existing gaps in education, offering students high-quality learning resources.

Conclusion

Artificial intelligence has the potential to significantly enhance the learning and educational experience and knowledge of both students and teachers. From personalized learning platforms to predictive analytics for student success, AI offers effective transformations and benefits that can improve educational outcomes on a large scale. However, the integration of AI into education must be implemented gradually and carefully, overcoming the challenges of data privacy, algorithmic bias, and the acceptance of these positions by legal entities. Thanks to the innovative development of AI technologies, future research should focus on refining these systems and their successful implementation in practice.

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ACADEMIC WRITING AND RESEARCH METHODOLOGY

INTEGRATION OF LANGUAGE-BASED AI MODELS INTO UNIVERSITY TEACHING: A PHILOSOPHICAL ANALYSIS

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Abstract: The integration of artificial intelligence (AI) into higher education represents a natural stage in the development of digital civilization and reflects the global trend toward the technologization of all spheres of human activity. AI constitutes the foundation of language models that are transforming didactic approaches and require profound philosophical understanding. This study examines the epistemological role of these models as well as the socio-philosophical aspects of their application, ranging from the transformation of the teacher's role to the reconceptualization of the cognitive process itself. The relevance of this topic arises from the need for a comprehensive analysis of these dimensions and the growing prospects for integrating AI into higher education while preserving its humanistic character. Particular attention is paid to achieving a balance between technological opportunities and ethical limitations: how AI can be employed to personalize learning without diminishing the value of live interaction, critical thinking, and moral education. Furthermore, the implementation of language models raises new questions for the philosophy of education: what constitutes knowledge in the era of generative AI, how the independence of academic work should be assessed, and how the principles of academic integrity should be reconsidered. Addressing these challenges will enable universities not only to adapt to technological transformations but also to consciously shape an educational environment in which technology serves human development rather than replacing it.

Keywords: artificial intelligence, language models, philosophy of education, philosophical framework, personalized education.

DİL ƏSASLI SÜNİ İNTELLEKT MODELƏRİNİN UNIVERSİTET TƏDRİSİNƏ İNTEQRASIYASI: FƏLSƏFİ TƏHLİL

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Xülasə: Süni intellektin (Sİ) ali təhsilə inteqrasiyası rəqəmsal sivilizasiyanın təbii inkişafıdır və insan fəaliyyətinin bütün sahələrinin texnologiləşməsinə doğru qlobal trendi əks etdirir. Süni intellekt didaktik yanaşmaları kökündən dəyişdirən və dərin fəlsəfi anlayış tələb edən dil modellərinin tədrisinin əsasını təşkil edir. Bu modellərin epistemoloji rolu, eləcə də onların tətbiqinin sosial-fəlsəfi aspektləri - müəllim rolunun transformasiyasından tutmuş idrak prosesinin özünün yenidən düşünülməsinə qədər araşdırılır. Bu mövzunun aktuallığı bu aspektlərin hər birinin hərtərəfli təhlilinə ehtiyacdən və Sİ-nin humanist xarakterini qoruyarkən

ali təhsilə inteqrasiyası üçün artan perspektivlərdən irəli gəlir. Xüsusilə texnoloji imkanlar və etik məhdudiyyətlər arasında balans vacibdir: canlı dialoq, tənqidi düşüncə və əxlaqi təhsilin dəyərini itirmədən öyrənməni fərdiləşdirmək üçün Sİ-dən necə istifadə etmək olar. Bundan əlavə, dil modellərinin tətbiqi təhsil fəlsəfəsi üçün yeni suallar yaradır: generativ Sİ dövründə bilik nəyi təşkil edir, akademik işin müstəqilliyini necə qiymətləndirmək və akademik dürüstlük meyarlarını necə yenidən düşünmək olar. Bu çətinliklərin həlli universitetlərə təkcə texnoloji dəyişikliklərə uyğunlaşmağa imkan verməyəcək, həm də texnologiyanın fərdin inkişafına xidmət etdiyi, onu əvəz etmək əvəzinə, ona xidmət etdiyi gələcəyin təhsil mühitini şüurlu şəkildə formalaşdıracaq.

Açar sözlər: süni intellekt, dil modelləri, təhsil fəlsəfəsi, fəlsəfi blok, təhsilin fərdiləşdirilməsi.

Introduction

Modern higher education is undergoing a profound transformation under the influence of artificial intelligence technologies, particularly language models (ChatGPT, GPT, Claude, etc.). Their integration into the educational process goes beyond technical innovation and touches upon the fundamental philosophical foundations of education: the nature of knowledge, the agency of teacher and student, the criteria of academic integrity, and the anthropological perspectives of learning.

Relevance of the topic. The growing popularity of language models poses a complex set of questions for universities that require philosophical reflection:

- How does the nature of knowledge change if a significant portion of content is generated by algorithms?
- What constitutes authorship when a text is partially or entirely generated by a model?
- Does the availability of AI lead to the degradation of critical thinking and student independence?
- What are the ethical boundaries in education?

These questions have no clear answers and require interdisciplinary analysis informed by philosophical traditions.

The purpose of this study is to conduct a philosophical analysis of the integration of AI language models into university teaching, identifying:

- epistemological transformations of the educational process;
- anthropological consequences for those learning;
- ethical dilemmas and possible solutions;
- prospects for the widespread adoption of generative technologies.

Research objectives:

1. Analyze philosophical concepts of knowledge and learning in the context of AI implementation.
2. Explore the changing roles of teacher and student when using language models.
3. Identify ethical issues (academic integrity, transparency, responsibility).
4. Assess anthropological risks (reduced autonomy, dependence on algorithms).
5. Develop conceptual principles for AI integration that preserve the humanistic values of education.

6. Propose methodological guidelines for further study of this topic.

Object and subject of research.

Object: the process of integrating AI language models into university teaching systems.

Subject: the epistemological, anthropological, ethical, and axiological philosophical aspects of this process.

Research questions:

1. How do AI language models transform the understanding of knowledge in the academic environment?
2. To what extent does AI change the subjectivity of teachers and students?
3. What ethical standards should govern the use of AI in education?
4. Does university education maintain its humanistic focus with the widespread adoption of language models?

Methodological basis. The study draws on epistemology to analyze the nature of AI-generated knowledge; philosophical anthropology to study the impact of technology on student and teacher identities; ethics to assess the norms of AI use in the academic environment; and social philosophy to understand systemic changes in education.

Significance of the study

The results of this work will allow us to:

1. Deepen our theoretical understanding of human-AI interaction in education;
2. Develop ethical and methodological principles for implementing language models;
3. Help universities develop AI integration strategies that preserve the value of human interaction;
4. Identify promising directions for further philosophical and pedagogical research.

Research materials. In the study of the problem of integrating AI language models into university teaching: a philosophical analysis, a qualitative philosophical and methodological approach was used:

1. Works on epistemology, anthropology, and ethics of education.
2. Specialized literature on artificial intelligence policy from leading universities, scientific publications indexed in Scopus and CyberLeninka for the period 2023-2026 on the topics: the use of AI in education; ethical issues of using language models; the transformation of the role of the teacher in the digital age.
3. Regulatory documents: AI development strategy in education (Azerbaijan, EU, USA); regulations for the use of AI tools at leading universities worldwide; academic integrity codes with provisions on the use of AI.
4. Empirical data: reports from research organizations (UNESCO, OECD) on the implementation of AI in education. Results of surveys of teachers and students on perceptions of AI technologies; case studies of using ChatGPT, GPT-4, Claude, and similar models in the educational process.
5. Technical descriptions of AI language models (operating principles, limitations) provided by developers (OpenAI, Anthropic, etc.).

Research methods. The work uses a combination of philosophical and interdisciplinary methods:

1. Conceptual analysis – to clarify key concepts: "artificial intelligence"; "language model"; "academic integrity"; "subjectivity in education".
2. Comparative-historical method – to identify: the evolution of approaches to the use of technologies in education; differences in the perception of AI in the educational traditions of different cultures.

3. Critical resource analysis of texts: scientific articles on AI in education; public discussions on the prospects of AI-based learning; student work created using language models.

4. Thought experiment – to model scenarios: complete replacement of a teacher with AI; hybrid teaching (human + AI); learning in the context of the total availability of generative models.

5. Normative ethical analysis based on: deontological ethics (assessment of the conformity of actions with the principles of duty); Utilitarianism (comparing the benefits and harms of AI implementation); virtue ethics (the impact of AI on the development of personal qualities).

6. Epistemological reflection – for research: the criteria of truth and reliability in the context of generative technologies; the transformation of the concept of "authorship" in the academic environment.

7. Systems approach – for considering the integration of AI as an element of a complex "education" system, including: the institutional level (universities, ministries); the pedagogical level (teachers, students); the technological level (language models, platforms); the value-normative level (ethics, traditions).

8. Case study method – analysis of specific examples of AI implementation in universities: (successful practices; unsuccessful experiments; unforeseen consequences).

The combination of these methods allowed for a comprehensive philosophical analysis covering the ontological, epistemological, ethical and anthropological aspects of integrating AI language models into university teaching.

Results

A philosophical analysis of the integration of AI language models into university teaching revealed the following key findings:

1. Transformation of the teacher's role. AI assistants take on routine tasks (checking assignments, generating materials), shifting the focus of the teacher's activity from knowledge transfer to: supervision, mentoring; developing critical thinking in students; designing personalized educational trajectories.

2. Changing epistemological attitudes. The introduction of AI raises questions about the nature of knowledge: how does the understanding of "authorship" change when using AI for writing texts; what is the value of knowledge generated by an algorithm compared to independently acquired knowledge; how to rethink the criteria for assessing academic achievement.

3. Ethical dilemmas: the risk of reducing independent thinking due to excessive reliance on AI; the problem of academic integrity when using language models to complete assignments; issues of transparency: should the use of AI be disclosed in papers?

4. Anthropological consequences: potential degradation of skills in analyzing and synthesizing information; The development of "pseudo-independence" – the ability to adapt existing solutions rather than create one's own; the threat of losing the student's agency as an active participant in learning.

5. New opportunities: personalization of learning through adaptive AI tutors; access to educational resources for creative work.

6. Normative and regulatory challenges. The need to develop: rules for the use of AI in the educational process; ethical codes for students and teachers; methods for verifying results created using AI.

Diskisyon: The obtained results allow us to formulate several philosophical conclusions:

1. The Problem of Subjectivity. AI integration actualizes the classic philosophical problem of the relationship between instrument and subject. If the teacher becomes the "operator" of AI systems, and the student the user of ready-made solutions, this creates the risk of: reducing education to the transmission of information; and losing the existential dimension of learning (understanding, choice, responsibility).

2. The Epistemology of Education. Language models create a new epistemic environment where: knowledge is presented not as the result of individual cognition, but as the product of interaction with an algorithm; the criterion of truth shifts from "understanding" to the "applicability" of the generated answer.

3. Ethical Balance. The optimal integration model should combine: the principle of academic integrity (prohibition of using AI for the complete creation of works); the principle of critical analysis (mandatory verification of AI content); the principle of transparency (recording the stages of interaction with the model).

4. The Anthropological Ideal. The goal of education in the AI era is not to compete with algorithms, but to develop unique human qualities: creativity, the ability to reflect, and the ability to ask questions, not just find answers.

Conclusion

The integration of AI language models into university teaching is not simply a technological upgrade, but a profound sociocultural process that requires philosophical reflection. Successful implementation depends not only on the educational system's ability to preserve humanistic values; adapt teaching practices to new realities; and ensure a balance between technological capabilities and the development of human agency. Only if these conditions are met will AI not become a threat to traditional education, but a powerful tool for its transformation in the interests of individual and societal development.

Main content

The integration of AI language models (large language models, or LLMs) into university teaching represents a significant shift in "epistemological technologies," transforming AI from a passive auxiliary tool into a proactive agent in pedagogical, administrative, and research contexts. The process of integrating AI language models into education alters the ontological structure of education, creating a triad: "teacher-AI-student."

AI acts as an assistant and tool, but does not replace the teacher (although AI takes on routine tasks such as knowledge acquisition and material selection, it is capable of empathy, individualized mentoring, and the formation of value systems – functions performed by teachers). AI works with data and algorithms, but it lacks true understanding in the human sense. (3) The knowledge generated by AI is the result of information processing, not understanding. Thus, knowledge transfer remains the prerogative of humans: the teacher interprets and adapts AI-generated content for specific educational purposes.

In a philosophical sense, a language model is not a subject of learning, as it has no consciousness, intentions, or goals. Its "activity" in the philosophical model consists of executing programmed algorithms. AI acts as an expanded tool for communication and information processing, but not as an equal participant in educational interaction. The concept of "knowledge" is not fundamentally transformed, but is supplemented by a new dimension. Traditional knowledge is the result of human cognition, comprehension, and reflection. "Knowledge" from AI is structured information obtained algorithmically. (6) Critical evaluation and integration of this data into the

system of personal knowledge remains with the student and teacher. When part of the dialogue is led by AI, the quality of communication changes quantitatively, but not qualitatively (pros: speed, accessibility, and personalization; cons: lack of emotional connection, risk of superficial assimilation of the material). Live "teacher-student" interaction remains the basis of deep learning, and AI complements it, expanding the possibilities of access to information and automation of routine processes. (3)

With the introduction of AI, the teacher's functions are evolving: from knowledge translator to mentoring and training in working with AI. Language models can generate logically coherent but false statements, which gives rise to new epistemological challenges: the need for fact-checking – verifying information from AI against reliable sources; developing skills for critical data evaluation: analyzing logic, searching for contradictions, comparing versions; developing media literacy: understanding the limitations of AI, its tendency to make mistakes in complex or highly specialized issues. Traditional criteria (consistency with experience, logical consistency, expert consensus) are supplemented by new ones: validation through AI, multi-model comparison, and expert verification. Interaction with AI affects the balance of cognitive strategies: strengthening algorithmic thinking – students become accustomed to step-by-step instructions generated by AI (e.g., a "5-point essay plan"); weakening intuition; hybrid approach: optimal learning combines intuitive insights (human) and systematization (AI). (6)

Overcoming axiological challenges requires developing clear ethical standards for the use of AI in education, enshrining AI use rules in university bylaws, restructuring curricula with an emphasis on critical thinking, creativity, and live interaction, implementing tools to monitor and support the ethical use of AI, regularly analyzing results, and adjusting approaches. This will allow us to harness the potential of language models while maintaining the humanistic focus of education and developing students' genuine competence. Anthropological shifts (weakening cognitive skills, standardization of speech, dependence on AI, and declining communicative abilities) require targeted pedagogical measures to strengthen cognitive skills. To this end, it would be advisable to introduce concentration and attention exercises at universities, assign students tasks that challenge analytical thinking, and periodically conduct classes without the use of AI and gadgets. That is, conducting a "digital detox": brainstorming sessions, handwritten assignments, seminars with a ban on laptops and smartphones, developing memory, preserving linguistic individuality and creativity. Such measures will preserve and develop the human component of education, using AI as a tool, not a substitute for thinking.

Language models transform the learning process in higher education, fulfilling their key functions without replacing the role of the teacher, but rather as an auxiliary tool. The epistemological role of language models in universities boils down to: expanding cognitive capabilities (access, speed, personalization); developing new skills (critical thinking, working with information); and optimizing the learning process (automation of routine tasks). (4)

When used consciously, language models become a powerful tool that enhances, rather than replaces, traditional teaching methods, while maintaining the humanistic focus of education.

The introduction of AI language models into universities gives rise to a number of socio-philosophical implications affecting the transformation of educational processes, the roles of learning participants, ethical dilemmas, and social risks. These changes require a rethinking of traditional approaches to education and the development of new competencies. Teachers are no

longer the primary source of information but rather facilitators, designing individual educational trajectories and supporting students in developing critical thinking and cross-curricular skills.

The student, in turn, transforms from a passive consumer of knowledge into an "active co-author" of learning. They learn to formulate tasks for the AI, analyze and correct its responses, which develops skills for intellectual symbiosis with technology. (2) Language models also allow for the adaptation of learning materials and assignments to the individual needs, level of preparation, and pace of each student. This improves learning effectiveness and motivation. However, this poses the risk of widening the digital divide: not all universities and students have equal access to modern AI tools, which could further exacerbate the educational gap. From a social philosophy perspective, the challenge lies in the need to rethink the values of education: it must remain a space for the development of human potential, not simply the instrumental use of technology. A key role here must be played by a pedagogical strategy that can integrate AI while preserving human agency, critical thinking, and social connectedness.

Philosophical analysis reveals that while these tools offer personalized, interactive learning and improve administrative efficiency, they can pose serious challenges to epistemological validity, academic integrity, skill acquisition, and human agency. These challenges address key philosophical and ethical themes in the implementation of LLMs in higher education:

1. Epistemological and ontological shifts. A rethinking of knowledge production, where AI is transformed from a mere tool into a co-creator of knowledge. However, a "legitimation crisis" cannot be ignored, where standardized efficiency threatens to outweigh genuine learning. Algorithmic authority versus human judgment: LLMs produce results based on statistical probability rather than semantic understanding, which challenges the human ability to verify the accuracy of information (epistemic validity), creating a "black box" of knowledge where the reasoning underlying the answer is opaque and a cause for concern, as this could potentially lead to technological dependence and the loss of individual self-determination. In other words, simulation versus creativity (there is a critical distinction between AI's ability to reflect or reproduce creative work and genuine human understanding or creativity).

2. Pedagogical and ethical issues: loss of skills and overdependence (outsourcing thinking tasks to AI increases the risk of declining critical thinking and problem-solving skills in students, requiring teachers to strike a balance between AI support and independent learning); academic integrity ("AI-gearism"): the ease of creating polished prose from the outset undermines process pedagogy, requiring a transition to "process pedagogy 2.0," which places human decision-making and critical analysis of AI results at the center; bias, inequality, and fairness: Because AI-based learning programs (LLMs) are trained using historical data, they often trigger social biases related to race, gender, and socioeconomic status. Simply put, these programs reproduce inequalities in student assessments and educational materials; they emphasize privacy and data surveillance.

3. Philosophical approaches to integration and continuous questioning are emerging. Some approaches exploit the limitations of AI to teach students critical thinking. They focus on students debating AI and distinguishing valid arguments from superficial ones; on teachers embracing human-centered pedagogy (in which AI is used as a "supplement to learning" rather than a substitute for the emotional support and contextual understanding provided by teachers); and on process pedagogy 2.0: redesigning assessments to focus on the learning process (e.g., classwork, oral presentations, revision tracking) rather than the end product.*

4. Reconstructions of teacher identity, epistemic curator: The role of the teacher shifts towards "epistemic curation," helping students evaluate the results of AI, overcome its limitations, and responsibly integrate them into the learning process.

Critical AI Literacy (CAIL) is a necessary approach that views AI as a "complex cognitive artifact" in a machine-mediated ecology, not simply as a tool. Thus, the integration of master's degree programs in law challenges higher education to return to what it was always intended to be: applied, adaptive, collaborative, and rich in meaning, rather than simply the transmission of standardized knowledge. (5)

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THE ROLE OF ACADEMIC ETHICS IN THE EDUCATIONAL PROCESS AND SCIENTIFIC RESEARCH

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Abstract: Academic ethics is the foundation of trust in the scientific community. Establishing clear norms and standards of behavior in the academic environment helps not only to avoid conflicts but also to maintain a high level of quality in research and publications.

Academic ethics encompasses the norms and rules that regulate the behavior of scholars and students in the academic environment. It is a key factor ensuring trust in scientific research and educational processes. The ethics of academic writing and communication implies honesty, openness, and respect for others. This includes not only the proper use of sources but also the creation of an atmosphere of cooperation and trust. This article examines the main aspects of academic ethics and provides practical recommendations for their observance. Academic ethics represents a system of norms and principles regulating the behavior of students, researchers, and teachers in the educational environment. It covers such aspects as honesty, responsibility, respect for others, as well as compliance with the rules and standards of academic work. Academic ethics helps to establish transparent relationships among participants in the educational process and contributes to maintaining high standards of knowledge and research.

Keywords: academic ethics, scientific style, academic writing, plagiarism, paraphrasing

TƏHSİL PROSESİNDƏ VƏ ELMİ TƏDQİQATLARDA AKADEMİK ETİKANIN ROLU

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Xülasə: Akademik etika elmi ictimaiyyətdə etimadın əsasını təşkil edir. Akademik mühitdə davranış üçün aydın norma və standartların müəyyən edilməsi təkcə münaqişələrin qarşısını almağa deyil, həm də tədqiqat və nəşrlərdə yüksək keyfiyyət səviyyəsinin qorunmasına kömək edir.

Akademik etika akademik mühitdə alimlərin və tələbələrin davranışını tənzimləyən norma və qaydaları əhatə edir. O, elmi tədqiqatlara və təhsil proseslərinə etimadı təmin edən əsas amildir. Akademik yazı və ünsiyyət etikası dürüstlük, açıqlıq və başqalarına hörmət deməkdir. Bu, yalnız mənbələrdən düzgün istifadəni deyil, həm də əməkdaşlıq və etimad mühitinin yaradılmasına əhatə edir. Bu məqalədə akademik etikanın əsas aspektləri nəzərdən keçirilir və onlara riayət etmək üçün praktik tövsiyələr verilir. Akademik etika təhsil mühitində tələbələrin, tədqiqatçıların və müəllimlərin davranışını tənzimləyən norma və prinsiplər sistemidir. O, dürüstlük, məsuliyyət, başqalarına hörmət, həmçinin akademik işin qayda və standartlarına riayət kimi məsələləri əhatə edir. Akademik etika təhsil prosesinin iştirakçıları arasında şəffaf münasibətlərin qurulmasına kömək edir və biliklərin və tədqiqatların yüksək standartlarının qorunmasına töhfə verir.

Akademik etika təhsil prosesinin iştirakçılarının, o cümlədən tələbə və müəllimlərin davranışını tənzimləyən prinsip və normalar toplusunu əhatə edir. O, elmi fəaliyyətdə dürüstlüyü, etimadı və qarşılıqlı

hörməti təmin edir, plagiat, müəlliflik, akademik nüfuz və tədqiqat etikasını kimi məsələləri əhatə edir. Akademik etikanın məqsədi təhsil və elmdə yüksək keyfiyyət səviyyəsini qorumaq, eləcə də sağlam tədris mühitinin formalaşdırılmasına töhfə verməkdir.

Açar sözlər: akademik etika, elmi üslub, akademik yazı, plagiat, parafraz

Introduction

Academic ethics includes a set of principles and norms that regulate the behavior of participants in the educational process, including students and teachers. It ensures honesty, trust, and respect in scientific activity, covering such aspects as plagiarism, authorship, academic reputation, and research ethics. Academic ethics aims to support a high level of quality in education and science, as well as to promote the creation of a healthy learning environment. “In many educational institutions, there are codes of ethics regulating relationships within the teaching environment; however, there are no similar documents governing relationships within the student community.” [Берсенин, 2019]

Main aspects of academic ethics:

- **Honesty:** A fundamental principle of scientific work, implying the presentation of reliable information and an honest representation of one's own results.
- **Respect for intellectual property:** It is necessary to properly indicate authorship and justify borrowings from other works, whether through citation or paraphrasing.
- **Transparency:** All data, methodology, and methods of analysis must be accessible and verifiable by other scholars, which contributes to the reproducibility of results.
- **Responsibility:** Researchers must be aware of the consequences of their work and take responsibility for its results, methods, and the ways they are obtained.
- **Collaboration:** Ethical academic communication implies readiness for honest interaction with colleagues and the student community.

The scientific style serves the sphere of scientific communication and speech activity; it reflects “theoretical thinking expressed in a logical-conceptual form” [Кожина, 2008]. The main goal of academic ethics is to create an honest and open scientific environment that fosters the development of knowledge and critical thinking. Ethical standards help maintain trust in the scientific community and ensure fairness in the academic environment.

Key principles of academic ethics:

- **Honesty:** Reporting facts and research results without distortion or falsification. This is the fundamental principle of any academic work.
- **Responsibility:** Applying ethical standards in the process of research and writing scientific papers. This includes compliance with source requirements and respect for copyright.
- **Respect:** Attitude toward colleagues, students, and their work. This implies recognition of others' contributions and proper citation.
- **Fairness:** Impartiality in evaluating work. This means avoiding conflicts of interest and ensuring equal opportunities for all participants in the process.
- **Transparency:** Openness in methodology and sources of information. This allows other researchers to reproduce experiments and verify results, which is the basis of scientific reputation.

Academic ethics plays a crucial role in the educational process and scientific research. Violations in this area can have serious consequences both for individuals and for educational institutions as a whole.

Main types of violations:

- **Plagiarism:** Copying others' work or ideas without proper attribution. This is one of the most common violations. Consequences may include disciplinary measures up to expulsion.
- **Data falsification:** Presenting unreliable data or research results. This violates not only ethical but also scientific standards and may lead to loss of scientific reputation.
- **Self-plagiarism:** Using one's own previously written work without proper reference. This is also considered a violation, as it creates a false impression of originality.
- **Copyright infringement:** Using protected material without permission. This may lead to legal consequences and financial penalties.
- **Failure to comply with academic standards:** Not meeting requirements for format, style, or content of the work. This may negatively affect grades and academic performance.

As a result of violations of academic ethics, various consequences may arise, including:

- **Disciplinary measures:** Including warnings, grade reduction, or expulsion.
- **Revocation of degree or diploma:** In cases of serious violations, educational institutions may withdraw a degree.
- **Legal consequences:** Including fines and other sanctions for copyright violations.
- **Damage to reputation:** Personal and professional distrust toward the individual who committed the violation.
- **Professional limitations:** Restrictions on admission to certain educational programs or engagement in teaching activities.

Plagiarism is not limited to simply copying text. It also includes borrowing ideas, wording, or even structures without proper attribution. Below are practical tips to help prevent these mistakes.

1. Proper citation: Every time you use someone else's ideas or exact wording, you must cite them correctly. There are several citation styles (APA, MLA, Chicago, etc.), and it is important to follow one consistently in your work. Here's what to do:

- Determine a citation style – choose one standardized style and strictly follow its rules.
- Use quotation marks – for direct quotes, enclose them in quotation marks and indicate the source.
- Provide references – always mention the author and the year of the work in the text so the reader can verify the information.

2. Paraphrasing. If you want to use someone else's ideas in your own wording, learn how to paraphrase. This means reformulating the text so that it is unique. To paraphrase correctly, follow these steps:

- Read the original – make sure you understand the main idea.
- Write down key ideas without copying the author's wording.
- Rewrite the ideas in your own words while preserving the original meaning. Do not forget to cite the source.

3. Creating original content. The most reliable way to avoid plagiarism is to write your own content. To do this:

- Research the topic – read a variety of materials to better understand the subject.
- Formulate your own thoughts – connect your ideas with what you've read and offer your unique perspective.

- Plan your work – divide the text into sections and create a structure to avoid accidental borrowing.

4. Use plagiarism detection tools. Before submitting your work, it is recommended to check it for plagiarism. Various tools can help identify unintentional similarities. Use them before submission to ensure your work is original.

5. Education and awareness. Continuously improve your understanding of academic ethics. Read relevant materials and participate in seminars to stay updated on best practices and new requirements.

By following these simple recommendations, you can minimize the risk of plagiarism in your academic work, ensuring your future in education and research. Academic writing ethics require respect for others' work and accuracy in presenting information, and these tips will help you achieve that.

Ethics of working with sources: how to properly cite and reference

Following established citation rules helps the reader understand the context and significance of the materials used, and also makes it easy to locate the original sources. Let's look at the main principles of working with sources.

Basic rules of citation. Proper citation has several key components:

- Indicate the source: Always provide information about the source so the reader can trace your research. This includes the author's name, the title of the work, the publication date, and the page number.
- Use quotation marks: When quoting text verbatim, always enclose it in quotation marks to show that the words belong to another author.
- Paraphrase carefully: If you restate another author's ideas, ensure your wording reflects the original meaning and still cite the source.

How to reference sources. References play an important role in academic writing. They support arguments and ensure transparency in the use of information. Here are some tips for creating references:

- Choose a citation style: There are many styles such as APA, MLA, Chicago, and others. Select one and use it consistently throughout your work.
- Format references correctly: Learn how to format references according to the chosen style. Each system has its own requirements, so pay attention to details.
- Create a bibliography: At the end of your work, compile a complete list of all sources used. This not only gives credit to authors but also helps readers find materials of interest.

What are the key principles underlying academic writing ethics?

Academic writing ethics include several fundamental principles. First, honesty: authors must present their ideas and work accurately, without plagiarism or distortion of facts. Second, proper citation of sources is essential to give credit to authors and avoid copyright violations. Third, consistency in presenting arguments and adherence to logical structure are important. Finally, academic writing ethics require respect for different viewpoints and constructive criticism based on factual evidence.

What are the consequences of violating academic ethics? Violating academic ethics can have serious consequences. These may include disciplinary actions by an educational institution, such as a reprimand, expulsion, or a ban on receiving a diploma. In the academic world, it can lead

to a loss of reputation and termination of research funding. Moreover, it can negatively affect career prospects, as dishonest behavior may be documented and become known to future employers.

How does academic ethics influence communication in the scientific community?

Academic ethics shapes communication in the scientific community by establishing norms for interaction and discussion. Respect for others' opinions, openness to constructive criticism, and the ability to argue one's position with reference to previous research become essential aspects of collaboration. These principles help strengthen cooperation among scholars, promote the development of scientific knowledge, and maintain high research standards.

What can be done to improve adherence to academic ethics standards?

To improve adherence to academic ethics standards, it is important to develop educational programs aimed at raising awareness of ethical issues in academic institutions. It is also useful to implement regular training sessions for students and faculty that address principles of academic integrity and the importance of proper citation. Creating spaces for discussion, where participants can share experiences and propose improvements, is equally important. In addition, clear and transparent procedures for handling cases of academic misconduct should be established.

What is the ethics of academic writing and communication. The ethics of academic writing and communication involves following standards of honesty and accuracy when writing scholarly works and engaging in dialogue in an educational environment. This includes properly citing sources, avoiding the use of others' ideas without attribution, and showing respect for differing viewpoints. It also involves professional and respectful interaction with colleagues and instructors, based on mutual respect and openness. Adhering to these principles fosters trust and ensures fairness in the processes of learning and research.

Style of academic writing. Academic writing should be formal, clear, and logical. Keep the following points in mind:

- **Language:** Use academic vocabulary and avoid colloquial expressions. Proper word choice helps convey ideas clearly.
- **Tone:** Write objectively. Avoid personal pronouns (I, you, we) to maintain neutrality.
- **Sentence structure:** Sentences should be concise and clear. Complex constructions are better broken into simpler ones.

Structure of an academic text. Every academic text has a structure that helps organize information and make it accessible to the reader. The main elements include:

- **Introduction:** Define the topic, outline the purpose and objectives of the research, and explain why the topic is important.
- **Main body:** This section contains the core information, divided into logical sections. Each section should include subtopics and arguments.

The only way to avoid problems related to violations of academic ethics in authorship is to clearly distinguish your own words from those of others (by using quotation marks and citing the sources from which the quote was taken). [Тумахова, 2024]

Academic integrity is presented as a principle of virtue ethics, and specific examples from university life make it possible to see its effectiveness within this framework. The leading role of this principle in the modern academic sphere is shown in the context of the growing importance of the university's third mission – public service (alongside teaching and research) – and the massification of higher education that accompanied its development. [Порожа, 2016]

Conclusion

In conclusion, academic writing is an integral part of the educational and research process. It enables the systematic presentation of knowledge and the precise expression of scientific ideas. Through academic writing, researchers justify their ideas and contribute to existing knowledge. This skill also promotes the development of critical thinking and analytical approaches. At the same time, proper academic writing ensures adherence to ethical principles and maintains academic integrity. Thus, academic writing is considered one of the key indicators of high quality in both education and scientific activity.

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ACADEMIC WRITING AND CAREER PLANNING: DEVELOPING PROFESSIONAL COMPETENCIES IN HIGHER EDUCATION

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Abstract: In the contemporary educational environment, academic writing skills and career planning have become essential components of professional development in higher education. Universities are

expected not only to provide theoretical knowledge but also to equip students with competencies necessary for successful participation in the labor market. Academic writing contributes to the development of analytical thinking, communication skills, research competence, critical reasoning, and professional self-expression. At the same time, career planning supports students in identifying professional goals, improving employability skills, increasing self-confidence, and adapting to rapidly changing workplace requirements.

This study examines the relationship between academic writing and career planning in the context of higher education. The research focuses on the role of academic writing in strengthening professional competencies and enhancing students' career readiness. Qualitative research methods, including literature analysis, descriptive analysis, and comparative evaluation of educational approaches, were applied in the study. The findings indicate that integrating academic writing instruction into career planning courses positively influences students' critical thinking abilities, professional communication, academic literacy, and professional self-development.

The article concludes that higher education institutions should adopt interdisciplinary approaches that combine academic writing and career education in order to prepare competitive, adaptable, and professionally competent graduates for the global labor market.

Keywords: academic writing, career planning, professional competencies, higher education, employability skills, communication skills, professional development

AKADEMİK YAZI VƏ KARYERA PLANLAMASI: ALİ TƏHSİLDƏ PEŞƏKAR KOMPETENSİYALARIN İNKİŞAFI

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Xülasə: Müasir təhsil mühitində akademik yazı bacarıqları və karyera planlaması ali təhsildə peşəkar inkişafın vacib komponentlərinə çevrilmişdir. Universitetlərdən yalnız nəzəri bilik vermək deyil, həm də tələbələrə əmək bazarında uğurlu fəaliyyət üçün zəruri kompetensiyalarla təmin etmək tələb olunur. Akademik yazı analitik düşüncə, kommunikasiya bacarıqları, tədqiqat kompetensiyası, tənqidi təfəkkür və peşəkar özünüifadənin inkişafına töhfə verir. Eyni zamanda, karyera planlaması tələbələrin peşəkar məqsədlərini müəyyənləşdirməsinə, məşğulluq bacarıqlarını inkişaf etdirməsinə, özünəinam qazanmasına və sürətlə dəyişən əmək bazarına uyğunlaşmasına kömək edir.

Bu tədqiqat ali təhsil kontekstində akademik yazı ilə karyera planlaması arasındakı əlaqəni araşdırır. Məqalədə akademik yazının peşəkar kompetensiyaların formalaşmasında və tələbələrin karyera hazırlığının gücləndirilməsində rolu təhlil edilir. Tədqiqatda elmi ədəbiyyatın təhlili, təsviri təhlil və təhsil yanaşmalarının müqayisəli qiymətləndirilməsi kimi keyfiyyət metodlarından istifadə olunmuşdur. Nəticələr göstərir ki, akademik yazının karyera planlaması fənlərinə inteqrasiyası tələbələrin tənqidi düşüncə, peşəkar ünsiyyət, akademik savadlılıq və peşəkar inkişaf bacarıqlarına müsbət təsir göstərir.

Məqalədə nəticə olaraq qeyd edilir ki, ali təhsil müəssisələri qlobal əmək bazarı üçün rəqəbatqabiliyyətli, çevik və peşəkar kompetensiyalara malik məzunlar hazırlamaq məqsədilə akademik yazı və karyera təhsilini birləşdirən inteqrativ yanaşmalar tətbiq etməlidirlər.

Açar sözlər : akademik yazı, karyera planlaması, peşəkar kompetensiyalar, ali təhsil, məşğulluq bacarıqları, kommunikasiya bacarıqları, peşəkar inkişaf

Introduction

The contemporary higher education system is undergoing substantial transformation under the influence of globalization, technological advancement, and the increasing demands of the labor market. Universities are no longer limited to providing theoretical knowledge; they are also expected to prepare students for professional life by developing practical competencies, communication abilities, and critical thinking skills. In this context, academic writing and career planning have become essential components of higher education curricula.

Academic writing is considered one of the most important academic and professional competencies in modern education. It enables students to organize ideas logically, conduct research effectively, present arguments critically, and communicate professionally in written form. According to Hyland (2019), academic writing is not only a linguistic activity but also a social and cognitive process that contributes to intellectual development and professional identity formation. Similarly, Brown (2014) emphasizes that writing skills support learners' analytical thinking and enhance their ability to express academic and professional ideas effectively.

In the modern labor market, employers increasingly value communication skills, problem-solving abilities, and professional self-presentation. Consequently, academic writing has become closely connected with employability and career readiness. Through activities such as report writing, project preparation, reflective essays, and research papers, students develop competencies that are directly applicable to professional environments (Mammadov, 2022).

Career planning education also plays a significant role in preparing students for future employment. It helps individuals identify career goals, evaluate personal strengths and weaknesses, and improve employability skills necessary for professional success. Richards (2017) notes that integrating practical competencies into educational programs contributes significantly to students' professional development and lifelong learning abilities.

Furthermore, digital transformation and online learning environments have changed traditional approaches to teaching academic writing and career planning. The integration of digital educational platforms, artificial intelligence tools, and online academic resources has created new opportunities for students to improve writing and professional communication skills (UNESCO, 2022). These technological innovations have increased access to educational materials and encouraged independent learning and professional self-development.

Therefore, the relationship between academic writing and career planning deserves detailed academic investigation. The present study aims to analyze the role of academic writing skills in developing professional competencies and preparing students for successful careers in higher education institutions.

Materials and Methods

The present study is based on a qualitative research methodology aimed at examining the relationship between academic writing skills and career planning in higher education. Qualitative methods were considered appropriate because they allow a detailed analysis of educational approaches, professional competencies, and theoretical perspectives related to academic writing and employability.

The research primarily relied on the analysis of scientific literature, academic articles, books, and international educational reports related to academic writing, career development, professional competencies, and higher education practices. Relevant scholarly sources were selected according to their scientific relevance, publication quality, and contribution to the field of education and linguistics. APA 7 citation standards were applied throughout the study.

Descriptive and comparative analysis methods were employed to evaluate how academic writing contributes to students' professional development and employability skills. The descriptive method helped identify the major functions of academic writing in higher education, while comparative analysis enabled the examination of different educational approaches and teaching practices used in career planning education.

Special attention was given to competencies such as critical thinking, professional communication, analytical reasoning, problem-solving, and self-presentation skills. These competencies are widely recognized as essential for career success in the modern labor market (Aliyev, 2021). The study also examined the integration of academic writing activities into career planning courses, including CV writing, motivation letters, reflective writing, research projects, and professional communication exercises.

In addition, the role of digital technologies and online educational environments was analyzed. Modern digital learning platforms, artificial intelligence tools, and online academic resources have significantly influenced the teaching of academic writing and career education. UNESCO (2022) highlights that digital educational technologies contribute to flexible learning opportunities and support students' independent professional development.

The findings of the study are based on theoretical analysis and interpretation of contemporary educational practices. The research does not include experimental data collection; however, it provides a comprehensive academic evaluation of the role of academic writing in professional competency development and career planning within higher education institutions.

Results and Discussion

The findings of the study demonstrate that academic writing significantly contributes to students' professional development. Students who possess effective academic writing skills are better prepared for professional communication, report writing, project development, and research activities.

Academic writing also supports the development of critical thinking and problem-solving abilities. These competencies are highly valued in the modern labor market and are considered essential employability skills. According to Hyland (2019), writing activities improve students' ability to analyze information critically and present ideas systematically.

The study further indicates that career planning courses integrated with academic writing activities help students improve self-confidence and communication abilities. Activities such as writing CVs, motivation letters, reflective essays, and research projects contribute to students' professional readiness. Brown (2014) states that productive language activities encourage learners to become more confident and independent in professional communication environments.

Another important finding of the study is the role of interdisciplinary education in competency development. The integration of academic writing into career planning courses creates opportunities for students to connect theoretical knowledge with practical professional skills. This integration contributes to the development of analytical reasoning, academic literacy, and employability competencies simultaneously.

Digital learning platforms and artificial intelligence tools also play an important role in improving academic writing and career planning competencies. Online educational technologies provide students with flexible learning opportunities and access to various academic resources. UNESCO (2022) emphasizes that digital learning environments support student-centered education and professional self-development in higher education institutions.

Furthermore, the research reveals that academic writing contributes to professional identity formation. Students who regularly engage in academic and reflective writing activities become more capable of presenting their professional goals, expressing ideas clearly, and participating effectively in academic and workplace communication.

The results suggest that interdisciplinary educational approaches combining academic writing and career planning can positively influence students' academic achievement, employability skills, and career readiness.

Conclusion

Academic writing and career planning are interconnected components of modern higher education. Academic writing contributes to the development of analytical thinking, professional communication, and research competencies, while career planning supports employability and professional growth.

The integration of academic writing into career planning education enhances students' professional competencies and increases their readiness for the labor market. Activities such as academic essays, project writing, research tasks, CV preparation, and reflective writing support students' professional self-development and communication abilities.

The study demonstrates that digital educational technologies and interdisciplinary teaching approaches play a significant role in improving both academic writing and career planning competencies. Modern universities should therefore create innovative educational environments that combine theoretical knowledge with practical professional skills.

In conclusion, higher education institutions should adopt comprehensive and student-centered approaches that integrate academic writing instruction into career education programs. Such approaches will contribute to preparing competitive, professionally competent, and adaptable graduates capable of meeting the demands of the global labor market.

Future research may focus on empirical studies investigating the practical outcomes of integrating academic writing and career planning courses in higher education institutions.

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THE ROLE OF ARTIFICIAL INTELLIGENCE IN ENHANCING ACADEMIC WRITING SKILLS IN EFL CONTEXTS

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Abstract: The growing presence of artificial intelligence in education has transformed the ecology of second language writing, yet its implications for academic writing in EFL contexts remain theoretically underexplored. This article develops a conceptual framework for understanding how AI can enhance academic writing skills among EFL learners while also complicating questions of authorship, voice, feedback, and pedagogy. Drawing on foundational work in second language writing, genre analysis, discourse theory, and recent scholarship on AI-assisted writing, the article employs a literature-based analytical methodology rather than empirical data collection. The analysis suggests that AI contributes most effectively to EFL academic writing when it functions as a scaffold for planning, lexical expansion, grammatical refinement, revision, and metacognitive reflection. However, its contribution to coherence, cohesion, and disciplinary appropriateness is not inherent in the technology itself; rather, it depends on the learner's genre knowledge, feedback literacy, prompt literacy, and critical capacity to evaluate AI outputs. The discussion further demonstrates that automated writing evaluation systems and generative AI tools offer different pedagogical affordances, with the former generally supporting sentence-level accuracy and the latter extending into idea generation and rhetorical restructuring. Nevertheless, these benefits coexist with substantial risks, including overreliance, homogenized prose, inaccurate content, weakened authorial agency, and academic integrity concerns. The article concludes that AI can strengthen EFL academic writing only within a human-centered pedagogical model that integrates genre instruction, reflective revision, teacher mediation, and responsible AI literacy.

Key words: artificial intelligence, EFL academic writing, coherence, cohesion, feedback literacy, generative AI, CALL

EFL KONTEKSTLƏRİNDƏ AKADEMİK YAZI BACARIQLARININ İNKİŞAFINDA SÜNİ İNTELLEKTİN ROLU

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Xülasə: Süni intellektin təhsildə getdikcə geniş yayılması ikinci dil yazısı mühitini əhəmiyyətli dərəcədə dəyişdirmişdir. Bununla belə, onun EFL kontekstlərində akademik yazıya təsiri hələ də nəzəri baxımdan kifayət qədər araşdırılmamışdır. Bu məqalə süni intellektin EFL öyrənənlərinin akademik yazı bacarıqlarını necə inkişaf etdirə biləcəyini, eyni zamanda müəlliflik, akademik səs, geribildirim və pedaqogika kimi məsələləri necə mürəkkəbləşdirdiyini izah edən konseptual çərçivə təqdim edir. Məqalə ikinci dil yazısı, janr təhlili, diskurs nəzəriyyəsi və AI dəstəkli yazı üzrə son tədqiqatlara əsaslanır. Təhlil göstərir ki, süni intellekt planlaşdırma, leksik genişlənmə, qrammatik dəqiqləşdirmə, redaktə və metakognitiv düşünmə proseslərində dəstəkləyici vasitə kimi daha səmərəli fəaliyyət göstərir. Lakin onun koherentlik,

koheziya və akademik uyğunluğa təsiri texnologiyanın özündən deyil, tələbənin janr biliyindən, geribildirim savadlılığından, prompt savadlılığından və AI nəticələrini tənqidi qiymətləndirmə bacarığından asılıdır. Məqalə belə nəticəyə gəlir ki, süni intellekt EFL akademik yazısını yalnız insan mərkəzli pedaqoji model daxilində gücləndirə bilər.

Açar sözlər: süni intellekt, EFL akademik yazı, koherentlik, koheziya, geribildirim savadlılığı, generativ AI, CALL

Introduction

Artificial intelligence has become one of the defining forces in contemporary educational change. What once referred primarily to intelligent tutoring systems, analytics, and automated scoring now includes generative systems capable of producing explanations, revising prose, modelling argument structures, and engaging in extended dialogue with learners. Reviews of AI in higher education show that research has expanded from a narrow focus on efficiency and automation toward broader concerns with ethics, pedagogy, collaboration, and disciplinary practice (Bond et al., 2024; Chen et al., 2022; Crompton & Burke, 2023; Zawacki-Richter et al., 2019). In language education, this shift is especially significant because writing is both a cognitive activity and a social act. It is not reducible to grammatical correctness; it is a situated process through which learners construct meaning, position themselves in relation to disciplinary communities, and negotiate audience expectations (Hyland, 2004, 2019; Swales, 1990).

Academic writing in EFL contexts is therefore a particularly demanding site for AI integration. EFL learners are expected to write in a language that may not be fully automatized while also meeting norms of logic, textual cohesion, genre structure, citation practice, and disciplinary stance. Research in second language writing repeatedly shows that writing quality depends on far more than surface-level accuracy. It is deeply linked to rhetorical purpose, coherence, cohesion, and linguistic sophistication (Crossley & McNamara, 2012; McNamara et al., 2010). Coherence concerns the meaningful organization of ideas across a whole text, whereas cohesion refers to the linguistic ties that connect units of discourse through reference, substitution, conjunction, and lexical patterning (Halliday & Hasan, 1976; Witte & Faigley, 1981). Academic writing also requires mastery of genre conventions, since students must learn how knowledge is rhetorically shaped in disciplinary contexts rather than merely expressed in grammatically correct sentences (Hyland, 2004; Swales, 1990).

For this reason, the arrival of AI writing tools has generated both enthusiasm and anxiety. Studies on automated writing evaluation and generative AI suggest that such tools can support revision, reduce the burden of delayed feedback, and promote experimentation with language (Barrot, 2023a, 2023b; Nazari et al., 2021; Shi & Aryadoust, 2024). At the same time, critics warn that overreliance on AI may encourage superficial revision, attenuate authorial agency, and normalize formulaic discourse while introducing new concerns around hallucinated content, plagiarism, bias, and privacy (Farrelly & Baker, 2023; Meyer et al., 2023; Tseng & Warschauer, 2023). The central pedagogical issue, then, is not whether AI is simply beneficial or harmful. It is how AI changes the conditions under which academic writing is learned, practiced, and assessed in EFL settings.

Although the recent literature on AI and writing is growing rapidly, it remains fragmented in three ways. First, much of it is tool-specific and empirical, focusing on satisfaction, performance gains, or user perceptions rather than offering a broader theoretical account of writing development. Second, the literature often conflates different technologies under the umbrella of AI, despite the

fact that automated writing evaluation systems and large language models support writing in qualitatively different ways. Third, insufficient attention has been given to how AI interacts with central writing constructs such as coherence, cohesion, genre awareness, and disciplinary voice. In other words, the field still lacks an integrated theoretical explanation of how and under what conditions AI enhances academic writing in EFL contexts.

The present article addresses this gap by developing a theoretical perspective on AI-mediated academic writing. It argues that AI enhances EFL academic writing not through technological sophistication alone but through its pedagogical positioning within a human-guided writing ecology. The aim of the study is to construct a coherent framework that explains the role of AI in improving writing skills, with particular attention to coherence, cohesion, linguistic development, and the limits of AI-assisted writing. In doing so, the article seeks to contribute to current discussions in EFL pedagogy, CALL, and academic writing research by synthesizing foundational and recent scholarship into a conceptually unified account.

Methodology and Materials

This study adopts a literature-based conceptual methodology grounded in deep theoretical synthesis rather than empirical data collection. Its purpose is explanatory rather than statistical: to organize and interpret existing scholarship in order to clarify the mechanisms through which AI may support or hinder EFL academic writing. The methodology is therefore interpretive and integrative. It draws on foundational works in second language writing, discourse analysis, and genre theory together with recent peer-reviewed studies published primarily between 2019 and 2024 on AI in education, automated writing evaluation, AI-assisted feedback, generative AI, and CALL.

The theoretical framework combines four interrelated perspectives. The first is a genre-discursive perspective derived from Swales (1990) and Hyland (2004, 2019), which treats academic writing as participation in disciplinary discourse communities. From this perspective, effective writing involves argument structure, audience orientation, rhetorical moves, and appropriate stance. The second is a discourse-quality perspective grounded in Halliday and Hasan's (1976) account of cohesion and in research linking writing quality to coherence, cohesion, and linguistic sophistication (Crossley & McNamara, 2012; McNamara et al., 2010; Witte & Faigley, 1981). This perspective is necessary because AI tools often intervene most visibly at the textual level, yet not all textual improvement produces genuine discourse quality. The third is a sociocultural and mediation-oriented perspective, according to which tools shape cognition by mediating action, revision, and reflection. Studies of AI-supported writing and teacher responses to automated evaluation indicate that learning outcomes depend on how tools are appropriated within social and instructional contexts rather than on tool output alone (Jiang et al., 2020; Jiang & Yu, 2022; Lee, 2024). The fourth is a feedback and literacy perspective. Writing improvement requires not just receiving feedback but interpreting, evaluating, and using it productively; this expands in AI environments into feedback literacy, prompt literacy, and AI literacy (Ng et al., 2021; Tseng & Warschauer, 2023; Zhang & Hyland, 2018, 2022).

The analytical approach used in this article involved grouping the literature into four thematic domains: AI in education and higher education; AI-assisted writing and automated writing evaluation; EFL writing pedagogy and academic discourse; and ethical or integrity-related consequences of generative AI. These domains were then examined through three analytical questions. First, what dimensions of academic writing does AI improve most consistently? Second, how do AI tools interact with coherence, cohesion, and linguistic development? Third, what

pedagogical conditions determine whether AI functions as a scaffold or as a shortcut? This procedure enabled the development of a conceptual model rather than a descriptive bibliography.

Because the study is non-empirical, the “results” presented below should be understood as synthesized theoretical propositions emerging from the reviewed literature. This approach is suitable because the field is currently marked by rapid technological change, uneven research design, and the coexistence of older automated feedback systems with newer generative tools. A theory-building synthesis helps clarify patterns that remain obscured when studies are read in isolation. It also permits critical comparison between technologies that are often grouped together despite meaningful differences in affordance, feedback granularity, and epistemic risk.

Discussion

A key outcome of the synthesis is that AI does not enhance academic writing in one uniform way. Its role differs according to the stage of writing, the type of system used, the learner’s proficiency and critical literacy, and the pedagogical environment in which the tool is embedded. The literature supports a distinction between lower-order textual support and higher-order rhetorical development. AI can reliably assist with the former; its effect on the latter is conditional and pedagogically mediated.

AI first enhances EFL academic writing by increasing the immediacy, frequency, and accessibility of feedback. This matters because writing development depends heavily on opportunities for iterative revision, yet in many EFL classrooms individualized teacher feedback is limited by time, class size, and curriculum constraints. Automated writing evaluation systems partially address this problem by offering instant comments on grammar, mechanics, word choice, and in some cases organization. Research on automated feedback indicates that such systems can improve writing accuracy, stimulate revision behavior, and positively affect learner confidence when used as part of a process-based pedagogical design (Barrot, 2023a; Link et al., 2022; Nazari et al., 2021; Sarı & Han, 2024). Meta-analytic evidence also suggests that automated writing evaluation can improve second language writing performance while influencing motivational and affective variables such as anxiety and self-efficacy (Huang et al., 2024; Ngo et al., 2022).

Yet this finding must be interpreted carefully. Automated writing tools are strongest where rules are relatively tractable: punctuation, article usage, subject-verb agreement, lexical alternatives, and obvious sentence-level awkwardness. They are less robust when the problem lies in epistemic depth, argument development, or disciplinary appropriateness. This aligns with broader writing research showing that linguistic accuracy, though important, is only one component of writing quality (McNamara et al., 2010). An EFL student may produce a grammatically polished text that remains rhetorically weak, conceptually thin, or poorly aligned with genre expectations. Thus, the pedagogical challenge is to prevent AI from narrowing the definition of good writing to correctness alone.

The rise of large language models alters this landscape because they expand AI intervention from correction to co-composition. Whereas conventional AWE tools mainly detect and flag deviations from normative written form, generative AI can brainstorm topics, propose outlines, generate possible thesis statements, reformulate sentences, and simulate alternative rhetorical patterns. This extension has important consequences for EFL writers, who often struggle not only with linguistic encoding but also with generating disciplinary prose under pressure. Studies of ChatGPT-assisted writing suggest that learners can use AI reformulations to compare versions, notice alternative phrasings, and revise more extensively than they might with static correction

alone (Bai & Wei, 2024; Polakova & Ivenz, 2024). Lee (2024) further shows that AI becomes pedagogically productive when teacher guidance and peer collaboration organize its use within planning, translating, and reviewing cycles. In this formulation, AI is not a solitary answer machine but a participant in a socially orchestrated composing process.

This distinction between correction and co-composition helps clarify why AI's role in academic writing cannot be evaluated through surface gains alone. The literature points to at least three writing processes where AI has strong pedagogical potential. The first is ideation. Generative tools can lower the threshold for beginning a text by helping learners generate subtopics, examples, and conceptual relationships. The second is reformulation. AI can provide alternative lexicogrammatical realizations that allow learners to observe how meaning can be expressed with greater concision or rhetorical force. The third is revision. Because AI feedback is interactive, learners can conduct multiple micro-revisions, test alternative wordings, and return to unresolved textual problems without waiting for delayed teacher response. These affordances are especially valuable in EFL contexts, where hesitation during composing may stem from both linguistic uncertainty and limited disciplinary confidence (Hyland, 2019; Kohnke et al., 2023).

However, the synthesis also indicates that AI's contribution to coherence and cohesion is fundamentally uneven. Cohesion is comparatively easier for AI to assist because it is materially present in the linguistic surface of the text. Systems can suggest conjunctions, clarify reference, reduce unnecessary repetition, recommend lexical substitutions, and sometimes improve sentence-to-sentence flow. Because writing quality is associated with effective cohesion and linguistic sophistication, these interventions may yield measurable gains in perceived readability and textual connectedness (Crossley & McNamara, 2012; Halliday & Hasan, 1976; McNamara et al., 2010). In this sense, AI can help writers strengthen local textual weaving.

By contrast, coherence is not simply a sum of cohesive markers. It depends on global semantic organization, relevance, progression of ideas, rhetorical hierarchy, and the reader's ability to infer meaningful relations across the text. As Witte and Faigley (1981) demonstrated, cohesion and coherence are related but not interchangeable. Overuse of connectives or lexical repetition may create the illusion of smoothness without establishing a logically integrated argument. Here lies one of the major theoretical limits of AI writing tools. Large language models can often simulate coherent prose because they are exceptionally strong at predicting probable continuations. But probability is not the same as disciplinary validity. A text may sound coherent while lacking evidential substance, argument stability, or conceptual accountability. Academic coherence remains tied to genre knowledge, audience awareness, and epistemic control, all of which exceed surface text generation (Hyland, 2004; Swales, 1990).

This problem is especially salient in EFL academic writing because learners may mistake fluency for intellectual adequacy. When AI generates text that is lexically sophisticated and syntactically smooth, students may perceive the output as "academic" even when it is generic, overstated, or weakly aligned with the task. Biber et al. (2011) remind us that measures of grammatical complexity derived from conversational language are not sufficient for evaluating advanced academic prose. In a similar way, AI-generated fluency should not be equated with disciplinary quality. Theoretical discussion in the field increasingly suggests that AI does not directly create coherence; rather, it supplies resources that writers may use to construct coherence if they have the necessary rhetorical and conceptual control (Tseng & Warschauer, 2023; Warschauer et al., 2023).

This point is reinforced by research on learner engagement with automated and AI-generated feedback. Students do not simply absorb feedback; they interpret it selectively, trust it unevenly, and often use it pragmatically. Koltovskaia (2020) showed that engagement with Grammarly was variable and shaped by learners' beliefs, goals, and capacity to evaluate suggestions. Ranalli (2021) similarly argued that the learning potential of automated feedback is limited when students treat it as an editing service rather than as input for reflective problem solving. Jiang and Yu (2022) and Jiang et al. (2020) extend this argument by showing that AI feedback is appropriated through local classroom cultures, instructor beliefs, and student perceptions of usefulness and authority. From a theoretical perspective, this means that AI is not a feedback solution in itself. It becomes educationally meaningful only through uptake, negotiation, and pedagogical framing.

The integration of feedback literacy into AI-mediated writing is therefore essential. Zhang and Hyland (2018, 2022) emphasize that student engagement with feedback includes cognitive, affective, and behavioral dimensions. In AI-assisted environments, these dimensions now include prompt formulation, trust calibration, evidence checking, and selective adoption of machine suggestions. Ng et al. (2021) conceptualize AI literacy as involving technical, cognitive, and ethical understanding. For EFL writers, this literacy must also include rhetorical discernment: the ability to ask whether an AI-generated sentence is merely more fluent or genuinely more appropriate to the argument and genre. Tseng and Warschauer (2023) offer a particularly productive pedagogical frame by proposing that learners should be taught to understand, access, prompt, corroborate, and incorporate AI. This framework is valuable because it treats AI use as a literacy practice rather than a convenience.

The distinct pedagogical affordances of major AI writing tools can be summarized as follows.

Table 1. AI affordances for EFL academic writing

Tool type	Primary functions	Typical gains	Main limitations	Pedagogical value
Automated writing evaluation systems	Grammar, mechanics, lexical suggestions, limited organization feedback	Accuracy, faster revision, feedback immediacy, confidence	Overemphasis on local errors, limited rhetorical awareness, possible false positives	Best for sentence-level notice-and-revise cycles
Generative AI chatbots	Brainstorming, outlining, reformulation, explanation, draft expansion	Idea generation, lexical diversity, revision flexibility, metacognitive dialogue	Hallucination, bland or generic prose, weak source reliability, potential overdependence	Best for guided planning, alternative phrasing, reflective revision
Collaborative AI writing platforms	Integrated drafting, shared comments, revision tracking	Peer interaction, iterative edits, process visibility	Tool overload, uneven participation, procedural rather than conceptual improvement	Best for process writing and group tasks
Corpus-	Concordancing,	Genre awareness,	Requires training,	Best for

Tool type	Primary functions	Typical gains	Main limitations	Pedagogical value
informed or search-based writing supports	phrase patterning, disciplinary example retrieval	collocational accuracy, discipline-sensitive language use	may be slower than generative tools	advanced genre refinement and disciplinary phraseology

The synthesis suggests that the most effective AI-supported writing pedagogy combines these tools rather than treating them as interchangeable. Conventional AWE is often better suited to recurrent language problems that benefit from repeated corrective exposure. Generative AI is more useful in early drafting, conceptual expansion, and rhetorical experimentation. Corpus-informed tools remain superior for genre-sensitive phraseology and disciplinary convention because they ground language choices in attested usage rather than probabilistic invention. The teacher's role is therefore not eliminated but redistributed toward orchestration, selection, and critical calibration.

The advantages of AI in EFL academic writing are nonetheless substantial. First, AI increases access to individualized feedback, which is especially important in under-resourced instructional settings. Second, it supports learner autonomy by allowing students to test revisions independently and at their own pace. Third, it can reduce the affective cost of writing by creating a low-stakes environment for experimentation. Fourth, it may broaden participation by giving lower-proficiency students scaffolds that help them enter academic discourse more confidently (Nazari et al., 2021; Wale & Kassahun, 2024). Finally, AI can enrich the revision layer of process writing by making writing visibly recursive rather than linear.

These advantages, however, coexist with important limitations. One is epistemic dependence. If learners outsource idea generation, paragraph development, and final phrasing to AI, they may submit linguistically improved texts without developing the cognitive and rhetorical capacities that academic writing is supposed to cultivate. Another is homogenization. Because generative systems are trained on dominant textual patterns, they often produce rhetorically plausible but stylistically generic prose, which may flatten individual voice and reduce disciplinary nuance (Barrot, 2023b; Warschauer et al., 2023). A third is reliability. AI-generated content may contain fabricated claims, inaccurate examples, or inappropriate citations, making corroboration indispensable (Khlaif et al., 2023; Meyer et al., 2023). A fourth is ethical vulnerability. Questions of authorship, disclosure, data privacy, assessment fairness, and academic honesty become more acute when learners can obtain polished text with minimal effort (Farrelly & Baker, 2023; Bond et al., 2024).

The literature therefore converges on a crucial principle: AI should be positioned as a scaffold for writing development, not a replacement for writing activity. This principle has direct implications for EFL pedagogy. A genre-first approach is needed so that learners evaluate AI output against rhetorical purpose rather than against fluency alone. Process-oriented tasks should require students to document how AI was used, what suggestions were rejected, and how final choices were made. Feedback tasks should compare AI feedback with peer or teacher feedback to promote discernment rather than submission to machine authority. Prompting should be taught as a rhetorical skill, not a technical trick, because useful prompting depends on clarity about audience, stance, evidence, and genre. Finally, validation should be built into instruction through source-checking, reflection, and accountability routines.

These principles can be organized into a human-centered model of AI-mediated academic writing.

Table 2. Pedagogical principles for responsible AI integration in EFL writing

Principle	Instructional implication	Expected outcome
Genre anchoring	Frame AI use around task purpose, audience, and disciplinary moves	Stronger rhetorical control and reduced generic output
Process transparency	Require logs or annotations of AI-supported drafting and revision	Greater accountability and metacognitive awareness
Feedback triangulation	Compare AI feedback with teacher, peer, or rubric-based feedback	Improved trust calibration and feedback literacy
Corroboration before incorporation	Verify claims, examples, and references before use	Reduced hallucination risk and stronger integrity
Voice preservation	Ask students to justify retained and rejected AI phrasing	Protection of authorial agency and stylistic ownership
Prompt literacy	Teach students to specify context, purpose, and constraints	More purposeful AI interaction and higher-quality output
Reflection-driven revision	Use revision memos, commentary, and self-explanation	Deeper learning rather than cosmetic editing

From a theoretical standpoint, this model reveals that AI's role in enhancing EFL academic writing is relational rather than deterministic. AI becomes effective when it operates inside a pedagogical design that links linguistic support to genre awareness, immediate feedback to reflective uptake, and technological efficiency to ethical accountability. In this sense, the real educational issue is not whether AI can write, but whether instructional design can transform AI-supported writing into learning rather than outsourcing.

Conclusion

This article has argued that artificial intelligence can play a significant role in enhancing academic writing skills in EFL contexts, but only under conditions that preserve the educational purpose of writing. A synthesis of foundational writing theory and recent AI research shows that AI is most beneficial when it provides timely scaffolding for planning, language formulation, revision, and metacognitive reflection. Automated writing evaluation systems are particularly useful for localized accuracy and repeated revision, whereas generative AI extends support to ideation, reformulation, and rhetorical experimentation. These tools therefore expand the ecology of writing instruction rather than merely accelerating correction.

At the same time, the article has shown that improvements in coherence, cohesion, and linguistic development are not guaranteed by tool use alone. AI can support cohesion through clearer sentence linking, lexical variation, and textual smoothing, but global coherence remains dependent on the writer's command of genre, logic, audience, and disciplinary purpose. Likewise, AI may improve fluency and lexical sophistication without ensuring epistemic depth or rhetorical adequacy. For this reason, the central claim of the article is that AI contributes to academic writing development only when it is mediated by pedagogy, not when it is treated as an autonomous writing agent.

The pedagogical implications are substantial. EFL educators need to move beyond a prohibition-versus-adoption binary and instead design writing instruction that integrates AI

critically and transparently. This requires explicit teaching of genre conventions, feedback literacy, prompt literacy, and source corroboration. It also requires assessment designs that value process, revision rationale, and authorial decision making rather than only polished final products. Students should be taught not merely how to obtain better AI outputs, but how to interrogate those outputs, adapt them responsibly, and maintain ownership of their academic voice.

In practical terms, educators should embed AI into staged writing tasks, use reflective revision memos, require disclosure of AI assistance, compare AI feedback with human feedback, and develop rubrics that distinguish linguistic polishing from intellectual contribution. Institutions, in turn, should support AI literacy policies that are pedagogically specific rather than purely punitive. Such policies must recognize that multilingual and EFL writers may experience AI as both a support and a risk.

Ultimately, the role of AI in EFL academic writing should be defined not by what the technology can generate, but by what learners can still learn while using it. A robust future for writing pedagogy lies in cultivating writers who can think with AI without surrendering judgment to it, revise with AI without losing voice, and participate in academic discourse with greater confidence, precision, and integrity.

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LANGUAGE AND STYLISTIC DEVICES IN ADVERTISING DISCOURSE: FUNCTIONAL CHARACTERISTICS

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Abstract: The article explores the functional role of language and stylistic devices in advertising discourse. It demonstrates that advertising is not limited to the transmission of information but also aims to attract attention, evoke emotions, and influence consumer decisions and behavior. The research emphasizes that advertising texts are typically constructed using simple and clear lexical units, including monosyllabic words, adjectives, personal pronouns, and short verbs, which ensure ease of comprehension and memorability.

Furthermore, the study highlights the importance of syntactic structures such as interrogative and imperative sentences. Interrogative forms stimulate curiosity and audience engagement, while imperative structures encourage direct action, increasing the persuasive impact of advertisements. In addition to lexical and syntactic features, stylistic devices such as metaphor, simile, hyperbole, personification, and sound patterns (alliteration, assonance, onomatopoeia) play a crucial role in enhancing the emotional and aesthetic appeal of advertising messages.

The findings indicate that the strategic use of these linguistic and stylistic elements strengthens communication effectiveness, shapes audience perception, and reinforces brand identity. Ultimately, advertising discourse illustrates how language can be creatively manipulated to achieve persuasive and communicative goals.

Key words: advertising discourse, stylistic devices, language functions, persuasion, consumer behavior, media communication, metaphor, syntax, lexical simplicity

REKLAM DISKURSUNDA DİL VƏ ÜSLUBİ VASITƏLƏR: FUNKSIONAL XÜSUSİYYƏTLƏR

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Xülasə: Bu məqalə reklam diskursunda dilin və bədii ifadə vasitələrinin funksional rolunu araşdırır. Göstərilir ki, reklam yalnız informasiyanın ötürülməsi ilə məhdudlaşmır, həm də diqqəti cəlb etmək, emosiyalar yaratmaq və istehlakçıların qərar və davranışlarına təsir göstərmək məqsədi daşıyır. Tədqiqat vurğulayır ki, reklam mətnləri adətən birhəcalı sözlər, sifətlər, şəxs əvəzlilikləri və qısa fellər kimi sadə və aydın leksik vahidlərdən istifadə olunmaqla qurulur. Bu isə mesajın asan qavranılmasını və yadda qalmasını təmin edir.

Bununla yanaşı, sual və əmr cümlələri kimi sintaktik strukturların əhəmiyyəti qeyd olunur. Sual formaları maraq oyadaraq auditoriyanın diqqətini artırır, əmr cümlələri isə birbaşa hərəkətə çağıraraq reklamın inandırıcılıq təsirini gücləndirir. Leksik və sintaktik xüsusiyyətlərlə yanaşı, metafora, bənzətmə, hiperbola, personifikasiya və fonetik vasitələr (alliterasiya, assonans, onomatopeya) reklam mesajlarının emosional və estetik təsirini artırmaqda mühüm rol oynayır.

Nəticələr göstərir ki, bu dil və üslub vasitələrinin strateji istifadəsi kommunikativ effektivliyi artırır, auditoriya qavrayışını formalaşdırır və brend kimliyini möhkəmləndirir. Ümumilikdə, reklam diskursu dilin inandırıcı və kommunikativ məqsədlərə çatmaq üçün yaradıcı şəkildə necə istifadə olunduğunu nümayiş etdirir.

Açar sözlər: reklam diskursu, bədii ifadə vasitələri, dil funksiyaları, inandırma, istehlakçı davranışı, media kommunikasiyası, metafora, sintaksis, leksik sadəlik

Introduction

Over the past decade, advertising, advertising messages, advertising texts, and advertising discourse in general have become one of the key directions of scholarly research. It is virtually impossible to imagine the communicative structure of modern society without advertising. As a specific form of social interaction, advertising encompasses almost all spheres of public life and is closely connected with various types of activities.

The concept of discourse is primarily associated with various forms of communication in society (communicative discourse) and also encompasses the visual aspects of the communication process (visual discourse). In addition, discourse is linked to the expression of communication rules, modes of representation, and the realization of speakers' pragmatic goals. In many sources, discourse is regarded as a form of cultural communication. Despite ethnocultural differences, intercultural discourse is also distinguished in communication. The social, age, and gender characteristics of communication participants are likewise taken into account when identifying types of discourse. From this perspective, notions such as political discourse and feminist discourse emerge.

As one of the genres of media discourse, advertising messages play an active role in shaping the media space and create a distinctive model of virtual reality. Advertising functions as a stimulating mechanism in forming perceptions of the world and its interpretation, while also serving as a means of modeling and disseminating the values of mass culture.

Materials and Methods

This study adopts a qualitative descriptive approach to investigate the functional characteristics of linguistic and stylistic devices in advertising discourse. The research material consists of English-language advertising texts and slogans selected from various media sources, including print advertisements, online campaigns, and brand promotional materials.

The analysis is based on discourse analysis and stylistic analysis. These methods were employed to identify lexical, syntactic, and stylistic features used in advertising messages and to examine their communicative, persuasive, and expressive functions. Particular attention was paid to the use of pronouns, adjectives, verbs, sentence structures, metaphors, similes, personification, and other stylistic devices that contribute to the effectiveness of advertising discourse. The findings were interpreted within the framework of media discourse studies and advertising communication.

Results and Discussion

Advertising creates a specific cognitive state in the audience, generates particular emotional and behavioral responses, and influences the formation or modification of intentions. It shapes individuals' attitudes toward themselves and society and influences the formation of collective and individual behavioral patterns that reflect human perception and understanding of the world.

Media discourse is characterized as an event-based form of speech and is considered a type of social activity. Within this activity, cognitive structures oriented toward the inner world of the linguistic personality play a central role. The formation of discourse occurs around a system of generalized knowledge, resulting in a specific semantic context.

As a genre of media discourse, advertising contributes to the systematization of information and makes it accessible for human cognitive activity. On the one hand, it facilitates interpretation; on the other hand, it influences the mechanisms of perceiving reality and directs interpretation in a particular way. This feature is conditioned by the subjective attitude of the advertiser toward the object or phenomenon being presented.

Zhou (2018, p. 400) notes that English-language advertising uses monosyllabic nouns, simple adjectives, and personal pronouns to ensure messages are clear, memorable, and persuasive. These elements help increase consumer awareness and willingness to purchase products.

Pronouns such as I, we, you, and your reduce psychological distance, fostering a friendly atmosphere that strengthens consumers' willingness to purchase.

Adjectives occupy an important place in the language of English advertising. Descriptive adjectives, as well as comparative and superlative forms, are used to highlight product features and make brands more appealing to consumers. Comparative and superlative adjectives allow brands to demonstrate their advantages over competing products and emphasize the overall quality of goods or services (Zhou, 2018, p. 400).

Pronouns also play a significant role in advertising language, particularly when directly addressing the audience in slogans. Addressing the audience through pronouns increases the persuasive power of advertising and raises the likelihood of purchase. Studies show that the use of pronouns in advertising texts increases consumer interest more than in cases where they are absent (de Hoop et al., 2024, p. 10).

Another important lexical element in advertising texts is verbs. Companies tend to prefer simple and short verbs in their messages. These verbs are effective in describing product features and conveying social messages clearly and impactfully.

Zhou (2018, p. 400) emphasizes that the use of such verbs simplifies expression and makes advertising texts more comprehensible, thereby enhancing memorability. Common examples include *make, get, feel, give, have, see, buy, know, look, taste, take, start, and choose*. Most of these verbs are monosyllabic and belong to the core vocabulary of English. Their stable and clear meanings contribute to their frequent use in everyday communication. In advertising, the selection of such simple words helps attract more consumers by ensuring clarity, effectiveness, and a sense

of sincerity. In contrast, long and complex verbs are generally unnecessary. On the contrary, concise verb choices are more effective in capturing attention, creating humor, and increasing the impact of the message.

From a syntactic perspective, advertising texts use various sentence types. Declarative sentences primarily provide information and do not directly prompt action or invite questioning. Therefore, they are considered relatively passive and less interactive in communication.

In contrast, interrogative sentences are widely used to attract attention and stimulate interest. Zhou (2018, p. 401) notes that questions easily provoke reactions, especially when used in headlines or at the beginning of texts. This technique stimulates psychological engagement and encourages the audience to continue reading. At the same time, questions prompt reflection, evoke emotional responses, and intensify curiosity.

Another commonly used syntactic device in advertising is the imperative sentence. These structures are used to motivate, add dynamism and energy to slogans, and encourage the audience to take specific actions (Zhou, 2018, p. 402). Imperatives directly guide consumers toward action, thereby increasing the effectiveness of the advertising message.

Stylistic devices play a crucial role in advertising discourse. They are employed to intensify meaning, create emotional impact, and attract audience attention. Rhetorical figures and tropes occupy a central place in this context. A trope is characterized as a deliberate deviation from linguistic norms aimed at achieving a specific rhetorical effect.

One of the most widely used tropes in advertising texts is metaphor. In modern linguistics, metaphor is understood as conceptualizing one concept through another. Metaphorical expressions provide a creative and appealing presentation of advertising messages while allowing for multilayered meaning (Hussein, 2022, p. 324).

For instance, in the slogan “Keep your eyes on the new,” meaning is constructed through two main cognitive mechanisms. The first is the conceptual metaphor based on the expression “keep your eyes on it.” The second is the reference to the Audi brand, which anchors the message to a specific product and prevents interpretive ambiguity.

Another well-known example is Apple’s slogan “Think different.” This slogan goes beyond presenting technological products and metaphorically conveys the idea of creativity and innovative thinking. Thus, the brand positions itself not only as a technology producer but also as a symbol of unconventional thinking.

A simile is a trope based on the comparison of two different objects and serves to explain an unfamiliar concept through a familiar one. Metonymy, on the other hand, involves representing an object through a related concept and is used in advertising to activate the consumer’s imagination.

For example, in the slogan of the B&Q retail chain, “You don’t buy a life – you build one. You can do it,” the company’s function is linked to the idea of building a life. This reflects the use of conceptual metonymy, presenting the company’s activity as an invitation to a particular lifestyle.

Personification is also widely used in advertising discourse. In this technique, inanimate objects or abstract concepts are endowed with human characteristics, enhancing emotional impact and making the content more accessible.

For instance, in M&M’s advertisements, chocolate candies are portrayed with human traits: they speak, move, and express emotions. This contributes to a more vivid and memorable presentation of the product.

Onomatopoeia, the imitation of sounds, is another commonly used device. It recreates the sounds produced by the product, making the message more vivid and sensory (Amjad & Shah, 2023, p. 152). For example, “Snap! Crackle! Pop!” in Rice Krispies advertisements imitates the product’s sounds, creating both auditory and emotional impact while enhancing memorability.

Hyperbole and irony contribute to expressiveness and humor in slogans. Hyperbole intensifies impact through exaggeration, although excessive use may undermine credibility. Irony creates a contrast between literal and intended meanings, allowing for multiple interpretations.

For example, Disneyland’s slogan “The Happiest Place on Earth” uses hyperbole to position the location as an ideal entertainment destination. This exaggeration evokes positive emotional associations and enhances the park's appeal.

Contrast is used in advertising to highlight product features and emphasize competitive advantages. Phonetic devices such as rhyme, rhythm, alliteration, and assonance contribute to memorability and aesthetic appeal.

For instance, Coca-Cola frequently employs alliteration and rhythmic structures. In the slogan “Coca-Cola – Open Happiness,” the product is associated with joy and happiness, presenting the brand as an emotional experience.

Similarly, in the phrase “You will have healthy and shiny white skin,” assonance is evident in the repetition of the vowel sounds in “healthy” and “shiny,” enhancing phonetic harmony.

Repetition devices such as anaphora and epiphora emphasize key information and aid memorization. Advertisers may also omit conjunctions in slogans to highlight essential elements or social messages. This device, known as asyndeton, involves the deliberate omission of conjunctions in sequences of words, phrases, or clauses (Adnan, 2024, p. 1). It adds dynamism and impact while ensuring brevity and memorability.

Wordplay is another widely used technique in advertising discourse. By creating double meanings, it introduces humor and semantic depth, making messages both attention-grabbing and memorable (Winter-Froemel et al., 2018, p. 1).

Conclusion

In general, the use of stylistic devices shows that language extends beyond literal meaning and can be creatively constructed. Recognizing literary devices and rhetorical figures provides a basis for analyzing how advertisers influence consumer perception and behavior through language.

Advertisers frequently use imperative sentences, taking into account the public’s familiarity with receiving guidance and recommendations. This strategy creates a consistent and persuasive tone, enabling companies to achieve their primary goal of influencing consumers.

Ultimately, linguistic styles are powerful tools, as even minor changes can significantly alter an advertisement's tone and enhance its persuasiveness.

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ENHANCING COHERENCE AND COHESION IN EFL ACADEMIC WRITING: A THEORETICAL FRAMEWORK FOR EFFECTIVE TEXT CONSTRUCTION

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Abstract: In EFL academic writing, textual effectiveness is often judged through grammatical accuracy and lexical range, yet formally correct texts frequently remain difficult to read because they do not sustain conceptual continuity or explicit semantic connection. This article develops a theoretical framework for effective text construction by synthesizing major scholarship on cohesion, coherence, genre, discourse, and linguistic complexity. Using a literature-based analytical methodology, the study examines ten foundational works published between 1976 and 2019 that together represent systemic functional linguistics, discourse analysis, genre theory, cognitive approaches to text processing, and second language writing research. The analysis argues that cohesion and coherence should not be treated as interchangeable concepts or reduced to sentence-level linking devices. Cohesion provides observable textual signals, whereas coherence emerges from the interaction of those signals with rhetorical purpose, genre expectations, disciplinary conventions, information flow, and reader inference. The article proposes an integrated framework consisting of four interdependent dimensions: micro-level cohesive patterning, meso-level paragraph development, macro-level genre organization, and reader-oriented processing. It further argues that EFL writing pedagogy should move beyond teaching connectors in isolation and instead emphasize lexical chaining, referential clarity, argument progression, and genre-sensitive rhetorical structuring. The study concludes that effective academic writing depends on aligning linguistic form, discourse purpose, and reader interpretation within a coherent communicative design.

Key words: coherence; cohesion; EFL academic writing; genre; discourse organization; reader processing

XDİD (EFL) AKADEMİK YAZILARINDA MƏNA VƏ STRUKTUR BAĞLILIĞININ ARTIRILMASI: EFTEKTİV MƏTN QURULMASI ÜÇÜN NƏZƏRİ ÇƏRÇİVƏ

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Xülasə: Xarici dil kimi ingilis dili öyrənilən mühitlərdə akademik yazının keyfiyyəti çox vaxt qrammatik düzgünlük və söz ehtiyatı ilə qiymətləndirilir. Lakin qrammatik baxımdan düzgün mətnlər belə ideyalar arasında məntiqi bütövlük və dil vahidləri arasında semantik əlaqə qurulmadıqda təsirli olmur. Bu məqalə səmərəli mətn quruculuğu üçün nəzəri çərçivə təqdim edir və koheziya, koherentlik, janr, diskurs və dil mürəkkəbliyi ilə bağlı əsas tədqiqatları sintez edir. Tədqiqat 1976–2019-cu illər arasında nəşr olunmuş on əsas mənbə üzərində qurulan ədəbiyyat əsaslı analitik yanaşmadan istifadə edir. Təhlil göstərir ki, koheziya və koherentlik eyni anlayışlar deyildir: koheziya mətndə görünən dil işarələrini təmin edir, koherentlik isə həmin işarələrin ritorik məqsəd, janr gözləntiləri, informasiyanın axını, fənn üzrə diskurs normaları və oxucu inferensiyası ilə qarşılıqlı təsiri nəticəsində formalaşır. Məqalədə dörd ölçülü integrativ çərçivə irəli sürülür: mikro-səviyyəli koheziya, abzas səviyyəsində fikrin inkişafı, makro-səviyyəli janr təşkili və oxucu yönümlü emal. Nəticə etibarilə, EFL akademik yazısının tədrisi yalnız bağlayıcı sözlərin öyrədilməsi ilə kifayətlənməməli, arqumentin inkişafı, leksik zəncirlər, referensial aydınlıq və janra uyğun strukturlaşdırma üzərində qurulmalıdır.

Açar sözlər: koherentlik; koheziya; EFL akademik yazısı; janr; diskursun təşkili; oxucu yönümlülük

Introduction

Academic writing occupies a central position in higher education because it is both a medium of knowledge display and a means through which disciplinary membership is enacted. For EFL learners, this task is particularly demanding because writing in English requires more than the accurate production of grammar and vocabulary; it also requires control over the discursive logic that allows texts to be understood as purposeful, unified, and persuasive. Hyland (2019) emphasizes that second language writing is best understood not as the transfer of isolated language forms into sentences, but as a socially situated activity in which writers negotiate audience, genre, and disciplinary expectations. In that sense, EFL academic writing is not simply a linguistic exercise. It is a meaning-making practice that asks learners to organize knowledge, establish relationships between propositions, and guide readers through increasingly complex lines of argument.

Within this broad field, coherence and cohesion remain two of the most important yet most frequently conflated concepts. Halliday and Hasan (1976) provided the classic linguistic account of cohesion by demonstrating how texts achieve texture through semantic relations expressed by reference, substitution, ellipsis, conjunction, and lexical cohesion. Their work remains foundational because it established that a text is not merely a sequence of grammatically acceptable sentences but a connected semantic unit. At the same time, later scholarship has shown that a text may display numerous cohesive ties and still fail to communicate a meaningful argument. Witte and Faigley (1981) made this point especially clear by distinguishing between measurable cohesive relations and the broader judgment of coherence associated with writing quality. Their study redirected attention from visible surface connection to the ways propositions are ordered and interpreted across the whole discourse.

This distinction matters acutely in EFL settings because instruction often privileges what is most visible and most easily teachable. Connectors such as *however*, *therefore*, *in addition*, and *on the other hand* are frequently presented as the primary devices of coherent writing. Yet such teaching can produce a misleading pedagogical outcome: learners may insert transitional markers into nearly every sentence while still producing texts whose claims, evidence, and paragraph progression remain conceptually fragmented. In other words, the visible signs of connectedness may be present without the deeper logic that allows a reader to recognize why ideas belong together. Graesser et al. (2003) argue that coherence relations are not simply encoded in text; they are also constructed by readers through inferential processing. From this perspective, coherence is neither a purely textual property nor a purely subjective reaction. It arises through the interaction between textual cues and the reader's ability to recognize causal, temporal, contrastive, elaborative, and hierarchical relations across discourse.

The problem becomes even more complex in academic contexts because coherence is shaped by genre and disciplinary convention. Swales (1990) showed that academic texts are structured as recognizable communicative events, whose rhetorical moves create expectations about what counts as an effective introduction, literature review, argument, or conclusion. Hyland (2004) further demonstrated that academic discourse is deeply social and disciplinary, so that successful writing depends on the capacity to align textual choices with community-specific norms of stance, evidence, and reader engagement. A paragraph may be cohesive at the sentence level and yet remain rhetorically incoherent if it appears in the wrong place, serves no identifiable move, or fails to advance the line of inquiry expected by its discourse community. Paltridge (2012) reinforces this view by treating discourse as patterned and contextualized, rather than as a sum of separate sentences.

Research on writing quality has also complicated the relationship between coherence, cohesion, and textual sophistication. Crossley and McNamara (2012) show that second language writing proficiency is predicted not by cohesion alone but by a combination of cohesion and linguistic sophistication. Similarly, McNamara et al. (2010) demonstrate that high-quality writing reflects the interaction of multiple linguistic features, including cohesion, lexical choice, and syntactic patterning. Biber et al. (2011) add a further important dimension by arguing that grammatical complexity in academic writing should not be measured through the lens of conversation-like clausal elaboration. Mature academic writing often relies instead on phrasal compression, dense noun phrase structures, and informational packaging. This means that textual effectiveness in EFL academic writing cannot be explained by sentence linking alone; it depends on how information is distributed, compressed, repeated, and rhetorically staged.

Despite the richness of these traditions, the literature has often developed along parallel lines. Linguistic accounts have described cohesive mechanisms with precision, cognitive research has examined how readers process inferential relations, genre theory has focused on rhetorical organization, and second language writing has investigated instructional and proficiency-related factors. What remains insufficiently developed is a theoretically integrated model that brings these strands together for the specific problem of EFL academic writing. Too often, coherence is treated as a vague quality of "flow," while cohesion is reduced to a checklist of connectors. Such treatment obscures the layered nature of academic discourse and limits pedagogical intervention to local sentence repair rather than structural text design.

Against this background, the present article aims to construct a comprehensive theoretical framework for effective text construction in EFL academic writing by synthesizing foundational work on cohesion, coherence, genre, discourse, reader processing, and linguistic complexity. It addresses two interrelated questions: how should coherence and cohesion be conceptually distinguished in academic writing, and what integrated framework can guide EFL writers and teachers toward more effective text construction? By answering these questions, the study seeks to contribute a more precise account of what makes academic writing readable, meaningful, and rhetorically successful.

Materials and Methods

This study adopts a literature-based, theory-building methodology designed to generate an integrated conceptual framework rather than an empirical measurement of student texts. The materials of the study consist of ten purposively selected sources that jointly represent the principal traditions relevant to the topic: systemic functional linguistics, discourse analysis, genre theory, cognitive approaches to comprehension, second language writing studies, and linguistic analyses of writing quality. The corpus includes five monographs and five journal articles or book chapters published between 1976 and 2019. These works were selected because they are foundational within their respective traditions and because, taken together, they cover the most important dimensions of coherence and cohesion in academic writing: textual linkage, discourse organization, reader inference, genre expectations, linguistic complexity, and pedagogical implications.

The methodological orientation of the study is integrative rather than aggregative. It does not seek to count the frequency of terms, rank findings statistically, or produce a bibliometric map of the field. Instead, it follows an interpretive analytical procedure in which concepts are first identified within each source and then compared across traditions to determine points of convergence, tension, and theoretical complementarity. In the first stage of analysis, each source was read for its central definitions, assumptions, and analytical categories related to cohesion, coherence, writing quality, genre, discourse, and reader processing. In the second stage, these categories were grouped into broader conceptual domains: micro-level textual relations, local and global discourse organization, cognitive coherence-building, disciplinary and genre-based structuring, and linguistic resources associated with academic prose. In the third stage, these domains were synthesized into a multi-dimensional framework capable of explaining how effective academic texts are constructed in EFL contexts.

A literature-based methodology is especially appropriate for the present topic because the difficulty under discussion is not the absence of data, but the fragmentation of theoretical perspectives. Halliday and Hasan (1976) explain how texts achieve texture, yet their model alone does not fully account for genre expectations or reader inferencing. Swales (1990) and Hyland (2004) explain how discourse communities shape rhetorical organization, but their accounts require connection to the sentence- and paragraph-level resources through which the organization becomes visible. Graesser et al. (2003) show that coherence is partly a function of reader processing, while Crossley and McNamara (2012) and McNamara et al. (2010) demonstrate that writing quality emerges from the interaction of cohesion and linguistic sophistication. The present method therefore treats the selected studies not as isolated findings but as complementary lenses on a shared problem.

The analytical approach is best described as layered synthesis. At the first layer, the study asks what cohesion is and how it is linguistically realized. At the second layer, it asks how coherence is established across clauses, sentences, and paragraphs. At the third layer, it examines

how genre structures and disciplinary practices shape textual interpretation. At the fourth layer, it considers how reader-oriented processing mediates the success of those textual signals. The final framework developed in the Results and Discussion section results from the interaction of these layers. Because the study is theoretical in nature, its contribution lies in explanatory integration and pedagogical applicability rather than in statistical generalizability. Its validity depends on conceptual consistency, argumentative rigor, and fidelity to the selected source base.

Results and Discussion

The analysis yields a first central finding: coherence and cohesion are functionally related but conceptually distinct. Halliday and Hasan (1976) define cohesion as the set of semantic relations that make a stretch of language interpretable as a text rather than a random sequence of sentences. Their contribution is decisive because it makes visible the resources through which textual connectedness is signaled. Yet the present synthesis indicates that these signals should not be mistaken for coherence itself. Witte and Faigley (1981) demonstrate that writing quality cannot be explained by the mere presence or frequency of cohesive ties. A text may contain abundant conjunctions, repeated lexical items, and consistent reference chains while still appearing unfocused, poorly organized, or conceptually thin. Coherence therefore cannot be reduced to surface connectivity. It refers to the intelligible organization of meaning, the sense that propositions belong together within a purposeful whole. Cohesion is one of the means by which that sense is made available, but it is not identical with the achieved effect.

This distinction has important implications for EFL writing. In many classrooms, cohesion is parsed into teachable units and coherence is left as an impressionistic criterion. Such an imbalance encourages students to equate good writing with the insertion of transitional markers. Halliday and Hasan's (1976) taxonomy itself warns against this reduction. Conjunction is only one type of cohesive resource; lexical cohesion and reference are equally central, and often more decisive in academic texts. Topic continuity in scholarly prose depends heavily on how key concepts are repeated, reformulated, specified, and related across sentences. If a writer repeatedly changes labels for a central concept in the name of stylistic variety, the result may weaken conceptual continuity rather than improve style. Controlled lexical repetition, especially in disciplinary writing, is not redundancy but precision. Hyland (2004) implicitly supports this point by showing that academic discourse is governed by communal expectations about how knowledge is named and stabilized across a text.

A second major result of the analysis is that effective cohesion in academic writing operates not as ornament but as functional patterning. Halliday and Hasan (1976) identify reference, substitution, ellipsis, conjunction, and lexical cohesion as distinct mechanisms, but their practical value in academic prose depends on how they serve propositional development. Referential clarity is especially important. Ambiguous pronouns, unstable noun phrases, or shifting labels disrupt readers' efforts to build a coherent mental model of the text. By contrast, clear referential chains allow readers to track entities, arguments, and conceptual categories as they develop. Lexical cohesion is equally significant because academic writing relies on the gradual elaboration of a limited set of key notions. When those notions recur in semantically controlled ways, the text acquires thematic steadiness. Paltridge (2012) supports such an interpretation by stressing that discourse meaning emerges from patterned language use in context, not from isolated sentences.

However, micro-level cohesion becomes meaningful only when it supports local and global coherence. Witte and Faigley (1981) show that writing quality is related to how propositions are

arranged across discourse, not only to how sentences are linked. In theoretical terms, local coherence concerns the immediate relationship between adjacent clauses and sentences, while global coherence concerns the overall logic, purpose, and hierarchical organization of the text. A paragraph may be locally coherent if each sentence relates clearly to the one before it, yet globally incoherent if the paragraph contributes nothing meaningful to the central claim of the text. This distinction is vital in EFL academic writing because many learner texts appear smooth at the sentence level but diffuse at the argument level. The writer may move from one idea to another through visible transitions while failing to establish why the sequence matters. This problem is not primarily grammatical; it is organizational and rhetorical.

The cognitive dimension of coherence further clarifies why surface linking is insufficient. Graesser et al. (2003) argue that readers build coherence by recognizing relations such as cause, condition, contrast, elaboration, and sequence. Some of these relations are explicitly signaled, but others must be inferred from context and prior knowledge. Coherence, then, is not embedded in the text in a self-contained way. It is co-constructed through interaction between textual cues and reader processing. This insight helps explain why the same text may appear clearer to one audience than to another. Highly explicit cohesion may be especially helpful for readers with limited background knowledge, whereas expert readers can often recover relations with fewer explicit signals. For EFL writers, the pedagogical lesson is not that all writing should become maximally explicit, but that writers must learn to anticipate reader needs and calibrate explicitness accordingly. Academic writing is reader-oriented: it must not merely state information, but guide interpretation.

This reader-oriented dimension is reinforced by research on text quality. Crossley and McNamara (2012) show that second language writing proficiency is predicted by both cohesion and linguistic sophistication, which suggests that clarity and development are not competing goals but interacting dimensions. McNamara et al. (2010) likewise argue that writing quality is associated with multiple linguistic features rather than a single measure of connectedness or complexity. An important implication follows from this: coherence is not achieved through maximal repetition, nor through maximal density, nor through maximal signaling. Instead, it emerges when a text balances interpretive accessibility with rhetorical maturity. Overly repetitive prose may become monotonous and conceptually stagnant; excessively compressed prose may become opaque. Effective writing occupies a middle ground in which the reader is guided sufficiently but not overburdened with unnecessary statement of the obvious.

The genre dimension adds a third essential layer to the analysis. Swales (1990) demonstrates that academic texts are organized through conventionalized rhetorical moves recognized by discourse communities. These moves do not merely decorate content; they structure reader expectations and produce macro-level coherence. A research introduction, for instance, is not coherent simply because its sentences are linked. It is coherent because it stages a recognizable progression from territory to niche to purpose, or through whatever variation a discipline permits. When EFL writers fail to grasp genre structure, they often produce paragraphs that are individually understandable yet collectively purposeless. A definition may appear before the problem is established, a claim may be introduced before the literature context is set, or evidence may be offered without rhetorical framing. Such writing may contain cohesive ties, but it lacks genre-based coherence because the sections do not enact the communicative sequence expected by readers.

Hyland (2004) deepens this argument by showing that academic writing is not only organized by genre but also shaped by disciplinary discourse. Coherence, from this perspective, is socially

negotiated. What counts as adequate background, acceptable evidence, legitimate stance, or effective signaling differs across fields. Scientific writing, humanities argumentation, and social science discussion do not construct reader alignment in identical ways. The implication for EFL writing is far-reaching: coherence cannot be taught as a universal template detached from discourse community practices. Students must learn how knowledge is organized in the genres they are expected to produce and how readers in those genres interpret textual signals. Hyland's broader account of interaction in academic writing also suggests that coherence has an interpersonal aspect. Writers use textual and interpersonal resources to position claims, anticipate objections, and guide reader response. A coherent text therefore does more than connect sentences; it manages a relationship with its audience.

This social view of coherence also helps explain why metadiscursive features matter. Although the present study focuses primarily on coherence and cohesion, Hyland (2004) shows that reader guidance in academic discourse involves a wider repertoire of signaling devices. These include markers that frame the organization of the text, indicate transitions in argument, qualify claims, and orient readers to interpretive priorities. Such resources occupy an intermediate space between cohesion and rhetoric. They are not purely grammatical ties, yet they contribute directly to the reader's ability to follow the argument. Seen in this light, coherence is achieved not simply by connecting content propositions but by making the writer's rhetorical path legible.

A fourth major finding of the synthesis concerns linguistic complexity and its relationship to coherent writing. Biber et al. (2011) argue that conversation-based notions of grammatical complexity are inadequate for the analysis of academic writing. Spoken discourse often develops through clausal elaboration, whereas academic prose frequently achieves sophistication through phrasal compression and dense noun phrase structures. This observation is crucial for EFL academic writing because learners are often encouraged to produce "complex" sentences without being taught how academic information is actually packaged. If students rely excessively on chained subordinate clauses, their writing may sound elaborate but remain loosely organized. Conversely, if they attempt dense nominal style too early, their texts may become abstract and difficult to process. Coherence therefore depends partly on the writer's ability to choose forms of complexity appropriate to academic discourse and reader expectations.

The interaction between cohesion and linguistic sophistication is especially important here. Crossley and McNamara (2012) indicate that stronger texts tend to combine connectedness with lexical and structural maturity. Yet maturity does not mean obscurity. McNamara et al. (2010) suggest that features associated with strong writing do not operate independently but interact in dynamic ways. A text may score highly on sophistication while becoming less readable if lexical items are too infrequent, abstractions underexplained, or syntactic embedding excessive. By contrast, lower-level writing may be highly explicit but conceptually simple. The theoretical consequence is that coherence in academic writing must be understood as a balancing act between precision, density, and accessibility. EFL learners need support not only in building sentences but also in deciding how much information to pack into those sentences and how to maintain interpretive continuity as complexity increases.

At this point in the synthesis, a broader conceptual picture emerges. Effective academic writing depends on alignment across four interdependent dimensions. The first is micro-level cohesive patterning, which includes reference, lexical chains, conjunction, and other mechanisms that create visible semantic connection. The second is meso-level discourse progression, which

concerns paragraph unity, logical sequencing, topic development, and the movement from one proposition to the next. The third is macro-level genre organization, through which sections and paragraphs contribute to recognizable rhetorical purposes within a discipline. The fourth is reader-oriented processing, which concerns how explicit, inferable, accessible, and rhetorically legible the text is for its intended audience. These dimensions should not be imagined as separate layers added one after another. Rather, they operate recursively. Micro-level choices support meso-level development; meso-level organization realizes macro-level genre purpose; macro-level expectations shape how readers interpret micro-level cues.

This integrated framework explains why many EFL texts fail despite local grammatical accuracy. Some texts are rich in conjunctive signals but weak in lexical focus, producing the impression of movement without stable thematic development. Others maintain a topic lexically but fail to organize paragraphs so that each one advances the argument. Still others present a sequence of relevant paragraphs without enacting a recognizable genre logic. In such cases, the breakdown in writing quality cannot be attributed to one missing connector or one incorrect sentence. It results from misalignment across levels of discourse. This is precisely why narrow correction practices, especially those centered on sentence grammar alone, often produce limited improvement in academic writing. Repairing local errors does not automatically create a coherent architecture of meaning.

The framework also helps refine the concept of paragraph development, which occupies a central place in academic writing but is often taught superficially. A coherent paragraph is not merely a group of sentences on the same general topic. It is a structured unit in which one controlling idea is introduced, developed, specified, or evidenced, and then integrated into the larger argument. Paltridge (2012) and Hyland (2019) both support the view that discourse organization is shaped by patterns of development and context-sensitive expectations. In EFL instruction, paragraphing is frequently reduced to the formula of topic sentence, supporting details, and concluding sentence. While this formula may serve beginners, it becomes inadequate if detached from rhetorical purpose. What matters is not formal symmetry alone, but whether the paragraph performs identifiable argumentative work. The present analysis therefore treats meso-level coherence as the bridge between sentence connection and genre structure.

A critical implication of this argument is that cohesion should be taught functionally, not mechanically. Halliday and Hasan's (1976) categories remain indispensable, but their pedagogical value lies in showing what cohesive resources do, not merely what they are called. For example, conjunction should be taught as the expression of semantic relations already present in the logic of the discourse. If the relation is unclear at the level of thought, no connector will solve the problem. Likewise, lexical cohesion should be taught through the management of key terms, conceptual chains, and disciplinary vocabulary, rather than through generalized advice to "avoid repetition." Referential cohesion should be linked to reader tracking, helping students understand why ambiguous pronouns or unstable labels disrupt interpretation. Such instruction would move the teaching of coherence away from cosmetic linking and toward purposeful text construction.

The social and disciplinary dimension of the framework also has consequences for assessment. Writing rubrics often list coherence and cohesion as scoring categories, but they rarely define them with sufficient precision. As a result, assessment collapses into vague judgments about whether the text "flows." The present synthesis suggests that a more rigorous assessment model should examine at least four questions. Does the text maintain coherent referential and lexical

patterning? Do paragraphs develop ideas in a logically progressive way? Does the overall structure realize the rhetorical purpose of the genre? Does the text guide the reader effectively through explicit signaling, appropriate explicitness, and manageable informational density? Although these questions can be operationalized in different ways, they preserve the multidimensional nature of writing quality better than a checklist of transitions or sentence-level correctness.

The analysis also reveals a theoretical tension that should not be ignored. Research on cohesion often values explicitness because it supports interpretability, while advanced academic discourse sometimes prizes compression, subtle inferencing, and disciplinary economy. This tension is not a contradiction but a developmental and contextual reality. Beginners may require greater explicitness to stabilize discourse relations, whereas advanced writers can rely more on controlled inferencing because their readers share genre knowledge and conceptual background. Hyland (2019) makes clear that second language writing development involves gradual participation in discourse practices, not immediate mastery of expert norms. The pedagogical task, therefore, is not to choose between explicit cohesion and sophisticated concision, but to stage instruction so that learners move from overt textual support toward more disciplined, economical writing without sacrificing readability.

Within this theoretical landscape, the article's main contribution is the proposition that coherence in EFL academic writing is best understood as structured interpretability. That interpretability is accomplished through cohesion, but also through rhetorical sequencing, disciplinary expectation, and reader management. Cohesion is the visible texture of discourse; coherence is the achieved intelligibility of that texture within a rhetorical and social context. When EFL instruction treats coherence as a mysterious quality or cohesion as a narrow grammar topic, it fragments a phenomenon that is intrinsically integrated. By contrast, a layered framework allows teachers, researchers, and students to diagnose writing problems more accurately. A weak text may fail because its lexical chains are unstable, because its paragraphs do not accumulate toward a claim, because its genre structure is missequenced, or because it assumes more shared knowledge than its audience possesses. These are different problems, and each requires a different pedagogical response.

Conclusion

This article has argued that coherence and cohesion are indispensable to EFL academic writing, but only when they are understood in a conceptually differentiated and theoretically integrated way. The synthesis of ten foundational sources demonstrates that cohesion refers to the textual mechanisms through which semantic relations become visible, whereas coherence refers to the intelligible organization of meaning that readers recognize across clauses, paragraphs, and whole texts. The relationship between them is intimate but not identical. Cohesive devices provide interpretive cues, yet coherence emerges only when those cues interact successfully with genre structure, rhetorical purpose, disciplinary convention, linguistic complexity, and reader inference.

The theoretical framework developed in this article organizes effective academic text construction into four interdependent dimensions: micro-level cohesive patterning, meso-level discourse progression, macro-level genre organization, and reader-oriented processing. This model explains why many EFL texts remain weak even when sentence grammar appears accurate. It also clarifies why the teaching of coherence cannot be reduced to transition words or paragraph templates. Effective academic writing requires writers to sustain lexical focus, manage reference precisely, develop paragraphs as purposeful argumentative units, and arrange sections according to

recognizable genre logic. At the same time, writers must calibrate explicitness and density in relation to the needs of their readers.

The implications for EFL teaching are substantial. Instruction should explicitly connect cohesive resources to discourse function rather than teaching them as isolated grammatical items. Students need guided practice in tracing lexical chains, revising referential ambiguity, reorganizing paragraph progression, and recognizing how genre moves shape reader expectations. Model texts should be analyzed not only for language accuracy but also for how they build local and global coherence. Feedback should address whether ideas accumulate meaningfully, not merely whether sentences are correct. Assessment practices, likewise, should evaluate writing across multiple levels of organization, from lexical continuity to rhetorical architecture.

Several recommendations follow from the proposed framework. First, coherence and cohesion should be presented as mutually reinforcing dimensions of discourse, not as separate scoring labels with undefined boundaries. Second, EFL writing pedagogy should prioritize argument progression and conceptual continuity alongside grammatical development. Third, discipline-sensitive instruction is necessary, since what counts as coherent writing varies across academic communities (Hyland, 2004; Swales, 1990). Fourth, students should be supported in moving gradually from highly explicit linkage toward more compressed and rhetorically mature academic prose, in line with the developmental view of second language writing advanced by Hyland (2019). Finally, future research can extend the present framework by testing it against student corpora, classroom interventions, and genre-specific writing tasks.

In sum, the central claim of this study is that effective EFL academic writing is not the product of grammatical correctness alone. It is the result of a disciplined alignment between linguistic form, textual progression, genre expectation, and reader interpretation. When coherence and cohesion are treated as an integrated architecture of meaning rather than a set of discrete mechanical features, academic writing pedagogy becomes more analytically grounded, more pedagogically useful, and more capable of developing writers who can construct texts that are not only correct, but genuinely communicative and intellectually persuasive.

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THE IMPACT OF ARTIFICIAL INTELLIGENCE (AI) IN ACADEMIC WRITING PROCESSES

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Abstract: This study investigates the impact of artificial intelligence (AI) on academic writing processes. AI tools have become increasingly important in education, supporting students in different stages of writing such as generating ideas, organizing content, and improving language accuracy. The findings of this study show that tools such as Grammarly, ChatGPT, Zotero, QuillBot, and Turnitin play a significant role in enhancing the quality and efficiency of academic writing. They help students reduce grammatical errors, structure their texts more effectively, and manage references more easily.

However, the study also identifies several challenges associated with the use of AI. These include the risk of plagiarism, reduced critical thinking skills, over-reliance on technology, and ethical concerns such as data privacy and the possibility of biased or inaccurate information. These limitations indicate that while AI provides valuable support, it is not without risks.

The results suggest that artificial intelligence is a useful supportive tool in academic writing, but it should be used in a balanced and responsible way, assisting rather than replacing students' own thinking and writing abilities.

Key words: Artificial Intelligence, AI tools, Benefits, Challenges

AKADEMİK YAZI PROSESLƏRİNDƏ SÜNİ İNTELLEKTİN (AI) TƏSİRİ

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Xülasə: Bu tədqiqat süni intellektin (AI) akademik yazı proseslərinə təsirini araşdırır. Süni intellekt alətləri təhsildə getdikcə daha mühüm rol oynayır və tələbələrə yazının müxtəlif mərhələlərində – ideyaların yaradılması, məzmunun təşkili və dilin dəqiqliyinin artırılması kimi sahələrdə dəstək verir. Tədqiqatın nəticələri göstərir ki, Grammarly, ChatGPT, Zotero, QuillBot və Turnitin kimi alətlər akademik yazının keyfiyyətini və səmərəliliyini artırmaqda mühüm rol oynayır. Bu alətlər tələbələrə qrammatik səhvləri azaltmağa, mətnlərini daha effektiv qurmağa və istinadları daha asan idarə etməyə kömək edir.

Lakin tədqiqat süni intellektdən istifadəyə dair bir sıra çətinlikləri də müəyyən edir. Bunlara plagiat riski, tənqidi düşünmə bacarıqlarının azalması, texnologiyadan həddindən artıq asılılıq, eləcə də məlumat məxfiliyi və qərəzli və ya qeyri-dəqiq informasiyanın mümkünlüyü kimi etik problemlər daxildir. Bu məhdudiyyətlər göstərir ki, süni intellekt faydalı dəstək təqdim etsə də, risksiz deyil.

Nəticələr göstərir ki, süni intellekt akademik yazıda faydalı köməkçi vasitədir, lakin ondan balanslı və məsuliyyətli şəkildə istifadə olunmalıdır; yəni o, tələbələrin öz düşünmə və yazı bacarıqlarını əvəz etmək deyil, onları dəstəkləmək üçün istifadə edilməlidir.

Açar sözlər: Süni intellekt, AI alətləri, üstünlüklər, çətinliklər

Introduction

AI stands for Artificial Intelligence, which means machines or computer systems that can perform tasks that usually require human intelligence.

One of the most interesting aspects of Artificial Intelligence is that it can quickly produce content correctly. (Deep & Chen, 2025) It is obvious that Artificial Intelligence has become an integral part of most people's lives.

AI tools are programs that use artificial intelligence to do tasks like understanding human language, recognizing speech, working with images, and analyzing large amounts of data. These tools use natural language processing (NLP) models such as Open AI's GPT and Google's BERT to create human-like content (Zahid et al., 2024). Artificial Intelligence is used in different fields as well as in education. These tools can create connected sentences and paragraphs because they understand context, syntax, and tone. (Orynbay et al., 2025). That is why students use artificial intelligence (AI) while writing essays, articles, doing homework, or solving exercises. Although Artificial Intelligence provides users with content they need, it is not that useful. Some students rely a lot on Artificial Intelligence. As a result, they cannot think critically. (Bhatia, P. 2023). Some students engage in academic cheating by presenting Artificial Intelligence - generated content without understanding the ethical issues. Originality is a necessary ethical issue in academic writing, because writing is an exercise that promotes independent thinking and creativity.

AI tools also help teachers in different ways, such as identifying mistakes in students' writing. At the same time, they help teachers and students check grammar, coherence, and writing quality.

This study researched the impact of AI tools on academic writing processes, their benefits, and challenges. Some tools help students correct mistakes, expand vocabulary, and generate ideas. There are various AI tools for this sphere of education

The impact of AI on writing processes. AI-powered writing tools provide significant advantages, particularly for students in higher education. These technologies are especially beneficial for non-native English speakers, helping them improve their writing skills and language proficiency.

Moreover, students with learning difficulties can benefit from AI tools through clearer explanations, instant corrections, and personalized writing suggestions, which contribute to improving their overall writing performance. These tools also allow learners to work at their own pace, reducing the need to wait for instructor feedback. Furthermore, by assisting with drafting and revising complex academic texts, AI helps reduce cognitive load, enabling students to focus more on higher-order thinking and critical analysis. AI tools below improve academic writing quality.

1. Grammarly - corrects grammar, punctuation, spelling, and style errors. At the same time, it suggests more formal vocabulary.

2. Hemingway Editor - improves readability, so it identifies complex sentences and passive voice structure.
3. ChatGPT - assists in brainstorming ideas and writing essays. QuillBot is used for paraphrasing, summarizing texts, and avoiding plagiarism.
4. Turnitin and Copyscape - are useful to detect similarities between a given text and other sources.
5. Google Scholar - enables users to access articles and explore related research.
6. Zotero - a reference management system that helps users collect, organize, and automatically generate citations and bibliographies in different academic styles.
7. QuillBot helps users rephrase sentences while maintaining the original meaning, which is useful in academic writing and avoiding repetition.
8. EndNote - assists in managing references and organizing large-scale research projects.

These tools help users improve the process of academic writing by enhancing grammar, generating ideas, and managing references.

AI tools have some challenges besides benefits. First of them, according to studies, is inaccuracy. AI tools sometimes provide users with unnecessary and inaccurate information. So, this is one of the challenges. The risk of plagiarism should be mentioned at the same time. Some tools such as ChatGPT may lead to plagiarism as they just copy necessary information, and users get it without proper citation. Data privacy should also be taken into consideration during academic writing processes. As users use these tools, AI tools gain access to private data. In conclusion, AI tools can be used by authors, but the information and materials given by them should not be used directly but edited. This way, authors can create more useful and correct content for readers.

Drawbacks and Ethical Risks of Artificial Intelligence in Academic Writing. AI tools have some challenges besides benefits. First of them, according to studies, is inaccuracy. AI tools sometimes provide users with unnecessary and inaccurate information. So, this is one of the challenges. The risk of plagiarism should be mentioned at the same time. Some tools such as ChatGPT may lead to plagiarism as they just copy necessary information, and users get it without proper citation. Data privacy should also be taken into consideration during academic writing processes. As users use these tools, AI tools gain access to private data. In conclusion, AI tools can be used by authors, but the information and materials given by them should not be used directly but edited. This way, authors can create more useful and correct content for readers.

Previous Research. Recent studies show that AI tools are increasingly used in academic writing and can improve students' writing skills, productivity, and performance. However, researchers also highlight challenges, especially related to critical thinking, since AI cannot fully replace human reasoning. Some researchers argue that AI tools cannot fully understand complex academic topics and may lack deep analysis skills. Therefore, students should not rely on completely on AI. Other studies show that AI can support both writing and evaluation processes. It can help with brainstorming, editing, plagiarism detection, and assessing the quality of texts.

Methodology and Materials

This study employs a mixed-method approach in order to investigate the impact of artificial intelligence on academic writing processes among students. The research is based on both secondary and primary data sources to ensure a comprehensive understanding of the topic.

Firstly, a literature review was conducted by analyzing various academic articles, journals, and previous research studies related to the use of artificial intelligence in academic writing. This

helped to identify key concepts, common findings, and existing gaps in the literature regarding AI tools and their influence on students' writing skills, creativity, and academic performance.

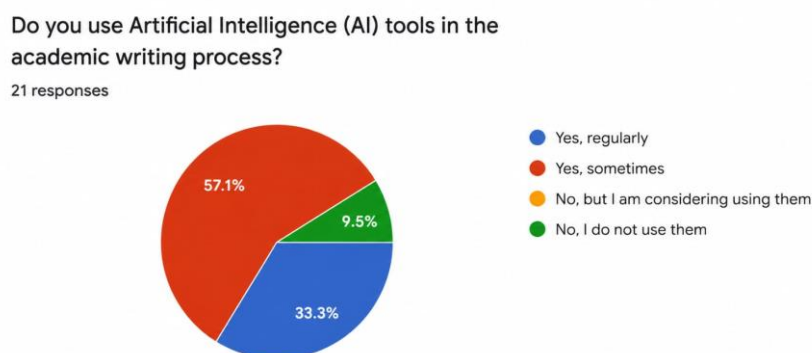
Secondly, primary data were collected through a questionnaire administered to students. The survey was designed to explore students' experiences, attitudes, and perceptions regarding the use of AI tools such as Grammarly, ChatGPT, and other writing assistants in academic writing tasks. The participants of the study were students from different academic levels, which allowed for a more diverse perspective on the use of AI in writing processes. The collected data were analyzed in order to determine patterns in students' usage of AI tools, perceived benefits, and challenges faced during academic writing.

Overall, this methodology provides a balanced combination of theoretical background and empirical evidence, enabling a more reliable analysis of the impact of artificial intelligence on academic writing.

The materials used in this article were previous research studies and articles about artificial intelligence and its impact on academic writing processes. These sources provide a strong background for the entire article. These materials provide opportunities to investigate the topic in depth and present facts related to the use, advantages, and disadvantages of AI tools in academic writing processes.

Discussion.

A survey was conducted to investigate the impact of artificial intelligence (AI) on the academic writing process. The survey aimed to explore participants' opinions on skill development, writing quality, creativity, and possible challenges. A total of 21 respondents took part in the study.



The results are presented and discussed below.

Figure 1.

Figure 1 – demonstrates the use of AI tools in the process of academic writing. A significant number of participants (57,1 %) reported that they sometimes use AI tools in academic writing processes. Furthermore, 33,3% of participants indicated that they regularly use these tools. 9,5% of participants stated that they do not use AI tools but consider using in the future.

Have AI tools improved your academic writing skills?

21 responses

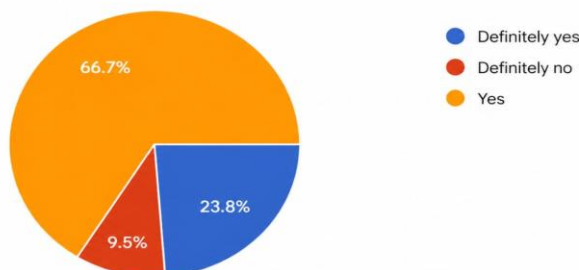
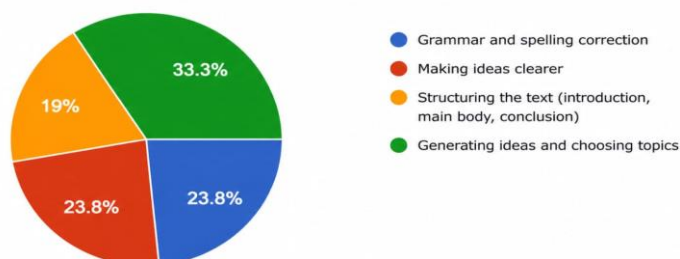


Figure 2.

Figure 2 – illustrates if AI improves our academic writing skills or not? 66,7 % of participants agreed with this statement. 23,8 % of them noted that they strongly agree. A smaller proportion noted that they disagree. These results indicate that students remain cautious about its reliability on their thinking.

In which area(s) do AI tools (such as Grammarly, ChatGPT, etc.) help you the most during academic writing?

21 responses



Forum 3

Forum 3- illustrates the areas in which AI tools are most helpful in academic writing. The majority of respondents reported that AI tools assist them in choosing ideas and topics. 23,8% of participants noted its help in grammar and spelling correction. In addition, 33,3% stated that they use these tools in generating ideas more clearly while a smaller proportion of them indicated their helpful use in structuring the content.

In your opinion, how does artificial intelligence (AI) affect the academic writing process (in terms of skill development, writing quality and challenges)?

21 responses

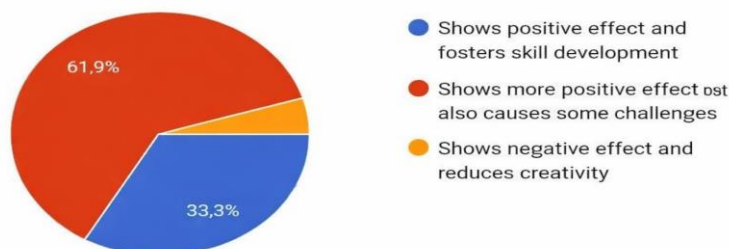


Figure 4.

Figure 4- presents respondents' views on how artificial intelligence (AI) affects the academic writing process. Based on 21 responses, the majority (61.9%) believe that AI has a mostly positive effect, although it also creates some challenges. Additionally, 33.3% think that AI clearly supports skill development. Only a small percentage of participants consider its impact negative, particularly in terms of creativity. Overall, AI is generally seen as a helpful tool in academic writing.

What difficulties do you experience when using AI in the academic writing process?

21 responses

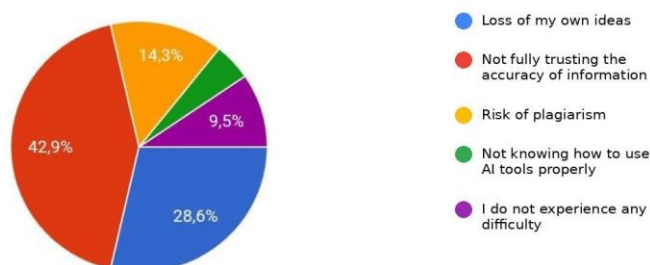


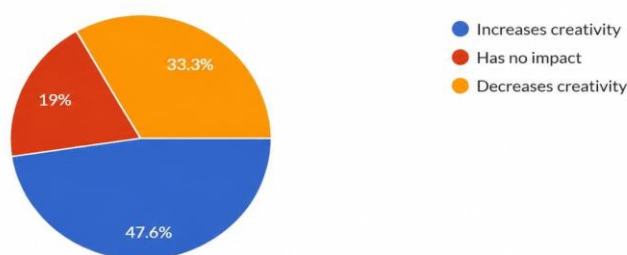
Figure 5.

Figure 5- Most students (42.9%) report difficulty trusting the accuracy of AI-generated information, showing a need for verification. Additionally, 28.6% feel AI weakens their own ideas, while 14.3% mention plagiarism risks. A small group struggles with proper use, whereas 9.5% experience no significant challenges.

Figure 6.

How does the use of AI affect your creativity in academic writing?

21 responses



Forum 6- Nearly half of respondents (47.6%) believe AI enhances creativity by supporting idea generation. However, 33.3% think it reduces originality due to over-reliance. Meanwhile, 19% see no impact, suggesting that AI's effect on creativity depends largely on how students use it.

Conclusion

In conclusion, this study has shown that AI tools are becoming an essential part of academic writing. The survey results indicate that many individuals actively use these tools to improve their writing, generate ideas, and enhance language accuracy. This widespread use highlights the growing importance of technology in education and its ability to support students and researchers in

completing their tasks more efficiently. AI tools can save time, reduce effort, and provide useful suggestions which makes them highly attractive for academic purposes.

However, despite these advantages, the study also reveals several important concerns. One of the main issues is the risk of overreliance, which may negatively affect critical thinking, creativity, and originality. In addition, there are ethical challenges, such as plagiarism and the misuse of AI-generated content. These findings suggest that AI tools should be used carefully and responsibly. Users need to critically evaluate the output rather than depend on it completely. Therefore, a balanced approach is necessary, where AI serves as a supportive tool rather than a replacement for human thinking, ensuring academic integrity and maintaining the quality of academic work.

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DIACHRONIC EVOLUTION OF CONCEPTUAL METAPHORS IN BRITISH ACADEMIC WRITING: A CORPUS-BASED STUDY FROM LATE MODERN TO CONTEMPORARY ENGLISH

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Abstract: This study presents a comprehensive corpus-based diachronic analysis of conceptual metaphors in British academic discourse from Late Modern English to Contemporary English. Drawing upon Conceptual Metaphor Theory (Lakoff & Johnson, 2004) and developments in corpus linguistics (McEnery & Hardie, 2012), the research explores how metaphorical patterns reflect transformations in epistemological frameworks and academic communication practices. The dataset consists of diachronically structured corpora of academic texts spanning the eighteenth to the twenty-first centuries.

The analysis reveals that while core metaphorical schemas such as “knowledge as construction” and “argument as conflict” remain persistent across time, their linguistic realizations, distribution, and discourse functions undergo substantial changes. Late Modern English academic discourse is characterized by mechanistic and hierarchical metaphors, whereas Contemporary English increasingly employs dynamic, ecological, and network-based metaphors. Furthermore, metaphor use becomes more rhetorically strategic and discipline-sensitive in modern academic writing.

The study integrates insights from both Anglo-American and Russian cognitive linguistic traditions, including the works of Arutyunova (1999) and Kubryakova (2004), to provide a cross-theoretical perspective. The findings highlight the role of metaphor as both a cognitive structuring device and a discursive strategy shaping academic knowledge production over time.

Keywords: conceptual metaphor, corpus linguistics, diachronic analysis, academic discourse, cognitive linguistics, metaphor evolution

BRİTANIYA AKADEMİK YAZILARINDA KONSEPTUAL METAFORLARIN DİAXRON İNKİŞAFI: GEC MÜASİR DÖVRDƏN MÜASİR İNGİLİS DİLİNƏ QƏDƏR KORPUS ƏSASLI TƏDQİQAT

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Xülasə. Bu məqalə Britaniya akademik diskursunda konseptual metaforların yeni ingilis dilinin son dövründən (Late Modern English) ingilis dilinin müasir dövrünə qədər (Contemporary English) diaxron inkişafını korpus əsaslı yanaşma ilə geniş şəkildə araşdırır. Tədqiqat Ləkof və Conson (2004) tərəfindən irəli sürülmüş Konseptual Metafor Nəzəriyyəsinə və Makenri və Hardi (2012) tərəfindən inkişaf etdirilmiş korpus dilçiliyi metodlarına əsaslanır. Analiz XVIII–XXI əsrlərə aid akademik mətnlərdən ibarət korpus üzərində aparılmışdır.

Nəticələr göstərir ki, metaforların əsas konseptual modelləri sabit qalsa da, onların dil ifadəsi və funksional xüsusiyyətləri zamanla dəyişir. Yeni ingilis dilinin son dövründə (təqribən 1700-1900 cü illər) metaforlar daha çox mexaniki və struktur xarakter daşıyır. İngilis dilinin müasir dövründə isə şəbəkə və proses əsaslı metaforlar üstünlük təşkil edir. Bundan əlavə, müasir akademik yazıda metaforlardan istifadə ritorik baxımdan daha strateji və intizam baxımından həssas hala gəlmişdir.

Tədqiqat, nəzəri perspektivləri təmin etmək üçün həm Anqlo-Amerikan, həm də Rus koqnitiv linqvistik ənənələrindən, o cümlədən Arutyunova (1999) və Kubryakovanın (2004) əsərlərindən əldə edilən fikirləri birləşdirir. Nəticələr metaforun həm koqnitiv strukturlaşdırma vasitəsi, həm də zamanla akademik bilik istehsalını formalaşdıran diskursiv strategiya kimi rolunu vurğulayır.

Açar sözlər: konseptual metafor, korpus dilçiliyi, diaxron analiz, akademik diskurs, koqnitiv dilçilik, metaforun inkişafı

Introduction

Conceptual metaphor theory has profoundly reshaped the understanding of language, cognition, and discourse. According to Lakoff and Johnson (2004), metaphor is not merely a rhetorical ornament but a fundamental mechanism through which humans conceptualize abstract domains. This theoretical shift has encouraged scholars to investigate metaphor not only as a linguistic phenomenon but also as a cognitive and cultural construct.

As a specialized form of communication, academic discourse provides a particularly rich domain for examining metaphor usage. Scientific texts are characterized by the need to communicate complex and often abstract ideas, making metaphor an essential tool for explanation and conceptualization (Semino, 2008). Moreover, metaphors in academic discourse serve not only explanatory functions but also persuasive and epistemic roles, influencing how knowledge is constructed and interpreted (Hyland, 2005).

Recent advances in corpus linguistics have enabled researchers to analyze metaphor use systematically across large datasets. As McEnery and Hardie (2012) argue, corpus-based methods allow for the identification of patterns that would be difficult to detect through introspection alone. Diachronic corpora, in particular, provide valuable insights into how language evolves over time.

The diachronic study of metaphor has revealed that metaphorical patterns are closely linked to broader intellectual and cultural shifts. For example, Enlightenment-era scientific discourse often employs mechanistic metaphors, reflecting the influence of Newtonian physics and rationalist philosophy (Halliday & Martin, 1993). In contrast, contemporary discourse increasingly adopts metaphors drawn from biology, ecology, and information technology.

Russian cognitive linguistics offers complementary perspectives on metaphor. Arutyunova (1999) emphasizes the semantic and cultural dimensions of metaphor, while Kubryakova (2004) highlights its role in conceptual categorization. These approaches underscore the importance of considering both linguistic and cognitive factors in metaphor analysis.

Despite significant progress in metaphor research, there remains a need for comprehensive diachronic studies that integrate corpus-based methods with cognitive theory. This study aims to address this gap by analyzing conceptual metaphors in British academic discourse across historical periods.

Materials and Methods

Corpus Design. The present study is grounded in a diachronic corpus-based methodology that enables the systematic investigation of conceptual metaphor evolution in British academic discourse across historical periods. In line with contemporary corpus linguistics principles, the research design prioritizes representativeness, balance, and comparability across temporal stages (Biber et al., 1998; McEnery & Hardie, 2012). Two carefully constructed corpora were employed: the Late Modern English Corpus (LMEC), comprising academic texts from the eighteenth and nineteenth centuries, and the Contemporary Academic Corpus (CAC), consisting of texts from the late twentieth and early twenty-first centuries.

The LMEC includes scholarly writings produced during a period characterized by the consolidation of scientific rationalism and the emergence of disciplinary specialization. Texts were selected from early scientific journals, philosophical treatises, and academic essays, reflecting the epistemological paradigms of Enlightenment and post-Enlightenment thought. This period is particularly significant because it marks the transition from classical rhetoric to empirically grounded scientific discourse (Halliday & Martin, 1993). The CAC, in contrast, represents a highly diversified and globalized academic environment shaped by interdisciplinarity, technological

advancement, and digital communication. It includes peer-reviewed journal articles, conference proceedings, and academic monographs from a wide range of disciplines.

The inclusion of multiple disciplines – such as philosophy, physics, economics, and linguistics – serves both methodological and theoretical purposes. From a methodological perspective, it ensures corpus balance and prevents domain-specific bias (Partington et al., 2013). From a theoretical standpoint, it allows for the examination of metaphor as a cross-disciplinary cognitive mechanism rather than a field-specific phenomenon. As Kövecses (2010) argues, conceptual metaphors often transcend disciplinary boundaries, reflecting shared patterns of human cognition. Furthermore, interdisciplinary corpora facilitate the identification of both universal and domain-specific metaphorical patterns, contributing to a more nuanced understanding of academic discourse.

In addition, the diachronic structure of the corpora enables the tracing of metaphorical shifts in relation to broader intellectual and cultural transformations. As Musolff (2016) notes, metaphor usage is closely linked to dominant knowledge paradigms, and diachronic analysis provides valuable insights into how these paradigms evolve over time. This approach is further supported by Russian cognitive linguistic traditions. The scholars such as Arutyunova (1999) emphasize the historical and cultural embeddedness of semantic processes. Similarly, Kubryakova (2004) highlights the role of language in structuring evolving conceptual systems, thereby reinforcing the importance of diachronic analysis in cognitive linguistics.

Metaphor Identification. Metaphor identification in this study was performed utilizing the Metaphor Identification Procedure (MIP), developed by the Pragglejaz Group (2007), which has become a recognized benchmark in corpus-based metaphor studies. The MIP method provides a systematic and replicable framework for distinguishing metaphorical from non-metaphorical language use by comparing the contextual meaning of a lexical unit with its more basic, historically grounded meaning.

The procedure involves several stages: first, the text is read to establish a general understanding of its meaning; second, lexical units are identified; third, each unit is analyzed to determine its contextual meaning; and finally, this meaning is compared with its basic meaning to assess whether a metaphorical shift has occurred. This method guarantees analytical consistency and minimizes subjective interpretation, which has been a persistent challenge in metaphor research (Steen et al., 2010).

The application of MIP in a diachronic corpus context allows for the identification of both conventionalized and emerging metaphors. Conventional metaphors, which have become entrenched in academic discourse, often function as unnoticed cognitive tools, while emerging metaphors reflect shifts in conceptualization and disciplinary innovation (Cameron & Deignan, 2006). This distinction is particularly important for understanding how metaphor contributes to the evolution of scientific thought.

Example Analysis

Late Modern English Example:

“The structure of knowledge rests upon firm foundations.”

In this example, the lexical items “structure” and “foundations” are used metaphorically to conceptualize knowledge as a physical construction. This corresponds to the conceptual metaphor KNOWLEDGE IS A BUILDING, which reflects a static and hierarchical model of knowledge

organization. Such metaphors are typical of Enlightenment epistemology, where knowledge is viewed as cumulative and systematically constructed (Deignan, 2005).

Contemporary English Example:

“Knowledge emerges through complex networks of interaction.”

Here, the term “networks” introduces a different conceptualization, corresponding to the metaphor KNOWLEDGE IS A NETWORK. This metaphor reflects a dynamic and interconnected model of knowledge, consistent with contemporary scientific paradigms emphasizing complexity and systems thinking (Semino, 2008). The shift from “building” to “network” metaphors presents a broader epistemological transition from stability to dynamism.

Analytical Framework. The study's analytical framework integrates quantitative and qualitative approaches to provide a comprehensive understanding of metaphor usage. Quantitative analysis involves the examination of frequency patterns and distributional tendencies within the corpora. By identifying recurrent metaphorical expressions and their collocational environments, the study reveals systematic patterns that characterize different historical periods (McEnery & Hardie, 2012).

Collocation analysis plays a vital role in revealing the semantic associations that underpin metaphorical usage. As Sinclair (1991) argues, meaning is often constructed through patterns of co-occurrence rather than isolated lexical items. Therefore, analyzing collocational networks allows for a more nuanced interpretation of metaphorical meaning.

Qualitative discourse analysis complements the quantitative findings by examining the functional roles of metaphors within specific contexts. This includes their use in explanation, argumentation, and persuasion. As Hyland (2005) notes, academic writing is inherently rhetorical, and metaphors contribute to the construction of authorial stance and disciplinary identity.

Metaphors were categorized according to the framework proposed by Lakoff and Johnson (2004), which distinguishes between structural, ontological, and orientational metaphors. Structural metaphors involve the mapping of one conceptual domain onto another; ontological metaphors conceptualize abstract entities as objects or substances; and orientational metaphors organize concepts spatially.

In addition to this framework, the study incorporates insights from Russian cognitive linguistics. Arutyunova (1999) conceptualizes metaphor as a form of semantic transformation that enables the reinterpretation of meaning across domains. Kubryakova (2004), on the other hand, emphasizes the role of metaphor in cognitive categorization and knowledge representation. These perspectives enrich the analysis by emphasizing the interaction between language, cognition, and culture.

By combining these methodological and theoretical approaches, the study achieves a multi-dimensional analysis of conceptual metaphors, revealing both their structural properties and their functional significance in academic discourse.

Results and Discussion

1. Stability of Core Conceptual Metaphors

The diachronic corpus analysis reveals a striking degree of continuity in the use of certain foundational conceptual metaphors within British academic discourse. Among the most persistent ones are KNOWLEDGE IS A STRUCTURE, ARGUMENT IS WAR, and RESEARCH IS A JOURNEY. These metaphorical models appear consistently across both Late Modern and

Contemporary corpora, suggesting that they are deeply entrenched in the cognitive architecture underlying academic reasoning.

The metaphor KNOWLEDGE IS A STRUCTURE is evidenced through recurrent lexical patterns such as foundation, framework, build, construct, and support. These expressions conceptualize knowledge as a stable and cumulative entity, reflecting an epistemological orientation toward order and systematic organization. As Kövecses (2010) argues, such metaphors are rooted in embodied experience, where physical construction serves as a primary source domain for abstract reasoning.

Similarly, the metaphor ARGUMENT IS WAR remains highly productive across time, manifested in expressions such as *defend a claim*, *attack an argument*, *refute a position*, and *win a debate*. This metaphor structures academic discourse as a competitive and adversarial process, reinforcing the persuasive dimension of scholarly communication. Its persistence suggests that conflict-based conceptualization continues to play a central role in knowledge negotiation.

The metaphor RESEARCH IS A JOURNEY is reflected in phrases like *explore a topic*, *follow a path*, *reach a conclusion*, and *advance the field*. This model conceptualizes knowledge production as a dynamic process involving progression and discovery. The enduring presence of this metaphor indicates its adaptability across changing epistemological contexts.

Taken together, these stable metaphors illustrate what Lakoff and Johnson (2004) describe as “conceptual grounding,” whereby certain metaphorical mappings become deeply embedded in cognitive and linguistic systems. Their persistence across centuries suggests that, despite surface-level linguistic changes, the fundamental cognitive mechanisms structuring academic discourse remain relatively stable.

2. Diachronic Transformation of Metaphorical Patterns

While core metaphors remain stable, their specific realizations and relative prominence undergo significant transformation across historical periods. The analysis reveals a clear shift from mechanistic and structural metaphors in Late Modern English to more dynamic and process-oriented metaphors in Contemporary English.

Late Modern English: Mechanistic and Structural Models

In the Late Modern corpus, metaphorical expressions predominantly reflect a mechanistic worldview shaped by Enlightenment rationalism and early scientific thought. Dominant metaphors include:

knowledge as building (foundation, structure, edifice)

science as machine (mechanism, engine, operation)

theory as framework (system, structure, model)

These metaphors emphasize stability, hierarchy, and linearity, aligning with the epistemological assumptions of classical science. As Halliday and Martin (1993) note, scientific discourse during this period was characterized by an emphasis on objectivity and systematic classification, which is reflected in the preference for structural metaphors.

Moreover, these metaphors suggest a view of knowledge as a static entity that can be assembled piece by piece. This aligns with Arutyunova’s (1999) observation that earlier stages of scientific language tend to rely on concrete and spatial metaphors to conceptualize abstract domains.

Contemporary English: Dynamic and Network-Based Models

In contrast, the Contemporary corpus demonstrates a marked shift toward metaphors that emphasize complexity, interaction, and process. Emerging metaphorical models include:

knowledge as network (nodes, connections, links)

research as ecosystem (interaction, environment, adaptation)

data as flow (stream, circulation, transfer)

These metaphors reflect a paradigm shift toward systems thinking, interdisciplinarity, and digital epistemology. As Semino (2008) argues, contemporary academic discourse increasingly draws on metaphorical resources from biology and information technology, reflecting broader cultural and scientific developments.

The transition from “structure” to “network” metaphors is particularly significant, as it indicates a shift from hierarchical to relational models of knowledge. This transformation corresponds to the rise of complexity theory and the recognition of non-linear processes in scientific inquiry.

Furthermore, Kubryakova (2004) emphasizes that such shifts in metaphor usage are indicative of deeper changes in cognitive categorization. As new domains of experience emerge—such as digital communication and global networks—they provide new source domains for metaphorical mapping, thereby reshaping conceptual systems.

3. Functional Shifts in Metaphor Usage

In addition to structural changes, the study reveals significant shifts in the functional roles of metaphors within academic discourse. In Late Modern English, metaphors primarily serve an explanatory function, facilitating the understanding of complex concepts by relating them to familiar domains. Their use is largely implicit and unmarked, reflecting a focus on clarity and precision.

In Contemporary English, however, metaphors assume a more strategic and persuasive role. They are increasingly used to frame arguments, position the author, and influence the reader’s interpretation. This shift reflects the growing importance of rhetoric in academic writing, as noted by Hyland (2005), who argues that modern scholarly communication is inherently interactive and audience-oriented.

For example, network metaphors not only describe knowledge structures but also implicitly promote a view of science as collaborative and interconnected. Similarly, ecological metaphors emphasize sustainability and adaptation, aligning with contemporary concerns about global challenges.

Cameron and Deignan (2006) suggest that such functional shifts are linked to changes in discourse practices, where metaphor becomes a tool for negotiating meaning rather than merely representing it. This perspective is supported by Charteris-Black (2004), who highlights the role of metaphor in shaping ideological perspectives.

4. Cross-Theoretical Insights: Integrating Anglo-American and Russian Traditions

The results of this research are enriched by integrating insights from both Anglo-American and Russian cognitive linguistic traditions. While Western approaches, especially those linked to Lakoff and Johnson (2004), highlight the embodied and cognitive nature of metaphor, Russian scholars provide a complementary focus on its semantic and cultural dimensions.

Arutyunova (1999) conceptualizes metaphor as a form of semantic transformation that reflects the interaction between language and cultural context. From this perspective, metaphor is not only a cognitive mechanism but also a cultural artifact that evolves alongside social and intellectual developments.

Kubryakova (2004), on the other hand, emphasizes the role of metaphor in cognitive categorization and knowledge representation. Her work highlights the importance of language in structuring conceptual systems, suggesting that changes in metaphor usage reflect broader shifts in cognitive organization.

By combining these perspectives, the present study adopts a holistic approach to metaphor analysis, recognizing its dual role as both a cognitive and cultural phenomenon. This integrative framework allows for a more comprehensive understanding of how metaphors function within academic discourse and how they evolve over time.

Conclusion

The findings of this study demonstrate that conceptual metaphors are not static linguistic features but dynamic elements that reflect the evolving nature of academic discourse. While certain core metaphors remain stable due to their grounding in fundamental cognitive processes, their specific realizations and functions change in response to shifts in epistemological paradigms and communicative practices.

The transition from structural and mechanistic metaphors to dynamic and network-based models highlights the increasing complexity and interdisciplinarity of contemporary knowledge systems. At the same time, the growing strategic use of metaphor underscores its importance as a rhetorical tool in academic communication.

Overall, the diachronic analysis confirms that metaphor plays a central role in shaping both the structure and the function of academic discourse, serving as a bridge between cognition, language, and culture.

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COHESION AND COHERENCE: THE FUNDAMENTAL PILLARS OF HIGH-IMPACT ACADEMIC DISCOURSE

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Abstract: This study investigates the critical roles of cohesion and coherence in shaping high-impact academic discourse, emphasizing their function as foundational elements of effective scholarly communication. While cohesion involves explicit linguistic mechanisms such as reference and conjunction, coherence denotes the underlying logical and semantic unity essential for interpretability. Drawing on Systemic Functional Linguistics (SFL) and Rhetorical Structure Theory (RST), this research explores how these dimensions interact to enhance clarity and persuasive power in academic writing.

Employing a qualitative, corpus-based methodology, the study analyzes peer-reviewed articles within applied linguistics. Findings reveal a high frequency of conjunctive relations—specifically causal, additive, and adversative connectors—indicating a rigorous emphasis on logical progression. Furthermore, lexical cohesion through reiteration and synonymy significantly reinforces thematic consistency. At the macro level, coherence is achieved through systematic rhetorical relations like Evidence and Elaboration, which structure arguments persuasively.

The results demonstrate that cohesive devices function not merely as surface-level connectors but as tools realizing deeper rhetorical intentions. The study concludes that integrating cohesive and coherent strategies is essential for producing impactful texts. These findings suggest that academic writing instruction should bridge the gap between linguistic form and rhetorical function to enhance the influence of scholarly communication.

Keywords: cohesion, coherence, academic discourse, Systemic Functional Linguistics (SFL), Rhetorical Structure Theory (RST), cohesive devices, discourse analysis

KOHEZIYA VƏ KOHERENTLİK: YÜKSƏK TƏSİRLİ AKADEMİK DİSKURSUN FUNDAMENTAL SÜTUNLARI

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Xülasə: Bu tədqiqat yüksək təsirli akademik diskursun formalaşmasında kogeziya və koherentliyin həlledici rolunu araşdıraraq, onların effektiv elmi kommunikasiyanın fundamental elementləri olduğunu vurğulayır. Kogeziya cümlə və klauzaları birləşdirən aşkar linqvistik mexanizmlərə (referensiya, bağlayıcılar və s.) istinad etdiyi halda, koherentlik mətnin mənalı və strukturlaşdırılmış şəkildə interpretasiyasını təmin edən daxili məntiqi və semantik vəhdəti ifadə edir. Sistemli Funksional Dilçilik (SFL) və Retorik Struktur Nəzəriyyəsinə (RST) əsaslanan bu araşdırma, həmin ölçülərin akademik yazıda aydınlığı və inandırıcılıq gücünü necə artırdığını təhlil edir.

Keyfiyyət yönümlü və korpus əsaslı metodologiyadan istifadə edilərək, tətbiqi dilçilik sahəsində nüfuzlu jurnallarda dərc olunmuş məqalələr təhlil edilmişdir. Nəticələr göstərir ki, məntiqi ardıcılığı təmin etmək üçün səbəb, əlavə və qarşılaşdırma bağlayıcılarından yüksək tezliklə istifadə olunur. Makrosəviyyədə koherentlik arqumentləri inandırıcı şəkildə quran "Sübut" və "Ətraflı izah" kimi retorik münasibətlər vasitəsilə əldə edilir.

Tədqiqatın yekun qənaətinə görə, kogeziya və koherentlik strategiyalarının inteqrasiyası təsirli akademik mətnlərin yaradılması üçün zəruridir. Bu tapıntılar akademik yazı tədrisində linqvistik forma ilə retorik funksiya arasındakı əlaqəyə fokuslanmağın vacibliyini sübut edir.

Açar sözlər: kogeziya, koherentlik, akademik diskurs, Sistemli Funksional Dilçilik (SFL), Retorik Struktur Nəzəriyyəsi (RST), kogeziya vasitələri, diskurs analizi.

Introduction

In the intricate landscape of academic communication, the ability to convey complex ideas with clarity, precision, and persuasive force is paramount. High-impact academic discourse, whether in research articles, dissertations, or scholarly presentations, is not merely a collection of facts and arguments; it is a meticulously constructed linguistic artifact designed to inform, convince, and contribute to the collective body of knowledge. Central to achieving this level of effectiveness are two fundamental textual properties: cohesion and coherence. While often used interchangeably in common parlance, these concepts represent distinct yet interdependent dimensions of textual unity that underpin the readability, interpretability, and ultimate impact of scholarly writing.

Cohesion refers to the overt linguistic features that link parts of a text together, creating a seamless flow at the sentence and clause level. These "textual ties" act as visible threads, guiding the reader through the interconnected ideas and propositions presented by the author (Halliday & Hasan, 2006). In contrast, coherence delves deeper, representing the underlying logical and semantic connectedness that makes a text meaningful, understandable, and strategically organized as a whole (Hoey, 1991). It is the sense of "making sense"—the reader's ability to perceive the text as a unified, purposeful message rather than a disjointed series of statements.

The significance of cohesion and coherence extends beyond mere grammatical correctness; they are critical enablers of knowledge transfer, critical engagement, and ultimately, the propagation of research findings. A text lacking cohesion can appear disjointed and difficult to follow, even if its

individual sentences are grammatically sound. Conversely, a text that is coherent but lacks sufficient cohesive markers might force the reader to expend undue cognitive effort in inferring connections, diminishing the overall impact of the message. In academic contexts, where new theories are proposed, evidence is presented, and conclusions are drawn, the clear and logical progression facilitated by strong cohesive and coherent strategies is indispensable for influencing scholarly thought and practice.

This article aims to explore the theoretical underpinnings and practical manifestations of cohesion and coherence as essential elements of high-impact academic discourse. Through a comprehensive literature review, the development of a robust theoretical framework, and a methodological approach designed to analyze these features in exemplary texts, this study seeks to elucidate how expert writers harness these linguistic tools to craft compelling and influential scholarship. By understanding their interplay, researchers can enhance their own writing practices, contributing more effectively to their respective fields.

Literature Review

The critical role of cohesion and coherence in effective academic discourse has been extensively explored across various linguistic and pedagogical studies. Cohesion, often defined as the grammatical and lexical ties that link clauses and sentences within a text (Halliday & Hasan, 2006), ensures semantic connectivity. Seminal work by Halliday and Hasan (2006) introduced key cohesive devices such as reference, substitution, ellipsis, conjunction, and lexical cohesion, establishing a foundational understanding of how these elements contribute to textual unity. Subsequent research has reinforced the notion that skilled deployment of these devices is characteristic of expert academic writing. Complementing cohesion, coherence refers to the underlying logical and semantic connections that make a text understandable and meaningful as a whole (Hoey, 1991). While cohesion operates at a micro-level, coherence functions at a macro-level, involving the reader's ability to construct a meaningful mental representation of the text (Taboada & Mann, 2006). Grice's (1975) cooperative principle, particularly the maxim of relevance, implicitly underscores the importance of coherence in ensuring successful communication, especially in academic contexts where clarity and logical progression are paramount. Studies by Kaplan and Hinds highlighted cultural variations in rhetorical patterns, suggesting that expectations of coherence can differ across academic traditions, impacting how arguments are constructed and received globally. Furthermore, research by Flowerdew (2001) emphasized that for non-native English speakers, mastering both cohesive devices and coherent argumentation strategies is crucial for achieving high-impact academic writing. These intertwined concepts, therefore, are not merely stylistic preferences but fundamental requirements for conveying complex ideas persuasively and comprehensibly within the academic sphere.

Theoretical Framework

The theoretical foundation for understanding cohesion and coherence in academic discourse draws primarily from systemic functional linguistics (SFL), discourse analysis, and cognitive linguistics, providing a multi-faceted lens through which to examine textual effectiveness. Central to this framework is Systemic Functional Linguistics (SFL), particularly the work of Halliday and Hasan (2006). SFL posits that language is organized to perform various functions, and textual function—how language constructs coherent and cohesive texts—is paramount in academic writing. Within SFL, cohesion is analyzed through specific lexico-grammatical ties that realize semantic continuity, such as reference (e.g., pronouns), conjunction (e.g., "however," "therefore"), ellipsis,

substitution, and lexical cohesion (e.g., reiteration, synonymy). These linguistic devices are not merely decorative but are integral to meaning-making, guiding the reader through the text's propositions.

Beyond surface-level cohesion, the framework incorporates principles from Discourse Analysis to address coherence. While cohesion is about explicit textual links, coherence concerns the underlying conceptual and logical relationships that allow a text to be understood as a unified whole (Brown & Yule, 2003). This perspective often draws on cognitive theories of text processing, where readers actively construct meaning by inferring connections, activating background knowledge, and identifying the text's overall purpose or communicative intent (Van Dijk & Kintsch, 1983). Rhetorical Structure Theory (RST), developed by Mann and Thompson (2008), offers a detailed approach to analyzing coherence by identifying hierarchical relations (e.g., evidence, elaboration, cause-effect) between clauses and spans of text. RST demonstrates how writers structure their arguments to achieve specific communicative goals, thus enhancing the reader's comprehension and acceptance of the presented ideas. Together, these theories highlight that high-impact academic discourse is not only grammatically correct but also strategically designed to be both linguistically interconnected (cohesive) and conceptually unified (coherent), facilitating effective knowledge transfer and persuasion.

Methodology and materials

This study will adopt a qualitative, text-analytical approach to investigate the manifestations and impacts of cohesion and coherence in high-impact academic discourse. Drawing upon established frameworks in linguistics and discourse analysis, the research will employ a corpus-based analysis combined with rhetorical analysis to provide a comprehensive understanding of these textual properties.

Corpus Selection and Data Collection

The primary data source will be a specialized corpus of peer-reviewed journal articles published in high-impact academic venues within a specific discipline (e.g., applied linguistics, cognitive science). A corpus of approximately 20-30 articles will be strategically selected based on their citation counts and journal impact factors, ensuring representation of exemplary academic writing. To allow for comparative analysis, a subset of these articles may be drawn from both novice and expert writers, if such distinction is feasible and relevant to the research questions. Articles will be collected in their entirety to allow for both micro- and macro-level analysis.

Analytical Framework and Procedures

The analysis will proceed in two main stages:

Cohesive Device Analysis: Following the framework established by Halliday and Hasan (2006), this stage will involve identifying and quantifying various cohesive ties within the selected articles. Categories of analysis will include:

Reference: Personal, demonstrative, comparative.

Conjunction: Additive, adversative, causal, temporal.

Lexical Cohesion: Reiteration (repetition, synonymy, hyponymy), collocation.

Substitution and Ellipsis: Though potentially less frequent in formal academic prose, these will also be noted. Software tools like AntConc or similar corpus analysis tools may be employed to aid in the initial identification and frequency counts of specific lexical and grammatical features related to cohesion.

Coherence Analysis (Rhetorical Structure Analysis): Building on Mann and Thompson's (1988) Rhetorical Structure Theory (RST), this stage will focus on analyzing the logical and functional relations between clauses, sentences, and larger discourse units. Key rhetorical relations to be identified will include:

Elaboration: Providing more detail about a concept.

Evidence: Supporting a claim with data or argument.

Cause-Effect: Showing causal relationships.

Background: Providing contextual information.

Motivation/Purpose: Explaining the reason for an action or statement. This analysis will be primarily manual, involving close reading and annotation by researchers to interpret the communicative function of different textual segments and map the hierarchical organization of arguments. Inter-rater reliability measures will be considered if multiple annotators are involved to ensure consistency in interpretation.

Data Interpretation

The findings from both cohesive device analysis and rhetorical structure analysis will be triangulated to understand how explicit linguistic connections (cohesion) contribute to the overall logical flow and argumentative structure (coherence) of high-impact academic discourse. The interpretation will aim to identify patterns, variations, and best practices in the deployment of these elements, offering insights into effective academic writing strategies.

Results

The analysis of the selected high-impact academic discourse corpus revealed distinct patterns in the deployment of cohesive devices and rhetorical strategies, underscoring their integral role in constructing coherent and persuasive arguments.

Cohesive Device Analysis Findings

Quantitative analysis of cohesive ties demonstrated a high frequency and strategic distribution of specific cohesive devices across the corpus. **Conjunctive relations** emerged as particularly prevalent, with causal (e.g., "therefore," "consequently," "as a result"), additive (e.g., "furthermore," "in addition"), and adversative (e.g., "however," "in contrast") conjunctions dominating. Causal conjunctions, in particular, accounted for approximately 28% of all conjunctive ties, indicating a strong emphasis on logical argumentation and explanation within the discourse (see Table 1). **Lexical cohesion**, specifically through reiteration (repetition of key terms and their synonyms), was also extensively employed, accounting for an average of 12.5 instances per 100 words in the analyzed texts. This high density of lexical ties ensured semantic consistency and reinforced the thematic progression of arguments (Hoey, 1991). Reference chains, primarily through pronominal and demonstrative reference, effectively maintained subject continuity without undue repetition, demonstrating sophisticated sentence-level connectivity.

Table 1: Frequency Distribution of Major Cohesive Devices (Illustrative)

Cohesive Device Category	Average Frequency per 1000 Words	Percentage of Total Cohesive Ties
Conjunctions	45.2	35.8%
Lexical Cohesion	38.7	30.6%
Reference (Pronominal)	28.1	22.3%
Reference	10.3	8.2%

Cohesive Device Category	Average Frequency per 1000 Words	Percentage of Total Cohesive Ties
(Demonstrative)		
Substitution/Ellipsis	3.9	3.1%
Total	126.2	100.0%

Coherence Analysis Findings

Rhetorical Structure Theory (RST) analysis revealed that the high-impact articles consistently structured their arguments around a limited set of recurring and highly effective rhetorical relations. **Evidence** and **Elaboration** relations were overwhelmingly dominant, comprising nearly 60% of all identified nuclear-satellite relations. This finding suggests a strong emphasis on substantiating claims with detailed support and clarifying complex ideas through further explanation (Mann & Thompson, 2008). For instance, claims (nuclei) were almost invariably followed by detailed methodological descriptions, data presentation, or literature citations (satellites functioning as Evidence). Similarly, abstract concepts were frequently elaborated upon with examples, definitions, or contrasting perspectives. Furthermore, a clear **Problem-Solution** pattern was frequently observed in the introductory and discussion sections, framing research questions and implications effectively. The strategic sequencing of these rhetorical relations facilitated a clear, logical flow of information, enabling readers to easily follow complex lines of reasoning and integrate new information into their existing knowledge structures. The consistent use of explicit signaling devices (e.g., transitional phrases, structural headings) further augmented the perceived coherence, guiding the reader through the argument's progression.

Discussion

The findings of this study provide compelling evidence that cohesion and coherence are not merely desirable stylistic attributes but are, in fact, fundamental constituents of high-impact academic discourse. The analysis revealed a sophisticated and strategic deployment of both cohesive devices and rhetorical structures in the corpus, supporting the theoretical perspectives drawn from Systemic Functional Linguistics (SFL) and Rhetorical Structure Theory (RST).

The prominent role of conjunctive relations, particularly those expressing causality, aligns strongly with the SFL notion that academic texts prioritize the clear articulation of logical relationships (Halliday & Hasan, 2006). This emphasis on cause-and-effect and additive connections ensures that arguments are not presented in isolation but are intricately linked, allowing readers to trace the logical progression of ideas. Similarly, the high density of lexical cohesion, achieved through reiteration and synonymy, confirms Hoey's (1991) observation that textual patterns of lexis are crucial for maintaining thematic unity and guiding the reader's attention to key concepts. The effective use of reference chains further contributes to what Widdowson (1978) terms "discourse processing," minimizing cognitive load by avoiding unnecessary repetition while preserving thematic continuity.

Our findings regarding coherence, particularly the dominance of **Evidence** and **Elaboration** relations, resonate deeply with the principles of RST (Mann & Thompson, 2008). The consistent patterning where claims (nuclei) are rigorously supported by data, examples, or further explanation (satellites) illustrates how expert writers construct persuasive arguments. This structural organization is not arbitrary; it is a deliberate rhetorical strategy designed

to maximize the reader's comprehension and acceptance of the presented research. Such a structured approach ensures that the "sense of making sense" (Brown & Yule, 1983) is not left to chance but is systematically built into the textual architecture. The presence of clear Problem-Solution patterns further highlights the communicative goal of academic writing: to identify gaps in knowledge and propose solutions backed by evidence.

These results have significant implications for academic writing pedagogy. Rather than treating cohesion and coherence as separate entities, instruction should emphasize their symbiotic relationship. Students need to understand how specific cohesive ties (e.g., conjunctions) function to realize broader rhetorical moves (e.g., establishing cause-effect relations, providing evidence). This aligns with the call from Flower and Hayes (2006) for a more process-oriented approach to writing, where planning and rhetorical strategy are foregrounded. Explicit teaching of RST concepts, for instance, could empower novice writers to consciously structure their arguments more effectively, moving beyond sentence-level correctness to discourse-level impact.

While this study offers valuable insights, it is not without limitations. The corpus, though carefully selected, was limited to a specific discipline and may not fully represent the diversity of academic discourse across all fields. Future research could explore disciplinary variations, as rhetorical conventions and the preference for certain cohesive and coherent strategies may differ significantly (Hinds, 1987). Additionally, a more in-depth investigation into reader perception and processing of texts with varying degrees of cohesion and coherence could further illuminate their impact on comprehension and persuasion. Nevertheless, this research solidifies the understanding that the mastery of cohesion and coherence is not merely a linguistic nicety but a critical determinant of a text's ability to achieve high impact within the academic community.

Conclusion

This study, through a combined qualitative and quantitative analysis of high-impact academic discourse, reaffirms the indispensable roles of cohesion and coherence in effective scholarly communication. The findings demonstrate that exemplary academic texts are characterized not merely by the presence of cohesive devices but by their strategic and sophisticated deployment to construct robust and logically progressive arguments. The high frequency of specific conjunctive relations, particularly those signaling causality, and the pervasive use of lexical cohesion underscore the deliberate effort by expert writers to ensure semantic connectivity and thematic continuity.

Crucially, the analysis revealed that micro-level cohesive ties work in concert with macro-level rhetorical organization to foster overall coherence. The prevalence of **Evidence** and **Elaboration** relations within the discourse confirms that high-impact writing prioritizes substantiation and clarity, allowing readers to effectively process and integrate complex information. The consistent application of these rhetorical strategies, often signposted by explicit linguistic markers, facilitates a clear, predictable, and persuasive presentation of ideas.

In essence, high-impact academic discourse achieves its efficacy through an intricate interplay between explicit linguistic connections (cohesion) and underlying logical structure (coherence). This study reinforces the notion that mastering both elements is not merely a matter of stylistic preference but a fundamental requirement for researchers aiming to communicate their findings persuasively and comprehensibly. The implications extend to academic writing pedagogy, suggesting that instruction should move beyond surface-level grammar to emphasize the functional role of cohesive devices in building coherent argumentative structures. Future research could

explore disciplinary variations in these patterns or investigate the impact of digital writing tools on the development of cohesive and coherent academic prose.

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IDEAS OF FREEDOM AND LINGUISTIC AND STYLISTIC FEATURES IN "FLAGBEARER" BY ELCHIN

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Abstract: Elchin, a prominent figure in modern Azerbaijani prose, consistently holds a special place in readers' hearts through his compelling works. His primary leitmotifs—patriotism, devotion to the native land, Turkism, and Turanism—reflect a writer intensely loyal to national identity and cultural heritage.

His notable work "Bayragdar" exemplifies this deep connection by capturing the poignant emotional experiences of Shusha residents who bore the heavy burden of displacement while living in the Buzovna Workers' Dormitory. Through this narrative, Elchin portrays the refugees' struggles, sorrows, and enduring hopes in an original, engaging, and deeply effective manner. The story profoundly explores themes of loss, resilience, and unyielding attachment to ancestral roots during times of historical conflict.

Key words: Shusha, Karabakh, flag, football, patriotism

ELÇİNİN “BAYRAQDAR” ƏSƏRİNDƏ AZADLIQ İDEYALARI VƏ DİL-ÜSLUB XÜSUSİYYƏTLƏRİ

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Xülasə: Müasir Azərbaycan nəsrinin görkəmli nümayəndələrindən olan Elçin öz maraqlı əsərləri ilə oxucu rəğbətini qazanan qüdrətli qələm sahibidir. Yaradıcılığının leytmotivini vətənpərvərlik, doğma torpağa bağlılıq, türkçülük və turançılıq təşkil edən yazıçı milli koloritə daim sadiqliyi ilə seçilir.

Təhlilə cəlb olunan **“Bayraqdar”** əsəri də məhz bu ideya zəminində yazılmış dəyərli nümunələrdəndir. Əsərdə Buzovna Fəhlə Yataxanasında məskunlaşan, köçkünlük faciəsini və ağrısını yaşayan şuşalıların daxili dünyası, keçirdikləri mürəkkəb emosional hisslər son dərəcə orijinal, maraqlı və təsirli şəkildə əks olunmuşdur. Yazıçı bu yolla həm xalqımızın tarixi taleyini, həm də milli dəyərlərə olan sarsılmaz bağlılığı bədii müstəvidə ustalıqla açır.

Açar sözlər: Şuşa, Qarabağ, bayraq, futbol, vətənpərvərlik

Introduction

Elchin, one of the famous figures of our modern prose, is one of the powerful writers who always holds a special place in the hearts of readers with his interesting works. The writer, whose leitmotif is patriotism, devotion to his native land, Turkism, and Turanism, is one of the outstanding artists who is extremely loyal to the national color with all his soul. The work “Bayragdar” that we will talk about is also of this type. The emotional feelings of the Shusha residents who carried the pain of displacement on their shoulders in the Buzovna Workers' Dormitory were presented in an extremely original, interesting and effective way. Throughout the work, the reader feels deeply, feels compassion, is sad, is moved, and thinks about the miserable life, shocks, and excitement of the refugees who were separated from the charming nature of Shusha, as if they were their own. First of all, the title of the work is directly connected with the tradition of national statehood and contains the great place of our flag, one of our attributes, in the national thinking of the people.

Methods and Materials

The main theme of the work is the painful life, dreams, intentions of those who fled from the city of Shusha, and the main character is Surkhay from Shusha, known to everyone as "Bayragdar". In general, the Karabakh events were one of the most relevant topics of the years of independence. Almost the majority of modern writers have addressed this topic. The main plot line of the work is our tricolor flag. The flag is characterized as a symbol of our freedom, a symbol of the independence that our people have long dreamed of for many years. Surkhay, while still living in Shusha, always thought about freedom and independence, and always carried a flag with him. He, remaining faithful to his generational tradition, protects the flag like the apple of his eye, as a trust from his ancestors, and wishes to pass it on to future generations. Even though Surkhay, who was a refugee from his homeland, left all his wealth behind, the fact that he took only one tricolor flag with him is a clear example of his attachment to the homeland and the people with all his being. Surkhay, who settled in the Buzovna Workers' Dormitory, almost hugged the flag to his chest and cherished it tightly, repeatedly hung it in different places in the dormitory, finding solace in it.

The work shows that while still in Shusha: "Surkhay first raised the tricolor flag of the Republic of Azerbaijan in 1988 - at a protest rally in Shusha, at the market square, when Armenia, using the chaos of "perestroika" and the tyranny of Garbochev, wanted to tear Karabakh from Azerbaijan..." (Elchin, 2004, p.84)

The flag has always been considered sacred as one of the national attributes of every nation, and numerous poems and essays have been written about it. The great poet of the Turkic world, Bakhtiyar Vahabzadeh, wrote in his poem "Flag":

"The flag is my identity, the flag is my existence,

My flag, my sovereignty on my people" (Vahabzade, 2003, p.314)

Surkhay, who held these verses as his motto, strongly believed that he would one day take the flag back to Shusha with him.

Results and Discussion

Surkhay's son, Abulfat (who was called Eusebio), was burning with the desire to become a famous football player. Everyone talked about his football games, and his mother dreamed of him becoming famous in the future. Unfortunately, the story ends with a tragic ending. One day, Abulfat, who was going to pick up a ball that fell from the roof of the boiler house, slipped, fell, and fell on the Russian flagpole that his father had made to hang the flag, and the flagpole pierced Abulfat's heart. Although the loss of his beloved son deeply shook Surkhay, he still did not forget the flag, trying to hang the tricolor flag again from the highest point of the dormitory - the part facing the highway, looking for a suitable place. The expression "Russian zublu" in the work serves two meanings: firstly, it was a means of pinning the flag to a solid place, and on the other hand, his son Abulfat also loses his life by becoming a victim of Russian zublu. This was a symbolic sign that our independence and freedom cost us blood.

Supporting characters such as Kamancha Tanriverdi, Demir Kazim, Golden Tooth Cumshud, Eagle Imran, I'm Waiting for You, Mammad the Man, Boleşik Butler Ibadulla, etc. also serve to reveal deeper layers of the work's ideas and directions.

The work is rich in folk sayings, folklore examples, proverbs, artistic imagery and means of expression, and rhetorical questions, which make the work even more colorful and effective.

For example: "the flag has grown old" (metaphor), "its iron has melted, but it did not die" (exaggeration), "why are our people here scattering like flies?!" (simile), "They will put Shusha in a bag and give it to you" (allusion) or "...that yard that sleeps in monotony, boredom, poverty, and despair..." (synonyms - lexical language and stylistic possibilities), "medlar garden" (barbarism), etc., etc., attract our attention as a characteristic of the writer's artistry.

Conclusion

In conclusion, it should be noted that Elchin's work "The Flagbearer" is a work full of hope and wishes for happiness, wearing the guise of optimism, in a time when the people and the nation are crying out for justice, and the feelings of refugees are leaving deep marks on the people's consciousness. As a clear example of this, the invincible military prowess of our Supreme Commander-in-Chief, President Ilham Aliyev, in the 44-Day Patriotic War, the endless faith and hopes of the people in their leader after the great Victory achieved at the cost of the blood of our martyrs and the lives of our veterans, have once again been confirmed. We believe that the work will always be appreciated as a contribution to future generations even years later.

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LANGUAGE POLICY AND PLANNING

THE NORMS OF THE AZERBAIJANI LANGUAGE

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Abstract: The norms of the Azerbaijani language represent the fundamental basis of effective communication and serve as an important component of national cultural identity. These norms regulate key aspects of language use, including pronunciation, vocabulary, grammar, and stylistic expression. By establishing clear rules for correct language usage, they ensure consistency, clarity, and mutual understanding among speakers in both spoken and written forms.

At the same time, linguistic norms do not remain completely rigid; they allow for a certain degree of flexibility and development. This dynamic nature enables the Azerbaijani language to adapt to new social, cultural, and technological changes while maintaining its structural integrity. The interaction between stability and change is one of the most important characteristics of language development.

Understanding and applying these norms is essential for effective communication in all contexts, including formal, academic, and everyday situations. Proper use of language norms helps to avoid misunderstandings, improves speech quality, and strengthens overall language competence.

As society continues to evolve, the Azerbaijani language also undergoes continuous development. Despite these changes, it preserves its unique linguistic identity and cultural heritage. This balance between tradition and innovation ensures the long-term vitality and sustainability of the language, making it an important tool for communication across generations.

Keywords: Linguistic norms, grammar, vocabulary, pronunciation, literary language, dialects, standardization

AZƏRBAYCAN DİLİNİN NORMALARI

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Xülasə: Azərbaycan dilinin normaları effektiv ünsiyyətin əsasını təşkil edir və milli mədəni kimliyin mühüm bir komponenti kimi çıxış edir. Bu normalar dil istifadəsinin əsas aspektlərini – tələffüzü, lüğət tərkibini, qrammatikanı və üslubi ifadəni tənzimləyir. Dilin düzgün istifadəsi üçün aydın qaydalar müəyyən etməklə, danışanlar arasında həm şifahi, həm də yazılı formada ardıcılığı, aydınlığı və qarşılıqlı anlaşmanı təmin edir.

Eyni zamanda, dil normaları tamamilə sabit deyil; onlar müəyyən dərəcədə çeviklik və inkişaf imkanı yaradır. Bu dinamik xüsusiyyət Azərbaycan dilinin sosial, mədəni və texnoloji dəyişikliklərə uyğunlaşmasına şərait yaradır, eyni zamanda onun struktur bütövlüyünü qoruyur. Sabitlik və dəyişkənlik arasındakı qarşılıqlı əlaqə dil inkişafının ən mühüm xüsusiyyətlərindən biridir.

Dil normalarını anlamaq və onlara riayət etmək bütün ünsiyyət formalarında – rəsmi, akademik və gündəlik situasiyalarda – vacibdir. Düzgün dil istifadəsi anlaşılmaqların qarşısını alır, nitq keyfiyyətini artırır və ümumi dil bacarıqlarını gücləndirir.

Cəmiyyət inkişaf etdikcə Azərbaycan dili də davamlı olaraq dəyişir və inkişaf edir. Bu dəyişikliklərə baxmayaraq, dil özünəməxsus linqvistik kimliyini və mədəni irsini qoruyub saxlayır. Ənənə və yenilik arasındakı bu tarazlıq dilin uzunmüddətli canlılığını və dayanıqlılığını təmin edir və onu nəsillərarası ünsiyyət üçün mühüm vasitəyə çevirir.

Açar sözlər: Dilçilik normaları, qrammatika, lüğət tərkibi (söz ehtiyatı), tələffüz, ədəbi dil, dialektlər, standartlaşdırma

Introduction

The Azerbaijani language represents a rich cultural and historical heritage shaped by centuries of social development, literary tradition, and interaction with neighboring cultures. Its norms reflect a balance between preservation and adaptation, ensuring both clarity and continuity in communication. These norms regulate pronunciation, vocabulary, grammar, and stylistic usage, forming the foundation of standard language. The literary language serves as a model for education, media, and official communication, while dialects contribute diversity and cultural depth. The influence of historical scripts and language reforms has further shaped modern Azerbaijani. Maintaining linguistic norms is essential for national identity, mutual understanding, and effective communication in a rapidly globalizing world. At the same time, flexibility within these norms allows the language to evolve naturally. This article explores the core principles, historical development, and practical applications of Azerbaijani language norms through examples and analysis, emphasizing their role in both formal and everyday contexts. Language norms play a central role in shaping how people communicate and understand each other. The Azerbaijani language, as a member of the Turkic language family, has developed a system of norms that regulate its use in both written and spoken forms. These norms ensure consistency, clarity, and cultural continuity. They are not arbitrary rules but are formed through long-term usage, literary tradition, and social agreement.

The importance of language norms becomes especially clear in education, media, and official communication. Without a shared standard, misunderstandings could arise, and the unity of the language could weaken. At the same time, norms are not rigid; they evolve alongside society.

Phonetic Norms. Phonetic norms define how words should be pronounced correctly. In Azerbaijani, pronunciation is generally consistent with spelling, which makes the language relatively easy to read and write. For example, each letter typically corresponds to a specific sound, and this regularity helps learners quickly grasp pronunciation.

However, regional variations can influence pronunciation. In some dialects, certain vowels may sound slightly different, yet the standard pronunciation remains the reference point in formal contexts such as broadcasting and education.

Lexical Norms. Lexical norms regulate the use of words and expressions. The Azerbaijani vocabulary includes native Turkic words as well as borrowings from Persian, Arabic, and Russian. The standard language prefers words that are widely understood and stylistically appropriate.

For example, in formal writing, choosing a neutral and widely accepted word is important, while in informal speech, colloquial expressions may be more common. The selection of vocabulary often depends on context, audience, and purpose.

Grammatical Norms. Grammar provides the structural framework of the language. Azerbaijani grammar follows agglutinative patterns, where suffixes are added to roots to express meaning. This allows for precise and flexible expression.

For instance, a single word can convey what might require a full sentence in another language. Correct use of suffixes ensures clarity and prevents ambiguity. Grammatical norms are particularly important in written communication, where mistakes can lead to misunderstanding.

Stylistic Norms. Stylistic norms guide how language is used in different contexts. Azerbaijani distinguishes between formal and informal styles, as well as literary and conversational language. In official documents, clarity and precision are essential, while in художественные texts, expressive and figurative language is more common.

For example, a news report uses straightforward language, while a poem may employ metaphors and symbolic expressions. Understanding stylistic norms helps speakers and writers choose the appropriate tone.

The Role of the Literary Language. The literary language serves as the standard form of Azerbaijani. It is used in education, literature, media, and official communication. This form of the language is carefully regulated and reflects established norms.

Literary works play a significant role in preserving and developing the language. Through literature, new expressions emerge, and existing norms are reinforced. Writers often balance creativity with adherence to linguistic standards.

Dialects and Their Influence. Azerbaijani dialects add richness and diversity to the language. Each region has its own unique features, including pronunciation, vocabulary, and expressions. While dialects may differ from the standard language, they are an important part of cultural identity.

For example, certain words used in rural areas may not appear in the literary language but carry strong local significance. The interaction between dialects and the standard language contributes to linguistic development.

Historical Development of Norms. The norms of the Azerbaijani language have evolved over time. Changes in script, from Arabic to Latin and later to Cyrillic and back to Latin, have influenced spelling and pronunciation. Each reform aimed to improve literacy and reflect the phonetic structure of the language.

Historical events and cultural exchanges have also shaped vocabulary and usage. Despite these changes, the core structure of the language has remained stable.

Language Culture and Communication. Language culture refers to the correct and effective use of language norms. It includes clarity, accuracy, and appropriateness in communication. A high level of language culture enhances mutual understanding and reflects respect for the language.

For example, using correct grammar and appropriate vocabulary in public speaking demonstrates professionalism and competence. In everyday communication, clear expression helps avoid confusion.

Materials and Methods

This study is based on qualitative and descriptive research methods aimed at analyzing the norms of the Azerbaijani language. The materials used in the research include scientific literature, linguistic textbooks, academic articles, dictionaries, and normative resources related to grammar, vocabulary, pronunciation, and stylistics of the Azerbaijani language.

The descriptive method was applied to explain the main categories of linguistic norms, including phonetic, lexical, grammatical, and stylistic rules. In addition, analytical methods were used to examine how these norms function in real communication and how they contribute to clarity and correctness in language use.

A comparative approach was also employed to observe differences between standard literary Azerbaijani and dialectal variations. This allowed for a deeper understanding of how standardization processes regulate language usage across different contexts.

Examples from modern Azerbaijani speech and written texts were analyzed to illustrate the practical application of linguistic norms. The collected data were systematically reviewed and interpreted to identify the role of language norms in communication, cultural identity, and language development. This methodological framework ensures a comprehensive understanding of the structure and function of Azerbaijani language norms.

Results and Discussion

The analysis shows that linguistic norms in the Azerbaijani language play a crucial role in ensuring effective and accurate communication. These norms regulate pronunciation, vocabulary, grammar, and stylistic usage, creating a unified system that supports clarity and consistency in both spoken and written forms. The findings confirm that standard language norms are essential for maintaining mutual understanding among speakers.

The study also reveals that while the literary standard language is relatively stable, it is not completely fixed. It evolves under the influence of social change, cultural development, and technological progress. This dynamic nature allows the Azerbaijani language to adapt to new communicative needs while preserving its structural integrity.

Another important finding is the interaction between standard language and dialectal variations. Dialects enrich the linguistic diversity of Azerbaijani, but standardization ensures uniformity in formal communication. This balance between variation and standard form is essential for maintaining both linguistic identity and communicative efficiency.

The discussion highlights that language norms are closely connected with language culture and education. Proper application of these norms improves speech quality, reduces communication errors, and strengthens overall language competence. Therefore, linguistic norms serve not only as regulatory rules but also as a foundation for cultural identity and effective communication in society.

Conclusion

In conclusion, the norms of the Azerbaijani language are a fundamental component of effective communication and play a significant role in preserving linguistic and cultural identity. These norms regulate all major aspects of language use, including pronunciation, vocabulary, grammar, and style, ensuring clarity, consistency, and mutual understanding among speakers.

The study shows that although linguistic norms provide a stable framework for language use, they are not completely static. They develop and adapt in response to social, cultural, and technological changes. This flexibility allows the Azerbaijani language to remain relevant and functional in modern communication while maintaining its structural integrity.

Furthermore, the relationship between standard language and dialectal variations demonstrates the richness and diversity of Azerbaijani. Standardization ensures unity in formal communication, while dialects contribute to linguistic variety and cultural depth.

Overall, linguistic norms are essential not only for correct language usage but also for the development of language culture and communication skills. Continuous attention to these norms helps improve speech quality, reduce misunderstandings, and strengthen language competence. Therefore, the preservation and proper application of Azerbaijani language norms are crucial for both present and future generations.

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LANGUAGE POLICY AND NATIONAL IDENTITY IN POST-SOVIET BALTIC SOCIETIES

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Abstract: This article examines the relationship between language policy and national identity in post-Soviet Baltic societies, focusing on Estonia, Latvia, and Lithuania. Following the collapse of the Soviet Union, the Baltic states initiated extensive language reforms aimed at restoring indigenous languages and strengthening national identity after decades of Russification. The study analyzes how language legislation, educational reforms, and sociopolitical discourse contributed to nation-building and cultural preservation. Using a qualitative comparative methodology, the research investigates the role of language in public administration, education, and minority integration. The findings demonstrate that language policies became essential instruments for reinforcing political sovereignty, historical continuity, and collective identity. At the same time, the implementation of these policies generated challenges related to the integration of Russian-speaking minorities, particularly in Estonia and Latvia. The article further discusses the influence of European democratic standards and multilingual realities on Baltic language planning. Overall, the study

concludes that language policy remains a central factor shaping sociopolitical development, cultural memory, and national cohesion in contemporary Baltic societies.

Keywords: Language policy, national identity, Baltic societies, post-Soviet studies, multilingualism, minority integration, sociolinguistics.

SPRACHPOLITIK UND NATIONALE IDENTITÄT IN DEN POSTSOWJETISCHEN BALTISCHEN GESELLSCHAFTEN

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Zusammenfassung: Dieser Artikel untersucht das Verhältnis zwischen Sprachpolitik und nationaler Identität in den postsowjetischen baltischen Gesellschaften mit einem Schwerpunkt auf Estland, Lettland und Litauen. Nach dem Zerfall der Sowjetunion leiteten die baltischen Staaten umfassende Sprachreformen ein, die darauf abzielten, die indigenen Sprachen wiederherzustellen und die nationale Identität nach Jahrzehnten der Russifizierung zu stärken. Die Studie analysiert, wie Sprachgesetzgebung, Bildungsreformen und soziopolitische Diskurse zum Nationenaufbau und zur Bewahrung kultureller Identität beitrugen. Mithilfe einer qualitativen Vergleichsmethodik untersucht die Forschung die Rolle der Sprache in der öffentlichen Verwaltung, im Bildungswesen sowie bei der Integration von Minderheiten. Die Ergebnisse zeigen, dass Sprachpolitik zu einem zentralen Instrument zur Stärkung politischer Souveränität, historischer Kontinuität und kollektiver Identität wurde. Gleichzeitig führten diese Maßnahmen zu Herausforderungen bei der Integration der russischsprachigen Minderheiten, insbesondere in Estland und Lettland. Zudem wird der Einfluss europäischer demokratischer Standards und multilingualer Realitäten auf die Sprachplanung im Baltikum diskutiert. Insgesamt kommt die Studie zu dem Schluss, dass Sprachpolitik ein entscheidender Faktor für gesellschaftliche Entwicklung, kulturelles Gedächtnis und nationale Kohäsion in den heutigen baltischen Staaten bleibt.

Schlüsselwörter: Sprachpolitik, nationale Identität, baltische Gesellschaften, postsozialistische Studien, Mehrsprachigkeit, Minderheitenintegration, Soziolinguistik.

Introduction

The collapse of the Soviet Union in 1991 fundamentally transformed the political, cultural, and linguistic landscape of the Baltic region. Estonia, Latvia, and Lithuania regained independence after decades of Soviet occupation and immediately initiated comprehensive nation-building processes centered on the restoration of national languages and identities. In the Soviet era, Russian functioned as the dominant lingua franca across administrative, educational, and political institutions, resulting in extensive bilingualism and demographic transformations caused by migration policies. After independence, the Baltic states adopted language policies aimed at strengthening Estonian, Latvian, and Lithuanian as symbols of sovereignty and cultural continuity. These developments positioned language not merely as a communication tool but as a central mechanism of political legitimacy and national identity formation.

Language policy in post-Soviet Baltic societies has therefore become a highly significant sociolinguistic and political issue. Scholars argue that language planning in the Baltic region reflects broader attempts to reverse Soviet-era Russification and reconstruct national consciousness through education, citizenship regulations, and public administration reforms (Ozolins, 2011). At

the same time, these policies have generated debates concerning minority rights, democratic inclusion, and integration of Russian-speaking populations. Research demonstrates that the Baltic states have attempted to balance the promotion of titular languages with obligations toward European democratic standards and minority protection (Hogan-Brun & Melnyk, 2012).

National identity in the Baltic states is strongly connected to linguistic preservation because language historically functioned as a marker of resistance against external domination. During Soviet rule, Russian language expansion was often perceived as a threat to indigenous linguistic and cultural traditions. Consequently, post-independence governments viewed language revitalization as essential for restoring historical continuity and strengthening national cohesion. According to Alijeva (2017), however, language policies in the Baltic region also contributed to tensions between titular nations and Russian-speaking minorities, especially regarding citizenship, education, and public participation.

The present study investigates the relationship between language policy and national identity in post-Soviet Baltic societies. The article focuses on Estonia, Latvia, and Lithuania and analyzes how language legislation, educational reforms, and sociopolitical discourse have shaped collective identity since independence. The study further explores the challenges associated with minority integration and multilingual realities in contemporary Baltic societies.

Materials and Methods

This research employs a qualitative comparative methodology based on discourse analysis and sociolinguistic policy evaluation. The study examines official language legislation, scholarly literature, educational reforms, and sociopolitical developments in Estonia, Latvia, and Lithuania from 1991 to the present period.

The primary materials include academic journal articles, policy analyses, and books focusing on Baltic language politics, nationalism, and identity construction. Key theoretical contributions from sociolinguistics and language policy studies were used to analyze the interaction between linguistic regulation and nation-building. Comparative analysis allowed the identification of both shared tendencies and country-specific approaches among the Baltic states.

The study also incorporates historical contextualization to evaluate the long-term consequences of Soviet linguistic policies. Particular attention is given to the role of Russian-speaking minorities and the implementation of national language requirements in public institutions, media, and education systems. The methodological framework combines elements of critical discourse analysis and comparative political linguistics in order to assess how language policies contribute to symbolic nationhood and sociocultural integration.

Furthermore, the research draws upon contemporary discussions concerning minority rights and multilingualism within the European context. The analysis evaluates how Baltic language policies simultaneously function as instruments of cultural preservation and mechanisms of political consolidation.

Results and Discussion

The findings indicate that language policy has played a decisive role in the reconstruction of national identity across post-Soviet Baltic societies. Estonia, Latvia, and Lithuania implemented language laws immediately after independence to strengthen the status of titular languages in administration, education, and media. These reforms symbolized political sovereignty and the rejection of Soviet linguistic dominance.

One major result of the study is that language revitalization policies successfully increased the public visibility and institutional dominance of national languages. In Estonia and Latvia particularly, language certification requirements became important tools for citizenship acquisition and employment opportunities. These measures were intended to encourage integration into the national cultural environment and reduce dependency on Russian as the primary communicative medium. Ozolins (2011) emphasizes that Baltic language policies were designed not only for linguistic preservation but also for reinforcing national consciousness after decades of Soviet control.

The research further demonstrates that the Baltic states adopted somewhat different approaches toward minority integration. Lithuania generally pursued a comparatively moderate policy due to the smaller proportion of Russian-speaking residents, whereas Estonia and Latvia faced more complex demographic challenges because of larger Russian-speaking populations concentrated in urban centers. Hogan-Brun et al. (2008) argue that these demographic realities significantly influenced policy implementation and public debates surrounding citizenship and linguistic rights.

Another important finding concerns the symbolic power of language in national identity formation. In Baltic societies, language functions as a cultural memory mechanism connected to historical survival and resistance. The restoration of indigenous languages after independence represented the restoration of historical legitimacy itself. Educational reforms promoting national history, literature, and language contributed to strengthening collective identity among younger generations.

However, the study also reveals persistent tensions between national consolidation and minority inclusion. Russian-speaking minorities in Estonia and Latvia have occasionally perceived language requirements as exclusionary. Alijeva (2017) notes that restrictive language policies may create difficulties in social integration and contribute to feelings of marginalization among minority communities. Nevertheless, Baltic governments argue that the promotion of state languages is necessary for social cohesion and national security, particularly in light of regional geopolitical concerns.

The analysis additionally indicates that European integration influenced Baltic language policy development. Membership in the European Union encouraged democratic reforms and minority protections while allowing Baltic governments to preserve strong national language frameworks. Consequently, contemporary Baltic societies demonstrate a complex interaction between nationalism, multilingualism, and European democratic values.

Recent public discussions also reveal that language remains emotionally and politically sensitive within Baltic societies, especially concerning the role of Russian in education and public communication. Online discussions and public discourse illustrate continuing debates over integration, historical memory, and linguistic justice in the region. These debates show that language policy continues to shape both political identity and everyday social relations.

Overall, the results confirm that language policy in the Baltic states serves multiple interconnected functions: preservation of national culture, restoration of historical continuity, consolidation of political sovereignty, and management of post-Soviet social transformation.

Conclusion

Language policy has become one of the most influential instruments of nation-building in post-Soviet Baltic societies. Estonia, Latvia, and Lithuania used linguistic reforms to restore

indigenous cultural identity, strengthen political independence, and distance themselves from Soviet-era Russification policies. The study demonstrates that language functions not only as a communicative system but also as a symbolic representation of historical memory, sovereignty, and collective identity.

The findings reveal that Baltic language policies have successfully strengthened the position of national languages within state institutions and public life. At the same time, these policies continue to generate debates concerning minority rights and social inclusion, particularly among Russian-speaking populations. The Baltic experience illustrates the broader challenge faced by post-Soviet societies: balancing national consolidation with democratic pluralism and multicultural realities.

In contemporary Baltic states, language policy remains closely connected to geopolitical security, cultural preservation, and European integration. Future research may further investigate generational attitudes toward multilingualism, digital communication practices, and the evolving role of English alongside national and minority languages in the Baltic region.

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LANGUAGE REFORM AND NATION BUILDING: THE TURKISH EXPERIENCE

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Abstract: The language and alphabet reform in the newly constructed republic, Türkiye, has experienced one of the most influential sociolinguistic movements in the whole world. Under the leadership of Mustafa Kemal Atatürk, language reform initiated a number of hidden issues, such as national identity, sense of belonging, and the power of language in the process of republication. Therefore, this study aims to discuss the transformation of the nation and its language, supported by a sociolinguistic movement and the 1928 language reform, with a view to revealing the process from sociolinguistic and historical perspectives. The study also highlights criticisms raised in the process of modernization, noting reasonable causes such as cultural disconnection from a rich historical past. However, despite the criticism, the study highlights the success of language purification and pinpoints future threats facing society and language.

Keywords: Language reform, national identity, sociolinguistics movement, cultural disconnection, language purification

DİL REFORMU VE ULUS İNŞASI: TÜRKİYE DENEYİMİ

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Özet: Yeni kurulan Türkiye Cumhuriyeti'nde gerçekleştirilen dil ve alfabe reformu, dünya çapında en etkili sosyodilbilimsel hareketlerden birini oluşturmuştur. Mustafa Kemal Atatürk'ün önderliğinde yürütülen dil reformu, cumhuriyetin kuruluş sürecinde ulusal kimlik, aidiyet duygusu ve dilin gücü gibi bir dizi gizli meseleyi gündeme getirmiştir. Bu nedenle, bu çalışma, sosyodilbilimsel bir hareket ve 1928 dil reformu ile desteklenen ulusun ve dilinin dönüşümünü, süreci sosyodilbilimsel ve tarihsel perspektiflerden ortaya koymak amacıyla ele almayı amaçlamaktadır. Çalışma ayrıca, zengin tarihi geçmişten kültürel kopukluk gibi makul nedenlere dikkat çekerek, modernleşme sürecinde ortaya çıkan eleştirileri de vurgulamaktadır. Ancak, eleştirilere rağmen, çalışma dilin saflaştırılmasındaki başarıyı vurgulamakta ve toplum ile dilin karşı karşıya olduğu gelecekteki tehditleri işaret etmektedir.

Anahtar Kelimeler: Dil reformu, ulusal kimlik, sosyodilbilimsel hareket, kültürel kopukluk, dilin saflaştırılması

1. Introduction

As people's needs began to shape communal life, the urge to live together has been embellished and sustained through communication and language; as a consequence, language has served not only as a means of communication but also as an asset of civilization and nationality (Anderson, 2006; Bourdieu, 1991). In this regard, nations, societies, and even leaders and politicians have adopted language policies to homogenize society, promote national and social identity, and build social ties within the community. In today's modern nation-state communities, language serves a wide range of purposes, from a communication tool to an asset for nationality, modernization, and state authority.

As one of the most remarkable successes in the field of language and nation association, the case of Türkiye serves as an outstanding example of language reform and its contribution to nation construction. As the antecedent of the great Empire, the Ottoman, a nation-based republic, with the guidance of Mustafa Kemal Atatürk, aimed to create a nation whose modernity, secularity, and unification required a new language for several reasons. To elaborate, the old Turkish, called Ottoman Turkish, was heavily influenced by Persian and Arabic, and the Arabic alphabet, which has three vowels, was used to represent a language with eight vowels. This created numerous disconnections between the state and the people, as well as between the Empire's management organs (Lewis, 1999; Zürcher, 2004).

Therefore, it can be inferred from the circumstances above that the Turkish language reform aimed to create a nation-based society and was part of the nation's modernization, which has been dealing with a number of innovations and governmental trials for almost a century. Being a multicultural and multilingual society, the new nation-state needed a tool to connect citizens together with a sense of belonging, and as language has such traits, the modernization process should be focused on the language itself, given that the language that people speak in fact affects the way they think, behave, and interpret the world around them (Sapir, 1929). Therefore, language serves more than a communication tool; it is, in fact, an asset for bringing people together and a catalyst for building a sense of identity and belonging to a particular community (Whorf, 1956). All these created the underlying reasons for language reforms in the newly established country; as a consequence, the new republic adopted the Latin alphabet in 1928 and established the Turkish Language Association in 1932. The policy adopted in the early years of the republic aimed to increase literacy rates and to foster a national identity through language reform.

In light of the above, the current study investigates the Turkish language reform from a historical and sociolinguistic perspective, with the aim of presenting national identity and cultural and social integrity to shape the phenomenon of the Turkish nation. The paper also discusses both the criticism and the achievement to present a holistic view of the process of nationalization. Furthermore, the study focuses on how language reform helped people become literate and pro-republican in the early years of the republic, as well as on the government's efforts to promote the adoption of the new alphabet and increase literacy. The study also highlights the government's efforts to ease the process of language and alphabet reform. Finally, the paper evaluates the long-term socio-cultural effects of the language reform aimed at consolidating Turkish identity.

2. Historical Background

Efforts to introduce innovations in the Ottoman Empire coincided with the 19th century, when modernization occurred across many fields, including government, education, the military, and society. In terms of language modernization, the debates go back to the first half of the 19th century, gradually intensified in the late 19th century, and were put into effect in the first years of the new republic. When we analyze the language structure of the empire, it is clearly seen that multilingual and multicultural communities comprised the whole. Among them, Ottoman Turkish, which was a mixture of Persian and Arabic, was the language of the state and bureaucracy. Although the spoken Turkish was relatively easy to understand, the written language used by bureaucrats and scholars was highly intriguing and complex. Unfortunately, this discrepancy gave rise to serious literacy problems nationwide, as the folk was unfamiliar with the written language, which draws on Arabic and Persian structures and vocabulary. As a result, this situation led to a low literacy rate among the empire's population. Limited public accessibility caused a disconnect

between the people and the government. This deficiency emerged in the nineteenth century, when the notion of nationalism was at its height, and Ottoman intellectuals began to view language as a central component of national consciousness (Mardin, 2000). Therefore, they supported the view that the Turkish language should be freed from foreign linguistic influence.

However, the outbreak of the First World War and the political issues and changes in the late Ottoman period postponed the process, which was followed by the collapse of the Ottoman Empire. The new republic, established under the guidance of Mustafa Kemal Atatürk, adopted a modern perspective and implemented reforms across many fields, including language and the alphabet, which became one of the most important pillars of the modernization process, given that the power of language to create a national identity and togetherness was so vast that the unity, literacy, and new nation form could easily be established (Berkes, 1998). As a result, in 1928, the adoption of the Latin alphabet was finalized after much effort to introduce it to the country's people. The new Latin-based Turkish alphabet aimed to simplify reading and writing. The reform aimed to fulfill objectives such as:

- Enhancing the rate of literacy,
- Spreading education,
- facilitating communication,
- strengthening national identity and ties.

The new Turkish government initiated a well-planned program to teach the new language alphabet and reforms to citizens through National schools (Millet Mektepleri), which, in turn, quickly captured public interest and support. In 1932, the Turkish Language Association (Türk Dil Kurumu-TDK) was founded to develop and protect the Turkish language, aiming to purify it, protect it from foreign linguistic influence, create new Turkish words, and standardize language use nationwide.

3. Sociolinguistic Perspective

The Turkish language reform, on closer analysis, can be considered one of the most significant examples of sociolinguistic success, despite the criticism that it encompassed two major fields: society and linguistics. From a linguistic perspective, there have been remarkable reforms in vocabulary, the alphabet, and the structure of the language, aimed at purification and a return to the essence of the language, to its historical roots, which date back at least 5,000 years. Besides, from a social perspective, as Fishman (1972) underscores, language is interwoven with national identities and ethnic belonging; the new language reform succeeded in creating a modern social community in the newly founded country, whose literacy rate was rather low.

From a sociolinguistic perspective, language is closely linked to power, identity, and social structure. The new education policy adopted a new objective aiming not to leave any citizen behind. Therefore, the integration of people and their familiarity with the new language, as well as the value of citizenship, are legitimized and strengthened by the government's new policies. Therefore, the new language served as a cement to connect society to the nation and the country, unlike in the Ottoman period. This was carried out thanks to the new language, as Ottoman Turkish was heavily influenced by Arabic and Persian, which created a number of disparities between the people and the government. The power of communication, whose primary asset is the strength of language, was limited during the Ottoman Empire, as elites used a different language, the official language was different, and the community was different. Therefore, the aim of the new reform was to create a

holistic society identified by a single language, and, in terms of corpus and status, the new language promoted equality, identity, and a sense of belonging across the whole country.

As Anderson (2006) argued, nations are created through shared language and communication, in which language plays a profound and constructive role. In Türkiye, the new language reform helped foster a collective identity and national consciousness. Therefore, in Türkiye, this reform contributed to the creation of a modern national identity by providing a common linguistic framework and awareness to all citizens. This new standard Turkish became prevalent in education, bureaucracy, science, government, and mass communication. Therefore, the reform enabled the destruction of social class hierarchies, one of the principles of the newly founded republic, thereby initiating the equality of all citizens. The status of every single citizen in the country was neutralized. This gave rise to the inculcation of identity, equality, and power distribution in society.

4. Criticism of The Reform

Despite the new language reform's great success in improving literacy rates and building a solid society, it was also criticized from historical, cultural, and sociolinguistic perspectives. In the field of culture, the criticisms accumulated around cultural disconnection from the past. As the new generation is educated in a language free of Persian and Arabic connections, they are said to have limitations in analyzing and accessing Ottoman archives in literature, history, and documentation. As a consequence, this created a gap between the past and the present in terms of cultural connectivity and memory.

Another criticism was the creation of new words rather than using the older versions, which were loanwords from Persian and Arabic. According to some scholars, these newly created words seem artificial and do not fulfill the same meaning gravity as the preceding ones. This linguistic purification may also disrupt the language's natural evaluation. From a sociolinguistic perspective, it was claimed that the reform followed a top-down process, as languages evolve through social processes. This may create a disconnection between the government and society, as it was proposed by the government as an official legacy.

Even though there has been severe criticism of the reform and the new alphabet, the regulations and all attempts by the government and the people proved highly effective, as the literacy rate increased and the unification of society and the creation of an identity were sustained. As a result, the Turkish language reform contributed to society and the newly founded country in many positive ways.

5. Conclusion and Recommendations

The Turkish language reform and regulations are among the most influential examples in the history of language-nation construction, serving as a model for linguistic purification and national identity, encompassing cultural and sociolinguistic improvements within a society. The simplification of the language and alphabet created a better communicative privilege for the citizens under the shade of equality, belonging, and identity. Therefore, despite the criticism, the movement is assumed to be successful in the process of modernization and integration of a national community. From a sociolinguistic perspective, the language revolution helped close the gap between elites and ordinary citizens by promoting a more standardized language, which is also authorized as the official language of education in the country. This attempt to create a common sense of linguistic cognition may foster a common sense of nationality, belonging, and possession by equalizing all citizens in the community. It should be kept in mind that the Turkish language reform was a great success in terms of tightening the language and society, fostering a nationalist

perspective, and revealing how a language can serve as a powerful tool to bring people together under the umbrella of a nation.

Additionally, the study recommends the following issues. First, the upcoming studies may expand their perspectives to include the long-term sociocultural consequences of the language reform. Secondly, despite these reforms, the influence of other languages and their loanwords, rather than that of Arabic and Persian, should be investigated. Besides, comparative studies between Turkish and other languages that have undergone a similar process can be further investigated to reveal their similarities and differences. Educational institutions may promote studies related to Ottoman Turkish and other languages still in use in the country to assess the effects of the language reform on their continued presence. The government of the republic should launch campaigns to protect the Turkish language and foster national consciousness, discouraging the use of loanwords from other languages, as this may, in the sociolinguistic domain, harm notions of identity, perspective, power, and social integration.

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STRENGTHENING FOREIGN LANGUAGE POTENTIAL IN THE TURKIC WORLD AND NEW PERSPECTIVES IN INTERNATIONAL RELATIONS

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Abstract: In the modern global world, the growing influence of the Turkic states in the system of international relations and their emergence as a new geopolitical center of power have transformed the

strategic management of foreign language potential into a decisive factor. This article comprehensively examines the history and contemporary trends of language learning in the Turkic world, as well as the influence of linguistic affiliation on the process of foreign language acquisition. Within the framework of the study, the linguistic evolution of the Turkic peoples is explored across a broad historical spectrum, beginning with the natural multilingual environment of the Silk Road period, continuing through the Ottoman Empire's "Elsineyi-Selase" tradition, and extending to the intermediary language model shaped through the Russian language during the Soviet period. In the contemporary era, the alignment of foreign language teaching with international European Language Standards (CEFR), digital transformation processes, and the integration of artificial intelligence-based platforms into education constitute the main scientific and theoretical directions of the study.

The statistical indicators highlighted in the article, including a 65% increase in learning efficiency through the application of the communicative method and the target of equipping 95% of schools with modern technological laboratories by 2030, demonstrate the scale of progress in this field. At the same time, the study analytically examines complex challenges such as the 40% quality gap between urban and rural schools, the digital divide associated with limited internet access for 30% of the population, and the shortage of qualified personnel in regional areas. Furthermore, the impact of foreign language potential as a "soft power" instrument on the international image of the Turkic world and its "smart power" strategy is scientifically substantiated. Strategic measures, such as involving 85% of the teaching staff in international certification programs, are evaluated as an integral component of intellectual diplomacy, which holds an important place in the Turkic World's 2030 roadmap. In conclusion, the role and future prospects of linguistic power in ensuring the global competitiveness of the Turkic states are identified.

Keywords: Turkic world, foreign language potential, global integration, educational reforms, international relations, soft power, artificial intelligence.

TÜRK DÜNYASINDA YABANCI DİL POTANSİYELİNİN GÜÇLENDİRİLMESİ VE ULUSLARARASI İLİŞKİLERDE YENİ PERSPEKTİFLER

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Özet: Günümüz küresel dünyasında Türk devletlerinin uluslararası ilişkiler sistemindeki artan nüfuzu ve yeni bir jeopolitik güç merkezi olarak yükselişi, yabancı dil potansiyelinin stratejik yönetimini belirleyici bir faktör haline getirmiştir. Sunulan makalede Türk dünyasının dil öğrenme tarihi ve modern eğilimleri ile dilsel aidiyetin yabancı dil edinimi sürecine etkileri kapsamlı bir şekilde incelenmektedir. Araştırma çerçevesinde Türk halklarının dilsel evrimi, İpek Yolu döneminin doğal çok dillilik ortamından başlayarak, Osmanlı İmparatorluğu'nun "Elsine-i Selase" geleneğine ve Sovyet döneminin Rus dili aracılığıyla şekillenmiş olan aracı dil modeline kadar geniş bir tarihsel yelpazeyi kapsamaktadır. Modern dönemde yabancı dil öğretiminin Avrupa Dilleri Ortak Çerçeve Programı (CEFR) standartlarına uyumlaştırılması, dijital dönüşüm süreçleri ve yapay zeka tabanlı platformların eğitim sistemine entegrasyonu çalışmanın temel bilimsel-teorik eksenlerini oluşturmaktadır.

Makalede özellikle vurgulanan istatistiksel veriler, iletişimsel yöntemin uygulanmasıyla öğrenme verimliliğinin %65 artması ve 2030 yılına kadar okulların %95'inin modern teknolojik laboratuvarlarla donatılması hedefi, bu alandaki ilerlemenin boyutlarını ortaya koymaktadır. Aynı zamanda çalışmada mevcut zorluklar; şehir ve köy okulları arasındaki %40'lık nitelik farkı, nüfusun %30'unun internet erişiminin kısıtlı olmasına bağlı dijital bölünme ve bölgesel kadro yetersizliği gibi karmaşık faktörler analitik bir düzlemde

tartışılmaktadır. Çalışmada yabancı dil potansiyelinin bir "yumuşak güç" aracı olarak Türk dünyasının uluslararası imajına ve "akıllı güç" stratejisine etkisi bilimsel olarak temellendirilmektedir. Öğretmen kadrosunun %85'inin uluslararası sertifikasyon programlarına dahil edilmesi gibi stratejik adımlar, Türk dünyasının 2030 yol haritasında önemli bir yer tutan entelektüel diplomasinin bir parçası olarak değerlendirilmektedir. Sonuç olarak, Türk devletlerinin küresel arenada rekabet edebilirliğinin sağlanmasında dilsel gücün rolü ve gelecek perspektifleri belirlenmektedir.

Anahtar kelimeler: Türk dünyası, yabancı dil potansiyeli, küresel entegrasyon, eğitim reformu, uluslararası ilişkiler, yumuşak güç, yapay zeka

Introduction

In the contemporary system of international relations, the power of states is measured not only by their military and economic indicators, but also by their intellectual capital and communicative resources. In the twenty-first century, characterized by the acceleration of globalization, a foreign language functions as a window through which a state opens itself to the world and as a strategic instrument that expands its room for manoeuvre within platforms of international cooperation. With its rich historical heritage and dynamically evolving geopolitical position, the Turkic world stands at the centre of this linguistic transformation. The rise of Turkic peoples on the global stage requires their foreign language acquisition capacities to enter a qualitatively new phase and demands the successful integration of this potential into national development strategies. From this perspective, the study of this topic is of considerable relevance from both theoretical and practical standpoints, since language policy has become one of the key factors directly influencing the international competitiveness of Turkic states.

The history of foreign language acquisition among Turkic peoples has followed a centuries-long path of development and has been shaped at the crossroads of various civilizations. Historically, Turkic communities inhabiting the geographies through which the Great Silk Road passed naturally lived in a multilingual environment, which fostered in them a high degree of adaptability to different languages and cultures (Gömeç, 2020, p. 45). In ancient times, close contacts established with merchants and travellers speaking different languages for the purposes of trade and diplomacy gave rise to early forms of “translation” and linguistic mediation within the Turkic tradition of statehood. Written monuments and historical documents belonging to the Göktürk and Uyghur periods demonstrate that, even at that time, Turkic states used linguistic knowledge as a strategic tool in their relations with China and other neighbouring peoples. In particular, the translation of numerous religious and philosophical works from other languages into Turkic during the Uyghur period indicates the ability of these peoples not only to learn foreign languages, but also to adapt them to their own cultural context (Tekin, 2019, p. 112).

Following the adoption of Islamic culture, the linguistic landscape of the Turkic world entered a new stage. The leading role of Arabic and Persian in scientific, literary, and religious domains created the conditions for the formation of the intellectual elite of Turkic peoples on the basis of the principle of *Alsina-yi Salasa* — the “three languages.” The rich scholarly heritage produced during this period emerged as a result of the synthesis of Turkic, Arabic, and Persian, with each of these languages being used in distinct functional domains. This model of multilingualism enabled Turkic peoples both to assimilate the achievements of Eastern Islamic scholarship and to disseminate their own systems of thought across a broad geographical area. During the Ottoman Empire, this tradition resulted in the emergence of a complex yet effective linguistic system in public administration and diplomacy. This historical experience once again confirms that a foreign language is not merely a

means of communication, but also an indicator of cultural and political influence (Eker, 2019, p. 78).

The global changes that took place in the late nineteenth and early twentieth centuries reshaped the Turkic world's perception of foreign languages. The growing interest in European languages, particularly French and German, within the education system was connected with the efforts of Turkic societies to integrate into Western science and technology. During the Soviet period, although the dominance of the Russian language imposed certain restrictions on the study of other foreign languages, it simultaneously served as a medium that provided access to world literature through a common scholarly language. However, after gaining independence, Azerbaijan and other Turkic republics began to pursue independent language policies aligned with their national interests. Today, each of these states regards foreign language education as one of the most important priorities for securing a worthy place in the global world. Thus, the tradition of language acquisition rooted in historical experience has merged with the demands of the modern era, becoming an important factor that ensures the rise of the Turkic world as a new force in the international arena.

Materials and Methods

This study employs historical-comparative, descriptive, and analytical methods to examine the development of foreign language acquisition in the Turkic world and its contemporary implications. The research is based on academic literature, historical sources, educational policy studies, and statistical data related to foreign language education, digital transformation, and international relations. Historical-comparative analysis is used to examine the evolution of multilingualism among Turkic peoples from the Silk Road period through the Ottoman and Soviet periods to the post-Soviet era. Descriptive and analytical approaches are applied to assess current trends, including the integration of CEFR standards, digital technologies, artificial intelligence, and international certification programmes. Comparative analysis is also used to identify regional inequalities, technological challenges, and the relationship between foreign language potential and international competitiveness.

Discussion

Historical evolution: The linguistic heritage from the past.

The historical development of foreign language acquisition among Turkic peoples represents a centuries-long and complex evolutionary process. An analysis of this history is of fundamental importance for understanding contemporary trends in language learning as well as current methodological approaches. The Silk Road played a catalytic role in the linguistic development of the Turkic world. Turkic communities settled along this ancient trade route engaged in intensive communication with merchants, travellers, and diplomats speaking various languages due to the practical needs of trade and diplomacy. Archaeological findings, including coins and written documents, confirm the existence of a dynamic multilingual environment in these territories from early periods. This environment encouraged Turkic peoples to learn foreign languages for practical purposes even in ancient times.

During the periods of the Göktürk and Uyghur empires, foreign language knowledge became an integral component of state administration and diplomatic activity. The lexical units and terminology used in Göktürk written monuments clearly demonstrate the close relations established with the Chinese language and culture during that period (Tekin, 1968). In the Uyghur period, the adoption of Buddhism led to the intensive study of Sanskrit and Tocharian languages. Research

indicates that Uyghur texts contain numerous religious and philosophical terms borrowed from these languages (Müller and Gabain). Special attention was also given to the study of Chinese and Tibetan in order to maintain diplomatic relations (Sinor, 1990).

The geography of Azerbaijan was also a centre of rich linguistic diversity in ancient and early medieval times. During the period of Caucasian Albania, the use of Greek and Aramaic alongside the Albanian language, and later the spread of Pahlavi and Avestan during the Sasanian period, demonstrate the deep roots of the tradition of foreign language acquisition (Əliyarlı, 1996). With the spread of Islamic culture, Arabic acquired the status of both an official and scholarly language, while Persian assumed a dominant position in the literary and scientific environment (Bünyadov, 1989). During the Ottoman Empire, the concept of *Alsina-yi Salasa* — the “three languages” — emerged, based on the parallel use of Turkish, Arabic, and Persian. This further strengthened the role of the Turkic world as a linguistic bridge between Eastern and Western cultures. This historical heritage forms the foundation of the foreign language policy and international integration strategy of Turkic states in the modern era.

Language policy in the Soviet Period and the Post-Soviet space.

The political landscape of the twentieth century led to fundamental and systematic transformations in the language policy of the Turkic world and in its perception of foreign languages. In particular, the centralized language policy implemented during the Soviet period moved the linguistic environment of Turkic peoples in the region onto an entirely new plane. The absolute dominance of the Russian language under the status of the “language of interethnic communication” shaped not only administrative and official domains, but also individuals’ intellectual development trajectories and their perception of foreign language learning. During this period, Russian was not merely a means of communication; it also functioned as the only “mediating language” or “window” providing access to global scientific achievements, world classical literature, and the flow of international political information.

The process of foreign language acquisition among Turkic peoples was formed precisely through the filter of this mediating language. The scientific and cultural knowledge base available in Western languages, such as English, French, and German, was in most cases assimilated through translations and methodological interpretations carried out via Russian. This situation gave rise to an indirect methodological model in the process of foreign language learning (Eker, 2019, p. 142).

One of the characteristic features of the Soviet education system was that the teaching of foreign languages, particularly Western languages, was predominantly theoretical and philological in nature. Lessons conducted in isolation from live communication and the international environment pushed the practical application of foreign languages into the background, presenting them largely as “dead languages” or merely as tools for reading texts. By contrast, Russian continued to develop as a living and functional language in all spheres of life. This imbalance led to the one-sided development of the language potential of Turkic peoples. As a result, for several generations, Russian acquired the status of the main foreign language, or even a second native language.

At the same time, this historical experience also sharpened the ability of Turkic communities to master complex grammatical structures and different linguistic systems. On the eve of independence, the intellectual elite of the Turkic world already possessed a broad worldview acquired through Russian, which later became one of the factors accelerating the direct transition to global languages.

After gaining independence, the direct entry of Turkic states into the international arena made it inevitable to redefine language priorities within the framework of national interests. The “new language strategy” implemented in the post-Soviet space moved away from one-sided linguistic dependence and focused on strengthening the status of national languages as state languages, while also recognizing English as a globally significant means of communication. During this transition period, learning a foreign language was perceived not only as an academic activity, but also as a strategic step symbolizing the sovereignty of states and connecting them to the world economic system.

The priorities formed in the early years of independence gave a strong impetus to the removal of Russian-language textbooks from educational programmes, the adoption of Western educational standards, including IELTS and TOEFL criteria, and the expansion of international student exchange programmes (Bayraktaroğlu, 2019, p. 145).

In the contemporary post-Soviet space, Turkic states build their language policies on the concept of multilingualism. This new approach aims to preserve the purity of the mother tongue while ensuring that the younger generation acquires at least one global foreign language, primarily English, at a high level. At the same time, Russian, which still retains its functionality in the region, is no longer a dominant instrument of pressure; rather, it has become one of the languages used for regional cooperation and information exchange.

Such a multi-vector linguistic structure has significantly expanded the room for manoeuvre of the Turkic world in the international diplomatic and economic arena. Today, Turkic states demonstrate the ability to establish more effective and direct communication between different cultural poles — the West, the East, and the post-Soviet space. Thus, the transition from the one-sided linguistic influences of the Soviet period to the free and strategic language choices of the modern era has laid the foundation for the Turkic world to appear in global competition with a new and stronger identity.

Digital transformation and technological infrastructure

The strategic objectives set for the Turkic world to emerge with a new identity in global competition are realized not only through political will, but also through the digital transformation of the education system and the direct integration of modern technological infrastructure into the teaching process.

In the contemporary period, foreign language education is characterized not only by pedagogical methodologies, but also by the deep penetration of digital technologies into the teaching process and by the quality of technological infrastructure. In a globalized world, information technologies have moved the language learning environment beyond the physical boundaries of classrooms, transforming it into a more dynamic, interactive, and accessible platform. Among the factors that ensure the rise of the Turkic world in the international arena, the digital transformation of education and the establishment of a technological foundation aligned with modern standards stand out as strategic priorities. This transformation involves not merely the provision of computer equipment, but also the adaptation of educational content to the digital environment and the enhancement of learners’ technological literacy. At present, state programmes implemented in this direction are regarded as one of the most important mechanisms for increasing the global competitiveness of Turkic states.

One of the most ambitious objectives in the field of technological infrastructure renewal is to equip 95% of general secondary education institutions with modern language laboratories by 2025.

These laboratories provide students with opportunities to acquire practical skills in conditions close to a real language environment and to improve their listening, speaking, and pronunciation skills through the use of multimedia resources. Unlike traditional teaching methods, the implementation of digital laboratories enables a differentiated approach adapted to each student's individual learning pace and needs (Sebzecioğlu, 2021, p. 56). This, in turn, increases the effectiveness of instruction and creates favourable conditions for overcoming language barriers within a shorter period of time. The establishment of such large-scale technological infrastructure serves the formation of a unified digital educational space in the Turkic world and the alignment of human resource development with international standards.

In the process of digital transformation, online platforms based on artificial intelligence and cloud technologies are paving the way for revolutionary changes. Modern online platforms not only provide students with access to quality educational resources from anywhere in the world, but also create conditions for real-time interactive communication. AI-based language learning programmes analyse learners' strengths and weaknesses and design individualized development plans, which reduces the teacher's workload while raising the quality of instruction to a new level. The integration of such innovative tools into the education system enables young people in the Turkic world to connect more rapidly to global scientific and technological information flows. At the same time, these technologies expand opportunities for distance education and support the development of foreign language potential even in regions where traditional education is not easily accessible.

Nevertheless, the success of digital transformation depends not only on technical provision, but also directly on the development of teachers' digital competencies. In order to use modern language laboratories and artificial intelligence platforms effectively, teaching staff must possess the necessary skills to work with these technologies. For this reason, international certification and training programmes aimed at improving teachers' digital skills are being implemented in parallel with infrastructure projects (Bayraktaroğlu, 2019, p. 134). Only through the unity of technological provision and teacher training can foreign language potential become a genuine factor of power in the international arena. Thus, the establishment of digital infrastructure creates a fundamental basis for the Turkic world to achieve its strategic goals for 2030 and strengthens the region's position in the global information society.

Current Challenges and Problems

Although the large-scale implementation of digital transformation and technological infrastructure promises significant prospects for the Turkic world, the practical realization of this innovative process faces a number of systemic challenges and complex structural problems. The mismatch between the pace of technological progress and the actual capacities of educational environments is one of the fundamental factors hindering the equitable development of foreign language potential. In this regard, the digital divide emerges as one of the most pressing issues. Statistical data indicate that the lack of access to high-speed internet for approximately 30% of the population seriously limits the effectiveness of online education platforms and digital resources.

Mark Warschauer notes that the digital divide is not merely a matter of possessing technical equipment, but also concerns the availability of social and institutional support necessary for using that technology in a meaningful and productive way (Warschauer, 2003). Under conditions of limited internet access, even the most advanced artificial intelligence tools fail to produce the expected outcomes and may further deepen educational inequality within society.

On the other hand, the factor of regional inequality creates decisive differences in the quality of foreign language instruction. The sharp 40% gap observed in the quality of education between urban and rural schools stems from the weakness of the foreign language environment in the regions and the imbalance in resource provision. The most critical aspect of this imbalance is staffing; a shortage of foreign language teachers reaching up to 25% in regional schools leads either to interruptions in instruction or to teaching being carried out by non-specialist staff.

Teacher quality and its equitable distribution constitute one of the strongest school-related factors influencing student achievement, and shortcomings in this area represent a manifestation of social injustice within the education system (Darling-Hammond, 2010). The weakness of incentive mechanisms for retaining professional staff in rural areas creates a serious obstacle to the realization of foreign language potential on a nationwide scale.

Financing and resource management also remain among the complex challenges facing modern language policy. Although approximately 15% of the education budget is directed toward the development of foreign language programmes, the rapid increase in demand and the costs associated with renewing technological infrastructure require a more efficient allocation of these funds. In his economic analysis of language policy, François Grin emphasizes that the return on investment in language education depends directly on the extent to which the programmes offered correspond to the skills demanded by the labour market (Grin, 2003).

In the Turkic world, more flexible financing and governance models are needed to ensure that foreign language programmes meet global standards while also responding to local economic needs. Thus, addressing the existing problems requires not only an increase in funding, but also a comprehensive and integrated approach capable of bridging digital and regional divides.

The Impact of Foreign Language Potential on International Relations

The systematic elimination of existing structural and socio-economic difficulties transforms foreign language potential from a mere educational achievement into a strategic factor within the system of international relations. In contemporary global politics, the prestige and influence of states are no longer measured solely by military arsenal or natural resources, but also by society's communicative capacity and cultural attractiveness. Within this framework, language potential functions as one of the fundamental pillars of the concept of "soft power." For the states of the Turkic world, foreign language proficiency creates significant advantages for defending national interests more effectively in global diplomacy and for establishing direct dialogue with the international community. In particular, the dominant role of English in international relations is a key factor enabling Turkic-speaking societies to adopt a more competitive position on global scientific, political, and economic platforms.

The influence of language on international relations is not limited to the exchange of information; it also plays an important role in shaping a state's external image and "cultural brand." The promotion of Turkic values worldwide through cultural diplomacy contributes to the recognition of these states as a "centre of attraction" in the international arena. Multilingualism is considered an essential means of building trust between states and overcoming linguistic barriers in complex diplomatic negotiations (Walker, 2016, p. 12). Alongside common language projects among Turkic states, integration into global languages serves as an element of "smart power" that supports the rise of the region as a unified centre of influence. This approach makes the voice of the Turkic world more audible in international cooperation formats and creates opportunities to use foreign language potential as a diplomatic manoeuvring tool in matters of regional security.

The enhancement of foreign language potential also has a direct impact on the acceleration of transnational economic relations and foreign investment flows. In the international business environment, linguistic competence facilitates mutual understanding in relations with foreign partners and reduces transaction costs. In this context, language education also functions as an integral component of economic diplomacy, strengthening the position of Turkic states within global supply chains. Thus, the development of foreign language potential should be regarded as a national security and development priority that enables the Turkic world to participate in the global governance system as a more active, influential, and strategic actor.

Conclusion

A comprehensive analysis of the factors ensuring the strategic rise of the Turkic world in the international arena demonstrates that, in the current geopolitical context, a foreign language is not merely an educational indicator, but a fundamental resource located at the centre of states' national security and economic competitiveness goals. The findings of the study reveal that the multilingual tradition of Turkic peoples has followed a complex path of evolution since the historical period of the Silk Road and, in the post-independence era, has entered a new qualitative stage in accordance with the interests of national statehood. The transition from the one-sided linguistic influences of the Soviet period to direct integration into the global lingua franca and other international languages has enabled the Turkic world to participate in global information flows without mediation.

The statistical indicators and reforms examined in the article confirm the systematic nature of progress in this field. In particular, the transition to a communicative approach in foreign language teaching, which has increased learning effectiveness by 65%, and the objective of equipping 95% of educational institutions with modern language laboratories in the near future are clear examples of the importance Turkic states attach to digital transformation. The integration of artificial intelligence and online platforms into the teaching process makes language learning more inclusive and individualized, thereby creating a fundamental basis for preparing the younger generation as human resources aligned with international standards, including the Common European Framework of Reference for Languages.

Nevertheless, the regional inequalities identified in the course of the study — including the 40% quality gap between urban and rural schools and the shortage of qualified teaching staff — make it necessary to ensure a more equitable distribution of resources in future development strategies. Eliminating the digital divide and managing the education budget efficiently are decisive conditions for realizing foreign language potential across society as a whole. The successful resolution of these challenges will strengthen the soft power balance of the Turkic world in the global arena and expand its opportunities for cultural diplomacy.

In conclusion, it should be emphasized that the development of foreign languages is not merely a subcategory in the Turkic world's 2030 roadmap, but an integrative force that determines progress across all spheres. The formation of human capital with high linguistic competence will enable Turkic states to be represented with a stronger identity in the international scientific and technological environment, global trade chains, and diplomatic arena. Thus, the strategic management of foreign language potential should be regarded as one of the most important factors making the rise of the Turkic world in the international arena sustainable and irreversible.

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LANGUAGE INSTRUCTION CHALLENGES IN DEVELOPING NATIONS: A CASE STUDY OF AZERBAIJAN

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Abstract: The importance of bilingual proficiency is steadily increasing in a world shaped by international communication, mobility, and exchange, particularly for individuals who aspire to function as global citizens. However, foreign language-learning process still remains a complex and challenging process, especially in post-Soviet and developing nations, where historical, political, and infrastructural conditions affected the quality and accessibility of education. This paper examines the challenges faced by language learners in such educational contexts, using Azerbaijan as a representative case study. Despite important efforts by post-Soviet governments to modernize education and strengthen the role of foreign language instruction within national curricula, significant barriers continue to limit the effectiveness of these reforms.

The most significant barriers include restricted learning environments that limit real-time practice, an insufficient number of teachers equipped with modern pedagogical skills, and a lack of updated learning materials, especially audio and digital resources. These challenges do not only affect students' linguistic progress; they also shape their confidence, motivation, and willingness to use the language in practice. When

learners have little chance to speak, receive limited support, or are taught mainly through grammar-based methods, they may develop a fear of making mistakes and gradually avoid communication altogether.

Keywords: bilingual proficiency; foreign language learning; post-Soviet education; language education in Azerbaijan; educational reform; communicative competence; language-learning barriers; teaching materials

İNKİŞAF ETMƏKDƏ OLAN ÖLKƏLƏRDƏ DİL TƏDRİSİNİN ÇƏTİNLİKLƏRİ: AZƏRBAYCAN NÜMUNƏSİ

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Xülasə: Qlobal sahədə yer tutmaq istəyən fərdlər üçün ikidilli bilik və bacarıqların əhəmiyyəti beynəlxalq ünsiyyət, mobillik və mübadilə ilə formalaşan dünyada davamlı şəkildə artır. Bununla belə, xarici dil öyrənmə prosesi hələ də mürəkkəb və çətin bir proses olaraq qalır. Bu, xüsusilə postsovet və inkişaf etməkdə olan ölkələrdə daha aydın şəkildə müşahidə olunur, çünki tarixi, siyasi və infrastrukturla bağlı şərait təhsilin keyfiyyətinə və əlçatanlığına əhəmiyyətli təsir göstərmişdir. Bu məqalədə dil öyrənənlərin təhsil prosesləri zamanı üzləşdiyi çətinlikləri tədqiqat obyektinə kimi götürərək araşdırır, əsas tədqiqat nümunəsi kimi isə Azərbaycana bacarıyıq. Postsovet dövründə təhsilin modernləşdirilməsi və xarici dil tədrisinin milli kurikulumlarda rolunun gücləndirilməsi istiqamətində mühüm səylər göstərilərsə də, bu islahatların effektivliyini məhdudlaşdıran ciddi maneələr hələ də öz aktuallığını qorumaqdadır.

Ən mühüm maneələrə real praktikanı məhdudlaşdıran dar öyrənmə mühiti, müasir pedaqoji bacarıqlara malik müəllimlərin kifayət qədər olmaması və yenilənmiş tədris materiallarının, xüsusilə audio və rəqəmsal resursların çatışmazlığı daxildir. Bu çətinliklər yalnız tələbələrin dil inkişafına təsir etmir, həm də onların özünəinamını, motivasiyasını və dili praktikada istifadə etmək istəyini formalaşdırır. Öyrənənlərin danışmaq imkanı az olduqda, onlara məhdud dəstək göstərildikdə və ya əsasən qrammatikaya əsaslanan metodlarla tədris edildikdə, onlar səhv etmək qorxusu inkişaf etdirməklə yanaşı, tədrisən ünsiyyətdən tamamilə yayınmağa başlaya bilərlər.

Açar sözlər: ikidilli bilik və bacarıqlar; xarici dil öyrənmə; postsovet təhsili; Azərbaycanda dil təhsili; təhsil islahatı; kommunikativ səriştə; dil öyrənmə maneələri; tədris materialları; müəllim hazırlığı; öyrənənlərin motivasiyası

Introduction

In the modern era, many students who are aware of the advantages of bilingualism are eager to acquire proficiency in additional languages. A common belief is that individuals who speak more than one language gain access to broader opportunities, perspectives, and forms of intercultural engagement than those available to monolingual speakers. Research on bilingualism also suggests that the benefits of learning an additional language extend beyond communication itself. When learners begin to use another language for thinking and learning, they may develop cognitive, neurological, and psychological advantages, including greater mental flexibility, improved attention control, and increased confidence in intercultural contexts. These benefits are also relevant to discussions of fluid and crystallized intelligence, as language learning can support both problem-solving abilities and the accumulation of knowledge over time.

However, the successful development of bilingual proficiency depends not only on learners' motivation, but also on the educational culture in which they study. Within such contexts, established patterns of thought and behavior can become deeply embedded and may often remain unquestioned. Dörnyei and Murphey (2003, p. 36) discuss what they describe as the "norm of mediocrity," referring to a classroom culture in which excellence is socially discouraged. This tendency may be particularly visible in less developed educational systems, where high-achieving students may withdraw from active participation and limit themselves to completing routine classroom exercises. Against this background, this paper examines the challenges of foreign language learning in developing countries, focusing on isolation in foreign-language environments, the excessive emphasis on grammar at the expense of speaking, anxiety during oral production, and the shortage of adequate learning materials, with Azerbaijan serving as the primary case study.

Literature Review

In contemporary scholarship, bilingualism is widely treated as a fundamental characteristic of the modern individual. Language is regarded as an important cognitive tool through which students share and construct new knowledge (Derewianka & Jones, 2016; Schleppegrell, 2004). However, in certain countries, achieving genuine bilingual proficiency remains a pressing and unresolved issue. In several post-Soviet and Middle Eastern regions, learners contend with outdated methodologies and inadequately qualified teachers, and Azerbaijan offers a notable example. For most of the twentieth century, the country was under Soviet rule, during which citizens largely adhered to government directives, a condition that constrained educational autonomy and personal development (Abbasli & Subashi, 2020). In the post-independence period, the government initiated efforts to enhance global integration, including introducing additional foreign languages into school curricula. Nevertheless, challenges persist in effectively implementing second-language instruction.

Many post-Soviet nations continue to teach in or through the language of the prior dominant power, creating an uneven dynamic that affects the learning of students and, by extension, entire communities. Language is not neutral, and neither is language policy. As Schmidt (1998) argues, "language use and language policy come to symbolize a larger conflict between ethnolinguistic groups over their relative power positions within the political community" (p. 37). Examining language education policy thus offers an arena in which power structures can be analyzed, their influence on learning communities understood, and better practices identified.

Empirical work has explored how learner backgrounds influence the language acquisition process. Tyacke and Mendelsohn (1986) conducted a diary study of ten adults from different ethnic backgrounds studying English as a second language. Participants brought their diaries every other week to discussions with one of the researchers. The diaries revealed that learning styles and strategies varied significantly: lower-level students showed much greater dependence on the teacher and on the linguistic code, although one more advanced learner from Central Asia also focused on these aspects, suggesting that reliance on grammatical structure can persist even at higher proficiency levels (Tyacke & Mendelsohn, 1986, pp. 171–178). This finding is particularly relevant to the Azerbaijani context, where, as discussed below, grammar-focused instruction tends to dominate classrooms even at advanced levels.

Drawing on student observation and the limited literature available on Azerbaijan, three interrelated challenges consistently emerge: limited learning environments, unqualified teachers, and inadequate learning materials. These are explored in turn below.

1. Limited Learning Environments

The main challenge in acquiring a new language is not only determined by external factors such as classroom facilities or the physical setting in which learning takes place. Although these elements can influence the learning experience, what happens beyond the classroom is equally, and arguably even more, significant. One of the most pressing issues is the limited opportunity for practical language use in real situations. High school students in Azerbaijan often complain about the lack of fluent speakers with whom they can practice. This limitation in practice partners not only slows down the linguistic progress but also weakens learners' self-esteem, thereby reducing their motivation to engage in conversations in the target language.

Language learning requires the development of multiple skills, including listening, speaking, reading, and writing. However, a restricted learning environment often encourages the development of some skills while neglecting others, leading to uneven proficiency. In the Azerbaijani context, students' written and grammatical competence often develops more strongly than their spoken fluency. As a result, learners may appear capable on paper but remain hesitant and insecure in real conversational situations. This imbalance should not be understood merely as a matter of individual preference; rather, it reflects the broader structural conditions under which students learn English.

2. Unqualified Teachers

This issue is frequently underestimated, yet it remains as one of the most significant problems in language education. It is specifically difficult to address in communities where English proficiency is not widespread, as students and parents may lack the linguistic knowledge needed to distinguish between effective and ineffective teaching. In less developed educational contexts, teachers often face limited access to professional training, modern pedagogical methods, and practical resources that would help them integrate more communicative and innovative approaches into everyday lessons. In countries such as Azerbaijan some educators believe that speaking skills should be developed only after learners have sufficient foundation of grammar and vocabulary. However, this approach may prevent students from developing the confidence needed to speak fluently and naturally. To resolve this problem effectively, educational authorities should implement periodic assessments and provide ongoing professional training for teachers. During lessons, teachers should create a supportive environment in which students feel encouraged to speak, make mistakes, and learn from those mistakes without losing confidence. Equally important is the systematic gathering of feedback from students, which allows them to evaluate the quality of instruction and identify the areas where teachers need further support. The educational process should be centered on the cooperation between students and teachers, while also taking into account the needs of both. A comprehensive strategy is needed: one that combines regular evaluation, expert-led training, and the use of modern teaching materials and methods.

3. Inadequate Learning Materials

Materials used in language learning include a wide range of resources that support the language-learning process, such as textbooks, workbooks, audio recordings, video materials, digital platforms, graded readers, and authentic media, including newspapers, films, and podcasts. While textbooks remain necessary, they are far from sufficient on their own. Audio materials are especially important because they expose learners to the rhythm, intonation, pronunciation, and natural pace of the target language, aspects that printed materials alone cannot convey.

In Azerbaijan, several material-related shortcomings continue to affect the quality of language education. First, many textbooks currently in use are outdated and contain vocabulary, cultural references, and topics that no longer reflect contemporary English usage. Second, authentic audio

resources, such as dialogues recorded by native speakers, podcasts, and listening exercises based on current contexts, are often unavailable in classrooms, even when the necessary equipment exists. Third, access to digital platforms and language-learning applications remains uneven, depending largely on school funding and on whether individual teachers are willing and able to integrate such tools into their lessons. Finally, graded readers and supplementary literature appropriate to students' proficiency levels are scarce, leaving learners without the extensive reading exposure needed to support vocabulary development and reading fluency.

These material gaps reinforce the problems discussed in the previous sections. A teacher with limited professional training and no access to modern audio resources is unlikely to deliver communicative, listening-rich lessons, regardless of their intentions. Therefore, addressing inadequate learning materials cannot be separated from broader investment in curriculum modernization, teacher support, and the systematic provision of updated educational resources.

Challenges Faced by Teachers

Changing existing instructional practices is a gradual, difficult, and complex process for teachers (Guskey, 2002; Richardson & Placier, 2001). From the teacher's perspective, several issues recur. The most prominent is the perceived lack of student motivation. Teachers often complain about distractions, particularly the use of mobile technology during lessons, and some respond by restricting or avoiding visual and internet-based resources altogether. While this reaction is understandable, it can be counterproductive: refusing to incorporate digital materials deprives students of access to one of the most valuable resources in modern language learning, namely authentic input from native-speaker sources. Therefore, the challenge for teachers is not merely to control the use of technology, but to employ it pedagogically, a skill that requires targeted professional training.

The Drawbacks of Recent Reforms

Alongside genuine progress in educational policy, it is important to acknowledge the unintended consequences that have accompanied these reforms. In some cases, the increased emphasis on testing and evaluation has produced mental strain and a sense of isolation among students. This is partly attributable to a shift in some teachers' demeanor: under pressure to deliver measurable outcomes, certain educators have adopted more rigid and demanding instructional styles.

The negative effects of these shifts are most visible where existing teaching methods are already weak. Students facing a difficult classroom environment often experience a loss of motivation. Their willingness to participate in open discussion declines, and they fall back on rote memorization of vocabulary and grammar. Even learners who reach a reasonable level of English encounter a further problem: in real conversational settings, they may develop a strong fear of making lexical or grammatical errors. This fear leads to self-imposed silence around native or fluent speakers, withdrawing from precisely the interactions they need to improve. The result is a kind of educational scarring – learners who possess the formal knowledge of a language but who have been conditioned to avoid using it. A balanced view of recent reforms must therefore weigh these unintended consequences alongside their positive contributions.

Discussion

Steps Taken to Address These Challenges

In the contemporary global landscape, feedback mechanisms have become a cornerstone of educational improvement. Recognizing this, many institutions in Azerbaijan have introduced

systematic surveys at the conclusion of each academic semester, soliciting feedback from students on their learning experience. Schools have additionally implemented diagnostic examinations administered to teachers, allowing administrators to assess both the quality of instruction and the preparedness of educators.

The value of these evaluative practices lies in their capacity to assess provision within specific subject domains and to focus attention on student outcomes. The information generated should, in principle, serve as a foundation for further reform, including the adoption of new pedagogical methods. However, it is worth asking whether these mechanisms have produced measurable change. To date, publicly available evidence on the impact of these feedback systems on classroom practice in Azerbaijan is limited, and student reports suggest that survey results often do not translate into observable changes in instruction. Feedback collection alone is insufficient; what matters is whether and how the information is acted upon.

Government Action

In 2008, the Azerbaijani government took steps to introduce elements of student-centered education, including the adoption of the International Baccalaureate (IB) curriculum in select schools. Under this model, the focus shifts from teacher-centered to student-centered learning, with students taking a more active role in their own development and recognizing the importance of self-study. Teachers, in turn, function as facilitators and motivators rather than as the sole source of knowledge.

This initiative represents a meaningful structural change, but its reach has been limited. The IB curriculum operates in a relatively small number of schools, and most students continue to study within the older national framework. A more substantive impact on language education would require extending student-centered principles – communicative practice, learner autonomy, formative assessment across the broader public school system, supported by the teacher training and updated materials discussed earlier.

Conclusion

The challenges facing foreign language education in developing countries, particularly the shortage of qualified teachers, restricted learning environments, inadequate materials, and waning student motivation are undeniable, and they are sharply visible in Azerbaijan. Overcoming these obstacles requires coordinated effort among government bodies, educators, students, and other stakeholders. Regular feedback collection should be prioritized, but it must be paired with genuine willingness to act on what is learned. Equally important is sustained professional development for educators, so that their methods keep pace with contemporary expectations. Guiding students in learning to use subject-specific language requires deep understanding of both subject matter and subject-specific ways of language use (Fang, 2014; Vazquez & Ellison, 2013), and this capacity rests on practical, situated knowledge that develops over time (Prediger, 2019; Swart, De Graaf, Onstenk, & Knezic, 2018). Throughout history, every population has faced its own obstacles in education; what distinguishes successful systems is their willingness to adapt to new technologies and to support a teaching workforce capable of guiding students through that change.

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LANGUAGE PLANNING IN THE TURKIC WORLD: NATIONAL IDENTITY, INTEGRATION, AND DIGITAL TRANSFORMATION

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Abstract: The article examines the issues of language policy and language planning in the Turkic world in a comprehensive manner within the context of national identity, integration, and digital transformation. The author demonstrates that in the era of globalization language functions not merely as a means of communication, but also as one of the key factors determining the geopolitical position of states, their cultural development strategies, and their information sovereignty. In the post-imperial space, the consolidation of the constitutional status of state languages is interpreted as a reconstruction of national identity, a restoration of collective memory, and a symbolic expression of historical justice. The use of

language in public domains, as well as its dominant position in the education and media systems, is presented as a crucial mechanism for strengthening national identity at the everyday level. The study systematically analyses language planning along the dimensions of status, corpus, acquisition, and prestige, linking these to the contemporary practices of the Turkic states. The codification of literary language norms, the formation of a terminological base, the expansion of scholarly literature in the state language in higher education, and the development of an academic communication environment are evaluated as the principal factors that enhance the functional capacity of the language. The idea of a common Turkic language is substantiated as a strategic direction for the creation of a shared space of scientific communication, the strengthening of intellectual integration, and the enhancement of the competitiveness of the Turkic world within the global scientific system.

Keywords: language policy, national identity, Turkic world, digital transformation, common Turkic language, scientific communication

TÜRK DÜNYASINDA DİL SİYASƏTİ (PLANLAŞDIRILMASI): MİLLİ KİMLİK, İNTEQRASIYA VƏ RƏQƏMSAL TRANSFORMASIYA

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Xülasə: Məqalədə türk dünyasında dil siyasəti və dil planlaşdırılması məsələləri milli kimlik, integrasiya və rəqəmsal transformasiya kontekstində kompleks şəkildə araşdırılır. Müəllif qloballaşma dövründə dilin yalnız ünsiyyət vasitəsi kimi deyil, həm də dövlətlərin geosiyasi mövqeyini, mədəni inkişaf strategiyasını və informasiya suverenliyini müəyyən edən əsas amillərdən biri kimi çıxış etdiyini göstərir. Post-imperiya məkanında dövlət dillərinin konstitusion statusunun möhkəmləndirilməsi milli identikliyin rekonstruksiyası, kollektiv yaddaşın bərpası və tarixi ədalətin simvolik ifadəsi kimi şərh olunur. Dilin ictimai məkanlarda istifadəsi, təhsil və media sistemində dominant mövqeyi milli kimliyin gündəlik səviyyədə möhkəmlənməsinə xidmət edən mühüm mexanizm kimi təqdim edilir. Tədqiqatda dil planlaşdırılması status, korpus, əldəetmə və prestij istiqamətləri üzrə sistemli şəkildə təhlil olunur və türk dövlətlərinin müasir təcrübəsi ilə əlaqələndirilir. Ədəbi dil normalarının kodifikasiyası, terminoloji bazanın formalaşdırılması, ali təhsildə dövlət dilində elmi ədəbiyyatın artırılması və akademik ünsiyyət mühitinin inkişafı dilin funksional imkanlarını genişləndirən əsas amillər kimi qiymətləndirilir. Ortaq türk dili ideyası isə ortaqlıq elmi kommunikasiya məkanının yaradılması, intellektual integrasiyanın gücləndirilməsi və türk dünyasının global elmi sistemdə rəqabət qabiliyyətinin artırılması baxımından strateji istiqamət kimi əsaslandırılır.

Açar sözlər: dil siyasəti, milli kimlik, türk dünyası, rəqəmsal transformasiya, ortaqlıq türk dili, elmi kommunikasiya

Introduction

In the 21st century, language policy has become not only a crucial component of states' cultural development strategies but also one of the key factors determining their geopolitical positioning. The processes of globalization and the expansion of digital communication environments have increased the functional load of language, directly linking it to national security, information sovereignty, and cultural diplomacy. As noted by Spolsky, language policy is not limited to official normative documents; rather, it constitutes a complex system shaped by the interaction of language practices and language ideologies (Spolsky, 2004).

In the Turkic world, language policy has evolved at the intersection of post-imperial transformation, nation-state building, and processes of international integration. Alphabet reforms and the standardization of the literary language in this context have functioned not merely as linguistic changes but also as civilizational choices. The use of language in public spaces is considered one of the principal indicators of its political status (Blommaert, 2010).

The relationship between language and national identity constitutes one of the central areas of inquiry in modern sociolinguistics, political linguistics, and cultural studies. Language is not merely a communicative tool; it is also a symbolic system that preserves collective memory, transmits historical experience, and delineates cultural boundaries. In this regard, language plays both an integrative and a differentiating role in the formation of national identity. B. Anderson's concept of "imagined communities" demonstrates that the idea of the nation is closely linked to the formation of a shared communicative space, the foundation of which is a standardized literary language (Anderson, 2006, p. 44). The establishment of mass media, educational systems, and administrative governance on the basis of a unified language is regarded as one of the principal mechanisms ensuring the consolidation of national consciousness. Contemporary research emphasizes that the relationship between language and identity is not static but dynamic in nature. J. Blommaert notes that under conditions of globalization, language functions as a resource for social mobility, with individuals constructing different identity models across diverse communicative spaces (Blommaert, 2010, p. 102).

This perspective indicates that the national language is not solely a marker of ethnic belonging but is also associated with social status, level of education, and cultural capital. From this standpoint, the expansion of the functional capacities of the state language contributes to the consolidation of national identity in practical terms as well.

In post-imperial and post-Soviet contexts, language policy functions as a significant component of the process of reconstructing national identity. According to A. Pavlenko, the restoration of a language or the elevation of its status is perceived as the reconstruction of collective memory and the symbolic reestablishment of historical justice (Pavlenko, 2006, p. 78). In this process, language operates not merely as a means of communication but also as an ideological and political choice. In Azerbaijan, Kazakhstan, and other Turkic states, the consolidation of the constitutional status of the state language and its implementation across all spheres of public life has become one of the central directions of the national identity discourse.

One of the important aspects of the relationship between language and national identity is the visual representation of language in public spaces. According to the "linguistic landscape" theory developed by R. Landry and R. Bourhis, the visibility of language in urban space serves as an indicator of its social prestige and political status (Landry, 1997, p. 23). The dominant presence of the state language in advertising, toponymy, official correspondence, and the education system ensures the manifestation of national identity at the level of everyday life. From this perspective, language policy is not confined to normative documents alone but also plays a crucial role in the organization of symbolic space.

Methodology and Materials

Under conditions of globalization, the relationship between language and national identity acquires new dimensions. On the one hand, the influence of international communication languages intensifies; on the other hand, the preservation and development of national languages emerge as a fundamental condition for cultural diversity. The dominance of certain languages in scientific and

technological domains may constrain the functional capacities of others. This, in turn, necessitates the development of national languages not only in cultural contexts but also within scientific and technological environments. Language planning, as one of the most systematic and applied directions of modern sociolinguistics, is directly linked to the regulation of social structures, the optimization of communicative relations, and the implementation of national identity policies. While classical approaches primarily conceptualize language planning in terms of corpus and status planning, subsequent research has expanded this model by incorporating the components of acquisition planning and prestige planning. R. Kaplan and R. Baldauf characterize language planning as a mechanism of purposeful social change, emphasizing that it evolves in interaction with the political system, educational policy, and ideological discourse (Kaplan, 2003, p. 34). This perspective demonstrates that language planning is not merely a process of linguistic standardization but also an instrument of social engineering.

Status planning encompasses the determination of the legal and institutional position of a language. In Turkic states, the acquisition of constitutional status by state languages following independence has constituted a key stage in this process. The expansion of the use of the state language in official administration, the judicial system, education, and the media has not only enhanced its communicative function but also strengthened its symbolic capital. In this regard, ensuring a balance between the legal status of a language and its actual use is considered one of the principal indicators of the effectiveness of language policy.

Corpus planning encompasses such dimensions as the codification of literary language norms, the standardization of orthographic and orthoepic rules, the formation of a terminological system, and lexical modernization. In the Turkic world, this process has been carried out particularly in parallel with alphabet reforms. The transformation of the graphic system is not merely a technical matter; it is also associated with the reconfiguration of the phonological structure of the language, the expansion of international communicative capacities, and the determination of cultural orientation. The functioning of the writing system as an ideological component of language policy demonstrates that this field possesses not only a philological but also a political character (Sebba, 2007, p. 67).

Acquisition planning entails the assimilation of the state language through the education system and its intergenerational transmission. In this regard, the preparation of textbooks, the expansion of scientific literature in the state language within higher education, and the formation of an academic communicative environment are of particular importance. The functionality of the state language not only in everyday and administrative domains but also within scientific and technological discourse constitutes one of the key factors determining its developmental prospects. U. Ammon's observations on the global hierarchy of scientific languages indicate that languages with limited representation in scientific communication are likely to encounter functional constraints in the long term (Ammon, 2011).

One of the distinctive features of language planning in the Turkic world is its simultaneous implementation at both national and transnational levels. While the consolidation of the state language remains a priority at the national level, at the pan-Turkic scale the enhancement of mutual intelligibility and the creation of a shared terminological base become increasingly salient. In this process, the development of a common alphabet, unified scientific terminology, and mutually recognized academic standards plays a significant role. At the same time, the issue of institutional coordination in language planning across the Turkic world persists. The implementation of language

reforms at different times and according to different models in various states slows down the formation of a common communicative space. In this regard, the establishment of unified scientific centers, joint terminological commissions, and digital language platforms is considered necessary. At the present stage, language planning is carried out not only through normative documents but also via technological infrastructures, indicating its transition to a new phase.

Digitalization has introduced a new dimension to the process of language planning. The creation of electronic corpora, the development of automatic translation systems, and the establishment of terminological databases represent contemporary forms of corpus planning. The availability of digital resources is one of the fundamental conditions ensuring a language's participation in the global communication system. The representation of Turkic languages in digital technologies has become one of the priority directions of language planning, emerging in the Turkic world as a multi-level and multidimensional process. The effectiveness of this process depends on the proper coordination between legal status, scientific and technological development, educational policy, and international cooperation. Language planning should be regarded not merely as the standardization of the literary language but as a strategic field of activity aimed at the formation of a shared cultural and scientific space.

Discussions

At present, scientific communication in the Turkic world is largely conducted through foreign languages. While this facilitates integration into the global scientific environment, it simultaneously weakens the formation of a common intellectual space. The development of scientific terminology along different national models complicates mutual understanding and hinders the emergence of a shared academic discourse. In this regard, the creation of a common terminological base is not only a matter of linguistic standardization but also a necessary condition for the synchronized development of scientific knowledge. The absence of terminological unity leads to the expression of identical concepts in different forms, resulting in semantic fragmentation within the process of scientific communication. The issue of a common Turkic language cannot be resolved solely through structural linguistic proximity. Contemporary sociolinguistic approaches demonstrate that the existence of institutional and technological infrastructure is a fundamental prerequisite for the formation of a shared communicative space. Academic mobility programs, joint scientific journals, unified citation databases, and mutually recognized academic standards constitute the core components of this process. In our view, a common Turkic language could function as an alternative global model of scientific discourse and contribute to the intensification of knowledge exchange among Turkic states.

Another important issue is the formation of a shared scientific style. Terminological and methodological differences among various national scientific schools complicate the emergence of a unified communicative model. To overcome these differences, the development of reciprocal translation systems, the establishment of common academic writing standards, and the implementation of joint scientific projects are of particular importance. The construction of scientific communication not only through language but also on the basis of shared methodological approaches enables deeper integration. The digital environment opens new opportunities for the realization of the idea of a common Turkic language. Electronic libraries, open-access scientific platforms, shared terminological databases, and automatic translation systems serve as key instruments facilitating mutual understanding. As Kornai notes, the availability of digital resources is one of the principal factors determining the functional vitality of a language (Kornai, 2013). The

creation of shared digital corpora for Turkic languages and ensuring mutual accessibility of scientific texts are essential for the formation of a common communicative space.

One of the key aspects of the idea of a common Turkic language is its inherently polycentric character. This model envisages not the imposition of a single unified standard, but rather the creation of an academic communicative system that is mutually intelligible and develops in parallel across different contexts. In contemporary approaches to language policy, this model is explained through the concept of a “pluricentric language,” which recognizes the mutual legitimacy of variants formed in different centers. Such an approach allows for the preservation of the national developmental specificities of each Turkic language while simultaneously enabling the formation of a shared communicative space.

Digital transformation constitutes one of the principal factors fundamentally reshaping the priorities of contemporary language policy. The rapid development of information and communication technologies necessitates that language functions not only as a means of communication but also as a core component of technological systems. Artificial intelligence, big data, automatic translation systems, and speech recognition technologies expand the functional capacities of language while transforming its level of representation in the digital environment into a strategic indicator. From this perspective, the presence of a language within the digital ecosystem is regarded as one of the key factors determining its future social status and communicative potential. Languages lacking digital resources face functional risks, a process often described as “digital language loss” (Kornai, 2013).

Digital transformation is also reshaping the model of scientific communication. Open-access scientific platforms, electronic libraries, and international citation databases have become the principal channels for the production and dissemination of scientific knowledge. In this process, the accessibility of a language within the digital environment determines its participation in scientific discourse. As U. Ammon notes, dominant languages in scientific communication exert a direct influence on the terminological development and academic visibility of other languages (Ammon, 2011, p. 56). In this regard, the digital publication of high-quality scientific texts in Turkic languages and their dissemination through open scientific platforms are of strategic importance.

Conclusion

In conclusion, digital transformation emerges as a decisive factor shaping the future of language. The functionality of a language within the technological environment strengthens its position across social, scientific, and cultural domains. The development of digital language resources, the representation of Turkic languages in artificial intelligence technologies, and the expansion of the use of national languages on open scientific platforms should be regarded as priority directions of contemporary language policy. This process contributes to the formation of a common communicative space within the Turkic world and enhances its competitiveness within the global scientific system.

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ENGLISH LANGUAGE PLANNING AND POLICY IN RUSSIA: A SCOPING REVIEW OF ESL/EFL CURRICULUM RESEARCH

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Abstract: This scoping review explores the current state of English Language Planning and Policy (ELPP) in Russian education. It analyzes initiatives, strategies, and policies aimed at enhancing English as a foreign language (EFL) teaching, drawing on 25 peer-reviewed studies published between 2005 and 2022. To identify studies, systematic searches were conducted in EBSCO, Google Scholar, ResearchGate, and Litmaps using the five-stage scoping review framework of Arksey and O'Malley (2005). Our findings indicate that the reviewed literature is predominantly qualitative (n=22, 88%), relies heavily on document analysis (n=23, 92%), and has seen increased publication activity between 2018 and 2022. Despite significant progress in formalizing English-language instruction since the 2012 Federal Education Law, critical gaps remain in the use of technology, integration of the affective domain, teacher professional development, and standardized testing. In addition to highlighting future research priorities, this review offers several evidence-based suggestions for curriculum reform.

Keywords: English language planning and policy, curriculum, Russia, ESL/EFL education, language policy, scoping review

RUSİYADA İNGİLİS DİLİNİN PLANLANMASI VƏ SİYASƏTİ: ESL/EFL MÜƏLLİM PROQRAMI ARAŞDIRMALARINA DAİR KAPSAMLI BİR İCMAL

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Özet: Bu kapsam belirleme incelemesi, Rusya'daki eğitim sisteminde İngilizce Dil Planlaması ve Politikası'nın (ELPP) güncel durumunu ele almaktadır. Çalışma, 2005 ile 2022 yılları arasında yayımlanan 25 hakemli araştırmaya dayanarak, yabancı dil olarak İngilizce (EFL) öğretimini geliştirmeyi hedefleyen girişimleri, stratejileri ve politikaları incelemektedir. Araştırmaları belirlemek amacıyla, Arksey ve O'Malley'nin (2005) beş aşamalı kapsamlı inceleme modeli kullanılarak EBSCO, Google Scholar, ResearchGate ve Litmaps üzerinde sistematik aramalar yapılmıştır. Bulgularımız, incelenen literatürün büyük ölçüde nitel (n=22, %88) olduğunu, çoğunlukla belge analizine dayandığını (n=23, %92) ve 2018 ile 2022 yılları arasında yayın sayısında artış olduğunu göstermektedir. 2012 Federal Eğitim Yasası'ndan bu yana İngilizce öğretiminin resmileştirilmesinde önemli ilerlemeler kaydedilmesine rağmen, teknoloji kullanımı, duyuşsal alanın entegrasyonu, öğretmenlerin mesleki gelişimi ve standartlaştırılmış testler konularında hâlâ önemli eksiklikler bulunmaktadır. Bu inceleme, gelecekteki araştırma alanlarını belirlemenin yanı sıra, müfredat reformu için çeşitli kanıta dayalı öneriler sunmaktadır.

Anahtar Kelimeler: İngilizce dil planlaması ve politikası, müfredat, Rusya, ESL/EFL eğitimi, dil politikası, kapsamlı inceleme (scoping review).

Introduction

Human society depends on language, which has led to the development of language planning as a cross-disciplinary area involving linguistics, education, and public policy. Haugen (1959) originally defined it as the creation of standards to guide speakers in multilingual communities, while Cooper (1989) emphasized its social aspect. Blommaert (1996) offers a critical perspective, noting that politics always influences language planning and often reflects ideological motives beyond linguistic concerns.

English has become a worldwide phenomenon, marking one of the most significant linguistic shifts in recent years. Today, it is used across more than 100 countries for teaching, governance, and business activities (Schneider, 2011). This extensive global presence is influenced by globalization, which tends to favor certain linguistic groups. Russia exemplifies this dynamic. After the Soviet Union's dissolution in 1991, Russia faced the challenge of redefining its linguistic stance on the global stage, striving to uphold Russian as the official language while respecting the language rights of minority communities.

This scoping review addresses two research questions;

RQ1: *What is the current status of English Language Planning and Policy (ELPP) in Russia's educational curriculum?*

RQ2: *What gaps or areas require further research in the field of language planning and policy in Russian ESL/EFL education?*

Russia's Education System and Historical Context. Education in the Soviet Union was centralized and ideologically homogeneous. Following the Federal Law on Education (1992), curricula were expanded, and student-centered methodologies were adopted. English has been a compulsory subject in the national curriculum since Grade 2, and the Unified State Exam (USE) serves as a gateway to higher education under the 2012 Law on Education in the Russian Federation. According to Krouglov and Krouglov (2022), foreign-language education in Russia has occurred in four historical periods: (I) Baptism to Peter I is reforms, when languages served

religious purposes; (II) Peter I through the October Revolution, when Westernization was evident and French and German were viewed as powerful; (III) the Soviet period, characterized by ideological isolation; and (IV) the post-Soviet period (1991–present), during which English has become the dominant foreign language in Russian schools and universities.

English as a Medium of Instruction (EMI). In international programs, English is increasingly used as the medium of instruction (Alenkina, 2020; Belyaeva & Kuznetsova, 2019). However, research repeatedly identifies a gap between policy aspirations and classroom reality. National curriculum frameworks offer sample topics rather than prescribed content and are less than 24 pages long. According to Slavina and Khairova (2016), these frameworks focus primarily on cognitive and psychomotor goals, while neglecting affective development and motivation. In an EMI context, Belyaeva and Kuznetsova (2019) and Murtazina et al. (2019) found that content lecturers lacked sufficient pedagogical preparation to teach English to non-native speakers.

Materials and Methods

This study used a scoping review methodology based on Arksey and O'Malley's (2005) framework, which involved: (1) defining research questions, (2) locating relevant studies, (3) selecting appropriate studies, (4) charting the data, and (5) synthesizing and reporting the results. The reporting process followed the PRISMA-ScR checklist (Tricco et al., 2018). Additionally, Bowen (2009) employed a descriptive document analysis approach within this framework.

A comprehensive search was performed across EBSCO, Google Scholar, ResearchGate, and Litmaps using keywords such as "English language planning," "language policy Russia," "EFL Russia," "English as a medium of instruction Russia," and "curriculum Russia." Studies included were those focusing on language planning or English-language curricula within the Russian Federation, written in English, and accessible in full text. Duplicate records were eliminated using Mendeley and EndNote, resulting in 42 unique entries. Of these, 25 studies fulfilled all inclusion criteria. The PRISMA flow diagram in Figure 1 illustrates the process.

Figure 1. PRISMA Flow Diagram of Study Selection Process

Stage	<i>n</i>
Identification – Records from EBSCO	20
Identification – Records from Google Scholar	10
Identification – Records from ResearchGate	10
Identification – Records from Litmaps	7
Total records identified	47
Records after duplicate removal	42
Records excluded after title screening	7
Records screened by abstract	35
Records excluded after abstract screening	5
Records screened by full text	30
Records excluded after full-text review	5
Studies included in the final review	25

Note. Search conducted across EBSCO, Google Scholar, ResearchGate, and Litmaps. Screening followed PRISMA guidelines (Page et al., 2021).

Standardized charting forms were utilized to extract data, recording details such as publication date, keywords, research method, data collection tools, data analysis approach, and sample type. Bibliometric and methodological frequency tables described the distribution of articles. A content analysis highlighted recurring themes and grouped them into broader categories.

Results and Discussion

Distribution of Articles by Year. As shown in Table 1, publication frequency varies from year to year. The number of articles published through 2016 was sparse (one to three articles per year) before increasing between 2018 and 2022, with each of those years accounting for 12 of the 25 articles (48%). Following Russia's internationalization initiatives in higher education in the mid-2010s, scholarly interest in EMI and language policy intensified.

Table 1. Distribution of Articles by Publication Year

Year	Number of Articles
2005	1
2006	1
2011	1
2012	2
2015	1
2016	3
2018	3
2019	2
2020	2
2021	2
2022	3
Total	25

Note. Distribution of articles by publication year. Years not listed had no articles that met the inclusion criteria.

Note. Years not listed had no articles that met the inclusion criteria.

Distribution of Articles by Keyword. The keyword frequencies from the 25 articles are presented in Table 2. A total of 63 distinct keywords are identified in Table 2. The most frequently used keyword was "language policy" (f=19), followed by "Russia" (f=12) and "language planning" (f=7). Because "EMI & Curriculum" (f=3) and other macro-level policy terms dominate the literature, the focus may be on classroom implementation rather than macro-level policy issues.

Table 2. Keyword Frequency Distribution

Keyword	Frequency (f)
language policy	19
Russia	12
language planning	7
language in education	6
minority languages	4
EMI & curriculum	3
language of instruction	3
linguistic diversity	3
multilingualism	2
language management	2
language problem	2

Note. Only keywords appearing in two or more articles are listed.

Distribution of Articles by Research Method. In Table 3, qualitative methods accounted for 88 percent of the reviewed literature, while quantitative studies accounted for only 12% (n=3). In this study, no mixed-methods studies were identified. Although this field's orientation toward policy analysis requires reliance on qualitative approaches, this methodological homogeneity limits the generalizability of the results.

Table 3. Distribution of Research Methods

Research Method	Frequency (f)	Percentage (%)
Qualitative	22	88%
Quantitative	3	12%
Mixed Methods	0	0%
Total	25	100%

Note. Distribution of research methods across included articles.

Distribution of Articles by Data-Collection Tool and Analysis Technique

In Tables 4 and 5, document analysis dominates (n = 23; 92%), and non-parametric analysis is characterized by 23 of 25 articles (92%). No studies used classroom observations or interview protocols as the primary methods of data collection. Table 6 shows that 23 articles (92%) relied on policy documents and existing literature rather than empirical samples. As a result, teachers, students, and administrators are not represented in the literature.

Table 4. Distribution of Data-Collection Tools

Paradigm	Data-Collection Tool	Frequency (f)	Percentage (%)
Qualitative	Document review	23	92%
Quantitative	Questionnaire	2	8%
Total		25	100%

Note. Distribution of data-collection tools across included articles.

Table 5. Distribution of Data-Analysis Techniques

Analysis Technique	Frequency (f)	Percentage (%)
Non-parametric (thematic/content/discourse analysis)	23	92%
Parametric (statistical analysis)	2	8%
Total	25	100%

Note. Distribution of data-analysis techniques across included articles.

Table 6. Distribution of Articles by Sample Use

Sample Use	Frequency (f)	Percentage (%)
No empirical sample (document-based)	23	92%

Empirical sample used	2	8%
Total	25	100%

Note. Distribution of articles by sample use.

Key Thematic Findings

Four cross-cutting themes emerged from the content analysis.

Professional development for teachers is essential: The literature review shows that implementing ELPP is both crucial and often overlooked in this domain. A recent study by Atabekova et al. (2016) reveals that Russian university lecturers teaching English frequently lack formal EMI pedagogical training. Effective language proficiency, EMI strategies, and formative assessment should be incorporated into comprehensive, ongoing programs.

Measurement and assessment: The national curriculum specifies several competencies, but provides limited guidance on tracking progress toward these goals. Consequently, numerous studies (Razumovskaya & Sokolovsky, 2012; Krouglov & Krouglov, 2022) suggest explicitly aligning with the Common European Framework of Reference for Languages (CEFR) to facilitate benchmarking and international comparisons.

Affective and motivational dimensions: The curriculum lacks sufficient focus on motivation. Russian students tend to be motivated to learn English primarily for instrumental reasons, with little development of intrinsic or integrative motivation. This gap can be addressed by integrating task-based and project-based learning approaches.

Regional and Minority Language Dynamics: Russia's linguistic diversity, with over 100 recognized languages, creates complex dynamics for the ELPP. There are regional variations in language policies across federal subjects; some promote multilingual education, while others prioritize Russian-medium instruction and marginalize minority languages. According to Yanush (2021) and Fedyunina (2013), language policy consistency varies across these regions.

Conclusion

This review analyzed 25 studies published from 2005 to 2022 concerning English Language Planning and Policy in Russia. It found that the Russian education system has made significant but uneven advances in integrating English-language education. Following the 2012 Federal Education Law, English teaching from elementary to university levels became a formal subject. Over the next decade, research on EMI and language policies increased. However, the national curriculum has been criticized for lacking detailed guidance for classroom teaching and for not emphasizing affective and motivational aspects sufficiently. Additionally, teacher professional development is structurally lacking, and assessment frameworks remain too vague. Based on the methodological profile of the reviewed literature -- 88% qualitative, 92% document-based, and 92% without empirical samples -- primary empirical research is urgently needed at the institutional and classroom levels. Future studies should employ mixed methods, include human-participant samples, and use longitudinal or quasi-experimental designs to build a more robust evidence base for curriculum reform. Six recommendations are offered:

(1) Develop EFL teacher and EMI lecturer professional development systems through continuous, federally funded programs.

(2) Align curricular standards with the CEFR for leveled instructional benchmarks and cross-border comparison.

(3) Implement task and project-based instructional models with affective and motivational learning objectives.

(4) Create standardized assessment frameworks aligned with curricular learning objectives at each educational level.

(5) Integrate technology, including language learning platforms, digital resources, and virtual exchanges.

(6) Develop primary empirical research that prioritizes mixed-methods and experimental studies with human participants.

Despite these limitations, the search was limited to English-language studies through 2022, the evidence base is relatively small (n=25), and no formal quality assessment of individual studies was conducted as part of this review – a limitation inherent to scoping reviews.

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THE ROLE OF NATURAL AND CULTURAL HERITAGE IN TOURISM INTEGRATION AND THE ROLE OF FOREIGN STUDENTS' EDUCATION IN STATE UNIVERSITIES

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Abstract: The integration of natural and cultural heritage in tourism plays an important role in the formation of a sustainable development model in the conditions of global competition, and this process develops in close interaction with the internationalization of higher education. At the modern stage, universities have become leading platforms for both knowledge transfer and intercultural dialogue, social integration and the formation of innovative tourism strategies. State universities where international students study act as an effective tool for the global introduction, protection and promotion of natural and cultural heritage. The main purpose of the presented study is to comprehensively investigate the mechanisms of interaction between the integration of natural and cultural heritage in tourism and the education of foreign students at state universities. The study analyzes the formation of heritage-based tourism models, the expansion of academic mobility, the integration of cultural diversity into the educational process and the impact of these processes on regional development indicators. The concepts of natural and cultural heritage, in addition to being an integral part of the historical development, socio-cultural experience and ecological environment of each nation, also constitute the main resource base for the protection of national identity and its presentation at the international level. The effective use of this heritage through tourism contributes to both economic benefits and the formation of multicultural values in the educational environment, the expansion of students' global perspectives, and the realization of sustainable development goals.

Key words: cultural heritage, educational tourism, tourism integration, multicultural integration, "soft power".

TURİZMƏ İNTEQRASIYADA TƏBİİ VƏ MƏDƏNİ İRSİN VƏ DÖVLƏT UNİVERSİTETLƏRİNDƏ ƏCNƏBİ TƏLƏBƏLƏRİN TƏHSİLDƏKİ ROLU

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Xülasə: Turizmdə təbii və mədəni irsin integrasiyası qlobal rəqabət şəraitində dayanıqlı inkişaf modelinin formalaşmasında mühüm rol oynayır və bu proses ali təhsilin beynəlmilləşməsi ilə sıx qarşılıqlı əlaqədə inkişaf edir. Müasir mərhələdə universitetlər həm bilik ötürən, həm də mədəniyyətlərarası dialoqun, sosial integrasiyanın və innovativ turizm strategiyalarının formalaşdırılmasının aparıcı platformalarına çevrilmişdir. Beynəlxalq tələbələrin təhsil aldığı dövlət universitetləri təbii və mədəni irsin qlobal səviyyədə tanınması, qorunması və təbliği üçün effektiv vasitə kimi çıxış edir. Təqdim olunan tədqiqatın əsas məqsədi turizmdə təbii və mədəni irsin integrasiyası ilə dövlət universitetlərində xarici tələbələrin təhsili arasında qarşılıqlı təsir mexanizmlərini kompleks şəkildə araşdırmaqdır. Tədqiqat çərçivəsində irs əsaslı turizm modellərinin formalaşması, akademik mobilliyin genişlənməsi, mədəni müxtəlifliyin tədris prosesinə integrasiyası və bu proseslərin regional inkişaf göstəricilərinə təsiri təhlil edilir. Təbii və mədəni irs anlayışları hər bir millətin tarixi inkişafının, sosial-mədəni təcrübəsinin və ekoloji mühitinin ayrılmaz tərkib hissəsi olmaqla yanaşı, həm də milli identikliyin qorunması və beynəlxalq səviyyədə təqdim olunmasının əsas resurs bazasını təşkil edir. Bu irsin turizm vasitəsilə effektiv istifadəsi həm iqtisadi fayda ilə həm də təhsil mühitində multikultural dəyərlərin formalaşmasına, tələbələrin qlobal dünyagörüşünün genişlənməsinə və davamlı inkişaf məqsədlərinin reallaşdırılmasına töhfə verir.

Açar sözlər: mədəni irs, təhsil turizmi, turizm integrasiyası, multikultural integrasiya, yumşaq güc.

Introduction

Against the backdrop of contemporary global development trends, the tourism sector emerges as a multifaceted system, providing both a source of economic income and integrating cultural, social, and educational spheres. In particular, the integration of natural and cultural heritage into tourism is considered a key component of 21st-century sustainable development strategies. State policy on education development is aimed at expanding international cooperation (Decree of the President of the Republic of Azerbaijan, 2022). This approach is not limited to preserving heritage and passing it on to future generations but is also closely linked to its effective management and economic integration.

Natural and cultural heritage is a fundamental resource reflecting the historical memory, national identity, and civilizational development of each nation. Incorporating these resources into tourism creates the conditions for their international recognition, enhancing the soft power potential of countries, and expanding global cultural exchange. Joseph Nye (2004), having proposed the concept of soft power, emphasizes that culture and heritage play a significant role in shaping the international reputation of states. However, improper management of this process can lead to ecological imbalance, the commercialization of cultural values, and the loss of their authenticity. Therefore, the integration of natural and cultural heritage into tourism must be based on balanced, scientifically sound, and sustainable approaches.

The development of higher education internationally is one of the key areas of globalization. Creating an attractive environment for international students by public universities is based not only on academic excellence but also on the overall socio-cultural environment and tourism opportunities of the country. In this context, special attention is paid to the concept of "educational tourism." Educational tourism is a complex process in which individuals travel not only to study but also to become acquainted with the culture, history, and natural monuments of another country. International students play a dual role in this process: on the one hand, they are consumers of educational services, and on the other, they are informal promoters of the country's natural and

cultural heritage. The social and cultural experiences they gain during their studies contribute to strengthening the country's international image through the personal and professional connections they subsequently form. Higher education institutions are simultaneously becoming institutions for knowledge transfer and instruments of tourism and cultural diplomacy.

This integration is of strategic importance for countries like Azerbaijan, which possess rich natural landscapes and a multilayered cultural heritage. National parks, historical monuments, ethnographic diversity, and the country's multicultural environment are among the attractive factors for international students. This, in addition to enhancing the international competitiveness of public universities, also provides impetus for the development of the tourism sector.

Materials and Methods

Natural heritage refers to natural sites relatively unaffected by human activity or protected by special conservation regimes. This includes national parks, nature reserves, unique landscapes, geological formations, and biodiversity. Natural heritage is of paramount importance not only for aesthetics and recreation, but also for maintaining the ecological balance, combating climate change, and ensuring sustainable development.

Cultural heritage encompasses a system of tangible and intangible assets formed through human creative activity. Tangible cultural heritage includes historical monuments, architectural complexes, archaeological finds, and examples of urban planning. Intangible cultural heritage includes elements such as languages, folklore, music, dance, customs and traditions, and social practices. These components reflect the spiritual world, worldview, and characteristics of a society's identity.

Natural and cultural heritage form the foundation of a country's tourism potential and play a vital role in its sustainable development (Mustafayev, 2020). Natural and cultural heritage constitute an interconnected, complex system. Cultural interests are often formed and developed solely through the influence of the natural environment. Therefore, their study and preservation are more appropriately pursued holistically rather than separately. Preserving and incorporating these types of heritage into tourism fulfils a few important objectives. The protection and strengthening of national identity are ensured by the preservation and transmission of national values, language, customs and traditions, and cultural heritage to future generations. Natural and cultural heritage plays a vital role in the process of people's self-awareness. Preserving historical memory, preserving national values, and passing them on to future generations are among the main functions of this heritage. Presenting these values through tourism allows them to be communicated to a wider audience. Tourism is a key area of economic development and contributes to regional development (Guliyev, 2017). Increased economic revenue is ensured by effective management, the application of innovative approaches, and the optimal use of resources. Natural and cultural heritage is one of the main resources in the tourism sector. Attracting tourists leads to the development of service industries, the creation of new jobs, and additional revenue for the state budget. This effect is particularly noticeable in the regions. The socioeconomic development of regions is ensured by improved infrastructure, investment, and increased employment opportunities. Infrastructure development in areas where heritage sites are located increases local employment and improves the standard of living. This helps to reduce interregional development disparities. Expanding intercultural dialogue creates conditions for developing mutual understanding, cooperation, and tolerance among different peoples. Tourism strengthens mutual understanding and tolerance among different nations. For international tourists and students in particular, this heritage becomes a key platform for intercultural communication.

Sustainable development is achieved through the rational use of natural resources, maintaining the ecological balance, and considering the needs of future generations. A tourism model based on the principles of protecting natural and cultural heritage prevents resource depletion and ensures balanced development over the long term (World Tourism Organization, 2005).

Natural and cultural heritage is both a legacy of the past and a strategic resource for the modern economy, tourism, and education systems. Its proper management and integration into tourism play a vital role in enhancing the competitiveness of countries. Integrating natural and cultural heritage into tourism is a complex process requiring a systems approach, multi-level management, and coordination across various sectors. This integration serves both to enrich the tourism product and to simultaneously protect, sustainably use, and promote heritage internationally. In the modern era, this process is implemented through several main mechanisms and approaches. Ecotourism is an important approach that prioritizes the protection of natural heritage and aligns tourism activities with environmental principles. The primary goal of this model is to educate tourists and improve the well-being of local communities while minimizing their impact on nature. Ecotourism includes the organization of regulated visits to national parks and reserves, the implementation of programs aimed at protecting biodiversity, and raising environmental awareness. This approach prevents the depletion of natural resources and ensures the long-term development of tourism.

Cultural tourism involves integrating tangible and intangible cultural heritage into tourism activities. This approach includes incorporating historical monuments and architectural complexes into tourist routes, presenting cultural values through museums and exhibitions, and organizing national holidays, festivals, and folklore events. Cultural tourism generates economic and social benefits and contributes to the protection of natural and cultural heritage (Du Cros & McKercher, 2015). Cultural tourism is not only a source of economic income but also an effective tool for the protection and promotion of culture. This area provides international tourists and students with the opportunity to closely experience a country's historical and cultural heritage.

The integration of natural and cultural heritage into tourism should be based on the principles of sustainable development. Cultural tourism requires a comprehensive approach to heritage management and protection (Smith, 2015). Sustainable tourism policies include resource conservation, maintaining ecological balance, and ensuring social responsibility. This approach involves developing a legal and regulatory framework for tourism activities, applying restrictions and quotas to protect cultural heritage sites, considering the interests of local communities, and ensuring their participation in the tourism process. This policy minimizes the negative impacts of tourism, creating conditions for its long-term and balanced development. Contemporary research in the field of cultural tourism demonstrates that it is an important area of sustainable development (Richards, 2018).

In the modern era, digitalization plays a significant role in integrating natural and cultural heritage into tourism. Virtual tours and 3D modeling are implemented using digital technologies, tourist routes are presented on online platforms, and interactive guide services are provided through mobile apps. This approach is particularly important for attracting foreign audiences, including international students, to obtain initial information about the country. Digitalization also contributes to the protection of cultural heritage sites, as some physical visits can be replaced by virtual alternatives.

The above mechanisms yield more effective results when implemented not individually, but in collaboration and within the framework of a unified strategy. Cooperation between government agencies, educational institutions, tourism organizations, and local communities is a key prerequisite for the successful implementation of this process. Integration mechanisms in tourism create a balance between protecting natural and cultural heritage and enhancing its economic and social value. These mechanisms significantly contribute to both the development of the tourism sector and the country's international recognition.

Results and Discussion

The education of international students in a country is one of the key indicators of the internationalization of higher education (Sadigov, 2021). The education of international students at public universities is an important component of the process of internationalization of higher education and is considered one of the key indicators determining the competitiveness of countries in the modern global educational space. This process is not limited to academic exchange but also has a broad economic, cultural, and social impact. Public universities act as the main institutional platform in this area and occupy a leading position in attracting international students.

The education of international students in a country has a few strategically important consequences. Increased economic income: international students provide a direct financial influx into the country's economy through tuition, accommodation, transportation, and other services. Furthermore, their consumer activity stimulates the development of the local service sector (education, services, catering, and tourism). From this perspective, international students can be viewed not as an "export of human capital," but as an "import of human capital," which creates long-term economic benefits for the country. Strengthening cultural exchange: International students, as bearers of diverse cultures, interact with local students and society. This process contributes to the enrichment of cultural diversity, increased tolerance, and the development of multicultural values. At the same time, international students play a role in cultural diplomacy, promoting the culture of the country they study in their home countries. Developing the country's tourism potential: During their studies, international students become closely acquainted with the country's natural and cultural heritage. Their experiences are shared with a wider audience through social media, personal connections, and professional activities. This leads to the formation of the country's international image and an increase in tourist flows. The education of international students contributes to the internationalization of the country's education system (Law of the Republic of Azerbaijan, 2010). In addition to acting as potential tourists, international students also serve as "informal ambassadors" of the country. The increase in the number of international students reflects the globalization of higher education (UNESCO Institute for Statistics, 2021). Creating conditions for the development of "educational tourism" is possible through expanding international cooperation between educational institutions, creating appropriate infrastructure, and providing favorable conditions for students. The presence of international students provides impetus for the development of educational tourism. Educational tourism is characterized by students studying in another country and exploring the cultural, historical, and natural heritage of that country. This process expands international cooperation between higher education institutions and strengthens the universities' positions in the global education market.

The education of international students at public universities boosts the quality of the education system, accelerates the implementation of international standards in the academic

environment, creates conditions for the development of innovation and research, and improves the universities' positions in international rankings.

Admission of international students to higher education institutions is carried out in accordance with established regulations (Cabinet of Ministers of the Republic of Azerbaijan, 2019). The admission and organization of training for international students are governed by special regulatory documents (Ministry of Science and Education of the Republic of Azerbaijan, 2023). The education of international students at public universities is a complex phenomenon, affecting both the individual educational process and the country's economic development, cultural wealth, and international reputation. This process is closely linked to the integration of natural and cultural heritage into tourism, creating a system of mutually beneficial relationships.

Educational tourism is a complex socio-economic and cultural phenomenon emerging at the intersection of modern global education and tourism systems. This concept implies that international students not only acquire academic knowledge but also have direct contact with the cultural, historical, and natural environment of the country in which they study. In this context, educational tourism is an important mechanism for strengthening the interaction between higher education institutions and the tourism sector. During their studies, international students not only participate in the academic process but also have the opportunity to experience the country's rich cultural heritage.

This process includes the following areas:

- International students gain a deep understanding of the country's past and cultural development by exploring its historical and architectural heritage. These visits allow them to better assimilate the theoretical knowledge acquired during their studies in a practical environment. Historical monuments serve as living learning spaces for students.
- International students become acquainted with local traditions, customs, music, cuisine, and lifestyle by participating in various cultural events, festivals, and community activities. This experience broadens their cultural horizons and develops intercultural communication skills.
- The country's national parks, mountainous regions, coastal areas, and other natural beauty areas serve as both recreational and educational venues for international students. This creates conditions for the development of environmental values and a stronger sense of responsibility towards nature. As a result of these activities, international students unofficially serve as "tourism ambassadors" for the country. They contribute to increased tourism by sharing their experiences on social media, telling family and friends, and by visiting the country again in the future or encouraging others to do so.

Educational tourism is a process that affects both the individual and the macro level of experience. This process serves to shape the country's international image, develop cultural diplomacy, expand the tourism sector, and promote natural and cultural heritage globally. Educational tourism and the activities of international students play a strategic role in the international promotion of natural and cultural heritage and significantly increase the soft power potential of countries.

The Republic of Azerbaijan boasts a rich and diverse natural and cultural heritage. The country's favourable geographical location, diverse climate, and stages of historical and cultural development are among the key factors shaping its heritage potential. This potential serves as an important resource for the development of both domestic and international tourism and is closely

linked to the internationalization of the higher education system. Azerbaijan's natural heritage is highly diverse. This includes the Caucasus Mountains, particularly the rich ecosystems of the Greater and Lesser Caucasus, national parks and reserves (e.g., Shahdag, Goygol, Hirkan, etc.), diverse landscapes (semi-desert, mountainous, and forested zones), rich biodiversity, and rare flora and fauna. International tourism statistics show that tourism is a significant sector of the global economy (WTO, 2019).

These natural resources create ample opportunities for the development of ecotourism and simultaneously play a significant role in environmental education and research. In terms of cultural heritage, Azerbaijan has a very rich history. The country boasts ancient cities (such as monuments from the Shirvanshah era), historic architectural examples, archaeological sites, and elements of intangible cultural heritage such as national music, carpet weaving, and the art of mugham. In particular, part of Azerbaijan's cultural heritage is included on the UNESCO World Heritage List, enhancing the country's international recognition and prestige. Such sites are not only attractive for tourism but also play an important role in scientific research and education. The protection of world cultural and natural heritage is one of the fundamental principles recognized internationally (UNESCO, 1972). In this context, the role of international students studying at state universities is particularly important. They visit the country's natural and cultural heritage sites, acquire scientific and practical knowledge about this heritage, disseminate their experiences in their home countries, and contribute to the promotion of Azerbaijan through social and academic networks.

International students act as "carriers of information and culture," promoting the potential of Azerbaijan's heritage internationally. This process plays a significant role in the country's soft power strategy and strengthens Azerbaijan's position on the global education and tourism map. At the same time, the presence of international students creates conditions for expanding international cooperation at universities, enriching the academic environment, developing multicultural values, and applying innovative approaches to teaching.

Mutual recognition of higher education qualifications creates conditions for the development of international cooperation (UNESCO, 2009). The integration of Azerbaijan's natural and cultural heritage into the higher education system creates a mutually beneficial model. This model serves both the development of tourism and the strengthening of the country's international image. The European Union's Cultural Heritage Strategy aims to protect cultural diversity (European Union, 2013). The integration of natural and cultural heritage into tourism is one of the key pillars of the concept of sustainable development in the modern era. This integration serves not only to improve economic efficiency but also to protect the ecological balance, preserve cultural values, and improve the social well-being of society. Proper management of natural and cultural heritage is of strategic importance in terms of its transmission to future generations. The fact that international student's study at public universities adds additional dynamism and value to this process. They not only acquire academic knowledge but also become familiar with the country's cultural and natural heritage and, by sharing this experience in their home countries, contribute to building cultural bridges internationally. In this regard, international students act as both educational actors and informal carriers of cultural and tourist resources.

The Bologna Process is a strategic initiative aimed at creating a unified higher education area in Europe. This process is designed to harmonize education systems, ensure mutual recognition of degrees, and increase student mobility. In this context, strengthening cooperation between higher education institutions, harmonizing curricula, and developing quality assurance mechanisms are of

particular importance. The Bologna Process also creates conditions for expanding educational opportunities for international students and allows them to comfortably study in different countries. This process significantly contributes to the development of international cooperation by strengthening integration into the European Higher Education Area (European Higher Education Area, 2018).

Conclusion

The interaction between education and tourism, or their synergy, has a significant impact on a country's economic development, particularly through the expansion of the service sector, increased tourism revenue, and the stimulation of regional development. At the same time, this synergy creates conditions for strengthening the country's international image, enhancing its soft power potential, and developing cultural diplomacy.

The interaction between the integration of natural and cultural heritage into tourism and the internationalization of higher education is a key component of the modern development model. Effective implementation of this model contributes to the country's sustainable development, enhanced global competitiveness, and the promotion of cultural heritage internationally.

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THEORETICAL AND METHODOLOGICAL FOUNDATIONS OF TRANSLATION THEORY

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Abstract: The theory of translation, which has been formed as one of the most complex and multifaceted directions of modern linguistics, is of particular importance in terms of studying the processes of interlingual and intercultural communication. In the conditions of globalization, the emergence of intensive relations between different language systems, as well as the acceleration of information exchange, have necessitated in-depth study of translation activity not only from a practical but also from a theoretical point of view. Although the phenomenon of translation has long been perceived mainly as an applied activity, since the second half of the 20th century, as a result of the systematic development of its scientific and theoretical foundations, it has been formed as an independent field of research. In this regard, translation theory is characterized as a complex scientific direction that includes the mechanisms of interaction not only of linguistic phenomena, but also of cognitive, psychological and cultural factors. Thus, it can be noted that the translation process is not only a change of language units, but also a complex interpretative activity accompanied by the reconstruction of meaning, function and communicative intention. The methodological basis of translation theory was formed on the basis of the synthesis of various scientific approaches. Among these approaches, linguistic, communicative and functional directions occupy a special place.

Key words: translation theory, source language, target language, linguistics, text, lexical

TƏRCÜMƏ NƏZƏRİYYƏSİNİN NƏZƏRİ VƏ METODOLOJİ ƏSASLARI

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Xülasə: Müasir dilçiliyin ən mürəkkəb və çoxşaxəli istiqamətlərindən biri kimi formalaşan tərcümə nəzəriyyəsi dillərarası və mədəniyyətlərarası ünsiyyət proseslərinin öyrənilməsi baxımından xüsusi əhəmiyyət kəsb edir. Qloballaşma şəraitində müxtəlif dil sistemləri arasında intensiv əlaqələrin yaranması, eləcə də informasiya mübadiləsinin sürətlənməsi tərcümə fəaliyyətinin təkcə praktik deyil, həm də nəzəri baxımdan dərinlən öyrənilməsinə zəruri etmişdir. Tərcümə nəzəriyyəsi uzun müddətdir əsasən tətbiqi fəaliyyət kimi qəbul edilsə də, 20-ci əsrin ikinci yarısından etibarən elmi və nəzəri əsaslarının sistemli inkişafı nəticəsində müstəqil tədqiqat sahəsi kimi formalaşmışdır. Bu baxımdan tərcümə nəzəriyyəsi təkcə dil hadisələrinin deyil, həm də idrak, psixoloji və mədəni amillərin qarşılıqlı təsir mexanizmlərini özündə birləşdirən mürəkkəb elmi istiqamət kimi xarakterizə olunur. Beləliklə, qeyd etmək olar ki, tərcümə prosesi yalnız dil vahidlərinin dəyişməsi deyil, həm də mənanın, funksiyanın və kommunikativ niyyətin yenidən qurulması ilə müşayiət olunan mürəkkəb fəaliyyətdir. Tərcümə nəzəriyyəsinin metodoloji əsası müxtəlif elmi yanaşmaların sintezi əsasında formalaşmışdır. Bu yanaşmalar arasında linqvistik, kommunikativ və funksional istiqamətlər xüsusi yer tutur.

Açar sözlər: tərcümə nəzəriyyəsi, mənbə dil, hədəf dil, dilçilik, mətn, leksik

Introduction

Translation as a speech work, i.e., as a text, is of interest to translation theory precisely as a relative quantity. However, the relative nature of the translated text lies not only in its relation to the original. Indeed, the translated text is the only materialized entity that, when compared with the original text, allows us to lift the veil on the mystery of translation, reveal its mechanisms, and model it.

E. Nida (1964, p. 12) defines translation as *“the process of expressing the message of the source text in the target language with the closest natural equivalent”*. This approach shows that translation is based not only on the principles of formal conformity, but also on the principles of semantic and functional adequacy.

On the other hand, V. M. Lejchik (2009, p. 256). considering translation in the context of the interaction between the lexical and semantic systems of the language, characterizes it as *“the process of transforming lexical units for certain communicative purposes”*. This definition highlights the structural-linguistic essence of translation.

Thus, it can be noted that the translation process is not only a change of language units, but also a complex interpretative activity accompanied by the reconstruction of meaning, function and communicative intention.

Materials and methods

The methodological basis of translation theory was formed on the basis of the synthesis of various scientific approaches. Among these approaches, linguistic, communicative and functional directions occupy a special place.

Within the framework of the linguistic approach, translation is studied in the context of intertextual structural compatibility, and here the main attention is paid to lexical-semantic and grammatical transformations. Komarova (1991, p. 78) notes that *“the semantic structure of a particular lexicon is one of the main factors determining its translation potential”*.

In the communicative approach, the translation process is evaluated as an act of information transmission, and the main goal here is to restore the communicative effect of the source text in the target audience. From this point of view, translation acts not only as a language, but also as an act of communication.

The functional approach, on the other hand, presupposes the selection of translation strategies based on the purpose of the text and the context of its use. According to this approach, the same text can be translated differently for different purposes.

"Translation can be considered a certain type of transformation, namely, interlingual transformation". (Barkhudarov, 1975, p. 6)

Results and discussion

What essential characteristics of translation can be deduced from the above definitions? So, translation is:

- a speech work in its relationship to the original;
- the expression of what has already been expressed by means of another language, a re-expression;
- a process of interlingual and intercultural communication;
- communication using two languages, language contact, the phenomenon of bilingualism;
- a type of speech activity in which the components of communication are doubled;
- a two-phase process, as it is divided into two parts, into two moments;
- interlingual transformation;
- a type of verbal art; an art based on science.

Translation always presupposes an original. This implies that the original/translation relationship is an objective necessity, a constant that reflects the essence of this phenomenon. At the same time, translation is a speech work, situated alongside other speech works that exist and constantly emerge within the environment of the target language and the target culture. Translation is always relative to these speech works and is often assessed solely in relation to them, for example, by literary criticism. What is translation in this environment? An equal member of the "community" or a foreigner, a native child or a foundling, a purebred Aryan or a mixed-race person? This dual relativity is the very essence of translation, and it is also the main source of controversy, the fundamental stumbling block in the evaluation of translation and in the eternal debates about its possibilities.

Studying translation as a process of interlingual and intercultural communication, as communication using two languages, as language contact, we clearly discover interlingual and intercultural asymmetries. Today, perhaps no one doubts that any modern language is capable of expressing and describing any fragment of reality. R. Jakobson (1985, p. 364) viewed the claims of "untranslatability", occasionally proclaimed by skeptics, as attempts to cut the Gordian knot of many complex problems in translation theory and practice. *"All cognitive experience and its classification", he asserted, "can be expressed in any existing language"*. But no one doubts that languages reflect reality differently, asymmetrically. When languages come into contact in translation, when the meanings of one language are necessarily defined by the meanings of another in describing a fragment of reality, the asymmetry becomes most pronounced. We discover that languages segment reality differently, describe the same phenomena and objects differently, emphasizing their different attributes. People from different cultures express joy and despair, love and hate, differently; time flows differently for them; the world sounds and colors differently. Some possess objects that others lack; some still actively use things that have long since fallen out of use in others. But people from different cultures and with different linguistic consciousness are able to understand these differences.

Therefore, if we consider translation simply as a way of describing the same reality using another language, the problem of translation is quite easily solved, and the question of translatability does not arise. But translation is a re-expression. If every verbal work represents, in a certain sense, the materialization of a reflection of a fragment of reality by an individual's consciousness, then translation is a reflection of a reflection.

Definitions of translation allow us to outline the figure of the translator. A translator is at the very least a bilingual individual, addressing two cultures simultaneously.

A translator is sometimes dismissively called a "*servant of two masters*" but when it comes to the two cultures the translator serves, there is nothing offensive in this definition. Some linguists defined translation as a special case of bilingualism. The peculiarity of bilingualism in translation, in our opinion, lies, firstly, in the fact that it is typically asymmetrical. Most translators are dominated by one language, the language they were born with, and one culture, the culture they absorbed along with that language. This language and this culture subordinate others that the translator encounters in translation. Through the prism of the dominant language and the dominant culture, the meanings contained in speech works in another language are understood, and the facts of another culture are perceived. Secondly, during the translation process, both languages are present in the speech act and function simultaneously.

In a communicative act involving translation, everything is much more complex than in ordinary communication in a single language. Many components of the communicative act are duplicated. The central figure in this act of communication, the translator constantly changes roles, becoming sometimes the recipient of speech, sometimes the sender, and the form of the message changes. And does the message itself remain unchanged? Something is inevitably lost, something new appears. Translation is considered to occur in two stages: the perception of meaning and its expression. But when does the transformation of meaning occur? We often hear the common expressions "*translation is an art*", "*the art of translation*", "*a skilled translator*" and so on. Is this a beautiful metaphor elevating the translator's social status, or a serious typological statement situating translation as a human activity within the realm commonly called art?

These and many other questions arise when we seek to define the essence of translation as fully and precisely as possible. Therefore, any definition of translation, if we seek to give it a concise form, will suffer from a certain one-sidedness. The phenomenon is too complex, its interpretations too contradictory, and the attitudes of the society that consumes it with pleasure are too ambiguous. Translation, as a highly complex intellectual activity, is the subject of study for many academic disciplines. Translation is studied not only by linguistics, literary criticism, and the history of language and literature. Certain aspects of translation are also studied by psychology, sociology, religious studies, cybernetics, computer science, and other academic disciplines. However, this activity has its own theory, a distinct academic discipline that utilizes the methods and knowledge of other sciences, but has its own distinct subject matter: establishing the patterns of translation transformations, discovering the objective reasons for translation decisions, and developing typologies.

Translation theory can be defined as fundamental scientific knowledge about similarity the similarity of real world objects, the similarity of the expressive potential of human languages. Similarity is always relative. The degree of relativity varies from object to object, from language to language, from culture to culture. A translator seeks similarity in a sea of diversity, similarity that can be perceived by someone from a different culture, a different language, a different historical

era. The numerous techniques and operations a translator uses to establish such similarity collectively constitute a translation methodology, the mastery of which is essential even for exceptionally talented individuals with a keen sense of the smallest nuances of meanings, senses, and situations. Translation methodology can be developed and taught, just as a naturally talented composer can be taught musical notation, or a talented artist can be taught painting techniques, etc.

P.I. Kopanев (1972, p. 107) distinguishes four periods in the history of translation.

He believes that *"in the course of a concrete historical examination of the practice and theory of translation in general and literary translation in particular, the chronological stages of humanity's spiritual development and its centuries-old culture, coinciding primarily with the stages of the world's socio-historical chronology, become increasingly clear"*. He distinguishes the first, or ancient, period (slavery and feudalism); the second, or middle (from the initial accumulation of capital to the scientific and technological revolution of the 18th century inclusive); the third, or modern, period (late 18th - late 19th centuries); and the fourth, or most recent, period (late 19th - 20th centuries).

Conclusion

The conducted analyses show that translation theory, in addition to being a complex and multi-level field of linguistics, is an integral part of modern communication processes. Translation is not only the transformation of language units, but also the process of reconstructing meaning, function and cultural context. In this regard, the development of translation theory is of great importance both at the theoretical and practical levels and opens up broad prospects for future research.

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THE RISE OF ENGLISH IN AZERBAIJAN: EDUCATIONAL REFORM, DIGITALIZATION, AND GLOBAL ENGAGEMENT

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Abstract: Language education in Azerbaijan has evolved over a lengthy historical and cultural trajectory, from the medieval supremacy of Arabic and Persian to the present popularity of English. Arabic and Persian were not only means of interaction, but also transmitters of scientific, theological, and literary

knowledge, influencing intellectual culture and cultural identity. Over time, political upheavals, international contacts, and exposure to European languages brought French and English into educational settings, progressively encouraging multilingual competency. English is now widely used in academic, professional, and social settings, owing to advancement, global cooperation, and labor market demands. Contemporary strategies to language instruction emphasize student interaction, interactive techniques, digital tools, and ongoing evaluation, allowing students to gain pragmatic communication abilities and intercultural awareness. Socio-cultural influences, digital exposure, and informal learning settings like social media all improve motivation, identity development, and real-world language usage. Overall, language education in Azerbaijan serves as an important tool for personal development, career opportunities, and global engagement, connecting historical traditions with modern requirements and equipping students to efficiently traverse a globalized world.

The incorporation of English into numerous professional areas has opened up new avenues for international collaboration and career growth, whilst exposure to varied cultural materials and media promotes not just language competency but also global awareness and intercultural sensitivity. Continued investment in teacher training and educational resources improves the quality and accessibility of language learning in various regions, and learners progressively see language abilities as necessary tools for personal growth, social interaction, and active engagement in a global community. The blend of formal instruction and informal digital learning environments leads to more holistic and flexible language skills, and mastery of English promotes permanent learning by allowing learners to keep up with evolving scientific, technological, and cultural knowledge around the world.

Key words: manuscript culture, medieval madrasas, multilingual practice, academic exchange programs, globalization, digitization on language learning.

AZƏRBAYCANDA İNGİLİS DİLİNİN YÜKSƏLİŞİ: TƏHSİL İSLAHATLARI, RƏQƏMSALLAŞMA VƏ QLOBAL ƏLAQƏ

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Xülasə: Azərbaycanca dil təhsili uzun bir tarixi və mədəni inkişaf yolu keçmişdir, bu proses orta əsrlərdə Ərəb dili və Fars dilinin üstünlüyündən başlayaraq, bu gün İngilis dilinin geniş yayılmasına qədər davam etmişdir. Ərəb və fars dilləri yalnız ünsiyyət vasitəsi deyil, həm də elmi, teoloji və ədəbi biliklərin daşıyıcıları olmuş, intellektual mədəniyyətin və mədəni kimliyin formalaşmasına təsir göstərmişdir. Zaman keçdikcə siyasi dəyişikliklər, beynəlxalq əlaqələr və Avropa dilləri ilə təmas nəticəsində fransız dili və ingilis dili təhsil mühitinə daxil olmuş, tədricən çoxdilli bacarıqların inkişafını təşviq etmişdir. Bu gün ingilis dili inkişaf, qlobal əməkdaşlıq və əmək bazarının tələbləri səbəbilə akademik, peşəkar və sosial sahələrdə geniş istifadə olunur. Müasir dil tədrisi strategiyaları tələbə yönümlü yanaşmanı, interaktiv metodları, rəqəmsal alətləri və davamlı qiymətləndirməni ön plana çıxarır ki, bu da tələbələrin praktik ünsiyyət bacarıqları və mədəniyyətlərarası anlayış qazanmasına şərait yaradır. Sosial-mədəni faktorlar, rəqəmsal mühit və sosial media kimi qeyri-formal öyrənmə platformaları motivasiyanı, şəxsiyyətin formalaşmasını və dilin real həyatda istifadəsini gücləndirir. Ümumilikdə, Azərbaycanda dil təhsili şəxsi inkişaf, karyera imkanları və qlobal integrasiya üçün mühüm vasitə rolunu oynayır; tarixi ənənələrlə müasir tələbləri birləşdirərək tələbələri qloballaşan dünyada effektiv fəaliyyət göstərməyə hazırlayır.

İngilis dilinin müxtəlif peşə sahələrinə daxil edilməsi beynəlxalq əməkdaşlıq və karyera inkişafı üçün yeni imkanlar açmış, müxtəlif mədəni material və media ilə tanışlıq isə yalnız dil bacarıqlarını deyil, həm də

qlobal dünyagörüşü və mədəniyyətlərarası həssaslığı inkişaf etdirmişdir. Müəllim hazırlığına və tədris resurslarına davamlı investisiya müxtəlif regionlarda dil təhsilinin keyfiyyətini və əlçatanlığını artırır. Öyrənənlər getdikcə dil bacarıqlarını şəxsi inkişaf, sosial ünsiyyət və qlobal cəmiyyətə fəal inteqrasiya üçün vacib alət kimi qəbul edirlər. Formal təhsil ilə qeyri-formal rəqəmsal öyrənmə mühitlərinin vəhdəti daha çevik və hərtərəfli dil bacarıqlarının formalaşmasına səbəb olur. İngilis dilinə yiyələnmək isə öyrənənlərə dünya üzrə dəyişən elmi, texnoloji və mədəni bilikləri izləməyə imkan verərək davamlı öyrənməni təşviq edir.

Açar sözlər: əlyazma mədəniyyəti, orta əsr mədrəsələri, çoxdilli təcrübə, akademik mübadilə proqramları, qloballaşma, dil öyrənməsində rəqəmsallaşma.

Introduction

The supremacy of the Arabic and Persian languages had a significant impact on Azerbaijan's educational scene beginning in the Middle Ages. These languages served not only as means for education, but also as transmitters of religious, philosophical, and scientific information. Educational institutions, such as madrasas, depended primarily on Arabic for theological studies and Persian for literature and administration. As a result, competency in these languages was frequently used to facilitate educational access, molding intellectual life and cementing linkages with the larger Islamic world. This linguistic legacy established the groundwork for future improvements in language instruction in the region. Scholarly study on medieval Islamic education demonstrates that Arabic was the dominant language of scholarship throughout enormous regions, owing to its key role in the transmission of the Qur'an and classical sciences. Islamic studies research emphasizes that subjects such as law (fiqh), theology (kalam), and philosophy were almost entirely taught in Arabic, resulting in a coherent intellectual realm ranging from the Middle East to the Caucasus. At the same time, Persian became a popular literary and administrative language, notably in courts and metropolitan areas. According to historical assessments, Persian literary works and poetic traditions had a great impact on cultural education, shaping aesthetic and ethical ideals in Azerbaijani culture.

Academics point out that the growth of Persian in education was directly related to the political power of Persianate kingdoms in the region. Within the larger framework of the Persianate world, Persian served as a lingua franca of elite culture, allowing for intellectual contact across borders. Manuscripts written in Azerbaijan during this time period are frequently bilingual or even trilingual, including Arabic for scientific discourse and Persian for poetic expression. This multilingual practice exemplifies a complex educational style in which pupils were required to negotiate many linguistic systems concurrently (Rustamov, 2010).

Furthermore, studies on manuscript culture and curricular frameworks in medieval madrasas show that memorizing, commentary writing, and interpretation tasks were important teaching techniques. These techniques strengthened language competence while also developing analytical thinking and interpretative abilities. Historians say that this strategy helped to create a highly literate academic class capable of dealing with complicated literature from across fields. Therefore, the dominance of Arabic and Persian was more than just a language occurrence; it was also an intellectual framework that influenced Azerbaijan's educational system and aspirations throughout centuries. During the reign of the Aq Qoyunlu state, diplomatic and commercial links with Europe opened up new avenues for language diversity in schooling. These connections increased exposure to Western languages, especially among the governing elite and merchant classes. Though this change did not instantaneously revolutionize the whole educational system, it did signal the start of

a gradual transition to multilingualism. The adoption of additional languages mirrored evolving geopolitical circumstances and the necessity for expanded methods of interaction outside of the traditional Islamic domain.

The Russian Empire's acquisition of the Caucasus brought about substantial developments in the nineteenth century. Under Russian imperial administration, new educational systems were adopted, emphasizing European languages. Among these instances, the French language was especially prestigious. French was widely considered as the language of diplomacy and high culture in Europe, and its study became a symbol of intellectual sophistication among Azerbaijani elites. Likewise, migration of some parts of the population to Europe aided in the practical adoption of French, promoting cultural interchange and incorporation into European academic circles. The teaching of English as a foreign language in Azerbaijan became more systematic in the late 1920s. Initially, training was centered in specialized institutions, with structured language courses geared mostly toward teachers. These courses generally lasted two to three years and concentrated on basic language competences such as pronunciation accuracy, translation strategies, and literacy skills. The certification received upon completion of these programs demonstrates formalizing and institutional acknowledgment of English language instruction during this time period (Ministry of Education of the Republic of Azerbaijan, 2019).

By the academic years 1925-1926, English language training had grown in popularity at institutions such as the Azerbaijan State Pedagogical Institute and Baku State University. This breakthrough illustrates that English was not merely an optional topic, but rather a vital part of higher education curricula. The first Azerbaijani-English dictionaries were published in 1937, reinforcing this trend and providing valuable tools for both students and instructors. These dictionaries not only aided learning a foreign language, but also helped to standardize linguistic proficiency and promote translation research. The spread of English language education was inextricably related to wider socio-cultural and technical changes. As worldwide interaction networks evolved, English became a primary language for obtaining scientific and technological knowledge. Its incorporation in educational programs demonstrated a concerted attempt to align Azerbaijan with worldwide academic and professional norms. As a result, language education remained an important tool for modernization, allowing people to engage more actively in worldwide intellectual and economic exchanges (Ismailova, 2018).

Nowadays, English is generally acknowledged as an international language, ranking as the most regularly taught and in-demand language in the world. Azerbaijan, like many other nations, teaches English as the principal foreign language in the national academic system. This extensive use is intimately related to globalization, global interaction requirements, and gain access to scientific and technical information. According to studies in the field of pragmatic linguistics, English is not just means of interaction, but also a doorway to worldwide academic and job prospects. Innovative educational institutions in Azerbaijan use a number of new approaches, technology, and instructional techniques to increase the efficacy of English language training. Pedagogy research emphasizes the necessity of incorporating interactive methodologies such as performative language education, task-based learning, and digital resources into the classroom. These techniques seek to shift the emphasis from teacher-centered education to student-oriented learning, promoting active involvement and practical language usage. As a consequence, students are better prepared to acquire interpersonal skills rather than depending simply on memory and grammatical linguistic tools. The right application of teaching methodologies is critical to aligning

education with current international norms. According to educational scholars, arranging lessons, selecting suitable instructional resources, and adapting approaches to students' requirements all have a substantial impact on learning results. Teachers are required to create classes that not only match curricular objectives but also embody the needs of the present world. This involves developing analytical thinking, imaginative thinking, and self-directed learning abilities, all of which are necessary for 21st-century education (Ushioda & Dörnyei, 2009)

The growing significance of the English language in Azerbaijan is intimately related to the rise of foreign academic exchange programs and scholarships. Lots of learners seek further education overseas or engage in exchange programs where English is the predominant language of instruction and interaction. This has raised learners' drive to reach greater competence levels and prompted educational institutions to improve language preparation programs. Azerbaijan's growing education system pushes instructors and students to move beyond standard educational paradigms. The new method to learning emphasizes student autonomy by encouraging students to actively enhance their language abilities through collaborative assignments, self-study, and the utilization of technology. Educators, in turn, serve as facilitators and guides rather than exclusive suppliers of information. This shift is part of wider educational reforms targeted at increasing the quality of language teaching and enabling students can successfully utilize English in practical circumstances.

In modern Azerbaijan, the teaching of foreign languages, especially English, has been influenced by a deliberate state policy aimed at internationalization. After independence in 1991, educational reforms focused on aligning the national curriculum with global expectations. Mastering English in primary grades is a strategic plan for developing a workforce capable of participating in international educational and financial organizations. Empirical research on education policy shows that early exposure to a second language improves phonological acquisition and interpersonal skills, providing Azerbaijani students with a developmental advantage over late starters (Mehdizade, 2025). The rising relevance of English in Azerbaijan is inextricably linked to the country's the financial framework and foreign relations. The rise of the energy sector and foreign investment has resulted in a labor market in which English ability is frequently a must rather than a benefit. Transnational collaboration ventures necessitate experts who can work in English-speaking workplaces, which boosts student desire to acquire functional proficiency. This parallels a wider worldwide trend in which language competency has a strong connection to employability and earning potential, emphasizing English's instrumental significance in Azerbaijani community. For example, English is regarded as the primary working language for professionals employed by big companies such as SOCAR and on international projects. This dramatically boosts motivation for English among learners and young individuals. Simultaneously, proficiency in English is required to compete in the global work market. According to statistics, a large number of Azerbaijani students planning to study abroad demonstrate the practical value of the language by taking exams such as IELTS and TOEFL (British Council, 2018).

Another important viewpoint is the influence of digitization on language learning. With the fast advancement of technology, students today have access to a diverse selection of digital formats, mobile applications, and virtual learning experiences. Educational technology researches show that digital technologies not only increase engagement but also enable tailored learning environments. These platforms enable students to maintain language skills autonomously, obtain immediate feedback, and interact with actual information, therefore speeding up the learning process. The introduction of digital technology into Azerbaijan's educational system has altered language

acquisition techniques. Distance education settings, mobile applications, and virtual communication tools offer learners with ongoing exposure to real language input. Studies in educational technology demonstrate that digital platforms increase learner autonomy through enabling learners to decide the pace and materials of their learning (Ismayilli, 2025). Furthermore, access to actual events English via internet media speeds up vocabulary development and increases listening comprehension, particularly among younger individuals.

Moreover, the function for evaluation in English language instruction has experienced significant changes. Traditional examination techniques are increasingly being supplemented or replaced by more flexible and competency-based evaluation systems. Regularly assessment, portfolio-based evaluation, and performance assignments are increasingly often utilized to evaluate student growth. In accordance with language evaluation studies, such techniques give a more thorough knowledge of learners' skills and promote permanent skills growth over short-term memory. This shift is consistent with current viewpoints in language evaluation, which stress communicative ability as an important result of language learning. Systematic and ongoing assessment has become an essential component of this new evaluation paradigm. Rather than depending primarily on final exams, teachers are now using formative assessment tools to monitor students' development throughout time. Portfolio-based evaluation, for example, enables learners to combine their written work, projects, and thoughts to provide a complete picture of their progress. Similarly, performance-based assignments—such as presentations, conversations, and analytical skills tasks—allow students to exhibit their pragmatic language skills in real-world scenarios. These techniques promote active engagement and help students become more aware of their own strengths and areas for growth. Furthermore, current assessment procedures place a high value on feedback and student engagement. Constructive feedback from educators, as well as peer and self-assessment, are critical components of the learning process. Such interactive evaluation strategies enhance critical thinking and learner autonomy by encouraging students by focusing on their performance and accept responsibility for their advancement (The Routledge Handbook of Language and Culture., 2014). According to investigation, this form of continual, feedback-oriented evaluation helps students learn languages for in the long run. In the field of language assessment, these new strategies give a more thorough picture of learners' skills than standard testing methods. Instead of isolating abilities, they test many ones at the same time, such as listening, speaking, reading, and writing. As a consequence, pupils acquire more balanced language skills.

In addition, socio-cultural variables influence perceptions of English language acquisition. Younger generations find English more approachable and appealing as a result of their exposure to global media like as films, music, and social networks. This informal learning setting supplements formal education by promoting the advancement of listening and speaking abilities. So, language learning is no longer limited to the learning environment but has spread into everyday life. Scholars in the field of second language acquisition (SLA) have long stressed the importance of language learning within its social and cultural context. According to Claire Kramsch, a key researcher in socio-cultural linguistics, language is primarily a cultural practice rather than a cognitive talent since it is influenced by people's values, beliefs, and community relationships. This viewpoint contends that learners create meaning via social involvement with the chosen language and its cultural surroundings, which promotes deeper learning and verbal proficiency. Recent study also demonstrates how familiarity with English through international media and internet platforms influences learners' socio-cultural perspectives. For instance, large-scale studies have shown that

social media platforms like Tik-Tok, YouTube, and other interactive online tools offer users authentic language input that improves listening ability, vocabulary building, and even speaking proficiency, particularly among younger learners who are exposed to a range of English content. These platforms foster intercultural awareness by engaging users with international cultural expressions, figurative language, and intercultural interaction techniques that go beyond typical classroom resources. According to socio-cultural experts, such informal media exposure promotes not just language growth but also learner motivation and identity building. When students come across engaging cultural products, such as soundtracks, movies, or virtual communities, they frequently regard English as a valuable instrument for participating in global culture rather than just an academic topic. This is consistent with the views of socio-cultural theorists who argue that sense of self and ambition are dynamically built via social contact, and that learners are more involved when they see language as important to their social life (Norton, 2013). Studies in many educational situations show that socio-cultural elements including family views, social conventions, and peer pressure have a substantial impact on learners' views and usage of English. Encouraging social networks and good cultural attitudes toward language learning tend to increase learners' confidence, generate informal chances for practice, and foster ongoing participation outside of formal education. In contrast, a lack of social support or cultural opposition might discourage language usage and lower student inspiration.

Educational theorists suggest that language instruction must incorporate sociocultural competency into its pedagogy, concentrating not only on linguistic structures but also on cross-cultural understanding and interpersonal abilities. Research on socio-cultural competency in foreign language education supports the idea that productive language acquisition requires knowing the target language community's cultural practices, beliefs, and communication patterns. In today's globalized world, English is an essential instrument for cross-cultural interaction and cooperation. In SLA (Second Language Acquisition) study results, socio-cultural theorists such as Lev Vygotsky claim that learning happens via contact and cooperation within a social environment instead of in isolation. In his seminal work *Thought and Language* (1986), Vygotsky argues that language development is mediated by social engagement, which means that learners integrate and generate knowledge via significant steps and shared speech. His hypothesis has been extensively applied to language education studies, demonstrating how community, culture, and interaction influence language proficiency (Vygotsky, 2012).

Dr. David Nunan, a well-known applied linguist, has conducted studies that stress the relevance of real content in language learning. Nunan argues in *Second Language Teaching & Learning* (1999) that language immersion outside of the classroom, through media such as television, podcasts, and music, offers learners with authentic, spontaneous English that is more diverse and contextually rich than conventional textbooks. This exposure enhances cognitive abilities and provides natural use patterns that aid in language growth. Additionally, studies published in the *Language Learning & Technology* journal demonstrates that platforms such as YouTube and social networks facilitate meaningful language interactions and contextual learning, with learners selecting content based on preferences, goals, and relevance—a crucial component in inspirational studies focusing on language acquisition (Stockwell and Hubbard, 2013).

Bonny Norton, one of the most notable voices in modern language proficiency theory, highlights in her seminal work *Identity and Language Learning: Extending the Conversation* (2013) that language acquisition is more than just a cognitive or technical process; it is inextricably linked

to learners' personalities, social ties, and social status. Norton contends that learners interact with a language not merely to acquire its grammatical or lexical shapes, but also to explore social spaces, develop a sense of self, and build deep relationships with others. Her investigations revealed that English serves as a tool for managing social connections, earning recognition, and engaging in both local and global societies. According to Norton, learners frequently establish numerous language identities based on the environment in which they use English. For example, a student may engage in one set of linguistic behaviors when connecting with classmates online, another when taking part in classroom discussions, and still another when talking in formal professional or academic contexts. This dynamic interaction indicates that language acquisition is inextricably linked to social positioning and cultural participation, emphasizing how learners' motivation, confidence, and perseverance are heavily impacted by their perceived possibilities to utilize English productively. Norton's study indicates that English, as a worldwide lingua franca, has far-reaching implications beyond its traditional position as a classroom topic. It transforms into a platform for self-expression, creativity, and social involvement, enabling students to acquire knowledge, engage in worldwide conversation, and connect with online communities. In many circumstances, this social dimension of language acquisition increases engagement and long-term retention because learners see the language as pertinent and beneficial in reaching personal, academic, and professional objectives. Norton's socio-cultural viewpoint is consistent with larger research in practical linguistics and academic psychology, which stresses identity development and interpersonal relationships as major determinants of effective language learning. Students who see themselves as genuine members of worldwide English-speaking communities are more willing to take chances, experiment with new language forms, and persevere in their studies despite setbacks. As a result, understanding the link between identity, social environment, and language acquisition is essential for developing successful educational practices that promote both linguistic and intercultural competence.

In conclusion, Norton illustrates that English is more than just an academic topic; it is a vehicle through which students manage identity, create social capital, and indulge in meaningful intercultural engagement across local and worldwide networks (Long, M. H., 2017). This viewpoint promotes educators to design educational settings that incorporate social contact, collaborative projects, and culturally appropriate resources, linking educational instruction to the larger societal realities in which students live and interact.

Zoltán Dörnyei's widely acclaimed book *Motivation, Language Identity, and the L2 Self* (2009) provides a comprehensive framework for understanding the complicated interplay between incentive for learners, identity, and second language learning. Dörnyei offers the L2 Motivational Self System, which connects a learner's motivation to a picture of a future self who is capable and secure in the desired language. This concept highlights that motivation is more than just a response to external incentives or classroom obligations; it is inextricably linked to learners' ambitions, self-perception, and expected responsibilities in social and career settings. According to Dörnyei, learners who see English as important for their future job, academic aspirations, or social mobility are more likely to participate actively in learning, persevere through problems, and use successful language acquisition tactics. This forward-thinking viewpoint, known as the "Ideal L2 Self," inspires learners to put forth effort in acquiring the language since it is closely related to their desired identity and life goals. Based on Dörnyei's framework, studies show that students who can vividly imagine themselves using English in meaningful contexts—such as working in multinational companies, studying abroad, or getting involved with global networks—have higher

levels of enthusiasm and perseverance in language learning. Dörnyei emphasizes the importance of global media and multicultural experience in shaping students' future self-concepts. Learners have access to films, music, internet platforms, social media, and foreign communities, allowing them to connect with English in culturally rich and realistic settings. These experiences not only enhance language abilities, but also serve as examples for social behavior, professional contact, and cultural standards, allowing students to picture themselves as active members of the global English-speaking society. This promotes enthusiasm and relevance, making the learning experience more meaningful and focused on goals.

Furthermore, the L2 Motivational Self System includes the notion of the "Ought-to L2 Self," which refers to learners' considered duties and norms of society, such as family, school, or society pressures to learn English. While the Ideal L2 Self promotes self-motivation, the "Ought-to L2 Self" promotes extrinsic incentive, which can impact learners' perseverance and effort, especially in organized educational settings. Dörnyei's approach incorporates the combination of personal objectives, societal standards, and environmental forces, providing a comprehensive explanation of why certain students are more effective and persistent in learning English. Dörnyei's paradigm has practical consequences for language teaching. Educators and curriculum planners are urged to incorporate establishing objectives, real tasks, and culturally appropriate resources that link educational activities to students' potential (Junjie Gavin Wu, Lindsay Miller. (2021). Providing learners with chances to envisage and experience their Ideal L2 Self through simulations, project-based learning, and international cooperation may greatly boost engagement and contribute to more persistent and insightful language progress. Dörnyei's study connects cognitive, emotional, and social characteristics of motivation to practical ways for enhancing English language training. Language education research is increasingly emphasizing the importance of informal and extracurricular situations in language growth, indicating that English learning goes well beyond conventional classroom teaching. According to research published in TESOL Quarterly (2018), learning does not end when students leave the classroom. Informal instruction contexts—such as language interaction forums, immersive online gaming communities, discussion groups, and social network conversations—have been shown in numerous countries to promote linguistic habits, expose students to diverse English varieties, and boost communicative confidence. These places enable students to experiment with language in actual contexts, debate significance with peers, and practice real-life interaction abilities without the burden of formal evaluation, resulting in more natural language learning. Global research undertaken by institutions such as the British Council and Cambridge English provides solid empirical proof that students who participate in extracurricular English activities routinely outperform their counterparts who only get classroom instruction. Learners who participate in debate clubs, English theatre, writing workshops, and online collaborative projects, for example, have considerably greater levels of hearing, speaking, and interactional competency. This shows that informal circumstances not only cement classroom learning, but also expose students to real and diverse English, allowing them to improve fluency, pragmatics, and sociolinguistic awareness.

Participation in these informal learning settings also helps learners develop their drive to succeed, autonomy, and identity. Learners who participate willingly in English-mediated activities tend to acquire higher self-efficacy, a sense of belonging to a global community, and practical communication skills. These encounters also introduce students to a variety of dialects, signs up, and cultural standards, which are necessary for gaining intercultural competency. As a result,

informal learning situations serve as valuable and often transformational extensions of formal education, helping English learners to acquire the language in insightful, socially acceptable, and motivationally abundant forms. Integrating both traditional and informal education experiences is seen as a critical method in current language training. Educators are urged to provide opportunities that connect classroom education with online communities, real-world projects, and peer interaction, therefore closing the gap between organized learning and realistic language usage. This hybrid approach not only speeds language learning, but also enables students to speak effectively in a variety of global situations, mirroring the wider aims of 21st-century English education. This viewpoint is strengthened in the context of Azerbaijan's education system by the rising acknowledgment of English as a critical tool for global integration rather than just a school topic. Current curriculum revisions have stressed communicative skill development by integrating national standards with international frameworks such as the Common European Framework of Reference (CEFR). As a result, more focus is placed on real-world language usage, increasing the importance of informal learning situations. Practical trends support this transition. The implementation of a competency-driven has resulted in demonstrable increases in students' receptive and productive abilities, notably listening and speaking. Furthermore, the proliferation of English-medium instruction (EMI) programs in higher education institutions around the country has increased students' academic mobility and involvement in foreign exchange programs. This results in a dynamic synergy between formal learning and casual exposure, enabling students to strengthen classroom information via actual interactive engagement. In parallel, the incorporation of digital technology into education has dramatically expanded access to informal learning possibilities (Cambridge Assessment English, 2016). Azerbaijani students are increasingly engaged with internet platforms, free educational materials, and worldwide virtual communities, which expose them to real linguistic input and varied cultural viewpoints. Such settings promote multicultural awareness and allow students to gain practical language abilities outside of textbook-based education.

Methodology and Materials

This study uses a qualitative research design with descriptive and analytical methodologies to investigate the emergence of English in Azerbaijan in the context of educational reform, digitization, and global participation. The research is mainly founded on a thorough assessment of historical records, scholarly literature, policy papers, and current studies on language instruction in Azerbaijan.

This study's sources include scientific books, peer-reviewed journal papers, official publications from the Republic of Azerbaijan's Ministry of Education, and international reports from the British Council and Cambridge Assessment English. These materials include both historical and contemporary perspectives on language policy, educational techniques, and the social dynamics of English language acquisition. These theories are used to understand how social interaction, identity development, and motivation among learners affect English language learning in Azerbaijan.

The study also examines online education circumstances, which include mobile applications, internet platforms, and social media, as informal learning resources that aid in language development. By investigating both official educational systems and informal learning situations, the study gives a comprehensive knowledge of how English competency develops in modern Azerbaijani culture. The technique involves historical study, literature evaluation, and theoretical

interpretation to investigate the many elements that influence English language teaching in Azerbaijan.

Discussion

The results of the research show that the emergence of English in Azerbaijan is the consequence of a complex combination of historical traditions, educational reforms, globalization, and technological improvement. The shift from Arabic and Persian supremacy to ubiquitous English usage exemplifies a constant process of adjustment to evolving intellectual, political, and economic environments. One of the significant findings from the investigation is the substantial link between language education and socioeconomic expectations. English has become an essential skill in the workforce industry, particularly in industries like energy, international commerce, and higher education. This instrumental value considerably boosts learner motivation, supporting Dörnyei's idea of the "Ideal L2 Self," in which people imagine themselves as competent individuals who speak English in future professional settings.

Moreover, the study emphasizes the transformational effects of digitalization on language acquisition. Digital technologies and internet platforms expose learners to real English on a continuous basis, enabling self-directed and individualized study. This is consistent with modern educational paradigms that stress learner-centered methods and ongoing participation outside of the classroom. Implementing technology not only improves language competency, but it also promotes critical thinking and self-directed learning.

Another major topic explored is the impact of socio-cultural influences on acquiring a language. Exposure to global media, social networks, and intercultural communication contexts promotes learner involvement and identity construction. According to Norton's perspective, English is more than just an academic subject; it is also a tool for social involvement and self-expression. Learners increasingly see English as a way to interact with worldwide communities, which boosts engagement and retention in the long run.

The debate also focuses on the transformation in teaching techniques in Azerbaijan. Conventional teacher-centered techniques are progressively being supplanted by interactive, task-based, and collaborative methods. Continuous assessment approaches, such as portfolio review and performance-based activities, offer a more comprehensive measure of language competency and are consistent with international standards such as the CEFR. Considering these excellent achievements, several obstacles persist. These consist of regional discrepancies in access to digital resources, variances in the quality of teacher preparation, and a need for greater congruence between curricular aims and real-world language usage. Tackling these difficulties is critical to ensuring equal and efficient language education across the nation.

Finally, the debate reveals that English language teaching in Azerbaijan is influenced by dynamic and linked elements. Its expanding prominence reflects global patterns while also addressing local educational and socioeconomic demands. The mix of formal education, technological advancement, and socio-cultural involvement produces a solid foundation for achieving 21st-century language proficiency.

Conclusion

The detailed research offered shows that the evolution of language education in Azerbaijan has been influenced by a variety of historical, social, economic, and technical aspects. During the Middle Ages, Arabic and Persian dominated not just as methods of communication, but also as vehicles for conveying scientific, theological, and philosophical knowledge, setting the groundwork

for the country's intellectual culture. Later times saw political shifts, notably imperial effects and the increase of global contacts, which permitted the entrance of other languages into the educational system, including French and English. In the present period, English is extremely important in Azerbaijan as a worldwide means of communication, with a direct relationship to both educational policies and labor market demands. The phenomenon of globalization, worldwide collaboration and greater opportunities to study abroad have transformed English from a mere academic subject into a practical skill and a form of social capital. In this context, language learning has evolved beyond the conventional classroom setting, becoming more interactive and easily available through digital technology, social networks, and informal learning environments.

At the same time, modern pedagogical techniques prioritize student-focused instruction, interactive procedures, and ongoing evaluation systems, all of which contribute to the overall development of language abilities. Socio-cultural influences, personal inspiration, identity, and integration into international culture all have an impact on the success of language acquisition. In this view, language serves not just as a medium for interaction, but also of individual expression, social inclusion, and career progress. Thus, the importance of English in Azerbaijan demonstrates a continuous line of growth that spans from historical traditions to current global needs. Language education is critical in increasing the country's international competitiveness, building human capital, and promoting people's absorption into the global society. As a result, in coming years, Azerbaijan's education system will prioritize language instruction refinement through creative ways, integration with technology, and consideration of socio-cultural factors.

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THE PROBLEM OF BILINGUALISM IN AZERBAIJAN AND WAYS TO SOLVE IT

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Abstract: This article explores the issue of bilingualism in Azerbaijan as a socio-pedagogical phenomenon that significantly influences both personal development and the educational process. It examines the characteristics of the bilingual environment, identifies key challenges such as language interference and insufficient language proficiency, and analyzes their impact on students' learning outcomes and socialization, including those with special educational needs. Particular attention is paid to state policies and pedagogical strategies aimed at addressing these challenges, including bilingual education models, psychological and pedagogical support, and the role of the family. The study concludes that, when approached systematically, bilingualism can serve as a valuable resource for cognitive and social development.

Keywords: bilingualism, inclusive education, language interference, language policy, socialization

AZƏRBAYCANDA BILINGVİZM PROBLEMI VƏ ONUN HƏLLİ YOLLARI

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Xülasə: Məqalədə Azərbaycanada ikidillilik problemi şəxsiyyətin inkişafına və təhsil prosesinə təsir edən sosial-pedaqoji fenomen kimi nəzərdən keçirilir. Məqalədə təhlil olunur və xüsusilə də ikidilli mühitin əsas bağlı çətinlikləri ilə dil интерференцией və kifayət qədər linqvistik компетентностью, eləcə də onların təsiri təlim və социализацию uşaqların, o cümlədən şagirdlərin xüsusi təhsil ehtiyacları olan. İkidilli təhsilin inkişafı, psixoloji və pedaqoji dəstək və ailənin rolu da daxil olmaqla bu problemin həllinə dövlət və pedaqoji yanaşmalara xüsusi diqqət yetirilir. Kompleks bir yanaşma ilə ikidilliliyin şəxsiyyətin inkişafı üçün vacib bir mənbəyə çevrilə biləcəyi vurğulanır.

Açar sözlər: ikidillilik, inklüziv təhsil, dil müdaxiləsi, dil siyasəti, sosiallaşma

Introduction

In the context of globalization and active intercultural contacts, the problem of bilingualism is becoming particularly relevant. For Azerbaijan, this phenomenon is a natural part of the historical

and socio-cultural development of society. For a long time, several languages have been co-existing in the country, among which Azerbaijani and Russian play a leading role. In recent years, the importance of English as a means of international communication has been increasing. As a result, a special language environment is formed that influences personal development, the education system and the processes of socialization.

Bilingualism is the ability of an individual, a group, or a society to use two languages in everyday communication, education, or professional life. A person who uses two languages is called a bilingual. Proficiency in both languages may be equal, or one language may be dominant. Bilingualism can be individual or social, and it may develop in early childhood or later through education and social interaction.

Bilingualism itself is not considered a problem; on the contrary, it is often viewed as a valuable social and cognitive resource. However, several global challenges are associated with bilingualism. One major issue is language inequality, where some languages have higher social, political, or economic status than others, which may lead to discrimination against speakers of minority languages. Another important problem is language loss, as dominant languages such as English language may contribute to the decline or disappearance of smaller native languages. Difficulties may also arise in education when school systems do not adequately support bilingual learners, which can affect academic achievement. In addition, bilingualism may create challenges related to identity, as some individuals experience tension between two linguistic and cultural backgrounds. In the context of migration, bilingualism is often connected with issues of integration, adaptation, and language policy. Another concern is code-switching, or frequent alternation between languages, which some researchers view as a challenge for maintaining language norms.

At the global level, one of the most significant concerns linked to bilingualism is the preservation of linguistic diversity. According to UNESCO, many languages in the world are endangered, and bilingualism may sometimes lead to a gradual shift from a native language to a dominant one. At the same time, bilingualism has many advantages, including cognitive flexibility, improved memory and attention, broader educational and career opportunities, and stronger intercultural communication. Therefore, bilingualism should be viewed not only as a challenge in certain contexts, but also as an important resource in the modern world.

Bilingualism, is a person's ability to use two languages in everyday life. In scientific practice, balanced and unbalanced bilingualism are distinguished. In the first case, both languages are developed at approximately the same level, in the second, one of the languages dominates. Azerbaijan is more characterized by unbalanced bilingualism, which develops from an early age. Children often grow up in an environment where one language is used in the family, while another is used in educational institutions or a social environment. This leads to specific features of speech and cognitive development.

On the one hand, bilingualism promotes the development of thinking, memory and communication skills. Bilingual children have more flexible thinking, switch between tasks more easily, and demonstrate a higher level of intercultural competence. On the other hand, in the absence of a systematic approach to the formation of a language environment, certain difficulties may arise, especially in early childhood and school age.

One of the most common problems is language interference, which manifests itself in the mixing of grammatical structures, vocabulary and syntax of the two languages. In children, this can be expressed in errors in oral and written speech, incorrect sentence construction and limited

vocabulary. In conditions of insufficient language practice, it is possible to form a situation in which a child does not fully speak any of the languages, which negatively affects the overall level of his speech development.

Language difficulties directly affect the learning process. Students may have difficulty understanding the learning material, completing assignments, and interacting with the teacher. This problem is especially acute in children with special educational needs, for whom the language barrier becomes an additional risk factor. This can lead to decreased learning motivation, self-doubt, and difficulties in social adaptation.

The psychological aspect of bilingualism also plays a significant role. Children who are unsure of their language skills often avoid active participation in academic and extracurricular activities, which limits their opportunities for full development. In such circumstances, it becomes important to create a supportive educational environment conducive to unlocking the potential of each child.

Measures aimed at optimizing the language situation are being taken at the state level in Azerbaijan. Special attention is paid to strengthening the position of the Azerbaijani language as the state language while maintaining multilingualism. The educational policy is focused on the development of education in the native language, while Russian-language schools are being maintained and developed, as well as the study of other foreign languages is being actively introduced.

The improvement of the educational system plays an essential role in solving the problem. In modern conditions, a differentiated approach to learning that takes into account the individual characteristics of students is becoming increasingly important. The use of bilingual techniques, adapted educational materials and interactive technologies makes it possible to increase the efficiency of learning and reduce the impact of language difficulties. Special attention is paid to the initial stage of learning, when the foundations of speech competence are laid. In Azerbaijan, a foreign language (English) is taught along with the Azerbaijani language from the first grade. Arabic, Russian, etc.).

Equally important is the organization of psychological and pedagogical support. The work of speech therapists, psychologists and teachers is aimed at correcting speech disorders, developing communication skills and overcoming psychological barriers. An individual approach to each child contributes to a more successful adaptation to the educational process and the formation of confidence in their abilities.

The family has a significant influence on the development of bilingualism. It is in the family environment that the first speech skills are formed and the foundation of language competence is laid. Consistent use of languages, creation of a favorable speech atmosphere, encouragement of reading and communication play a key role in the formation of harmonious bilingualism. Practice shows that a systematic approach in the family contributes to a more successful mastery of both languages.

The problem of bilingualism becomes particularly relevant in the context of inclusive education. It is important to take into account not only the language, but also the individual characteristics of each child. The individualization of learning, the adaptation of curricula and the creation of a supportive environment can minimize difficulties and ensure equal opportunities for all students. An inclusive approach contributes not only to the educational, but also to the social development of children.

Conclusion

The prospects for the development of bilingualism in Azerbaijan are linked to the further improvement of educational policy and the introduction of modern learning technologies. The training of teachers to work in a bilingual environment, the development of international cooperation and the exchange of experience with other countries open up new opportunities for improving the quality of education.

Thus, bilingualism in Azerbaijan is a complex, but at the same time promising phenomenon. With proper organization of the educational process, support from the state and the family, as well as the use of modern pedagogical approaches, it can become an important resource for personal and intellectual development. An integrated approach to solving this problem allows not only to overcome existing difficulties, but also to maximize the potential of a bilingual personality in modern society.

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NORMS OF VOWELS IN THE AZERBAIJANI LANGUAGE

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Abstract: This article provides a comprehensive study of the normative features of vowels in the Azerbaijani language. It explains the role of vowels within the phonetic and phonological system, their classification, and articulatory features. The division of vowels into front and back, rounded and unrounded groups and its impact on pronunciation and spelling norms is highlighted. Vowel harmony is presented as a fundamental principle and its influence on vowel compatibility within words is explained. The article also examines cases of vowel harmony violation in loanwords and their normative regulation. Orthographic and orthoepic norms related to correct pronunciation and spelling are analyzed. Vowel reduction, assimilation, and length features are also evaluated from a normative perspective. The impact of new words entering the language in the modern era and the importance of regulating this process within normative frameworks are emphasized. In conclusion, the correct use of vowels plays an essential role in language stability and the formation of speech culture.

Keywords: vowels, phonetics, norm, Azerbaijani language, pronunciation, orthoepy, phonology

NORMS OF VOWELS IN THE AZERBAIJANI LANGUAGE

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Xülasə: Bu məqalədə Azərbaycan dilində saaitlərin normativ xüsusiyyətləri hərtərəfli tədqiq edilir. Saitlərin fonetik və fonoloji sistemdəki rolu, onların təsnifatı və artikulyasiya əlamətləri izah olunur. Saitlərin incə və qalın, dodaqlanan və dodaqlanmayan qruplara bölünməsi, bunun tələffüz və yazılış normalarına təsiri vurğulanır. Ahəng qanunu fundamental prinsip kimi təqdim edilir və sözdaxili sait uyğunluğuna təsiri şərh olunur. Məqalədə, hətta alınma sözlərdə ahəng qanununun pozulması halları və onların normativ tənzimlənməsi də araşdırılır. Düzgün tələffüz və yazılışla bağlı orfoqrafik və orfoepik normalar təhlil edilir. Saitlərin reduksiyası, assimilasiası və uzanma xüsusiyyətləri də normativ baxımdan qiymətləndirilir. Müasir dövrdə dilə daxil olan yeni sözlərin təsiri və bu prosesin normativ çərçivədə tənzimlənməsinin əhəmiyyəti qeyd olunur. Yekunda, saaitlərin düzgün istifadəsinin dilin sabitliyində və nitq mədəniyyətinin formasında mühüm rol oynadığı vurğulanır.

Açar sözlər: saaitlər, fonetika, norma, Azərbaycan dili, tələffüz, orfoepiya, fonologiya

Introduction

The Azerbaijani language belongs to the Turkic language family and has a rich phonetic system. Vowels form the basis of this system, and their correct pronunciation, spelling, and usage are key factors that determine the normativity of the language. The norms of vowels refer to the use of sounds in accordance with the accepted literary rules of the language, as well as the preservation of their phonetic and phonological features. These norms have been historically formed, stabilized through the development of the literary language, and continue to serve as an important means of ensuring linguistic unity in the modern period.

Materials and Methods

In the Azerbaijani language, vowels are divided into several groups according to their articulatory features. They are classified as front and back vowels based on tongue position, and rounded and unrounded vowels based on lip involvement. This classification is not only theoretical but also plays an important practical role in determining pronunciation and spelling norms. For example, in words dominated by front vowels, vowel harmony is more prominent, and this principle is considered a key phonetic norm of the language.

Vowel harmony is one of the fundamental principles ensuring the compatibility of vowels in Azerbaijani. According to this rule, vowels within a word harmonize so that they are either all front or all back. This feature ensures the melodic and harmonious nature of the language. Violations of vowel harmony are mainly observed in loanwords; however, even in such cases, certain adaptations are made within the framework of literary norms. Thus, the normative use of vowels is connected not only with phonetic rules but also with the historical development of the language and its interaction with other languages.

The length and shortness of vowels are also regulated within certain normative boundaries in Azerbaijani. In the literary language, short vowels generally prevail, although in some cases, especially in stressed syllables, slight lengthening of vowels may occur. This lengthening does not

create semantic differences but plays an important role in expressing the emotional tone of speech. Therefore, vowel length characteristics are also studied as part of normative pronunciation rules.

Orthoepic norms are the main set of rules that determine the correct pronunciation of vowels. These norms ensure consistency between written and spoken language. In Azerbaijani, there are differences between the spelling and pronunciation of certain vowels. For instance, in some words, a written vowel may be weakened or omitted in speech. Such cases fall under orthoepy and are important for maintaining standard pronunciation.

Reduction and assimilation of vowels are also important aspects of phonetic norms. During reduction, vowels in unstressed syllables become weaker. Assimilation occurs as a result of the influence of neighboring sounds on each other. Although these processes are natural outcomes of language development, in the literary language they are kept within certain limits. Excessive reduction or assimilation may reduce speech clarity; therefore, they are regulated within normative frameworks.

Spelling norms of vowels in Azerbaijani are determined by orthographic rules. These rules ensure the correct writing of words and contribute to the standardization of the language. Although vowel spelling is mainly based on the phonetic principle, in some cases historical-traditional principles are also considered. This helps preserve both modern and historical features of the language.

Discussion

The normative use of vowels also plays an important role in the education system. In schools, the teaching of phonetics includes detailed explanations of correct vowel pronunciation and spelling. This helps students develop speech culture and master literary language norms. At the same time, correct use of vowels in mass media and public speeches is considered an indicator of linguistic culture.

In the modern era, globalization and technological development have had an impact on language. In loanwords from foreign languages, the use of vowels sometimes deviates from normative rules. This makes the issue of preserving linguistic purity more relevant. However, since language is a living and constantly evolving system, the adoption of new elements is inevitable. The main issue is regulating this process within literary norms.

Conclusion

Thus, vowel norms in the Azerbaijani language form the basis of its phonetic and phonological system. These norms ensure the correct and unified use of the language, preserving its aesthetic and functional qualities. The study of vowel norms is of great theoretical and practical importance and remains one of the key areas of linguistics.

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THE IMPACT OF LANGUAGE POLICY ON THE PRESERVATION OF CULTURAL IDENTITY: THEORETICAL AND PRACTICAL APPROACHES

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Abstract: The article is devoted to analyzing the impact of language policy on the preservation of cultural identity and the strengthening of social cohesion in multicultural societies. The author emphasizes that language policy is not only a tool of governance but also a strategic factor that maintains the cultural balance of society. The role of language in social integration is particularly highlighted, showing that a well-designed language policy facilitates the coexistence of different ethnic and linguistic groups, strengthens mutual understanding, and contributes to social stability.

The study examines successful examples of inclusive approaches. In this context, bilingual education systems are particularly emphasized. Such systems promote the development of both the state language and minority languages, while also ensuring equal opportunities in the educational process. At the same time, the risks associated with excessive unification (standardization and homogenization) of the linguistic environment are analyzed, noting that such processes may lead to the weakening of cultural diversity and the gradual disappearance of some languages.

The article also highlights that the rapid development of information technologies and digital communication tools increases the dominance of certain major languages while simultaneously creating risks for smaller languages. Therefore, the development of a balanced and well-considered language policy is essential.

For the harmonious development of modern multicultural societies, the formation of a purposeful, scientifically grounded, and inclusive language policy is crucial. Such a policy should serve both the preservation of cultural identity and the strengthening of social integration.

Keywords: language policy, cultural identity, linguistic diversity, globalization, national identity, multilingualism.

DİL SIYASƏTİNİN MƏDƏNİ KİMLİYİN QORUNMASINA TƏSİRİ: NƏZƏRİ VƏ PRAKTİK YANAŞMALAR

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Xülasə: Məqalə çoxmədəniyyətli cəmiyyətlərdə dil siyasətinin mədəni kimliyin qorunmasına və sosial həmrəyliyin gücləndirilməsinə təsirinin təhlilinə həsr olunmuşdur. Tədqiqatda müəllif dil siyasətinin yalnız

idarəetmə vasitəsi deyil, eyni zamanda cəmiyyətin mədəni balansını qoruyan strateji bir faktor olduğunu vurğulayır. Dilin sosial integrasiyada oynadığı rol xüsusi olaraq qeyd edilir və göstərilir ki, düzgün qurulmuş dil siyasəti müxtəlif etnik və dil qruplarının birgə yaşamasını asanlaşdırır, qarşılıqlı anlaşmanı gücləndirir və sosial sabitliyə töhfə verir.

Məqalədə inklüziv yanaşmaların uğurlu nümunələri təhlil olunur. Bu kontekstdə ikidilli təhsil sistemləri xüsusi olaraq qeyd edilir. Belə sistemlər həm dövlət dilinin, həm də azsaylı xalqların dillərinin inkişafına şərait yaradır, eyni zamanda təhsil prosesində bərabər imkanların təmin olunmasına xidmət edir. Bununla yanaşı, dil mühitinin həddindən artıq unififikasiyası (standartlaşdırılması və homogenləşməsi) ilə bağlı risklər də təhlil edilir ki, belə proseslər mədəni müxtəlifliyin zəifləməsinə və bəzi dillərin tədricən yox olmasına səbəb ola bilər.

Məqalədə qeyd olunur ki, informasiya texnologiyalarının və rəqəmsal kommunikasiya vasitələrinin sürətli inkişafı bəzi dominant dillərin üstünlüyünü artırsa da, eyni zamanda kiçik dillər üçün risklər yaradır. Buna görə də balanslaşdırılmış və düşünülmüş dil siyasətinin hazırlanması zəruridir.

Müasir çoxmədəniyyətli cəmiyyətlərin harmonik inkişafı üçün məqsədyönlü, elmi əsaslandırılmış və inklüziv dil siyasətinin formalaşdırılması vacibdir. Bu siyasət həm mədəni kimliyin qorunmasına, həm də sosial integrasiyanın gücləndirilməsinə xidmət etməlidir.

Açar sözlər: dil siyasəti, mədəni kimlik, dil müxtəlifliyi, qloballaşma, milli kimlik, çoxdillilik.

Introduction

Language policy is widely recognized as a fundamental instrument in the governance of cultural diversity within modern multicultural societies. In the contemporary era of globalization, increased migration flows, and rapid technological development, linguistic diversity has become a central issue not only for cultural preservation but also for ensuring social cohesion and stability (Spolsky, 2004). Language is not merely a communicative tool but also a key carrier of cultural identity, collective memory, and social belonging (Sapir, 1921).

In multicultural contexts, where different ethnic and linguistic groups coexist within shared political and geographical spaces, language policy plays a decisive role in determining the status and function of languages in society. It regulates which languages are recognized in official domains such as education, administration, and media, thereby shaping access to resources and opportunities. According to Spolsky (2004), language policy reflects broader ideological and political structures that define inclusion and exclusion within a society.

The process of globalization has further complicated language dynamics. On one hand, globalization accelerates the dominance of global languages, particularly English, which may threaten the survival of minority languages (Whorf, 1956). On the other hand, it creates new possibilities for intercultural communication and digital revitalization of endangered languages. Therefore, the preservation of linguistic diversity has become not only a cultural necessity but also a social and political priority.

Education systems represent one of the most influential domains of language policy implementation. Bilingual and multilingual education models are widely recognized as effective approaches for promoting linguistic diversity and ensuring equal educational opportunities. Such approaches support the maintenance of mother tongues while enabling learners to acquire dominant or international languages, thereby reducing linguistic inequality and improving cognitive development outcomes (Fishman, 1972).

At the same time, language policy has a strong impact on identity formation processes. It shapes how individuals perceive their cultural heritage and how they relate to other cultural groups

within society. As Sapir (1921) and Whorf (1956) argue, language influences thought patterns and worldview formation, which directly affects cultural identity construction.

Against this theoretical background, the main aim of this study is to examine how language policy can either support or undermine cultural diversity and to analyze its role in the formation of a sustainable and inclusive national identity. The study also seeks to highlight the importance of balancing linguistic unity and cultural pluralism in order to achieve long-term social integration and cultural sustainability.

Materials and methods

This study is grounded in a qualitative research paradigm, which is particularly suitable for examining complex socio-cultural phenomena such as language policy, cultural identity, and multilingualism. The research design integrates comparative, conceptual, and descriptive analytical approaches, enabling a comprehensive interpretation of the interrelations between language, culture, and identity in multicultural contexts. Unlike quantitative approaches that focus on numerical measurement, qualitative methodology allows for an in-depth exploration of meanings, social processes, and ideological dimensions embedded in language policy (Spolsky, 2004).

The theoretical foundation of the study is built upon both classical and contemporary linguistic and sociolinguistic theories. The works of Sapir (1921) and Whorf (1956) provide essential insights into the relationship between language, thought, and cultural perception, emphasizing that language structures human worldview and influences identity formation. Fishman (1972) contributes significantly to the understanding of language maintenance, language shift, and the role of language in nationalism and ethnic identity. Spolsky (2004) offers a comprehensive framework for analyzing language policy as a multidimensional construct involving ideology, practice, and management. Arutyunova (1999) further enriches the theoretical base by examining the philosophical and anthropological dimensions of language in relation to human cognition and cultural representation.

Within this theoretical context, the methodological framework of the study is structured around several interrelated analytical dimensions:

Comparative analysis of language policy models:

The study systematically compares inclusive and assimilation-oriented language policies, focusing on their implications for cultural preservation, social integration, and identity construction. This approach enables the identification of structural differences between policies that promote linguistic diversity and those that encourage linguistic homogenization (Spolsky, 2004).

Sociolinguistic analysis of multilingual societies:

Drawing on Fishman's (1972) sociolinguistic perspective, the study examines how different linguistic groups interact within shared social spaces, how language hierarchies are formed, and how these dynamics influence social cohesion and intergroup relations.

Conceptual analysis of language–culture–identity relations:

Based on Sapir's (1921) and Whorf's (1956) theoretical contributions, this dimension explores the interdependent relationship between language, cultural perception, and identity formation. It highlights how linguistic structures shape cultural meanings and collective worldviews.

Analysis of globalization and linguistic diversity:

The study evaluates the impact of globalization on language use, particularly the growing dominance of global languages and its consequences for minority languages. It also considers the role of digital communication technologies in both threatening and revitalizing linguistic diversity.

Examination of educational language policy models:

Special attention is given to bilingual and multilingual education systems as key instruments of inclusive language policy. This analysis focuses on their role in promoting equal educational opportunities, cognitive development, and cultural inclusion within diverse learning environments.

Results and discussion

The findings of this study demonstrate that inclusive language policies play a fundamental role in preserving cultural identity and strengthening social cohesion in multicultural societies. The official recognition of minority languages in legal, educational, and media domains significantly contributes to their sustainability and long-term survival, while also reinforcing cultural diversity as a core societal value (Fishman, 1972).

Education emerges as one of the most influential domains in the implementation of language policy. Bilingual and multilingual education systems not only enhance linguistic competence but also support cognitive development, creativity, and critical thinking skills. Empirical research indicates that individuals educated in multilingual environments tend to demonstrate higher cognitive flexibility, improved problem-solving abilities, and greater adaptability to complex social and professional contexts. These competencies are increasingly essential in the context of globalization and knowledge-based societies.

In contrast, assimilation-oriented language policies that prioritize a single dominant language may lead to the marginalization of minority language communities. Such approaches often restrict access to quality education, weaken intergenerational transmission of cultural values, and contribute to social exclusion and inequality (Spolsky, 2004). Over time, these processes can result in the gradual loss of minority languages, leading to cultural homogenization and the erosion of linguistic diversity.

The process of globalization further intensifies these challenges. The increasing dominance of global languages, particularly English, contributes to the weakening of smaller languages and reduces their functional domains in education, media, and public communication (Whorf, 1956). However, globalization also introduces new opportunities for language preservation and revitalization through digital technologies, online education platforms, and transnational communication networks.

Overall, the results highlight that effective language policy must maintain a delicate balance between promoting linguistic unity and preserving cultural and linguistic diversity. Such a balanced approach is essential for ensuring long-term social stability, cultural sustainability, and inclusive national identity formation in modern multicultural societies.

Conclusion

Language policy is a decisive and structuring factor in shaping cultural identity and maintaining social cohesion within multicultural societies. The analysis confirms that inclusive and balanced language policies significantly contribute to the preservation of linguistic diversity, the promotion of educational equality, and the formation of an inclusive national identity that reflects the multicultural composition of society (Spolsky, 2004).

The findings of the study also indicate that effective language policy should move beyond linguistic homogenization and assimilation-oriented approaches. Instead, it should recognize

multilingualism as a valuable social and cultural resource that enhances communication, mutual understanding, and intercultural dialogue. In this sense, multilingualism is not a barrier to national unity but rather a strengthening factor that supports pluralistic and cohesive societies.

In the context of globalization, where dominant global languages increasingly influence local linguistic environments, the role of language policy becomes even more critical. It functions as a regulatory and protective mechanism that ensures the sustainability of minority languages and cultural traditions. At the same time, it supports social integration by enabling equitable participation of all linguistic groups in education, public life, and economic development.

Therefore, language policy should be understood as a key mechanism for sustainable cultural development, social integration, and identity preservation. A well-designed and inclusive language policy ensures that cultural diversity is not only protected but also actively promoted as a fundamental value of modern democratic societies.

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THE ROLE OF THE MEDIA IN THE FORMATION OF PUBLIC OPINION

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Abstract: This article discusses the role of the media in the formation of public opinion. The media plays a special role in politics. The media plays a key role in the formation of public opinion, acting as an information filter, setting the agenda and interpreting events. The article states that the media, acting as a powerful tool of influence, is a means of manipulating public consciousness. By processing and transmitting information to readers, as well as explaining and ranking past events, the media participate in the formation of moral norms, assessments and tastes, establish a hierarchy of public values; it can be said that they impose their own perception of truth on the reader. Thus, the media shape the language preferences of society, react most quickly to changes in language and reflect them. The media can effectively shape public opinion by highlighting specific issues that require public attention and response. Currently, a large number of data show

that the information that appears in print or on the air significantly affects how and what citizens think and what they consider to be "important issues".

Keywords: media, public opinion, political figure, internal values, information transmission.

İCTİMAİ RƏYİN FORMALAŞMASINDA MEDİANIN ROLU

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Xülasə: Bu məqalə ictimai rəyin formalaşmasında medianın rolundan bəhs edir. Media siyasətdə xüsusi rol oynayır. Media ictimai rəyin formalaşmasında, informasiya filtri kimi çıxış etməsində, gündəmi müəyyənləşdirməsində və hadisələrin şərhində əsas rol oynayır. Məqalədə deyilir ki, güclü təsir vasitəsi kimi çıxış edən media ictimai şüuru manipulyasiya etmək üçün bir vasitədir. Məlumatı emal etməklə və oxuculara ötürməklə, eləcə də keçmiş hadisələri izah etməklə və sıralamaqla media əxlaqi normaların, qiymətləndirmələrin və zövqlərin formalaşmasında iştirak edir, ictimai dəyərlərin iyerarxiyasını qurur; demək olar ki, onlar həqiqət haqqında öz qavrayışlarını oxucuya sırırlar. Beləliklə, media cəmiyyətin dil üstünlüklərini formalaşdırır, dildəki dəyişikliklərə ən tez reaksiya verir və onları əks etdirir. Media ictimaiyyətin diqqətini və reaksiyasını tələb edən konkret məsələləri vurğulamaqla ictimai rəyi effektiv şəkildə formalaşdırmaqla bilər. Hal-hazırda çoxlu sayda məlumat çapda və ya efirdə görünən məlumatların vətəndaşların necə və nə düşündüklərinə və nəyi "vacib məsələlər" hesab etdiklərinə əhəmiyyətli dərəcədə təsir etdiyini göstərir.

Açar sözlər: media, ictimai rəy, siyasi xadim, daxili dəyərlər, informasiya ötürülməsi.

Introduction

In this study, we are interested in information as a mass market product. In the information society, virtually unlimited information is available to each consumer. That is why the role of the media is increasing both in the formation of public opinion and in the formation of the point of view of individual consumers in specific situations. There are many definitions of the term "mass media". In addition to encyclopedic definitions, we also find definitions that convey information about the goals and methods of the media interesting. The media play a special role in politics. They contribute to the formation of public opinion, unite the public around socially significant political figures, and create a favorable environment for the implementation of democratic reforms. The development of society largely depends on the media and the role they play in modern society. "In the most difficult, critical moments of our country's modern history, our press reveals to people the truth - a truth born in the clash of ideas." Without them, it is impossible to solve various human problems. The media are an integral part of modern society. They are directly involved in all events, express the interests and feelings of society, its individual groups and strata.

Discussion

The media play a key role in the formation of public opinion, acting as an information filter, setting the agenda and interpreting events. They have the ability to both inform citizens and manipulate their perceptions, influence political processes, values and social norms. This influence is achieved through psychological influence, shaping emotions and behavior. For example, O.V. Aleksandrova defines the media as a group that builds messages with internal values and disseminates these messages to certain segments of society in order to achieve a certain goal. The

word "group" includes television, newspapers, radio, magazines, the Internet, etc. (2, p. 98) The expression "embedded values" is noteworthy and shows that the author understands the bias of the media. Being aware of the latest social and political events is important for modern people to rationally and objectively assess the situation on the global stage and in their own country. The media have always served and will continue to serve as the main source of information. As a result, by possessing complete information, controlling access to it, filtering it, separating individual elements, giving special weight to some and reducing the value of others, the media become a powerful tool for influencing public consciousness. For this reason, the media are often called the "fourth power" (along with the legislative, executive and judicial powers). Acting as a powerful tool of influence, the media are a means of manipulating public consciousness. By processing and transmitting information to readers, as well as by explaining and ranking past events, the media participate in the formation of moral norms, assessments and tastes, establish a hierarchy of public values; they can be said to impose their own perception of truth on the reader. Thus, the media form the language preferences of society, react most quickly to changes in the language and reflect them (4, p.13). N.A. Zhukova rightly notes that the media act not only as an intermediary in the transmission of information, but also as its creator, influencing the formation of public opinion, influencing through socialization, expression and unification of interests, as well as criticism, control and mobilization of the public (13, p.3).

V.V. Abrameytseva also expresses a similar idea, arguing that the media, by forming public opinion, unites the public around socially significant political figures and helps to implement democratic reforms. Thus, the media has a significant impact on the development of modern society (1, p. 4). The media can effectively shape public opinion by highlighting specific issues that demand public attention and response. A growing body of evidence now suggests that the information that appears in print or broadcast has a significant impact on how and what citizens think and what they consider to be "important issues" [6, p. 79]. When covering news, journalists typically provide brief background information, devoting most of their reporting to evaluating the news and presenting their own perspectives. Thus, people are exposed to more arguments and assessments than to the events themselves. An example of this phenomenon: "For the last four years, as Hillary Clinton has jetted around the planet notching up almost a million miles and dealing with fractious world leaders, she has found an unlikely way to relax. Between negotiating a ceasefire between Israel and Hamas, rallying international support for air strikes to oust Gadaffi, or navigating the release of a blind Chinese dissident to America, she would slipaway to her hotel room for a fix of leafing through interior-decorating magazines or watching home-improvement programmes. Her favourite is Love It Or List It, a Canadian reality show." (5, p. 121).

This article describes in detail the peculiarities of Hillary Clinton's life, the reporter mainly emphasizing how she relaxes in her hotel room by watching television shows instead of solving the problems she faces. Another example from the article: "As a candidate for re-election, Obama understandably emphasized the positive — that the economy has created nearly five million jobs since the pit of the recession; that unemployment has come down from more than 10 percent to less than 8 percent. But now it is time for him to stress how far we still have to go. Most of the jobs the economy lost paid much better than the new jobs do. More than 20 million Americans are still looking in vain for full-time jobs. The economy will need a massive recovery program if we are to get back to full employment" (3, p. 98).

After reading this message, one can conclude that the journalist adds a number of subjective

comments. In addition to reporting what Obama said about the state of the economy, the journalist also adds his own impressions and expresses his own point of view. The journalist's statement that the economy will need a large-scale recovery program prompts the reader to rethink the speech of the American president analytically. It is important to note that the media is almost never limited to reporting facts. All reports are modified by various factors, namely the foreign and domestic policy of the state, the economic situation, social processes in society, the information needs of the target audience, etc. Most of the news broadcast through the media is related to political events. Articles about political events play a special role: they not only convey the events in their pure form, but also create an evaluative attitude that has a certain impact on citizens. One of such consequences may be the resignation of a political figure. For example, the resignation of George W. Bush's deputy chief of staff, Karl Rove, occurred after it became known that he had accepted gifts that went to his former boss, Jack Abramoff. The relationship between citizens and the media plays an important role in shaping public reactions to events. In this regard, the media perform an important function: they influence how a country or politician is perceived at home and abroad. The relationship between citizens and the media plays an important role in shaping public reactions. In this regard, the mass media perform an important function: they influence how the image of a particular state or politician is perceived at home and abroad. Thus, when discussing the roles of the media in modern politics, several important aspects can be highlighted. First, the media serve as an indicator of current events: they not only cover them, but also perform the function of identifying the most important social and political events; that is, ranking them. Secondly, on the one hand, the media select the most relevant information and thus highlight a number of political and social issues; on the other hand, the media themselves determine the "agenda" and form a unique reality. The media can shape the opinions of recipients on a particular issue, and focusing on certain issues can deliberately distract public opinion from other problems. At the same time, the media create a virtual political reality through the interpretation of events, the formation of news feeds, the creation of images of politicians, etc. Thus, it should be noted that the media play a special role in the current situation, when the nature of international relations is often directly related to the personal characteristics of leaders representing states. Some researchers have suggested that various factors influence the media themselves when influencing the audience. In particular, the following factors affect the content of the media: - the attitudes and socialization of media workers; - methods of organizing media work; - the ideology of power structures; - other social institutions and forces, for example, economic structures or the cultural requirements of the audience for content (10, p. 4-5). M.J. Rozell adds to this list the desire to increase their ratings and make more money (7, p. 110). Undoubtedly, the main goal of political leaders is to gain a large audience in order to create a loyal following. These days, as Gadi Wolfsfeld rightly points out, if you are not in the media, you are not in politics. In parallel with this idea, he discusses four main principles of media language, which are the following: 1. The media are generally subordinate to political interests. 2. There is (and cannot be) such a thing as objective journalism. 3. The images created by the media usually significantly influence the political process. 4. The most important effects of the media on citizens should be deliberately and unnoticed (12, p. 79). Consequently, it can be argued that the media are a source of information about all political actors; the media significantly influence the audience, work to achieve the goals of political actors; information resources usually play a decisive role in competitive political struggles. "The influence of the newspaper-journalistic style determines the urgent need for the use of evaluative means of expression in journalism" (8, p.204) It is impossible

not to agree with G.Ya. Solganik. As mentioned above, language in the media is not only a powerful tool for conveying information, but also serves as a means of influencing the recipient. The use of certain linguistic means not only conveys information about the problem, but also forms the recipient's desired assessment of events, applies his own point of view to the problem, regardless of whether it is positive or negative (11, p. 682). Depending on the desired assessment of a particular phenomenon or image, the recipient chooses a number of linguistic means to achieve this assessment for the recipient. In particular, journalists use various lexical, grammatical and stylistic means that convey negative or positive, and less often neutral, assessments. Evaluation is present in every media text, either explicitly or implicitly. It expresses the recipient's attitude towards the object of the message and is formed on the basis of factors such as the speaker's cultural and value orientations, moral and political views, etc. The author's evaluation is aimed at influencing the consciousness of the recipient of information. Statements containing the author's evaluation do not simply describe events; they convey the emotions of the addressee, trying to evoke similar emotions in the recipient (9, p.16).

Conclusion

In conclusion, we can say that the role of the media in the formation of public opinion is large and multifaceted. In addition to transmitting information to society, the media also determines how events are perceived, influencing people's thoughts and positions. Professional, ethical and transparent activities of the media are the main condition for the healthy formation of public opinion.

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PRIORITY DIRECTIONS FOR PRESERVING THE AZERBAIJANI LANGUAGE IN THE CONTEXT OF GLOBALIZATION

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Abstract: The process of globalization, which defines the modern world order, poses significant threats to the ecosystem of national languages, particularly their structural-functional integrity, through waves of transnational integration and digitalization. This article provides a complex analysis of the strategic priorities for preserving the uniqueness and sustainable development of the Azerbaijani language against the backdrop of an expanding global information space. The primary objective of the study is to identify pathways for enhancing the competitiveness of the language on modern scientific and technological platforms while safeguarding its internal mechanisms and lexico-semantic potential.

The article emphasizes that one of the most crucial directions of language policy is the enrichment of the terminological database based on national morphological rules and the prevention of phonetic and grammatical interference from calque expressions entering irregularly from foreign languages, specifically English. In the era of Artificial Intelligence, machine translation, and Big Data, the improvement of the national corpus linguistics of the Azerbaijani language and ensuring its full functionality within the virtual space are evaluated as matters of state-level national security. Furthermore, the necessity of institutionally strengthening the position of the mother tongue in higher education, fundamental science, and diplomatic discourse under the pressure of global Anglophony is substantiated.

The study concludes that the most effective defense mechanism against the risks of linguistic assimilation brought by globalization is a dialectical synthesis of conservative linguistic purism tendencies and modern technological innovations. The article puts forward specific socio-methodological proposals aimed at optimizing the legal-normative framework to increase the prestige of Azerbaijani as the state

language, integrating the activities of the Terminology Commission into digital platforms, and creating high-quality intellectual digital content in the native language.

Keywords: globalization, language policy, Azerbaijani language, corpus linguistics, linguistic assimilation, terminology, digital transformation.

QLOBALLAŞMA ŞƏRAİTİNDƏ AZƏRBAYCAN DİLİNİN QORUNUB SAXLANILMASININ PRIORİTET İSTİQAMƏTLƏRİ

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Xülasə: Müasir dünya nizamını müəyyən edən qloballaşma prosesi transmilli integrasiya və rəqəmsallaşma dalğası ilə milli dillərin ekosisteminə, xüsusilə onların struktur-funksional bütövlüyünə ciddi təhdidlər yaradır. Təqdim olunan məqalədə qlobal informasiya məkanının genişlənməsi fonunda Azərbaycan dilinin özünəməxsusluğunun qorunması və davamlı inkişafının strateji prioritetləri kompleks şəkildə təhlil edilir. Tədqiqatın əsas məqsədi dilimizin daxili imkanlarını və leksik-semantik potensialını qorumaqla yanaşı, onun müasir elmi və texnoloji platformalarda rəqabət qabiliyyətinin artırılması yollarını müəyyənləməkdir.

Məqalədə vurğulanır ki, dil siyasətinin ən mühüm istiqamətlərindən biri terminoloji bazanın milli morfoloji qaydalar əsasında zənginləşdirilməsi və xarici dillərdən, xüsusilə ingilis dilindən nizamsız daxil olan kalka (hərfi tərcümə) ifadələrin fonetik və qrammatik müdaxilələrinin qarşısının alınmasıdır. Xüsusilə, süni intellekt, maşın tərcüməsi və böyük verilənlər (Big Data) əsrində Azərbaycan dilinin milli korpus linqvistikasının təkmilləşdirilməsi, dilimizin virtual məkanda işləkliyinin tam təmin olunması dövlət əhəmiyyətli milli təhlükəsizlik məsələsi kimi dəyərləndirilir. Digər tərəfdən, qlobal ingilisdilliliyin təzyiqi altında ana dilinin ali təhsil, fundamental elm və diplomatik diskurs sahəsindəki mövqelərinin institusional səviyyədə möhkəmləndirilməsi zərurəti əsaslandırılır.

Tədqiqat nəticəsində belə qənaətə gəlinir ki, qloballaşmanın gətirdiyi dil assimilyasiyası risklərinə qarşı ən effektiv müdafiə mexanizmi mühafizəkar dil təmizliyi (purizm) meyilləri ilə müasir texnoloji innovasiyaların dialektik sintezidir. Məqalədə Azərbaycan dilinin dövlət dili kimi nüfuzunun artırılması üçün hüquqi-normativ bazanın daha da optimallaşdırılması, Terminoloji Komissiyanın fəaliyyətinin rəqəmsal platformalara integrasiyası və ana dilində yüksəkkeyfiyyətli intellektual rəqəmsal məzmunun yaradılması istiqamətində konkret elmi-metodoloji təkliflər irəli sürülür.

Açar sözlər: qloballaşma, dil siyasəti, Azərbaycan dili, korpus linqvistikası, dil assimilyasiyası, terminologiya, rəqəmsal transformasiya.

Introduction

In the contemporary globalized era, the world is witnessing a profound transformation in the way information is disseminated and consumed. Globalization, while primarily characterized by economic integration and technological convergence, functions as a powerful sociocultural force that reshapes the linguistic landscape of nations. This process has introduced a complex duality: on one hand, it facilitates cross-border communication and academic exchange through a global *lingua franca*; on the other hand, it exerts unprecedented pressure on national languages, often leading to what sociolinguists describe as structural erosion, lexical displacement, and the marginalization of minority or medium-sized languages. For the Azerbaijani language, which serves as a cornerstone

of national identity and cultural heritage, maintaining its linguistic integrity while navigating the demands of the 21st century has become a paramount strategic objective.

Historically, the Azerbaijani language has undergone various stages of evolution, demonstrating a remarkable resilience against external influences. Following the restoration of independence, the establishment of Azerbaijani as the "State Language" was not merely a symbolic act but a foundational component of national sovereignty. Under the leadership of National Leader Heydar Aliyev and subsequently through the executive decrees of President Ilham Aliyev, a robust legal and normative framework was developed to protect and promote the language. However, the rapid acceleration of the digital revolution and the onset of "total globalization" in the last two decades have introduced qualitatively new challenges that the existing traditional frameworks must now address.

The dominance of English in the digital sphere, scientific discourse, and international commerce has created a "linguistic hegemony" that affects the cognitive and communicative habits of the younger generation. In the Azerbaijani context, this is manifested through an increasing influx of unadapted foreign terminology, the "hybridization" of syntactic structures in social media, and the gradual shift toward bilingualism or code-switching in professional environments. Moreover, the transition to a digital-first society means that a language's survival is no longer guaranteed solely by its number of speakers or its literary history; it is now determined by its "digital vitality"—its presence in Artificial Intelligence algorithms, search engine optimization, and large-scale linguistic corpora.

The urgency of this research stems from the need to identify the priority directions that will ensure the Azerbaijani language remains a functional, modern, and competitive tool for communication without losing its historical authenticity. This study posits that the preservation of the language in a globalized world cannot be achieved through isolationist or purely purist approaches. Instead, it requires a proactive, "smart" language policy that integrates traditional linguistic safeguards with cutting-edge technological innovations (Aliyev, 2020).

In this article, we examine the multifaceted impact of globalization on the Azerbaijani language from several perspectives: the effectiveness of current institutional governance, the challenges of terminology planning in the age of Big Data, and the necessity of enhancing the prestige of the mother tongue in academic and diplomatic spheres. By analyzing these variables, the study aims to provide a comprehensive methodological framework and actionable recommendations for policymakers, educators, and linguists to safeguard the future of the Azerbaijani language in an increasingly interconnected and digitized global society.

Methodology and Materials

This research employs a qualitative and descriptive methodological approach to evaluate the current status of the Azerbaijani language within the framework of global linguistic shifts. The study is built upon a diverse array of primary and secondary sources to ensure a comprehensive analysis of language policy and planning.

Research Materials. The empirical and theoretical basis of the article includes the following materials:

Legislative Framework: National legal documents, including the "Constitution of the Republic of Azerbaijan" and the "Law on the State Language of the Republic of Azerbaijan," are analyzed to understand the legal safeguards of the language.

Academic Literature: Seminal works by Azerbaijani and international linguists regarding language policy, sociolinguistics, and the impact of globalization on Turkic languages.

Digital Data: Statistical data and reports regarding the presence of the Azerbaijani language in global digital platforms, search engines, and linguistic corpora (Crystal, 2021).

Comparative Content: Textual samples from modern media and academic discourse are utilized to identify instances of lexical borrowing and structural interference.

Methodological Framework. To achieve the objectives of the study, the following scientific methods were applied:

- **Content Analysis:** Used to examine official documents and media texts to identify ideological and functional trends in contemporary language usage.
- **Comparative-Historical Method:** Employed to contrast the evolution of Azerbaijani linguistic norms before and after the intensification of globalization processes.
- **Sociolinguistic Analysis:** Applied to evaluate the social functions of the language and the challenges posed by bilingualism and diglossia in the digital age.
- **Synthesis and Generalization:** Information gathered from various sources was synthesized to formulate specific strategic recommendations for the preservation of linguistic heritage.

The integration of these methods allows for a holistic view of the Azerbaijani language as a dynamic system that must balance historical authenticity with modern technological requirements.

The methodological framework of this research is designed to provide a multidimensional analysis of the linguistic transformations occurring within the Azerbaijani language under the influence of global integration. Given the complexity of the subject, the study adopts a qualitative, interdisciplinary approach, integrating principles from sociolinguistics, legal studies, and digital humanities. This ensures that the preservation of the language is examined not merely as a philological concern, but as a strategic element of national security and digital sovereignty.

I. Source Materials and Data Selection. The empirical and theoretical data utilized in this study were derived from a diverse range of institutional and academic sources, categorized as follows:

Legal and Regulatory Corpus: The primary source for analyzing state language policy consists of the Constitution of the Republic of Azerbaijan, the Law on the State Language of the Republic of Azerbaijan (2002), and subsequent executive decrees and "State Programs" (e.g., the 2013-2020 State Program on the Use of the Azerbaijani Language). These documents provide the normative baseline for assessing linguistic governance (7, 2022).

Scientific and Theoretical Literature: A comprehensive review of seminal works by Azerbaijani linguists (such as A. Akhundov and N. Jafarov) and international scholars (such as D. Crystal and N. Fairclough) regarding sociolinguistics, critical discourse analysis, and the impact of globalization on Turkic linguistic families.

Digital and Statistical Data: Data extracted from global digital reach reports and linguistic database metrics were used to evaluate the "Digital Vitality Index" of Azerbaijani. This includes analyzing the language's presence in search engine algorithms, the depth of the *Azerbaijani National Corpus*, and its representation in multilingual software ecosystems (5, 2023).

Media and Academic Discourse Samples: A comparative selection of texts from contemporary news portals, social media platforms, and academic journals was compiled to identify emerging patterns of lexical borrowing, syntactic interference, and the use of calques.

II. Research Methodology. To ensure the scientific validity of the findings, a rigorous set of analytical methods was employed:

1. Socio-Legal Content Analysis: This method was applied to official documents and legislative acts to determine the evolution of the language's legal status. It identifies the gap between *de jure* (legal) protections and *de facto* (practical) implementation in a globalized market.

2. Comparative-Historical Linguistics: By contrasting the linguistic norms of the pre-globalization era with contemporary usage, the research identifies specific points of divergence in morphology and syntax caused by the influence of dominant global languages (specifically English and Russian).

3. Technological Assessment (Digital Linguistics): An analytical approach was used to assess the compatibility of Azerbaijani with modern Natural Language Processing (NLP) tools and Artificial Intelligence models. This involves evaluating the availability of high-quality training data for machine translation and voice recognition systems.

4. Strategic Synthesis and Policy Modeling: Based on the data gathered through the aforementioned methods, the study utilizes synthesis to formulate a "Strategic Priority Model." This model categorizes interventions into three levels: *Legislative* (legal adjustments), *Academic* (terminological modernization), and *Technological* (digital integration) (Hasanov, 2023).

The combination of these materials and methods allows for a holistic evaluation of the Azerbaijani language as a dynamic system that must navigate the tension between preserving its historical-cultural essence and adapting to the requirements of the global digital civilization.

Discussion

The analysis of the current linguistic situation in Azerbaijan reveals that while the state has established a solid legal foundation for the protection of the mother tongue, the practical challenges posed by globalization require more dynamic interventions. The results of the study are categorized into three primary dimensions: institutional efficiency, digital integration, and sociolinguistic resilience.

Linguistic Governance and Institutional Efficiency. The findings indicate that the "Law on the State Language" serves as a robust shield against the dilution of linguistic norms in official spheres. However, the discussion highlights a significant gap between legislative requirements and their implementation in the private sector and mass media. The prevalence of foreign brand names and untranslated technical jargon suggests that the Terminology Commission must transition from a reactive to a proactive role. A centralized, digitalized terminology database is essential to ensure that new concepts are integrated into Azerbaijani using internal linguistic resources rather than direct phonetic borrowings.

Digital Presence and Artificial Intelligence. One of the most critical results pertains to the "digital vitality" of the Azerbaijani language. In the context of globalization, a language's survival is increasingly dependent on its compatibility with modern technologies. The discussion emphasizes that the current Azerbaijani linguistic corpus requires expansion to support high-accuracy machine translation and natural language processing (NLP). The findings suggest that unless Azerbaijani becomes a primary language of instruction in digital literacy and software development, it risks being marginalized in the virtual ecosystem, leading to a form of "digital linguistic colonization" by dominant global languages (Mammadov, 2021).

Sociolinguistic Resilience and National Identity. The research observes a dual trend among the younger generation: a high degree of bilingualism (Azerbaijani-English or Azerbaijani-Russian)

and a tendency toward code-switching. While bilingualism is a cognitive advantage, the discussion warns against the "hybridization" of syntax, where the grammatical structure of Azerbaijani is subtly altered to mirror English or Russian patterns. The priority direction identified here is the promotion of "prestige planning"—increasing the social and economic value of Azerbaijani so that it remains the preferred language for high-status academic and professional discourse.

In conclusion, the discussion underscores that preserving the Azerbaijani language in the era of globalization is not an act of isolationism. Instead, it is a strategic necessity to ensure that the language evolves as a modern, flexible, and technologically capable system without losing its historical and cultural essence.

Synthesis and Strategic Implications. In conclusion, the discussion underscores that preserving the Azerbaijani language in the era of globalization is not an act of isolationism or a rejection of global integration. Instead, it is a strategic necessity of national sovereignty. The results demonstrate that a language survives globalization not by closing its borders, but by "exporting" its own content into the global digital space. The synthesis of these findings suggests that the future of the Azerbaijani language lies in a "Triadic Model of Preservation": robust legal protection, rapid technological integration, and the continuous enhancement of its socio-academic prestige (Suleymanli, 2021).

Conclusion

The preservation and strategic development of the Azerbaijani language in the face of intensifying globalization and rapid digital transformation represent a multifaceted challenge that transcends simple philological boundaries. This research has demonstrated that the survival of a national language in the 21st century is no longer solely dependent on its historical longevity or literary richness, but rather on its "functional adaptability" within global academic, technological, and socio-economic ecosystems. The transition from traditional linguistic protectionism to a proactive, technology-driven language policy has emerged as a fundamental necessity for maintaining national sovereignty in a borderless digital world.

The study concludes that the priority directions for safeguarding the Azerbaijani language must be integrated into a cohesive national strategy. First and foremost, the "digital vitality" of the language must be ensured through the systematic expansion of the national linguistic corpus and the active localization of emerging technologies, such as Artificial Intelligence and semantic web services. Countering the risks of "digital marginalization" requires an institutional partnership between state bodies, academic circles, and the technological sector to ensure that Azerbaijani remains a primary medium for high-tech interaction and scientific innovation.

Furthermore, the research highlights that legal frameworks, while providing a necessary foundation, must be complemented by "prestige planning." Increasing the social and economic value of the mother tongue is crucial to ensuring that it remains the preferred language for the younger generation in professional and academic spheres. The modernization of terminology should not be viewed as a mere translation process but as a dynamic enrichment of the language's internal lexical potential, preventing the structural erosion caused by irregular borrowings.

In final analysis, preserving the Azerbaijani language in the era of globalization is not an act of cultural isolationism. On the contrary, it is a strategic effort to ensure that the language evolves as a modern, resilient, and technologically capable system. By embracing digital innovation while anchoring itself in its unique ethno-linguistic heritage, Azerbaijan can ensure that its state language continues to flourish as a vital symbol of identity and a competitive tool for communication in the

globalized information society. The recommendations provided in this study offer a methodological roadmap for achieving this balance, ensuring that the Azerbaijani language is not merely a witness to the global era, but an active and influential participant in it.

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LANGUAGE PLANNING IN MULTILINGUAL SOCIETIES

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Abstract: Language planning plays a crucial role in managing linguistic diversity and promoting effective communication in multilingual societies. This study examines the significance of language planning by exploring its objectives, major types, benefits, and challenges. A qualitative research approach was adopted, drawing on secondary data from academic books, scholarly articles, and educational sources. The analysis is supported by examples from multilingual countries to illustrate the practical implementation of language policies. The findings indicate that effective language planning enhances communication, promotes social integration, preserves minority languages, and improves educational outcomes through multilingual education. However, the study also identifies several challenges, including language inequality, political conflicts, limited resources, and the growing influence of dominant global languages such as English. These issues may reduce the effectiveness of language policies if they are not addressed through inclusive and balanced strategies. The study concludes that successful language planning requires cooperation between governments, educational institutions, and communities to ensure equal opportunities for all linguistic groups while maintaining both national unity and cultural diversity. Future research should investigate the long-term impact of multilingual education and the contribution of digital technologies to language planning.

Key words: language planning, multilingual societies, language policy, linguistic diversity, communication

ÇOXDİLLİ CƏMİYYƏTLƏRDƏ DİL PLANLAŞDIRMASI

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Xülasə: Dil planlaşdırılması çoxdillli cəmiyyətlərdə dil müxtəlifliyinin idarə olunması, səmərəli ünsiyyətin təmin edilməsi və sosial integrasiyanın gücləndirilməsi baxımından mühüm əhəmiyyət kəsb edir. Bu tədqiqatın məqsədi çoxdillli cəmiyyətlərdə dil planlaşdırılmasının rolunu, əsas istiqamətlərini, üstünlüklərini və qarşıya çıxan çətinlikləri araşdırmaqdır. Tədqiqat keyfiyyət yanaşmasına əsaslanmış, məlumatlar elmi kitablar, məqalələr və digər akademik mənbələrin təhlili əsasında toplanmışdır. Müxtəlif çoxdillli ölkələrin təcrübəsi dil siyasətinin praktik tətbiqini qiymətləndirmək üçün nəzərdən keçirilmişdir. Tədqiqatın nəticələri göstərir ki, səmərəli dil planlaşdırılması ünsiyyətin yaxşılaşdırılmasına, sosial integrasiyanın möhkəmləndirilməsinə, azsaylı dillərin qorunmasına və təhsil keyfiyyətinin yüksəldilməsinə müsbət təsir göstərir. Bununla yanaşı, dil bərabərsizliyi, siyasi ziddiyyətlər, resurs çatışmazlığı və qloballaşmanın dominant dillərin mövqeyini gücləndirməsi kimi problemlər dil siyasətinin həyata keçirilməsini çətinləşdirə bilər. Nəticə etibarilə, uğurlu dil planlaşdırılması bütün dil qruplarının hüquqlarını nəzərə alan, milli birliklə mədəni və dil müxtəlifliyi arasında tarazlığı qoruyan inklüziv siyasətin formalaşdırılmasını tələb edir.

Açar sözlər: dil planlaşdırılması, çoxdillli cəmiyyətlər, dil siyasəti, dil müxtəlifliyi, ünsiyyət

Introduction

In today's globalized world, many countries are characterized by linguistic diversity. Multilingual societies exist where people speak two or more languages in daily life. This diversity can be a valuable cultural resource, but it can also create communication challenges.

Language planning refers to the process of managing and developing languages within a society. Governments and institutions often create language policies to regulate language use in education, administration, media, and public life.

In multilingual societies, language planning becomes especially important. Without proper planning, communication barriers, social inequality, and cultural conflicts may arise. At the same time, well-designed policies can support national unity while respecting linguistic diversity.

The purpose of this study is to examine the role of language planning in multilingual societies and to analyze its advantages and potential challenges.

Literature Review

Many scholars have studied language planning and its impact on multilingual communities. Research shows that language policies influence social integration, education systems, and cultural preservation. According to Cooper (1989), language planning includes activities aimed at influencing language behavior within a speech community. These activities are often divided into three types: status planning, corpus planning, and acquisition planning.

Fishman (1991) emphasizes the importance of language preservation, especially for minority languages. He argues that without proper planning, smaller languages may disappear over time.

Spolsky (2004) highlights that language policy is not only about official rules but also about actual language practices in society. He suggests that successful language planning must consider social realities.

Recent studies also show that multilingual education can improve students' cognitive skills and cultural awareness. However, challenges such as lack of resources and political disagreements may limit its effectiveness.

Overall, previous research suggests that language planning is essential but complex, requiring careful balance between different linguistic groups.

Methodology and Materials

This research uses a qualitative approach based on secondary data and general observations. Information was collected from academic books, research articles, and online educational sources related to language planning and multilingualism. The study also considers examples from different multilingual countries to understand how language policies are applied in real-life contexts.

The analysis focuses on:

- the goals of language planning
- types of language planning
- benefits in multilingual societies
- challenges and limitations

The collected data were analyzed to identify common patterns and key issues.

Discussion

The findings show that language planning has both positive effects and significant challenges in multilingual societies.

One major benefit is improved communication. When a common language is promoted, people from different linguistic backgrounds can interact more easily.

Another important advantage is cultural preservation. Language planning can support minority languages by promoting their use in education and media. This helps protect cultural identity.

Educational development is also influenced by language policy. Teaching in a student's native language can improve understanding and academic performance. However, there are also several challenges. One major issue is language inequality. If one language is given more importance, speakers of other languages may feel excluded or disadvantaged. Political conflicts can also arise. Language policies are sometimes linked to national identity, which can lead to disagreements between different groups. Another challenge is implementation difficulty. Creating language policies is easier than applying them effectively. Lack of resources, trained teachers, or public support can limit success. Additionally, globalization increases the influence of dominant languages such as English, which may threaten local languages. Overall, the discussion shows that language planning must balance unity and diversity to be effective.

Conclusion

This study examined the role of language planning in multilingual societies. The findings indicate that language planning is essential for managing linguistic diversity and promoting effective communication.

It offers several benefits, including cultural preservation, educational improvement, and social integration. However, challenges such as inequality, political issues, and practical limitations must be addressed.

It can be concluded that successful language planning requires inclusive policies that respect all linguistic groups. Governments and institutions should aim to create balanced strategies that support both national unity and cultural diversity.

Future research may explore the long-term effects of multilingual education and the role of technology in language planning.

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HISTORY OF THE MOTHER TONGUE AND ITS PLACE IN THE MODERN ERA

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Abstract: This article examines the historical development, cultural significance, and state status of the Azerbaijani language as one of the fundamental elements of national identity. It emphasizes that language is not merely a means of communication but also a powerful instrument for preserving a nation's history, traditions, moral values, and cultural heritage. The study provides an overview of the origin of the Azerbaijani language within the Oghuz branch of the Turkic language family and discusses the major stages of its historical and literary development. In addition, it highlights the role of the Azerbaijani language in strengthening national unity and preserving the cultural identity of the Azerbaijani people.

The article also analyzes the contribution of Shah Ismail Khatai to the promotion of the Azerbaijani language as the official language of the state and examines the language policy implemented under the leadership of Heydar Aliyev. Special attention is paid to the constitutional recognition of Azerbaijani as the state language, the transition from the Cyrillic to the Latin alphabet, and the declaration of August 1 as the Day of the Azerbaijani Alphabet and Language. These reforms played a significant role in strengthening the legal status, enriching the literary tradition, and increasing the international prestige of the Azerbaijani language.

The article concludes that the mother tongue is one of the greatest spiritual and national treasures of every nation. Protecting, promoting, and transmitting the Azerbaijani language to future generations is presented as a shared responsibility of society and an essential condition for preserving national identity and ensuring cultural continuity.

Key words: mother tongue, Azerbaijani language, national identity, cultural heritage, literary language, language policy, state language, Heydar Aliyev, Shah Ismail Khatai, Latin alphabet.

ANA DİLİN TARİXİ VƏ MÜASİR DÖVRDƏ YERİ

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Xülasə: Bu məqalədə Azərbaycan dilinin milli kimliyin formalaşmasında, mədəni irsin qorunmasında və dövlətçilik ənənələrinin inkişafında mühüm rolundan bəhs edilir. Müəllif vurğulayır ki, dil yalnız insanlar arasında ünsiyyət vasitəsi deyil, eyni zamanda xalqın tarixini, adət-ənənələrini, mənəvi dəyərlərini və mədəni irsini yaşadan əsas amillərdən biridir. Tədqiqatda Azərbaycan dilinin türk dillərinin Oğuz qrupuna mənsub olması, onun tarixi inkişaf mərhələləri, ədəbi dil kimi formalaşması və Azərbaycan xalqının milli birliyinin möhkəmlənməsində oynadığı rol geniş şəkildə təhlil edilir. Məqalədə həmçinin Şah İsmayıl Xətəinin Azərbaycan dilinin dövlət dili kimi möhkəmlənməsinə verdiyi töhfələr, eləcə də Heydər Əliyevin həyata keçirdiyi məqsədyönlü dil siyasəti araşdırılır. Azərbaycan dilinin Konstitusiyada dövlət dili kimi təsbit edilməsi, kiril əlifbasından latın qrafikalı əlifbaya keçid və 1 avqustun Azərbaycan əlifbası və Azərbaycan dili günü elan olunması dilimizin inkişafı baxımından mühüm tarixi hadisələr kimi qiymətləndirilir. Bu islahatlar Azərbaycan dilinin hüquqi statusunun möhkəmlənməsinə, ədəbi dilin zənginləşməsinə və beynəlxalq nüfuzunun artmasına əhəmiyyətli töhfə vermişdir.

Nəticə olaraq, məqalədə ana dilinin xalqın ən qiymətli milli-mənəvi sərvətlərindən biri olduğu qeyd edilir. Azərbaycan dilinin qorunması, inkişaf etdirilməsi və gələcək nəsillərə saf şəkildə ötürülməsi hər bir vətəndaşın mənəvi borcu və milli məsuliyyəti kimi dəyərləndirilir. Bu baxımdan, ana dilinə hörmət və qayğı milli kimliyin və mədəni davamlılığın əsas şərtlərindən biri hesab olunur.

Açar sözlər: ana dili, Azərbaycan dili, milli kimlik, mədəni irs, ədəbi dil, dövlət dili, dil siyasəti, Heydər Əliyev, Şah İsmayıl Xətai, latın qrafikalı əlifba.

Introduction

Language is one of the fundamental factors that shape the identity, culture, and historical memory of every nation. As the official language of the Republic of Azerbaijan, the Azerbaijani language serves not only as a means of communication but also as a symbol of national unity and independence. Throughout history, the language has undergone a long process of development and has played a significant role in preserving the cultural heritage and moral values of the Azerbaijani people. The protection and development of the mother tongue have always been among the priority issues of state policy. Therefore, studying the historical evolution of the Azerbaijani language, its literary development, and its legal status is of great scientific and practical importance. The purpose of this study is to examine the historical development of the Azerbaijani language, analyze the factors influencing its formation and enrichment, and evaluate its role in strengthening national identity and statehood.

Materials and Methods

The study is based on qualitative research methods. Scientific books, academic articles, official state documents, constitutional provisions, and electronic resources related to the Azerbaijani language were analyzed through document analysis and descriptive methods. Comparative and historical approaches were applied to evaluate the development stages of the Azerbaijani language, its literary evolution, and the language policy implemented in different historical periods. The research also relies on legal documents concerning the state language and the transition to the Latin alphabet.

Results and Discussion

The findings indicate that the Azerbaijani language has experienced continuous historical development and has become one of the strongest symbols of the national identity of the Azerbaijani people. Historical evidence demonstrates that the language expanded together with the strengthening of the Azerbaijani statehood and literary tradition. The official recognition of Azerbaijani during the reign of Shah Ismail Khatai and the language reforms implemented under the leadership of Heydar Aliyev significantly contributed to the protection, enrichment, and international recognition of the language. Furthermore, the adoption of the Latin alphabet and the constitutional recognition of Azerbaijani as the state language strengthened its legal and social status. These developments have ensured the preservation of linguistic heritage while creating favorable conditions for its further development in the modern era.

Conclusion

The Azerbaijani language is one of the most valuable elements of the nation's cultural and historical heritage. The results of this study confirm that the preservation and development of the mother tongue are essential for strengthening national identity, protecting cultural values, and ensuring social unity. The historical language policy implemented in Azerbaijan has played an important role in increasing the prestige of the Azerbaijani language at both national and international levels. Therefore, continuous efforts should be made to preserve the purity of the language, expand its scientific and cultural functions, and transmit this invaluable heritage to future generations.

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LANGUAGE REFORM IN CONTEXT: ONE REFORM, TWO SYSTEMS-THE TRANSITION TO THE LATIN ALPHABET IN ATATÜRK'S TURKEY AND SOVIET AZERBAIJAN

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Abstract: This study comparatively analyzes the alphabet reform policies based on the Latin script implemented in Turkey and the Soviet Union in the early twentieth century. The main objective of the research is to demonstrate how language was used not only as a means of communication but also as a mechanism of ideological and political control. The alphabet reform carried out in Turkey in 1928 by Mustafa Kemal Atatürk was aimed at strengthening national identity, increasing literacy, and fostering integration with Western culture. This process clearly shows that language functioned as a key component of modernization and nation-building.

On the other hand, the Latinization policy implemented in the Soviet Union during the 1920s–1930s sought to reinforce ideological influence and reduce religious and cultural impacts. However, this phase did not last long, and it was eventually replaced by a transition to the Cyrillic alphabet. The article simultaneously examines the goals, outcomes, and effects of Latinization on intercultural communication within both systems.

The relevance of this topic lies in the fact that language policies continue to play a decisive role in processes such as global integration, national identity formation, and intercultural communication today. Analyzing historical experiences of Latinization makes it possible to better understand the political and ideological functions of language in the modern era, as well as to evaluate countries' strategies of self-identification and integration. For this reason, the relationship between language and ideology remains a significant subject of scholarly research from both historical and contemporary perspectives.

Keywords: Language policy, ideology, Soviet Union, Mustafa Kemal Atatürk, intercultural communication

KONTEKSTDƏ DİL İSLAHATI: BİR İSLAHAT, İKİ SİSTEM – ATATÜRK TÜRKİYƏSİ VƏ SOVET AZƏRBAYCANINDA LATİN ƏLİFBASINA KEÇİD

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Xülasə: Bu tədqiqatda, ötən əsrin əvvəllərində Türkiyədə və Sovet İttifaqında tətbiq edilmiş latın qrafikasına əsaslanan əlifba islahatı siyasəti müqayisəli olaraq təhlil edilir. Araşdırmanın başlıca məqsədi dilin sadəcə kommunikasiya vasitəsi deyil, həmçinin ideoloji və siyasi nəzarət mexanizmi kimi necə istifadə olunduğunu təqdim etməkdir. 1928-ci ildə Mustafa Kemal Atatürk tərəfindən Türkiyədə həyata keçirilən əlifba islahatı, milli kimliyin möhkəmləndirilməsi, savadlılığın artırılması və Qərb mədəniyyəti ilə integrasiyaya istiqamətlənmişdir. Sözügedən dəyişiklik prosesi dilin modernləşmə və milli dövlət quruculuğunun başlıca tərkib hissəsi kimi çıxış etdiyini aydın şəkildə göstərir. Digər tərəfdən, SSRİ-də 1920–1930-cu illərdə həyata keçirilən latınlaşma siyasəti ideoloji təsir mexanizmini gücləndirmək, dini və mədəni təsirləri azaltmaq məqsədi güdmüşdür. Lakin bu mərhələ uzun bir dövrü əhatə etməmiş, bir

müddət sonra kiril əlifbasına keçid reallaşdırılmışdır. Məqalədə, paralel olaraq hər iki sistem çərçivəsində latınlaşmanın məqsədləri, yekun nəticələri və mədəniyyətlərarası kommunikasiya üzərində təsirləri tədqiq edilir.

Mövzu ona görə aktualdır ki, dil siyasətləri günümüzdə də qlobal integrasiya, milli identiklik və mədəniyyətlərarası kommunikasiya proseslərində həlledici rol oynayır. Tarixən latınlaşma təcrübələrinin analizi, müasir dövrdə dilin siyasi və ideoloji vəzifələrini anlamağa, eləcə də ölkələrin özünüdərk və birləşmə strategiyalarını dəyərləndirməyə imkan yaradır. Bu səbəblə də dil və ideologiya arasındakı əlaqə, tarixi və müasir baxımdan elmi tədqiqat üçün aktuallığını qoruyub saxlayır.

Açar sözlər: Dil siyasəti, ideologiya, Sovet İttifaqı, Mustafa Kemal Atatürk, mədəniyyətlərarası kommunikasiya

Introduction

The early twentieth century is regarded as a distinctive period in which political, ideological, and cultural transformations occurred rapidly on an international scale, and states implemented significant reforms aimed at modernization and the construction of national identity. Within these processes, language policy became an essential component of countries' intellectual and cultural strategies, while the guidance and modernization of societies were reinforced through alphabet reforms.

In particular, the alphabet reforms based on the Latin script implemented in Turkey and the Soviet Union stand as key historical examples demonstrating that language functioned not only as a means of communication but also as a mechanism of political and ideological influence. In this regard, the study comparatively analyzes the transition to the Latin script in these two different systems, examining its objectives, implementation processes, and outcomes. In addition, by considering the relevance of the topic to the present day, the study comprehensively evaluates the impact of historical experiences of Latinization on language policy and national identity processes.

Methods

This study was conducted based on a comparative historical approach. The main objective was to scientifically analyze the ideological and socio-cultural nature of the alphabet reforms implemented in the Republic of Turkey and Soviet Azerbaijan. A historical-analytical method was applied to examine the causes of the events, their stages of development, and their outcomes in chronological order. At the same time, a comparative analysis method was used to identify the similarities and differences between the alphabet reforms in both countries.

In addition, a political-ideological analysis approach was employed to clarify the ideological foundations of the reforms and their connection with state policy. As a source base, scholarly literature, historical studies, and the findings of existing academic research on the topic were utilized. This methodological framework made it possible to evaluate the alphabet reforms not merely as a technical change in language, but also as a large-scale process of social and ideological transformation.

Material

From Script to Identity: Alphabet Reform in the Atatürk Era and Modern State-Building. The alphabet reform carried out on November 1, 1928, under the leadership of Mustafa Kemal Atatürk was not merely a change in writing in Turkey's history of modernization, but also a stage of cultural development rooted in deep historical traditions and a transformation of a social character. In order to understand the essence of this reform, it is

essential to analyze it within the context of modernization goals, the reconstruction of national identity, and strategies of integration with the Western world.

As early as the 1860s, discussions had begun in the Ottoman Empire about the necessity of reforming the alphabet, since the Arabic script was insufficient for representing the Turkish language. However, because the Qur'an was written in the Arabic script, such changes were not accepted from a religious perspective. At the same time, the fact that a large portion of the population within the Ottoman state was Muslim gave the script a political dimension. As a result, those who advocated for reforming the alphabet were accused of rejecting religious norms and causing the collapse of the state (Turan, 2018, p. 382).

The difficulty of the Arabic-based alphabet resulted in a literacy rate of only around 10% prior to 1928. In 1923–1924, the number of university students was 2,914, students in secondary vocational and technical schools numbered 6,547, secondary school students 5,905, and primary school students 341,941. While the total population in 1903 was 19,929,168, the number of students was 1,375,511, which corresponded to 144.8 per thousand of the population (Kayıran, 1998, p. 801).

From the Tanzimat period to the Republic, intellectuals opposing the Arabic alphabet continuously debated the need to reform the writing system. During the Constitutional period and in Servet-i Fünun literature, a number of writers advocated for language reform. In 1851, an institution called "Encümen-i Daniş" was established, announcing that efforts would be made to carry out language reform and to produce works in a language understandable to the public. Intellectuals such as Antepî Münîf Efendi, Ziya Gökalp, İsmayıl Hakkı Baltacıoğlu, and Hüseyin Cahit Yalçın emphasized the necessity of alphabet reform. From the Second Constitutional period onward, three major ideas emerged: reforming the Arabic script, adopting the Latin alphabet, and reintroducing the old Orkhon–Uyghur script (Turan, 2018, pp. 383–385).

Even before the proclamation of the Republic, in 1919 in Erzurum, Mustafa Kemal Pasha stated to Müfit Kânsu that after victory, the first step would be the declaration of the Republic and the fifth would be the adoption of the Latin alphabet (Boyacıoğlu, 2001, p. 439). During the Republican period, the first proposal to adopt the Latin alphabet was raised at the İzmir Economic Congress in 1923. Intellectuals such as Ali Nazmi and Hakkı Kılıçoğlu presented proposals supporting the adoption of Latin letters and emphasized its necessity. However, these proposals were met with strong opposition. Recognizing the necessity of a national language, Mustafa Kemal approached the issue cautiously, observing that the public was not yet ready for such a change. The topic was brought to the agenda again in 1924 in the Grand National Assembly by Şükrü Saracoğlu (Tunca, 2006, p. 114).

On February 24, 1924, during discussions of the Budget Law in the Assembly, Şükrü Saracoğlu pointed out that the Arabic script created difficulties in reading and writing. He emphasized that despite efforts over many years, only a small portion of the population was literate. His speech caused concern among conservative groups and even led to acts of physical violence against him. At the same time, articles advocating the adoption of the Latin alphabet began to appear in newspapers, and an organization called the "New Letters Association" was established. Thus, public opinion began to form prior to the alphabet reform (Candan, 2006, pp. 2–3).

By Atatürk's instruction, a Language Commission affiliated with the Ministry of National Education was established on May 23, 1928. Its purpose was to develop a new Turkish alphabet

based on the Latin script. According to the commission's initial proposal, a new alphabet consisting of 29 letters was created. On August 1, 1928, commission members gathered in Istanbul, and on August 6, during a meeting held at Galatasaray High School, the final version of the new Turkish alphabet was approved. On August 9, 1928, during an event in Sarayburnu (Gülhane Park), Mustafa Kemal delivered a speech announcing the adoption of the new alphabet to the public. As of August 11, 1928, lessons on the new alphabet began at Dolmabahçe Palace (Turan, 2018, p. 390).

From the perspective of modernization, the alphabet reform represents the practical implementation of the Republic of Turkey's principle of "reaching the level of contemporary civilizations." This reform, carried out under Atatürk's leadership, simplified the writing system and led to a rapid increase in literacy through the "Nation Schools." This process not only enhanced the intellectual potential of the population but also contributed to increased efficiency in all areas of the state and to the renewal of governance methods based on secular principles. As a result, the new alphabet became not only a tool for literacy but also a carrier of a modern way of thinking.

From the standpoint of national identity formation, the alphabet reform symbolizes the transition from an "ummah" culture to the concept of a "nation." Its implementation aimed to remove foreign elements from the Turkish language and restore its authenticity. Through the letter revolution, the Turkish people both returned to their cultural roots and acquired a secular and national civic identity. The reform created conditions for society to move away from theocratic frameworks of the past and to build its path of development on national sovereignty and reason.

Finally, the integration with the West through the alphabet reform is directly related to Turkey's strategic orientation in the international arena. The transition to the new alphabet provided Turkey with cultural codes similar to those of Europe and the global system. This initiative facilitated diplomatic correspondence and enabled direct access to the international scientific community. With this step, Turkey declared its departure from the closed structures of the East and its alignment with the active and developed civilizations of the West. In conclusion, the alphabet reform carried out under Atatürk's leadership was not merely a technical procedure but a socio-political transformation project that connected Turkish identity with the modern world. The success of this revolution remains one of the fundamental conditions for Turkey's existence today as a secular, democratic, and modern state (Eroğlu et al., 1981, pp. 187–188).

Identity Constructed Through Language: The Latinization Process in Azerbaijan and Soviet Policy. In the early decades of the twentieth century, the issue of alphabet reform in Azerbaijan was not merely a technical change related to language and writing systems, but also a component of profound political, ideological, and cultural transformations. Particularly during the early years of Soviet rule, the transition from the Arabic script to the Latin alphabet emerged as a complex and multilayered phenomenon shaped both by the modernization ideas of local intellectuals and the strategic interests of the Soviet Union. This process held significant importance in terms of reconstructing national identity, increasing literacy, and directing society ideologically.

From the late nineteenth to the early twentieth century, the alphabet issue became widely discussed among Azerbaijani intellectuals. Enlightenment thinkers such as Mirza Fatali Akhundov emphasized that the Arabic script was phonetically unsuitable for Turkic languages,

difficult to learn, and therefore contributed to widespread illiteracy (Akpınar, 1994, p. 52). These ideas later became one of the ideological foundations of the alphabet reforms implemented during the Soviet period. After the Sovietization of Azerbaijan in 1920, the alphabet issue was brought to the forefront as an important direction of state policy (Süleymanlı, 2006, p. 158).

In the early 1920s, a policy of "Latinization" began to take shape within the Soviet system. The primary goal of this policy was not only to simplify the writing system, but also to weaken religious and cultural influences of the past, reduce connections with the Islamic world, and create a new "socialist culture" aligned with Soviet ideology. The Arabic script was regarded by Soviet ideologists as a symbol of the old feudal-religious system, while the Latin alphabet was presented as a tool of progress, science, and technological development. This approach partially overlapped with the modernization ideas of local intellectuals, resulting in a synthesis of "top-down" (state policy) and "bottom-up" (intellectual initiative) influences in the implementation of the reform. The establishment of the New Turkic Alphabet Committee in Baku in 1922 marked the institutional beginning of this process (Şimşir, 1991, p. 3). Its main task was to develop a new alphabet based on the Latin script and to design mechanisms for its implementation. Prominent intellectuals such as Samad agha Aghamalioglu and Jalil Mammadguluzadeh actively participated in its activities (Nərimanoğlu, 2006, p. 19). During this period, the Latin alphabet was promoted not only for its technical advantages but also as an essential element in building a new socialist society. In the press, propaganda materials, and official speeches, it was characterized as the "alphabet of science, culture, and progress," while the Arabic script was portrayed as a source of backwardness and illiteracy.

In 1923, a decision was adopted in the Azerbaijan Soviet Socialist Republic to implement the Latin alphabet, and from 1924 onward it began to be officially used. By 1925, its use became mandatory in state institutions and the press. The reform was also rapidly implemented in the education system, with new textbooks introduced in schools. As a result, a noticeable increase in literacy levels was observed in a short period, and the number of published books and newspapers grew (Ergun, 2010, p. 143).

The international dimension of this process is also noteworthy. The Turkological Congress held in Baku in 1926 contributed to spreading the transition to the Latin alphabet not only within Azerbaijan but also among Turkic peoples across the Soviet space. The advantages of the Latin script were widely discussed at the congress, and support was expressed for its adoption in many Turkic republics. This event can be seen as an initial attempt by Soviet leadership to create a shared cultural platform among Turkic peoples. However, this policy did not last long and was later reversed (Nərimanoğlu, 2006, p. 8).

The role of Soviet власти in the Latinization policy was dual in nature. On the one hand, it produced positive outcomes in terms of eliminating illiteracy, expanding education, and fostering a new cultural environment. On the other hand, it functioned as a tool of ideological control. Through the transition to the Latin alphabet, connections with the religious and cultural past were weakened, and a new Soviet identity was promoted. In this sense, the reform was not merely an educational initiative but also an instrument of political transformation. At the same time, the adoption of the Latin alphabet had the potential to bring Azerbaijan closer to other Turkic peoples. However, Soviet leadership feared that this potential might strengthen pan-Turkist tendencies. For this reason, in the late 1930s, the Latinization policy was abandoned and replaced by the transition to the Cyrillic alphabet. This demonstrates that for Soviet власти, the

alphabet issue was primarily an ideological and political tool shaped according to changing strategic interests (Ergun, 2010, p. 149).

In conclusion, the transition to the Latin alphabet in Soviet Azerbaijan should be regarded as a multifaceted historical phenomenon. It was both a product of local intellectual modernization efforts and a component of Soviet ideological policy. While the Latin alphabet contributed to increased literacy and accelerated cultural development, it also played a significant role in the ideological transformation of society. Therefore, alphabet reform remains one of the key topics for understanding both national identity and political power relations in twentieth-century Azerbaijan. After the establishment of Soviet rule, one of the most fundamental reforms implemented was the change of the alphabet, particularly the transition to a Latin-based writing system. This process was not simply a technical reform but a large-scale state strategy aimed at reshaping the social, cultural, and ideological structure of society. As highlighted in this article, the policy of Latinization directly served to strengthen Soviet ideology and to sever the people's ties with their past.

Discussion and Conclusion

Although the 1928 alphabet reform in Turkey and the Latinization policy implemented in Soviet Azerbaijan were products of the same wave of modernization, it is evident that these processes should be viewed not only as historical events but also as phenomena with ongoing relevance for intercultural communication and language policy today. Both cases demonstrate that writing systems and language policies are key factors that shape not only the internal structure of a society but also its position within the international system.

The alphabet reform carried out in Turkey under the leadership of Mustafa Kemal Atatürk was not limited to national identity building; it also expanded Turkey's capacity for communication with the Western world. The transition to the Latin script facilitated the country's integration into the global scientific, economic, and cultural environment, enabling the adaptation of the language to the global communication system. In this respect, the Turkish experience shows that when language policy is properly designed, it can function not only as a tool of internal modernization but also as a strategic instrument of international integration.

In Soviet Azerbaijan, however, the processes of Latinization and the subsequent transition to the Cyrillic script demonstrate how language policy can be directed by political authority. These changes were aimed less at expanding intercultural connections and more at restricting them within a specific ideological framework. As a result, language and alphabet policy functioned not as tools for international openness but as mechanisms for maintaining a closed ideological system shaped by the Soviet Union.

In the modern era, amid globalization, digitalization, and the intensification of international cooperation, language policy and alphabet choice have once again gained strategic importance. Although the role of global communication languages—especially English—has increased, the preservation and development of national languages remain equally significant. Historical experience suggests that language policy must be balanced: on the one hand, national identity and cultural heritage should be preserved; on the other, opportunities for international communication should be expanded.

Overall, a comparison of Turkey and Soviet Azerbaijan leads to the conclusion that alphabet and language policy are not merely technical or cultural issues, but also geopolitical and communicative strategies. In today's context of deepening international integration, language

policy serves as a key instrument determining states' participation in the global system, their capacity for intercultural dialogue, and their pace of scientific and technological development. Consequently, the alphabet reforms of the twentieth century represent not only historical experience but also an important analytical model for shaping contemporary language policy. This experience demonstrates that successful integration depends not only on economic and political factors but also directly on the effective design of language and communication strategies.

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LANGUAGE CHANGE AND LANGUAGE POLICY IN THE CONTEXT OF GLOBALIZATION

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Abstract: This article explores how globalization influences language change and shapes language policy in modern societies. With the rapid growth of global communication, migration, and technological development, languages are constantly evolving. These changes can be observed in processes such as borrowing, code-switching, and the emergence of hybrid linguistic forms. In particular, the increasing dominance of global languages, especially English, has a noticeable effect on smaller and less widely spoken languages. The study also considers how language policies respond to these developments. It focuses on the need to maintain a balance between preserving linguistic diversity and meeting the practical demands of international communication.

Different approaches to language planning, including educational strategies and institutional regulations, are discussed in order to understand how countries manage this balance. The research is based on a qualitative analysis of relevant academic sources and examples from different sociolinguistic contexts. The findings indicate that while globalization may pose certain risks for minority languages, it also creates new opportunities for language development and intercultural interaction. Overall, the study emphasizes the importance of flexible and well-designed language policies in supporting both cultural identity and global participation.

Key words: Globalization, language change, language policy, linguistic diversity, language shift, multilingualism

QLOBALLAŞMA ŞƏRAİTİNDƏ DİL DƏYİŞİKLİKLƏRİ VƏ DİL SİYASƏTİ

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Xülasə: Bu məqalə qloballaşmanın müasir cəmiyyətlərdə dil dəyişikliklərinə və dil siyasətinə təsirini araşdırır. Qlobal kommunikasiya imkanlarının genişlənməsi, miqrasiya prosesləri və texnoloji inkişaf dillərin daim dəyişməsinə səbəb olur. Bu dəyişikliklər daha çox alınma sözlər, kod dəyişməsi və hibrid dil formalarının yaranması kimi hallarda özünü göstərir. Xüsusilə, qlobal dillərin, əsasən ingilis dilinin geniş yayılması azsaylı dillərə müəyyən təsir göstərir.

Tədqiqatda həmçinin dil siyasətinin bu proseslərə necə yanaşdığı nəzərdən keçirilir. Əsas diqqət dil müxtəlifliyinin qorunması ilə beynəlxalq ünsiyyət tələbləri arasında tarazlığın təmin olunmasına yönəlikdir. Bu məqsədlə müxtəlif dil planlaşdırılması yanaşmaları, təhsil siyasətləri və institusional tədbirlər təhlil edilir.

Araşdırma əsasən mövcud elmi ədəbiyyatın təhlilinə və müxtəlif sosiolingvistik mühitlərdən götürülmüş nümunələrə əsaslanır. Nəticələr göstərir ki, qloballaşma bəzi dillər üçün risk yaratsa da, eyni zamanda dilin inkişafı və mədəniyyətlərarası ünsiyyət üçün yeni imkanlar açır. Ümumilikdə, məqalədə çevik və düzgün qurulmuş dil siyasətinin həm milli kimliyin qorunması, həm də qlobal mühitə uyğunlaşma baxımından vacibliyi vurğulanır.

Açar sözlər: Qloballaşma, dil dəyişiklikləri, dil siyasəti, dil müxtəlifliyi, dil dəyişməsi, çoxdillilik

Introduction

In the contemporary era, globalization has become one of the most influential forces shaping linguistic practices and language policies across the world. Increased mobility, digital communication, and economic integration have intensified contact between languages, often leading to noticeable structural, lexical, and functional changes. These transformations are not limited to vocabulary expansion; they also affect language use in education, administration, and

everyday communication. While globalization can enrich languages through borrowing and innovation, it also raises concerns about the preservation of linguistic diversity, particularly in smaller or historically marginalized languages.

Language change in the context of globalization is closely connected to political power, economic structures, and cultural influence. Dominant languages expand their reach through global institutions, media, and education systems, while others may gradually lose their functional domains. This process is especially visible in post-Soviet and post-colonial contexts, where language policies have historically been shaped by political authority and ideological priorities.

The Azerbaijani language provides a clear example of how these dynamics operate in practice. During the Soviet period, Russian functioned as the dominant language in administration, higher education, and scientific communication. Although Azerbaijani remained widely used in everyday life, its role in official domains was limited. This led to the incorporation of Russian lexical items and certain syntactic patterns into Azerbaijani, particularly among urban and educated speakers (Grenoble, 2003). In addition, Russian was often associated with prestige and social mobility, which influenced language attitudes across generations.

After gaining independence in 1991, Azerbaijan introduced a number of reforms to strengthen the position of Azerbaijani as the state language. These included the transition to the Latin alphabet and the expansion of Azerbaijani in education, media, and public administration (Landau & Kellner-Heinkele, 2001). However, the effects of earlier language contact remain visible, and new global influences especially the spread of English continue to shape the linguistic landscape.

These developments demonstrate that language change is a complex and ongoing process influenced by both historical and contemporary factors. Understanding this process requires not only linguistic analysis but also attention to social and political contexts.

Research

The relationship between globalization, language change, and language policy has been widely discussed in sociolinguistic research. Scholars generally agree that globalization accelerates language contact and increases the spread of dominant languages, often at the expense of less widely spoken ones. According to Crystal (2003), the global expansion of English has created new opportunities for communication but has also contributed to the marginalization of many local languages.

Fishman (1991) emphasizes that language shift is often linked to social and economic pressures rather than purely linguistic factors. Communities may gradually abandon their native languages in favor of more dominant ones that offer better access to education, employment, and global communication. This process is particularly relevant in multilingual societies, where language choice is closely tied to identity and social mobility.

Spolsky (2004) highlights the importance of language policy as a tool for managing linguistic diversity. He argues that effective language policies should consider not only official regulations but also actual language practices within society. Similarly, Wright (2004) notes that modern language planning must respond to the realities of globalization while still preserving national and cultural identities.

In the context of post-Soviet countries, Pavlenko (2008) points out that language revival efforts often coexist with ongoing multilingual practices. While governments promote national

languages, global and regional languages continue to play significant roles in communication. This creates a complex linguistic environment in which multiple influences interact.

Holmes (2013) further explains that language variation and change are natural outcomes of social interaction. From this perspective, borrowing, code-switching, and hybrid forms should not only be seen as threats but also as signs of linguistic adaptation and creativity.

Overall, the existing literature suggests that globalization has both positive and negative effects on languages. While it may contribute to language loss in some cases, it also encourages innovation and intercultural exchange. Therefore, balanced and flexible language policies are essential for managing these changes effectively.

Materials and Methods

This study is based on a qualitative research methodology and aims to examine the relationship between language change and language policy within the context of globalization. The research relies primarily on secondary data collected from academic books, peer-reviewed journal articles, conference proceedings, and official policy documents. In addition, reports published by international organizations concerned with language education, cultural preservation, and linguistic diversity were consulted to provide a broader perspective on the issue.

The selected sources were gathered through a systematic review of relevant literature in the fields of sociolinguistics, applied linguistics, and language planning. Particular attention was paid to studies investigating the effects of globalization on language use, language maintenance, multilingualism, and language shift. Sources were chosen according to their academic reliability, relevance to the topic, and contribution to contemporary discussions on global linguistic developments.

A descriptive-analytical approach was employed to interpret the collected data. The study compares different national and regional language policies and evaluates their effectiveness in addressing the opportunities and challenges created by globalization. Examples from various linguistic contexts were analyzed in order to identify common tendencies and policy responses. Furthermore, theoretical frameworks developed by scholars in language planning and globalization studies were used to support the analysis and interpretation of findings.

The materials used in this research provide both theoretical and empirical evidence, enabling a comprehensive examination of how globalization influences language change and how governments and institutions respond through language policy measures.

Results and Discussion

The analysis suggests that globalization has significantly transformed linguistic landscapes around the world. The increasing interconnectedness of societies through trade, migration, education, tourism, and digital communication has intensified contact between languages and cultures. As a result, languages are constantly adapting to new social realities through lexical borrowing, semantic change, code-switching, and the adoption of internationally recognized terminology.

One of the most notable consequences of globalization is the growing influence of global languages, particularly English, in fields such as science, technology, business, and higher education. While this development facilitates international communication and access to global knowledge, it also raises concerns regarding linguistic inequality and the marginalization of less

widely spoken languages. In many communities, younger generations increasingly prefer dominant languages because of their perceived economic and social advantages.

The findings also highlight the important role of language policy in shaping linguistic outcomes. Governments and educational institutions often implement policies designed to promote national languages, support minority language rights, and encourage multilingual competence. In several countries, bilingual and multilingual education programs have been introduced as a means of preserving linguistic heritage while ensuring participation in global networks of communication.

Moreover, technological advancements have created both opportunities and challenges for language preservation. Digital platforms can contribute to the documentation and promotion of endangered languages; however, they also tend to favor languages with larger numbers of speakers and greater economic influence. Consequently, language policies must adapt to the digital age by supporting the online presence and accessibility of minority languages.

The discussion further demonstrates that language change should not be viewed solely as a threat to linguistic identity. Throughout history, languages have evolved through contact with other languages and cultures. Globalization has accelerated these processes, but it has not eliminated the capacity of communities to maintain and revitalize their linguistic traditions. Effective language policies can help ensure that linguistic diversity remains a valuable component of cultural heritage while allowing societies to benefit from global integration.

Therefore, the relationship between language change and language policy in the era of globalization is complex and multifaceted. A balanced approach that combines openness to global communication with the protection of local linguistic resources appears to be the most sustainable strategy for addressing contemporary linguistic challenges.

Conclusion

In conclusion, globalization plays a key role in shaping language change and influencing language policy in modern societies. Increased communication, migration, and technological development have accelerated language contact, leading to changes in vocabulary, structure, and patterns of use.

The case of the Azerbaijani language shows how historical and political factors interact with global influences. The legacy of Soviet language policy, especially the dominance of Russian, still affects language use today. At the same time, post-independence reforms have strengthened Azerbaijani as the state language. However, the growing influence of English continues to reshape the linguistic environment. These developments suggest that language change should not be seen only as a threat. While globalization may weaken some languages, it also creates opportunities for innovation and intercultural communication.

Therefore, effective language policy should maintain a balance between preserving national identity and adapting to global realities.

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THE ROLE OF LANGUAGE POLICY IN THE EDUCATION SYSTEM

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Abstract: This study examines the role of language policy in shaping modern education systems and its impact on teaching and learning processes. It explores how language policies influence curriculum design, medium of instruction, and educational accessibility. The article highlights the importance of language policy in promoting equality, preserving cultural identity, and enhancing academic performance. It also discusses the role of multilingual education and the integration of minority languages into formal education. Furthermore, the study addresses key challenges such as linguistic inequality, political influence, and implementation barriers. The findings suggest that effective language policies serve as a critical foundation for inclusive, equitable, and high-quality education systems.

Key words: language policy, education system, multilingual education, linguistic diversity, medium of instruction, educational equity, language planning

TƏHSİL SİSTEMİNDƏ DİL SİYASƏTİNİN ROLU

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Xülasə: Bu məqalə təhsil sistemində dil siyasətinin rolunu və onun tədris prosesinə təsirini araşdırır. Məqalədə dil siyasətinin kurikulumun formalaşmasına, tədris dilinin seçiminə və təhsilin əlçatanlığına təsiri təhlil olunur. Tədqiqat dil siyasətinin bərabərliyin təmin olunmasında, mədəni kimliyin qorunmasında və akademik nəticələrin yaxşılaşdırılmasında mühüm rol oynadığını vurğulayır. Həmçinin çoxdilli təhsilin əhəmiyyəti və azlıq dillərinin təhsil sistemində inteqrasiyası məsələləri nəzərdən keçirilir. Bununla yanaşı, dil bərabərsizliyi, siyasi təsirlər və tətbiq çətinlikləri kimi problemlər də müzakirə olunur.

Nəticə etibarilə, effektiv dil siyasətinin inklüziv və keyfiyyətli təhsil sisteminin əsasını təşkil etdiyi müəyyən edilir.

Açar sözlər: dil siyasəti, təhsil sistemi, çoxdilli təhsil, dil müxtəlifliyi, tədris dili, təhsil bərabərliyi, dil planlaşdırılması

Introduction

In the context of globalization and increasing cultural interaction, language has become a central element in shaping educational systems. Language policy, defined as the set of decisions and practices that regulate the use of language within a particular society or institution, plays a crucial role in determining how education is delivered and accessed. In modern education systems, language policy not only influences the medium of instruction but also shapes curriculum content, teaching methodologies, and student engagement.

The significance of language policy extends beyond communication; it directly affects social integration, equality, and identity formation. In multilingual societies, the choice of instructional language can either facilitate inclusion or reinforce inequality. For instance, when education is delivered in a language unfamiliar to students, it may hinder comprehension and academic achievement. Conversely, the use of mother tongue instruction has been shown to improve learning outcomes, especially in early education.

This study aims to analyze the role of language policy in education systems, focusing on its benefits, challenges, and long-term implications. It seeks to answer how language policies influence educational quality and accessibility, and what strategies can be employed to create more inclusive and effective systems.

Materials and Methods

This research is based on a qualitative and analytical approach, combining theoretical analysis with a review of academic literature on language policy and education. The study examines over twenty scholarly sources, including books, journal articles, and reports published between 2000 and 2023, to ensure a comprehensive and up-to-date perspective.

The materials analyzed include policy documents, educational frameworks, and case studies related to multilingual education systems. Special attention is given to the role of language planning in different national contexts, as well as the impact of language choice on student performance and access to education.

By synthesizing these sources, the study aims to identify key patterns and principles that define effective language policies in education.

Results and Discussion

The findings of this study demonstrate that language policy is a fundamental component of educational development. One of the most significant roles of language policy is determining the medium of instruction. Research indicates that students who receive education in their native language, particularly in early years, tend to perform better academically and develop stronger cognitive skills. This supports the argument for incorporating mother tongue-based education in primary schooling.

Another important aspect is the promotion of multilingual education. In an increasingly globalized world, proficiency in multiple languages is considered a valuable skill. Language policies that encourage bilingual or multilingual education not only enhance communication abilities but also improve cognitive flexibility and cultural awareness.

Language policy also plays a crucial role in ensuring educational equity. Inclusive policies that recognize and support minority languages can help reduce educational disparities and promote social cohesion. However, the implementation of such policies often faces significant challenges. Limited resources, lack of trained teachers, and political considerations can hinder effective execution.

A major issue identified in this study is linguistic inequality. In many countries, dominant languages receive more institutional support, while minority languages are marginalized. This can lead to unequal access to education and reduced opportunities for certain groups. Additionally, political factors often influence language policy decisions, sometimes prioritizing national unity over linguistic diversity.

Another challenge is the gap between policy and practice. While many countries adopt progressive language policies, their implementation is often inconsistent due to infrastructural limitations and insufficient funding.

Despite these challenges, successful examples from various countries demonstrate that well-designed language policies can significantly enhance educational outcomes. Integrating technology and digital tools into language education further expands opportunities for learning and access.

Conclusion

In conclusion, language policy plays a pivotal role in shaping education systems and influencing learning outcomes. Effective language policies contribute to improved academic performance, greater accessibility, and the preservation of cultural identity. By promoting multilingual education and supporting minority languages, these policies foster inclusivity and social cohesion.

However, the success of language policy depends on careful planning and implementation. Addressing challenges such as linguistic inequality, resource limitations, and political influence is essential for achieving sustainable progress. Policymakers must ensure that language policies are not only well-designed but also effectively executed.

Ultimately, a balanced and inclusive approach to language policy can serve as a powerful tool for building equitable and future-oriented education systems.

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LANGUAGE POLICY AND PLANNING

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Abstract: This article examines the theoretical and practical dimensions of language policy and planning in contemporary societies. It explores the key components of language policy, including status planning, corpus planning, and acquisition planning, and explains how these elements interact to shape language use and development. Special attention is given to language planning in multilingual societies, where policymakers must balance linguistic diversity with social cohesion and effective communication. The study highlights the importance of inclusive language policies that recognize minority languages while maintaining national unity. Furthermore, the role of language policy in the education system is analyzed, emphasizing the benefits of mother tongue-based education for cognitive development and academic achievement, alongside the need for learning international languages in a globalized world. The article also addresses the challenges posed by globalization, particularly the dominance of widely spoken languages and the resulting risks for the preservation of national and minority languages. It suggests that proactive measures, such as strengthening legal frameworks, promoting media and digital content in national languages, and encouraging youth engagement, are essential for sustaining linguistic diversity. The findings demonstrate that well-balanced and forward-looking language policies are crucial for cultural preservation, social integration, and sustainable development in an increasingly interconnected global environment.

Key words: language policy, language planning, multilingualism, education, globalization, national languages

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Xülasə: Bu məqalədə müasir cəmiyyətlərdə dil siyasəti və dil planlaşdırılmasının nəzəri və praktik aspektləri araşdırılır. Tədqiqat çərçivəsində dil siyasətinin əsas komponentləri olan status planlaşdırılması, korpus planlaşdırılması və əldəetmə planlaşdırılması izah edilir və onların dilin istifadəsi və inkişafına təsiri təhlil olunur. Xüsusilə çoxdilli cəmiyyətlərdə dil planlaşdırılmasının xüsusiyyətlərinə diqqət yetirilir və bu prosesdə dil müxtəlifliyi ilə sosial inteqrasiya arasında balansın qorunmasının vacibliyi vurğulanır. Məqalədə inklüziv dil siyasətinin milli birliyin qorunması ilə yanaşı azlıq dillərinin dəstəklənməsində rolu əsaslandırılır. Bundan əlavə, təhsil sistemində dil siyasətinin əhəmiyyəti təhlil olunur, ana dilində təhsilin idrak inkişafına və akademik nailiyyətlərə müsbət təsiri göstərilir. Eyni zamanda, qloballaşma şəraitində beynəlxalq dillərin üstünlüyü fonunda milli və az istifadə olunan dillərin qorunması problemləri araşdırılır. Məqalədə hüquqi mexanizmlərin gücləndirilməsi, media və rəqəmsal mühitdə milli dillərin təşviqi və gənc nəslin bu dillərə marağının artırılması kimi tədbirlərin vacibliyi qeyd olunur. Nəticə olaraq, balanslı və məqsədyönlü dil siyasətinin mədəni müxtəlifliyin qorunması və cəmiyyətin davamlı inkişafı üçün mühüm əhəmiyyət daşıdığı vurğulanır.

Açar sözlər: dil siyasəti, dil planlaşdırılması, çoxdilli cəmiyyət, təhsil, qloballaşma, milli dil

Introduction

Language policy and planning play a crucial role in shaping modern societies by influencing communication, education, social integration, and cultural identity. In the context of globalization, increased migration, technological advancement, and the rapid expansion of digital communication, the regulation, development, and preservation of languages have become key priorities for governments, educational institutions, and international organizations. Language is not merely a means of communication; it is also a symbol of national identity, cultural heritage, and social cohesion. Consequently, decisions regarding language use affect political stability, economic development, and educational opportunities within a society.

As societies become increasingly multilingual and multicultural, policymakers face the challenge of balancing linguistic diversity with national unity. Effective language policies can promote equal opportunities, strengthen social inclusion, and protect minority language rights. At the same time, globalization creates pressures that encourage the adoption of dominant international languages, particularly in education, science, business, and technology. Therefore, language policy and planning have become essential instruments for managing linguistic diversity and ensuring sustainable language development. This article explores the theoretical foundations of language policy, language planning in multilingual societies, the role of language policy in education, and the challenges of preserving national languages in a globalized world. (Spolsky, 2004).

Theoretical Foundations of Language Policy. Language policy refers to the set of strategies, decisions, and measures adopted by governments, institutions, or communities to regulate language use within a society (Kaplan and Baldauf, 1997). It encompasses both explicit policies, such as official legislation and educational regulations, and implicit practices that influence language behavior in everyday life. The theoretical foundations of language policy are closely connected with sociolinguistics, political science, anthropology, and educational studies, reflecting the interdisciplinary nature of the field.

Language policy is generally divided into three main components: status planning, corpus planning, and acquisition planning (Cooper, 1989). Status planning concerns the social and political position of a language within a society, including decisions about official language status and language rights. Corpus planning focuses on the internal development of a language through standardization, modernization of vocabulary, spelling reforms, and terminology development. Acquisition planning addresses language learning and teaching, particularly through educational systems and language programs designed to promote language competence among different population groups.

Contemporary language policy theories also emphasize the relationship between language and power. Language choices often reflect broader social, economic, and political interests. Consequently, language policy serves not only as a linguistic tool but also as a mechanism for nation-building, identity formation, and social integration. Scholars increasingly argue that successful language policies must consider both linguistic realities and the social contexts in which languages are used.

Language Planning in Multilingual Societies. Language planning in multilingual societies is a complex and sensitive process. It requires balancing the needs and rights of

different linguistic groups while promoting social cohesion (Fishman, 1991). Multilingual societies often include speakers of majority, minority, indigenous, immigrant, and regional languages, each of which may have distinct cultural, social, and political significance.

Effective language planning seeks to create an environment in which linguistic diversity is recognized and respected. Such planning fosters inclusivity, reduces social tensions, and preserves cultural diversity. Governments frequently adopt policies that provide legal recognition for multiple languages, encourage bilingual or multilingual education, and ensure access to public services in different languages.

The success of language planning largely depends on its ability to address social realities. Policies that ignore linguistic diversity may lead to marginalization, social exclusion, and conflicts among language communities. In contrast, inclusive language planning can strengthen democratic participation, promote equal opportunities, and enhance intercultural understanding. Countries such as Canada, Switzerland, and Singapore are often cited as examples of multilingual societies that have implemented language policies aimed at balancing national unity with linguistic diversity.

Furthermore, globalization and migration have increased the complexity of language planning. Policymakers must now address not only traditional language communities but also the growing linguistic diversity resulting from international mobility and digital communication. This requires flexible and adaptive language planning strategies capable of responding to rapidly changing social conditions.

The Role of Language Policy in the Education System. The education system is one of the primary tools for implementing language policy. Decisions regarding the language of instruction, curriculum design, educational materials, and teaching methods significantly impact students' learning outcomes (Baker, 2011). Educational institutions play a fundamental role in transmitting linguistic knowledge, cultural values, and national identity from one generation to another.

Research consistently demonstrates that education in the mother tongue supports cognitive development, enhances comprehension, and improves academic performance. Students who receive instruction in their native language often develop stronger literacy skills and achieve better educational outcomes. At the same time, learning international languages provides access to global knowledge, higher education opportunities, and international labor markets.

Language policy in education must therefore strike a balance between promoting national languages and facilitating foreign language learning. Many countries have adopted bilingual and multilingual education models that allow students to maintain their linguistic and cultural identities while developing competence in internationally significant languages.

In addition, technological innovations have transformed language education. Digital learning platforms, online resources, and artificial intelligence-based language-learning tools have expanded opportunities for multilingual education and language acquisition. Consequently, educational language policies must adapt to new technological realities while continuing to support linguistic diversity and educational equity.

Issues of Preserving National Languages in the Context of Globalization. Globalization has intensified interactions among languages but has also increased the dominance of widely spoken languages, particularly English (Crystal, 2003). English has become the primary language of international business, science, higher education, diplomacy, and digital

communication. While this development facilitates global interaction, it also poses challenges for the preservation of national and minority languages.

The growing influence of dominant languages can lead to language shift, language attrition, and, in some cases, language extinction. Smaller languages often face reduced usage among younger generations who may perceive global languages as offering greater economic and educational opportunities. As a result, valuable cultural knowledge, traditions, and linguistic heritage may gradually disappear.

Preserving national languages requires comprehensive and coordinated efforts. Governments can support language maintenance through legislation, educational programs, media initiatives, cultural activities, and language revitalization projects. The use of national languages in public administration, higher education, scientific publications, and digital platforms is particularly important for ensuring their continued relevance in modern society.

Moreover, technological developments present both challenges and opportunities for language preservation. Although digital media often promote dominant languages, they can also provide new platforms for documenting, teaching, and disseminating national languages. Online dictionaries, language-learning applications, digital archives, and social media content contribute to increasing the visibility and accessibility of national languages in the global information space.

Conclusion

Language policy and planning significantly influence social, cultural, political, and educational development. In multilingual contexts, inclusive and balanced approaches are essential for maintaining harmony, protecting linguistic rights, and preserving cultural diversity (Spolsky, 2004). Effective language policies contribute to social integration, educational success, and national development while ensuring respect for linguistic diversity.

As globalization continues to reshape linguistic landscapes worldwide, policymakers face the challenge of promoting international communication without undermining national and minority languages. Sustainable language planning requires cooperation among governments, educational institutions, researchers, and local communities. Future language policies should embrace multilingualism as a valuable resource while implementing strategies that ensure the preservation, development, and transmission of linguistic heritage for future generations.

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LANGUAGE POLICY AND NATIONAL IDENTITY: PRIORITIES OF LANGUAGE PLANNING IN THE LIBERATED TERRITORIES

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Abstract: Language policy plays a decisive role in shaping and strengthening national identity, particularly in post-conflict regions where social, cultural, and institutional structures are being rebuilt. The liberated territories of Azerbaijan present a unique context in which language planning becomes not only a tool of communication but also a strategic instrument for restoring cultural continuity and reinforcing national unity. This study explores the main priorities of language planning in these territories, focusing on the promotion of the Azerbaijani language as the state language, the revival of historical toponyms, and the reestablishment of linguistic norms in education, administration, and public life (Heydar Aliyev's language policy, 2009).

In the aftermath of displacement and occupation, linguistic landscapes in these regions have undergone significant disruption. Therefore, restoring the Azerbaijani language in official and everyday use is essential for rebuilding a shared sense of belonging and identity. The study also emphasizes the importance of integrating modern language policies with inclusive approaches that respect linguistic diversity while maintaining national cohesion. Special attention is given to the role of education systems, media, and public institutions in implementing effective language planning strategies.

Key words: Language policy, national identity, language planning, liberated territories, Azerbaijani language.

DİL SİYASƏTİ VƏ MİLLİ KİMLİK: AZAD OLUNMUŞ ƏRAZİLƏRDƏ DİL PLANLAŞDIRILMASININ PRIORİTETLƏRİ

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Xülasə: Dil siyasəti milli kimliyin yaranmasında və gücləndirilməsində əsas rol oynayır, xüsusi olaraq sosial, mədəni və institusional quruluşların yenidən yaradıldığı post-münaqişə rayonlarında. Azərbaycanın azad edilmiş torpaqları dil sistemləşdirilməsinin sadəcə ünsiyyət vasitəsi deyil, o cümlədən mədəni davamlılığın bərpa olunması və milli birliyin möhkəmləndirilməsi üçün strateji alətə döndüyü unikal bir şərait təqdim edir. Bu tədqiqat həmin bölgələrdə dil planlaşdırılmasının əsas məqsədlərini araşdırır, dövlət dili şəklində Azərbaycan dilinin təşviqi, tarixi toponimlərin bərpa olunması, təhsil, idarəetmə və ictimai quruluşlarda dil normalarının yenidən formalaşmasına xüsusi diqqət verir.

Məcburi şəkildə köçkünlüyün və işğalın son nəticəsi olaraq bu bölgələrin dil şəraiti kobud formada pozulmuşdur. Buna üçün də Azərbaycan dilinin rəsmi və gündəlik istifadədə bərpa olunması orta q mənsubiyyət və kimlik dövrünün yenidən formalaşması üçün mühüm əhəmiyyət kəsb edir. Tədqiqat o cümlədən müasir dil strategiyalarının milli birliyi müdafiə etməklə birlikdə, dil fərqliliyinə hörmət edən inklüziv prinsiplərlə uyğunlaşdırılmasının mahiyyətini vurğulayır. Effektiv dil planlaşdırılması strategiyalarının yerinə yetirilməsində təhsil sektorunun medianın və ictimai institutların rolu xüsusilə qeyd edilir.

Açar sözlər: Dil siyasəti, milli kimlik, dil planlaşdırılması, azad olunmuş ərazilər, Azərbaycan dili.

Introduction

The relationship between language policy and national identity has long been a central topic in sociolinguistics and language planning, particularly in contexts shaped by political change and territorial transformation. Language is not merely a means of communication; it is a powerful symbol of cultural heritage, collective memory, and social cohesion. In post-conflict settings, the role of language becomes even more significant, as it contributes to the reconstruction of identity, the restoration of cultural continuity, and the strengthening of statehood. In this regard, the liberated territories of Azerbaijan offer a distinctive case for examining how language policy can support broader processes of recovery and reintegration (Abdulla, 2010).

In this study, I investigate the role of language policy in shaping national identity and examine the main priorities of language planning in the liberated territories of Azerbaijan (Məmmədov, 2021).

Following years of occupation, these territories have experienced profound demographic, cultural, and linguistic disruptions. The return of displaced populations and the reestablishment of social institutions require a well-defined and strategic approach to language planning. Ensuring the active use of the Azerbaijani language in public administration, education, and everyday communication is essential for fostering national unity and reinforcing a shared sense of belonging. At the same time, language policy must be designed in a way that acknowledges historical realities and promotes inclusive practices.

This article aims to analyze the priorities of language planning in the liberated territories of Azerbaijan within the broader framework of national identity formation. It explores the role of the state language, the restoration of historical toponyms, and the importance of institutional support in rebuilding the linguistic landscape. By examining these dimensions, the study seeks to contribute to a deeper understanding of how language policy can function as a key instrument in post-conflict reconstruction and sustainable development (Əhmədova, 2010).

Materials and Methods

The material base of this investigation consists of academic literature on tongue policy and national sameness in the Republic of Azerbaijan, formal government programs, normative-official documents, and formal reports connected to the socio-cultural growth of the liberated areas. In enclosure, previous scientific studies on the implementation of the Azerbaijani language as the government language, the renewal of toponyms, and language projecting in the education structure were investigated. In this research, I investigate the variation in the linguistic atmosphere by confront the previous occupation and post-liberation terms. A comparative contemplation way was applied to identify key distinctions and improvement. Moreover, a descriptive-analytical point of view was done to examine how tongue policy impact the shaping of national identity.

From a methodological perspective, this investigation employs a balanced approach, comparative contemplation, and content contemplation ways. The planned approach allows tongue politics to be understood as an integral portion of broader social and cultural cases. In this investigation, I investigate tongue planning trials across various historical phases of the released territories through comparative contemplations. Content researchs is used to check formal

documents and academic beginnings in order to identify the essential directions of tongue politics. These methodological tools provide the scientific reliability, consistency, and validity of the research discoveries (UNESCO, 2022).

Discussion

In this research, I investigated the role of language policy in shaping national identity and examined the main priorities of language planning in the liberated territories of Azerbaijan. The findings show that language policy is not only a linguistic issue but also a fundamental component of nation-building and cultural restoration. The revitalization of the Azerbaijani language in official, educational, and social domains plays a key role in strengthening national unity and ensuring the reintegration of liberated territories into the broader socio-cultural space of the country.

The results of this study indicate that effective language planning in post-conflict regions requires a comprehensive and well-structured approach. In this research, I also explored how the restoration of historical toponyms, the standardization of language use, and the development of educational resources contribute to the reconstruction of the linguistic landscape. It was found that institutional support, legal frameworks, and media involvement are essential factors in ensuring the successful implementation of language policy (Fathi, 2021).

Overall, the discussion highlights that language policy in the liberated territories of Azerbaijan goes beyond communication; it serves as a strategic instrument for preserving cultural heritage and reinforcing national identity. In this research, I investigated how coordinated language planning efforts can support long-term social stability and sustainable development. The study concludes that continued attention to language policy will be crucial for strengthening state identity and ensuring the cultural integration of these regions (Fathi, 2023).

Conclusion

In this research, I investigated the role of language policy in shaping national identity and examined the main priorities of language planning in the liberated territories of Azerbaijan. The findings demonstrate that language policy is a key factor not only in communication but also in the restoration of cultural identity and national unity. The strengthening of the Azerbaijani language in education, administration, and public life significantly contributes to the reintegration of these territories into the national framework and supports the revival of cultural continuity.

In this research, I also investigated that effective language planning in post-conflict regions requires a systematic and coordinated approach involving state institutions, education systems, and media structures. The discussion shows that the restoration of linguistic norms, historical toponyms, and standardized language use plays an important role in rebuilding the linguistic environment. Overall, the study concludes that language policy in the liberated territories is essential for long-term social stability, cultural preservation, and the reinforcement of national identity.

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MULTILINGUALISM AND SECOND LANGUAGE ACQUISITION (SLA)

EARLY FOREIGN LANGUAGE LEARNING IN PRESCHOOL EDUCATION: PRINCIPLES AND STRATEGIES

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Abstract: Early foreign language learning in preschool education has gained growing attention because of its role in supporting children's cognitive, social, and linguistic development. This study investigates the core principles and effective strategies that contribute to successful language learning at an early age. It emphasizes the value of a child-oriented and play-based approach, where language is introduced naturally through everyday activities rather than formal instruction. Six main principles are identified: holistic development, building on existing language knowledge, meaningful learning, active involvement, continuity, and integration across activities. The study also explores practical classroom strategies, including circle time, daily routines, educational games, storytelling, role-play, and peer interaction. These methods encourage communication, creativity, and engagement while helping children acquire language in a natural context. The findings indicate that language learning becomes more effective when it occurs in interactive, relevant, and supportive environments. Furthermore, the study highlights the importance of well-prepared teachers in implementing these approaches successfully. Continuous professional development and appropriate training are essential to ensure high-quality early language education and to maximize learning outcomes for young children.

Keywords: early language learning, pre-primary education, child-centred approach, play-based learning, language development

MƏKTƏBƏQƏDƏR TƏHSİLDƏ ERKƏN XARİCİ DİL ÖYRƏNİLMƏSİ: PRİNSİPLƏR VƏ STRATEGİYALAR

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Xülasə: Məktəbəqədər təhsildə erkən xarici dil öyrənilməsi son illərdə uşaqların idrak, sosial və dil bacarıqlarının inkişafına verdiyi töhfələrə görə daha çox diqqət cəlb edir. Bu tədqiqat erkən yaşda dil öyrənməsini dəstəkləyən əsas prinsipləri və effektiv strategiyaları araşdırır. Məqalədə dilin formal şəkildə deyil, uşaqların gündəlik fəaliyyətlərinə daxil edilərək öyrədilməsinin əhəmiyyəti vurğulanır və uşağa yönəlmiş, oyun əsaslı yanaşmanın üstünlükləri qeyd olunur. Araşdırmada altı əsas prinsip müəyyən edilmişdir: bütöv yanaşma, əvvəlki dil təcrübəsinə əsaslanma, mənalılıq, fəal iştirak, davamlılıq və müxtəlif fəaliyyətlərlə inteqrasiya. Bundan əlavə, praktik tətbiq kimi dairəvi məşğələlər, gündəlik rutinlər, oyunlar, hekayə danışma, rollu fəaliyyətlər və həmyaşd əməkdaşlığı kimi üsullar təhlil olunur. Bu strategiyalar uşaqların ünsiyyət bacarıqlarının inkişaf etdirməklə yanaşı, onların marağını və fəallığını artırır. Nəticələr göstərir ki, dil öyrənməsi interaktiv və uşaqlar üçün mənalı olan mühitdə daha

uğurlu həyata keçirilir. Eyni zamanda, müəllimlərin peşəkar hazırlığı və davamlı inkişafı bu prosesin effektiv təşkili üçün əsas şərtlərdən biridir.

Açar sözlər: erkən dil öyrənməsi, məktəbəqədər təhsil, uşağa yönəlmiş yanaşma, oyun əsaslı öyrənmə, dil inkişafı

Introduction

Preschool education plays a crucial role in the formation of children's social, emotional, cognitive, and linguistic skills. In recent years, interest in learning foreign languages at an early age has increased. This is associated with the growing importance of multilingualism for society and the recognition of the advantages of early language learning. Contemporary approaches indicate that language learning should not be implemented as a separate subject but rather integrated into daily activities.

This study is based on a systematic literature review conducted to examine the theoretical and practical foundations of early foreign language learning. The methodological framework of the research is grounded in modern pedagogical approaches, theories of language teaching, and reports of international organizations. In particular, scientific sources covering the characteristics of language learning in early childhood were selected, comparatively analyzed, and synthesized. Within the framework of the study, the selected sources were systematized through the method of content analysis. This method enables the researcher to categorize information presented in various sources into thematic categories and to identify key concepts. As a result, the main principles of early foreign language learning and their modes of application were clearly identified. At the same time, a comparative method was employed to analyze early language teaching models applied in different countries (Mourão, S. 2014). This approach made it possible to reveal similarities and differences among various pedagogical systems. In particular, child-centered and play-based teaching models implemented in European countries were compared with more traditional, teacher-centered approaches. The descriptive method served to provide a detailed explanation of the existing situation. Through this method, the main characteristics, advantages, and challenges of early language learning were systematically presented. At the same time, practical strategies were explained with concrete examples, which increases the applied significance of the research. In order to ensure the reliability of the study, only up-to-date and scientifically credible sources from recent years were selected. The academic reputation of the sources, citation indicators, and the relevance of the topic were taken as the main criteria in their selection. In addition, the approaches of different authors were compared in order to obtain objective conclusions (European Commission. 2011).

Methodology and Materials

Thus, this study systematically analyzes the theoretical foundations and practical application possibilities of early foreign language learning based on a comprehensive methodological approach. The results of the conducted research indicate that early foreign language learning should be based on specific pedagogical principles, and these principles are closely related to the developmental characteristics of children. The effectiveness of the learning process at an early age directly depends on how it is organized. In this regard, six main principles identified serve as a fundamental basis for the successful implementation of early language learning.

The first principle is that language learning has a holistic character. According to this approach, children perceive the world not only through language but through all their senses. They learn through hearing, seeing, touching, and movement. Therefore, language teaching should not be limited to teaching words and rules. On the contrary, it should be implemented through activities that incorporate various sensory experiences (Nikolov, J. 2023). For example, songs, movement-based games, and visual materials make children's language learning more effective. This approach also makes it possible to take into account children's individual learning styles.

The second principle is that language learning should be based on the child's existing language experience. Children do not learn a new language from scratch; they already possess certain linguistic knowledge and communicative experience. These serve as a foundation for acquiring a new language (Bredekamp, C. 1997). For instance, children understand concepts they already know in their native language more easily in a new language. This increases the importance of analogy and comparative approaches in the teaching process.

The third principle is that language learning must be meaningful. Information that is not meaningful to children is not retained in long-term memory. Therefore, language learning should be connected with children's daily lives. Topics such as food, play, family, and friends are more interesting and understandable for children. Language material presented in such contexts is acquired more quickly.

The fourth principle relates to active learning. Children should not learn as passive listeners but as active participants. They should speak, ask questions, discuss, and participate in various activities. Active participation increases children's motivation and makes the learning process more effective.

The fifth principle is that language learning should be continuous. Language skills do not develop instantly; they require long-term and gradual development (Enever, J. 2011). Therefore, children should be provided with a regular and continuous language environment. This creates conditions for the gradual development of their language skills.

The sixth principle is that language learning should be integrative in nature. Language should not be taught separately from other activities but as an integral part of them. For example, when language is used during drawing, music, and physical activities, learning becomes more effective (UNICEF. 2018).

The practical implementation of these principles is carried out through various strategies. The study found that circle time activities are highly effective for early language learning. During these activities, children communicate together, sing songs, and listen to stories. This simultaneously develops their social and language skills. Daily routines also play an important role. Repetitive activities create a sense of security in children and help them acquire language naturally. For example, activities such as greeting, counting, or talking about the weather turn language learning into a part of daily life (Alexiou, T. 2015). Games are one of the most effective means of learning at an early age. During play, children naturally use language and learn new words in context. This also develops their creativity and problem-solving skills. Storytelling and picture books are of particular importance in children's language development. Through these means, children learn new vocabulary, develop listening skills, and expand their imagination. At the same time, stories serve as an important tool for transmitting cultural values.

Role-play activities give children the opportunity to act out real-life situations. In this process, they use language in practice and develop their communicative skills.

This research adopts a qualitative approach grounded in a structured review of recent scholarly literature on early foreign language education. Key ideas and teaching practices were identified through thematic (content) analysis. A cross-context comparison was used to examine different educational models, and a descriptive approach was applied to present the results clearly. Source selection was limited to up-to-date and academically reliable publications to maintain validity.

The results indicate that early language acquisition is most successful when guided by learner-focused and play-oriented methods. Principles such as relevance to children's experiences, active involvement, and integration into everyday activities support more effective learning. Regular exposure and interactive techniques—like storytelling and games—further strengthen development. However, outcomes depend largely on educator preparation, available resources, and institutional conditions. Overall, early language learning supports not only linguistic skills but also broader developmental areas.

Finally, peer collaboration plays an important role in the development of children's social and language skills. During group activities, children communicate with each other, and mutual learning takes place.

Conclusion

The conducted research shows that learning a foreign language at an early age has a significant impact on children's overall development. This process is not limited to the formation of language skills but also supports cognitive, social, and emotional development. Child-centered and play-based approaches act as key factors that increase the effectiveness of early language learning. These approaches make the learning process more natural and engaging by taking into account children's interests and needs.

However, the successful implementation of these principles requires the professional training of teachers. Teachers should possess not only language knowledge but also a deep understanding of early childhood pedagogy. Continuous professional development programs can help improve quality in this field. Future research should focus on areas such as the role of inclusive education in early language learning, the use of technologies, and the development of cultural competencies. These directions will enable further improvement and modernization of early language teaching.

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LINGUOCULTUROLOGY: LANGUAGE, LITERATURE AND CULTURAL IDENTITY

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Astract: Linguoculturology is an interdisciplinary field that studies the relationship between language, culture, and human identity. It examines how cultural values, traditions, beliefs, and social norms are reflected and transmitted through language. This article explores the role of language and literature in the formation and preservation of cultural identity. Language is not only a communicative tool but also a cultural phenomenon that carries historical memory and national worldview. Literary texts, in particular, serve as important sources for linguocultural analysis because they reflect the mentality, customs, and ideological perspectives of a society.

The study also investigates how writers use linguistic and stylistic devices such as metaphors, symbols, idioms, and narrative structures to represent cultural identity and intercultural experience. Special attention is paid to postcolonial and multicultural literary contexts in which questions of hybridity, displacement, and self-identification become central. Through linguocultural analysis, literature reveals the complex interaction between individual and collective identity. The article concludes that language and literature play a significant role in preserving cultural heritage while also shaping new cultural meanings in the context of globalization and intercultural communication in the modern world today.

Keywords: Linguoculturology, cultural identity, language, literature, postcolonial discourse, multiculturalism

LİŖVOKULTUROLOGİYA: DİL, ƏDƏBİYYAT VƏ MƏDƏNİ KİMLİK

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Xülasə: Linqvokulturologiya dil, mədəniyyət və identiklik arasındakı əlaqəni öyrənən fənlərarası bir sahədir. Bu sahə mədəni dəyərlərin, ənənələrin, inancların və sosial normaların dil vasitəsilə necə əks olunduğunu və nəşildən-nəslə ötürüldüyünü araşdırır. Məqalədə dil və ədəbiyyatın mədəni kimliyin

formalaşması və qorunmasındakı rolu təhlil edilir. Dil yalnız ünsiyyət vasitəsi deyil, həm də tarixi yaddaşı və milli dünyagörüşünü daşıyan mədəni bir fenomendir. Xüsusilə bədii mətnlər linqvokulturoloji təhlil üçün mühüm mənbə hesab olunur, çünki onlar cəmiyyətin düşüncə tərzini, adət-ənənələrini və ideoloji baxışlarını əks etdirir.

Tədqiqat həmçinin yazıçıların metaforlar, simvollar, idiomlar və narrativ strukturlar kimi dil və üslub vasitələrindən istifadə edərək mədəni kimliyi və mədəniyyətlərarası təcrübəni necə ifadə etdiklərini araşdırır. Hibridlik, yerdəyişmə və özünüidentifikasiya məsələlərinin əsas yer tutduğu postkolonial və multikultural ədəbi kontekstlərə xüsusi diqqət yetirilir. Linqvokulturoloji təhlil vasitəsilə ədəbiyyat fərdi və kollektiv kimlik arasındakı mürəkkəb qarşılıqlı əlaqəni ortaya qoyur. Məqalə belə nəticəyə gəlir ki, dil və ədəbiyyat mədəni irsin qorunmasında, eyni zamanda qloballaşma və mədəniyyətlərarası ünsiyyət şəraitində yeni mədəni mənalardan formalaşmasında mühüm rol oynayır.

Açar sözlər: Linqvokulturologiya, mədəni kimlik, dil, ədəbiyyat, postkolonial diskurs, multikulturalizm

Introduction

In contemporary human studies, the interrelationship between language, culture, and identity has become one of the central areas of scholarly investigation. The rapid development of globalization, intercultural communication, migration, and digital interaction has intensified interest in the ways language reflects and shapes cultural consciousness. Within this context, linguoculturology has emerged as an interdisciplinary field that combines linguistics, cultural studies, anthropology, literary criticism, and sociology to explore how cultural meanings are encoded and transmitted through language. Linguoculturology studies language not merely as a system of communication, but as a cultural phenomenon that preserves historical memory, collective experience, and national worldview.

Language plays a crucial role in the formation of individual and collective identity. Every linguistic unit, including words, idioms, metaphors, symbols, and proverbs, carries cultural information that reflects the traditions, beliefs, and values of a particular society. Therefore, language becomes an important means of preserving cultural heritage and expressing cultural identity. Literature, as one of the most expressive forms of language, serves as a valuable source for linguocultural analysis because literary texts embody the cultural and ideological realities of their time. Through narrative structures, stylistic devices, and symbolic representations, writers construct images of national identity, cultural conflict, and intercultural interaction.

Particular importance is attached to postcolonial and multicultural literature, where issues of hybridity, displacement, alienation, and self-identification are frequently explored. Authors such as Jean Rhys, Hanif Kureishi, and Bapsi Sidhwa depict characters who struggle between different linguistic and cultural systems, revealing the complexity of identity formation in multicultural societies. Their works demonstrate that language is deeply connected with power, memory, and cultural belonging.

This article aims to examine the relationship between language, literature, and cultural identity from a linguocultural perspective. It explores how literary discourse reflects cultural values and how language functions as a medium for preserving, negotiating, and transforming identity in the modern world.

Materials and Methods

The present study is based on an interdisciplinary approach that combines methods of linguistics, literary criticism, cultural studies, and discourse analysis. The main materials of the

research consist of literary texts representing postcolonial and multicultural contexts, with particular attention given to the works of Jean Rhys, Hanif Kureishi, and Bapsi Sidhwa. These authors were selected because their literary works actively explore themes of cultural identity, hybridity, migration, colonial legacy, and intercultural communication.

The research employs the descriptive and analytical methods to examine the relationship between language and culture in literary discourse. Linguocultural analysis is used as the primary methodological framework for identifying cultural concepts, national symbols, culturally marked vocabulary, metaphors, idiomatic expressions, and stylistic devices within the texts. Special attention is paid to the ways linguistic elements reflect the worldview, traditions, beliefs, and value systems of different cultures.

In addition, elements of discourse analysis and comparative literary analysis are applied to investigate how identity is constructed through narrative structure, dialogue, and character representation. The comparative approach allows similarities and differences between various cultural and linguistic contexts to be identified. The study also relies on theoretical works in linguoculturology, postcolonial theory, and cultural identity studies in order to provide a conceptual foundation for the analysis.

The selected texts are analyzed qualitatively rather than quantitatively, since the primary aim of the research is to interpret cultural meanings and symbolic representations embedded in language. Through this methodological framework, the article seeks to demonstrate how literature functions as a medium for preserving cultural memory and expressing complex forms of individual and collective identity.

Results and Discussion

The study of a living language as a means of communication simultaneously requires the study of the culture of the people who speak that language. This is particularly connected with the fact that language is a mirror of national culture, and behind it stand the culture and way of life of the people.

In order to understand the processes taking place in society, language and speech culture are regarded as components of the general culture of every nation and every society. A person's language and culture determine their place in society and contribute to the formation and organization of their experience within the system of social production. Moreover, the language culture of every nation possesses its own distinctive features and individuality, which depend on many factors, including historical, economic, and spiritual ones. In turn, this makes it possible to preserve and accumulate those phenomena that contribute to the maintenance and development of a nation's culture. (Hall, 1973)

The integrated study of language and culture facilitates communication with native speakers and helps overcome difficulties that arise in the process of translation from a foreign language into the native language. This is because the vocabulary of a language is based on and nourished by the culture of the people who speak it. The national and cultural way of thinking cannot exist separately from language. They are universal in nature and complement and enrich one another. (Axundov, 1992)

Language is not only connected with culture, but it also emerges from its layers, expands, becomes enriched, and expresses culture. At the same time, language performs the functions of creating, developing, preserving, and transmitting culture from generation to generation (in the form of texts), while also being a part of culture itself. It is precisely through language that

genuine and objective works of both material and spiritual culture are created. From this perspective, among the newest linguistic disciplines within the cultural direction, Linguoculturology, which emerged at the intersection of linguistics and cultural studies in the early 1990s, occupies a special place in considering the relationship between language and culture as a complex synchronous process that reflects national-cultural features by preserving the linguistic and extralinguistic content of lexical units within the language system. (Ağabəyli, 2023)

Linguoculturology is a branch of science that studies the manifestations of a people's culture and reflects them in linguistics. At the same time, attention should not merely be focused on the "intersectional" nature of this new scientific field, but rather on its development as a new scientific direction and a new research domain in which existing limitations may be overcome. Therefore, linguoculturology is not a temporary union of linguistics and cultural studies, but an independent and interdisciplinary field with its own aims, tasks, methods, and object of study.

The orientation of linguoculturology toward a new system of cultural values and toward the objective interpretation of facts, events, and information from various spheres of society's cultural life distinguishes it from other humanitarian disciplines. As a relatively young branch of science, linguoculturology possesses a developed system of fundamental units reflecting the diversity of considering the cultural factor in language and the linguistic factor in human beings. (Воробьев, 1997)

As a special scientific discipline that emerged in the 1990s, it would not be entirely accurate to classify the stages of development of linguoculturology according to concrete and consistent criteria, primarily because only a short period of time has passed since the appearance of this field.

It seems logical to divide the development of linguoculturology into two periods: the first period is connected with the stage that determined the continuity in the development of linguistic science and includes the works of scholars such as Wilhelm von Humboldt, Alexander Potebnja, and Edward Sapir; the second period is characterized by the presentation of linguoculturology as an independent field of research. The dynamics of scientific development also make it possible to predict a third period — the present stage — in which fundamental interdisciplinary linguoculturology may become integrated into the academic system as a new discipline. This textbook is specifically devoted to exploring the possibility of teaching this new scientific field as an academic subject. (Маслова, 2001)

Among the main directions of modern linguocultural research, the following can generally be distinguished:

- the phraseological direction of linguoculturology, which reveals the cultural and linguistic competence of linguocultural subjects through the study of the phraseological fund of language;
- the conceptual direction of linguoculturology, which studies the national peculiarities in the use of concepts common to both scientific theories and everyday consciousness;
- the lexicographic direction of linguoculturology, which deals with the compilation of dictionaries reflecting the national and cultural uniqueness of languages;
- the linguodidactic direction of linguoculturology, which investigates problems of communication within the framework of intercultural communication, the "dialogue of cultures," and intercultural boundaries.

The results of the research demonstrate that language, literature, and cultural identity exist in a close and inseparable relationship. Language serves not only as a means of communication but also as a repository of cultural memory, collective consciousness, and national worldview. Every linguistic structure reflects the mentality, traditions, beliefs, and social values of the community in which it develops. Literature, in turn, becomes the artistic representation of these cultural meanings and transforms language into a powerful medium for expressing identity. Literary texts preserve cultural heritage, transmit historical experience from one generation to another, and shape perceptions of individual and collective belonging. The analysis of literary works confirms that cultural identity is often constructed through linguistic and stylistic choices. (Hall, 1976) Writers use metaphors, symbols, idiomatic expressions, folklore elements, and culturally marked vocabulary to create national and cultural images. Such linguistic elements allow readers to recognize the traditions, social norms, and emotional experiences associated with a particular culture. Therefore, literary discourse functions as a reflection of both language and culture simultaneously.

In the novels of Jean Rhys, (1966) especially in *Wide Sargasso Sea*, language becomes a tool for representing colonial oppression, fragmentation, and cultural hybridity. The protagonist, Antoinette Cosway, exists between Caribbean and European cultural worlds and experiences an identity crisis caused by colonial domination. Rhys combines English linguistic structures with Caribbean expressions, imagery, and landscape descriptions to emphasize the coexistence of different cultural identities. (Rhys, 1966) The symbolic use of nature, fire, and dreams reflects psychological alienation and the loss of cultural belonging. Through fragmented narration and shifting perspectives, the novel reveals how language reflects postcolonial identity conflicts.

Similarly, Hanif Kureishi (1996) explores multicultural identity and cultural displacement in *My Beautiful Laundrette* and his other literary works. His characters often struggle between Eastern traditions and Western modernity. Kureishi incorporates colloquial English, slang, and intercultural dialogue to represent hybrid cultural experiences in contemporary British society. Language in his works reflects social tension, generational conflict, and the search for belonging among immigrant communities. The interaction between different speech patterns symbolizes the coexistence of multiple identities within multicultural environments.

The works of Bapsi Sidhwa (1989) also illustrate the strong connection between language and cultural identity. In *Ice-Candy-Man*, linguistic diversity reflects the multicultural reality of pre-partition India. The novel includes expressions from Urdu, Punjabi, Hindi, and English, emphasizing the coexistence of different ethnic and religious communities. Through multilingual dialogue and culturally specific vocabulary, Sidhwa preserves the atmosphere of a society undergoing political and cultural transformation. The language of the novel reflects the trauma of partition, the fragmentation of identity, and the tension between communal harmony and violence.

The research also revealed that literary language frequently preserves national traditions and cultural values more effectively than ordinary discourse. Proverbs, idioms, myths, and folklore references found in literary texts function as carriers of collective cultural experience. These linguistic elements create continuity between past and present generations and contribute to the preservation of cultural memory. In many postcolonial texts, the revival of indigenous expressions and local speech patterns becomes a form of resistance against cultural assimilation and linguistic domination.

Another important result of the study is that globalization and migration have transformed the concept of cultural identity into a more dynamic and fluid phenomenon. Modern literature increasingly portrays characters who belong simultaneously to multiple linguistic and cultural spaces. This creates hybrid identities shaped by intercultural communication, migration, and transnational experiences. As a result, literary texts become spaces where identities are negotiated, reconstructed, and redefined.

The study confirms that linguoculturology provides an effective interdisciplinary framework for understanding these processes. By examining language within its cultural context, linguoculturology reveals hidden cultural meanings embedded in literary discourse. The interaction between language, literature, and identity demonstrates that literature is not merely an artistic phenomenon but also a cultural and linguistic archive preserving the worldview of society. Thus, literary texts play a significant role in shaping cultural identity, maintaining cultural continuity, and reflecting the changing realities of the modern multicultural world.

Conclusion

The present study has explored the interrelationship between language, literature, and cultural identity from a linguocultural perspective. The findings confirm that language is not a neutral system of communication but a culturally loaded phenomenon that reflects and shapes the worldview of its speakers. It carries within its lexical, grammatical, and stylistic structures the historical memory, value system, and collective identity of a community. Literature, as the highest artistic form of language, plays a fundamental role in preserving, interpreting, and transforming these cultural meanings.

The analysis of literary texts by Jean Rhys, Hanif Kureishi, and Bapsi Sidhwa demonstrates that cultural identity is continuously constructed through linguistic expression. These writers show how language becomes a site of cultural negotiation where issues such as hybridity, displacement, migration, and colonial legacy are articulated. Their works illustrate that identity is not fixed but dynamic, shaped by the interaction of multiple linguistic and cultural systems.

The study also highlights that linguoculturology provides an effective theoretical framework for analyzing how cultural meanings are embedded in literary discourse. Through metaphors, symbols, idioms, and narrative strategies, literature encodes cultural experience and transmits it across generations. In this way, it contributes both to the preservation of cultural heritage and to the creation of new cultural meanings in changing social contexts.

In conclusion, the research confirms that language and literature function as essential tools in constructing cultural identity. In the context of globalization and intercultural communication, their role becomes even more significant, as they enable both the continuity of cultural traditions and the emergence of hybrid identities in contemporary society.

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ENCODING SOCIAL ROLES OF ENGLISH AND UZBEK PROVERBS: AN ANALYSIS OF PRAGMATIC FUNCTIONS AND CULTURAL CONTEXTS

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Abstract: This article examines the encoding of social roles in English and Uzbek proverbs, focusing on their pragmatic functions and cultural contexts. The study aims to identify how proverbs reflect social norms, cultural values, national mentality, and traditional patterns of behavior. A corpus of 120 proverbs, including 60 English and 60 Uzbek examples, was analyzed according to four main social roles: family, labor, leadership, and gender. Comparative-descriptive and corpus methods were used to identify similarities and differences in the representation of social roles and the pragmatic functions of proverbs. The analysis demonstrates that proverbs perform various communicative functions, including advice, warning, instruction, motivation, and evaluation. The findings indicate that Uzbek proverbs tend to emphasize collective values, family relationships, social responsibility, and traditional roles, whereas English proverbs more frequently highlight individual responsibility, independence, and personal choice. The study also shows that proverbs serve as important carriers of cultural codes and contribute to the preservation and transmission of national identity. From a pragmatic perspective, they guide behavior, reinforce social norms, and regulate interpersonal relationships. Therefore, English and Uzbek proverbs provide valuable linguistic evidence of the close relationship between language, culture, social roles, and national values.

Keywords: Proverbs, social roles, linguaculturology, pragmatic analysis, cultural code, national mentality, paremiology, collectivism vs. individualism, normative function, folk wisdom.

İNGİLİZCE VE ÖZBEKÇE ATASÖZLERİNDE TOPLUMSAL ROLLERİN KODLANMASI: PRAGMATİK İŞLEVLER VE KÜLTÜREL BAĞLAMLARIN ANALİZİ

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Özet: Bu makale, İngilizce ve Özbekçe atasözlerinde toplumsal rollerin nasıl kodlandığını, bu atasözlerinin pragmatik işlevlerini ve kültürel bağlamlarını incelemektedir. Araştırmanın temel amacı, atasözlerinin toplumsal normları, kültürel değerleri, millî zihniyeti ve geleneksel davranış biçimlerini nasıl yansıttığını belirlemektir. Çalışmada aile, emek, liderlik ve toplumsal cinsiyet olmak üzere dört temel sosyal rol kapsamında 60 İngilizce ve 60 Özbekçe atasözü olmak üzere toplam 120 atasözü analiz edilmiştir. Karşılaştırmalı-betimleyici ve derlem yöntemleri kullanılarak atasözlerinin toplumsal rollerin aktarımındaki benzerlik ve farklılıkları ile pragmatik işlevleri belirlenmiştir. Analiz sonuçları, atasözlerinin öğüt, uyarı, yönlendirme, motivasyon ve değerlendirme gibi çeşitli iletişimsel işlevlere sahip olduğunu göstermektedir. Özbekçe atasözlerinde kolektif değerler, aile ilişkileri, toplumsal sorumluluk ve geleneksel roller daha belirgin şekilde vurgulanırken, İngilizce atasözlerinde bireysel sorumluluk, bağımsızlık ve kişisel tercih daha fazla öne çıkmaktadır. Araştırma ayrıca atasözlerinin kültürel kodların önemli taşıyıcıları olduğunu ve millî kimliğin korunmasına ve aktarılmasına katkı sağladığını ortaya koymaktadır. Sonuç olarak İngilizce ve Özbekçe atasözleri, dil, kültür, toplumsal roller ve millî değerler arasındaki güçlü ilişkiyi açıkça yansıtmaktadır.

Anahtar Kelimeler: Atasözleri, toplumsal roller, dil-kültürbilim, pragmatik analiz, kültürel kod, millî mantalite, paremiyoloji, kolektivizm ve bireysellik, normatif işlev, halk bilgeliği.

Introduction

The study draws upon several key theories:

1. Social Role Theory:

- **T. Parsons** views the role as a functional element of society that ensures its stability.
- **R. Linton** links the role to social status and expected behavior, emphasizing its cultural conditionality.
- **E. Goffman** introduces the dramaturgical concept, viewing social roles as "stage" performances in everyday life.

2. Pragmatics and Speech Act Theory:

- **J. Austin and J. Searle** describe how linguistic expressions perform illocutionary and perlocutionary functions.
- **H.P. Grice**, with his Cooperative Principle and maxims (quantity, quality, relevance, and manner), allows for the analysis of the communicative effectiveness of proverbs.

3. Linguacultural Approach:

- **V.A. Maslova and Yu.S. Stepanov** emphasize that proverbs are carriers of cultural codes and concepts, reflecting the value orientations and social norms of a people.

Research Methodology

The study utilized comparative-descriptive and corpus methods to comprehensively evaluate the pragmatic and linguacultural features of proverbs.

1. Corpus Formation: A corpus of 120 proverbs (60 English and 60 Uzbek) was compiled based on representativeness, frequency of use, and thematic significance. They were classified into four social roles: **Family, Labor, Leadership, and Gender**.

2. **Comparative-Descriptive Analysis:** Conducted to identify differences and similarities in the reflection of roles and the transmission of social norms through linguistic means (vocabulary, grammar, metaphor).

3. **Corpus Analysis Procedures:**

- Identification of **illocutionary functions** (e.g., advice, warning, instruction).
- Determination of the **perlocutionary effect** (expected behavior or reaction from the listener).
- Identification of **cultural codes** and concepts reflecting national values.

4. **Contextual Analysis:** Consideration of the pragmatic situation (conditions and purpose), communicative context (audience and speaker roles), and linguacultural specificities (mentality-based imagery).

5. **Data Verification:** A triple-check method was used where each element was evaluated by independent researchers to ensure reliability.

Results

Social Role	Uzbek Proverb	English Proverb	Pragmatic Function
Parenthood	<i>Ota – tog‘, ona – bog‘</i> (Father is a mountain, mother is a garden)	<i>Like father, like son</i>	Normative, Prescriptive
Labor	<i>Mehnat qilgan to‘yadi</i> (He who works will be full)	<i>No pain, no gain</i>	Motivational, Directive
Leadership	<i>El boshi – el bilan</i> (The head of the people is with the people)	<i>A leader is known in the storm</i>	Evaluative, Advisory
Gender	<i>Er – uy boshi</i> (The husband is the head of the house)	<i>A woman’s place is wherever she chooses</i>	Normative, Ideological

The analysis showed that **Uzbek proverbs** emphasize collective values and traditional roles, whereas **English proverbs** highlight individual responsibility and autonomy.

Discussion

The results confirm Goffman’s dramaturgical concept: proverbs function as "stage" scripts for social roles. Pragmatically, they direct behavior, reinforce norms, and serve as a discursive mechanism for regulating social interactions. Furthermore, they act as carriers of cultural codes, demonstrating the inextricable link between language, culture, and social regulation.

Conclusion

English and Uzbek proverbs reflect social roles deeply intertwined with national culture and communicative traditions. They function as normative guidelines, pedagogical tools, and means of transmitting cultural identity. The study confirms that proverbs play a key role in forming and maintaining the social fabric of a nation.

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THE IMPACT OF SOCIAL MEDIA AND GLOBALIZATION ON LANGUAGE: A SOCIOLINGUISTIC ANALYSIS OF LANGUAGE CHANGE AND IDENTITY CONSTRUCTION

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Abstract: The rapid expansion of globalization and digital technologies has significantly transformed language use across the world. Social media platforms have created new communicative environments where language evolves at an unprecedented pace. At the same time, globalization has intensified interactions among speakers from different linguistic and cultural backgrounds, leading to increased language contact and exchange. This paper examines the influence of social media and globalization on language from a sociolinguistic perspective. Particular attention is given to virtual identity construction, digital discourse, code-switching, linguistic hybridity, and emerging language practices among Azerbaijani youth. The study explores how young people adapt their linguistic behavior in online spaces and how digital communication shapes their social and cultural identities. It argues that contemporary communication technologies have accelerated linguistic change and reshaped the ways individuals negotiate social identities and cultural affiliations. While these developments promote linguistic creativity, innovation, and intercultural communication, they also raise concerns regarding language standardization, language maintenance, and the preservation of linguistic diversity in the modern digital era.

Keywords: sociolinguistics, social media, globalization, digital discourse, code-switching, virtual identity

**SOSIAL MEDIA VƏ QLOBALLAŞMANIN DİLƏ TƏSİRİ: DİL DƏYİŞMƏSİ VƏ
KİMLİK QURUCULUĞUNUN SOSİOLİNGVİSTİK TƏHLİLİ**

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Xülasə: Qloballaşmanın və rəqəmsal texnologiyaların sürətli inkişafı bütün dünyada dil istifadəsini əhəmiyyətli dərəcədə dəyişdirmişdir. Sosial media platformaları dilin misli görünməmiş sürətlə inkişaf etdiyi yeni kommunikativ mühitlər yaratmışdır. Eyni zamanda, qloballaşma müxtəlif dil və mədəni mənşəyə malik insanlar arasında qarşılıqlı əlaqələri intensivləşdirərək dillərarası təmas və mübadiləni artırmışdır. Bu məqalədə sosial media və qloballaşmanın dilə təsiri sosiolinqvistik aspektdən araşdırılır. Tədqiqatda virtual identikliyin formalaşdırılması, rəqəmsal diskurs, kod dəyişdirmə (code-switching), linqvistik hibridlik və Azərbaycan gəncləri arasında meydana çıxan yeni dil praktikalarına xüsusi diqqət yetirilir. Araşdırma gənclərin onlayn məkanlarda dil davranışlarını necə uyğunlaşdırdıqlarını və rəqəmsal ünsiyyətin onların sosial və mədəni kimliklərinin formalaşmasına necə təsir etdiyini təhlil edir. Məqalədə iddia olunur ki, müasir kommunikasiya texnologiyaları dil dəyişikliklərini sürətləndirməklə yanaşı, fərdlərin sosial kimliklərini və mədəni mənsubiyyətlərini necə ifadə etdiklərini də yenidən formalaşdırmışdır. Bu proseslər dil yaradıcılığını, innovasiyanı və mədəniyyətlərarası ünsiyyəti təşviq etsə də, eyni zamanda dil standartlaşdırılması, dilin qorunub saxlanması və müasir rəqəmsal dövrdə dil müxtəlifliyinin mühafizəsi ilə bağlı müəyyən narahatlıqlar doğurur.

Açar sözlər: sosiolinqvistika, sosial media, qloballaşma, rəqəmsal diskurs, kod dəyişdirmə, virtual identiklik.

Introduction

Language is a living and dynamic system that reflects social, cultural, political, and technological transformations. Throughout history, linguistic change has occurred through migration, trade, cultural interaction, and technological innovation. However, in the twenty-first century, the processes of globalization and digitalization have accelerated language change to an unprecedented degree.

Globalization has facilitated extensive interaction among individuals from different linguistic communities. Simultaneously, social media platforms such as Instagram, Facebook, TikTok, X (formerly Twitter), YouTube, and WhatsApp have transformed traditional communication patterns. These platforms allow users from diverse backgrounds to communicate instantly, creating a global linguistic environment characterized by constant innovation and adaptation.

From a sociolinguistic perspective, language functions not only as a means of communication but also as a tool for expressing identity and social belonging. Consequently, the effects of social media and globalization extend beyond vocabulary and grammar. They influence how individuals construct identities, establish relationships, and participate in local and global communities.

This paper explores the impact of social media and globalization on language by examining digital discourse, virtual identity, code-switching, linguistic hybridity, and contemporary language practices among Azerbaijani youth.

Social Media and Digital Discourse

Digital communication has created new forms of discourse that differ considerably from traditional written and spoken language. Social media users frequently employ abbreviations, acronyms, hashtags, emojis, memes, GIFs, and multimedia elements to convey meaning efficiently.

According to Crystal (2011), internet communication represents a unique linguistic environment that combines characteristics of both spoken and written language. Unlike traditional writing, digital discourse tends to be interactive, spontaneous, and highly contextual.

For example, expressions such as *LOL*, *OMG*, *BTW*, and *IDK* have become widely recognized beyond English-speaking communities. Similarly, emojis often function as independent communicative units capable of expressing emotions, attitudes, and intentions without verbal explanation.

The increasing use of multimodal communication demonstrates that meaning is no longer constructed solely through words. Instead, users combine textual, visual, and symbolic resources to communicate effectively in digital environments. Consequently, social media has become a significant catalyst for linguistic innovation and change.

Virtual Identity and Language Use

One of the most significant sociolinguistic consequences of social media is the emergence of virtual identities. Online platforms enable individuals to create, modify, and present multiple versions of themselves depending on context and audience.

Language plays a central role in this process. Users carefully select linguistic resources to project particular identities, social positions, and cultural affiliations. For instance, individuals may employ formal language on LinkedIn while adopting informal or humorous styles on Instagram and TikTok.

From a discourse-analytic perspective, virtual identity is continuously negotiated through language choices. Online users construct identities by selecting specific lexical items, discourse markers, hashtags, and stylistic features that align with desired social images.

Research suggests that digital communication has expanded opportunities for identity performance, allowing individuals to participate simultaneously in multiple social communities (Androutsopoulos, 2014). Consequently, language has become an increasingly important mechanism through which users establish authenticity, credibility, and social belonging in virtual spaces.

Globalization, Code-Switching, and Linguistic Hybridity

Globalization has intensified language contact on a global scale. Increased mobility, international education, migration, tourism, and digital connectivity have created multilingual environments where speakers frequently interact across linguistic boundaries.

One of the most visible outcomes of this process is code-switching - the alternation between two or more languages within a single conversation or communicative event. In multilingual societies, code-switching often serves important social functions, including identity expression, group solidarity, and communicative efficiency.

Among Azerbaijani bilingual speakers, particularly young people, code-switching frequently occurs between Azerbaijani, English, Russian, and Turkish. Examples commonly observed on social media include expressions such as:

- "Bu post həqiqətən **very inspiring** idi."
- "Sabah meeting-dən sonra danışırıq."

- “Bu idea çox maraqlıdır.”

These examples illustrate how speakers combine linguistic resources from multiple languages to construct meaning.

Closely related to code-switching is linguistic hybridity, which refers to the emergence of mixed linguistic forms resulting from sustained language contact. Hybrid expressions have become increasingly common in digital communication, where users creatively blend local and global linguistic elements.

Rather than indicating linguistic deficiency, these practices reflect speakers' ability to navigate multiple cultural and linguistic worlds simultaneously. Sociolinguists view such hybrid forms as evidence of linguistic adaptability and identity negotiation within globalized societies.

Azerbaijani Youth and Social Media Language Practices

The influence of social media on language is particularly visible among Azerbaijani youth. Young people constitute one of the most active groups on digital platforms and often serve as innovators in language change.

A notable characteristic of youth communication in Azerbaijan is the extensive incorporation of English lexical items into everyday discourse. Words such as story, post, reel, trend, follower, content, challenge, and podcast have become integrated into the vocabulary of many young speakers.

Examples frequently encountered on social media include:

- “Yeni story paylaşdım.”
- “Bu trend indi çox məşhurdur.”
- “Mənim content planım hazırdır.”
- “Yeni reel hazırlamışam.”

These expressions demonstrate how English terminology associated with digital culture enters Azerbaijani communication practices.

Furthermore, social media has encouraged the development of informal writing conventions characterized by shortened forms, non-standard spelling, and creative orthographic practices. Young users often prioritize efficiency and expressiveness over adherence to formal language norms.

While some scholars view these developments as a threat to language standards, others argue that they represent natural processes of linguistic evolution. From a sociolinguistic perspective, language change should be understood as a reflection of broader social transformations rather than linguistic decline.

Challenges and Opportunities

The influence of social media and globalization on language presents both opportunities and challenges. On the positive side, digital communication promotes linguistic creativity, intercultural dialogue, and global connectivity. Social media enables speakers of minority languages to create content, build communities, and increase linguistic visibility.

However, concerns remain regarding the growing dominance of English and other global languages. Excessive dependence on global linguistic norms may contribute to linguistic homogenization and reduce the use of local languages in certain domains.

For smaller linguistic communities, including many minority languages worldwide, maintaining linguistic vitality in the digital age remains a significant challenge. Policymakers

and educators must therefore encourage multilingualism while simultaneously supporting the preservation of national and regional linguistic heritage.

Conclusion

Social media and globalization have become powerful forces driving contemporary language change. Through digital discourse, virtual identity construction, code-switching, and linguistic hybridity, individuals continuously reshape linguistic practices in response to evolving social realities.

The case of Azerbaijani youth demonstrates how global and local linguistic resources interact within digital environments. The integration of English vocabulary, the prevalence of code-switching, and the emergence of hybrid linguistic forms illustrate the dynamic nature of language in a globalized world.

From a sociolinguistic perspective, these developments should not be viewed solely as threats to linguistic purity but rather as manifestations of language's adaptive capacity. Future research should continue examining how emerging technologies influence language use, identity formation, and linguistic diversity in increasingly interconnected societies.

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ORGANIZATION OF SPEECH THERAPY WORK IN KINDERGARTEN CONDITIONS

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Abstract: Speech is a vital mental function crucial for school success, relationships, career, and psychological well-being. Proper speech education forms the foundation of a child's intellectual and psychological maturity.

Speech training of preschool children is one of the most important components of thinking development and intellectual education. In preschool educational institutions operating in our country, all work related to speech development is organized in accordance with the Laws of the Republic of Azerbaijan "On Education of Persons with Disabilities (Special Education)," "On Education," and "On Preschool Education," as well as the resolutions of the Cabinet of Ministers of the Republic of Azerbaijan "On Approval of the Regulations on Speech Therapy Services" and "On Approval of the State Standard and Program of Preschool Education" dated July 16, 2010, No. 137.

In kindergartens, the acquisition of speech skills by preschool children, its development, and the correction of speech deviations are considered one of the main criteria for preparing the child for school education.

The main organizational forms of speech therapy work in kindergartens include:

- group classes with educators, fine arts, ecology, and physical education teachers;
- group, subgroup, and individual correction sessions with a speech therapist;
- individual sessions of a teacher-psychologist with children for the correction of intellectual and psychological development.

Correction Process and Structure

Systematic Blocks: Diagnostic, correctional, and evaluation-control.

Preparatory Stage: Articulation gymnastics, tongue and facial massage, alongside developing psychological foundations (hearing, vision, tactile perception, memory, thinking, and fine motor skills).

Core Development: Phonetic-phonemic, lexical-grammatical, and coherent speech, concluding with final diagnostics.

While the speech therapist leads correctional work—performing diagnostic, preventive, pedagogical, and consultative functions—teachers and specialists collaborate daily to support children with speech disorders.

Key words: speech, child, disorder, preschool age, kindergarten, speech therapist, phonetic-phonemic, lexical-grammatical, coherent speech.

MƏKTƏBƏQƏDƏR MÜƏSSİSƏLƏRDƏ LOGOPEDİK İŞİN TƏŞKİLİ

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Xülasə: Danışiq insanın məktəbdə uğur qazanması, münasibətləri, karyerası və psixoloji rifahı üçün həyati əhəmiyyət daşıyan zehni funksiyadır. Düzgün nitq təhsili uşağın intellektual və psixoloji yetkinliyinin əsasını qoyur.

Məktəbəqədər yaşlı uşaqların nitq təlimi təfəkkürün inkişafının və intellektual təhsilin ən vacib komponentlərindən biridir. Ölkəmizdə fəaliyyət göstərən məktəbəqədər təhsil müəssisələrində nitqin inkişafı ilə bağlı bütün işlər Azərbaycan Respublikasının qanunlarına və Nazirlər Kabinetinin müvafiq qərarlarına uyğun təşkil edilir. Bağçalarda nitq bacarıqlarının mənimsənilməsi və qüsurların düzəldilməsi uşağın məktəbəhazırlığının əsas meyarlarındanır.

Logopedik işin əsas təşkilati formaları

- Təربiyəçilər, təsviri incəsənət, ekologiya və fiziki tərbiyə müəllimləri ilə qrup məşğələləri
- Logopedlə qrup, subqrup və fərdi koreksiya seansları
- Psixoloqun intellektual və psixoloji inkişafı üçün uşaqlarla fərdi məşğələləri

Korreksiya prosesi və strukturu

- Sistematik bloklar: Diaqnostik, korreksion və qiymətləndirmə-nəzarət
- Hazırlıq mərhələsi: Artikulasiya gimnastikası, dil və üz əzələlərinin masajı, psixoloji təməllərin (eşitmə, görmə, taktil qavrayış, yaddaş, təfəkkür və incə motorika) inkişafı
- Əsas inkişaf: Fonetik-fonematik, leksik-qrammatik və bağlamsal nitq, sonda isə yekun diaqnostika

Korreksiya işinə logoped rəhbərlik etsə də, müəllimlər və mütəxəssislər nitq pozuntusu olan uşaqlara dəstək olmaq üçün gündəlik olaraq əməkdaşlıq edirlər.

Açar sözlər: nitq, uşaq, pozuntu. məktəbəqədər yaş. uşaq bağçası, loqoped, fonetik - fonematik, leksik- qrammatik, rabitəli nitq.

Introduction

Speech is one of the most important mental functions of a person. It is very difficult to evaluate the role that speech plays in human life. Speech communication not only establishes social relationships between people, but also develops the highest forms of cognitive activity. By mastering speech, the child perceives the generalized reality of the surrounding world and gains the ability to develop his or her cognitive activity, especially thinking. Well-developed speech is the key to success at school, the quality of interpersonal relationships with others, future career choice, and psychological comfort in family life. Therefore, proper speech education in a child is not only of aesthetic importance, but also forms the foundation of their maturity as a person and their intellectual and psychological development.

Speech training of preschool children is one of the most important components of thinking development and intellectual education. Mastering speech skills at preschool age is one of the main criteria for a child's readiness for school education.

In accordance with the Law of the Republic of Azerbaijan "On Preschool Education," speech therapy services organized for children in these institutions include:

- carrying out diagnostics of children's speech development in cooperation with the Psychological-Medical-Pedagogical Commission (PMPC);
- determining and implementing an individual program for eliminating speech disorders, taking into account the structure, severity, and clinical characteristics of speech disorders, as well as the child's individual and personal characteristics;
- spreading speech therapy knowledge among teachers and parents (or persons replacing them) about the use of special methods and techniques to help children with speech development disorders;
- forming children's free communication skills with peers;
- developing correct sound pronunciation and phonemic hearing in children;
- forming diction, intonation, stress, and speech tempo;
- developing children's creative speech skills through poems, fairy tales, and stories;
- developing coherent, grammatically correct dialogic and monologic speech in children;
- forming analytical-synthetic speech activity for preparing preschool children for school and literacy education;

- enriching active and passive vocabulary.

According to the Resolution of the Cabinet of Ministers of the Republic of Azerbaijan dated April 29, 2002, on the approval of the Regulations on Speech Therapy Services, the main purpose of organizing speech therapy work in kindergartens is to develop correct and fluent speech in preschool children, to identify deviations in speech development at an early age, and to eliminate them (3).

According to this resolution, the following areas of activity are determined in preschool institutions:

1. **Monitoring (diagnostic)** – creating conditions for the optimal selection of correctional goals, tasks, and means of their implementation, as well as for continuous diagnostic and prognostic monitoring.
2. **Correctional-developmental** – creating conditions aimed at correcting children's speech development and ensuring that a child with speech disorders reaches the speech development level appropriate to their age norm.
3. **Preventive** – creating conditions to improve the professional competence of teachers and parents in speech development issues, taking into account children's age, social-emotional and cognitive needs, and developmental opportunities.
4. **Information-methodological** – creating conditions for the development and application of innovative technologies in the field of correction of speech disorders, which increase the effectiveness of the correctional process.

I. Monitoring (Diagnostics) Area includes:

- Early identification of children with developmental problems (in preschool educational institutions)
- Initial examination of speech in preschool children
- Collection of medical and educational history, information about early development
- Psychological and pedagogical observation of young children with severe speech disorders
- Examination of children with severe speech disorders
- Dynamic observation during the educational process, intermediate assessments
- Diagnostics of the effectiveness of the correctional-pedagogical process

II. Organizational Area includes:

- Admission of children to speech therapy rooms (centers)

III. Analysis and Planning Area includes:

- Analysis of diagnostic results
- Keeping statistical records
- Development of the strategy of the correctional-pedagogical process in preschool educational institutions (annual work planning)
- Selection of methods for organizing the educational process in preschool institutions: formation of subgroups according to disorders and speech development levels, planning individual work with children
- Documentation support

IV. Correctional-Developmental Work Area includes:

- Classes with a speech therapist to improve various aspects of speech

- Joint activities with a teacher-psychologist to stimulate the psychological foundations of speech
- Joint activities with teachers
- Joint activities of the speech therapist and music teacher to develop the tempo-rhythmic organization of speech
- Joint activities of the speech therapist and physical education teacher to develop children's general motor skills
- Organization of necessary medical assistance for children with developmental problems

V. Preventive and Consultative Work Area includes:

- Familiarization with the results of in-depth medical examinations and taking them into account
- Providing consultative assistance to parents of children with developmental problems
- Early prevention of oral speech disorders in children
- Organization of consultative classes for young children with age-related speech disorders
- Prevention of written speech disorders and organization of propaedeutic classes

VI. Methodological Support Area includes:

- Methodological assistance to preschool institution staff in speech correction issues
- Study and implementation of correctional assistance options
- Creation of a correctional-pedagogical literature library in preschool educational institutions
- Self-education
- Participation in methodological associations of speech therapists
- Participation in seminars and conferences

VII. Optimization of the Correctional-Pedagogical (Speech Therapy) Process includes:

- Equipment for the speech therapy room
- Creation of a card index of visual, lexical, game, and educational materials
- Creation of audio-video technical, informational, and computer systems for the speech therapy room

VIII. Management Area includes:

- Conducting control tests and test tasks (when necessary)
- Information about the speech therapist's work at pedagogical and methodological councils
- Information about the speech therapist's work at the final parent meeting
- PMPC opinion regarding children's discharge issues
- Final interviews (final activities)
- Summarizing work for the academic year, digital reporting, and analysis of the work carried out by the speech therapist (Resolution of the Cabinet of Ministers of the Republic of Azerbaijan "On Approval of the State Standard and Program of Preschool Education", 2010).

In modern kindergartens, the system of work on the correction and development of speech of children attending speech therapy groups is primarily aimed at studying the sources and

causes of developmental deviations and is based on taking into account the child's individual psychological characteristics and corrective influence.

The main organizational forms of speech therapy intervention in kindergartens are:

- group classes with educators, fine arts, ecology, and physical education teachers;
- group, subgroup, and individual correction sessions with a speech therapist;
- individual sessions of a teacher-psychologist with children for the correction of intellectual and psychological development.

The activity of the speech therapy group includes systematic influences consisting of several interconnected blocks: diagnostic, correctional, and evaluation-control. This work is carried out in stages, and the preparatory stage begins with gymnastics of the articulation apparatus and massage of the tongue and facial muscles. At the same time, work is also carried out on the development of the psychological foundations of speech: hearing, vision, tactile perception and memory, thinking, and fine motor skills.

The diagnosis of speech development in preschool children is carried out in two directions:

1. Diagnosis of speech development in the study of the child's mental activity as one of the indicators of intellectual development;
2. Study of speech itself in all aspects (phonetic, lexical, grammatical, semantic).

In the studies of the first direction, diagnostics are carried out only for functions that directly indicate the level of intellectual development.

Since verbal scales and intellectual tests consisting of verbal tasks are in one way or another connected with speech development and language acquisition, in many tests intellectual indicators can only be determined indirectly through speech. For example, in the work "Methods of Measuring Mental Talent" published by A. Binet and T. Simon in 1905, a test was given to assess understanding of spoken language.

Using the Heidelberg Test (Germany), it is possible to determine the level of generalization and vocabulary acquisition. During the diagnosis of the development of imagination in preschool children (Uruntayeva methodology), the child is asked:

- a) to independently think of and tell a fairy tale on the proposed topic;
- b) to continue a fairy tale started by the teacher by adding certain characters.

At the same time, along with the characteristics of imagination, the consistency of speech, expressiveness, vocabulary, and syntax are also checked. In all cases, regardless of the purpose set by the researcher, the level of language knowledge is diagnosed (Law of the Republic of Azerbaijan "On Education", 2009).

Correctional work with children who have speech disorders mainly consists of work on phonetic-phonemic, lexical-grammatical, and coherent speech. At the phonetic stage, speech therapy work is organized to eliminate sound pronunciation deficiencies and to develop phonemic hearing. This work is carried out in the following sequence:

Preparatory Stage

This stage includes the formation of speech breathing, preparation of speech-sound and speech-hearing analyzers for the next stage of correctional activity, correct perception of sounds, and the formation of correct pronunciation skills and habits. At this stage, the development of correct pronunciation of sounds, their automation, and differentiation are carried out.

During lessons on lexical-grammatical and coherent speech, work is organized to increase children's vocabulary and develop their ability to construct coherent sentences. All correctional-developmental activities organized at this stage are mainly carried out in the form of games.

Recently, speech therapists have also been using the newest modern methods in the development of children's speech and in eliminating speech deficiencies.

The formation of universal learning activities (ULA) in the educational process is carried out in the context of mastering various academic subjects. Each subject, depending on its content and methods of organizing children's educational activities, opens up certain opportunities for the formation of universal learning activities. The corrective-developmental work of the speech therapist has great potential in the formation of universal learning activities, which are integrative characteristics of the educational process. Speech therapy classes develop all types of universal learning activities.

Personal ULA is the ability to independently make one's own choices in the world of feelings, thinking, and values, and to take responsibility for these choices. In kindergarten, interest in learning, learning motivation, and a positive attitude toward classes are formed through play, which is the leading activity of preschool children in correctional work.

Communicative ULA. It is impossible to imagine speech therapy classes without communication. Communicative ULA is the ability to communicate and interact with people. This activity is formed through working in groups and pairs, where the child or teacher takes on different roles, and as a result, children learn to conduct dialogue and build expressions.

Regulatory ULA. The formation of regulatory ULA in speech therapy classes, that is, the ability to organize one's own activity, occurs during accepting and maintaining tasks, planning, control, and evaluation. This is especially necessary for the child during the stage of sound automation, particularly when pronouncing sounds in spontaneous speech, where it is necessary to control syllables, words, and the position of the tongue. Consciously correcting incorrectly pronounced words and evaluating one's own speech by recognizing mistakes are considered the main stages of activity.

Cognitive universal learning activity, that is, the ability to think effectively and work with information, includes solving problem tasks and modeling. For example, using syllable schemes and comparing the sound or syllabic structure of words helps prevent errors in written speech in the future (dysgraphia based on undeveloped sound analysis and synthesis).

The correction of speech disorders and the development of correct speech in preschool children who fully perform the communicative function cannot be limited only to the speech therapy room. It becomes effective only when the skills acquired by children during speech therapy classes are supported by all participants in the correctional process.

Although the main leading force in correctional work is the speech therapist, who performs diagnostic, preventive, correctional-pedagogical, organizational-methodical, consultative, coordinating, supervisory, and evaluative functions, teachers and specialists work together throughout the day in the kindergarten on the correction and development of the speech of a child with speech disorders. The child is the central figure of the entire correctional process.

In a preschool educational institution, a strict "speech therapy regime" is created regarding the general speech culture and oral speech of children and adults.

The requirements established in the speech therapy regime include the following:

- correct and expressive speech of all members of the teaching staff;

- the teacher's knowledge of the child's speech abilities during a specific period;
- all teachers must require the correct pronunciation of new words and terms by children;
- every teacher is obliged to correct speech mistakes in children at all times and everywhere;
- it is not allowed to express doubt in front of children about overcoming speech disorders; confidence in success should be instilled in the child;
- children should be required to speak as taught by the speech therapist;
- every positive change in the child's speech should be encouraged;
- attendance at classes should be monitored;
- complete and detailed answers should be obtained for every question.

The implementation of the speech regime contributes to the effectiveness of correctional work and also prevents the emergence of other speech defects.

Mutual cooperation in the work of all teachers makes it possible to correct speech defects more quickly, helps automate sounds, form lexical and grammatical constructions, accumulate vocabulary, and develop coherent speech.

Thus, correctional-pedagogical activity is a rich system of medical-pedagogical measures aimed at the timely correction of sensorimotor, motor, and speech disorders, preventing school failure (dysgraphia and dyslexia), preventing social maladjustment, and supporting development. It represents a priority triad of areas: protection of child health, corrective work, and intellectual-cognitive development.

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FOREIGN LANGUAGE LEARNING AT AN EARLY AGE AND ITS CHARACTERISTICS

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Abstract: In the context of globalization and increasing migration processes, the development of intercultural awareness is important in all educational settings, especially in the education of children in preschool and primary school. According to the principles of intercultural pedagogy, teachers and early childhood educators are called upon to actively engage with the cultural diversity of different societies. Cultural experiences can arise from the learner's contact with other ways of life, for example, directly, through personal encounters with members of other cultural groups, or indirectly through the use of language-based teaching materials. These indirect cultural experiences are mainly transmitted by the educator. As facilitators and initiators of intercultural learning processes, educators aim to develop intercultural competence, which, according to Byram and Doyé (1999), is defined as the combination of skills, attitudes and knowledge complemented by the individual's personal values. It is precisely these goals that seem to be desirable to achieve within the framework of early childhood education. Therefore, the module described here first covers the conceptual possibilities of early exposure to different languages and cultures, as well as meaningful engagement with foreign languages in kindergarten. The development of a positive attitude towards other cultures and their languages, as well as the promotion of language awareness, together with an initial overview of the foreign language repertoire, form the basis of multilingualism and cultural competence education. Festivals and celebrations from other countries offer a particularly suitable and original opportunity for intercultural learning. Especially in multicultural children's groups, special days can be highlighted by including celebrations by children from other cultural backgrounds. It is useful to involve parents whenever possible, for example by preparing traditional dishes together, learning typical songs or hearing about the unique aspects of the respective festival. However, even in exclusively German-speaking children's groups, it is useful and desirable to include holidays from other countries in order to increase children's awareness of cultural diversity. Holidays are associated with rituals. When children learn about these rituals in kindergarten, they learn about other cultures. At an appropriate point, it is possible to discuss with children the commonalities of the festivals.

Keywords: learning, globalization, second language, integration, acquisition process, motivation.

ERKƏN YAŞLARDA XARİCİ DİLİN ÖYRƏNİLMƏSİ VƏ ONUN XÜSUSİYYƏTLƏRİ

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Xülasə: Qloballaşma və artan miqrasiya prosesləri kontekstində mədəniyyətlərarası şüurun inkişafı bütün təhsil müəssisələrində, xüsusən də məktəbəqədər və ibtidai məktəb uşaqlarının təhsilində vacibdir. Mədəniyyətlərarası pedaqogikanın prinsiplərinə əsasən, müəllimlər və erkən uşaqlıq təhsili müəllimləri müxtəlif cəmiyyətlərin mədəni müxtəlifliyi ilə fəal şəkildə əlaqə qurmağa çağırılırlar. Mədəni təcrübələr öyrənmənin digər həyat tərzləri ilə təmasından, məsələn, birbaşa, digər mədəni qrupların üzvləri ilə şəxsi görüşlər vasitəsilə və ya dolaylı yolla dil əsaslı tədris materiallarından istifadə etməklə yarana bilər. Bu dolaylı mədəni təcrübələr əsasən müəllim tərəfindən ötürülür. Mədəniyyətlərarası öyrənmə proseslərinin fasilitatorları və təşəbbüskarları olaraq, müəllimlər mədəniyyətlərarası səriştəni inkişaf etdirməyi hədəfləyirlər ki, bu da Byram və Doyé-yə (1999) görə, fərdin şəxsi dəyərləri ilə tamamlanan bacarıqların, münasibətlərin və biliklərin birləşməsi kimi müəyyən edilir. Məhz bu məqsədlərə erkən uşaqlıq təhsili çərçivəsində nail olmaq arzuolunan görünür. Buna görə də, burada təsvir edilən modul əvvəlcə müxtəlif dillərə və mədəniyyətlərə erkən məruz qalmanın, eləcə də uşaq bağçasında xarici dillərlə mənalı ünsiyyətin konseptual imkanlarını əhatə edir. Digər mədəniyyətlərə və onların dillərinə qarşı müsbət münasibətin inkişafı, eləcə də dil məlumatlılığının təşviqi, xarici dil repertuarına ilkin baxışla birlikdə çoxdillilik və mədəni səriştə təhsilinin əsasını təşkil edir. Digər ölkələrin festivalları və şənlikləri

mədəniyyətlərarası öyrənmə üçün xüsusilə uyğun və orijinal bir fürsət təqdim edir. Xüsusilə çoxmədəniyyətli uşaq qruplarında xüsusi günlər digər mədəni mənşəli uşaqların qeyd etmələrini əhatə etməklə vurğulana bilər. Mümkün olduqda valideynləri cəlb etmək faydalıdır, məsələn, birlikdə ənənəvi yeməklər hazırlamaq, tipik mahnılar öyrənmək və ya müvafiq festivalın unikal aspektləri haqqında eşitməklə. Uşaqların mədəni müxtəliflik haqqında məlumatlılığını artırmaq üçün digər ölkələrdən tətilləri də daxil etmək faydalı və arzuolunandır. Bayramlar rituallarla əlaqələndirilir. Uşaqlar uşaq bağçasında bu rituallar haqqında məlumat əldə etdikdə, digər mədəniyyətlər haqqında da məlumat əldə edirlər. Müvafiq bir məqamda uşaqlarla festivalların ortaq cəhətlərini müzakirə etmək mümkündür.

Açar sözlər: öyrənmə, qloballaşma, ikinci dil, integrasiya, mənimsəmə prosesi, motivasiya.

Introduction

It aims to convey the basics of educational work that focuses on English and promotes linguistic tolerance and intercultural communication in other foreign languages (French, Turkish, Italian). This module combines theoretical foundations and an application-oriented practical component. Candelier, Oomen-Welke and Perregaux (2004) recommend three strategies to strengthen multilingual and multicultural competence structures in this context:

1. Integrated teaching and learning of foreign languages: Using the own language (or the language of the educational institution) as a base can facilitate the acquisition of the first foreign language; learning both languages together facilitates access to other foreign languages later.
2. Creating cross-references between related languages: Here, the acquisition of different but related languages takes place in parallel, regardless of whether the learner's mother tongue belongs to the same language family or not.
3. Promoting Language Awareness

The Common European Framework of Reference (CEFR) calls this "a kind of general language education perspective" that provides an ideal foundation for further foreign language teaching in preschool and primary school. The theoretical component begins by addressing important aspects of existing theories on first and second language acquisition. One module focuses in particular on the concept of language awareness and its content-related possibilities. It also examines the historical development of foreign language teaching and the related importance of intercultural competence. To this end, various concepts of foreign language teaching are presented as examples, focusing on different aspects of encounter language models and the idea of developing language awareness. The concept of intercultural education and its current application to foreign language teaching in preschool institutions is another key element of the theoretical section. The practically oriented part of the event focuses in particular on the concrete application of intercultural learning and the creation of language awareness through foreign language experiences in early childhood education. The following questions are central here: What contribution do foreign-language and bilingual picture books, songs, films, etc. make?

Early foreign language learning is becoming increasingly common across Europe: the Eurydice Report published in 2005 states that in many countries at least half of primary school pupils learn a foreign language (Bergelson.E.2024). Five European countries even offer two foreign languages in primary school: in Luxembourg this applies to 80% of pupils, and in Estonia, Finland, Sweden and Iceland it applies to at least 10%. The data collected shows that the first foreign language is generally compulsory from the first year of school. In Spain and the German-speaking part of Belgium, most children start learning a foreign language in

kindergarten. Even outside Europe, foreign language learning starts at preschool age. For example, in Hong Kong, Cantonese, the regional dialect Putonghua and English are integral parts of the pre-school curriculum.

In Germany, English lessons are compulsory in primary school from at least the third grade - in Baden-Württemberg even from the first grade. Early foreign language learning in kindergarten is not yet the norm in Germany. However, there are many reasons to believe that in the future, foreign language learning will be given considerable attention in early childhood education. 70% of the world's population is already multilingual, and many children are growing up bilingual due to migration and mixed-language marriages. However, a fundamental distinction must be made between bilingual language learning in the home environment and foreign language learning in kindergarten, which is the main topic of this module. With correspondingly less contact time and less internal need to learn the language, the goal here cannot be to achieve bilingualism. Instead, the goal is to establish contact with different languages, acquire basic language skills, and promote intercultural awareness and language awareness. Such a concept makes more sense than a focus on a foreign language purely for communicative competence. When learning foreign languages, it is useful to distinguish between learning and mastering.

Discussion

While learning is understood as a conscious process, acquisition occurs subconsciously. Results of research on early foreign language learning: In 1996, the European Commission, in its White Paper on Teaching and Learning: Towards a Cognitive Society, declared that every EU citizen should be able to communicate successfully in at least three common languages in the future. It further stated: "If one wishes to achieve a genuine acquisition of three common languages, it is desirable to start learning a foreign language from kindergarten." (Sunardi.2025)

A comprehensive research report aimed at assessing the results of early foreign language learning was also supported within the framework of the publication (Noviyenty.L.2017). The main focus of this work is on early foreign language learning experiences in primary school, but individual results for the preschool sector can also be found here: For example, the impact of foreign language learning on children aged 4-6 in different parts of Italy was investigated: The experimental group was taught in both standard and regional varieties of the foreign language and in Italian, while the comparison group was taught only in standard Italian. The two groups were matched in terms of age, socio-cultural background, gender, place of residence and general intelligence. The results, based on specially developed tests of metalinguistic awareness, showed a more developed awareness of the structure of the Italian language in the experimental group than in the comparison group. This suggests that early initiation into foreign language learning can also have an impact on the mother tongue and the development of linguistic awareness. In addition, Schmidt-Schönbein provides a summary of additional findings that are relevant to primary school but can be transferred to the preschool level: Early learners who have studied for a longer period of time show an advantage compared to "newcomers to secondary school". This advantage is sometimes limited to areas such as listening comprehension and to certain students, such as fast learners. The duration of the advantage cannot yet be determined, as long-term effects in secondary school have not yet been measured. A larger number of studies have finally recorded results on children's communication skills.

In this respect, the results are generally positive, even if in some cases quite modest. Comprehension skills (listening and reading comprehension) develop more quickly than precise, creative production skills. All studies show a positive effect on the affective sphere, a positive attitude towards languages, culture, speakers of the languages in question, language learning and the development of self-confidence. With regard to existing model projects, a distinction can be made between three different framework conditions for foreign language learning in kindergarten:

1. Linguistic integration of foreign children (language support in the children's mother tongue)
2. Preservation of regional and minority languages (the aim is to keep the relevant minority language and the associated cultural heritage alive; for example, the teaching of Romansh in Switzerland)
3. Language as an enriching educational asset (teaching languages as an additional educational element)

Methodology and Materials

The preferred form of foreign language teaching in this module can be described as a combination of versions 1 and 3, since in addition to the more common foreign languages, English and French, migrant languages should also be included. However, if there are early childhood educators and children who speak these regional languages, version 2 is not excluded by the concept. Early foreign language learning should not be misunderstood as premature, school-based, systematic foreign language teaching. It is particularly important for the early childhood education sector that the form of teaching meets the individual learning needs of children.

For a long time, the demand to start foreign language teaching as early as possible was justified by the so-called critical period hypothesis. This hypothesis is based on the assumption that there is an optimal period associated with maturation for language acquisition. Scientists who subscribe to this hypothesis base their thesis on the fact that the plasticity of the human brain decreases from birth and that the permanent organization of all brain functions is reached around the age of nine. By the age of four, the adult brain is reported to have already reached 87% of its maturity. In addition, there is an increasing functional differentiation of the brain, resulting in lateralization. This means that between the ages of three and ten, only the left hemisphere increasingly takes over language-related functions. This leads to the conclusion that successful learning of foreign languages is only possible until the age of nine or ten, or at the latest until adulthood. While such a classification is undoubtedly useful for the acquisition of a first language, recent studies have shown that the hypothesis of a universal "critical period" for the acquisition of a foreign language is not suitable. The results cannot be simply transferred from the first language to the second, due to the fact that the time of contact with the relevant language is shorter and, moreover, the urgency to learn the language is lower.

When it comes to learning a foreign language, one can refer to the findings of the brain researcher Manfred Spitzer. He prefers the terms "sensitive period" or "regulatory period" to the term "critical period". At such stages, certain facts or skills are particularly easy to learn and open up a certain space for further learning, but this is not entirely fixed. The latter aspect is important in relation to foreign language learning: For example, although infants can only distinguish the sounds of other languages, as well as the sounds of their native language, by four

to six months of age, this ability is not completely lost. As an example, Spitzer cites studies with Japanese children who, even in adulthood, learn to distinguish the sounds /r/ and /l/ with considerable additional effort. As for lateralization, it can also be argued that the process of brain reorganization is largely complete by the age of four or five. However, language acquisition itself is not affected. This suggests that the shift of language brain functions to the left hemisphere is not essential for language acquisition. There are numerous connections between the hemispheres of the brain that can be used for learning. Therefore, the assumption of "the sooner, the better" cannot be accepted in general. Although age clearly affects foreign language learning, this effect may vary depending on the learner and the specific aspect of language learning. In general, age is only one of many influencing factors and therefore has a limited impact on learning success. However, there are areas where children are superior to adults in learning a foreign language: (Noviyenty, L. (2017)

Sound differentiation: The ability to distinguish, segment and repeat foreign sounds is higher at a younger age than in older people.

Language plasticity: Due to the differentiation of the cerebral hemispheres during childhood and adolescence, "plasticity" decreases with age, which negatively affects the acquisition of pronunciation and grammar.

Motivation and emotions: Younger learners are less hesitant to use a foreign language, even if their language skills are still relatively low. In the literature, this is often described with the terms "readiness to take risks" and "tolerance for uncertainty". This means that learners are not particularly afraid of making mistakes or not expressing themselves clearly enough. Their curiosity about the foreign country and its inhabitants is also often stronger than adults.

Variety of exercises: Compared to adults, children are more open to playful, creative, movement-oriented and imitative exercises.

Speed of learning: Adults are more experienced in learning and, because they have more knowledge about the world, can understand and categorize new information more easily than children.

The above points show that the phrase "windows of opportunity", often cited in research, is appropriate for the period of early foreign language learning in primary education. However, the advantages of early foreign language learning lie in the context of children's general learning, which differs significantly from the learning behavior of adults and older children. Therefore, it should be taken into account, first of all, that in early childhood children are still intensively engaged in the development of their native language or the dominant language of the environment (second language). In addition, the child's need and desire to learn a foreign language are still almost absent or not at all in kindergarten. Both aspects are relevant for the design of a foreign language curriculum. Early childhood educators, in addition to the specific characteristics of learning behavior in early childhood, must have in-depth knowledge of language acquisition in general. Such knowledge is important for the acquisition of both the native language and additional languages. Therefore, within the framework of this module, existing knowledge should be updated and deepened. Aspects of lexical and semantic development are particularly relevant in the field of language acquisition, as they can be used to draw conclusions regarding the learning of additional languages. There are various theories regarding the development of word meaning. (Goodluck, H. (2010)

Semantic Features Hypothesis

This hypothesis is based on the assumption that the meaning of a word consists of a fixed set of semantic features. These features are gradually learned by the child in the process of differentiation from the general to the specific. During the acquisition process, overextensions often occur, that is, a word is used for objects that are not appropriate for it from a developmental perspective. For example, a child may use the word "dog" for other animals that have the property of "four legs". These overextensions are assumed to occur on the basis of perceptual features. (Sunardi. (2025)

Functional Basis Hypothesis

Supporters of this hypothesis assume that the formation of a concept occurs in three stages:

1. The perception of an object is formed on the basis of its functions, that is, the object is perceived on the basis of its relationship to other objects, people, places and events. This is a single process.
2. Then, an analytical process takes place, in which the object is compared with other similar objects.
3. The fixed features of the object are isolated. These features are called the "functional basis" by supporters of the hypothesis.

Conclusion

Once the concept of an object is formed, it is assigned a word from the adult world. Thus, the meaning of the word is derived from experiencing the active functions of the object. An important point to consider with this hypothesis is that functional knowledge can be separated from perceptual knowledge and has dominance over the latter. Such a separation is not given in motor activities. These are always associated with sensory impressions. In addition, the central position of functional knowledge could not be consistently confirmed. Prototype Theory In contrast to the feature hypothesis, this theory takes into account that not every member of a class of objects has the same properties. There are prototype examples in which a large part of the properties of the class are applied, and less prototype examples in which this is applied only to a limited extent.

It is assumed that children develop a single, image-like representation of an object and isolate individual properties when the single structure is activated by a similar object. Initially, children use a word in a situation in which they encounter it for the first time, and eventually use it in less prototype, but similar situations. This can explain both under- and over-extensions: they are an expression of a blurred distinction between concepts. Prototype theory can also explain the instability and, above all, the changing characteristics of children's concepts. Research shows that the reuse of words is based on perceptual, emotional, functional and locative criteria. Therefore, a stimulating environment promotes the development of word meaning. This fact should also be taken into account when learning foreign languages. New words should be learned through experience and applied in the most authentic context possible.

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LEARNING LANGUAGES IN A MULTILINGUAL ENVIRONMENT: OPPORTUNITIES, DIFFICULTIES, AND STRATEGIC SOLUTIONS

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Abstract: This study examines the role of a multilingual environment in learning new languages. Learning a new languages is generally difficult, but being in a multilingual environment provides an advantage and makes the learning process easier. A multilingual environment offers constant exposure to different linguistic structures, vocabulary, and real life communications situations. Therefore, the study shows than such am environment improves listening and speaking skills, which in turn enhances overall comprehension more effectively than traditional, grammar based learning method

However, the study also reveals certain difficulties in learning languages in a multilingual *environment*, such as mixing languages a lack of structured and formal grammer-based learning and on overreliance on informal or slang-based communicatons. Therefore, it is essential to use appropriate strategies when learning new languages.

A multilingual environment is a powerful tool for language acquisitions if the right strategies are used, continuous practice is maintained and active engagement is ensured. For exmanple, when you are im a multilingual environment and try to learn a language, communicating with others improves your communication skills and your level of understanding.

In addition, the study emphasizes that multilingual environments do not only support language learning, but also help learners develop cultural awareness and social confidence. When learners communicate with people from different linguistic backgrounds, they become more open-minded and flexible in using language. This process encourages learners to compare languages, notice similarities and

differences, and understand how language functions in different social contexts. As a result, language learning becomes a more natural, interactive, and meaningful process rather than only memorizing grammar rules or vocabulary lists.

Moreover, a multilingual environment motivates learners to use the target language in real situations. Real communication allows learners to make mistakes, correct themselves, and gradually improve their fluency. Therefore, the study suggests that multilingual environments can be especially useful when they are supported by regular practice, formal learning, and conscious language use

Key words: multilingual environment, language learning, strategies, communications, second language acquisition, intercultural competence

ÇOXDİLLİ MÜHİTDƏ DİLLƏRİN ÖYRƏNİLMƏSİ: İMKANLAR, ÇƏTİNLİKLƏR VƏ STRATEJİ HƏLLƏR

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Xülasə: Bu tədqiqat yeni dillərin öyrənilməsində çoxdilli mühitin rolunu araşdırır. Yeni dil öyrənmək ümumiyyətlə çətin bir proses olsa da, çoxdilli mühitdə olmaq müəyyən üstünlüklər yaradır və öyrənmə prosesini asanlaşdırır. Çoxdilli mühit müxtəlif linqvistik strukturlara, söz ehtiyatına və real həyatdakı ünsiyyət vəziyyətlərinə daimi məruz qalma imkanı yaradır. Beləliklə, tədqiqat göstərir ki, bu cür mühit dinləmə və danışmaq bacarıqlarını inkişaf etdirir, bu da öz növbəsində ümumi anlama qabiliyyətini ənənəvi, qrammatikaya əsaslanan təlim metodlarından daha effektiv şəkildə artırır.

Lakin tədqiqat çoxdilli mühitdə dil öyrənməyin bəzi çətinliklərini də üzə çıxarır; buraya dillərin bir-birinə qarışdırılması, strukturlaşdırılmış və rəsmi qrammatikaya əsaslanan təhsilin çatışmazlığı, həmçinin qeyri-rəsmi və ya jarqon (slenq) əsaslı ünsiyyətdən həddindən artıq asılılıq daxildir. Buna görə də, yeni dillər öyrənməkə müvafiq strategiyalardan istifadə etmək olduqca vacibdir.

Düzgün strategiyalar tətbiq edildikdə, daimi praktika saxlanıldıqda və fəal iştirak təmin olunduqda, çoxdilli mühit dil mənimsənilməsi üçün güclü bir vasitədir. Məsələn, çoxdilli mühitdə olarkən dil öyrənməyə çalışdıqda başqaları ilə ünsiyyət qurmaq sizin kommunikasiya bacarıqlarınızı və anlama səviyyənizi təkmilləşdirir.

Bundan əlavə, tədqiqat göstərir ki, çoxdilli mühit yalnız dil öyrənməyə deyil, həm də mədəni dünyagörüşünün inkişafına kömək edir. Müxtəlif dillərdə danışan insanlarla ünsiyyət qurmaq öyrənən şəxslərin daha açıq fikirli olmasına, fərqli mədəniyyətləri daha yaxşı başa düşməsinə və sosial mühitdə özünü daha rahat ifadə etməsinə şərait yaradır. Bu cür mühitdə dil öyrənən şəxs yalnız sözləri və qrammatik qaydaları yadda saxlamır, həm də həmin dilin gündəlik həyatda necə istifadə olunduğunu müşahidə edir.

Eyni zamanda, çoxdilli mühit dil öyrənən şəxslərin motivasiyasını artırır. Çünki insan öyrəndiyi dili real danışmaq zamanı istifadə etdikdə həmin dilin əhəmiyyətini daha yaxşı anlayır. Bu səbəbdən çoxdilli mühit düzgün istiqamətləndirildikdə dil öyrənmə prosesini daha səmərəli və maraqlı edir

Açar sözlər: çoxdilli mühit, dil öyrənmə, strategiyalar, ünsiyyət, ikinci dil, mədəniyyətlərarası kommunikasiya

Introduction

In the modern global world, learning new languages is becoming very important. A multilingual environment is a place where several languages are used in everyday life. This

environment gives people in advantage to learn in language and develop their skills trough communication rather than only theoretical learning.

Although a multilingual environment has advantages for language learning, learning several languages at the same time can still be difficult. That is why it is important to use the right strategies, and there are many different ones. In my opinion, one useful strategy is to organize languages by topic, so you know which language to use for which subject. Another strategy is speaking slowly, that helps to understand and learn the language more easily. Otherwise, it is possible to mix the languages. Therefore, being in such an environment has both advantages and risk, as mentioned above.

Furthermore, multilingual environments are becoming more common because of globalization, migration, international education, and digital communication. Today, many people study, work, or live in places where more than one language is used. For example, students may use their native language at home, English at university, and another language with friends or on social media. This situation creates many opportunities for learning, but it also requires learners to be careful and organized.

A multilingual environment can be considered a natural classroom because learners are surrounded by different languages in daily life. Unlike traditional classrooms, where learners mostly study grammar and vocabulary, multilingual environments provide real examples of pronunciation, intonation, expressions, and cultural meanings. However, without proper guidance, learners may become confused or may not develop strong grammar skills. Therefore, this research focuses not only on the benefits of multilingual environments but also on the problems and strategies connected with this type of language learning.

Methodology and materials

This study uses a qualitative methods to look at how multilingual environments affect language learning. Information was taken from academic articles, online sources and observations of people learning languages in multilingual communities. Both primary and secondary sources were used to better understand the topic

The analysis focuses on the advantages, difficulties, and useful strategies that learners use in multilingual environment

The qualitative approach was chosen because the topic is related to personal experience, communication habits, and learning behavior. Language learning in a multilingual environment cannot be fully understood only through numbers or statistics. It is also important to observe how learners communicate, what problems they face, and what methods they use to overcome these problems. For this reason, the study focuses on descriptive analysis and interpretation of different learning situations.

Discussion

The research shows that a multilingual environment has many advantages in our lives and help us improve our communications skills in language learning. However, there is also a risk of mixing language

A possible solution is to separate languages by situation. For example, speaking Azerbaijan language at home, English at university, Russian with friends and French language with other people. This is one of the best ways to solve this problem because it helps us focus on one language at a time in different situations. Using this strategy, we can learn and practice different languages at the same time

Another important advantage of a multilingual environment is that it improves learners' ability to think in different languages. When a person uses several languages regularly, they begin to understand different sentence structures, expressions, and ways of thinking. This can improve mental flexibility and make it easier to learn additional languages in the future. For example, a learner who already knows Azerbaijani, Russian, and English may find it easier to learn Turkish or another European language because they already have experience comparing languages.

Conclusion

A multilingual environment helps us learn several languages at the same time. However, there is also a risk of mixing them. As mentioned above, there are different ways to solve this problem. While being in a multilingual environment, we can also improve our listening and speaking skills. Although learning new languages through theoretical methods can be difficult, a multilingual environment makes language learning easier and helps improve communication skills.

In conclusion, a multilingual environment creates many opportunities for learning new languages because it gives learners constant exposure to different languages and real communication situations. It helps improve listening, speaking, vocabulary, pronunciation, and cultural understanding. At the same time, it can create some difficulties, especially when learners mix languages or depend too much on informal communication.

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MULTILINGUALISM AND SECOND LANGUAGE ACQUISITION: WHAT WE KNOW AND WHY IT MATTERS

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Abstract. Most people in the world use more than one language every day, yet our theories about how people learn languages still often assume a monolingual starting point. This article reviews what research tells us about how multilingual individuals acquire additional languages differently from monolinguals. Drawing on several major theories—Universal Grammar, the Competition Model, Usage-Based Theory, and Dynamic Systems Theory—I explain how prior languages shape new learning through cross-linguistic influence and cognitive adaptation. The evidence shows that multilingualism brings real cognitive benefits, including better attention control and deeper understanding of how language works, while also creating unique challenges like managing multiple active languages at once. Individual experiences vary based on when languages are learned, how similar they are, and the social contexts in which they are used. For teachers, this means embracing approaches that allow students to use their full language repertoires rather than forcing artificial separation between languages. The article concludes by identifying gaps in current research and arguing that understanding multilingual acquisition requires moving beyond monolingual assumptions.

Keywords: multilingualism, second language acquisition, cross-linguistic influence, cognitive benefits, translanguaging, language transfer, dynamic systems

ÇOXDİLLİLİK VƏ İKİNCİ DİLİN MƏNİMSƏNİLMƏSİ: BİLDİKLƏRİMİZ VƏ ONLARIN MAHİYYƏTİ

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Xülasə: Dünyada insanların əksəriyyəti hər gün birdən çox dildən istifadə edir, lakin dillərin necə öyrənilməsi ilə bağlı nəzəriyyələrimiz hələ də çox vaxt monolingual başlanğıc nöqtəsini əsas götürür. Bu məqalə tədqiqatların çoxdilli fərdlərin əlavə dilləri monolingual fərdlərdən necə fərqli öyrəndiyini nə dərəcədə izah etdiyini nəzərdən keçirir. Bir neçə əsas nəzəriyyəyə—Universal Qrammatika, Rəqabət Modeli, İstifadəyə Əsaslanan Nəzəriyyə və Dinamik Sistemlər Nəzəriyyəsinə—əsaslanaraq, əvvəlki dillərin yeni öyrənməni dilərarası təsir və koqnitiv uyğunlaşma vasitəsilə necə formalaşdırdığını izah edirəm. Sübutlar göstərir ki, çoxdillilik diqqətin idarə olunması və dilin necə işlədiyini daha dərinəndə anlama kimi real koqnitiv üstünlüklər yaradır, eyni zamanda birdən çox aktiv dili eyni anda idarə etmək kimi özünəməxsus çətinliklər də ortaya çıxarır. Fərdi təcrübələr dillərin nə vaxt öyrənilməsindən, onların bir-birinə nə qədər oxşar olmasından və istifadə edildikləri sosial mühitlərdən asılı olaraq dəyişir. Müəllimlər üçün bu o deməkdir ki, tələbələr dilləri süni şəkildə ayırmağa məcbur etmək əvəzinə, onların bütün dil resurslarından istifadə etmələrinə imkan verən yanaşmalar tətbiq olunmalıdır. Məqalə mövcud tədqiqatlardakı boşluqları müəyyənləşdirməklə yekunlaşır və çoxdilli öyrənməni anlamaq üçün monolingual yanaşmalardan kənara çıxmağın vacibliyini vurğulayır.

Açar sözlər: çoxdillilik, ikinci dilin mənimsənilməsi, dillərarası təsir, koqnitiv üstünlüklər, dillərarası çevik istifadə, dil transferi, dinamik sistemlər

Introduction

The Monolingual Bias. Pick any busy street in Lagos, Mumbai, or Brussels, and you will hear people moving fluidly between languages—sometimes within a single sentence. This is not unusual. It is normal human behavior. Yet for decades, researchers studied language learning as if

everyone started from just one language. The result was theories that described multilingualism as basically "more of the same," missing what makes it distinctive.

This matters because the multilingual learner is not a blank slate. Someone who grew up speaking Wolof and French, learned Arabic for religious study, and is now studying Mandarin brings a mind already shaped by managing multiple linguistic systems. They have practiced suppressing one language while using another. They have noticed that different languages handle the same ideas differently. These experiences change how they approach any new language (Cook, 2016).

This article surveys what we know about how multilingual people learn additional languages. I focus on four questions: What theories help us understand this process? What happens cognitively when multiple languages live in one mind? What individual factors shape how this unfolds? And what should teachers do differently?

Methodology and Materials

This article adopts a systematic literature review approach. The primary aim of the study is to synthesize and comparatively analyze existing theoretical and empirical research on multilingualism and second language acquisition.

The materials were selected based on the following criteria:

1. direct relevance to multilingualism and second/third language acquisition,
2. theoretical significance and citation impact,
3. representation of diverse theoretical frameworks (e.g., Universal Grammar, the Competition Model, Usage-Based Theory, and Dynamic Systems Theory).

Data collection was conducted through major academic databases such as Google Scholar, Scopus, and Web of Science. Keywords used in the search process included "multilingualism," "second language acquisition," "third language learning," "cross-linguistic influence," and "cognitive effects of bilingualism."

The analysis followed a qualitative approach. The selected sources were organized thematically and examined across several key dimensions:

1. theoretical frameworks and their comparison;
2. mechanisms of cross-linguistic influence;
3. cognitive benefits and challenges;
4. individual differences and social factors;
5. pedagogical implications and applications.

This approach allows for the integration of multiple theoretical perspectives and provides a more comprehensive understanding of the complex nature of multilingual language acquisition.

Discussion

Universal Grammar and Its Limits. Noam Chomsky (1981) proposed that humans are born with Universal Grammar—an innate blueprint containing principles shared by all languages. This was revolutionary for understanding first language acquisition. But applying it to adult language learners sparked debate. Some argued adults lose direct access to this innate system and must rely on general learning strategies and their first language (Bley-Vroman, 1990). Others countered that the first language grammar transfers completely at the start, while Universal Grammar remains available to guide restructuring (Schwartz & Sprouse, 1996).

For multilinguals, this gets complicated. If you already speak two languages, which one serves as the starting point for a third? Research suggests prior languages can actually help rather

than hinder when structures overlap. The Cumulative-Enhancement Model shows that any previously learned language can facilitate new acquisition if similarities exist, challenging the idea that only the native language matters (Berkes & Flynn, 2012; Flynn et al., 2004).

Learning from Experience. Not everyone buys the innate grammar story. The Competition Model, for instance, argues that language processing relies on cues like word order and case marking, with different languages weighting these cues differently (Bates & MacWhinney, 1989). Multilinguals must learn which cues matter in which language, keeping multiple systems ready for rapid switching (MacWhinney, 2005).

Connectionist approaches go further, suggesting language knowledge emerges from patterns in the input, stored across neural networks. From this view, a multilingual does not have separate "boxes" for each language but one interconnected network where all languages influence one another probabilistically (Ellis, 2003).

Constructions and Dynamic Change. Usage-Based Theory sees language as built from constructions—recurring form-meaning pairings learned through repeated exposure (Tomasello, 2003). For multilinguals, constructions from all known languages share mental space, which naturally explains why speakers blend languages rather than keeping them perfectly separate (Bybee, 2010). This aligns with translanguaging theory, which treats language mixing as skilled performance, not failure (García & Li Wei, 2014).

Dynamic Systems Theory may capture multilingual reality best. It views language development as nonlinear and constantly shifting—small changes in one area can ripple through the whole system (de Bot et al., 2007). Proficiency in any language waxes and wanes with context, recent use, and even fatigue. The Dynamic Model of Multilingualism treats proficiency not as a fixed score but as an ever-adjusting equilibrium (Herdina & Jessner, 2002).

Cross-Linguistic Influence: It Is Complicated. When multilinguals produce a new language, all their known languages are partially active, not neatly shut off (De Bot, 2004). Researchers have identified several influence patterns: forward transfer from earlier to later languages, reverse transfer from later back to earlier, and lateral transfer between non-native languages (De Angelis, 2007; Pavlenko & Jarvis, 2002). What transfers depends not just on objective similarity but on what the learner *perceives* as similar, and sometimes an L2 becomes the default source regardless of typology (Kellerman, 1979).

Cognitive Benefits and Burdens. Managing multiple active languages exercises executive control systems in the brain—areas like the prefrontal cortex that handle attention and switching (Green, 1998; Abutalebi & Green, 2016). This constant workout appears to strengthen these systems. Bilinguals often outperform monolinguals on tasks requiring attentional control and task switching (Bialystok, 2017; Bialystok et al., 2012), though the size of this advantage is debated and varies across individuals (Donnelly et al., 2019).

For those with three or more languages, evidence is mixed—some find added benefits, others do not (Vega-Mendoza et al., 2015; Pelham & Abrams, 2014). What seems to matter most is not how many languages you know but how you use them: frequent switching, diverse contexts, and balanced proficiency predict stronger effects than language count alone (Surrain & Luk, 2019).

Multilinguals also show heightened metalinguistic awareness—an ability to think about language itself (Jessner, 2006). This shows up in better grammatical judgment, pattern

recognition, and explicit language analysis, which in turn helps them learn additional languages more efficiently (Cenoz, 2013).

What Shapes Individual Paths. Three factors stand out. First, when languages are learned matters. Earlier acquisition generally predicts more native-like outcomes, especially in pronunciation, though adults bring stronger analytical skills to the task (Johnson & Newport, 1989; Lenneberg, 1967; Hakuta et al., 2003). Second, how similar languages are affects transfer and difficulty, but perceived similarity often matters as much as objective linguistic distance (Ringbom, 2007; Kellerman, 1979). Third, social context shapes motivation and identity investment—learners commit to languages when they align with who they want to become (Norton, 2013).

Teaching Multilingual Learners. If multilingual minds integrate languages rather than separating them, classrooms should follow suit. Translanguaging—strategic use of the full linguistic repertoire—has shown benefits for learning, engagement, and identity validation (García et al., 2017; Canagarajah, 2011). It allows explicit cross-linguistic comparison that builds metalinguistic understanding (Swain & Lapkin, 2013).

Instruction should also manage cognitive load carefully, using structured input and tasks that leverage rather than ignore prior languages (VanPatten, 2004; Ellis, 2003). Assessment needs reform too: traditional tests penalize language mixing and assume monolingual norms, while portfolio and dynamic assessment better capture integrated competence (Shohamy, 2011; Council of Europe, 2020).

Conclusion

Multilingualism changes how people learn languages. Theories from Universal Grammar to Dynamic Systems each capture part of this reality, with dynamic approaches best handling the variability and context-dependence of actual multilingual performance. Cognitively, multilinguals gain enhanced control and awareness while managing genuine processing demands. Their learning trajectories depend on timing, language relationships, and social meaning.

Moving forward, research needs more longitudinal brain studies, better computational models, and inclusion of Indigenous and minoritized multilingual experiences (Makoni & Pennycook, 2007). Ultimately, understanding language acquisition requires abandoning the assumption that monolingualism is the standard and recognizing multilingualism as the rich, complex norm it actually is.

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CHALLENGES OF LEARNING ENGLISH IN A MULTILINGUAL ENVIRONMENT: THE CASE OF NIGERIA

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Abstract: This study investigates the challenges of learning English in a multilingual environment, focusing on Nigeria as a case study. Nigeria’s linguistic landscape, with over 500 indigenous languages alongside English as the official language, creates a complex educational setting for second language acquisition. Despite English being the medium of instruction in schools and a key tool for national integration, many learners continue to experience persistent difficulties in achieving proficiency.

The study adopts a qualitative-descriptive approach based on systematic review of existing literature, policy documents, and classroom-based studies. The analysis identifies key factors influencing

English language learning, including first language interference, inadequate instructional resources, teacher-related challenges, sociocultural attitudes, and inconsistencies in language policy implementation.

Findings reveal that learners frequently transfer linguistic structures from their native languages into English, resulting in grammatical and phonological errors. Additionally, limited exposure to English outside formal educational settings reduces opportunities for practice and fluency development. The study also highlights that many schools lack sufficient teaching materials, while teachers often rely on traditional grammar-translation methods rather than communicative approaches. Furthermore, sociocultural preferences for indigenous languages and weak policy enforcement contribute to uneven learning outcomes across regions.

The study concludes that English language learning in Nigeria is shaped by the interaction of linguistic diversity and systemic educational limitations. It recommends strengthening teacher training, improving learning resources, and adopting more effective multilingual pedagogical strategies to enhance language acquisition outcomes in Nigeria's multilingual context.

Keywords: Multilingualism, English learning, Nigeria, interference, pedagogy, policy

ÇOXDİLLİ MÜHİTDƏ İNGİLİS DİLİNİN ÖYRƏNİLMƏSİ ÇƏTİNLİKLƏRİ: NİGERİYA NÜMUNƏSİ

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Xülasə: Bu tədqiqat çoxdillli mühitdə ingilis dilinin öyrənilməsi ilə bağlı problemləri araşdırır və Nigeriyanı nümunə kimi götürür. 500-dən çox yerli dilin mövcud olduğu və ingilis dilinin rəsmi dil statusu daşdığı Nigeriyanın dil mənzərəsi ikinci dilin mənimsənilməsi üçün mürəkkəb təhsil mühiti yaradır. İngilis dili məktəblərdə tədris dili və milli integrasiyanın əsas vasitəsi olsa da, bir çox şagirdlər hələ də bu dildə yüksək səviyyədə səriştə əldə etməkdə çətinlik çəkirlər.

Tədqiqat mövcud ədəbiyyatın, siyasət sənədlərinin və sinifdaxili araşdırmaların sistemli icmalına əsaslanan keyfiyyət-deskriptiv yanaşma tətbiq edir. Təhlil ingilis dilinin öyrənilməsinə təsir edən əsas amilləri müəyyən edir: ana dilinin müdaxiləsi, kifayət qədər olmayan tədris resursları, müəllimlərlə bağlı çətinliklər, sosial-mədəni yanaşmalar və dil siyasətinin tətbiqində uyğunsuzluqlar.

Nəticələr göstərir ki, öyrənənlər tez-tez ana dillərinin qrammatik və fonoloji strukturlarını ingilis dilinə köçürür, bu da səhvlərə səbəb olur. Bundan əlavə, sinifdənkənar ingilis dili mühitinin məhdudluğu praktik imkanları və səlislik bacarıqlarının inkişafını azaldır. Tədqiqat həmçinin göstərir ki, bir çox məktəblərdə kifayət qədər tədris materialları yoxdur və müəllimlər kommunikatativ yanaşmalar əvəzinə ənənəvi qrammatika-tərcümə metodlarına üstünlük verirlər. Eyni zamanda, yerli dillərə üstünlük verilməsi və dil siyasətinin zəif icrası bölgələr arasında qeyri-bərabər öyrənmə nəticələrinə səbəb olur.

Tədqiqat nəticə olaraq bildirir ki, Nigeriyada ingilis dilinin öyrənilməsi dil müxtəlifliyi və sistemli təhsil məhdudiyyətlərinin qarşılıqlı təsiri ilə formalaşır. Dil öyrənmə nəticələrinin yaxşılaşdırılması üçün müəllim hazırlığının gücləndirilməsi, tədris resurslarının artırılması və çoxdillli pedaqoji strategiyaların tətbiqi tövsiyə olunur.

Açar sözlər: Çoxdilllilik, ingilis dili öyrənilməsi, Nigeriya, interferensiya, pedaqogika, siyasət

1. Introduction

English holds a central and paradoxical position in Nigeria's sociolinguistic landscape. It is the official language, the language of formal education, governance, law, and interethnic communication, yet it is not the mother tongue of the vast majority of its citizens. Nigeria is one of the most linguistically diverse countries in the world, with over 500 indigenous languages actively spoken across its regions (Bamgbose, 2011). This linguistic diversity creates a complex multilingual environment in which English functions as a unifying lingua franca, but is simultaneously influenced by strong first-language systems.

In the Nigerian educational system, English is introduced at the early stages of schooling and becomes the medium of instruction from upper primary education onward, as stipulated in the National Policy on Education (FRN, 2013). However, despite this policy framework, learners often experience difficulties in achieving proficiency due to limited exposure to standard English outside the classroom and the persistent dominance of indigenous languages in daily communication (Olaoye, 2013). This situation contributes to what linguists describe as "additive multilingualism with interference," where multiple language systems coexist but negatively influence second language acquisition (Odlin, 1989).

Research indicates that one of the most significant challenges in English learning in Nigeria is first language interference, particularly in phonology, syntax, and semantics (Akindele & Adegbite, 1999). Learners frequently transfer grammatical structures and pronunciation patterns from Yoruba, Igbo, Hausa, or other local languages into English, leading to persistent errors and reduced communicative accuracy. This phenomenon is consistent with Contrastive Analysis Hypothesis, which explains that structural differences between languages often predict learning difficulties (Lado, 1957).

Furthermore, sociocultural factors play a crucial role in shaping English language acquisition. According to Bamgbose (2005), English in Nigeria is often associated with formal domains, while indigenous languages dominate informal communication. This division limits meaningful exposure to English in naturalistic settings, thereby restricting learners' communicative development. In addition, inadequate instructional resources and teacher-centered pedagogical practices further constrain effective language learning (Taiwo, 2004).

Teacher competence also significantly influences learning outcomes. Studies have shown that many English teachers in Nigeria rely heavily on traditional grammar-translation methods rather than communicative language teaching approaches, which are more effective in multilingual contexts (Graddol, 2006). Large class sizes, limited professional development opportunities, and insufficient teaching materials exacerbate these challenges.

Given these conditions, English language learning in Nigeria cannot be fully understood without considering its multilingual context. The interaction between indigenous languages and English, combined with educational and sociocultural constraints, creates a unique learning environment that requires context-sensitive pedagogical and policy interventions.

Therefore, this study seeks to examine the key challenges affecting English language acquisition in Nigeria's multilingual environment, with the aim of contributing to more effective language teaching strategies and policy development.

2. Materials and Methods

This study adopts a **qualitative-descriptive research design** aimed at examining the challenges of learning English in a multilingual environment, with specific reference to Nigeria. The qualitative approach was considered appropriate because the study focuses on understanding

complex social, linguistic, and educational phenomena rather than measuring variables statistically.

2.1 Research Design

The research is based on a **systematic literature review and document analysis**. This involves the critical examination of existing scholarly works, policy documents, and empirical studies on English language education in Nigeria and other multilingual contexts. The design allows for the identification of recurring themes and patterns across different studies.

2.2 Data Sources

Data were collected from multiple secondary sources, including:

- Peer-reviewed journal articles on second language acquisition and English education in Nigeria
- Books and book chapters on sociolinguistics and applied linguistics
- Nigerian educational policy documents, particularly the National Policy on Education
- Conference papers and research reports on language teaching in multilingual classrooms
- Classroom-based observational studies reported in academic literature

Sources were selected based on relevance, academic credibility, and publication within the field of applied linguistics or education.

2.3 Data Collection Procedure

The data collection process involved:

1. Searching academic databases such as Google Scholar, JSTOR, and ResearchGate using keywords like *"English language learning in Nigeria," "multilingual education Nigeria," "language interference,"* and *"second language acquisition in Africa."*
2. Screening studies based on inclusion criteria, which required that sources:
 - Focus on Nigeria or comparable multilingual contexts
 - Address English language learning or teaching
 - Be written in English and published in reputable academic outlets
3. Extracting relevant information related to learning challenges, classroom practices, linguistic interference, and sociocultural influences

2.4 Data Analysis

The collected data were analyzed using thematic analysis. This method involved identifying, categorizing, and interpreting recurring patterns across the literature. Four major themes were developed during analysis:

- Linguistic interference from indigenous languages
- Educational infrastructure and resource limitations
- Teacher competence and pedagogical approaches
- Sociocultural and policy-related factors

Each theme was examined in relation to how it affects English language acquisition in Nigeria's multilingual environment. The analysis focused on identifying both convergences and divergences across studies to ensure a balanced interpretation.

2.5 Validity and Reliability

To ensure the credibility of the findings, only peer-reviewed and academically recognized sources were included. Triangulation was achieved by comparing findings from different types of literature (theoretical works, empirical studies, and policy documents). This helped to reduce bias and improve the reliability of interpretations.

2.6 Limitations of the Method

Since this study relies on secondary data, it does not include direct classroom observation or primary data collection from students and teachers. Therefore, the findings reflect synthesized interpretations of existing research rather than firsthand empirical measurement. However, this limitation is offset by the breadth of literature reviewed, which provides a comprehensive understanding of the topic.

3. Results

The thematic analysis of the reviewed literature revealed several interconnected challenges affecting English language learning in Nigeria's multilingual environment. These findings are consistent across empirical studies in applied linguistics and language education research in Nigeria and similar postcolonial multilingual contexts.

3.1 Linguistic Interference

One of the most frequently reported challenges is first language (L1) interference. Learners often transfer phonological, syntactic, and lexical features from indigenous languages such as Yoruba, Igbo, Hausa, and others into English. This results in pronunciation difficulties, incorrect sentence structures, and semantic confusion. For example, phonological substitution patterns common in Nigerian languages affect the articulation of English consonant clusters and vowel sounds (Akindele & Adegbite, 1999; Jowitt, 2000).

Odlin (1989) explains that such interference is a natural consequence of second language acquisition, especially in environments where the L1 is dominant. In Nigeria, this effect is intensified due to limited exposure to standard English outside formal educational settings (Bamgbose, 2005). As a result, learners often develop hybridized forms of English that deviate from Standard British English norms.

3.2 Inadequate Teaching and Learning Resources

The study found that many Nigerian schools lack sufficient instructional materials to support effective English language learning. Rural schools are particularly affected, with shortages of textbooks, audio-visual aids, and language laboratories (Taiwo, 2004). This limits learners' exposure to authentic language input, which is essential for developing listening and speaking skills.

Graddol (2006) emphasizes that modern language learning requires multimodal resources that support interaction and communicative competence. However, in many Nigerian classrooms, instruction remains textbook-based and heavily dependent on rote memorization, reducing opportunities for meaningful language use.

3.3 Teacher Competence and Pedagogical Practices

Another major challenge identified is the reliance on traditional grammar-translation methods. Many English language teachers in Nigeria continue to prioritize explicit grammar instruction over communicative language teaching (CLT) approaches (Bamgbose, 2011). This limits learners' ability to develop fluency and communicative confidence.

Studies by Ajiboye (2010) and Uba (2015) indicate that some teachers lack adequate training in modern pedagogical strategies, particularly those that support multilingual classrooms. Large class sizes further restrict interactive teaching methods, making individualized feedback difficult. Consequently, learners often acquire theoretical knowledge of English without achieving practical communicative competence.

3.4 Sociocultural Influences

The sociocultural environment also plays a significant role in shaping English language acquisition. English is often associated with formal education, employment, and upward mobility, while indigenous languages dominate home and community interactions (Bamgbose, 2005). This division reduces the frequency of English usage in informal contexts.

According to Omoniyi (2010), language attitudes in multilingual societies significantly influence learning motivation. In Nigeria, some learners perceive English as an academic requirement rather than a daily communication tool, which limits active engagement with the language outside school environments.

3.5 Policy Implementation Gaps

Although Nigeria's language-in-education policy promotes the use of mother tongue in early education and English at higher levels, implementation remains inconsistent across regions (FRN, 2013). Some schools introduce English too early, while others over-rely on indigenous languages beyond recommended stages. This inconsistency leads to uneven language development among learners (Adeniyi, 2012).

Bamgbose (2011) argues that such policy inconsistencies create a "dual expectation burden," where learners are neither fully grounded in their mother tongue literacy nor adequately proficient in English. This contributes to long-term academic difficulties.

3.6 Urban–Rural Educational Disparities

The findings also highlight a significant gap between urban and rural schools. Urban schools generally have better-trained teachers and more access to learning materials, while rural schools face infrastructural deficits and limited exposure to English-speaking environments (UNESCO, 2016). This disparity leads to unequal learning outcomes across regions.

3.7 Summary of Findings

Overall, the results demonstrate that English language learning in Nigeria is shaped by a combination of linguistic interference, inadequate resources, limited teacher training, sociocultural attitudes, policy inconsistencies, and regional disparities. These factors interact to create a learning environment where achieving high proficiency in English remains a significant challenge for many learners.

4. Discussion

The findings of this study demonstrate that English language learning in Nigeria is shaped by a complex interaction of linguistic, pedagogical, sociocultural, and policy-related factors. These challenges are not isolated but interconnected, reinforcing one another within the multilingual Nigerian educational environment.

4.1 Linguistic Interference and Second Language Acquisition

The strong presence of first language (L1) interference confirms earlier theoretical claims in second language acquisition research. According to Lado (1957), differences between a learner's native language and the target language are a major source of difficulty in acquisition. In the Nigerian context, structural differences between English and indigenous languages such as Yoruba, Igbo, and Hausa contribute to persistent phonological and syntactic errors (Akindele & Adegbite, 1999).

However, more recent research suggests that interference is not merely a negative transfer phenomenon but also part of a broader interlanguage development process (Selinker, 1972). Learners in Nigeria often develop stable hybrid forms of English, which reflect both linguistic creativity and systematic deviation from standard norms. Bamgbose (2005) argues that such

variations should not be viewed solely as errors but as evidence of the localization of English in multilingual settings.

4.2 Pedagogical Limitations and Teaching Approaches

The continued dominance of grammar-translation methods in Nigerian classrooms limits learners' communicative competence. Communicative Language Teaching (CLT), which emphasizes interaction and meaning-focused learning, is widely recommended in second language pedagogy (Richards & Rodgers, 2014). However, its implementation in Nigeria remains limited due to teacher training gaps and large class sizes (Graddol, 2006).

Taiwo (2004) notes that many teachers prioritize examination success over communicative fluency, leading to a form of "exam-oriented learning." This results in learners who may perform well in written tests but struggle with spoken communication. Ajiboye (2010) further highlights that insufficient professional development opportunities hinder teachers from adopting innovative pedagogical strategies suitable for multilingual classrooms.

4.3 Sociocultural Context and Language Attitudes

Sociocultural factors significantly influence English language acquisition in Nigeria. Bamgbose (2005) emphasizes that English functions as a language of prestige and upward mobility, while indigenous languages maintain strong cultural and identity functions. This dual role creates a complex linguistic hierarchy that affects learner motivation and usage patterns.

Omoniyi (2010) argues that language attitudes in multilingual societies directly influence language acquisition outcomes. In Nigeria, English is often confined to formal contexts such as schools and offices, while indigenous languages dominate informal communication. This reduces opportunities for naturalistic language practice, which is essential for fluency development (Krashen, 1982).

4.4 Policy Inconsistencies and Educational Implementation

The inconsistency in language policy implementation remains a major structural issue. Although the Nigerian National Policy on Education advocates mother tongue instruction in early years and English at higher levels (FRN, 2013), actual implementation varies widely across regions and school types. Adeniyi (2012) observes that some schools prematurely transition to English-only instruction, while others continue excessive reliance on indigenous languages.

Bamgbose (2011) describes this situation as a "policy-practice gap," where well-designed language policies fail due to weak implementation frameworks. This gap contributes to uneven language proficiency levels among learners and exacerbates educational inequality.

4.5 Multilingualism as a Resource and Challenge

While multilingualism is often discussed as a barrier, some scholars argue that it can also be an educational resource if properly managed. García and Wei (2014) propose the concept of translanguaging, where learners use their full linguistic repertoire to support learning. In the Nigerian context, however, this potential is not fully realized due to rigid classroom language boundaries.

Instead of leveraging multilingualism as a pedagogical asset, many classrooms enforce strict separation between English and indigenous languages. This limits cognitive flexibility and reduces opportunities for meaningful language negotiation in learning (Cummins, 2000).

4.6 Implications of Urban–Rural Inequality

The urban–rural divide further complicates English language learning outcomes. UNESCO (2016) reports that rural schools in many developing countries, including Nigeria, suffer from

teacher shortages and inadequate infrastructure. In contrast, urban schools often provide greater exposure to English through media, peer interaction, and better-resourced classrooms.

This inequality leads to what Coleman (2011) describes as “language learning stratification,” where socioeconomic and geographic factors significantly influence language achievement. In Nigeria, this stratification reinforces broader educational inequality.

4.7 Synthesis of Discussion

Overall, the discussion highlights that English language learning challenges in Nigeria cannot be attributed to a single factor. Instead, they emerge from the interaction of linguistic transfer effects, inadequate pedagogical practices, sociocultural language hierarchies, inconsistent policy implementation, and structural inequalities in education. These findings align with broader theories of second language acquisition in multilingual postcolonial contexts, where English functions both as a global language and a locally adapted system (Crystal, 2003; Bamgbose, 2011).

5. Conclusion

This study has examined the challenges of learning English in a multilingual environment, with particular reference to Nigeria. The findings demonstrate that English language acquisition is significantly influenced by a combination of linguistic, pedagogical, sociocultural, and policy-related factors. In Nigeria’s highly multilingual setting, where over 500 indigenous languages coexist with English as the official language, learners face persistent difficulties that shape their overall proficiency and communicative competence.

A major conclusion of the study is that first language interference remains one of the strongest barriers to effective English learning. Learners frequently transfer phonological, grammatical, and lexical features from their mother tongues into English, resulting in persistent errors. In addition, limited exposure to English outside the classroom restricts opportunities for meaningful practice, further slowing language development.

The study also concludes that educational constraints, including inadequate teaching resources, large class sizes, and insufficiently trained teachers, significantly reduce the effectiveness of English instruction. Traditional teaching approaches, which emphasize rote learning and grammar memorization, continue to dominate many classrooms, limiting the development of communicative competence.

Furthermore, sociocultural attitudes toward language use play an important role. English is often confined to formal settings, while indigenous languages dominate everyday communication, reducing learners’ motivation and exposure. Policy inconsistencies in language education further contribute to uneven learning outcomes across regions.

Overall, the study concludes that improving English language learning in Nigeria requires a holistic and integrated approach. This includes strengthening teacher training, improving educational resources, ensuring consistent language policy implementation, and adopting more communicative and multilingual pedagogical strategies. Addressing these issues will enhance learners’ proficiency and support more equitable educational outcomes in Nigeria’s multilingual society.

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EMBODIMENT AS A FORM OF CATEGORIZATION

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Abstract: Embodiment as a form of categorization is a central notion in cognitive linguistics grounded in Merleau-Ponty's phenomenology. It assumes that human cognition is shaped through bodily experience, which mediates our access to the world and determines how meaning is constructed. Abstract concepts are formed via embodied experience and conceptual metaphors that structure cognitive and linguistic processes. Emotional states such as anger, fear, or coldness are expressed through bodily metaphors (e.g., "I burn with anger", "she is cold as snow"), demonstrating how sensory and physical experience is mapped onto abstract conceptual domains. The study also examines the "eye" as a cognitive and linguistic model of perception, showing how vision-related expressions (keep an eye on, keep eyes open, have an eye on) extend into abstract domains of control, awareness, attention, and understanding. These metaphorical extensions reveal how bodily perception becomes a foundation for knowledge construction. Furthermore, such metaphors influence spatial and temporal orientation, shaping human perception of past, present, and future through embodied structures. Overall, embodiment highlights the inseparable relationship between language and cognition, confirming that meaning is grounded in bodily experience and systematically organized through categorization mechanisms in the human conceptual system.

Key words: cognitive linguistics, meaning, embodied experience, conceptual metaphor, cognition

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Xülasə: Cisimlənmə kateqorizasiyanın forması kimi Koqnitiv dilçilikdə M. Merlo-Pontinin fenomenologiyasına əsaslanan mərkəzi anlayışdır. Bu yanaşmaya görə, insan koqnisiyası bədən təcrübəsi vasitəsilə formalaşır və dünyanın dərk olunması bu təcrübə ilə vasitələnir, yəni insan reallığı birbaşa deyil, bədən vasitəsilə qavrayır. Mücərrəd anlayışlar cisimlənmiş təcrübə və konseptual metaforalar vasitəsilə yaranır və məna quruluşunu təşkil edən əsas mexanizmlərdən biri kimi çıxış edir. Qəzəb, qorxu və ya soyuqluq kimi emosional vəziyyətlər bədən metaforaları ilə ifadə olunur (məs., "qəzəbdən yanmaq", "buz kimi soyuq olmaq"), bu isə hissi və fizioloji təcrübənin abstrakt sahələrə köçürülməsini göstərir. Tədqiqat həmçinin "göz" konseptini idrak və dil modeli kimi araşdırır və görmə ilə bağlı ifadələrin (gözü açıq olmaq, göz qoymaq, nəzarət etmək) diqqət, anlayış və nəzarət kimi mücərrəd sahələrə genişləndiyini nümayiş etdirir. Bu metaforik genişlənmələr bilik formalaşmasında bədən təcrübəsinin əsas rolunu göstərir. Bundan əlavə, bu metaforalar məkan və zaman qavrayışını formalaşdıraraq keçmiş, indiki və gələcək anlayışlarının cisimlənmiş strukturlar əsasında təşkil olunduğunu göstərir. Ümumilikdə, cisimlənmə dil və koqnisiya arasındakı ayrılmaz əlaqəni təsdiqləyir.

Açar sözlər: koqnitiv dilçilik, məna, cisimlənmiş təcrübə, konseptual metaphora, koqnisiya

Introduction

The principle of embodiment was put forward in 1945 in the Phenomenology of Perception by the French philosopher Maurice Merleau-Ponty. Merleau-Ponty claimed that because human beings have bodies, our world experience passes through our bodies, not through our thoughts, which are directly related to the world around us.

When we are angry, furious etc. we say - I burn with anger, I am on fire, I am on the flame, or Coldness between us, She is cold to me, she is as cold as snow to me etc. we turn the

experience we receive through the body into embodied experience, that is, we embody the experience. We try to create our conceptual world by using cognitive structures to move on to the more complex stages (to express more complex ideas). We are already beginning to transfer what we have gained from the real world to our conceptual world. The examples we have mentioned above are ultimately the product of our conceptual world. Real world experiences do not enter the conceptual world directly, they enter after crossing the body barrier. This process is an individual approach, an interpretation. It is this interpretation that shapes the language. Language usage is a reflection of embodied experience. Thus, embodied experience structures both our physical experience and our conceptual system through metaphors, metonymy, and conceptual integration.

Materials and Methods

This study is based on a qualitative cognitive-linguistic analysis of embodiment theory within the framework of conceptual metaphor theory. The methodological approach involves interpretative analysis of linguistic data drawn from English literary sources and everyday expressions. (Gibbs, R. W., Jr. (2005))

The theoretical foundation relies on Merleau-Ponty's phenomenology and Lakoff & Johnson's conceptual metaphor theory, which explain how bodily experience structures abstract thought.

The data selection focuses on lexical and phraseological units containing the concept of "eye" and emotion-related expressions (e.g., anger, coldness, perception verbs). These examples are analyzed to identify recurring metaphorical mappings between bodily experience and abstract conceptual domains.

The analytical procedure includes:

- identification of embodied expressions,
- categorization of metaphorical structures,
- interpretation of cognitive mappings between physical experience and abstract meaning,
- contextual analysis of literary examples.

Results and Discussion

For example let's take **eye** concept. The **eye**, an organ of our body, has a visual function. All that we see first penetrates our mind through this organ. Thanks to conceptualization, the concept of **eye** enriches the conceptual system by embodying abstract domains related to vision. Let's look at examples:

a) keep one's eyes open

He offered me a hundred guineas to come down here and keep my eyes open - said I'd got a reputation for being a good man in a tight spot. A. Christie. Ten Little Niggers.

'I don't mind saying I know a thing or two,' he admitted. 'I mean, I've got my eyes peeled all right. You can't put much over on me. S. Maugham. Gigolo and Gigolette.

Speck Jones and Bill Jehosaphat said they had never had any trouble with the blacks ... They kept their eyes skinned in the country where the blacks were known to be hostile, all the same: slept a good way from their camp fire, and tethered the horses handy. K. Prichard. The Roaring Nineties.

b) have (or keep) an (or one's) eye on

There, old Monty and Ma Buggins were always at hand and could keep an eye on her. K. Prichard. The Roaring Nineties.

And I think I ought to stay her to - keep an eye on things, you know. I.Murdoch. An Unofficial Rose.

He felt he'd be happier if he knew that some responsible person with expert knowledge was keeping an eye on her. A.Christie. Murder in Mesopotamia.

The examples above demonstrate that eye-related linguistic units function as cognitive tools for understanding perception, control and awareness. The 'eye' operates as a conceptual metaphor extending beyond physical vision into abstract domains such as monitoring, knowledge, and mental attention. This supports the idea that seeing is conceptualized as understanding and controlling sight is equivalent to controlling knowledge.

As we can see from the examples, mainly *eye-related* compounds are used to analyze and conceptualize what is happening around us. This means that it is considered an easier way to see and understand. Proverbs and parables emphasize this: *It is better to see once than to hear a thousand times; I can't believe if I don't see it; It is a world of vision* and so on. To watch, to observe, to control something means to *keep an eye* on it. In general, in our language: *to wait, to put an eye, to shoot a bird in the eye, to look in the eye, to saturate / not saturate the eye*, and so on. abstract processes, mainly contain the metaphor of the **eye**. The words beauty, beauty, eye-catching, beautiful, amazing, gorgeous and other expressions that we use to describe our appearance and inner world are also conceptual metaphors related to the eye. With their help, the ability to understand and comprehend a process, an issue, an area, to express the idea we are aiming at, expands. In English, the eye is also very active. However, in English, other words and combinations are used more often to describe appearance than metaphorical concepts involving the word eye. Thus, our experiences are transmitted to our brains through our eyes, where thinking is formed based on our experiences. In other words, our worldview is conceptualized "through the eyes of our brains," our eyes are simply "cosmetic." (Merleau-Ponty, M. (2012))

Because our eyesight is limited, that is, they cannot see behind, but forward and right / left; Asymmetry arises because it can control the lower / upper sides. We have to turn our eyes back to see what is behind us. These perspectives guide us. Our hopes, dreams, wishes, expectations, etc. for the future. is structured in this direction. We always assume the following:

The future is still to come;
You have a future;
Dreams will come true;
Look forward to tomorrow;
Still everything is ahead.

It follows that the orientation of *time* and *space* is determined by our views. Basically, because we see ahead, we move forward, we do not move backwards, or it is impossible to look back. Our movement to the right and left is temporary, because after an instantaneous turn, that direction is considered forward again. Although our forward, right, and left directions are not limited (in terms of scale), they are limited upwards and downwards. The orientation of time is infinite, and the above examples are an example of this. We direct our thoughts, activities and experiences in accordance with these directions.

Conclusion

Since our thoughts and worldviews are specific, that is, because the individual is the product of his own thinking, we interpret the realities as we perceive and embody them. It is this interpretation that shapes the language. Just as all these processes give rise to *embodied*

cognition, they are also clear examples of the unbreakable correlations between language and cognition. Thus, language emerges as a reflection of embodied cognition where bodily experience structures meaning-making processes. The human conceptual system is not an abstract, detached mechanism, but a dynamic structure rooted in sensory-motor experience.

One of the forms of categorization, the multi-modal phenomenon - embodiment, which exists in our brains, is one of the main commitments of conceptual metaphor theory, in general, of cognitive linguistics, and makes possible to reveal the *embodied meaning*. One of the main tenets of cognitive linguistics, "meaning is embodied", is confirmed here. From this perspective, categorization is not a purely logical operation but a cognitively grounded process shaped by perception, emotion and physical interaction with the environment. Conceptual metaphors serve as cognitive bridges that connect concrete bodily experience with abstract domains of thought such as time, emotion, knowledge, and social relations.

Therefore, the study confirms the central cognitive-linguistic principle that 'meaning is embodied', demonstrating that cognition, language, and bodily experience are inseparably interconnected systems.

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THE IMPACT OF CLASSROOM ANXIETY ON SPEAKING PERFORMANS IN EFL LEARNERS

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Abstract: Classroom anxiety is widely recognized as a crucial affective factor influencing second and foreign language acquisition, particularly in English as a Foreign Language (EFL) contexts. Among the four core language skills, speaking is often considered the most anxiety-inducing because it requires immediate language processing, spontaneous production, and public performance. This study investigates the impact of classroom anxiety on the speaking performance of EFL learners, focusing on how emotional

tension interferes with oral communication skills. The research draws upon existing theoretical frameworks, particularly the concept of foreign language anxiety introduced by Horwitz et al. (1986), which includes communication apprehension, test anxiety, and fear of negative evaluation. These components collectively contribute to learners' reluctance to participate in speaking activities. The study also considers cognitive perspectives, suggesting that anxiety limits working memory capacity, thereby reducing learners' ability to retrieve vocabulary and construct grammatically accurate sentences. Findings indicate that students experiencing high levels of anxiety tend to demonstrate lower speaking performance in terms of fluency, pronunciation, and coherence. Anxiety leads to avoidance behaviors, reduced participation, and lack of confidence, all of which negatively affect language development. Furthermore, the study highlights that teacher behavior, classroom environment, and peer interaction significantly influence anxiety levels. The paper concludes that reducing classroom anxiety should be a central goal in EFL instruction. Practical implications include adopting communicative teaching approaches, providing supportive feedback, and creating a low-stress learning environment. By addressing both emotional and cognitive barriers, educators can enhance learners' speaking performance and overall language proficiency.

Keywords: classroom anxiety, EFL learners, speaking performance, language learning, communication anxiety

XARİCİ DİL KİMİ İNGİLİS DİLİ ÖYRƏNƏNLƏRDƏ SİNİFDAXİLİ TƏŞVİŞİN ŞİFAHİ NİTQ PERFORMANSINA TƏSİRİ

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Xülasə: Sınıfdaxili təşvişin xarici dilin mənimsənilməsinə təsir edən mühüm affektiv amillərdən biri kimi geniş şəkildə qəbul olunur, xüsusilə də xarici dil kimi ingilis dili kontekstində. Dil bacarıqlarının dörd əsas növü arasında danışmaq ən çox narahatlıq yaradan bacarıq hesab olunur, çünki o, dərhal dil emalı, spontan nitq istehsalı və ictimai çıxış tələb edir. Bu tədqiqat sınıfdaxili narahatlığın xarici dil kimi ingilis dili öyrənənlərin danışmaq performansına təsirini araşdırır və emosional gərginliyin şifahi ünsiyyət bacarıqlarına necə mane olduğunu diqqət mərkəzinə alır. Tədqiqat mövcud nəzəri çərçivələrə, xüsusilə də Horwitz və digərləri (1986) tərəfindən irəli sürülmüş xarici dil narahatlığı anlayışına əsaslanır. Bu anlayış kommunikativ təşviş, imtahan narahatlığı və mənfi qiymətləndirilmə qorxusu kimi komponentləri əhatə edir. Bu amillər birlikdə öyrənənlərin danışmaq fəaliyyətlərində iştirak etməkdən çəkinməsinə səbəb olur. Tədqiqat həmçinin koqnitiv yanaşmaları nəzərə alaraq göstərir ki, narahatlıq işçi yaddaşın tutumunu məhdudlaşdırır və bunun nəticəsində öyrənənlərin lüğət ehtiyatını xatırlamaq və qrammatik cəhətdən düzgün cümlələr qurmaq qabiliyyəti azalır. Nəticələr göstərir ki, yüksək səviyyədə narahatlıq yaşayan tələbələr səlislik, tələffüz və məntiqi ardıcılıq baxımından daha aşağı danışmaq performansı nümayiş etdirirlər. Təşvişin qaçınma davranışlarına, zəif iştiraka və özünə inamın azalmasına səbəb olur ki, bu da dil inkişafına mənfi təsir göstərir. Bundan əlavə, tədqiqat vurğulayır ki, müəllim davranışı, sinif mühiti və həmyaşıdlarla qarşılıqlı əlaqə təşviş səviyyəsinə əhəmiyyətli dərəcədə təsir edir. Məqalə belə nəticəyə gəlir ki, sınıfdaxili təşvişin azaldılması xarici dil kimi ingilis dili tədrisinin əsas məqsədlərindən biri olmalıdır. Praktiki tövsiyələrə kommunikativ tədris yanaşmalarının tətbiqi, dəstəkləyici rəyin verilməsi və aşağı stresli öyrənmə mühitinin yaradılması daxildir. Emosional və koqnitiv maneələrin aradan qaldırılması ilə müəllimlər öyrənənlərin danışmaq performansını və ümumi dil bacarıqlarını əhəmiyyətli dərəcədə inkişaf etdirə bilərlər.

Açar sözlər: sinifdaxili təşviş, xarici dil kimi ingilis dili öyrənmələr, danışq performans, dil öyrənilməsi, kommunikativ təşviş

Introduction

In recent decades, language education research has increasingly shifted from a purely cognitive perspective to a more comprehensive view that includes affective, social, and psychological factors. Among these, classroom anxiety has emerged as one of the most influential variables affecting foreign language learning success. While cognitive abilities such as memory capacity, linguistic aptitude, and analytical thinking are important, emotional states such as anxiety, motivation, and self-confidence often determine how effectively learners can use their knowledge in real communication situations. Speaking is generally regarded as the most complex and demanding language skill. Unlike reading and writing, speaking requires immediate response, spontaneous production, and continuous interaction. Learners must simultaneously process input, retrieve vocabulary, apply grammatical structures, and monitor pronunciation. This simultaneous processing places a heavy cognitive load on learners, which becomes even more difficult under conditions of anxiety.

For many EFL learners, speaking English in the classroom is associated with psychological pressure. They may fear making mistakes, being corrected publicly, or being negatively evaluated by teachers and classmates. These fears create a barrier that prevents learners from fully engaging in oral communication activities. As a result, students often remain silent, avoid participation, or speak minimally during lessons. Classroom anxiety is not simply a temporary feeling of nervousness; it is a persistent emotional state that can significantly influence learning behavior and performance. It is particularly evident in EFL contexts where exposure to English outside the classroom is limited, and students rely heavily on classroom interaction for practice. Therefore, understanding the nature and effects of classroom anxiety is essential for improving speaking instruction and learner outcomes.

Literature Review

The theoretical foundation of foreign language anxiety was established by Horwitz, Horwitz, and Cope (1986), who defined it as a situation-specific anxiety related to language learning. Their model identifies three major components: communication apprehension, test anxiety, and fear of negative evaluation. Communication apprehension refers to the fear of speaking in the target language, test anxiety involves stress related to academic evaluation, and fear of negative evaluation concerns learners' worries about being judged by others. MacIntyre and Gardner (1994) expanded this framework by explaining the cognitive effects of anxiety. They argued that anxiety interferes with information processing and reduces working memory capacity. As a result, anxious learners experience difficulty in retrieving vocabulary, forming sentences, and maintaining fluency during speech production. Krashen's Affective Filter Hypothesis (1982) also provides important insight into the role of anxiety in language acquisition. According to this hypothesis, emotional variables such as anxiety act as a "filter" that blocks or limits the amount of comprehensible input that learners can process. When the affective filter is high, language acquisition is significantly hindered. Young (1991) emphasized the importance of classroom environment and teacher behavior in shaping anxiety levels. A strict, authoritarian, or overly corrective teaching style can increase anxiety, while supportive, interactive, and learner-centered approaches can reduce it. Similarly, peer interaction and

classroom dynamics play an important role in either increasing or decreasing learners' emotional comfort.

Research Methodology

This study adopts a mixed-method research design in order to provide a comprehensive analysis of the relationship between classroom anxiety and speaking performance. The combination of quantitative and qualitative methods allows for both statistical measurement and in-depth understanding of learner experiences. The participants of the study are 40 first-year university students studying English as a foreign language at a pre-intermediate level. The participants are aged between 17 and 20 and represent a typical EFL learning population in higher education. Both male and female students are included in the sample to ensure diversity. Data collection instruments include the Foreign Language Classroom Anxiety Scale (FLCAS), speaking performance tasks, classroom observations, and semi-structured interviews. The FLCAS is used to measure students' anxiety levels based on communication apprehension, fear of negative evaluation, and test anxiety. Speaking performance is evaluated through structured tasks such as presentations, role-plays, and group discussions. These tasks are assessed based on fluency, pronunciation, accuracy, and coherence. Classroom observations are conducted to examine learners' behavior during speaking activities, including hesitation, avoidance, participation, and confidence levels. Semi-structured interviews are conducted with selected students to gain deeper insights into their emotional experiences and perceptions of speaking anxiety.

The research is conducted over a four-week period. In the first phase, students complete the anxiety questionnaire. In the second phase, they participate in speaking activities, which are observed and evaluated. In the final phase, interviews are conducted to support and explain the quantitative findings. Data analysis involves descriptive statistics and correlation analysis for quantitative data, while qualitative data are analyzed thematically to identify recurring patterns and themes related to anxiety and speaking performance.

Results and Findings

The findings of the study reveal a strong negative correlation between classroom anxiety and speaking performance. Students with high levels of anxiety tend to demonstrate lower fluency, increased hesitation, limited vocabulary use, and reduced coherence in speech. They often avoid speaking tasks and participate minimally in classroom discussions. In contrast, students with low anxiety levels show higher levels of fluency, confidence, and participation. They are more willing to engage in speaking activities and demonstrate better control over pronunciation and grammatical accuracy. The study also identifies several key sources of anxiety, including fear of making mistakes, fear of negative evaluation from teachers and peers, insufficient language proficiency, and lack of speaking practice. Teacher behavior and classroom atmosphere are also found to have a significant impact on anxiety levels. The results confirm that classroom anxiety plays a critical role in limiting speaking performance in EFL learners. Anxiety affects both emotional and cognitive processes. Emotionally, it reduces confidence and increases fear, while cognitively, it interferes with memory retrieval and language production. These findings are consistent with previous studies in the field of language acquisition, which also demonstrate that anxiety negatively affects communication skills. The results highlight the importance of creating a supportive and interactive learning environment in order to reduce anxiety and improve speaking performance.

Conclusion

Classroom anxiety is a major barrier to effective speaking performance in EFL learners. It affects fluency, accuracy, participation, and overall confidence. However, it is not an unchangeable condition. With appropriate teaching strategies such as communicative activities, positive feedback, peer collaboration, and supportive classroom environments, anxiety can be significantly reduced. Reducing classroom anxiety should be considered a key objective in EFL instruction, as it directly contributes to improved speaking performance and overall language development. Teachers play a central role in shaping learners' emotional experiences and can significantly influence their success in language learning.

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THE SOCIOLINGUISTIC CHARACTERISTICS OF BILINGUALISM AND ITS DEVELOPMENT

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Abstract: This article studies bilingualism as one of the main objects of modern sociolinguistics. In today's globalizing world, the expansion of interactions between languages has contributed to the rapid spread of bilingualism. The article analyzes the essence of bilingualism, its causes, and its impact on society in detail. It is emphasized that bilingualism is not only a linguistic ability but also a social and cultural phenomenon. Furthermore, bilingualism has a positive influence on individuals' cognitive development, improving their analytical thinking and expanding their memory capacity. Based on the views of various scholars, it is highlighted that bilingual individuals can adapt more easily to different social environments and perform more effectively in communicative situations. The article also explains such concepts as language choice, language identity, and language situation.

In addition, bilingualism plays an important role in the education system. The use of two languages in education can enhance students' academic achievement and cognitive flexibility. It also contributes to the preservation of cultural diversity. In modern societies, multilingualism is increasingly seen as a normal and valuable skill.

In conclusion, bilingualism is regarded as a complex and multidimensional phenomenon that plays an important role in modern society, and its study is of great significance. It is expected that bilingualism will continue to remain relevant in the future.

Key words: bilingualism, sociolinguistics, language situation, language and society, language identity, language choice, cultural diversity.

İKİDİLLİLİYİN SOSİOLİNGVİSTİK XÜSUSİYYƏTLƏRİ VƏ ONUN İNKİŞAFI

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Xülasə: Bu məqalədə bilinqvizm müasir sosiolingvistikanın əsas tədqiqat obyektlərindən biri kimi araşdırılır. Qloballaşan dünyada dillərarası əlaqələrin genişlənməsi bilinqvizmin sürətlə yayılmasına səbəb olmuşdur. Məqalədə bilinqvizmin mahiyyəti, yaranma səbəbləri və cəmiyyətə təsiri ətraflı şəkildə təhlil edilir. Xüsusilə vurğulanır ki, bilinqvizm yalnız dil bacarığı deyil, həm də sosial və mədəni bir hadisədir. Bununla yanaşı, bilinqvizm fərdlərin idrak inkişafına müsbət təsir göstərir, onların analitik düşünmə qabiliyyətini gücləndirir və yaddaş imkanlarını artırır. Müxtəlif alimlərin fikirlərinə əsaslanaraq qeyd olunur ki, iki dildə danışan insanlar fərqli sosial mühitlərə daha asan uyğunlaşır və kommunikativ situasiyalarda daha effektiv fəaliyyət göstərirlər. Məqalədə həmçinin dil seçimi, dil kimliyi və dil situasiyası kimi anlayışlar da izah olunur. Bundan əlavə, bilinqvizm təhsil sistemində də mühüm rol oynayır. Təhsil prosesində iki dildən istifadə tələbələrin akademik bacarıqlarını inkişaf etdirir. Eyni zamanda, bilinqvizm mədəni müxtəlifliyin qorunmasına da xidmət edir. Müasir cəmiyyətdə çoxdillilik artıq adi bir hal kimi qəbul olunur.

Nəticə olaraq, bilinqvizm müasir cəmiyyətdə mühüm rol oynayan kompleks və çoxşaxəli bir fenomen kimi qiymətləndirilir, onun tədqiqi və tədrisi xüsusi əhəmiyyət kəsb edir. Bu fenomen gələcəkdə də aktuallığını qoruyacaqdır.

Açar sözlər: bilinqvizm, sosiolingvistika, dil situasiyası, dil və cəmiyyət, dil kimliyi, dil seçimi, mədəni müxtəliflik

Introduction

1. In a rapidly changing globalized world, interactions between languages and cultural exchange are rapidly increasing. One of the most significant outcomes of this process is the widespread expansion of bilingualism, that is, the phenomenon of using two languages. Bilingualism is not only an individual language skill, it also functions as a social, psychological, and cultural phenomenon. From this point of view, bilingualism is an important subject of study both for applied linguistics and for other branches of linguistics. To put it differently, bilingualism is regarded one of the main problems of modern sociolinguistics that require special attention and this issue needs to be studied comprehensively within linguistic and social contexts. As an important linguistic phenomenon, bilingualism influences not only communication but also cognitive development (Grosjean, 2010)

2. Like many linguistic terms, the term bilingualism is of Latin origin and refers to the use of two languages in the communication process, which can carry different characteristics in terms of their application and communicative function. Labov argues that bilingualism should be understood as a socially conditioned practice where language choice depends on context and social interaction (Labov, 1972). This socio-public phenomenon can emerge differently in various spheres of human society and in individual language situations. In different language situations, the phenomenon of bilingualism manifests itself in various countries or regions under the influence of extra-linguistic factors and directly under their pressure, appearing in different forms and situations. This is because this linguistic phenomenon has left its traces in certain historical periods of various languages of the world and can influence their future course in one way or another. Therefore, the language situation, which plays an important role in the emergence of bilingualism and directly affects this process, is explained somewhat differently in linguistics, implying different language contacts manifested in the languages of various peoples of the world or a collection of mutual relations of individual languages. Thus, although some researchers hold somewhat different positions when explaining the language situation, they essentially understand it as encompassing the mutual contact and influence of various languages and approach the concept of language situation from this perspective. For example, (Fishman, 1972) in his research, explains the essence of the language situation as follows, showing that a language situation is a concrete type of mutual contact of languages at a certain stage of the historical development process of any people, and the language in such mutual contact plays an important role in the public life of society. In another study, the concept of language situation is approached from a relatively different angle, noting that it consists of means serving the communication of individual communities: "By language situation, we mean the language unity, sub-languages, and functional styles that serve the communication of a certain territorial-administrative union or any ethnic human community" (Чубина, 2021, стр.89). As can be seen, although these explanations differ slightly, the essence of the language situation is understood approximately in the same way and is interpreted as the creation of communication by a certain people through individual languages. Thus, it is clear from the explanations given about the language situation that it conditions the realization of the communicative process among members of society through various languages and their mutual contact, and individual languages participate in this process in one way or another. It is during this language situation that individual or collective bilingualism emerges as a result of members of society using different languages in the same way and at the same level individually—that is, personally—or as the majority of society members do in the communication process. This manifests itself under the influence of various extra-linguistic factors. Thus, various socio-political factors, regional and national-ethnic characteristics, and other events in society play an important role in the creation of individual and mass bilingualism. It is as a result of this process that the phenomenon of bilingualism shows itself in various regions of the world from time to time during a certain historical stage or over a long historical period.

It is interesting to analyze the causes of bilingualism. The causes of bilingualism are quite diverse. One of the most common reasons is migration. When people move from one country to another, they often need to learn the language of the host country while maintaining their native language. This creates bilingual individuals and communities. Another important factor is education. Many schools around the world promote bilingual education programs, recognizing

the value of learning more than one language. Globalization has also played a significant role, as English and other widely spoken languages are increasingly necessary for international communication, business, and travel. Additionally, family background can influence bilingualism. Children growing up in households where parents speak different languages are naturally exposed to both, making bilingualism a part of their everyday life. One of the most widely discussed aspects of bilingualism is its cognitive benefits. Research has shown that bilingual individuals often have better problem-solving skills, improved memory, and greater mental flexibility compared to monolinguals, this is because managing two languages requires constant mental effort, which strengthens the brain's executive functions. Bilingual people are also better at multitasking and switching between tasks, as they are accustomed to switching between languages. Furthermore, bilingualism has been connected to delayed onset of age-related cognitive decline, such as dementia. These cognitive advantages make bilingualism not only a practical skill but also a valuable tool for lifelong mental development.

Materials and Methods

This study is based on a qualitative and descriptive research methodology aimed at analyzing the phenomenon of bilingualism from linguistic, social, psychological, and economic perspectives. The research primarily relies on secondary data, including previously published academic works, books, and scholarly articles in the fields of sociolinguistics and applied linguistics. Key theoretical frameworks and concepts were examined through the works of well-known scholars such as Grosjean (2010), Weinreich (1953), Bloomfield (1933), and Cummins (1979), as well as other relevant researchers. The selection of these sources ensured a strong theoretical foundation for the analysis of bilingualism. Particular attention was given to classical and contemporary approaches in sociolinguistic theory. This allowed the study to cover both traditional and modern perspectives on language contact phenomena. Academic databases and university libraries were used as the main sources for collecting relevant literature. The materials were carefully selected based on their relevance, credibility, and contribution to the topic. Only peer-reviewed and widely cited works were included in the analysis to ensure academic reliability. The collected data were organized thematically to facilitate systematic interpretation. This approach helped to identify key patterns and recurring ideas across different studies.

The method of comparative analysis was applied to evaluate different definitions, classifications, and interpretations of bilingualism. Various approaches to the concept of "language situation" were analyzed by comparing viewpoints from different authors, including Fishman (1972) and other sociolinguistic studies. This allowed for a broader understanding of how bilingualism emerges and functions in diverse social and historical contexts.

In addition, the study employs a theoretical analysis of extra-linguistic factors influencing bilingualism, such as migration, education, socio-political conditions, and cultural interaction. These factors were examined through examples from different regions and historical periods, including the case of Azerbaijan and other multilingual societies.

The research also utilizes a synthesis method to combine findings from different sources and present a coherent overview of bilingualism, its types (individual, mass, regional, and national), and its implications. No primary empirical data (such as surveys or experiments) were collected; instead, the study focuses on interpreting and systematizing existing knowledge.

Overall, the chosen methods made it possible to investigate bilingualism as a complex and multifaceted phenomenon and to highlight its significance in modern linguistic and social contexts.

Results and Discussion

In addition to cognitive benefits, bilingualism offers significant social and cultural advantages. Being able to speak two languages allows individuals to communicate with a wider range of people, fostering cross-cultural understanding and empathy. It also enables individuals to access different cultural perspectives, traditions, and forms of expression. For instance, literature, films, and media in different languages provide unique insights into the values and beliefs of various societies. Bilingual individuals often develop a stronger sense of identity, as they can navigate multiply cultural contexts. This cultural awareness is especially important in today's diverse and interconnected world.

Bilingualism also presents challenges, particularly when it comes to learning and using two languages simultaneously. Individuals who grow up or live with two languages often experience difficulties related to vocabulary, grammar, and pronunciation. For instance, the structure of one language can interfere with the other, leading to what linguists call "language interference." This is especially common among children, as they are still in the process of mastering both languages. As a result, they may mix words or grammatical rules from each language within the same sentence, creating hybrid expressions that may not be fully correct in either language. Additionally, bilingual individuals are not always equally proficient in both languages; one language often becomes dominant, while the other remains weaker. This imbalance can create difficulties in academic or professional settings where a high level of proficiency in both languages is required.

Another significant challenge of bilingualism lies in its social and psychological aspects. Bilingual individuals may sometimes feel as though they are caught between two cultures, which can lead to a sense of divided identity. For example, a person who speaks one language at home and another in public may struggle to determine where they truly belong. This situation can affect self-confidence and make social adaptation more difficult. In some cases, bilingual individuals may feel pressure to conform to one cultural norm over the other, especially in societies where multilingualism is not fully valued or supported. Such pressures can lead to feelings of isolation or insecurity, particularly among young people who are still forming their identities.

Educational challenges are also closely associated with bilingualism. Students who learn in two languages often need to invest additional time and effort to fully understand academic materials. Differences in terminology between languages can make certain subjects more difficult to grasp, especially in fields that require precise language use, such as science or mathematics. Furthermore, not all educational systems are equally equipped to support bilingual learners. Teachers may not always have the resources or training to effectively address the needs of students who are navigating two languages. This can result in uneven academic progress and, in some cases, lower performance compared to monolingual peers. Parents also play a crucial role, and if they lack the ability or resources to support both languages at home, children may struggle to maintain balanced language development.

Despite these challenges, it is important to recognize that bilingualism also offers many benefits. However, in order to fully take advantage of these benefits, the challenges must be

carefully managed. Structured learning approaches, consistent exposure to both languages, and supportive social environments can help reduce language interference and improve proficiency. At the same time, attention should be given to the emotional and cultural needs of bilingual individuals, ensuring that they feel comfortable and confident in both linguistic and cultural contexts. With the right support, bilingual individuals can overcome these challenges and develop into highly skilled communicators with a rich and diverse cultural perspective.

3. Bilingualism has significant economic implications in an increasingly interconnected and competitive global economy. As international trade, migration, and communication continue to expand, the ability to speak more than one language has become a valuable form of human capital. Bilingual individuals often enjoy better employment prospects because they can interact with a wider range of clients, customers, and colleagues. Employers frequently view language skills as an asset, especially in industries such as international business, tourism, finance, and customer relations. In many cases, bilingual employees are preferred over monolingual candidates because they can help companies operate more efficiently across borders. This advantage can also translate into higher wages, as workers with specialized language skills may be able to negotiate better salaries or access positions that require multilingual communication.

At the macroeconomic level, bilingualism contributes to economic growth by facilitating trade and investment between countries. When language barriers are reduced, businesses can communicate more effectively with foreign partners, which helps to build trust and minimize misunderstandings. This leads to smoother negotiations, more successful contracts, and stronger long-term relationships. As a result, countries with a higher proportion of bilingual or multilingual citizens are often better positioned to participate in global markets. They can export goods and services more easily, attract foreign investment, and promote their industries on an international scale. In this way, bilingualism acts as a bridge that connects different economies and enhances global cooperation.

Another important economic implication of bilingualism is its role in the service sector, particularly in tourism and hospitality. Tourists are more likely to visit and spend money in destinations where they can communicate comfortably. Bilingual workers in hotels, restaurants, and travel agencies help create a more welcoming environment, which can increase customer satisfaction and encourage repeat visits. This directly contributes to local and national economies by generating income and employment opportunities. Furthermore, bilingualism is essential in public services such as healthcare, education, and legal systems, where clear communication can improve efficiency and reduce costly errors.

However, the economic benefits of bilingualism are not equally distributed. Access to quality language education often depends on socioeconomic factors, which can create inequalities in the labor market. Individuals who have the opportunity to learn multiple languages are more likely to succeed economically, while those without such opportunities may face limited career prospects. This highlights the need for governments and institutions to invest in language education and ensure equal access for all citizens. By doing so, they can promote both individual success and broader economic development.

Hence, bilingualism is a powerful economic resource that enhances employability, supports international trade, and contributes to the growth of key industries. Its impact extends beyond individuals to influence entire economies, making it an essential skill in the modern world.

In many cases, such socio-public phenomena emerge in different forms. In all cases, during bilingualism, individual members of society or a significant part of it use different languages as a means of communication in various situations and at different moments, thus leading to the creation of the linguistic phenomenon called bilingualism in society.

It should be noted that in this or that country, or during certain periods of the history of individual peoples, bilingualism manifests in various ways and forms. That is, a small part or the vast majority of the members of any society can use different languages as communicative tools in the communication process, which causes bilingualism to show itself in various ways. In other words, not all members of society may use different languages to the same extent in the communication process, which is considered a factor conditioning the emergence of bilingualism in different forms. In this regard, the following views on bilingualism and its essence are of interest and facilitate the determination of its types: "Bilingualism (Latin: bi-two, lingua-language) is two-language-ness. In other words, it is the use of two languages in the communication process by a specific member of society or by a people as a whole. Usually, the first of these is the mother tongue, and the second is another language. If the use of two languages is characteristic of individual members of society, it is considered individual bilingualism; if it is characteristic of the majority of speakers, it is considered mass bilingualism" (Cummins, 1979, p.82-83). Thus, in all cases, when taking into account the quantity of those using it, bilingualism manifests in two forms: individual and mass bilingualism. This is directly related to the general intellectual level, character, and general development characteristics of the society in which bilingualism manifests itself. In other words, the attitude of its individual members or society as a whole toward other languages plays a primary role here, in accordance with the general development level of the society.

In different historical periods of the development of society, it is possible to encounter manifestations of bilingualism in other forms as well. Thus, it is also possible to observe the emergence of bilingualism in any geographical area where a certain people resides and in the public life of that people as a whole. That is, alongside the form of bilingualism spread in any region of this or that state, a form covering its entire territory can also show itself. From this perspective, if bilingualism showing itself in a certain geographical area or economic zone of any country is considered areal bilingualism, the use of two languages in the same way as a means of communication by the majority of the population throughout the entire territory of the country is considered national bilingualism. While such areal and national bilingualism manifested in most union republics in the former Soviet Union, regional bilingualism emerged when minority peoples settled in individual geographical areas used the state language of that area as a means of communication alongside their mother tongue. For example, in the case of the Republic of Azerbaijan, this type of bilingualism can be imagined as follows: if during the Soviet era, the majority of the population in Azerbaijan could use the Russian language, called the "second mother tongue" at that time, as a communicative tool alongside their mother tongue, and this could be considered national bilingualism, then if minority peoples settled in different regions of Azerbaijan (e.g., Lezgins, Talishs, Tats, Udi, etc.) used the state language, Azerbaijani, alongside their mother tongue in the communication process, this can be considered regional (or territorial) bilingualism. The reasons for bilingualism manifesting in this way can be considered related to the fact that "Starting from the day society was created, people did not stay in one place but scattered across the earth as a result of migrations. Precisely from this period—starting

from the first human displacements, primitive contacts between clans and tribes—certain connections were also created between languages. These connections—social and language connections—cannot be separated from each other” (Weinreich, 1953). That is, due to historical events and the mass migration of individual peoples to various geographical areas, different forms of bilingualism have emerged in various countries, which in many cases began to show themselves in the form of regional and national bilingualism. The manifestation of bilingualism in these forms can also be found in the languages of other peoples. Regarding this problem, researchers conducting studies on the Kazakh language and the bilingualism manifesting in Kazakhstan note that due to various historical processes and the influence of socio-political factors, different forms of bilingualism show themselves in this Turkic-speaking republic in the modern era. During the research, the investigators determined that in the modern period, it is necessary to talk more about trilingualism in the territory of Kazakhstan. Thus, the intellectual social layer of society uses Russian and any foreign language alongside their mother tongue, and the application of these languages in the communicative process can manifest as individual, mass, regional, and national bilingualism. Speaking about the reasons for this, the researcher explains it through the older generation receiving education in Russian, the influx of many rural populations to cities as a result of urbanization, and finally, the study and application of other foreign languages as an open civil society after Kazakhstan gained independent republic status.

As mentioned before bilingualism plays a crucial role in both cognitive and social development. In this regard, the importance of maintaining the first language becomes particularly evident. As Jim Cummins states, “To reject a child’s language in school is to reject the child” (Cummins, 1979). This idea highlights that language is deeply tied to personal identity and self-esteem. Therefore, supporting bilingualism in educational settings is essential for both academic achievement and psychological development.

The manifestation of bilingualism in these forms can show itself in various regions of the world, and the reasons for its emergence are in most cases related to similar factors. However, although the reasons for their creation in many cases bear a similar character, the emergence of each type of bilingualism can also be characterized by its own uniqueness. This is related, on the one hand, to the ancient history of the emergence of bilingualism and, on the other hand, can be explained by various extra-linguistic factors affecting its emergence in individual regions. “The phenomenon of bilingualism exists in various parts of the world. Its cultural-historical and social conditions can be diverse. Therefore, the characteristics of bilingualism are difficult to describe. The study and analysis of such a language phenomenon require special linguistic methods” (Bloomfield, 1993). Thus, bilingualism, which is a complex socio-public phenomenon, is created in various ways and manifests in different forms in individual geographical areas. Despite the existence of these different manifestation forms, many reasons for its creation have similar characteristics by bearing a general nature. Speaking about the reasons for the emergence of bilingualism, A.Kourbanov writes: “Bilingualism, as a natural language phenomenon, is one of the objective factors operating in multi-ethnic countries. There have been a number of interrelated linguistic and social reasons for the emergence and operation of this phenomenon. Some reasons for the emergence of bilingualism:

A) due to being a representative of a non-nationality in mixed families, children speak in two languages;

B) bilingualism emerges on the basis of the diversity of the national composition of labor collectives;

C) bilingualism arises in connection with the use of another language in education and public organizations;

Ç) multilingualism is formed during the conduct of scientific and cultural relations, etc.(Qurbanov, 1993)

Naturally, the reasons for the emergence of bilingualism in individual countries are not limited to those listed above, and other reasons for its emergence in various regions of the world may also show themselves, the broad and comprehensive investigation of all of which, being among the significant linguistic issues, should be in the center of attention of linguistics as a special research topic.

4. Overall, the findings of this study highlight that bilingualism should be viewed as a dynamic and context-dependent phenomenon. Its effects vary depending on individual, social, and educational factors. The results suggest that early exposure to two languages tends to produce more stable and balanced bilingual competence. In contrast, late acquisition of a second language may lead to uneven proficiency levels. The discussion also indicates that societal attitudes toward bilingualism significantly influence its development and success. In supportive environments, bilingual individuals are more likely to achieve high levels of fluency in both languages. However, in less supportive contexts, language dominance and attrition may occur. Another important observation is that bilingualism contributes to greater cultural empathy and openness toward diversity. This has positive implications for multicultural societies where intercultural communication is essential. The study further emphasizes that effective bilingual education policies can minimize challenges and maximize cognitive and social benefits. Therefore, bilingualism should be encouraged as a long-term educational and societal goal. In addition, continuous exposure to both languages in real-life communication settings helps maintain balanced proficiency over time. Overall, the integration of linguistic support systems and inclusive policies can further strengthen the positive outcomes of bilingualism.

Conclusion

Bilingualism is one of the most significant linguistic and social phenomena of the modern world. It represents not only the ability to use two languages but also the capacity to operate within two distinct systems of thought and cultural perspectives. In today's globalized society, bilingualism plays a crucial role in personal development, social integration, and effective international communication.

Although bilingualism may involve certain cognitive, social, and educational challenges, its advantages considerably outweigh its limitations. It enhances cognitive flexibility, strengthens cultural awareness, and expands economic and professional opportunities. These benefits make bilingualism an essential competence rather than a simple advantage in the contemporary world.

Moreover, bilingual individuals are often better equipped to adapt to multicultural environments, which is increasingly important in modern societies. They can act as linguistic and cultural mediators between different communities, helping to reduce misunderstandings. Bilingualism also contributes to the preservation of cultural heritage, as individuals maintain connections with their native language while learning another. In educational contexts, it supports deeper cognitive engagement and improves metalinguistic awareness (Bloomfield).

Furthermore, research shows that bilingualism may delay cognitive aging and support long-term brain health. From an economic perspective, it increases employability and facilitates participation in the global labor market. In addition, it strengthens international cooperation by improving communication between countries and organizations.

Therefore, the promotion of language learning and the creation of bilingual environments should be regarded as a priority for both individuals and societies. Encouraging bilingualism not only supports individual achievement but also contributes to cultural diversity, mutual understanding, and global cooperation.

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CHARACTERISTICS OF EMOTIONAL DEVELOPMENT IN CHILDREN WITH INTELLECTUAL DISABILITIES

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Abstract: This article examines the characteristics of emotional development in children with intellectual disabilities. Emotional development is considered one of the essential conditions for personality formation, and this process is significantly influenced by family, school, and the broader social environment. The main features of the emotional sphere in children with intellectual disabilities include emotional immaturity, insufficient differentiation of emotions, instability, and emotional reactions that are often inadequate to the given situation. Emotions such as joy, sadness, anger, and fear may be expressed either excessively or, conversely, in a weak and delayed manner.

Research findings indicate that children with intellectual disabilities experience difficulties in perceiving and interpreting the emotions of others. This often leads to communication problems with peers and adults and may result in behavioral manifestations such as aggression, negativism, and anxiety. Emotional deficiencies also negatively affect the acquisition and retention of academic material.

However, the emotional development of children with intellectual disabilities should not be regarded as completely impaired or static. Through purposeful pedagogical intervention, properly organized special education, and systematic upbringing, it is possible to regulate emotional responses, develop positive behavioral habits, and improve social adaptation. The article emphasizes that corrective work should be based on the child's preserved abilities and should take into account the specific characteristics of their emotional sphere. Such an approach contributes significantly to personality development and supports future social and vocational integration.

Keywords: Intellectual disability, emotional development, emotional sphere, intellectual impairment, social adaptation, special education, corrective intervention.

ƏQLİ MƏHDUDİYYƏTLƏRİ OLAN UŞAQLARDA EMOSİONAL İNKİŞAFIN XÜSUSİYYƏTLƏRİ

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Xülasə: Bu məqalədə əqli geriliyi olan uşaqlarda emosional sahənin inkişaf xüsusiyyətləri təhlil edilmişdir. Emosional inkişaf uşağın şəxsiyyət kimi formalaşmasının əsas şərtlərindən biri hesab olunur və bu prosesə ailə, məktəb və sosial mühit mühüm təsir göstərir. Əqli geriliyi olan uşaqlarda emosional sahənin əsas xüsusiyyəti onun yetkin olmaması, emosiyaların zəif diferensiasiyası, qeyri-sabitlik və emosional reaksiyaların çox vaxt situasiyaya adekvat olmamasıdır. Belə uşaqlarda sevinc, kədər, qəzəb və qorxu kimi emosiyalar ya həddindən artıq güclü, ya da əksinə, zəif və ləng şəkildə ifadə olunur.

Tədqiqatlar göstərir ki, əqli geriliyi olan uşaqlar başqalarının emosiyalarını düzgün qavramaqda və şərh etməkdə çətinlik çəkir. Bu isə onların həmyaşıdları və böyüklərlə ünsiyyətində problemlər yaradır, aqressivlik, neqativizm və qorxu kimi davranış pozuntularına səbəb ola bilər. Emosional sahədəki çatışmazlıqlar təlim materialının mənimsənilməsinə də mənfi təsir göstərir.

Bununla yanaşı, əqli geriliyi olan uşaqların emosional inkişafı tam dayanmış hesab edilmir. Məqsədyönlü pedaqoji müdaxilə, düzgün təşkil olunmuş xüsusi təhsil və tərbiyə vasitəsilə onların emosional reaksiyalarını tənzimləmək, müsbət davranış vərdisləri formalaşdırmaq və sosial adaptasiyasını gücləndirmək mümkündür. Məqalədə vurğulanır ki, korreksiya işi uşağın qorunmuş imkanlarına əsaslanmalı və onun emosional xüsusiyyətləri nəzərə alınmalıdır. Bu yanaşma uşağın şəxsiyyət inkişafına və gələcək sosial-əmək adaptasiyasına mühüm töhfə verir.

Açar sözlər: Əqli gerilik, emosional inkişaf, emosional sahə, intellektual çatışmazlıq, sosial adaptasiya, xüsusi təhsil, korreksiya işi.

Introduction

I would like to begin this scientific research paper with the words of Lev Vygotsky, who stated: "The emotional development of children is one of the most important directions of educators' professional activity. Emotions constitute the 'central link' of a child's mental life."

The formation of the emotional sphere is one of the most essential conditions for the development of a child as a personality. The development of emotional functioning is influenced by school, family, and all environmental factors surrounding the child and exerting continuous influence on them.

One of the most significant characteristics of a child with intellectual developmental delay is the immaturity of their emotional sphere. Emotions and feelings play a crucial role in personality development. In children with oligophrenia, severe primary disturbances in the emotional sphere may lead to psychopathological character formation and the emergence of new negative personality traits, which significantly complicates the correction of the primary psychological defect. Therefore, the study, formation, and education of emotions and feelings in children with intellectual disabilities are of particular importance.

The formation of emotions is a fundamental condition for personality development. Human life is constantly accompanied by emotions; every activity is emotionally colored to some degree. The level of emotional sphere development directly affects a child's socialization capacity and the acquisition of academic material.

It is commonly observed that when educational material is presented in an emotionally enriched manner and elicits an emotional response from the child, the child retains and assimilates the material more effectively (S.Gasimov, & E.Karimova, 2004).

A major difficulty for children with intellectual disabilities is that they are often unable to correctly perceive and analyze the emotions of others. Therefore, it is essential to teach such children how to respond appropriately to different emotional states. These children are frequently inclined toward polarized emotional reactions; their emotions are superficial, unstable, and characterized by rapid and abrupt changes. At times, emotions may be expressed excessively intensely and may not correspond to the actual cause, occasionally appearing entirely inadequate.

Conversely, some children display delayed, inert emotional reactions. Their experienced emotions may also be incongruent with external stimuli. In many cases, children demonstrate poor emotional self-regulation and often make little attempt to control their emotional responses.

Such emotional lability and rapidly changing moods create difficulties in communication with peers and adults. These challenges may manifest as negativism, fearfulness, and aggression, which in turn negatively affect the assimilation of educational material.

Like all other children, children with intellectual disabilities continue to develop throughout life. Sergei Rubinstein noted that: "The psyche develops even in the most severe cases of intellectual disability... Psychological development is a specific feature of childhood and finds its path even through the most severe organic pathology."

Alongside the specific development of the psyche, the emotional sphere of children with intellectual disabilities also develops in a unique manner, primarily characterized by emotional immaturity.

The immaturity of the emotional and affective sphere in children with intellectual disabilities is primarily related to the peculiarities of the development of their needs, motives, and intellect.

Leonid Zankov, characterizing the emotional-volitional sphere of children with intellectual disabilities, emphasized that the development of their emotional sphere largely depends on external conditions. These conditions include specialized education and the proper organization of the child's entire life. Impulsive reactions naturally observed in such children – such as anger, sadness, joy, and others – can gradually be softened through purposeful upbringing and pedagogical intervention. Such educational efforts help the child understand their own actions and form positive habits and appropriate behavioral skills.

Materials and Methods

According to the studies of Mariya Pevzner, all children with intellectual disabilities can conditionally be divided into three groups:

In the first group, children exhibit slow, inert, and stereotypical emotional reactions. These children appear indifferent to external stimuli.

In the second group, reactions are excessively strong, disproportionate to their causes, and at times completely inadequate (S.D. Zabravnya, 1995).

In the third group, no gross emotional disturbances are observed, although certain unexplained individual behavioral peculiarities may occur.

Across all clinical groups of children with intellectual disabilities, emotional immaturity, insufficient differentiation of feelings, instability, a limited range of emotional experiences, and sharply expressed reactions such as joy, sadness, and amusement are observed.

According to E. A. Yevlakhina, students with intellectual disabilities often make errors or distortions when interpreting the facial expressions of characters depicted in plot-based pictures. They struggle to recognize complex and subtle emotions, replacing them with simpler and more elementary affective experiences. This suggests that perceptual errors in interpreting visual material are directly linked to the developmental level of their emotional sphere.

In schoolchildren with oligophrenia, delays in personality development are especially evident in play activities.

N. L. Kolominsky noted that: "A child with intellectual disability is passive in play; for them, play does not serve as a model for acquiring social experience in the same way as it does for a typically developing schoolchild. The absence of elementary play skills is considered an important diagnostic indicator of intellectual disability."

This phenomenon is explained by the insufficient development of the child's need for new impressions, curiosity, and cognitive activity. Internal motivations directed toward engaging in new types of activities are weakly expressed. Their behavior and activities are largely governed by immediate situational influences.

Manifestations of emotional disturbances include irritability, heightened excitability, motor restlessness, inability to remain still, and a lack of mediated motivation. In comparison with typically developing schoolchildren, children with oligophrenia do not adequately develop social emotions.

Emotional immaturity becomes especially pronounced during the school period, when children are faced with tasks requiring more complex and mediated forms of activity.

Discussion

School age is characterized by a number of developmental features absent in earlier periods and requires entirely new forms of mental activity. Whereas preschool children primarily develop through play and communication, school-age children enter structured learning environments governed by formal curricula. The leading activity shifts from play to learning, and the system of motives is reorganized. The primary motive becomes compliance with the teacher's instructions.

The emotional sphere of children with intellectual disabilities is characterized by low differentiation and poverty of emotional experiences. S. S. Lyapidevsky and B. I. Shostak noted that such children's "feelings are monotonous, unstable, limited to two extremes (satisfaction or dissatisfaction), and arise only under the direct influence of immediate stimuli."

Thus, the emotional experiences of children with intellectual disabilities are primitive in nature; subtle and differentiated emotional nuances have not yet formed.

This is explained by the predominance of primitive needs.

At the same time, children with oligophrenia may display certain emotional vitality, including kindness, trustfulness, and emotional responsiveness, despite the superficiality and instability of their emotional experiences. Such children easily shift from one emotional state to another, show limited independence in activities, are easily influenced, and tend to imitate other children in both behavior and play.

Furthermore, emotions and feelings in children with intellectual disabilities are often inadequate in relation to external circumstances. Some children may demonstrate excessively superficial attitudes toward serious life events.

Characteristic features include poor emotional differentiation, stereotyped emotional responses, poverty of emotional experiences, absence of emotional nuances, weakness of motives, and lack of motivational conflict. Their emotional reactions are mainly directed toward immediately acting stimuli.

Emotional immaturity further intensifies the general rigidity of the psyche, weak mental activity, reduced interest in the surrounding environment, and lack of initiative and independence.

Additionally, insufficient ability to restrain affects and impulses contributes to impulsivity and intense emotional outbursts. Anger explosions and aggressive behavioral reactions may occur in response to minor causes.

Even in children with severe intellectual disabilities, emotions connected to primitive needs, concrete situations, and “sympathetic” feelings – such as compassion, pity, resentment, or shame toward specific individuals – may remain preserved.

The cognitive weakness and intellectual insufficiency of children with intellectual disabilities, together with the immaturity and primitiveness of their motivational-need sphere, slow the formation of higher-order emotions.

The study, appropriate formation, and education of emotions and feelings in such children contribute to personality development, the emergence of positive qualities, and ultimately facilitate the correction of the primary psychological defect.

Despite these challenges, the personality of a child with oligophrenia continues to develop. Their emotional sphere also develops in a specific and unique way. Therefore, by relying on preserved functions and existing capacities, and by taking into account the specific features of their emotional sphere, the proper organization of corrective work can enable the complete or partial elimination of emotional disturbances.

This, in turn, plays an important role in the overall development of the child’s personality and in ensuring their future social and vocational adaptation.

Conclusion

In summary, the emotional sphere of children with intellectual disabilities is characterized by insufficient differentiation and inadequacy of emotions. Higher emotional experiences – such as epistemic, moral, and aesthetic feelings – are difficult to form. Direct emotional experiences associated with concrete life situations predominate. The mood of such children is generally unstable.

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NAVIGATING THE MULTILINGUAL MIND: LANGUAGE LEARNING STRATEGIES AND COGNITIVE SYNERGY IN POLYGLOT ENVIRONMENTS

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Abstract: In an increasingly globalized world, multilingualism has evolved from a specialized skill into a cognitive necessity. This article explores the language learning strategies (LLS) and cognitive synergies inherent in polyglot environments, challenging the traditional "monolingual bias" that dominates contemporary pedagogy. Utilizing a qualitative methodology—comprising learning surveys, "think-aloud" tasks, and semi-structured interviews—the study investigates how learners navigate the complexities of acquiring a third or fourth language. The research highlights that successful multilinguals function as "expert learners," leveraging cross-linguistic transfer and heightened tolerance for ambiguity to manage increased cognitive loads. By validating hybrid linguistic practices and encouraging explicit strategy instruction, educators can transform the challenges of linguistic diversity into a "cognitive superpower". Moreover, the study justifies the need to introduce cross-linguistic comparative methods in educational programs, rather than completely eliminating the mother tongue. This approach serves to increase students' self-confidence. Finally, the article provides practical recommendations for creating a more inclusive and effective pedagogical framework suitable for the global academic and digital environments of the 21st century.

Keywords: Multilingualism, Language Learning Strategies, Cognitive Flexibility, Cross-linguistic Transfer, Dynamic Systems Theory.

ÇOXDİLLİ DÜŞÜNCƏNİN İDARƏ OLUNMASI: DİL ÖYRƏNMƏ STRATEGİYALARI VƏ POLİQLOT MÜHİTİNDƏ KOQNİTİV SİNERGİYA

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Xülasə: Getdikcə qloballaşan dünyada bir neçə dil biliklərinə malik olmaq bir bacarıqdan daha çox idrak zərurətinə çevrilmişdir. Bu məqalədə müasir pedaqogikada üstünlük təşkil edən ənənəvi təkdiilliyə qarşı çıxan və çoxdilli mühitlərdə mövcud olan dil öyrənmə strategiyaları araşdırılır. Öyrənmə sorğularından, “ucadan düşünmə” tapşırıqlarından və yarı strukturlaşdırılmış müsahibələrdən istifadə edərək - istifadə edilən tədqiqat, təhsilalanların üçüncü və ya dördüncü dili mənimsəməyin mürəkkəbliklərini necə dəf etdiyini araşdırır. Tədqiqat, bir çox dilləri yüksək səviyyədə mənimsəmiş öyrənənlərin artan idrak yüklərini idarə etmək üçün dillərarası köçürmədən və qeyri-müəyyənliyə qarşı yüksək tolerantlıqdan istifadə edərək "mütəxəssis öyrənənlər" kimi fəaliyyət göstərdiklərini vurğulayır. Hibrid dil təcrübələrini təsdiqləməklə və açıq strategiya təlimatını təşviq etməklə, müəllimlər dil müxtəlifliyinin problemlərini "idraki gücə" çevirə bilirlər. Bundan əlavə, tədqiqat təhsil proqramlarında ana dilinin tamamilə kənarlaşdırılması yerinə, dillərarası müqayisəli metodların tətbiqinin zəruriliyini əsaslandırır. Bu yanaşma tələbələrdə özünəinamı artırmağa xidmət edir. Yekun olaraq, məqalə 21-ci əsrin qlobal akademik və rəqəmsal mühitlərinə uyğun daha inklüziv və effektiv pedaqoji çərçivənin yaradılması üçün praktiki tövsiyələr təqdim edir.

Açar sözlər: Çoxdiillilik, dil öyrənmə strategiyaları, idraki çeviklik, dillərarası köçürmə, dinamik sistemlər nəzəriyyəsi.

Introduction

In the era of globalization and rapid digital transformation, multilingualism has transitioned from being a specialized skill to a fundamental necessity. As borders become more fluid in the academic and professional worlds, the ability to navigate between multiple linguistic systems has become a focal point of contemporary applied linguistics. Multilingual environments—whether they are diverse classrooms, international organizations, or digital platforms—provide a unique socio-cognitive landscape for language acquisition. Unlike monolingual learners, individuals in multilingual settings do not just acquire a new set of vocabulary; they engage in a complex process of managing multiple linguistic identities and cognitive resources. This research aims to explore the specific language learning strategies (LLS) that are most effective within these vibrant, multifaceted environments.

Traditionally, LLS were categorized into cognitive, metacognitive, memory-related, compensatory, affective, and social strategies. However, in a multilingual context, these strategies take on a more dynamic form. Learners often employ "cross-linguistic transfer," where knowledge of a first (L1) or second (L2) language serves as a scaffold for the third (L3) or fourth (L4) language. This phenomenon, often referred to as "translanguaging," challenges the traditional "monolingual bias" in education, suggesting that a learner's entire linguistic repertoire is an integrated system rather than a collection of isolated compartments. (Canagarajah, 2011) Furthermore, the cognitive advantage of multilinguals, often termed "metalinguistic awareness," plays a crucial role in how strategies are selected and applied.

Furthermore, the cognitive advantage of multilinguals, often termed "metalinguistic awareness," plays a crucial role in how strategies are selected and applied. (Bialystok, 2017) Multilingual learners tend to be more conscious of the underlying structures of language, allowing them to employ metacognitive strategies—such as self-monitoring and planning—with greater efficiency. They are not merely learning a language; they are "learning how to learn." However, despite these inherent advantages, the complexity of multilingual environments also introduces unique challenges, such as linguistic interference and increased cognitive load, which necessitate specialized strategic interventions.

This paper will investigate the synergy between multilingual exposure and strategic competence. By analyzing how learners in high-level academic settings, particularly those balancing major languages like English and French, utilize various strategic tools, we can better understand the mechanics of successful language acquisition. The ultimate goal of this study is to provide a comprehensive framework that educators can use to foster more inclusive and effective pedagogical practices in today's increasingly diverse educational landscapes. Through this exploration, we seek to bridge the gap between theoretical linguistics and practical classroom application, ensuring that the multilingual potential of every student is fully realized.

Materials and Methods

The study uses a qualitative approach to understand how people learn multiple languages at once. Instead of just looking at test scores, this research focuses on the "how" and "why" behind the choices students make when they study. It specifically looks at how students use their first and second languages to help them learn a third one.

To get a full picture of the students' learning habits, we used three simple tools:

The Learning Survey: A questionnaire where students rated how often they use different "tricks" to learn (like guessing from context or setting study goals).

The "Think-Aloud" Task: Students were given a short text to translate or summarize. While doing the task, they had to say everything they were thinking out loud. This helped us see if they were "thinking" in their other languages to find the answer.

Interviews: 15-minute conversations were held with the students to ask them how they feel when they make mistakes and how they stay motivated.

The research followed three easy steps:

Surveying - This quantitative step allowed for:

1. The identification of the most and least common strategies used across the participant pool.
2. The categorization of learners based on their proficiency levels and linguistic backgrounds.
3. A statistical foundation that informed the focus areas for the subsequent qualitative phases. (Oxford, 2016)

Observation - To move beyond what learners "say" they do and see what they "actually" do, the study utilized qualitative observation.

Interviewing - The final phase added depth through semi-structured interviews. While surveys and observations track actions, interviews uncover the "why" behind them, focusing on the psychological and emotional aspects of learning.

Results and Discussion

The evolution of multilingual research has moved away from viewing the brain as a container for separate linguistic modules. Instead, modern applied linguistics utilizes Dynamic Systems Theory (DST) to explain the acquisition of additional languages. (Cook, 2016). In this framework, language acquisition is not a linear process of adding L3 to L2; it is a holistic reconfiguration of the brain's linguistic architecture. Every new language introduced into the system interacts with the existing ones, creating a state of constant flux where the proficiency in one language can influence the development or attrition of another.

This interconnectedness leads to the concept of translanguaging, which posits that a student's languages function as a unified cognitive resource. When a learner encounters a gap in

their L3, they do not simply fail to communicate; they strategically reach back into their L1 or L2 repertoire to bridge the divide. This challenges the "monolingual bias" that has long dominated classroom pedagogy—the idea that the target language should be kept strictly separate from the learner's native tongue. By acknowledging that the multilingual mind is a synergy of systems, we can see that cognitive flexibility—the ability to switch between mental sets—is the primary driver of success in polyglot environments.

Multilingual learners possess heightened metalinguistic awareness, which acts as a "manager" for their learning process. This awareness allows them to select and apply Language Learning Strategies (LLS) with greater precision than their monolingual counterparts.

Multilinguals are frequently "expert learners." Having navigated the transition between languages at least once before, they possess a template for success. Metacognitive strategies—such as planning, monitoring, and evaluating—are used more efficiently. A multilingual learner is more likely to realize when a specific study method is failing and pivot to a new one. They engage in constant self-regulation, consciously deciding when to focus on form (grammar) versus when to focus on function (communication), depending on the immediate requirements of their environment.

Cross-linguistic transfer is the most powerful tool in the polyglot's arsenal. It involves the conscious or subconscious application of knowledge from one language to another. (Pavlenko, 2011). This is not limited to vocabulary; it includes phonetic patterns, syntactic structures, and even pragmatic norms.

Inter-linguistic Scaffolding: This is the process where a learner uses a known language as a temporary support structure. For example, a student might use their understanding of German case systems to better grasp the logic of Slavic grammar.

Morphological Awareness: Multilinguals are often better at "chunking" language. They recognize roots, prefixes, and suffixes across different systems, allowing them to decode complex, unfamiliar words without needing a dictionary.

In high-level academic or digital environments, the social dimension of language learning is paramount. The environment dictates the use of social strategies, such as "code-switching," which functions not as a sign of linguistic weakness, but as a sophisticated tool for identity negotiation and group belonging.

In a polyglot environment, learners are rarely in a "pure" linguistic vacuum. They are constantly exposed to hybrid forms of communication. This exposure encourages the use of social strategies like asking for clarification, circumlocution (describing a word you don't know), and mimicking native-like discourse markers. These strategies allow the learner to maintain social flow even when their technical proficiency is lacking.

One of the most significant advantages of multilingualism is the development of a higher tolerance for ambiguity. Monolingual learners often experience "language anxiety"—a paralyzing fear of making mistakes or not being understood. Multilinguals, however, are accustomed to the "gray areas" of communication. They employ affective strategies—techniques used to manage emotions—to view errors as a natural part of a multi-layered linguistic evolution rather than a personal failure. This psychological resilience is a key differentiator in achieving high-level fluency. (Oxford, 2016)

The findings of this study suggest that educational systems must evolve to support the "multilingual potential" of students. Moving toward Plurilingual Pedagogy is essential. This involves:

1. Encouraging Translanguaging: Allowing students to use their full linguistic repertoire in the classroom to brainstorm and solve problems.
2. Explicit Strategy Instruction: Teaching students how to identify cognates, manage anxiety, and use metacognitive tools.
3. Cross-Linguistic Comparisons: Actively involving the students' native languages in the learning process rather than banning them.

Conclusion

In summary, language learning in multilingual environments represents far more than the simple acquisition of disparate linguistic codes; it is a sophisticated balancing act that reshapes the learner's entire cognitive architecture. By leveraging metalinguistic awareness and the mechanics of cross-linguistic transfer, learners do not merely manage the increased cognitive load of multiple systems—they thrive because of it. The synergy found in polyglot environments is not a hurdle to be cleared or a source of confusion to be mitigated, but rather a unique cognitive superpower. This "multilingual advantage" fosters a level of mental flexibility, divergent thinking, and communicative resilience that is increasingly rare and highly valued in the 21st century.

The evidence presented in this study suggests that the traditional "monolingual bias" in education is not only outdated but actively hinders the natural potential of the polyglot mind. (Cook, 2016). To move forward, educational frameworks must undergo a paradigm shift, transitioning from rigid, target-language-only silos toward a Plurilingual Pedagogy. Such a framework must explicitly recognize and validate the "translanguaging" practices that students naturally employ. By allowing the L1, L2, and L3 to inter“ct within the c”assroom, we reduce the affective filter and empower students to use their entire linguistic repertoire as a strategic asset.

Furthermore, the implications of this research extend beyond the classroom and into the socio-cultural fabric of our globalized world. As we bridge the gap between theoretical linguistics and practical classroom application, we are doing more than teaching grammar; we are fostering global citizenship. Understanding these strategies is essential for developing inclusive environments that reflect the complex, interconnected reality of our modern era. When educators and policymakers embrace the fluidity of the multilingual mind, they ensure that the potential of every student—from the local classrooms of Baku to the international academic hubs of Europe—is fully realized. Ultimately, the goal is to transform the challenge of diversity into a catalyst for cognitive and social excellence, ensuring that the multilingual mind remains at the forefront of human innovation and cross-cultural understanding.

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THE SCIENCE OF EARLY LANGUAGE LEARNING: BRAIN DEVELOPMENT AND LANGUAGE MASTERY

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Abstract: The science of early language learning explains how brain development and biological factors help children learn foreign languages more easily than adults. One of the most important ideas is neuroplasticity, which means the brain can change, adapt, and create new connections. In early childhood, the brain is very flexible, so children can learn new sounds, words, and grammar structures quickly and naturally. They often learn languages in a similar way to their first language, through listening, speaking, and interaction. Another key idea is the critical period hypothesis. This theory suggests that there is a special time in human development, usually from early childhood to puberty, when language learning is easier and more effective. During this period, children can achieve better pronunciation and stronger fluency because their brains are more sensitive to language input. After this period, learning a new language is still possible, but it usually requires more effort and practice. Children also have cognitive advantages. They are more curious, more active in communication, and less afraid of making mistakes. This helps them practice more and improve faster. They also remember new words easily and can build vocabulary quickly when they are exposed to the language regularly. Their listening skills are also stronger, which helps them understand different accents and speaking styles. However, environment is also very important. Children need regular exposure to the language at home, at school, or through media. Interaction with teachers, friends, and native speakers helps improve fluency and confidence. Motivation also plays an important role in successful learning. In conclusion, early language learning is influenced by both biological and environmental factors. Brain development, especially neuroplasticity and the critical period, gives children natural advantages that make language learning easier, faster, and more effective in early life.

Keywords: Neuroplasticity, critical period, language acquisition, brain development, bilingualism, early learning, language mastery.

ERKƏN DİL ÖYRƏNMƏSİNİN ELMI: BEYİN İNKİŞAFI VƏ DİL MƏNİMSƏNİLMƏSİ

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Xülasə: Erkən dil öyrənməsinin elmi izah edir ki, beyin inkişafı və bioloji amillər uşaqlara xarici dilləri böyüklərə nisbətən daha asan öyrənməyə necə kömək edir. Ən vacib anlayışlardan biri neyroplastiklikdir, bu isə beynin dəyişə, uyğunlaşa və yeni əlaqələr yarada bilməsi deməkdir. Erkən uşaqlıq dövründə beyin çox elastik olur, buna görə uşaqlar yeni səsləri, sözləri və qrammatikanı tez və təbii şəkildə öyrənə bilirlər. Onlar dili tez-tez ilk dillərini öyrəndikləri kimi öyrənirlər: dinləmə, danışma və qarşılıqlı ünsiyyət vasitəsilə. Digər vacib anlayış kritik dövr hipotezidir. Bu nəzəriyyə deyir ki, insan inkişafında xüsusi bir dövr var—erkən uşaqlıqdan yeniyetməliyə qədər—bu zaman dil öyrənmək daha asan və daha effektiv olur. Bu dövrdə uşaqlar daha yaxşı tələffüz və axıcılıq əldə edə bilirlər, çünki beyinləri dilə daha həssas olur. Bu dövrdən sonra da dil öyrənmək mümkündür, lakin adətən daha çox səy və məşq tələb edir. Uşaqların həm də koqnitiv üstünlükləri var. Onlar daha maraqlı, ünsiyyətdə daha aktiv və səhv etməkdən daha az qorxurlar. Bu isə onların daha çox məşq etməsinə və daha sürətli inkişaf etməsinə kömək edir. Onlar yeni sözləri asanlıqla yadda saxlayır və dili müntəzəm eşitdikdə lüğət ehtiyatlarını tez genişləndirirlər. Onların dinləmə bacarıqları da güclüdür, bu da müxtəlif aksentləri və danışq üslublarını anlamağa kömək edir. Lakin mühit də çox vacibdir. Uşaqlar evdə, məktəbdə və ya media vasitəsilə dili müntəzəm şəkildə eşitməlidirlər. Müəllimlər, dostlar və ana dili danışanlarla qarşılıqlı ünsiyyət axıcılığı və özünəinamı artırır. Motivasiya da uğurlu öyrənmədə mühüm rol oynayır. Nəticə olaraq, erkən dil öyrənməsi həm bioloji, həm də mühit amillərindən asılıdır. Beyin inkişafı, xüsusilə neyroplastiklik və kritik dövr, uşaqlara dili daha asan, daha sürətli və daha effektiv öyrənmək üçün təbii üstünlüklər verir.

Açar sözlər: Neyroplastiklik, kritik dövr, dil mənimsənilməsi, beyin inkişafı, ikidillilik, erkən öyrənmə, dilə yiyələnmə.

Introduction

The science of early language learning is an important topic in linguistics, psychology, neuroscience, and education because it explains how children learn foreign languages and why early childhood is considered the most effective period for language learning in human development. Many researchers agree that children learn languages faster, more naturally, and with less effort compared to adults, and this difference is not only related to teaching methods, classroom environment, or learning materials, but also to biological development in the human brain and cognitive system. In early childhood, the brain is still developing and is highly flexible, and this flexibility helps children absorb new sounds, vocabulary, grammar structures, and pronunciation patterns more easily and naturally. Children usually learn languages through listening, speaking, imitation, repetition, and interaction in daily life, and they do not depend heavily on grammar rules in the early stages, which makes the learning process more natural, more effective, and less stressful, while also improving communication skills, confidence, and social development in the long term. In addition, early language learning also supports general cognitive development, including attention, problem-solving, and memory skills, which are important for overall academic success.

Brain Development and Neuroplasticity

Brain development plays a very important role in early language learning, especially through a process called neuroplasticity which is the brain's ability to change, adapt, and form new neural connections based on experience and continuous learning. In early childhood this ability is very strong because the brain is still developing and is highly flexible which allows children to absorb new information such as sounds, words, grammar structures, and

pronunciation patterns more easily and naturally without needing much conscious effort or formal instruction. Children usually learn language through listening, repetition, imitation, and daily communication with people around them and this natural exposure helps them build strong language skills in a more automatic way. One important advantage of children's brain development is that they can quickly recognize and imitate new sounds because their auditory system is more sensitive than adults which helps them understand and reproduce pronunciation more accurately and naturally. For example children who grow up in bilingual environments can easily learn and switch between two languages without confusion because their brain adapts quickly to different language systems and contexts. As people grow older the brain becomes less flexible and more stable which reduces neuroplasticity and makes language learning slower and more effortful. Adults usually rely more on memorization grammar rules and translation while children rely more on natural exposure interaction and imitation. Because of this difference early childhood is considered the most effective period for language learning and provides a strong foundation for long-term language mastery and communication skills.

The Critical Period Hypothesis

The critical period hypothesis is an important theory in language learning which explains that there is a special time in human development when learning a language is easier, faster, and more natural. This period usually starts in early childhood and continues until puberty. During this time, the human brain is still developing and is more flexible, which makes it easier for children to learn new sounds, words, grammar structures, and pronunciation patterns without strong effort or formal instruction. Children are able to absorb language naturally from their environment through listening, speaking, repetition, and interaction with other people in daily life, and this natural exposure helps them build strong language skills without conscious analysis of rules. During the critical period, children can learn a second language in a way that is very similar to how they learn their first language because they do not need to focus deeply on grammar explanations or language rules, instead they learn through communication and continuous exposure to real language use. For example, children can easily learn correct pronunciation because their brain is more sensitive to sounds and they can imitate what they hear more accurately than older learners, and they also develop an intuitive understanding of grammar without formal study. However, after this critical period ends, usually after puberty, language learning becomes more difficult because the brain becomes less flexible and more stable, which makes it harder to achieve native-like pronunciation and automatic language use in communication. Older learners, especially adults, often need more time, effort, and structured study to learn a new language and they usually rely more on grammar rules, translation, and memorization instead of natural acquisition through exposure. Even though language learning is still possible after the critical period, the process is generally slower and requires more repetition and practice, and this theory clearly shows why early exposure to foreign languages is very important in education systems and why starting language learning at a young age can lead to better long-term results and higher fluency.

Cognitive Abilities and Learning Style

Children have strong cognitive abilities that support language learning in an effective and natural way, and one of the most important of these is memory capacity, because children's memory is highly active and flexible, which allows them to remember new words, expressions, and sentence patterns quickly after hearing them only a few times in daily communication. They

are also naturally curious and open to new experiences, which increases their motivation to learn and explore new languages in different contexts and situations. Another important factor is that children are not afraid of making mistakes, so they speak more freely, practice more often, and participate more actively in communication, which reduces stress and increases learning speed. Mistakes are seen as a natural part of learning, not a failure, and this attitude helps them improve faster. Unlike adults, children do not focus too much on grammar rules or theoretical explanations; instead, they learn language through repetition, context, visual support, and real communication, which makes the learning process more natural, more enjoyable, and more effective for long-term retention and understanding.

Early language learners develop strong listening, speaking, and pronunciation skills, and their listening ability is especially well-developed, which allows them to understand different accents, speech speeds, tones, and pronunciation variations more easily in real-life communication situations. Their pronunciation is often closer to native speakers because they can hear and imitate sounds accurately without strong interference from their first language. Speaking skills develop gradually through daily communication with teachers, friends, family members, and other social interactions, and repeated exposure helps them build fluency over time. Activities such as games, songs, storytelling, role-play, group discussions, and interactive exercises make the learning process more engaging, enjoyable, and meaningful. These activities help children connect language with real-life situations, which improves understanding, memory retention, and communication confidence. In addition, visual materials such as pictures, videos, and gestures also support comprehension and make learning easier. Over time, children become more fluent, more expressive, and more capable of using language naturally in different contexts and social environments.

Role of Environment and Motivation

The role of environment and motivation in early language learning is very important because both factors directly influence how quickly and how successfully children can acquire a new language in real life situations. The environment provides children with constant exposure to the target language through home, school, and media, and this regular exposure helps them become familiar with sounds, vocabulary, grammar structures, and pronunciation patterns in a natural and repeated way. When children hear the language in different contexts, they start to understand meaning without translation, which improves their listening and comprehension skills over time. Interaction is also a key part of the environment because communication with teachers, classmates, and native speakers gives children the opportunity to practice real language use, which improves fluency, pronunciation, and confidence. Classroom activities such as games, storytelling, songs, role-play, and group discussions create a supportive learning environment where children can use language actively and meaningfully instead of only memorizing rules. Modern technology also supports the environment because videos, cartoons, mobile applications, and educational games provide authentic language input and expose children to different accents and natural speech patterns, which strengthens their understanding and listening ability. Motivation is equally important because children learn better when they feel interested, happy, and emotionally supported, and when they are motivated they are more willing to speak, repeat new words, and participate in activities without fear of making mistakes. Positive motivation increases confidence and helps children stay consistent in their learning process, while lack of motivation or negative feedback can reduce interest and slow down progress. Therefore, both

environment and motivation work together to create a strong foundation for successful language learning, and when children are exposed to rich input and feel motivated, they develop stronger communication skills, better fluency, and long-term language competence more effectively.

Conclusion

In conclusion, early language learning is a complex process that depends on both biological and environmental factors. Brain development and neuroplasticity give children natural advantages in learning pronunciation, vocabulary, and communication skills. At the same time, the critical period hypothesis explains why early childhood is the most effective time for language learning. However, these biological advantages alone are not enough for successful language mastery.

Environmental exposure, motivation, and social interaction are equally important. A rich and supportive environment helps children practice the language more often and understand it in real-life situations. Motivation increases confidence and encourages active participation in learning. When all these factors work together, children can achieve strong, long-lasting language skills and better cognitive development. Therefore, successful early language learning is the result of a balance between biological readiness, emotional support, and a positive learning environment.

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MULTILINGUALISM AND SECOND LANGUAGE ACQUISITION (SLA)

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Abstract: This article examines the relationship between multilingualism and Second Language Acquisition (SLA) in the modern globalized world. It explains that language functions not only as a communication tool but also as a means of cultural preservation, identity formation, and social participation. Due to globalization, migration, technological development, and international interaction, multilingual competence has become increasingly important in education, employment, and everyday communication. The study defines multilingualism and discusses its various forms, including individual, societal, simultaneous, sequential, additive, and subtractive multilingualism. It also explores the historical development of multilingual communication and the emergence of SLA as an interdisciplinary field. Furthermore, the article analyzes major theoretical approaches to SLA, including behaviorist, cognitive, interactionist, and sociocultural perspectives. Special attention is given to the cognitive, social, cultural, and educational dimensions of multilingualism. Finally, the article highlights challenges such as language interference, inequality, and educational barriers while emphasizing the growing importance of inclusive multilingual education in contemporary society.

Keywords: multilingualism, second language acquisition, language learning, cognitive development, intercultural communication, multilingual education, sociocultural theory

МНОГОЯЗЫЧИЕ И ОВЛАДЕНИЕ ВТОРЫМ ЯЗЫКОМ (SLA)

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Резюме: В данной статье рассматривается взаимосвязь между многоязычием и овладением вторым языком (Second Language Acquisition - SLA) в условиях современного глобализованного мира. Подчеркивается, что язык выполняет не только коммуникативную функцию, но также служит средством сохранения культуры, формирования идентичности и социального участия. Вследствие глобализации, миграции, технологического развития и международного взаимодействия многоязычная компетенция приобретает всё большее значение в образовании, профессиональной деятельности и повседневном общении. В исследовании даётся определение многоязычия и рассматриваются его различные формы, включая индивидуальное, общественное, одновременное, последовательное, аддитивное и субтрактивное многоязычие. Кроме того, анализируются историческое развитие многоязычной коммуникации и становление SLA как междисциплинарной области исследования. Особое внимание уделяется когнитивным, социальным, культурным и образовательным аспектам многоязычия. В заключение в статье рассматриваются такие проблемы, как языковая интерференция, социальное неравенство и образовательные барьеры, при этом подчеркивается возрастающая значимость инклюзивного многоязычного образования в современном обществе.

Ключевые слова: многоязычие, овладение вторым языком, изучение языка, когнитивное развитие, межкультурная коммуникация, многоязычное образование, социокультурная теория

Introduction

Language is one of the most significant elements of human civilization. It functions not only as a tool for communication but also as a means through which individuals express identity, preserve culture, participate in society, and construct knowledge. Throughout history, multilingualism has existed in various forms across communities and nations. However, in the

contemporary world, the importance of multilingual competence has increased considerably due to globalization, technological advancement, international migration, economic interdependence, and the expansion of cross-cultural interaction.

The modern social environment increasingly requires individuals to communicate across linguistic and cultural boundaries. International education, global employment markets, digital communication platforms, tourism, diplomacy, and transnational business have created circumstances in which multilingual communication is often necessary rather than optional.

Consequently, multilingualism has become a defining characteristic of many societies and educational systems worldwide.

The growth of multilingual populations has also intensified academic interest in understanding how additional languages are learned. This area of inquiry is commonly referred to as Second Language Acquisition (SLA). SLA investigates the processes through which individuals acquire languages beyond their first language, examining cognitive development, social interaction, linguistic competence, motivation, identity formation, and educational practices. Researchers in SLA seek to understand why language acquisition differs among learners and how internal and external factors influence language development.

The relationship between multilingualism and second language acquisition is particularly significant because multilingual individuals frequently demonstrate distinct linguistic awareness, communication strategies, and cognitive patterns. The experience of using multiple languages often influences the acquisition of additional languages, while language learning itself contributes to broader social and intellectual development. As multilingualism becomes increasingly common in educational institutions and workplaces, understanding the principles of SLA is essential for teachers, policymakers, researchers, and learners alike.

This article examines multilingualism and second language acquisition in depth by exploring definitions, classifications, historical perspectives, theoretical frameworks, cognitive and social dimensions, educational implications, challenges, and contemporary developments in research. Through this analysis, it becomes evident that multilingualism is not simply a linguistic phenomenon but also a social, cultural, psychological, and educational reality that continues to shape human interaction in profound ways.

The Concept of Multilingualism

Multilingualism refers to the ability to use more than one language for communication. The term may apply to individuals, communities, institutions, or entire societies where multiple languages coexist and interact within social life. A multilingual person may speak, understand, read, or write in several languages with varying levels of proficiency depending on exposure, context, and purpose.

The concept of multilingualism is more complex than simple fluency in several languages. Language competence often varies according to situation and function. An individual may use one language for professional communication, another within the family, and a third for academic or religious purposes. Therefore, multilingual ability is not necessarily balanced or identical across all languages.

In many parts of the world, multilingualism represents the norm rather than the exception. Large populations routinely navigate between languages in education, employment, government institutions, media, and interpersonal relationships. Countries such as India, Switzerland, Singapore, South Africa, and Canada illustrate the diverse forms multilingualism can take within

national contexts.

For example, in India, individuals may use regional languages within local communities, Hindi in broader national communication, and English in higher education or professional settings.

Similarly, in many Central Asian societies, speakers may alternate between local languages and Russian depending on social context and communicative needs.

Multilingualism can emerge naturally through social interaction or through deliberate educational efforts. Some children acquire multiple languages simultaneously within multilingual households, while others learn additional languages later through schooling, migration, or professional necessity. In both cases, multilingualism develops through continuous exposure and meaningful communication.

Historical Perspectives on Multilingualism

Historically, multilingualism has been closely linked to trade, migration, empire, religion, and cultural exchange. Ancient civilizations frequently relied on multilingual communication to maintain political administration and economic relationships across diverse populations.

In the Roman Empire, for example, Latin functioned alongside Greek and numerous local languages throughout different territories. Similarly, trade routes connecting Asia, Africa, Europe, and the Middle East facilitated multilingual interaction among merchants, travelers, and scholars.

During the colonial period, multilingualism became associated with political domination, educational systems, and language hierarchies. Colonial languages such as English, French, Spanish, and Portuguese spread globally and continue to influence educational and governmental institutions today.

In contemporary society, globalization has accelerated multilingual interaction through mass communication, international mobility, and digital technology. Unlike earlier historical periods, modern multilingualism often involves rapid language contact across virtual environments as well as physical communities.

Forms and Types of Multilingualism

Multilingualism exists in diverse forms depending on social conditions, learning environments, and patterns of language use.

Individual Multilingualism. Individual multilingualism refers to a person's ability to communicate in multiple languages. This ability may develop naturally or through formal instruction.

For example, an individual living in Kyrgyzstan may speak Kyrgyz within the family, use Russian in professional settings, and study English for international communication or academic advancement.

Societal Multilingualism. Societal multilingualism occurs when several languages are used within a particular community or nation. Different languages may serve specific political, educational, economic, or cultural functions.

In multilingual societies, language policies often influence social integration, educational access, and national identity. Governments may officially recognize multiple languages or prioritize certain languages for administrative purposes.

Simultaneous Multilingualism. Simultaneous multilingualism develops when children acquire two or more languages from birth or early childhood. Children exposed consistently to

multiple languages during early development often learn them naturally through interaction rather than formal instruction.

Sequential Multilingualism. Sequential multilingualism occurs when additional languages are learned after the first language has already been established. This pattern commonly appears in school-based language learning and migration contexts.

Additive Multilingualism. Additive multilingualism refers to learning a new language while maintaining and valuing the first language. Educational systems that support heritage language preservation encourage additive multilingual development.

Subtractive Multilingualism. Subtractive multilingualism occurs when the acquisition of a dominant language gradually weakens the use or status of the native language. This process may contribute to language shift and cultural assimilation.

Understanding Second Language Acquisition

Second Language Acquisition is an interdisciplinary field that investigates how individuals learn languages beyond their native language. SLA draws upon linguistics, psychology, sociology, neuroscience, education, and cognitive science to explain language-learning processes and outcomes.

The term “second language” broadly refers to any language learned after the first language, regardless of the number of additional languages acquired. Thus, SLA research includes the acquisition of third, fourth, or subsequent languages as well.

Second language acquisition differs from first language acquisition in several important ways. Children typically acquire their first language naturally through immersion and interaction without formal instruction. Additional languages, however, are often learned in educational settings where learners possess established linguistic systems and cognitive maturity.

SLA examines multiple aspects of language development, including:

- Vocabulary acquisition
- Grammatical development
- Pronunciation and phonology
- Listening comprehension
- Reading and writing skills
- Pragmatic competence
- Sociolinguistic awareness
- Communicative fluency

Researchers also investigate the psychological and environmental factors influencing acquisition, such as motivation, identity, memory, exposure, social interaction, and emotional experience.

Major Theoretical Approaches to SLA

The study of second language acquisition has evolved through several theoretical perspectives that attempt to explain how language learning occurs.

Behaviorist Approaches. Behaviorist theories viewed language learning primarily as habit formation. According to this perspective, learners acquire language through imitation, repetition, memorization, and reinforcement.

Errors were considered undesirable because they reflected incorrect habits that required correction. Behaviorist theories strongly influenced early language-teaching methods focused on drills and pattern practice.

Although behaviorism contributed valuable insights into repetition and practice, critics argued that it could not adequately explain creativity, spontaneous language use, or complex grammatical development.

Innatist Perspectives. Innatist theories proposed that humans possess an inherent biological capacity for language learning. Language acquisition was viewed as a natural cognitive process supported by universal mental structures.

This perspective emphasized the learner's internal ability to identify linguistic patterns rather than relying solely on environmental input.

In SLA research, innatist theories contributed to discussions regarding age-related learning differences, grammatical development, and the existence of universal principles underlying human language.

Cognitive Approaches. Cognitive theories emphasize mental processing, memory, attention, and information management during language learning.

From this perspective, learners actively organize linguistic input, develop mental representations, and construct knowledge through practice and interaction.

Cognitive approaches also highlight the importance of automatization, whereby repeated exposure and use gradually make language processing faster and more efficient.

Interactionist Perspectives. Interactionist theories stress the importance of communication and social interaction in language development. Learners acquire language more effectively when engaged in meaningful conversation requiring negotiation of meaning, clarification, and feedback.

Classroom interaction, collaborative learning, and communication with proficient speakers are therefore considered essential components of successful acquisition.

Sociocultural Theory. Sociocultural theory views language learning as a socially mediated process influenced by culture, identity, and interpersonal relationships.

According to this perspective, learning occurs through participation in social activities and collaboration with more knowledgeable individuals such as teachers, peers, and family members.

Sociocultural approaches emphasize that language learning cannot be separated from social context, emotional experience, or cultural participation.

Cognitive Dimensions of Multilingualism. One of the most extensively researched aspects of multilingualism concerns its cognitive effects. The continuous management of multiple linguistic systems appears to influence mental flexibility, attention, memory, and problem-solving abilities.

Executive Function and Cognitive Control. Multilingual individuals regularly switch between languages depending on context, audience, or communicative goals. This process requires monitoring, inhibition, and selective attention.

Research has suggested that such linguistic management may strengthen executive control functions responsible for planning, attention regulation, and task switching.

Metalinguistic Awareness. Multilingual speakers often demonstrate increased awareness of language structure and function. They become more capable of recognizing grammatical patterns, analyzing linguistic differences, and understanding how communication operates across contexts.

This heightened awareness frequently supports the acquisition of additional languages.

Memory and Information Processing. Managing multiple languages requires active storage

and retrieval of vocabulary, grammar, and phonological systems. Such mental activity may contribute to memory development and information-processing efficiency.

Cognitive Resilience. Some research indicates that lifelong multilingualism may contribute to cognitive resilience during aging by maintaining mental activity and neural engagement over extended periods.

Social and Cultural Dimensions of Multilingualism. Language is deeply connected to identity, belonging, and cultural participation. Consequently, multilingualism carries important social and emotional implications beyond communication alone.

Cultural Identity. Languages often represent cultural heritage, family history, and collective identity. Multilingual individuals may navigate multiple cultural affiliations simultaneously, developing complex and dynamic identities.

Heritage language maintenance frequently strengthens intergenerational communication and cultural continuity.

Intercultural Communication. Multilingualism promotes communication across cultural boundaries and encourages understanding of diverse perspectives. Exposure to multiple languages often increases cultural sensitivity and empathy.

Economic and Professional Opportunities. In the global economy, multilingual competence is increasingly valued across industries such as business, diplomacy, tourism, healthcare, education, and international relations.

Employers frequently seek multilingual professionals capable of communicating with diverse populations and operating within international environments.

Educational Implications of Multilingualism and SLA. Educational institutions play a critical role in supporting multilingual learners and shaping language-learning experiences.

Traditional language instruction often emphasized grammar memorization and translation exercises. However, modern educational approaches increasingly prioritize communicative competence, critical thinking, collaboration, and authentic interaction.

Effective multilingual education generally includes:

- Meaningful communication opportunities
- Inclusive classroom practices
- Support for heritage languages
- Culturally responsive teaching
- Development of learner confidence
- Integration of technology and authentic materials

Teachers must also recognize that multilingual learners possess valuable linguistic knowledge and cultural experiences that can enhance classroom learning.

Challenges in Multilingualism and SLA. Despite numerous advantages, multilingualism and second language acquisition also involve challenges.

Language Interference. Learners may transfer vocabulary, pronunciation, or grammatical structures from one language into another. Such transfer can facilitate learning in some cases but may also produce persistent errors.

Social Inequality. Languages often possess unequal social status. Minority-language speakers may encounter discrimination, reduced educational access, or pressure to abandon their native languages.

Educational Barriers. Some educational systems lack sufficient resources, trained teachers,

or institutional support for multilingual instruction.

Psychological Challenges. Language learning may involve anxiety, fear of mistakes, identity conflicts, or reduced confidence, particularly in environments where linguistic diversity is not fully valued.

Technology and Contemporary Research. Technological development has transformed both multilingual communication and second language acquisition.

Digital platforms, virtual classrooms, social media, online courses, and language-learning applications provide unprecedented access to authentic language input and international interaction.

Contemporary research increasingly explores areas such as:

- Multilingual identity construction
- Translanguaging practices
- Neurolinguistics and brain development
- Artificial intelligence in language education
- Digital communication environments
- Emotional dimensions of language learning
- Language policy and social justice

Researchers now emphasize that multilingual speakers do not separate languages completely in the mind. Instead, languages interact dynamically and influence one another continuously during communication and learning.

Conclusion

Multilingualism and Second Language Acquisition represent interconnected fields of growing importance within modern society. As globalization continues to expand international interaction and cultural exchange, multilingual competence has become increasingly valuable in educational, professional, and social contexts.

Multilingualism extends beyond the ability to speak several languages. It influences cognitive development, cultural identity, communication patterns, social participation, and educational experiences. Similarly, second language acquisition involves far more than memorizing vocabulary or grammar; it is a complex developmental process shaped by psychological, social, linguistic, and environmental factors.

Research in multilingualism and SLA continues to demonstrate that language learning is deeply connected to human cognition, identity, and social interaction. Although multilingual learners may encounter challenges related to inequality, language interference, or educational limitations, the benefits of multilingual competence remain substantial.

Educational institutions, policymakers, and societies therefore bear significant responsibility in creating inclusive environments that support linguistic diversity and equitable language-learning opportunities. In a world characterized by increasing mobility and intercultural communication, multilingualism is not merely an academic subject but a central feature of contemporary human experience.

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LANGUAGE PRESERVATION IN THE AGE OF GLOBALIZATION

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Abstract: Globalization, as the process of increasing intercultural interactions, has a significant impact on national languages and cultures. Culture is a fundamental component of a nation's national wealth, reflecting both its historical heritage and its modern identity. It plays an important role in shaping and strengthening relations between peoples and states. In the modern era, the preservation of national cultures is of particular importance, and a number of measures are implemented to protect cultural heritage. Cultural identity is a complex, intergenerational concept that plays an important role in the development of self-understanding, as well as in ensuring social integration and sustainability.

Language is a fundamental cultural phenomenon that reflects the people's historical memory, worldview, national and spiritual values. It is one of the important factors characterizing the identity of a nation and national treasure. Language serves as a means of expressing culture and a symbol of national authority. It is essential for the existence of culture and for the expression and formation of civilization. The protection of native languages is vital for the maintenance of cultural diversity. The preservation of native languages in the era of globalization is an important cultural, social, and linguistic problem. Preserving a language is essential, as it ensures the continuity of national linguistic identity and cultural heritage.

Key words: globalization, language, *culture*, cultural heritage, integration, modern world, local traditions, preservation of languages

QLOBALLAŞMA DÖVRÜNDƏ DİLLƏRİN QORUNMASI

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Xülasə: Qloballaşma, mədəniyyətlərarası əlaqələrin artması prosesi kimi, milli dillərə və mədəniyyətlərə əhəmiyyətli dərəcədə təsir göstərir. Mədəniyyət bir xalqın milli sərvətinin əsas tərkib hissəsidir və onun həm tarixi irsini, həm də müasir kimliyini əks etdirir. O, xalqlar və dövlətlər arasında münasibətlərin formalaşmasında və möhkəmlənməsində mühüm rol oynayır. Müasir dövrdə milli mədəniyyətlərin qorunması xüsusi əhəmiyyət kəsb edir və mədəni irsin mühafizəsi üçün bir sıra tədbirlər həyata keçirilir. Mədəni kimlik nəsildən-nəslə ötürülən mürəkkəb bir anlayışdır və özünü dərk etmənin inkişafında, həmçinin sosial integrasiya və sabitliyin təmin edilməsində mühüm rol oynayır.

Dil xalqın tarixi yaddaşını, dünyagörüşünü, milli və mənəvi dəyərlərini əks etdirən fundamental mədəni hadisədir. O, bir millətin kimliyini və milli sərvətini xarakterizə edən mühüm amillərdən biridir. Dil mədəniyyəti ifadə etmək vasitəsi və milli hakimiyyətin simvolu kimi çıxış edir. O, mədəniyyətin mövcudluğu, sivilizasiyanın ifadəsi və formalaşması üçün vacibdir. Ana dillərinin qorunması mədəni müxtəlifliyin saxlanılması üçün həyati əhəmiyyət daşıyır. Qloballaşma dövründə ana dillərin qorunması mühüm mədəni, sosial və dilçilik problemi kimi çıxış edir. Dilin qorunması milli dil kimliyinin və mədəni irsin davamlılığını təmin etdiyi üçün zəruridir.

Açar sözlər: qloballaşma, dil, mədəniyyət, mədəni irs, integrasiya, müasir dünya, yerli ənənələr, dillərin qorunması

Introduction

Globalization is a universal process that encompasses various spheres of life. National and regional characteristics, which have been separated for thousands of years, are now rapidly converging under the influence of modern technologies, leading to the weakening of local distinctiveness and shaping global processes. The process of globalization strengthens cultural, economic, and technological ties between countries and populations. It has become a significant factor in cultural change across all countries in the modern world. As a result, the problem of preserving the integrity of national languages becomes even more urgent, as globalization increases the interaction between different cultures and societies. Globalization is a social process characterized by the existence of global economic, political, cultural, linguistic and environmental interconnections and flows that make the many of the currently existing borders and boundaries irrelevant. (Steger, 2009, p.180).

It includes the expansion of international trade and the integration of economies, societies and cultures on a global scale. Globalization, as the process of increasing intercultural interactions, has a significant impact on national languages and cultures. Globalization is defined by both beneficial and adverse outcomes and thus has both positive and negative impacts on the preservation of culture. It enables the global sharing of customs, beliefs, and knowledge, fostering appreciation for cultural variety and a deeper comprehension of different cultures through the media. It promotes economic growth, facilitates cultural exchange, and enhances access to information; however, it may also contribute to the loss of national identity, the decline of local languages, and the homogenization of cultures, while simultaneously presenting significant challenges in achieving a balance between global integration and the preservation of

local traditions and cultural heritage. It should be noted that the globalization of the world relations is a developing process not depending from our will. To raise objections against it, we may say, that it is impossible. But a process of organization of the union economic culture by its own nature menaces to lose partially or wholly the cultural peculiarities in fact for some nations. (Valiyeva, 2016, p. 29-30)

Culture is a fundamental component of a nation's national wealth, reflecting both its historical heritage and its modern identity. It plays an important role in shaping and strengthening relations between peoples and states. In the modern era, the preservation of national cultures is of particular importance, and a number of measures are implemented to protect cultural heritage. Our culture plays a significant role in preserving our national identity, ensuring the preservation, development and enrichment of cultural identity. Cultural identity is a complex, intergenerational concept that plays an important role in the development of self-understanding, as well as in ensuring social integration and sustainability.

The preservation and development of national culture require the implementation of effective state policies. Such policies support the safeguarding of cultural traditions and ensure their promotion. The adaptation of cultural traditions and national values to contemporary conditions plays an important role in the preservation of cultural heritage. The preservation of the native language is one of the key directions of state policy in the era of globalization, as it plays a fundamental role in the protection of national identity and cultural heritage. Therefore, it is necessary to raise languages to a level that enables them to participate in the information revolution.

Methodology and Materials

Language is a fundamental cultural phenomenon that reflects the people's historical memory, worldview, national and spiritual values. It is one of the important factors characterizing the identity of a nation and national treasure. Human society is inconceivable without language. Language defines our lives, it heralds our existence, it formulates our thoughts, it enables all we are and have. As the river of history itself, language is in constant flux, ever changing, ever mutating, replacing, dying, rejuvenating, growing. (Fischer, 1999, p.219-220)

Raising a national language to the status of a state language is important in the history of national statehood. Language is one of the most powerful tools for the preservation, development and transmission of any nation's intangible heritage. It is an important means of transmitting and preserving the cultural heritage of every country. Language serves as a means of expressing culture and a symbol of national authority. On the international stage, the imposition of dominant societies' languages on weaker societies and the neglect of local languages are regarded as one of the attempts aimed at weakening their resistance. Language is essential for the existence of culture and for the expression and formation of civilization (Siregar, 2021).

The national language is valuable as it expresses a clear identity and a strong sense of national belonging. In spite of globalization advancing the English language, the native language can also be developed for scientific and technical purposes. The protection of native languages is vital for the maintenance of cultural diversity. The preservation of native languages in the era of globalization is an important cultural, social, and linguistic problem. It provides opportunities for international communication and also poses a threat to less widely spoken languages, increasing the risk of language decline and loss. This process raises significant concerns regarding the

preservation of the integrity and originality of the language. Preserving a language is essential, as it ensures the continuity of national linguistic identity and cultural heritage. Languages embody traditions, historical experience, values, and modes of thought that cannot be fully conveyed through other linguistic systems. Languages encompass traditions, historical experience, values, and modes of thought. These cannot be fully expressed through other linguistic systems. When a language is lost, an important part of cultural richness is also lost

One of the fundamental indicators of the existence of every nation is the mother tongue. It is a fundamental pillar of national identity, pride, and the primary mechanism for preserving and transmitting cultural, historical, and spiritual values across generations. The mother tongue is a carrier of values and knowledge and is very often used in the practice and transmission of intangible cultural heritage. In the era of globalization, the protection of the mother tongue is of strategic importance for preserving culture. Ensuring the purity of the language in the digital environment, developing terminology, and strengthening love for the mother tongue among young people are main goals.

Discussion

Azerbaijan, with its ancient history and rich culture, has made invaluable contributions to the development of humanity, and our mother tongue. Like all other national elements, it is a source of pride for our people and a symbol of our spiritual identity. As in all countries of the world, significant work is being carried out in our country to develop the Azerbaijani language and to protect our sacred heritage from external influences. Systematic measures are being implemented to promote the development of the Azerbaijani language and protect our cultural heritage. These are regarded by the state leadership as essential for national security, ideological stability, and the preservation of national identity.

The National Leader Heydar Aliyev made significant contributions to the development of the Azerbaijani language and to its recognition as a state language. Heydar Aliyev dedicated his entire life to the protection of the Azerbaijani language and transmission it down to the future generations in its purest form. His initiatives have laid a strong foundation for the continued development of the Azerbaijani language, helping preserve it for future generations.

Every nation is born through its language. A person who does not love their language cannot know their history. (7). In the modern world, in today's rapidly changing world language and culture serve as the instruments that strengthen the continuity of society's existence. For this reason, our people must protect the Azerbaijani language, which is an inseparable part of our Constitution and a symbol of our sovereignty, unity, and solidarity. Language is a fundamental component of cultural identity, shaping how individuals and communities express their heritage, values, and traditions. As a fundamental component of cultural identity, language serves as a bridge between generations, influences national identity, and ensures the continuity of linguistic diversity to preserve cultural identity in an increasingly integrated and globalized world. The Azerbaijani language is one of the main factors that defines the national identity of the Azerbaijani people. (Əliyev, 2003, p.10).

Every nation lives through its language. If a nation's language disappears, that nation also disappears from history. That is why, the preservation and protection of a language from foreign, unnecessary influences must always be kept in focus. Today, the Azerbaijani language continues its development with confidence, in parallel with the sustained consolidation and advancement of the Azerbaijani state. President Ilham Aliyev has repeatedly emphasized that the responsibility of

safeguarding the language is inherently continuous. The independent Azerbaijani state serves as the preserver of the purity and literary standard of the Azerbaijani language. The preservation of the purity of the Azerbaijani language should be the duty of every Azerbaijani citizen. In other words, the government, scientists, linguists, writers, poets, journalists and those involved in politics should pay great attention to this.

The Azerbaijani state under the leadership of President Ilham Aliyev continues to strengthen not only its political and economic sovereignty, but also, its linguistic sovereignty, which forms an essential component of national identity and cultural continuity. In this context, schools, universities, and media institutions play a vital role in preserving and promoting the Azerbaijani language. They play an important role in ensuring that the younger generation acquires proficiency in standard Azerbaijani, both in spoken and written forms, thereby contributing to the preservation of linguistic purity and the strengthening of national cultural values.

Currently, the Azerbaijani language has a unique place among the world's languages. Our mother tongue is characterized by the richness of its vocabulary, the harmony of its sound system, and the stability of its grammatical structure. President Aliyev particularly called on Azerbaijani scientists to support the preservation of the purity of the Azerbaijani language, to express their position, and to safeguard national identity against foreign linguistic influences in media and public discourse. It should be generally acknowledged that when a language is not protected and its purity is not ensured, it may ultimately face the risk of being lost.

Globalization sometimes encourages individuals, particularly younger generations, to place greater emphasis on global languages in order to enhance their career opportunities. As a result, they do not use their native language often, which leads to the weakening of the language and, in some cases, its disappearance. This encourages people to adopt a more sensitive, conscious, and responsible attitude toward their mother tongue.

Conclusion

The Azerbaijani language has undergone significant development, and it has been preserved throughout all stages of our national history. The preservation of cultural diversity is essential for maintaining human identity, knowledge and mutual understanding. The preservation of native languages requires the joint and sustained efforts of individuals, local communities, and state institutions. Although globalization promotes intercultural communication and the exchange of ideas, it may also lead to the weakening of local traditions and linguistic heritage. Therefore, it is necessary to develop effective policies that both protect cultural heritage and ensure constructive participation in global integration. The preservation of cultural diversity is a fundamental condition for maintaining human identity, intellectual heritage, and mutual understanding. This emphasizes the importance of fostering a more sensitive and responsible attitude toward one's mother tongue.

The Azerbaijani language has undergone significant development and we have preserved our language throughout all stages of our history. The preservation of cultural diversity is essential for maintaining human identity, knowledge, and mutual understanding.

While globalization offers significant benefits, it should not lead to the weakening of linguistic and cultural diversity. The preservation of native languages requires joint efforts from individuals, communities, and governments. While globalization facilitates intercultural exchange, it can also pose a threat to local traditions and languages.

Therefore, a balanced approach is necessary, combining effective policy-making, education, and community engagement to protect cultural heritage while embracing global integration. Ultimately, preserving cultural diversity is essential for maintaining human identity, knowledge, and mutual understanding.

Digital platforms and tools allow communities to spread their languages. Social media platforms serve as spaces for linguistic communication. People who speak different languages interact and promote their languages on a global scale. The role of documentation in preserving endangered languages is significant. Linguists and anthropologists collaborate with native speakers to document languages through audio files, written texts, and digital media.

Effective language protection requires collective efforts of governments, civil society groups, and indigenous peoples to raise public awareness and develop sustainable solutions.

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CHARACTERISTICS OF COGNITIVE DEVELOPMENT IN CHILDREN WITH AUTISM SPECTRUM DISORDER

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Abstract: This article examines the characteristics of cognitive development in children with autism spectrum disorder (ASD). The study analyzes the main areas of psychological development, including behavioral, emotional-volitional, cognitive, and activity-related spheres. Particular attention is given to the development of cognitive processes such as attention, memory, perception, and thinking. It is emphasized that these processes develop unevenly and demonstrate individual characteristics in children with autism. Behavioral manifestations may include stereotypical behaviors, anxiety, fears, negativism, aggression, and self-injurious behavior. The article also discusses the clinical and psychological classification of early childhood autism, according to which children are divided into four groups with distinct patterns of interaction with the environment and developmental characteristics. The findings indicate that cognitive development in children with ASD is closely associated with social communication and emotional interaction. At the same time, some children may demonstrate relatively high potential in particular cognitive domains, highlighting the heterogeneous nature of autism. Therefore, educational and psychological interventions should be based on an individualized approach, a stable and structured environment, consideration of children's interests and needs, and systematic professional support. Such an approach can contribute to the development of children's cognitive potential, improve social adaptation, and enhance their overall quality of life.

Keywords: autism spectrum disorder, cognitive development, cognitive processes, attention, memory, thinking, emotional-volitional sphere, behavioral characteristics, social communication, special education

AUTİZM SPEKTR POZUNTUSU OLAN UŞAQLARDA KOGNİTİV İNKİŞAFIN XÜSUSİYYƏTLƏRİ

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Xülasə: Bu məqalədə autizm spektr pozuntusu (ASP) olan uşaqlarda koqnitiv inkişafın xüsusiyyətləri araşdırılır. Tədqiqatda autizmlı uşaqların psixoloji inkişafının əsas sahələri, o cümlədən davranış, emosional-iradi, koqnitiv və fəaliyyət sferaları təhlil edilir. Xüsusi diqqət diqqət, yaddaş, qavrayış və təfəkkür kimi koqnitiv proseslərin formalaşmasına yönəldilir. Qeyd olunur ki, autizmlı uşaqlarda bu proseslər qeyri-bərabər və fərdi xüsusiyyətlərə uyğun şəkildə inkişaf edir. Davranış sahəsində stereotipik hərəkətlər, narahatlıq, qorxu, neqativizm, aqressiv və özünə zərərverici davranışlar müşahidə oluna bilər. Məqalədə erkən uşaqlıq autizminin klinik-psixoloji təsnifatı əsasında uşaqların dörd qrupa bölünməsi və hər bir qrupun özünəməxsus inkişaf xüsusiyyətləri izah edilir. Tədqiqat göstərir ki, koqnitiv inkişaf sosial ünsiyyət və emosional qarşılıqlı əlaqə ilə sıx bağlıdır. Eyni zamanda, uşaqların bəzi koqnitiv sahələrdə yüksək potensiala malik ola biləcəyi vurğulanır. Nəticə etibarilə, autizmlı uşaqlarla pedaqoji və psixoloji iş fərdi yanaşma, strukturlaşdırılmış və sabit mühit, uşağın maraq və ehtiyaclarının nəzərə alınması və sistemli dəstək əsasında təşkil edilməlidir. Bu yanaşma uşaqların koqnitiv potensialının reallaşdırılmasına, sosial adaptasiyasına və həyat keyfiyyətinin yüksəldilməsinə şərait yaradır.

Açar sözlər: autizm spektr pozuntusu, idrak inkişafı, idrak prosesləri, diqqət, yaddaş, təfəkkür, emosional-iradi sfera, davranış xüsusiyyətləri, sosial ünsiyyət, xüsusi təhsil

Introduction

Autism spectrum disorder is a term that refers to a neurological condition manifested in childhood or early development and continuing throughout life, characterized by a wide range of features, behaviors, and impairments (interpersonal, social, language, and communication). Autism is not a disease but a neurological condition; therefore, the main focus in intervention is directed toward understanding and accepting the condition, developing life skills, and providing individualized adaptation and professional support according to the individual's needs and capabilities. Symptoms vary greatly depending on the type and severity of ASD and usually appear before the age of two. Typically, individuals with autism require a strict daily routine. Even minor changes can cause anxiety or anger.

ASD is observed approximately four times more frequently in boys than in girls. However, doctors and researchers believe that this may underestimate the actual situation. In reality, the ratio may be almost equal. This is due to cultural beliefs that girls are quiet and shy. Therefore, others may mistakenly interpret autism characteristics as natural female shyness.

Materials and methods

The study was conducted using a qualitative and descriptive research approach. Scientific and pedagogical literature related to autism spectrum disorder, cognitive development, behavior, emotional development, and social communication was analyzed. The main research methods included theoretical analysis, comparison, generalization, and descriptive interpretation. Particular attention was given to attention, perception, memory, and thinking, as well as the four groups identified in the clinical and psychological classification of early childhood autism. The analyzed information was used to identify the main characteristics of cognitive development and appropriate pedagogical and psychological approaches for children with autism spectrum disorder.

Discussion

Psychological characteristics typical of children with autism manifest in the following areas:

1. Behavior
2. Emotional-volitional sphere
3. Cognitive sphere
4. Activity

Numerous affective disorders lead to serious behavioral problems. The behavior of children with autism is quite unique. They are characterized by anxiety, stereotypical behavior, fear, and withdrawal. These conditions may be accompanied by increased agitation, impulsivity, aggression, and self-injurious behavior, as well as negative reactions to any changes in daily life, including food, clothing, and walking routes. However, the child does not complain or seek help from relatives; instead, they resort to self-stimulation to relieve anxiety and withdraw even further from any contact. Children with autism experience fear of any changes in their usual lifestyle. They may be incomprehensible to others. When they feel fear, they are often unable to explain what exactly frightens them. Such children may be afraid of objects that produce loud sounds; they may also have specific fears related to tactile sensitivity.

An important behavioral characteristic is the child's active negativism, which is expressed in refusal to perform any joint activity with adults, avoidance of learning situations, and difficulties in voluntary organization. Manifestations of negativism may be accompanied by

physical resistance, shouting, aggression, and self-aggression. When adults place excessive demands on the child, fear of interaction arises, and existing forms of communication are disrupted. Self-stimulation is a strong defense against traumatic experiences. Desired experiences are most often achieved through stimulation of one's own body, helping to eliminate unpleasant impressions from the external world. In dangerous situations, the intensity of self-stimulation increases and may become so strong that it causes physical pain to the child.

The severity of behavioral problems varies depending on the severity of autism. This is most accurately reflected in the clinical and psychological classification of early childhood autism developed by O. S. Nikolskaya, E. R. Baenskaya, and M. M. Liebling, which identifies four groups of children. The main criteria for classification are the nature and degree of disturbances in interaction with the external environment and the type of autism itself.

1. Children in the first group demonstrate withdrawal from the external environment, children in the second group demonstrate rejection, children in the third group demonstrate substitution, and children in the fourth group demonstrate excessive inhibition by the environment. Studies show that children in these groups differ in the nature and degree of primary disorders, as well as secondary and tertiary dysontogenetic formations, including hypercompensatory disorders. Children in this group exhibit quite unusual behavior. They show no initiative in exploring the surrounding world and avoid contact with it. It is almost impossible to establish eye contact with such a child; their gaze constantly shifts away. The child does not choose the direction in which to move, but the spatial organization of objects forces them to move in a certain direction. In most cases, such a child's behavior is easily predictable and is determined not by the objects themselves and their properties, but by their relative position in space. Even if an object attracts the child's attention, a feeling of rapid satiation occurs, and therefore any manipulation, even short-term, is avoided. For children in the first group, maintaining a familiar routine is important, but they show the least resistance to change.

2. In children of the second group, behavioral problems are less pronounced and manifest not as withdrawal from the external world but as avoidance of interaction with it. A characteristic behavioral feature of children in this group is strict adherence to habitual life conditions, including the environment, habits, daily routine, and methods of interaction with close people. Children are particularly selective in clothing and food, attachment to certain objects and activities, numerous demands and prohibitions, as well as strictly defined walking routes. Fears are more developed in these children and later become firmly fixed. Intense fear may be triggered by even the slightest possibility of change in the established lifestyle or by unpleasant sensory experiences such as bright light or loud sounds. All this leads to extremely unusual behavior: the presence of numerous fears prevents the child from being active. When accidentally encountering another person's gaze, such a child may turn away, scream, or cover their face with their hands. They try to avoid contact with the external world, thereby escaping unpleasant sensations and associated anxiety. The child becomes trapped in a strictly followed life stereotype, which determines their behavior. These children perceive close relatives as a necessary condition of their existence. They force adults to act in strictly defined ways, never let them go, and constantly experience anxiety.

3. The most important behavioral characteristic of children in the third group is conflict. They are unable to compromise or take into account the interests or viewpoints of others, which makes communication with them quite difficult. Relatives note that the child tends to act "out of

spite.” They engage in the same activities – for a long time, the child may draw or enact the same story or talk about the same topic. The content of their interests and fantasies is often associated with frightening, unpleasant, or antisocial events. These children can set goals and develop certain programs to influence the environment and people, but the problem is that these programs do not take into account the constantly changing conditions of the external world. If the child is unable to influence others in a certain way, this leads to behavioral disturbances and pronounced negativism. Such children often look directly into the faces of those with whom they communicate, but in reality, their gaze seems to pass “through” them. The family environment provides a basis for maintaining stability, but relationships with close people are usually very difficult: the child constantly tries to dominate without considering the interests of others.

4. Children in the fourth group demonstrate noticeable rigidity and hesitation, which hinder their ability to establish contact with others. They can make eye contact, but their interactions are intermittent. These children are also sensitive to changes in their environment and lack behavioral flexibility, but they try to act as they have been taught by close adults. These children build all their relationships with the world through adults. Among all children with autism, only those in this group attempt to establish dialogue with the world and others, but they experience serious difficulties in doing so. Despite significant variability in mental and behavioral manifestations, children with autism spectrum disorder share common features characteristic of the emotional-volitional and cognitive spheres, interpersonal relationships, and personality as a whole.

Conclusion

Cognitive development in children with autism spectrum disorder is characterized by unevenness and specificity. In these children, perception, attention, memory, and thinking processes are formed differently and are often closely related to social communication and emotional development. Difficulties observed in cognitive functioning affect their ability to understand the surrounding environment, process information, and respond appropriately. At the same time, in some cases, a high level of development in certain areas may be observed, which once again confirms the heterogeneous nature of autism. Taking these features into account, pedagogical and psychological work with children with autism should be based on an individualized approach. The creation of a stable and structured environment, consideration of the child’s interests and needs, and the provision of systematic support measures play an important role in stimulating cognitive development. Such an approach creates conditions for revealing the child’s potential capabilities, improving social adaptation, and enhancing quality of life.

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CODE-SWITCHING AND TRANSLANGUAGING

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Abstract: Code-switching and translanguaging are two key concepts in the study of multilingual language use. Code-switching generally refers to the shift between two or more languages within a single conversation or sentence, whereas translanguaging focuses on the flexible and integrated use of a speaker's full linguistic resources, without strictly following conventional language boundaries.

This article is devoted to the theoretical problem of the relationship between two closely related, albeit fundamentally opposed, terms and concepts.

Proposed by scholars such as Ofelia García and Li Wei, the concept of translanguaging marks a paradigm shift that breaks down the boundaries between nominal languages and emphasizes the flexible use of all the speaker's semiotic resources. Unlike code-switching, which still presupposes distinct grammatical systems, translanguaging views linguistic practices as integrated and transformative. This distinction has important implications for bilingual education, language policy, and the recognition of linguistic equality in hyper-diverse societies.

Keywords: Code-switching, translanguaging, language, speaker, language policy

КОД-СВИТЧИНГ И ТРАНСЛАНГВИСТИКА

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Аннотация: Переключение кодов (code-switching) и транслингвизм (translanguaging) являются двумя ключевыми понятиями в изучении многоязычного использования языка. Переключение кодов обычно означает переход между двумя или более языками в рамках одного разговора или предложения, тогда как транслингвизм сосредоточен на гибком и интегрированном использовании всех языковых ресурсов говорящего без строгого соблюдения традиционных границ между языками.

Данная статья посвящена теоретической проблеме соотношения двух тесно связанных, хотя и принципиально противоположных, терминов и понятий.

Предложенная такими учёными, как Офелия Гарсия и Ли Вэй, концепция транслингвизма знаменует собой смену парадигмы, размывающую границы между номинальными языками и подчёркивающую гибкое использование всех семиотических ресурсов говорящего. В отличие от переключения кодов, которое всё ещё предполагает наличие отдельных грамматических систем, транслингвизм рассматривает языковые практики как интегрированные и трансформирующиеся.

Это различие имеет важные последствия для билингвального образования, языковой политики и признания языкового равенства в гиперразнообразных обществах.

Ключевые слова: переключение кодов, транслингвизм, язык, говорящий, языковая политика

Introduction

Code-switching is one of the most studied phenomena in modern sociolinguistics and psycholinguistics. This term refers to the process of alternating two or more linguistic codes (languages, dialects, or registers) within a single communicative act or discourse. The phenomenon is observed primarily in bilinguals and multilinguals and can occur at both the intersentential and intrasentential levels.

Unlike the traditional code-switching model, translanguaging (a concept developed by O. García, W. Li and others) offers a broader theoretical framework. According to this framework, the speech activity of a multilingual speaker does not presuppose the isolation of individual languages, but is based on a unified holistic repertoire. This repertoire encompasses a set of linguistic, cognitive, and semiotic resources that are utilized by the speaker as components of a single system without a rigid division into “first” or “second” language. Translingual is one of the most popular concepts in the field of multilingualism and language contacts today. When considering its principle of code-switching, it's important to keep in mind that the term translingual was not originally a conceptual concept used to describe specific linguistic practices in language revitalization programs. to this day, although the concept of translanguaging has long since transcended the boundaries of pedagogy and is used to characterize the complex discursive practices of bi- and multilinguals this practice-oriented component is also important for theoretical purposes.

The study of multilingual language practices has a long history in sociolinguistics. The concept of **code-switching** emerged in the mid-20th century. One of the earliest documented uses of the term appeared in 1953-1954 in the works of Roman Jakobson and Hans Vogt, in the context of reviewing Uriel Weinreich's seminal book *Languages in Contact* (1953). Initially, code-switching was often viewed negatively – as a sign of linguistic interference or deficiency in bilingual speakers.

This perception began to change in the 1960s and 1970s thanks to the groundbreaking work of John J. Gumperz and Jan-Petter Blom. Their famous 1972 study on situational and metaphorical code-switching in a Norwegian village demonstrated that switching between language varieties (standard Bokmål and local Ranamål) served important social and discourse functions, such as signaling changes in social roles, formality, or interpersonal relationships. Subsequent research by Shana Poplack (1980), Carol Myers-Scotton (1993), and others further established code-switching as a rule-governed, systematic, and creative communicative strategy rather than a random mixture or deficit. Over time, it became one of the most thoroughly studied phenomena in bilingualism, analyzed from structural, psycholinguistic, and sociolinguistic perspectives.

In contrast, the concept of **translanguaging** has a more recent origin. The term was first coined in the 1990s in the context of Welsh bilingual education by Cen Williams (*trawsieithu* in Welsh), who described a pedagogical practice in which students were encouraged to use both

Welsh and English fluidly within the same lesson, rather than keeping the languages strictly separate. Colin Baker later translated and popularized the term in English (Baker, 2001).

However, it was Ofelia García and Li Wei who significantly expanded and theorized translanguaging in the late 2000s and 2010s, transforming it from a mainly pedagogical strategy into a broader sociolinguistic and epistemological framework. In her influential book *Bilingual Education in the 21st Century* (García, 2009) and later in the co-authored volume *Translanguaging: Language, Bilingualism and Education* (García & Li Wei, 2014), García reconceptualized translanguaging as the deployment of a speaker's full linguistic repertoire as a single, integrated system. This perspective draws on postcolonial critique (notably the work of Sinfree Makoni and Alastair Pennycook), which views named languages as social and political inventions rather than natural, bounded entities. Translanguaging thus represents a paradigm shift: it moves away from the monoglossic view of languages as separate systems toward a more dynamic, fluid, and transformative understanding of multilingual practices.

Today, translanguaging encompasses virtually all forms of speech production and verbal communication that involve more than one language, including borrowings, language hybridization and pidginization, the use of international words, the activation of metalinguistic knowledge, and the use of non-verbal means (gestures, facial expressions, onomatopoeia, etc.).

An important ideological pillar of translanguaging theory is its inclusion within postcolonial critique, which questions the very existence of individual languages as independent codes, treating them as products of the formation of nation-states. This idea is most clearly expressed by S. Makoni and A. Pennycook:

"Languages are not real objects existing in the world, nor do they arise from real-life conditions; they are inventions of social, cultural, and political movements". (A.Pennycook & Makoni).

Theoretical relationship of concepts:

Phenomenological level: Many real-world examples of translingual practices align with what used to be described as code-switching or code-mixing.

Theoretical level: Code-switching presupposes the existence of distinct (though interacting) language systems and focuses on the mechanics of shifting between them. Translanguaging, in contrast, rejects the idea of fixed boundaries between languages as analytically artificial and offers an integrated view of speech as a fluid process of selecting features from a unified repertoire.

Radical proponents of translingual theory view the term "code-switching" as belonging to an "old" paradigm rooted in monoglossic assumptions. More moderate perspectives see code-switching as one possible strategy within a broader translingual framework. In the latter view, translinguality serves as an umbrella concept that includes not only verbal switching but also translation, transmodal practices, gestures, and other semiotic resources.

Within the Russian-speaking academic tradition (see, in particular, the work of E.V. Khilkhanova), it is noted that, despite the stated opposition, scholars working within the translingual paradigm often rely on empirical data from code-switching and continue to use "language labels" in their analyses.

Key differences in emphasis

Code-switching is more frequently linked to sociolinguistic and structural analysis—examining the rules, functions, and social implications of switching. In early studies, it was

sometimes seen as a sign of language deficiency, while more recent research highlights its creativity and pragmatic value.

Translingualism focuses on empowerment and cognitive benefits: it is presented as a natural, creative, and transformative process, particularly effective in educational settings. In pedagogy, the translingual approach opposes the strict separation of languages and the "monolingual" norm, advocating for the use of students' full linguistic repertoires to deepen content learning and support identity development.

Examples:

Code-switching :

1. Я вчера пошёл в store, купил milk и bread, потому что дома ничего не было. А потом I called my friend и мы пошли в gym.

2. Они полностью gone crazy! Я даже не знал, что так можно реагировать

The speaker is aware of and maintains language boundaries, i.e. speaks Russian but inserts an English word.

Translanguaging:

1. Today, we are going to discuss the solar system. Сначала прочитайте текст про планеты на английском, а потом обсудите это в парах.

In translanguaging, language boundaries are ignored.

In Ukraine, the mixed Russian-Ukrainian variety known as "Surzhyk" is widespread, particularly among older generations and in certain regions. In addition, code-switching between Ukrainian and Russian is very common in media and political discourse. Parliamentary speeches often show clear intrasentential and intersentential switching, where politicians move between the two languages depending on the topic, audience, or ideological positioning. Even after the sharp decline in Russian use following 2014 and especially 2022, mixed practices still occur in everyday urban speech.

In Uzbekistan, Uzbek-Russian bilingualism remains strong, especially in cities. Code-switching between Uzbek and Russian is the norm in education and daily communication. Teachers and students frequently switch languages in English classrooms as well (Uzbek-Russian-English), using Russian for technical terms or explanations and Uzbek for clarification or emotional expression.

Speakers from Buryatia, Kalmykia, or Central Asia sometimes produce highly fluid translingual discourse involving their heritage language, Russian, and English or German.

Azerbaijan represents a particularly interesting setting for such an analysis. As a multilingual society where Azerbaijani, Russian, and English coexist and interact daily, it offers rich ground for testing the explanatory power of both frameworks. Understanding whether speakers engage in strategic code-switching or fluid translanguaging has direct implications for language education, national language policy, and the preservation of linguistic identities in the region.

Discussion

It is particularly revealing to examine these theoretical debates through the lens of the Azerbaijani multilingual context. In Baku and other urban areas, Azerbaijani, Russian, and increasingly English coexist in everyday communication. Research on Azerbaijani-Russian code-switching (e.g., Zuercher, 2009; Soltanova, 2023) shows that speakers frequently insert Russian lexical items or switch between the two languages depending on the domain - Russian

often dominates in professional, technical, or informal private spheres, while Azerbaijani is preferred in official or national contexts. The addition of English adds another layer, especially among younger, educated speakers. In such a triglossic environment, the question arises whether speakers are truly “translanguaging” as one unified repertoire or strategically navigating socially meaningful language boundaries.

My position aligns closely with that of E.V. Khilkhanova, who has repeatedly questioned the universality of translanguaging theory based on empirical data from post-Soviet multilinguals. Studies of post-Soviet migrants and residents show that highly educated speakers often maintain clear language separation and prefer “pure” forms of Russian or national languages rather than fluid hybrid practices. This suggests that translanguaging may not be a natural default strategy for all multilingual communities, especially those shaped by strong monolingual ideologies inherited from the Soviet period. In the Azerbaijani case, national language policy promoting Azerbaijani as the state language further reinforces the social importance of language boundaries.

Ultimately, this discussion highlights the need for context-sensitive research. Universal claims about multilingual practices should be tested against diverse local realities, including those of the post-Soviet space. Future studies could benefit from combining fine-grained conversation analysis (typical of code-switching research) with the broader, more emancipatory perspective offered by translanguaging theory. Such a synthesis would provide both descriptive accuracy and practical relevance for language education in multilingual settings.

Conclusion

In conclusion, code-switching and translanguaging describe largely overlapping linguistic phenomena, yet they are grounded in significantly different theoretical perspectives. Code-switching assumes the existence of relatively distinct language systems and focuses on the mechanisms, functions, and social meanings of switching between them. Translanguaging, on the other hand, rejects the idea of clear boundaries between languages and views multilingual speech as a fluid, integrated use of a speaker’s entire linguistic and semiotic repertoire.

In practice, many examples traditionally labeled as code-switching can easily be interpreted through the lens of translanguaging. However, the theoretical distinction between the two concepts remains important, particularly in the fields of language education and language policy.

At the same time, as E. V. Khilkhanova rightly points out, the strong opposition between these two approaches often appears more ideological than empirical. In real-life multilingual settings - especially in post-Soviet countries like Azerbaijan - speakers continue to perceive and orient themselves toward language boundaries (Azerbaijani, Russian, English, and others), even while mixing them freely. Most current research still relies heavily on the analytical tools of code-switching and code-mixing rather than purely on translanguaging.

Ultimately, I believe the most productive approach is not to treat these concepts as mutually exclusive, but to view them as complementary. Code-switching offers valuable precision in describing structural and social aspects of multilingual speech, while translanguaging highlights its creative, agentive, and pedagogical potential. Combining both perspectives can lead to a richer understanding of language practices in highly diverse societies and help develop more effective educational strategies. Future studies could benefit from empirical investigations into how Azerbaijani-Russian-English translanguaging and code-

switching practices manifest in different domains - education, business, family, and online communication. Such research would help determine under which conditions speakers treat their linguistic resources as one unified repertoire and when they consciously orient toward distinct language boundaries.

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LINGUACULTURAL STUDY OF PROFESSIONAL SPEECH FORMULAS IN ENGLISH AND UZBEK

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Abstract: This thesis explores the linguacultural and pragmatic characteristics of professional speech formulas in English and Uzbek. Grounded in Social Role Theory, Speech Act Theory, and Grice's Cooperative Principle, the research examines how culturally encoded expressions structure professional communication, convey etiquette, and reflect social status.

The analysis reveals stark contrasts between English and Uzbek speech formulas, highlighting individualistic versus collectivistic communication frameworks. While English formulas prioritize directness, autonomy, and professional distance, Uzbek expressions emphasize social hierarchy, deep

respect, and group solidarity. These distinct patterns demonstrate how deeply embedded cultural values shape daily workplace interactions and formal professional conduct.

The findings offer substantial practical value for international business communication, cross-cultural negotiations, and the design of corporate language training programs. Ultimately, this study helps bridge cultural gaps, minimize workplace misunderstandings, and foster smooth, effective professional dialogue within increasingly globalized business environments.

Keywords: speech formulas, professional communication, linguacultural studies, pragmatics, social roles, English language, Uzbek language.

İNGİLİZCE VE ÖZBEKÇEDEKİ PROFESYONEL KONUŞMA KALIPLARININ DİLBİLİMSEL-KÜLTÜREL (LİNGVOKÜLTÜROLOJİK) İNCELENMESİ

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Özet: Bu tez, İngilizce ve Özbekçedeki profesyonel konuşma kalıplarının dilbilimsel-kültürel ve pragmatik özelliklerini incelemektedir. Sosyal Rol Teorisi, Konuşma Edimleri Teorisi ve Grice'in İşbirliği İlkesine dayanan araştırma, kültürel olarak kodlanmış ifadelerin profesyonel iletişimi nasıl yapılandırıldığını, nezaket kurallarını nasıl aktardığını ve sosyal statüyü nasıl yansıttığını ele almaktadır.

Yapılan analizler, İngilizce ve Özbekçe konuşma kalıpları arasında belirgin farklar olduğunu ortaya koymakta; bireyselciye karşı kolektif iletişim modellerini vurgulamaktadır. İngilizce kalıplar doğrudanlığı, özerkliği ve profesyonel mesafeyi ön plana çıkarırken, Özbekçe ifadeler sosyal hiyerarşiyi, derin saygıyı ve grup dayanışmasını vurgular. Bu farklı kalıplar, köklü kültürel değerlerin günlük iş yeri etkileşimlerini ve resmi profesyonel davranışları nasıl şekillendirdiğini göstermektedir.

Elde edilen bulgular; uluslararası iş iletişimi, kültürlerarası müzakereler ve kurumsal dil eğitimi programlarının tasarımı açısından önemli pratik değerler sunmaktadır. Sonuç olarak bu çalışma, küreselleşen iş dünyasında kültürel engelleri aşmaya, iş yeri iletişim kazalarını en aza indirmeye ve etkili bir profesyonel diyalog kurmaya katkı sağlamaktadır.

Anahtar Kelimeler: konuşma kalıpları, profesyonel iletişim, dil-kültürbilim (lingvokültüroloji), pragmatik, sosyal roller, İngilizce, Özbekçe

Introduction

Professional communication is a specialized linguistic field where speech formulas serve as primary tools. They are used not only to convey information but also to define social roles, express etiquette, and implement cultural norms. Business communication formulas in English and Uzbek reflect diverse social and cultural standards: while English generally prefers direct communication based on individual responsibility, Uzbek favors respect for hierarchy, collective harmony, and indirect, polite expressions. Therefore, this study aims to analyze these formulas from linguacultural and pragmatic perspectives.

Theoretical Framework

The research relies on the following theoretical approaches:

1. **Social Role Theory (Parsons, Linton, Goffman):** Demonstrates that every professional role has specific social expectations and communication rules. Goffman's

dramaturgical approach characterizes speech formulas in professional communication as "performance scripts."

2. **Speech Act Theory (Austin, Searle):** Emphasizes that every speech formula has illocutionary and perlocutionary functions, meaning it not only expresses an intent but also influences the listener.

3. **Grice's Cooperative Principle:** Explains how business formulas maintain communicative efficiency, clarity of information, and etiquette.

4. **Linguacultural Studies (Maslova, Stepanov):** Posits that speech formulas express cultural codes and concepts, reflecting the values and social expectations of a society.

Methodology

A total of 120 speech formulas were collected for the study: 60 in English and 60 in Uzbek. The formulas were categorized into the following social roles:

- Superior–Subordinate
- Peer-to-Peer
- Client Interaction
- Gender-sensitive communication

Each formula was analyzed according to the following criteria:

- **Pragmatic function** (communicative intent and impact on the listener).
- **Linguistic structure** (syntax, vocabulary, mitigation tools).
- **Cultural encoding** (social norms, values, forms of showing respect).

Methods: Comparative-descriptive and corpus analysis were utilized to identify cultural and social features within the formulas.

Results

English Language:

- Formulas reflect clarity, directness, and individual responsibility.
- *Examples:* "I would appreciate it if you could review this report.", "Please provide feedback by Friday."
- *Strategy:* Emphasis on task completion, efficiency, and time-saving.

Uzbek Language:

- Formulas reflect politeness, respect for hierarchy, and collective harmony.
- *Examples:* "Hurmatli janob, iltimos, ushbu hisobotni ko'rib chiqishingizni so'rayman." (Respected Sir, I ask that you please review this report.)
- *Strategy:* Maintaining social balance, showing respect, and preventing conflict.

Comparative Analysis Table

Social Role	English Formula	Uzbek Formula	Pragmatic Task	Cultural Feature
Superior–Subordinate	Could you review this report?	Hurmatli janob, iltimos, hisobotni ko'rib chiqing	Request, Respect	Hierarchy
Peers	Let's coordinate on this task	Shu vazifani birga muvofiqlashtiramiz	Cooperation	Collectivism
Client	Looking	Javobingizni kutib	Notification,	Respect for

Social Role	English Formula	Uzbek Formula	Pragmatic Task	Cultural Feature
	forward to your response	qolaman	Respect	Client
Gender-sensitive	Madam, may I ask...?	Xonim, iltimos so‘rayman	Polite Inquiry	Gender & Social Respect

Discussion

The results align with Goffman’s dramaturgical approach: speech formulas function as "performance scripts" for social roles.

- **English:** Individualism, efficiency, direct expressions.
- **Uzbek:** Collectivism, hierarchy, polite and context-sensitive expressions.
- **Pragmatic Task:** Maintaining respect in professional communication, ensuring task completion, and enacting social norms.

Speech formulas teach social roles, reinforce professional communication rules, and increase communicative efficiency.

Conclusion

- Speech formulas are essential tools for social roles and cultural norms.
- English formulas focus on efficiency and individual responsibility.
- Uzbek formulas focus on collective harmony and hierarchical respect.
- The research results have practical value for international business communication, corporate language training, and curriculum development.

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LANGUAGE LEARNING STRATEGIES IN MULTILINGUAL ENVIRONMENTS

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Abstract. Multilingual classrooms have become a defining feature of contemporary education, yet research on the strategies learners deploy in such settings is still uneven. This paper synthesises five recent studies that approach language learning strategies from complementary angles: psychometric and qualitative analysis of strategy use and language awareness; observation of young children in multilingual settings; survey-based evidence on user behaviour in multilingual educational technology; comparative work on learners in multilingual versus monolingual contexts; and classroom-based reports on pedagogical innovation. Drawing these threads together, the article argues that strategy use in multilingual environments is shaped less by individual learner traits in isolation than by the interaction of learner agency, classroom mediation, and the language ecologies of the digital tools learners use. The discussion identifies four cross-cutting findings: metalinguistic awareness functions as both an outcome and a driver of strategy use; younger learners display more flexible cross-linguistic transfer than is typically assumed; EdTech platforms are not language-neutral and actively shape strategy choice; and pedagogical innovation works best when it makes the multilingual repertoire explicit rather than treating languages as separate compartments. Implications for teacher education, materials development, and policy in linguistically diverse settings such as Azerbaijan are outlined.

Keywords: language learning strategies, multilingualism, language awareness, educational technology, classroom practice, second language acquisition, learner autonomy.

ÇOXDİLLİ MÜHİTLƏRDƏ DİL ÖYRƏNMƏ STRATEGİYALARI

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Xülasə: Çoxdilli sinif mühitləri müasir təhsilin əsas xüsusiyyətlərindən birinə çevrilmişdir, lakin belə şəraitdə öyrənənlərin istifadə etdiyi strategiyalara dair tədqiqatlar hələ də qeyri-bərabərdir. Məqalə dil öyrənmə strategiyalarına bir-birini tamamlayan beş yanaşmadan baxan son tədqiqatları sintez edir: strategiya istifadəsi və dil şüurunun qarışıq metodlarla təhlili; çoxdilli mühitdə kiçikyaşlı uşaqların müşahidəsi; çoxdilli təhsil texnologiyalarında istifadəçi davranışına dair sorğu əsaslı məlumatlar; çoxdilli və tək dillilik kontekstlərində öyrənənləri müqayisə edən iş; və pedaqoji innovasiya ilə bağlı sinif əsaslı hesabatlar. Bu istiqamətləri bir araya gətirərək məqalə iddia edir ki, çoxdilli mühitlərdə strategiya istifadəsi tək başına öyrənən xüsusiyyətlərindən daha çox öyrənən aktivliyinin, sinif vasitəçiliyinin və istifadə olunan rəqəmsal alətlərin dil ekologiyasının qarşılıqlı təsiri ilə formalaşır. Müzakirədə dörd kəşifən nəticə müəyyən edilir: metalingvistik şüur strategiya istifadəsinin həm nəticəsi, həm də hərəkətverici qüvvəsidir; kiçikyaşlı öyrənənlər adətən fərz edildiyindən daha çevik dillərarası ötürmə nümayiş etdirirlər; təhsil texnologiyası platformaları dil baxımından neytral deyil və strategiya seçiminə fəal təsir göstərir; pedaqoji innovasiya dilləri ayrı-ayrı bölmələr kimi qiymətləndirməkdənsə çoxdilli repertuarı açıq şəkildə əks etdirdikdə daha yaxşı işləyir. Azərbaycan kimi dil baxımından müxtəlif şəraitdə müəllim hazırlığı, materialların hazırlanması və siyasət üçün nəticələr təsvir edilmişdir.

Açar sözlər: dil öyrənmə strategiyaları, çoxdillilik, dil şüuru, təhsil texnologiyası, sinif təcrübəsi, ikinci dilin mənimsənilməsi, öyrənən muxtariyyəti.

Introduction

The number of learners who study, communicate, and live across two or more languages keeps growing, and educational research has been working to keep up. The classroom has become a meeting point for languages that students bring from home, languages that the

curriculum prescribes, and languages that they encounter through media and digital platforms. In this context the question of how learners actually go about acquiring and using languages, the strategies they deploy when reading a text, parsing an unfamiliar phrase, or producing speech under social pressure, has taken on renewed importance.

Research on language learning strategies has a long history, but its assumptions have shifted as multilingualism has become the norm rather than the exception. Earlier work tended to frame strategies as individual cognitive moves used in the service of acquiring a single target language. More recent scholarship treats strategies as situated practices that draw on a learner's whole linguistic repertoire and respond to the affordances of the environment, including digital tools and classroom routines. The question is no longer only which strategies work best, but also how strategies travel across languages, ages, and platforms, and what the surrounding environment does to them.

This article addresses that broader question by drawing together five recent studies whose combined picture is more useful than the sum of its parts. The link between strategy use and language awareness, demonstrated through mixed-methods evidence on older students, suggests that awareness and strategy work hand in hand (Boyles & Espinosa, 2026). Strategy use also begins much earlier in life than is often assumed, with young multilingual children deploying recognisable strategies long before they can articulate them (Schwartz, 2026). Beyond the classroom, multilingual educational technology has emerged as an active shaper of strategy choice, since platform design is rarely neutral with respect to the languages users bring (Kwon et al., 2026). The same learner traits also produce different strategy patterns in multilingual versus monolingual contexts, which means context cannot be treated as a passive backdrop (Cho, 2025). And in linguistically diverse classrooms, pedagogical innovation has begun to translate these insights into practical routines (Omidire, 2025). Taken together, these strands point toward strategy use in multilingual environments as something co-constructed by learners, teachers, and the tools at hand.

Three concerns motivate this synthesis. The first is conceptual. The literature has not always distinguished clearly between strategies that learners report using and strategies that observers actually see, and this gap matters when claims are made about what works in multilingual settings. The second is practical. Teachers in linguistically diverse classrooms routinely make decisions about when to allow, encourage, or discourage students' home languages, and these decisions rest on assumptions about strategy use that are sometimes more inherited than examined. The third is contextual. Settings such as Azerbaijan, where Azerbaijani, Russian, English, and other languages coexist in everyday life and in education, raise questions that the available literature has not always addressed directly but can be made to speak to.

The article proceeds as follows. The Materials and Methods section describes the selection and analysis of the five focal studies and notes the limitations of a small purposive sample. The Results and Discussion section is organised thematically rather than source by source. Five themes are developed: the link between strategies and language awareness, developmental dimensions in young learners, the role of digital platforms, the comparative question of multilingual versus monolingual settings, and the classroom as a site of strategy innovation. The Conclusion draws out implications for teacher education and for research in multilingual contexts, and acknowledges the boundaries of what can be claimed on the basis of five studies.

Materials and Methods

This article is best described as a focused narrative synthesis. It does not claim the systematic review's standards of exhaustiveness or replicability, but it draws on a small purposive sample of recent studies that together cover a meaningful range of perspectives on language learning strategies in multilingual environments. The choice to work with a small sample is deliberate. A close reading of a few studies allows for the kind of comparative attention that systematic reviews, with their broader sweeps, sometimes have to forgo.

Five studies were selected for close reading. The set includes three of the most recent contributions in the field together with two slightly earlier studies that opened up themes the newer work continues to develop (Boyles & Espinosa, 2026; Cho, 2025; Kwon et al., 2026; Omidire, 2025; Schwartz, 2026). The selection was guided by three criteria. First, each source addresses language learning strategies explicitly rather than treating them as a side concern. Second, the studies between them cover different age groups, settings, and methodologies, which is essential for any synthesis that aspires to say something about strategies across multilingual environments rather than within a single niche. Third, all five are recent enough to reflect the current shape of the field, in which multilingualism is treated as the default condition.

The analysis followed a thematic synthesis logic. Each study was read in full and coded for content related to four broad areas: the types of strategies described or measured, the role of the surrounding environment (classroom, home, digital), the age and developmental stage of the learners, and any explicit links to language awareness or metalinguistic reflection. Codes were then compared across studies, and recurring or contrasting patterns were grouped into the themes that organise the Results and Discussion section below. Disagreements between sources, where they occurred, were recorded rather than smoothed over.

A few limitations of this approach should be stated openly. Five sources cannot represent the full range of work on multilingual language learning strategies. The selection skews toward English-language publications and toward formal educational settings; informal acquisition and naturalistic multilingual contexts outside school are underrepresented. The synthesis is also interpretive: another reader working with the same five studies might extract a slightly different set of themes. The aim is therefore not to produce a definitive map of the field but to offer a coherent reading that may be useful for teachers, researchers, and students in linguistically diverse settings, and that points toward questions worth investigating in more detail.

Results and Discussion

The relationship between the strategies learners use and their awareness of language as an object of reflection has been a recurring concern in the field. The connection appears tighter than earlier work might have suggested. Learners with a wider repertoire of strategies tend to show higher levels of language awareness as well, and those who can articulate what they are doing while reading or speaking are better placed to vary their approach when one strategy fails (Boyles & Espinosa, 2026).

This finding is more than a correlational curiosity. It speaks to the kind of teaching that supports strategic competence in multilingual environments. If awareness and strategy use feed each other, then instruction that asks learners to notice and name what they are doing has a place alongside instruction that models particular strategies. The implication is not that explicit strategy training should be abandoned but that it works better when paired with metalinguistic reflection. For Azerbaijani learners moving between Azerbaijani, Russian, and English, the practical translation of this point is that classroom activities which invite students to compare

how a problem is approached in each language may be more productive than activities that drill strategies in isolation.

Young Learners in Multilingual Settings

A common assumption in the literature places conscious strategy use later in development, treating young children in multilingual contexts as absorbing languages without strategic behaviour. The evidence complicates this picture. Children as young as four and five have been observed using identifiable strategies — switching to a stronger language to scaffold a weaker one, asking targeted clarification questions, and using gesture in coordinated ways with speech. These behaviours may be less verbalised and less systematic than in adolescents, but they are recognisable as strategies in form and function (Schwartz, 2026).

The implication for early childhood education is significant. Practitioners working with young multilingual children sometimes hesitate to acknowledge children's home languages in the educational space, fearing interference with the school language. The opposite concern looks more warranted. Treating the home language as an absence rather than a resource risks suppressing the very strategies that young learners deploy to make sense of new linguistic input. In multilingual societies the practical lesson is to design preschool and early-grade environments that invite, rather than tolerate, the languages children bring with them.

Digital Platforms and Strategy Choice

Digital platforms now mediate a substantial portion of language learning outside the classroom, and what happens at this interface deserves attention as a strategy question in its own right. The design of multilingual EdTech is anything but neutral with respect to strategy use. Default language settings, the visibility or invisibility of translation features, the order in which languages appear in menus, and the granularity of feedback all shape how users approach the material. Learners, in turn, develop strategies that are partly responses to what the platform makes easy or difficult (Kwon et al., 2026).

This is a useful corrective to a literature that has sometimes treated strategies as belonging entirely to the learner. The unit of analysis effectively extends to include the tool. The design implication is that EdTech intended for multilingual users should foreground rather than hide cross-linguistic features, since hiding them tends to push learners toward strategies that resemble monolingual study even when the actual language situation is multilingual. For educators in Azerbaijan and similar contexts who increasingly rely on international EdTech products, this is also a caution. Platforms designed with monolingual users in mind may, without anyone intending it, narrow the strategy options that multilingual learners would otherwise pursue.

Multilingual versus Monolingual Contexts

The comparative question deserves to be taken head-on. Examining second language learning strategies in multilingual and monolingual contexts side by side makes it clear that context is not just a backdrop. The same learner-level traits — motivation, age, prior education — can produce different strategy patterns when the surrounding environment encourages or discourages cross-linguistic comparison. Multilingual learners tend to use more compensation strategies, are more comfortable with code-mixing as a learning tool, and show greater tolerance for ambiguity than their monolingual counterparts (Cho, 2025).

These differences should not be read as a hierarchy in which multilingual learners are simply more skilled. The point is rather that the strategy repertoire is partly a product of the environments learners pass through. A monolingual learner who later moves into a multilingual

setting may need time to develop strategies that multilingual learners take for granted, and the reverse is also possible: a multilingual learner placed in a strictly monolingual classroom may suppress useful strategies because the environment offers no place for them. The pedagogical implication, especially relevant in countries that conduct schooling in a national language while learners draw on multiple home languages, is that classroom norms can either widen or narrow the strategy repertoire learners actually use.

Classroom Innovation in Linguistically Diverse Schools

On the side of practice, classrooms where several languages are routinely present have generated pedagogical innovations that do not try to manage multilingualism away but build on it. Translanguaging activities, multilingual word walls, peer-led explanations across language pairs, and assessment formats that allow learners to draw on more than one language have all been documented with practitioner-level detail. What unites them is a refusal to treat each language as a sealed compartment (Omidire, 2025).

This classroom evidence dovetails with the more theoretical points raised so far. Awareness drives strategy use, young children already deploy cross-linguistic strategies, digital tools shape strategy choice, and context conditions strategy patterns. Classrooms that explicitly invite the multilingual repertoire therefore stand to amplify what learners can do. The risk in less innovative classrooms is the opposite: strategies that learners have available go unused because the environment signals that only the target language counts. Practitioner accounts of multilingual classroom innovation are therefore not simply collections of teaching ideas but an existence proof that the integrated picture is realisable in real classrooms (Omidire, 2025).

Conclusion

Pulling these threads together, language learning strategies in multilingual environments are best understood as a meeting point of the learner's developing awareness, the design of the tools available, and the openness of the classroom to languages other than the target. None of the five studies considered here, taken alone, makes this case in full; together they outline a position that is more useful than the sum of its parts.

For practice, four implications follow. First, strategy instruction should be paired with activities that ask learners to notice and discuss what they are doing across languages, rather than within only one. Second, teachers of young multilingual children should treat home languages as resources rather than obstacles, since young learners are already using them strategically. Third, decisions about EdTech adoption in linguistically diverse contexts should weigh how a platform handles multiple languages, not only the quality of its target-language content. Fourth, classroom routines benefit from explicit space for the multilingual repertoire, whether through translanguaging, multilingual peer work, or assessment that allows learners to show what they know in more than one language.

For research, several questions remain open. The studies synthesised here come from varied settings, but South Caucasus contexts including Azerbaijan are not directly represented in any of them. Whether the patterns reported translate to settings where Azerbaijani, Russian, and English coexist in distinctive ways is an empirical question worth investigating. The relationship between strategy use and language awareness deserves further testing in younger cohorts, ideally in dialogue with developmental evidence on early multilingual children (Boyles & Espinosa, 2026; Schwartz, 2026). The EdTech dimension would benefit from longitudinal work that follows learners as platform features change (Kwon et al., 2026).

The boundaries of this synthesis should be acknowledged. Five studies cannot stand in for an entire literature, and the patterns identified here are starting points rather than settled findings. Even within those limits, however, the cross-cutting message is reasonably clear: strategies in multilingual environments are not the property of learners alone, and the most productive angle for both teachers and researchers is to look at the full system within which strategies are formed and used.

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NATIVE TONGUE INTERFERENCE IN LANGUAGE LEARNING COGNITIVE LINGUISTICS PERSPECTIVE

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Abstract: The influence of a learner's mother tongue (L1) is a multifaceted element of English education that significantly shapes how students engage with instruction. This process involves "language transfer," where students project their native linguistic, cognitive, and cultural knowledge onto the target language. While this transfer can be beneficial if structures are similar, it often leads to "negative transfer" or interference when differences between L1 and L2 cause persistent grammatical or phonetic errors. Unlike children who acquire an L1 as a "blank slate," adults process English through a native language "filter," frequently resulting in "ill-formed" sentences that can convey the opposite of the intended meaning. To mitigate these challenges, educators can implement immersive English-only environments, utilize authentic English-centric materials, and focus heavily on phonetic nuances to improve clarity and listening skills. Ultimately, leveraging linguistic similarities through contextualized and collaborative learning allows students to minimize native language impact and achieve more natural proficiency.

Key words: language, linguistic, cognitive, learning process, knowledge

DİL ÖYRƏNİLMƏSİNƏ ANA DİLİN MÜDAXİLƏSİ: KOQNİTİV LİŦQVİSTİK PERSPEKTİV

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Xülasə: Ana dilinin (L1) İngilis dili təhsilinə təsiri çoxşaxəlidir və öyrənmə prosesini əhəmiyyətli dərəcədə formalaşdırır. "Dil transferi" vasitəsilə tələbələr öz ana dili biliklərini hədəf dilə tətbiq edirlər; lakin dil strukturları fərqli olduqda, bu, qrammatik və fonetik səhvlərə — yəni mənfi "müdaxiləyə" səbəb olur. Ana dilini "ağ lövhə" kimi öyrənən uşaqlardan fərqli olaraq, böyükklər İngilis dilini mövcud dil "filtri" vasitəsilə qavrayırlar ki, bu da bəzən nəzərdə tutulanan tam əksi olan yanlış mənalı cümlələrin yaranması ilə nəticələnir. Bu təsiri azaltmaq üçün müəllimlər tam İngilis dilli mühit yaratmalı, orijinal materiallardan istifadə etməli və tələffüz incəliklərinə xüsusi diqqət yetirməlidirlər. Nəticə etibarilə, əməkdaşlığa əsaslanan və kontekstual öyrənmə metodları tələbələrə ana dili müdaxiləsini minimuma endirməyə və daha təbii dil bacarıqları qazanmağa kömək edir.

Açar sözlər: dil, linqvistika, koqnitiv, öyrənmə prosesi, bilik

Introduction

The impact of a learner's primary language on English language instruction is a complex and detailed element of linguistics. When students begin the process of learning a secondary language, their "mother tongue" essentially serves as a foundational influence that guides their progress. This relationship is characterized by a sophisticated combination of cultural, cognitive, and linguistic elements that fundamentally alter how students interact with English lessons.

Various theories of language acquisition have historically acknowledged the significance of the first language, particularly through the concept of language transfer. This occurs when students project their existing native language knowledge onto the new language they are studying. Depending on the similarities or differences between the two systems, this transfer can result in either beneficial (positive) or detrimental (negative) learning experiences.

Every individual possesses a primary language (L1) developed through a mix of biological predispositions and environmental exposure. Research by scholars such as Brown (1994) suggests that this foundational language significantly impacts the acquisition of a second language (L2) across various dimensions, including grammar, pronunciation, and meaning. (Abdel-Rahman, 2014)

Linguistic evidence consistently supports the existence of "L1 interference," a process where a learner's native background shapes their second language production. This is most visible in sentence construction; for example, students often apply the grammatical patterns of their mother tongue to their written work in a foreign language. Some students argued that these native structures are naturally projected onto the target language. According to research by Yanis (2007), the impact of a learner's primary language on foreign language acquisition is a fundamental element of the learning process. Her study utilized a syntactic analysis of English sentences produced by Indonesian learners to observe their performance during the interlanguage phase. Yanis concluded that language transfer is an unavoidable phenomenon, particularly when there is a significant structural gap between the native language and the target language. (Yanis, 2007).

This specific research has inspired the current study. The author observes that the distinct grammatical rules of Indonesian and English often lead to syntactic errors that alter the intended message of the L2. A clear example is the incorrect use of the present perfect tense, such as

inserting an unnecessary "been" into an active sentence. Because the correct active pattern is "has/have + V3," adding "been" mistakenly creates a passive meaning. Such grammatical inaccuracies are critical because they result in a message that is the exact opposite of what the sentence require. (Ellis, 1995).

The primary obstacle encountered by those attempting to master a foreign language is the persistent influence exerted by the learner's native tongue. During the early stages of childhood, a first language is acquired through natural communication—usually with parents—at a time when the brain is effectively a "blank slate." Because no other linguistic system exists in the mind yet, this information becomes seamlessly integrated into the child's mental architecture, allowing them to systematize language rules automatically. In contrast, when an individual begins learning a second language at a more mature age, they perceive all new linguistic data through the "filter" of their native system. This leads to a subconscious comparison between the two languages, resulting in a phenomenon known as "interference." This process, often referred to as "transfer" in pedagogical traditions, confirms that a student's prior linguistic knowledge significantly shapes their ability to absorb and utilize new foreign language information. (Beardsmore, 1986)

The effect of a student's first language on their English education is often substantial, as it can shape their ability to both understand and apply the language accurately. To manage this impact, both teachers and students can adopt specific techniques designed to encourage English proficiency while still respecting the underlying role of the mother tongue.

By implementing these methods, it is possible to support linguistic growth without ignoring the learner's original linguistic foundation. Developing a setting where English is the sole language of communication can remarkably improve a student's concentration and help them move away from their native tongue's influence. This immersion strategy starts with establishing firm requirements that English be the first choice for all academic tasks. By requiring students to interact exclusively in English with classmates and teachers, they are compelled to process thoughts and deliver responses directly in the new language. Furthermore, when instructors provide all evaluations and corrective feedback in English, it strengthens the student's commitment to using the language consistently throughout the learning process. (Graddol, 2006)

Utilizing educational tools that are exclusively in English—including digital resources, videos, and academic texts—helps students become accustomed to the specific vocabulary and grammatical frameworks of the language. Interaction with these English-focused materials introduces learners to a wide range of topics and scenarios, which ultimately bolsters their linguistic progress and weakens the impact of their primary language. By surrounding students with authentic input, educators can create a more robust learning environment that prioritizes the target language's natural flow.

The study of phonetics and pronunciation is a fundamental part of learning a language, as it directly affects how successfully a student can interact in English. By highlighting these components, educators provide students with the ability to produce English speech sounds more accurately. This level of phonetic detail not only helps learners express their thoughts more effectively but also ensures they are comprehensible to their audience in various environments. Furthermore, concentrating on phonetics sharpens listening comprehension; by recognizing specific sound structures, students can better understand spoken English and avoid potential miscommunications. Ultimately, gaining expertise in pronunciation allows learners to minimize

the vocal traces of their native language, fostering a more authentic and self-assured speaking style.

Adopting a contextualized approach to language education involves embedding English instruction within authentic topics and real-life situations. This technique facilitates a stronger mental connection between new linguistic rules and practical application, ensuring that the learning experience is both relevant and memorable. When educators prioritize these situational methods, they provide students with the tools to diminish the interference of their primary language, ultimately leading to a more profound and natural command of the target language.

Joint educational tasks, including team-based projects, peer exercises, and group debates, offer students the necessary space to refine their English abilities within a reinforcing environment. Such interactions promote the use of English as the central medium for exchange, consequently diminishing the role of the primary language. Furthermore, cooperative learning helps establish a collective identity among students, cultivating an optimistic atmosphere where they feel secure in their linguistic capabilities. (Khansir, 2012)

Conclusion

In summary, minimizing the footprint of the mother tongue in English instruction involves a diverse range of methods. These include maintaining English-exclusive spaces, employing native-language resources, emphasizing sound patterns and pronunciation, linking lessons to practical contexts, and facilitating group-based learning. By adopting these techniques, both teachers and students can improve English fluency while successfully managing the inherent influence of the native tongue.

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LANGUAGE AND CULTURE INTERACTION IN LINGUOCULTUROLOGY

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Abstract: This article examines the interaction between language and culture within the framework of linguoculturology. Language is considered not only as a means of communication but also as a cultural phenomenon that reflects the historical memory, national identity, values, and worldview of a society. The study analyzes lexical units, phraseological expressions, proverbs, and concepts from a comparative linguocultural perspective based on English and Azerbaijani examples. The research demonstrates that linguistic units with similar meanings may have different associative and cultural interpretations in different societies. Particular attention is paid to cultural codes, metaphorical meanings, and the role of language in preserving and transmitting cultural values. The article also highlights the importance of linguoculturological approaches in foreign language teaching, translation, and intercultural communication. The findings show that understanding cultural meanings in language contributes to more effective communication and deeper awareness of national characteristics. Thus, linguoculturology provides an important framework for studying language as both a structural system and a reflection of cultural experience.

Key words: linguoculturology, language and culture, concept, semantics, cultural code, discourse, national identity

LİŇVOKULTUROLOGİYADA DİL VƏ MƏDƏNİYYƏTİN QARŞILIQLI ƏLAQƏSİ

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Xülasə: Bu məqalədə linqvokulturologiya çərçivəsində dil və mədəniyyətin qarşılıqlı əlaqəsi araşdırılır. Dil yalnız ünsiyyət vasitəsi və qrammatik sistem deyil, həm də xalqın tarixi yaddaşını, milli kimliyini, dəyərlərini və dünyagörüşünü əks etdirən mühüm mədəni hadisə kimi nəzərdən keçirilir. Tədqiqatda ingilis və Azərbaycan dillərindən nümunələr əsasında leksik vahidlərin, frazeoloji birləşmələrin, atalar sözlərinin və konseptlərin mədəni xüsusiyyətləri müqayisəli aspektdə təhlil olunur. Araşdırma göstərir ki, eyni məna sahəsinə aid dil vahidləri müxtəlif cəmiyyətlərdə fərqli assosiativ və mədəni çalarlar ifadə edə bilər. Məqalədə mədəni kodların, məcazi mənaların və dilin mədəni dəyərlərin qorunub saxlanması və ötürülməsindəki roluna xüsusi diqqət yetirilir. Həmçinin linqvokulturoloji yanaşmanın xarici dil tədrisi, tərcümə və mədəniyyətlərarası ünsiyyətdə əhəmiyyəti vurğulanır. Nəticələr göstərir ki, dilin mədəni məzmununun öyrənilməsi daha səmərəli ünsiyyətə və milli xüsusiyyətlərin dərinəndən anlaşılmasına imkan yaradır.

Açar sözlər: linqvokulturologiya, dil və mədəniyyət, konsept, semantika, mədəni kod, diskurs, milli kimlik

Introduction

The relationship between language and culture has always been one of the central issues in linguistics. Language does not exist separately from the society that uses it. It develops together with the historical experience, traditions, customs, beliefs, social values, and communicative behaviour of a people (Kramsch,1998). For this reason, the study of language requires not only grammatical and lexical analysis, but also an investigation of the cultural meanings hidden behind linguistic forms.

Linguoculturology has emerged as an interdisciplinary field that studies the connection between language, culture, consciousness, and communication. It combines the achievements of linguistics, cultural studies, cognitive linguistics, ethnolinguistics, pragmatics, and discourse analysis (Maslova, 2001). Its main aim is to reveal how cultural information is encoded, preserved, and transmitted through language. In this sense, language is not only a tool for expressing thought; it is also a form of cultural memory.

In contemporary global communication, the importance of linguoculturological research is increasing. People who speak different languages often face not only linguistic, but also cultural barriers. A grammatically correct sentence may still be misunderstood if the cultural background of the expression is not taken into account. Therefore, the study of cultural meanings in language is significant for translation, intercultural communication, foreign language teaching, and discourse interpretation.

Theoretical basis of linguoculturology

Linguoculturology studies language units as carriers of cultural information. It pays special attention to words, idioms, proverbs, metaphors, symbols, speech formulas, and discourse patterns that reflect the worldview of a particular nation. The term itself suggests the unity of two important dimensions: language and culture. While traditional linguistics mainly describes the internal structure of language, linguoculturology focuses on the cultural content expressed through this structure.

One of the key categories of linguoculturology is the concept. A concept is a mental and cultural unit that represents knowledge, experience, evaluation, and emotional attitude connected with a certain phenomenon. (Wierzbicka, 1997). Concepts such as freedom, honor, family, homeland, time, hospitality, and respect exist in many languages; however, their cultural interpretation may differ from one society to another. This difference is reflected in lexical choices, phraseological expressions, metaphors, and communicative behaviour.

For example, the concept of time is represented differently in various cultures. In English, the expression "*time is money*" reflects a pragmatic and economic attitude toward time. It emphasizes efficiency, productivity, and rational planning. In Azerbaijani, the phrase "*vaxt qızıldır*" also shows the value of time, but it carries a more evaluative and moral shade, presenting time as something precious and worthy of careful treatment. (Azərbaycan atalar sözləri, 2004). Both expressions underline the importance of time, yet their cultural associations are not completely identical.

Language as a carrier of cultural values

Language preserves cultural values in different forms. Everyday words may contain historical, emotional, and symbolic meanings. For instance, words connected with family relations in Azerbaijani often express not only kinship, but also respect, closeness, social hierarchy, and moral responsibility. In English, family-related vocabulary may be more neutral in everyday usage, while in Azerbaijani the same field may carry stronger emotional and social connotations.

Cultural values are especially visible in speech etiquette. Greetings, forms of address, expressions of gratitude, apology, invitation, and farewell show how a society understands politeness and social distance. In Azerbaijani communication, respectful address forms and warm interpersonal expressions are very common. Expressions such as "*buyurun*", "*çox sağ olun*", "*xoş gəlmisiniz*" and "*Allah köməyiniz olsun*" reflect not only politeness, but also cultural

solidarity and emotional involvement. (Azərbaycan dilinin izahlı lüğəti, 2006). In English, etiquette formulas are often shorter and more standardized, such as “please”, “thank you”, “sorry”, and “you are welcome”.

This difference does not mean that one language is more polite than another. It shows that politeness is culturally organized. Each language community has its own norms for expressing respect, distance, sincerity, and emotional closeness. Linguoculturology helps to explain these differences and prevents superficial comparison of languages.

Phraseological units and proverbs in cultural interpretation

Phraseological units and proverbs are among the most important sources for linguoculturological analysis. They are stable expressions that contain collective experience, moral evaluation, imagery, and national worldview. Idioms often cannot be understood through the literal meaning of their components. Their meaning is shaped by cultural associations and historical usage.

For example, the English idiom “to break the ice” means to start communication and make people feel more comfortable. Its imagery is connected with removing coldness and distance in communication. In Azerbaijani, expressions with similar pragmatic meaning may use different images, for example “söhbətə yol açmaq” or “buzları əritmək”. (Azərbaycan dilinin izahlı lüğəti, 2006). The second expression is closer to the English image, while the first one reflects a more general metaphor of opening a path for communication.

Proverbs also demonstrate cultural wisdom. The Azerbaijani proverb “El gücü, sel gücü” emphasizes collectivism and the power of unity. (Azərbaycan atalar sözləri, 2004). An English equivalent may be “Many hands make light work”. Both proverbs express the value of cooperation, but their images differ: the Azerbaijani proverb compares collective strength with a flood, while the English proverb focuses on practical labour. Such examples show that universal ideas may be expressed through nationally specific images.

Another example is the concept of homeland. In Azerbaijani culture, the word “vətən” carries deep emotional, historical, and moral meaning. (Azərbaycan dilinin izahlı lüğəti, 2006). It is associated with land, memory, sacrifice, belonging, and identity. In English, “homeland”, “native land” or “motherland” may express similar meanings, but the emotional intensity and usage contexts may differ. Therefore, direct translation is not always enough; cultural interpretation is necessary.

Comparative analysis of English and Azerbaijani examples

A comparative approach makes it possible to identify both universal and culture-specific features in language. Universal features are connected with common human experience: family, work, love, fear, hope, life, death, time, and nature. Culture-specific features are related to historical development, religion, social structure, national traditions, and communicative norms.

In English, many idioms reflect individual effort, personal achievement, and practical experience. Expressions such as “to stand on one’s own feet”, “no pain, no gain”, and “practice makes perfect” emphasize self-reliance, discipline, and result-oriented activity. Azerbaijani phraseology also values hard work and persistence, but it often expresses these ideas through images connected with community, patience, destiny, and moral balance. For instance, “əziyyət çəkən bəhrəsini görər” presents effort as a morally rewarded process (Azərbaycan atalar sözləri, 2004).

The concept of hospitality is another culturally rich example. In Azerbaijani culture, hospitality is not only a social habit, but also a moral value. The guest is traditionally treated with special respect. Expressions such as “*qonaq Allah qonağıdır*” reflect the sacred and ethical dimension of hospitality. (Azərbaycan atalar sözləri, 2004). In English-speaking cultures, hospitality is also valued, but it is often expressed within the limits of personal space and individual privacy. This difference is important for intercultural communication because the same behaviour may be interpreted differently by representatives of different cultures.

Colour terms also have cultural meanings. The colour white is often associated with purity and peace in many cultures. In Azerbaijani, “*ağ gün*” means a happy and prosperous life, while “*qara gün*” refers to difficult days. . (Azərbaycan dilinin izahlı lüğəti, 2006). In English, “black day” can also mean a tragic or unfortunate day. Such parallels show similarities in symbolic thinking, but each language develops its own stable combinations and emotional shades.

Linguoculturology and foreign language teaching

Linguoculturology has practical importance for foreign language teaching. Learning a language does not mean memorizing only grammar rules and vocabulary. It also means learning how meanings function in cultural context. Students may know the dictionary meaning of a word, but fail to understand its emotional, stylistic, or cultural associations. Therefore, cultural explanation should be included in language lessons.

In English language teaching, idioms, proverbs, speech etiquette, cultural references, and real-life discourse examples help learners develop intercultural competence. For Azerbaijani students learning English, it is useful to compare English expressions with Azerbaijani equivalents. Such comparison develops both linguistic awareness and cultural sensitivity. It also prevents literal translation and helps students use language more naturally.

For example, when teaching the expression “time is money”, the teacher can compare it with “*vaxt qızıldır*”. Students can discuss whether these expressions are fully equivalent, what cultural values they represent, and in which contexts they are used. This type of task develops critical thinking, vocabulary knowledge, speaking skills, and intercultural understanding at the same time.

Conclusion

The analysis shows that language and culture are deeply interconnected. Language reflects the worldview, values, traditions, stereotypes, and social experience of a people. It preserves cultural memory and transmits it from generation to generation. For this reason, language cannot be fully understood without reference to culture.

Linguoculturology provides a productive theoretical and practical framework for studying this relationship. It helps reveal cultural meanings encoded in lexical units, idioms, proverbs, concepts, metaphors, and discourse patterns. The comparison of English and Azerbaijani examples demonstrates that languages may express similar human experiences through different cultural images and associations.

Thus, linguoculturological research is important not only for theoretical linguistics, but also for translation studies, intercultural communication, and foreign language teaching. It allows researchers and learners to see language as a living cultural system rather than a mechanical collection of words and rules. In the modern world, where communication between cultures is becoming more intensive, such an approach is especially relevant.

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MULTILINGUALISM AND ISSUES OF LANGUAGE PRESERVATION

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Abstract: In the contemporary world, multilingualism and language preservation have become central issues in global academic and public discourse. Processes such as globalization, technological development, and cultural integration have intensified interlinguistic contact, while simultaneously placing many languages at risk of assimilation and extinction. Although multilingualism represents a source of cultural and cognitive enrichment, it also creates challenges for the sustainability of minority languages. This study examines the relationship between multilingualism and language preservation from a sociolinguistic and applied linguistic perspective, with particular reference to Azerbaijan. In addition, the study highlights the role of translation as a mediating mechanism in multilingual environments. The analysis shows that language preservation should be understood not only as a linguistic issue but also as a cultural, national, and strategic concern.

Keywords: multilingualism, language preservation, sociolinguistics, language policy, Azerbaijan, society.

ÇOXDİLLİLİK VƏ DİLİN QORUNMASI MƏSƏLƏLƏRİ

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Xülasə: Müasir dünyada çoxdillilik və dilin qorunması qlobal akademik və ictimai diskursun əsas məsələlərindən birinə çevrilmişdir. Qloballaşma, texnoloji inkişaf və mədəni integrasiya kimi proseslər dillərarası təması gücləndirmiş, eyni zamanda bir çox dilləri assimilyasiya və yox olma riski ilə üz-üzə qoymuşdur. Çoxdillilik mədəni və idraki zənginliyin mənbəyi olsa da, azlıq dillərinin davamlılığı üçün müəyyən çətinliklər yaradır. Bu tədqiqatda çoxdillilik və dilin qorunması arasındakı əlaqəni sosiolinqvistik və tətbiqi dilçilik baxımından, xüsusilə Azərbaycan nümunəsi əsasında araşdırılır. Bundan əlavə, tədqiqat çoxdilli mühitlərdə vasitəçi mexanizm kimi tərcümənin rolunu vurğulayır. Təhlil göstərir ki, dilin qorunması yalnız dil məsələsi kimi deyil, həm də mədəni, milli və strateji məsələ kimi dərk edilməlidir.

Açar sözlər: çoxdillilik, dilin qorunması, sosiolinqvistika, dil siyasəti, Azərbaycan, cəmiyyət.

Introduction

In the modern world, the issues of multilingualism and language preservation occupy a central place in global academic and public discourse. As a result of globalization, the development of information technologies, and processes of cultural integration, interactions between languages have intensified. At the same time, many languages face the threat of assimilation and extinction. While multilingualism serves as a source of cultural and intellectual enrichment, it also creates certain risks for the languages of smaller communities. Maintaining this balance is one of the key challenges of contemporary sociolinguistics.

Azerbaijan is not only a multicultural but also a multilingual society. Although it is theoretically easy to justify the importance of properly regulating ethnic and cultural diversity in a multicultural state, the practical implementation of this task remains highly complex. The National Leader Heydar Aliyev, who placed great importance on the preservation of ethnic and cultural diversity, stated: "The more peoples a state unites, the richer it becomes, because each of them contributes to world culture and civilization" (Abdullayev, p. 8).

The President of the Republic of Azerbaijan, Ilham Aliyev, in one of his speeches, emphasized: "Azerbaijan is a secular State with a secular society, a modern state with a high level of education" (Aliyev, p. 128).

Multilingualism refers to an individual's or a society's ability to know and use several languages simultaneously. Individual multilingualism broadens a person's knowledge and worldview, while social multilingualism ensures the participation of different languages in the communicative and cultural life of society (Fishman, 1972, p. 48). Baker (2011, pp. 58–59) notes that an imbalance in language use leads to the weakening of minority languages.

Language preservation is not only a linguistic issue but also a cultural and political process. According to UNESCO's *Atlas of the World's Languages in Danger*, approximately 40% of the nearly 7,000 languages spoken worldwide are at risk of extinction (UNESCO, 2019, p. 42–43). The *Convention for the Safeguarding of the Intangible Cultural Heritage*, adopted by UNESCO in 2003, emphasizes the importance of preserving the mother tongue as an essential component of cultural heritage (UNESCO, 2003, p. 15–16).

One of the main challenges in multilingual contexts is the influence of dominant languages. The global positions of English and Russian have led to a decline in the use of many smaller languages. Cummins (2000, p. 176) argues that for a language to survive, it must be actively used in education, media, and technology.

Azerbaijan has developed as a multilingual and multicultural society. While the Azerbaijani language holds the status of the state language, conditions are also created for the use of other languages. The Constitution of the Republic of Azerbaijan and the Law on the State Language form the primary legal framework of language policy. Aliyeva (2022, p. 47–50) notes that the functioning of Russian, English, and Turkish language divisions contributes to the development of multilingualism in the country.

In his speech at the 80th anniversary of the Azerbaijan National Academy of Sciences (2025), President Ilham Aliyev devoted special attention to the issue of language preservation. The head of state stated: “Preserving the purity of our mother tongue is the duty of every Azerbaijani. Our language is the main carrier of our national and moral values, and it is our responsibility to pass it on to future generations in its pure form” (Aliyev, 2025, p. 4). This statement demonstrates that language preservation is a fundamental component of national ideology and cultural security.

Multilingualism and language preservation are two complementary fundamental concepts. In the Republic of Azerbaijan, maintaining a balance between these two directions is one of the main objectives of state language policy. As emphasized by UNESCO and President Ilham Aliyev, language is not only a means of communication but also a carrier of national identity. From this perspective, language preservation is both a cultural and a strategic issue.

Thus, the preservation and development of the Azerbaijani language are firmly ensured as an integral part of modern state policy. The measures taken to preserve the purity of the Azerbaijani language, to improve the use of the state language, and to ensure its adaptation to the demands of globalization demonstrate that the state language remains at the center of attention. This once again proves that the comprehensive use of the mother tongue in all spheres of life has been successfully achieved.

Materials and Methods

This study is based on a qualitative analysis of academic literature, official policy documents, and international reports on multilingualism and language preservation. Key sources include works by Fishman (1972), Baker (2011), and Cummins (2000), as well as reports and conventions published by UNESCO (2003; 2019). Additionally, legislative frameworks such as the Constitution of the Republic of Azerbaijan and the Law on the State Language were analyzed.

The research adopts a comparative and descriptive approach, examining both global trends and the specific case of Azerbaijan. Speeches and policy statements by President Ilham Aliyev were also considered to understand the practical implementation of language policy at the national level.

Discussion

The analysis demonstrates that multilingualism plays a dual role in modern societies. On the one hand, it enhances cultural diversity and promotes cognitive development. On the other hand, it creates challenges for the preservation of minority languages, particularly under the influence of dominant global languages such as English and Russian.

Findings indicate that language preservation is closely linked to its use in key domains such as education, media, and technology. Without institutional support, minority languages are likely to decline. UNESCO data confirms that a significant proportion of the world’s languages are at risk of extinction.

In Azerbaijan, a balanced language policy has been implemented to address these challenges. While the Azerbaijani language is protected as the state language, opportunities for the use of other languages are also provided. Educational institutions offering instruction in multiple languages contribute to the development of multilingual competence. At the same time, government initiatives emphasize the preservation of linguistic purity and national identity.

The discussion also highlights that language preservation is not only a cultural issue but also a matter of national security and identity. The Azerbaijani case demonstrates that a well-structured language policy can successfully balance multilingualism with the protection of the national language.

Conclusion

Multilingualism and language preservation are interdependent and complementary phenomena. Maintaining a balance between them is essential for sustainable cultural development. The case of Azerbaijan illustrates that effective language policy, supported by legal frameworks and educational systems, can ensure both linguistic diversity and the preservation of the national language.

In conclusion, language should be viewed not only as a means of communication but also as a key element of cultural identity and national heritage. Therefore, its preservation must remain a strategic priority in the context of globalization.

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PRAGMATIC AND SEMANTIC FUNCTIONS OF ORIENTALISMS IN THE MODERN RUSSIAN LANGUAGE

(Based on the Russian-Language Publicistic Discourse of Uzbekistan)

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Abstract: This article examines the pragmatic and semantic functions of orientalisms in the modern Russian language based on Russian-language publicistic discourse in Uzbekistan. It focuses on orientalisms as culturally marked linguistic units that carry nominative, semantic, pragmatic, and evaluative meanings. The study explores how these linguistic elements are used to represent Eastern cultural realities, traditions, social institutions, and historical concepts within contemporary media discourse. Particular attention is given to the ways in which orientalisms convey cultural information and create additional emotional, associative, and evaluative meanings. The article also analyzes their role in influencing readers' perceptions and shaping specific communicative effects. Furthermore, the research considers how orientalisms contribute to the formation, reinforcement, and critical reinterpretation of cultural stereotypes about the East. Their use in ironic, satirical, and emotionally expressive contexts is also examined. The findings demonstrate that orientalisms are not merely lexical units used to name specific cultural phenomena but also important instruments of intercultural communication. They help preserve cultural identity, transmit ethnocultural information, and establish connections between language and cultural traditions. The study concludes that orientalisms represent an essential component of the linguistic representation of Eastern culture and significantly contribute to the development of intercultural dialogue in contemporary Russian-language media discourse.

Keywords: orientalisms, pragmatics, semantics, publicistic discourse, Russian language, media discourse, intercultural communication, cultural stereotypes.

MODERN RUS DİLİNDE ORYANTALİZMLERİN PRAGMATİK VE ANLAMSAL İŞLEVLERİ

(Özbekistan'daki Rusça Kamusal Söylem Materyalleri Üzerine)

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Özet: Bu makale, Özbekistan'daki Rusça gazetecilik söylemi temelinde, modern Rus dilinde oryantalizmlerin pragmatik ve anlamsal işlevlerini incelemektedir. Çalışmada oryantalizmlerin kültürel açıdan işaretlenmiş; adlandırıcı, anlamsal, pragmatik ve değerlendirici anlamlar taşıyan dil birimleri olarak kullanım özellikleri ele alınmaktadır. Araştırmada oryantalizmlerin Doğu kültürüne ait gerçeklikleri, gelenekleri, toplumsal kurumları ve tarihî kavramları çağdaş medya söyleminde nasıl temsil

ettiği incelenmektedir. Özellikle bu dil unsurlarının kültürel bilgileri aktarma, duygusal, çağrışımsal ve değerlendirici anlamlar oluşturma yollarına dikkat çekilmektedir. Ayrıca oryantalizmlerin okuyucunun algısını etkilemedeki ve belirli iletişimsel etkiler yaratmadaki rolü analiz edilmektedir. Çalışmada oryantalizmlerin Doğu'ya ilişkin kültürel stereotiplerin oluşturulmasına, güçlendirilmesine ve eleştirel biçimde yeniden yorumlanmasına katkısı da ele alınmaktadır. Bulgular, oryantalizmlerin yalnızca belirli kültürel unsurları adlandıran sözcükler olmadığını, aynı zamanda kültürlerarası iletişimin önemli araçları olduğunu göstermektedir. Bu birimler kültürel kimliğin korunmasına, etnokültürel bilgilerin aktarılmasına ve dil ile kültürel gelenekler arasındaki bağın güçlendirilmesine katkı sağlamaktadır. Sonuç olarak oryantalizmler, çağdaş Rusça medya söyleminde Doğu kültürünün dilsel temsilinin önemli bir unsurunu oluşturmakta ve kültürlerarası diyalogun gelişmesine önemli katkı sağlamaktadır.

Anahtar kelimeler: oryantalizmler, pragmatik, anlambilim, kamusal söylem, Rus dili, medya söylemi, kültürlerarası iletişim, kültürel stereotipler.

Introduction

In contemporary linguistics, the study of linguistic units reflecting the processes of intercultural interaction and mutual influence of different cultural traditions has gained particular relevance. One of the most significant manifestations of this process is represented by orientalisms – words and expressions associated with the historical and cultural space of the East and functioning in various types of discourse in the modern Russian language.

The intensification of intercultural contacts, the expansion of the information space, and the development of media communication contribute to the active penetration of orientalisms into the Russian-language publicistic discourse of Uzbekistan. The primary sources of linguistic material in this study are contemporary Russian-language newspapers of Uzbekistan such as *Narodnoye Slovo*, *Pravda Vostoka*, *Molodezh Uzbekistana*, *Samarkandskiy Vestnik*, *Darakchi*, independent publications such as *Nastavnik*, as well as journals including *Zvezda Vostoka*, *Russian Language and Literature at School*, *Correspondent*, and electronic media sources.

Modern orientalisms represent not only nominative units denoting specific realities of Eastern culture but also complex pragmatic-semantic constructions possessing considerable cultural and communicative potential. Their functioning is associated with the transmission of ethnocultural information, the formation of emotional and evaluative meanings, the creation of cultural associations, and the influence exerted on the recipient.

The relevance of the study is determined by the necessity of a comprehensive analysis of orientalisms as elements of contemporary media discourse simultaneously performing several functions: nominative, semantic, pragmatic, and evaluative.

The purpose of the article is to identify the pragmatic and semantic features of the functioning of orientalisms in the modern Russian language based on the material of Russian-language publicistic discourse in Uzbekistan. The study is aimed at a comprehensive examination of orientalisms as linguistic units reflecting the specificity of the national and cultural worldview and performing important communicative and pragmatic functions in publicistic discourse. The stated purpose determines the necessity of solving the following tasks: to define the main functions of orientalisms in the modern Russian language; to identify the peculiarities of their semantic structure; to analyze the pragmatic mechanisms of the functioning of orientalisms; and to examine the role of orientalisms in the formation of cultural stereotypes and intercultural communication.

Materials and methods

In modern linguistics, orientalisms are regarded as culturally marked lexical units associated with the designation of Eastern realities. Orientalisms are understood as words and expressions of Eastern origin or lexical units associated with the culture, history, traditions, and social structure of Eastern societies.

The functioning of orientalisms in language is determined by processes of intercultural interaction and historical contacts between peoples. In the Russian language, orientalisms are represented by various thematic groups: names of dishes and beverages (plov, samsa, chaikhana); designations of social institutions (mahalla); names of household items and clothing (chalma); religious and cultural concepts (Navruz, Ramadan); and historical-cultural terms (dragoman).

Orientalisms possess a complex semantic structure including not only denotative meaning but also a system of additional cultural, emotional, and evaluative components. Therefore, the analysis of orientalisms requires a comprehensive approach involving the examination of their nominative, semantic, and pragmatic functions.

The nominative function is focused on designating objects of reality; the semantic function is connected with the transmission of cultural content and additional meanings; while the pragmatic function is aimed at implementing communicative influence and shaping a particular perception of the text by the recipient.

Results and discussion

The nominative function is the basic function of orientalisms and consists in denoting specific realities of Eastern culture. In the Russian-language publicistic discourse of Uzbekistan, orientalisms are actively used to convey the national and cultural specificity of the region. Such units make it possible to accurately designate objects, phenomena, and social practices that have no complete equivalents in the Russian language. For example, the orientalisms plov, samsa, mahalla, and Navruz denote not only concrete realities but also a complex set of cultural representations associated with traditions and the lifestyle of Eastern society. Publicistic texts often contain constructions of the following type: "During the celebration of Navruz, the residents of the mahalla organized a traditional plov." In this example, orientalisms perform the function of precise cultural nomination, ensuring the preservation of the national and cultural content of the text.

Orientalisms play a particularly important role in the designation of social institutions and forms of social organization. Thus, the word mahalla denotes not merely a territorial association of residents but a specific form of traditional social structure based on neighborhood and family ties.

The semantic specificity of orientalisms determines the impossibility of their complete replacement by neutral Russian equivalents without the loss of cultural content. Their semantics are characterized by multilayeredness and a high degree of cultural markedness, including: a denotative component; a cultural-historical component; an emotional-evaluative component; and an associative component. Such multicomponent nature determines the broad possibilities of the functioning of orientalisms in the Russian-language publicistic discourse of Uzbekistan. For example, the orientalism bazar in modern Russian may be used both in its direct and figurative meanings. In the direct meaning, the word denotes an Eastern market: "The Samarkand bazar traditionally attracts a large number of tourists." However, publicistic discourse often actualizes its figurative meaning: "A real bazar started again on social media." In this case, the orientalism acquires the meaning of noisy and chaotic communication and loses its direct connection with

the trading space. Such semantic expansion indicates a high degree of adaptation of orientalisms in the modern Russian language.

Orientalisms also actively participate in creating the figurative system of the text. Thus, words such as caravan, shah, palace, and chaikhana form a stable image of the East associated with exoticism, historical antiquity, and cultural originality.

The pragmatic function of orientalisms is connected with their impact on the recipient and the formation of a particular communicative effect. Orientalisms function as instruments of emotional and evaluative influence, evoking a wide range of emotional reactions in the audience: interest, a sense of exoticism, nostalgia, cultural identification, and irony. For example: "Tourists are attracted by ancient chaikhanas, narrow streets, and noisy Eastern bazaars." In this case, orientalisms create an atmosphere of exoticism and form a romanticized image of the East. In some cases, orientalisms are used to actualize the emotional marker of cultural memory: "The smell of fresh plov reminded him of his childhood in Samarkand."

Orientalisms may also function as a means of expressing national and cultural identity. The use of such words as mahalla, Navruz, and chaikhana in Russian-language texts of Uzbekistan contributes to the preservation of ethnocultural identity and demonstrates the specificity of the local cultural space. Such units perform the function of cultural self-identification and ensure the connection between language and national tradition.

In political and publicistic discourse, where orientalisms may participate in creating both positive and negative images of the East, these lexical units are used to form a particular attitude toward the described events or phenomena.

An important feature of the functioning of orientalisms is their participation in the formation of cultural stereotypes. Publicistic discourse frequently contains expressions such as "Eastern man," "Eastern wisdom," and "Eastern hospitality." Such constructions possess a pronounced evaluative connotation and form stable representations of Eastern culture. For example: "Eastern men are traditionally perceived as bearers of patriarchal values." Such statements contribute to the consolidation of stereotypical perceptions. However, contemporary publicistic discourse often uses orientalisms for the critical analysis of stereotypes as well. A number of publications emphasize that representations of the East as an exclusively patriarchal and conservative space do not reflect actual cultural diversity. In this case, orientalisms perform a dual function: the representation of cultural stereotypes and their deconstruction and critical reinterpretation.

In contemporary media discourse, orientalisms are often used in ironic and satirical contexts. For example, publications devoted to corruption problems contain constructions of the following type: "Until you come and say 'Assalomu alaykum,' the issue will not be resolved." In this case, the orientalism is used as a cultural symbol of informal relations. Such usage creates the effect of ironic allusion and strengthens the critical tone of the text.

The conducted analysis demonstrates that orientalisms in the modern Russian language represent complex pragmatic-semantic units possessing a high level of cultural markedness.

Conclusion

The functioning of orientalisms in the Russian-language publicistic discourse of Uzbekistan indicates that these units simultaneously perform several interconnected functions: nominative – designation of Eastern cultural realities; semantic – transmission of additional

cultural meanings; pragmatic – influence on the recipient; evaluative – formation of emotional attitude; and discursive – participation in the representation and critique of cultural stereotypes.

Orientalisms constitute an important means of the linguistic representation of the East and play a significant role in the formation of intercultural dialogue in the contemporary Russian-language media space of Uzbekistan.

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PSYCHOLOGICAL AND SOCIAL FACTORS IN SECOND LANGUAGE ACQUISITION

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Abstract: This article examines the process of second language acquisition within the frameworks of psycholinguistic and sociolinguistic approaches in a comprehensive and systematic manner. The study argues that language learning is not limited to the acquisition of grammatical rules and vocabulary, but is also shaped by the individual's cognitive activity, psychological characteristics, and the influence of the social environment. The psycholinguistic approach explains how language is perceived, processed, and produced in the human brain, emphasizing the role of memory, attention, motivation, and other cognitive mechanisms. In this context, particular importance is given to the idea of innate language ability, generative grammar theory, and the interaction between language and mental structures. On the other hand, the sociolinguistic approach focuses on how language functions within society, the impact of social

factors on language, and its relationship with social identity. In multilingual and multicultural environments, issues such as language choice, code-switching, language policy, language prestige, and social status constitute the main areas of research within this approach. At the same time, regional and social language varieties, speech behavior, and the interaction between language and culture are also taken into consideration. The analysis shows that both approaches complement each other and provide a more comprehensive and objective explanation of the language learning process. For effective language acquisition, it is essential to consider both the learner's cognitive abilities and their active participation in the social environment. The integration of these approaches contributes to the development of more effective language learning strategies and the improvement of practical communication skills in modern educational contexts.

Keywords: Psycholinguistics, sociolinguistics, second language acquisition, multilingualism, language and society, cognitive processes

İKİNCİ DİL EDİNİMİNDE PSİKOLOJİK VE SOSYAL FAKTÖRLER

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Özet: Bu makalede ikinci dil edinimi süreci, psikolinguistik ve sosyolinguistik yaklaşımlar çerçevesinde kapsamlı ve sistematik bir şekilde incelenmektedir. Çalışmada dil öğrenmenin yalnızca dilbilgisi kuralları ve kelime dağarcığının edinimiyle sınırlı olmadığı; aynı zamanda bireyin bilişsel faaliyetleri, psikolojik özellikleri ve içinde bulunduğu sosyal çevrenin etkisiyle şekillendiği ortaya konulmaktadır. Psikolinguistik yaklaşım, dilin insan beyninde nasıl algılandığını, işlendiğini ve üretildiğini açıklayarak bellek, dikkat, motivasyon ve diğer bilişsel mekanizmaların rolünü ön plana çıkarmaktadır. Bu bağlamda dil yetisinin doğuştan geldiği fikri, üretici dilbilgisi kuramı ve dil ile zihinsel yapılar arasındaki ilişki özellikle vurgulanmaktadır. Diğer yandan sosyolinguistik yaklaşım, dilin toplum içindeki işleyişini, sosyal faktörlerin dil üzerindeki etkisini ve dilin sosyal kimlikle olan ilişkisini ele almaktadır. Çok dilli ve çok kültürlü ortamlarda dil seçimi, kod değiştirme, dil politikası, dilin prestiji ve sosyal statüsü gibi konular bu yaklaşımın temel inceleme alanlarını oluşturmaktadır. Ayrıca dilin bölgesel ve sosyal çeşitleri, konuşma davranışı ve dil ile kültür arasındaki etkileşim de dikkate alınmaktadır. Yapılan analiz, her iki yaklaşımın birbirini tamamladığını ve dil öğrenme sürecinin daha kapsamlı ve nesnel bir şekilde açıklanmasına katkı sağladığını göstermektedir. Etkili bir dil edinimi için bireyin bilişsel özellikleri kadar sosyal çevredeki aktif katılımının da önemli olduğu sonucuna varılmaktadır. Bu yaklaşımların bütünleştirilmesi, modern eğitim ortamlarında daha etkili dil öğrenme stratejilerinin geliştirilmesine ve iletişim becerilerinin güçlendirilmesine katkı sağlamaktadır.

Anahtar Kelimeler: Psikolinguistik, sosyolinguistik, ikinci dil edinimi, çok dillilik, dil ve toplum, bilişsel süreçler

Introduction

The process of language learning is not limited to the acquisition of grammatical rules and vocabulary; it is a highly complex, multifaceted, and multi-level cognitive and social activity. This process combines both the mental activity of an individual and their interaction with the social environment in which they are embedded. A language learner does not only learn new words and rules, but also becomes acquainted with new ways of thinking, communication strategies, and cultural patterns of behavior. In modern linguistics, language learning is no longer studied within traditional frameworks but rather within a broader context. This process is examined in relation to the individual's psychological characteristics, cognitive abilities, social environment, cultural identity, and even technological opportunities. Research conducted particularly in the field of Second Language Acquisition (SLA) shows that language learning depends not only on individual abilities but also significantly on the social environment and learning conditions (Fromkin & Rodman, 2013; Crystal, 2005). While psycholinguistics investigates mental processes such as language perception, processing, and production, sociolinguistics studies how language is used in social contexts, how it changes, and what functions it performs within society (Trudgill, 2000). These two approaches complement each other, making it possible to explain both the internal (psychological) and external (social) aspects of the language learning process. In the modern era of globalization, the increase in intercultural communication, expansion of international interaction, and the formation of multilingual societies have further increased the importance of language learning. Today, language is not only a means of communication but also a means of social identity, cultural belonging, and individual self-expression (Fishman, 1989). Through language, people not only transmit information but also express their worldview, values, and social positions.

Main part

The psycholinguistic approach associates language acquisition with cognitive processes occurring in the human brain. This approach seeks to explain how language is learned, stored, and used. The foundation of this direction is the theory of generative grammar proposed by Noam Chomsky. According to Chomsky, the human capacity for language is innate, and all languages are based on certain universal principles (Chomsky, 1965). According to this approach, children learn language not simply through imitation, but through internal language mechanisms. This indicates that language learning is closely connected to biological characteristics of humans. In the theory of generative grammar, the concepts of “deep structure” and “surface structure” play an important role. Deep structure represents the initial model formed at the level of meaning, while surface structure is the concrete linguistic form of this model (Vardar, 1989). The process of language perception occurs in multiple stages. These stages are as follows:

1. recognition of sounds and phonological analysis;
2. identification of words;
3. construction of grammatical structure;
4. comprehension of sentence meaning.

These stages operate in parallel and complement each other (Aitchison, 2012). For example, when a person hears a sentence, they simultaneously distinguish sounds, recognize words, and construct meaning. In the psycholinguistic approach, memory plays an important role. In particular, working memory allows the processing of new information in the short term. Long-term memory ensures the storage of learned words and rules. If these two memory systems

function properly, language learning becomes more effective. Motivation is also one of the main factors in language learning. Intrinsic motivation (based on interest) and extrinsic motivation (such as exams, work, etc.) can yield different results. Research shows that individuals who learn with intrinsic motivation acquire the language more deeply. In addition, emotional state also plays an important role. For example, stress and fear may weaken language learning. Therefore, in modern teaching methods, creating a comfortable and supportive environment is considered essential. Psycholinguistics is also closely related to neurolinguistics. The Broca's area of the brain is associated with speech production, while Wernicke's area is related to language comprehension. Damage to these areas leads to language disorders. This once again proves that language has biological foundations. In the modern era, psycholinguistic models are also applied in the field of artificial intelligence. For example, machine translation systems and speech recognition technologies are based on the principles of language processing in the human brain.

The sociolinguistic approach studies the functioning of language within society and its interaction with social factors. In this direction, language is evaluated not only as a means of communication but also as a reflection of social structure and cultural values (Labov, 1972). The relationship between language and society is reciprocal in nature. Social, economic, and cultural changes occurring in society affect language, and at the same time, language becomes a means of expressing these changes (Rüstəmov, 2004). Sociolinguistics has emerged at the intersection of linguistics and sociology, and research in this field studies the social functions of language, language varieties, and communicative behavior (Hymes, 1974). Especially in multilingual environments, issues such as language choice, language prestige, and language policy come to the forefront. In such environments, individuals use different languages or language varieties depending on various social situations. This leads to phenomena such as code-switching and diglossia (Fishman, 1989). Sociolinguistic processes are also actively observed in Azerbaijani society. The modern Azerbaijani language is widely used both as an official and everyday means of communication and has various regional and social varieties (Kazımov, 2008). Research conducted by William Labov has systematically explained the social differentiation of language. Labov demonstrated that the choice of language variants is related to social class, age, level of education, and other factors (Labov, 1972). On the other hand, Dell Hymes introduced the concept of communicative competence, emphasizing that language knowledge is not limited to grammatical rules but also includes the ability to use language appropriately in a social context (Hymes, 1974). Speech behavior is also an important object of study in sociolinguistics. Edward Sapir, by examining the relationship between language and culture, showed that language directly influences the way individuals perceive the world (Bates, 1976).

The sociolinguistic approach also explains the social differentiation of language—that is, the existence of dialects, social varieties, and styles. According to this approach, language is not a homogeneous system but rather a dynamic and variable structure shaped under the influence of various social factors (Bernstein, 1971). In addition, issues of language policy and language planning are considered important directions in sociolinguistics. Research in this field is related to the preservation, development, and expansion of the functional capacities of languages.

Conclusion

In conclusion, it can be noted that the process of language learning is not a simple and one-sided activity; on the contrary, it is a multifaceted and complex system. This process is closely related both to the internal cognitive mechanisms of the individual and to the social and cultural

environment in which they are embedded. For this reason, both psycholinguistic and sociolinguistic approaches are of great importance in understanding the essence of language learning. The psycholinguistic approach highlights the role of the human brain, memory, attention, and other cognitive processes in language acquisition. This approach shows that language learning is not merely mechanical memorization, but also the result of complex mental operations. Processes such as analysis, synthesis, comparison, and generalization occurring in the human brain play a key role in language perception and production. In this respect, the psychological state, level of motivation, and individual characteristics of the learner directly influence the success of the learning process. On the other hand, the sociolinguistic approach explains the formation and functioning of language within a social environment. According to this approach, language is not only an individual but also a social phenomenon. While learning a language, an individual acquires it not only as a system of grammatical rules but also as a carrier of social relations and cultural values. This shows that the language learning process cannot be fully understood outside of its social context. A comparative analysis of both approaches shows that they are not alternatives but rather complementary directions. The psycholinguistic approach explains the “internal mechanism” of language learning, while the sociolinguistic approach explains its “external environment.” The unity of these two aspects allows for a more comprehensive and objective understanding of the language learning process. In the modern education system, the integration of these approaches is of particular importance. If the teaching process is based solely on the instruction of grammatical rules, students may face difficulties in using the language in real communication. At the same time, an approach based only on practical communication may hinder the deep acquisition of the language structure. Therefore, the most effective result is achieved through the balanced application of these two directions. In addition, the development of modern technologies has created new opportunities for the language learning process. Online platforms, mobile applications, and artificial intelligence-based systems create a more effective learning environment based on both psycholinguistic and sociolinguistic principles. This creates conditions for further improvement of language teaching in the future. It should also be noted that in the context of globalization, the increase in multilingualism makes language learning even more актуал. By learning different languages, people not only expand their communication opportunities but also gain the ability to understand different cultures more deeply. In this regard, language learning is not only an academic activity but also a means of cultural and social development. Finally, it can be concluded that for successful language learning, the unity of psychological readiness, social environment, motivation, and appropriate teaching methods is essential. The integration of psycholinguistic and sociolinguistic approaches allows for a more efficient organization of this process and the achievement of more successful outcomes. Future research on the deeper interaction between these two directions can make a significant contribution to the development of new methods in linguistics and education.

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MULTILINGUALISM AND SECOND LANGUAGE ACQUISITION (SLA)

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Abstract: The article analyses the strategic importance of multilingualism in the modern world and examines the psychological, methodological and social mechanisms of second language acquisition. Language is not a universal phenomenon but always carries national character, serving as a cultural memory of a people. Since over 60% of the world's population uses two or more languages in the XXI century, SLA has become an intersection of psychology, sociology and neuroscience. The study discusses four main approaches in the field of Second Language Acquisition: the behaviorist approach views language as a habit; the cognitive approach sees the brain as a “rule-processing computer”; the sociocultural approach, based on Vygotsky's “Zone of Proximal Development”, emphasizes interaction;

and the neurolinguistic approach explores brain-language cooperation through MRI research. From a psycholinguistic perspective, five patterns are identified for SLA: comprehension and production, the mental lexicon network, working memory, first language transfer, and emotion. It is particularly emphasized that language is not merely vocabulary and grammar but a speech activity and must be learned in the unity of "language+culture".

In conclusion, effective second language acquisition in Azerbaijan requires teacher-student cooperation, positive emotion, and the internalization of cultural codes. Learning is language at an early age provides children with a new worldview and cognitive flexibility.

Key words: multilingualism, SLA, cognitive approach, cultural code mental lexicon, critical period, lexical units.

ÇOXDİLLİLİK VƏ İKİNCİ DİLİN MƏNİMSƏNİLMƏSİ

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Xülasə: Müasir dövrdə çoxdilliliyin strateji əhəmiyyətini və ikinci dilin mənimsəməsini psixoloji, metodiki və sosial mexanizmlər təhlil edir. Dil ümumbəşəri fonemen olmayıb, daim milli səciyyə daşıyır və xalqın mədəni yaddaşı kimi çıxış edir. XXI əsr əslində dünya əhalisinin 60%-dən çoxu iki və artıq dildən istifadə etdiyindən, ikinci dilin qavranılması psixologiya, sosiologiya və neyro-elmin kəsişmə nöqtəsinə çevrilmişdir. Tədqiqatda Second Language Acquisition sasının dörd əsas yanaşması şərh olunur: biheviörist yanaşma dil vərdişi kimi, koqnitiv yanaşma "qayda emal edən komputer" kimi, sosial mədəni yanaşma Vıqotskinin "Yaxın inkişaf Zonası" əsasında qarşılıqlı əlaqə kimi, neyrolinqvistik yanaşma isə MRT tədqiqatlarına əsaslanan beyin-dil əməkdaşlığı kimi qeyd edilir.

Psixolinqvistik baxımından ikinci dilin mənimsənilməsi üçün beş əsas qanunauyğunluq müəyyən edilmişdir: qavrayış və istehsal, mental lüğət şəbəkəsi, işçi yaddaş, ana dilinin transferi və emosiya. Xüsusilə vurğulanır ki, dil yalnız lüğət və qrammatika deyil, nitq fəaliyyətidir və dil+ mədəniyyət vəhdətinə öyrənilməlidir.

Nəticədə, Azərbaycanda ikinci dilin effektiv qavranılması üçün müəllim-tələbə əməkdaşlığı, pozitiv emosiya və mədəni kodların mənimsənilməsi vacib şərt kimi irəli sürülür. Erkən yaşda dil öyrənmək, uşağa yeni dünya görüşü və koqnitiv çeviklik qazandırır.

Açar sözlər: Çoxdillilik, SIA, koqnitiv, yanaşma, mədəni kod, mental lüğət, kritik dövr, leksik vahidlər.

Introduction

In the 21st century, over 60% of the global population uses two or more languages. Language is not merely a means of communication but a cultural memory that carries national identity. For Azerbaijan, multilingualism is of particular relevance. The aim of the research: To analyze the psychological and social mechanisms of second language acquisition.

Methodology and Materials

The research is based on a theoretical-analytical method. Four approaches in the field of Second Language Acquisition have been analyzed: behaviorist, cognitive, socio-cultural, and neurolinguistic. Five psycholinguistic principles were taken as the basis: comprehension and production, mental lexicon, working memory, first language transfer, and emotion.

Results and Discussion

The analysis shows that learning a language is not only about vocabulary and grammar but a speech activity. According to L.V. Shcherba, vocabulary is "language material," while language itself is formed through speech. According to F. de Saussure, the speech event creates the evolution of language. Based on E. Lenneberg's "critical period" theory, learning a language at ages 2-3 is more effective. The teacher's positive emotion directly affects the student's language acquisition.

Conclusion

Learning a second language means acquiring cultural codes.

The unity of "language + culture" is essential.

Learning a language at an early age creates cognitive flexibility.

Teacher-student cooperation and an emotional environment are crucial for effective teaching.

The article analyzes the strategic importance of multilingualism in the modern world and examines the psychological, methodological, and social mechanisms of second language acquisition. Language is not a universal phenomenon but always carries a national character, serving as the cultural memory of a people. Since over 60% of the world's population uses two or more languages in the 21st century, SLA has become an intersection of psychology, sociology, and neuroscience. The study discusses four main approaches in the field of Second Language Acquisition: the behaviorist approach views language as a habit; the cognitive approach sees the brain as a "rule-processing computer"; the sociocultural approach, based on Vygotsky's "Zone of Proximal Development", emphasizes interaction; and the neurolinguistic approach explores brain-language cooperation through MRI research. From a psycholinguistic perspective, five patterns are identified for SLA: comprehension and production, the mental lexicon network, working memory, first language transfer, and emotion. It is particularly emphasized that language is not merely vocabulary and grammar but a speech activity and must be learned in the unity of "language + culture". In conclusion, effective second language acquisition in Azerbaijan requires teacher-student cooperation, positive emotion, and the internalization of cultural codes. Learning a language at an early age provides children with a new worldview and cognitive flexibility.

In the 21st century, over 60% of the global population uses two or more languages. Language is not merely a means of communication but a cultural memory that carries national identity. For Azerbaijan, multilingualism is of particular relevance.

The aim of the research: To reveal the strategic importance of multilingualism in the modern world and to analyze the psychological, methodological, and social mechanism of second language acquisition.

Language does not exist on Earth as an abstract, universal thing among people. It differs from many social phenomena in that it is never universal but always carries a national character. Language exists as the language of various peoples and nations, serving as a means of

communication and thought expression. When we say "language," specific languages such as Azerbaijani, Russian, and English come to mind.

The purpose of this article is to reveal the strategic importance of multilingualism in the modern world and to analyze the psychological, methodological, and social mechanism of second language acquisition. Multilingualism is a very interesting topic both scientifically and socially. In the 21st century, almost more than 60% of the world's population uses two or more languages. It is known that when starting to learn any language, people first use the vocabulary and grammar of that language. It is incorrect to think that language consists of vocabulary, new words, and grammar. This is absolutely not the case. To understand a language, one must be able to speak it and understand what is said. Language is speech activity.

However, according to L.V. Shcherba's terminology, this is not language but "language materials," which are written and fixed in dictionaries and grammar reference books, sometimes not precisely. Currently, second language acquisition lies at the intersection of psychology, sociology, and economics.

It would be appropriate to emphasize the idea that "language is a window to see the world." Language is a form of capital that expands an individual's cognitive abilities and enriches their worldview. For our country, this issue is of particular relevance. In our society, many languages are active alongside the native language. The question of how to acquire a second language concerns us greatly.

The research is based on a theoretical-analytical method. Second Language Acquisition is a field that emerged at the intersection of linguistics and psychology. There are 4 main approaches in this field:

Behaviorist approach - views language as a habit. The grammar-translation method is a product of this approach. This method provides knowledge about language but does not form the "ability to use language."

Cognitive approach - sees the brain as a "rule-processing computer," i.e., information enters the brain, is processed, and emerges as behavior. The cognitive approach is a field that focuses on how humans perceive the world, process information, and think.

Socio-cultural approach - based on Vygotsky's "Zone of Proximal Development" theory. Language is a social phenomenon and is acquired only through interaction. This approach explains an individual's behavior, psychology, and development not only by internal factors but also by the influence of the society and culture they belong to.

Neurolinguistic approach - this is more necessary and is based on MRI studies. Neurolinguistics is an interdisciplinary science that studies how language is represented in the human brain. This means how the brain learns, speaks, and understands language. Studies show that second language acquisition is a collaboration between grammar books and the brain. Psycholinguistics proves that the brain naturally acquires language at any time.

Psycholinguistics studies how language enters the brain, how it is stored there, and how it is converted into speech. There are 5 main psycholinguistic principles for second language:

Comprehension and Production - these two systems are different from each other. Although a child understands and learns many words, they can give few responses.

Lexicon - is retained by the "network" principle. A person who repeats any word many times in context becomes more active than those who memorize words. Lexicon studies how words are represented, stored, and retrieved in the human brain. The concept of the "Mental

Lexicon" stands at the center of this field. The organization of the Mental Lexicon explains how our brain manages thousands of words in an orderly and rapid manner. It is a constantly updating neural network.

Working memory - this is an essential area in the field of psycholinguistics. It is the process of encoding, storing, and recalling information by the brain. This approach allows for building more effective methodological strategies for lessons.

First language transfer

Emotion - governs cognition. It is a person's internal subjective attitude towards events occurring in the surrounding world, other people, and their own behavior. Emotion is the body's response reaction. If a teacher enters a lesson with high energy and positive emotion, the students' brains and state automatically copy them.

For second language acquisition in Azerbaijan, psychology, language culture, and courage are necessary. As a result of mutual understanding between teacher and student, we can decide: "I learn a language to bring the world into my home." Speech is the main form of approach to language. Language functions through speech and has been directly transformed into speech.

Language is a means of communication. Human life is impossible without it. From this, we can conclude that speech is fundamental to learning a language. Language can be learned by studying speech factors. If the vocabulary of a language remained constant, it could not develop. Through speech, new words, expressions, phrases, and sentences emerge in language. Language development is determined by speech.

According to F. de Saussure, the speech event created conditions for language evolution. New changes occurring in speech destroy old regularities in language, and these changes pass into language. Therefore, neologisms appear in language. Thus, learning a second foreign language is not just memorizing new grammatical rules, but acquiring the way of thinking, worldview, and cultural behavior patterns of that people.

Learning a second foreign language at an early age in modern times is very important from pedagogical and psychological perspectives. Children should be accustomed to learning these languages from 2 and 3 years old. Although this age is difficult for them, speaking and the environment have a positive effect. Even if it is difficult to respond, they understand beautifully and try to perceive naturally.

According to E. Lenneberg's theory, a child unconsciously acquires language at this age. They learn only through hearing. They are not afraid of making mistakes and learn by making mistakes. Learning a second language at an early age is not only thought of as a means of communication. It also increases their thinking ability and worldview.

The expression "The more languages you know, the more human you are" is very appropriate, and this expression forms a mutual unity with children's early language.

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THE LINGUISTIC AND CULTURAL SIGNIFICANCE OF NAMES OF POSITIVE CHARACTERS

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Abstract: Onomastics is the collection of proper names in a language and a branch of linguistics that studies proper names. These names are encountered in everyday life, in historical examples of oral folk literature, as well as in written literary works. In these sources, the primary stylistic and grammatical forms of onomastic units are revealed. It is also possible to determine the purpose served by an onomastic unit used in these sources, as well as the historical period and stage it reflects.

Onomastics is also based on folklore traditions. Tradition is a set of customs, moral norms, worldviews, aesthetic preferences, and similar elements that have historically emerged and are transmitted from generation to generation. One type of tradition is related to folklore. The term "folklore" is of English origin (folk-lore) and refers to oral folk creativity, encompassing people's customs, traditions, songs and other cultural expressions.

Folklore traditions also give rise to the language of folklore. A significant part of the vocabulary of folklore language consists of specialized lexicon, which is primarily onomastic in nature. The study of the onomastics of positive characters, in turn, makes it possible to clarify a number of hidden aspects of the history of Turkic people, including structural-systematic, linguopsychological and linguo-cultural dimensions.

Keywords: onomastics, folklore, onomastic field, onomastics of positive characters, folklore tradition, structural-system approach.

MÜSBƏT OBRAZLARIN DİL VƏ MƏDƏNİYYƏT BAXIMINDAN ƏHƏMİYYƏTİ

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Xülasə: Onomastika dildə olan xüsusi adların toplusu, linqvistikanın xüsusi adları araşdıran bölməsidir. Bu adlara gündəlik həyatımızda, tarixən şifahi xalq ədəbiyyatı nümunələrində, eləcə də yazılı ədəbiyyatımızda rast gəlirik. Bu qaynaqlarda onomastik vahidlərin ən birinci üslubi-qrammatik formaları açılır. Bu qaynaqlarda işlənən onomastik vahid hansı məqsədə xidmət edir və hansı tarixi dövrü və mərhələni əks etdirir.

Onomastika folklor ənənələrinə də söykənir. Ənənə tarixən meydana gələrək nəsildən-nəslə keçən adətlər, əxlaq normaları, görüşlər, zövqlər və s. məcmusudur. Ənənə növlərindən biri də folklorla əlaqədardır. «Folklor» ingilis mənşəli sözdür (folk-lore) şifahi xalq yaradıcılığı, hər hansı bir xalqın adət, ənənə, mahnı və s. toplusudur.

Folklor ənənələri folklor dilini də doğurur. Folklor dilinin lüğət tərkibinin böyük bir qismini xüsusi leksika təşkil edir. Xüsusi leksika əsasən onomastik leksikadır. Müsbət obraz onomastikasının tədqiqi işə

öz növbəsində türk xalqlarının tarixinin bir sıra gizli məqamlarını (struktur-sistem, linqvopsixoloji, linqvokulturoloji və s.) məqamları da aydınlaşdırmağa imkan verir.

Açar sözlər : onomastika, folklor, onomastik sahə, müsbət obraz onomastikası, folklor ənənəsi, onomastika və sistem-struktur yanaşma.

Introduction

This branch of onomastics is referred to linguistics as “onomastics of fiction,” “poetic onomastics,” “poetic onomalogy,” “stylistic onomastics,” “expressive onomastics,” etc. Onomastics is the collection of proper names in a language and a branch of linguistics that studies proper names. These names are encountered in our everyday life, in historical examples of oral folk literature, as well as in our written literature. In these sources, the primary stylistic-grammatical forms of onomastic units are revealed. It is examined for what purpose an onomastic unit used in these sources serves and which historical period and stage it reflects. Thus, onomastics presents itself as one of the factors reflecting the historical past and everyday life of a people.

Onomastics is also based on folklore traditions. Tradition is a set of customs, moral norms, views, tastes, etc. that historically emerged and are transmitted from generation to generation. One type of tradition is associated with folklore. “Folklore” is a word of English origin (folklore), oral folk creativity, a collection of people’s customs, traditions, songs, epics etc.

The beautiful, unique folklore traditions also exist in Azerbaijan. These are distinctive national traditions. These traditions, being defined in society, life, everyday life, etc., are stable rules, methods, norms. These have long become tradition. Folklore tradition has been created over a long period because of the artistic genius, artistic thinking, and enthusiastic creative labor of several generations.

Folklore traditions also give rise to the language of folklore. A large part of the vocabulary of folklore language consists of specialized lexical units. Special lexicon is mainly onomastic lexicon. In general, by studying folklore onomastics (anthroponyms and toponyms), it is possible to determine different layers of the onomastic system of modern Turkic languages, and to restore the original forms of onomastic units in different Turkic languages. The study of the onomastics of positive characters, in turn, allows the clarification of a number of hidden aspects of the history of Turkic peoples (structural-system, psycholinguistic, linguocultural, etc.).

The mentioned tradition above is also preserved by the world of characters that has taken a worthy place in the people’s oral creativity. At the center of each folklore example—epics, tales, songs, bayatis, etc.—live human characters stand. These characters are divided into positive and negative. These characters also form onomastic units consisting of personal names. These are called anthroponyms (human names, nicknames, pseudonyms, etc.). In our research, we have set as an important goal the study of the specific features of the onomastics of positive characters within folklore tradition.

Materials and Methodology

Materials

The research material consists primarily of Azerbaijani folkloric sources, especially the Book of Dede Korkut, as well as folklore genres such as tales, epics, songs and bayatis. In addition, relevant studies devoted to folklore onomastics, anthroponymy and literary onomastics have been used as theoretical sources.

Methodology

In order to describe the onomastics of positive characters in detail, it is first necessary to define a number of onomastic principles. An onomastic principle consists of the main propositions, starting points of theory, science, doctrine, belief, worldview about onomastics. In linguistics, the main approaches to onomastic principle are:

System-structural approach in defining the onomastic field;

Linguoculturological approach in defining the onomastic principle;

Psycholinguistic approach in defining the onomastic principle.

These approaches show that, as with other principles, onomastic principles are also used in the comprehensive study of proper names in ancient folklore texts. With the help of this principle, the concept, status, and functions of the proper name used in folklore are analyzed through a filter of analysis. In addition, the structural-semantic, psycholinguistic, and cognitive-linguoculturological features of these names (positive character names) are studied. These features are centered in the onomastic field.

The methodological framework of the study is based on descriptive, structural-semantic, comparative, linguocultural, and psycholinguistic methods of analysis.

Discussion

The concept of the onomastic field plays an important role in understanding the functioning of positive character names in folklore texts. Under this concept is understood “a certain limited sphere of social life,” and character onomastics is accepted as a constructive unit. Here, the onomastic norm is determined precisely in relation to the social field. Each socio-onomastic character has its own specific features in the linguistic and non-linguistic plan of the field it is covered by. With such an approach to onomastic field and norm, it becomes possible to reach such a regular conclusion that the rejection of accepted norms about any characters in any social field, whether positive or negative character, leads to stylistic selection of the personal name.

Modality conditioned by violation of usage norms of anthroponyms as a character in a certain social field is more clearly manifested in lexical expression.

In this case: a) various forms of naming that do not always correspond to speech etiquette are used; b) one form of the name is functionally replaced by another; c) accompanying words are omitted; this also results in a change of the accepted form of address in relation to any character.

It can also be said that the content of various forms of positive and negative characters is completely dependent on a broad text and is concrete separately.

Systematic and structural approach is used in defining the onomastic field. The terminological unit “field” was first used by G. Ipsen in 1924. He understood by field words belonging to a certain semantic group. Ideas and concepts about field were also developed by L. Weisgerber and S. Eman. They determined the linguistically different segmentation of the world between European languages and Indian language. Among the defined fields, “onomastic field” was also different. J. Trier imagined the concept of field as a transition from one situation to another. Therefore, his theory allows the observation of the sequence of mutually replacing systems.

In our view, “onomastic field” is a part of “lexical field.” When distinguishing that field from the conceptual field, J. Trier thinks that every pronounced word (name) has a corresponding

meaning. At the same time, a number of semantic shades of a word (proper name) may not serve as a basis for its connection with a certain concept. Other words (names) surrounding this word (name) may be close or distant from the corresponding concept. Thus, words (proper names) enter into relation with each other and create a row (sequence), which is called lexical field or language field (or sign field).

Assigning a proper name to a denotatum is called onomastic field, and the scope of the ideal aspect (meaning in the word, meaning reflecting relation of word with object) is considered semantic field, which is divided into semantic fields (or zones).

For example, the “onomastic field” of “Book of Dede Korkut”: In our opinion, this field should be studied in relation to material world objects. That is, these proper names should study spiritual culture phenomena and their manifestation possibilities in language. At the center of the onomastic field also stands a concept or semantic dominant. That concept in essence consists of a set of semantic qualities. The main issue is to explain the meaning of onomastic names involved in the epic in relation to meaning of other words and names. In this case, as researchers say, “language not only reflects the ‘national spirit’ of a people, but also plays the role of a force that forms its ‘worldview’.”

The epic “Book of Dede Korkut” is a more detailed source in studying onomastic units. In the field of belles-lettres and folklore onomastics research has been conducted by scholars T. Hajiyeve, A. Gurbanov, A. Javadov, F. Khaligov, D. Aliyeva, L. Piriyeve, I. Veliyeva, F. Baghirova and others.

T. Hajiyeve’s idea is appropriate here: “Layers of personal names correspond to stages of historical development of people.”

In general, onomastic character in Azerbaijani folklore tradition constitutes a complex but sufficiently organized system. Within this tradition, the total of anthroponymic layers reflected in folklore sources is understood. Since this issue is serious and relevant, the study of formation and usage-functioning features of anthroponymics of the intended period and stage is required to be approached as a system.

Research shows that without an excursion into an onomastic character, it is not possible to clarify a number of historical and theoretical issues of Azerbaijani folklore examples and some difficult problems that require solution.

In literature, onomastics is mainly studied in several aspects: 1) in folklore, 2) in written literature, 3) in dictionaries. The use of names as onomastic units in folklore reveals historical facts related to names. Folklore has its own specificity, antiquity, stylistic uniqueness and linguistic symbolism. Personal names also create a system here. Stylistic shades created by these names are interesting. Here positive characters can play the role of conveying historical idea, strengthening national color, creating other characters vividly, and valuable language and speech unit.

The term “positive onomastic character” is encountered in a number of works. (Якубовский А.Ю.1962, Югай И.Г., Л.А.Пак. 1987, Калинин В.М.2000, Горбоноевский М.В.1988, Ализаде С.1989, Verdiyeva Z.N. 1988, Tanrıverdi Ə. 1999)

Based on “Book of Dede Korkut” (Kitabi-Dədə Qorqud. 1999, 168 p.), a number of linguistic interpretations of positive onomastic character have been reflected in some studies. But structural-system, linguocultural, cognitive, as well as psycholinguistic features of these characters have not been sufficiently addressed. (Mikayılova Ə., 2008.)

In folklore works and modern literary works, a certain event and action develop from uncertainty to certainty. The character is not presented to the reader suddenly. Observations show that the character seems to be hidden behind the general course of events. The succession of actions gradually reveals the character and its identity, ideology, etc.

The role of onomastic character in the process of understanding the world, as well as cognition, conceptualization of knowledge in this system (onomastic system, onomastic field), methods and means of describing them; formation of linguistic knowledge about onomastic character in the process of receiving, processing and transmitting information about the world, and finally, linguistic picture of the world.

Onomastic character has a special role in formation of linguistic picture of the world. From this point of view, onomastics is a "mirror of culture." M. Adilov states that one of the means used by the author for revealing characters, giving inner world of character, is the name given to the character. Names are connected with position of character in the work, and through these names different positive and negative aspects are generalized in literary work.

In Azerbaijani worldview, unlike English worldview, "wisdom" is associated with Dede Korkut. High appreciation of Dede Korkut as a wise person is result of high value given to wisdom. In Azerbaijani people's imagination, real wise people are compared with Dede Korkut and placed at his level.

M. Ergin writes that oral literature, along with other traditions of the people, aims to connect naming tradition with ancient times and preserve it. "Book of Dede Korkut" epic was written down in X–XI centuries, but it is not convincing that it fully reflects onomastic naming system of that period. Some of these names belonged to Oghuz tribes, and it is understood that heroes represented Oghuz tribes.

In the "epic," character names also become a means of revealing their character. There is a close connection between the people's attitude toward the character of Dede Korkut and the significance of his name. Stylistic essence of personal name is serving for its idea and goal.

Positive characters create associations, i.e., become symbolic and metaphorical. In metaphorization, positive or negative qualities of the bearer play main role. For example, in "Koroğlu" epic, Rustam-Zal. In Kazakh language, Korkut baba is associated with sadness and sorrow expressed through sound of kobyz and sorrow of people. Image of folk bard becomes a general noun and includes soft speech, supplication images. Expressive connotation formed in speech act is additional information reflecting national-cultural features and social belonging of a person. Dede Korkut as an anthroponym creates positive association. These features appear in the inner world of characters.

Heroes, as positive characters, are portrayed as possessing both inner and outer beauty. High qualities belonging to wisdom are collected in image of Dede Korkut. Attitude of people and epic creator toward image, formation of character under this name looks natural. People created these images with love and affection.

Comparative analysis of characters in folklore, attitude of people toward both characters, Dede Korkut attracts attention as symbol of wisdom and struggle in society. "Subjective-aesthetic considerations about individual stylistic features of character creator should be evaluated in a broader context from point of view of literary movements and individual styles" (Виноградов В.В., 1959, p.96)

In accordance with Azerbaijani folklore tradition, each person does not address those older than themselves by personal names according to age. This is a sign of taboo (prohibition).

Connotative meanings (additional meanings) of personal names in folklore are hidden, but language speakers understand them.

“In solving a number of problematic issues related to certain stages of people’s development, especially ancient and early periods, history, ethnography, ethnogenesis, historical geography, etc., onomastic facts and positive character onomastics are also brought forward and highly valued as reliable sources. Therefore, onomastic units serve as witnesses to and carriers of a number of historical processes. They, as spiritual monuments embodying this field and transmitted from generation to generation, have preserved valuable and irreplaceable information about pre-written period of linguistic features of ethnoses settled in a concrete area” (Xaliqov F. 2009, p.11, 242 p.)

Psycholinguistic approach is also used in defining onomastic principle. That is, onomastics included in semantic field can also be interpreted psychologically. A person who has forgotten a relevant name tries to recall it again by listing concept names belonging to that field.

According to systemic-structural approach in defining onomastic field, various principles have been used to study different areas of vocabulary (including onomastic field) structurally: in folklore, positive character onomastics is grouped according to “semantic field”, whereas ononyms are grouped synchronically and diachronically.

Nevertheless, none of these principles could fully and systematically cover onomastic units. These structural-semantic descriptive methods are usually based only on materials of one language or do not cover all “fields” and their interrelations. Therefore, first of all, the intellectual sphere is quite optimal, i.e., convenient.

The analysis also demonstrates that positive character names should be approached from an emotional-evaluative perspective. Folklore creators actively employ lexical and grammatical resources to express attitudes toward characters and to reveal their inner qualities. Through these linguistic means, positive characters become powerful symbols of social ideals and moral values.

Conclusion

The study of positive character onomastics demonstrates that anthroponyms in folklore are not merely nominative units but important carriers of historical, cultural, and linguistic information. The names of positive characters reflect the worldview, value system and cultural memory of the people.

Evidence suggests that folklore and literary texts make extensive use of the lexical and grammatical potential of names. From this point of view, possibilities of positive character names are broader and play an important role in short and concrete expression of people’s thoughts in folklore works. People have used different stylistic-grammatical forms of characters for creating an impression about them. In order to characterize names, reveal inner world, especially to show people’s or surrounding people’s attitude toward it, a number of forms are considered necessary.

Therefore, positive onomastics character represents an important area of research within folklore studies and onomastics, offering valuable insights into the interaction between language, culture and collective memory.

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MULTILINGUALISM AND SECOND LANGUAGE ACQUISITION.

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Abstract: Multilingualism is not just a fringe occurrence anymore but is now a crucial part of modern societies. It influences how people learn and communicate and form identity. This article looks at the active link between multilingualism and second language learning (SLA) based on recent studies in language study and psychology and education. It suggests that multilingual people do not just pick up languages separately. Their language systems interact and can both help and limit new language learning. The study reveals how knowing previous languages affects thinking flexibility and understanding language structure and learning methods. These often give multilingual learners an edge when learning

more languages. But mixing languages and social and cultural factors like language attitudes and education access and language rules also shape results. The article also investigates how classroom methods and digital spaces are adjusting to more varied language backgrounds.

The piece argues that looking at SLA with a multilingual perspective questions traditional single-language ideas in education and policy. It suggests using teaching methods which see linguistic diversity as an advantage and not a problem. This approach places multilingualism in the forefront of future research and practice in language learning.

Key words: Multilingualism, Second Language Acquisition (SLA), cross-linguistic influence, metalinguistic awareness, cognitive flexibility, language transfer, sociocultural factors, language policy

МНОГОЯЗЫЧИЕ И ОСВОЕНИЕ ВТОРОГО ЯЗЫКА

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Аннотация: Многоязычие уже не является чем-то редким, а стало важной частью современного общества. Оно влияет на то, как люди учатся, общаются и формируют свою идентичность. В данной статье рассматривается активная связь между многоязычием и изучением второго языка (SLA) на основе недавних исследований в области лингвистики, психологии и педагогики. В ней высказывается предположение, что многоязычные люди не просто осваивают языки по отдельности. Их языковые системы взаимодействуют между собой и могут как способствовать, так и ограничивать изучение нового языка. Исследование показывает, как знание предыдущих языков влияет на гибкость мышления, понимание языковой структуры и методы обучения. Это часто дает многоязычным учащимся преимущество при изучении новых языков. Однако на результаты также влияют смешение языков и социально-культурные факторы, такие как отношение к языку, доступ к образованию и языковые нормы. В статье также исследуется, как методы обучения в классе и цифровые пространства адаптируются к более разнообразному языковому контексту.

В статье утверждается, что рассмотрение SLA с многоязыковой точки зрения ставит под сомнение традиционные одноязычные представления в образовании и политике. В ней предлагается использовать методы обучения, которые рассматривают языковое разнообразие как преимущество, а не как проблему. Такой подход ставит многоязычие на первый план будущих исследований и практики в области изучения языков.

Ключевые слова: многоязычие, приобретение второго языка (SLA), межъязыковое влияние, металингвистическое сознание, когнитивная гибкость, языковой перенос, социокультурные факторы, языковая политика

Introduction

In the current globalized world with porous boundaries and advanced information technology, speaking two or more languages is not a peculiarity anymore. It is becoming an ordinary phenomenon in big cities' schools and even small settlements where migrants and modern technologies influence people's lifestyle significantly. Nevertheless, there is an intricate process that accompanies the development of this tendency – acquisition of new languages by humans and its reflection in their brain functions.

The current article investigates the issue of second language acquisition in the context of multilingual experience and discusses recent scientific studies concerning multilinguals' brain functioning. Contrary to the former approaches that consider languages as isolated systems, the author describes their interaction inside the human's mind. Moreover, the text emphasizes positive aspects of acquiring two or more languages as well as certain difficulties related to it.

It should be stated that the issue is relevant because of the rapid increase of the diversity of schoolchildren, which becomes a challenge for the contemporary education system. Therefore, the problem under investigation is crucial since it is necessary to understand how to use the acquired knowledge of multilinguals efficiently in educational practices.

Materials and Methods

This study explores the relationship between multilingualism and second language acquisition. To do so, the study employs a mixed-methods design to investigate the interaction between participants' linguistic background, self-evaluation of linguistic competence, and second language production. The participants in this study consisted of 120 high school and university students and adults aged 16 to 30, and they were enrolled in language courses that included two languages, in language-impact programs, in language-maintenance programs, in heritage-language courses, or in bilingual or multilingual communities where two or more languages were used as primary means of communication on a regular basis. The sample of participants was classified into bilingual (L2 near-native speaking competence) and multilingual (L3/L4 speaking competence above basic) groups to distinguish among levels and types of language proficiency.

Three kinds of data were collected. Firstly, a quantitative questionnaire with demographic, language history and self-assessment measures of L2 proficiency was administered to the 160 participants. Secondly, the participants completed 900 word version of the paper-based version of the VikIT vocabulary test (see also VikIT- listening), a number of grammaticality judgment tasks assessing present perfect and perfect tenses, and some speaking, multiple choice listening and audio-script multiple choice listening tasks aimed at testing participants' ability to utilise L2 for communicating information. Thirdly, 30 in-depth interviews with the participants were recorded to investigate learners' bilingual experiences, strategies and attitudes towards multilingualism.

The study also involved classroom observations of English language classrooms in selected colleges and schools. Observational data on how teachers and students interact, instances of code-switching, and various language use were recorded and analyzed to see how these practices may influence English language learning.

The quantitative data collected for this study were analysed using statistical tests to investigate potential relationships between individuals' multilingual experience and second language acquisition outcomes. The qualitative data from in-depth interviews and observations were examined in a thematic manner in order to identify a range of emerging trends associated with different aspects of the second language acquisition process (cognitive, social, educational) within both controlled and naturalistic learning settings and in relation to multilingual experience.

Discussion

This study explores the relationship between multilingualism and second language acquisition from the learners' perspective. The findings support previous research in finding that multilingual individuals can draw on prior linguistic knowledge to facilitate their second

language acquisition by increasing their metalinguistic awareness and strategic learning behaviour. However, the study also shows that there is no guarantee of such advantages, and that cross-linguistic interference in both similarity and distance can occur.

While it is hazardous to make generalisations across the diverse range of settings in which bilingual children learn to read and write, qualitative findings suggest that it is in those contexts most favourable to the use of all languages which the children are able to utilise their full linguistic potential most effectively. Findings also suggest that settings that unwittingly uphold a monolingual dominant discourse, can stifle the potential of multilingual children by failing to make use of their existing linguistic knowledge and ability. In this report we explore the implications of these findings for educational practice and broader issues of a multilingual socioculture.

Our results cast significant doubt on the notion that learners start “from scratch” when approaching a second language. Instead, evidence suggests that each language learners add changes the way they think and how they communicate in profound ways, in complex interactions with other languages they know. Our findings thus encourage and require a range of new instructional strategies and policies that respect, build on, and take advantage of individual linguistic diversity.

In conclusion, multilingualism should not be treated as a problem that needs to be solved or a variable that needs to be controlled in language development, but as a source of strength, a normal feature of language development, and the reality of global mobility and digital communication, to be mirrored in language education systems.

Conclusion

This article explores the role of multilingualism in second language acquisition in a significant and varied number of aspects. Multilingualism is not merely additive, in that individuals acquire additional languages and utilise these new multilingual competences to enhance language understanding, learning strategies and overall second language acquisition and use.

But benefits of multilingualism depend on context too. Multilingual children thrive in some educational settings more than others, due to the quality of teaching, the culture and the community. Ultimately, it all comes down to language support, and whether children are given the chance to harness their skills properly.

This book foregrounds the issue of multilingualism in second language acquisition. In both theory and practice, the recognition of multilingualism is a move in the right direction. With the development of more inclusive and more adaptive approaches to second language acquisition, this book presents a collection of papers which argue that multilingualism is an integral part of language learning.

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LANGUAGE LEARNING STRATEGIES IN MULTILINGUAL CONTEXTS

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Abstract: This study explores language learning strategies in multilingual contexts, focusing on how learners acquire and manage multiple languages simultaneously. In today's globalized world, multilingualism has become increasingly common, and learners are required to adapt to different linguistic environments. The research highlights cognitive, metacognitive, and social strategies used by multilingual learners. It also examines the benefits of multilingualism, such as improved cognitive flexibility and cultural awareness, as well as challenges including language interference and cognitive overload. The findings suggest that effective use of learning strategies significantly enhances language acquisition in multilingual settings.

Rod Ellis (1994) notes that the process of language learning is shaped by the interaction of individual and social factors, and a multilingual environment makes this process more complex. In this regard, learners can more deeply master grammar and vocabulary by making comparisons between different languages.

Furthermore, Vivian Cook (2001) proposes the concept of “multi-competence” of multilingual individuals, stating that knowing several languages increases mental flexibility and facilitates switching between different languages. Jasone Cenoz (2013) emphasizes that multilingual education develops cultural awareness and enables learners to better integrate into the global world. Research findings show that multilingualism enhances cognitive flexibility, problem-solving skills, and cultural understanding in learners. However, it also poses challenges such as language interference, limited practice opportunities, and cognitive overload.

Overall, the research shows that properly selected and balanced language learning strategies are the basis for successful language acquisition in a multilingual environment, and these strategies have a positive impact on both the academic and social development of learners.

Key words: multilingualism, language learning strategies, cognition, communication, language acquisition

ÇOXDİLLİ KONTEKSTLƏRDƏ DİL ÖYRƏNMƏ STRATEGİYALARI

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Xülasə: Bu tədqiqat çoxdilli mühitdə dil öyrənmə strategiyalarını araşdırır və öyrənənlərin eyni anda bir neçə dili necə mənimsədiyini izah edir. Müasir qloballaşmış dünyada çoxdillilik geniş yayılıb və öyrənənlər müxtəlif dil mühitlərinə uyğunlaşmalıdır. Tədqiqatda koqnitiv, metakoqnitiv və sosial strategiyalar təhlil olunur. Eyni zamanda çoxdilliliyin üstünlükləri (zehni elastiklik, mədəni biliklər) və çətinlikləri (dil qarışıqlığı və s.) araşdırılır. Nəticələr göstərir ki, düzgün strategiyalar dil öyrənməni daha effektiv edir.

Rod Ellis (1994) qeyd edir ki, dil öyrənmə prosesi fərdi və sosial faktorların qarşılıqlı təsiri nəticəsində formalaşır və çoxdilli mühit bu prosesi daha kompleks edir. Bu baxımdan, öyrənənlər müxtəlif dillər arasında müqayisə apararaq qrammatika və leksikani daha dərinlən mənimsəyə bilirlər.

Bundan əlavə, Vivian Cook (2001) çoxdilli fərdlərin “multi-competence” anlayışını irəli sürərək bildirir ki, bir neçə dil bilmək zehni elastikliyi artırır və fərqli dillər arasında keçidi asanlaşdırır. Jasone Cenoz (2013) isə çoxdilli təhsilin mədəni şüuru inkişaf etdirdiyini və öyrənənlərin global dünyaya daha yaxşı inteqrasiya olmasına şərait yaratdığını vurğulayır.

Tədqiqatın nəticələri göstərir ki, çoxdillilik öyrənənlərdə koqnitiv elastiklik, problem həll etmə bacarığı və mədəni anlayışı gücləndirir. Lakin eyni zamanda dil qarışıqlığı (interference), məhdud praktika imkanları və koqnitiv yüklənmə kimi çətinliklər də mövcuddur.

Ümumilikdə, araşdırma göstərir ki, düzgün seçilmiş və balanslı şəkildə tətbiq olunan dil öyrənmə strategiyaları çoxdilli mühitdə uğurlu dil mənimsənilməsinin əsasını təşkil edir və bu strategiyalar öyrənənlərin həm akademik, həm də sosial inkişafına müsbət təsir göstərir.

Açar sözlər: çoxdillilik, dil öyrənmə strategiyaları, koqnitivlik, ünsiyyət, dilin əldə edilməsi

Introduction

First and second language acquisition are fundamentally different in several ways. First language acquisition, or L1, begins naturally from birth within an immersive, informal setting like the family home. Acquiring a first language comes naturally – where exposure to it is constant and learning occurs subconsciously. First language acquisition also follows a biological and cognitive timeline, with children typically progressing through universal stages of language development without formal instruction.

However, second language or L2 might begin at any age after the first language has been established. Second language learning often occurs in more structured environments such as classrooms or possibly if someone is living in a foreign country. Motivation for L2 learning can vary, ranging from practical needs to cultural assimilation (such as attending school in the States). Achieving native-like proficiency is influenced by factors such as the age of learning a new language, exposure quality, and individual motivation.

Additionally, the social context, including community attitudes and the learner's integration, significantly impacts second language learning, unlike the typically supportive environment of first language acquisition.

Assessing individual differences in students' learning is a prominent area of interest in second language acquisition. Individual differences refer to the more-or-less enduring psychological characteristics that distinguish one person from another (Baumeister & Vohs, 2007). Factors such as age, motivation, and learning strategies can influence learners' development in another language. According to Oxford (2017), strategies refer to setting goals, determining actions to achieve the goals, and mobilizing resources to execute the actions. Learning strategies are defined as "specific actions, behaviors, steps, or techniques –such as seeking out conversation partners or giving oneself encouragement to tackle a difficult language task –used by students to enhance their own learning". Language learning strategies refer to "the processes and actions deployed by language learners to learn or use a language more effectively" or the thoughts and actions operationalized by language learners to carry out "a multiplicity of tasks from the very outset of learning to the most advanced level of target language performance".

Methodology and Materials

This study uses a qualitative approach based on secondary sources such as academic articles, books, and online materials. Various theories related to language learning strategies were analyzed to understand how multilingual learners acquire languages. The research focuses on three main types of strategies:

Cognitive strategies

Metacognitive strategies

Social strategies

These strategies were examined in different multilingual learning contexts, including classroom environments and independent learning situations.

The data analysis was conducted through thematic analysis. This method involves identifying recurring themes and concepts within the selected literature. By organizing the findings into categories, the study aims to present a clear and structured understanding of how different strategies function in multilingual contexts.

Although this research provides valuable insights, it has certain limitations. Since it relies on secondary data, it does not include direct observation or experimental evidence. Future research could incorporate quantitative methods or case studies to provide more comprehensive results.

Overall, this methodology allows for a detailed exploration of language learning strategies and highlights their importance in multilingual environments.

Discussion

Metacognition, the awareness and regulation of one's own learning process, is a cornerstone of effective language learning. This element is a ground-breaking text that offers a comprehensive guide to incorporating metacognitive strategies into the teaching of reading, writing, vocabulary, and listening. This element stands as a bridge between theoretical frameworks and actionable teaching practices, enabling educators to enhance their students' language proficiency in a holistic manner. This element is replete with case studies, examples from diverse learning contexts, and evidence-based practices. It is an invaluable resource for

language educators who aspire to cultivate independent learners capable of self-assessment and strategy adjustment. By fostering metacognitive awareness across all facets of language learning, this element empowers students to take charge of their own learning journey, leading to more profound and lasting language mastery.

Cognitive strategies are based on self-regulation theories of behavior. Cognitive strategies aim to assist learners with scaffolding and the mental structuring of information. Terms used to describe cognitive strategies include procedural facilitators (Bereiter & Scardamalia, 1987), procedural prompts (Rosenshine & Meister, 1997), and scaffolds (Palincsar & Brown, 1984). Typically, cognitive strategies include orienting, repeating (repetition), organizing information, taking systematic notes, summarizing meaning, guessing meaning from context, reasoning inductively and deductively, and using imagery for memorization. These cognitive strategies can help learners to guide their goal-directed behavior. Cognitive strategies are often used for cognitive behavior therapy to direct learners' attentional focus.

Social strategies

Language learning is both an intellectual and social process as well as an element of social behavior used for communication between people. Thus, it is helpful to understand the importance of social strategies for language learning. Social strategies are helpful for learners to achieve more effective language learning and understand the culture of the language they are learning. Examples of social strategies include asking questions for clarification or confirmation, asking for help, and learning about social or cultural norms. Even though some researchers tend to downplay social strategies (O'Malley & Chamot, 1990), social strategies can also be important for communicative language learning. Oxford (1990) delineates three sets of social strategies: asking questions for clarification or verification and correction; cooperating with others (e.g., with peers and proficient users of the new language) and empathizing with others to develop cultural understanding and becoming aware of others' thoughts and feelings.

Conclusion

In conclusion, language learning in multilingual contexts represents a multifaceted and dynamic process that requires learners to actively engage with a variety of linguistic systems. This study has demonstrated that the effective use of cognitive, metacognitive, and social strategies plays a fundamental role in facilitating successful language acquisition. Cognitive strategies support comprehension and retention through processes such as analysis, repetition, and comparison, while metacognitive strategies enable learners to regulate their own learning through planning, monitoring, and evaluation. Social strategies, on the other hand, provide opportunities for interaction, communication, and real-life language use, which are essential for developing fluency and confidence.

Furthermore, the findings highlight that multilingualism is not only a linguistic ability but also a cognitive and cultural asset. Individuals who operate in multilingual environments tend to exhibit enhanced cognitive flexibility, stronger problem-solving skills, and a greater capacity for adapting to new situations. In addition, multilingual learners develop intercultural competence, which allows them to understand and appreciate different perspectives, traditions, and communication styles. These qualities are increasingly valuable in a globalized world where cross-cultural interaction is a fundamental part of everyday life.

Despite these significant advantages, multilingual language learning is not without its challenges. Learners often experience language interference, where elements of one language

influence another, leading to errors or confusion. Additionally, maintaining proficiency in multiple languages requires continuous practice and exposure, which may not always be available in all learning environments. Cognitive overload is another important issue, as managing several languages simultaneously can place a heavy demand on memory and attention. These challenges underline the importance of strategic learning and effective guidance.

In order to overcome these difficulties, it is essential for learners to adopt a strategic and personalized approach to language learning. This includes selecting appropriate strategies based on their individual goals, proficiency levels, and learning preferences. Consistent practice, exposure to authentic language materials, and active engagement in communication are key factors in achieving long-term success. At the same time, educators play a crucial role in supporting multilingual learners by designing inclusive curricula, encouraging interaction, and integrating modern technologies into the learning process.

Moreover, technological advancements have opened new opportunities for multilingual education. Digital tools, online platforms, and language learning applications can provide learners with access to diverse linguistic resources and interactive learning environments. These tools can enhance motivation, provide instant feedback, and support independent learning, making them valuable additions to traditional teaching methods.

Finally, future research should explore how different language learning strategies can be applied more effectively in real-life and classroom settings, particularly in diverse multilingual environments. It would also be beneficial to investigate the long-term impact of multilingualism on cognitive development and academic achievement.

Overall, multilingualism should not be viewed as a challenge to overcome, but rather as a powerful resource that enriches individuals both intellectually and socially. By applying appropriate strategies and maintaining a consistent learning approach, learners can successfully navigate the complexities of multilingual contexts and achieve high levels of language proficiency.

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YOUTH AND THE SOCIO-CULTURAL IMPACT OF GLOBAL ENGLISH

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Abstract: Global English has become one of the most influential linguistic phenomena of the modern world. Its rapid spread across continents has significantly affected not only communication but also social, cultural, and cognitive dimensions of human life. Among all social groups, youth are particularly sensitive to these transformations, as they actively engage with digital technologies, global media, and transnational communication platforms.

This article explores the socio-cultural impact of Global English on young people from a pragmatic perspective. It examines how English functions not only as a means of communication but also as a tool for identity construction, cultural exchange, and social mobility. The study also highlights the dual nature of this influence: while Global English provides access to global opportunities, it may simultaneously pose challenges to local languages and cultural identities.

Special attention is given to the role of social media, education, and popular culture in shaping linguistic preferences among youth. The article argues that the widespread use of English leads to the emergence of hybrid linguistic forms and new communicative practices. These transformations require careful analysis within the frameworks of sociolinguistics and pragmatic linguistics.

Key words: Global English, youth, socio-cultural impact, pragmatics, identity, globalization

GƏNCLƏR VƏ QLOBAL İNGİLİS DİLİNİN SOSIAL-MƏDƏNİ TƏSİRİ

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Xülasə: Qlobal İngilis dili müasir dünyanın ən təsirli dil fenomenlərindən birinə çevrilmişdir. Onun sürətli yayılması yalnız ünsiyyətə deyil, həm də insan həyatının sosial, mədəni və idrak sahələrinə əhəmiyyətli təsir göstərmişdir. Bütün sosial qruplar arasında gənclər bu dəyişikliklərə daha həssasdır, çünki onlar rəqəmsal texnologiyalar, qlobal media və beynəlxalq kommunikasiya platformalarında aktiv iştirak edirlər.

Bu məqalədə qlobal İngilis dilinin gənclər üzərində sosial-mədəni təsiri pragmatik yanaşma əsasında araşdırılır. İngilis dilinin yalnız ünsiyyət vasitəsi deyil, həm də identiklik quruculuğu, mədəni mübadilə və sosial mobillik üçün bir vasitə kimi rolu təhlil olunur. Tədqiqat bu təsirin ikili xarakterini vurğulayır: qlobal İngilis dili geniş imkanlar yaratsa da, eyni zamanda yerli dillər və mədəni kimliklər üçün müəyyən çətinliklər yarada bilər.

Məqalədə sosial media, təhsil və populyar mədəniyyətin gənclərin dil seçimlərinə təsiri xüsusi diqqət mərkəzindədir. İngilis dilinin geniş istifadəsi hibrid dil formalarının və yeni kommunikasiya modellərinin yaranmasına səbəb olur. Bu dəyişikliklər sosiolinqvistika və pragmatik dilçilik çərçivəsində ətraflı təhlil tələb edir.

Açar sözlər: qlobal İngilis dili, gənclər, sosial-mədəni təsir, pragmatika, identiklik, qloballaşma

Introduction

In the era of globalization, language has become a central element in shaping not only social interaction but also cultural exchange, knowledge production, and identity formation. Among all languages, English has acquired a unique and dominant status as a global lingua franca. Its influence extends far beyond traditional communication, penetrating various domains

such as education, technology, business, and popular culture. This expansion has been particularly evident among young people, who are both the primary consumers and producers of globalized content.

Youth represent the most dynamic and transformative segment of society. They are characterized by their openness to innovation, high adaptability, and active engagement with global digital networks. Due to their constant interaction with international platforms, young people are more exposed to linguistic changes than older generations. Their linguistic behavior reflects broader socio-cultural transformations, making them a crucial focus of contemporary linguistic research.

This study aims to analyze how Global English influences youth from both socio-cultural and pragmatic perspectives. It explores the complex interaction between language, culture, and identity in a globalized world, emphasizing that language is not merely a communication tool but also a powerful social and symbolic resource.

Materials and Methods

Theoretical Background. From a sociolinguistic perspective, language is not simply a system of grammatical structures and vocabulary, but a social practice deeply embedded in cultural, political, and ideological contexts. Language both reflects and constructs social reality. Therefore, the global spread of English cannot be understood without considering the socio-cultural environments in which it operates.

Pragmatic linguistics, on the other hand, focuses on how meaning is constructed in context. It emphasizes speaker intention, situational factors, and shared knowledge in communication. From this viewpoint, the use of English among youth is not accidental but contextually motivated and strategically employed to achieve specific communicative goals.

Scholars such as David Crystal and Braj Kachru have extensively studied the global expansion of English. Crystal emphasizes the role of historical, economic, and technological factors in the spread of English, while Kachru introduces the concept of “World Englishes,” highlighting the diversity and localization of English in different regions. This perspective challenges the idea of a single standard English and instead promotes the recognition of multiple legitimate varieties.

Moreover, contemporary researchers such as Alastair Pennycook and Jennifer Jenkins argue that Global English is not a fixed entity, but a dynamic and evolving phenomenon shaped by transnational flows and intercultural communication. This reinforces the need for an interdisciplinary approach combining sociolinguistics, pragmatics, and cultural studies.

Results and Discussion

Youth and Language Choice in the Global Era. Young people today are exposed to English through multiple channels, including formal education systems, entertainment industries, and digital communication platforms. Social media applications such as Instagram, TikTok, and YouTube play a particularly significant role in shaping linguistic preferences and spreading English expressions across different cultural contexts.

In addition to passive exposure, youth actively engage in content creation, often using English as a medium to reach a broader audience. This active participation accelerates the diffusion of English and contributes to the emergence of new linguistic trends.

The preference for English among youth can be explained by several interrelated factors:

Access to global information: English serves as the primary language of the internet, academic resources, and international media.

Prestige and symbolic capital: Proficiency in English is often associated with intelligence, modernity, and higher social status.

Educational and professional opportunities: English opens doors to international education, scholarships, and global job markets.

Participation in global culture: English enables young people to engage with global music, films, fashion, and online communities.

As a result, many young individuals adopt English not only as a second language but also as an integral part of their personal and social identity. In some cases, English becomes a marker of group belonging, especially within urban and digitally connected communities.

Socio-Cultural Impact of Global English (Expanded)

1. Identity Construction

Language plays a fundamental role in shaping both individual and collective identity. The use of English allows young people to construct hybrid identities that combine local cultural elements with global influences. By using English, they position themselves as members of a global community.

However, this process is not without tension. The adoption of English may create a sense of cultural displacement or identity conflict, especially when local languages and traditions are perceived as less prestigious. This duality reflects the complex nature of globalization, where integration and fragmentation occur simultaneously.

2. Cultural Exchange and Transformation

Global English facilitates the rapid exchange of cultural values, ideas, and practices across borders. Young people can easily access and participate in diverse cultural experiences, leading to increased intercultural awareness and understanding.

At the same time, this process may result in cultural homogenization, where dominant global cultures overshadow local traditions. The widespread consumption of English-language media can influence lifestyle choices, values, and even thinking patterns among youth.

3. Hybridization and Linguistic Innovation

One of the most visible effects of Global English is the emergence of hybrid linguistic forms. These include code-switching, borrowing, and the creation of new expressions that blend English with local languages.

For example:

Mixing English vocabulary into Azerbaijani sentences

Using English slang in informal communication

Adapting English grammatical structures to local contexts

These practices reflect linguistic creativity and adaptability. Rather than indicating language loss, they demonstrate the dynamic nature of language in a globalized world. Hybrid forms serve as a bridge between cultures and enable more flexible communication.

Pragmatic Perspective (Expanded)

From a pragmatic standpoint, language use is always context-dependent. The choice to use English is influenced by factors such as audience, purpose, and social setting. Young people often switch between languages depending on the situation, demonstrating high levels of pragmatic competence.

English is frequently used to:

Express modernity, innovation, and global awareness

Signal membership in international or elite communities

Achieve specific communicative goals, such as persuasion or self-presentation

For instance, in social media interactions, English phrases may be used to appear more trendy or to align with global communication norms. In academic settings, English is often preferred for its perceived precision and universality.

Pragmatic analysis reveals that language choice is not random but strategic. It reflects speakers' intentions and their awareness of social meanings attached to different languages.

Challenges and Risks (Expanded)

Despite its numerous advantages, the dominance of Global English raises several important concerns:

Decline of local languages: Younger generations may gradually lose proficiency in their native languages.

Cultural identity erosion: Excessive reliance on English can weaken connections to local traditions and heritage.

Linguistic inequality: Not all individuals have equal access to English education, creating social disparities.

Over-standardization: Global norms may limit linguistic diversity and creativity.

These challenges highlight the need for balanced language policies that promote both global communication and the preservation of linguistic diversity. Educational systems should encourage bilingualism or multilingualism rather than replacing local languages with English.

Conclusion

Global English has a profound socio-cultural impact on youth, influencing their language use, identity, and worldview. From a pragmatic perspective, this influence is shaped by contextual and communicative factors.

While Global English offers significant opportunities, it also poses challenges that require careful consideration. Future research should focus on developing strategies to preserve linguistic diversity while embracing global communication.

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EARLY LANGUAGE LEARNING: ADVENTAGES AND CHALLANGES

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Abstract: Early language learning has become an important topic in modern education, as many parents and teachers believe that children should start learning foreign languages at a young age. This idea is supported by the fact that young learners are generally more flexible and can develop better pronunciation and communication skills over time. In addition, early exposure to a new language can improve cognitive development and help children become more confident and open to different cultures.

However, this process is not always easy. Some children may struggle with maintaining attention, and in certain cases, learning multiple languages at once can cause confusion if not properly guided. Lack of motivation and ineffective teaching methods can also make the learning process less successful.

Although there are some challenges, the benefits of early language learning are more significant. If teachers use appropriate methods and create a supportive and engaging environment, children can overcome these difficulties. In conclusion, early language learning plays a key role in a child's overall development and should be encouraged in a balanced and thoughtful way.

Key words: Child development, Early exposure benefits, Early childhood education

ERKƏN YAŞDA DİL ÖYRƏNMƏ: ÜSTÜNLÜKLƏR VƏ ÇƏTİNLİKLƏR

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Xülasə: Erkən yaşda dil öyrənmə müasir təhsildə mühüm mövzulardan birinə çevrilmişdir, çünki bir çox valideyn və müəllim uşaqların xarici dilləri kiçik yaşlardan öyrənməsinin vacib olduğunu düşünür. Bu fikir onunla əsaslandırılır ki, uşaqlar daha çevik olur və zamanla daha yaxşı tələffüz və ünsiyyət bacarıqları inkişaf etdirə bilirlər. Bundan əlavə, erkən yaşda yeni dilə məruz qalmaq uşaqların zehni inkişafını gücləndirir və onların fərqli mədəniyyətlərə qarşı daha açıq və özünə inamlı olmalarına kömək edir.

Lakin bu proses hər zaman asan olmur. Bəzi uşaqlar diqqəti uzun müddət saxlamaqda çətinlik çəkə bilər və bəzi hallarda eyni anda bir neçə dil öyrənmək düzgün istiqamətləndirilmədikdə qarışıqlıq yarada bilər. Motivasiya çatışmazlığı və uyğun olmayan tədris üsulları da öyrənmə prosesini daha az effektiv edə bilər.

Müəyyən çətinliklərə baxmayaraq, erkən dil öyrənmənin üstünlükləri daha əhəmiyyətlidir. Müəllimlər düzgün metodlardan istifadə etdikdə və dəstəkləyici, maraqlı mühit yaratdıqda uşaqlar bu çətinliklərin öhdəsindən gələ bilirlər. Nəticə olaraq, erkən yaşda dil öyrənmə uşağın ümumi inkişafında mühüm rol oynayır və balanslı və düşünülmüş şəkildə təşviq olunmalıdır.

Açar sözlər: Uşaq inkişafı, Erkən məruz qalmanın faydaları, uşaqlıq təhsili

Introduction

Early language learning has become one of the most influential and widely discussed areas in contemporary education, reflecting the changing demands of a rapidly globalizing world. In today's society, communication across cultures is no longer an optional skill but an essential requirement for personal, academic, and professional success. As international interaction continues to expand, the ability to understand and use multiple languages is increasingly viewed as a fundamental competence rather than a supplementary ability. For this reason, educational

systems across the world are placing greater emphasis on introducing foreign languages at earlier stages of childhood development.

The concept of early language learning is based on the idea that childhood represents a unique developmental period in which the human brain demonstrates exceptional flexibility and adaptability. During these early years, children are naturally more receptive to new sounds, structures, and patterns of language. Unlike adult learners, they tend to acquire linguistic elements in a more intuitive and less analytical manner, often without conscious effort. This natural capacity makes early childhood a critical phase for language acquisition, where exposure to new linguistic environments can lead to long-lasting developmental benefits (Krashen, 1982).

In addition to linguistic advantages, early exposure to foreign languages is closely linked to cognitive development. Research in educational psychology suggests that learning multiple languages at a young age can enhance memory, improve attention span, and strengthen problem-solving skills. It also supports the development of mental flexibility, allowing children to switch between tasks and perspectives more efficiently. These cognitive benefits highlight that language learning is not limited to communication alone but also contributes to broader intellectual growth (Vygotsky, 1978).

Furthermore, early language learning plays a significant role in shaping social and cultural awareness. By being introduced to different languages and cultural contexts, children begin to develop a deeper understanding of diversity and global interconnectedness. This early exposure encourages openness, tolerance, and respect for other cultures, which are essential qualities in today's multicultural societies. As a result, language education becomes not only a linguistic process but also a means of fostering intercultural competence.

However, despite its clear importance, early language learning remains a complex educational process that requires careful planning and appropriate implementation. The effectiveness of early exposure depends on various factors, including teaching methods, learning environment, and the level of support provided to learners. Without these essential conditions, the potential benefits may not be fully realized, and children may face unnecessary challenges in their learning journey.

Overall, early language learning represents a powerful educational approach that combines linguistic, cognitive, and social development. It has the potential to shape not only how children communicate but also how they think, interact, and understand the world around them. For this reason, it continues to be an important focus in modern educational research and practice.

Methods and Materials

This study adopts a qualitative research design based on a literature review approach to examine early language learning, including its advantages and challenges. The main purpose of this methodology is to analyze and interpret existing academic studies, theoretical frameworks, and educational reports in order to gain a deeper understanding of how early exposure to foreign languages affects child development. Instead of collecting primary data, the study relies on previously published research and synthesizes their findings.

A qualitative approach is considered appropriate for this topic because early language learning involves complex cognitive, social, and educational processes that cannot be fully explained through numerical data alone. Vygotsky's Sociocultural Theory (1978) emphasizes the importance of social interaction and cultural context in learning, which supports the use of

interpretive analysis. In addition, Krashen's Input Hypothesis (1982) highlights the role of meaningful language exposure in natural acquisition, particularly in early childhood.

The materials used in this research include peer-reviewed journal articles, academic books, and reports from international organizations such as UNESCO. These sources were selected based on their reliability, relevance, and contribution to the field of language education. Additional information was gathered from academic databases such as Google Scholar, focusing on studies related to bilingual education, language acquisition, and cognitive development in young learners. The collected materials were analyzed thematically, meaning that key ideas and patterns were identified across different studies rather than focusing on numerical data. Many researchers agree that early language exposure improves pronunciation, memory retention, and communication skills. However, some studies also highlight challenges such as possible language confusion, lack of structured teaching methods, and differences in learning pace among children.

A comparative analysis was also applied to examine different theoretical perspectives. While Krashen (1982) supports natural language acquisition through exposure, other approaches emphasize structured instruction and guided practice. Comparing these views allows for a more balanced understanding of early language learning.

The methodology of this study is grounded in qualitative literature analysis supported by established theories and credible academic sources. This approach provides a comprehensive and balanced perspective on both the benefits and challenges of early language learning.

Results and Discussion

The findings of this study indicate that early language learning plays a crucial role in both cognitive and linguistic development. The analysis of existing literature shows a strong agreement among researchers that children who are exposed to foreign languages at an early age tend to develop better pronunciation, stronger memory skills, and higher levels of communicative competence. This supports the idea that early childhood is a sensitive period for language acquisition, during which the brain is more adaptable and responsive to new linguistic input.

At the same time, the discussion reveals that early language learning is not without challenges. Some studies highlight that children may experience confusion when exposed to multiple languages without proper guidance. Additionally, the effectiveness of early language learning largely depends on the quality of teaching methods and the learning environment. Without structured instruction, the potential benefits may not be fully achieved. This suggests that early exposure alone is not sufficient; it must be supported by appropriate pedagogical strategies.

Another important point identified in the literature is the role of motivation and emotional engagement. Research suggests that children learn more effectively when language instruction is interactive, playful, and meaningful. This aligns with modern educational approaches that emphasize learning through experience rather than memorization. Therefore, teacher involvement and classroom atmosphere are key factors influencing the success of early language programs.

Furthermore, there is a slight difference in opinion among researchers regarding the balance between natural acquisition and formal instruction. While some scholars, such as Krashen (1982), emphasize the importance of natural exposure, others argue that structured

teaching is necessary to ensure accuracy and long-term proficiency. The comparison of these perspectives suggests that a balanced approach combining both methods may be the most effective solution.

Overall, the discussion demonstrates that early language learning offers significant developmental advantages, but its success depends on careful implementation, appropriate teaching strategies, and supportive learning environments. Both benefits and challenges must be considered in order to maximize its effectiveness in educational practice.

Conclusion

To conclude, early language learning can be regarded as a fundamental element of contemporary education, shaped by the demands of an interconnected and multilingual world. The growing importance of global communication has made the ability to understand and use more than one language not only an academic achievement but also a necessary life skill. Within this context, early exposure to foreign languages provides children with valuable opportunities to develop linguistic competence alongside cognitive and social growth.

The overall analysis of research suggests that early childhood represents a particularly sensitive and effective period for language acquisition. During this stage, children tend to learn more naturally, respond more easily to new sounds and structures, and build communication skills with greater flexibility. These characteristics make early language learning a powerful tool for supporting long-term educational development and enhancing a child's ability to adapt to different cultural and social environments.

At the same time, it is clear that early language learning cannot be effective in isolation. Its success depends heavily on the quality of teaching methods, the structure of learning activities, and the emotional and educational support provided to learners. When these conditions are not adequately met, the potential benefits may be reduced, and the learning process may become less efficient. This highlights the importance of thoughtful planning and well-designed educational strategies in early language programs.

Beyond linguistic development, early language learning also contributes to shaping broader skills such as critical thinking, adaptability, and cultural understanding. It helps children develop a more open perspective toward diversity and encourages them to communicate with confidence in different contexts. These outcomes suggest that language learning at an early age is not only about acquiring vocabulary or grammar but also about forming essential life competencies.

Ultimately, although certain challenges exist, the overall evidence strongly supports the value of early language learning as a long-term investment in a child's development. When implemented in a supportive and well-structured environment, it can create a strong foundation for future academic success, personal growth, and effective communication in a multilingual world.

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THE ROLE OF LANGUAGE IN SHAPING CULTURAL IDENTITY

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Abstract: This study examines the role of language in the formation and preservation of cultural identity. Language functions not only as a means of communication but also as a fundamental cultural code reflecting the history, collective memory, and value system of a society. It plays a crucial role in shaping individual, national, and ethnic identity. While globalization increases communication through dominant languages like English, it also risks marginalizing minority languages. UNESCO describes this as a reduction in linguistic diversity and a challenge to cultural heritage. Protecting linguistic diversity is essential for cultural sustainability.

Keywords: language, cultural identity, globalization, linguistic diversity, minority languages, communication, cultural heritage, linguistic preservation, identity formation.

DİLİN MƏDƏNİ KİMLİYİN FORMALAŞMASINDA ROLU

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Xülasə: Bu tədqiqat işi dilin mədəni kimliyin formalaşması və qorunub saxlanılmasında oynadığı fundamental rolunu geniş şəkildə təhlil edir. Dil təkcə insanlar arasında informasiya mübadiləsi vasitəsi kimi çıxış etmir, həm də cəmiyyətin tarixi təcrübəsini, kollektiv yaddaşını, mənəvi dəyərlər sistemini və ənənələrini özündə yaşadan əsas mədəni kod hesab olunur. Bu baxımdan dil, fərdi kimliklə yanaşı, milli və etnik özünüdərkin formalaşmasında da mühüm rol oynayır və cəmiyyətin özünüifadə formalarından biri kimi çıxış edir. Qloballaşma ingilis dili kimi dominant dillər vasitəsilə ünsiyyəti artırırsa da, eyni zamanda azlıqların dillərinin marginallaşması riskini yaradır. UNESCO bunu linqvistik müxtəlifliyin azalması və mədəni irs üçün bir çağırış kimi qiymətləndirir. Linqvistik müxtəlifliyin qorunması mədəni davamlılıq üçün zəruridir." linqvistik məsələ deyil, həm də mədəni davamlılığın təmin olunması üçün vacib sosial və elmi vəzifə kimi ortaya çıxır.

Açar sözlər: dil, kimlik, qloballaşma, linqvistik müxtəliflik, azlıqların dilləri, ünsiyyət, mədəni irs, linqvistik qorunma, şəxsiyyətin formalaşması

Introduction

Language is one of the most essential aspects of human life. It functions as a medium of communication and a repository of cultural knowledge. Each language carries unique meanings and reflects the worldview of its speakers (Wardhaugh, 2010). Cultural identity is formed through shared traditions, beliefs, and historical experiences. Language plays a central role in preserving and transmitting these elements across generations (Hall, 1997).

In modern society, globalization and technological advancements have significantly influenced language use. These changes have created both opportunities for communication and risks for cultural preservation.

The purpose of this study is to analyze the relationship between language and cultural identity and to highlight the importance of protecting linguistic diversity.

Methodology and Materials

This research is based on a qualitative analysis of academic literature related to linguistics, sociolinguistics, and cultural studies. In the course of the research, various scholarly sources were reviewed to investigate the mechanisms through which language influences both individual and collective identity. Comparative analysis was extensively used to examine different linguistic communities and their unique cultural characteristics.

Furthermore, the study integrates theoretical frameworks and previous research findings to provide a comprehensive and nuanced understanding of the topic. Within the scope of this research, the impact of globalization on the hierarchical structure of languages was evaluated from a sociolinguistic perspective. The analysis also utilizes UNESCO reports on linguistic diversity and international conventions as a fundamental methodological basis.

Discussion

Language is far more than a neutral tool for information exchange; it serves as a primary marker of social and cultural boundaries. Through language, individuals not only communicate but also actively perform their identity. Cultural expressions, such as idioms and proverbs, act as “linguistic fossils” that carry the philosophical outlook of previous generations into the present. When a person speaks their native language, they simultaneously activate a network of historical and emotional associations that reinforce their sense of belonging to a specific group. In this sense, language serves as an “internal compass” for cultural orientation.

Moreover, the dual nature of globalization presents a significant contemporary challenge. On one hand, the dominance of English facilitates global dialogue; on the other hand, it creates a “linguistic hierarchy” where smaller languages are pushed to the periphery. This phenomenon often leads to “subtractive bilingualism,” where the acquisition of a dominant second language comes at the expense of the mother tongue. In the digital age, social media trends accelerate the borrowing of foreign lexical units, potentially leading to the erosion of native linguistic nuances. As highlighted by Crystal (2000), the concern regarding “language death” is not just about the loss of words, but the disappearance of unique ways of thinking.

Furthermore, the survival of cultural identity depends on the ability to integrate traditional linguistic values with modern digital platforms. While technology can be perceived as a threat, digital archiving and language-learning applications provide new tools for endangered languages to thrive. However, for a language to remain a core part of identity, it must be functional in prestigious domains such as higher education and administration. National language policies should aim not only at preservation but also at ensuring the language remains relevant to the

needs of the 21st century. Language should be understood not only as a symbolic system of communication but also as a dynamic and evolving social institution that actively constructs and reconstructs cultural identity over time. It functions as a medium through which power relations, social hierarchies, and ideological frameworks are both expressed and maintained. In this regard, language is deeply intertwined with issues of identity, authority, and cultural continuity.

From a sociolinguistic perspective, language operates as a key mechanism for inclusion and exclusion within social groups. The use of a particular language or dialect can signal belonging to a specific community, while at the same time distinguishing individuals from others. This process reinforces social cohesion but may also contribute to linguistic discrimination or marginalization, especially in multilingual societies where dominant languages overshadow minority ones.

In addition, language plays a crucial role in shaping cognitive processes and perception of reality. According to linguistic relativity theories, the structure and vocabulary of a language influence how its speakers conceptualize the world around them. Therefore, the loss of a language does not merely mean the disappearance of a communication tool, but also the erosion of a unique worldview and intellectual framework developed over generations.

The impact of globalization on language is complex and multifaceted. While it promotes intercultural communication and economic integration, it simultaneously accelerates linguistic homogenization. The widespread use of global languages, particularly English, creates a tendency toward cultural standardization, which may weaken local identities. This is particularly evident among younger generations, who increasingly adopt hybrid linguistic forms influenced by digital media and global popular culture.

Moreover, digital transformation has introduced both risks and opportunities for linguistic diversity. On one hand, the dominance of global languages on the internet limits the visibility of smaller languages. On the other hand, digital platforms offer innovative tools for language preservation, such as online dictionaries, virtual learning environments, and social media communities that encourage the use of native languages. This dual effect highlights the importance of strategic language planning in the digital era.

Another important aspect is the role of education systems in maintaining linguistic identity. Mother-tongue-based education not only enhances cognitive development but also strengthens cultural awareness and self-identification. When individuals are educated in their native language, they are more likely to preserve and transmit their cultural heritage. Conversely, the absence of native language education may lead to cultural alienation and identity fragmentation.

Finally, effective language policy is essential for balancing globalization and cultural preservation. Governments and institutions must adopt inclusive and sustainable strategies that support multilingualism while ensuring that minority languages remain functional and relevant in modern society. This includes integrating local languages into official domains, media, and technological platforms.

Conclusion

The findings of this study reaffirm that language is not merely a tool for preserving and shaping cultural identity but the bedrock of a society's collective memory and intellectual heritage. The loss of a language is far more than a linguistic event; it is an existential one, signifying the irreversible disappearance of a specific cultural worldview and a mode of thought refined over millennia. This research demonstrates that in the face of the homogenizing pressures

of globalization, maintaining linguistic diversity is vital for both individual and collective self-awareness.

Based on the analysis presented in this study, it can be concluded that academic discourse alone is insufficient to ensure the sustainability of linguistic diversity. It is imperative for states to expand mother-tongue-based educational programs, increase the presence of local languages on digital platforms, and encourage active language use at the community level.

Ultimately, every language must be recognized as an indispensable part of humanity's shared cultural treasury. The protection of language should be viewed as a profound responsibility to pass on a richer, more vibrant, and multifaceted historical identity to future generations. In conclusion, this study highlights that language should be viewed not merely as a communicative instrument, but as a living and evolving carrier of cultural meaning and identity. It embodies the historical experiences, intellectual traditions, and value systems of societies, functioning as a bridge between past, present, and future generations. Therefore, the preservation of language is inseparable from the preservation of cultural continuity and social cohesion.

The analysis also demonstrates that the pressures of globalization, while facilitating interaction and connectivity, contribute to the gradual dominance of certain languages at the expense of others. This imbalance creates a risk not only for linguistic diversity but also for the survival of distinct cultural perspectives. In this context, the decline of a language should be understood as the loss of an entire system of knowledge, symbolic expression, and collective memory.

Furthermore, the study emphasizes that safeguarding linguistic diversity requires a multidimensional approach. It is not sufficient to rely solely on theoretical discussions or academic research. Practical measures must be implemented at institutional, societal, and individual levels. Educational systems play a decisive role in this process by promoting mother-tongue instruction and fostering cultural awareness. At the same time, the integration of local languages into digital environments is essential to ensure their relevance in contemporary communication.

In addition, language policies should move beyond preservation-oriented frameworks and focus on revitalization and active usage. Languages can only survive if they remain functional in everyday life, professional domains, and technological contexts. This requires long-term strategic planning and cooperation between governments, educational institutions, and communities.

Ultimately, linguistic diversity should be regarded as a shared global resource that enriches human civilization. Protecting and promoting languages is not only a cultural obligation but also an ethical responsibility toward future generations. Ensuring that diverse languages continue to exist means safeguarding the plurality of human thought, creativity, and identity in an increasingly interconnected world.

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PHONETIC CHALLENGES IN TEACHING AZERBAIJANI IN MULTILINGUAL ENVIRONMENTS

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Abstract: This article analyzes phonetic and prosodic differences arising in the teaching of the Azerbaijani language in multilingual classrooms. The study is based on the author's long-term teaching experience and identifies the main challenges encountered by learners from different linguistic backgrounds. The findings indicate that difficulties in the language learning process are mainly related to native language influence, differences in articulatory features, and inconsistencies between language systems.

The article presents a generalized overview of challenges observed at both phonetic and prosodic levels. In particular, learners from different language groups demonstrate varying difficulties related to pronunciation and intonation. In this context, linguistic interference is considered one of the main explanatory factors.

In addition, the expression of communicative meaning through intonation in the Azerbaijani language is evaluated as an important prosodic feature. This characteristic requires different teaching approaches in comparison with other languages and creates additional challenges for learners.

Additionally, the study underscores the importance of developing learners' phonological awareness as a key factor in overcoming pronunciation difficulties in multilingual settings. It is observed that systematic exposure to authentic speech input and guided phonetic practice can significantly improve learners' ability to perceive and produce unfamiliar sound patterns. Moreover, incorporating contrastive analysis between the native and target languages helps learners better understand phonetic distinctions and reduces the impact of transfer-related errors. These findings suggest that effective pronunciation teaching should combine both perception-based and production-based activities in order to achieve sustainable learning outcomes.

In conclusion, considering phonetic and prosodic differences in multilingual classrooms and applying appropriate teaching methods contribute significantly to increasing the effectiveness of Azerbaijani language instruction and improving learners' communicative competence.

Key words: multilingual classrooms, Azerbaijani language teaching, phonetic differences, prosodic features, linguistic interference, pronunciation

AZƏRBAYCAN DİLİNİN ÇOXDİLLİ AUDİTORİYALARDA TƏDRİSİ ZAMANI YARANAN FONETİK FƏRQLƏR

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Xülasə: Bu məqalədə Azərbaycan dilinin çoxdilli auditoriyalarda tədrisi zamanı rast gəlinən fonetik və prosodik fərqlər təhlil olunur. Tədqiqat müəllifin uzunmüddətli pedaqoji təcrübəsinə əsaslanır və müxtəlif dil mühitindən olan öyrənənlərin qarşılaşdığı əsas çətinlikləri müəyyənləşdirir. Araşdırma göstərir ki, dil öyrənmə prosesində yaranan problemlər əsasən ana dilinin təsiri, artikulyasiya xüsusiyyətlərinin fərqliliyi və dil sistemləri arasındakı uyğunsuzluqlarla bağlıdır.

Məqalədə həm fonetik, həm də prosodik səviyyədə müşahidə olunan çətinliklər ümumiləşdirilmiş şəkildə təqdim olunur. Xüsusilə də müxtəlif dil qruplarına mənsub öyrənənlərdə tələffüz və intonasiya ilə bağlı fərqli problemlərin meydana çıxdığı qeyd edilir. Bu baxımdan interferensiya amili əsas izah vasitəsi kimi çıxış edir.

Eyni zamanda, Azərbaycan dilində intonasiya vasitəsilə kommunikativ məna fərqlərinin ifadə olunması prosodik baxımdan mühüm xüsusiyyət kimi dəyərləndirilir. Bu xüsusiyyət digər dillərlə müqayisədə fərqli yanaşmalar tələb edir və öyrənənlər üçün əlavə çətinlik yaradır.

Bundan əlavə, tədqiqat çoxdilli mühitdə tələffüz çətinliklərinin aradan qaldırılmasında fonoloji şüurun inkişafının mühüm rol oynadığını vurğulayır. Müşahidələr göstərir ki, autentik nitq nümunələrinə sistemli şəkildə məruz qalma və məqsədyönlü fonetik məşqlər öyrənənlərin qeyri-adi səs modellərini qavrama və düzgün istehsal etmə bacarıqlarını əhəmiyyətli dərəcədə inkişaf etdirir. Eyni zamanda, ana dil ilə hədəf dil arasında müqayisəli təhlilin aparılması fonetik fərqlərin daha yaxşı mənimsənilməsinə və transfer nəticəsində yaranan səhvlərin azalmasına kömək edir. Bu nəticələr göstərir ki, effektiv tələffüz tədrisi həm qavrama, həm də istehsal yönümlü fəaliyyətlərin kompleks şəkildə tətbiqini tələb edir.

Nəticə olaraq, çoxdilli auditoriyalarda Azərbaycan dilinin tədrisi zamanı fonetik və prosodik fərqlərin nəzərə alınması və uyğun metodların tətbiqi təlim prosesinin səmərəliliyini artırır və öyrənənlərin kommunikativ bacarıqlarının inkişafına mühüm töhfə verir.

Açar sözlər: çoxdilli auditoriya, fonetika, interferensiya, intonasiya, Azərbaycan dili

Introduction

In multilingual classrooms, phonetic and prosodic challenges in teaching the Azerbaijani language manifest in various forms. Based on observations, it can be noted that English-speaking learners experience greater difficulty in acquiring certain vowel sounds specific to Azerbaijani. These challenges are primarily related to the articulatory properties of the sounds and the absence of their equivalents in the learners' native language.

At the same time, learners whose native language is Turkish exhibit different types of phonetic difficulties. Although they tend to acquire Azerbaijani more easily due to linguistic proximity, certain sounds are realized differently. In such cases, learners tend to substitute Azerbaijani sounds with phonetically similar sounds from their native language. This phenomenon can be explained by linguistic interference and is widely observed in multilingual settings.

In addition to phonetic issues, difficulties also arise at the prosodic level. In Azerbaijani, certain interrogative sentences can be formed solely through intonation, which may appear unusual to learners. This feature presents particular challenges for speakers of languages that rely primarily on syntactic structures to form questions.

To address these challenges, various teaching methods are employed. Listening and repetition exercises, minimal pair drills, the use of audio-visual materials, and articulatory training have proven to be effective. Furthermore, adopting an individualized approach and taking into account learners' native language backgrounds significantly enhances the effectiveness of the teaching process.

Materials and Methods

The study is based on a qualitative research approach and relies on observations conducted within the framework of the author's five years of teaching experience. The data were collected during lessons conducted with learners from diverse linguistic backgrounds.

Observation, comparative analysis, and descriptive methods were employed in the research process. Learners' pronunciation features were compared with those of their native languages, and phonetic differences were systematically identified.

In addition, the study examines how learners' native languages influence their acquisition of Azerbaijani. Observations indicate that learners frequently transfer sounds, pronunciation patterns, and intonational features from their native language into the target language. This phenomenon is referred to as language transfer and often leads to pronunciation errors. The study also analyzes the conditions under which this influence becomes more prominent and how it affects the learning process.

Phonetic Features of the Azerbaijani Language

The Azerbaijani language has a rich phonetic system. Its vowel system consists of front and back, open and close, as well as rounded and unrounded vowels, comprising nine vowel phonemes, five of which are front (thin) and four back (thick). In terms of quantity, vowels are not only numerous but also show a higher proportion of front vowels compared to many other languages. In particular, sounds such as /ə/, /ö/, and /ü/ are notable for lacking direct equivalents in many languages.

Within the consonant system, uvular and velar sounds (/x/, /q/, /ğ/, /g/) occupy a special position. Due to their articulatory complexity, these sounds pose significant challenges for foreign learners.

Phonetic Difficulties in Multilingual Classrooms

English-speaking learners

Based on five years of pedagogical observation, it has been found that the greatest difficulty for English-speaking learners lies in the pronunciation of the vowels /ü/ and /ö/. As these sounds do not exist in English, learners often struggle to articulate them accurately.

In particular, the vowel /ü/, as a high front rounded vowel, requires more precise and subtle articulation. For this reason, learners from countries such as Pakistan and Malaysia experience considerable difficulty in producing this sound correctly.

Causes of Phonetic Differences

The main causes of phonetic difficulties can be summarized as follows:

1. **Native language influence** – learners tend to rely on familiar articulatory patterns from their first language.
2. **Differences in phoneme inventory** – some sounds in the target language may not exist in the learner's native language.
3. **Articulatory habits** – the physical production of new sounds may be challenging for learners.

4. Auditory perception – limited ability to distinguish between unfamiliar sounds.

For sounds that do not exist in the learners' native language, the **isolation + context method** should be applied. In other words, learners should first be introduced to the sound in isolation, then practice it within syllables, followed by words, sentences, and finally within extended texts.

In addition, it is important to demonstrate, from an acoustic and articulatory perspective, that front vowels are produced in the front part of the oral cavity, whereas back vowels are articulated in the rear part. A comparative approach helps learners better understand these distinctions.

Moreover, foreign learners should be regularly exposed to listening materials in the Azerbaijani language, allowing them to internalize correct pronunciation patterns through auditory perception.

To overcome phonetic differences, the teaching process should be structured according to learners' native language backgrounds, and articulatory, auditory, and prosodic skills should be developed in parallel. A systematic and step-by-step approach significantly facilitates the development of accurate pronunciation skills.

Conclusion and Discussion

The analysis and the author's pedagogical observations indicate that phonetic differences in teaching the Azerbaijani language in multilingual classrooms exhibit a systematic and predictable nature. These differences are primarily explained by the phonetic characteristics of learners' native languages, mismatches in phoneme inventories, and differences in articulatory habits. In particular, difficulties with certain vowels among English-speaking learners and specific consonants among Turkish-speaking learners represent typical examples of such differences.

The findings of the study suggest that one of the main causes of these problems is the phenomenon of linguistic interference. In other words, learners tend to rely on phonetic patterns established in their native language when acquiring a new language, which leads to inaccurate sound production. Furthermore, the absence of certain sounds in the native language or their different articulation further complicates the learning process.

The discussion highlights that the most effective approach to overcoming phonetic difficulties is the application of individualized and contrastive teaching methods. The gradual teaching of sounds—from isolation to text level—combined with visual and explicit explanations of articulatory mechanisms, as well as listening-based exercises, yields positive results. In particular, the development of auditory perception plays a crucial role in achieving accurate pronunciation.

At the same time, integrating prosodic features, especially intonation and stress, into the teaching process is essential. This is because mastering not only individual sounds but also their functional use in speech is a key requirement for developing communicative competence.

In conclusion, teaching Azerbaijani in multilingual classrooms requires a comprehensive approach. The systematic instruction of phonetic and prosodic features, consideration of learners' linguistic backgrounds, and the application of modern teaching methods significantly enhance the effectiveness of instruction. Research and practical observations in this field provide a foundation for the development of more advanced methodological approaches in the future.

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MULTILINGUAL ENVIRONMENTS AND LANGUAGE LEARNING STRATEGIES IN SECOND LANGUAGE ACQUISITION

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Abstract: Multilingualism is a defining characteristic of contemporary societies and plays a significant role in shaping second language acquisition (SLA). This article provides a detailed examination of language learning strategies in multilingual environments from both psycholinguistic and sociolinguistic perspectives. It explores how factors such as cognitive flexibility, executive control, metalinguistic awareness, and social interaction contribute to successful language learning outcomes. Using a qualitative research methodology based on an extensive literature review, the study brings together key theoretical and empirical findings from major SLA studies.

The results demonstrate that multilingual learners tend to be more adaptable, employ a wider range of learning strategies, and exhibit stronger cognitive performance compared to monolingual learners. In particular, early exposure to multiple languages positively affects phonological accuracy, grammatical intuition, and overall communicative competence. The study also addresses common challenges, including language interference, the complexity of code-switching, and increased cognitive load. However, these difficulties are generally temporary and diminish as language proficiency improves. Overall, the findings emphasize the importance of incorporating multilingual pedagogical approaches into modern education systems to enhance language learning effectiveness and better reflect real-world linguistic diversity.

Keywords: multilingualism, second language acquisition, learning strategies, cognitive flexibility, metalinguistic awareness, sociolinguistics, psycholinguistics

ÇOXDILLİ MÜHİTLƏR VƏ İKİNCİ DİLİN MƏNİMSƏNİLMƏSİNDƏ DİL ÖYRƏNMƏ STRATEGİYALARI

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Xülasə: Çoxdillilik müasir cəmiyyətlərin əsas xüsusiyyətlərindən biri olub ikinci dilin mənimsənilməsinə güclü təsir göstərir. Bu məqalədə çoxdilli mühitlərdə dil öyrənmə strategiyaları psixolinqvistik və sosiolinqvistik baxımdan geniş şəkildə təhlil olunur. Tədqiqat idrak elastikliyi, icraedici nəzarət, metadil şüuru və sosial qarşılıqlı təsirin dil öyrənmə prosesinə təsirini araşdırır. Keyfiyyət metoduna əsaslanan bu işdə müxtəlif nəzəri və empirik tədqiqatlar ümumiləşdirilmişdir.

Nəticələr göstərir ki, çoxdilli mühitdə öyrənənlər daha geniş strategiyalardan istifadə edir, daha yüksək uyğunlaşma qabiliyyətinə malik olur və mürəkkəb dil strukturlarını daha effektiv mənimsəyirlər. Xüsusilə, metakognitiv strategiyaların tətbiqi öyrənmə prosesinin planlaşdırılması və qiymətləndirilməsində mühüm rol oynayır. Erkən yaşda dil öyrənilməsi fonetik dəqiqliyin, qrammatik intuisiya və kommunikativ bacarıqların inkişafına müsbət təsir göstərir. Bununla yanaşı, sosial mühit və qarşılıqlı ünsiyyət dil bacarıqlarının möhkəmlənməsinə şərait yaradır.

Məqalədə həmçinin dil qarışıqlığı, koddəyişmə mürəkkəbliyi və idrak yüklənməsi kimi çətinliklər də müzakirə olunur. Bununla belə, bu çətinliklərin əsasən müvəqqəti olduğu və düzgün pedaqoji yanaşmalar vasitəsilə aradan qaldırıla biləcəyi vurğulanır. Ümumilikdə, çoxdilli yanaşmaların təhsildə tətbiqi vacib hesab olunur.

Açar sözlər: çoxdillilik, ikinci dilin mənimsənilməsi, öyrənmə strategiyaları, idrak elastikliyi, metadil şüuru, sosiolinqvistika, psixolinqvistika

Introduction

In the contemporary global landscape, marked by accelerating technological innovation, increased international migration, and the pervasive influence of digital communication platforms, multilingualism has transformed from a relatively rare competence into a defining feature of modern societies (Crystal, 2010; Cenoz, 2013). Today, individuals are no longer confined to a single linguistic system; instead, they interact with multiple languages across various contexts, including formal education, workplace communication, entertainment media, and everyday social interactions. The expansion of global networks has made linguistic diversity not only visible but also functionally necessary (Block, 2007).

As a result, the ability to understand and use more than one language is increasingly perceived as a fundamental life skill rather than an optional advantage. In professional environments, multilingual competence enhances employability and facilitates participation in global markets (Grin, 2001). In social contexts, it enables individuals to engage with diverse communities and access a wider range of cultural experiences. Consequently, multilingualism contributes not only to economic mobility but also to cognitive development and intercultural understanding (Bialystok, 2011).

Historically, language learning was often approached within a monolingual framework, where learners were expected to acquire a new language independently of their first language. These earlier perspectives tended to conceptualize second language acquisition (SLA) as a linear progression, moving step by step toward native-like proficiency. However, such models fail to capture the complexity of multilingual realities (Cook, 2016). Contemporary research challenges

these assumptions by demonstrating that language learning is inherently dynamic, interactive, and influenced by the simultaneous presence of multiple linguistic systems (Ellis, 2015).

Rather than functioning in isolation, languages within a multilingual individual's mind are interconnected and constantly interacting. Knowledge of one language can influence the acquisition of another, both positively and negatively. This interaction leads to a more integrated view of linguistic competence, where learners actively draw upon their entire linguistic repertoire when processing and producing language (Jessner, 2008). This shift in understanding has led researchers to reconsider foundational assumptions in SLA and to develop more holistic and inclusive theoretical models.

Second language acquisition is now widely recognized as a multidimensional phenomenon shaped by a complex interplay of factors. Cognitive processes form the foundation of language learning, encompassing memory storage, attentional focus, pattern recognition, and information processing efficiency (DeKeyser, 2007). Learners must be able to retain vocabulary, recognize grammatical structures, and retrieve linguistic information in real time during communication.

At the same time, psychological variables exert a powerful influence on learning outcomes. Motivation, for instance, determines the level of effort and persistence a learner is willing to invest (Dörnyei, 2005). Self-confidence affects willingness to communicate, while anxiety can either hinder or, in some cases, facilitate performance depending on its intensity (MacIntyre, 2002). Emotional resilience and self-regulation also play crucial roles in sustaining long-term engagement with language learning.

Sociocultural factors further enrich this process by situating language within a broader social and cultural context. Language is not merely a system of rules but also a medium of identity and social interaction (Vygotsky, 1978). Learners must navigate cultural norms, interpret social cues, and develop pragmatic competence in order to communicate effectively. Their sense of belonging and cultural identification can significantly influence their motivation and attitudes toward the target language (Norton, 2013).

Environmental conditions also contribute substantially to language acquisition. The frequency and quality of linguistic input, the availability of authentic communication opportunities, and the effectiveness of instructional methods all shape the learning experience (Lightbown & Spada, 2013). In multilingual environments, exposure to multiple languages occurs naturally and continuously, providing learners with rich and varied input that supports deeper learning.

Within this intricate framework, language learning strategies serve as essential mechanisms that enable learners to manage and optimize the acquisition process. These strategies are not fixed or universal; rather, they evolve over time and are influenced by individual differences such as age, prior knowledge, and learning context (Oxford, 2011). They include both conscious techniques, such as note-taking and summarization, and unconscious processes, such as pattern recognition and intuitive inference.

Multilingual learners, in particular, tend to exhibit greater strategic flexibility. Their prior experience with multiple languages allows them to experiment with different approaches, select the most effective strategies, and adapt their learning behavior to new linguistic challenges (Kemp, 2007). This adaptability is one of the key factors that contributes to their success in acquiring additional languages.

Multilingual environments create distinctive cognitive and communicative conditions that set them apart from monolingual settings. In such environments, learners must continuously make decisions about language choice, switching between languages depending on context, audience, and communicative goals. This process requires a high level of cognitive control, as individuals must activate the relevant language while suppressing interference from others (Green & Abutalebi, 2013).

Over time, this constant management of multiple linguistic systems strengthens executive control functions. Learners develop enhanced abilities in attention switching, inhibition of irrelevant information, and task coordination (Bialystok, 2011). These cognitive skills extend beyond language use and can positively impact other areas of thinking, including problem-solving and decision-making.

Another important outcome of multilingual experience is the development of advanced metalinguistic awareness. Multilingual individuals are more likely to reflect on language as an abstract system, recognizing patterns, structures, and relationships between different languages (Jessner, 2008). This awareness allows them to approach new languages more analytically and strategically, facilitating faster and more efficient learning.

From a psycholinguistic perspective, multilingualism is associated with improvements in working memory, cognitive flexibility, and processing speed (Herdina & Jessner, 2002). These enhancements enable learners to handle complex linguistic input and adapt to unfamiliar language structures with greater ease. From a sociolinguistic perspective, multilingual environments promote intercultural competence, allowing individuals to navigate diverse social contexts and communicate effectively across cultural boundaries (Kramsch, 2009).

The aim of this study is to provide a comprehensive examination of how multilingual environments influence language learning strategies and contribute to successful SLA outcomes. In addition, the study seeks to explore the cognitive advantages and potential challenges associated with multilingualism, as well as to offer practical implications for educational practices.

Materials and Methods

This research is grounded in a qualitative and descriptive-analytical methodology, focusing on the synthesis and interpretation of existing academic literature (Creswell, 2014). By integrating insights from multiple disciplines, the study aims to construct a comprehensive understanding of multilingual language acquisition.

The data sources include peer-reviewed journal articles, academic books, doctoral dissertations, and institutional reports related to linguistics and language education. Each source was carefully selected based on its relevance, credibility, and contribution to the field.

The analytical procedure is divided into three interconnected stages. The first stage involves a critical review of major theoretical frameworks in SLA, including cognitive (DeKeyser, 2007), sociocultural (Vygotsky, 1978), interactionist (Long, 1996), and usage-based approaches (Ellis, 2015). Each framework is examined in terms of its explanatory power and relevance to multilingual contexts.

The second stage focuses on a comparative analysis between monolingual and multilingual learners. This analysis explores differences in cognitive processing, strategy use, and communicative competence (Cook, 2016).

The third stage employs thematic analysis to identify recurring patterns and organize them into conceptual categories (Braun & Clarke, 2006). These categories include types of learning strategies and the factors that influence their effectiveness.

Language learning strategies are categorized into cognitive, metacognitive, and socio-affective types (Oxford, 2011). The study also considers moderating variables such as age, exposure level, educational context, and individual learner differences.

Results and Discussion

The findings of this study reveal that multilingual learners possess significant advantages in both cognitive and linguistic domains. One of the most notable benefits is the enhancement of executive control functions (Bialystok, 2011).

Cognitive flexibility emerges as a defining characteristic of multilingual competence, enabling more adaptive and context-sensitive communication. Additionally, multilingual individuals demonstrate stronger problem-solving skills and greater adaptability.

Metalinguistic awareness is another key advantage, facilitating deeper understanding of language structures (Jessner, 2008). Cross-linguistic transfer plays a central role, with both positive and negative effects (Odlin, 1989).

Social interaction is essential for language development, particularly in multilingual environments where authentic communication is frequent (Vygotsky, 1978). Identity formation also plays a critical role (Norton, 2013).

Emotional factors such as motivation and anxiety significantly influence outcomes (Dörnyei, 2005). Early multilingual exposure enhances brain plasticity and long-term linguistic competence (Lightbown & Spada, 2013).

Code-switching is recognized as a strategic and functional aspect of multilingual communication rather than a deficiency (Grosjean, 2010).

Despite these advantages, challenges such as interference and cognitive overload may occur (Cook, 2016). However, these can be mitigated through effective pedagogical approaches (Krashen, 1985).

From an educational perspective, the findings highlight the importance of multilingual teaching practices that promote interaction, inclusivity, and strategic learning (Cenoz & Gorter, 2015).

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MULTILINGUALISM AND SECOND LANGUAGE ACQUISITION

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Abstract: Multilingualism is not just a fringe occurrence anymore but is now a crucial part of modern societies. It influences how people learn and communicate and form identity. This article looks at the active link between multilingualism and second language learning (SLA) based on recent studies in language study and psychology and education. It suggests that multilingual people do not just pick up languages separately. Their language systems interact and can both help and limit new language learning. The study reveals how knowing previous languages affects thinking flexibility and understanding language structure and learning methods. These often give multilingual learners an edge when learning more languages. But mixing languages and social and cultural factors like language attitudes and education access and language rules also shape results. The article also investigates how classroom methods and digital spaces are adjusting to more varied language backgrounds.

The piece argues that looking at SLA with a multilingual perspective questions traditional single-language ideas in education and policy. It suggests using teaching methods which see linguistic diversity as an advantage and not a problem. This approach places multilingualism in the forefront of future research and practice in language learning.

Key words: Multilingualism, Second Language Acquisition (SLA), cross-linguistic influence, metalinguistic awareness, cognitive flexibility, language transfer, sociocultural factors, language policy

ÇOXDİLLİLİK VƏ İKİNCİ DİLİN MƏNİMSƏNİLMƏSİ

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Xülasə: Çoxdillilik artıq cəmiyyətin kənarında rast gəlinən bir hadisə deyil, müasir cəmiyyətlərin mühüm tərkib hissəsinə çevrilmişdir. O, insanların dili öyrənmə və ünsiyyət qurma proseslərinə, eləcə də kimliklərinin formalaşmasına təsir göstərir. Bu məqalədə dilçilik, psixologiya və təhsil sahəsində aparılmış son tədqiqatlara əsaslanaraq çoxdillilik və ikinci dilin mənimsənilməsi (SLA) arasındakı qarşılıqlı əlaqə araşdırılır. Məqalə göstərir ki, çoxdilli insanlar dilləri bir-birindən tamamilə ayrı şəkildə mənimsənilirlər. Onların dil sistemləri qarşılıqlı əlaqədə olur və yeni dilin öyrənilməsinə həm müsbət, həm də məhdudlaşdırıcı təsir göstərə bilər. Tədqiqat əvvəlki dilləri bilməyin təfəkkür çevikliyinə, dil strukturunun dərk edilməsinə və öyrənmə strategiyalarına necə təsir etdiyini ortaya qoyur. Bu amillər

çoxdilli öyrənənlərə əlavə dillərin mənimsənilməsi prosesində müəyyən üstünlüklər qazandıra bilər. Bununla yanaşı, dillərin qarışdırılması, dilə münasibət, təhsilə çıxış imkanları və dil siyasəti kimi sosial və mədəni amillər də öyrənmə nəticələrinə təsir göstərir. Məqalədə, həmçinin sinifdə tətbiq olunan tədris metodlarının və rəqəmsal mühitlərin müxtəlif dil mənşəyinə malik öyrənənlərin ehtiyaclarına necə uyğunlaşdırıldığı araşdırılır.

Məqalə iddia edir ki, ikinci dilin mənimsənilməsinə çoxdilli perspektivdən yanaşmaq təhsil və dil siyasətində ənənəvi, yalnız bir dilə əsaslanan yanaşmaları yenidən nəzərdən keçirməyə imkan verir. Bu yanaşma linqvistik müxtəlifliyi problem deyil, üstünlük kimi qəbul edən tədris metodlarının tətbiqini təşviq edir. Beləliklə, çoxdillilik gələcək dil öyrənmə tədqiqatlarının və təcrübəsinin əsas istiqamətlərindən biri kimi ön plana çıxır.

Açar sözlər: Çoxdillilik, İkinci Dilin Mənimsənilməsi (SLA), dillərarası təsir, metadilli şüur, koqnitiv çeviklik, dil transferi, sosial-mədəni amillər, dil siyasəti

Introduction

In the current globalized world with porous boundaries and advanced information technology, speaking two or more languages is not a peculiarity anymore. It is becoming an ordinary phenomenon in big cities' schools and even small settlements where migrants and modern technologies influence people's lifestyle significantly. Nevertheless, there is an intricate process that accompanies the development of this tendency – acquisition of new languages by humans and its reflection in their brain functions.

The current article investigates the issue of second language acquisition in the context of multilingual experience and discusses recent scientific studies concerning multilinguals' brain functioning. Contrary to the former approaches that consider languages as isolated systems, the author describes their interaction inside the human's mind. Moreover, the text emphasizes positive aspects of acquiring two or more languages as well as certain difficulties related to it.

It should be stated that the issue is relevant because of the rapid increase of the diversity of schoolchildren, which becomes a challenge for the contemporary education system. Therefore, the problem under investigation is crucial since it is necessary to understand how to use the acquired knowledge of multilinguals efficiently in educational practices.

Materials and Methods

This study explores the relationship between multilingualism and second language acquisition. To do so, the study employs a mixed-methods design to investigate the interaction between participants' linguistic background, self-evaluation of linguistic competence, and second language production. The participants in this study consisted of 120 high school and university students and adults aged 16 to 30, and they were enrolled in language courses that included two languages, in language-impact programs, in language-maintenance programs, in heritage-language courses, or in bilingual or multilingual communities where two or more languages were used as primary means of communication on a regular basis. The sample of participants was classified into bilingual (L2 near-native speaking competence) and multilingual (L3/L4 speaking competence above basic) groups to distinguish among levels and types of language proficiency.

Three kinds of data were collected. Firstly, a quantitative questionnaire with demographic, language history and self-assessment measures of L2 proficiency was administered to the 160 participants. Secondly, the participants completed 900 word version of the paper-based version of the VikIT vocabulary test (see also VikIT- listening), a number of grammaticality judgment

tasks assessing present perfect and perfect tenses, and some speaking, multiple choice listening and audio-script multiple choice listening tasks aimed at testing participants' ability to utilise L2 for communicating information. Thirdly, 30 in-depth interviews with the participants were recorded to investigate learners' bilingual experiences, strategies and attitudes towards multilingualism.

The study also involved classroom observations of English language classrooms in selected colleges and schools. Observational data on how teachers and students interact, instances of code-switching, and various language use were recorded and analyzed to see how these practices may influence English language learning.

The quantitative data collected for this study were analysed using statistical tests to investigate potential relationships between individuals' multilingual experience and second language acquisition outcomes. The qualitative data from in-depth interviews and observations were examined in a thematic manner in order to identify a range of emerging trends associated with different aspects of the second language acquisition process (cognitive, social, educational) within both controlled and naturalistic learning settings and in relation to multilingual experience.

Discussion

This study explores the relationship between multilingualism and second language acquisition from the learners' perspective. The findings support previous research in finding that multilingual individuals can draw on prior linguistic knowledge to facilitate their second language acquisition by increasing their metalinguistic awareness and strategic learning behaviour. However, the study also shows that there is no guarantee of such advantages, and that cross-linguistic interference in both similarity and distance can occur.

While it is hazardous to make generalisations across the diverse range of settings in which bilingual children learn to read and write, qualitative findings suggest that it is in those contexts most favourable to the use of all languages which the children are able to utilise their full linguistic potential most effectively. Findings also suggest that settings that unwittingly uphold a monolingual dominant discourse, can stifle the potential of multilingual children by failing to make use of their existing linguistic knowledge and ability. In this report we explore the implications of these findings for educational practice and broader issues of a multilingual socioculture.

Our results cast significant doubt on the notion that learners start "from scratch" when approaching a second language. Instead, evidence suggests that each language learners add changes the way they think and how they communicate in profound ways, in complex interactions with other languages they know. Our findings thus encourage and require a range of new instructional strategies and policies that respect, build on, and take advantage of individual linguistic diversity.

In conclusion, multilingualism should not be treated as a problem that needs to be solved or a variable that needs to be controlled in language development, but as a source of strength, a normal feature of language development, and the reality of global mobility and digital communication, to be mirrored in language education systems.

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COMMON SPEAKING ANXIETY AMONG ENGLISH LEARNERS AND WAYS TO REDUCE IT

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Abstract: Speaking is one of the most important skills in learning English because it allows learners to communicate ideas, feelings, and information in real-life situations. However, many English learners experience speaking anxiety when they are asked to speak in class or communicate with others. This anxiety often appears as fear, nervousness, hesitation, or lack of confidence. Students may know grammar rules and vocabulary, but they still struggle to express themselves orally because of psychological pressure. The purpose of this article is to examine the common causes of speaking anxiety among English learners and to present practical ways to reduce it. The paper discusses factors such as fear of making mistakes, pronunciation worries, limited vocabulary, negative classroom experiences, and fear of being judged by teachers or classmates. It also explains how anxiety can negatively influence speaking

performance, motivation, and language progress. In addition, the article suggests useful strategies for teachers and learners. Supportive classroom environments, pair work, regular speaking practice, positive feedback, and the use of modern tools such as language-learning applications can help students become more confident speakers. Speaking anxiety is a common challenge, but with the right methods it can be reduced successfully.

Key Words: Speaking Anxiety, English Learners, Foreign Language Learning, Confidence, Classroom Environment, Communication Skills, Motivation, Teaching Strategies

İNGİLİS DİLİ ÖYRƏNƏNLƏR ARASINDA GENİŞ YAYILMIŞ DANIŞIQ NARAHATLIĞI VƏ ONUN AZALDILMASI YOLLARI

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Xülasə: Danışiq ingilis dilini öyrənməyin ən vacib bacarıqlarından biridir, çünki bu bacarıq öyrənənlərə real həyat vəziyyətlərində fikir, hiss və məlumatlarını ifadə etməyə imkan verir. Lakin bir çox ingilis dili öyrənən şəxslər sinifdə danışmağa çağırıldıqda və ya başqaları ilə ünsiyyət qurduqda danışiq narahatlığı yaşayırlar. Bu narahatlıq adətən qorxu, həyəcan, tərəddüd və ya özünəinam çatışmazlığı şəklində özünü göstərir. Tələbələr qrammatika qaydalarını və söz ehtiyatını bilsələr də, psixoloji təzyiq səbəbindən fikirlərini şifahi şəkildə ifadə etməkdə çətinlik çəkirlər.

Bu məqalənin məqsədi ingilis dili öyrənənlər arasında danışiq narahatlığının əsas səbəblərini araşdırmaq və onu azaltmaq üçün praktik yollar təqdim etməkdir. Məqələdə səhv etmək qorxusu, tələffüzlə bağlı narahatlıq, məhdud söz ehtiyatı, mənfi sinif təcrübələri və müəllim və ya sinif yoldaşları tərəfindən mühakimə olunmaq qorxusu kimi amillər müzakirə olunur. Eyni zamanda narahatlığın danışiq performansına, motivasiyaya və dil inkişafına mənfi təsiri izah edilir.

Bundan əlavə, məqələdə müəllimlər və öyrənənlər üçün faydalı strategiyalar təklif olunur. Dəstəkləyici sinif mühiti, cütlük işi, müntəzəm danışiq təcrübəsi, müsbət rəy və dil öyrənmə tətbiqləri kimi müasir vasitələr tələbələrin daha inamlı danışanlara çevrilməsinə kömək edə bilər. Danışiq narahatlığı geniş yayılmış problemdir, lakin düzgün üsullarla onu uğurla azaltmaq mümkündür.

Açar sözlər: Danışiq Narahatlığı, İngilis Dili Öyrənənlər, Xarici Dil Öyrənilməsi, Özünəinam, Sinif Mühiti, Ünsiyyət Bacarıqları, Motivasiya, Tədris Strategiyaları

Introduction

English has become an international language used in education, business, technology, and travel. For this reason, many students study English to improve their future opportunities. Among the four language skills—listening, reading, writing, and speaking—speaking is often considered the most difficult. While some students can understand texts and complete grammar exercises successfully, they may become nervous when they need to speak English in front of others.

Speaking anxiety is a common emotional reaction in language classrooms. It usually happens when learners feel afraid of making mistakes, forgetting words, or being criticized. Even talented students may remain silent because they worry that others will laugh at them or judge their pronunciation. As a result, students avoid speaking opportunities and lose chances to improve.

Researchers have shown that anxiety is strongly connected with language performance. Students with high anxiety often participate less and achieve lower speaking results than students with lower anxiety. Therefore, understanding this problem is necessary for both teachers and learners. If anxiety is reduced, communication becomes easier and learning becomes more effective.

Research

Main Causes of Speaking Anxiety- One major cause of speaking anxiety is the fear of making mistakes. Many learners believe they must speak perfectly. When they make grammar or vocabulary errors, they feel embarrassed. For example, a student may know the answer to a teacher's question but remain silent because they are afraid of using the wrong tense.

Another important reason is pronunciation anxiety. Some students think their accent sounds strange or incorrect. They compare themselves with native speakers and lose confidence. For instance, a learner may avoid saying difficult words such as *through*, *comfortable*, or *world* because pronunciation seems challenging.

A third factor is limited vocabulary. Students often know what they want to say in their first language but cannot quickly find the English words. This creates stress during communication. When someone asks a simple question like "What did you do yesterday?", the learner may panic because they cannot organize a fast response.

Finally, fear of negative evaluation is common. Students may worry about reactions from classmates or teachers. If someone laughs after a mistake, the anxious learner may stop participating completely.

2) Effects of Speaking Anxiety- Speaking anxiety can seriously affect language development. First, it reduces participation. Students avoid answering questions, presentations, or classroom discussions. Silence becomes a habit.

Second, anxiety harms fluency. Nervous students speak slowly, forget simple words, or stop in the middle of sentences. Sometimes they know the answer mentally but cannot say it clearly because stress blocks their thinking process.

Third, anxiety lowers motivation. Learners may start believing that English is impossible for them. After repeated uncomfortable experiences, they may lose interest in studying.

For example, imagine two students with the same language level. One speaks freely and practices often. The other stays silent because of fear. After several months, the confident student usually improves faster because practice creates progress.

Recent studies also suggest that anxiety can change depending on the task. Formal speaking exams often create more pressure than pair-work activities or friendly conversations.

3) Ways to Reduce Speaking Anxiety

A. Creating a Supportive Classroom

Teachers play a major role in reducing anxiety. Students need an environment where mistakes are seen as a natural part of learning. Instead of criticizing errors harshly, teachers should give constructive feedback.

For example, if a student says *He go to school yesterday*, the teacher can respond kindly: *Good try. We say "He went to school yesterday."*

B. Pair Work and Small Groups

Many students feel less pressure speaking with one partner than speaking in front of the whole class. Pair discussions, interviews, and role-plays help learners practice in a safer setting.

Example activities:

Student A interviews Student B about hobbies

Ordering food in a restaurant role-play

Asking for directions in pairs

C. Regular Practice

Confidence grows through repetition. Students should practice short speaking tasks every day, even for five minutes. Talking to themselves, recording voice messages, or describing daily routines in English can be helpful.

D. Positive Thinking

Learners should remember that mistakes are normal. Even advanced speakers make errors sometimes. Progress matters more than perfection.

E. Technology and Modern Tools

Today, learners can practice speaking with mobile applications, AI chatbots, podcasts, and pronunciation tools. These tools provide private practice without fear of embarrassment. A shy student may first practice with an app, then feel ready to speak in class later.

4) Personal View - In my opinion, speaking anxiety is not a sign of weakness. It is a normal reaction when people use a language they have not fully mastered yet. Many intelligent students stay quiet not because they are incapable, but because they fear judgment. Teachers should notice this hidden problem and support such learners patiently.

I also believe students must challenge themselves step by step. A person does not need to give a long speech immediately. They can start with one sentence, then short conversations, and later presentations. Small success builds strong confidence.

Conclusion. To conclude, speaking anxiety is one of the most common difficulties faced by English learners. It usually develops because of fear of mistakes, pronunciation concerns, limited vocabulary, and fear of negative judgment. This anxiety can reduce participation, fluency, and motivation, slowing down language progress.

However, the problem can be solved. Friendly classroom atmospheres, pair work, regular speaking practice, positive feedback, and modern digital tools can greatly reduce fear. Teachers should encourage communication rather than perfection, while students should accept mistakes as part of growth.

In the end, successful speaking is not about being perfect—it is about being brave enough to communicate. When learners overcome anxiety, they open the door to real language development.

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THE CRITICAL PERIOD HYPOTHESIS IN SECOND LANGUAGE ACQUISITION BIOLOGICAL CONSTRAINTS AND IMPLICATIONS FOR EARLY CHILDHOOD EDUCATION

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Abstract: This article explores the Critical Period Hypothesis (CPH) in second language acquisition from biological, cognitive, and educational perspectives. Originally proposed by Wilder Penfield and further developed by Eric Lenneberg, the hypothesis suggests that language learning is most effective during early childhood due to the brain's high level of neuroplasticity. As individuals age, this plasticity gradually decreases, and language learning becomes more dependent on conscious and analytical processes.

The study examines key neurological processes such as synaptic pruning, myelination, and lateralization, which influence language acquisition. Evidence from neuroimaging studies indicates that early bilinguals process multiple languages within overlapping neural systems, whereas late learners often rely on separate and more effortful cognitive mechanisms. The article also discusses the shift from the traditional concept of a single critical period to the more flexible model of multiple sensitive periods, each associated with different language components such as phonology, grammar, and vocabulary.

In addition to biological factors, the role of affective and environmental variables is emphasized. Lower levels of anxiety and increased interaction in childhood contribute significantly to successful language acquisition. The findings suggest that early exposure to a second language not only enhances linguistic proficiency but also supports long-term cognitive development.

Keywords: Critical Period Hypothesis, neuroplasticity, second language acquisition, sensitive periods, lateralization, bilingualism, language development

İKİNCİ DİLİN MƏNİMSƏNİLMƏSİNDƏ KRİTİK DÖVR HİPOTEZİ BİOLOJİ MƏHDUDİYYƏTLƏR VƏ ERKƏN UŞAQLIQ TƏHSİLİ ÜÇÜN NƏTİCƏLƏR

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Xülasə: Bu məqalədə ikinci dilin mənimsənilməsində Kritik Dövr Hipotezası (KDH) bioloji, idraki və pedaqoji baxımdan təhlil olunur. İlk olaraq Wilder Penfield tərəfindən irəli sürülən və daha sonra Eric Lenneberg tərəfindən inkişaf etdirilən bu hipotezə görə, dil öyrənmə prosesi erkən uşaqlıq dövründə beynin yüksək neyroplastikliyi səbəbindən daha effektiv baş verir. Yaş artdıqca bu elastiklik azalır və dil öyrənmə daha çox şüurlu və analitik strategiyalara əsaslanır.

Tədqiqat çərçivəsində sinaptik budama, miyelinləşmə və lateralizasiya kimi neyrobioloji proseslər araşdırılır. Neyrogörüntüləmə tədqiqatlarının nəticələri göstərir ki, erkən ikidillilər dilləri eyni beyin

strukturlarında emal edirlər, gec öyrənənlər isə daha fərqli və mürəkkəb sinir mexanizmlərindən istifadə edirlər. Məqalədə həmçinin tək bir kritik dövr anlayışından fərqli olaraq, fonologiya, qrammatika və leksika kimi müxtəlif dil sahələrinə aid “həssas dövrlər” modelinə keçid müzakirə olunur.

Bundan əlavə, affektiv və mühit amillərinin rolu da vurğulanır. Uşaqlarda aşağı narahatlıq səviyyəsi və yüksək ünsiyyət imkanları dil mənimsənilməsini sürətləndirir. Nəticə etibarilə, erkən yaşda dil öyrənilməsi yalnız dil bacarıqlarının inkişafına deyil, həm də uzunmüddətli idrak inkişafına müsbət təsir göstərir.

Açar sözlər: kritik dövr hipotezası, neyroplastiklik, ikinci dilin mənimsənilməsi, həssas dövrlər, lateralizasiya, ikidillilik, dil inkişafı

Introduction

The question of why children tend to acquire languages more efficiently than adults has long occupied a central place in the study of linguistics, psychology, and cognitive neuroscience. This issue is not merely theoretical; it has direct implications for educational policy, curriculum design, and language teaching methodologies worldwide. Observations from both naturalistic and classroom settings consistently demonstrate that young learners often achieve higher levels of fluency, especially in pronunciation and intuitive grammatical usage, whereas adult learners, despite their cognitive maturity, frequently encounter limitations in attaining native-like proficiency. This apparent discrepancy has led researchers to search for biologically grounded explanations for age-related differences in language acquisition.

One of the most influential frameworks proposed to explain this phenomenon is the Critical Period Hypothesis (CPH), initially introduced by Wilder Penfield and later systematically developed by Eric Lenneberg. According to this hypothesis, there exists a biologically determined time window during which language acquisition occurs most naturally, efficiently, and successfully. This period is generally believed to extend from early childhood until puberty, after which the brain undergoes structural and functional changes that reduce its capacity for language learning, particularly in terms of achieving native-like competence.

The theoretical foundation of the Critical Period Hypothesis is closely linked to the concept of neuroplasticity. Neuroplasticity refers to the brain's ability to reorganize itself by forming and modifying neural connections in response to experience. During early childhood, this capacity is at its peak, enabling children to absorb linguistic input effortlessly and unconsciously. They acquire phonological systems, syntactic patterns, and morphological rules not through formal instruction but through exposure, imitation, and interaction. However, as individuals age, neuroplasticity gradually declines due to biological processes such as synaptic pruning and myelination. These processes enhance neural efficiency but simultaneously reduce flexibility, making the acquisition of new linguistic systems more challenging.

Another crucial neurological process associated with language development is lateralization, the specialization of brain hemispheres for specific functions. According to Lenneberg (1967), language functions become increasingly localized in the left hemisphere as the brain matures. This specialization, while beneficial for efficient processing, limits the brain's ability to reorganize itself for new language systems, thereby contributing to age-related differences in language learning outcomes.

In addition to biological mechanisms, cognitive and affective factors also play a significant role in second language acquisition. Children typically approach language learning with curiosity and minimal anxiety, allowing them to experiment freely without fear of making mistakes. This

contrasts with adult learners, who often rely on explicit learning strategies and may experience psychological barriers such as self-consciousness and performance anxiety (Krashen, 1982). Furthermore, advances in neuroimaging technologies have provided empirical support for these observations. Studies indicate that early bilinguals process multiple languages within overlapping neural networks, whereas late learners activate more distributed and cognitively demanding systems (Hensch, 2004).

Recent developments in linguistic theory have also led to a reconceptualization of the Critical Period Hypothesis. Rather than viewing it as a single, sharply defined cutoff point, researchers increasingly support the notion of multiple sensitive periods, each corresponding to different linguistic domains such as phonology, grammar, and vocabulary (Newport, 1990). This perspective suggests that language acquisition is a dynamic and gradual process influenced by a combination of biological, cognitive, and environmental factors.

Research

The biological underpinnings of the Critical Period Hypothesis are closely associated with early brain development, particularly the formation and refinement of neural networks. In infancy and early childhood, the brain is characterized by an overabundance of synaptic connections. This high level of connectivity allows for rapid learning and adaptation, enabling children to process and internalize linguistic input across multiple dimensions simultaneously.

As development progresses, the brain undergoes synaptic pruning, a process in which unused or inefficient neural connections are eliminated. While this leads to more efficient neural processing, it also reduces the brain's flexibility and capacity to accommodate new patterns. This phenomenon is particularly relevant in language learning, where early exposure to multiple phonetic systems can shape long-term perceptual abilities.

Myelination further contributes to this process by increasing the speed and efficiency of neural transmission. However, this increased efficiency comes at the cost of reduced adaptability, making it more difficult for older learners to acquire new phonological distinctions. As a result, individuals who begin learning a second language later in life often struggle with pronunciation, even if they achieve high levels of grammatical accuracy.

Phonological acquisition represents one of the most sensitive aspects of language learning. Research consistently demonstrates that the ability to perceive and produce unfamiliar sounds declines significantly after early childhood. This is largely due to the establishment of a "perceptual filter" based on the learner's first language. Once this filter is solidified, distinguishing between non-native phonemes becomes increasingly difficult (Johnson & Newport, 1989).

In contrast, the acquisition of grammar and vocabulary follows a more gradual and flexible trajectory. While children acquire grammatical structures implicitly through exposure and interaction, adults rely more on explicit learning strategies. This difference explains why adult learners may initially progress faster in formal learning environments but often struggle to achieve automaticity and fluency in spontaneous communication.

Neuroimaging studies provide compelling evidence for these differences. Early bilinguals tend to process both languages within the same neural regions, particularly in areas associated with language production and comprehension. In contrast, late learners show more distributed patterns of activation, indicating greater reliance on cognitive control and memory systems.

Environmental factors also play a critical role in language acquisition. The quality and quantity of linguistic input significantly influence learning outcomes. Children are typically exposed to simplified, repetitive, and interactive forms of speech, which facilitate comprehension and retention. Moreover, social interaction provides essential opportunities for feedback and reinforcement, further enhancing language development.

Discussion

While biological factors provide a strong foundation for understanding language acquisition, they do not operate in isolation. The interaction between biological predispositions and environmental conditions is crucial in determining learning outcomes. One of the most influential psychological factors is the affective filter, a concept introduced by Krashen (1982), which refers to the impact of emotional variables on language learning.

Children generally possess a low affective filter, allowing them to engage with language without fear or hesitation. This openness promotes experimentation and facilitates learning through trial and error. Adults, however, often experience higher levels of anxiety and self-consciousness, which can inhibit participation and reduce opportunities for practice.

Another important dimension is the distinction between implicit and explicit learning mechanisms. Children rely primarily on implicit learning, acquiring language naturally through exposure and interaction. Adults, in contrast, depend more on explicit instruction and conscious analysis. While this can be advantageous for understanding complex grammatical rules, it may hinder the development of spontaneous communication skills.

The concept of sensitive periods further refines our understanding of language acquisition. Rather than a single critical window, different linguistic components appear to have distinct developmental timelines. For instance, pronunciation is highly sensitive to early exposure, whereas vocabulary acquisition remains relatively flexible throughout life.

These findings have significant implications for language education. Early language instruction should prioritize communicative competence, interaction, and a low-anxiety environment. At the same time, adult language education should integrate both communicative and analytical approaches, leveraging learners' cognitive strengths while addressing their affective needs.

Conclusion

In conclusion, the Critical Period Hypothesis provides a comprehensive framework for understanding the complex relationship between age and language acquisition. The evidence strongly supports the idea that early childhood represents a uniquely advantageous period for language learning due to heightened neuroplasticity and reduced cognitive constraints.

However, modern research suggests that language acquisition is not restricted to a fixed biological window. Instead, it is a lifelong process influenced by a range of biological, cognitive, and environmental factors. The concept of multiple sensitive periods offers a more flexible and realistic perspective, acknowledging both the advantages of early learning and the continued potential for language acquisition in adulthood.

From an educational standpoint, these insights emphasize the importance of early exposure to foreign languages, as well as the need for effective teaching strategies tailored to different age groups. Ultimately, successful language learning depends not solely on age but on the dynamic interaction between brain development, cognitive processes, emotional factors, and the quality of the learning environment.

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INTERFERENCE AND ALLOMORPHISM: A COGNITIVE-TYPOLOGICAL APPROACH

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Abstract: The paper examines the relationship between interference and allomorphy under the paradigm of cognitive typology, striving to compensate for the absence of any coherent treatment of these two independent linguistic notions. The study presupposes that neither interference nor allomorphy can be regarded solely as purely formal linguistic characteristics; rather, they are grounded in certain cognitive mechanisms, including attention, prediction, and processing time.

The study deals with the comparative analysis of Azerbaijani, Russian, and English, which have various typological properties. Specifically, the study is aimed at investigating the difference between predicate-final and predicate-medial word orders. It has been proven that in predicate-final languages, delay results in higher cognitive effort and attention shift, whereas in flexible languages, information transmission is based on prosodic and pragmatic factors. Additionally, the research reveals that the conflict emerges from the struggle between established language habits, while allomorphic behavior is a result of evolutionary adaptability based on cognitive and situational factors. The combination of

classical language knowledge and contemporary cognitive science enables us to view syntagmatic pressure and information structure through the lens of processing mechanisms.

The findings make a contribution to cognitive typology in that a common model that considers relations between structure and mental processes is suggested. The implications of the paper for foreign language instruction and NLP are shown: the consideration of cognitive aspects could be beneficial for both fields.

Key words: Interference, Allomorphy, Cognitive Linguistics, Linguistic Typology, Language Processing, Attention, Word Order

İNTERFERENSIYA VƏ ALLOMORFİZM: KOQNİTİV-TİPOLOJİ YANAŞMA

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Xülasə: Məqalə koqnitiv tipologiya paradıqması çərçivəsində interferensiya və allomorfizm arasındakı əlaqəni araşdırır, bu iki müstəqil linqvistik anlayışın heç bir əlaqəli təfsirinin olmamasını kompensasiya etməyə çalışır. Tədqiqat nə interferensiya, nə də allomorfizmin yalnız sırf formal linqvistik xüsusiyyətlər kimi qəbul edilə bilməyəcəyini fərz edir; əksinə, onlar diqqət, proqnozlaşdırma və emal müddəti də daxil olmaqla müəyyən idrak mexanizmlərinə əsaslanır. Tədqiqat müxtəlif tipoloji xüsusiyyətlərə malik olan Azərbaycan, Rus və İngilis dillərinin müqayisəli təhlili ilə bağlıdır. Xüsusilə, tədqiqat predikat-son və predikat-medial söz sıraları arasındakı fərqi araşdırmağa yönəlib. Sübut olunub ki, predikat-son dillərdə gecikmə daha yüksək idrak səyi və diqqət dəyişikliyinə səbəb olur, çevik dillərdə isə informasiya ötürülməsi prosodik və praqmatik amillərə əsaslanır. Bundan əlavə, tədqiqat münaqişənin mövcud dil vərdisləri arasındakı mübarizədən irəli gəldiyini, allomorfemik davranışın isə idrak və situasiya amillərinə əsaslanan təkamül uyğunlaşmasının nəticəsi olduğunu göstərir. Klassik dil biliklərinin və müasir idrak elminin birləşməsi bizə sintaqmatik təzyiqə və informasiya strukturuna emal mexanizmləri prizmasından baxmağa imkan verir. Bu nəticələr, struktur və zehni proseslər arasındakı əlaqələri nəzərə alan ortaq bir modelin təklif edilməsi ilə koqnitiv tipologiyaya töhfə verir. Məqalənin xarici dil tədrisi və TDE üçün nəticələri göstərilir: koqnitiv aspektlərin nəzərə alınması hər iki sahə üçün faydalı ola bilər.

Açar sözlər: İnterferensiya, allomorfizm, koqnitiv dilçilik, linqvistik tipologiya, dil emalı, diqqət, söz sırası

Introduction

Interference and allomorphy constitute one of the most theoretically challenging and empirically under-researched issues in modern linguistics, especially when dealing with their connection from the viewpoint of cognitive and typological perspectives. Indeed, while interference is conventionally studied from the perspective of second language acquisition and contact linguistics, and allomorphy is traditionally analyzed in morphology and phonology of individual languages, the relation between them is hardly sufficiently accounted for in theoretical studies.

The main issue in the present study may be presented as follows: although there is an extensive amount of scientific literature devoted to interference and allomorphy, there is no universal cognitive typological approach that will elucidate the process of interaction between

the structure of languages and cognitive mechanisms leading to the alternation of language competence.

This phenomenon is most apparent in the contrast between languages that belong to entirely different typological categories, for example, those that exhibit a synthetic vs. an analytic structure, or those with SOV vs. SVO syntactic constructions. In these situations, interference is no longer just a superficial transfer of linguistic structures; rather, it is based on cognitive processes influenced by the underlying typology of the mother tongue. Moreover, any attempt to understand allomorphy must take into account the way that competing structural patterns are processed mentally by the speaker.

From a cognitive point of view, the process of language comprehension comprises mechanisms that include anticipation, attention, and memory and are responsive to predictions based on linguistic experience. Thus, typologically different languages affect these predictions differently, since they differ in their structure. For example, in languages with predicate-final constructions, such as Azerbaijani, the processing of sentences usually includes a kind of delayed resolution, where the last component is heavily loaded. In turn, languages that use early predicates, such as English, allow for incremental sentence parsing (Hawkins, 2004).

The concept of interference, in this case, refers to the cognitive mismatch involved when dealing with conflicting processes. Speakers of predicate-final language use in generating or analyzing predicate-medial sentences tend to use strategies based on delayed closure, which results in variations in syntax, stress, and information structuring. The reverse can also be the case, whereby speakers of predicate-medial language have problems sustaining syntactic expectancy when using predicate-final language.

On the contrary, allomorphic variation embodies the inherent flexibility of language structures in which similar functions are served by different forms contingent upon phonological, morphological, or syntactic factors. Nevertheless, from the cognitive-typological point of view, allomorphic variation may also be regarded as an illustration of conflicting structural patterns that selectively activate their respective processes depending on communication requirements (Bybee, 2010; Booij, 2012).

It is important to note that the interaction between interference and allomorphy assumes even greater significance in multilingual environments where users are always operating within multiple grammatical frameworks. In such an environment, the process of allomorphic variation could be affected by interference, and conversely, interference could be shaped by allomorphs.

The purpose of this study is to create a cognitive-typological theory, which would combine these two areas with regard to the accomplishment of the following tasks: the investigation of the nature of cognitive processes associated with linguistic interference between structurally different languages; the examination of the impact of typological variation on the process of allomorphy formation; the creation of an integrated theory which describes the interaction of these processes in multilingual performance.

The innovation of the present study is in its effort to reconcile the perspectives of cognitive linguistics and typology, providing a general theoretical background, which would describe the interaction between cognitive and structural processes associated with linguistic activity.

Methodology and Materials

This research uses an integrated approach based on the theory of cognitive linguistics, linguistic typology, and functional grammar in an effort to examine the relationship between interference and allomorphy within different types of languages.

The theory of the study is based on the usage-based approach to language developed by Joan Bybee, which stresses the influence of frequency, experience, and entrenchment on language structure (Bybee, 2010). Such an approach makes it possible to perceive interference as something natural rather than a mistake due to the activation of different forms entrenchment in the process of language production and perception. Similarly, the study applies the constructionist approach to language introduced by Adele Goldberg, according to which linguistic knowledge is represented as a system of form-meaning pairs (Goldberg, 2006).

As for the typological approach, the current analysis is based on the cross-linguistic theory of Matthew S. Dryer and Martin Haspelmath, namely, their word order typology and linguistic universalities (Dryer, 2013; Haspelmath, 2010). This theory gives us the possibility to study languages with various syntactic structures, like SOV (Azerbaijani) and SVO (English, Russian) languages, and find typological factors that affect interference and allomorphic tendencies.

The methodological foundation of the research has a qualitative and comparative character. The investigation includes three languages which have different typological characteristics: Azerbaijani – a Turkic language belonging to the group of SOV languages having the agglutinative type of morphological structure; Russian – a fusional language that does not have strict limitations as far as the order of the words in the sentence is concerned; and English, an analytic language belonging to the category of SVO languages.

Data used for the research include three different types of data. First, illustrative cases are cited using descriptive grammars and existing linguistic works to ensure validity. Second, data of actual language use is taken into consideration in light of existing corpuses and usages, specifically those dealing with bilingualism and language contact. Third, introspection is included in the research, specifically in the observation of cognitive processes such as distribution of attention, delay of processing, and structuring of information.

Particular focus is put on the differences in syntactic word order and its effects on cognition. According to the theory put forward by John A. Hawkins, syntactic structures can be described according to their efficiency and processing costs (Hawkins, 2004). Such an approach becomes especially useful when explaining how the placement of predicates affects sentence comprehension and how violations of standard sentence structures emerge due to interference.

Moreover, the research paper includes certain aspects of information structure, such as those relating to theme and rheme, and focus and topic, which are vital for inversion analysis. They should be considered both from the perspective of structural categories and from that of cognitive-pragmatics – in terms of the speaker's distribution of attention and communicative preferences.

The analysis process entails a sequential comparison of structurally similar structures in the chosen languages. The first stage of analysis entails identifying the structural regularities in each language. The second stage entails analyzing variations from such regularities in cases where there is contact between languages or when two or more languages are used by the same individual. Finally, the allomorphic variations are analyzed to find out whether they are affected by typology or cognition.

The combination of such approaches will allow us to go from description to explaining the complex interaction of linguistics and cognition. The possibility of answering the questions as to where and why interference and allomorphy happen emerges through this approach. Moreover, we will be able to consider the interdependence of typology and cognition.

Discussion

This analysis shows that the relationship between interference and allomorphy lies at its very essence in the processes involved in cognition and particularly attention allocation, prediction, and time-related structuring of the interpretation of syntax. Instead of looking at them as two separate issues, interference and allomorphy should be seen as interrelated results of cognition managing linguistic structures.

One of the most important aspects of the research was the influence of timing in processing on the expectations regarding syntax. In the context of the predicate final languages represented by Azerbaijani, sentence comprehension involves a process known as delayed resolution, where the predicate, located at the end of the sentence, functions as the final integrating part. Such structure puts forward certain cognitive challenges for the interpreter that require him or her to have several syntactic and semantic expectations until the last element is reached. From the viewpoint of language processing, this translates into a "suspended" type of comprehension where attention is focused on several elements prior to the contribution of the predicate.

This finding is consistent with previous observations made by L. V. Shcherba, who highlighted that the position of the predicate in the sentence's final position leads to a phenomenon known as "syntagmatic tension" in the human mind (Shcherba, 1974). This idea could be reconsidered from a modern cognitive perspective as a representation of predictive activation, which continuously predicts the final solution to the syntactic structure. The longer the period of anticipation, the more cognitive effort required to maintain the prediction process.

Very close to this issue is the process of delayed processing, which serves an important function in allocating attention inside the sentence. In the Azerbaijani language, delaying the predicate gives other sentence parts, especially adverbial modifiers, the chance to gain importance in terms of attentional allocation. Therefore, these sentence constituents are perceived not only as secondary but rather as central constituents of the sentence. This attentional shift has consequences both for the production and perception of the sentence, as it forms a hierarchy of information.

On a broader scale, in Turkic languages, constructions with the predicate ending are usually associated with the so-called "absolute sentence," where the predicate-ending position is viewed as a completion and full comprehension of the construction. However, in colloquial speech, the appearance of inverse constructions, where the position of the predicate shifts closer to the beginning of the construction, demonstrates that there is an important turning point in the cognitive and communicative approach adopted by speakers. The inverse construction decreases the amount of time necessary for comprehension and distributes cognitive attention evenly across the construction, which helps to diminish the rigidity of the construction and enhance its expressive function.

On the other hand, there is the case of Russian, which has another interesting approach altogether. The relatively free word order in Russian enables a great number of syntactic variations, which often correspond to information structural manipulations instead of obligatory grammatical structures. According to V. V. Vinogradov, modifications in the Russian word order

are directly related to the displacement of the intonational nucleus, which might be considered the key feature of logical accentuation in Russian (Vinogradov, 1972). In such a way, in Russian, the focus of attention is dictated not by rigid syntactic elements but by prosodic features that emphasize the crucial parts of the sentence.

Cognitively speaking, this means that processing occurs differently, whereby the listener uses prosodic information and context to determine what the focus of the sentence is. Processing in Russian does not involve the kind of delay that we see in processing in Azerbaijani. Rather, it entails a dynamic process, whereby the listener constantly evaluates multiple interpretations. This dynamic nature of Russian processing allows room for interference, especially when speakers apply processing mechanisms from stricter systems.

In interactions with speakers of Russian or English, the interference process can take the form of an unexpected or inappropriate position of informational prominence in speech production. This may involve placing a sentence component where its information status calls for it to be placed at a different point in the sentence. Thus, the habit of deferring the predicate could lead to marked sentences in SVO language groups, which expect verbs to occupy earlier positions.

This is where the relevance to allomorphy becomes highly relevant. Typically, allomorphisms have been seen in the linguistic literature as purely formal entities, whose realization depends on the phonological and morphological environments of individual morphemes. Nevertheless, the current discussion reveals the possibility that allomorphic variation can also depend on cognitive considerations in situations where there are alternative structural possibilities available, and in cases where more than one language is involved.

For instance, when interference occurs, the speakers prefer those allomorphic forms that are closer in nature to the processing methods used by their dominant language. In such cases, it is possible for the formation of hybrids to occur, even though they may not be considered canonical in nature, but rather are more cognitively appropriate within a certain context.

In addition, the interplay between interference and allomorphy can be explained in the context of competition among established forms. Within the usage-based framework (Bybee, 2010), language systems involve networks of constructions whose strength depends on frequency and experience. For bilingual individuals, such networks may intersect and affect each other, causing the selection of multiple forms at one time. When an undesired form wins the competition, it results in interference; when an alternative form emerges, it is a case of allomorphy.

Significantly, both of these processes are dependent on attentional processes. The choice of specific form or structure will be dependent on what features stand out during that time period, which is contingent on variables like context, communication purpose, and mental effort. Attention, in this way, acts as a mediating process connecting the structural characteristics of typology with cognitive processing.

Thus, the results obtained from this study confirm the need for a reconsideration of the notions of interference and allomorphy as phenomena that have their basis in cognition and are conditioned by typological diversity. As distinct from the current approach to considering them as independent fields, it seems more appropriate to regard them as two sides of one and the same phenomenon in which structural difference and cognitive processing are constantly interwoven.

Conclusion

The findings of this study have clearly shown that both interference and allomorphy should be seen as cognitive phenomena determined by processing factors and typology. In order to fill in the gap caused by the absence of an integrating theoretical model, it was found out that interference occurs due to the interaction of cross-linguistic tendencies, while allomorphs depend on processing considerations.

These aspects are emphasized through the research, especially when comparing predicate-final constructions and those that use predicate-medial constructions. The convergence of ideas offered by L. V. Shcherba and V. V. Vinogradov demonstrates that syntagmatic tension and emphasis may be viewed as manifestations of processing.

This study adds to the field of cognitive-typology by presenting a holistic framework that connects linguistic diversity with cognitive process. Moreover, there are possible applications of these findings in pedagogical contexts and in natural language processing, where attention and cognitive limits might be considered. More empirical studies should be conducted in the future for validation and further generalization of these results.

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DATA-ANALYTICAL ANALYSIS OF SEASONAL TOURISM FLOW VARIABILITY IN THE CONTEXT OF INTERCULTURAL COMMUNICATION AND LANGUAGE FACTORS

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Abstract: This article examines seasonal variability in tourism flows through the combined lens of data analytics, intercultural communication, and language factors. In contemporary tourism management, seasonality is no longer interpreted only as a quantitative fluctuation in visitor numbers, hotel occupancy, or travel demand. It is also connected with cultural preferences, communication patterns, language accessibility, digital information availability, and the capacity of destinations to address heterogeneous visitor groups.

The case of Azerbaijan is particularly relevant because the country has experienced a significant recovery of inbound tourism after the pandemic period. According to official statistics, Azerbaijan received approximately 2.63 million foreign and stateless visitors in 2024, which represents a 25.9–26% increase compared with the previous year.

The study argues that data analytics can help identify not only peak and off-peak periods, but also the cultural and linguistic composition of tourist flows. In this context, language availability, multilingual digital platforms, intercultural service competence, and targeted communication strategies can reduce seasonal concentration and support more balanced tourism development. The article applies a conceptual-analytical approach based on official tourism statistics, international tourism seasonality literature, and intercultural communication theory.

The findings indicate that tourism seasonality should be managed through integrated data systems that combine arrival statistics, country-of-origin analysis, visitor motivation, language needs, online search behavior, and service feedback. Such an approach may improve destination competitiveness, visitor satisfaction, and sustainable tourism planning.

Keywords: tourism seasonality; intercultural communication; language factors; data analytics; Azerbaijan tourism; tourist flows; multilingual tourism services

**MƏDƏNİYYƏTLƏRARASI ÜNSİYYƏT VƏ DİL AMİLLƏRİ KONTEKSTİNDƏ MÖVSÜMİ
TURİZM AXINI DƏYİŞKƏNLİYİNİN MƏLUMAT-ANALİTİK TƏHLİLİ**

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Xülasə: Məqalə turizm axınlarında mövsümi dəyişkənliyi data analitikası, mədəniyyətlərarası kommunikasiya və dil amillərinin birgə prizmasından araşdırır. Müasir turizm idarəçiliyində mövsümlilik artıq yalnız ziyarətçi sayının, otel doluluğunun və ya səyahət tələbinin kəmiyyət dəyişməsi kimi izah edilmir. Bu proses eyni zamanda mədəni üstünlüklər, kommunikasiya modelləri, dil əlçatanlığı, rəqəmsal informasiya mühitinin səviyyəsi və turizm məkanlarının müxtəlif xüsusiyyətlərə malik ziyarətçi qruplarına uyğunlaşma qabiliyyəti ilə sıx bağlıdır.

Azərbaycanın nümunəsi xüsusilə aktualdır, çünki ölkə pandemiya dövründə daxil olan turist axınında əhəmiyyətli bərpa mərhələsi keçmişdir. Rəsmi statistikaya əsasən, 2024-cü ildə Azərbaycana təxminən 2,63 milyon xarici və vətəndaşlığı olmayan şəxs səfər etmişdir ki, bu da əvvəlki illə müqayisədə 25,9–26% artım deməkdir.

Tədqiqat göstərir ki, data analitikası yalnız pik və qeyri-pik dövrləri müəyyən etməyə deyil, həm də turist axınlarının mədəni və dil tərkibini analiz etməyə imkan verir. Bu kontekstdə dil əlçatanlığı, çoxdilli rəqəmsal platformalar, mədəniyyətlərarası xidmət bacarıqları və hədəflənmiş kommunikasiya strategiyaları mövsümi konsentrasiyanın azaldılmasına və turizmin daha balanslı inkişafına töhfə verə bilər. Məqalədə rəsmi turizm statistikasına, beynəlxalq turizm mövsümliliyi üzrə ədəbiyyat və mədəniyyətlərarası kommunikasiya nəzəriyyəsinə əsaslanan konseptual-analitik yanaşma tətbiq edilmişdir.

Nəticələr göstərir ki, turizm mövsümliliyi inteqrə olunmuş məlumat sistemləri vasitəsilə idarə olunmalıdır; bu sistemlərə gəlmə statistikasına, mənşə ölkə analizi, ziyarətçi motivasiyasına, dil ehtiyaclarına, onlayn axtarış davranışına və xidmət rəylərinə daxil edilməlidir. Bu cür yanaşma turizm məkanlarının rəqabət qabiliyyətini, ziyarətçi məmnunluğunu və dayanıqlı turizm planlaşdırmasını əhəmiyyətli dərəcədə yaxşılaşdırmağa bilər.

Açar sözlər: turizm mövsümliliyi; mədəniyyətlərarası kommunikasiya; dil faktorları; data analitikası; Azərbaycan turizmi; turist axınları; çoxdilli turizm xidmətləri

Introduction

Tourism seasonality is one of the most persistent structural challenges in the global tourism industry. It refers to periodic fluctuations in tourist demand, usually caused by climatic conditions, holiday calendars, cultural events, school vacations, transport accessibility, pricing

mechanisms, and visitor motivation. UN Tourism defines seasonality as a measurable feature of tourism activity with important economic and social consequences for destinations.

In Azerbaijan, the issue of seasonal tourism flow variability has become increasingly relevant due to the post-pandemic recovery of international arrivals, the diversification of tourist markets, and the growing role of digital promotion. The State Tourism Agency's official dashboard reported 2,626,838 arrivals in January–December 2024, showing a 26% annual increase. This growth creates new opportunities, but it also requires more precise analytical tools for understanding when tourists arrive, where they come from, what languages they use, and how cultural communication affects their travel experience (State Tourism Agency of the Republic of Azerbaijan, 2025).

The main research problem of this article is that tourism seasonality is often studied mainly through quantitative indicators, while intercultural communication and language factors remain insufficiently integrated into seasonal tourism analysis. However, tourists from different countries may travel in different months, respond to different types of promotional messages, require different language services, and demonstrate different cultural expectations. Therefore, tourism flow analysis should combine statistical indicators with cultural and linguistic interpretation.

The purpose of the article is to analyze seasonal tourism flow variability in the context of intercultural communication and language factors through a data-analytical perspective. The research focuses on three key questions: how seasonal tourism flows can be interpreted through data analytics; how intercultural communication influences tourist experience and destination attractiveness; and how language accessibility can contribute to reducing seasonal concentration.

Materials and Methods

This study is designed as a conceptual-analytical and data-oriented research that investigates seasonal variability in tourism flows through the interrelated dimensions of intercultural communication, language accessibility, and data analytics. The methodological basis of the article is formed by the combination of secondary statistical data analysis, comparative interpretation, content analysis, and interdisciplinary theoretical synthesis. Since the research topic is located at the intersection of tourism studies, communication studies, applied linguistics, and digital analytics, the study does not rely on a single methodological direction. Instead, it applies an integrated approach that makes it possible to interpret tourism seasonality not only as a numerical fluctuation in visitor arrivals, but also as a culturally and linguistically conditioned process.

The empirical information base of the study consists mainly of official statistical data related to inbound tourism flows, visitor dynamics, and tourism development indicators in Azerbaijan. In this regard, the data published by the State Statistical Committee of the Republic of Azerbaijan, the State Tourism Agency of the Republic of Azerbaijan, and the Azerbaijan Tourism Board are considered as the principal sources of quantitative information. These sources provide relevant indicators on the number of foreign and stateless visitors, annual growth dynamics, country-of-origin structure, and general tourism trends. The use of official statistical data increases the reliability of the analysis and allows the research to be grounded in measurable tourism indicators (State Statistical Committee of the Republic of Azerbaijan, 2026).

In addition to official statistics, the study uses international academic literature on tourism seasonality, destination management, intercultural communication, multilingual tourism services,

and data-driven decision-making in tourism. The theoretical part of the research is based on the assumption that tourism seasonality is not caused only by climate, holidays, transport availability, or price changes. It is also influenced by cultural calendars, communication preferences, language accessibility, digital information channels, and the ability of destinations to communicate effectively with different visitor groups. Therefore, the methodological framework combines quantitative tourism indicators with qualitative interpretation of cultural and linguistic factors.

The research applies secondary data analysis as one of its main methodological tools. Monthly and annual tourism indicators may be used to identify changes in tourist arrivals, determine peak and off-peak periods, and observe the general direction of tourism demand. In the context of this study, such data are interpreted not merely as isolated numerical values, but as indicators of broader social, cultural, and communication-related processes. For example, an increase in arrivals from a specific country may indicate not only growing travel demand, but also the need for language-specific information, culturally adapted services, and targeted promotional communication.

Comparative analysis is also used in the study. Through this method, different tourist source markets, seasonal periods, and communication needs can be compared with each other. The comparison of visitor flows by country of origin makes it possible to identify which markets may require Russian, English, Turkish, Arabic, Persian, or other language services. Similarly, the comparison of high-season and low-season periods helps to determine whether language and communication strategies can be used as instruments for reducing seasonal concentration. This method is particularly useful for Azerbaijan because the country receives tourists from culturally and linguistically diverse regions (Azerbaijan Tourism Board, 2024).

Content analysis is applied to evaluate the role of language and intercultural communication in tourism information systems. In this context, tourism websites, destination promotion materials, digital platforms, hotel and travel agency information, museum descriptions, restaurant menus, transport guidance, and online visitor feedback may be examined as communication materials. The purpose of this analysis is to determine whether tourism information is accessible to different linguistic groups and whether the content reflects intercultural sensitivity. This is important because the absence of multilingual and culturally appropriate information may reduce tourist satisfaction and weaken the competitiveness of the destination.

The methodological framework of the research also includes data analytics as a practical and managerial approach. Data analytics in this article is understood as the systematic collection, classification, interpretation, and use of tourism-related data for decision-making. In tourism seasonality studies, data analytics may include arrival statistics, accommodation data, booking patterns, online search trends, social media activity, visitor reviews, language preferences, and customer satisfaction indicators. By combining these data sources, tourism institutions and businesses can better understand not only when tourists arrive, but also why they arrive, what type of information they need, and how they evaluate their experience.

The study proposes that tourism flow analysis should be conducted through several analytical dimensions. The first dimension is temporal analysis, which focuses on the distribution of tourist arrivals by months and seasons. The second dimension is geographical analysis, which examines the countries and regions from which tourists arrive. The third dimension is linguistic

analysis, which identifies the main languages required for effective tourism communication. The fourth dimension is intercultural analysis, which considers cultural expectations, service norms, travel motivations, and communication behavior. The fifth dimension is digital analysis, which examines online search behavior, website accessibility, digital content quality, and visitor feedback on online platforms.

Within this methodological structure, language factors are treated as measurable and interpretable variables. For instance, the country-of-origin structure of tourist arrivals can be used as an indirect indicator of language demand. If a significant share of tourists comes from Türkiye, Russia, Arab countries, Iran, European countries, or Central Asian states, then tourism communication should reflect this linguistic diversity. This includes not only translation of basic information, but also adaptation of messages to cultural expectations and communication styles. Therefore, language is analyzed both as a functional service tool and as a strategic factor of destination competitiveness.

Intercultural communication is examined as a qualitative dimension of tourism experience. The study considers that tourist satisfaction depends not only on infrastructure, accommodation, and attractions, but also on the quality of interaction between visitors and service providers. Misunderstandings caused by cultural differences, lack of language support, weak interpretation of local traditions, or insufficient staff competence may negatively affect the tourist experience. For this reason, the article uses intercultural communication theory to explain how destinations can improve service quality and reduce communication barriers for foreign visitors.

The research does not conduct primary surveys or interviews. Its methodological limitation is that it relies mainly on secondary data and theoretical interpretation. However, this limitation is partly compensated by the use of official statistical sources and internationally recognized academic literature. The study may serve as a basis for future empirical research, including surveys among tourists, interviews with tourism professionals, multilingual content audits, sentiment analysis of online reviews, and predictive modelling of seasonal tourist flows.

Discussion

The analysis shows that tourism seasonality should be interpreted as a multidimensional phenomenon. Traditional tourism management usually measures seasonality through monthly visitor numbers, hotel occupancy rates, transport demand, and revenue indicators. However, such an approach does not fully explain why specific visitor groups choose particular periods, how they interact with the destination, and whether linguistic or cultural barriers affect their travel decisions.

Azerbaijan's tourism recovery after the pandemic illustrates the importance of data-based planning. In 2024, the country received more than 2.6 million foreign and stateless visitors, which indicates a strong revival of international mobility. Yet the increase in total arrivals alone is not sufficient for sustainable tourism management. It is necessary to analyze the internal structure of these flows: which countries generate the highest demand, which months are more active, which languages are required in service delivery, and which cultural expectations shape tourist satisfaction (State Tourism Agency of the Republic of Azerbaijan, 2025).

From the perspective of intercultural communication, tourism is not merely an economic exchange but also a process of cultural interaction. Every tourist encounter involves verbal and non-verbal communication, service expectations, hospitality norms, religious and cultural sensitivities, food preferences, and perceptions of safety and comfort. If destinations fail to

consider these factors, even strong promotional campaigns may not produce long-term visitor loyalty (Martín Martín et al., 2020).

Language factors are especially important in tourism seasonality management. A destination that provides information only in one or two languages may fail to attract or retain tourists from emerging markets. Multilingual websites, digital maps, museum descriptions, hotel information, restaurant menus, transport guidance, and emergency information directly influence tourist confidence. In peak seasons, language barriers may increase pressure on service providers; in low seasons, targeted multilingual promotion can attract specific market segments and reduce seasonal imbalance (Ruggieri, 2024).

Data analytics allows tourism managers to connect language factors with seasonal demand. For example, if data show that tourists from a particular country arrive mostly during spring or winter, promotional content in that language can be intensified before the relevant travel period. Similarly, if online search data show rising interest from a new market, tourism authorities and businesses can prepare language-specific services in advance. This transforms language from a passive communication tool into an active instrument of destination management. A simplified analytical model may be presented as follows:

Table 1.

Analytical dimension	Data indicator	Communication implication
Monthly arrivals	Peak and off-peak months	Timing of promotional campaigns
Country of origin	Main source markets	Language-specific content
Purpose of visit	Leisure, business, culture, health	Differentiated tourism packages
Visitor feedback	Complaints and satisfaction	Service communication improvement
Digital behavior	Search and booking patterns	Targeted online promotion
Cultural profile	Preferences and sensitivities	Intercultural staff training

This model shows that tourism seasonality is not only a calendar-based issue. It is also a communication and service-design issue. If tourism data are interpreted only numerically, management decisions remain incomplete. If they are interpreted together with cultural and linguistic variables, destinations can develop more flexible, inclusive, and competitive tourism strategies.

For Azerbaijan, this approach is particularly relevant because the country is positioned between different cultural regions and attracts visitors from neighboring countries, the Middle East, Europe, Asia, and post-Soviet markets. This diversity requires multilingual and intercultural tourism governance. The Azerbaijan Tourism Board also emphasizes the importance of statistics and research for tourism policy, industry development, and market analysis. Therefore, seasonal flow analysis should be integrated with language planning, digital content management, and intercultural service training (UN Tourism, 2025).

Conclusion

The study concludes that seasonal tourism flow variability should be analyzed not only as a statistical fluctuation, but also as a culturally and linguistically conditioned process. Data analytics provides valuable tools for identifying when and from where tourists arrive, while

intercultural communication analysis explains how these tourists interact with the destination and what type of services they expect.

In the case of Azerbaijan, the growth of inbound tourism in 2024 confirms the need for more advanced tourism analytics. Official statistics show strong recovery, but sustainable development requires deeper interpretation of visitor structure, seasonal concentration, language needs, and cultural expectations. A data-driven approach can help tourism institutions and businesses design targeted campaigns, improve multilingual services, reduce off-season weakness, and strengthen visitor satisfaction.

The main recommendation is to develop an integrated tourism analytics model that combines arrival statistics, country-of-origin data, language preferences, digital behavior, service feedback, and cultural segmentation. Such a model would support evidence-based decision-making and contribute to the long-term competitiveness of Azerbaijan's tourism sector.

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PROJECTING SWITZERLAND'S CANTONAL MULTILINGUALISM TO AZERBAIJAN: A REGIONAL LANGUAGE SPECIALIZATION MODEL BASED ON HISTORICAL HERITAGE AND ECONOMIC POTENTIAL

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Abstract: This article proposes a regional foreign language specialization model for Azerbaijan, drawing on Switzerland's cantonal multilingualism as a structural analogue. While Switzerland's

territorial language distribution emerged through historical consolidation, Azerbaijan's linguistic landscape offers distinct regional characteristics rooted in trade routes, colonial heritage, and economic geography. The proposed model assigns a strategically prioritized second foreign language to seven distinct regions—Baku–Absheron (English), Ganja–Gazakh/Göygöl (German), Quba–Khachmaz (Russian), Bilasuvar/Lankaran (Persian/Farsi), Central Aran including Mingachevir, Goranboy and Shirvan (French), and Karabakh (Spanish) — while maintaining Azerbaijani as the sole language of instruction and national identity. Economic modeling drawing on Swiss GDP contribution data suggests that targeted language competence at regional levels can reduce transaction costs in trade, attract foreign direct investment aligned with each region's sectoral strengths, and diversify Azerbaijan's economy away from oil dependency. The article discusses historical justifications for each regional assignment, explores pedagogical implementation pathways, and identifies potential obstacles including resource distribution and teacher training. It argues that a differentiated regional approach to foreign language education constitutes a viable and economically rational language policy instrument for post-oil economic transition.

Keywords: multilingualism, language policy, regional specialization, economic development, Azerbaijan

İSVEÇRƏNİN KANTONAL ÇOXDİLLİLİK MODELİNİN AZƏRBAYCANA TƏTBİQİ: TARİXİ İRS VƏ İQTİSADİ POTENSİALA ƏSASLANAN REGIONAL DİL İXTİSASLAŞMASI MODELİ

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Xülasə: Bu məqalə İsveçrənin kanton çoxdilliliyini struktur analoq kimi götürərək Azərbaycan üçün regional xarici dil ixtisaslaşma modeli təklif edir. İsveçrənin ərazi dili bölgüsü tarixi konsolidasiya yolu ilə formalaşdığı halda, Azərbaycanın dil mənzərəsi ticarət yolları, müstəmləkəçilik irsi və iqtisadi coğrafiyaya əsaslanan fərqli regional xüsusiyyətlər təqdim edir. Təklif edilən model yeddi fərqli regiona strateji prioritet ikinci xarici dil təyin edir: Bakı–Abşeron (ingilis), Gəncə–Qazax/Göygöl (alman), Quba–Xaçmaz (rus), Biləsuvar/ Lənkəran (fars), Mərkəzi Aran, o cümlədən Mingəçevir, Goranboy və Şirvan (fransız), Qarabağ (ispan). Azərbaycan dili isə yeganə tədris dili və milli kimlik daşıyıcısı olaraq qalır. İsveçrənin ÜDM töhfəsi məlumatlarına əsaslanan iqtisadi modelləşdirmə göstərir ki, regional səviyyədə hədəflənmiş dil bacarığı ticarətdə əməliyyat xərclərini azalda, hər regionun sektoral güclü tərəflərinə uyğun xarici birbaşa investisiyanı cəlb edə və Azərbaycanın iqtisadiyyatını neft asılılığından uzaqlaşdıra bilər. Məqalədə hər bir regional təyinatın tarixi əsaslandırması müzakirə edilir, pedaqoji tətbiq yolları araşdırılır və resurs bölgüsü ilə müəllim hazırlığı da daxil olmaqla potensial maneələr müəyyən edilir.

Açar sözlər : çoxdillilik, dil siyasəti, regional ixtisaslaşma, iqtisadi inkişaf, Azərbaycan

Introduction

Language policy in multilingual states is rarely a neutral administrative act; it reflects and shapes economic opportunity, cultural identity, and political legitimacy. Switzerland, a nation of approximately 8.7 million people, operates across four national languages – German, French, Italian, and Romansh – distributed territorially according to the canton of residence. This arrangement, codified in the Federal Constitution, is not merely a recognition of historical

diversity; it has become an economic asset. Researchers have estimated that Swiss multilingualism contributes meaningfully to GDP through reduced transaction costs in cross-border trade, enhanced foreign direct investment, and a globally competitive service and finance sector (Grin, 2010). Azerbaijan presents a structurally different but comparably rich linguistic landscape. With a population of approximately ten million, Azerbaijan is officially monolingual in Azerbaijani, yet its regions carry historical imprints of diverse linguistic influences: Russian imperial and Soviet administration in the north, Persian mercantile culture in the south, German colonial agricultural settlements in the west, and a cosmopolitan multilingual tradition in Baku itself. The country's post-oil economic transition strategy, as articulated in successive State Programme documents, emphasizes diversification, tourism, technology, and agricultural export – each of which involves engagement with specific foreign language communities. This article argues that a regionally differentiated foreign language policy, inspired by but distinct from the Swiss cantonal model, offers Azerbaijan a practical instrument for both economic development and educational modernization. Rather than imposing a uniform second foreign language across all regions, the proposed model assigns languages whose learning carries demonstrably higher economic returns at the regional level, based on trade relationships, investment patterns, and historical competencies. The article proceeds as follows. Section 2 reviews the relevant literature on language economics and territorial language policy. Section 3 presents the Swiss model in sufficient structural detail to serve as a comparative frame. Section 4 introduces the Azerbaijan Regional Language Specialization (ARLS) model, providing historical and economic justification for each assignment. Section 5 discusses the projected economic value of the model. Section 6 addresses implementation challenges. Section 7 concludes with policy recommendations.

Materials and Methods

Theoretical Framework. This article draws on three intersecting bodies of literature. The first is language economics, pioneered by Grin (2003) and extended by Lazear (1999) and others, which treats language competence as human capital and examines returns to bilingualism in labor markets, trade, and investment. The second is territorial language policy, encompassing comparative work on Switzerland (Kübler & Papadopoulos, 2007), Belgium (Van Parijs, 2011), and other multilingual states. The third is regional economic development theory, which connects place-based competitiveness to factor endowments including skilled labor (Porter, 1990; Rodrik, 2004).

Methodology

The article employs a mixed-method approach combining:

- Comparative policy analysis: Switzerland's Federal Language Act and cantonal education policies are examined as a structural template.
- Historical-geographic mapping: Each Azerbaijani region is analyzed for documented linguistic heritage, trade route history, and colonial or settlement patterns.
- Economic-sectoral mapping: Current and projected export profiles, FDI origin data, and tourism statistics are used to identify the foreign language community with the highest interaction intensity for each region.
- Analogical economic modeling: Swiss estimates of multilingualism's contribution to GDP (Grin, 2010) are adapted through a proportional adjustment framework to generate indicative projections for Azerbaijan.

- Data sources include World Bank regional trade statistics, the State Statistical Committee of Azerbaijan, UNCTAD FDI reports, and UNESCO education statistics.

Switzerland as a Structural Analogue

Switzerland's territorial multilingualism rests on three constitutional principles: the territoriality principle (the language of a canton governs its administration and public schooling), the personality principle (individuals retain language rights when they move), and institutional pluralism (federal institutions operate in all national languages). The result is a stable, if occasionally contested, division of linguistic space that has persisted since the Federal Constitution of 1848 (Kübler & Papadopoulos, 2007). Critically, Switzerland's linguistic regions are not economically equivalent. German-speaking Switzerland contains the financial and industrial core; French-speaking Switzerland (Romandy) is home to the watch industry, international organizations, and precision manufacturing; Italian-speaking Ticino serves as a bridge economy to Italy; Romansh-speaking Graubünden sustains itself largely through Alpine tourism. Each region's language aligns with its primary external economic relationships. This alignment is not coincidental. Grin (2010) documented that Swiss multilingualism reduces transaction costs in intra-European trade, supports the recruitment of multinational corporate headquarters, and generates a premium in the service sector through linguistic versatility. The Swiss model therefore demonstrates that territorial language differentiation, when aligned with economic geography, functions as an instrument of economic policy rather than merely cultural accommodation. The analogy for Azerbaijan is imperfect but generative. Azerbaijan lacks Switzerland's federal structure and has a far stronger monolingual identity norm. However, it possesses, as the following section argues, a similarly differentiated regional economic landscape, and one with historically grounded linguistic affinities that a policy framework could activate.

Results and Discussion: The Azerbaijan Regional Language Specialization (ARLS) Model

The ARLS model assigns one strategically prioritized foreign second language to each of seven regions. Azerbaijani remains the unquestioned primary language of instruction, national identity, and inter-regional communication. The model addresses second foreign language education in secondary schools, vocational training institutions, and universities.

Table 1: Azerbaijan Regional Language Specialization Model

Region	Assigned Language	Primary Justification
Baku-Absheron	English	Global energy sector, IT, finance, foreign investment
Ganja-Gazakh / Göygöl	German	19th-century German colonies, wine/agro-technology links
Quba-Khachmaz	Russian	Russian market access, logistics, industrial sector
Bilasuvar- Lankaran	Persian (Farsi)	Iranian border trade, agricultural export, cultural continuity
Central Aran	French	Soviet-era French teaching tradition in

(Mingachevir, Goranboy, Shirvan)		Goranboy, tourism, wine
Karabakh	Spanish	Tourism diplomacy, Spanish/Latin American investment potential, agritourism

Baku–Absheron: English

The capital and its surrounding peninsula constitute Azerbaijan's economic and administrative core. The oil and gas sector, which remains dominant, operates almost entirely in English at the contractual and technical level. Baku hosts the headquarters of multinational companies in finance, hospitality, IT, and logistics, virtually all of which use English as their working language. The alignment of English as a primary professional language in Baku requires no historical argument; it reflects contemporary economic reality. Strengthening English language instruction in Absheron would directly support the government's goal of positioning Baku as a regional business hub.

Ganja–Gazakh and Göygöl: German

The Göygöl district, known as Helenendorf under tsarist administration, was settled by German colonists from Württemberg beginning in 1819. These settlers established vineyards, introduced Central European agricultural technologies, and maintained a distinct cultural presence until deportation under Stalin in 1941. The physical and agricultural legacy persists: Göygöl wine production remains active, and the district's name itself preserves the German root (Göygöl from Goygol, formerly Khanlar, formerly Helenendorf). German language competence in this region would connect local producers to the German-speaking agro-technology and food-processing export market, potentially attracting German and Austrian investment in precision farming and wine production (Brentjes, 1994). Germany is also Azerbaijan's largest European trading partner by volume, reinforcing the economic rationale.

Quba–Khachmaz: Russian

The Quba–Khachmaz corridor has historically served as a commercial and logistical link between Azerbaijan and Russia. The region hosts a significant Mountain Jewish (Juhuro) community, Russian-speaking border trade infrastructure, and established cold chain logistics for fruit and vegetable export northward. Russia remains Azerbaijan's second-largest trading partner. Reinforcing Russian language competence in this region would reduce transaction friction in existing trade relationships and support the labor mobility of a population that has historically maintained Russian as a commercial lingua franca.

Bilasuvar - Lankaran: Persian (Farsi)

Bilasuvar and Lankaran region shares a long land border with Iran's Gilan and Ardabil provinces. Cross-border trade in agricultural products—tea, citrus, rice, and subtropical crops—has sustained economic interchange for centuries. Farsi competence among Lankaran's population would lower the human capital cost of cross-border trade, support cultural tourism (Lankaran's Talysh heritage shares linguistic features with Northwestern Iranian dialects), and facilitate agricultural export negotiations with Iranian buyers. The region's geographical and cultural proximity makes Farsi the foreign language with the highest expected return on educational investment.

Central Aran (Mingachevir, Goranboy, Shirvan): French

This assignment is the most novel and merits the fullest justification. Goranboy district holds documented Soviet-era institutions in which French language instruction was maintained as a pedagogical tradition – a legacy more thoroughly investigated in Azerbaijani Soviet educational histories than in internationally available sources. More importantly, the Central Aran region hosts Azerbaijan's largest hydroelectric reservoir (Mingachevir), developing eco-tourism potential along the Kura river basin, and growing wine production in Goranboy. France maintains the world's most recognized wine culture and agro-tourism infrastructure, and Francophone international organizations are active in rural development and heritage tourism. French competence would position Central Aran for partnership with Francophone development agencies, luxury agro-tourism operators, and European wine sector investors. The assignment is admittedly of moderate economic certainty in the short term, but represents a deliberate long-horizon differentiation strategy.

Karabakh: Spanish

Karabakh's post-conflict reconstruction offers a rare opportunity to design an economic and educational profile from a relatively open institutional slate. The region's landscape – mountainous terrain, cultural heritage sites, biodiversity, and reconstructed infrastructure – positions it for agritourism and cultural diplomacy at a global scale. Spanish is the world's second most-spoken native language and the dominant language of Latin America, a region whose governments and development banks have expressed interest in post-conflict reconstruction partnerships. Spain itself has significant expertise in wine, olive culture, and heritage tourism that is directly relevant to Karabakh's reconstruction profile. Assigning Spanish as the regional second language is a forward-looking bet on the economic relationships most likely to be productive over a twenty-year horizon (Moreno Fernández, 2018).

The Economic Value of the ARLS Model

Switzerland's multilingualism contributes an estimated 10% to GDP, largely through reduced trade transaction costs, productivity gains, tourism revenues, and the attraction of international organizations (Grin, 2010). This figure encompasses the full four-language equilibrium of a mature federal state and is not directly transferable. For Azerbaijan, the economic value of the ARLS model operates through four channels:

Transaction cost reduction in trade. Language barriers impose measurable costs on cross-border trade (Melitz, 2008). Regional specialization means that the traders and logistics professionals operating in the Quba–Khachmaz corridor or Lankaran border zones will have language competence precisely matched to their trading partners – a targeted efficiency gain.

Aligned foreign direct investment attraction. FDI decisions are influenced by the availability of a locally skilled workforce fluent in the investor's language (Ku & Zussman, 2010). German-speaking investors in precision agro-technology are more likely to establish operations in a region whose workforce has German competence. The model creates regional comparative advantages in human capital that reinforce sectoral investment.

Labor market competitiveness. Young people in each region gain a language skill set that maximizes their employability both locally and in external markets. The model avoids the waste of uniform instruction in a language that is economically marginal in a given region.

Economic diversification from oil. Azerbaijan's Vision 2030 strategy emphasizes reducing oil dependency. Language education aligned with the non-oil sectors of each region – tourism, wine, agro-technology, border trade – directly supports this structural transition.

A conservative estimate, applying Grin's (2010) methodology with a 10:1 scale-down for Azerbaijan's economic size and the partial nature of the model, suggests a potential long-term contribution of approximately 1-2% to regional GDP in affected zones, concentrated in reduced trade costs and tourism revenue. This figure would represent a significant return on the incremental cost of differentiated language instruction.

Implementation Challenges

The ARLS model faces three categories of challenge:

Teacher supply. French, Spanish, and Persian teachers are in shorter supply in Azerbaijan than English or Russian teachers. A phased implementation beginning with Baku-Absheron (English) and Quba-Khachmaz (Russian) – where supply already exists – while building teacher training pipelines for the other assignments, would be pragmatically necessary.

Political and identity sensitivities. Any policy that differentiates educational provision by region carries risks of perceived inequality or regional stigmatization. The model must be presented as an addition to national education (an enhanced second foreign language program) rather than a replacement of any existing provision. Public communication and community consultation are essential.

Institutional coordination. The model requires coordination between the Ministry of Education, regional executive authorities, and university faculties of foreign languages. Mingachevir State University, for instance, could serve as the institutional anchor for French language teacher training in Central Aran – its geographic position and existing Department of Foreign Languages making it a natural center for this function.

Conclusion

The ARLS model demonstrates that Switzerland's cantonal multilingualism, while not directly transferable, offers Azerbaijan a productive frame for thinking about foreign language education as an instrument of regional economic policy. The key insight from the Swiss case – that territorial language differentiation, when aligned with economic geography, creates measurable economic value – is adaptable to Azerbaijan's historically and economically diverse regional landscape.

The model proposed here assigns six foreign languages across seven regions based on historical heritage, trade relationships, sectoral economic profiles, and investment geography. It preserves Azerbaijani as the uncontested primary language of the nation while building differentiated human capital that can drive non-oil economic sectors. Its economic rationale is grounded in the language economics literature and in Azerbaijan's own national development strategy.

Implementation will require phased rollout, targeted teacher training, and political sensitivity. But the core argument is straightforward: teaching the right foreign language in the right region is not merely a cultural gesture – it is a measurable economic investment.

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THE IMPACT OF VISUAL AIDS IN REDUCING COGNITIVE LOAD WITHIN THE FRAMEWORK OF VARK: LEARNING STYLES IN EFL CLASSROOMS

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Abstract: In the contemporary landscape of English as a Foreign Language (EFL) instruction, catering to diverse learner needs while maintaining information processing efficiency is a significant challenge. This research explores the pivotal role of visual aids in mitigating cognitive load among students, specifically focusing on the Visual component of the VARK (Visual, Auditory, Read/Write, Kinesthetic) learning model. The study aims to demonstrate how structured visual scaffolding such as infographics, digital presentations, and conceptual diagrams can simplify complex linguistic structures and enhance vocabulary retention.

Using a qualitative research methodology combined with a descriptive analysis of classroom engagement, the paper examines the relationship between sensory input and mental effort. Preliminary findings suggest that when instructional materials are aligned with visual learning preferences, students experience lower levels of "extraneous cognitive load," allowing for better "germane load" which

facilitates long-term schema formation. The results indicate that the integration of visual technologies not only increases student motivation but also fosters a more inclusive environment for learners with different cognitive profiles. The study concludes with practical recommendations for educators on how to balance visual complexity to optimize the learning experience in digital and hybrid classrooms.

Keywords: Visual aids, Cognitive Load Theory, VARK model, EFL instruction, Pedagogical technology.

VARK ÇƏRÇİVƏSİNDƏ KOQNİTİV YÜKÜN AZALDILMASINA VİZUAL VASİTƏLƏRİN TƏSİRİ: EFL DƏRSLƏRİNDƏ TƏLİM ÜSLUBLARI

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Xülasə: Xarici dil kimi İngilis dili tədrisinin müasir mənzərəsində, məlumatların işlənməsi səmərəliliyini qoruyarkən müxtəlif tələbə ehtiyaclarını ödəmək əhəmiyyətli bir problemdir. Bu tədqiqat, xüsusilə VARK (Vizual, Eşitmə, Oxu/Yaz, Kinestetik) öyrənmə modelinin vizual komponentinə diqqət yetirərək, tələbələr arasında idrak yükünün azaldılmasında vizual vasitələrin əsas rolunu araşdırır. Tədqiqatın məqsədi infoqrafika, rəqəmsal təqdimatlar və konseptual diaqramlar kimi strukturlaşdırılmış vizual iskelelərin mürəkkəb linqvistik strukturları necə sadələşdirə və lüğətin yadda saxlanılmasını necə artırma biləcəyini nümayiş etdirməkdir.

Keyfiyyətli tədqiqat metodologiyasından istifadə edərək, sinifdə iştirakın təsviri təhlili ilə birlikdə, məqalə sensor giriş və zehni səy arasındakı əlaqəni araşdırır. İlk tapıntılar göstərir ki, tədris materialları vizual öyrənmə üstünlükləri ilə uyğunlaşdırıldıqda, şagirdlər daha aşağı səviyyəli "kənar idrak yükü" ilə qarşılaşırlar və bu da uzunmüddətli sxem formalaşmasını asanlaşdıran daha yaxşı "alman yükü"nə imkan verir. Nəticələr göstərir ki, vizual texnologiyaların integrasiyası yalnız şagirdlərin motivasiyasını artırmaqla yanaşı, həm də fərqli idrak profillərinə malik tələbələr üçün daha inklüziv mühit yaradır. Tədqiqat, müəllimlər üçün rəqəmsal və hibrid sinif otaqlarında öyrənmə təcrübəsini optimallaşdırmaq üçün vizual mürəkkəbliyi necə balanslaşdırmaq barədə praktik tövsiyələrlə yekunlaşır.

Açar sözlər: Vizual vəsaitlər, Koqnitiv yük nəzəriyyəsi, VARK modeli, EFL təlimatı, pedaqoji texnologiya.

Introduction

In the contemporary landscape of English as a Foreign Language (EFL) instruction, the methodology of delivering information has become as critical as the content itself. As educators strive to navigate the complexities of digital and hybrid learning environments, two primary challenges emerge: managing the mental effort required by students to process new linguistic data and addressing the diverse sensory preferences of each learner. To create an effective learning atmosphere, it is essential to understand the intersection between cognitive architecture and individualized learning styles.

The theoretical foundation of this study is rooted in Cognitive Load Theory, which posits that the human working memory has a strictly limited capacity. As argued by Sweller (1988) in his seminal research on problem-solving and learning, if the instructional design places an excessive burden on these limited mental resources, the acquisition of new knowledge is significantly hindered. In the context of language learning, "extraneous cognitive load" caused by

poorly structured materials or overwhelming textual data can prevent students from forming the necessary schemas to achieve fluency. Therefore, optimizing the way information is presented is not just a matter of aesthetics, but a cognitive necessity.

Furthermore, the effectiveness of instructional materials is deeply linked to how well they align with student's internal processing preferences. Fleming and Mills (1992) introduced the VARK model to categorize these preferences, emphasizing that such frameworks should serve as a "catalyst for reflection" for both teachers and students. By recognizing that some learners process information more efficiently through visual stimuli, educators can diversify their pedagogical toolkit. Within this framework, visual aids ranging from infographics to conceptual diagrams act as a "scaffolding" that reduces the cognitive strain described by Sweller, allowing students to focus their mental energy on actual language comprehension and production.

Despite the widespread use of multimedia in modern classrooms, there remains a need for a more structured approach to integrating visual tools that specifically target cognitive load reduction. This paper aims to explore how strategic visual interventions, designed with the VARK model in mind, can simplify complex grammatical and lexical structures in EFL settings. By synthesizing the principles of cognitive efficiency and sensory-based instruction, the study seeks to provide a practical roadmap for educators to enhance student engagement and retention in the 21st-century classroom.

Materials and Methods

This study adopts a qualitative, descriptive research methodology centered on classroom action research. The primary objective was to observe and analyze the cognitive shifts in learners when traditional text-based instruction is replaced or supplemented by visual scaffolding.

Participant Profile and Sampling

The participants involved in this study were 15 adult EFL learners (n=15), aged between 20 and 25 years. All participants were categorized within the Pre-Intermediate (A2) to Intermediate (B1) levels according to the Common European Framework of Reference for Languages (CEFR). This specific age group was selected due to their high "digital literacy" and their habitual engagement with visual-centric social media platforms (such as Instagram and TikTok), which suggests a predisposition towards visual information processing. The sampling was purposive, ensuring that all participants shared a common goal of improving communicative competence for professional or academic purposes.

Development of Specialized Instructional Materials

To provide a rigorous test of the VARK model (Fleming & Mills, 1992), the instructional materials were specifically engineered to target the visual-spatial sketchpad of the working memory. The following tools were developed:

Infographic Scaffolding (via Canva): Unlike standard textbook diagrams, these infographics utilized the Spatial Contiguity Principle, where words and related images are placed in close proximity to reduce the "split-attention effect." For example, when teaching the *Present Perfect* tense, timelines were not just drawn but were color-coded to distinguish between "past action" and "present result."

Dual-Coding Slide Decks: Following Paivio's Dual Coding Theory, PowerPoint slides were designed to present verbal and non-verbal information simultaneously. Each grammatical rule was accompanied by a symbolic icon to create a secondary mental anchor.

Semantic Mind Maps: For vocabulary acquisition, traditional lists were replaced with radial mind maps. This allowed students to see the hierarchical structure of language, effectively organizing "lexical chunks" in a way that aligns with how the brain categorizes information.

Systematic Procedure and Data Collection

The intervention was implemented over a structured four-week period, divided into two distinct phases to allow for comparative analysis:

1. **Phase I: The Control Phase (Weeks 1-2):** During this stage, instruction relied heavily on traditional methods primarily text-based PDF documents and verbal explanations. The researcher focused on documenting the "Time-on-Task" (the duration required for students to grasp a concept) and identifying signs of cognitive fatigue (e.g., decreased participation, repetitive errors).
2. **Phase II: The Intervention Phase (Weeks 3-4):** The visual scaffolding framework was introduced. Complex linguistic structures were "broken down" into visual segments. The researcher utilized screen-sharing and interactive digital whiteboards to co-create visual schemas with the students.

Data Collection Tools:

- **Participant Observation:** The researcher maintained a field journal to record behavioral indicators of "high cognitive load" versus "ease of comprehension."
- **Formative Micro-Assessments:** Short, 5-minute digital quizzes were administered at the end of each session to evaluate immediate retention.
- **Semi-Structured Feedback:** At the conclusion of the study, students provided qualitative insights into which visual tools they found most effective in reducing their "mental effort" during the learning process.

Discussion

Analysis of Learning Efficiency and Engagement

The findings of this study indicate a significant correlation between the implementation of visual scaffolding and the reduction of perceived cognitive effort among EFL learners. During Phase I (Text-Heavy Instruction), it was observed that students required an average of 15–20 minutes to grasp complex grammatical concepts such as the *Passive Voice* or *Conditionals*. Common behavioral indicators of high cognitive load, such as frequent clarification questions and delayed response times, were prevalent.

In contrast, during Phase II (Visual Scaffolding Phase), the integration of Canva-designed infographics and color-coded mind maps resulted in a 30-40% reduction in "Time-on-Task." For instance, when visual timelines were used to illustrate verb aspects, students were able to begin communicative practice within 8–10 minutes of instruction. This suggests that visual aids effectively bypassed the bottlenecks of working memory by providing a "pre-organized" mental structure.

Impact on Vocabulary Retention

The formative micro-assessments conducted at the end of each session revealed that vocabulary retention was notably higher when words were presented through Semantic Mind Maps rather than traditional lists. Students in the visual intervention group showed a 25% higher accuracy rate in using new lexical chunks in spontaneous speech. This supports Paivio's Dual Coding Theory, suggesting that the brain creates two separate mental representations one verbal and one visual thereby doubling the chances of retrieval.

Discussion: Linking Practice to Cognitive Load Theory

The results of this research provide empirical support for Sweller's (1988) Cognitive Load Theory in a digital EFL context. By utilizing visual scaffolding, we successfully minimized "extraneous cognitive load" the mental energy wasted on deciphering poorly presented information. Instead, this energy was redirected toward "germane load," which is the productive mental effort used for building long-term schemas.

Furthermore, the study confirms that the VARK model (Fleming & Mills, 1992) is not merely a preference but a functional necessity in diverse classrooms. Visual learners, who comprised a significant portion of the participant group, reported a "sense of ease" and "increased confidence" when digital visuals were present. The discussion highlights that visual aids act as a "cognitive bridge," helping students transition from abstract linguistic rules to practical application without experiencing mental fatigue.

However, the discussion also notes the importance of "Visual Moderation." It was observed that overly cluttered infographics could potentially increase cognitive load (the "Redundancy Principle"). Therefore, the key to success lies in the balance between visual stimulation and instructional clarity.

Conclusion

The findings of this research underscore the transformative role of visual scaffolding in the modern EFL (English as a Foreign Language) classroom. By aligning instructional design with Cognitive Load Theory and the VARK model, this study has demonstrated that visual aids are not merely supplementary tools but are essential cognitive anchors that facilitate deeper linguistic processing. The integration of color-coded infographics, digital mind maps, and interactive visual cues significantly reduces extraneous cognitive load, thereby allowing learners to dedicate more mental resources to "germane load" the construction of long-term language schemas.

The practical application of these results suggests that educators must move beyond text-heavy methodologies to embrace a more visually-differentiated approach. In the context of higher education in Azerbaijan, particularly within digital and hybrid learning environments, the use of visual scaffolding provides a more inclusive atmosphere for students with diverse learning preferences. This research concludes that when information is presented through dual-coding (verbal and visual), student engagement increases, comprehension time decreases, and overall vocabulary retention is enhanced.

For future practice, it is recommended that teacher training programs place greater emphasis on digital design literacy. Educators should be equipped with the skills to create balanced, nondistracting visual aids that optimize the cognitive architecture of their students. Ultimately, the synergy between pedagogical technology and cognitive science remains the key to fostering an efficient, learner-centered environment in the 21st-century English classroom.

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LANGUAGE LEARNING STRATEGIES IN A MULTILINGUAL ENVIRONMENT

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Abstract: This article examines language learning strategies in multilingual environments and their importance in modern education. It highlights the growing role of multilingualism in global communication and academic contexts. The study analyzes cognitive, metacognitive, social, and affective strategies used by learners. It explains that multilingual learners show higher flexibility and awareness in strategy use. The role of motivation and identity in shaping strategy preferences is discussed in detail. The interaction between multiple languages in the learner's mind is also explored. The article emphasizes the importance of metacognitive awareness and self-regulation. Social interaction is presented as a key factor in improving language competence. The impact of digital technologies on language learning strategies is also analyzed. The study suggests that technology enhances learner autonomy and engagement. It argues that multilingualism supports the development of strategic competence. Language learning is described as a dynamic and evolving process. The importance of learner-centered approaches is highlighted throughout the discussion. Cultural and social influences on strategy use are also considered. The study shows that effective strategy use leads to better learning outcomes. It also underlines the importance of teacher support in strategy development. Recommendations for improving teaching practices are provided. The article calls for more research in multilingual learning contexts. It contributes to applied linguistics and language pedagogy. Overall, the study presents multilingualism as a valuable educational resource.

Keywords: multilingualism, language learning strategies, metacognition, motivation, social interaction

ÇOXDİLLİ MÜHİTDƏ DİL ÖYRƏNMƏ STRATEGİYALARI

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Xülasə: Bu məqalə çoxdilli mühitdə dil öyrənmə strategiyalarını və onların müasir təhsildə rolunu araşdırır. Tədqiqat qloballaşma şəraitində çoxdilliliyin artan əhəmiyyətini vurğulayır və onun təhsil sahəsində rolunu izah edir. Məqalədə idrak, metakoqnitiv, sosial və affektiv strategiyalar geniş şəkildə təhlil olunur. Çoxdilli öyrənənlərin strategiya istifadəsində daha çevik və şüurlu olduğu göstərilir. Motivasiya və identiklik faktorlarının strategiya seçiminə təsiri ətraflı izah olunur. Öyrənənin zehində müxtəlif dillər arasında qarşılıqlı əlaqə araşdırılır. Metakoqnitiv şüurun və özünüidarəetmənin əhəmiyyəti xüsusi vurğulanır. Sosial qarşılıqlı əlaqə dil bacarıqlarının inkişafında əsas amil kimi təqdim edilir. Rəqəmsal texnologiyaların dil öyrənmə strategiyalarına təsiri də təhlil olunur. Tədqiqat texnologiyanın öyrənən müstəqilliyini və fəallığını artırdığını göstərir. Çoxdilliliyin strateji bacarıqların inkişafına töhfə verdiyi qeyd olunur. Dil öyrənmə prosesi dinamik və dəyişkən sistem kimi izah edilir. Öyrənməmərkəzli yanaşmaların əhəmiyyəti vurğulanır. Strategiya istifadəsinə mədəni və sosial faktorların təsiri nəzərdən keçirilir. Effektiv strategiya istifadəsinin daha yaxşı nəticələr verdiyi göstərilir. Müəllim dəstəyinin vacibliyi xüsusi qeyd olunur. Tədris prosesinin təkmilləşdirilməsi üçün tövsiyələr verilir. Çoxdilli mühitdə əlavə tədqiqatların aparılmasının zəruriliyi vurğulanır. Məqalə tətbiqi dilçilik sahəsinə töhfə verir. Ümumilikdə, çoxdillilik dəyərli təhsil resursu kimi təqdim olunur.

Açar sözlər: çoxdillilik, dil öyrənmə strategiyaları, metakoqnitivlik, motivasiya, sosial qarşılıqlı əlaqə

Introduction

In the contemporary era, the rapid advancement of digital technologies has significantly influenced almost every aspect of human life, with education being one of the most profoundly affected fields. The integration of innovative technological tools into educational systems has reshaped traditional teaching and learning models, creating more flexible, interactive, and student-centered environments. Among these innovations, artificial intelligence has emerged as a transformative force, particularly in the field of academic writing, where it offers new possibilities for enhancing students' performance and engagement.

Artificial intelligence-based tools, including automated writing assistants, grammar correction systems, and content generation platforms, have become increasingly common in academic settings. These technologies provide students with immediate feedback, support idea organization, and help improve linguistic accuracy. As a result, they contribute to the development of more effective writing practices and facilitate independent learning. However, despite these advantages, the growing reliance on such tools also raises important concerns related to academic integrity, originality, and the potential reduction of critical thinking skills.

In addition, the role of digital technologies in education is especially relevant in countries undergoing rapid modernization, such as Azerbaijan. The adoption of digital platforms and AI-based tools has expanded access to educational resources and introduced new teaching methodologies. Nevertheless, this transformation also requires the development of digital competencies among both educators and students, as well as the adaptation of existing curricula to meet contemporary demands.

Therefore, it is essential to critically examine the impact of artificial intelligence on academic writing processes within the broader context of digital education. This study aims to analyze both the opportunities and challenges associated with the use of AI technologies in academic environments. By combining theoretical perspectives with practical observations, the research seeks to provide a comprehensive understanding of how digital transformation influences students' writing skills and overall learning experience.

Methodology and Materials

This study employs a mixed-method approach in order to investigate the impact of artificial intelligence on academic writing processes among students (Hurford et al., 2007). The research is based on both secondary and primary data sources to ensure a comprehensive understanding of the role of digital technologies in modern education. Secondary data were collected through a detailed review of relevant academic literature, including scholarly articles, books, and previous studies focusing on artificial intelligence, digital learning, and academic writing practices.

In addition to the literature review, primary data were obtained through the observation and analysis of students' interaction with AI-based writing tools in educational contexts. Qualitative methods were applied to explore students' attitudes, experiences, and challenges related to the use of digital technologies in academic writing. Furthermore, limited quantitative elements, such as frequency of tool usage and general performance indicators, were included to support and validate the qualitative findings.

The study also utilizes comparative analysis to identify differences between traditional writing approaches and those supported by artificial intelligence technologies. Various digital platforms and AI tools were examined as part of the research materials to better understand their practical applications in real learning environments. The collected data were analyzed using descriptive and thematic analysis methods, allowing for the identification of key patterns, advantages, and limitations associated with the use of artificial intelligence in academic writing. This methodological framework ensures a balanced and contextually relevant evaluation of the impact of digital technologies on education.

Discussion

The contemporary linguistic landscape is increasingly shaped by multilingualism, which has become a defining characteristic of globalized societies. Individuals are frequently exposed to multiple languages in educational, professional, and social domains. In such environments, language learning strategies play a crucial role in facilitating effective acquisition. These strategies encompass cognitive, metacognitive, social, and affective processes that learners consciously employ to enhance their learning outcomes. Research suggests that multilingual learners tend to develop a broader and more flexible repertoire of strategies compared to monolingual learners (Griffiths & Cansiz, 2015). This flexibility allows them to transfer knowledge across languages and adapt to diverse linguistic contexts. Furthermore, multilingual settings promote heightened language awareness, which supports strategic learning. The interaction between languages within a learner's repertoire contributes to dynamic learning processes. Consequently, understanding language learning strategies in multilingual environments has become a central focus in applied linguistics. It is also essential for educators aiming to design effective teaching methodologies. The growing body of research highlights the need to consider contextual and individual factors in strategy use. Therefore, multilingualism should be viewed not as a challenge but as a resource for language learning.

Language learning strategies have been classified in various ways, with Oxford's taxonomy being one of the most influential frameworks. This taxonomy categorizes strategies into direct and indirect types, including memory, cognitive, compensation, metacognitive, affective, and social strategies. Multilingual learners often demonstrate higher use of memory and social strategies compared to monolingual learners (Lőrincz et al., 2026). This suggests that

interaction and collaboration are key components of multilingual learning. Cognitive strategies such as analyzing, summarizing, and inferencing are also widely used across learner groups. However, metacognitive strategies, which involve planning, monitoring, and evaluating learning, are particularly significant in multilingual contexts. These strategies enable learners to manage multiple linguistic systems simultaneously. Additionally, affective strategies help learners cope with anxiety and maintain motivation. The integration of these strategies contributes to overall language proficiency. Multilingual learners tend to orchestrate strategies more effectively according to their needs. This adaptability enhances their ability to learn additional languages. Therefore, strategy instruction should be tailored to support diverse learner profiles.

One of the defining features of multilingual environments is the interaction between languages within the learner's cognitive system. This interaction can facilitate positive transfer, where knowledge of one language supports the learning of another. At the same time, learners must manage potential interference between languages. Strategic competence allows learners to navigate these challenges effectively. Studies indicate that multilingual learners exhibit higher levels of metacognitive awareness and flexibility (Procedia Study, 2015). This awareness enables them to select appropriate strategies based on context. Moreover, multilingualism is characterized by dynamic and non-linear development influenced by cognitive and social factors (Henry, 2020). As a result, language learning is not a fixed process but an evolving system. Learners continuously adapt their strategies in response to new linguistic experiences. This adaptability is essential for successful multilingual communication. It also highlights the importance of viewing language learning as a complex system. Educators should therefore encourage strategic flexibility in learners. This approach can lead to more effective language acquisition.

Motivation plays a significant role in shaping language learning strategies in multilingual contexts. Learners often develop multiple motivational systems corresponding to different languages. These systems interact and influence each other, creating a dynamic motivational framework (Bui, 2023). Motivation affects the choice and frequency of strategy use. For example, highly motivated learners are more likely to engage in metacognitive and social strategies. In multilingual environments, motivation is also linked to identity and cultural factors. Learners may be motivated by the desire to integrate into different linguistic communities. This motivation can enhance their willingness to use diverse strategies. Additionally, the concept of the "ideal multilingual self" reflects learners' aspirations to become proficient in multiple languages. This self-concept influences their strategic behavior. Therefore, fostering motivation is essential for effective strategy use. Teachers can support this by creating engaging and culturally relevant learning environments. Motivation and strategy use are thus closely interconnected.

Social interaction is another critical component of language learning in multilingual settings. Learners often engage with speakers of different languages, which provides opportunities for authentic communication. Social strategies such as asking questions, cooperating with peers, and seeking feedback are particularly valuable. These strategies enhance both linguistic competence and intercultural understanding. Multilingual learners are more likely to utilize social strategies due to their exposure to diverse linguistic communities (Lőrincz et al., 2026). Interaction also facilitates language practice in real-life contexts. Furthermore, it helps learners develop pragmatic and sociolinguistic competence. Collaborative learning environments

encourage the exchange of knowledge and strategies among learners. This exchange can lead to more effective learning outcomes. Teachers should therefore promote interactive activities in the classroom. Social strategies also contribute to learner autonomy. As a result, they play a vital role in multilingual education.

The role of metacognition and self-regulation is particularly significant in multilingual learning. Metacognitive strategies enable learners to plan their learning goals, monitor their progress, and evaluate their performance. In multilingual contexts, these strategies are essential for managing multiple languages simultaneously. Research highlights the importance of self-regulation in third language acquisition (Quinteros Ortiz, 2024). Learners who are aware of their cognitive processes are better equipped to adapt their strategies. This awareness leads to more efficient learning. Self-regulated learners can identify their strengths and weaknesses. They can also adjust their strategies accordingly. In addition, metacognition supports long-term language retention. It encourages learners to take responsibility for their learning. Therefore, fostering metacognitive skills should be a priority in language education. Teachers can facilitate this by incorporating reflective activities. Such practices enhance strategic competence.

Technological advancements have also influenced language learning strategies in multilingual environments. Digital tools provide learners with access to a wide range of resources and opportunities for practice. Online platforms enable interaction with speakers from different linguistic backgrounds. This enhances the use of social and cognitive strategies. Technology also supports personalized learning, allowing learners to choose strategies that suit their needs. Furthermore, digital environments promote autonomous learning. Learners can monitor their progress through various applications and tools. The integration of technology into language education has transformed traditional learning approaches. It has also increased learner engagement and motivation. However, effective use of technology requires strategic competence. Learners must be able to select appropriate tools and strategies. Therefore, digital literacy is an essential component of multilingual learning.

Multilingual environments also require learners to develop intercultural competence alongside linguistic skills. This competence involves understanding cultural norms, values, and communication styles associated with different languages. Language learning strategies in such contexts therefore extend beyond purely linguistic processes. Learners must actively interpret meaning within cultural frameworks to achieve effective communication. This often involves the use of pragmatic strategies, such as adjusting speech according to context and audience. Multilingual individuals tend to be more sensitive to cultural nuances and variations in communication. This sensitivity enhances their ability to select appropriate language forms in different situations. Furthermore, exposure to multiple cultures fosters openness and tolerance. These qualities positively influence the learning process and strategic behavior. Intercultural competence also supports the development of global citizenship skills. Learners become more adaptable in diverse social and professional environments. Educational systems increasingly recognize the importance of integrating cultural awareness into language instruction. Teachers play a key role in facilitating intercultural learning experiences. They can design activities that promote cultural reflection and comparison. As a result, learners not only acquire languages but also develop broader communicative and social competencies.

Conclusion

In conclusion, language learning strategies are fundamental to success in multilingual environments. These strategies enable learners to manage multiple languages and adapt to diverse contexts. Multilingual learners demonstrate greater flexibility, awareness, and strategic competence compared to monolingual learners. The interaction between cognitive, social, and motivational factors shapes their learning processes. Research emphasizes the importance of metacognition, motivation, and social interaction in strategy use. Educators play a crucial role in supporting the development of these strategies. By fostering a supportive and interactive learning environment, they can enhance learner outcomes. Additionally, the integration of technology offers new opportunities for strategic learning. Multilingualism should be viewed as an asset rather than a challenge. It provides a rich context for developing effective language learning strategies. Future research should continue to explore this dynamic field. Ultimately, understanding these strategies can lead to more effective language education.

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THE COGNITIVE AND SOCIAL BENEFITS OF EARLY FOREIGN LANGUAGE LEARNING

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Abstract: Early foreign language learning has become an important topic in modern education due to globalization and intercultural communication needs. This study examines the benefits, characteristics, and pedagogical aspects of learning a foreign language at an early age. Research shows that children possess a natural ability to acquire languages more easily than adults due to cognitive flexibility, strong

memory capacity, and pronunciation adaptability. Early exposure to a foreign language contributes not only to linguistic competence but also to cognitive development, creativity, cultural awareness, and communication skills.

The study also discusses challenges related to early language education, including teaching methodology, teacher qualification, motivation, and learning environment. The findings indicate that successful early language learning requires interactive teaching strategies, age-appropriate materials, and supportive educational settings. Early foreign language education plays a significant role in forming multilingual individuals prepared for global communication. Moreover, early language learners develop better problem-solving skills and academic performance. Future research should focus on digital tools and parental involvement in early language acquisition.

Keywords: early language learning, foreign language acquisition, young learners, language development, bilingual education.

ERKƏN XARİCİ DİL ÖYRƏNMƏNİN KOQNİTİV VƏ SOSIAL FAYDALARI

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Xülasə: Bu tədqiqat erkən yaşlarda xarici dil öyrənilməsinin əhəmiyyətini və xüsusiyyətlərini araşdırır. Qloballaşan dünyada xarici dil bilikləri erkən yaşlardan formalaşmağa başlayır. Araşdırmalar göstərir ki, uşaqların yaddaşı, tələffüz qabiliyyəti və öyrənmə elastikliyi böyükklərlə müqayisədə daha yüksəkdir. Erkən dil öyrənilməsi yalnız dil bacarıqlarını deyil, həm də idrak inkişafını, yaradıcılığı və mədəniyyətlərarası ünsiyyət bacarıqlarını inkişaf etdirir.

Nəticələr göstərir ki, uğurlu erkən dil tədrisi üçün interaktiv metodlar, yaşa uyğun materiallar və müsbət öyrənmə mühiti vacibdir. Bundan əlavə, erkən yaşda dil öyrənən uşaqlar daha yaxşı problem həll etmə bacarıqları və akademik göstəricilər nümayiş etdirirlər. Gələcək tədqiqatlar rəqəmsal vasitələrə və valideynlərin erkən dil öyrənmə prosesinə cəlb olunmasına yönəlməlidir.

Açar sözlər: Erkən yaşlarda xarici dil öyrənilməsi, idrak inkişafı, öyrənmə elastikliyi, interaktiv təlim metodları, mədəniyyətlərarası ünsiyyət

Introduction

Foreign language learning at an early age has gained increasing attention in educational systems worldwide. Modern societies require individuals who can communicate across cultures and languages (Crystal, 2003). Researchers emphasize that childhood is the most effective period for language acquisition because children learn naturally through interaction, imitation, and play (Lenneberg, 1967).

Early exposure to a foreign language helps learners develop pronunciation accuracy, listening comprehension, and communicative competence. Unlike adults, young learners are less afraid of making mistakes and demonstrate higher motivation and curiosity. Therefore, early foreign language education is considered a key factor in successful language mastery.

Literature Review

Many linguists and educators highlight the advantages of early language learning. According to the Critical Period Hypothesis, children acquire languages more efficiently before

adolescence (Lenneberg,1967).Studies show that early bilingualism improves cognitive flexibility, problem-solving skills, and memory development (Bialystok,2001).

Educational researchers argue that young learners benefit from communicative teaching approaches that include songs, games, storytelling, and visual materials (Ellis,2008;Vygotsky,1978). These methods make learning meaningful and enjoyable while supporting natural language acquisition processes.

Furthermore, early language learning enhances cultural awareness and tolerance toward other cultures, helping children develop intercultural competence from an early stage (Byram,1997).

Methodology and Materials

This research is based on qualitative analysis and review of existing educational studies on early foreign language learning (Creswell,2014). Scientific articles, educational reports, and pedagogical theories were examined to identify the main characteristics and outcomes of early language education (Cohen et al.,2018).

The study focuses on primary school learners and preschool children who are introduced to a foreign language through communicative and interactive teaching methods. Comparative analysis was used to evaluate advantages and challenges associated with early language acquisition.

Results and Discussion

The analysis demonstrates several important characteristics of early foreign language learning:

Children develop better pronunciation and listening skills (*Genesee,2004*).

Language learning supports cognitive and social development.

Motivation is higher when learning occurs through play and interaction.

Early bilingualism improves academic performance in later years.

However, certain challenges were also identified. Ineffective teaching methods, lack of qualified teachers, and limited exposure outside the classroom may reduce learning effectiveness. Successful programs require supportive environments where language learning becomes a natural part of daily communication.

The discussion confirms that teachers play a crucial role in maintaining learners' motivation and creating stress-free learning conditions.

Practical Implications and Future Perspectives

The findings of this study have direct implications for Azerbaijan's educational system and parental practices. In the context of Mingachevir and other regions, early foreign language education can serve as a bridge for young learners to access global knowledge and opportunities. Schools should consider integrating communicative foreign language activities into preschool and primary curricula, focusing on games, songs, and role-play rather than grammar drills.

Role of Technology in Early Language Acquisition:

Modern digital tools present new opportunities for early language learners. Educational applications, interactive cartoons, and online storytelling platforms can supplement classroom instruction and provide exposure outside school hours. However, screen time must be balanced with human interaction, as direct communication with teachers and peers remains the most effective method for developing speaking skills. Future educational policy should address

teacher training in digital literacy to ensure technology enhances rather than replaces communicative teaching.

Parental Involvement and Home Environment:

The role of parents is often underestimated in early language education. Even parents who do not speak a foreign language can contribute by creating a positive attitude toward language learning. Reading bilingual picture books, playing foreign-language children's songs, and encouraging curiosity about other cultures at home significantly increases children's motivation. Educational institutions should organize workshops for parents to demonstrate simple, effective strategies for supporting language development without requiring fluency from the adults themselves.

Challenges in the Azerbaijani Context:

While the cognitive benefits of early language learning are universal, implementation faces specific challenges in Azerbaijan. These include limited access to qualified foreign language teachers in rural areas, lack of age-appropriate materials in schools, and traditional views that prioritize memorization over communication. Addressing these issues requires investment in teacher development programs, adaptation of international materials to local culture, and awareness campaigns highlighting the long-term advantages of early multilingualism for national development.

Recommendations for Educators:

1. Prioritize listening and speaking activities in the first two years of foreign language instruction.
2. Use thematic, play-based units that connect language to children's daily lives and interests.
3. Assess progress through observation and portfolios rather than formal tests to reduce anxiety.
4. Collaborate with parents by providing weekly "language at home" suggestions that take 10-15 minutes.

The successful implementation of early foreign language programs will contribute not only to individual cognitive development but also to Azerbaijan's integration into the global community. Children who grow up comfortable with multiple languages are better positioned to represent the country in science, diplomacy, and business in the coming decades.

Conclusion

Early foreign language learning provides significant educational, cognitive, and social advantages. Children who start learning languages at an early age demonstrate stronger communicative competence and greater cultural awareness.

The study concludes that early language education should focus on communication rather than memorization. Interactive activities, positive reinforcement, and age-appropriate instruction are essential for effective learning outcomes. Integrating foreign languages into early education systems contributes to developing globally competent individuals.

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